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PRINCIPLES OF PLASMA PHYSICS

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PREFACE

This textbook was developed to provide graduate students in physics and astronomy at the University of Maryland with a comprehensive background in plasma physics. It was designed for a two-semester course, normally taken in the second year of graduate study by students who have completed the usual sequence of graduate courses in classical and statistical mechanics, electromagnetism, and mathematical methods. It is a basic introduction to the field and assumes no background in plasma physics. Although it was intended as a graduate text, it could be used for an advanced undergraduate course by proper selection of material.

The basic plan of the book is to begin with the exact statistical description of a many-body system and then to obtain various reduced descriptions of the plasma state. The most reduced description, called fluid theory, or macroscopic theory, is developed first and is used to explore a wide range of plasma phenomena and problems involving equilibrium, waves, and instabilities. Macroscopic theory, together with transport phenomena in plasmas (Chapter 6), are typically covered in the first semester of the course.

Next, the book takes up a less reduced description, the Vlasov or microscopic theory, based on a continuous distribution function of velocity and configuration space, and again considers problems of plasma equilibrium, waves, and instabilities, in addition to nonlinear interactions of waves and the plasma distribution. This "Vlasov-Maxwell" description of the plasma state predicts phenomena that depend on the details of the velocity-space distribution function, including, for example, Landau damping, velocity-space instabilities, velocity-space diffusion, etc. The Vlasov results for these plasma properties are compared with the results of the somewhat simpler fluid treatment of the earlier chapters.

Correlations, fluctuations, and radiation, being phenomena that depend on discreteness, are not included in either of the above reduced descriptions. A test-particle approach is used to calculate these quantities. Further discrete

properties of plasmas are obtained from higher-order reduced descriptions of the exact statistical equations, leading to the Fokker-Planck or Balescu-Lenard equations and calculations of transport coefficients for plasmas.

The theory of the orbits of single-charged particles in static and time-varying electric and magnetic fields is included as an appendix, consistent with our objectives of trying to be complete, and at the same time not devote extensive text space to treatment of topics that are simple and adequately treated in many readily available reference sources.

One unique feature of this book is an extensive introductory chapter intended to eliminate the "jargon gap" that we found to be a major stumbling block for students first encountering plasma physics. Most of our students have an adequate background in mathematics and classical physics, but many of the concepts and the terminology associated with plasma physics are unfamiliar to them, creating an artificial difficulty.

This introductory chapter is really a short course (three to four lectures) in the concepts and terminology of plasma physics. As such it is useful to anyone who desires a qualitative introduction to the field. Another unique feature is a chapter on the equilibrium statistical mechanics of plasmas. The purpose of this chapter is to provide the student with the proper perspective of plasma physics as a many-body problem and the plasma state as a statistical system. This chapter calculates the thermodynamic properties of the equilibrium plasma, and sets the stage for the rest of the book, which deals with the many realistic situations in which the plasma is not in a state of thermodynamic equilibrium.

With regard to references, we selected primarily books and review papers that we found of particular value to students. It is our opinion that a proper grasp of the physics is the overriding concern, to which the historical aspects of the field are subordinate.

Recent advances in the use of digital computers to simulate a plasma have provided considerable insight into plasma processes. Some simulation results are included where they amplify or extend a particular analytical result, or contribute to a better understanding of plasma properties.

There are many problems in each chapter, varying in difficulty from routine algebraic manipulations leading to well-known results to challenging problems that will help the student interested in research in plasma physics obtain a better feeling for this diverse and intriguing field.

Gaussian-cgs units are used throughout this book, since their use is widespread in the plasma physics research literature. A conversion table from Gaussian-cgs units to standard practical units for quantities such as resistivity, capacitance, electric field strength, etc., is included in Appendix III, along with a list of the most frequently occurring symbols (such as plasma frequency, cyclotron frequency, etc.).

A project as extensive as this text naturally involves the contribution of a large number of people.

We thank our friends and colleagues in plasma physics with whom we have worked and learned about this subject. We thank the students who took the plasma physics course at Maryland during the period this book was being written. Their conscientious help and advice in the development of the course and the textbook was very valuable to us. We are particularly grateful to Ted Northrop, Derek Tidman, and Allan Kaufman for their early suggestions and advice on how to develop and teach a graduate course in plasma physics. We benefited greatly from their previous experience in teaching plasma physics. We appreciate the advice, encouragement, and support of our colleagues at Maryland, particularly Hans Griem, Ron Davidson, and Bob Pechacek. We are grateful to our Department Chairman, Howard Laster, for allowing us to concentrate our efforts on teaching and developing the plasma physics course on which this book is based. We appreciate the careful reading of one or more chapters of the manuscript by Ron Davidson, Alan DeSilva, Seishi Hamasaki, John Hey, Walt Jones, Chris Kapetanakos, Hank Klein, Paulett Liewer, Marvin Schwartz, Don Spero, Derek Tidman, and Maria Zales-Caponi. Our special thanks go to Mrs. Mary Ann Ferg and Mrs. Clara Rodriguez for their excellent work in typing the various drafts of the manuscript, and to Mrs. Barbara Hornady for preparing the Indexes.

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Nicholas A. Krall
Alvin W. Trivelpiece

1

INTRODUCTION TO PLASMA PHYSICS

Plasma physics is the study of charged particles collected in sufficient number so that the long-range Coulomb force is a factor in determining their statistical properties, yet low enough in density so that the force due to a near-neighbor particle is much less than the long-range Coulomb force exerted by the many distant particles. It is the study of low-density ionized gases. The term "plasma" was first used to describe a collection of charged particles by Tonks and Langmuir,[1] in 1929, in their studies of oscillations in electric discharges. However, the most characteristic aspect of the plasma state, the fact that because of the long range of the Coulomb force the charged particles exhibit a collective behavior, was known much earlier, and was probably first described by Lord Rayleigh,[2] in 1906, in his analysis of electron oscillations in the Thomson model of the atom.

The term "fourth state of matter," often used to describe the plasma state, was coined by W. Crookes[3] in 1879 to describe the ionized medium created in a gas discharge. The term fourth state of matter follows from the idea that as heat is added to a solid, it undergoes a phase transition to a new state, usually liquid.

[1] L. Tonks and I. Langmuir, Oscillations in Ionized Gases, *Phys. Rev.*, **33**:195 (1929).
[2] Lord Rayleigh, *Phil. Mag.*, **11**:117 (1906).
[3] W. Crookes, *Phil. Trans.*, **1**:135 (1879).

If heat is added to a liquid, it undergoes a phase transition to the gaseous state. The addition of still more energy to the gas results in the ionization of some of the atoms. At a temperature above 100,000°K most matter exists in an ionized state; this ionized state of matter is called the *fourth state*. A plasma state can exist at temperatures lower than 100,000°K provided there is a mechanism for ionizing the gas, and if the density is low enough so that recombination is not rapid.

Although 99.9 percent of the apparent universe exists in a plasma state, there is very little in the way of natural plasma here on earth because the low temperature and high density of the earth and its near atmosphere preclude the existence of plasma. This means that plasma must be created by experimental means to study its properties. However, in the upper atmosphere (ionosphere), plasma does exist, created by photoionization of the tenuous atmosphere. Farther out from the earth, plasma is trapped in the earth's magnetic field in the near vacuum of space. Plasma streams toward the earth from the sun (the solar wind), and fills many regions of interstellar space, forming the medium through which outer space is viewed.

Plasma physics generally involves the well-known physics of classical mechanics, electromagnetism, and nonrelativistic statistical mechanics. The challenge of plasma physics comes from the fact that many plasma properties result from the long-range Coulomb interaction, and therefore are collective properties that involve many particles interacting simultaneously.

In its simplest form, a plasma is a collection of protons and electrons at sufficiently low density so that binary (short-range) interactions are negligible. Many-body theory, or the many-body problem, is the study of the properties of such a medium. When a collection of protons and electrons coexist in an equilibrium state, the properties of this state are described by equilibrium statistical mechanics with the appropriate Gibbs ensemble. However, most of the interesting features of plasmas occur for nonequilibrium situations.

Revived interest in plasma physics in the United States began in 1952 with the attempts of a program, then classified, known as Project Sherwood,[1] to develop a controlled thermonuclear fusion reactor. Similar programs were started in England, France, and the U.S.S.R. at about the same time. These programs have grown substantially since that time, and now there are many nations with major research programs in this field. Although the development of a controlled fusion reactor is one of the more challenging practical applications of plasma physics, it is only one of the many areas in which plasma physics plays a role. Plasma physics has played a major role in the development of much

[1] A. S. Bishop, "Project Sherwood," Addison-Wesley, Reading, Mass., 1958.

of contemporary physics, and it is important in the study of problems in such areas as astrophysics, atomic physics, chemistry, life sciences, molecular physics, magnetohydrodynamic power generation, and atmospheric physics.

Plasma physics has its own vocabulary and set of ideas. The main purpose of this chapter is to review plasma physics on an elementary level, provide a background sketch of the familiar concepts of the field, identify many of the terms used repeatedly in discussing the plasma state, review some of the schemes by which plasma is produced in the laboratory, and review some of the methods by which plasma properties are measured.

PART ONE: PLASMA CONCEPTS AND TERMINOLOGY

1.1 EQUILIBRIUM AND METAEQUILIBRIUM

The term "equilibrium" is often loosely used in plasma physics to describe a quasi-steady-state condition that persists only until the plasma particles collide with each other. Frequently, plasma studies are made by investigating small perturbations about such a *metaequilibrium* state.

Thermodynamic equilibrium means that the ions and the electrons are each described by a maxwellian distribution characterized by the same single parameter, the temperature. In this situation, the medium is in equilibrium with its surroundings, and it radiates and absorbs energy at the same rate. The spectrum of emitted radiation is *blackbody*.

In many of the theoretical and experimental situations of interest in plasma physics, the ions and electrons are neither at the same temperature nor in thermodynamic equilibrium with their surroundings. The term metaequilibrium is used to describe situations that eventually will be altered by binary collisions.

1.2 DEBYE LENGTH

The electrostatic potential of an isolated particle of charge q is

$$\phi = \frac{q}{r} \qquad (1.2.1)$$

In a plasma, electrons are attracted to the vicinity of an ion and shield its electrostatic field from the rest of the plasma. Similarly, an electron at rest repels other electrons and attracts ions. This effect alters the potential in the vicinity of a charged particle. The potential of a charge at rest in a plasma is given by

$$\phi = \frac{q}{r} e^{-r/\lambda_D} \qquad (1.2.2)$$

where λ_D is the Debye length originally defined in the Debye-Hückel theory of electrolytes. For an electron-proton plasma

$$\lambda_D = \left(\frac{\kappa T}{8\pi n e^2}\right)^{1/2} = 4.9\left(\frac{T}{n}\right)^{1/2} \qquad (1.2.3)$$

where n = density of electrons (or ions), cm^{-3}
T = temperature, °K
κ = Boltzmann's constant ($= 1.38 \times 10^{-16}$ ergs/°K)

The Debye length is a measure of the sphere of influence of a given test charge in a plasma. In general, the Debye length depends on the speed of the test charge with respect to the plasma.

1.3 PLASMA PARAMETER

The plasma parameter g indicates the number of plasma particles in a Debye sphere, and is defined by

$$g = \frac{1}{n\lambda_D{}^3} \qquad (1.3.1)$$

For Debye shielding to occur, and for the description of a plasma to be statistically meaningful, the number of particles in a Debye sphere must be large; that is, $g \ll 1$. The assumption $g \ll 1$ is called the *plasma approximation*. The plasma parameter is also a measure of the ratio of the mean interparticle potential energy to the mean plasma kinetic energy. An ideal gas corresponds to *zero* potential energy between the particles. In many situations the plasma parameter is small and the plasma is treated as an ideal gas of charged particles, that is, a gas that can have a charge density and electric field but in which no two *discrete* particles interact.

To ensure that $n\lambda_D{}^3$ be large, the density must be *low*, since

$$g = \frac{1}{n\lambda_D{}^3} \propto \frac{n^{1/2}}{T^{3/2}}$$

Because the collision frequency decreases with density n, and also decreases with increasing temperature T, the condition $g \to 0$ corresponds to a decreasing collision frequency.

Problem 1.3.1 Show that, if ratio of mean kinetic energy to mean interparticle potential energy is much greater than 1, that is, if

$$\frac{\langle KE \rangle}{\langle PE \rangle} \gg 1$$

the number of particles per Debye sphere $n\lambda_D{}^3$ must also be much larger than 1. ////

The plasma parameter g is one of the more important dimensionless parameters associated with a plasma, and may be interpreted as a measure of the degree to which plasma or collective effects dominate over single-particle behavior. The plasma state is described by equations obtained from an expansion of the exact many-body equations in powers of g.

A distinctive constrast between the statistical mechanics of a plasma and that of a neutral gas is that in a plasma the expansion parameter g is small when *many* particles interact at the same time, since $\lambda_D{}^3$ is essentially the volume of the interaction region; for a neutral gas, the atomic radius R is a measure of the interaction region and nR^3 ($\ll 1$) is the expansion parameter. The plasma behaves as a nearly ideal gas *in spite of* the presence of many interacting particles; the reason is that the strength of the interaction between individual particles is so weak, as shown in Prob. 1.3.1.

1.4 DISTRIBUTION FUNCTION

The most detailed description of a plasma gives the location and velocity of each plasma particle as a function of time. It is impossible to obtain such a description of a real plasma, except in some recent "experiments" which involve the use of digital computers to follow the position and velocity of a large number of ions and electrons. Therefore it is customary to use the distribution function f to describe a plasma. The distribution function is the number of particles per unit volume in six-dimensional velocity-configuration phase space. From the Boltzmann H theorem[1] it is known that under the action of binary collisions an ideal gas relaxes to a maxwellian distribution of velocities.

$$\bar{n}f(v) = \bar{n}\left(\frac{m}{2\pi\kappa T}\right)^{3/2} e^{-mv^2/2\kappa T} \qquad (1.4.1)$$

where $\bar{n} = N/V$, with N the number of particles of a certain type (e.g., ions or electrons) in the system and V the volume of the system. Although laboratory plasmas probably never achieve exactly a maxwellian distribution, they may approach it closely, and it is useful in many theoretical treatments to assume that the plasma is described by a maxwellian velocity distribution.

[1] S. Chapman and T. G. Cowling, "The Mathematical Theory of Non-uniform Gases," Cambridge, London, 1952.

Because ions and electrons behave rather differently on a time scale less than ion-electron relaxation times (a binary collision process), it is usually necessary to define a separate distribution function for each charge species α, so that

$$\bar{n}_\alpha f_\alpha(\mathbf{x}, \mathbf{v})\, d\mathbf{x}\, d\mathbf{v} = \text{number of particles of type } \alpha \text{ in a volume element } d\mathbf{x}\, d\mathbf{v} \text{ centered about } \mathbf{x}, \mathbf{v}$$

with the normalization

$$\int \bar{n}_\alpha f_\alpha\, d\mathbf{x}\, d\mathbf{v} = N_\alpha = \text{total number of particles of type } \alpha \text{ in the system}$$

1.5 TEMPERATURE AND OTHER MOMENTS OF THE DISTRIBUTION FUNCTION

In the maxwellian distribution, the parameter T, which defines the distribution function, is the temperature of the plasma.

$$\tfrac{3}{2} n\kappa T = \frac{1}{2}\int \bar{n} m \mathbf{v}^2 f\, d\mathbf{v} \qquad (1.5.1)$$

Even in cases where laboratory plasmas are not accurately described by a maxwellian, it is customary to relate the experimental distribution function to the maxwellian distribution which best fits the data, by describing the system as a plasma at temperature T, with T *defined* by (1.5.1). If the distribution function has no relation to a maxwellian distribution, it is poorly described by merely specifying the temperature. Instead, a complete description must either specify the distribution function itself or specify *all* the moments of the distribution.

$$n_\alpha = \bar{n}_\alpha \int f_\alpha\, d\mathbf{v} \qquad \text{density} \qquad (1.5.2)$$

$$\mathbf{V}_\alpha = \frac{\int \mathbf{v} f_\alpha\, d\mathbf{v}}{\int f_\alpha\, d\mathbf{v}} \qquad \text{mean velocity} \qquad (1.5.3)$$

$$\mathsf{P}_\alpha = \frac{n_\alpha m_\alpha \int (\mathbf{v} - \mathbf{V}_\alpha)(\mathbf{v} - \mathbf{V}_\alpha) f_\alpha\, d\mathbf{v}}{\int f_\alpha\, d\mathbf{v}} \qquad \text{pressure tensor} \qquad (1.5.4.)$$

$$\vdots$$

These moments are associated with the macroscopic parameters of the system such as the density, mean velocity, pressure, etc. Note that for an isotropic system at rest, the pressure tensor reduces to a scalar,

$$\mathsf{P}_\alpha = \begin{bmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{bmatrix}$$

where $p = n_\alpha \kappa T_\alpha$.

1.6 MAGNETIC PRESSURE

Frequently plasma is imbedded in, or surrounded by, a magnetic field. In a static magnetic field there is a pressure force $B^2/8\pi$ dynes/cm^2 across a surface tangent to the magnetic flux surface, at a point. Characteristically, a plasma is diamagnetic, and thus it excludes a magnetic field suddenly applied to its boundary. This means that a magnetic pressure force $B^2/8\pi$ can balance the kinetic pressure $p = n\kappa T$ of the plasma at a boundary between plasma and magnetic field. If $B^2/8\pi > p$, the plasma may be compressed to a higher density and temperature.

In situations where plasma and magnetic field coexist, a frequently used parameter is the local ratio of plasma pressure to magnetic pressure, given the name *plasma beta*; i.e.,

$$\beta = \frac{n_i \kappa T_i + n_e \kappa T_e}{B^2/8\pi} \qquad (1.6.1)$$

This β should not be confused with the relativistic $\beta = v/c$. At a plane plasma magnetic field interface in equilibrium, pressure balance requires

$$\nabla\left(n\kappa T + \frac{B^2}{8\pi}\right) = 0 \qquad (1.6.2)$$

Problem 1.6.1 What is the magnetic pressure in atmospheres of a 100,000-G magnetic field? What magnetic field strength is required to confine a plasma of density $n = 10^{12}$ cm^{-3} at a kinetic temperature of 100 keV, if $\beta = 1.0$ percent? ////

1.7 PARTICLE DRIFTS

Most of the interesting plasma properties result from collective effects, as previously mentioned. However, some plasma properties can be obtained by considering the motion of individual charged particles in various combinations of electric, magnetic, and gravitational fields, which may be time-varying and inhomogeneous. In a uniform magnetic field in the absence of collisions, a charged particle moves on a helical trajectory. The particle rotates at the cyclotron frequency $\omega_c = eB/mc$ perpendicular to the magnetic field with a

radius of rotation $a_L = v_\perp/\omega_c$, called the *Larmor*[1] *radius*, or the *gyro radius*. Its motion along the magnetic field is unimpeded. The cyclotron frequency is given in radians per second. For example,

$$\omega_c = \frac{eB}{mc} = 1.76 \times 10^7 B \text{ (Gauss)} \qquad \text{rad/s, for electrons} \qquad (1.7.1)$$

Problem 1.7.1 Find the orbit of a particle of charge q and mass m started from rest in a crossed electric and magnetic field. Write the orbit in terms of a cyclotron orbit and a drift, and show that the drift is $V_{DE} = cE/B$. Show that one half of this particle's kinetic energy is in drift motion and one half is orbital energy. ////

This cyclotron rotation is the basic motion of a charged particle perpendicular to a magnetic field, but there are departures from this motion when other fields are involved. For instance, an $\mathbf{E} \times \mathbf{B}$ drift occurs when a steady electric and magnetic field both act on a charged particle. The combined action of these two fields causes, in addition to the cyclotron rotation, a drift motion perpendicular to both the electric and magnetic fields, at a velocity given by

$$\mathbf{V}_{DE} = c\frac{\mathbf{E} \times \mathbf{B}}{B^2} \qquad (1.7.2)$$

This drift motion is independent of either the mass or charge of the particle involved, and is superposed on the cyclotron rotation. Frequently the cyclotron motion is much more rapid than the drift motion, and the motion is well described by specifying the orbit of the *guiding center*,[2] defined as the center of the particle's circular orbit. Thus $\mathbf{V}_{DE}$ is referred to as a drift velocity of the guiding center.

If instead of an electric field acting on the particle there is a gravitational field, the drift velocity is

$$\mathbf{V}_{DG} = \frac{mc}{q}\frac{\mathbf{g} \times \mathbf{B}}{B^2} \qquad (1.7.3)$$

Another drift motion that is important in plasma problems is the gradient B drift that occurs when the magnetic field is inhomogeneous. The gradient drift is

[1] This radius of gyration should be called the "cyclotron radius," to be consistent with conventional terminology; however, it is referred to as the Larmor radius in most of the contemporary plasma physics literature.

[2] The concept of a guiding center for charged-particle motions was introduced in 1940 by H. Alfvén, who used the guiding-center motions in a theory of magnetic storms and aurora. He collected his works and ideas in "Cosmical Electrodynamics," Oxford, New York, 1950, and in 1970 he was awarded the Nobel prize in physics for his pioneering work in plasma physics.

also superposed on the cyclotron rotation. For weak gradients ($\nabla B/B < 1/a_L$), the drift speed of the guiding center (perpendicular to $\mathbf{B}$ and to ∇B) is

$$V_{D,\text{grad}} = \frac{w_\perp c}{qB^2}\nabla_\perp B \qquad (1.7.4)$$

where $w_\perp$ ($=\frac{1}{2}mv_\perp^2$) is the kinetic energy perpendicular to $\mathbf{B}$. The details of particle-orbit theory are given in Appendix I, where curvature drift and other drift motions are treated.

1.8 PLASMA FREQUENCY

Because of the long-range forces between plasma particles, a plasma behaves in some situations as a system of coupled oscillators. One basic characteristic oscillator frequency of the plasma state is the plasma frequency ω_p, defined as

$$\omega_p = 2\pi f_p = \left(\frac{4\pi ne^2}{m}\right)^{1/2} \qquad \text{rad/s}$$

$$f_p \approx 10^4\sqrt{n} \qquad \text{Hz, for electrons} \qquad (1.8.1)$$

where n is the number of particles per cubic centimeter, and m is the electron mass. It is customary to call ω_p the plasma "frequency," even though it has units of radians per second.

The plasma frequency is often used as a means of specifying the electron density in a plasma. It is also a measure of the length of time required for an electron or ion moving with thermal speed to travel a Debye length.

If a group of electrons in a two-component plasma are displaced slightly from their equilibrium position $\mathbf{x}_0$, they will experience a force that seeks to return them to $\mathbf{x}_0$. When they arrive at the equilibrium position, they will have a kinetic energy equal to the potential energy of their initial displacement, and will continue past $\mathbf{x}_0$ until they have reconverted their kinetic energy back to potential energy. The frequency of this simple period harmonic motion will be at ω_p, the plasma frequency. This phenomenon is known as a *plasma oscillation*.

1.9 WAVES IN PLASMAS

There are a large number of possible oscillations in a system of coupled oscillators with as many degrees of freedom as a plasma, and wavelike disturbances, e.g., electric fields $\mathbf{E} = \mathbf{E}_0 \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$, will propagate in a plasma. The frequency ω and the wave number k are functionally related to one another by a

dispersion relation obtained from the plasma equations. A knowledge of the dispersion characteristics $\omega(k)$ for the propagating waves is certainly necessary for an understanding of the plasma state. The phase velocity of a wave is $v_p = \omega/k$, and the group velocity of a wave is $v_g = \partial\omega/\partial k$.

A plasma can propagate both linear and nonlinear waves. *Linear* refers to the simplifying approximations that are possible for small-amplitude waves, and *nonlinear* refers to shock waves and large-amplitude phenomena not predicted by linear models.

Four examples of linear waves (ion-sound waves, electromagnetic waves, drift waves, and plasma waves) are discussed below.

1.9.1 Ion-sound Waves

Sound waves in an ideal gas are dispersionless longitudinal waves that propagate below the gas collision frequency with speed $V_s = (\gamma\kappa T/m)^{1/2}$. A plasma also propagates a low-frequency acoustic, or *ion-sound wave* by a mechanism in which the perturbation charge-separation electric field provides the coupling between the electrons and the ions. These waves propagate with speed

$$C_s = \left(\frac{Z\gamma_e \kappa T_e + \gamma_i \kappa T_i}{m_i}\right)^{1/2} \qquad (1.9.1)$$

The symbols γ_e and γ_i denote the usual ratio of specific heats, and Z is the ionic charge expressed in electron units. These waves are strongly damped (unlike sound waves in an ordinary gas) unless the electron temperature is much higher than the ion temperature and the wave frequency is much higher than the collision frequency. In this latter case these longitudinal waves propagate as acoustic waves. The electron pressure provides the restoring force, and the ion mass provides the inertial effect.

1.9.2 Electromagnetic Waves

Homogeneous plane electromagnetic waves propagate through a dielectric medium with dispersion given by

$$k^2 = \frac{\omega^2 \varepsilon}{c^2} \qquad (1.9.2)$$

For a cold homogeneous isotropic plasma of mobile electrons and immobile ions, the high-frequency dielectric constant is

$$\varepsilon = 1 - \frac{\omega_p{}^2}{\omega^2} \qquad (1.9.3)$$

Problem 1.9.1 Derive the dielectric constant as given in (1.9.3) for a plasma of cold electrons and stationary ions by computing the field arising from a displacement of electrons. ////

The corresponding dispersion relation for electromagnetic waves in a plasma is

$$k^2 = \frac{\omega^2 - \omega_p{}^2}{c^2} \qquad (1.9.4)$$

If the wave frequency ω is less than the plasma frequency ω_p, the wave number is imaginary and the waves are evanescent. Above the plasma frequency ($\omega > \omega_p$), waves propagate, and at very high frequencies, the free electrons of the plasma only slightly influence the electromagnetic wave. Effects such as finite size, a steady magnetic field, or plasma inhomogeneities modify this picture considerably.

1.9.3 Waves in Magnetized or Nonuniform Plasmas

In a homogeneous steady magnetic field many additional wave motions are possible. For example, a slow dispersionless electromagnetic wave or magnethydrodynamic wave, called an *Alfvén wave*, propagates through the plasma at frequencies below the ion-cyclotron frequency with speed

$$V_A = \frac{B}{\sqrt{4\pi\rho_m}} \qquad (1.9.5)$$

where $\rho_m = nm_i$ is the mass density.

As a second example, drift waves propagate in inhomogeneous plasmas as a result of particle drifts and plasma currents associated with plasma density gradients. This extra freedom introduced by the density gradient allows a new wave motion in which

$$\omega \approx \frac{\kappa T}{m}\frac{1}{\omega_c}\frac{\nabla n_0}{n_0}k \qquad (1.9.6)$$

where κ is the Boltzmann constant, and T is the temperature.

1.9.4 Plasma Waves

These waves are also known as *space-charge waves*, *electrostatic waves*, or *Langmuir waves*. They propagate only if there is a distribution of electron velocities or if the electrons have an average velocity in the observer's frame.

They are the previously described plasma oscillations, at frequency ω_p, aroused by an initially displaced clump of charge. The dispersion relation for these waves is $\omega = \omega_r + i\omega_i$.

$$\omega_r^2 = \omega_p^2 + \frac{3\kappa T}{m} k^2$$
$$\omega_i = -\sqrt{\frac{\pi}{8}} \frac{\omega_p}{(k\lambda_{De})^3} \exp[-\tfrac{1}{2}(k\lambda_{De})^{-2} - \tfrac{3}{2}] \tag{1.9.7}$$

The finite temperature causes the medium to be both dispersive and absorptive. The dispersion is evidenced by the first of the two equations. The absorption is evidenced by the second equation, and vanishes at $T \to 0$. Equation (1.9.7) is valid only when the absorption is weak, $\omega_i \ll \omega_r$; $\lambda_{De} = (kT_e/4\pi ne^2)^{1/2}$.

1.10 LANDAU DAMPING

A new physical effect is introduced into the theory of plasma waves by the addition of finite temperatures, indicated by the presence of the imaginary part of the frequency ω_i in (1.9.7). This is *Landau damping*, or collisionless damping. This means that a plasma disturbance is damped as it propagates away from its point of origination, even though there are no short-range binary collisions in the plasma. This phenomenon is made plausible by observing that in a thermal distribution such as is shown in Fig. 1.10.1, at any particular wave speed ($v = \omega/k$), the number of plasma electrons moving slightly slower than the wave is greater than the number moving slightly faster than the wave. Electrons moving faster than the wave can give energy to it, while those moving slower than the wave can take energy from it. In a monotonically decreasing distribution function, there always are more electrons taking energy from the wave than those giving energy to it; hence the wave damps.

1.11 PLASMA STABILITY AND CONTROLLED THERMONUCLEAR FUSION

The argument that is used to make plausible Landau damping has a more dramatic consequence for nonthermal distributions (e.g., Fig. 1.11.1). A wave moving with speed $\omega/k = v_0$ sees more electrons moving slightly faster than the wave than it sees moving slightly slower than the wave. Consequently, the wave should grow, taking more energy from the faster ones than it gives to the slower ones.

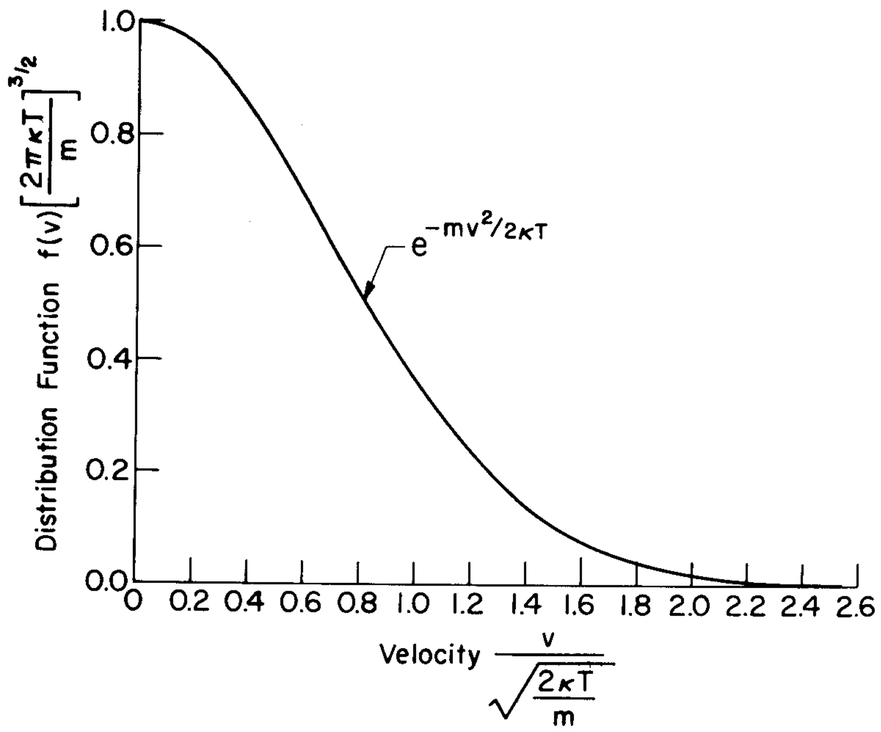


FIGURE 1.10.1
Maxwell-Boltzmann distribution function for an isotropic plasma.

This growth of Langmuir waves for a nonmaxwellian distribution such as shown in Fig. 1.11.1 is an example of the plasma phenomenon called *instability*. An instability is a process in which a small perturbation in a plasma equilibrium state grows larger with time. In a general sense, an instability represents the ability of a plasma to relax from a nonthermal state by collective processes in a time much less than the binary collision time. In a practical sense, instability often represents the ability of a plasma to escape from a configuration of fields that would otherwise contain a single charged particle indefinitely.

The objective of much plasma research has been to devise configurations for the containment of a high-temperature plasma at a density and for a period of time sufficient for nuclear fusion reactions to take place, liberating more energy than was required to create and confine the plasma. Instabilities tend to disassemble a confined plasma. Thus plasma-stability considerations began with some of the early confinement experiments. Without plasma instabilities, it is possible that the controlled-fusion problem might have been a short-term developmental problem. For example, a large electric current passed through an ionized mixture of deuterium and tritium creates a large magnetic field which exerts a magnetic pressure that causes the discharge column to collapse, or

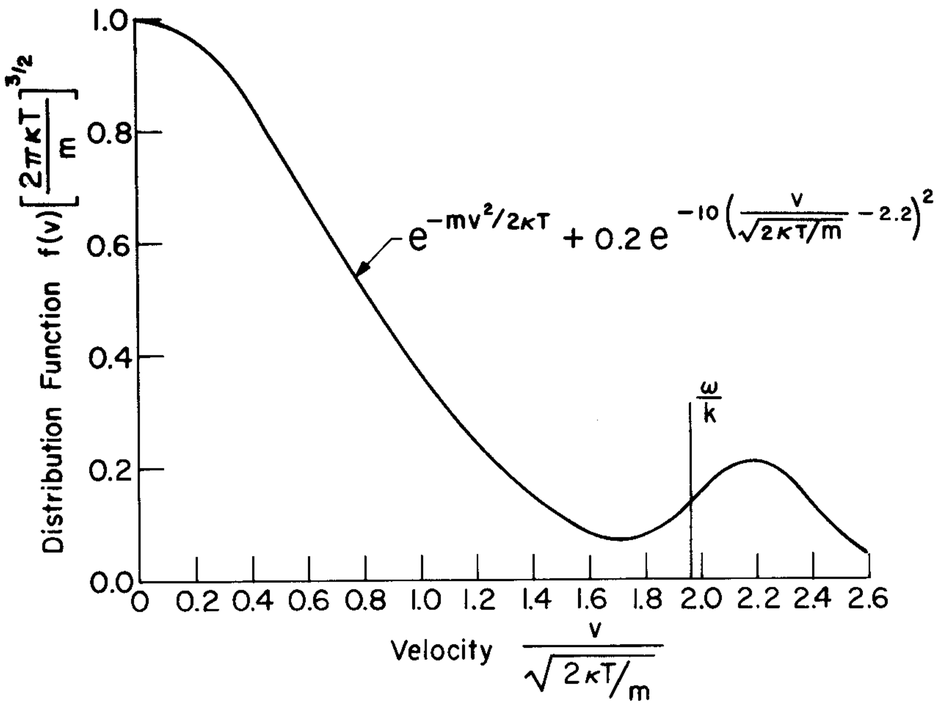


FIGURE 1.11.1
Distribution function of a plasma with a nonthermal feature: a drifting beam of electrons in the tail of a Maxwell-Boltzmann distribution.

pinch, thus increasing the temperature and density to the level required for thermonuclear reactions.

However, early experiments showed that the plasma had a means by which it could escape from the magnetic field faster than the binary collision rates would suggest, and indeed faster than the rate at which the compression process was increasing its temperature. This demonstrated that the magnetic field–plasma configuration described is inherently unstable, with a very short time constant. Stability theory thus is a central feature of plasma research.

During the past decade a better understanding of the mechanisms that cause instability and of the means to suppress these instabilities has been achieved. Dr. Edward Teller has described the attempts to contain a plasma as somewhat similar to attempts to contain a ball of jello in a basket of stretched rubber bands. The jello would rather be on the outside, with the rubber bands on the inside. This represents a lower energy state. So it is with a plasma localized in space by a magnetic field configuration. The plasma and magnetic field would be in a lower potential-energy state if the plasma were on the outside and the magnetic field were on the inside. One classification scheme for plasma instabilities is based on the fact that plasma may be unstable if it is unfavorably localized

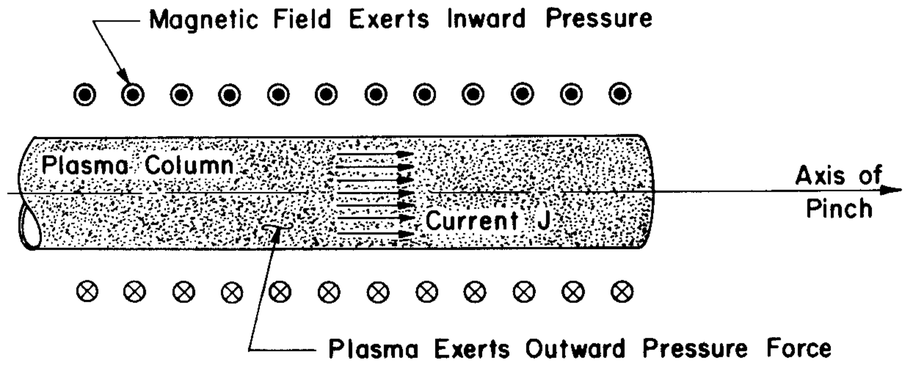


FIGURE 1.11.2
Schematic of a plasma pinch. If inward magnetic pressure exceeds outward particle pressure, the plasma column "pinches" to a smaller radius.

in physical space or in velocity space. Each type of localization has many interesting modes of instability.

1.11.1 Configuration-space Instabilities

An example of a configuration-space instability occurs in a plasma pinch, which is depicted schematically in Fig. 1.11.2. A plasma pinch consists of a radially bounded current carrying plasma surrounded by a magnetic field that balances the plasma pressure and holds in the plasma. This configuration is unstable to various types of perturbations that deform the plasma–magnetic field interface. Such perturbations or deformations of the interface grow exponentially with time, and in an experiment the plasma is soon transported to the walls. This final state is a lower potential-energy state than the state for which the plasma is

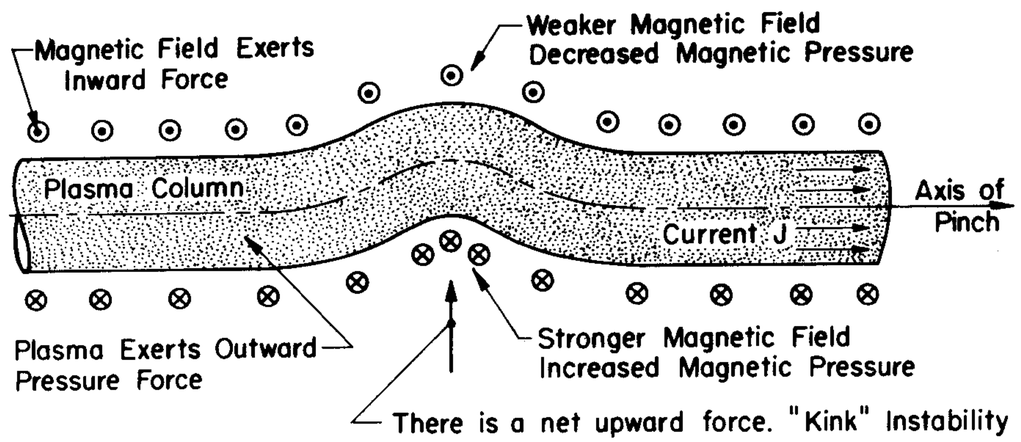


FIGURE 1.11.3
Schematic of a plasma pinch showing a kink instability.

localized. Figure 1.11.3 illustrates a "kink" instability. Since the reservoir of energy that derives the instabilities comes from spatial localization, these instabilities are called *configuration-space instabilities.*

1.11.2 Velocity-space Instabilities

One velocity-space instability was described at the start of this section; it was the unstable analog of Landau damping. Another simple velocity-space instability occurs when a beam of monoenergetic electrons is injected into a cold plasma. The dielectric constant for the higher-frequency response for an electron gas neutralized by a uniform density of background ions was given by (1.9.3).

The beam-plasma instability is analyzed by first modifying the high-frequency dielectric constant for a plasma to include the two types of electrons, electrons at rest and electrons moving with an average drift speed V. The dielectric constant for the moving electrons is obtained by a doppler shift $\omega \to \omega - kV$. The dielectric constant for the combined system of moving and stationary electrons becomes

$$\varepsilon = 1 - \frac{{\omega_{p1}}^2}{(\omega - kV)^2} - \frac{{\omega_{p2}}^2}{\omega^2} \qquad (1.11.1)$$

where ω_{p1} is plasma frequency for the drifting beam, and ω_{p2} is the plasma frequency for the stationary component of the plasma. The algebraic equation obtained by setting the dielectric constant equal to zero is the dispersion relation for longitudinal electrostatic waves. Solving this expression for real ω, letting k be complex, gives for the imaginary part of the spatial growth constant

$$k_i = \frac{\omega_{p1}}{V} \frac{1}{\sqrt{{\omega_{p2}}^2/\omega^2 - 1}} \qquad (1.11.2)$$

Since disturbances grow with distance as $e^{k_i z}$, large growth of signals or noise in a short physical distance might be expected for frequencies slightly less than the plasma frequency. Of course, such disturbances cannot develop more energy than originally present in the nonthermal feature of the distribution, in this case $W_B = \frac{1}{2} n_1 m V^2$, and in any event, the linearized plasma equations used to derive the dispersion relation are not valid at large wave amplitudes.

1.12 SHOCK WAVES AND SOLITARY WAVES

In plasmas, as in other gases, large-amplitude waves can propagate at speeds above the speed of sound. These waves steepen as they propagate and form

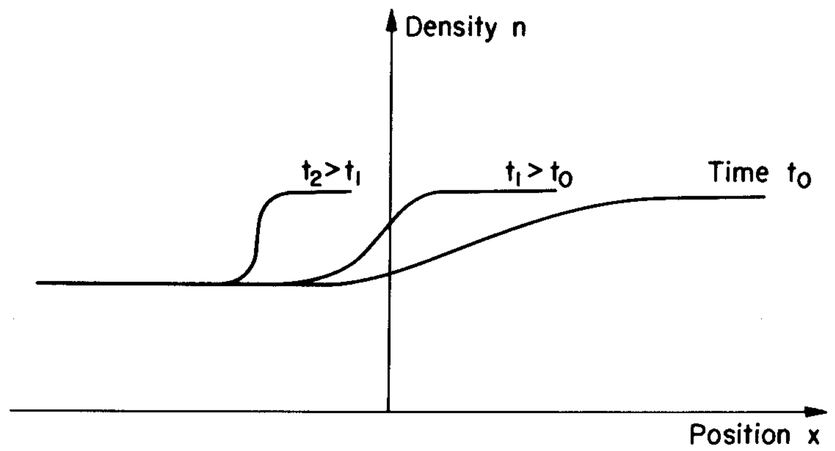


FIGURE 1.12.1
Schematic of a shock wave moving from right to left and steepening as a function of time.

shocks, i.e., thin layers, propagating through the system, separating regimes of different density and temperature (Fig. 1.12.1). In neutral gas there is a unique sound speed, the steepening process is balanced by diffusion across the layer, and binary collisions provide the mechanism for the change of state across the transition layer. In the plasma there are various "sound waves," and any of these can form the basis of a plasma shock wave. In a plasma, the steepening process can be balanced by diffusion, but it can also be balanced by dispersion: shorter wavelengths, needed to build a thin front, go more slowly and fall behind as, for example, if

$$\frac{\omega}{k} = V(1 - k^2 a^2)$$

for the sound wave $p = p_0 \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$ on which the shock is based.

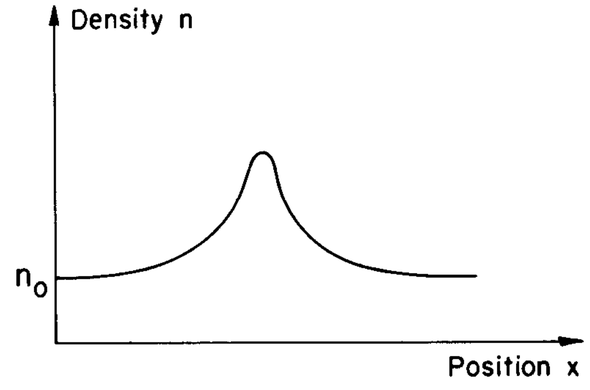


FIGURE 1.12.2
A solitary wave in a plasma.

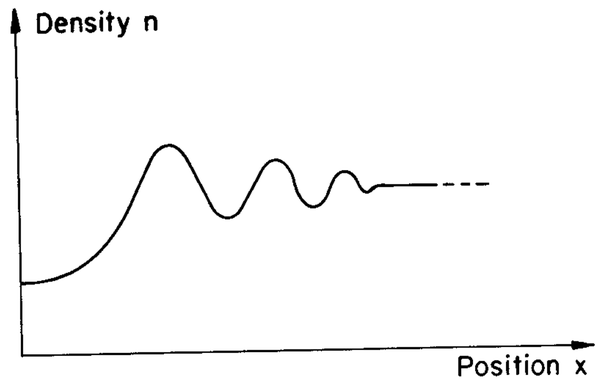


FIGURE 1.12.3
Collisionless shock wave in a plasma.

A class of waves in which steepening is balanced by dispersion, called *solitary waves*, propagate as isolated finite-amplitude disturbances in plasma (Fig. 1.12.2). A change of state across a solitary wave can be provided by binary collisions, by Landau damping, by instabilities, or by trapping and reflection of particles by the waves, which thus convert the solitary-wave structure to a shock-like structure (Fig. 1.12.3).

1.13 COLLISIONS

In a plasma, both elastic and inelastic collisions occur among plasma particles. In a partially ionized gas, for example, there are electron-electron collisions, electron-ion collisions, electron-neutral collisions, etc. Collision processes are conveniently described by the concept of a cross section. However, care must be exercised, since some of the simple cross sections are modified by the plasma. For example, in the case of an electron or proton impinging on an electron or proton, the cross section (known as the *Rutherford cross section*) is logarithmically divergent at large impact parameters, if the particles are free. This divergence is a result of the long-range nature of the Coulomb force. In a plasma the logarithmic divergence is avoided because the potential around a particular charge falls off as $1/r$ only within a Debye length, and then it decreases exponentially with radius [Eq. (1.2.2)]. It is customary to replace the upper limit of the total-cross-section integral with the Debye length: the cross section is then known as the *Coulomb cross section*. It is still larger than the cross sections for nuclear reactions[1] (that is, $d + d \rightarrow p + t$, etc., which are of interest in problems

[1] S. Glasstone and R. H. Lovberg, "Controlled Thermonuclear Reactions," Van Nostrand, New York, 1960.

of controlled thermonuclear fusion reactors). The Coulomb cross section for 90° scattering due to many cumulative small-angle scattering events is, approximately,

$$\sigma_{90^\circ} \approx \frac{2.6 \times 10^{-18}}{W_{\text{keV}}{}^2} \quad \text{cm}^2 \qquad (1.13.1)$$

This is much larger than the cross section for fusion of two deuterons, that is,

$$\sigma_{d+d} = 10^{-26}\ \text{cm}^2 \qquad \text{for } W = 100\ \text{keV}$$

The fact that the reaction cross section is so much smaller than the scattering cross section means that plasma particles will escape from a confinement configuration before they undergo nuclear fusion reactions. The long-range nature of the Coulomb force is such that the cumulative effect of many small-angle scatterings of a plasma particle produces a net particle deflection of 90° long before a single large-angle scattering produces such a deflection.

Useful concepts related to the cross section are the mean free path and the collision frequency. If the collision cross section is σ, an incoming particle travels a distance

$$l_{mfp} = \frac{1}{n\sigma} \qquad (1.13.2)$$

known as the *mean free path*, before it suffers a collision. If the incoming particle were traveling with speed v, relative to the target particle, it would travel one mean free path in a time τ_c. The reciprocal of this time interval is the collision frequency

$$\nu_c = n\sigma v \qquad (1.13.3)$$

In most cases the cross section is a function of the velocity v, and the velocities of the target particles are thermally distributed. The effective collision frequency is the average value of the product of cross section and velocity; i.e.,

$$\nu_c = n\langle\sigma(v)v\rangle \qquad (1.13.4)$$

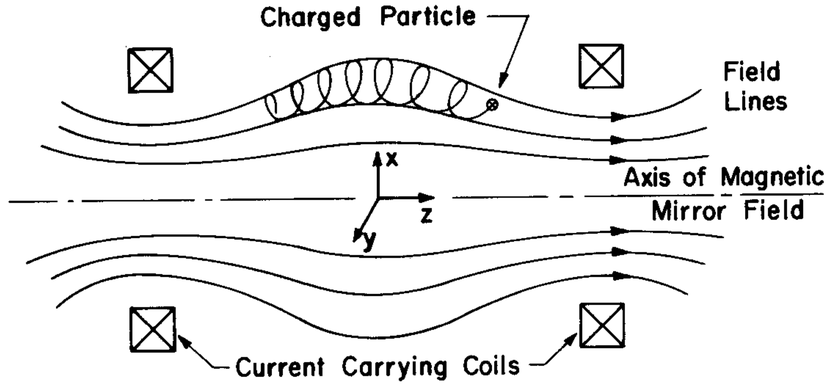
This is the collision rate per particle. For many incident particles, the total number of events per cubic centimeter per second is

$$R = n_1 n_2 \langle\sigma v\rangle_{1,2} \qquad (1.13.5)$$

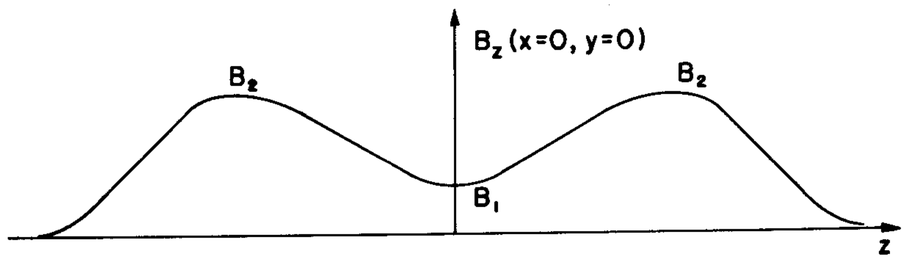
Problem 1.13.1 Show that the electrical conductivity of a fully ionized gas can be expressed in terms of the collision frequency and plasma frequency as follows:

$$\sigma = \frac{\omega_p{}^2}{4\pi\nu_c} \qquad (1.13.6)$$

////



(a). Magnetic Mirror Coils and Magnetic Mirror Field Showing one Transit of a Trapped Charged Particle.



(b). Magnetic Flux Density Along the Axis of the Magnetic Mirror.

FIGURE 1.14.1
Schematic of a magnetic-mirror field configuration used for the confinement of plasma.

1.14 DIFFUSION AND BOHM DIFFUSION

Whenever there is a gradient in the density in a neutral gas, there is a possibility for a net flow of particles attempting to reduce the gradient. This process is called *diffusion*, and in a neutral gas is *impeded* by collisions. In a plasma, collisions can sometimes also *cause* diffusion. This is because magnetic fields are often used to prevent the free streaming of plasma particles (which causes the diffusion in inhomogeneous neutral gases), and thus to restrict diffusion of the plasma. Figure 1.14.1 shows such a field configuration. In this case collisions, which allow the particle the freedom to cross the magnetic field, *restore* the diffusion process.

The flux of particles Γ per unit area per unit time is equal to the product of the particle density and the mean velocity of the particles. The resultant flux of particles caused by a density gradient is a measure of the diffusion coefficient D, defined through

$$\Gamma = n\langle v\rangle = -D\frac{\partial n}{\partial x} \qquad (1.14.1)$$

Problem 1.14.1 Show that the classical diffusion coefficient for a gas of neutral particles is

$$D \approx \frac{\kappa T}{m\nu_c} = \frac{\langle v^2\rangle}{3\nu_c} \qquad (1.14.2)$$

////

Even in the absence of magnetic fields, the diffusion coefficient for a plasma of ions and electrons is altered by the electrical properties of the plasma. For example, in a tenuous plasma diffusing through a dense neutral background, the more mobile electrons will try to outrun the ions. This causes a charge-separation electric field that acts to retard the diffusion of the electrons and to enhance the diffusion of the ions, thus maintaining charge neutrality. The enhanced diffusion of ions is called *ambipolar diffusion*. The ambipolar-diffusion coefficient for a plasma in which the electron and ion temperatures are equal is twice the free-diffusion coefficient for ions alone; i.e.,

$$D_A \approx 2D_i = \frac{2\langle v_i^2\rangle}{3\nu_c} \qquad (1.14.3)$$

where ν_c is the frequency of collision of ions with the neutral background.

A steady magnetic field causes the charged particles to move along helical trajectories. If there is a transverse density gradient of particles in a magnetic field, then as a result of collisions between the electrons and ions, there will be diffusion of charged particles across the magnetic field. The diffusion coefficient in this case is

$$D_\perp = D\frac{\nu_c^2}{\omega_c^2} \qquad (1.14.4)$$

This shows that the free-diffusion coefficient is reduced by $1/B^2$, and that more collisions mean *more* diffusion. This $1/B^2$ reduction of diffusion across magnetic field lines suggests that high-B fields will help confine a plasma for long times. Unfortunately, most experiments to study the magnetic containment of plasmas indicate a diffusion rate in excess of that predicted by (1.14.4) if ν_c is determined by the cross section for two-body collisions. Even worse, the experiments show that the diffusion is not even proportional to $1/B^2$. Diffusion proportional to $1/B$ in a plasma is called *Bohm diffusion*, after David Bohm, who proposed that collective processes in a plasma, such as instabilities, might lead to a diffusion coefficient

$$D_B \approx \frac{c\kappa T}{16eB} \qquad (1.14.5)$$

This diffusion coefficient was *proposed*, not derived. However, the fastest rate at which a charged particle would move across a magnetic field occurs, comparing

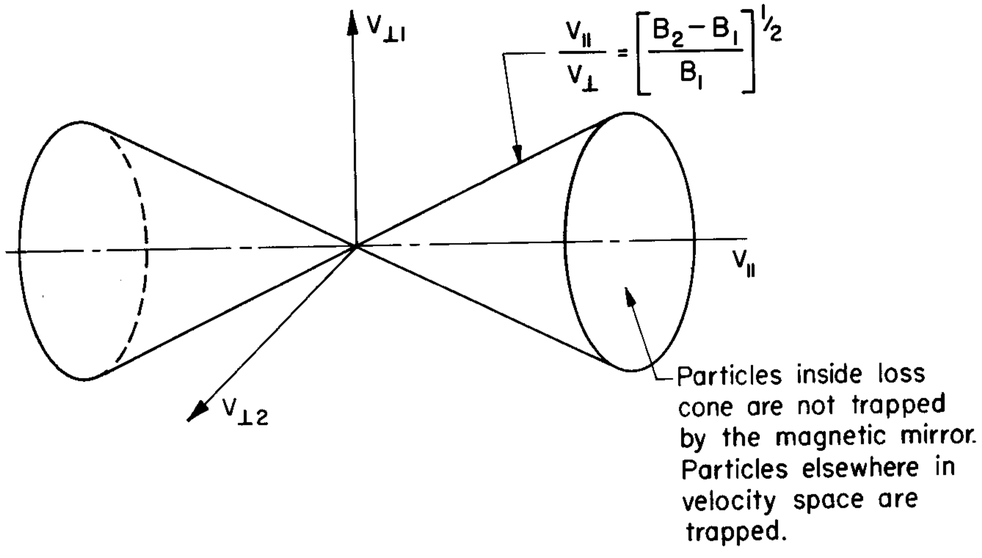


FIGURE 1.14.2
Midplane velocity-space loss cones for a magnetic-mirror field configuration.

Eqs. (1.14.2) and (1.14.4), where $\nu_c/\omega_c \approx 1$. Bohm diffusion is obtained by assuming this optimum value for ν_c. Bohm may have surmised that instabilities would increase the effective collision frequency to this value. When $\nu_c = \omega_c$, a particle moves, due to collisions, a distance across the field equal to its Larmor radius ($a_L = v/\omega_c$) each cyclotron period. Some of the current literature concerning experiments on plasma containment by magnetic fields refers the containment time in terms of Bohm times, i.e., the number obtained when the experimental diffusion coefficient is divided by the Bohm diffusion coefficient.

In addition to the configuration-space diffusion just described, there is also velocity-space diffusion. This type of diffusion also occurs as a result of collisions among the particles, and seeks to reduce any anisotropic or other entropically unfavorable features in the velocity-distribution function. An example of velocity-space diffusion occurs in a plasma confined in a magnetic mirror field (Figs. 1.14.1 and 1.14.2). Particles are trapped in this magnetic-mirror-field configuration if their velocity coordinates lie outside the so-called mirror loss cone, i.e., if

$$\frac{v_{\parallel 0}}{v_{\perp 0}} \leq \left(\frac{B_2 - B_1}{B_1}\right)^{1/2}$$

at the midplane. Particles in the loss cone pass over the peak of magnetic field, while particles outside the velocity-space loss cone are trapped. In the absence of collisions, they remain trapped indefinitely in the magnetic mirror field. As

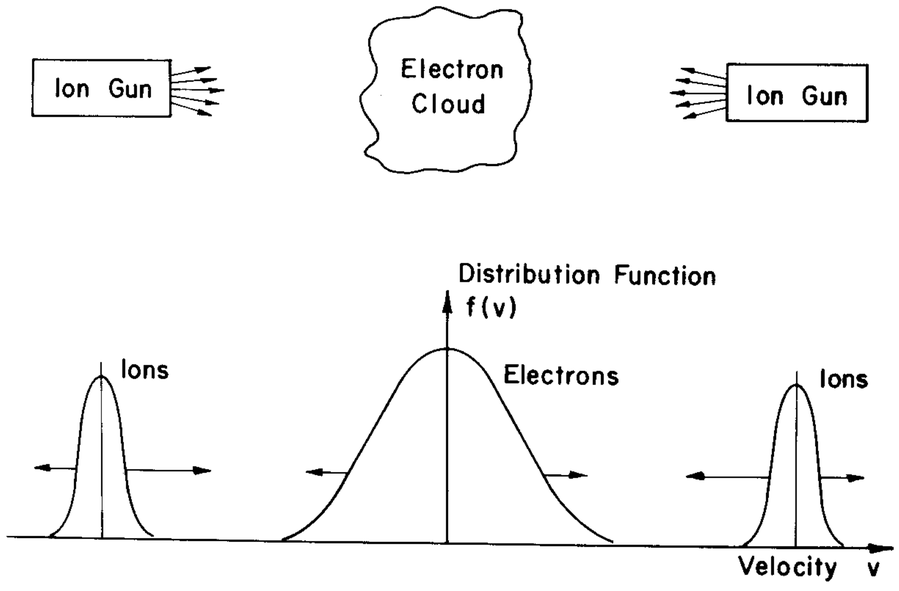


FIGURE 1.14.3
Schematic of counterstreaming plasmas and corresponding initial velocity-space distribution function. Arrows indicate direction of diffusion caused by weak turbulence generated by the plasma instability.

a result of elastic scattering, the velocity-space coordinates of a given particle change as a function of time, in a random manner. The random walk in velocity space of such a particle causes it eventually to pass into the loss cone. On the next trip in the B_2 region it escapes from the magnetic mirror. In this example particles do not diffuse across field lines, but by changing their velocity coordinate while on a given field line, they escape from the magnetic "bottle." Instabilities may also cause diffusion in velocity space. For example, consider a distribution function that has counterstreaming groups of particles, such as shown in Fig. 1.14.3. This distribution function is unstable, and as a result waves of various types are generated in the plasma. This causes particles to diffuse in velocity space, in the direction indicated by the arrows in Fig. 1.14.3, i.e., toward a more stable distribution function. When the wave energy remains small, the process by which this diffusion takes place is termed *weak turbulence*. Weak turbulence has been widely studied, and considerable analysis is available.

1.15 PLASMA RADIATION

There are several mechanisms that cause plasmas to absorb or emit radiation. These can be grouped into two categories: (1) radiation from emitting atoms and

(2) radiation from accelerated charges. The main interest in plasma radiation comes from the fact that emitted radiation can be used to measure plasma properties; in the case of confined plasma, radiation represents an energy loss which must be accounted for in power-balance considerations related to controlled thermonuclear reactors.

1.15.1 Bremsstrahlung

Strictly speaking, the term Bremsstrahlung includes most sources of radiation. In plasma physics it is generally identified with the free-free transition radiation which occurs when one particle encounters another and makes a sudden change in direction. This radiation intensity for a hydrogenic plasma is

$$P_B = 5.35 \times 10^{-24} n^2 T^{1/2} \qquad \text{ergs/(cm}^3\text{)(s)} \qquad (1.15.1)$$

where T is in kilo-electron volts (keV). This is the lowest radiation rate that a plasma can ever have, and the calculation to obtain this rate assumes no reabsorption of emitted radiation. This would represent a negligible loss of energy from a confined thermonuclear plasma. That is, the power generated by fusion reactions at a plasma temperature of 100 keV exceeds the power lost due to Bremsstrahlung.

1.15.2 Blackbody radiation

If a plasma were in thermodynamic equilibrium, it would radiate according to the Stefan-Boltzmann law.

$$P_{bb} = \alpha T^4 \qquad \text{ergs/(cm}^2\text{)(s)} \qquad (1.15.2)$$

where $\alpha = 5.7 \times 10^{-5}$ ergs/(cm^2)(s)(°K^4). In contrast to Bremsstrahlung, at plasma temperatures of interest for a controlled-fusion reactor, the blackbody-radiation losses would greatly exceed the power generated through fusion reactions. This is an irrelevant consideration for a laboratory plasma because of the large size of a plasma needed for the plasma to radiate as a blackbody: $R_{bb} = 10^{23} T^{7/2}/Z^2 n_i n_e$, where T is in degrees Kelvin.

Problem 1.15.1 Compare radiation rates for Bremsstrahlung and blackbody radiation for a 3-cm-diameter plasma with 10-eV temperature and a density of 10^{14} particles per cubic centimeter. What fraction of the total energy does this radiation loss amount to in 1 s? What is the minimum radius for a plasma which radiates as a blackbody? ////

1.15.3 Impurity Radiation

Intermediate between the two limits of blackbody radiation and Bremsstrahlung is the radiation of the excited states of atoms or ions in a partially ionized plasma. Such atoms or ions, are almost always present in laboratory plasma, and are called *impurities*; hence the radiation they emit is called *impurity radiation.* This impurity radiation is frequently of interest in plasma diagnostics since it can be used to measure electron temperature and concentration. Although it is a serious-energy-loss mechanism, it is one that can be controlled by improved vacuum technology, and it is not regarded as a barrier for controlled thermonuclear fusion.

1.15.4 Synchrotron or Cyclotron Radiation

A point-charged particle moving in a circular orbit, such as an electron in a magnetic field, radiates at the cyclotron frequency and its harmonics. For a magnetically confined plasma at temperatures of interest for controlled fusion, the plasma electrons are mildly relativistic, and as a result the maximum radiation occurs at a harmonic of the rest-mass cyclotron frequency ($n \approx \gamma^2$, where $\gamma = 1/\sqrt{1 - v^2/c^2}$). There is more energy loss from all the harmonics than from the fundamental. The total energy loss for synchrotron radiation from a magnetically confined plasma usually exceeds the power generated by fusion reactions, if electrons and ions are at the same temperature. This problem has not been resolved, but there are plasma-confinement schemes in which the electrons are at a lower temperature than the ions.

1.15.5 Cherenkov Radiation

Whenever a charged particle moves through a medium at a speed greater than the velocity of light in the medium, it loses energy. This energy is deposited in the medium as radiation, called *Cherenkov radiation.* Fast-moving particles in a plasma lose energy by this mechanism, exciting plasma waves and electromagnetic waves. For plasma waves, excitation occurs when the particle speed exceeds the plasma wave speed.

PART TWO: PLASMA PRODUCTION

Experimental studies of plasma are usually complicated by the fact that it is necessary to first create the plasma, since it does not exist as a normal state of

matter on earth. The methods of creating laboratory plasmas are quite varied, and depending on the method of creation, the plasma may be high- or low-density, high- or low-temperature, steady-state or transient, stable, unstable, and so on. Thus the kinds of laboratory studies that can be done depend to a great extent on the plasma-production scheme itself. Similarly, an understanding of the objectives of theoretical plasma studies requires some appreciation of the problems of plasma production and measurement.

Before describing some of the experiments used in the study of plasmas, a brief description of some of the successful practical uses of plasmas will be given.

The most widespread practical use of plasmas is in the field of lighting. Arc lamps are used in movie projectors, street lamps, searchlights, etc. Fluorescent lamps are low-pressure cold-cathode discharges of electricity through plasma that produce copious quantities of ultraviolet radiation which excite a fluorescent coating on the inside of the discharge tube. Neon (etc.) lamps are cold-cathode discharges that produce a line-spectra light by the collisional excitation of neutral atoms or ions, which then emit radiation. Arc welders are plasma devices that are used to fuse various metals. Ignitrons and thyratrons, and related devices, are high- and low-current switches used to control machinery (motors, spot welders, capacitor energy-storage systems, etc.) in practically every industrial process. The list of uses of plasma devices is long and varied. The reader is encouraged to discover some for himself.

In the following sections, some of the schemes used to produce plasmas for laboratory studies are described. There are many plasma-production schemes other than those described here or elsewhere in this book.

Discussion topic For each of the references given below prepare a brief summary of the principles of operation, advantages, limitations, and properties of the plasma-production scheme.

1 Shock-tube plasmas: A. C. Kolb and H. R. Griem, High Temperature Shock Waves, chap. 5 in "Atomic and Molecular Processes," Academic, New York, 1962.

2 Penning ion-gauge discharges: J. C. Helmer and R. L. Jepsen, Electrical Characteristics of Penning Discharges, *Proc. IRE*, **49**:1920 (1961).

3 Electron-beam-produced discharges: W. D. Getty and L. D. Smullin, Beam Plasma Discharges: Build-up of Oscillations, *J. Appl. Phys.*, **34**:3421 (1963).

4 Laser-produced plasmas: A. F. Haught and D. H. Polk, *Phys. Fluids*, **9**:2047 (1966).

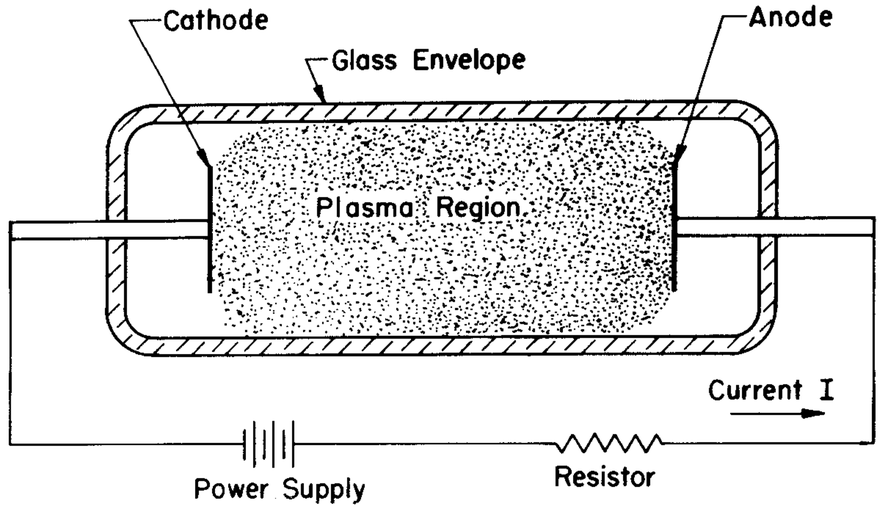


FIGURE 1.16.1
Schematic of a low-pressure, cold-cathode discharge and related electric circuitry.

1.16 THE LOW-PRESSURE COLD-CATHODE DISCHARGE

Electrodes sealed in an evacuated vessel, as in Fig. 1.16.1, can be used to initiate and maintain a low-pressure, steady-state (or pulsed) arc discharge in the vessel by applying a voltage as shown. The plasma produced in this device is low-density ($n \approx 10^{10}$ cm^{-3}) and low-temperature ($T_e < 5000$°K). These devices were used in some of the early plasma studies and in studies of atomic spectroscopy. They have many interesting features[1] and are still used in many plasma studies. Glow-discharge devices (small neon tubes) are used as panel lights, voltage-regulator tubes, etc.

1.17 THE THERMIONIC-ARC DISCHARGE

An improvement on the cold-cathode discharge for laboratory plasma studies is obtained by using a thermionic oxide–coated cathode (or Phillips dispenser cathode) as a source of electrons for a discharge tube. A typical version

[1] For a full description see L. B. Loeb, "Fundamental Processes of Electrical Discharges in Gases," Wiley, New York, 1955.

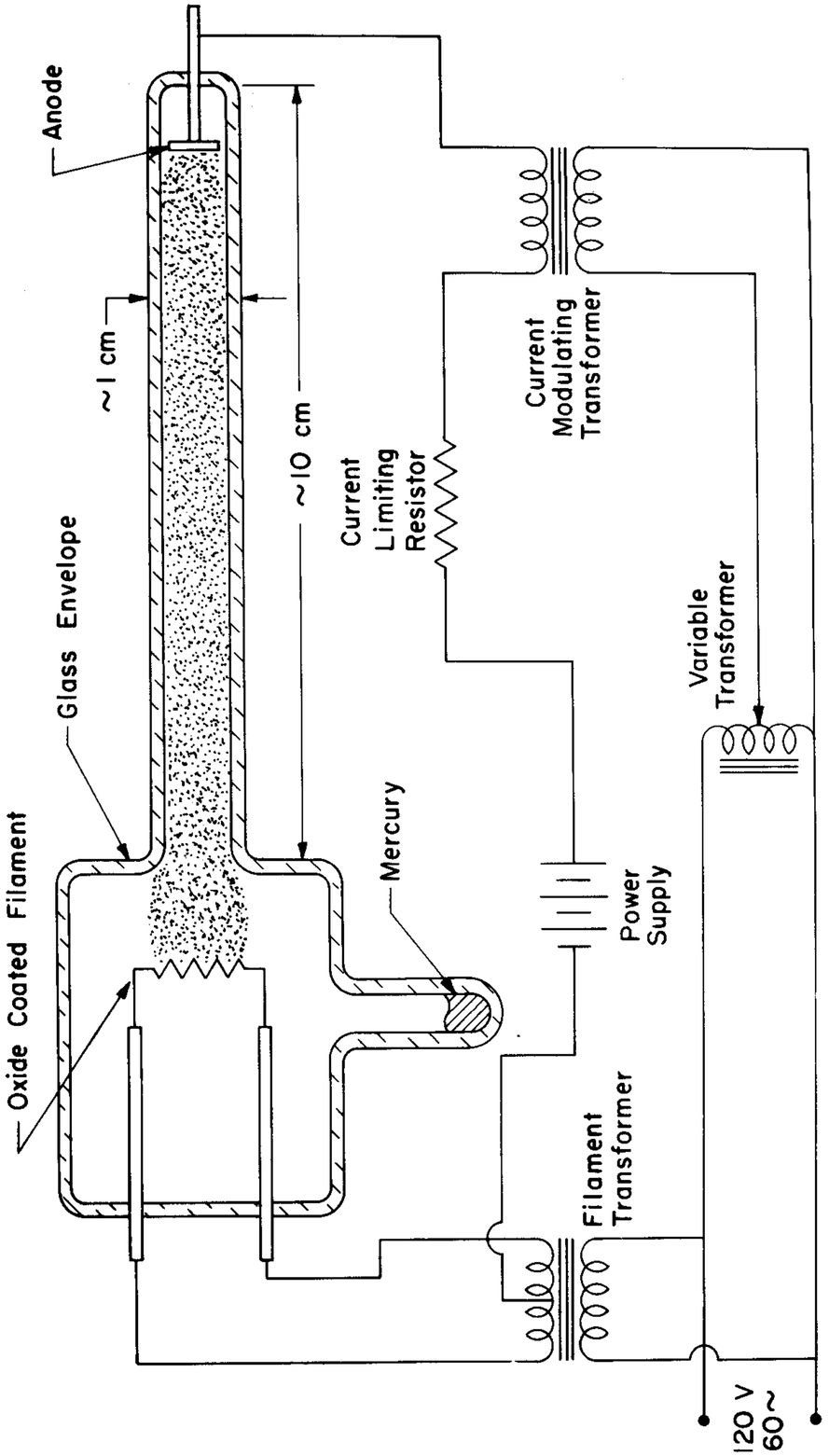


FIGURE 1.17.1
Schematic of a thermionic-arc discharge tube and related electric circuitry.

of this device is depicted schematically in Fig. 1.17.1. These discharge devices are frequently operated at the room-temperature vapor pressure of mercury ($\sim 10^{-3}$ Torr), which provides plasma densities in a range from $n_e = 10^9$ cm^{-3} to $n_e = 10^{13}$ cm^{-3} for 1-cm-diameter 10- to 20-cm-long discharge tubes. The tube is constructed by conventional glass-blowing techniques. A drop of mercury is deposited in the tube, and the cathode is activated after the pressure in the tube is reduced to less than 10^{-6} Torr by means of a vacuum pump. One means of handling the mercury during evacuation and activation of the cathode is to freeze the mercury at liquid nitrogen temperature (77°K) in a small appendage on the tube. This procedure keeps the vapor pressure of the mercury within acceptable limits during activation.

A typical setup for this discharge tube is shown in Fig. 1.17.1. The thermionic cathode (or bare heater) is supplied with power by a transformer with suitable insulation to stand off the supply voltage. Although the discharge operates with only a few volts-per-centimeter drop along the discharge, a high-voltage supply is usually required to initiate the discharge. A Tesla coil is sometimes needed to provide an initial supply of free electrons to initiate the discharge. The external resistance stabilizes the discharge current, which would become very large (limited only by the power-supply internal resistance) without it. That is, this arc discharge has a negative resistance characteristic. The transformer shown in the discharge circuit provides a means of modulating the discharge current. Since the plasma density is proportional to the discharge current, this provides a means of varying the plasma frequency while keeping other parameters fixed. The usefulness of this method of varying plasma density is seen in the discussion of plasma resonances in Chap. 4. An early and thorough study of the density and temperature of such a thermionic discharge was made by Klarfeld.[1] His useful study makes possible the design laboratory discharges with known parameters. This type of discharge tube was also used in experiments to study plasmas waves and oscillations in finite plasmas.[2,3]

By a slight mechanical modification of this discharge tube it is possible to inject a small-diameter (1-mm) electron beam (100 to 1,000 V) into and along the discharge tube. This type of beam-plasma system was used by Boyd, Field, and Gould[4] and by Gould and Trivelpiece[5] to investigate beam-plasma instabilities.

[1] B. Klarfeld, *J. Phys.* (*U.S.S.R.*). **5**:155 (1941).
[2] A. W. Trivelpiece, "Slow Wave Propagation in Plasma Waveguides," San Francisco Press, San Francisco, 1967.
[3] P. E. Vandenplas, "Electron Waves and Resonances in Bounded Plasmas," Wiley, New York, 1968.
[4] G. D. Boyd, L. M. Field, and R. W. Gould, *Phys. Rev.*, **109**:1393 (1958).
[5] R. W. Gould and A. W. Trivelpiece, "A New Mode of Propagation on Electron Beams," Electric Waveguides Polytechnic Institute of Brooklyn, New York, 1958.

1.18 PLASMA GUNS

In many plasma experiments it is desirable to be able to inject plasma into the region where the measurements of plasma properties are to be made. There are several schemes by which a plasma can be created in one place and propelled into another.

A poorly understood and yet widely used source that has this feature is called an *occluded gas source*, or *deuterated titanium washer stack source*. This source has been developed to its present refined state almost entirely by empirical, cut-and-try experiments.[1] Figure 1.18.1 shows one version of this plasma gun, together with a schematic of a typical system of electrical apparatus needed to operate it.

The operation of this plasma source is based on the fact that metallic titanium has the property that, when heated to a temperature of around 1000°C in hydrogen or deuterium, it will have an exothermic reaction with the hydrogen and form titanium hydride. This process, if done with limited quantities of hydrogen, will "load" the titanium washer with about one hydrogen atom for each titanium atom. Although the titanium can accept more hydrogen, it becomes structurally weak and crumbles if much more than a 1:1 loading is attempted.

The loaded titanium washer is placed as indicated in Fig. 1.18.1. A small trigger electrode at the back of the source provides an initial quantity of electrons, which are then accelerated by the voltage (>1.0 kV) applied across the source. Some of these electrons strike the deuterated washer, depositing an energy of about 1 MW/cm^2 for a microsecond or so. This energy heats the surface of the titanium, releasing the trapped deuterium or hydrogen, which is then ionized in the discharge. The plasma thus formed is accelerated (probably by the potential well caused by an electron precursor) out through the washer hole into the region where the experiment is to be performed. These sources supply a 2- to 10-μs burst of 10^{17} plasma particles at energies of about 1.0 keV for the ions. The properties of the plasma produced by these sources are still being studied.

These sources are used in many of the magnetic-mirror compression experiments. The source is usually located along the magnetic axis outside of one of the magnetic mirrors. Injection takes place early in the compression cycle, when the magnetic field is small and rising.

These sources are also used in other injection experiments, such as multipole devices, where it is desired to study the decay of plasma after its injection into the containment region. One of the advantages of these sources is that

[1] F. H. Coensgen, W. F. Cummins, and A. E. Sherman, *Phys. Fluids*, **2**:350 (1959).

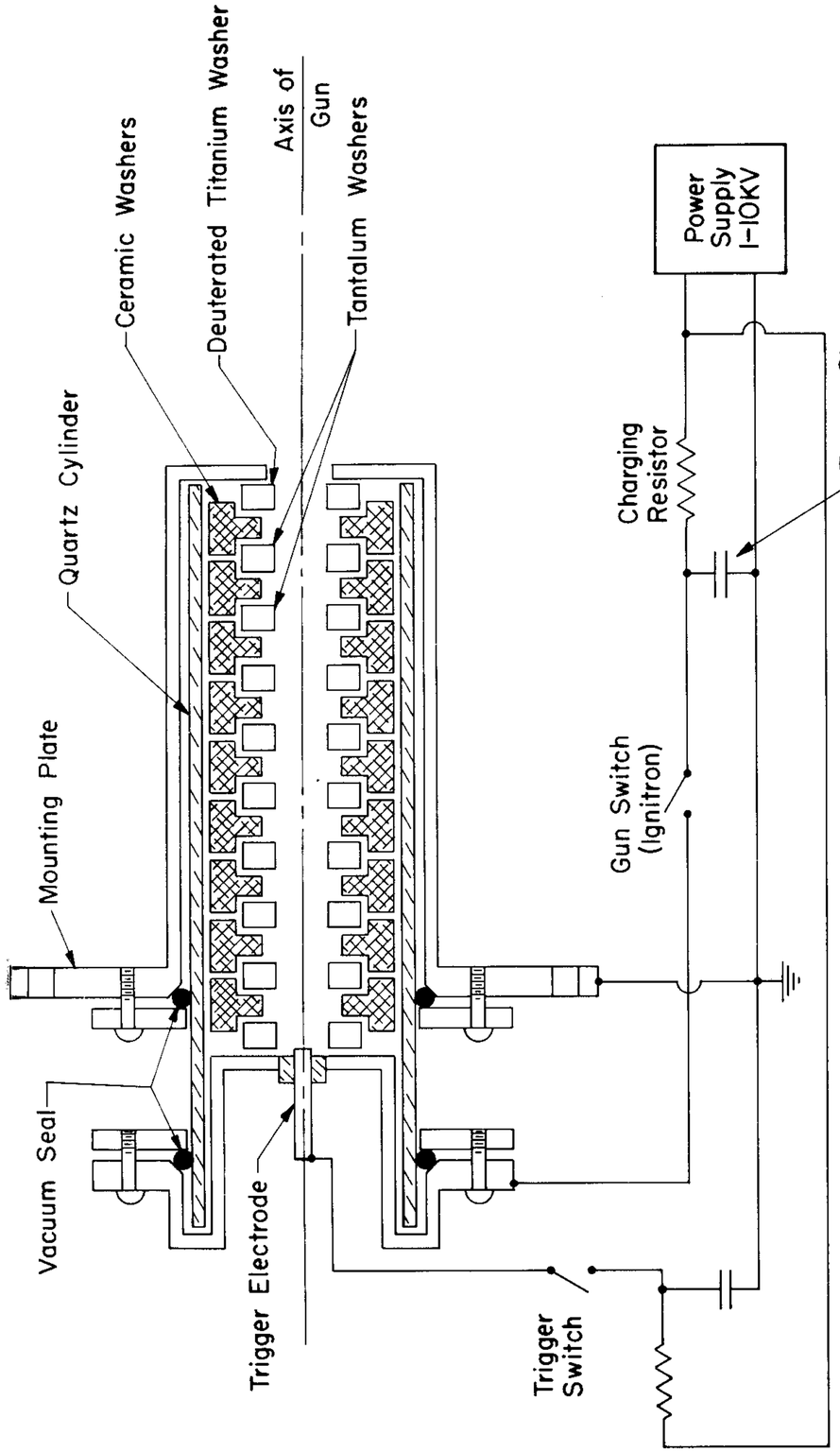


FIGURE 1.18.1
Schematic of occluded-gas plasma gun and related electric circuitry. The trigger switch is usually closed at the same time as the gun switch.

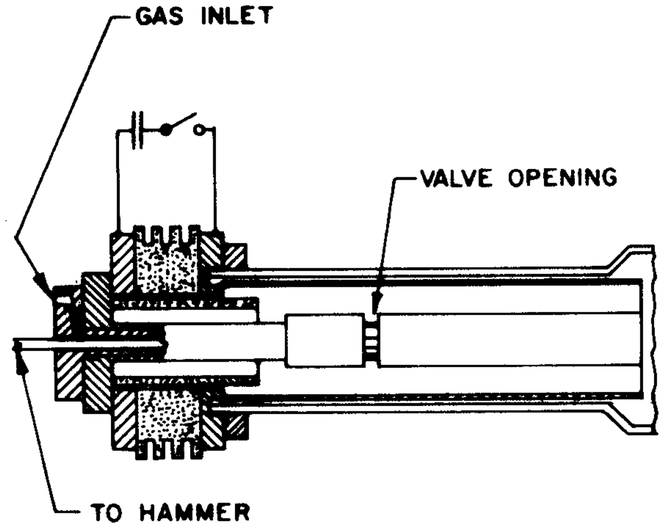


FIGURE 1.18.2
Schematic of coaxial plasma-gun source (Marshall gun) and related electric circuitry.

they can be used in high-vacuum systems, since they can be discharged without the presence of an initial-gas background; i.e., the gas is stored in the deuterated washer. Also, this means that only a small amount of neutral gas is injected into the experiment along with the plasma.

Another type of plasma gun is shown schematically in Fig. 1.18.2. This type of coaxial-accelerating plasma source is called a *Marshall*[1] *gun*. Its operation is as follows: A voltage is applied between the coaxial-insulated conductors by a capacitor bank. A small quantity (10^{18} atoms) of neutral gas is admitted into the region between the two conductors. The voltage exceeds the breakdown voltage of the gas, and an arc is formed. The current flow caused by the arc generates a magnetic field behind the arc, as shown. This magnetic field exerts an accelerating force on the arc, propelling it down the coaxial system toward the open end of the plasma gun. The plasma is thus ejected from the gun with a velocity of about 10^8 cm/s. These sources have been used in injection studies in much the same way as the deuterated washer stack source. Its advantages include the possibility of producing plasmas of gases other than deuterium or hydrogen; it may also produce a higher-density and higher-temperature plasma. One of the disadvantages is the high percentage of neutral gas associated with the plasma.

[1] J. Marshall, *Phys. Fluids*, **3**:135 (1960). See also Proceedings of an International Symposium on Plasma Guns, *Phys. Fluids*, **7**:S1–S74 (1964).

1.19 ALKALI METAL VAPOR PLASMA—*Q* MACHINES

The term *Q machine* is used to denote a class of plasma experiments where the plasma has a very low kinetic temperature. Originally, *Q* stood for quiescent, but subsequent work showed that the plasma was not really quiescent but rather was rich in plasma phenomena, including several interesting instabilities.

Although others used various contact ionization schemes to make alkali metal vapor plasmas,[1] their widespread use in basic plasma studies was started with an experiment at Princeton.[2] This experiment is shown schematically in Fig. 1.19.1.

The basic idea behind this device is that contact ionization occurs when an atom of certain alkali metals, such as cesium, comes in contact with a heated tungsten plate. A cesium atom impinges on the tungsten plate and comes off from the plate as an ion. If the tungsten plate is raised to a sufficient temperature ($\sim$2000°C) to be an emitter of electrons, each cesium ion is accompanied, on the average, by an electron.

Thus, in Fig. 1.19.1, a small steady flow of cesium atoms on the heated plate at the left end of the experiment is converted to a plasma stream which is confined by the axial magnetic field. A hot plate at the other end reflects the plasma. These plasmas are usually a few centimeters in diameter, and the experiments vary in length from 10 cm to a couple of meters.

The plasma density can be varied from about 10^9 to 10^{13} cm^{-3}. Plasma temperatures of 1000°K are typical in *Q* machines. One of the important features of a *Q*-machine plasma is the fact that it is 99 percent ionized.

Q machines have proliferated greatly since the initial experiments, and many basic features of plasma theory, such as drift waves, diffusion, resistivity, beam-plasma instabilities, and sheath effects have been studied with them.

1.20 RF-PRODUCED PLASMAS

A gas at low pressure will break down under the action of an applied steady electric field. It will also break down and form a plasma if the applied electric field is alternating. There are some advantages to using such a scheme to form and maintain a discharge. One is the possibility of having an electrodeless discharge, as shown in Fig. 1.20.1. The alternating electric field penetrates the

[1] J. Y. Wada and R. C. Knechtli, *Proc. IRE*, **49**:1926 (1961).
[2] N. Rynn and N. D'Angelo, Device for Generating a Low Temperature, Highly Ionized Cesium Plasma, *Rev. Sci. Instrum.*, **31**:1326 (1960).

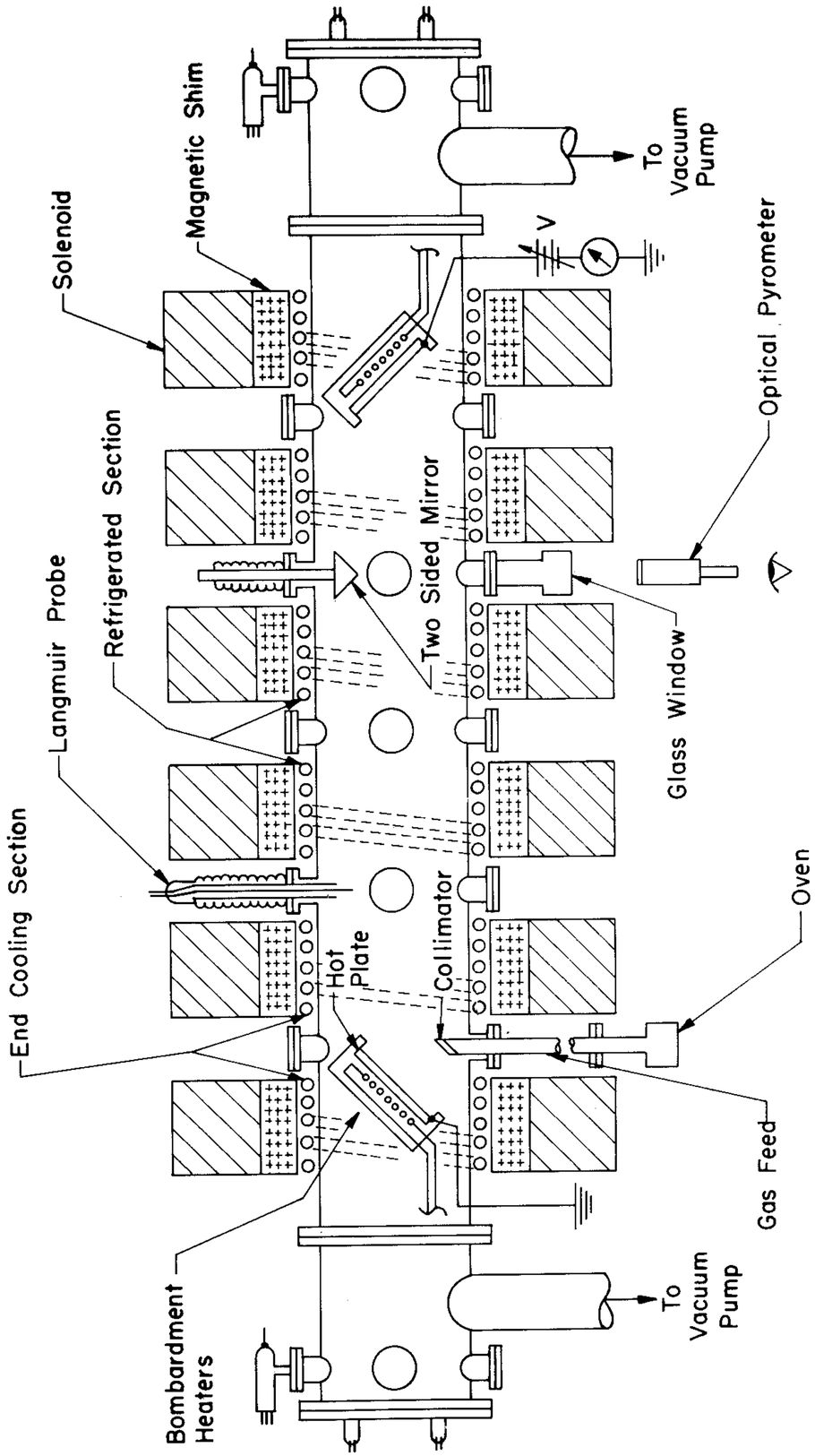


FIGURE 1.19.1
Schematic of a Q machine (Q-1). [*From N. Rynn, Improved Quiescent Plasma Source, Rev. Sci. Instrum.*, **35**:40 (1964).]

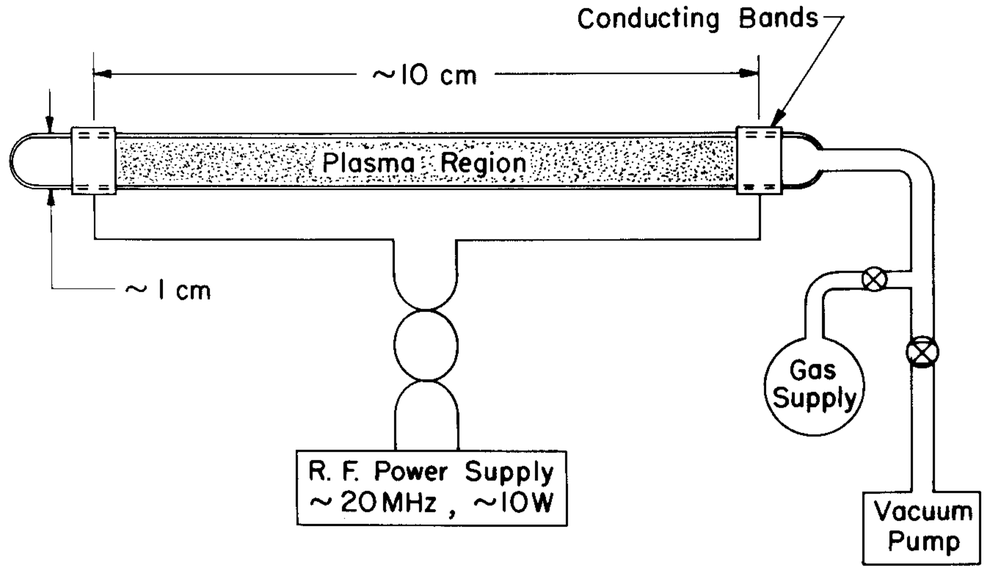


FIGURE 1.20.1
Schematic of an rf discharge plasma.

dielectric of the glass vessel with no difficulty and accelerates electrons in the vessel to energies above the ionization potential of the gas. The "steady-state" plasma density produced by the process of ionization and recombination usually is only a few percent of the background neutral-gas density. Plasma densities of 10^{10} cm^{-3} are typical.[1]

Sometimes an rf-produced plasma is created in a magnetic field. This modification can be used to produce plasmas with substantially higher energies than occur for no magnetic field. These are called cyclotron resonance plasmas when the exciting rf field has the same frequency as the cyclotron frequency for the electrons. Even with the modest rf power available from microwave sources (i.e., 3-cm-pulsed 10-kW magnetrons), plasmas with electron temperatures in the 20-keV range can be produced by this method.

More elaborate experiments in which an attempt is made not only to produce a cyclotron resonance plasma, but also to explore the limitations on the maximum plasma temperature and energy absorption have been done at various laboratories. The experiments involve high-power microwave sources (>100 kW) which couple power into cavity resonators (to increase the electric field strength and coupling to the plasma). Plasmas with mean electron energies in the million-electronvolt range and densities in the 10^{11} cm^{-3} range have been generated

[1] S. C. Brown, "Basic Data of Plasma Physics," Wiley, New York, 1967. Contains an extensive description of the properties of rf-produced plasmas.

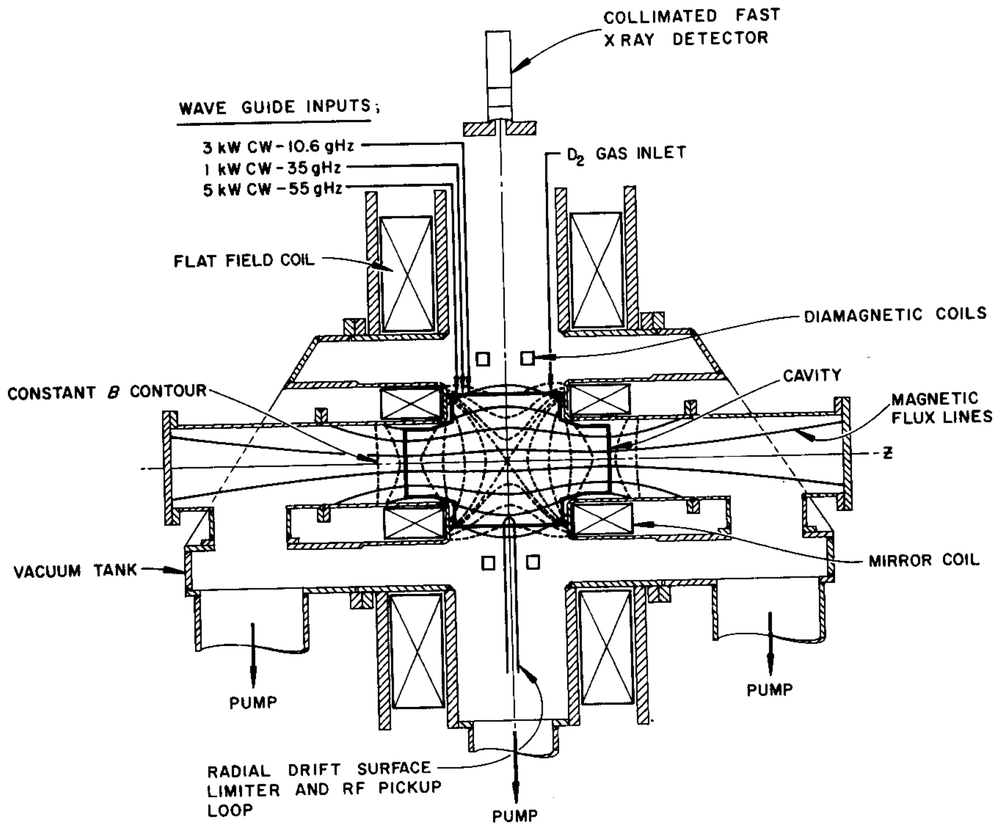


FIGURE 1.20.2
Schematic of the ELMO device used at the Oak Ridge National Laboratories to produce electron-cyclotron resonance-heated plasmas with mean electron energies of 0.5 MeV and greater. [*After R. A. Dandl et al., Phys. Fluids.* **9**:1498 (1966).]

with devices such as *ELMO* used at Oak Ridge National Laboratory. Similar experiments have been performed at the French laboratories in Saclay. The Oak Ridge version of such an experiment is shown in Fig. 1.20.2. This experiment produces a plasma with a volume of 2 liters in 20-kG magnetic mirror field by the application of 0.8 kW of microwave power at 8 mm wavelength. The plasma, formed by the rf breakdown of hydrogen, consists of three components: (1) cold background, $n_0 \approx 10^{12}$ cm^{-3}; (2) cool electrons, $n_e \approx 2 \times 10^{11}$ cm^{-3} at a temperature $T_e < 100$ eV; and (3) hot electrons, $n_e \approx 3 \times 10^{12}$ cm^{-3} at a mean energy of 0.5 MeV. The energetic electron plasma, which is in a shell 4 cm thick, 9 cm mean diameter, and 10 cm long, creates a magnetic well from its self-field.

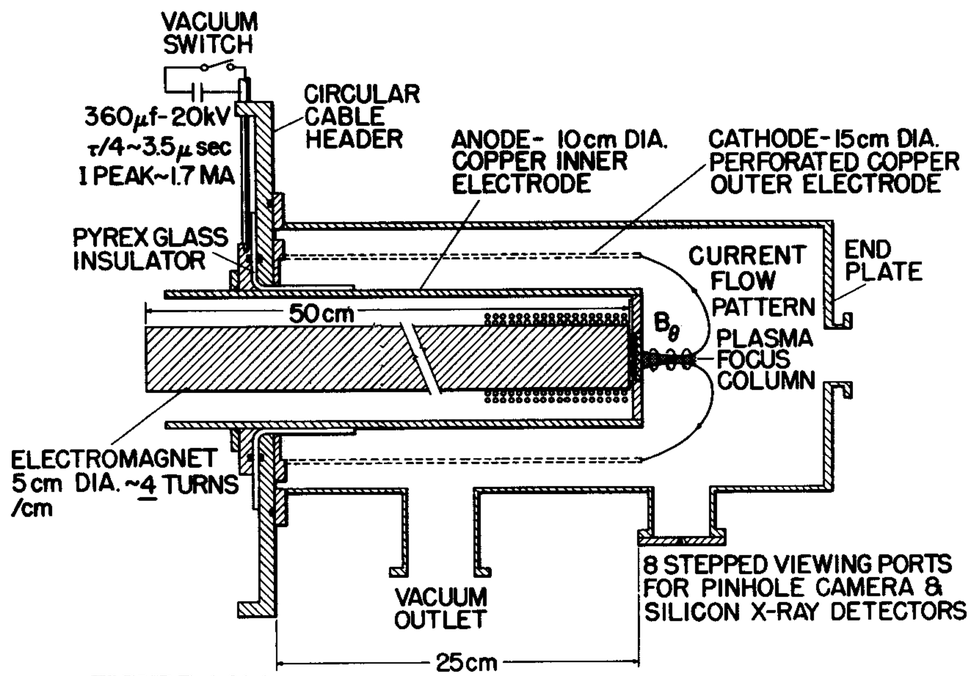


FIGURE 1.21.1
Schematic of dense-plasma-focus experiment used at Los Alamos Scientific Laboratories to study properties of high-energy density plasmas. [*After J. W. Mather, through the courtesy of the Los Alamos Scientific Laboratories, Los Alamos, New Mexico, and the U.S. Atomic Energy Commision, under whose auspices the work was performed.*]

1.21 DENSE-PLASMA FOCUS

Figure 1.21.1 is a schematic of an experiment used at the Los Alamos Scientific Laboratory to study the properties of a very dense ($n_e \approx 5 \times 10^{19}$ cm^{-3}), short-duration (FWHM[1] ≈ 100 ns) deuterium plasma whose temperature is large enough ($T_e \approx 3$ to 4 keV) to produce a significant neutron yield (10^{11} neutrons per pulse). The experiment combines the features of a coaxial plasma gun with the magnetic compression of a pinch plasma. The plasma is initially formed in the annular region near the header as a radial arc. This arc is accelerated by magnetic pressure down the coaxial system to a point where the coaxial conductors terminate. The arc then turns the corner and becomes a radially imploding pinch plasma whose final maximum parameters are given above. Figure 1.21.2 an x-ray pinhole photograph produced by the plasma focus

[1] Full width at half maximum.



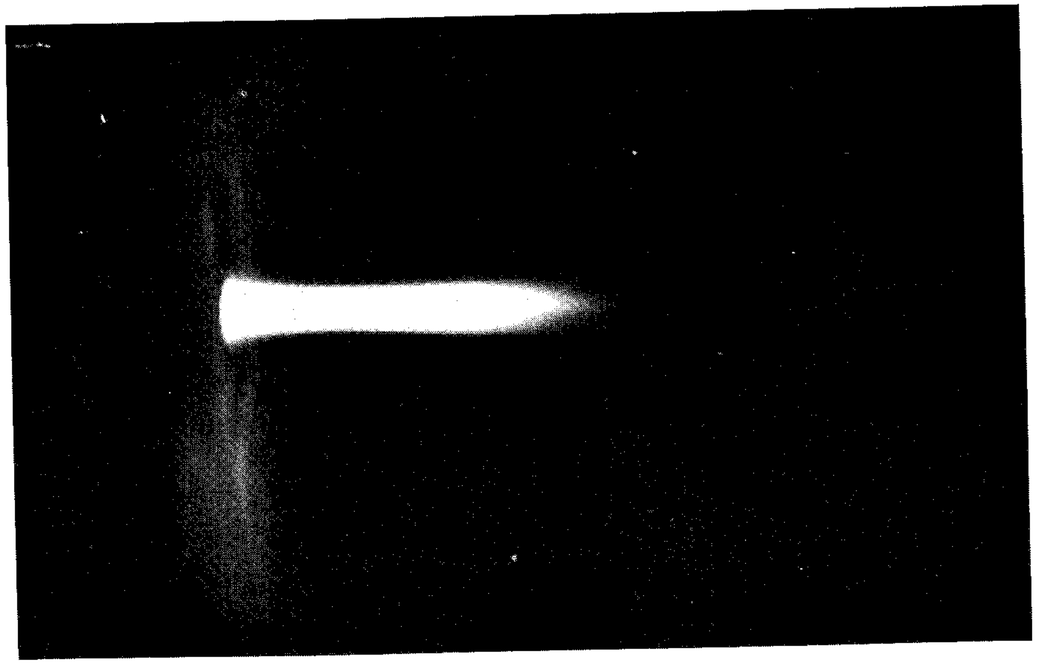


FIGURE 1.21.2
X-ray pinhole photograph of plasma produced by the Los Alamos Dense Plasma Focus (DPF III) for bias magnetic fields of (*a*) zero and (*b*) 600 G.

for two values of the magnetic field ($B = 0$ and $B = 600$ G) supplied by the electromagnetic shown in Fig. 1.21.1. As can be seen, the plasma properties are strongly dependent on this bias field, which helps focus the plasma as it turns the corner. In addition to its use in plasma studies, the plasma focus is a simple means of obtaining a supply of pulsed neutrons.

1.22 THE SOLAR PLASMA

The sun provides a supply of interesting problems on plasma equilibrium, instabilities, radiation, etc., which are studied through observations of sunspots, solar storms, and x-ray and visible spectroscopic studies of the solar chromosphere. In addition, the sun creates a plasma, the solar wind, that extends beyond the sun's boundary and well into space. This natural plasma is studied by means of satellites instrumented to measure the properties of the solar wind and its interaction with the earth and the moon. Such phenomena as the standing shock wave (*bow shock*) that occurs as a result of the earth's magnetic field interacting with the solar wind are studied.

A summary of the plasma properties at a distance of 1 astronomical unit (1 AU $= 93 \times 10^6$ mi $\approx$ average distance from the earth to the sun) from the sun are given in Table 1.22.1.

1.23 LASER-PRODUCED PLASMAS

The focused output of a high-power, Q-switched laser can be used to irradiate solids and compressed gases to produce dense, isolated, high-temperature plasmas for laboratory studies of confinement and stability. Linlor[1] used the focused spot from a 5-MW peak-power laser to irradiate a metallic target in vacuum and showed that the irradiated region formed a small dense plasma and that metallic ions with energies of about 1 keV were ejected from that region. Haught and Polk[2] produced a high-temperature plasma (100 eV) by irradiating 10- to 20-μm diameter lithium hydride pellets, which are suspended in an evacuated chamber by ac fields, with the focused output of a 20-MW Q-switched ruby laser. Recent studies (1972) of laser-produced plasma have been concerned with the possibility of heating a plasma to a sufficient temperature and density by laser irradiation to create a plasma in which thermonuclear

[1] W. I. Linlor, *Appl. Phys. Letters*, **3**:210 (1963).
[2] A. F. Haught and D. H. Polk, *Phys. Fluids*, **9**:2047 (1966).

Table 1.22.1 PROPERTIES OF THE SOLAR WIND: THE SOLAR PLASMA AT 1 AU

Frequency of appearance	Always present.
Degree of ionization	Believed to be fully ionized, but flux of neutrals not yet measured.
Composition	Mainly hydrogen, with a few percent helium. Abundance of heavier elements not measured.
Direction of bulk flow	Radially outward within experimental error of $\pm 10°$ in and normal to ecliptic plane.

	Quiet times	Disturbed times
Density	3–10 ion pairs/cm^3	20–40 ion pairs/cm^3
Bulk speed of plasma	300–400 km/s	800 km/s
Ion temperature	$\sim 8 \times 10^4$ °K	$\sim 3 \times 10^5$ °K
Proton energy associated with bulk-plasma speed	~ 600 eV	$\sim 3{,}000$ eV
Electron energy associated with bulk-plasma speed	~ 0.3 eV	~ 1.5 eV
Magnetic field intensity	$(3\text{–}8) \times 10^{-5}$ G	$(10\text{–}30) \times 10^{-5}$ G
Magnetic field direction	Average is at spiral angle of about 45° with sun-earth line, directed toward sun or away from sun.	
Electron-cyclotron frequency		~ 500 s^{-1}
Proton-cyclotron frequency		~ 0.3 s^{-1}
Energy density of magnetic field	60 eV/cm^3 (5γ field)	1,000 eV/cm^3 (30γ field)
Thermal-energy density	40–150 eV/cm^3	500–1,500 eV/cm^3
Energy density associated with bulk speed	6,000–10,000 eV/cm^3	100,000 eV/cm^3
Energy flux	~ 0.5 erg/(cm^2)(s)	~ 15 erg/(cm^2)(s)
Alfvén speed in the solar plasma	30–70 km/s	70–120 km/s
Ratio of plasma bulk speed to Alfvén wave speed	5 : 14	4 : 7

Derived parameters for typical conditions

Debye length	10 m
Plasma frequency	10^5 s^{-1}
Electron self-collision time	4×10^5 s
Mean free path (electrons and protons)	7×10^9 m

[1] SOURCE: Kinsey A. Anderson, *Ann. Rev. Nucl. Sci.*, **16**:291 (1966).

reactions occur.[1] These latter studies have been made possible through the development of lasers capable of delivering 1,000 J of energy in their output pulse. Such lasers have been used to produce plasmas with temperatures of about 5 keV.

PART THREE: MEASUREMENT OF PLASMA PROPERTIES

Once a plasma has been created (or even while it is being created) in the laboratory or in nature, it usually is desirable to be able to measure a variety of the properties of that plasma.

Very few laboratory plasmas behave initially in the manner predicted by their designer. The large number of degrees of freedom in a plasma makes it virtually impossible to take into account all the essential parameters in the theory or design of the experiment. There is a degree of opportunism in many plasma experiments. That is, sometimes a theoretical analysis or a plausible physical argument serves as the basis for design of a plasma experiment. The experiment may behave as predicted, but generally it will not. This discrepancy between the predicted and actual outcome of experiments has been one of the prime movers in plasma research. Frequently the outcome of the actual experiment suggests new theoretical models, or more importantly, results in the discovery of new phenomena, which can then be analyzed.

As a state of matter, plasma is a complex medium, and the measurement of its properties in a clear and unambiguous manner is correspondingly difficult. For example, the location of a solid, liquid, or ordinary gas is generally a trivial parameter to determine; however, for a plasma it is sometimes impossible to measure its location to the desired degree of accuracy. For instance, a pulsed magnetic mirror traps a low-density plasma in a weak magnetic field and then magnetically compresses it to higher density and temperature, well away from the walls of the vacuum chamber in which the initial plasma was formed. The plasma may last for only a few milliseconds, during which time it is not possible to insert probes into the plasma region to determine its shape and location. On the other hand, the location of a low-pressure thermionic-arc discharge (Sec. 1.17) is well known, since this steady-state plasma is confined by the glass walls of the discharge tube.

In addition to shape and location, the plasma properties to be measured

[1] This work is reviewed by M. J. Lubin and A. P. Fraas, *Sci. Amer.*, **224**:21 (June 1971).

include density, temperature, thermal conductivity, dielectric tensor, radiation rate, radiative-absorption coefficient, collision frequency, ordinary and anomalous resistivity, velocity- and configuration-space diffusion coefficients, distribution function, and the stability or instabilities of the plasma. Of course, some of these properties are related, and a measurement of one determines one or more of the others. There are many techniques for measuring plasma properties. Only a few of them can be described here.[1]

Plasma measurements are termed passive or active, depending on the technique used. *Passive* means that the plasma is not perturbed by the measurement, e.g., emitted radiation. *Active* means that the plasma is directly involved in the measurement, as when the plasma density is measured by the transmission of microwaves. For an active measurement, the plasma may be perturbed by the measurement itself, thus contributing to errors.

1.24 CURRENT AND VOLTAGE MEASUREMENTS IN PLASMAS

Since a plasma consists of mobile ions and electrons, the application of an electric field to a plasma causes these charged particles to move, resulting in an electric current. The ratio of electric field to current density is the resistivity of the plasma. Resistivity is one of the transport coefficients and depends on the collision frequency and density in simple plasma models. Thus the measured resistivity can be compared with theoretical values to verify or study plasma properties.

In some instances the current or voltage measured in the external circuit of the electrical apparatus used to create the plasma can provide useful information concerning the state of the plasma. An example of an experiment where the measured voltage provides useful information is the dynamic-pinch plasma, discussed in Sec. 3.11. In this experiment, depicted schematically in Fig. 1.24.1, the voltage is applied to the end electrodes, by means of a fast, high-current switch (ignitron), simultaneously with the initiation of a thin-cylindrical-shell

[1] Four useful books, with extensive bibliographies, on the techniques of measurement of plasma properties are:

R. H. Huddlestone and S. L. Leonard (eds.), "Plasma Diagnostic Techniques," Academic, New York, 1965.

M. A. Heald and C. B. Wharton, "Plasma Diagnostics with Microwaves," J. Wiley, New York, 1965.

H. R. Griem, "Plasma Spectroscopy," McGraw-Hill, New York, 1964.

H. R. Griem and R. H. Lovberg (eds.), "Methods of Experimental Physics: Plasma Physics," part A, Academic, New York, 1970.

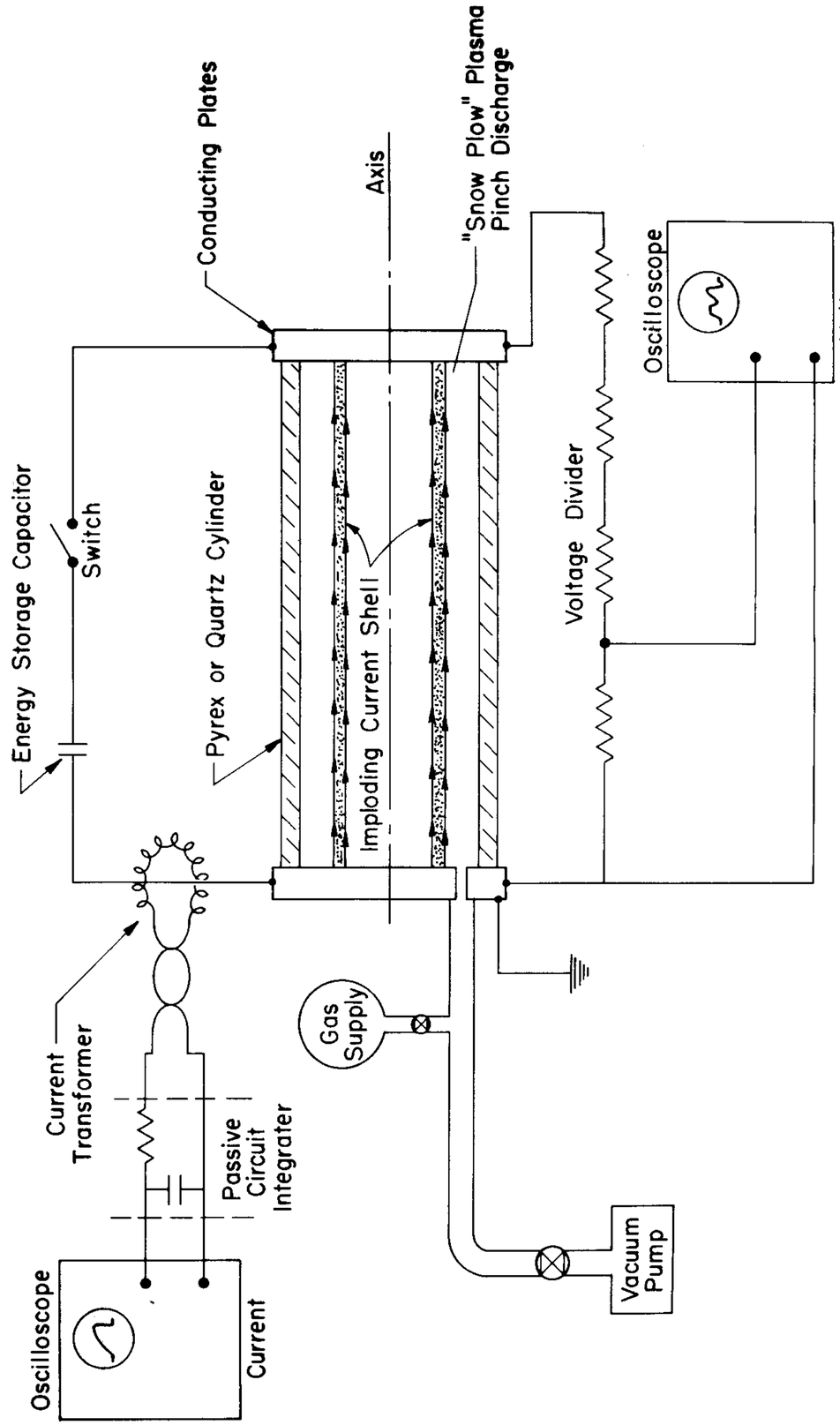


FIGURE 1.24.1
Schematic of a dynamic-pinch experiment showing method of measuring current and voltage associated with the discharge.

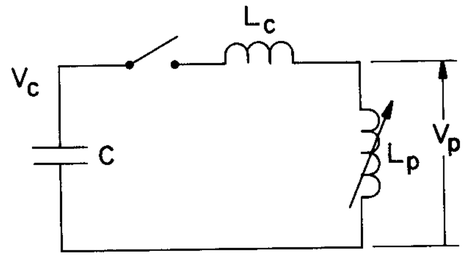


FIGURE 1.24.2
Equivalent circuit for a dynamic pinch.

discharge near the outer vacuum wall. The current shell, which is driven by the applied voltage, creates a magnetic field that exerts an inward force on the current shell, causing it to accelerate toward the axis. The rate at which the current shell collapses and the time at which it reaches the axis depend on such parameters as the initial gas pressure and applied voltage, as discussed, using a "snow-plow" model (Sec. 3.11). If the plasma behaved as predicted on the basis of this model, it could be thought of as a thin, perfect conductor moving inward according to the equation of motion (3.11.4), the solution of which is given in Fig. 3.11.3.

As far as the external electrical characteristics are concerned, the plasma pinch discharge can be replaced by a variable inductance whose value depends on the radius of the current shell. In some cases the pinch collapse time is much less than the oscillation period of the energy-storage capacitor, so that the applied voltage is constant (provided also that the pinch current is not too large) during the experiment, and in addition the resistance in the external circuit is negligible, so that the equivalent circuit for the dynamic-pinch discharge is as shown in Fig. 1.24.2. The L_c inductance represents the sum of the switch inductance, the transmission-line inductance, and whatever other external circuit feature contributes to the inductance.

The L_p inductance represents the plasma and depends on the plasma radius. If the discharge current varies slowly, as expected if $L_c > L_p$, the voltage across L_p, that is, the voltage measured across the plasma, is

$$V_p \approx I \frac{dL_p}{dt}$$

near the time of collapse. Thus the rapid deceleration of the discharge as it reaches the axis produces a large dL_p/dt and a correspondingly large voltage change. This measures the time at which the discharge reaches the axis. This time can be compared with theoretically predicted collapse times. The voltage and current for a typical pinch discharge[1] are shown in Fig. 1.24.3.

[1] For a more detailed discussion of equivalent-circuit representations of pinch plasmas, see S. Glasstone and R. H. Lovberg, "Controlled Thermonuclear Reactions," chap. 7, Van Nostrand, New York, 1960.

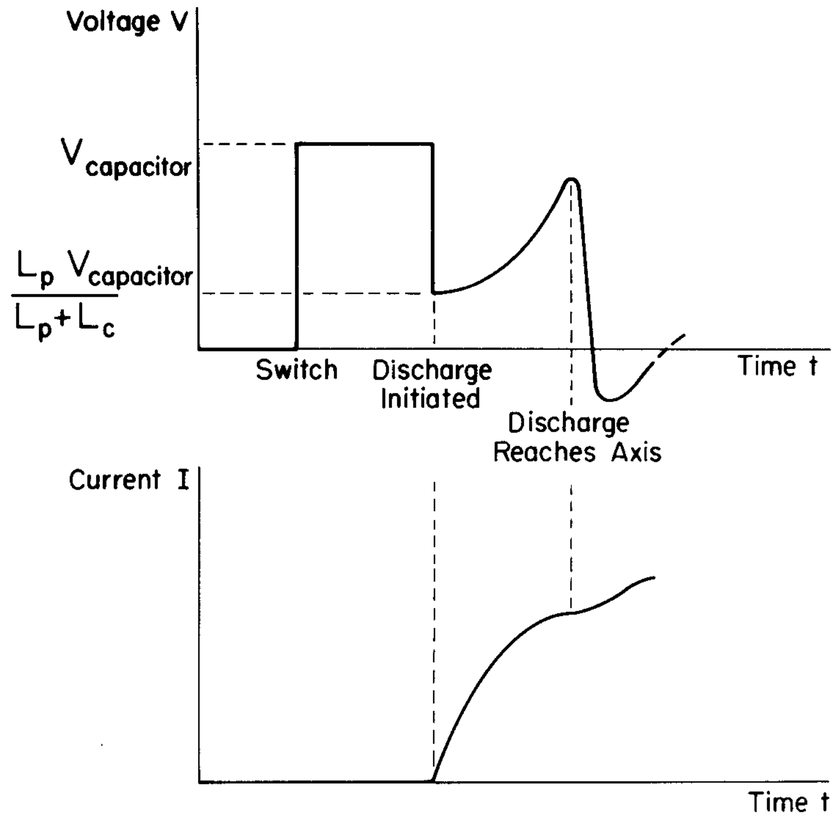


FIGURE 1.24.3
Voltage and current waveforms for a linear dynamic pinch.

The current in a plasma discharge is usually measured by means of a current transformer, which can be air-core, ferrite-core, or iron-core, depending on the rise-time requirements in a given experiment. A current transformer generates a voltage proportional to the derivative of the current through the loop. The voltage from the current transformer can be recorded directly on an oscilloscope and integrated graphically, or the signal can be integrated electronically by means of a passive (or active) circuit integrator before it is recorded. An air-core current transformer is shown schematically in Fig. 1.24.1, together with a passive-circuit integrator.

1.25 PLASMA PROBES

There are a variety of probes used to measure plasma properties. They all have the feature of being inserted into the plasma medium in order to sample plasma

properties in a local region. Most probes perturb the plasma in some way, and care must be exercised to ensure that plasma measured in the presence of the probe is the same as the plasma before the probe was introduced. The two simplest probes used in plasma measurements are electrostatic (or Langmuir) and magnetic probes.

Problem 1.24.1 In many cases the current in an electric circuit is measured by inserting a small resistance into the circuit and then measuring the voltage drop across it. Discuss the errors that would occur if this scheme were used to measure the current in a fast-pulse, high-current experiment. ////

Problem 1.24.2 What errors are likely to limit the accuracy of a current transformer in high-frequency applications (i.e., fast pulses)? ////

1.25.1 Electrostatic, or Langmuir, Probes

Figure 1.25.1 is a schematic of an electrostatic probe, together with the electric circuit needed to bias the probe to different voltages relative to the plasma while measuring the current collected by the probe. Langmuir and Mott-Smith[1] developed the theory of such probes and showed that they could be used to measure the electron density n_e, the ion density n_i, the electron temperature T_e, the plasma potential V_p, the plasma floating potential V_f (i.e., the potential of the probe for zero net current), and the random electron and ion current densities J_{er}, J_{ir}. It was through this work that such probes became known as Langmuir probes. Many improvements in their design and in the sophistication of electronic instrumentation associated with their use have been made over the years.[2] They are a simple and still useful tool in the study of laboratory plasmas.

Generally, a Langmuir probe is a small-diameter insulated tungsten wire with a small exposed region at its end to collect electrons or ions from the plasma, depending on the potential of the probe relative to the plasma. Figure 1.25.2 is a schematic of typical current-voltage characteristics obtained with a Langmuir probe in an unmagnetized plasma. If the potential of the probe is much larger than the local potential of the plasma, the probe attracts electrons and repels ions, forming a sheath region around the probe, which is electron-rich. This sheath region is a few Debye lengths thick and occurs for the same reason that a given charged particle in a plasma is shielded; namely, the range of the

[1] I. Langmuir and H. Mott-Smith, *Gen. Elec. Rev.*, **27**:449, 538, 616, 762, 810 (1924).

[2] A good review of the theory and applications of electrostatic probes is available in an article by F. F. Chen in R. H. Huddlestone and S. L. Leonard, (eds.), "Plasma Diagnostic Techniques," pp. 113–200, Academic, New York, 1965.

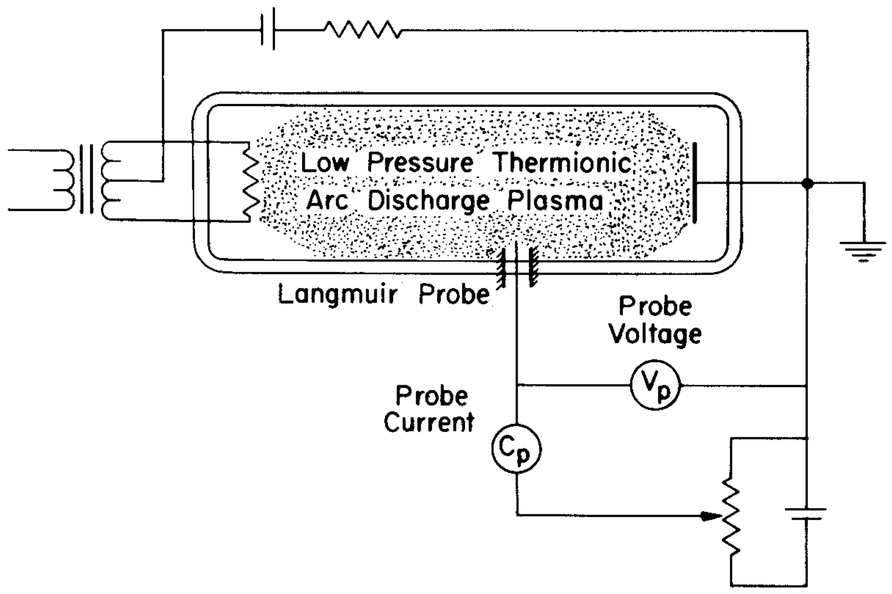


FIGURE 1.25.1
Schematic of a Langmuir electrostatic probe in a plasma showing biasing arrangement.

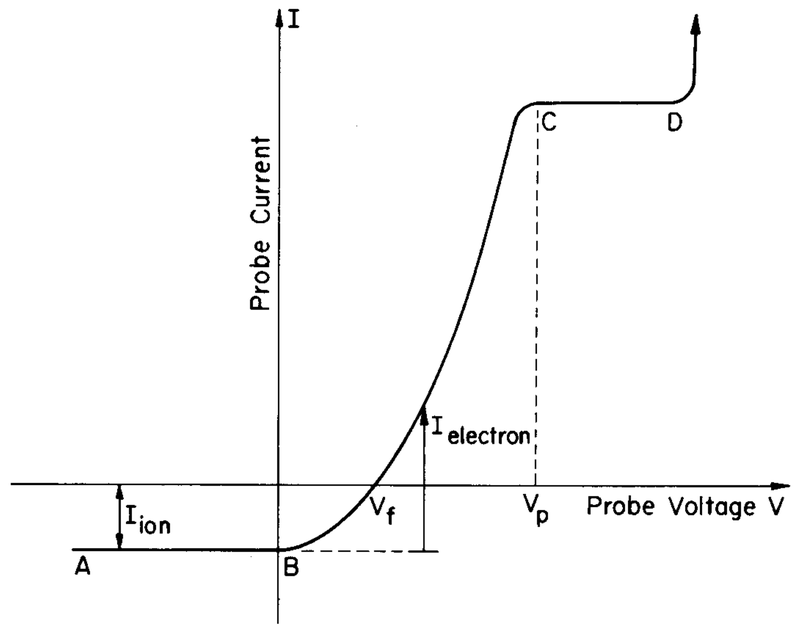


FIGURE 1.25.2
Volt-ampere characteristic of a probe in a low-density arc-discharge plasma. Diagram is not to scale since the maximum electron current is usually 1,000 times the maximum ion current.

force field (potential) is limited by the tendency of the particles of opposite sign to cluster around a given charge. Thus the influence of a probe in a plasma is limited to a region about one Debye length from the probe. The region AB in Fig. 1.25.2 represents saturated ion current for a probe biased negatively with respect to plasma floating potential. The curve BC is the true electron current, since the probe no longer repels electrons for potentials above B. The region CD is nearly horizontal, since at these probe potentials all the random electron current entering the probe sheath is collected. Above D, the voltage drop across the sheath is greater than the ionization potential, and the electrons moving toward the probe ionize the background gas, resulting in a probe arc.

1.25.2 Magnetic Probes

Magnetic probes are used to sample the magnetic fields in or around plasmas. The probe usually consists of a few turns of wire arranged in a loop, which may be a millimeter in diameter or larger, as required for the measurement. These magnetic probes operate on the principle that a time-changing magnetic field induces a voltage in the loops; the magnetic field can be determined from a measurement of the induced voltage. These magnetic probes, which sample the magnetic field in a given direction, are most generally used in fast-compression plasma experiments, such as the dynamic pinch. Figure 1.25.3 is a schematic of a magnetic probe as it might be used in a dynamic pinch. If the probe is oriented so as to be sensitive to the θ component of the magnetic field, the signal detected by the probe will be zero until the current sheet associated with the collapsing plasma passes the probe position. The magnetic field at the probe position then rapidly jumps to the value associated with the discharge current. If the probe is small and movable, it is possible to measure the location of the current shell as a function of time, and compare the results with the theoretically predicted collapse rate.

One difficulty connected with the use of magnetic probes is that they generate a voltage proportional to dB/dt rather than B. This problem is usually solved by integrating the probe signal with a passive circuit integrator, as shown in Fig. 1.25.3. The output signal of a probe system such as shown in Fig. 1.25.3 is

$$V = \frac{nA}{RC} B \times 10^{-8} \text{ volts}$$

n = number of turns in the loop
A = area of the loop, cm^2
B = magnetic field strength, Gauss
R = resistance, Ω
C = capacity, F

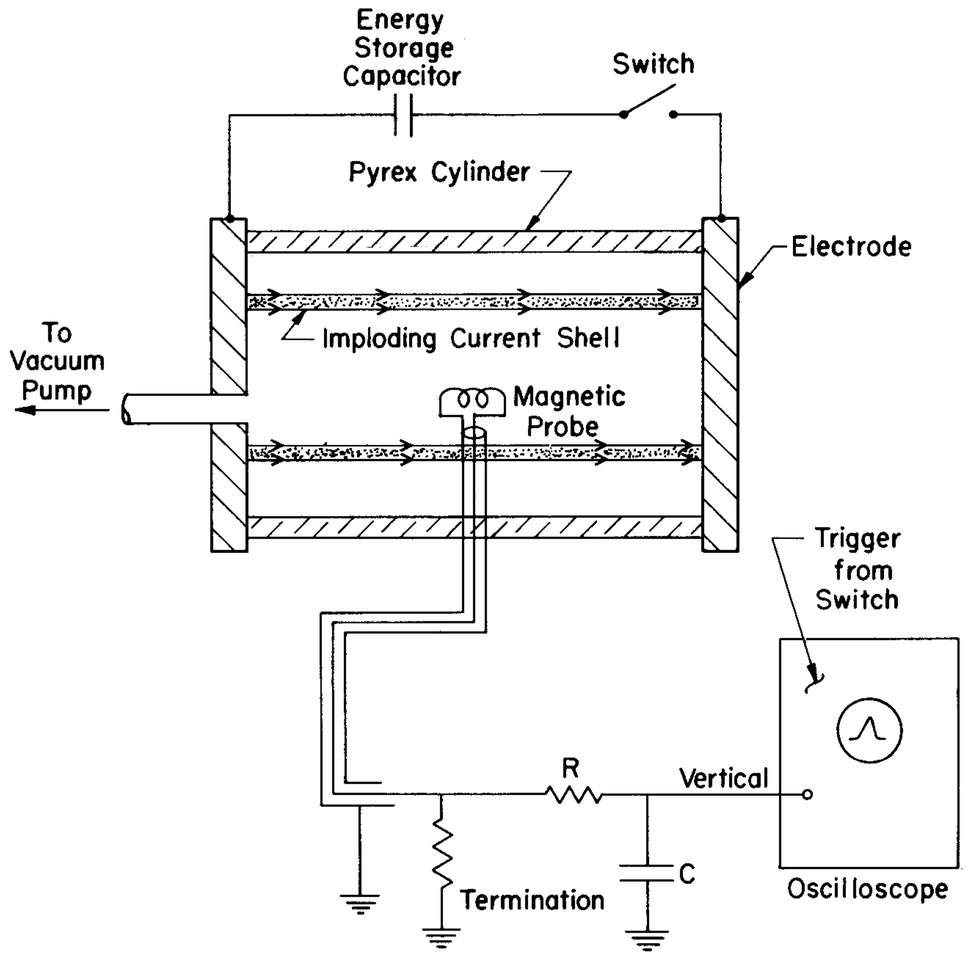


FIGURE 1.25.3
Schematic of a dynamic-pinch experiment showing a magnetic probe used to measure the time when the collapsing plasma passes a given point.

Care must be exercised to shield magnetic probes electrostatically so that electric fields associated with the discharge are not also detected. Such shielding limits the maximum frequency of magnetic field fluctuations that can be detected by the probe system. Many other details concerning magnetic probes must be understood in order to avoid errors in their use.[1]

Problem 1.25.1 Discuss the sources of errors in the use of magnetic probes to measure fields in fast and slow discharges. What errors occur as a result of the use of passive-circuit integration? ////

[1] A good review of the theory and application of magnetic probes is available in an article by R. H. Lovberg in R. H. Huddlestone and S. L. Leonard (eds.), "Plasma Diagnostic Techniques," pp. 69–112, Academic, New York, 1965.

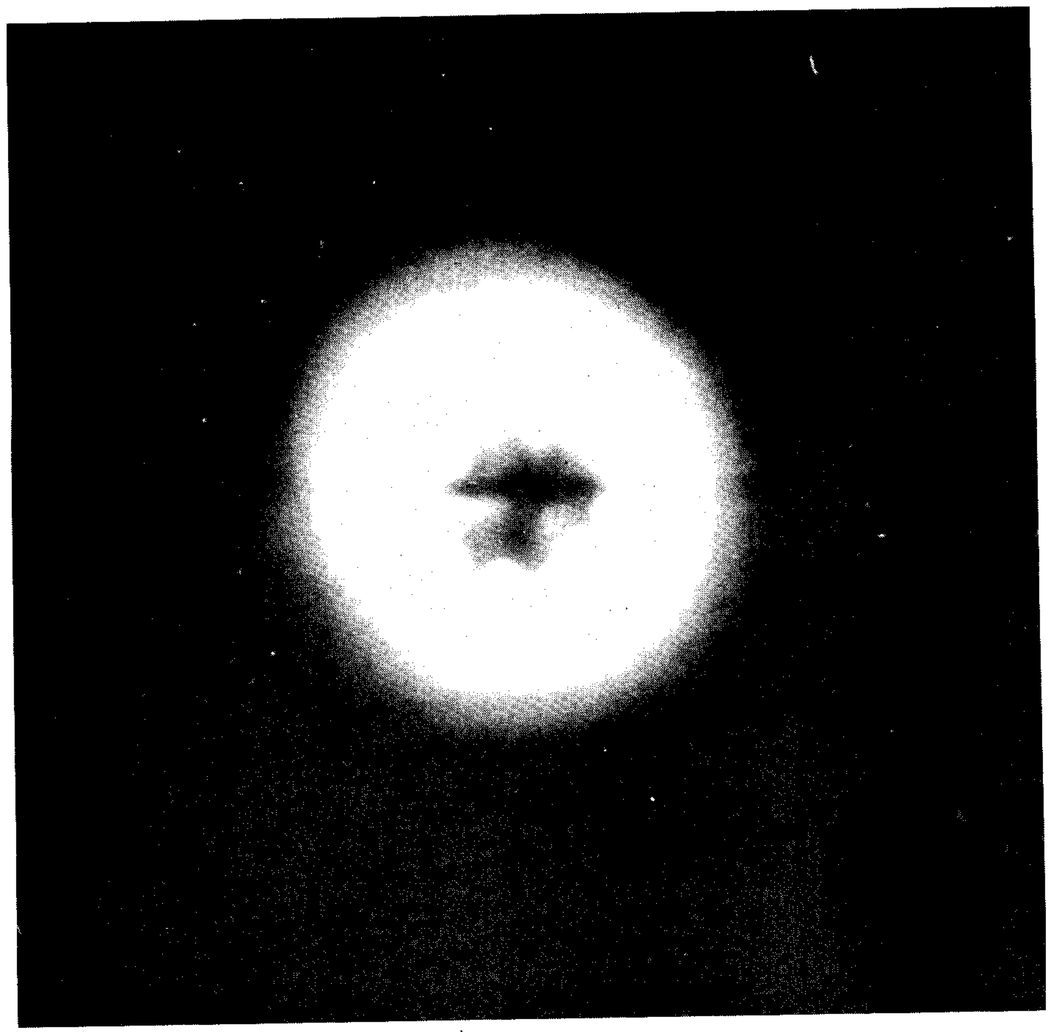


FIGURE 1.26.1
Photograph of a linear pinch, taken through a hollow electrode, looking down the hole in the center of the plasma cylinder. In this experiment an axial (directed out of the page) magnetic field was applied to "stiffen" the collapsing plasma. The inner ridges in the plasma formed when the pinch bounced against the compressed axial B field, and were associated with Rayleigh-Taylor instabilities (see Chap. 5).

1.26 OTHER METHODS OF MEASUREMENT OF PLASMA PROPERTIES

1.26.1 Photography and Atomic Spectroscopy

In many cases plasmas emit radiation from atomic transitions in the visible region of the electromagnetic spectrum with sufficient intensity (>0.1 erg/cm^2 at the film) so that the plasma can be photographed to determine its position and

motion. Figure 1.26.1 is an example where the time development of a plasma instability in a pinch experiment can be followed, and the modal pattern that develops compared with theory.[1]

The radiation that occurs from atomic transitions also occurs in vacuum ultraviolet and soft x-ray regions of the electromagnetic spectrum, depending on the temperature and density of the plasma. The presence of this line radiation and the profiles of the emission or absorption lines can be used to measure plasma density and temperature.[2]

1.26.2 Radiation Measurements

Plasmas also emit radiation in the microwave ($\lambda \approx 1$ cm) and millimeter-wave ($\lambda \approx 1.0$ mm) region of the electromagnetic spectrum. This radiation can be detected in a variety of ways, depending on its intensity and duration. For example, a magnetically confined hot electron plasma ($T_e \approx 50$ keV) has mildly relativistic electrons and emits radiation at the cyclotron frequency and its harmonics. The radiation can be detected with a conventional superheterodyne system with mixer, local oscillator, and intermediate-frequency amplifier or by a cryogenically cooled indium antimonide photodetector[3] and grating monochromator system. This latter system works best in the 8- to 0.1-mm-wavelength portion of the spectrum, and the superheterodyne system works best for wavelengths greater than 4 mm.

Lichtenberg et al.[4] used a cryogenic photodetector system to investigate the spectral distribution of synchrotron radiation from a magnetically confined plasma. Figure 1.26.2 is a plot of the absolute intensity of synchrotron radiation measured using a cryogenic photodetector system.

There are many other methods for the measurement of electromagnetic radiation emitted or absorbed by plasmas. This is an active area of experimental and theoretical research in plasma physics.[5]

1.26.3 Single-particle Measurements

Magnetically confined plasmas frequently emit charged or neutral particles, depending on the plasma density and temperature and the background gas

[1] D. Albares, N. A. Krall, and C. L. Oxley, *Phys. Fluids*, **4**:1031 (1961).
[2] The measurement of plasma properties using radiation from atomic transitions is treated thoroughly in H. R. Griem, "Plasma Spectroscopy," McGraw-Hill, New York, 1964.
[3] E. H. Putley, *Appl. Opt.*, **4**: 649 (1965).
[4] A. J. Lichtenberg, S. Sesnic, and A. W. Trivelpiece, *Phys. Rev. Letters* **13**:387 (1964).
[5] An excellent review of emission, absorption, and scattering of electromagnetic waves in plasmas is G. Bekefi, "Radiation Processes in Plasmas," Wiley, New York, 1966.

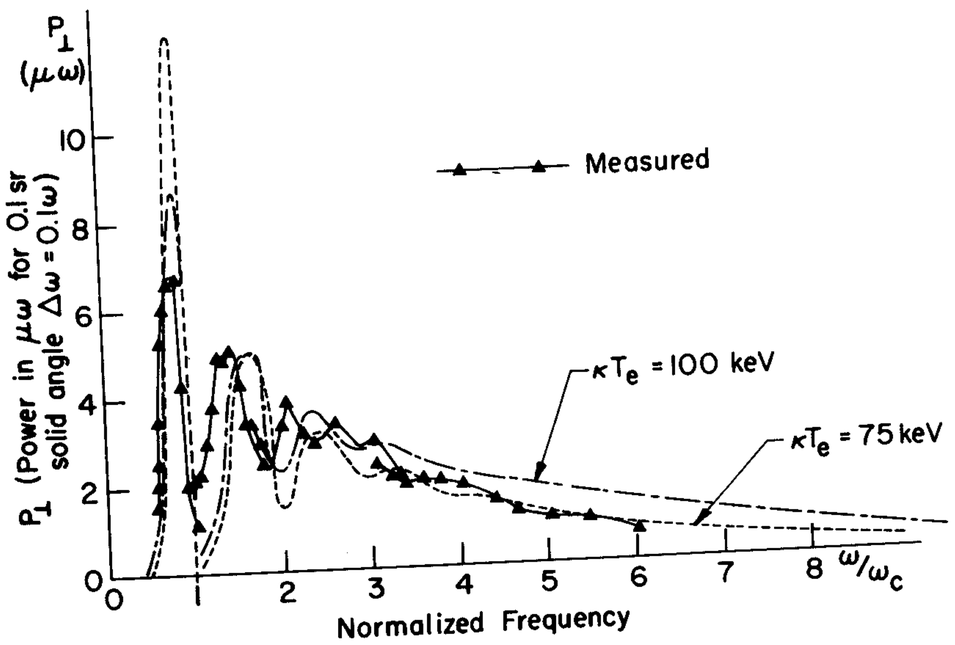


FIGURE 1.26.2
Spectrum of synchrotron radiation emitted by a hot-electron plasma in a 50-kG magnetic field, compared with theoretical spectra for a two-dimensional maxwellian distribution at 75 and 100 keV. [*After A. J. Lichtenberg, S. Sesnic, and A. W. Trivelpiece, Phys. Rev. Letters,* **13**:387 (1964).]

pressure in which the plasma resides. For example, a magnetic-mirror experiment (Chap. 5) has a velocity-space diffusion mechanism that drives particles into a loss cone, which results in their escape along the magnetic field lines. As they escape, the particles can be collected, energy-analyzed, and counted by a variety of methods, such as quadrupole mass spectrometers, crystal detectors, particle multipliers, etc. The properties of the escaping particles can be related to the properties of the plasma still in the magnetic mirror.

Neutral particles are also sometimes emitted by plasmas when conditions are right. Generally, low-density hot ions in a good vacuum lead to the proper conditions. The fast-escaping neutrals are created by the resonant charge-exchange collision of a fast ion and a slow neutral. These two particles exchange an electron, thus creating a fast neutral and a slow ion. If the mean free path for the fast neutral is greater than the size of the experiment, the fast neutral will escape where it can be analyzed and counted. This charge-exchange process is an energy-loss mechanism that must be minimized in any fusion reactor, but which leads to a useful measurement tool in the meantime.

Although there are some subtleties in the interpretation of plasma properties in terms of escaping single particles, the methods are generally quite useful and important.[1]

1.26.4 Neutrons

A high-temperature plasma of deuterium or other fusionable ions may produce high-energy neutrons.[2] These neutrons are then possible evidence of fusion reactions, and their time of emission, yield, and energy distribution can be used to provide information about the density and temperature of the plasma ion. Unfortunately, neutrons are also known to be produced by plasma instabilities, which accelerate a few deuterons to a high energy and then drive them into the wall of the experiment, generating neutrons. Photodisintegration neutrons are also produced in some hot-electron plasmas ($T_e > 1$ MeV) produced by cyclotron resonance.

1.26.5 Light Scattering

The invention and subsequent development of high-power lasers brought about a new means of measuring the density and temperature of plasmas. This new means is based on the scattering of electromagnetic waves by free electrons. The intensity of electromagnetic radiation scattered from a small-amplitude ($eE\lambda/2\pi m_0 c^2 \ll 1$) electromagnetic wave by a free electron is given by the Thomson cross section ($\sigma = \frac{8}{3}\pi r_0^2$). If the scattering electron is moving with a low velocity ($v \ll c$), the scattered radiation is doppler-shifted to a higher or lower frequency, depending on the angle of observation. This doppler shift of the scattered radiation is used to measure the temperature or velocity distribution of dense low-temperature ($T_e < 1.0$ keV) plasmas[3] (by measuring the doppler broadening of the laser line width for radiation scattered from a laser beam by a plasma). A maxwellian plasma-distribution function produces a gaussian-shaped spectral distribution of scattered radiation with a width proportional to the plasma temperature.

For plasmas with electron temperatures greater than 10 keV, the spectrum of scattered radiation is not gaussian in shape, but rather is skewed and shifted

[1] A good description of the useful methods of single-particle measurements for determining plasma properties is given by J. E. Osher, in R. H. Huddlestone and S. L. Leonard (eds.), "Plasma Diagnostic Techniques," Academic, New York, 1965.

[2] F. H. Coensgen, W. F. Cummins, R. E. Ellis, and W. E. Nexen, Decay Modes of a Dense Plasma in a Magnetic Well, Proceedings of the Conference on Plasma Physics and Controlled Nuclear Fusion Research, Novosibirsk, U.S.S.R., 1968, vol. II, p. 225.

[3] A good review of the methods of laser-scattering diagnostics of plasmas is given by H.-J. Kunze, in W. Lochte-Holtgreven (ed.), "Plasma Diagnostics," North Holland Publishing Company, Amsterdam, 1968.

to shorter wavelengths. The wavelength of radiation scattered by a given high-velocity electron in a particular direction depends only on that electron's velocity. However, in a laboratory-size hot-electron plasma ($T_e > 10$ keV), electrons will pass completely through the scattering region during the observation time, and as a result there is a correction to the intensity of scattered radiation that must be taken into account in determining the number of electrons in each velocity class.[1]

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2

THERMODYNAMICS AND STATISTICAL MECHANICS OF EQUILIBRIUM PLASMAS

A plasma is a collection of a large number of charged particles. Collections of many particles are usually treated statistically as many-body systems. The simplest many-body system is a set of noninteracting point particles, i.e., an ideal gas. The equilibrium statistical mechanics of an ideal gas yields an equation of state

$$p = n\kappa T \tag{2.1}$$

where

n = density of particles
p = pressure
T = temperature
κ = Boltzmann's constant

and the mean internal energy per degree of freedom per particle

$$W = \frac{\kappa T}{2} \tag{2.2}$$

It is obvious that a rarefied real gas closely approximates an ideal gas because the interactions between particles are infrequent, the average spacing between particles is much greater than the atomic dimension, and the duration of

a collision is much less than the time between collisions. The equation of state of a real gas is usually expressed as a power series in the density:

$$p = n\kappa T[1 + nB(T) + n^2C(T) + \cdots] \tag{2.3}$$

where the second term includes the interaction of pairs of molecules, and so on. The coefficients B, C, ... are called the *virial coefficients*.

A plasma also closely resembles an ideal gas. This is not so obvious, since the electrostatic potential of each charged particle extends a Debye length λ_D, a distance much greater than the distance between particles ${n_0}^{-1/3}$, and consequently the particles are interacting at all times, rather than for only a brief collision period. Even so, a plasma does resemble an ideal gas in many ways. Although all the plasma particles interact all the time, the interaction energy $\langle W_{ij} \rangle$ between plasma particles is on the average so weak that the thermodynamic functions of a plasma can be expressed by virial expansion about the thermodynamic functions of an ideal gas with $\langle W_{ij} \rangle / \kappa T$ as the expansion parameter.

This chapter explores the ways in which the plasma gas differs from an ideal gas. These differences arise from the phenomenon of correlations between plasma particles, which contribute to the internal energy of the plasma, as well as affecting the free energy and the equation of state. The term correlations means, roughly, the tendency of particles of like charges to spend as little time near each other as possible, and the clustering tendency of unlike charges. In an ideal gas, particles are uncorrelated, since each particle is oblivious of its neighbors, while in real gases, particles are at least weakly correlated.

Throughout this text the plasma is assumed to be a classical system; i.e., it obeys Boltzmann statistics. This means that degeneracy is ignored and that the particle motion can be described within the framework of classical mechanics. The particles interact through the Coulomb force, $F_{12} = -\nabla W_{12}$, where the Coulomb potential energy of particles 1 and 2 is

$$W_{12} = \frac{q_1 q_2}{|\mathbf{x}_2 - \mathbf{x}_1|} \tag{2.4}$$

The classical approximation requires that the de Broglie wavelength λ be less than the mean spacing between the particles ${n_0}^{-1/3}$, and also requires that λ be less than the mean distance of closest approach for a gas of Coulomb particles with kinetic energy specified by a temperature T. The mean distance of closest approach is $e^2/\kappa T$. Thus

$$\lambda = \frac{h}{mv} \ll {n_0}^{-1/3}; \qquad \lambda \ll \frac{e^2}{\kappa T} \tag{2.5}$$

are the criteria for the classical approach. In Eq. (2.5), mv is particle momentum. If the first of these conditions is satisfied, the interaction of near-neighbor particles (e.g., binary collisions) can be treated classically. If the second condition is satisfied, degeneracy can be neglected, and Maxwell-Boltzmann statistics can be used in place of either Fermi-Dirac or Bose-Einstein statistics.

2.1 THE PLASMA PARAMETER

In practice it is impossible to determine the thermodynamic properties of a nonideal system unless there is a small parameter that describes the deviation of a nonideal system from an ideal system. In a plasma this small parameter is the ratio of the mean distance of closest approach to the average spacing between the particles. The plasma parameter, which is related to this ratio, is defined by

$$g = \left(\frac{8\pi e^2}{\kappa T}\right)^{3/2} {n_0}^{1/2} \tag{2.1.1}$$

Equation (2.1.1) can be written in terms of the Debye length.

$$\lambda_D = \left(\frac{\kappa T}{8\pi n e^2}\right)^{1/2}$$

Thus the plasma parameter can be expressed as

$$g = \frac{1}{n_0 {\lambda_D}^3} \tag{2.1.2}$$

This illustrates why the plasma approximation is referred to as "many particles in a Debye sphere." When there are many particles in a Debye sphere, g is small and the average potential energy of a plasma particle is much less than its average kinetic energy. To the extent that the interaction energy can be neglected, a plasma behaves as an ideal gas, and the thermodynamic functions can be expanded in terms of the plasma parameter g. It will now be demonstrated that this parameter is in fact the measure of the departure of the thermodynamic functions of a plasma from those of an ideal gas.

Problem 2.1.1 Show that the classical approximation fails at high as well as at low temperature. What is the temperature range in electron volts for the classical approximation to be valid in a plasma of density (*a*) $n = 10^{10}$ cm^{-3}, (*b*) 10^{13} cm^{-3}, and (*c*) $n = 10^{18}$ cm^{-3}? ////

2.2 GIBBS DISTRIBUTION AND CORRELATION FUNCTIONS

Consider a plasma of N particles, $N/2$ ions, and $N/2$ electrons. If the plasma is in thermal equilibrium with a heat bath at temperature T, the probability of finding the N particles at locations $(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N)$ is given by the Gibbs distribution

$$D(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) = \frac{1}{Z} \exp\left(- \frac{\sum_k \sum_{i>k} W_{ik}}{\kappa T} \right) \tag{2.2.1}$$

where

$$W_{ik} = \frac{q_i q_k}{|\mathbf{x}_i - \mathbf{x}_k|} + \phi_{\text{ext}}$$

is the potential,

$$Z = \int \exp\left(- \frac{\sum_k \sum_{i>k} W_{ik}}{\kappa T} \right) d\mathbf{x}_1 \cdots d\mathbf{x}_N$$

is the partition function, and ϕ_{ext} includes any potential energy due to external sources, e.g., the gravitational energy. D is the probability density of the entire system of particles. The probability density of finding a particle labeled 1 at $\mathbf{x}_1$ is given by

$$F_1(\mathbf{x}_1) = \int D \, d\mathbf{x}_2 \cdots d\mathbf{x}_N \tag{2.2.2}$$

In the absence of any external forces, there must be no preferred location for a particle, and F_1 is constant.

$$F_1 = \frac{1}{V}$$

where V is the volume of the system.

Similarly, the probability density of particle 1 being at $\mathbf{x}_1$ and particle 2 being at $\mathbf{x}_2$ is

$$F_2(\mathbf{x}_1, \mathbf{x}_2) = \int D \, d\mathbf{x}_3 \, d\mathbf{x}_4 \cdots d\mathbf{x}_N \tag{2.2.3}$$

and the s particle distribution is

$$F_s(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_s) = \int D \, d\mathbf{x}_{s+1} \cdots d\mathbf{x}_N \tag{2.2.4}$$

The distributions $F_1, F_2, \ldots, F_s$ are called *reduced distributions*. The joint probability density $F_{s>1}$ contains the effects of particle interactions. When

$W_{ik} \to 0$, the presence of a particle at $\mathbf{x}_2$ has no effect on the probability of a particle being at $\mathbf{x}_1$. In that case

$$F_s \xrightarrow{W_{ik}=0} F_1(\mathbf{x}_1) \, F_1(\mathbf{x}_2) \cdots F_1(\mathbf{x}_s) = V^{-s}$$

The Mayer[1] cluster expansion of F_s exhibits explicitly the effects of particle interactions:

$$\begin{aligned} F_2(\mathbf{x}_1, \mathbf{x}_2) &= [1 + P_{12}(\mathbf{x}_1, \mathbf{x}_2)] F_1(\mathbf{x}_1) F_1(\mathbf{x}_2) \\ F_3(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) &= [1 + P_{12}(\mathbf{x}_1, \mathbf{x}_2) + P_{12}(\mathbf{x}_2, \mathbf{x}_3) \\ &\quad + P_{12}(\mathbf{x}_1, \mathbf{x}_3) + T_{123}] F_1(\mathbf{x}_1) F_1(\mathbf{x}_2) F_1(\mathbf{x}_3) \\ &\vdots \end{aligned} \tag{2.2.5}$$

The function $P_{12}(\mathbf{x}_1, \mathbf{x}_2)$ is called the *two-particle correlation function.* It vanishes if the interactions vanish, and it contains the information that repelling particles have lower probability of being near neighbors than do attracting particles. Because the Coulomb force is a central force, it follows that

$$P(\mathbf{x}_1, \mathbf{x}_2) = P(|\mathbf{x}_2 - \mathbf{x}_1|)$$

In the plasma limit the interaction is weak; this suggests an approximation scheme in which

$$T_{123} \ll P_{12} \ll 1$$

This scheme is the basis of existing theories of the plasma state.

The complete Gibbs distribution depends on the position and velocity coordinates and is given by

$$\begin{aligned} &D^*(\mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N) \\ &\quad = \frac{1}{Z^*} \exp\left(- \frac{\sum_i \frac{1}{2} m_i v_i^2}{\kappa T} \right) \exp\left(- \frac{\sum_{i>k} \sum_k W_{ik}}{\kappa T} \right) \end{aligned}$$

However, this distribution is clearly separable in the velocity coordinates of the N particles.

$$D^* = \frac{\prod_i \exp(-m v_i^2 / 2\kappa T)}{[\int d\mathbf{v}_1 \exp(-m v_1^2 / 2\kappa T)]^N} D(\mathbf{x}_1, \ldots, \mathbf{x}_N)$$

This implies that there are no velocity-space correlations, so that the reduced distributions defined by Eqs. (2.2.2) to (2.2.4) are

$$\begin{aligned} F_1^*(\mathbf{x}_1, \mathbf{v}_1) &= f_1(\mathbf{v}_1) F_1(\mathbf{x}_1) \\ F_2^*(\mathbf{x}_1, \mathbf{x}_2, \mathbf{v}_1, \mathbf{v}_2) &= f_1(\mathbf{v}_1) f_1(\mathbf{v}_2) F_2(\mathbf{x}_1, \mathbf{x}_2) \\ &\vdots \end{aligned}$$

[1] J. E. Mayer and M. G. Mayer, "Statistical Mechanics of Fluids," Wiley, New York, 1940, and H. S. Green, *Nucl. Fusion*, 1:69 (1961).

where $$f_1(\mathbf{v}_1) = \left(\frac{m}{2\pi\kappa T}\right)^{3/2} \exp\left(-\frac{m{v_1}^2}{2\kappa T}\right)$$

Thus a nonrelativistic plasma is an ideal gas in velocity space and deviations from ideal-gas behavior are determined by the spatial correlations $F_2(\mathbf{x}_1, \mathbf{x}_2)$; this is the reason for developing the equilibrium statistical theory in terms of $D(\mathbf{x}, \mathbf{v})$. Note that, for a relativistic plasma, the magnetic fields produced by a moving charge cannot be ignored, and a velocity-dependent term appears in the interaction potential, giving rise to velocity-space correlations.[1]

2.3 TWO-PARTICLE CORRELATIONS IN AN EQUILIBRIUM PLASMA

The correlation functions P_{12} can be computed explicitly from the Gibbs ensemble by making the approximation $T_{123} \ll P_{12}$ and only taking the two-particle correlations into account in obtaining the higher distribution F_3. A chain of differential equations for F_s is obtained by differentiating (2.2.4).

$$\frac{\partial F_s}{\partial \mathbf{x}_1} = -\frac{1}{\kappa T}\sum_{k=2}^{s} F_s \frac{\partial W_{1k}}{\partial \mathbf{x}_1} - \frac{1}{\kappa T}\sum_{\alpha} N_\alpha \int \frac{\partial W_{1\alpha}}{\partial \mathbf{x}_1} F_{s+1}(\mathbf{x}_1, \ldots, \mathbf{x}_s, \mathbf{x}_{s+1}^{\alpha})\, d\mathbf{x}_{s+1}^{\alpha} \tag{2.3.1}$$

where $\sum_\alpha$ is over charge species (particle α can be an ion or an electron), and N_α is the number of particles of each species. The particle labeled $k = s + 1$ is indistinguishable from the other particles of the same species. Consequently, there are $N_\alpha - s$ identical integrals over $d\mathbf{x}_{s+1}$, where N_α denotes the number of particles of type α. The integration over $d\mathbf{x}_{s+2} \cdots d\mathbf{x}_N$ gives F_{s+1}. Thus the identity of particles of the same species has been used to replace a sum $\sum_{s+1}^{N}$ by $\sum_\alpha N_\alpha$, assuming $N_\alpha \gg s$. If the system includes $N/2$ ions and $N/2$ electrons, then $N_\alpha = N/2$ and $\sum_\alpha$ consists simply of two terms, one in which particle $s + 1$ is an ion and one in which particle $s + 1$ is an electron. A general case, however, might include $N/2$ electrons but more than one type of ion.

The $s = 2$ equation is (set $N_\alpha - s \approx N_\alpha$)

$$\frac{\partial F_2}{\partial \mathbf{x}_1} + \frac{1}{\kappa T} F_2 \frac{\partial}{\partial \mathbf{x}_1}\frac{q_1 q_2}{|\mathbf{x}_1 - \mathbf{x}_2|} + \frac{1}{\kappa T}\sum_{\alpha} N_\alpha \int F_3 \frac{\partial}{\partial \mathbf{x}_1}\frac{q_1 q_\alpha}{|\mathbf{x}_1 - \mathbf{x}_\alpha|}\, d\mathbf{x}_\alpha = 0 \tag{2.3.2}$$

[1] For further details on the Gibbs distribution, see L. D. Landau and E. M. Lifshitz, "Statistical Mechanics," Pergamon, London, 1958.

Equation (2.3.2) can be easily generalized to include a variety of ion types by introducing N_α, the number of particles of type α. Introducing the expansion of F_2 and F_3 given in Eq. (2.2.5), and approximating $T_{123} \ll P_{12} \ll 1$, Eq. (2.3.2) becomes

$$\frac{\partial P_{12}}{\partial \mathbf{x}_1} + \frac{1}{\kappa T}\frac{\partial}{\partial \mathbf{x}_1}\frac{q_1 q_2}{|\mathbf{x}_2 - \mathbf{x}_1|} + \frac{1}{\kappa T}\sum_{\alpha} \bar{n}_\alpha \int [P_{12}(\mathbf{x}_2, \mathbf{x}_\alpha) + P_{12}(\mathbf{x}_1, \mathbf{x}_\alpha)] \frac{\partial}{\partial \mathbf{x}_1} \cdot \frac{q_1 q_\alpha}{|\mathbf{x}_1 - \mathbf{x}_\alpha|}\, d\mathbf{x}_\alpha = 0 \tag{2.3.3}$$

where $\bar{n}_\alpha (= N_\alpha/V)$ is the mean density of particles of charge species α in the plasma. To solve Eq. (2.3.3), the Fourier transform method is used. For central forces, the correlation function depends only on the separation $|\mathbf{x}_1 - \mathbf{x}_2|$ of the particles. In that case, the Fourier transform of the pair correlation function can be defined as

$${P_{12}}^k \equiv \int \exp[i\mathbf{k} \cdot (\mathbf{x}_2 - \mathbf{x}_1)] P_{12}(|\mathbf{x}_1 - \mathbf{x}_2|)\, d\mathbf{x}_1$$

If it is assumed that the correlation function has the same magnitude for an electron-ion, ion-ion, or electron-electron pair, i.e., that

$$P_{12} = -\frac{q_1 q_2 |P_{12}|}{|q_1 q_2|} \tag{2.3.4}$$

the correlation function $|P_{12}|$ can be removed from the sum over particle types. This is a reasonable assumption, at least to lowest order in g, because P_{12} must be positive for unlike charged particles and negative for like charges. Further, Eq. (2.3.4) implies a symmetry in the correlations, which can be verified a posteriori from the solution, to the order of this calculation. The Fourier transform of Eq. (2.3.3) thus gives

$${P_{12}}^k + 4\pi \frac{1}{\kappa T}\frac{q_1 q_2}{k^2} + 4\pi {P_{12}}^k \frac{1}{k^2 \kappa T}\sum_{\alpha} \bar{n}_\alpha {q_\alpha}^2 = 0$$

If the plasma is charge-neutral, $\sum_\alpha \bar{n}_\alpha = 2n_e$, where n_e is the electron density. Then, inverting the transform gives

$$P_{12}(|\mathbf{x}_2 - \mathbf{x}_1|) = -\frac{q_1 q_2}{\kappa T}\frac{\exp\left(-\dfrac{|\mathbf{x}_2 - \mathbf{x}_1|}{\lambda_D}\right)}{|\mathbf{x}_2 - \mathbf{x}_1|} \tag{2.3.5}$$

where $$\lambda_D = \left(\frac{\kappa T}{8\pi n_e e^2}\right)^{1/2}$$

is again called the Debye length.[1] Equation (2.3.5) with λ_D as in Eq. (2.3.5*a*) gives the two-body correlation function and shielding parameter for a many-component plasma.

The correlation functions can be written in terms of the plasma parameter g, confirming that g is the expansion parameter when the cluster expansion is applied to a plasma.

$$P_{12} = -\frac{q_1 q_2}{q^2}\frac{g}{8\pi}\frac{\exp(-|\mathbf{R}|/\lambda_D)}{|\mathbf{R}|/\lambda_D} \qquad (2.3.6)$$

where $\mathbf{R} \equiv \mathbf{x}_2 - \mathbf{x}_1$.

Equation (2.3.6) verifies that the correlations are weak if g is small [that is, $P_{12} \approx 0(g)$]. This equation for pair correlations also shows that the correlation is proportional to the sign of the coupling between correlated particles, and that the correlations vanish if either $n \to 0$ or if $T \to \infty$. Equation (2.3.6) neglects terms of order g^2, but to order g the magnitude of P_{12} is symmetric, being the same for like or unlike particles of equal charge, verifying assumption (2.3.4).

So the probability of finding particle 1 at $\mathbf{x}_1$ and particle 2 at $\mathbf{x}_2$ is

$$F_2 = \frac{1}{V^2}\left[1 - \frac{q_1 q_2}{\kappa T}\frac{\exp\left(-\dfrac{|\mathbf{x}_2 - \mathbf{x}_1|}{\lambda_D}\right)}{|\mathbf{x}_2 - \mathbf{x}_1|}\right] \qquad (2.3.7)$$

Equation (2.3.7) is not valid when P_{12} is of order unity. However, Eq. (2.3.6) predicts $|P_{12}| > 1$ only within a small volume $R^3 \approx g^3 {\lambda_D}^3$. This volume is small compared with the interaction volume ${\lambda_D}^3$; exact knowledge of P_{12} within this volume is unnecessary for accurate calculations of the thermodynamic functions.

The correlation function (2.3.6) has the following properties:

1. $P_{12} > 0$ when particles 1 and 2 have opposite signed charges.
2. $P_{12} < 0$ when particles 1 and 2 have the same signed charge.
3. P_{12} is exponentially small when the particles are more than a Debye length apart.
4. The approximation $P_{12} \ll 1$ is valid as long as $R/\lambda_D \gg g$.

The first two properties simply remind us that oppositely charged particles attract and like charges repel each other. The third property is due to Debye shielding. Much beyond a separation λ_D, particle 2 is distributed independently of particle 1. This happens because all the *other* plasma particles are correlated

[1] A generalization of the Debye length is possible if the electrons and ions are each assumed to be distributed according to a Gibbs distribution but with the electron temperature not equal to the ion temperature. In that case

$$\lambda_D \equiv \left(\sum_\alpha \frac{4\pi \bar{n}_\alpha q_\alpha^2}{\kappa T_\alpha}\right)^{-1/2} \qquad (2.3.5a)$$

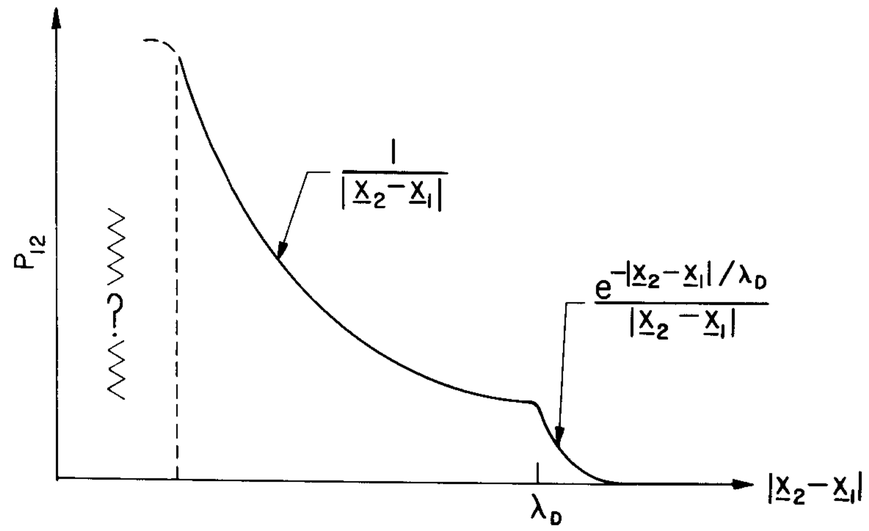


FIGURE 2.3.1
The two-particle correlation function P_{12} in a plasma as a function of relative separation of the particles.

with particle 1, which is surrounded by a cloud of radius λ_D, with a net charge in the cloud of $q = -q_1$.

The approximations used in terminating the Mayer expansion ($P_{12} \ll 1$) are certainly valid outside the Debye sphere. If the plasma parameter g is small, the correlations will continue to be small and the Mayer expansion valid, even when $|\mathbf{x}_1 - \mathbf{x}_2|/\lambda_D \ll 1$, and there will be a large volume within which the correlations are weak, but not exponentially weak.

$$P_{12} = -\frac{q_1 q_2}{\kappa T |\mathbf{x}_2 - \mathbf{x}_1|} \qquad |\mathbf{x}_2 - \mathbf{x}_1| \ll \lambda_D$$

The correlations are large, and the approximations used in obtaining P_{12} break down, when $|\mathbf{x}_1 - \mathbf{x}_2|$ is less than the average distance of closest approach for a gas of Coulomb particles at temperature T, $x_0 = e^2/\kappa T$. The ordering of lengths is shown in Fig. 2.3.1.

2.4 FREE ENERGY OF A PLASMA

The work done on a gas in an infinitesimal, isothermal, reversible change in its state equals the change ΔF in a quantity F, called the *free energy*, or the *Helmholtz energy*. F is related to the other thermodynamic quantities,

$$F = W - TS$$

where

W = energy of the system
T = temperature
S = entropy

In a state of thermodynamic equilibrium the free energy is a minimum.

The free energy of a plasma can be expressed in terms of the free energy of an ideal gas, with corrections proportional to the two-body correlation function.

The free energy per unit volume of a statistical system as given for a Gibbs distribution is[1]

$$F(\lambda) = -\frac{\kappa T}{V} \ln \int \exp\left(-\frac{\lambda}{\kappa T} \sum_k \sum_{i>k} W_{ik}\right) d\mathbf{x}_1 \cdots d\mathbf{x}_N \qquad (2.4.1)$$

where λ is a parameter which equals unity for a fully interacting plasma. The reason for introducing λ is that it enables one to imagine turning on the Coulomb interaction adiabatically.

Differentiating Eq. (2.4.1) with respect to λ,

$$\frac{\partial F(\lambda)}{\partial \lambda} = \frac{1}{Z^{(\lambda)} V} \int \sum_k \sum_{i>k} W_{ik} \exp\left(-\frac{\lambda}{\kappa T} \sum_k \sum_{i>k} W_{ik}\right) d\mathbf{x}_1 \cdots d\mathbf{x}_N$$

$$= \frac{1}{2V} \int \sum_{\alpha,\gamma} N_\alpha N_\gamma W_{\alpha\gamma}(|\mathbf{x}_\gamma - \mathbf{x}_\alpha|) F_2^{(\lambda)}(\mathbf{x}_\alpha, \mathbf{x}_\gamma)\, d\mathbf{x}_\alpha\, d\mathbf{x}_\gamma \qquad (2.4.2)$$

where the remaining sum is over charge species; the α or γ particle can be an ion or an electron. F_2 is the two-body distribution defined in Eq. (2.2.4); since $\sum_{\alpha,\gamma} W_{\alpha\gamma} = 0$, it follows that, explicitly,

$$\frac{\partial F}{\partial \lambda} = \sum_{\alpha,\gamma} \frac{\bar{n}_\alpha \bar{n}_\gamma}{2V} \int W_{\alpha\gamma} P_{12}{}^{(\lambda)}(\mathbf{x}_\alpha, \mathbf{x}_\gamma)\, d\mathbf{x}_\alpha\, d\mathbf{x}_\gamma \qquad (2.4.3)$$

where $P_{12}{}^{(\lambda)} = -(q_1 q_2 \lambda/\kappa T|\mathbf{x}|)\exp(-\sqrt{\lambda}|\mathbf{x}|/\lambda_D)$. Since $W_{\alpha\gamma}$ and P_{12} are functions only of $|\mathbf{x}_\alpha - \mathbf{x}_\gamma|$, the right-hand side of (2.4.3) becomes

$$\tfrac{1}{2} \sum_{\alpha,\gamma} \bar{n}_\alpha \bar{n}_\gamma \int W(\mathbf{x}) P_{12}(\mathbf{x}) d\mathbf{x}$$

The function F is obtained by integrating Eq. (2.4.3) over λ, from $\lambda = 0$ to $\lambda = 1$. Note that if $\lambda = 0$, $F(\lambda = 0) = F_0$, the free energy of an ideal gas, since $\lambda = 0$ corresponds to a noninteracting system. Then

$$F = F_0 + \frac{1}{2} \int_0^1 d\lambda \sum_{\alpha,\gamma} \bar{n}_\alpha \bar{n}_\gamma \int W_{\alpha,\gamma} P_{12}{}^{(\lambda)}(\mathbf{x}) d\mathbf{x} \qquad (2.4.4)$$

[1] L. D. Landau and E. M. Lifshitz, "Statistical Physics," Pergamon, London, 1958.

Equation (2.4.4) can be integrated using the previous calculation of P_{12}. That calculation was valid only for $|\mathbf{x}_\alpha - \mathbf{x}_\gamma| > e^2/\kappa T$; however, the error made in neglecting $\int W(\mathbf{x}) P_{12}(\mathbf{x}) d\mathbf{x}$ inside the volume $|x| < e^2/\kappa T$ is proportional to the ratio of the volume $(e^2/\kappa T)^3$ to that of the interaction region $\lambda_D{}^3$; this ratio is the plasma parameter g. So, integrating Eq. (2.4.4) gives

$$F = F_0 - \frac{8}{3}\sqrt{\frac{\pi}{2}} \frac{e^3 n_0{}^{3/2}}{\sqrt{\kappa T}} \qquad (2.4.5)$$

A gas with short-range forces (e.g., a neutral gas) has a free energy $F = F_0 - An^2$. The plasma correction is lower power in the density. This $n^{3/2}$ correction to the free energy of a gas with long-range forces (a plasma) is called the *Debye correction*.

Problem 2.4.1 Integrate (2.4.4) and verify (2.4.5). ////

Problem 2.4.2 Show that the next correction to the free energy of a plasma is proportional to n^2. ////

Problem 2.4.3 Evaluate the correction to the free energy for a typical interstellar plasma $T_e = 1$ eV, $n_e = 1$ cm^{-3}. ////

2.5 EQUATION OF STATE OF A PLASMA

For an ideal gas, $p = nkT$, where p is the pressure. For a plasma, the equation of state can be derived from the virial theorem.

$$\langle \text{KE} \rangle = \frac{1}{2} \left\langle \sum_i \mathbf{F}_i \cdot \mathbf{x}_i \right\rangle \qquad (2.5.1)$$

where KE is the kinetic energy; F, any forces; and $\langle\ \rangle$ means the ensemble average, that is, $\langle A \rangle$ means

$$\langle A \rangle \equiv \int A D_N d\mathbf{x}_1 \cdots d\mathbf{x}_N \qquad (2.5.2)$$

The virial includes interaction forces and pressure forces exerted on the particles by the walls of the containers.

$$\sum_i \mathbf{F}_i \cdot \mathbf{x}_i = \sum_l \sum_{k>l} \frac{\partial W_{kl}}{\partial |\mathbf{x}_k - \mathbf{x}_l|} \cdot |\mathbf{x}_k - \mathbf{x}_l| + 3pV \qquad (2.5.3)$$

where $\mathbf{x}_{kl} \equiv |\mathbf{x}_k - \mathbf{x}_l|$, and V is the volume of the system. The kinetic energy is

$$\langle \mathrm{KE} \rangle = \frac{3}{2}\left(\sum_\alpha N_\alpha\right)\kappa T \qquad (2.5.4)$$

where $\sum_\alpha$ is over particles species. Clearly [see the derivation of Eq. (2.4.3)],

$$p = 2n\kappa T - \sum_{\alpha,\gamma} \frac{n^2}{6} \int \mathbf{x}_{\alpha\gamma} \cdot \frac{\partial W_{\alpha\gamma}}{\partial \mathbf{x}_{\alpha\gamma}} P_{12}(|\mathbf{x}_\alpha - \mathbf{x}_\gamma|)\, d\mathbf{x}^{\alpha\gamma} \qquad (2.5.5)$$

Problem 2.5.1 Find the equation of state for a plasma and show that

$$p = 2n\kappa T\left(1 - \frac{g}{48\pi}\right) \qquad ////$$

Problem 2.5.2 Read T. O'Neil and N. Rostoker, *Phys. Fluids*, **8**:1109 (1965) and show that the next correction to the equation of state is $0(g^2 \ln g)$. This correction comes from *triplet* correlations.

2.6 THE PLASMA AS A FLUID

If the plasma parameter g is small, the correlations in the system are small and may often be ignored. In many cases of interest in plasma physics, g is small and the correlations are ignored. However, neglecting correlations is not the same as neglecting interactions entirely. That is, the potential ϕ at a point $\mathbf{x}_i$ in a plasma should include, in addition to any external potentials, the potential ϕ_p produced at $\mathbf{x}_i$ by all the particles in the system; i.e.,

$$\phi_p = \sum_j \frac{q_j}{|\mathbf{x}_i - \mathbf{x}_j|} \qquad (2.6.1)$$

Now consider a small volume $d\mathbf{x}_j$ such that $d\mathbf{x}_j \gg 1/n_\alpha$. This volume contains a large number of particles, which are assumed to be uniformly distributed to first order in g over this small volume. The density of particles of type α at the point $\mathbf{x}_j$ is $n_\alpha(\mathbf{x}_j)$. The electric potential at $\mathbf{x}_i$ due to particles in $d\mathbf{x}_j$ at x_j is on the average given by

$$\langle d\phi_i \rangle = \sum_\alpha \frac{d\mathbf{x}_j\, q_\alpha\, n_\alpha(\mathbf{x}_j)}{|\mathbf{x}_j - \mathbf{x}_i|} \qquad (2.6.2)$$

and the total potential at the point $\mathbf{x}_i$ is

$$\phi_p(\mathbf{x}_i) = \sum_\alpha \int \frac{d\mathbf{x}_j\, q_\alpha\, n_\alpha(\mathbf{x}_j)}{|\mathbf{x}_i - \mathbf{x}_j|} \qquad (2.6.3)$$

Equations (2.6.2) and (2.6.3) are not the exact potentials because all the particles $n_\alpha(\mathbf{x}_j)(d\mathbf{x}_j)$ in the small volume $d\mathbf{x}_j$ are not at exactly the same point $\mathbf{x}_j$, as assumed in (2.5.2) and (2.5.3). But if there are many particles in $d\mathbf{x}_j$ and if $|\mathbf{x}_i - \mathbf{x}_j| \gg (d\mathbf{x}_j)^{1/3}$, the average potentials (2.6.2) and (2.6.3) are good approximations to the real potentials. The ability to do this differential calculus in a discrete system depends on the assumption that there are many particles in the interaction sphere, $n\lambda_D{}^3 \gg 1$. In terms of this average potential, the Gibbs distribution is

$$D = \frac{1}{Z} \exp\left(-\frac{\sum_i q_i \phi(\mathbf{x}_i) + W_{\text{other}}}{\kappa T}\right)$$

where ϕ is the total electrostatic potential

$$\phi(\mathbf{x}_i) = \phi_p(\mathbf{x}_i) + \phi_{\text{ext}}(\mathbf{x}_i)$$

and W_{other} includes any nonelectrostatic potentials such as a gravitational potential. The potential ϕ_{ext} includes electrostatic potentials from other sources not included in the statistical description (e.g., capacitor plates, test particles, etc.). The density of particles of type α at the point $\mathbf{x}_i$ is given by

$$n_\alpha(\mathbf{x}_i) = \bar{n}_\alpha \exp\left(-\frac{q_\alpha \phi_p + q_\alpha \phi_{\text{ext}} + W_{\text{other}}}{\kappa T}\right) \qquad (2.6.4)$$

Equation (2.6.4) is valid in the plasma approximation, which neglects discreteness, but retaining the plasma potential retains the distinctive phenomenon of shielding of bare particles.

2.7 THE IDEAL PLASMA

The plasma approximation (i.e., interaction without correlation) represents exactly an idealized plasma which can, conceptually, be "built" from a real plasma. Imagine that each real-plasma particle is divided into P equal pieces. If the real particles have density n_0, charge q_0, mass m_0, and temperature T_0, the subdivided system has the following properties:

$$n^* = Pn_0 \qquad q^* = \frac{1}{P} q_0 \qquad m^* = \frac{m_0}{P} \qquad T^* = \frac{T_0}{P}$$

The characteristics of the subdivided system, given by the Debye length, plasma frequency, correlation function, plasma parameter, free energy, and equation of state are

$$\lambda_D^* = \lambda_{D0}$$

$$\omega_p^* = \omega_{p0}$$

$$g^* = g_0 \frac{1}{P}$$

$$P_{12}^* = (P_{12})_0 \frac{1}{P}$$

$$\rho^* = \text{charge density} = \rho_0$$

$$F^* = F_0 - \frac{1}{P}\frac{8}{3}\sqrt{\frac{\pi}{2}}\frac{{q_0}^3}{\sqrt{\kappa T_0}}\,{n_0}^{3/2} \qquad p = n\kappa T\left[1 + 0\left(\frac{1}{P}\right)g_0\right]$$

If the particles are subdivided many times so that $P \to \infty$, the correlation function, the plasma parameter, the correction to the free energy, and the corrections to the equation of state, all vanish. These are the single-particle properties of the plasma. The collective properties (plasma frequency, Debye length, charge density) are not changed by the subdividing process. This system interacts fully through its average properties, but has no correlations or other "discrete" properties. It is an ideal charged fluid.

This process of subdividing the plasma particles "turns off" the correlations between particles, but retains the interactions between large aggregates of particles. If, instead, the interaction were turned off by $q \to 0$, all distinctive plasma properties would disappear (that is, $\lambda_D \to \infty$, $\omega_p \to 0$, etc.).

2.8 POTENTIAL OF A TEST PARTICLE IN A PLASMA

The shielding property of a plasma, which is obvious from the form of the two-body correlation functions, is found in the fluid approximation by calculating the electrostatic potential produced by a charge q_T placed at $x = 0$. The charge density due to the plasma particles is given by

$$\rho_q = \sum_\alpha n_\alpha q_\alpha$$

where, because of the presence of the test charge q_T, the density $n_\alpha(\mathbf{x})$ is no longer a constant ($=N/V$), but is given by

$$n_\alpha(\mathbf{x}) = \frac{N_\alpha}{V}\exp\left(-\frac{q_\alpha\phi}{\kappa T}\right) \qquad (2.8.1)$$

where ϕ is the electrostatic potential ($=q_T/x + \langle\phi_p\rangle$) at the point x_α due to the presence of the test charge. Then, with $n_\alpha = N_\alpha/V$,

$$\nabla^2\phi = -4\pi q_T\,\delta(\mathbf{x}) - 4\pi\sum q_\alpha n_\alpha \exp\left(-\frac{q_\alpha\phi}{\kappa T}\right)$$

and if the test charge is not too large

$$\exp\left(-\frac{q_\alpha\phi}{\kappa T_\alpha}\right) \approx 1 - \frac{q_\alpha\phi}{\kappa T_\alpha} + \cdots$$

Thus Poisson's equation

$$\nabla^2\phi = -4\pi q_T\,\delta(\mathbf{x}) + 4\pi\sum_\alpha \frac{n_\alpha {q_\alpha}^2\phi}{\kappa T_\alpha}$$

has the solution

$$\phi = \frac{q_T}{|\mathbf{x}|}\exp\left(-\frac{|\mathbf{x}|}{\lambda_D}\right) \qquad (2.8.2)$$

where the shielding distance is the same as that found in calculating the two-body correlations.

$${\lambda_D}^{-2} = 4\pi\sum_\alpha \frac{n_{0\alpha}{q_\alpha}^2}{\kappa T_\alpha}$$

Near the test charge the potential is just $q_T/|\mathbf{x}|$, the potential of a bare charge. Farther out the potential is $\phi_{\text{test}} + \langle\phi_p\rangle$; $\langle\phi_p\rangle = (q_T/|\mathbf{x}|)[\exp(-x/\lambda_D) - 1]$ is the polarization charge produced by the cloud attracted around the test charge.

Problem 2.8.1 Note that $F_2(\mathbf{x}_1, \mathbf{x}_2)\,d\mathbf{x}_1\,d\mathbf{x}_2$, the joint probability of particle 1 being at $\mathbf{x}_1$ in the interval $d\mathbf{x}_1$ and particle 2 being at $\mathbf{x}_2$ in the interval $d\mathbf{x}_2$, is given by $F_2(\mathbf{x}_1, \mathbf{x}_2)\,d\mathbf{x}_1\,d\mathbf{x}_2 = F_1(\mathbf{x}_1)\,d\mathbf{x}_1\,F_c(\mathbf{x}_2|\mathbf{x}_1)\,d\mathbf{x}_2$, where F_c is the conditional probability of a particle being at $\mathbf{x}_2$ is $d\mathbf{x}_2$, given that a particle is at $\mathbf{x}_1$ in $d\mathbf{x}_1$. Using Eq. (2.8.2) for the potential of a charge placed at $\mathbf{x}_1 = 0$, show that

$$F_2(\mathbf{x}_1, \mathbf{x}_2) = F_1(\mathbf{x}_1)F_1(\mathbf{x}_2)\exp\left[-\frac{q_1q_2\exp(-|\mathbf{x}_2 - \mathbf{x}_1|/\lambda_D)}{\kappa T|\mathbf{x}_2 - \mathbf{x}_1|}\right]$$

and so deduce that

$$P_{12} = -1 + \exp\left[-\frac{q_1q_2}{\kappa T|\mathbf{x}_2 - \mathbf{x}_1|}\exp\left(-\frac{|\mathbf{x}_2 - \mathbf{x}_1|}{\lambda_D}\right)\right] \qquad (2.8.3)$$

which replaces Eq. (2.3.7) as a better approximation to the equation for the two-body correlation function. Note that (2.8.3) and (2.3.7) are not too different over most of the range of $|\mathbf{x}_2 - \mathbf{x}_1|$ but give $P_{12} = -1$ for the correlation between two electrons with $|\mathbf{x}_2 - \mathbf{x}_1| \to 0$.

For points near the point charge,

$$\phi_{\text{near}} = \frac{q_T}{|\mathbf{x} - \mathbf{x}_0|} - \frac{q_T}{\lambda_D}$$

The first term is the potential of the bare test charge, the second is the plasma polarization. Far from the test charge, $\phi \to 0$, showing again the shielding effect.

The plasma potential $\phi - \phi_{\text{test}}$ is clearly

$$\langle \phi_P \rangle = \frac{q_T}{|\mathbf{x}|}\left[\exp\left(-\frac{|\mathbf{x}|}{\lambda_D}\right) - 1\right]$$

and the total polarization charge is

$$q_P = \int_0^\infty 4\pi R^2 \left(\frac{\nabla^2 \phi_p}{-4\pi}\right) dR = -q_T$$

Now q_T can be considered to be *one* of the plasma particles, singled out as a *test particle*. The calculation here shows that each plasma particle is accompanied by a Debye cloud of equal but opposite charge and of radius λ_D. The plasma can be thought of as built up of a collection of these *dressed* test particles.

Problem 2.8.2 Read E. A. Frieman and D. L. Book, *Phys. Fluids*, **6**:1700 (1963), and discuss how they were able to include close encounters to find

$$P_{12} = -1 + \exp\left(-\frac{\lambda_D}{r}\frac{g}{8\pi}\right) + \frac{\lambda_D}{r}\frac{g}{8\pi}\left[1 - \exp\left(-\frac{r}{\lambda_D}\right)\right] \qquad (2.8.4)$$

Discuss the practical differences between (2.3.7), (2.8.3), and (2.8.4). ////

Problem 2.8.3 The distribution in Eq. (2.8.1) is not correctly normalized because the test charge was left out of the partition functions. Calculate the error so introduced. ////

2.9 OTHER EXAMPLES OF THE PLASMA AS A FLUID

Knowing n_α in terms of ϕ means that the equilibrium equations can be closed, forming a set of correlationless equilibrium equations.

$$\nabla^2 \phi_{\text{total}} = -4\pi\rho_{\text{ext}} - 4\pi \sum_\alpha q_\alpha n_\alpha(\mathbf{x})$$

$$\phi_{\text{total}} = \langle \phi \rangle + \phi_{\text{ext}} \qquad (2.9.1)$$

$$n_\alpha(x) = \bar{n}_\alpha \exp\left(-\frac{q_\alpha \langle \phi \rangle + q_\alpha \phi_{\text{ext}} + W_{\text{other}}}{\kappa T}\right)$$

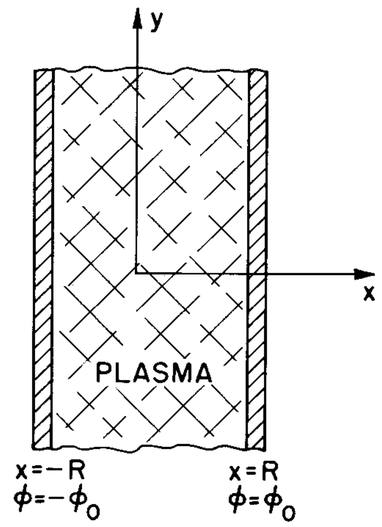


FIGURE 2.9.1
Schematic of a plasma capacitor.

W_{other} can be the potential energy due to gravity, for example; the approximations involved in this set of equations are (1) $\overline{n_1 n_2} = \bar{n}_1 \bar{n}_2$ and (2) the system is in thermodynamic equilibrium.

There are a number of simple problems that can be solved with Eq. (2.9.1), but only if further approximations are made. No matter how plasma theory is approached, there is a recurrent set of difficulties. Already, two of them have appeared:

1 The equations are correlated [leading to a chain of equations, such as Eq. (2.3.2)].
2 The equations are nonlinear (each term in the chain is nonlinear).

In Eq. (2.9.1) the theory has eliminated the first difficulty by setting $P_{12} \to 0$. The second difficulty (nonlinearity) is overcome by a weak-coupling approximation, as for a test charge, expanding the interaction potential.

$$\exp\left(-\frac{q\phi}{\kappa T}\right) \approx 1 - \frac{q\phi}{\kappa T} + \cdots$$

obtaining

$$\nabla^2 \phi = -4\pi\rho_{\text{ext}} - 4\pi \sum_\alpha q_\alpha n_{\alpha 0}\left(1 - \frac{q_\alpha \phi}{\kappa T_\alpha} - \frac{W_{\text{other}}}{\kappa T_\alpha}\right) \qquad (2.9.2)$$

If the plasma is neutral, $\sum_\alpha q_\alpha n_\alpha = 0$, the linear equation (2.9.2) can be solved in many cases. The characteristic length scale in Eq. (2.9.2) is again

$$\lambda_D = \left(\frac{8\pi q^2 n_0}{\kappa T}\right)^{-1/2}$$

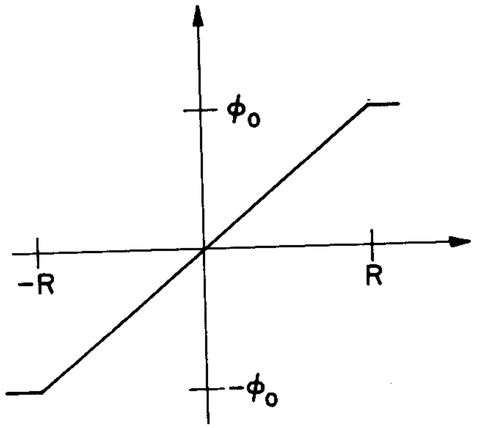


FIGURE 2.9.2
Potential as a function of position in a plasma capacitor where $R \ll \lambda_D$.

2.9.1 The Plasma Capacitor

Consider the plasma in an applied potential (Fig. 2.9.1). Inside the plasma $\rho_{ext} = 0$, and "other" potentials W_{other} are neglected. Further, consider a plasma which has the same number of electrons as ions. Then the solution to Eq. (2.9.2) with $\phi = \phi_p + \phi_0$ is

$$\phi = Ae^{x/\lambda_D} + Be^{-x/\lambda_D}$$

Defining $x = 0$ as the midplane, symmetry demands that $\phi(x = 0) = 0$; therefore $A = -B$. Including the boundary condition $\phi(x = R) = \phi_0$, the complete solution is

$$\phi = \phi_0 \frac{e^{x/\lambda_D} - e^{-x/\lambda_D}}{e^{R/\lambda_D} - e^{-R/\lambda_D}} \qquad (2.9.3)$$

If the plates are close together, $R \ll \lambda_D$, the potential is (see Fig. 2.9.2)

$$\phi_{R<\lambda_D} = \phi_0 \frac{x}{R}$$

In this limit there is no *shielding*.

If the plates are farther apart, $R \gg \lambda_D$, then

$$\phi_{R\gg\lambda_D} = \frac{e^{x/\lambda_D} - e^{-x/\lambda_D}}{e^{R/\lambda_D}} \phi_0$$

which is very small unless $x = \pm R$, because otherwise $\phi \approx \phi_0 e^{-R/\lambda_D}$, which is small when (see Fig. 2.9.3)

$$R \gg \lambda_D$$

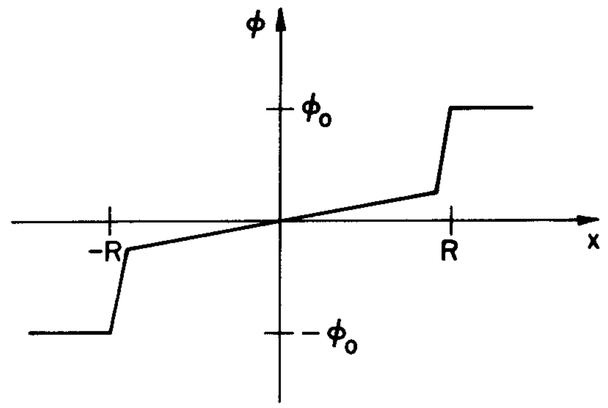


FIGURE 2.9.3
Potential as a function of position in a plasma capacitor where $R \gg \lambda_D$.

So, in a collisionless plasma (no correlations), an applied potential will be felt only within one Debye distance from the edge of the plasma. If an electric field applied to a thick plasma causes a uniform potential drop, or a uniform current flow, rather than a sheath, there must be some correlation (collisions) or other nonthermal process (instability, turbulence) operating inside the system. This ability of the plasma to shield out external potentials in the absence of nonthermal fluctuations is sometimes used as a diagnostic: If an electric field penetrates a collision-free plasma, it is an indication of the presence of strong fluctuations about the average plasma state, i.e., that the plasma is far from thermal equilibrium.

2.9.2 The Electric Field in the Ionosphere Due to Gravity

Next, consider an isothermal plasma under the influence of gravity, and take, in Eq. (2.9.2), $\rho_{ext} = \phi_0 = 0$, but take $W_{other} = -m_\alpha g R_E^2/x$, where m is the mass of the particle of type α, and R_E is the radius of the earth. The origin of coordinates has been taken at the center of the earth. Equation (2.9.2) becomes, in this case,

$$\nabla^2\phi = \frac{1}{\lambda_D^2}\phi - \sum_\alpha \frac{m_\alpha g R_E^2}{x q_\alpha \lambda_{D\alpha}^2}$$

$$\phi = \frac{1}{4\pi}\int \frac{d\mathbf{x}' \exp(-|\mathbf{x} - \mathbf{x}'|/\lambda_D)}{|\mathbf{x} - \mathbf{x}'|} \sum_\alpha \frac{m_\alpha g R_E^2}{q_\alpha \lambda_{D\alpha}^2 |\mathbf{x}'|}$$

where $\lambda_{D\alpha}{}^{-2} = 4\pi n_\alpha q_\alpha{}^2/\kappa T_\alpha$. Now since λ_D/x is very small, because $x > R_{\text{earth}} \gg \lambda_D$, then the integrand is sizable only over a small volume $\approx \lambda_D{}^3$ centered on x, and

$$\mathbf{E} = -\nabla\phi = \sum_\alpha \frac{m_\alpha g R_E{}^2}{2q_\alpha x^2}\,\hat{\mathbf{i}}_x \qquad (2.9.4)$$

The electric field in a plasma under gravity is due to pile-up of charge. If the plasma has $n_e = n_i$, then $\mathbf{E}$ points out from the earth. This is because ions pile up more due to their heavier mass, leaving a net positive charge near the earth. The field near the earth is

$$\mathbf{E} = \frac{m_i g}{2e}\,\hat{\mathbf{i}}_x$$

2.10 COULOMB ENERGY OF A PLASMA

The examples above (plasma capacitor, test-charge potential, plasma under gravity) are instructive, but the information they gave could have been obtained by viewing the plasma as a fluid with zero velocity, described by its charge density $n(x)$. Statistical theory was used only to see what approximation was necessary to neglect the plasma correlations.

There are plasma properties that have no analog in fluid theory: one example is the Coulomb energy of a plasma. This is a *correlation* property. If an equal number of electrons and ions were uniformly distributed, they would have no Coulomb energy, since no energy would be required to remove a plasma particle to infinity from such a neutral uncorrelated plasma. (In such a system an electron would be on the average as near another electron as an ion.) In fact, however, even in a neutral plasma an electron will on the average be closer to an attracting ion than a repulsive electron (correlations), and so it costs energy to remove the electron from an average potential well. The Coulomb energy (the assembly energy of a plasma) will be negative. This result will, for instance, alter the ionization energy of neutral atoms in a plasma.

To calculate the energy required to disassemble a plasma, again single out a test particle q_T at $x = 0$. The potential in the neighborhood of this charge is

$$\phi_T = q_T \frac{1}{x} e^{-x/\lambda_D}$$

The energy required to remove particle j located at x_j from this potential field is

$$W_{cj} = q_j q_T \int_{x_j}^{\infty} dx \left(\frac{d}{dx}\frac{1}{x}\, e^{-x/\lambda_D}\right)$$

The conditional probability that particle j is at x_j given a charge q_T at $x = 0$ is (from Prob. 2.8.1)

$$P_c(x_j|0) = \left(1 - \frac{q_j q_T}{x_j \kappa T}\, e^{-x_j/\lambda_D}\right)\frac{1}{V} \qquad (2.10.1)$$

Problem 2.10.1 Verify that the contribution to the Coulomb energy from inside $e^2/\kappa T$ is small, and that Eq. (2.10.1) is "nearly" correctly normalized. [Note that $\int_{\Delta V} P_c(x_1 \mid x_2)\, dx_2 \leq 1$ over any volume ΔV.] ////

Then the energy required to remove particle j to infinity from the potential of a test charge, averaged over the possible positions of the test charge, is

$$W_{cj} = q_j q_T \int d\mathbf{x}_j P_c(x_j|0) \int_{x_j}^{\infty} dR\, \frac{d}{dR}\left(\frac{1}{|R|}\, e^{-R/\lambda_D}\right) \qquad (2.10.2)$$

Since the integrals in Eq. (2.10.2) converge at $x = 0$, and since the volume $r_0{}^3 \leq (\lambda_D/n\lambda_D{}^3)^3$, in which the exact form of P_{12} is uncertain, is much smaller than the interaction volume $\lambda_D{}^3$, there is little error in extending the range of integration to include $x = 0$. The integrations are straightforward, and give

$$\overline{W}_{cj} = \frac{q_j q_T}{V}\left(4\pi\lambda_D{}^2 - \frac{4\pi q_j q_T}{\kappa T}\frac{\lambda_D}{2}\right)$$

$$= \frac{4\pi\lambda_D{}^2}{V}\, q_j q_T\left(1 - \frac{q_j}{q_T}\frac{1}{16\pi n\lambda_D{}^3}\right) \qquad (2.10.3)$$

There are N particles, j which could be removed from the field of particle q_T, and there are N particles which could have been chosen as the test particle. So the total Coulomb energy and the Coulomb energy per unit volume of a neutral plasma are (the $\frac{1}{2}$ occurs because each particle has been counted twice)

$$W_c = \text{energy} = \frac{1}{2}\, 4\pi \sum_\alpha N_\alpha q_\alpha \sum_\gamma N_\gamma q_\gamma \frac{\lambda_D{}^2}{V}\left(1 - \frac{q_\alpha}{q_\gamma}\frac{1}{n\lambda_D{}^3 16\pi}\right) \qquad (2.10.4)$$

$$\frac{W_c}{V} = \text{energy density} = -\frac{n_0 e^2}{2\lambda_D} \qquad (2.10.5)$$

where $n_0 = N/V$ and the sum is over ions and electrons. Note that for $n_e = n_i$, the leading contribution to Eq. (2.10.4) vanishes when summed over ions and electrons, a reminder that W (Coulomb) is a correlation energy. The second term is small, of order $1/n\lambda_D{}^3$, but does not cancel when summed. The Coulomb energy of a plasma with a large excess of one charge type is positive, while that of a neutral plasma is negative.

The ratio of electrostatic energy to thermal energy in a neutral plasma is, again,

$$\frac{W(\text{Coulomb})}{n_0 \kappa T} = \frac{1}{n_0 \lambda_D{}^3} = g$$

A plasma has many particles in an interaction sphere, but the interaction energy is small compared with the kinetic energy.

Problem 2.10.2 Calculate (or write down) the Coulomb energy for a system with $\lambda_D \ll q^2/\kappa T$, for example, an unshielded system. ////

Problem 2.10.3 At what plasma density will an H^- ion imbedded in a plasma at temperature 10^5 eV cease to be a bound state? ////

2.11 DISCUSSION

Plasma, whether in the laboratory or in nature, seldom exists in an equilibrium state. Still, considerations of the equilibrium statistical mechanics of plasma presented here are informative in that they reveal some aspects of plasmas that are reflected in all plasma behavior. First, the emphasis on plasma physics as a many-body problem involving long-range forces is clearly revealed in the equilibrium problem, and is also basic to the study of more complex plasma states. Second, the concept of the plasma as an ideal gas is demonstrated, along with the notion of the plasma as a fluid, by the study of the equilibrium problem that shows that *both* concepts are valid. In its thermodynamic properties (equation of state, free energy, etc.) the plasma is an ideal gas in the limit

$$n\lambda_D{}^3 \to \infty$$

In the same limit, however, the plasma appears dynamically as an ideal fluid, the average fields at each space point determining the motion of the (many) plasma particles present in a small-volume element. Third, the calculation of the plasma correlation function has been performed *only* for the case of the equilibrium plasma. It is *surmised* that the correlation remains small in a variety of non-equilibrium situations.

In subsequent chapters, the general statistical approach will not be abandoned, but it will be *reduced* by suitable averaging procedures to a set of tractable equations which describe the wide range of dynamic behavior possible for matter in the plasma state.

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3

MACROSCOPIC PROPERTIES OF PLASMAS

Although an exact knowledge of the state of a many-body system requires that the position and velocity of all the particles in the system be known, much of the behavior of such a system can be described in terms of macroscopic variables such as the density, the temperature, the average velocity, and the pressure. These quantities are related through the conservation laws and through the dynamical equations for momentum and energy transport. The purpose of this chapter is to define these variables and their relationships, starting from an exact many-body description, and then solve these equations to illustrate some macroscopic properties of plasmas. These variables and equations are a reduced description of the plasma, i.e., they contain less information than the complete many-body description. Therefore there are some plasma properties they cannot describe (for example, Landau damping and velocity-space instabilities; see Chap. 1). They *are* sufficient to describe a wide variety of plasma effects and applications.

The macroscopic variables for a plasma are defined in terms of velocity moments of the distribution function $f(\mathbf{x}, \mathbf{v}, t)$, which contains the statistical description of the system. The relationships between these macroscopic variables are derived from the differential equation for the distribution function.

3.1 THE DISTRIBUTION FUNCTION AND THE LIOUVILLE EQUATION

The complete statistical description of a system of N particles is given in terms of a distribution function

$$F(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N, \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N, t) \qquad (3.1.1)$$

where $\int F\, d\mathbf{x}_1 \cdots d\mathbf{x}_N\, d\mathbf{v}_1 \cdots d\mathbf{v}_N = 1$. For a plasma of $N/2$ ions and $N/2$ electrons in thermodynamic equilibrium, $F = D$, where D is the Gibbs distribution discussed in Chap. 2. For plasma that is not in thermodynamic equilibrium, F must in principle be calculated from the dynamics of the N particles. In practice, however, F can neither be calculated nor even estimated. Instead, the plasma is generally described in a reduced way, in terms of quantities obtained from and simpler than F. These reduced quantities can often be "calculated," which usually means finding how the system evolves from a known initial state. Reasonable initial values can often be assigned to the variables for a sufficiently reduced description. The most detailed description of a plasma that is still tractable is probably the *one-particle distribution*. Integrating F over the coordinates and velocities of all but one particle, of type α, and multiplying by the number of particles N_α of type α in the system defines the one-body distribution function, for particles of type α.

$$\bar{n}_\alpha f_\alpha^{(1)}(\mathbf{x}_1, \mathbf{v}_1, t)$$

$$= N_\alpha \int F(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N, \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N, t)\, d\mathbf{x}_2 \cdots d\mathbf{x}_N\, d\mathbf{v}_2 \cdots d\mathbf{v}_N \qquad (3.1.2)$$

where

$$\bar{n}_\alpha = \frac{N_\alpha}{V}$$

and V is the volume of the system. The quantity $\bar{n}_\alpha f_\alpha^{(1)}\, d\mathbf{x}_1\, d\mathbf{v}_1$ is the number of particles of type α at $\mathbf{x}_1$, $\mathbf{v}_1$, and t in the range $d\mathbf{x}_1\, d\mathbf{v}_1$. For a plasma in thermal equilibrium, $f_\alpha^{(1)}$ is a maxwellian distribution. In other cases it is often possible to suggest plausible initial values for $f_\alpha^{(1)}$, and it will be seen in later chapters that much plasma phenomena are qualitatively independent of the *exact* form of $f_\alpha^{(1)}$.

An entire set of reduced distributions can be defined from F. For example, integrating Eq. (3.1.1) over the coordinates and velocities of all but two particles gives the two-body distribution function

$$\bar{n}_\alpha \bar{n}_\gamma f_{\alpha\gamma}^{(2)}(\mathbf{x}_1, \mathbf{x}_2, \mathbf{v}_1, \mathbf{v}_2, t)$$

$$= N_\alpha N_\gamma \int F(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N, \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N)\, d\mathbf{x}_3\, d\mathbf{x}_4 \cdots d\mathbf{x}_N\, d\mathbf{v}_3\, d\mathbf{v}_4 \cdots d\mathbf{v}_N \qquad (3.1.3)$$

where α, γ can be the same type of particle or different species. The behavior of the reduced description is found by performing the reduction described by Eqs. (3.1.2) and (3.1.3) on the equation satisfied by F. The many-body distribution function (3.1.1) obeys the Liouville equation[1]

$$\frac{\partial F}{\partial t} + \sum_i \left(\frac{\partial F}{\partial \mathbf{x}_i} \cdot \mathbf{v}_i + \frac{\partial F}{\partial \mathbf{v}_i} \cdot \mathbf{a}_i{}^T \right) = 0 \qquad (3.1.4)$$

where $\mathbf{a}_i{}^T$ is the total acceleration of particle i due to external and interparticle forces. The Liouville equation, which is well known from statistical mechanics, can be written $dF/dt = 0$. This implies that F behaves as a hyperparticle with coordinates $(\mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N)$ at time t that moves about in hyperspace under the influence of the forces $F(\mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N)$. The value of F, that is, the probability density, remains constant at the location of the hyperparticle. This of course does *not* mean, in general, that F is constant in time at any *fixed* point $(\mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{v}_1, \ldots, \mathbf{v}_N)$, nor does it mean that F is the same at all spatial points.

The differential equation satisfied by the one-body distribution function is obtained by integrating the Liouville equation (3.1.4) over the coordinates and velocities of all but one particle; i.e.,

$$\frac{\partial \bar{n}_\alpha f_\alpha^{(1)}}{\partial t} + \mathbf{v}_1 \cdot \frac{\partial}{\partial \mathbf{x}_1} \bar{n}_\alpha f_\alpha^{(1)} + N_\alpha \int \mathbf{a}_1{}^T \cdot \frac{\partial F}{\partial \mathbf{v}_1} d\mathbf{x}_2 \cdots d\mathbf{x}_N \, d\mathbf{v}_2 \cdots d\mathbf{v}_N = 0 \qquad (3.1.5)$$

where

$$\int d\mathbf{x}_2 \frac{\partial}{\partial \mathbf{x}_2} \cdot \int \mathbf{v}_2 F \, d\mathbf{v}_2 = 0$$

has been used. This quantity represents the net flux of particles out of the system. Since, by assumption, N_α is constant, this flux vanishes. Similarly, $\int d\mathbf{v}_2 (\partial F/\partial \mathbf{v}_2) = 0$ by the condition $F(v = \infty) = 0$. If the force on the ith particle is due only to external forces, i.e., if the acceleration of the ith particle is given by

$$\mathbf{a}_i = \frac{q_i}{m_i} \mathbf{E} + \frac{q_i}{m_i} \frac{\mathbf{v}_i \times \mathbf{B}}{c} \qquad (3.1.6)$$

the third term in Eq. (3.1.5) becomes

$$N_\alpha \int \mathbf{a}_1 \cdot \frac{\partial F}{\partial \mathbf{v}_1} d\mathbf{x}_2 \cdots d\mathbf{x}_N \, d\mathbf{v}_2 \cdots d\mathbf{v}_N = \frac{q_1}{m_1} \left(\mathbf{E} + \frac{\mathbf{v}_1 \times \mathbf{B}}{c} \right) \cdot \frac{\partial}{\partial \mathbf{v}_1} \bar{n}_\alpha f_\alpha^{(1)} \qquad (3.1.7)$$

The kinetic equation for the one-body distribution function then becomes

$$\frac{\partial f_\alpha^{(1)}}{\partial t} + \mathbf{v}_1 \cdot \frac{\partial}{\partial \mathbf{x}_1} f_\alpha^{(1)} + \mathbf{a}^E \cdot \frac{\partial}{\partial \mathbf{v}_1} f_\alpha^{(1)} = 0 \qquad (3.1.8)$$

[1] D. ter Haar, "Elements of Statistical Mechanics," Rinehart, New York, 1954.

for the special case of particles which do not interact with each other. As with the Liouville equation, equation (3.1.8) shows that if there are no interparticle forces, the particle density in phase space is a constant along the trajectory of a single particle of type α acted on by the external force $\mathbf{F}^{(E)} = m_\alpha \mathbf{a}^E$.

The force on the ith particle, in fact, includes both external forces and interparticle forces; i.e.,

$$\mathbf{a}_i(\mathbf{x}_1, \ldots, \mathbf{x}_N) = \frac{q_i}{m_i} \mathbf{E} + \frac{q_i}{m_i} \frac{\mathbf{v}_i \times \mathbf{B}}{c} + \sum_j \mathbf{a}_{ij} \qquad (3.1.9)$$

where $\mathbf{a}_{ij}$ represents the acceleration from interparticle forces; then the third term in Eq. (3.1.5) is

$$\mathbf{a}_1{}^E \cdot \frac{\partial}{\partial \mathbf{v}_1} (\bar{n}_\alpha f_\alpha^{(1)}) + \int \sum_j N_\alpha \mathbf{a}_{1j} \cdot \frac{\partial F}{\partial \mathbf{v}_1} \, d\mathbf{x}_2 \cdots d\mathbf{x}_N \, d\mathbf{v}_2 \cdots d\mathbf{v}_N \qquad (3.1.10)$$

But since all particles of a single species β are identical, it follows that

$$\sum_j \int \mathbf{a}_{1j} F \, d\mathbf{x}_2 \cdots d\mathbf{x}_N \, d\mathbf{v}_2 \cdots d\mathbf{v}_N = \sum_\beta N_\beta \int \mathbf{a}_{1\beta} F \, d\mathbf{x}_2 \cdots d\mathbf{x}_N \, d\mathbf{v}_2 \cdots d\mathbf{v}_N$$

where $\mathbf{a}_1$ is the acceleration of particle 1 by a single particle of species β. Using the definition of $f_{\alpha\beta}^{(2)}$, expression (3.1.10) can be written

$$\mathbf{a}_1{}^E \cdot \frac{\partial}{\partial \mathbf{v}_1} (\bar{n}_\alpha f_\alpha^{(1)}) + \sum_\beta \int \mathbf{a}_{1\beta} \cdot \frac{\partial}{\partial \mathbf{v}_1} \bar{n}_\alpha \bar{n}_\beta f_{\alpha\beta}^{(2)}(\mathbf{x}_1, \mathbf{x}_\beta, \mathbf{v}_1, \mathbf{v}_\beta, t) \, d\mathbf{x}_\beta \, d\mathbf{v}_\beta \qquad (3.1.11)$$

With this result, it is clear that (3.1.5) is not a closed equation for $f^{(1)}$, and that it requires a knowledge of $f^{(2)}$. Some of the theoretical studies of plasma physics are devoted to finding ways to avoid actually calculating $f^{(2)}$ and instead resorting to some physical method of estimating the second term of (3.1.11). The method of treating or approximating the second term of Eq. (3.1.11) depends on the nature of the interparticle forces. For example, if the interparticle forces are strong and of short range, this term is primarily determined by the time-rate of change of the one-body distribution function due to binary collisions; i.e., the Liouville equation has the approximate form

$$\frac{df_\alpha^{(1)}}{dt} \equiv \frac{\partial f_\alpha^{(1)}}{\partial t} + \mathbf{v}_1 \cdot \frac{\partial}{\partial \mathbf{x}_1} f_\alpha^{(1)} + \mathbf{a}_1{}^E \cdot \frac{\partial}{\partial \mathbf{v}_1} f_\alpha^{(1)} = \left. \frac{\partial f_\alpha^{(1)}}{\partial t} \right|_{\text{due to interparticle interactions}}$$

where

$$\left. \frac{\partial f_\alpha^{(1)}}{\partial t} \right|_c = -\sum_\beta \bar{n}_\beta \int \mathbf{a}_{1\beta} \cdot \frac{\partial}{\partial \mathbf{v}_1} f_{\alpha\beta}^{(2)}(\mathbf{x}_1, \mathbf{x}_\beta, \mathbf{v}_1, \mathbf{v}_\beta, t) \, d\mathbf{x}_\beta \, d\mathbf{v}_\beta \qquad (3.1.12)$$

However, in a plasma, the interactions are long-range, and the interparticle force can be divided roughly into two parts: one part is the average force on a particle due to a large number of relatively distant particles, and the other part is

the force due to the nearest-neighbor particles. In a low-density system there are only a very few nearby particles, and in general the average force of the many distant particles far exceeds the "nearest-neighbor" force. It is convenient to lump the average forces of interparticle interactions together with the external forces, writing $\mathbf{a} = \mathbf{a}_{ext} + \langle \mathbf{a}_{int} \rangle$. This is useful because the average force from distant particles does not depend on the exact location of these particles, but upon the density of such particles at an average location. Forces that do not depend on the exact location of all particles have the appearance of *external* forces. The change of the distribution function due to the force of the nearest-neighbor particle (which is properly a *binary-collision rate*) is

$$\left.\frac{\partial f_\alpha^{(1)}}{\partial t}\right|_c = -\sum_\beta \int (\mathbf{a}_{1\beta} - \langle \mathbf{a}_{1\beta}{}^{int} \rangle) \cdot \frac{\partial}{\partial \mathbf{v}_1} f_{\alpha\beta}^{(2)}\, d\mathbf{x}_\beta\, d\mathbf{v}_\beta \qquad (3.1.13)$$

Thus a kinetic equation appropriate for the study of a plasma is

$$\frac{\partial}{\partial t} f_\alpha^{(1)} + \mathbf{v}_1 \cdot \frac{\partial}{\partial \mathbf{x}_1} f_\alpha^{(1)} + \frac{q_\alpha}{m_\alpha} \left\langle \mathbf{E} + \frac{\mathbf{v}_1 \times \mathbf{B}}{c} \right\rangle \cdot \frac{\partial}{\partial \mathbf{v}_1} f_\alpha^{(1)} = \left.\frac{\partial f_\alpha^{(1)}}{\partial t}\right|_c \qquad (3.1.14)$$

where $\langle \mathbf{E} \rangle$ and $\langle \mathbf{B} \rangle$ are the sum of the external and average internal fields, and satisfy the average Maxwell equations

$$\nabla \cdot \langle \mathbf{E} \rangle = 4\pi \langle \rho_q \rangle$$

and

$$\nabla \times \langle \mathbf{B} \rangle = \frac{1}{c} \frac{\partial \langle \mathbf{E} \rangle}{\partial t} + \frac{4\pi}{c} \langle \mathbf{J} \rangle$$

and the right-hand side is the effect of only the very few near-neighbor particles. If $\partial f/\partial t|_c$ is set to zero, i.e., if binary collisions are ignored, Eq. (3.1.14) becomes the plasma kinetic equation known as the *Vlasov equation*, or the *collisionless Boltzmann equation*. Beginning with Chap. 7, a hierarchy of statistical equations whose first term is the Vlasov equation is developed systematically. Plasma properties obtained from solutions of the Vlasov equation are described in Chaps. 7 to 10.

In this chapter, the plasma kinetic equation (3.1.14) is simply used as the basis for developing a reduced description of a plasma in terms of the macroscopic variables and their relationships.

3.2 MACROSCOPIC VARIABLES OF A PLASMA

The one-body distribution (3.1.2) is a reduced description of the plasma state; however, it is not a macroscopic observable of either a plasma or neutral gas. The macroscopically observable quantities are found from the velocity moments of the one-body distribution function.

3.2.1 Density

The number of particles of species α at point $\mathbf{x}$ in configuration space at time t is given by

$$n_\alpha(\mathbf{x}, t) = \int \bar{n}_\alpha f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v} \qquad (3.2.1)$$

where f_α is the one-body distribution function (3.1.2), and $\bar{n}_\alpha = N_\alpha/V$. Since only the one-body distribution function is used in this chapter, the superscript (1) is suppressed. The mass density $\rho_{m\alpha}$ and the charge density $\rho_{q\alpha}$ are similarly defined; i.e.,

$$\rho_{m\alpha}(\mathbf{x}, t) = \bar{n}_\alpha m_\alpha \int f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v} \qquad (3.2.2)$$

$$\rho_{q\alpha}(\mathbf{x}, t) = \bar{n}_\alpha q_\alpha \int f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v} \qquad (3.2.3)$$

3.2.2 Particle Flux and Velocity

The flux of particles of species α crossing a unit area in configuration space per unit time at the point $\mathbf{x}$ and at time t is

$$\begin{aligned} \boldsymbol{\Gamma}_\alpha(\mathbf{x}, t) &= \bar{n}_\alpha \int \mathbf{v} f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v} \\ &= n_\alpha(\mathbf{x}, t) \mathbf{V}_\alpha(\mathbf{x}, t) \end{aligned} \qquad (3.2.4)$$

where $\mathbf{V}_\alpha$ is the average velocity of particles of species α defined by

$$\mathbf{V}_\alpha(\mathbf{x}, t) = \frac{\int \mathbf{v} f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v}}{\int f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v}} = \frac{\bar{n}_\alpha}{n_\alpha(\mathbf{x}, t)} \int \mathbf{v} f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v} \qquad (3.2.5)$$

3.2.3 Current Density

The electric current density for charged particles of species α at the point $\mathbf{x}$ in configuration space at time t is

$$\begin{aligned} \mathbf{J}_\alpha(\mathbf{x}, t) &= q_\alpha \bar{n}_\alpha \int \mathbf{v} f_\alpha\, d\mathbf{v} \\ &= q_\alpha \boldsymbol{\Gamma}_\alpha(\mathbf{x}, t) \\ &= q_\alpha n_\alpha(\mathbf{x}, t) \mathbf{V}_\alpha(\mathbf{x}, t) \end{aligned} \qquad (3.2.6)$$

3.2.4 Heat Flux

The flux of kinetic energy for particles of species α crossing a unit area in configuration space per unit time at the point $\mathbf{x}$ and at time t is

$$\mathbf{H}_\alpha(\mathbf{x}, t) = \tfrac{1}{2} \bar{n}_\alpha m_\alpha \int \mathbf{v}(\mathbf{v} \cdot \mathbf{v}) f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v} \qquad (3.2.7)$$

3.2.5 Pressure Tensor

The pressure tensor for particles of species α at the point $\mathbf{x}$ in configuration space at time t is

$$\mathsf{P}_\alpha(\mathbf{x}, t) = \bar{n}_\alpha m_\alpha \int (\mathbf{v} - \mathbf{V}_\alpha)(\mathbf{v} - \mathbf{V}_\alpha) f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v} \qquad (3.2.8)$$

For a spherically symmetric velocity distribution this reduces to a diagonal pressure tensor

$$\mathsf{P}_\alpha = \begin{bmatrix} p_\alpha & 0 & 0 \\ 0 & p_\alpha & 0 \\ 0 & 0 & p_\alpha \end{bmatrix} \qquad (3.2.9)$$

and p_α is the scalar pressure,

$$p_\alpha = \frac{m_\alpha \bar{n}_\alpha}{3} \int (\mathbf{v} - \mathbf{V}_\alpha)^2 f_\alpha\, d\mathbf{v} = n_\alpha \kappa T_\alpha \qquad (3.2.10)$$

The lack of spherical symmetry of the pressure tensor is often related to collision rates[1] (e.g., to the *viscosity* of the fluid), and therefore the pressure tensor is normally broken up into two parts, one traceless, the other diagonal: $\mathsf{P}_\alpha = \mathsf{\Pi} + \mathsf{I}p$, where p is *defined* by (3.2.10), so that

$$\mathsf{\Pi} = \int \bar{n}_\alpha m_\alpha f_\alpha [(\mathbf{v} - \mathbf{V}_\alpha)(\mathbf{v} - \mathbf{V}_\alpha) - \tfrac{1}{3}\mathsf{I}(\mathbf{v} - \mathbf{V}_\alpha) \cdot (\mathbf{v} - \mathbf{V}_\alpha)]\, d\mathbf{v}$$

Additional moments of the distribution function are sometimes useful. However, the above are the only ones that have a simple physical interpretation.

Problem 3.2.1 Prove that P_α is a diagonal tensor if f_α is spherically symmetric in $\mathbf{v}$. Under what conditions is P_α diagonal but with $p_1 \neq p_2 \neq p_3$? ////

3.3 MACROSCOPIC EQUATIONS FOR A PLASMA; FLUID EQUATIONS

Just as the velocity moments of the distribution function give the macroscopic variables, so do the velocity moments of the plasma kinetic equation (3.1.14) give the equations that are satisfied by the macroscopic variables, and that describe

[1] See, for example, W. B. Thompson, "An Introduction to Plasma Physics," pp. 227–228, Addison-Wesley, Reading, Mass., 1962.

the time evolution of the plasma from a macroscopic point of view. Because the equations thus obtained are identical with the continuum hydrodynamic equations, the theories using the macroscopic equation are called *fluid theories.*

3.3.1 Continuity Equation

The integral of Eq. (3.1.14) over all velocity space is

$$\int \left(\frac{\partial f_\alpha}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{x}} f_\alpha + \frac{q_\alpha}{m_\alpha} \left\langle \mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c} \right\rangle \cdot \frac{\partial}{\partial \mathbf{v}} f_\alpha \right) d\mathbf{v} = \int \left. \frac{\partial f_\alpha}{\partial t} \right|_c d\mathbf{v} \qquad (3.3.1)$$

where $f_\alpha \equiv f_\alpha(\mathbf{x}, \mathbf{v}, t)$. The right side of Eq. (3.3.1) and the third term on the left-hand side of Eq. (3.3.1) both vanish, since both are of the form

$$\int \left[\frac{\partial}{\partial \mathbf{v}_1} \cdot \int \mathbf{a}_1(\mathbf{x}_1, \ldots, \mathbf{x}_N) F\, d\mathbf{x}_2 \cdots d\mathbf{x}_N\, d\mathbf{v}_2 \cdots d\mathbf{v}_N \right] d\mathbf{v}_1 \qquad (3.3.2)$$

which is zero, since $F(v_1 = \pm\infty) = 0$. This shows that collisions change the velocity of the particles in the system but do not directly affect the spatial density. Equation (3.3.1) is the continuity equation, which, after integrating $d\mathbf{v}$, is

$$\frac{\partial}{\partial t} n_\alpha(\mathbf{x}, t) + \nabla \cdot n_\alpha(\mathbf{x}, t) \mathbf{V}_\alpha(\mathbf{x}, t) = 0 \qquad (3.3.3)$$

Note that $\nabla \equiv \partial/\partial \mathbf{x}$ has been used in this macroscopic equation, along with the fact that

$$\int \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{x}} f\, d\mathbf{v} = \frac{\partial}{\partial \mathbf{x}} \cdot \int f\mathbf{v}\, d\mathbf{v} = \nabla \cdot n\mathbf{V}$$

This continuity equation represents conservation of particles. Source or sink terms on the right-hand side of Eq. (3.3.3) would represent ionization or recombination. The equation for mass continuity is obtained from Eq. (3.3.3) by multiplying through by the mass per particle m_α; that is,

$$\frac{\partial}{\partial t} (\rho_{m\alpha}) + \nabla \cdot (\rho_{m\alpha} \mathbf{V}_\alpha) = 0 \qquad (3.3.4)$$

where $\rho_{m\alpha} (= n_\alpha m_\alpha)$ is the mass density. The equation for continuity of charge is obtained from Eq. (3.3.3) by multiplying through by the charge per particle q_α; that is,

$$\frac{\partial}{\partial t} (\rho_{q\alpha}) + \nabla \cdot (\rho_{q\alpha} \mathbf{V}_\alpha) = 0 \qquad (3.3.5)$$

where $\rho_{q\alpha}$ $(= n_\alpha q_\alpha)$ is the electric charge density.

Problem 3.3.1 Discuss the following paradox: Although

$$\int (\partial f_\alpha/\partial t)|_c \, d\mathbf{v} = 0$$

implies that binary collisions do not change the plasma density, yet a system with a density gradient $n = n(x)$, but with an anisotropic pressure $p_x = 0$, $p_y \neq 0$, will diffuse in space, altering $n(x)$, only when collisions take place. ////

3.3.2 Momentum Transfer Equation

The integral over all velocity space of the product of the plasma kinetic equation (3.1.14) and the momentum $m_\alpha \mathbf{v}$ of a particle of species α is

$$\int m_\alpha \mathbf{v}\left(\frac{\partial f_\alpha}{\partial t} + \mathbf{v}\cdot\frac{\partial}{\partial \mathbf{x}} f_\alpha + \frac{q_\alpha}{m_\alpha}\left\langle \mathbf{E} + \frac{\mathbf{v}\times\mathbf{B}}{c}\right\rangle \cdot \frac{\partial}{\partial \mathbf{v}} f_\alpha\right) d\mathbf{v} = \int m_\alpha \mathbf{v} \left.\frac{\partial f_\alpha}{\partial t}\right|_c d\mathbf{v} \qquad (3.3.6)$$

With the aid of the continuity equation and in terms of previously defined variables, Eq. (3.3.6) reduces to the momentum transfer equation for particles of species α; that is,

$$n_\alpha m_\alpha \frac{\partial}{\partial t}(\mathbf{V}_\alpha) + n_\alpha m_\alpha \mathbf{V}_\alpha \cdot \nabla \mathbf{V}_\alpha - n_\alpha q_\alpha \left\langle \mathbf{E} + \frac{\mathbf{V}_\alpha \times \mathbf{B}}{c}\right\rangle + \nabla \cdot \mathsf{P}_\alpha$$

$$= m_\alpha \int \bar{n}_\alpha \mathbf{v} \left.\frac{\partial f_\alpha}{\partial t}\right|_c d\mathbf{v} \approx -\sum_\beta n_\alpha m_\alpha (\mathbf{V}_\alpha - \mathbf{V}_\beta)\langle \nu_{\alpha\beta}\rangle \qquad (3.3.7)$$

where $\langle \nu_{\alpha\beta}\rangle$ is a mean collision frequency for momentum transfer from all other types of plasma particles. By conservation of momentum, the right-hand side of (3.3.6) vanishes for collisions between like particles and, for instance, represents a momentum loss for a fast-moving population due to collisions with a slower-moving group of particles. The right-hand side of (3.3.7) is often a valid simplification of this resistive term. In (3.3.7), P_α is the pressure tensor, and the average fields $\langle \mathbf{E}\rangle$, $\langle \mathbf{B}\rangle$ must be consistent with the average plasma properties n_α, $\mathbf{V}_\alpha$, etc., through Maxwell's equations,

$$\nabla \cdot \langle \mathbf{E}\rangle = \sum_\alpha 4\pi n_\alpha q_\alpha + 4\pi \rho_{\text{ext}}$$

$$\nabla \times \langle \mathbf{B}\rangle = \frac{1}{c}\frac{\partial \langle \mathbf{E}\rangle}{\partial t} + \frac{4\pi}{c}\sum_\alpha n_\alpha q_\alpha \mathbf{V}_\alpha + \frac{4\pi}{c}\mathbf{J}_{\text{ext}} \qquad (3.3.8)$$

Problem 3.3.2 Supply the missing steps between (3.3.6) and (3.3.7). [*Hint:* $\nabla \cdot n_\alpha m_\alpha \mathbf{V}_\alpha \mathbf{V}_\alpha = n_\alpha m_\alpha \mathbf{V}_\alpha \cdot \nabla \mathbf{V}_\alpha + \mathbf{V}_\alpha(\nabla \cdot n_\alpha m_\alpha \mathbf{V}_\alpha)$.] ////

Problem 3.3.3 Show that the energy transfer equation, obtained by multiplying (3.1.14) by $\frac{1}{2}\bar{n}_\alpha m_\alpha v^2$ and integrating over velocity space is

$$\frac{\partial}{\partial t}\left(\frac{n_\alpha m_\alpha V_\alpha^2}{2} + \frac{3 n_\alpha \kappa T_\alpha}{2}\right) - n_\alpha \mathbf{F}_\alpha \cdot \mathbf{V}_\alpha + \nabla \cdot \mathbf{H}_\alpha = \frac{\partial}{\partial t}\left(\frac{n_\alpha m_\alpha V_\alpha^2}{2}\right)\Bigg|_c \qquad (3.3.9)$$

////

The first two terms in (3.3.7) combine to form a "comoving" derivative (that is, $\partial/\partial t + \mathbf{V}\cdot\partial/\partial \mathbf{x} \equiv D/Dt$). The combined terms represent the change of momentum per unit volume per unit time in the element moving with velocity $\mathbf{V}_\alpha$. The next term is the change of momentum per unit volume per unit time due to forces exerted by the average fields. The divergence of the pressure tensor gives the change of momentum per unit volume per unit time due to spatial inhomogeneities. This term includes the effect of viscosity. If particle collisions are sufficiently frequent, the pressure term reduces to the gradient of a scalar pressure (that is, $\nabla \cdot \mathsf{P} \to \nabla p$). This is a frequently used approximation.

The above set of moment equations are conservation laws for particle number density, momentum, and energy. They apply separately for each plasma species for which the plasma kinetic equation (3.1.14) is applicable. These equations do not constitute a closed set, in that each moment equation includes terms involving higher-velocity moments. A complete description requires *all* velocity moments; any finite set of equations does not completely specify plasma behavior. To find useful information from only a few of the moment equations, it is necessary somehow to close the chain of equations. An equation of state relating the higher moments, e.g., pressure, temperature, to the lower moments, e.g., density, is required. This is similar to the problem involved in finding the one-particle distribution function. The original description of the plasma through the Liouville equation contained detailed information about the behavior of the N-particle system. The kinetic equation (3.1.14) is a reduced description containing information about the one-particle distribution only, and is the first of an infinite set of equations relating $f^{(1)}$ to $f^{(2)}$ and $f^{(2)}$ to $f^{(3)}$, and so on. The set can be terminated at any order by neglecting higher correlations. The approach in this chapter is to terminate the chain with the one-particle kinetic equation by replacing the effect of correlations with $\partial f/\partial t|_c$, intending eventually to either neglect $\partial f/\partial t|_c$ or replace it by a collision model. In a similar way, the macroscopic description is a reduced description that contains information only about the behavior of the system in configuration space. This reduced description also consists of an infinite chain of equations in which n is related to $\mathbf{V}$, $\mathbf{V}$ to P, and so on. This chain is terminated either by neglecting higher-velocity moments or by an independent assumption of a relation between p and

n, that is, an equation of state. The choice of method for terminating the chain of macroscopic equations determines a model for the plasma. Sometimes an isothermal or adiabatic equation of state is used, and sometimes $p = 0$ is a good assumption; however, more elaborate models have been developed to solve certain plasma problems. For example, formal expansion of the one-particle distribution about a local maxwellian state leads to the Chapman-Enskog closure scheme.[1] In this text only simple closure schemes are used; even these lead to considerable complexity. It should be clear that the fluid theory, though of great practical use, relies heavily on the cunning of its user.

It is clearly advantageous to choose as simple a closure scheme as possible. However, the choice of model determines the plasma properties that can be studied, since the approximations used eliminate certain features of plasma behavior.

Even without specifying a closure scheme, there are two general approaches to a macroscopic description of a plasma. In one, the ions and electrons are treated as separate but interacting fluids, each having its own set of equations and properties. In the other, the plasma is described as *one* fluid with a net density, velocity, and current at each point. Although these two approaches are formally identical, they frequently suggest different approximation schemes.

3.4 TWO-FLUID PLASMA THEORY

The two-fluid equations treat the ions and the electrons each as conducting fluids that are coupled through momentum transfer collisions and by Maxwell's equations. The variables in this theory are the density, velocity, and pressure of each species, as defined by Eqs. (3.2.1), (3.2.5), and (3.2.8), related by the continuity and momentum transport equations.

Continuity:

$$\frac{\partial}{\partial t} n_\alpha + \nabla \cdot n_\alpha \mathbf{V}_\alpha = 0 \qquad (3.4.1)$$

Momentum:

$$n_\alpha m_\alpha \left(\frac{\partial}{\partial t} + \mathbf{V}_\alpha \cdot \nabla\right)\mathbf{V}_\alpha = n_\alpha q_\alpha \left(\mathbf{E} + \frac{\mathbf{V}_\alpha \times \mathbf{B}}{c}\right) - \nabla \cdot \Pi - \nabla p_\alpha - \int \bar{n}_\alpha m_\alpha \mathbf{v} \left.\frac{\partial f_\alpha}{\partial t}\right|_c d\mathbf{v} \qquad (3.4.2)$$

Maxwell:

[1] S. Chapman and T. G. Cowling, "The Mathematical Theory of Non-uniform Gases," 2d ed., Cambridge, London, 1952.

$$\nabla \cdot \mathbf{E} = 4\pi \sum_\alpha n_\alpha q_\alpha \qquad (3.4.3)$$

$$\nabla \times \mathbf{B} = \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} + \frac{4\pi}{c} \sum_\alpha n_\alpha q_\alpha \mathbf{V}_\alpha \qquad (3.4.4)$$

where

$$\Pi_\alpha = \mathsf{P}_\alpha - p\mathsf{I} \qquad (3.4.5)$$

$\Pi_\alpha = 0$ for a spherically symmetric distribution, and the approximation

$$\bar{n}_\alpha \int m\mathbf{v} \left.\frac{\partial f_\alpha}{\partial t}\right|_c d\mathbf{v} \approx -\sum_\beta n_\alpha m_\alpha (\mathbf{V}_\alpha - \mathbf{V}_\beta)\nu_{\alpha\beta} \qquad (3.4.6)$$

is often used.

Each fluid requires either an equation of state or some other restriction for closure. For example, an assumption about heat flow during a plasma process leads to either

$$p = An \qquad \text{isothermal} \qquad (3.4.7)$$

or

$$p = An^{5/3} \qquad \text{adiabatic} \qquad (3.4.8)$$

or even

$$p = 0 \qquad (3.4.9)$$

which is sometimes a good assumption for processes involving a strong magnetic field ($B^2/8\pi \gg p$). For a partially ionized plasma, there is also a neutral component that satisfies the momentum transfer equation and is coupled to the ions and electrons through momentum transfer collisions, but is not coupled through Maxwell's equations. The neutral fluid also requires a separate equation of state, and in some problems, such as magnetohydrodynamic (MHD) power generators,[1] the motion of the neutral component is very important.

Problem 3.4.1 Find a reference that gives the relation between Π and the collision rate for (*a*) a field-free plasma and (*b*) a plasma confined by a magnetic field. The purpose of this problem is, in part, to force the student to begin reading other plasma physics literature. ////

3.5 ONE-FLUID PLASMA THEORY: MAGNETOHYDRODYNAMICS

By combining the density and velocity of ions and electrons, fluid equations whose variables are the *total* mass density, center-of-mass velocity, electric

[1] A bibliography of papers on MHD power generators is available in W. B. Kunkel (ed.), "Plasma Physics in Theory and Application," chap. 11, McGraw-Hill, New York, 1966. (See also section 6.12.)

current, and charge density can be obtained. The one-fluid theory so obtained is a simpler starting point for many problems, and will be examined in detail.

3.5.1 One-fluid Variables

The mass density in the fluid is defined by

$$\rho_m(\mathbf{x}, t) = \sum_\alpha n_\alpha m_\alpha = n_e m_e + n_i m_i \tag{3.5.1}$$

The charge density is

$$\rho_q(\mathbf{x}, t) = \sum_\alpha n_\alpha q_\alpha = e(n_i - n_e) \tag{3.5.2}$$

The center-of-mass velocity is given by

$$\mathbf{V}(\mathbf{x}, t) = \frac{\sum_\alpha n_\alpha m_\alpha \mathbf{V}_\alpha}{\sum_\alpha n_\alpha m_\alpha} = \frac{n_e m_e \mathbf{V}_e + n_i m_i \mathbf{V}_i}{n_e m_e + n_i m_i} \tag{3.5.3}$$

The total current is

$$\mathbf{J} = \sum q_\alpha n_\alpha \mathbf{V}_\alpha$$

and the pressure of electrons and ions in the center-of-mass frame is

$$\mathsf{P}_\alpha^{\mathrm{CM}} = \bar{n}_\alpha m_\alpha \int (\mathbf{v} - \mathbf{V})(\mathbf{v} - \mathbf{V}) f_\alpha \, d\mathbf{v} \tag{3.5.4}$$

$$\mathsf{P} = \sum_\alpha \mathsf{P}_\alpha^{\mathrm{CM}} \tag{3.5.5}$$

Note that $\mathbf{V}$ in $\mathsf{P}_\alpha^{\mathrm{CM}}$ is the center-of-mass average velocity [Eq. (3.5.3)] and *not* $\mathbf{V}_\alpha$.

3.5.2 One-fluid Equations

The differential equations satisfied by the one-fluid variables are obtained by adding and subtracting the fluid equations (3.4.1) to (3.4.5), which separately describe the behavior of ions and electrons.

Equations for continuity of mass and charge density are simply obtained from (3.4.1). Multiplying (3.4.1) by m_α and adding the equations for ions and electrons gives

$$\frac{\partial \rho_m}{\partial t} + \nabla \cdot \rho_m \mathbf{V} = 0 \tag{3.5.6}$$

while multiplying (3.4.1) by q_α and adding the equations for ions and electrons gives

$$\frac{\partial}{\partial t} \rho_q + \nabla \cdot \mathbf{J} = 0 \tag{3.5.7}$$

Similarly, summing (3.4.2) over ions and electrons gives the equation for momentum transport in the one-fluid theory (see Prob. 3.5.1).

$$\rho_m \frac{\partial}{\partial t} \mathbf{V} + \rho_m(\mathbf{V} \cdot \nabla)\mathbf{V} = \rho_q \mathbf{E} + \frac{\mathbf{J} \times \mathbf{B}}{c} - \nabla \cdot \mathsf{P} \tag{3.5.8}$$

Problem 3.5.1 Derive (3.5.8). Note especially that

(*a*) $\mathsf{P} = \sum_\alpha \mathsf{P}_\alpha^{\mathrm{CM}}$, where $\mathsf{P}_\alpha^{\mathrm{CM}}$ is defined in the rest frame of the fluid as described in Eq. (3.5.5).

(*b*) The collision term vanishes by conservation of momentum *if there are no neutral particles*, i.e., if the plasma is fully ionized.

(*c*) The easiest way to derive (3.5.8) may be to start with (3.3.6) and later use the continuity equation (3.5.6) for the *one-fluid* mass density.

////

Problem 3.5.2 What is the equivalent of (3.5.8) and (3.5.9) for momentum transfer in a gas which is very weakly ionized? Assume that the more dense neutral background remains at rest and that the "one-fluid" description describes the electron-ion fluid.

What is the equivalent of (3.5.8) and (3.5.9) for an ionized gas in which the relatively few neutrals are moved about by the electron-ion fluid? First write a set of two-fluid equations to describe the system (let one fluid be composed of neutrals, the other fluid be composed of the electrons and ions). Next write a set of one-fluid equations in which the one fluid is composed of the ions, electrons, and neutrals. ////

A fourth equation for the one-fluid variables is obtained by multiplying (3.4.2) by q_α, dividing by m_α, and adding the ion and electron equations [summing (3.4.2) over charge species gave equations for $d(\rho_m \mathbf{V})/dt$; the idea now is to get an equation for $d\mathbf{J}/dt$]. The resulting equation is

$$\begin{aligned}\frac{\partial \mathbf{J}}{\partial t} + \nabla \cdot (\mathbf{V}\mathbf{J} + \mathbf{J}\mathbf{V} - \mathbf{V}\mathbf{V}\rho_q) &= \left(\sum_\alpha \frac{n_\alpha q_\alpha^2}{m_\alpha}\right)\mathbf{E} \\ &+ \left(\frac{e^2}{m_e c} + \frac{e^2}{m_i c}\right)\frac{\rho_m \mathbf{V} \times \mathbf{B}}{m_e + m_i} - \left(\frac{em_i}{cm_e} - \frac{em_e}{cm_i}\right)\frac{\mathbf{J} \times \mathbf{B}}{m_e + m_i} \\ &- \frac{e}{m_e} \nabla \cdot \left(\mathsf{P}_i^{\mathrm{CM}} \frac{m_e}{m_i} - \mathsf{P}_e^{\mathrm{CM}}\right) + \sum_\alpha \int q_\alpha \bar{n}_\alpha \mathbf{v} \left.\frac{\partial f_\alpha}{\partial t}\right|_c d\mathbf{v}\end{aligned} \tag{3.5.9}$$

for a plasma of ions and electrons. This equation is called the *generalized Ohm's law* for a plasma, since it relates the current to the electric field. The

collision term, which is $d\mathbf{J}/dt|_{\text{collisions}}$, is often estimated by a linear approximation

$$\sum_\alpha \bar{n}_\alpha q_\alpha \int \mathbf{v} \left.\frac{\partial f_\alpha}{\partial t}\right|_c d\mathbf{v} \approx -\nu \mathbf{J} \qquad (3.5.10)$$

where ν is an average collision frequency. In terms of ν, the resistivity and conductivity are defined by

$$\eta = \frac{\nu m_e}{ne^2}, \qquad \sigma = \frac{1}{\eta} \qquad (3.5.11)$$

in order that (3.5.9) reduce to Ohm's law, $\mathbf{E} = \eta \mathbf{J}$ in the $B = 0$, static, uniform-pressure limit.

Equations (3.5.6) to (3.5.10) constitute a set of equations for the one-fluid variables, but of course these equations are also not *closed*; i.e., there are more unknowns than equations. This situation is not remedied by taking higher moments of the kinetic equations (3.1.14), because each new equation contains an additional variable. Closure of (3.5.6) to (3.5.10) is generally achieved by assuming an equation of state, as in (3.4.7) to (3.4.9). Sometimes the choice of equation of state is plausible. In any event, the choice has a strong effect on the results found from the fluid equations, and should not be made arbitrarily.

3.6 APPROXIMATIONS COMMONLY USED IN ONE-FLUID THEORY

Even with a choice of an equation of state, the fluid equations are cumbersome. In practice, these equations are simplified by a set of assumptions valid for a wide range of phenomena. Since the assumptions do limit the applicability of the equations, it is important to take note of them before writing down the commonly used form of the fluid equations:

$$\text{Quasineutrality } \rho_{qe} = \rho_{qi} \qquad (3.6.1)$$

A major simplification is achieved by assuming that the charge density vanishes, $\rho_q = 0$. However, since the charge density does *not* in fact vanish, the assumption that $n_e = Zn_i$ represents an approximation and a loss of information. Yet, because the plasma as a whole is neutral ($N_e = ZN_i$), there must be some scale on which (3.6.1) is valid. This scale is the Debye length: an excess charge at rest is shielded, i.e., surrounded by a charge equal and opposite its own, in a distance λ_D. Thus, if slow motions of fluid elements of size greater than λ_D are studied, it may be assumed that $n_e = n_i$.

$$\frac{n_e - n_i}{n_e} \ll 1 \qquad \text{if } \frac{\omega_p^2}{\kappa T/m} L^2 \gg 1$$

where L is the length scale under study. If a system is being studied with a spatial resolution better than λ_D, or if a phenomenon involves spatial variations on scales $\sim\lambda_D$, then quasineutrality is a *bad* assumption. Note that, in general, the assumption (3.6.1) implies also

$$\nabla \cdot \mathbf{J} = 0 \qquad (3.6.2)$$

Another approximation frequently made is

$$ne\mathbf{V} \cdot \nabla \mathbf{V} = 0$$
$$\nabla \cdot (\mathbf{JV}) = 0 \qquad (3.6.3)$$

These simplifications are valid for a wide class of phenomena. For example, in problems involving perturbations about an initial state, the initial state will have $\mathbf{V} = 0$ in some frame, and in that frame terms $\mathbf{V} \cdot \mathbf{V}$ are *second-order* in the perturbation.

A trivial approximation is to neglect m_e/m_i, since

$$\frac{m_e}{m_i} \ll 1 \qquad (3.6.4)$$

However, when used in connection with (3.6.1), this implies further that

$$\rho_m = n_i m_i \qquad (3.6.5)$$

In many examples, only phenomena of very low frequency ω and very long spatial scale L are considered. In these cases, since $\mathbf{J} = (c/4\pi)(\nabla \times \mathbf{B})$ from Maxwell's equations, both $\mathbf{J} \times \mathbf{B}$ and $d\mathbf{J}/dt$ are neglected in Ohm's law:

$$\frac{1}{|\mathbf{J}|}\frac{1}{\omega_p}\frac{d\mathbf{J}}{dt} \approx \frac{\omega}{\omega_p} \ll 1$$
$$\frac{\mathbf{J} \times \mathbf{B}}{c} \approx \frac{B^2}{L} \ll ne\frac{\mathbf{V} \times \mathbf{B}}{c} \qquad (3.6.6)$$

Of course, there are problems in which pressure gradients dominate electric fields, and the $\mathbf{J} \times \mathbf{B}$ term will be important. (See, for example, Sec. 6.12.) Another frequent assumption is of isotropic pressure,

$$\nabla \cdot \mathsf{P}_\alpha = \nabla p_\alpha \qquad (3.6.7)$$

This approximation neglects viscosity effects; it is certainly valid when interparticle collisions are sufficiently frequent. This statement cannot be made quantitative without some theory for collisional effects. If the frequency of

ion-ion collisions is greater than the cyclotron frequency or any other frequency considered in the problem, one particular model, the Navier-Stokes theory, gives

$$R \equiv LV_0 \frac{m_i}{\kappa T} \nu_{ii} \gg 1 \qquad (3.6.8)$$

as the criterion for neglecting off-diagonal terms in the pressure tensor. R is called the Reynolds number. In (3.6.8) L is a characteristic length, and V_0 is a characteristic velocity of the medium under examination. Because of the simplifications it produces, the approximation (3.6.7) is used in the study of nearly collision-free plasma, where (3.6.8) is not satisfied. It is a fortunate fact that the plasma equations with $\nabla \cdot \mathsf{P} = \nabla p$ agree with a wide range of experiments, despite the lack of a clear basis for that approximation. It is presumed that the plasma frequency ω_p acts as a "collision frequency" in wiping out off-diagonal pressure-tensor elements.

When the terms $\mathbf{J} \times \mathbf{B}$, $\partial \mathbf{J}/\partial t$, and ∇p are all neglected, the simplified form of Ohm's law,

$$\mathbf{J} = \sigma\left(\mathbf{E} + \frac{\mathbf{V} \times \mathbf{B}}{c}\right) \qquad (3.6.9)$$

is obtained, with σ defined by (3.5.11). Even this simple form of Ohm's law leads to some interesting conclusions about the dynamics of a plasma in a strong magnetic field, which are discussed in Sec. 3.9. In discussing the validity of the simplified Ohm's law, it should be noted that dimensional analysis alone shows that the $\mathbf{J} \times \mathbf{B}$ term can be neglected when the characteristic length scale within which the field properties change is sufficiently long. The criterion for (3.6.9) can be expressed explicitly in terms of the length and velocity scales of interest for the problem. Let L be the length scale for spatial variation of the plasma parameters and V_0 be a characteristic velocity of the plasma fluid. Then the common approximations of MHD theory are valid under the following conditions:

$\frac{\mathbf{J} \times \mathbf{B}}{nec}$ can be neglected in Ohm's law if $\frac{L\omega_{pe}^2}{\omega_{ce}} \frac{V_0}{c^2} \gg 1$ (3.6.10)

$\frac{m_e}{ne^2} \frac{\partial \mathbf{J}}{\partial t}$ can be neglected if $\frac{L^2\omega_{pe}^2}{c^2} \gg 1$ (3.6.11)

∇p_e can be neglected in Ohm's law if $\frac{LV_0\,\omega_{ce}}{\kappa T_e/m_e} \gg 1$ (3.6.12)

In some cases the off-diagonal elements of P are related to the neglect of "finite Larmor radius"; i.e., in their orbits the ions and electrons see a different average force, because the size of their orbits is different. The condition under which this approximation of neglecting finite Larmor radius corrections is justified is

$$\frac{L\omega_{ci}}{\sqrt{\kappa T_i/m_i}} \gg 1 \qquad (3.6.13)$$

The above conditions all imply only very slow time variations of plasma properties. If time variations are not slow, a much more complicated set of conditions are needed to justify the simplified form of Ohm's law. It must be noted that $L \to \infty$ is sufficient to satisfy Eqs. (3.6.9) to (3.6.14), but that other limits such as $B \to \infty$ or $\langle \nu \rangle \to 0$ will weaken some of the inequalities and strengthen others. Thus the familiar form (3.5.9) of Ohm's law is valid only in a bounded region of parameter space. Ohm's law would be even simpler if the conductivity were infinite. This approximation is valid as follows:

$\frac{4\pi J}{\sigma}$ can be neglected in Ohm's law when $R_M \equiv \frac{4\pi\sigma LV_0}{c^2} \gg 1$ (3.6.14)

R_M is called the magnetic Reynolds number, and if $R_M \gg 1$, Ohm's law is

$$\mathbf{E} + \frac{\mathbf{V} \times \mathbf{B}}{c} = 0 \qquad (3.6.15)$$

3.7 SIMPLIFIED ONE-FLUID EQUATIONS AND THE MHD EQUATION

A simplified set of one-fluid equations can be obtained which describe plasma phenomena on a long space scale ($L \to \infty$). From the discussion in the preceding section, the approximation $n_e = n_i$ is valid on a long space scale. In addition, gradient terms (for example, ∇p, ∇B, $\nabla \cdot \mathbf{V}$, etc.) can be treated as small compared with field terms (for example, $\mathbf{B}$, $\mathbf{E}$, $\mathbf{J}$, $\mathbf{V}$, etc.). With these approximations, and the neglect of the ratio m_e/m_i compared with unity, the one-fluid equations become

$$\frac{\partial \rho_m}{\partial t} + \nabla \cdot \rho_m \mathbf{V} = 0 \qquad (3.7.1)$$

$$\rho_m \frac{D\mathbf{V}}{DT} = \frac{\mathbf{J} \times \mathbf{B}}{c} - \nabla(p_i + p_e) - [\nabla \cdot \mathsf{P}_i + \nabla \cdot \mathsf{P}_e - \nabla(p_i + p_e)] \qquad (3.7.2)$$

$$\mathbf{E} + \frac{\mathbf{V} \times \mathbf{B}}{c} = \eta \mathbf{J} + \frac{m_e}{ne^2} \frac{D\mathbf{J}}{Dt} \qquad (3.7.3)$$

$$\nabla \times \mathbf{B} = \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} + \frac{4\pi \mathbf{J}}{c}$$
$$\nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial B}{\partial t} \qquad (3.7.4)$$

along with the boundary condition[1]

$$\nabla \cdot \mathbf{B} = 0 \qquad (3.7.5)$$

and the definition of the comoving derivative

$$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + \mathbf{V} \cdot \nabla \qquad (3.7.6)$$

It is not inconsistent to neglect ∇p compared with $\mathbf{V} \times \mathbf{B}$ in Ohm's law (3.7.3) and still retain it in the momentum transport equation (3.7.2). This is because all terms on the right-hand side of (3.7.2) are proportional to gradients of the physical variable (since $4\pi\mathbf{J}/c \approx \nabla \times \mathbf{B}$). Equation (3.7.3), on the other hand, includes terms not proportional to gradients. These terms are treated as *large*, and gradients are neglected compared with them.

If in addition to the restriction that only long-spatial-scale phenomena are considered, a further restriction to *low-frequency* phenomena is imposed, the fluid equations are further simplified. Along with a restriction to low frequency it is often assumed that collisions (in collective effects) are sufficiently frequent to keep the system isotropic at all times. The set of equations then obtained are called the *MHD equations*, valid in the limit $\omega \to 0$, $L \to \infty$, $\nabla \cdot \mathsf{P} = \nabla p$.

Continuity:
$$\frac{\partial \rho_m}{\partial t} + \nabla \cdot \rho_m \mathbf{V} = 0 \qquad (3.7.7)$$

Momentum transport:
$$\rho_m \frac{\partial \mathbf{V}}{\partial t} = \frac{\mathbf{J} \times \mathbf{B}}{c} - \nabla p \qquad (3.7.8)$$

Ohm's law:
$$\mathbf{E} + \frac{\mathbf{V} \times \mathbf{B}}{c} = \eta \mathbf{J} \qquad (3.7.9)$$

Maxwell's equations:
$$\left.\begin{aligned} \nabla \times \mathbf{E} &= -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} &= \frac{4\pi \mathbf{J}}{c} \end{aligned}\right\} \qquad (3.7.10)$$

The set of equations (3.7.7) to (3.7.10) are closed by an equation of state or by some other model relating the pressure to the density. The usual approximation of magnetohydrodynamics is to use the assumption of

[1] Note that $\nabla \cdot \mathbf{E} = 0$ is not imposed as an additional equation on the fluid, and indeed is not a valid equation for compressional waves which emerge from the theory. The point is that $n_e = n_i$ has already been imposed (and is a good approximation on scales $L > \lambda_D$, as explained above), as an equation. If $\mathbf{E}$ is determined from Eqs. (3.7.1) to (3.7.6), Poisson's equation can be used to *determine* $n_e - n_i$ and verify that it is small.

$$\nabla \cdot \mathbf{V} = 0 \qquad \text{incompressible fluid}$$

or
$$\frac{d}{dt} p \rho_m^{-\gamma} = 0 \qquad \text{adiabatic fluid}$$

or
$$\frac{d}{dt}\frac{p}{\rho_m} = 0 \qquad \text{isothermal fluid}$$

or some other closure equation (perhaps some other aspect of the system is understood, e.g., perhaps J = constant).

The MHD theory implies a certain ordering of length scales and time scales in the plasma dynamics. The exact nature of this ordering is not clear from the equations themselves, except that they are *certainly* invalid if the frequency is too high or the wavelength too short. The neglect of charge separation certainly restricts the MHD equations to wavelengths $\lambda \gg \lambda_D$. The fact that $n_e = n_i$ implies that the electrons and ions move together adds another restriction to the time scale of MHD dynamics; that is, $T \gg (\omega_{ci})^{-1}$. This is because motions as rapid as cyclotron motions will affect the ions differently from the electrons. The assumption of a scalar pressure,

$$\nabla \cdot \mathsf{P} = \nabla p$$

is a more subtle approximation. It certainly is valid for a collision-dominated system. However, on a collisionless time scale, in a magnetic field, it is not clear what time and space scales are required to justify this approximation. Consequently, fluid models, in addition to the MHD model, have been developed for particular situations. It has been shown (Prob. 3.7.1) that for plasma in a magnetic field, the MHD theory implies, in addition to $L \gg \lambda_D$, the ordering

$$\left(\frac{a_i}{L}\right)^2 \ll \frac{1}{T\omega_{ci}} \ll 1 \qquad \text{MHD ordering}$$

where a_i is the gyroradius of ions. A fluid model called *finite Larmor radius* (FLR) *magnetohydrodynamics* has been developed for the study of plasma motions which have a different ordering of time and space scales. In particular, if

$$\frac{1}{T\omega_{ci}} \ll \left(\frac{a_i}{L}\right)^2 \ll 1 \qquad \text{FLR ordering}$$

the pressure tensor is modified and is no longer even a diagonal tensor. (See also Sec. 3.10 on a *further* possibility for a fluid equation.)

Problem 3.7.1 Read K. V. Roberts and J. B. Taylor, *Phys. Rev. Letters*, **8**:197 (1962) for an introduction and historical summary of finite Larmor radius magnetohydrodynamics. ////

3.8 PROPERTIES OF THE PLASMA DESCRIBED BY THE ONE-FLUID AND MHD MODELS

There are a number of plasma properties described in a simple way by the one-fluid equations.

3.8.1 Isobaric Surfaces

If the plasma is in a steady state, the momentum transport equation (3.7.2) becomes

$$\frac{\mathbf{J} \times \mathbf{B}}{c} = \nabla p \qquad (3.8.1)$$

where both the time derivatives and the off-diagonal pressure terms vanish in the steady state. From this it follows that both the current and the magnetic field are perpendicular to the pressure gradient.

$$\begin{aligned} \mathbf{B} \cdot \nabla p &= 0 \\ \mathbf{J} \cdot \nabla p &= 0 \end{aligned} \qquad (3.8.2)$$

This means that surfaces of constant pressure are also magnetic surfaces, i.e., can be made up of magnetic lines of force. No magnetic lines *pass through* an isobaric surface, and plasmas tend to extend themselves along the field. Note that the statement that the isobaric surface is also a magnetic surface does not imply that B^2 is constant on the surface; only that no lines cross the surface. Indeed, since in a steady state (using $\nabla \times \mathbf{B} = 4\pi\mathbf{J}/c$),

$$\frac{\mathbf{J} \times \mathbf{B}}{c} = \frac{(\nabla \times \mathbf{B}) \times \mathbf{B}}{4\pi} = -\nabla \frac{B^2}{8\pi} + \frac{1}{4\pi} \mathbf{B} \cdot \nabla \mathbf{B} \qquad (3.8.3)$$

it must be that

$$\nabla \frac{B^2}{8\pi} = \frac{\mathbf{B} \cdot \nabla \mathbf{B}}{4\pi} \qquad \text{on an isobaric surface} \qquad (3.8.4)$$

that is, $\hat{\mathbf{n}} \cdot [\nabla B^2/8\pi - (\mathbf{B} \cdot \nabla)\mathbf{B}/4\pi] = 0$, where $\hat{\mathbf{n}}$ is normal to the isobaric surface. If the field lines are everywhere straight, e.g., if $\mathbf{B} = B\hat{\mathbf{z}}$, then the condition $\nabla \cdot \mathbf{B} = 0$ requires that an isobaric surface be also a surface of constant magnetic pressure $B^2 =$ constant; this is not true if the field lines are curved.

Rewriting (3.8.1) in the form

$$\mathbf{J}_\perp = \frac{\mathbf{B} \times \nabla p}{B^2} c \qquad (3.8.5)$$

it is clear that in the one-fluid model, which assumes charge neutrality, no current flows across the magnetic field in a plasma unless there is a pressure gradient.

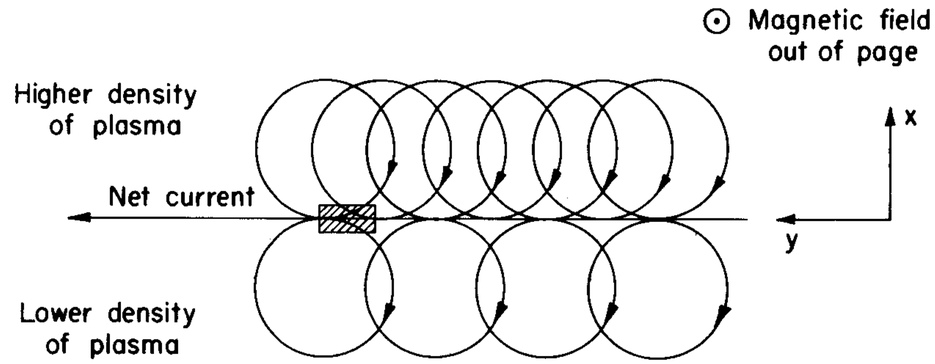


FIGURE 3.8.1
Diagram illustrating that a drift current in a plasma with a pressure gradient occurs without a net transport of particles.

It is instructive to notice that the existence of a current $\mathbf{J}$ does not necessarily mean a transport of particles across the magnetic field. Electric field and magnetic gradients produce particle drifts, but a gradient in the *pressure* does not. The current $\mathbf{J}_\perp$ in (3.8.5) does not arise from a motion of single particles, but rather from a gradient in the number (or velocity) of particles gyrating in a magnetic field. As shown in Fig. 3.8.1, the current J_y is present because more particles are moving to the left than to the right, at a given point $\mathbf{x}$, although each particle subsequently rotates and never actually makes any net progress to the left.

Problem 3.8.1 The current (3.8.5) gives rise to a magnetic gradient which does cause a real particle drift v_D, mentioned in Eq. (1.7.4) and in Appendix I. Show that v_D is small compared with $V_e \equiv J/ne$ when $\beta = n\kappa T/(B^2/8\pi)$ is small. ////

If there is an electric field in a plasma, there is a mass flow, even if the pressure is constant and the current is zero, since in a steady-state, current-free constant-pressure plasma Ohm's law (3.7.3) gives

$$\mathbf{E} + \frac{\mathbf{V} \times \mathbf{B}}{c} = 0 \qquad (3.8.6)$$

Vector multiplication of (3.8.6) by $\mathbf{B}$ gives

$$\mathbf{V}_\perp = \frac{\mathbf{E} \times \mathbf{B}}{B^2} c \qquad (3.8.7)$$

Conversely, motion of a plasma across a magnetic field gives rise to an electric field perpendicular to $\mathbf{V}$ and $\mathbf{B}$. This can be used to convert plasma flow energy to electric energy (see Chap. 6).

3.8.2 Magnetic Pressure

When there is a gradient in the plasma pressure, then under static equilibrium conditions there will be a gradient in the magnetic pressure. Eliminating **J** as in (3.8.3), the pressure and magnetic gradients are related by

$$\nabla\left(p+\frac{B^2}{8\pi}\right)=\frac{\mathbf{B}\cdot\nabla\mathbf{B}}{4\pi}$$

If the field lines are straight and parallel, this reduces to a simple statement of pressure balance,

$$\nabla\left(p+\frac{B^2}{8\pi}\right)=0 \qquad (3.8.8)$$

Clearly, from (3.8.8), a magnetic field exerts a pressure on the plasma, and $B^2/8\pi$ is called the magnetic pressure. However, all systems that satisfy (3.8.8) will not be static. Ohm's law shows that if the plasma is collisional, a gradient in B will be modified by magnetic field diffusion into the plasma. The condition $\sigma \to \infty$ is called the *infinite conductivity limit*.

3.8.3 Self-Confinement; The Equilibrium Pinch

A simple example of a plasma equilibrium is the pinch, in which plasma currents provide the magnetic pressure, which in turn confines the plasma. The pinch effect was first investigated by Bennett[1], in connection with the problem of self-focusing of a burst of fast electrons ejected by a solar flare into the low-density interplanetary plasma. The pinch effect occurs when the magnetic pressure $B^2/8\pi$ exceeds the particle pressure $n\kappa T$. Two phenomena are possible: (1) the magnetic field can diffuse in the plasma or (2) the plasma boundary can yield and move in a direction to increase the particle pressure. The first of these possibilities is called *resistive diffusion*; the second is the *dynamic-pinch effect*. Resistive diffusion is discussed in the next section.

It is possible to have a balance between particle pressure and magnetic pressure on a time scale shorter than the resistive-diffusion time scale. This is called the *equilibrium pinch*. The equilibrium pinch illustrates the use of the steady-state one-fluid MHD equations.

Consider a cylinder of plasma as shown in Fig. 3.8.2. The plasma is bounded with a radius $r = a$, and there are no fields except those produced by the plasma. The plasma carries a uniform current density $\mathbf{J} = J_z\hat{\mathbf{z}}$ for $r < a$, and of course $J = 0$ for $r > a$.

[1] W. H. Bennett, *Phys. Rev.*, **45**: 890 (1934).

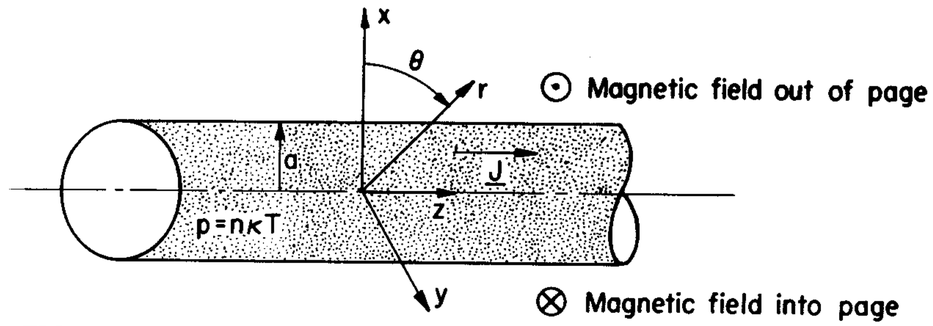


FIGURE 3.8.2
Schematic of a current-carrying plasma column that illustrates the plasma pinch effect. Plasma current J creates a magnetic field B_0 that exerts a radially inward force acting against the plasma pressure nkT.

If the plasma achieves a steady state, the one-fluid equations (3.7.1) to (3.7.4) for this cylindrically symmetric case reduce to

$$\frac{1}{r}\frac{d}{dr}r\rho_m V_r = 0 \qquad (3.8.9)$$

$$-\frac{J_z B_\theta}{c}=\frac{dp}{dr} \qquad (3.8.10)$$

$$E_r-\frac{V_z B_\theta}{c}=\frac{1}{ne}\frac{dp_i}{dr}, \qquad E_\theta = 0, \quad E_z = \eta J_z \qquad (3.8.11)$$

$$\frac{1}{r}\frac{d}{dr}(rB_\theta)=\frac{4\pi}{c}J_z \qquad (3.8.12)$$

where the dp_i/dr term in (3.8.11), neglected in the simplified one-fluid equation (3.7.3), is retained here. This is because, in the linear pinch, the ion pressure gradient may not be as small as E_r. It is important to note in the following material that the gross dynamics of the linear pinch do not depend on Eq. (3.8.11). See, however, the discussion in Sec. 3.8.4, where dp/dx is important.

Note that only two equations, (3.8.10) and (3.8.12), are required to determine the field **B** and the pressure p in terms of the (assumed given) axial current J_z. These can be solved to give the pressure distribution and radius of the equilibrium pinch, independent of the values of V_r, E_r, etc. The only restriction imposed on the solution has been azimuthal symmetry, $(\partial/\partial\theta)(B, \mathbf{V}, p, \ldots) = 0$. Since J_z = constant, Eq. (3.8.12) can be integrated, giving the magnetic field

$$B_\theta=\begin{cases}\dfrac{2\pi}{c}J_z r & r \le a\\[2ex] \dfrac{2\pi}{c}J_z\dfrac{a^2}{r} & r > a\end{cases} \qquad (3.8.13)$$

and the pressure distribution from (3.8.10) is

$$p(r) = \begin{cases} \dfrac{\pi J_z^2}{c^2}(a^2 - r^2) & r < a \\ 0 & r > a \end{cases} \tag{3.8.14}$$

These two equations describe the *equilibrium pinch*, and show that the radius of such a plasma is determined by the pressure on axis (tending to increase a) and the current (producing B_θ, which tends to reduce a), leading to an equilibrium radius

$$a = \sqrt{\frac{c^2 p(r=0)}{\pi}}\,\frac{1}{J_z} \tag{3.8.15}$$

Note that writing (since B_θ/r = constant for $r \le a$)

$$-\frac{B_\theta}{c} J_z = -\frac{B_\theta}{c}\left(\frac{c}{4\pi r}\frac{d}{dr} r B_\theta\right) = -\frac{1}{4\pi}\frac{d}{dr} B_\theta^2$$

the pressure-balance condition in the pinch is

$$p + \frac{B_\theta^2}{4\pi} = \text{constant} \tag{3.8.16}$$

which is also consistent with the general equilibrium condition

$$\nabla\left(p + \frac{B^2}{8\pi}\right) = \frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{4\pi}$$

since $\mathbf{B}\cdot\nabla\mathbf{B} = -\hat{\mathbf{r}}(B_\theta^2/r)$ for circular field lines $\mathbf{B} = B_\theta(r)\hat{\boldsymbol{\theta}}$.

Problem 3.8.2 Prove that

$$\mathbf{B}\cdot\nabla\mathbf{B} = -\frac{B_\theta^2}{r} = -\frac{d}{dr}\frac{B_\theta^2}{2}$$

for B_θ given by a uniform current density $\mathbf{J} = J_z\hat{\mathbf{z}}$. ////

The total pressure on axis is $p(0) = n_e\kappa T_e + n_i\kappa T_i$, and in terms of the total number of ions or electrons per unit length, N, the total current $I = J_z\pi a^2$ is (if $T_e = T_i$)

$$I^2 = (2N_e\kappa T)c^2\pi a^2$$

$$I(\text{amperes}) = 5.7\times 10^{-4} N^{1/2} T^{1/2}\ (\text{keV}) \tag{3.8.17}$$

There is still a lot of freedom left in the equilibrium state that satisfied (3.8.13) to (3.18.15), since these equations do not determine the radial electric field, the fluid velocity, or the *density* variation (only the product $n\kappa T$ is determined). These leftover variables are not determined by the condition that the pinch be in equilibrium, but by the way the plasma is produced.

Since, assuming electrons carry most of the current,

$$J = -neV_e$$

the fact that density in general varies with radius through Eq. (3.8.14) implies that either the current density or the electron drift velocity must also vary with radius. The example above assumed J = constant, but depending on how the current is produced, a more likely state might be V_e = constant, T_e = constant, T_i = constant. In that case the equilibrium equations give

$$\frac{1}{r}\frac{d}{dr}\frac{r}{n}\frac{dn}{dr} = -\frac{4\pi}{c^2}\frac{e^2 V_e^2 n}{\kappa(T_e + T_i)} \tag{3.8.18}$$

The solution of this differential equation with the boundary condition $dn/dr = 0$ at $r = 0$ is

$$n(r) = \frac{n(0)}{(1 + br^2)^2} \tag{3.8.19}$$

where

$$b = \frac{\pi}{2}\frac{e^2 V_e^2}{c^2}\frac{n(0)}{\kappa(T_e + T_i)}$$

Again it should be noted that the *assumption* T = constant was made in the derivation. The fluid equations require some assumption for closure. T = constant took the place of an equation of state.

Problem 3.8.3 What current would be required to confine a 1-cm-diameter 10-keV plasma with a density of 10^{14} cm^{-3}? ////

Problem 3.8.4 A cylinder of plasma of radius a carries a uniform current of density J_z along its axial direction. If the plasma is neutral, then as above, it will be radially confined ($\partial n/\partial t = 0$, $V_r = 0$) if the current is $J_z = (c/a)\sqrt{p(0)/\pi}$, where $p(0)$ is the total plasma pressure on axis. From similar arguments find: (*a*) What uniform current J_z is required to radially confine a nonneutral plasma in which N electrons and RN ions, both at the same temperature, are uniformly distributed within a cylinder of radius a and length L ($R < 1$)? And (*b*) If $R = 0$, can the electrons be self-confined by a physically realizable current J_z? ////

Problem 3.8.5 Verify (3.8.18) and (3.8.19). ////

3.8.4 Isothermal Atmosphere

The isothermal atmosphere considered as an example of the statistical approach in Chap. 2 can also be examined with the steady-state one-fluid equations. Consider a plasma including a gravitational force per unit mass $\mathbf{F}_g = -\mathbf{g}$. The momentum transport equation for a steady state is

$$\nabla p = \frac{\mathbf{J} \times \mathbf{B}}{c} - \rho_m \mathbf{g} \qquad (3.8.20)$$

while the appropriate Ohm's law for a plasma in a gravitational field, retaining dp_i/dx, is

$$\mathbf{E} + \frac{\mathbf{V} \times \mathbf{B}}{c} = \eta \mathbf{J} + \frac{1}{n_e e}(\rho_m \mathbf{g} + \nabla p_i) \qquad (3.8.21)$$

If the magnetic field B is small, and also if $p_i = p_e = p/2$, eliminating the pressure between (3.8.20) and (3.8.21) gives

$$\mathbf{E} = \eta \mathbf{J} + \frac{m_i}{2e}\mathbf{g} \qquad (3.8.22)$$

If there are no currents at the edge of the atmosphere, the electric field is

$$e\mathbf{E}_{\max} = \frac{m_i}{2}\mathbf{g} \qquad (3.8.23)$$

This shows that the vertical electric field reduces the net force on each ion to one-half its gravitational value (as if ions were *lighter* in a plasma atmosphere than in a neutral-gas atmosphere), and it provides a downward force on each electron equal to the downward force on the ions. This downward force on each electron far exceeds the gravitational force on the electron (as if the electron were half as heavy as an ion), by $m_i g/2$.

Problem 3.8.6 Justify (3.8.20) and (3.8.21) as the correct way to include gravitational force in the one-fluid equations. ////

Problem 3.8.7 What is the density distribution in the isothermal atmosphere? Using the $\mathbf{B}$ and $\mathbf{F}_g$ for the earth, is the neglect of B in (3.8.21) justified? ////

3.9 DYNAMIC PROPERTIES OF A PLASMA DESCRIBED BY THE ONE-FLUID AND MHD PLASMA THEORIES

Nonequilibrium solutions of the magnetohydrodynamic equations (3.7.1) to (3.7.6), together with an equation of state, are in general difficult to obtain. With the approximations described above, however, these equations can be solved to illustrate many plasma features. The fluid velocity can be expressed in terms of $\mathbf{B}$ and ∇p, using Maxwell's equations, giving

$$\rho_m \frac{\partial \mathbf{V}}{\partial t} = \frac{(\nabla \times \mathbf{B}) \times \mathbf{B}}{4\pi} - \nabla p \qquad (3.9.1)$$

This equation involves only neglect of the displacement current $(1/c)(\partial \mathbf{E}/\partial t)$, justified by $V_0^2 \ll c^2$, and is independent of collision frequencies. A second equation relating $\mathbf{B}$ and $\mathbf{V}$ is found by taking the curl of Ohm's law, noting that $\nabla \times \nabla p = 0$ and using $\nabla \times \mathbf{J} = (c/4\pi)\nabla \times \nabla \times \mathbf{B}$ to eliminate $\mathbf{J}$. Neglecting $\mathbf{J} \times \mathbf{B}$ and $\partial \mathbf{J}/\partial t$, the result is

$$\frac{c^2}{4\pi\sigma}\nabla^2 \mathbf{B} - \frac{\partial}{\partial t}\mathbf{B} = \mathbf{B}(\nabla \cdot \mathbf{V}) - (\mathbf{B} \cdot \nabla)\mathbf{V} + (\mathbf{V} \cdot \nabla)\mathbf{B} \qquad (3.9.2)$$

The electric field equation is obtained by taking $\partial/\partial t$ of Ohm's law, and using

$$\nabla \times \nabla \times \mathbf{E} = \frac{1}{c}\frac{\partial}{\partial t}\nabla \times \mathbf{B} = -\frac{4\pi}{c^2}\frac{\partial \mathbf{J}}{\partial t}$$

to write $\eta \mathbf{J}$ in terms of $\mathbf{E}$. The result is

$$\nabla^2 \mathbf{E} - \frac{4\pi\sigma}{c^2}\frac{\partial \mathbf{E}}{\partial t} = \frac{4\pi\sigma}{c^2}\frac{\partial}{\partial t}\frac{\mathbf{V} \times \mathbf{B}}{c} \qquad (3.9.3)$$

Apart from the choice of equation of state, the first two equations, (3.9.1) and (3.9.2), are simultaneous equations in the fluid velocity and the magnetic field.

3.9.1 Resistive Diffusion and Frozen Field Lines

A simple example of the use of the fluid equation to examine a nonequilibrium problem is that of the diffusion of magnetic field into a plasma at rest ($V = 0$). This problem is the same as the eddy-current problem in electrodynamics, except that the conducting medium is now a fluid instead of a rigid conductor. The magnetic field diffusion time is the longest that one might ever expect a magnetic field to confine a plasma or to move a plasma rigidly. For instance, the motion of magnetic field lines in the earth, the sun, or interstellar matter may be governed by a diffusion equation.

The equation for diffusion of the magnetic field is [from (3.9.2)]

$$\nabla^2 \mathbf{B} - \frac{4\pi\sigma}{c^2}\frac{\partial \mathbf{B}}{\partial t} = 0 \qquad (3.9.4)$$

Solutions to this equation have the form

$$B \propto e^{-t/\tau_m}$$

where $\tau_m = 4\pi\sigma L^2/c^2$ (L is the length which characterizes the magnetic gradients in the conducting object). For the earth's core, $\tau_m \approx 10^4$ years; for a copper sphere 1 m in radius, $\tau_m \approx 10$ s. Again defining V_0 as a characteristic plasma velocity and L as a characteristic gradient distance, (3.9.2) is often written $\partial\mathbf{B}/\partial t_n = \nabla_n \times \mathbf{V}_n \times \mathbf{B} + \nabla_n^2\mathbf{B}/R_M$, where $t_n = tV_0/L$, $V_n = V/V_0$, $\nabla_n = L\nabla$, and $R_M = 4\pi\sigma LV_0/c^2$ is the magnetic Reynolds number. If R_M is small, the field slips freely through the plasma. If the conductivity ($R_M \to \infty$) is infinite, the magnetic field does not diffuse through the medium. The field lines are "frozen" into the fluid medium and move with the fluid as it moves. Remember that this statement is restricted by the approximations made in simplifying Ohm's law. The equation for the time development of $\mathbf{B}$ in this *frozen-field-line* situation is

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{V} \times \mathbf{B}) \qquad (3.9.5)$$

Problem 3.9.1 Show that the change of $\mathbf{B}$ with time in a fluid element is related to changes of density by

$$\frac{D}{Dt}\frac{\mathbf{B}}{\rho} = \frac{\mathbf{B}}{\rho} \cdot \nabla \mathbf{V}$$

where

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{V} \cdot \nabla \qquad ////$$

3.9.2 Low-amplitude Waves; Alfvén and Ion Acoustic

Waves are a widely occurring nonequilibrium phenomenon in plasmas. The one-fluid equations can be solved for a variety of wave types, and will be treated in detail in Chap. 5. Here a few simple examples will be given to further illustrate MHD behavior. One well-known plasma wave is the Alfvén wave. In 1942 Alfvén recognized that a wave could propagate in an incompressible, perfectly conducting fluid immersed in a strong steady magnetic field, the magnetic field providing the restoring force and the fluid density providing the inertia. To see this take $\nabla \cdot \mathbf{V} = 0$, $p = 0$, and $\sigma = \infty$, in addition to neglecting $\mathbf{J} \times \mathbf{B}$ and $\partial\mathbf{J}/\partial t$ in Ohm's law. The result is

$$\rho_m \frac{\partial \mathbf{V}}{\partial t} = \frac{(\nabla \times \mathbf{B}) \times \mathbf{B}}{4\pi} \qquad (3.9.6)$$

$$\frac{\partial \mathbf{B}}{\partial t} = (\mathbf{B} \cdot \nabla)\mathbf{V} - (\mathbf{V} \cdot \nabla)\mathbf{B} \qquad (3.9.7)$$

To solve (3.9.6) and (3.9.7), assume the magnetic field to have a large uniform and steady part $\mathbf{B}_0$ plus a small oscillating part $\mathbf{B}_1$ (that is, $|\mathbf{B}_1| \ll |\mathbf{B}_0|$). The

average fluid velocity $\mathbf{V}_0$ is zero, and its perturbed velocity is $\mathbf{V}_1$. Neglecting terms that involve products of perturbation quantities gives

$$\rho_m \frac{\partial \mathbf{V}_1}{\partial t} = \frac{1}{4\pi}(\nabla \times \mathbf{B}_1) \times \mathbf{B}_0 \qquad (3.9.8)$$

$$\frac{\partial \mathbf{B}_1}{\partial t} = (\mathbf{B}_0 \cdot \nabla)\mathbf{V}_1 \qquad (3.9.9)$$

Now, for simplicity, take the steady magnetic field to lie in the z direction (that is, $\mathbf{B}_0 = \hat{\mathbf{k}}B_0$). One self-consistent solution of (3.9.8) and (3.9.9) has a perturbed magnetic field and velocity which are independent of x and y and have components only in the x and y directions. Thus (3.9.8) and (3.9.9) become

$$\rho_m \frac{\partial V_{1x,y}}{\partial t} = \frac{B_0}{4\pi}\frac{\partial B_{1x,y}}{\partial z} \qquad (3.9.10)$$

$$\frac{\partial B_{1x,y}}{\partial t} = B_0 \frac{\partial V_{1x,y}}{\partial z} \qquad (3.9.11)$$

which combine to give a wave equation for either velocity or magnetic field

$$\left(\frac{\partial^2}{\partial t^2} - V_A^2 \frac{\partial^2}{\partial z^2}\right)\begin{bmatrix} B_{1x,y} \\ V_{1x,y} \end{bmatrix} = 0 \qquad (3.9.12)$$

where $V_A = B_0/\sqrt{4\pi\rho_m}$. Equation (3.9.12) is a wave equation; its solution

$$B_{1x} = \bar{B}\sin(kz - \omega t) \qquad (3.9.13)$$

describes a transverse wave that propagates along a magnetic field in an incompressible conducting fluid, with its perturbation fields perpendicular to B_0 and with a speed $\omega/k = V_A$, independent of frequency. This wave is called an *Alfvén wave*, and the wave speed is called the *Alfvén velocity*; i.e.,

$$V_A = B_0/\sqrt{4\pi\rho_m} \qquad (3.9.14)$$

Alfvén waves and their properties are studied in more detail in Chap. 4. The purpose of this brief analysis is to illustrate the use of the one-fluid MHD equations in a nonequilibrium situation. An example of an incompressible conducting fluid is liquid mercury. However, Alfvén waves are a property of "ordinary" plasmas as well as of incompressible fluids (see Sec. 3.10).

Problem 3.9.2 Both the pressure and the resistivity were neglected in finding the Alfvén wave, although $p \to 0$ when $T \to 0$ while $\eta \to 0$ when $T \to \infty$. Find the frequency range which justifies the neglect of η for a plasma of ionospheric density and temperature. Also justify the neglect of pressure in the same parameter range with $B_0 = B_{\text{earth}}$. ////

An example of a nonequilibrium phenomenon that involves the plasma pressure is the ion-acoustic wave. If a uniform field-free plasma ($E_0 = B_0 = \nabla p_0 = V_0 = 0$) is slightly perturbed, for example, by a small density compression, the one-fluid equations predict the time history of the perturbation. As in the previous example, if the perturbed quantities are assumed small, products of the perturbed variable can be neglected, and Eqs. (3.7.1) to (3.7.6) become

$$\rho_{m0} \frac{\partial \mathbf{V}_1}{\partial t} = -\nabla p_1 \qquad (3.9.15a)$$

$$\frac{\partial}{\partial t} \rho_{m1} = -\rho_{m0} \nabla \cdot \mathbf{V}_1 \qquad (3.9.15b)$$

For simplicity, the perturbations can be taken one-dimensional, $\rho_{m1} = \rho_{m1}(x)$, $\nabla p = dp/dx$, $\mathbf{V}_1 = V_1 \hat{\mathbf{x}}$. Solution of (3.9.15) requires an equation of state, because otherwise there would be two equations and three unknown quantities. A general form for the equation of state is

$$p = \frac{p_0 \rho_m{}^{\gamma}}{\rho_{m0}{}^{\gamma}} \qquad (3.9.16)$$

with γ not yet specified. In terms of perturbed quantities,

$$p_1 = \gamma p_0 \frac{\rho_{m1}}{\rho_{m0}} \qquad (3.9.17)$$

The constant γ must be determined from considerations outside the framework of the fluid equations. Equations (3.9.15) can be combined to give

$$\frac{\partial^2 \rho_{m1}}{\partial t^2} = \frac{\gamma p_0}{\rho_{m0}} \frac{\partial^2 \rho_{m1}}{\partial x^2} \qquad (3.9.18)$$

The solution to (3.9.18) is a compression wave

$$\rho_{m1} = \bar{\rho}_{m1} \sin(kx - \omega t)$$

with speed

$$\frac{\omega}{k} = \left(\frac{\gamma p_0}{\rho_{m0}}\right)^{1/2} = \left(\kappa \frac{\gamma_e T_e + \gamma_i T_i}{m_i}\right)^{1/2} \qquad (3.9.19)$$

In Chap. 8 it is shown that unless $T_e \gg T_i$, the wave (3.9.19) is heavily Landau-damped. The constant γ can be determined by noting that with $T_e \gg T_i$, the wave (3.9.19) moves with a speed much less than the electron thermal speed; i.e.,

$$\frac{\omega}{k} = \left(\gamma_e \frac{\kappa T_e}{m_e}\right)^{1/2} \left(\frac{m_e}{m_i}\right)^{1/2} \ll \left(\frac{\kappa T_e}{m_e}\right)^{1/2}$$

Thus, in a time ω^{-1}, a thermal electron moves a distance $(m_i/m_e)^{1/2} k^{-1}$, which is much longer than a wavelength. This excellent communication means that the electron temperature must be constant over many wavelengths. This suggests that the process is isothermal, with

$$\gamma_e = 1 \qquad \text{isothermal compression}$$

The wave

$$\omega = k\sqrt{\frac{\kappa T_e}{m_i}} = kC_s \qquad (3.9.20)$$

is called the ion-acoustic wave, and C_s is the ion-sound speed.

3.9.3 Incompressibility as a Plasma Property

The derivation of the Alfvén wave used $\nabla \cdot \mathbf{V} = 0$ as an auxiliary equation, with the statement that it is appropriate for an incompressible fluid. Although there are conducting fluids (e.g., liquid mercury) that are incompressible, it is not suggested that a *plasma* is incompressible. Indeed, if $\nabla \cdot \mathbf{V} = 0$ had been used in the calculation of ion-acoustic waves, there would have been no wave [Eq. (3.15*b*) would give $\partial \rho_{m1}/\partial t = 0$; Eq. (3.9.17) would give $p_1 = 0$; and Eq. (3.9.15*a*) would give $V_1 = 0$]. The student may well ask what relation the Alfvén wave in an incompressible fluid has to the behavior of a plasma. The facts are that (1) the Alfvén wave does propagate in plasma with exactly the properties (3.9.12) to (3.9.14) and (2) $\nabla \cdot \mathbf{V} = 0$ is correct for many types of disturbances in a plasma, and is a great simplification when it is valid. The point is *not* that the plasma is incompressible; rather, that there are perturbations that do not compress the plasma. So the calculation of the Alfvén wave is valid even in the (compressible) plasma, describing a perturbation

$$V_{1x} = V_{1x}(z)$$
$$\rho_1 = 0$$

that does not compress the plasma. In this case $V_{1x}(z)$ produces, through $\mathbf{V} \times \mathbf{B}_z$ in Ohm's law, a perturbed electric field $E_y(z)$. The electric field in turn implies through [$\nabla \times \mathbf{E} = -(1/c)\, \partial \mathbf{B}/\partial t$] a perturbed magnetic field $\mathbf{B}_{1x}(z)$ and a current $\mathbf{J}_{1y} = (c/4\pi)\nabla \times \mathbf{B}$. The momentum transport equation again couples the force $\mathbf{J}_1 \times \hat{\mathbf{z}} B_0$ to the momentum change $\rho \partial V_{1x}/\partial t$. Thus no motion $V_{1z}(z)$ is induced by perturbations $V_{1x}(z)$, and $\nabla \cdot \mathbf{V} = 0$ is a *consistent* equation. Note that in the ion-acoustic-wave calculation, the perturbation was *chosen* to be $\mathbf{V}_1 = V_1(x)\hat{\mathbf{x}}$, and the calculation led to a compression wave. In the Alfvén wave calculation, instead of imposing $\nabla \cdot \mathbf{V} = 0$, the perturbation could have been chosen to be $\mathbf{V}_1 = V_1(z)\hat{\mathbf{x}}$, and the Alfvén waves would have emerged. Thus

the statement "we take the plasma to be incompressible, $\nabla \cdot \mathbf{V} = 0$," which is found in many calculations, is really a shorthand way of expressing something much more complicated.

3.9.4 Ion-acoustic-type Solitary and Shock Waves[1]

The ion-acoustic wave is used as the basis for a theory of nonmagnetic collision-free shock waves; this application shows the effect of some of the terms neglected in the simplified one-fluid equations (3.7.1) to (3.7.6), namely, $(\mathbf{V} \cdot \nabla)\mathbf{V}$ and $(n_e - n_i)/n_e$.

Consider a field-free uniform plasma ($\mathbf{E}_0 = \mathbf{B}_0 = p_0 = V_0 = 0$) of hot electrons and cold ions that is given a density perturbation. Then the *two-fluid* equations for the development of the perturbation are [the perturbation is taken to be one-dimensional with all quantities only functions of x; the perturbed velocity, appropriate to the ion-acoustic wave, is also taken to be in the x direction, that is, $\mathbf{V}_1 = V_1(x)\hat{\mathbf{x}}$]

$$\frac{\partial n_{i,e}}{\partial t} + \frac{\partial n_{i,e} V_{i,e}}{\partial x} = 0 \qquad (3.9.21)$$

$$\frac{\partial V_{i,e}}{\partial t} + V_{i,e}\frac{\partial V_{i,e}}{\partial x} = -\frac{q_{i,e}}{m_{i,e}}\frac{\partial \phi}{\partial x} - \frac{1}{(nm)_{i,e}}\frac{\partial p_{i,e}}{\partial x} \qquad (3.9.22)$$

$$\frac{\partial^2 \phi}{\partial x^2} = -4\pi e(n_i - n_e) \qquad (3.9.23)$$

With the approximations $n_e = n_i$ and $\mathbf{V} \cdot \nabla\mathbf{V} = 0$, these reduce to the one-fluid equations, which lead to solutions $V_1, \rho_1 \approx \sin k(x - C_s t)$, as in Eq. (3.9.19). If $n_e = n_i$ is assumed, but $\mathbf{V} \cdot \nabla\mathbf{V}$ is not discarded, the solution to (3.9.21) to (3.9.23) is a wave $\sin k(x, t)\,(x - C_s t)$, which steepens until, in a finite time, the gradients in ρ_1, V_1 become infinite, after which the profiles $\rho_1(x)$, $V_1(x)$ become double-valued,[2] as in Fig. 3.9.1, and there is no longer a wavelike solution to (3.9.21) to (3.9.23). This phenomenon is called *wave breaking*.

Problem 3.9.3 Show by iteration that the $\mathbf{V} \cdot \nabla\mathbf{V}$ term leads to gradients which steepen with time. ////

Problem 3.9.4 Show that the linearized *two-fluid* equations with $T_i = 0$ predict ion-sound waves with $\omega = kC_s(1 + k^2\lambda_{De}{}^2)^{-1/2}$. ////

[1] D. A. Tidman and N. A. Krall, "Shock Waves in Collisionless Plasmas," Wiley, New York, 1971, gives an exhaustive treatment of this and other types of collision-free shock waves.

[2] L. D. Landau and E. M. Lifshitz, "Fluid Mechanics," pp. 366–370, Pergamon, London, 1959.

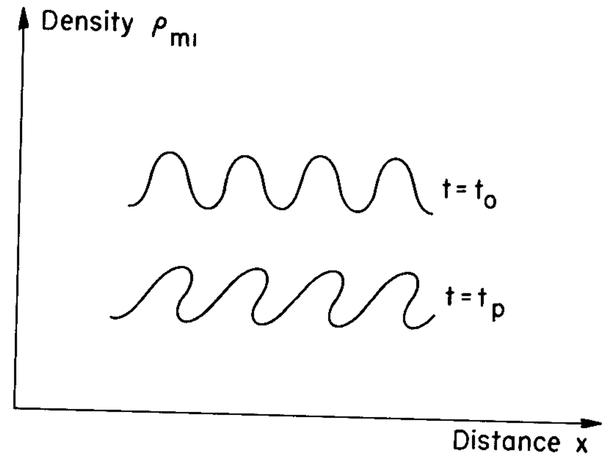


FIGURE 3.9.1
Sketch of perturbed density as a function of distance at two different times, illustrating how a shock wave becomes double-valued prior to the phenomenon of "breaking."

If the constraint $n_e = n_i$ is removed, for a low-amplitude wave the steepening is eventually balanced by dispersion, as discussed below. The steepening occurs because nonlinear terms in a wave equation couple higher harmonics $(2k, 2\omega)$ to the original wave (k, ω). If the higher harmonics travel at the same speed as the fundamental, this coupling is strong. The appearance of shorter wavelengths means steepening. If higher harmonics travel at slower (or faster) speeds than the fundamental, the coupling is weaker and *steepening* stops. In the case of ion-sound waves, when the wavelength approaches the Debye length, charge separation ($n_e \neq n_i$) affects the wave, and shorter wavelengths go more slowly.[1] This is called *dispersion*. The result of the competition between steepening and dispersion leads to a solution in which ρ_1 and V_1 are constant in a frame moving with speed U, of order C_s, as will now be demonstrated.

Because the electron mass is so small, the momentum equation for electrons [Eq. (3.9.22)] can be approximated by

$$0 = en_e\frac{\partial \phi}{\partial x} - \frac{\partial p_e}{\partial x} \qquad (3.9.24)$$

The approximation T_e = constant is a good one, as remarked earlier, because the electrons move so rapidly compared with ion waves. In that case the solution of (3.9.24) is

$$n_e = n_0\, e^{e\phi/\kappa T_e}$$

[1] The fact that things change on a scale $L \approx \lambda_D$ should be clear; see, for example, the discussion following (3.5.1); the explicit dependence of ω/k for ion waves on wavelength will be calculated explicitly in Chap. 8.

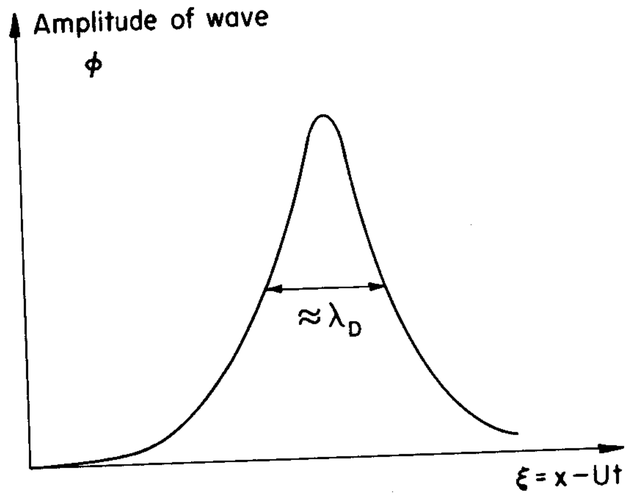


FIGURE 3.9.2
Amplitude of soliton as a function of distance, in a frame where the disturbance is at rest.

The ion dynamics are determined by

$$\frac{\partial V_i}{\partial t} + V_i \frac{\partial V_i}{\partial x} = -\frac{e}{m_i}\frac{\partial \phi}{\partial x}$$

$$\frac{\partial n_i}{\partial t} + \frac{\partial}{\partial x} n_i V_i = 0 \tag{3.9.25}$$

$$\frac{\partial^2 \phi}{\partial x^2} = -4\pi e(n_i - n_0 e^{e\phi/\kappa T_e})$$

Problem 3.9.5 The assumption $n_e = n_i$, $\phi = 0$, in (3.9.25), no longer leads to ion-acoustic waves. Why? ////

If a steady state is reached in a frame moving with speed U, all quantities must be functions of $\xi = x - Ut$. This leads at once to the steady-state equation for ϕ,

$$\frac{\partial^2 \phi}{\partial \xi^2} = -4\pi e\left(\frac{n_0}{\sqrt{1 - 2e\phi/m_i U^2}} - n_0 e^{e\phi/\kappa T_e}\right) \tag{3.9.26}$$

For small $e\phi/\kappa T_e$, this equation has a solution of the form shown in Fig. 3.9.2. This solution is a finite-amplitude wave propagating through a plasma. It is called a *solitary wave*, or *soliton*. Similar waves exist corresponding to other plasma waves of the sort $\omega = kV$.

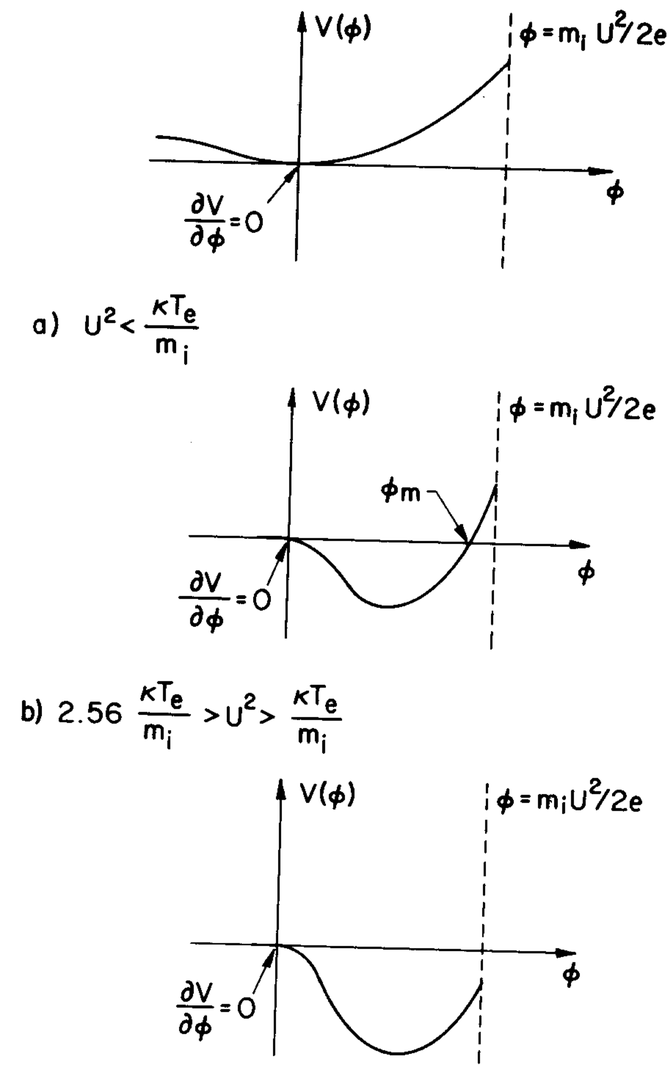


FIGURE 3.9.3
Solutions for the potential $V(\phi)$ as a function of ϕ for different choices of the speed U.

There are a variety of ways of finding a solution to (3.9.26) of the form shown in Fig. 3.9.2. For example, if ϕ is the "coordinate" of a particle, (3.9.26) has the form

$$\frac{dV(\phi)}{d\phi} = \frac{d^2\phi}{d\xi^2}$$

which is the equation of motion (ξ is the "time") of a particle in a potential well.

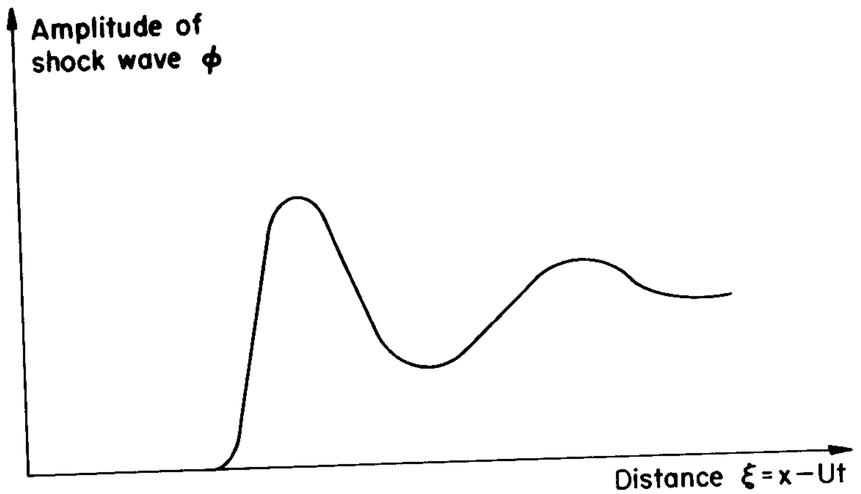


FIGURE 3.9.4
Schematic of a collisionless shock wave.

Problem 3.9.6 Show by expanding (3.9.26) for small ϕ that it has a solution $\phi = \phi_{max} \cosh^{-l}(\xi/\Delta)$. Find l, Δ, and U in terms of the wave amplitude ϕ_{max} for small ϕ. ////

Problem 3.9.7 Show directly from (3.9.26) that

$$U = -\left(\frac{\kappa T_e}{2m_i}\right)^{1/2} \frac{1 - e^{e\phi_m/\kappa T_e}}{(e^{e\phi_m/\kappa T_e} - 1 - e\phi_m/\kappa T_e)^{1/2}}$$

(*Hint:* $d\phi/d\xi = 0$ when $\phi = \phi_m$.) ////

Figure 3.9.3 shows $V(\phi)$ with different choices for the speed U. The "particle" starts at $\phi = 0$, and the solution has the form of Fig. 3.9.2 if

$$\sqrt{\kappa T_e/m_i} < U < 1.6\sqrt{\kappa T_e/m_i}$$

U increases with the amplitude ϕ_{max} of the solitary wave (see Problem 3.9.6 or Fig. 3.9.3 or Prob. 3.9.7), and this solitary wave does not exist if its amplitude is too large. In practice, the physical situation will dictate either U or ϕ_{max}. The relation $U = \sqrt{\kappa T_e/m_i}\,\mathcal{M}$ defines the Mach number $\mathcal{M}$. From Fig. 3.9.3 it is clear that no soliton exists for $\mathcal{M} < 1$ or for $\mathcal{M} > 1.6$.

If the effect of infrequent collisions or of the reflection of very high energy ions $E_i \approx e\phi/\kappa T$ from the potential ϕ were included, the symmetry of the solution would be broken, and the solution would have the form shown in Fig. 3.9.4. (Imagine ϕ as a damped oscillation in a potential well.)

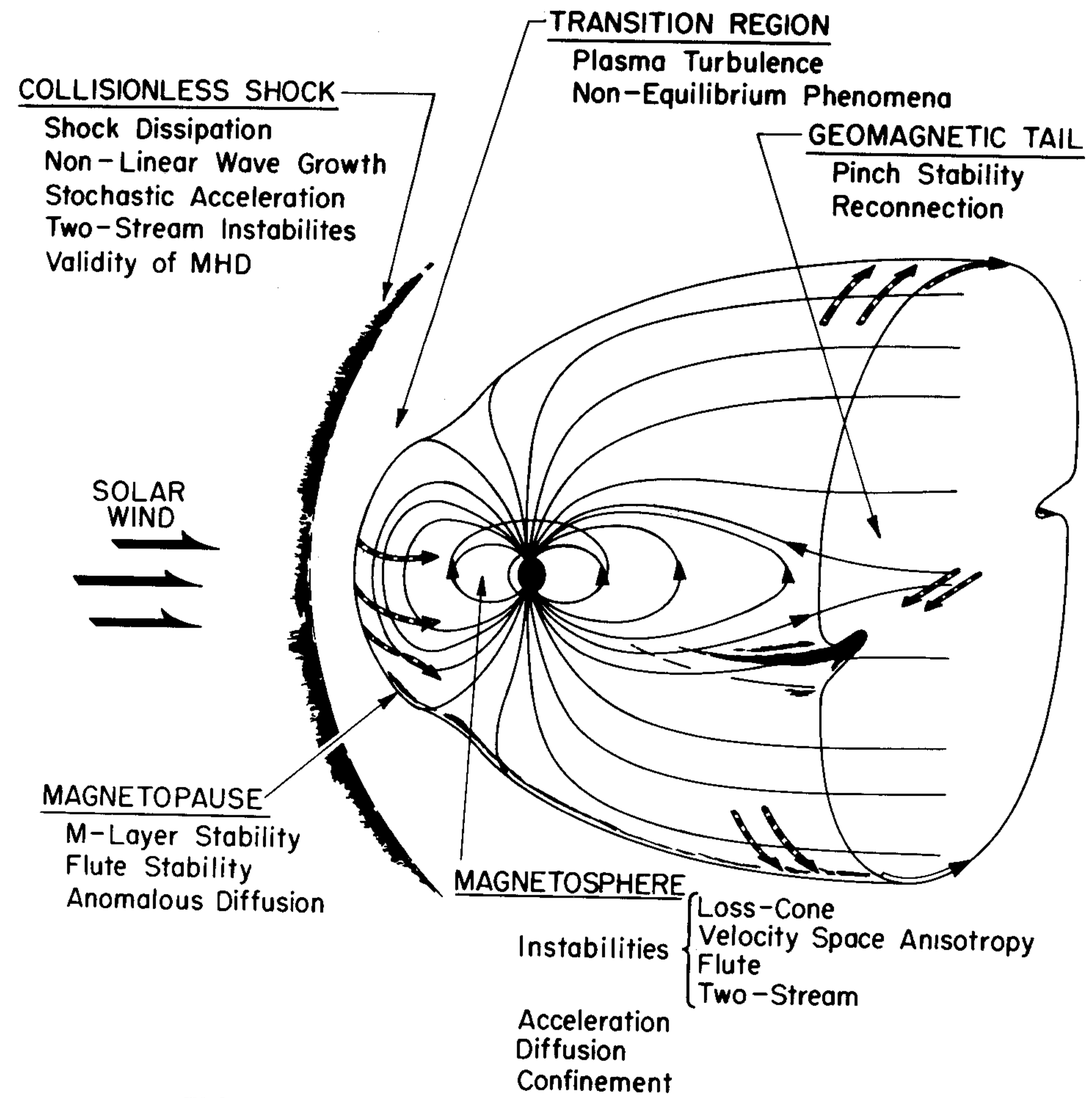


FIGURE 3.9.5
Sketch of solar wind impinging on the earth's magnetic field, showing the bow-shock region.

This change in plasma state across the distance λ_D, even though $\lambda_{collisions} \gg \lambda_D$, is called a *collisionless* shock wave. When the solar wind, streaming from the sun, encounters the earth's magnetosphere, it is converted from a cold unidirectional stream to a hot random plasma in a distance λ which is much less than a mean free path for collisions. The transition layer is called the *bow shock*, as shown in Fig. 3.9.5. The earth's bow shock[1] has properties that are related to the phenomenon described here.

[1] W. M. Hess (ed.), "Introduction to Space Science," pp. 363, Gordon and Breach, New York, 1965.

3.9.5 Nonneutral Plasma–The Magnetically Confined Electron Gas

A system in which even equilibrium properties are influenced by $\mathbf{V}\nabla\cdot\mathbf{V}$ and $n_e \neq n_i$ is the unneutralized electron gas. Electrical neutrality $\bar{n}_e = \bar{n}_i$ has been assumed in order to simplify the mathematics for the one-fluid plasma model. It is possible, however, to have a plasma of only one charge species. The properties of the equilibrium of a gas of electrons in a uniform steady magnetic field are demonstrated using the momentum transfer equation (3.4.2) for electrons, with zero pressure gradient and no collisions,

$$m\left(\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V}\cdot\nabla\mathbf{V}\right) + e\mathbf{E} + \frac{e\mathbf{V}\times\mathbf{B}}{c} = 0 \qquad (3.9.27)$$

the continuity equation (3.4.1) for electrons,

$$\frac{\partial n}{\partial t} + \nabla\cdot n\mathbf{V} = 0 \qquad (3.9.28)$$

and Maxwell's equations with $n_i = 0$,

$$\nabla\cdot\mathbf{E} = -4\pi e n \qquad (3.9.29)$$

$$\nabla\times\mathbf{B} = -\frac{4\pi e}{c} n\mathbf{V} \qquad (3.9.30)$$

All quantities refer to electrons, and so the subscripts have been omitted. These equations can be used to analyze the situation depicted schematically in Fig. 3.9.6. In cylindrical coordinates (r, θ, z) these equations are

$$\frac{\partial}{\partial t} V_r - \frac{V_\theta^2}{r} + (\mathbf{V}\cdot\nabla)V_r = -\frac{e}{m} E_r - \omega_c V_\theta \qquad (3.9.31)$$

$$\frac{\partial V_\theta}{\partial t} + \frac{2V_r V_\theta}{r} + (\mathbf{V}\cdot\nabla)\mathbf{V}_\theta = -\frac{eE_\theta}{m} + \omega_c V_r \qquad (3.9.32)$$

$$\frac{1}{r}\frac{\partial}{\partial t}(rE_r) + \frac{1}{r}\frac{\partial}{\partial\theta} E_\theta = -4\pi e n \qquad (3.9.33)$$

where $\omega_c (= eB/mc)$ is the cyclotron frequency. An equilibrium solution $(\partial/\partial t \to 0)$ is possible for constant density as a function of radius (n = constant) and no azimuthal variation $(\partial/\partial\theta \to 0)$. For constant density, the radial electric field is

$$E_r = -2\pi n e r \qquad (3.9.34)$$

The constancy of B_z, $(\partial B_z/\partial t) = 0$, implies $E_\theta = 0$, and so (3.9.32) is satisfied for $V_r = 0$. With these assumptions, (3.9.31) becomes

$$\frac{V_\theta^2}{r} + \frac{2\pi n_0 e^2}{m} r - \omega_c V_\theta = 0 \qquad (3.9.35)$$

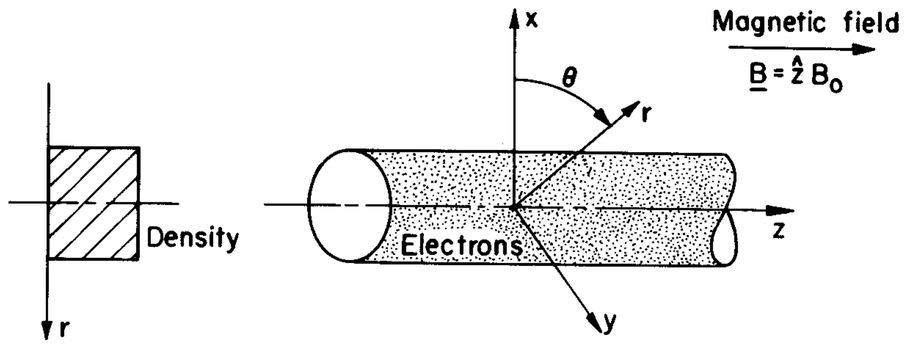


FIGURE 3.9.6
Cloud of electrons rotating as a rigid body in a uniform axial magnetic field. (Underscoring of symbol denotes vector.)

The solution of this quadratic equation is

$$\frac{V_\theta}{r} = \frac{\omega_c}{2}\left[1 \pm \left(1 - \frac{2\omega_p^2}{\omega_c^2}\right)^{1/2}\right] \qquad (3.9.36)$$

where $\omega_p^2 (= 4\pi n e^2/m)$ is the plasma frequency. There are two equilibrium "rigid-rotor" solutions to (3.9.36). These are shown in Fig. 3.9.7.

For low densities, $\omega_p^2/\omega_c^2 \ll 0.5$, one equilibrium involves rotation at nearly the cyclotron frequency, and the other involves rotation at a low angular frequency. Both of these equilibrium solutions involve a balance between the outward mechanical and electrostatic forces and the inward magnetic forces. At $\omega_p^2 = \omega_c^2/2$, the two solutions merge into one at an angular-rotation frequency of $\omega_c/2$. This equilibrium $(\omega_p^2 = \omega_c^2/2)$ is called *Brillouin flow*. It gives the least magnetic field for which there is an equilibrium flow of a drifting electron beam of a given charge density. The equilibrium solution is not self-consistent, in that it does not satisfy (3.9.30); i.e., the electron currents generate magnetic fields that in turn would alter the radial density distribution. This diamagnetic field is negligible if the electrostatic energy of the plasma is much less than the rest-mass energy density $E^2/8\pi \ll nm_0 c^2$, which is the case for nonrelativistic electron beams often used in the laboratory. There are now several experiments that study the equilibrium and stability of unneutralized plasmas. One of these experiments[1] creates an unneutralized electron ring plasma whose density is $\sim 10^{11}$ cm^{-3}, with a mean energy of 0.5 MeV in a pulsed 15-kG magnetic mirror field. The trapped plasma lasts for more than

[1] A. W. Trivelpiece, R. E. Pechacek, and C. A. Kapetanakos, *Phys. Rev. Letters*, **21**:1436 (1968).

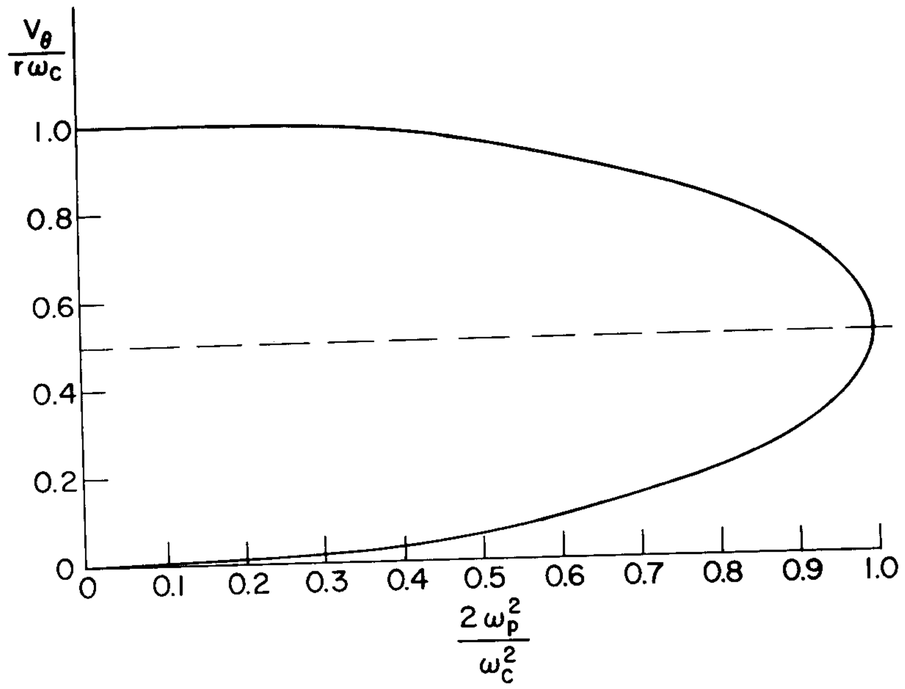


FIGURE 3.9.7
"Rigid-rotor" equilibrium angular velocity for an unneutralized uniform-density cloud of electrons in a uniform magnetic field showing the fast, or axis-encircling, mode (upper branch) and the slow, or drift, mode (lower branch) as a function of charge density.

15 ms. Other experiments in which such effects are important are the high-current (170-kA), high-energy (14-MeV), short-pulsed electron beams, which are being studied for the purpose of developing collective-effect acceleration.

3.10 DOUBLE-ADIABATIC THEORY

When the collision frequency is large enough to keep the plasma isotropic, and when the characteristic frequencies of interest are much less than the ion-cyclotron frequency, the one-fluid MHD equations, together with Maxwell's equations and an appropriate assumption to close the chain of moment equations, give an adequate description of the properties of a nonviscous, perfectly conducting, quasineutral plasma in a magnetic field.

However, if the magnetic field is large and collisions infrequent, the assumption of isotropy is no longer valid, and the plasma medium has different characteristics parallel and perpendicular to the magnetic field. Chew, Goldberger, and

Low[1] assumed negligible heat flow along the magnetic field and expanded the Vlasov equation in powers of $1/B$ to obtain a set of moment equations for the one-fluid macroscopic variables valid for situations where the phase velocity of plasma disturbances is greater than the mean thermal velocity of plasma particles, that is, $\omega/k \gg v_{\text{th}}$. The complete system of equations to describe a plasma using this model is

Continuity:
$$\frac{\partial}{\partial t}(\rho_m) + \nabla \cdot \rho_m \mathbf{V} = 0 \qquad (3.10.1)$$

Momentum transport:
$$\rho_m\left(\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V} \cdot \nabla \mathbf{V}\right) = -\nabla \cdot \mathsf{P} + \frac{\mathbf{J} \times \mathbf{B}}{c} \qquad (3.10.2)$$

Maxwell's equations:
$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t} \qquad (3.10.3)$$

$$\nabla \times \mathbf{B} = \frac{4\pi}{c}\mathbf{J} \qquad (3.10.4)$$

Ohm's law:
$$\mathbf{E} + \frac{\mathbf{V} \times \mathbf{B}}{c} = 0 \quad \text{(assuming } \sigma \to \infty\text{)} \qquad (3.10.5)$$

In the Chew-Goldberger-Low (CLG) theory the pressure is assumed locally isotropic in the plane perpendicular to the magnetic field; the pressure parallel to the local magnetic field is allowed to have an independent value $p_\parallel$. In other words, in a coordinate system whose z axis is aligned with $\mathbf{B}_0$ at a local point x_0, the pressure tensor has the form

$$\mathsf{P} = \begin{bmatrix} p_\perp & 0 & 0 \\ 0 & p_\perp & 0 \\ 0 & 0 & p_\parallel \end{bmatrix} \qquad (3.10.6)$$

In general, however, the magnetic field is curved, and there is no coordinate system in a fixed laboratory frame that has $\mathbf{B}$ aligned with a coordinate axis everywhere. In an arbitrary frame the pressure tensor (3.10.6) transforms to

$$\mathsf{P} = \begin{bmatrix} p_\perp + (p_\parallel - p_\perp)b_x b_x & (p_\parallel - p_\perp)b_x b_y & (p_\parallel - p_\perp)b_x b_z \\ (p_\parallel - p_\perp)b_y b_x & p_\perp + (p_\parallel - p_\perp)b_y b_y & (p_\parallel - p_\perp)b_y b_z \\ (p_\parallel - p_\perp)b_z b_x & (p_\parallel - p_\perp)b_z b_y & p_\perp + (p_\parallel - p_\perp)b_z b_z \end{bmatrix} \qquad (3.10.7)$$

where
$$\mathbf{b} = \frac{\mathbf{B}}{|B|}$$

[1] G. F. Chew, M. L. Goldberger, and F. E. Low, *Proc. Roy. Soc.* (*London*), **A236**: 112 (1956).

is a unit vector along the magnetic field. The tensor (3.10.7) can also be written in the form

$$\mathsf{P} = p_\perp \mathsf{I} + (p_\parallel - p_\perp)\mathbf{bb}$$

where I is the unit dyadic. For a curved field, $\mathbf{b} = \mathbf{b}(\mathbf{x})$, so that $\nabla \cdot \mathsf{P}$ contains not only the information on the variation of $p_\perp$ and $p_\parallel$ from point to point in the plasma, but also information on the change of the unit vector $\mathbf{b}$ from point to point.

If the magnetic field is nearly one-directional and uniform, that is, $\mathbf{B} = \mathbf{B}_0 + \mathbf{B}_1(\mathbf{x})$, as would be the case of low-amplitude waves propagating in a strong magnetic field, $\nabla \cdot \mathsf{P}$ simplifies by linearization in B_1. For example, if $\mathbf{B}_0 = B_0\hat{\mathbf{z}}$,

$$\nabla \cdot \mathsf{P} = \left[\frac{\partial}{\partial x} p_\perp + \frac{\partial}{\partial z}(p_\parallel - p_\perp)b_x\right]\hat{\mathbf{x}} + \left[\frac{\partial}{\partial y} p_\perp + \frac{\partial}{\partial z}(p_\parallel - p_\perp)b_y\right]\hat{\mathbf{y}}$$
$$+ \left[\frac{\partial}{\partial x}(p_\parallel - p_\perp)b_x + \frac{\partial}{\partial y}(p_\parallel - p_\perp)b_y + \frac{\partial}{\partial z} p_\perp + \frac{\partial}{\partial z}(p_\parallel - p_\perp)b_z^2\right]\hat{\mathbf{z}} \qquad (3.10.8)$$

Problem 3.10.1 Show that, in general,

$$(\nabla \cdot \mathsf{P})_\perp = \nabla_\perp p_\perp - (p_\parallel - p_\perp)(\mathbf{b} \cdot \nabla)\mathbf{b}$$

and

$$(\nabla \cdot \mathsf{P})_\parallel = (\mathbf{b} \cdot \nabla)p_\parallel + (p_\parallel - p_\perp)\nabla \cdot \mathbf{b} \qquad ////$$

There are two problems in the treatment of pressure common to all fluid theories. First is the question of isotropy, and whether it is reasonable to assume $\nabla \cdot \mathsf{P} = \nabla p$. It is in the treatment of this problem that the MHD theory differs most strongly from the CGL theory. Second, however, an equation of state must be selected to relate p to ρ, lest there be more unknowns than equations. The CGL theory requires two equations of state, one for $p_\perp$ and one for $p_\parallel$. For adiabatic processes in a collisionless gas, with isotropy in the plane perpendicular to $\mathbf{B}$ and no coupling of perpendicular and parallel motion (see Prob. 3.10.2),

$$\frac{p_\perp^2 p_\parallel}{\rho_m^5} = \text{constant} \qquad (3.10.9)$$

A second equation relating $p_\perp$, $p_\parallel$, and ρ_m is obtained by considering the dynamics of single-charged particles in a slowly time and space varying magnetic field. It is shown in Appendix I that the magnetic moment $\mu = mv_\perp^2/2B$ is nearly a constant of the motion. Thus (since $\overline{v_\perp^{-2}} \approx p_\perp/\rho_m$)

$$\frac{p_\perp}{\rho_m B} = \text{constant} \qquad (3.10.10)$$

is taken as the second "equation of state" required to close the CGL theory.

Problem 3.10.2 Show that for a collision-free ideal gas, if the motion in a plane ($\perp$) is independent of motion in the third direction ($\parallel$), the adiabatic equation of state is $p_\perp^2 p_\parallel \rho_m^{-5} = \text{constant}$. ////

Using again $(\mathbf{J} \times \mathbf{B})/c = (\nabla \times \mathbf{B}) \times \mathbf{B}/4\pi$ to eliminate the current from (3.10.2), using $\nabla \times \mathbf{E} = -(1/c)(\partial \mathbf{B}/\partial t)$ to eliminate $\mathbf{E}$, and using the results of Probs. 3.10.1, the set of equations generally called the *CGL theory*, or the *double-adiabatic theory*, is obtained,

$$\rho_m\left(\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V} \cdot \nabla\mathbf{V}\right)_\perp + \nabla_\perp\left(p_\perp + \frac{B^2}{8\pi}\right) - \frac{(\mathbf{B} \cdot \nabla)\mathbf{B}}{4\pi}\left(\frac{p_\perp - p_\parallel}{B^2/8\pi} + 1\right) = 0$$

$$\rho_m\left(\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V} \cdot \nabla\right)_\parallel + \nabla_\parallel p_\parallel + (p_\perp - p_\parallel)\left(\frac{\nabla B}{B}\right)_\parallel = 0$$

$$\nabla \times (\mathbf{V} \times \mathbf{B}) = \frac{\partial \mathbf{B}}{\partial t} \qquad (3.10.11)$$

$$\frac{\partial \rho_m}{\partial t} + \nabla \cdot \rho_m \mathbf{V} = 0$$

along with (3.10.9) and (3.10.10) as equations of state.

Since the plasma motions across the magnetic field are in some sense decoupled from the motion along the field, the double-adiabatic theory has the flexibility to cope with these effects.

The one-fluid MHD equations predicted low-frequency waves, known as Alfvén waves. The double-adiabatic theory provides a more complete description of their properties, and serves as an excellent example of the utility of the double-adiabatic approach. The method of analysis is the same as with the one-fluid MHD theory. Each quantity (except velocity) is assumed to have a steady value and a small time-dependent, wavelike perturbation,

$$\mathbf{V} = \hat{\mathbf{V}}_1 \exp\,[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$$
$$\rho_m = \rho_{m0} + \hat{\rho}_{m1} \exp\,[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$$
$$p_\perp = p_{\perp 0} + \hat{p}_{\perp 1} \exp\,[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$$
$$p_\parallel = p_{\parallel 0} + \hat{p}_{\parallel 1} \exp\,[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$$
$$\mathbf{B} = \mathbf{B}_0 + \hat{\mathbf{B}}_1 \exp\,[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$$

and the equations are linearized by neglecting the product of perturbed quantities. The resulting dispersion relation ($\mathbf{B}_0 = \hat{\mathbf{z}}B_0$) is a quadratic in ω^2 that has roots given by

$$\omega^2 = \frac{k^2}{2\rho_{m0}}\left(\left\{\frac{B^2}{4\pi} + p_\perp + 2p_\parallel \cos^2\theta + p_\perp \sin^2\theta\right\}\right.$$
$$\left. \pm \left\{\left[\frac{B^2}{4\pi} + p_\perp(1 + \sin^2\theta) - 4p_\parallel \cos^2\theta\right]^2 + 4p_\perp^2 \sin^2\theta \cos^2\theta\right\}^{1/2}\right) \qquad (3.10.12)$$

where θ is the direction of propagation measured from the steady magnetic field direction (i.e., from the z axis).

For propagation along the field, $\theta = 0$, and (3.10.12) has two solutions,

$$\omega^2 = \frac{k^2}{\rho_{m0}} \left(\frac{B^2}{4\pi} + p_\perp - p_\parallel \right) \quad (3.10.13)$$

and

$$\omega^2 = \frac{3k^2}{\rho_{m0}} p_\parallel \quad (3.10.14)$$

The first solution describes hydromagnetic waves in a low-density plasma in a strong magnetic field. When $p_\perp = p_\parallel$, Eq. (3.10.13) reduces to the Alfvén wave solution discussed from the MHD equation in Sec. 3.9. When $p_\perp \neq p_\parallel$, the speed of the Alfvén wave is altered. When $p_\parallel > p_\perp + B^2/4\pi$, a new phenomenon arises: ω^2 is *negative*, implying that the time history of the perturbation $B_1 e^{i\omega t}$ can be exponentially increasing or decreasing, $B_1 \approx e^{\pm \omega_i t}$, where $\omega_i = \mathrm{Im}(\omega)$. The Alfvén wave has divided into two branches, one damped, the other growing. This result shows that the hydromagnetic waves propagating along the magnetic field are *unstable* in the sense that a small perturbation B_1 about the steady-state value B_0 will grow with time. This example is called the *firehose instability*, and is one of many examples (see Chaps. 5 and 9 for detailed treatments) in which a plasma can relax away from a steady state which is not a state of thermodynamic equilibrium, on a time scale faster than the collision time scale.

The second solution (3.10.14) predicts ion-acoustic waves propagating parallel to the magnetic field.

For propagation perpendicular to the magnetic field, $\theta = \pi/2$, and (3.10.12) has *one* solution,

$$\frac{\omega^2}{k^2} = \frac{2}{\rho_{m0}} \left(\frac{{B_0}^2}{8\pi} + p_\perp \right) \quad (3.10.15)$$

This wave is a magnetic-sound wave, called the *magnetosonic wave*.

When the pressure is small, $\omega^2 = k^2 B^2/4\pi\rho$ for propagation at any angle. The waves are a mixture of magnetosonic and Alfvén waves.

Problem 3.10.3 Show that, for an arbitrary direction of propagation, there exist unstable modes for all values of θ less than θ_c, where θ_c is given by

$$\frac{{B_0}^2}{4\pi} + p_\perp(1 + \sin^2 \theta_c) = \frac{{p_\perp}^2}{3p_\parallel} \sin^2 \theta_c + p_\parallel \cos^2 \theta_c \quad (3.10.16)$$

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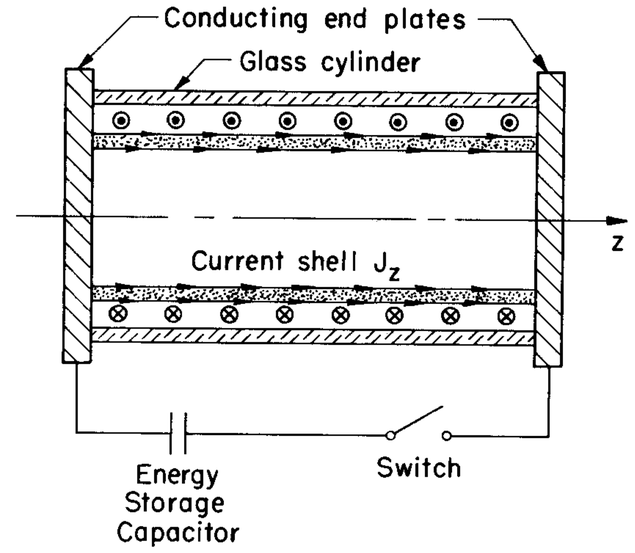


FIGURE 3.11.1
Schematic of linear, or Z, dynamic pinch.

3.11 THE DYNAMIC PINCH

The equilibrium pinch is an instructive example in the use of the equilibrium MHD equations. However, it is not representative of laboratory experiments that seek to confine plasma by the pinch principle. A typical laboratory pinch experiment involves the sudden compression of a low-density plasma by means of a large discharge current that lasts for only a few microseconds. During this time, several separate operations take place. First a low-density ($\sim 10^{14}$ cm^{-3}), low-temperature (~ 1 to 10 eV) plasma is created in the vacuum chamber by means of a high-frequency discharge, which is turned off prior to the pinch phase.

Second, a large voltage is applied suddenly to the ends of a plasma column in the case of a Z pinch (Fig. 3.11.1), or a large axial magnetic field is applied suddenly to the surface plasma column, in the case of a θ pinch (Fig. 3.11.2). The terms Z and θ pinch refer to the direction of current flow in the plasma. For both configurations, the compression magnetic field rises rapidly and $B^2/8\pi \gg \nabla p$. If the field B and the current J rise rapidly, then Ohm's law, assuming large conductivity, becomes

$$\frac{d}{dt} \left(B - \frac{c^2}{{\omega_p}^2} \frac{d^2 B}{dx^2} \right) = 0$$

The length c/ω_p is called the *collisionless skin depth*. The initial low-density plasma is assumed to have a large conductivity, and correspondingly, the skin

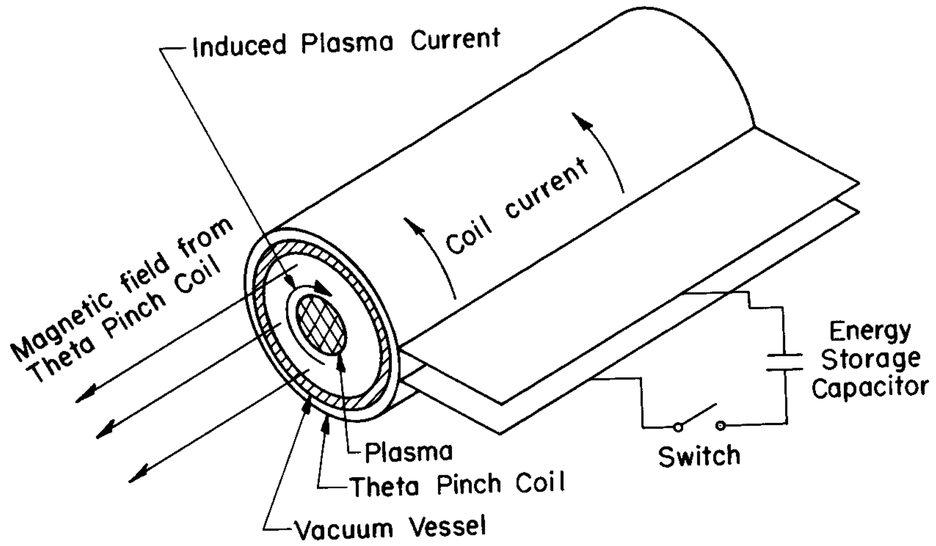


FIGURE 3.11.2
Schematic of θ dynamic pinch.

depth is much less than the radius of the plasma column. Thus the current flows in a thin shell on the surface of the plasma column. This current shell accelerates inward as a result of the magnetic pressure force, and is assumed to sweep up and capture all the particles it encounters on its travel to the axis. This is the so-called "snow-plow" model, which is useful to predict many of the gross features of the plasma pinch, but which, it must be realized, is only an approximate model. Experiments and theory have shown it to be a reasonable assumption from many plasma pinch experiments. The implosion or collapse converts part of the stored electric energy into kinetic energy associated with the inward radial velocity. When the collapsing plasma shell reaches the axis of the plasma column, it thermalizes, probably by means of a reflected shock wave that is generated as the shell approaches the axis.

Third, the resulting compressed ($n_e \approx 10^{17}$ cm^{-3}), high-temperature ($T_e = 100 - 1{,}000$ eV) pinch plasma is then studied to determine its density and temperature distribution, stability, radiation rate, and other properties.

The pinch plasma played a major role in the development of one area of controlled thermonuclear fusion research, and there are still many active experiments involving pinch plasmas. Pinches also generated theoretical interest in the physics of high-density, high-temperature plasmas; however, there are many regimes in which the imploding plasma is turbulent, and the snow-plow model is not adequate to describe the observed phenomena.

Recent experimental work has shown that large-current relativistic beams (up to 270 kA at 14 MeV) can be self-focused by the pinch effect and transported many meters without appreciable loss.

To illustrate the method of analyzing a situation where the compression forces are much larger than pressure forces, consider the Z pinch shown in Fig. 3.11.1. A complete analysis of the implosion of the current shell in the Z or θ pinch requires that the electric circuit parameters be included, since the current that flows in the plasma depends on the inductance of the circuit, which in turn depends on the location of the current shell. Thus the plasma equations and circuit equation are coupled. This problem is avoided here with the aid of the assumption that the plasma current is proportional to time [$I(t) = I_m t$].

For the Z-pinch configuration, the applied voltage is assumed to rise to its peak value instantaneously. If the plasma is assumed to be a perfect conductor on this time scale, the resulting plasma current flows in a thin shell on the plasma surface. This current shell causes a magnetic field B_θ outside it. The magnetic field inside is assumed to be zero. It is convenient to neglect the plasma kinetic pressure inside the current shell during the compression. This is usually a good approximation, since $B^2/8\pi \gg \nabla p$ until the very end of the implosion cycle.

The magnetic pressure outside the plasma acts to accelerate the inertial mass of the plasma as the snow-plow shell moves radially inward.

The one-fluid equations from (3.5.8) are

$$\frac{\partial}{\partial t}\rho_m \mathbf{V} + \mathbf{V}\cdot\nabla\rho_m\mathbf{V} + \rho_m\mathbf{V}\cdot\nabla\mathbf{V} = \frac{\mathbf{J}\times\mathbf{B}}{c} - \nabla p \qquad (3.11.1)$$

and

$$\nabla \times \mathbf{B} = \frac{4\pi\mathbf{J}}{c} \qquad (3.11.2)$$

The snow-plow model assumes that all particles in a thin shell move at the same speed $V_r(t)$. The current in the shell is

$$I = \int_0^\infty J_z 2\pi r\, dr$$

From (3.11.2)

$$B = \frac{2I}{rc}$$

outside the current shell. The total mass in the shell is (in length l)

$$M = \rho_m(0)\pi(a^2 - R^2)l$$

where R is the radial location of the shell at time t, and a is the location at $t = 0$. At the location R of the current shell, the fluid velocity is (elsewhere $V = 0$)

$$V = \frac{dR}{dt}$$

and

$$\frac{d}{dt}\rho_m(R)\frac{dR}{dt} = \frac{J_z B_\theta}{c}$$

But the volume of the current shell is vol $= 2\pi Rl\,\Delta R$, and so $\rho_m(R) = M/\text{vol}$ and $J_z = I/(2\pi R\,\Delta R)$. The equation of motion of the shell is

$$\frac{d}{dt} M \frac{dR}{dt} = \frac{IB}{c} \qquad (3.11.3)$$

In dimensionless variables the snow-plow-model equation of motion for the shell, assuming a linear current rise $[I(t) = (dI/dt)_0\, t]$, is

$$\frac{d}{d\tau}\left[(1 - x^2)\frac{dx}{d\tau}\right] = -\frac{\tau^2}{x} \qquad (3.11.4)$$

where the dimensionless variables are

$$x = \frac{r}{a} = \frac{\text{radius}}{\text{initial radius}}$$

$$\tau = \frac{t}{t_1} = \frac{\text{time}}{\text{time scale for compression}}$$

and $$t_1\left\{= \left[\frac{ac}{(dI/dt)_0}\right]^{1/2} M_0{}^{1/4}\right\} = \text{time scale for compression}$$

$M_0(=\pi a^2 \rho_m) =$ total mass of gas per unit length of plasma at beginning of compression

The initial conditions are $x(0) = 1$ and $(dx/d\tau)_0 = 0$.

With these initial conditions, (3.11.4) can be integrated numerically to give the result shown in Fig. 3.11.3. The radius of the pinch goes to zero at $\tau = 1.5$. Thus the collapse time is

$$t_c = 1.5\left[\frac{ac}{(dI/dt)_0}\right]^{1/2} M_0{}^{1/4} \qquad (3.11.5)$$

Problem 3.11.1 Verify that (3.11.3) can be obtained without using the fluid equations, from the dynamics of the current shell (still using the snow-plow approximation). ////

Obviously, neglecting the kinetic pressure of the plasma is a poor assumption near the end of collapse. Neglecting such effects as shock wave, radiation, ohmic heating, and ionization and assuming an adiabatic equation of state for a monatomic gas ($pV^{5/3} =$ constant), a pressure term can be included in the equation of motion for the current shell using the snow-plow model; i.e.,

$$\frac{d}{d\tau}\left[(1 - x^2)\frac{dx}{d\tau}\right] = -\frac{\tau^2}{x} + \alpha x^{-7/3} \qquad (3.11.6)$$

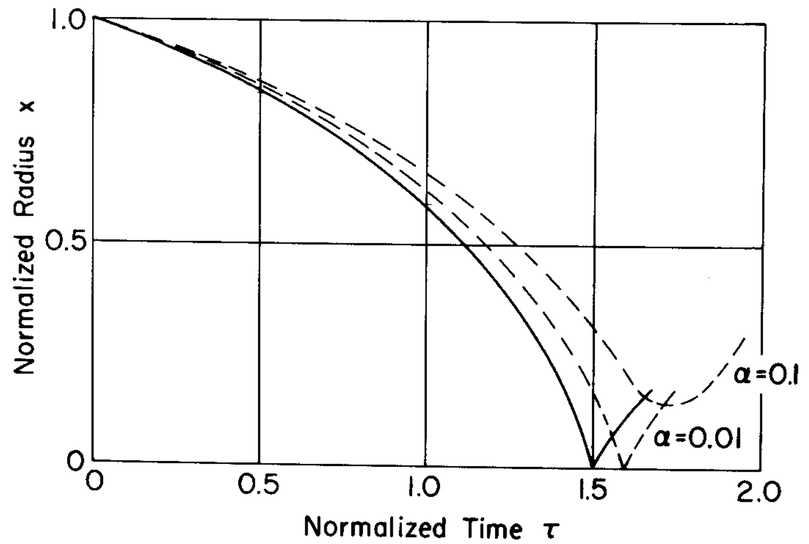


FIGURE 3.11.3
Normalized radius of dynamic-pinch plasma as a function of normalized time showing plasma collapse, or pinch effect (solid curve). Dashed curves show the effect of neutral-gas pressure on collapse.

where $\alpha = 2p_0 t_1{}^2/a^2\rho_{m0}$ is a dimensionless constant that measures initial pressure inside the shell. The initial conditions for this equation are $x = 1$ and $dx/d\tau = 0$ at $\tau = \sqrt{\alpha}$, since the initial pressure prevents the onset of compression until the right side of (3.11.6) changes sign. Curves for two values of α are shown in Fig. 3.11.3. This simplified analysis is based on several weak assumptions, but it shows that the collapse time predicted for $\alpha = 0$ is not altered greatly by the plasma kinetic pressure. Thus, for $\alpha > 0$, the plasma column collapses to some minimum radius, at which time the pressure is a maximum. The pressure is

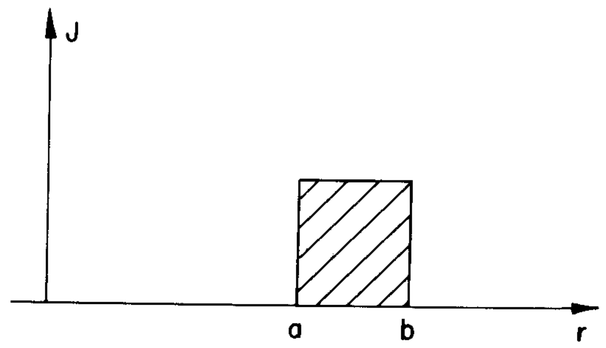


FIGURE 3.11.4
Current distribution in an equilibrium linear pinch.

greater than that for the equilibrium pinch with the same current, because the pressure includes the kinetic energy associated with the radial-implosion velocity, as well as the magnetic pressure.[1]

In this discussion the question of stability has been ignored. The dynamic pinch as described here is unstable to several perturbations. The pinch stability of the plasma is discussed in Chap. 4.

Problem 3.11.2 Assume a current J in an equilibrium pinch as shown in Fig 3.11.4, and assume no plasma for $r > b$.
Find $B(r)$ and $p(r)$ everywhere. Show that as $a \to b$; $B^2/8\pi$ at $r = b$ is equal to p at $r = 0$, while as $a \to 0$, $B^2/4\pi$ at $r = b$ is equal to p at $r = 0$.
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[1] A good review of this subject, including experimental results, is given in L. A. Artsimovich, "Controlled Thermonuclear Reactions," chap. 5, Fast High Power Discharges, Gordon and Breach, New York, 1964.

4

WAVES IN THE FLUID PLASMA

Although the plasma is first of all a gas, and nearly "ideal" as well, it has many properties in common with a conducting fluid, exhibiting coherent motions, i.e., waves, which reveal a great deal about the state and the nature of a plasma system. The rationale of the fluid approach was detailed in Chap. 3; in summary, if all the particles in a small volume element feel essentially the same forces, they will move in the same way, even though they are not coupled to their near neighbors by collisions. If a group of particles have widely varying thermal speeds, then of course they will not tend to remain together as a fluid element; the results of fluid treatments for a plasma are usually referred to, for example, as "waves in a *cold* plasma." The subject of plasma waves in regimes where thermal motions are important will be treated in Chap. 8, from a more microscopic point of view than is allowed in a fluid model. Even so, a surprisingly thorough view of plasma waves is obtained through fluid models.

As will be seen from material in this chapter, the plasma state is enormously rich in wave phenomena, and a variety of classification schemes have been devised to streamline the bookkeeping of plasma waves.[1] In this chapter, only a broad categorization of plasma waves is made according to the state of the

[1] T. H. Stix, "The Theory of Plasma Waves," McGraw-Hill, New York, 1962.

plasma that supports the waves (e.g., waves in a field-free plasma, waves in a magnetized plasma, etc.) because the normal modes of a plasma are so strongly influenced by the plasma and field configuration.

Only small-amplitude waves are discussed. The reason for this is that the fluid equations can then be *linearized* in the wave quantities. This is not to ignore the importance of nonlinear effects. Indeed, the large body of theory on collisionless shock waves in plasmas[1] rests on the nonlinear terms in the fluid equation (see Sec. 3.9).

4.1 DIELECTRIC CONSTANT OF A FIELD-FREE FLUID PLASMA ($\mathbf{E}_0 = \mathbf{B}_0 = 0$)

In some ways a plasma resembles a system of coupled oscillators. It has a spectrum of normal modes associated with the coupling of plasma particles and their inertia. However, a plasma also has electrical properties characteristic of a dispersive dielectric medium. Since the coupling between particles is electrical, both the mechanical and the dielectric properties of a plasma can be studied through its dielectric constant. That is, plasma waves are always associated with a time- and space-varying electric field, and the propagation properties of such fields are determined by the dielectric properties of the plasma, which in turn depend on the steady-state values of magnetic or electric fields in the plasma. A plasma may be both inhomogeneous and anisotropic, and this has a marked effect on its dielectric properties.

The dielectric constant can be calculated *for a cold plasma* from the fluid equations derived in Chap. 3. Assuming a harmonic time dependence,

$$\mathbf{E}_1 = \hat{\mathbf{E}}_1 e^{-i\omega t}$$
$$\mathbf{B}_1 = \hat{\mathbf{B}}_1 e^{-i\omega t}$$

for the (assumed small) perturbed fields and currents, the (linearized) fluid equations relate the charge density and current in the plasma to the electric and magnetic fields in the plasma. This allows Maxwell's equations to be written in the form

[1] D. A. Tidman, and N. A. Krall, "Shock Waves in Collisionless Plasmas," Wiley, York, 1971.

$$\nabla \times \hat{\mathbf{B}}_1 = -\frac{i\omega}{c}\hat{\mathbf{E}}_1 + \frac{4\pi}{c}\hat{\mathbf{J}}_1$$
$$= -\frac{i\omega}{c}\boldsymbol{\epsilon}\cdot\hat{\mathbf{E}}_1$$
$$\nabla \times \hat{\mathbf{E}}_1 = \frac{i\omega}{c}\hat{\mathbf{B}}_1 \tag{4.1.1}$$
$$\nabla \cdot \hat{\mathbf{B}}_1 = 0$$
$$\nabla \cdot \boldsymbol{\epsilon} \cdot \hat{\mathbf{E}}_1 = 0$$

where the quantity $\boldsymbol{\epsilon}$ is defined as the dielectric tensor. The solution of these equations for a particular $\boldsymbol{\epsilon}$ as determined from the plasma equilibrium conditions determines the normal modes or waves for that plasma equilibrium.

The two-fluid equations can be used [Eqs. (3.4.1) to (3.4.4)] to calculate the dielectric tensor of a cold plasma. For the simple case where there are *no steady-state fields* and where the *time-average velocity is zero*, the fluid equations are solved for small perturbations about a steady state:

$$\begin{aligned} n_\alpha &= n_{\alpha 0} + \hat{n}_{\alpha 1}(\mathbf{x})e^{-i\omega t} \\ \mathbf{V}_\alpha &= \hat{\mathbf{V}}_{\alpha 1}(\mathbf{x})e^{-i\omega t} \\ \mathbf{E} &= \hat{\mathbf{E}}_1(\mathbf{x})e^{-i\omega t} \\ \mathbf{B} &= \hat{\mathbf{B}}_1(\mathbf{x})e^{-i\omega t} \end{aligned} \tag{4.1.2}$$

In this case the *linearized* (neglect terms of the sort $V_1 B_1$, etc.) two-fluid equations become

Continuity:

$$\frac{\partial n_\alpha}{\partial t} + \nabla \cdot n_\alpha \mathbf{V}_\alpha = 0 \Rightarrow -i\omega\hat{n}_{\alpha 1} + n_{\alpha 0}\nabla \cdot \hat{\mathbf{V}}_{\alpha 1} = 0 \tag{4.1.3}$$

Momentum:

$$n_\alpha \frac{\partial \mathbf{V}_\alpha}{\partial t} = \frac{n_\alpha q_\alpha}{m_\alpha}\left(\mathbf{E} + \frac{\mathbf{V}_\alpha \times \mathbf{B}}{c}\right) \Rightarrow -i\omega n_{\alpha 0}\hat{\mathbf{V}}_{\alpha 1} = \frac{n_{\alpha 0} q_\alpha}{m_\alpha}\hat{\mathbf{E}}_1 \tag{4.1.4}$$

Maxwell's equations:

$$\nabla \times \mathbf{B} = \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} + \frac{4\pi \mathbf{J}}{c} \Rightarrow \nabla \times \hat{\mathbf{B}}_1 = -\frac{i\omega}{c}\hat{\mathbf{E}}_1 + \frac{4\pi}{c}\sum_\alpha n_{\alpha 0} q_\alpha \hat{\mathbf{V}}_{\alpha 1} \tag{4.1.5}$$

Using (4.1.4) to get $\hat{\mathbf{V}}_{\alpha 1}$ in terms of $\hat{\mathbf{E}}_1$, (4.1.5) can be rewritten

$$\nabla \times \hat{\mathbf{B}}_1 = -\frac{i\omega}{c}\left[1 - \frac{\omega_p{}^2(1 + m_e Z/m_i)}{\omega^2}\right]\hat{\mathbf{E}}_1 \tag{4.1.6}$$

where $$\omega_p^2 \equiv \frac{4\pi n_e e^2}{m_e}$$

and the ion charge is $q_i = Ze$. For this case of a field-free plasma, the dielectric constant is a scalar,

$$\varepsilon(\omega) \approx 1 - \frac{\omega_p^2}{\omega^2} \qquad (4.1.7)$$

where Zm_e/m_i is neglected, compared with 1.

Combining (4.1.6) with $\nabla \times \hat{\mathbf{E}}_1 = (i\omega/c)\hat{\mathbf{B}}_1$ gives

$$\nabla \times \nabla \times \hat{\mathbf{E}}_1 = \frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right)\hat{\mathbf{E}}_1 \qquad (4.1.8)$$

as the equation which determines the wave properties of a *cold field-free* plasma.

The main assumption in deriving (4.1.8) was the *cold-plasma approximation*. This approximation implies that

$$\sqrt{\frac{\kappa T_e}{m_e}} \ll \frac{\omega}{k} \qquad (4.1.9)$$

where $$k^{-1} \equiv \left|\frac{1}{E}\frac{dE}{dx}\right|$$

is the length scale over which the fields vary; reworded, the cold-plasma approximation implies that the particles have thermal speed so low that they do not move a "wavelength" in one wave period. Since the dielectric vanishes at

$$\omega = \omega_p \sqrt{1 + \frac{Zm_e}{m_i}} \qquad (4.1.10)$$

the plasma frequency is a resonant frequency of the plasma, and electrostatic oscillations ($\nabla \times \nabla \times \hat{\mathbf{E}}_1 = 0$) will persist at that frequency. Note that there is not a *separate* resonance at the *ion* plasma frequency $\omega_{pi} = \sqrt{4\pi ne^2/m_i}$; rather, the ion inertia shows up as a reduced mass in ω_p. [If the plasma is *warm*, then including ω_{pi} in the calculation of ε will, of course, give a resonance at $\omega = kC_s$, the ion-acoustic waves; see Eq. (3.9.14) et seq.]

The next sections examine wavelike solutions of (4.1.8).

Problem 4.1.1 Verify that the combination Zm_e/m_i in (4.1.6) is correct. ////

Problem 4.1.2 Can (4.1.6), (4.1.7), or (4.1.8) be derived from the *one-fluid* equations (3.7.1) to (3.7.4)? Do so, or else show directly that it cannot be done. Assume $Z = 1$. ////

4.2 PLASMA OSCILLATIONS

The cold-fluid equations gave (4.1.8) as the equation for the space and time development of small electric fields in a plasma. Taking the Fourier transform $\int d\mathbf{x} \exp(-i\mathbf{k}\cdot\mathbf{x})$ of that equation gives, for the amplitude of the Fourier component $\tilde{\mathbf{E}} \equiv \int d\mathbf{x} \exp(-i\mathbf{k}\cdot\mathbf{x})\hat{\mathbf{E}}(\mathbf{x})$,

$$-\mathbf{k} \times \mathbf{k} \times \tilde{\mathbf{E}} = \frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right)\tilde{\mathbf{E}} \qquad (4.2.1)$$

or taking $\mathbf{k} = k\hat{\mathbf{z}}$ (since there is nothing else in the problem to specify a coordinate system, $\mathbf{k} = k\hat{\mathbf{z}}$ does not limit the generality of the calculation), Eq. (4.2.1) can be written in matrix form

$$\begin{pmatrix} \omega^2 - \omega_p^2 - k^2c^2 & 0 & 0 \\ 0 & \omega^2 - \omega_p^2 - k^2c^2 & 0 \\ 0 & 0 & \omega^2 - \omega_p^2 \end{pmatrix} \begin{pmatrix} \tilde{E}_x \\ \tilde{E}_y \\ \tilde{E}_z \end{pmatrix} = 0 \qquad (4.2.2)$$

There are clearly three solutions to (4.2.2). Two correspond to electromagnetic waves and are treated in Sec. 4.5. The third solution is, for each mode $\mathbf{k}$,

$$\begin{aligned} \tilde{E}_x &= \tilde{E}_y = 0 \\ E_z &= \tilde{E}_z \exp(ikz - i\omega t) \\ \omega^2 &= \omega_p^2 \end{aligned} \qquad (4.2.3)$$

This solution corresponds to electromechanical oscillations at the plasma frequency. Such oscillations are called by a variety of names: *Langmuir oscillations*, *plasma oscillations*, and *space-charge waves* are three commonly used terms. Remembering that this calculation is based on a *cold-plasma approximation*, it is useful to list the properties of these oscillations:

1. Langmuir oscillations are dispersionless in a cold plasma; i.e., the eigenfrequency $\omega = \omega_p$ does not depend on the wavelength of the oscillation.
2. Langmuir oscillations do not *propagate* in a stationary cold plasma, as can be seen from the fact that the group velocity $v_g = \partial\omega/\partial k$ is zero, since $\omega = \omega_p$ does not depend on the wavelength. Even though these oscillations do not propagate, they have a phase velocity $v_p = \omega/k = \omega_p/k$. (When the wavelength $\lambda = 2\pi/k$ is too small, the "cold"-plasma approximation breaks down.)

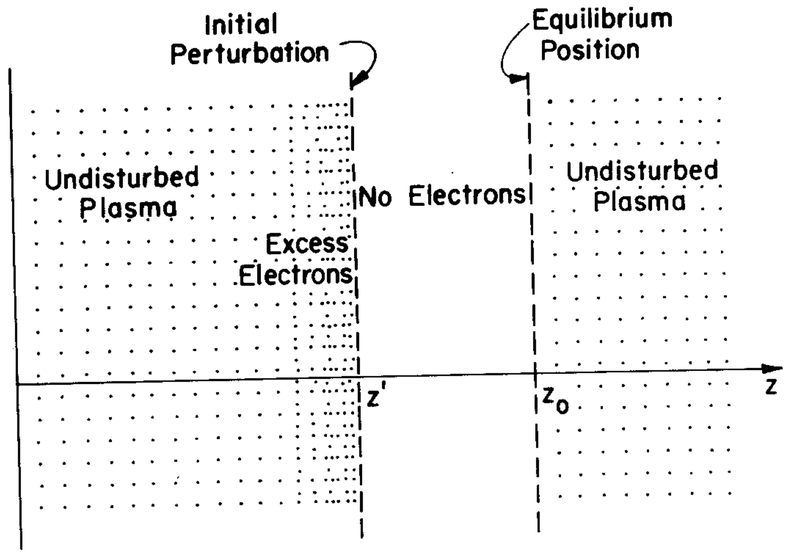


FIGURE 4.2.1
Schematic of a section of a "cold" plasma showing an initial displacement of electrons in a neutralizing background of immobile ions. The electrons, when released, oscillate at the plasma frequency $\omega_p = (4\pi ne^2/m)^{1/2}$.

3 These oscillations involve an interchange of the energy between the electric field and the kinetic energy of particle motion. There is no magnetic field energy associated with the oscillation, since the convection current density and displacement current density associated with these oscillations cancel each other, so that there is no net current in the plasma.

4 Langmuir oscillations are a space-charge disturbance. Their frequency can be derived from the following physical argument.

Imagine that all the electrons in a small slab are suddenly displaced, as shown in Fig. 4.2.1, so that there are no electrons between z' and z_0, and the electrons that were there are now to the left of z'. The amount of excess charge per square centimeter to the left of z' is

$$Q = -n_0 e(z_0 - z') \qquad (4.2.4)$$

where n_0 is the average number density of electrons; the charge density in the region vacated by the electrons is $n_0 e$. The electric field at z' tending to restore the displaced electrons to their equilibrium position is (integrating $dE/dz = 4\pi n_0 e$ from z_0, where $E = 0$, to z')

$$E_z(z = z') = 4\pi n_0 e(z' - z_0)$$

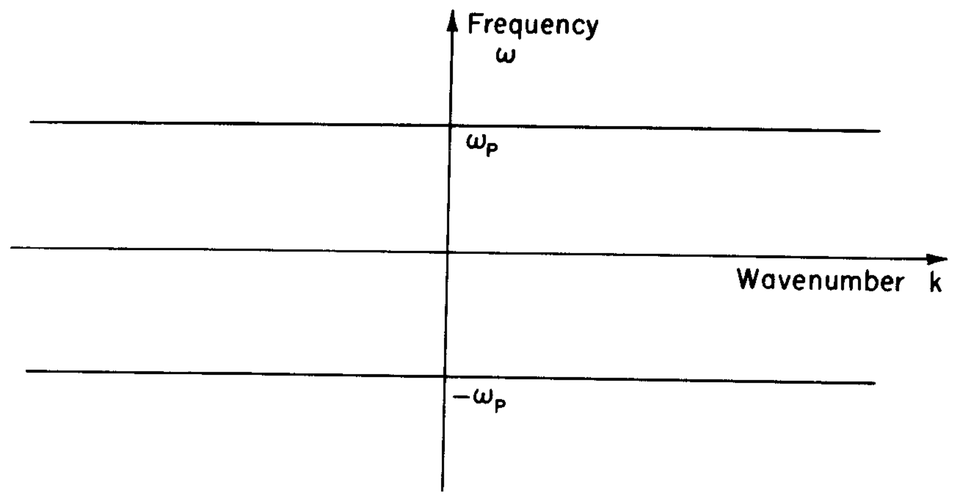


FIGURE 4.2.2
Frequency vs. wave-number diagram for Langmuir oscillations in a cold collisionless infinite plasma with immobile ions.

Using the force $F = -eE_z$, the equation of motion for an electron initially at z' is

$$\frac{d^2z'}{dt^2} = \frac{4\pi n_0 e^2}{m_e}(z_0 - z') \qquad (4.2.5)$$

Now let $\xi = z' - z_0$, and (4.2.5) becomes

$$\frac{d^2}{dt^2}\xi + \frac{4\pi n_0 e^2}{m_e}\xi = 0 \qquad (4.2.6)$$

The solution to the equation is

$$\xi = \hat{\xi}e^{\pm i\omega_p t}$$

(where $\omega_p^2 = 4\pi n_0 e^2/m_e$). Thus a group of electrons displaced in a plasma executes simple harmonic motion at the plasma frequency.

It is convenient to describe various types of plasma waves by plotting their eigenfrequency as a function of wave number k. For Langmuir oscillations in a cold plasma, the plot is especially simple (Fig. 4.2.2).

The effect of collisions on the properties of waves in plasmas is illustrated phenomenologically by retaining the collision term in the momentum transport equation for electrons. If the collision term is retained in the form

$$\int \mathbf{v}\left.\frac{\partial f_e}{\partial t}\right|_c \mathbf{dv} \approx \nu \mathbf{V}_e \qquad (4.2.7)$$

Eq. (3.4.2) leads to

$$-i\omega\hat{\mathbf{V}}_1 = -\frac{e}{m}\hat{\mathbf{E}}_1 - \nu\hat{\mathbf{V}}_1 \qquad (4.2.8)$$

instead of (4.1.4). Solving for $\hat{\mathbf{V}}_1$ gives

$$\hat{\mathbf{V}}_1 = \frac{(e/m_e)\hat{\mathbf{E}}_1}{i(\omega - i\nu)} \qquad (4.2.9)$$

The equivalent dielectric constant is now complex; i.e.,

$$\varepsilon = 1 - \frac{\omega_p{}^2}{\omega(\omega - i\nu)} \qquad (4.2.10)$$

and the plasma waves $\mathbf{E}_1 = \hat{\mathbf{E}}_1 e^{(\pm i\omega_p - \nu)t}$ are *collisionally damped.*

Although phenomenological models [such as replacing collision effects by Eq. (4.2.7)] are weak, they sometimes permit solutions for situations that are difficult to analyze with a complete theory, and thus show which areas might be profitably investigated with a complete theory. Phenomenological models are also sometimes useful for the purpose of analyzing experiments where the complications caused by the experimental configuration make a complete analysis intractable. In any event, the use of such models, and conclusions obtained using them, should be examined carefully.

4.3 PLASMA OSCILLATIONS IN A ONE-DIMENSIONAL DRIFTING PLASMA

Many plasma applications involve plasma drifting with respect to grids or other stationary reference objects. If space-charge disturbances in such a moving medium are excited, they have several properties in contrast to space-charge disturbances in a plasma at rest, and their study introduces some general aspects of plasma waves.

Consider a homogeneous, field-free plasma with a uniform background of immobile ($m_i \to \infty$) neutralizing ions, and assume the plasma is drifting with constant velocity in the z direction. The electrons are assumed to be "cold," in the sense discussed in the preceding section.

Plasma oscillations in this situation are also analyzed with the continuity equation for electrons in one-dimension (defining $V_z \equiv V_{ez}$),

$$\frac{\partial}{\partial t} n_e + \frac{\partial}{\partial z} n_e V_z = 0 \qquad (4.3.1)$$

the momentum transfer equation for electrons in one-dimension (neglecting collisions),

$$\frac{\partial}{\partial t} V_z + V_z \frac{\partial}{\partial z} V_z = -\frac{e}{m_e} E_z \qquad (4.3.2)$$

and Poisson's equation in one-dimension,

$$\frac{\partial E_z}{\partial z} = 4\pi \sum_\alpha n_\alpha q_\alpha \qquad (4.3.3)$$

Again assume that the density, velocity and electric field are perturbed slightly; in contrast with the preceding example, the term $V_z \partial V_z/\partial z$ in (4.3.2) must be retained since $V_0 \neq 0$. Equations (4.3.1) to (4.3.3) can be solved by neglecting terms of second order in the perturbed quantities and assuming a steady-state harmonic-time-dependent perturbation

$$\begin{aligned} n_e &= n_0 + \hat{n}_1 e^{i(kz-\omega t)} & n_1 &\ll n_0 \\ V_z &= V_0 + \hat{V}_1 e^{i(kz-\omega t)} & V_1 &\ll V_0 \\ E_z &= \hat{E}_1 e^{i(kz-\omega t)} \\ n_i &= n_0 \end{aligned} \qquad (4.3.4)$$

where the circumflex denotes a complex quantity.

Neglecting products of perturbation quantities as being second-order, (4.3.1) to (4.3.3) become

$$\begin{aligned} -i\omega\hat{n}_1 + ik(n_0 \hat{V}_1 + \hat{n}_1 V_0) &= 0 \\ -i\omega \hat{V}_1 + ikV_0 \hat{V}_1 &= -\frac{e}{m_e}\hat{E}_1 \\ ik\hat{E}_1 &= -4\pi e\hat{n}_1 \end{aligned} \qquad (4.3.5)$$

The equations are three homogeneous linear algebraic equations in three unknowns. The nontrivial solution of the equations is determined by setting the determinant of the coefficients equal to zero. The result is

$$(\omega - kV_0)^2 = \omega_p{}^2 \qquad (4.3.6)$$

This is the dispersion relation for plasma or space-charge waves associated with a drifting plasma or an electron beam. A coordinate transformation of the form $z \to z' - V_0 t'$ and $t \to t'$ shows that the frequency and wave number transform to $\omega \to \omega' + k'V_0$ and $k \to k'$, so that in the new frame $(\omega')^2 = \omega_p{}^2$. Thus, in the frame where the electrons are at rest, the disturbance is at the plasma frequency, recovering the plasma oscillation in a stationary plasma: plasma oscillations of a one-dimensional drifting plasma are the natural oscillations of the plasma transported at the drift velocity.

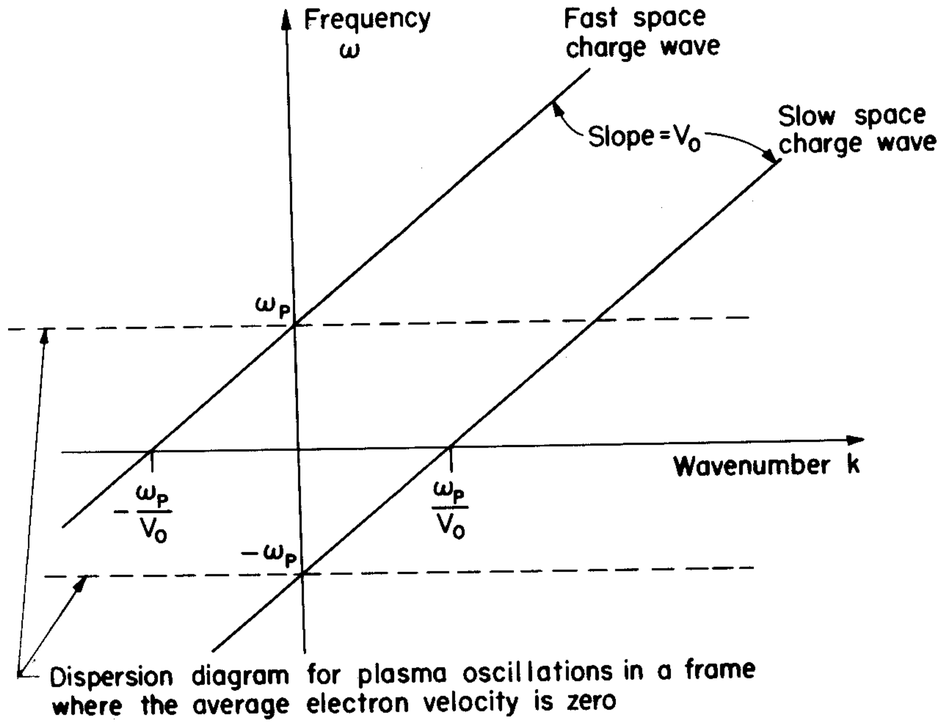


FIGURE 4.3.1
Frequency vs. wave-number diagram for one-dimensional plasma oscillations in a cold collisionless drifting plasma. The wave that propagates with speed greater than the average electron velocity is called the fast space-charge wave; the slower wave, called the slow space-charge wave, is a "negative" energy wave.

The dispersion relation (4.3.6) is shown in Fig. 4.3.1. These moving plasma oscillations are waves with a group velocity equal to the drift velocity ($\partial\omega/\partial k = V_0$), and thus they transport energy. The space-charge waves associated with drifting electrons are responsible for many plasma phenomena associated with growing waves or instabilities. An interesting property of the wave on a drifting plasma is the fact that the slow space-charge wave is a negative-energy wave (see below). This negative-energy property of the slow space-charge wave is responsible for the operation of many electronic devices, such as klystrons, traveling-wave tubes, etc., that generate or amplify high frequency ($f > 1{,}000$ MHz) electromagnetic energy.[1]

4.3.1 Wave Energy and the Dielectric Constant

The concept of a negative-energy wave simply means that if the system is in a state of energy W_0, then the energy of the system W ($= W_0 + W_1$) is *smaller* than W_0 when a wave of energy W_1 is launched. The total energy of the system

[1] J. R. Pierce "Traveling Wave Tubes," Van Nostrand, Princeton, N.J., 1950.

is, of course, never negative; the wave energy is composed of its electromagnetic energy $(E^2 + B^2)/8\pi$ and also the polarization energy W_p. If W_p is negative, the wave energy can in some cases also be negative. To see how W_p can be negative, consider a small perturbation (a push to the left) in the velocity of a localized clump of electrons,

$$\mathbf{V}_1 = -V_1\hat{\mathbf{z}} \qquad (4.3.7)$$

perhaps induced by an electric field associated with a wave. If the electrons were originally at rest, the energy change W_p would be positive

$$\Delta W \equiv W_p = \tfrac{1}{2}m_e V_1^2 \qquad \text{if } V_0 = 0 \qquad (4.3.8)$$

If the electrons were originally drifting with $\mathbf{V}_0 = V_0\hat{\mathbf{z}}$, the energy *change* due to the perturbation (4.3.7) would be *negative*, since

$$\Delta W \equiv W_p = \tfrac{1}{2}m_e(V_0 - V_1)^2 - \tfrac{1}{2}m_e V_0^2 \approx -m_e V_0 V_1 \qquad (4.3.9)$$

The "negative energy" arose from the fact that the electrons had a drift V_0; the existence of "negative-energy waves," in general, implies a drift or beam in the plasma.

In principle, the wave energy can be computed directly since the fluid equations relate $\mathbf{V}_1$ to $\mathbf{E}_1$. However, it is simpler to use directly a result from the theory of waves in dielectric media.[1] This result is that the time-average energy in a wave (kinetic plus electrostatic) can be expressed in terms of the dielectric constant as follows:

$$\overline{W_E} = \frac{\overline{E_1^2}}{8\pi}\frac{\partial}{\partial\omega}(\omega\varepsilon) \qquad (4.3.10)$$

where $\overline{W_E}$ is the time-average value (averaged over one oscillation period), the wave is electrostatic ($\nabla \cdot \varepsilon\mathbf{E}_1 = 0, \nabla \times \mathbf{E}_1 = 0$), and $\overline{E_1^2}$ is the time-average value of E_1^2, that is $E_1 = E_0 \sin\omega t$, $\overline{E_1^2} = \frac{1}{2}E_0^2$. The corresponding formula for a linearly polarized homogeneous plane wave is

$$\overline{W}_T = -\frac{\partial}{\partial\omega}\omega\left(\varepsilon - \frac{k^2c^2}{\omega^2}\right)\frac{\overline{E_1^2}}{8\pi} \qquad (4.3.11)$$

where $\nabla \cdot \mathbf{E}_1 = 0$, that is, for a purely transverse wave.

To apply this notion to waves on a drifting plasma, note that using (4.3.5) to calculate n_1 and V_1 in terms of E_1, and using $J_1 = -(n_0 eV_1 + n_1 eV_0)$, the dielectric constant for plasma response to high-frequency (this is what

[1] L. D. Landau and E. M. Lifshitz, "Electrodynamics of Continuous Media," pp. 253–256, Addison-Wesley, Reading, Mass., 1966.

$m_i \to \infty$ implies), long-wavelength (this is what "cold plasma" implies) field fluctuations is

$$\varepsilon = 1 - \frac{{\omega_p}^2}{(\omega - kV_0)^2} \qquad \text{(high-frequency dielectric constant for a cold plasma drifting at } V_0\text{)} \tag{4.3.12}$$

Using this dielectric, the positive- and negative-energy branches of the space-charge wave in a drifting plasma can be identified.

Problem 4.3.1 Indicate on Fig. 4.3.1 the positive- and negative-energy waves, using (4.3.10). ////

4.3.2 Positive and Negative Energy Waves in a Drifting Plasma

To show directly that one of the waves on a drifting plasma has a negative kinetic energy, it is useful to derive an energy-conservation theorem for these waves.

An energy-conservation theorem for space-charge and electromagnetic waves is obtained from Maxwell's equations.

$$\nabla \times \mathbf{E}_1 = -\frac{1}{c}\frac{\partial \mathbf{B}_1}{\partial t} \tag{4.3.13}$$

and

$$\nabla \times \mathbf{B}_1 = \frac{4\pi \mathbf{J}_1}{c} + \frac{1}{c}\frac{\partial \mathbf{E}_1}{\partial t} \tag{4.3.14}$$

Prescalar-multiply (4.3.13) by $\mathbf{B}_1$, and (4.3.14) by $\mathbf{E}_1$, subtract the second equation from the first, and use the identity

$$-\mathbf{A} \cdot \nabla \times \mathbf{B} + \mathbf{B} \cdot \nabla \times \mathbf{A} = \nabla \cdot (\mathbf{A} \times \mathbf{B})$$

to obtain

$$-\nabla \cdot \frac{c}{4\pi}(\mathbf{E}_1 \times \mathbf{B}_1) = \mathbf{E}_1 \cdot \mathbf{J}_1 + \frac{\partial}{\partial t}\left(\frac{{E_1}^2}{8\pi} + \frac{{B_1}^2}{8\pi}\right) \tag{4.3.15}$$

The quantity $\mathbf{P} = c(\mathbf{E}_1 \times \mathbf{B}_1)/4\pi$ is called the *Poynting vector* and gives the power flow in ergs/(cm^2)(s) associated with propagating disturbances. The quantities $W_E(={E_1}^2/8\pi)$ and $W_M(={B_1}^2/8\pi)$ are the time-dependent electric and magnetic energy densities associated with the propagating disturbance. The term $\mathbf{E}_1 \cdot \mathbf{J}_1$ gives the rate at which electrons extract energy from the field. If the electrons are allowed to move only in the z direction (for example, by applying a large magnetic field B_z), the $\mathbf{E}_1 \cdot \mathbf{J}_1$ term can be rewritten in a form which identifies the kinetic

power flow and kinetic energy density associated with propagating disturbances on drifting electron beams or plasmas. From (4.3.2)

$$\mathbf{E}_1 \cdot \mathbf{J}_1 = -\frac{m_e}{e}\left(\frac{\partial V_{1z}}{\partial t} + V_{0z}\frac{\partial}{\partial z}V_{1z}\right)J_{1z}$$

$$= -\frac{m_e}{e}\left[J_{1z}\frac{\partial V_{1z}}{\partial t} + \frac{\partial}{\partial z}(V_{0z}V_{1z}J_{1z}) - V_{0z}V_{1z}\frac{\partial J_{1z}}{\partial z}\right] \tag{4.3.16}$$

Using the perturbation current density

$$J_{1z} = -e(V_{0z}n_1 + n_0 V_{1z})$$

(4.3.16) can be written

$$\mathbf{E}_1 \cdot \mathbf{J}_1 = m_e\left[\frac{\partial}{\partial t}\left(\frac{n_0 {V_{1z}}^2}{2}\right) + V_{0z}n_1\frac{\partial V_{1z}}{\partial t} - \frac{1}{e}\frac{\partial}{\partial z}(V_{0z}V_{1z}J_{1z}) + V_{0z}V_{1z}\frac{\partial n_1}{\partial t}\right]$$

This equation can be written in the form

$$\mathbf{E}_1 \cdot \mathbf{J}_1 = \frac{\partial}{\partial z}P_K + \frac{\partial}{\partial t}W_K \tag{4.3.17}$$

where

$$W_K = m_e\left(\frac{n_0 {V_{1z}}^2}{2} + V_{0z}V_{1z}n_1\right) \tag{4.3.18}$$

is the kinetic energy density associated with a disturbance and

$$P_K = -\frac{m_e}{e}V_{0z}V_{1z}J_{1z} \tag{4.3.19}$$

is the kinetic power flow in the z direction associated with a propagating disturbance [compare (4.3.19) with 4.3.9)]. With this definition, (4.3.15) becomes

$$\nabla \cdot (\mathbf{P} + \mathbf{P}_K) + \frac{\partial}{\partial t}(W_E + W_M + W_K) = 0 \tag{4.3.20}$$

In the one-dimensional problem, in which none of the quantities vary with transverse coordinates, the energy-conservation theorem reduces to

$$\frac{\partial P_K}{\partial z} + \frac{\partial}{\partial t}(W_K + W_E) = 0 \tag{4.3.21}$$

since there is no perturbation magnetic field. For harmonic-time-dependent perturbation quantities, E_{1z}, V_{1z}, and n_1 of the form $n_1 = \hat{n}_1 \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t)$,

the time-average values of kinetic power flow, energy density, and electric energy density are

$$\bar{P}_K = \tfrac{1}{2}\,\mathrm{Re}[m_e V_{0z}\,\hat{V}_{1z}(n_0\,\hat{V}_{1z}^* + V_{0z}\,\hat{n}_1^*)]$$

$$\overline{W}_K = \tfrac{1}{2}\,\mathrm{Re}\left[m\left(n_0\,\frac{\hat{V}_{1z}\,\hat{V}_{1z}^*}{2} + V_{0z}\,\hat{n}_1\,\hat{V}_{1z}^*\right)\right] \qquad (4.3.22)$$

$$\overline{W}_E = \tfrac{1}{2}\,\mathrm{Re}\left(\frac{\hat{E}_1\hat{E}_1^*}{8\pi}\right)$$

where the asterisk denotes complex conjugate. The result (4.3.20) is valid for beams of finite cross section, provided the electrons are restricted to motion in the z direction only. The electromagnetic power flow is usually negligible compared with the kinetic power flow, and the magnetic energy density is usually negligible compared with the electric energy density.

Using the space-charge wave dispersion relation (4.3.6), the quantities become (the upper sign denotes the fast space-charge wave, and the lower sign denotes the slow space-charge wave)

$$\bar{P}_{K\pm} = \pm\frac{|\hat{E}_1{}^2|}{8\pi}\,\frac{\omega}{\omega_p}\,V_{0z}$$

$$\overline{W}_{E\pm} = \frac{|\hat{E}_1{}^2|}{16\pi} \qquad (4.3.23)$$

$$\overline{W}_{K\pm} = \frac{|\hat{E}_1{}^2|}{16\pi} - \frac{|\hat{E}_1{}^2|}{8\pi}\left(1 \mp \frac{\omega}{\omega_p}\right)$$

From these equations it is clear that the time-average kinetic power flow is the drift velocity times the sum of the time-average kinetic and electric stored-energy densities; i.e.,

$$\bar{P}_K = V_{0z}(\overline{W}_K + \overline{W}_E) \qquad (4.3.24)$$

For the slow space-charge wave on a drifting electron beam or plasma, the time-average kinetic power flow $\bar{P}_K$ and time-average energy density $\overline{W}_K + \overline{W}_E$ are clearly *negative*. This means that the total energy of the drifting electrons is reduced when a slow space-charge wave is set up on the beam.

Problem 4.3.2 (*a*) Show that the kinetic energy associated with the fast space-charge wave is positive for $\omega > \omega_p$.

(*b*) Find the frequency at which the kinetic energy associated with the fast space-charge wave is negative.

(*c*) Find the frequency at which total (wave plus kinetic) energy becomes negative. ////

Problem 4.3.3 If a drifting one-dimensional plasma passes through a pair of closely spaced grids that have a sinusoidal voltage applied between them, both the fast and the slow space-charge waves are excited and propagate on the drifting plasma. A linear combination of fast- and slow-space-charge wave solutions satisfies (4.3.5), so that beyond the exciting grids

$$V_1(z) = \hat{A}_f \exp\left[-\frac{i(\omega+\omega_p)z}{V_0}\right] + \hat{A}_s \exp\left[-\frac{i(\omega-\omega_p)z}{V_0}\right]$$

where $\hat{A}_f$ and $\hat{A}_s$ are the amplitudes of the fast and slow space-charge waves. If $\omega L/V_0 \ll 1$ (L is the grid spacing), find $\hat{A}_f$ and $\hat{A}_s$ by supplying the boundary conditions on the ac velocity and ac charge density at $z = 0$. Show that this type of excitation leads to "bunching," i.e., that the charge density has periodic maxima beyond the exciting grid. Determine the distance to the first maximum. Explain this result on the basis of electron trajectories. ////

4.4 SPACE-CHARGE WAVES IN A WARM PLASMA

The calculations in the two preceding sections showed that a stationary cold plasma can exhibit nonpropagating modes at the plasma frequency and that these oscillations become propagating modes if the plasma drifts at a constant speed. The question naturally arises as to what would be the properties of these modes if the electrons had a thermal distribution of velocities. A complete analysis of this situation is not possible within the framework of fluid theory, unless an independent means exists to determine $p(n)$ (an equation of state). This requires the use of the Vlasov equation (see Chap. 8). Even if an equation of state is chosen, phenomena which depend on the details of the velocity distribution, such as Landau damping, will of course not appear in a fluid description. Nevertheless, some of the general features of these waves can be studied with the fluid equations.

Again, consider a homogeneous field-free stationary plasma, except this time allow the electrons to have a finite temperature. To close the fluid equations, assume that the pressure is isotropic, and is related to the density by the equation of state (for electrons) $p/\rho^\gamma = \text{constant}$. The continuity equation for the perturbed density and velocity is

$$i\omega n_{i1} - i n_0\,\mathbf{k}\cdot\mathbf{V}_{i1} = 0$$
$$i\omega n_{e1} - i n_0\,\mathbf{k}\cdot\mathbf{V}_{e1} = 0 \qquad (4.4.1)$$

The momentum transport equation for the perturbed quantities, including the electron pressure, is (since $p_1 = p_0 \gamma \rho_{m1}/\rho_{m0}$, $\rho_m = nm$)

$$\begin{aligned} -i\omega \mathbf{V}_{i1} &= \frac{e}{m_i}\mathbf{E}_1 \\ -i\omega \mathbf{V}_{e1} &= -\frac{e}{m_e}\mathbf{E}_1 + \frac{i\mathbf{k}\gamma p_0}{n_0 m_e}\frac{n_{e1}}{n_0} \end{aligned} \tag{4.4.2}$$

Poisson's equation provides a third relation between n_1, $\mathbf{V}_1$, and $\mathbf{E}_1$.

$$i\mathbf{k}\cdot\mathbf{E}_1 = -4\pi e(n_{e1} - n_{i1}) \tag{4.4.3}$$

Eliminating $\mathbf{V}$ between (4.4.1) and (4.42) and then substituting n_{i1} and n_{e1} into (4.43) gives the result ($p = n\kappa T$)

$$\left(1 - \frac{{\omega_{pi}}^2}{\omega^2} - \frac{{\omega_{pe}}^2}{\omega^2}\frac{1}{1 - \dfrac{k^2}{\omega^2}\dfrac{\gamma\kappa T_e}{m_e}}\right)\mathbf{k}\cdot\mathbf{E}_1 = 0 \tag{4.4.4}$$

where γ has not yet been specified. For example, a reasonable choice of γ might be $\gamma = 3$ for one-dimensional adiabatic compression (high frequency), and $\gamma = 1$ for isothermal processes (low frequency). From (4.4.4) the equivalent dielectric constant for a plasma of warm electrons and cold ions is

$$\varepsilon = \left(1 - \frac{{\omega_{pi}}^2}{\omega^2} - \frac{{\omega_{pe}}^2}{\omega^2}\frac{1}{1 - \dfrac{k^2}{\omega^2}\dfrac{\gamma\kappa T_e}{m_e}}\right) \tag{4.4.5}$$

Instead of using Poisson's equation, Eqs. (4.4.1) to (4.4.2) can be combined with

$$\nabla \times \mathbf{B} = \frac{4\pi\mathbf{J}}{c} + \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t}$$

to give

$$\nabla \times \mathbf{B}_1 = -\frac{i\omega}{c}\varepsilon\mathbf{E}_1 \tag{4.4.6}$$

Now $\mathbf{k}$ dotted into (4.4.6) yields (4.4.4), singling out the electrostatic waves ($\mathbf{B}_1 = 0$). In addition, (4.4.6) includes electromagnetic waves ($\mathbf{k}\cdot\mathbf{E}_1 = 0$) (see Sec. 4.5).

There are two features to the solutions $\varepsilon = 0$ which appear because the pressure was not neglected. First, there are corrections to the $\omega = \omega_p$ oscillations, so that now

$$\omega^2 = {\omega_{pe}}^2 + k^2\frac{\gamma\kappa T_e}{m_e} \tag{4.4.7}$$

where m_e/m_i is neglected compared with unity.

The Langmuir oscillation in a *warm* plasma is dispersive (ω varies nonlinearly with wavelength) and is a propagating wave

$$v_g = \frac{\partial\omega}{\partial k} = \frac{\gamma\kappa T_e}{m_e}\frac{k}{\sqrt{{\omega_p}^2 + k^2\gamma\kappa T_e/m_e}}$$

The constant γ must be determined either with the aid of physical argument or directly from a solution of the Vlasov equation. In the spirit of calculating *corrections* to the cold-plasma theory, a reasonable assumption is that the electrons do not move a wavelength in one period $2\pi\omega^{-1}$, and that the compression must therefore be adiabatic. The appropriate γ for a one-dimensional adiabatic compression is $\gamma = 3$. Thus the dispersion relation (4.4.6) becomes

$$\omega^2 = {\omega_p}^2 + 3\frac{\kappa T_e}{m_e}k^2$$

$$\omega = \omega_p\left(1 + 3\frac{\kappa T_e}{m_e{\omega_p}^2}k^2\right)^{1/2} = \omega_p(1 + 3k^2{\lambda_{De}}^2)^{1/2}$$

$${\lambda_{De}}^2 \equiv \frac{\kappa T_e}{4\pi n e^2}$$

This is the same dispersion relation that will be obtained from the solution of the Vlasov equation (Chap. 8).

Problem 4.4.1 Find four roots $\varepsilon(\omega) = 0$ in (4.4.4). Show that there is a root near $\omega = kC_s/\sqrt{1 + k^2{\lambda_{De}}^2}$, where ${\lambda_{De}}^2 = \kappa T_e/4\pi n e^2$; this is the ion-acoustic wave. ////

This dispersion relation is still valid only for long-wavelength disturbances, i.e., for $k^2(\kappa T_e/m_e) \ll {\omega_p}^2$ (equivalent to $k^2{\lambda_{De}}^2 \ll 1$), so the propagation velocity

$$\frac{\partial\omega}{\partial k} \approx \frac{3\kappa T_e}{m_e\omega_{pe}}k = 3k\lambda_{De}\left(\frac{\kappa T_e}{m_e}\right)^{1/2} \tag{4.4.8}$$

is less than the electron thermal speed. This emphasizes the fact that this is a wave propagating *through* the plasma, in contrast to the case of a cold drifting plasma in which the wave "speed" was really the drift speed of the plasma.

In this case the wave is propagating even in the rest frame of the plasma. The dispersion relation for the warm Langmuir oscillation (4.4.7) is plotted in Fig. 4.4.1, emphasizing the range of validity of the calculation.

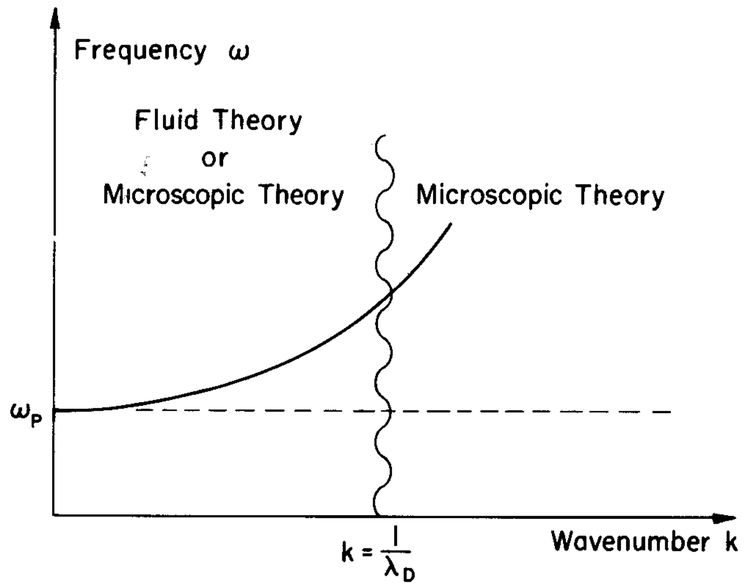


FIGURE 4.4.1
Frequency vs. wave-number diagram for Langmuir oscillations in a "warm" collisionless plasma, showing the region where macroscopic fluid theory, with $\gamma = 3$, agrees with the microscopic, or Vlasov, theory of plasma waves.

If $k\lambda_D \gg 1$, many electrons in a thermal distribution will move a wavelength or more in one period of the oscillation, and thus the compression will no longer be adiabatic.

A second feature of the waves in a warm plasma is the following: For the cold field-free plasma there was only one characteristic frequency [$\varepsilon(\omega) = 0$] at $\omega = \omega_p\sqrt{1 + m_e/m_i}$, with no separate resonance at the ion-plasma frequency. When the electrons are warm, there is a separate response at the ion-plasma frequency, as seen by the low-frequency solution of $\varepsilon = 0$ [Eq. (4.4.5)]. The fluid equations can be used to illustrate the features of the low-frequency resonances. Solving (4.4.4) in the low-frequency limit ($\omega^2 \ll \omega_{pe}{}^2$) gives

$$\omega^2 = \frac{k^2\gamma\kappa T_e/m_i}{1 + \gamma k^2\lambda_{De}{}^2} \qquad (4.4.9)$$
$$\lambda_{De}{}^2 \equiv \frac{\kappa T_e}{4\pi n e^2}$$

This is the ion-acoustic wave. The value of γ can be found by noting that since this wave is of low frequency, electrons *do* have time to stream a distance greater than the wavelength in one oscillation time. This suggests that the ion-sound wave is *isothermal*, and $\gamma = 1$ is appropriate (the Vlasov solution of this problem in Chap. 8 shows this to be the correct γ). With this argument Eq. (4.4.9) is replaced by

$$\omega = \frac{k\sqrt{\kappa T_e/m_i}}{\sqrt{1 + k^2\lambda_{De}{}^2}} \qquad (4.4.10)$$

The ion wave propagates with a speed (called the ion-sound speed)

$$\frac{\partial\omega}{\partial k} \approx C_S \equiv \left(\frac{\kappa T_e}{m_i}\right)^{1/2} \ll \left(\frac{\kappa T_e}{m_e}\right)^{1/2} \qquad (4.4.11)$$

somewhere between the electron and ion thermal speed. In this wave, energy is shared between the electrostatic energy E^2 and the mechanical energy of plasma particles oscillating in response to the low-frequency E fields. The division of energy between mechanical and electrical can be found by using (4.3.10) for the energy of a wave in a dielectric. Since, when evaluated at $\omega = kC_s/(1+k^2\lambda_{De}{}^2)^{1/2}$,

$$\frac{\partial(\omega\varepsilon)}{\partial\omega} \approx 2\left(1 + \frac{1}{k^2\lambda_{De}{}^2}\right) \qquad (4.4.12)$$

the energy in ion-acoustic waves [from (4.3.10)]

$$\overline{W}_{\text{wave}} = 2\left(1 + \frac{1}{k^2\lambda_{De}{}^2}\right)\frac{E^2}{8\pi} = \frac{E^2}{8\pi} + \frac{E^2}{8\pi}\left(1 + \frac{2}{k^2\lambda_{De}{}^2}\right)$$
$$= \overline{W}_{\text{electrostatic}} + \overline{W}_{\text{particle}}$$

is mostly in mechanical energy of the particles at the longer wavelengths, where $k^2\lambda_{De}{}^2 \ll 1$.

This discussion of the ion wave is accurate as far as it goes. However, in a warm plasma there will always be *some* particles moving at a speed comparable with the wave speed, and these resonant particles can affect the time history of the wave. These effects are outside the scope of fluid theories, and are treated from a microscopic (Vlasov) approach in Chaps. 8 and 9. It will be shown there that the cold-ion assumption made in the present derivation is required, and that

$$T_e \gg T_i \qquad (4.4.13)$$

is a necessary condition in order that the ion-acoustic wave not be heavily damped.

4.5 PLANE WAVES IN A COLD PLASMA

The fluid equations gave

$$\nabla \times \hat{\mathbf{B}}_1 = -\frac{i\omega}{c}\left(1 - \frac{\omega_{pi}{}^2}{\omega^2} - \frac{\omega_{pe}{}^2}{\omega^2}\right)\hat{\mathbf{E}}_1 \qquad (4.5.1)$$

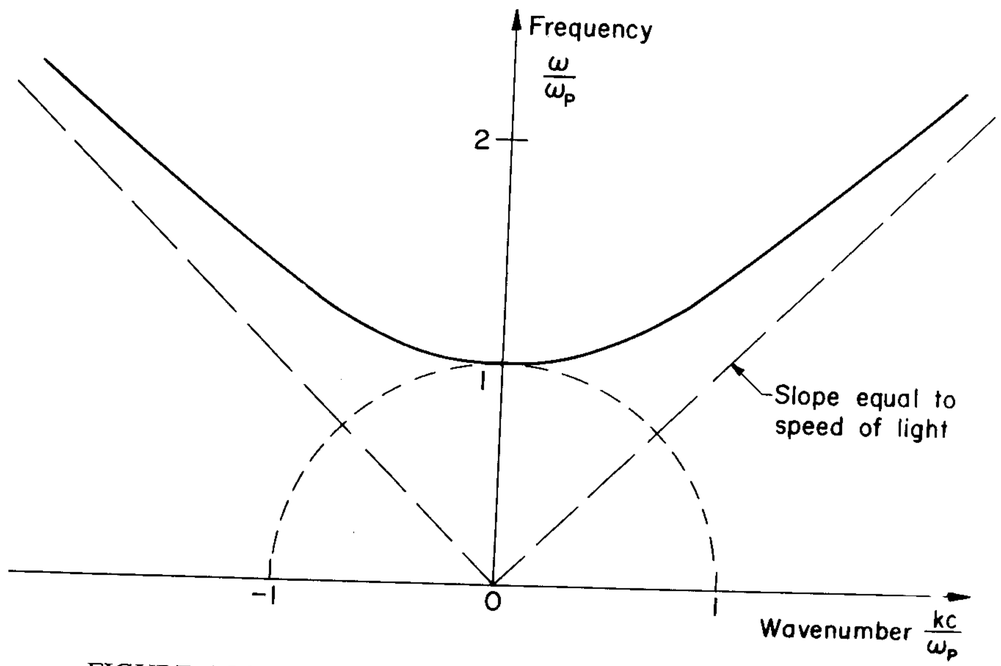


FIGURE 4.5.1
Frequency vs. wave-number diagram for plane electromagnetic waves in a cold collisionless plasma. For $\omega < \omega_p$, the waves are evanescent, with spatial damping coefficient shown by dashed semicircle.

for the development of a wave $\mathbf{E} = \hat{\mathbf{E}}_1 e^{-i\omega t}$ in a cold plasma. For the Langmuir wave, $\hat{\mathbf{B}}_1 = 0$; a second solution exists corresponding to electromagnetic waves propagating in a plasma. Using $\nabla \times \hat{\mathbf{E}}_1 = i\omega\hat{\mathbf{B}}_1/c$, (4.5.1) becomes

$$\frac{c^2}{\omega^2}\nabla \times \nabla \times \hat{\mathbf{E}}_1 = \left(1 - \frac{\omega_p{}^2}{\omega^2}\right)\hat{\mathbf{E}}_1 \qquad (4.5.2)$$

The electrostatic wave was found by taking the scalar product of $\mathbf{k}$ with (4.5.2). The electromagnetic-wave solution is found by taking the vector product of $\mathbf{k}$ with (4.5.2). This is the same as looking for a homogeneous-plane-wave solution of (4.5.2), giving a wave with its electric field perpendicular to its direction of propagation; i.e.,

$$\begin{aligned} \hat{\mathbf{E}}_1(\omega, \mathbf{x}) &= \tilde{\mathbf{E}}_1 \exp(i\mathbf{k}\cdot\mathbf{x}) \\ \mathbf{k}\cdot\tilde{\mathbf{E}}_1 &= 0 \end{aligned} \qquad (4.5.3)$$

From either point of view, (4.5.2) reduces to

$$(k^2c^2 + \omega_p{}^2 - \omega^2)\mathbf{k} \times \tilde{\mathbf{E}}_1 = 0 \qquad (4.5.4)$$

The nontrivial solution of Eq. (4.5.4) ($\tilde{\mathbf{E}}_1 \neq 0$) gives the dispersion relation

$$\omega^2 = k^2c^2 + \omega_p{}^2 \qquad (4.5.5)$$

These waves propagate in any direction in a plasma and, in the absence of plasma ($\omega_p{}^2 \propto n \to 0$), are light waves, $\omega = kc$. Figure 4.5.1 is the $\omega - k$ diagram for plane waves in a cold homogeneous isotropic plasma. This diagram shows that plane waves propagate in a plasma only for frequencies higher than the plasma frequency. In the vicinity of $\omega = \omega_p$, the wavelength becomes very large, and the effects of finite boundaries of the plasma can be expected to influence the dispersion relation for electromagnetic waves. For very high frequencies $\omega \gg \omega_p$, the electrons cannot respond, and cease to influence the propagation of the wave.

So these waves are just electromagnetic waves propagating in a dielectric medium. Their properties are quite different from those of the space-charge waves, and these properties are the key to some practical uses of this plasma mode of oscillation:

1 *Energy transport* These waves transport energy, even through a cold plasma at rest, as seen from the Poynting vector

$$\mathbf{P} = \frac{c\mathbf{E}_1 \times \mathbf{B}_1}{4\pi} = \frac{\mathbf{k}c^2}{4\pi\omega}E_1{}^2 \qquad (4.5.6)$$

2 *Group velocity* The speed at which these waves propagate information is given by

$$v_g = \frac{\partial\omega}{\partial k} = \frac{kc^2}{\omega} = \frac{c}{(1 + \omega_p{}^2/k^2c^2)^{1/2}} < c \qquad (4.5.7)$$

This group velocity is less than the velocity of light, and is also the speed at which energy is transported.

3 *Phase velocity* Since

$$\begin{aligned} v_p = \frac{\omega}{k} &= c\left(1 + \frac{\omega_p{}^2}{k^2c^2}\right)^{1/2} \\ &= \frac{c}{(1 - \omega_p{}^2/\omega^2)^{1/2}} > c \end{aligned} \qquad (4.5.8)$$

it is clear that the phase velocity of these waves in plasma exceeds their speed in vacuum. Nevertheless, energy is not transported at a speed exceeding c.

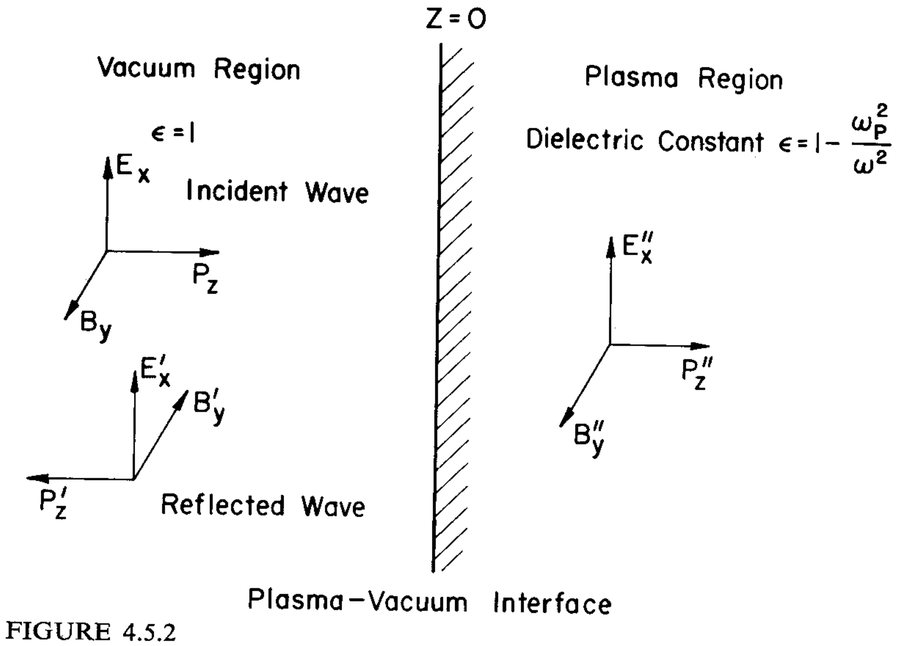


FIGURE 4.5.2
Schematic of a half-space ($z > 0$) of cold collisionless plasma, with an incident plane electromagnetic wave, showing the transmitted and reflected waves caused by the plasma-vacuum discontinuity.

4 *Zero density limit* As the plasma density is lowered, $\omega_p \to 0$, these waves become

$$\omega = kc$$

i.e., they are plane electromagnetic waves propagating in vacuum.

5 *High-density cutoff* From (4.5.5) it is clear that these waves propagate only for $\omega > \omega_p$. This has led to their use as a plasma diagnostic, as described in Sec. 4.6.

6 *Energy distribution in waves* The time-average energy in these waves is distributed among electric, magnetic, and mechanical energies

$$\overline{W} = \frac{\overline{E_1^2}}{8\pi} + \frac{\overline{B_1^2}}{8\pi} + \frac{1}{2} nm\, \overline{V_1^2}$$

Problem 4.5.1 Show that

$$\frac{\overline{E_1^2}}{8\pi} = \frac{1}{2}\overline{W}; \qquad \frac{\overline{B_1^2}}{8\pi} = \frac{1}{2}\overline{W}\left|1 - \frac{\omega_p^2}{\omega^2}\right|; \qquad \frac{1}{2} nm\, \overline{V_1^2} = \frac{1}{2}\overline{W}\frac{\omega_p^2}{\omega^2} \qquad ////$$

4.5.1 Reflection of Electromagnetic Waves from a Plasma Boundary

A commonly occurring situation in plasma measurements is that of electromagnetic waves incident on a plasma boundary. To illustrate how a plane wave reflects from a plane plasma interface, consider the situation shown in Fig. 4.5.2. Let the incident, reflected, and transmitted fields be given by

Incident:
$$E_x = E_{x0}\, e^{+ik_0 z}$$
$$B_y = E_{x0}\, e^{+ik_0 z} \qquad (4.5.9)$$

Reflected:
$$E'_x = RE_{x0}\, e^{-ik_0 z}$$
$$B'_y \doteq RE_{x0}\, e^{-ik_0 z} \qquad (4.5.10)$$

Transmitted:
$$E''_x = TE_{x0}\, e^{+ik_p z}$$
$$B''_y = T\frac{E_{x0} k_p c}{\omega} e^{+ik_p z} \qquad (4.5.11)$$

where
$$k_0^2 c^2 = \omega^2$$
$$k_p^2 c^2 = \omega^2\left(1 - \frac{\omega_p^2}{\omega^2}\right)$$

and where
R = reflection coefficient
T = transmission coefficient
E_{x0} = amplitude of incident wave

For a current-free, charge-free boundary, the necessary and sufficient conditions to ensure continuity of the electromagnetic field are that the tangential components of the field vectors be continuous. The reflection and transmission coefficients for a plane-plasma interface (homogeneous isotropic cold plasma) are obtained from this condition, with the result

$$R = \frac{k_0 - k_p}{k_0 + k_p} \qquad (4.5.12)$$

$$T = \frac{2k_0}{k_0 + k_p} \qquad (4.5.13)$$

For high-frequency waves ($\omega \gg \omega_p$), the reflection coefficient vanishes, and $T = 1$. For lower-frequency waves ($\omega \gtrsim \omega_p$), some fraction of the incident signal is reflected.

At $\omega = \omega_p$, $R = 1$, $T = 2$. At $\omega \ll \omega_p$, $R = -1$, $T = 0$; thus a plasma reflects waves below the plasma frequency with the same amplitude and phase as though the plasma were replaced by a perfect conductor.

4.5.2 Reactive Skin Depth

If the frequency of the incident wave is much less than the plasma frequency, the propagation constant is imaginary and waves decay in an evanescent manner away from the plasma-vacuum interface. That is,

$$\frac{\omega^2}{c^2}\left(1-\frac{\omega_p{}^2}{\omega^2}\right)_{\omega\ll\omega_p} \Rightarrow -\frac{\omega_p{}^2}{c^2} \qquad (4.5.14)$$

and

$$k_p \Rightarrow \pm i\frac{\omega_p}{c} \qquad (4.5.15)$$

The electric field inside the plasma decays with distance, as follows:

$$E_x(z) = E_x(0)e^{ikz} = E_x(0)e^{-\omega_p z/c} \qquad (4.5.16)$$

The plasma skin depth δ is defined as the distance in which the field is reduced by a factor $1/e$ (that is, $\omega_p\delta/c = 1$). Thus the reactive or collisionless skin depth in a plasma is

$$\delta = \frac{c}{\omega_p} \qquad (4.5.17)$$

It is called reactive because the waves are reflected from a medium which is a lossless dielectric in the absence of collisions.

Problem 4.5.2 Find the transmitted energy as a function of ω. Does the result $T = 2$ at $\omega = \omega_p$ imply energy transmission? Sketch the magnitude and angle θ for the reflection coefficient $R = |R|e^{i\theta}$ as a function of ω for $\omega \le \omega_p$. What does the angle θ correspond to? ////

Discussion topic The solution $k_p = \pm i\omega_p/c$ for low-frequency waves matching an incoming wave at a plasma-vacuum boundary might suggest $E_{\text{plasma}} \propto E_{x0}e^{+\omega_p z/c}$ rather than $e^{-\omega_p z/c}$ away from a boundary. How does one show that the former is not acceptable whereas the latter is acceptable? ////

Problem 4.5.3 Compute the skin depth in centimeters for a low-density plasma $n = 10^9$ cm^{-3} and a high-density plasma $n = 10^{16}$ cm^{-3}. Since the electromagnetic waves are reflected from the plasma, they must impart momentum to the plasma, and hence exert a body force on the plasma-vacuum interface. What power flow of electromagnetic waves would be required to balance the pressure force exerted by a plasma whose temperature is 10 keV and whose density is 10^{14} cm^{-3}? Is this a large or small power flow by contemporary standards? Compare this power flow with that from a ruby laser which delivers 6×10^7 ergs in 20×10^{-9} s over an area of 0.1 cm^2. ////

Problem 4.5.4 Show that no energy is lost as an electromagnetic wave with $\omega \ll \omega_p$ is reflected from a boundary, even though the wave dies out away from the boundary. ////

4.6 MICROWAVE TRANSMISSION METHOD OF MEASURING PLASMA PROPERTIES

A convenient way to measure the density of a laboratory plasma is to measure the phase shift of an electromagnetic wave as it propagates through the plasma. The fact that typical laboratory plasmas have densities in the range of 10^{10} to 10^{16} particles per cubic centimeter means that the plasma frequency for these experiments is in the range of 10^9 to 10^{13} Hz. This frequency range covers what is known as the microwave and millimeter wavelength region of the electromagnetic spectrum. For practical reasons having to do with size of experiments and the availability of sources of electromagnetic energy, this method is usually restricted to the 3-cm to 2-mm wavelength region, and it is called *plasma diagnostics with microwaves*.[1]

Apart from the fact that the frequency of the source is usually chosen to be equal to or slightly greater than the largest expected plasma frequency, this method of measuring plasma density always involves the same general collection of apparatus and the same general layout. There are many clever and sophisticated electronic embellishments on the basic scheme to improve the data reduction and the interpretation of the results of a given experiment; however, only the basic method will be outlined here.

Figure 4.6.1 is a schematic drawing of a microwave plasma diagnostic system which measures the plasma density by determining the frequency at which the transmitted signal goes to zero. For a steady-state plasma the frequency of the signal source is varied to find the cutoff frequency, while for a transient plasma the time (or *any* appropriate parameter) at which the transmission goes to zero is recorded. In either case, the plasma density is found by equating the known source frequency f to the transmission cutoff frequency,

[1] Two recommended sources of additional information on this subject are M. A. Heald and C. B. Wharton, "Plasma Diagnostics with Microwaves," Wiley, New York, 1965, and R. H. Huddlestone and S. L. Leonard, "Plasma Diagnostic Techniques," Academic, New York, 1965.

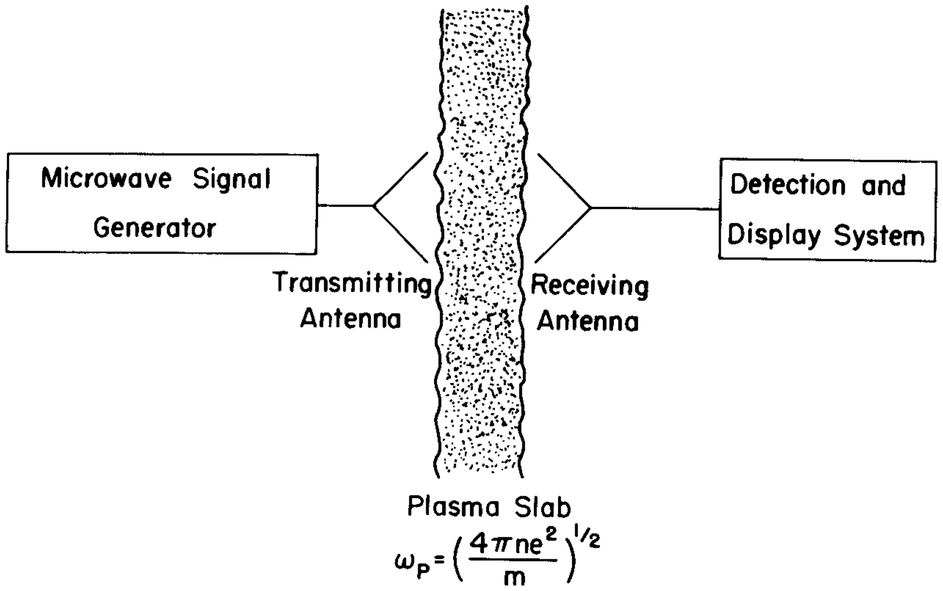


FIGURE 4.6.1
Schematic of a simple microwave system to measure plasma density by determination of plasma cutoff, that is, $\omega = \omega_p$.

$f_{\text{source}} = f_{\text{plasma}} \approx 10^4 \sqrt{n}$. This simple scheme suffers from the fact that if the density of plasma anywhere along the path traveled by the microwaves exceeds the critical density, cutoff occurs. Thus a plasma could have a small region in the middle where the density is large but have a low density elsewhere.

An improvement on this scheme that makes it possible to measure the density with frequencies of microwaves greater than the plasma frequency is shown schematically in Fig. 4.6.2. Some of the energy from the source is diverted into a reference path that is provided with a device to control the amplitude and phase of the reference wave. This reference wave is added to the wave transmitted through the plasma after it has been picked up by the receiving antenna. When no plasma is present in the system, the amplitude and phase of the reference wave are adjusted to give zero output in the detector; i.e., the wave transmitted through the plasma region is canceled out. Now, when plasma is introduced into the plasma region, the phase of the transmitted wave will change and will not cancel the signal from the reference path, thus giving an output in the detector. When the phase shift of an initially nulled system is increased by π, the two signals reinforce each other. When the phase shift through the plasma path is further increased to 2π, the signals again cancel. This repeats each π and 2π until cutoff of transmission occurs. This system is called a *microwave interferometer* and is very useful in recording the change of density in transient

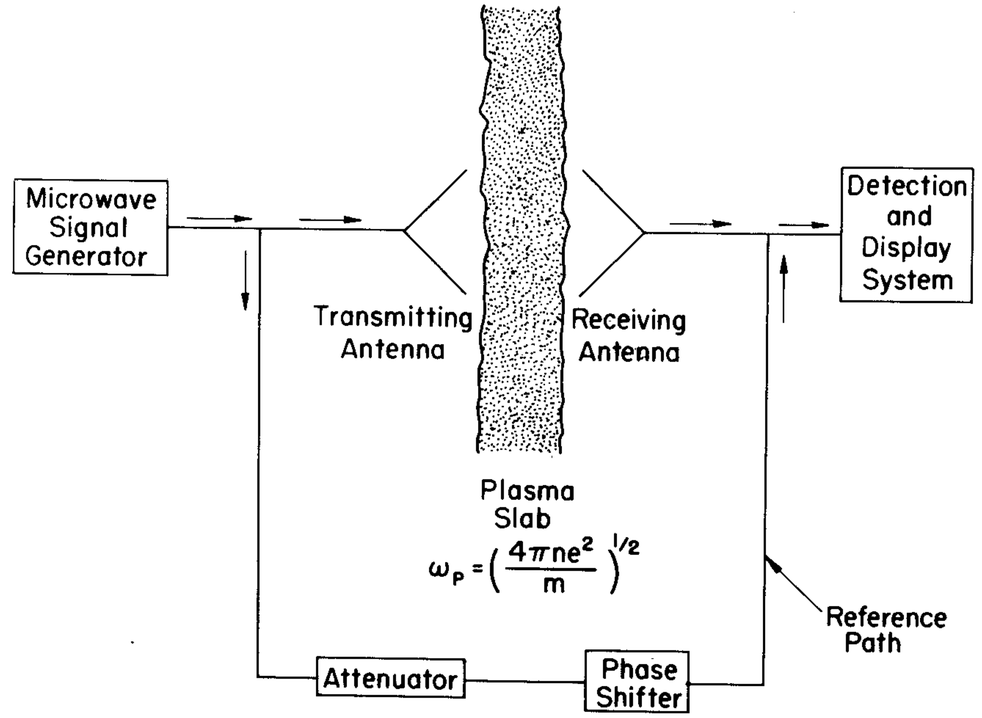


FIGURE 4.6.2
Schematic of microwave interferometer to determine plasma density from a measurement of the phase shift of a wave propagating through the plasma.

plasmas. A typical measurement of a transient plasma is shown in Fig. 4.6.3. As the plasma density increases, the output of the detector goes through successive nulls each time the phase shift through the plasma increases by 2π. The change in phase caused by the plasma depends on the plasma dielectric constant, which depends on density. For a uniform-density plasma slab of width L the change in phase as a function of density is

$$\Delta\phi = \frac{\omega}{c}\left[1 - \left(1 - \frac{\omega_p^2}{\omega^2}\right)^{1/2}\right]L \qquad (4.6.1)$$

By setting this expression equal to the measured change in phase, it is possible to determine the plasma density by solving the equation for the plasma frequency.

Notice that when the plasma density increases to the point where $\omega = \omega_p$, the transmitted signal goes to zero and the detected output becomes constant, equal to the reference-path signal. This provides a check point on the density measurement. As the plasma density decreases below the cutoff density, there is again transmission, and the relative phase shift "unwinds."

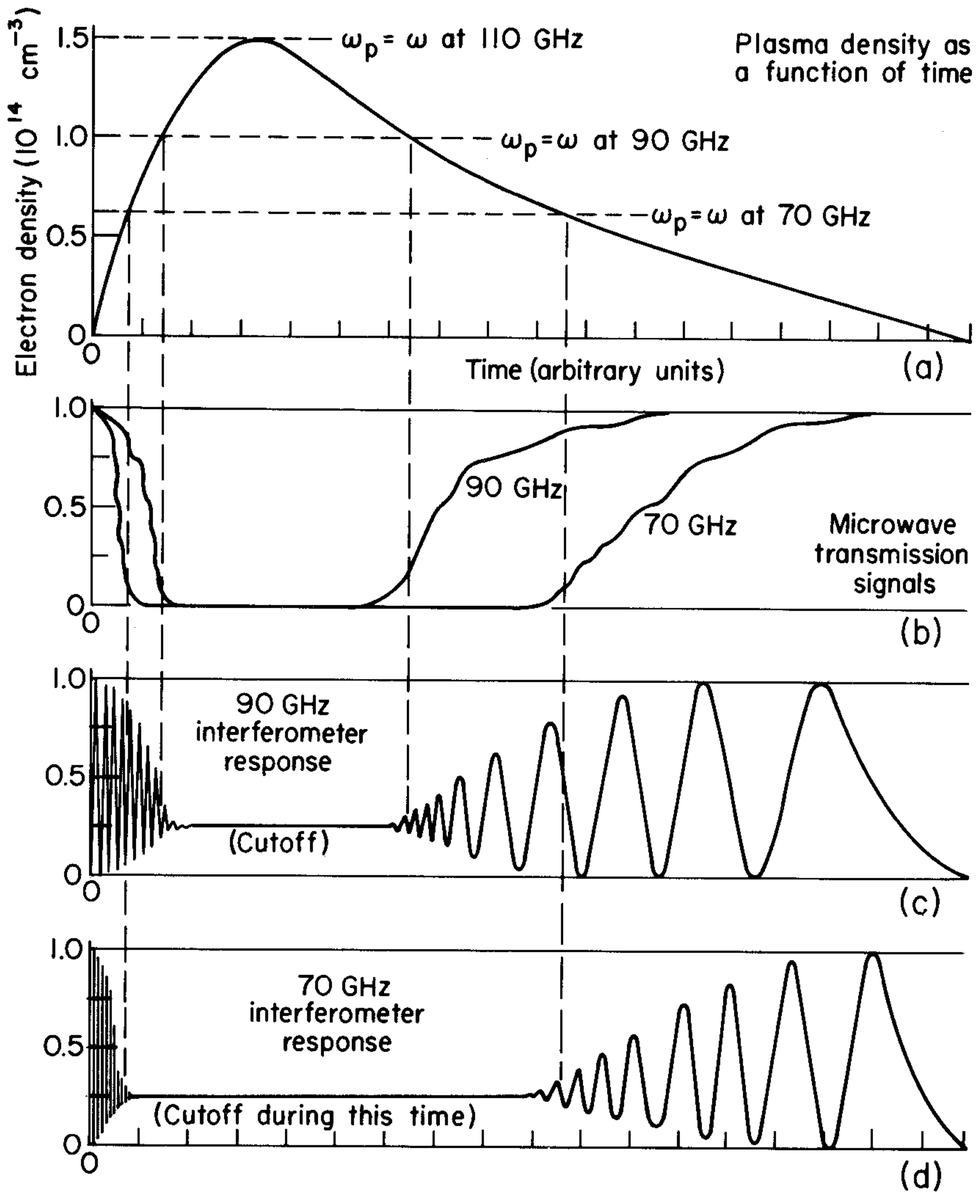


FIGURE 4.6.3
Transient plasma event. Microwave transmission and interferometer responses, using square-law video detectors at 70 and 90 GHz, are shown in relation to assumed electron-density variation with time. (*After M. A. Heald and C. B. Wharton, "Plasma Diagnostics with Microwaves," Wiley, New York*, 1965.)

This method of plasma measurement is also used for plasmas imbedded in magnetic fields, but in that case the phase shift must be computed from the appropriate dispersion relation that takes the tensor nature of the plasma dielectric constant into account (see Sec. 4.9).

4.7 PLASMA-COLUMN RESONANCES

A simple example of an experiment that can be understood by the concept of the plasma dielectric constant is that of the dipole (or higher-order-multipole) resonance of a plasma column. This experiment was first done by Tonks[1], who studied the properties of plasma discharge columns, and has been repeated by many workers in recent years. It has been very useful in stimulating work that confirmed the long-wavelength dispersion relation for a warm plasma [$\omega^2 = \omega_p^2 + (3\kappa T/m)k^2$], known as *Bohm-Gross dispersion.*[2]

The experiment is depicted schematically in Fig. 4.7.1. A wave from a signal source is coupled into a waveguide with dimensions corresponding to a cutoff wavelength of 10 cm. A thermionic-arc discharge (about 1 cm in diameter) column is situated at right angles to the incident waveguide electric field. Two direction couplers sample the amplitude of the incident wave and the amplitude of the reflected wave, as shown. The waveguide beyond the discharge column is terminated to eliminate spurious reflections. The experiment consists of measuring the ratio of the scattered power reflected by the plasma to the power incident on the plasma as a function of the density of the plasma. This ratio is called the *reflection coefficient*. For a typical experiment, the discharge column is a thermionic-arc discharge in mercury vapor at a pressure (10^{-3} Torr) such that the plasma electron density is proportional to the dc current in the discharge. In this regime, the mean free path of plasma electrons is much greater than the diameter of discharge, and so the plasma behaves as though it were collisionless. Thus one can measure the reflection coefficient at a given frequency of incident wave as a function of the plasma frequency ($\omega_p^2 \propto n_e \propto I_{dc}$).

To calculate what might be the expected result of this experiment, it is convenient to idealize the experiment by assuming that the incident electric field has no variation along the plasma column and that the discharge column exists without the confining glass cylinder which a laboratory experiment would have. The idealized model is shown in Fig. 4.7.2.

If the wavelength of the incident wave is much greater than the radius of the plasma column ($\lambda \gg a$), the electric field in the vicinity of the plasma column is nearly irrotational (that is, $\nabla \times \mathbf{E} \approx 0$), and the field distribution can be derived from a scalar potential

$$\mathbf{E} = -\nabla\phi \qquad (4.7.1)$$

This scalar potential must satisfy Poisson's equation, which, written in terms of the equivalent charge-free dielectric constant, is

$$\varepsilon\nabla^2\phi = 0$$

[1] L. Tonks, *Phys. Rev.* **37**:1458 (1931).
[2] D. Bohm and E. P. Gross, *Phys. Rev.* **75**:1851 (1949).

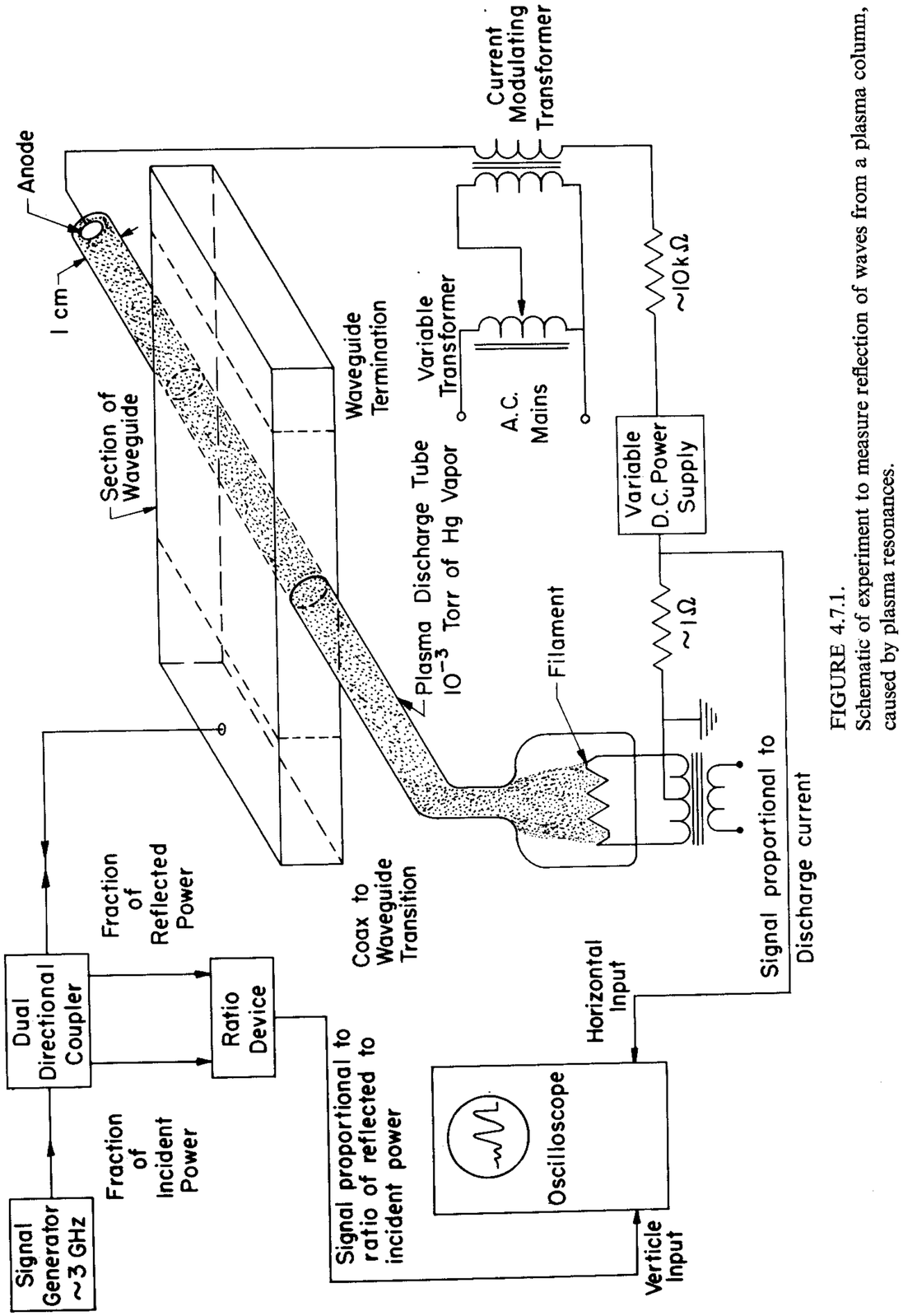


FIGURE 4.7.1
Schematic of experiment to measure reflection of waves from a plasma column, caused by plasma resonances.

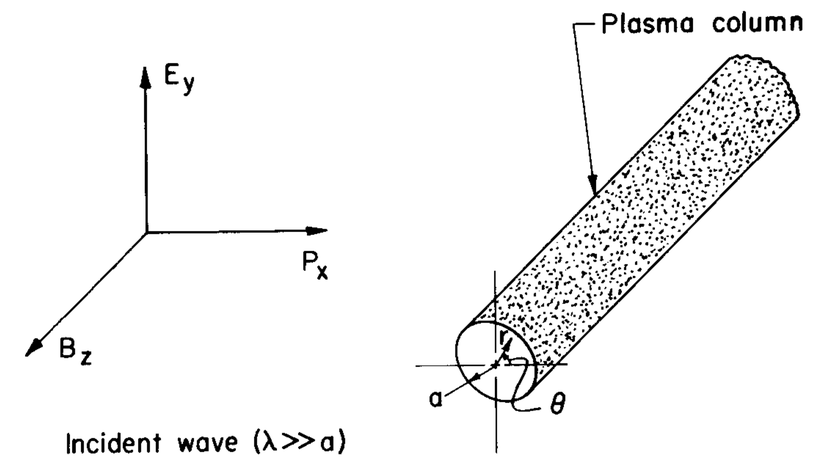


FIGURE 4.7.2
Idealized model to study scattering of electromagnetic waves by plasma-column resonances. The wavelength of the incident wave is assumed to be much longer than the radius of the column.

where $\varepsilon \neq \varepsilon(x)$; that is, the dielectric constant is assumed to be homogeneous. Thus ϕ must satisfy Laplace's equation with no z variation, both inside and outside the plasma. A complete set of solutions of $\nabla_\perp{}^2\phi = 0$ in cylindrical coordinates is made up of linear combinations of

$$\phi_n = (A_n r^n + B_n r^{-n})(C_n \cos n\theta + D_n \sin n\theta)$$

where n is any integer. The condition that ϕ vanishes at $r = 0$ means that inside the plasma column $B_n = 0$, while the geometry of the situation (see Fig. 4.7.2 to establish that $\theta = 0$ and $\theta = \pi$ must be at the same potential) dictates that $C_n = 0$. Then inside the column the potential must be given by

$$\phi_i = \sum_{n=1}^{\infty} A_n r^n \sin n\theta \qquad (4.7.2)$$

Similarly, outside the plasma column the potential at $r \to \infty$ must reduce to the incident potential. Thus the potential outside is

$$\phi_o = \sum_n B_n r^{-n} \sin n\theta + E_{\text{inc}} r \sin\theta \qquad (4.7.3)$$

The values of B_n and A_n must be determined by the boundary condition at the plasma surface. The boundary conditions at a dielectric-vacuum interface are

that the normal component of the displacement ($\mathbf{D} \equiv \varepsilon \mathbf{E}$) and tangential component of the electric field be continuous. These conditions are satisfied if (with the plasma surface at $r = a$)

$$\left(1 - \frac{\omega_p^2}{\omega^2}\right) \frac{\partial \phi_i}{\partial r}\bigg|_a = \frac{\partial \phi_o}{\partial r}\bigg|_a \qquad \varepsilon E_r \text{ continuous at } r = a \qquad (4.7.4)$$

$$\phi_i(a) = \phi_o(a) \qquad E_\theta \text{ continuous at } r = a \qquad (4.7.5)$$

This condition determines A_n and gives the potential inside the plasma in terms of the incident amplitude and frequency:

$$A_1 = \frac{2E_{\text{inc}}}{1 + \varepsilon}$$
$$(1 + \varepsilon)A_n = 0 \qquad n \geq 2 \qquad (4.7.6)$$

Thus the field in the plasma becomes large (resonant) when the frequency is such that $1 + \varepsilon = 0$, or

$$\omega = \frac{\omega_p}{\sqrt{2}} \qquad (4.7.7)$$

The fact that the plasma resonates at $\varepsilon = -1$ and not at $\varepsilon = 0$ is a consequence of the cylindrical geometry (see Prob. 4.7.2).

Since the plasma column is resonant at this frequency, the electrons in the column oscillate in response to the driving electric field. This motion reradiates, or scatters, the incident field in cylindrical waves. Since the amplitude of the motion of the electrons in the plasma is largest at resonance, the scattered power will be a maximum at resonance. If the plasma column extends across a waveguide as in Fig. 4.7.1, the reflection coefficient is a maximum at resonance.

Problem 4.7.1 Show that (in cylindrical coordinates) the condition that E_θ be continuous implies that ϕ also is continuous. ////

Problem 4.7.2 Show that a sphere of plasma with the same assumptions as above has a resonant mode at $\omega_{\text{res}} = \omega_p/\sqrt{3}$. ////

Figure 4.7.3 shows the results of this type of experiment. As can be seen, there is more than one resonance. Many workers tried to explain the additional peaks in terms of multiple resonances associated with the dielectric constant. However, the multiple resonances remained as an unsolved problem for a long time. The resonance at $\omega \approx \omega_p/\sqrt{2}$ (called the *main resonance*) is correctly

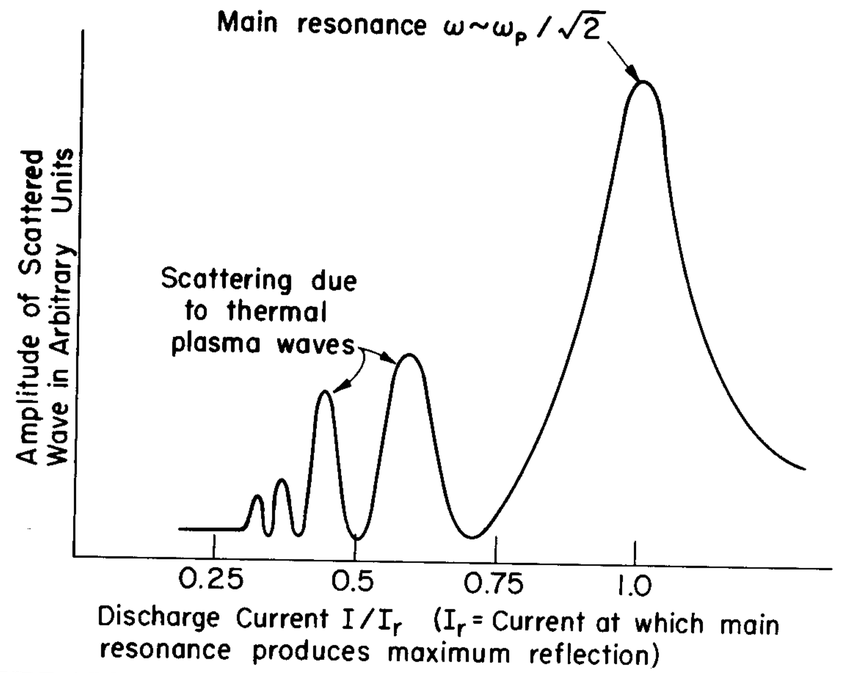


FIGURE 4.7.3
Amplitude of waves scattered from resonances of a plasma column as a function of discharge current (i.e., plasma density). Main resonance occurs for $\omega \sim \omega_p/\sqrt{2}$. The other resonances are due to scattering from thermal waves in the plasma.

identified with the largest scattering peak. In 1959[1] the other scattering peaks were identified as being due to thermal dispersion of plasma waves, which in a warm homogeneous plasma satisfy the Bohm-Gross dispersion relation (4.4.7),

$$\omega^2 = \omega_p^2 + \frac{3\kappa T}{m_e} k^2$$

The effect of thermal dispersion on scattering of waves from a plasma column can be studied with the aid of the fluid equations by including a scalar pressure. The equations are the two-fluid equations, with the ions considered too massive to respond to the electric field ($\mathbf{V}_i = 0$). The electron motion is determined by

$$\frac{\partial n}{\partial t} + \nabla \cdot n\mathbf{V} = 0 \qquad (4.7.8a)$$

$$\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla)\mathbf{V} = \frac{e}{m} \nabla \phi - \frac{1}{nm} \nabla p \qquad (4.78b)$$

[1] R. W. Gould, Proceedings of the Linde Conference on Plasma Oscillations, Indianapolis, Ind. 1959, and J. V. Parker, J. C. Nickel, and R. W. Gould, *Phys. Fluids*, 7:1489 (1964).

and

$$\nabla^2\phi = -4\pi e(n_i - n) \qquad (4.7.9)$$

$$p = \frac{p_0}{{n_0}^\gamma} n^\gamma \qquad (4.7.10)$$

where variables without subscripts refer to *electrons*. Linearizing these equations in the same manner as in preceding sections and combining them leads to an equation for the perturbed scalar potential

$$\nabla^2(\nabla^2 + k^2)\phi_1(r, \theta) = 0 \qquad (4.7.11)$$

where

$$k^2 = \frac{\omega^2 - {\omega_p}^2}{\gamma\kappa T/m} = \frac{1}{3{\lambda_D}^2}\left(\frac{\omega^2}{{\omega_p}^2} - 1\right) \qquad (4.7.12)$$

The value of $\gamma = 3$ is used for the same reason discussed in the section on waves in a warm plasma, namely, that the waves correspond to a one-dimensional adiabatic compression of the electrons.

The solutions of the differential equation for potentials that are finite on the axis and which have the property that $\phi(\theta = 0) = \phi(\theta = \pi)$ (see Fig. 4.7.2) are

$$\phi_i(r, \theta) = \sum_n \left[A_n\left(\frac{r}{a}\right)^n + B_n \frac{J_n(kr)}{J_n(ka)}\right] \sin n\theta \qquad (4.7.13)$$

Outside the plasma the solution is again

$$\phi_o = \sum_n C_n r^{-n} \sin n\theta + E_{\text{inc}} r \sin\theta \qquad (4.7.14)$$

In contrast to the case of scattering from a *cold* plasma, the two boundary conditions provided from Maxwell's equations are not enough to determine the solution, since there are now three constants A_n, B_n, C_n to be determined. There is nothing in the framework of the equations to uniquely determine these constants; the details of the coupling of the wave to the plasma particles are required to resolve this question, and these require a microscopic treatment.[1] What can be done is to consider various solutions of the equations, corresponding to different sorts of plasma motions, and wait for a microscopic treatment to determine which of the motions are most heavily excited by the incident wave. As an example, consider only perturbations for which the radial perturbation current is everywhere equal to zero at the boundary; that is $J_{1r}(a) = n_0 e V_{1r}|_{r=a} = 0$. From (4.7.8*b*) this gives a condition on ϕ_1

$$\frac{\partial}{\partial r}\phi_1\bigg|_{r=a} - \frac{3\kappa T}{m{\omega_p}^2}\frac{\partial}{\partial r}(\nabla^2\phi_1)\bigg|_{r=a} = 0 \qquad (4.7.15)$$

[1] D. E. Baldwin, *Phys. Fluids*, **12**:279 (1969).

which in turn gives a relation between A_n and B_n. The potential inside the plasma column becomes, for each harmonic n,

$$\phi_n(r, \theta) = B_n\left[\frac{J_n(kr)}{J_n(ka)} - \frac{ka}{n}\left(1 + \frac{(3\kappa T/m)k^2}{{\omega_p}^2}\right)\frac{J_n'(ka)}{J_n(ka)}\left(\frac{r}{a}\right)^n\right]\sin n\theta \qquad (4.7.16)$$

Again the electrostatic boundary conditions at the plasma-dielectric interface are used. The condition that $V_{1r} = 0$ implies, through the continuity equation (4.7.8*a*), that there is no surface charge on the boundary, so that the electrostatic boundary conditions are, in this case, that ϕ and $\partial\phi/\partial r$ be continuous at the plasma-vacuum interface. This requirement leads to the result that the amplitude of the nth mode can be nonzero only if

$$\frac{n}{ka}\frac{J_n(ka)}{J_n'(ka)} = \frac{\omega^2}{{\omega_p}^2} + \left(\frac{\omega^2}{{\omega_p}^2} - 1\right) \qquad (4.7.17)$$

where

$$ka = \frac{1}{\sqrt{3}}\left(\frac{a}{\lambda_D}\frac{\omega^2}{{\omega_p}^2} - 1\right)^{1/2} \qquad (4.7.18)$$

This gives the frequencies at which modes satisfying $V_{1r} = 0$ can be excited in a warm plasma. Because of the arbitrary restriction $V_{1r} = 0$, there is no guarantee that there are no *other* modes of the plasma which also could be excited. Equation (4.7.17) can be solved (graphically or numerically) for large a/λ_D. There is a main resonance at $n = 0$, at the same frequency as the resonance in the cold plasma,

$${\omega_0}^2 \approx \frac{{\omega_p}^2}{2} \qquad (4.7.19)$$

and a series of resonances at the plasma frequency and above, given by

$$\omega^2 = {\omega_p}^2\left(1 + \frac{3{\lambda_D}^2}{a^2}{x_n}^2\right) \qquad (4.7.20)$$

where $x_1 \approx 5.3$, $x_2 \approx 8.5$, and so on. Although the cold plasma resonated at a unique frequency, the inclusion of thermal effects in the plasma gives rise to a series of resonances.

Problem 4.7.3 Show that if the plasma is bounded, not by vacuum, but by a medium (e.g., glass) of dielectric constant K_e, the resonances occur at

$$\frac{n}{ka}\frac{J_n(ka)}{J_n'(ka)} = K_e\left(\frac{\omega^2}{{\omega_p}^2} - 1\right) + \frac{\omega^2}{{\omega_p}^2} \qquad (4.7.20a)$$

so that ${\omega_0}^2 = {\omega_p}^2/(1 + 1/K_e)$. ////

It should be noted that although this analysis predicts a main resonance at

$$\omega = \frac{\omega_p}{\sqrt{2}}$$

it is not predicting the same type of plasma motion associated with that resonance. The main resonance in the cold-plasma calculation gave a dipole field

$$\phi_1 \approx \bar{E}r \sin\theta + \sum_{n>1} E_n r^n \sin n\theta \qquad \omega = \frac{\omega_p}{\sqrt{2}}$$

while the warm-plasma calculation *with the boundary condition* $V_{1r}(r = a) = 0$ gave

$$\phi_1 \approx J_0(kr) \qquad \omega = \frac{\omega_p}{\sqrt{2}}$$

The difference between these two results is in the boundary condition satisfied by the perturbed field, and not in the "warmth" of the plasma. To see this, instead of the condition $V_{1r}(r = a) = 0$, impose the condition that $n_1 = 0$ *inside* the plasma. This means that $\nabla^2\phi_i = 0$, and requires, therefore, that in (4.7.13),

$$B_n = 0$$

If $n_1 = 0$ inside the plasma, then from (4.7.8) there must be a current V_{1r} inside the plasma,

$$-i\omega V_{1r} = \frac{e}{m}\frac{\partial\phi_1}{\partial r}$$

Since $V_{1r} = 0$ outside the plasma, Eq. (4.7.8*a*) shows that there must be a perturbed surface charge at $r = a$, and this is the difference between the plasma oscillation found with $V_{1r}(a) = 0$ and the oscillation described by (4.7.2) to (4.7.8). An alternative to that calculation (which calculated the plasma dielectric) would be to use Poisson's equation to obtain the jump in E_r across the boundary in terms of the charge on the boundary, as shown in Fig. 4.7.4.

$$A\left(\frac{\partial\phi_i}{\partial r} - \frac{\partial\phi_o}{\partial r}\right) = \int_{\text{vol}} 4\pi n_1 e\, dx = \frac{4\pi e}{i\omega} n_0 V_{1r}(r = a)A = \frac{4\pi n_0 e^2}{m\omega^2}\frac{\partial\phi_1}{\partial r} A \qquad (4.7.21)$$

using Gauss' law to write $\int_{\text{vol}} d\mathbf{x}\ \nabla\cdot\mathbf{V}_1 = \int_A d\mathbf{a}\ V_{1r} = AV_{1r}(r = a)$. Since $V_{1r} = 0$ outside the plasma, clearly (4.7.21) is equivalent to (4.7.4).

Although the analysis presented here predicts additional resonances, it does not correctly account for the relative spacing observed in the experiments. Experimentally, the frequency spacing was such that it appeared as if the radius

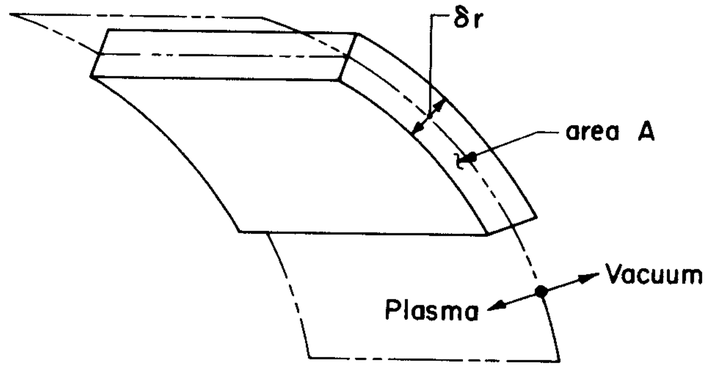


FIGURE 4.7.4
Schematic of plasma-vacuum interface showing gaussian pillbox used to calculate jump in electric field at plasma-vacuum boundary.

of the column were effectively smaller than its real size. The problem was resolved by assuming a radial distribution of the density in the plasma column, i.e., the plasma frequency vanishes outside the plasma, is equal to $\bar{\omega}_p^2$ on axis, and has intermediate values in between. Then, when waves are excited at a frequency below the on-axis plasma frequency, they will still propagate in the outer regions, where $\omega > \omega_p$. This is shown schematically in Fig. 4.7.5. Thus a

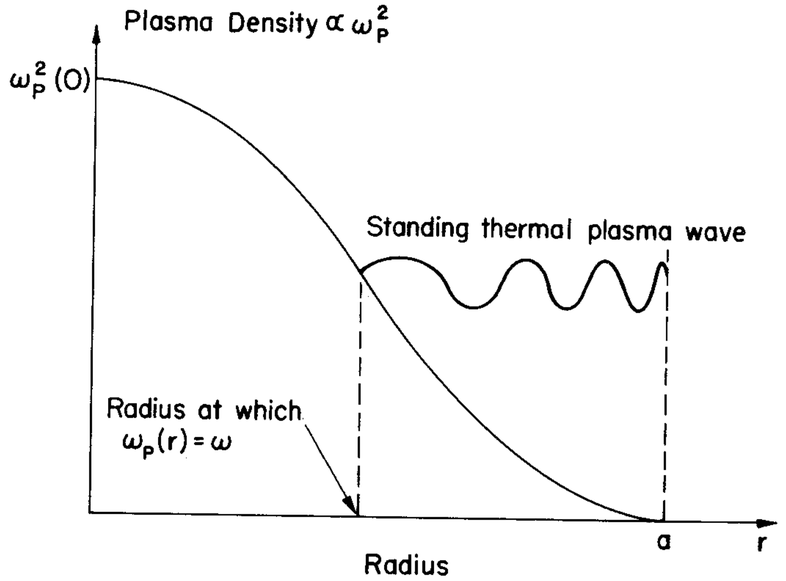


FIGURE 4.7.5
Diagram of variation of plasma density as a function of radius showing standing plasma wave between the wall of the discharge tube and the point on the plasma column where $\omega = \omega_p$.

complete analysis using a fluid model requires that thermal dispersion *and* a variation of density be included in the model. Parker, Nickle, and Gould[1] have made a comparison between theory and experiment, using the equations given in Prob. (4.7.5), together with a radial density variation appropriate to the plasma discharge columns used in their experiments.

Problem 4.7.4 Use the linearized fluid equations, assume the electric field is derivable from a scalar potential, and assume a one-dimensional adiabatic equation of state $p/n^\gamma = \text{constant}$ with $\gamma = 3$ in the radial direction. Show that the differential equation for a "warm" nonuniform plasma column is

$$\nabla^4\phi_1 - \frac{1}{\gamma}\left[\frac{\nabla n_0}{n_0}\cdot\nabla\right]\nabla^2\phi_1 + \left\{\frac{1}{\gamma\lambda_D^2(0)}\left[\frac{\omega^2}{\omega_p^2(0)} - \frac{n_0(r)}{n_0(0)}\right] - \frac{1}{\gamma}\nabla\cdot\frac{\nabla n_0}{n_0}\right\}\nabla^2\phi_1 - \frac{1}{\gamma\lambda_D^2(0)}\frac{\nabla\phi_1\cdot\nabla n_0}{n_0(0)} = 0 \qquad (4.7.22)$$

where

$\lambda_D(0)$ = Debye length at center of plasma column

$n_0(r)$ = radial-density distribution

$\omega_p(0)$ = plasma frequency at center of plasma column

Also show that requiring $V_{1r}(a) = 0$ leads to the following restriction in the potential at $r = a$:

$$\frac{\partial}{\partial r}(\nabla^2\phi_1) - \frac{1}{\gamma}\frac{1}{n_0}\frac{dn_0}{dr}\nabla^2\phi_1 - \frac{n_0(r)}{n_0(0)}\frac{1}{\gamma\lambda_D^2(0)}\frac{\partial}{\partial r}\phi_1 = 0 \qquad (4.7.23)$$

////

The results of a typical experiment are shown in Fig. (4.7.6). As can be seen, there is excellent agreement between experiment and theory. More recent work by Baldwin[2] has performed the microscopic analysis required to properly include thermal effects, especially including Landau damping, which could not come from a fluid theory. In either case the important point is that these "simple" scattering experiments demonstrate several important features of plasmas.

[1] J. V. Parker, J. C. Nickle, and R. W. Gould, *Phys. Fluids*, 7:1489 (1964).

[2] D. E. Baldwin, *Phys. Fluids*, 12:279 (1969).

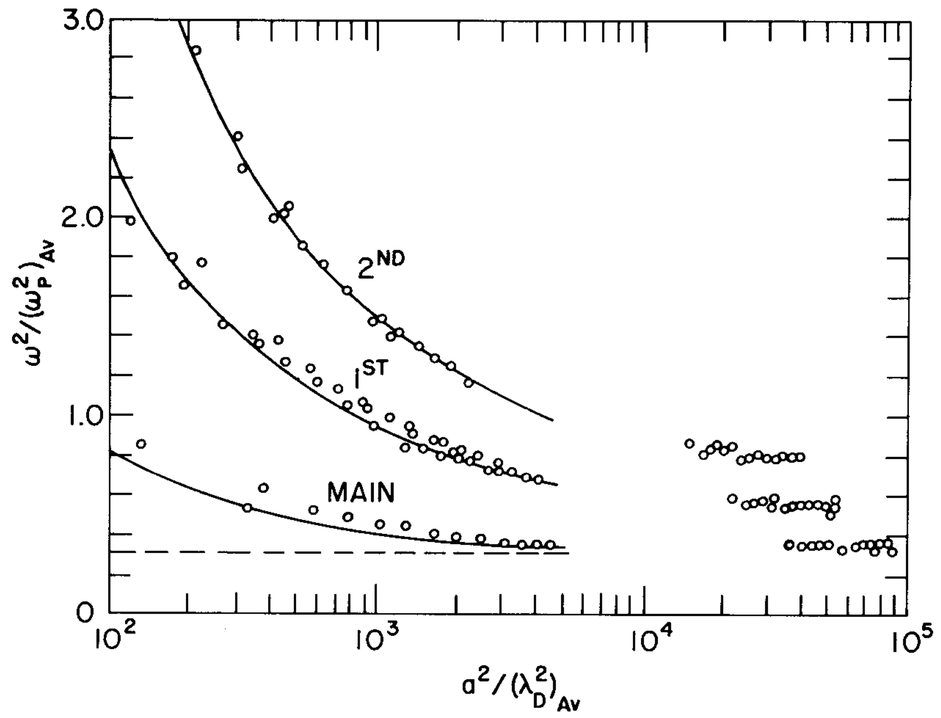


FIGURE 4.7.6
Comparison of theoretical dipole resonances of a plasma column, using macroscopic fluid theory, with experimental results. [*After J. V. Parker, J. C. Nickle, and R. W. Gould, Phys. Fluids*, 7: 1489 (1964).]

First, there is the dielectric nature of the cold plasma, which predicts the observed resonance at $\omega_p/\sqrt{2}$. Second, there are thermal effects, giving rise to longitudinal waves which obey the Bohm-Gross thermal dispersion relation $\omega^2 = \omega_p^2 + (3\kappa T/m)k^2$. And third, something not apparent from the discussion given here, there are the Landau damping properties associated with the longitudinal waves.

4.8 SPACE-CHARGE WAVES IN FINITE PLASMAS

The preceding section showed that although an infinite homogeneous plasma resonates at $\omega = \omega_p$, a cylindrical column resonates at $\omega_p/\sqrt{2}$ and a sphere at $\omega_p/\sqrt{3}$. There are other ways in which the finite size of a plasma system affects the properties of plasma waves. For example, consider the $\omega - k$ plot for Langmuir oscillations in Fig. 4.8.1 This figure shows that the phase velocity can be either greater or less than the velocity of light. At $k = 0$, $v_p = \infty$, and the

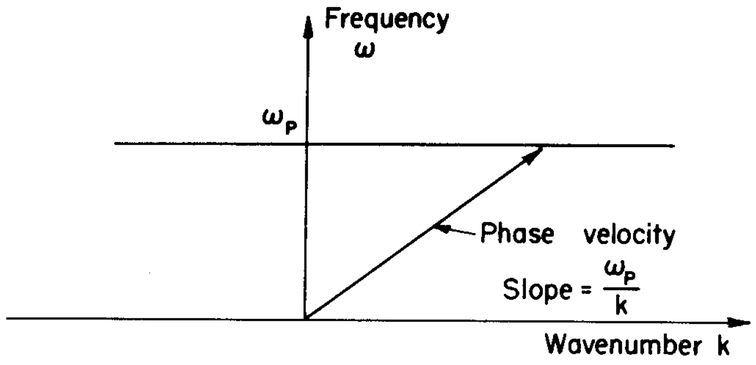


FIGURE 4.8.1
Frequency vs. wave-number diagram for plasma waves in a cold, collisionless plasma.

wave will sample even distant boundaries; the effects of finite size would be expected to influence the results of an experimental study of long-wavelength oscillations in a plasma. In this section two examples of plasma waves in bounded systems are presented.

4.8.1 Space-charge Waves in a Plasma-filled Cylinder in an Infinite Axial Magnetic Field

Consider a perfectly conducting cylinder of radius a filled with a cold plasma and immersed in an infinite axial magnetic field, as shown in Fig. 4.8.2.

In this example, the electrons are constrained by the magnetic field to move only in the z direction, and hence cannot interact with modes that have only transverse electric field components. This means that, for this case, the transverse electric waveguide modes are unaffected by the presence of the plasma. The transverse *magnetic* modes *have* an electric field component in the direction of the steady magnetic field, and they are affected by presence of the plasma.

The plasma will again be represented in terms of the dielectric tensor $\boldsymbol{\varepsilon}$, and for such an anisotropic medium, Maxwell's equations for the perturbation variables are

$$\nabla \times \mathbf{E}_1 = \frac{i\omega}{c}\mathbf{B}_1 \tag{4.8.1}$$

$$\nabla \times \mathbf{B}_1 = -\frac{i\omega}{c}\boldsymbol{\varepsilon}\cdot\mathbf{E}_1 \tag{4.8.2}$$

$$\nabla \cdot \mathbf{B}_1 = 0 \tag{4.8.3}$$

$$\nabla \cdot \boldsymbol{\varepsilon} \cdot \mathbf{E}_1 = 0 \tag{4.8.4}$$

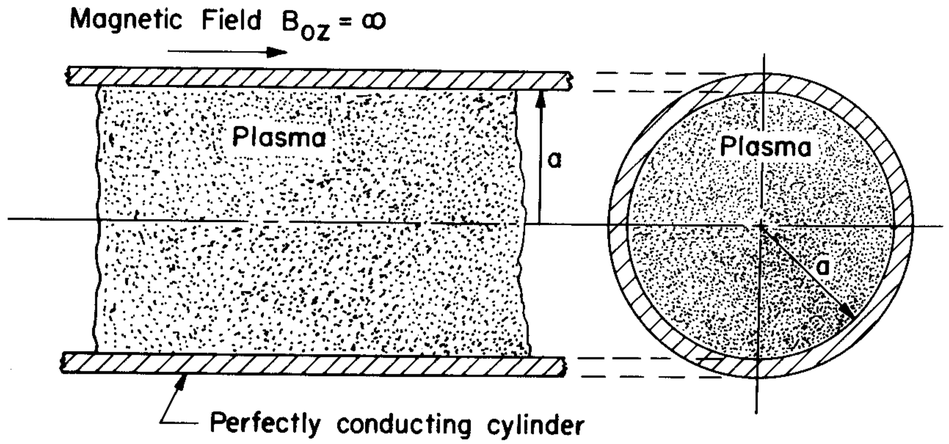


FIGURE 4.8.2
Schematic of plasma-filled waveguide in an infinite axial magnetic field.

where for $B_0 \to \infty$,

$$\boldsymbol{\varepsilon} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 - \dfrac{\omega_p^2}{\omega^2} \end{bmatrix} \tag{4.8.5}$$

The form (4.8.5) for $\boldsymbol{\varepsilon}$ should be clear, since for motions across $\mathbf{B}_0$, the plasma is like a vacuum, while along $\mathbf{B}_0$, the plasma behaves like the dielectric medium discussed in the preceding sections of this chapter.

Taking the curl of (4.8.1) and using (4.8.2) to eliminate $\nabla \times \mathbf{B}_1$ gives

$$\nabla(\nabla \cdot \mathbf{E}_1) - \nabla^2 \mathbf{E}_1 = \frac{\omega^2}{c^2}\boldsymbol{\varepsilon}\cdot\mathbf{E}_1 \tag{4.8.6}$$

$$\nabla^2 E_z + \frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right)E_z - [\nabla(\nabla \cdot \mathbf{E}_1)]_z = 0 \tag{4.8.7}$$

From (4.8.4) and (4.8.5),

$$\nabla \cdot \boldsymbol{\varepsilon} \cdot \mathbf{E} = \nabla \cdot \left[\hat{\mathbf{r}}E_r + \hat{\boldsymbol{\theta}}E_\theta + \hat{\mathbf{z}}E_z\left(1 - \frac{\omega_p^2}{\omega^2}\right)\right]$$

$$= \nabla \cdot \mathbf{E} - \frac{\partial}{\partial z}E_z\frac{\omega_p^2}{\omega^2} = 0 \tag{4.8.8}$$

Thus

$$[\nabla(\nabla \cdot \mathbf{E})]_z = \frac{\omega_p^2}{\omega^2}\frac{\partial^2 E_z}{\partial z^2} \tag{4.8.9}$$

Problem 4.8.1 Derive the form (4.8.5) from the fluid equations for the special case $B_0 \to \infty$! ////

Assuming a wave dependence of the form

$$E_z(r, \theta, z, t) = \hat{E}_z(r, \theta, k, \omega)\exp[i(kz - \omega t)] \qquad (4.8.10)$$

(4.8.7) becomes

$$\left[\nabla_T{}^2 + \left(\frac{\omega^2}{c^2} - k^2\right)\left(1 - \frac{\omega_p{}^2}{\omega^2}\right)\right]E_z = 0 \qquad (4.8.11)$$

where $\nabla_T{}^2$ is the transverse laplacian (in cylindrical coordinates for the examples considered here). The solution to the differential equation (4.8.11) which is finite at $r = 0$ is, by separation of variables,

$$E_z = \hat{A}J_n(Tr)\exp[i(kz - \omega t - n\theta)] \qquad (4.8.12)$$

where

$$T^2 = \left(\frac{\omega^2}{c^2} - k^2\right)\left(1 - \frac{\omega_p{}^2}{\omega^2}\right) \qquad (4.8.13)$$

is the separation constant. The boundary condition at the perfectly conducting waveguide wall requires that the tangential component of the electric field vanish at the wall; i.e., $E_z(r = a) = 0$. Thus

$$J_n(Ta) = 0 \qquad \text{and} \qquad Ta = p_{nv} \qquad (4.8.14)$$

where p_{nv} is the vth root of the nth-order Bessel's function of the first kind. From (4.8.13) and (4.8.14) the dispersion relation for these waves is

$$(ka)^2 = \frac{\omega^2 a^2}{c^2} - \frac{p_{nv}{}^2}{1 - \omega_p{}^2/\omega^2} \qquad (4.8.15)$$

Problem 4.8.2 Show that the dispersion relation for waves in a plasma of warm electrons and immobile ions, which fills a waveguide of radius a in an infinite axial magnetic field, is

$$k^2a^2 = \frac{\omega^2 a^2}{c^2} - \frac{p_{nv}{}^2}{1 - \dfrac{\omega_p{}^2}{\omega^2 - (3\kappa T/m)k^2}} \qquad (4.8.16)$$

Sketch ω versus k for an experiment $a \ll \lambda_D$, and for an experiment $a \gg \lambda_D$. ////

This dispersion relation is shown in Fig. 4.8.3. The upper passband in Fig. 4.8.3 represents the waveguide modes which, in the absence of plasma, would still propagate. The principal effect on the waveguide modes is the displacement to higher frequency of the waveguide cutoff frequency. That is, for

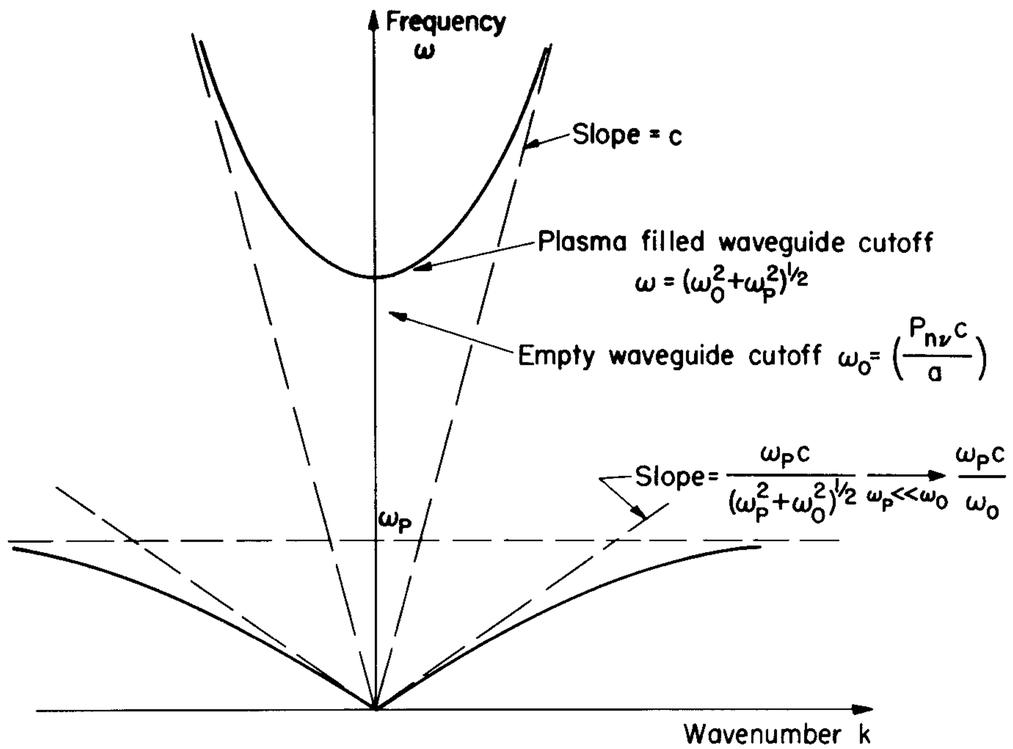


FIGURE 4.8.3
Frequency vs. wave-number diagram for the transverse magnetic modes in a plasma-filled waveguide in an infinite axial magnetic field. [*From A. W. Trivelpiece and R. W. Gould, J. Appl. Phys.* **30**: 1784 (1959).]

the empty waveguide, the cutoff frequency is $\omega = p_{nv}c/a$, while for the plasma-filled waveguide in an infinite axial magnetic field, the cutoff frequency is $\omega_\circ = [(p_{nv}c/a)^2 + \omega_p{}^2]^{1/2}$, so that each waveguide-mode cutoff is shifted upward.

Problem 4.8.3 At low frequencies, ion motions must be included in the plasma dielectric constant. Derive the dispersion relation for the same conditions described in Prob. 4.8.2, including ion motion. Sketch the resulting dispersion relation for all frequencies. ////

The lower passband between $\omega = 0$ and $\omega = \omega_p$ depends on the presence of the plasma, and it represents the plasma oscillations in a finite plasma. The finite boundaries have caused the system to be dispersive, and the Langmuir oscillations, which did not propagate in the absence of a uniform drifting motion or a thermal distribution of electrons, now propagate. One view of the new degree of freedom introduced by the finite boundaries is that the convection-current density and the displacement-current density no longer cancel. Thus

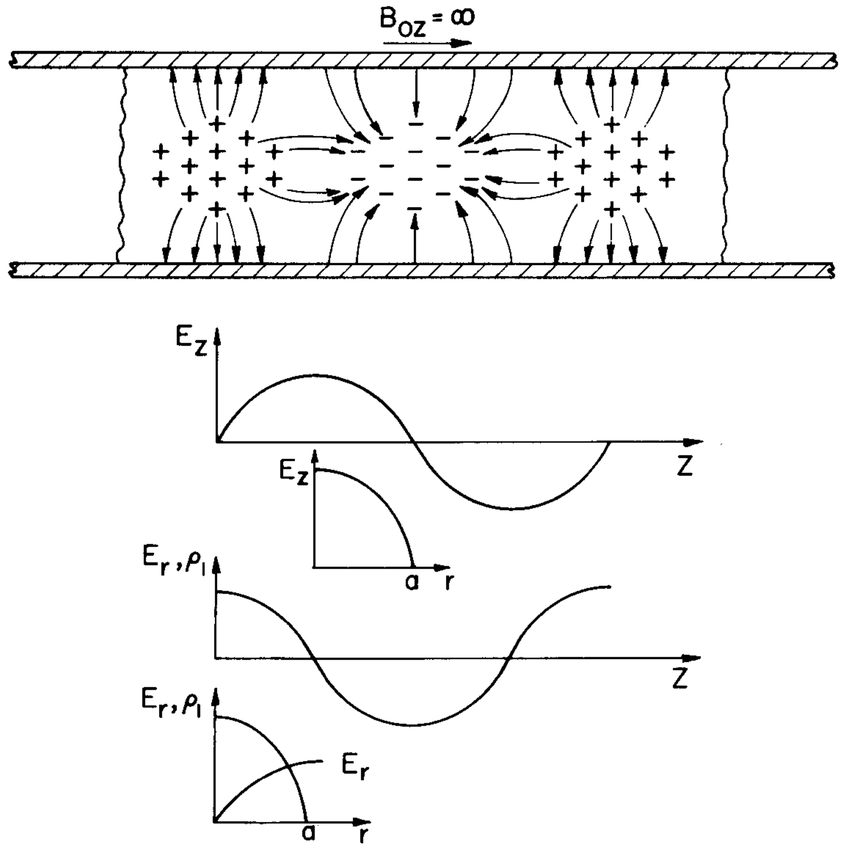


FIGURE 4.8.4
Electric-field charge-density distribution for space-charge waves in a plasma-filled waveguide in an infinite axial magnetic field for lowest azimuthally symmetric mode. One-quarter cycle later **E**, ρ are zero, and electrons have a velocity distribution proportional to E_z.

there is a net perturbation current, which in turn means there is a perturbed magnetic field, and therefore a real electromagnetic power flow, associated with these disturbances.

The maximum phase velocity of the space-charge waves occurs for low frequency, and is given by

$$\lim_{\omega \to 0} v_{\text{phase}} = \frac{c}{(1 + p_{nv}{}^2 c^2/a^2 \omega_p{}^2)^{1/2}} \qquad (4.8.17)$$

The thing to notice about the result (4.8.17) is that these long-wavelength space-charge waves in a finite-size plasma have a limiting phase velocity less than the velocity of light, in contrast to the situation in the unbounded plasma in which the phase velocity increased as k^{-1} for $k \to 0$. The field distribution for the space-charge waves is shown in Fig. 4.8.4.

Problem 4.8.4 Compute the power flow and the distribution of electric-, magnetic-, and kinetic-energy densities associated with the low-frequency space-charge waves in a cold uniform plasma-filled waveguide immersed in a magnetic field of infinite strength aligned along the waveguide axis. ////

Problem 4.8.5 Unlike waveguide modes which have a different cutoff frequency for each mode, the plasma wave modes all have the same passband ($0 < \omega < \omega_p$) for a radial or azimuthal variation. If the electric field at $z = 0$ is $E_z(r, \theta, 0, t) = E_0 \sin \omega t$, determine the amplitude of the propagating modes. ////

Problem 4.8.6 For a neutralized electron beam of radius a filling a conducting drift tube immersed in a steady infinite axial magnetic field, the perturbation electric field is given by $E_{1z} = AJ_n(p_{nv} r/a)\exp[i(kz - n\theta - \omega t)]$, with

$$\left(\frac{p_{nv}}{a}\right)^2 = \left(\frac{\omega^2}{c^2} - k^2\right)\left(1 - \frac{\omega_p{}^2}{(\omega - kV_{0z})^2}\right)$$

as the dispersion relation for the propagation of space-charge waves (small-amplitude perturbations), where p_{nv} are the roots $J_n(p_{nv}) = 0$. Starting with Maxwell's equations, show that the differential equation from which the above solution was obtained is

$$\frac{r}{R}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + r^2\left(\frac{\omega^2}{c^2} - k^2\right)\left[1 - \frac{\omega_p{}^2}{(\omega - kV_{0z})^2}\right] + \frac{1}{\Theta}\frac{d^2\Theta}{d\theta^2} = 0$$

where $E_{1z} = R(r)\Theta(\theta)\exp[i(kz - n\theta - \omega t)]$ has been used. ////

4.8.2 Surface Waves on an Unmagnetized Plasma Column

In the absence of a dc magnetic field, the dielectric tensor reduces to the scalar dielectric constant derived in Sec. 4.1, i.e.,

$$\varepsilon = 1 - \frac{\omega_p{}^2}{\omega^2} \qquad (4.8.18)$$

Consider an unmagnetized plasma column of radius a with the above dielectric constant in a conducting cylinder of radius b, as shown in Fig. 4.8.5. The

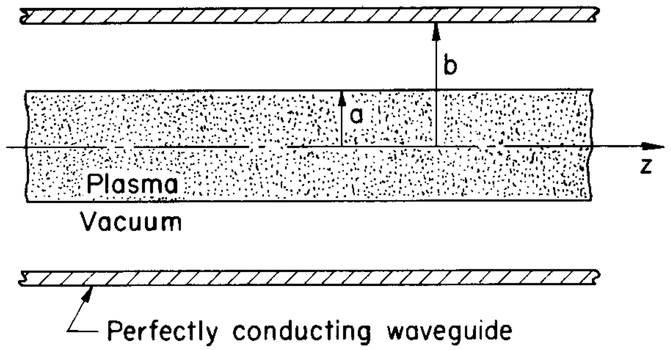


FIGURE 4.8.5
Schematic of a cylindrical waveguide partially filled with unmagnetized cold plasma.

differential equations for propagating modes in this system, obtained from Maxwell's equations, are

$$\begin{aligned} \left\{\nabla_T^2 + \left[\frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right) - k^2\right]\right\}\begin{bmatrix} E_{1z} \\ B_{1z} \end{bmatrix} = 0 \qquad & 0 < r < a \\ \left[\nabla_T^2 + \left(\frac{\omega^2}{c^2} - k^2\right)\right]\begin{bmatrix} E_{1z} \\ B_{1z} \end{bmatrix} = 0 \qquad & a < r < b \end{aligned} \tag{4.8.19}$$

where ∇_T^2 is the appropriate transverse laplacian

$$\nabla_T^2 = \frac{1}{r}\frac{\partial}{\partial r} r \frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} \qquad \text{in cylindrical coordinates}$$

Problem 4.8.7 Verify (4.8.19). ////

Problem 4.8.8 Show that all the transverse field components are derivable from E_{1z} and B_{1z} in the cylindrical system, shown in Fig. 4.8.5. What are these field components in this system? ////

Waves associated with the solution of the differential equation for E_{1z} are called *E modes*, while those associated with B_{1z} are called *B modes*. Although E_{1z} and B_{1z} satisfy the same differential equation, they lead to different eigenmodes, since the boundary conditions at the plasma-vacuum interface and the conducting wall influence these solutions in different ways. For azimuthally symmetric wave solutions, the boundary conditions at the plasma-vacuum interface do not couple the E and B modes, while with azimuthal dependence, these modes are coupled at the plasma-vacuum boundary, thus requiring a linear combination of E- and B-mode terms to satisfy the boundary conditions. For simplicity, only the azimuthally symmetric modes are considered here.

Problem 4.8.9 What are the properties of the azimuthally symmetric B modes for the system in Fig. 4.8.5? ////

The solution of (4.8.19) for azimuthally symmetric E modes propagating at less than the speed of light ($k^2 > \omega^2/c^2$) is

$$E_{1z} = \begin{cases} A\dfrac{I_0(\tau r)}{I_0(\tau a)} e^{i(kz-\omega t)} & 0 < r < a \qquad (4.8.20) \\ A\dfrac{I_0(\tau_0 r)K_0(\tau_0 b) - I_0(\tau_0 b)K_0(\tau_0 r)}{I_0(\tau_0 a)K_0(\tau_0 b) - I_0(\tau_0 b)K_0(\tau_0 a)} e^{i(kz-\omega t)} & a < r < b \qquad (4.8.21) \end{cases}$$

where

$$\tau_0^2 = k^2 - \frac{\omega^2}{c^2} \tag{4.8.22}$$

$$\tau^2 = k^2 - \frac{\omega^2}{c^2} + \frac{\omega_p^2}{c^2} \tag{4.8.23}$$

The functions I_0 and K_0 are modified Bessel's functions of the first and second kind, respectively. This solution has been chosen so that E_{1z} is finite on the axis ($r = 0$), continuous at the plasma boundary ($r = a$), and zero at the conducting boundary ($r = b$). The tangential magnetic field ($B_{1\theta}$) must also be continuous at the plasma boundary. Imposing this condition leads to the dispersion relation for waves,

$$\left(1 - \frac{\omega_p^2}{\omega^2}\right)\frac{1}{\tau a}\frac{I_0'(\tau a)}{I_0(\tau a)} = \frac{1}{\tau_0 a}\frac{I_0'(\tau_0 a)K_0(\tau_0 b) - I_0(\tau_0 b)K_0'(\tau_0 a)}{I_0(\tau_0 a)K_0(\tau_0 b) - I_0(\tau_0 b)K_0(\tau_0 a)} \tag{4.8.24}$$

where the prime denotes a derivative with respect to the argument.

Problem 4.8.10 Verify (4.8.24) ////

The solution of (4.8.24) for the case of $b/a \to \infty$ is shown in Fig. 4.8.6 for various values of $\omega_p a/c$, and E_{1z} as a function of radius is shown in Fig. 4.8.7. The quantity $\omega_p a/c$ is the ratio of the radius of the plasma column to the reactive skin depth ($\delta = c/\omega_p$) of penetration of waves into a cold lossless plasma. When $\omega_p a/c \ll 1$, the fields penetrate into the plasma, and at low frequency ($\omega \ll \omega_p$) the waves are nondispersive and propagate with phase velocity much less than the speed of light. As $\omega_p a/c$ increases, the fields become more concentrated at the surface of the plasma column and propagate with higher velocity, approaching the speed of light for $\omega_p a/c \gg 1$. For this latter condition, the fields do not penetrate appreciably into the plasma, and it behaves much like a conductor. The ratio of E_{1z}/E_{1r} outside the plasma becomes small, and the mode becomes like the transverse electromagnetic wave that propagates on a coaxial line. Since

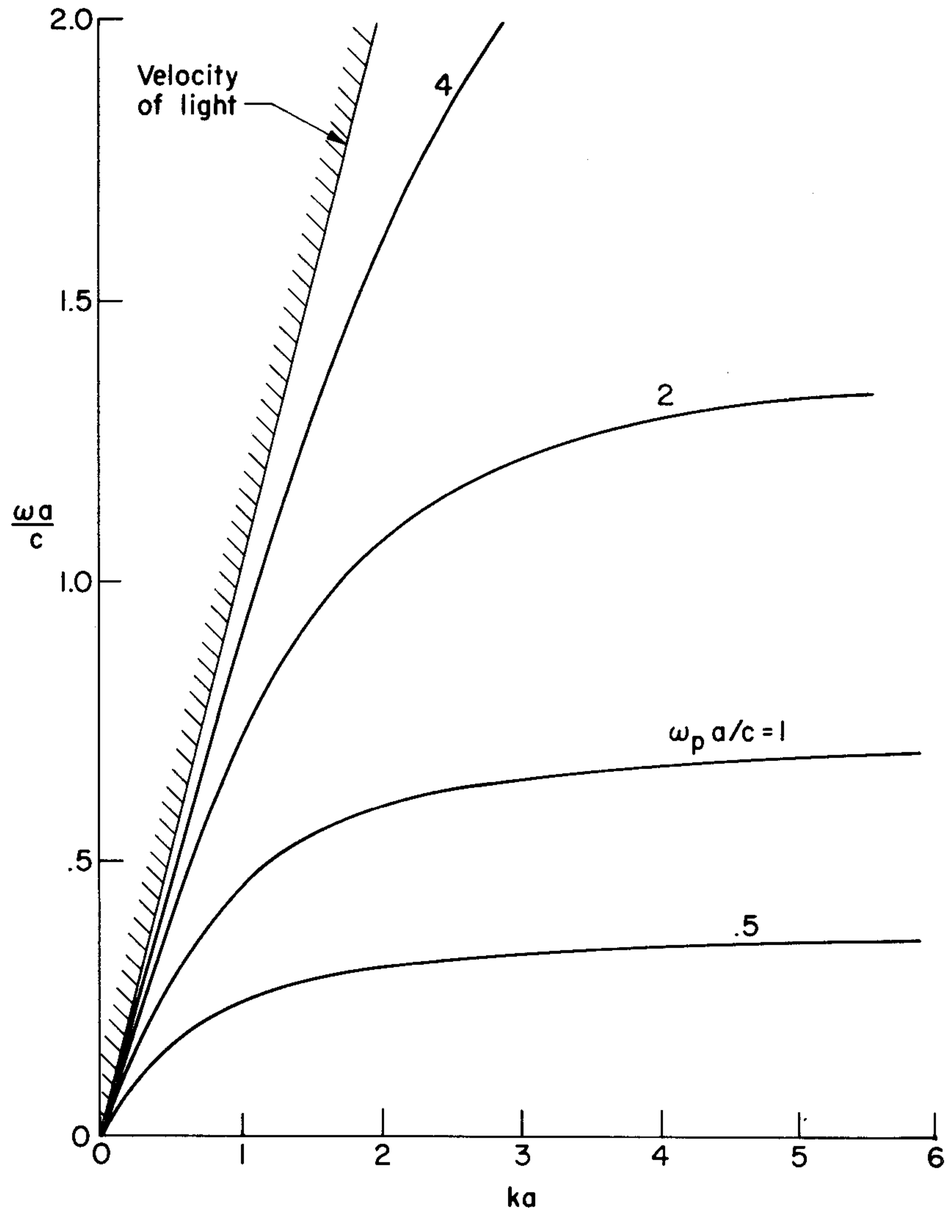


FIGURE 4.8.6
Frequency vs. wave-number diagram for surface wave on an unmagnetized plasma column of radius a. [*From A. W. Trivelpiece and R. W. Gould, J. Appl. Phys.* **31**: 1784 (1959).]

each component of the field is largest at the plasma-vacuum boundary and decreases away from the boundary, these waves are called *surface waves.*

There is no perturbed charge density associated with these waves. They propagate in a peristaltic manner with the field distribution as shown in Fig. 4.8.8.

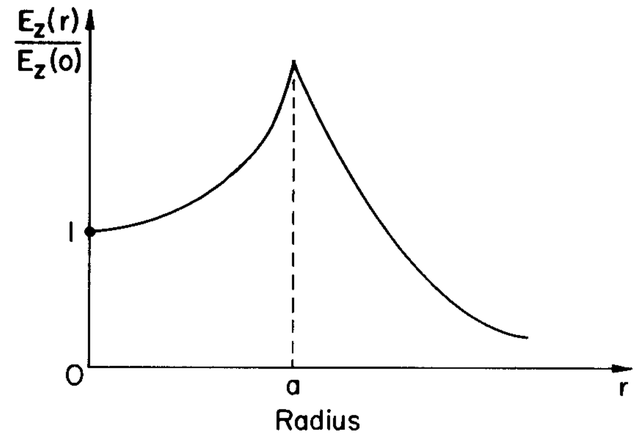


FIGURE 4.8.7
Electric field for the azimuthally symmetric surface wave mode as a function of radius for an unmagnetized plasma column in free space.

Problem 4.8.11 These waves are slow waves, in that the phase velocity is much less than the velocity of light. This means $\nabla \times \mathbf{E} \approx 0$ for these waves and that their properties can be investigated using a scalar potential ϕ, which satisfies $\nabla^2\phi = 0$. Show that the dispersion relation obtained using this scalar potential yields the following dispersion relation for surface waves on a plasma column of radius a in free space:

$$1 - \frac{\omega_p^2}{\omega^2} = \frac{I_0(ka)K_0'(ka)}{I_0'(ka)K_0(ka)}$$

Solve this dispersion relation to show that these surface waves have the phase characteristics shown in Fig. 4.8.9. Discuss the properties of these waves and compare them with the dispersion relation obtained using the complete Maxwell's equations. ////

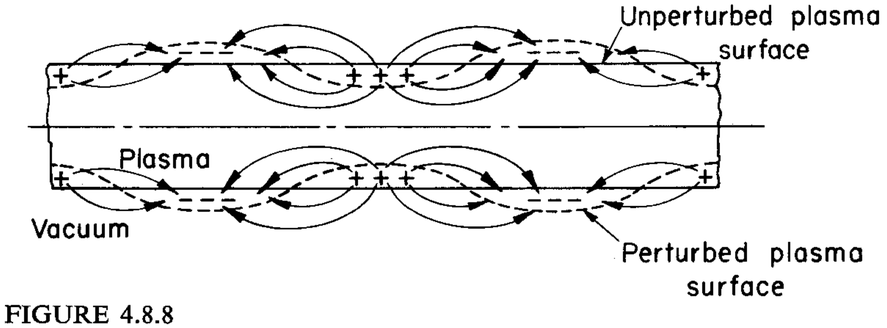


FIGURE 4.8.8
Electric field for the azimuthally symmetric surface wave mode on an unmagnetized plasma column in free space.

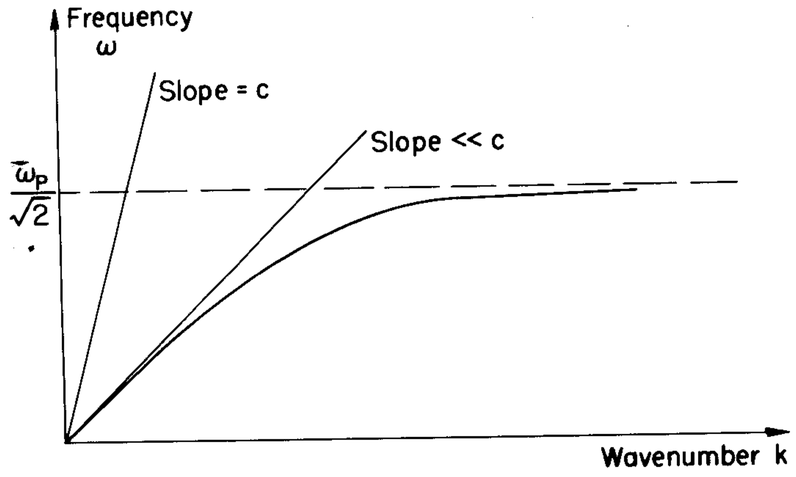


FIGURE 4.8.9
Frequency vs. wave-number diagram for waves on an unmagnetized plasma column in free space from quasistatic analysis.

4.9 DIELECTRIC CONSTANT OF A COLD MAGNETIZED PLASMA ($\mathbf{E}_0 = 0, \mathbf{B}_0 = B_0\hat{\mathbf{z}}$)

If plasma is immersed in a steady uniform homogeneous magnetic field, its equivalent dielectric constant is no longer the scalar function of frequency derived earlier in this chapter.

The plasma dielectric is found from the two-fluid equations (3.4.1) to (3.4.6), including the magnetic field. (The reason for using the two-fluid equations, and not the one-fluid equation, in studying wave phenomena is—see also the discussion of space-charge waves—that high-frequency oscillations will perturb the electrons and have a lesser effect on the ions, creating regions of local charge density ρ_q and making it inconvenient to use the exact set of one-fluid equations and incorrect to use the approximate set.) Consider a uniform plasma of electron density n_0 at rest in a uniform field $\mathbf{B} = B_0\hat{\mathbf{z}}$, with no electric fields or plasma currents present. Now slightly perturb this state

$$n = n_0 + n_1 e^{-i\omega t} \qquad n_1 \ll n_0$$

and calculate the perturbed quantities $\mathbf{B}_1$, $\mathbf{E}_1$, $\mathbf{V}_1$ from the fluid equations by neglecting any terms involving products of perturbed quantities (i.e., nonlinear terms). The equations give

Continuity:

$$-i\omega n_{\alpha 1} + n_{\alpha 0}\nabla \cdot \mathbf{V}_{\alpha 1} = 0 \qquad (4.9.1)$$

Momentum:

$$-i\omega \mathbf{V}_{\alpha 1} = \frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_1 + \frac{\mathbf{V}_{\alpha 1} \times \mathbf{B}_0}{c}\right) \qquad (4.9.2)$$

Maxwell:

$$\nabla \times \mathbf{E}_1 = \frac{i\omega}{c}\mathbf{B}_1 \qquad (4.9.3)$$

$$\nabla \times \mathbf{B}_1 = -\frac{i\omega}{c}\mathbf{E}_1 + \frac{4\pi}{c}\sum_\alpha q_\alpha n_{\alpha 0}\mathbf{V}_{\alpha 1} \qquad (4.9.4)$$

It is a tedious, but perfectly straightforward, exercise to write Maxwell's equations in the form

$$\begin{aligned} \nabla \times \mathbf{E}_1 &= \frac{i\omega}{c}\mathbf{B}_1 \\ \nabla \times \mathbf{B}_1 &= -\frac{i\omega}{c}\boldsymbol{\varepsilon} \cdot \mathbf{E}_1 \end{aligned} \qquad (4.9.5)$$

where

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 & +i\varepsilon_2 & 0 \\ -i\varepsilon_2 & \varepsilon_1 & 0 \\ 0 & 0 & \varepsilon_3 \end{bmatrix} \qquad (4.9.6)$$

with

$$\varepsilon_1 = 1 + \frac{\omega_{pe}^2}{\omega_{ce}^2 - \omega^2} + \frac{\omega_{pi}^2}{\omega_{ci}^2 - \omega^2} \qquad (4.9.7)$$

$$\varepsilon_2 = \frac{\omega_{ce}}{\omega}\frac{\omega_{pe}^2}{\omega_{ce}^2 - \omega^2} - \frac{\omega_{ci}}{\omega}\frac{\omega_{pi}^2}{\omega_{ci}^2 - \omega^2} \qquad (4.9.8)$$

$$\varepsilon_3 = 1 - \frac{\omega_{pe}^2}{\omega^2} - \frac{\omega_{pi}^2}{\omega^2} \qquad (4.9.9)$$

The generalization of these dielectric constants to a several-species plasma is obvious by their form. Note that by definition ω_{ce} and ω_{ci} are positive.

In many applications (4.9.7) to (4.9.9) can be simplified. For example, the plasma dielectric constant for high-frequency perturbations in an infinite cold collisionless homogeneous plasma immersed in a steady homogeneous magnetic field is given by (4.9.6) with

$$\varepsilon_1 = 1 + \frac{\omega_p^2}{\omega_{ce}^2 - \omega^2} \qquad (4.9.10)$$

$$\varepsilon_2 = \frac{\omega_{ce}}{\omega}\frac{\omega_p^2}{\omega_{ce}^2 - \omega^2} \qquad (4.9.11)$$

$$\varepsilon_3 = 1 - \frac{\omega_p^2}{\omega^2} \qquad (4.9.12)$$

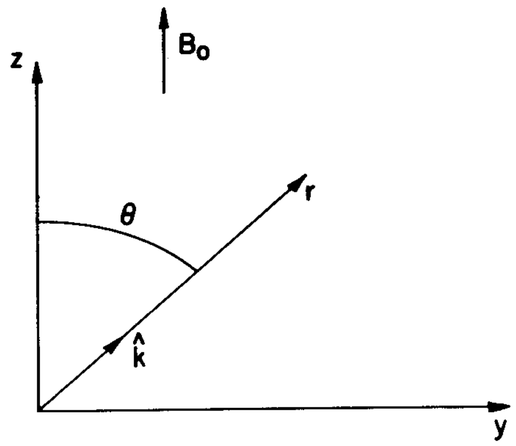


FIGURE 4.9.1
Coordinate system used in analysis of electromagnetic waves that propagate in magnetized plasma.

Problem 4.9.1 Verify (4.9.5) to (4.9.9). ////

It is most instructive, from the point of view of understanding the characteristic frequencies and modes of oscillation of magnetized plasma, to examine the solutions of Maxwell's equations in an unbounded plasma, and the next two sections will take up this problem. Equation (4.9.5) is a wave equation; explicitly, eliminating $\mathbf{B}_1$, Eq. (4.9.5) becomes

$$\nabla \times \nabla \times \mathbf{E}_1 = \frac{\omega^2}{c^2}\,\boldsymbol{\varepsilon} \cdot \mathbf{E}_1 \qquad (4.9.13)$$

The solutions of this equation have the form

$$\mathbf{E}_1(\mathbf{r}, t) = \hat{\mathbf{E}}_1(\mathbf{k}, \omega)\exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)] \qquad (4.9.14)$$

Thus (4.9.13) may be written

$$-\mathbf{k} \times (\mathbf{k} \times \hat{\mathbf{E}}_1) = \frac{\omega^2}{c^2}\,\boldsymbol{\varepsilon} \cdot \hat{\mathbf{E}}_1 \qquad (4.9.15)$$

or alternatively,

$$\hat{\mathbf{E}}_1 - \hat{\mathbf{k}}(\hat{\mathbf{k}} \cdot \hat{\mathbf{E}}_1) = \frac{1}{n^2}\,\boldsymbol{\varepsilon} \cdot \hat{\mathbf{E}}_1 \qquad (4.9.16)$$

where $\mathbf{k} = \hat{\mathbf{k}}\omega/v$ (v is the wave speed) and $n = c/v$ (n is the index of refraction) have been used. To simplify the algebra, a coordinate system where the z axis lies along the steady magnetic field $\mathbf{B}_0$ and where the vector $\mathbf{k}$ lies in the yz plane (Fig. 4.9.1) is chosen.

Problem 4.9.2 Verify (4.9.16). ////

Equation (4.9.16) is really three equations, one for each component of the electric field. These three equations can be written in matrix form as follows:

$$\begin{bmatrix} 1 - \dfrac{\varepsilon_1}{n^2} & -\dfrac{i\varepsilon_2}{n^2} & 0 \\ \dfrac{i\varepsilon_2}{n^2} & \left(\cos^2\theta - \dfrac{\varepsilon_1}{n^2}\right) & -\sin\theta\cos\theta \\ 0 & -\sin\theta\cos\theta & \sin^2\theta - \dfrac{\varepsilon_3}{n^2} \end{bmatrix} \begin{bmatrix} \hat{E}_{1x} \\ \hat{E}_{1y} \\ \hat{E}_{1z} \end{bmatrix} = 0 \qquad (4.9.17)$$

The nontrivial solution of this system of homogeneous algebraic equations comes from setting the determinant of the coefficients equal to zero. The determinant vanishes if

$$\tan^2\theta = -\frac{(1/n^2 - 1/\varepsilon_R)(1/n^2 - 1/\varepsilon_L)}{(1/n^2 - 1/\varepsilon_3)[1/n^2 - \frac{1}{2}(1/\varepsilon_R + 1/\varepsilon_L)]} \qquad (4.9.18)$$

where $\varepsilon_R \equiv \varepsilon_1 + \varepsilon_2$ and $\varepsilon_L \equiv \varepsilon_1 - \varepsilon_2$. (The dielectric constants ε_R and ε_L will later be associated with right and left circularly polarized wave components, respectively.)

Problem 4.9.3 Verify (4.9.18). ////

If the polarization of the eigenvectors is known, Maxwell's equations can be written as three scalar equations of the form $\varepsilon_\alpha E_\alpha = 0$. In many texts ε_α is referred to as the dielectric constant.[1]

Equation (4.9.18) is then the dispersion relation for waves propagating in a cold collisionless homogeneous plasma in a uniform magnetic field. It is sometimes called the Appleton-Hartree equation. For a given plasma frequency ω_p, a given cyclotron frequency ω_c, a given wave frequency ω, and a given direction of propagation θ, Eq. (4.9.18) can be solved for the index of refraction n, which determines the wave speed in that direction. As can be seen from the above result, many parameters are required to specify a wave in a magnetic field. In contrast, waves in a field-free plasma, described earlier in this chapter, were determined essentially by the density and temperature. The next sections study the simple cases of propagation parallel to and perpendicular to the steady magnetic field.

[1] C. Longmire, "Elementary Plasma Physics," eqs. (6.76) and (6.97), Interscience, New York, 1963.

Problem 4.9.4 Show for $\theta = 0$ that Maxwell's equations (4.9.17) can be written in the form

$$(n^2 - \varepsilon_R) E_R = 0$$
$$(n^2 - \varepsilon_L) E_L = 0$$
$$\varepsilon_3 E_3 = 0$$

(*a*) Find E_R, E_L, E_3.
(*b*) Show that $\varepsilon_R(\omega) = \varepsilon_L(-\omega)$.
(*c*) Show that

$$\varepsilon_R = 1 - \frac{\omega_{pe}^2}{\omega(\omega - \omega_{ce})} - \frac{\omega_{pi}^2}{\omega(\omega + \omega_{ci})}$$
$$\varepsilon_L = 1 - \frac{\omega_{pe}^2}{\omega(\omega + \omega_{ce})} - \frac{\omega_{pi}^2}{\omega(\omega - \omega_{ci})}$$
$$\varepsilon_3 = 1 - \frac{\omega_{pe}^2}{\omega^2} - \frac{\omega_{pi}^2}{\omega^2}$$

where

$$\omega_{pe}^2 = \frac{4\pi n_e e^2}{m_e} \qquad \omega_{pi}^2 = \frac{4\pi n_i e^2}{m_i}$$
$$\omega_{ce} = \frac{eB_0}{m_e c} \qquad \omega_{ci} = \frac{eB_0}{m_i c}$$

////

Problem 4.9.5 Show for $\theta = \pi/2$ that Maxwell's equations (4.9.17) can be written in the form

$$\varepsilon_E E_E = 0$$
$$\varepsilon_0 E_0 = 0$$

Find ε_E and ε_0.

////

4.10 WAVES THAT PROPAGATE PARALLEL TO THE MAGNETIC FIELD IN A COLD MAGNETIZED PLASMA ($\mathbf{E}_0 = 0$, $\mathbf{B}_0 = B_0\hat{\mathbf{z}}$)

4.10.1 High-frequency Waves

Setting $\theta = 0$ in (4.9.17) gives two solutions for the frequency of waves of a given wavelength, or equivalently, for the index of refraction $n = kc/\omega$. It is useful to explore these solutions by examining them in different ranges of frequency. For frequencies much higher than the ion gyrofrequency, $\omega \gg \omega_{ci}$, Eqs. (4.9.10) to

(4.9.12) can be used for ε_1, ε_2, and ε_3, and the two solutions of (4.9.17) are, in terms of the wave number k,

$$k_R = \frac{\omega}{c}\left(1 - \frac{\omega_p^2}{\omega(\omega - \omega_{ce})}\right)^{1/2} \tag{4.10.1}$$

$$k_L = \frac{\omega}{c}\left[1 - \frac{\omega_p^2}{\omega(\omega + \omega_{ce})}\right]^{1/2} \tag{4.10.2}$$

The solution (4.10.1) is associated with wave fields

$$\mathbf{E}^R = \hat{E}_1 e^{i(k_R z - \omega t)}(\hat{\mathbf{x}} + i\hat{\mathbf{y}}) \tag{4.10.3}$$

a circular polarized electromagnetic wave. Thus k_R is the wave number for a circularly polarized wave which drives electrons in the direction of their cyclotron motion, i.e., right circularly polarized waves; similarly, k_L is the wave number for a circularly polarized wave whose rotation sense is opposite to cyclotron motion of the electrons, i.e., left circularly polarized waves. These dispersion relations are shown on Fig. 4.10.1, where ω is a function of k (see also Fig. 4.10.6). The curve which intersects the ω axis at $\omega_1\{=\frac{1}{2}\omega_{ce}[(1 + 4\omega_p^2/\omega_{ce}^2)^{1/2} - 1]\}$ is the plot of the solution (4.10.2), and the curve which intersects the ω axis at $\omega_2\{=\frac{1}{2}\omega_{ce}[(1 + 4\omega_p^2/\omega_{ce}^2)^{1/2} + 1]\}$ is the plot of the solution (4.10.1). Each of these equations has a second branch, the left-hand wave approaching the frequency $\omega = -\omega_{ce}$, and the right-hand wave approaching $\omega = +\omega_{ce}$ as $k \to \infty$. As $k \to 0$, the lower-frequency branches are no longer well represented by (4.10.1) and (4.10.2), which assumed $\omega \gg \omega_{ci}$. The plot (4.10.1) assumes a weak magnetic field, such that $\omega_{ce} \ll \omega_p$. A feature of the parameter range is that there is a band of frequencies $\omega_{ce} < \omega < \omega_1$ in which *no* waves propagate along B. Note that as $\omega_p \gg \omega_{ce}$, $\omega_1 \to \omega_p$, so that this is just the cutoff of electromagnetic waves below the plasma frequency, discussed in Sec. 4.5. In contrast with the field-free plasma, electromagnetic waves in a magnetized plasma reappear at frequencies below the cyclotron frequency.

From Fig. 4.10.1, it is seen that below the cyclotron frequency only right-hand circularly polarized waves propagate, while for $\omega_2 > \omega > \omega_1$, only the left-hand circularly polarized wave propagates. For $\omega > \omega_2$ both circular polarized waves can propagate along the magnetic field. If their amplitudes are equal, the effect of superposition of a left and right circularly polarized wave is to produce a plane wave with a particular plane of polarization. Because the two polarizations propagate at different speed, as is clear from Fig. 4.10.1, the plane of polarization rotates as the wave propagates along $\mathbf{B}_0$. This is called *Faraday rotation*. In the limit $k \to \infty$, the two waves approach the same speed c, and the plane of polarization is essentially fixed. This is again the phenomenon

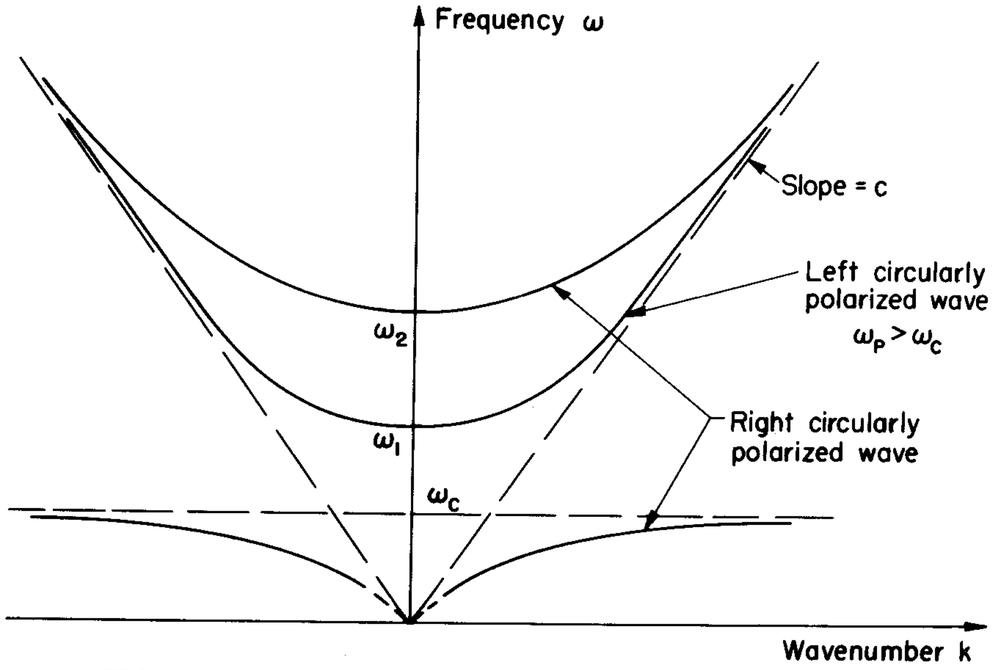


FIGURE 4.10.1
Frequency vs. wave-number diagram for plasma electromagnetic waves that propagate along the magnetic field in a cold homogeneous magnetized plasma.

that waves far above the plasma frequency simply ignore the plasma, and propagate as plane waves in a vacuum.

Problem 4.10.1 Obtain a plot of ω versus k (like that in Fig. 4.10.1) for the case $\omega_p < \omega_{ce}$, and discuss the cutoffs, etc. ////

4.10.2 Faraday Rotation

A linearly polarized wave can be decomposed into a pair of right- and left-hand circularly polarized waves (Fig. 4.10.2). For a plasma this decomposition is a natural one, since the two circular polarized waves each propagate at a different speed; the dielectric constant (ε_R, ε_L) seen by each of the circular polarized components is different. Assuming that $\hat{E}_{1z} = 0$ in (4.9.17), the solution associated with ε_R has an electric field whose x and y components are related by

$$\left(1 - \frac{\varepsilon_1}{n_R{}^2}\right)\hat{E}_{1xR} - i\frac{\varepsilon_2}{n_R{}^2}\hat{E}_{1yR} = 0 \qquad (4.10.4)$$

The total electric field can be written in terms one of the components, i.e.,

$$\mathbf{E}_{1R} = (\hat{\mathbf{x}} + i\hat{\mathbf{y}})E_{1R}\, e^{i(k_R z - \omega t)} \qquad (4.10.5)$$

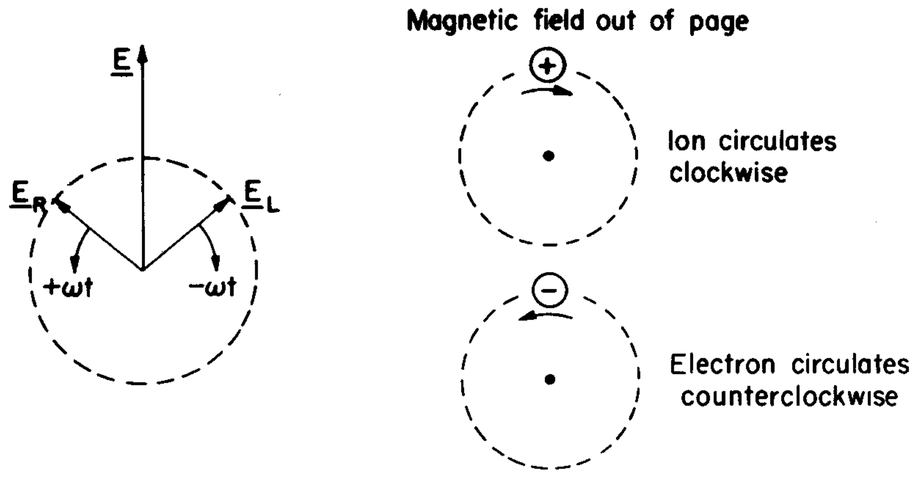


FIGURE 4.10.2
Diagram illustrating definition of right and left circularly polarized fields. Magnetic field out of page. (Single underscore below symbol signifies vector.)

The total electric field is the sum of the two circularly polarized components

$$\mathbf{E}_1 = [\hat{\mathbf{x}}(\hat{E}_{1R}\, e^{ik_R z} + \hat{E}_{1L}\, e^{ik_L z}) + i\hat{\mathbf{y}}(\hat{E}_{1R}\, e^{ik_R z} - E_{1L}\, e^{ik_L z})]e^{-i\omega t} \qquad (4.10.6)$$

In an isotropic medium, $k_R = k_L$ and the ratio of the x and y components of the field would remain fixed at space points and for all time. However, plasma in a magnetic field is anisotropic ($k_R \neq k_L$), and from (4.10.6), the ratio of the x and y components of the field is

$$\frac{E_x}{E_y} = -i\frac{1 + (E_{xL}/E_{xR})e^{i(k_L - k_R)z}}{1 - (E_{xL}/E_{xR})e^{i(k_L - k_R)z}} \qquad (4.10.7)$$

For $E_{xL}/E_{xR} = 1$, Eq. (4.10.7) reduces to

$$\frac{E_x}{E_y} = \cot\frac{k_L - k_R}{2}z \qquad (4.10.8)$$

Thus, if a plane wave is created at a location z_0, the plane of polarization *rotates* as a function of distance in the direction of propagation. The amount of rotation depends on the difference of the two wave numbers, which is determined by the plasma density and the magnetic field strength.

Faraday rotation can be used as a plasma probe. In a laboratory experiment this would be done by launching a plane wave along the magnetic field in a plasma. If the plane of polarization can be determined by the antenna design, a measurement of the plane of polarization well away from the source will determine at least the average electron density, since the magnetic field strength can usually be determined from magnetic probes. Using the expression for k_R and

k_L derived above, the angle of rotation for a high-frequency ($\omega \gg \omega_p$) wave propagating a distance z along $\mathbf{B}_0$ is

$$\phi(\omega \gg \omega_p) = \frac{e^3 B_0}{2\pi m^2 c^2 (\omega/2\pi)^2} n_e z \qquad \text{Faraday rotation} \qquad (4.10.9)$$

In astrophysical applications, radio frequency and optical observations can often be used to determine the polarization of the source of, for example, synchrotron radiation,[1] and Faraday rotation can be used to relate the electron density, the magnetic field along the line of sight, and the distance to the source. Since, in general, the fields and densities are space-dependent, (4.10.9) should be replaced by[2]

$$\phi = \frac{e^3}{2\pi m^2 c^2} \frac{1}{f^2} \int n_e \mathbf{B} \cdot d\mathbf{z} = \frac{2.35 \times 10^4}{f^2} \int n_e \mathbf{B} \cdot d\mathbf{z} \qquad (4.10.10)$$

where f is the frequency of the radiation in hertz.

4.10.3 Cyclotron Resonance

The right-hand circularly polarized electromagnetic wave rotates with the same sense as the electrons, and it should be no surprise that this wave exhibits resonances (Fig. 4.10.2). The dispersion relation

$$k_R = \frac{\omega}{c} \left(1 - \frac{\omega_p{}^2}{\omega(\omega - \omega_{ce})}\right)^{1/2} \qquad (4.10.11)$$

shows the cyclotron resonance with the electrons at $\omega = \omega_{ce}$. When $\omega \to \omega_{ce}$, the electrons can absorb a significant amount of energy from the wave and can attain large-amplitude orbits, which violate the restrictions used to obtain the "cold"-plasma dielectric tensor. The absorption of electromagnetic waves at electron-cyclotron resonance by a plasma may be a suitable means to heat plasma electrons. Chapter 10 investigates the nonlinear interaction between a wave and a charged particle.

4.10.4 Whistler Mode

For frequencies well below the electron-cyclotron frequency $\omega_{ci} < \omega \ll \omega_{ce}$, only the right circularly polarized component propagates. The dispersion relation in this region ($\omega \ll \omega_{ce}$) is

$$k_R = \frac{\omega_{pe}}{c} \sqrt{\frac{\omega}{\omega_{ce}}} \qquad (4.10.12)$$

[1] D. Wentzel, Magnetic Fields and Spiral Structure, *Ann. Rev. Astron. Astrophys.*, **1**:196 (1963).

[2] C. W. Allen, "Astrophysical Quantities," 2d ed., p. 31, Athlone Press, London, 1963.

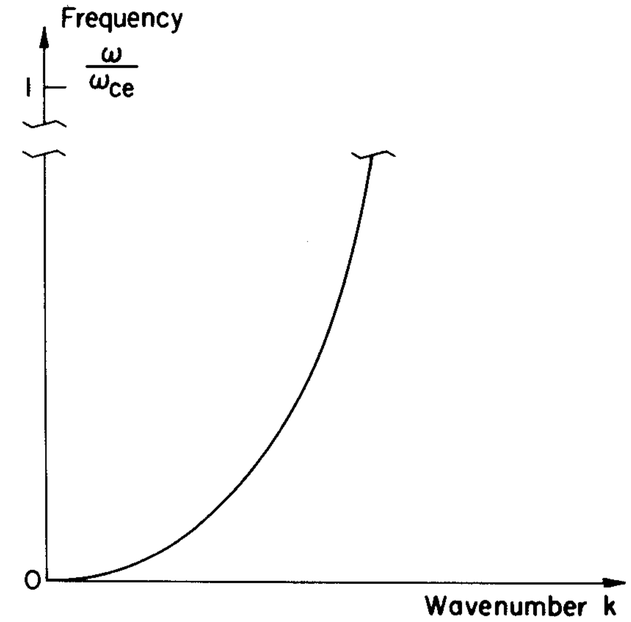


FIGURE 4.10.3
Frequency vs. wave-number diagram for a whistler wave. The $\omega \to 0$ range would be valid only if the ions were infinitely massive, but it is also valid if there are no ions at all in the system, as shown by R. C. Davidson and N. A. Krall, *Phys. Fluids*, **13**:1543 (1970).

The $\omega - k$ diagram for this wave, Fig. 4.10.3, shows that if a pulse containing many frequencies were to be launched in such a medium, the higher-frequency components of the pulse would travel along the field lines with a greater speed than the lower-frequency components. This means that a broad-band receiver placed far enough from the transmitter would detect the high frequencies first and the lower frequencies later.

This phenomenon was first observed by field radio operators in World War I using communication systems which operated at about 10 kHz. Occasionally, they would hear a short-duration signal whose frequency would start high and finish low, and it was assumed they were hearing the doppler-shifted noise from moving artillery shells. The correct explanation (due to Storey[1]) is that a lightning stroke on earth generates a pulse of electromagnetic waves, containing many frequencies, that impinges on the ionosphere and excites this low-frequency mode, which then propagates through the ionosphere, along the earth's magnetic field, to the point where the earth's magnetic field emerges from the ionosphere. Some of the wave energy is transmitted through the boundary of the ionosphere and impinges on the earth, where it is observed as a *whistler*

[1] L. R. O. Storey, *Phil. Trans. Roy. Soc. (London)*, **A246**:113 (1953).

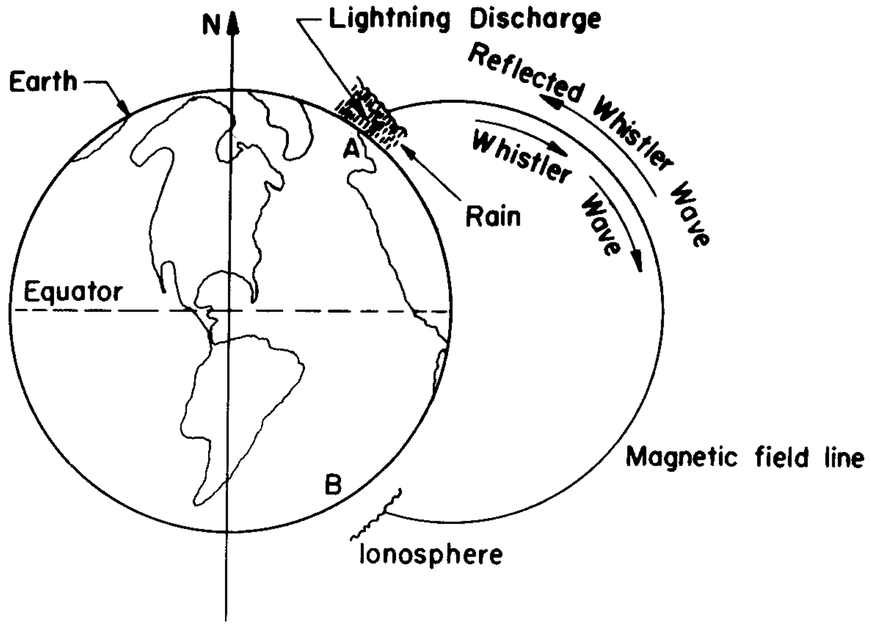


FIGURE 4.10.4
Schematic of whistler wave generated by lightning stroke propagating along an earth's magnetic field line to its conjugate point in the other hemisphere.

(Fig. 4.10.4). Some of the pulse is reflected and will make a trip back to the hemisphere where it started, be reflected there, and return again to the receiver, where it is heard as a second pulse. Sometimes four or five diminishing-amplitude whistlers that started from the same initial pulse can be detected.

A wave with similar dispersion which propagates in sodium at a low temperature is called the *helicon* wave. Because of the high density of electrons in sodium, the helicon wave has a very low velocity.

The time delay involved in the arrival of different frequencies ω can be used as a density diagnostic. Equation (4.10.12) gives the group velocity of whistlers of frequency ω and wavelength k as

$$\frac{\partial \omega}{\partial k} = \frac{2kc^2}{\omega_{pe}^{\ 2}}\,\omega_{ce} \qquad (4.10.13)$$

and the time of arrival [using (4.10.12) to write k in terms of ω] of frequency ω,

$$t = \int \frac{d\mathbf{s}}{d\omega/dk} = \int_{\text{path}} \frac{\omega_{pe}(x)}{2c\sqrt{\omega\omega_{ce}}}\,d\mathbf{s} \qquad (4.10.14)$$

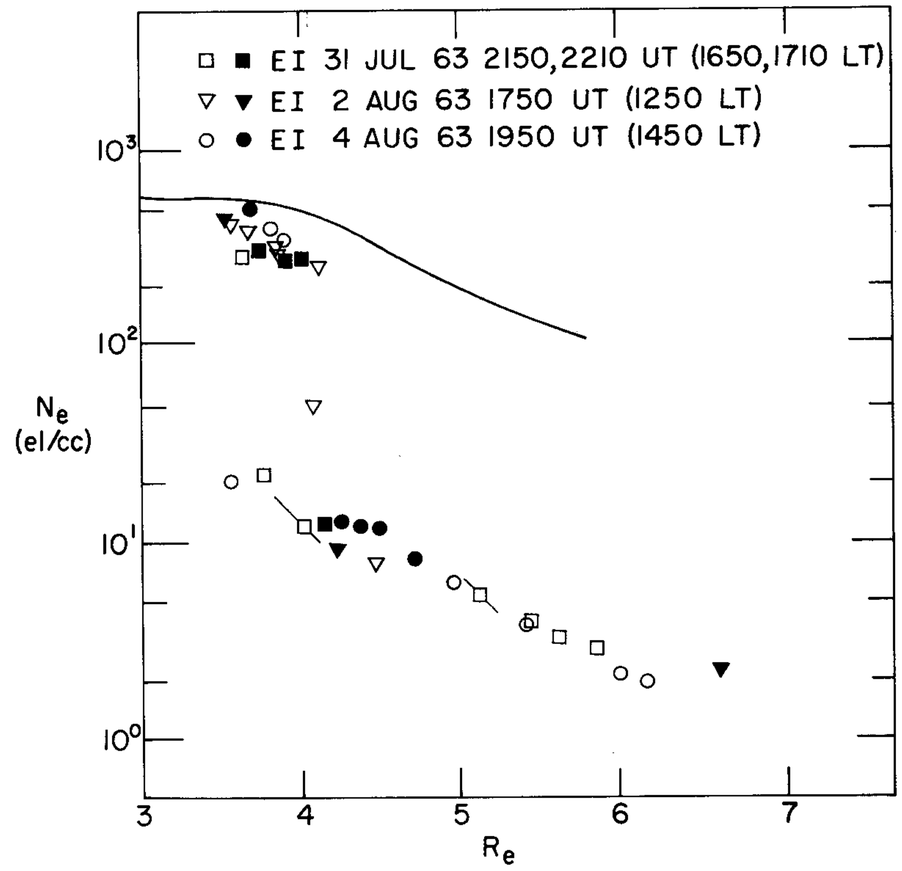


FIGURE 4.10.5
Profiles of electron density vs. geocentric distance in the equatorial plane, showing examples of the knee near $4R_E$ during the afternoon. [*After J. J. Angerami and D. L. Carpenter, J. Geophys. Res.*, **71**:693 (1966).]

For waves near the electron-cyclotron frequency, (4.10.14) should be modified to read

$$t = \frac{1}{2c}\int \frac{\omega_{pe}}{\sqrt{\omega}}\,\frac{2\omega - \omega_{ce}}{(\omega_{ce} - \omega)^{3/2}}\,d\mathbf{s} \qquad (4.10.15)$$

where $0 < \omega_{ce} - \omega \ll \omega_{ce}$.

Typical maps of electron density using this technique are shown in Fig. 4.10.5.[1] Note that the electron density decreases by an order of magnitude at about $4R_E$. This has been called the *knee* of the electron-density profile, and the region in which it occurs, the *plasmapause* (see also Fig. 4.10.6).

[1] S. D. Shawan, "Whistlers: Use for Determination of Composition and Temperature in Low Frequency Waves and Irregularitires in the Ionosphere," Springer-Verlag, New York, 1969.

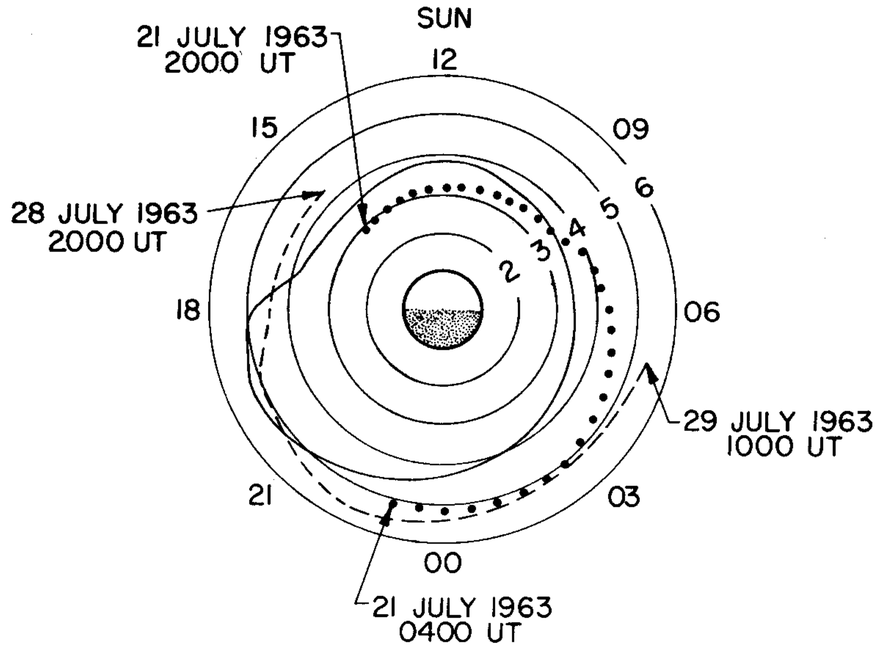


FIGURE 4.10.6
Equatorial radius of the knee vs. local time. The solid curve represents average behavior during periods of moderate, steady geomagnetic agitation ($K_p = 2$ to 4). The observations were made in July and August 1963 at Eights, Antartica. The dots show a particular example involving increasing magnetic agitation; the dashes, an example of decreasing agitation. [*After J. J. Angerami and D. L. Carpenter, J. Geophys. Res.*, **71**:693 (1966).]

Problem 4.10.2 A problem that might occur in studying the reflection of waves from the ionosphere is as follows: Given that a linearly polarized wave is incident on the ionosphere, what is the reflection coefficient? This problem can be idealized in the same manner as for the study of wave reflection from the isotropic plasma; i.e., consider a half-space filled with plasma, as shown in Fig. 4.10.7. The magnetic field is assumed to be in the z direction, and the wave is incident from the left. Show that the reflection coefficient, which is now complex, has components

$$R_x = \frac{k_R k_L - \omega^2/c^2}{(k_R + \omega/c)(k_L + \omega/c)}$$

$$R_y = i\frac{\omega}{c}\frac{k_R - k_L}{(k_R + \omega/c)(k_L + \omega/c)}$$

and that this reflection coefficient predicts that the reflected wave is elliptically polarized. The transmission coefficients (one for each solution)

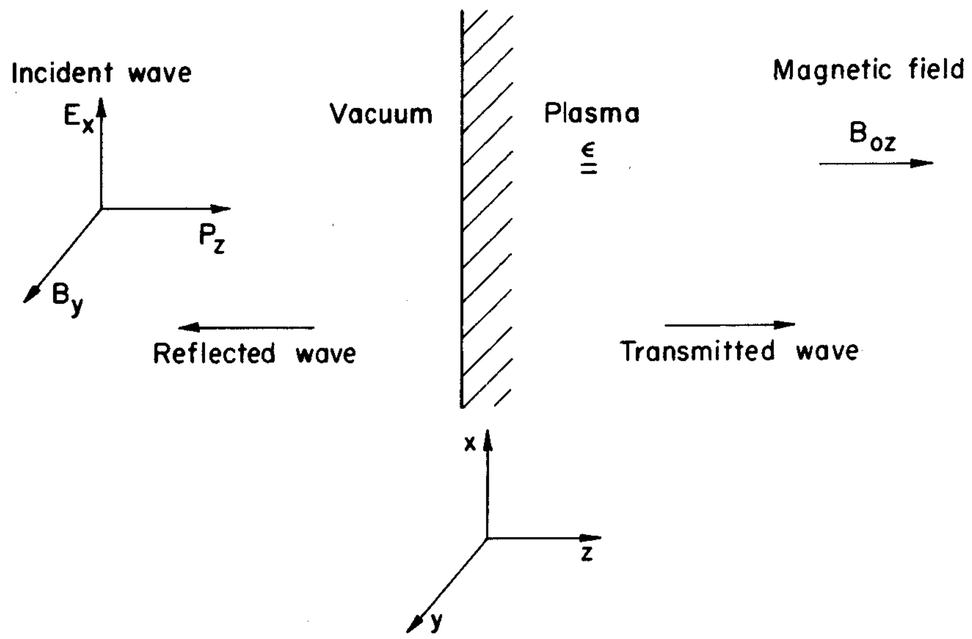


FIGURE 4.10.7
Schematic of electromagnetic wave incident on magnetized plasma that fills the half-space $z > 0$. The wave is propagating in z direction, and the steady magnetic field is in the z direction. (Double underscore below symbol signifies tensor.)

are also complex, and to obtain the above result, it will be necessary to use the symmetry conditions $T_{Rx} = -iT_{Ry}$ and $T_{Lx} = iT_{Ly}$. ////

Problem 4.10.3 Show that the phase shift caused by a plasma over a given transmission path is

$$\Delta_{\text{phase}} = -\int \left(k_p - \frac{\omega}{c}\right) dz$$

Consider an unmagnetized plasma where $k_p{}^2c^2 = \omega^2 - \omega_p{}^2$ with $\omega \gg \omega_p$ (i.e., low density). Show that for a path of length D in this plasma, the average density along the path $\langle n \rangle = (1/D)\int_0^D n(x)\,dx$ is

$$\langle n \rangle = \frac{m\omega c}{2\pi e^2 D}(\Delta_{\text{phase}}) \qquad ////$$

4.10.5 Intermediate-frequency Wave Propagating Parallel to B_0

At frequencies well below the electron-cyclotron frequency, the ions can respond to wave fields, and ion motion must be retained in calculating the behavior of such waves.

Ion motion complicates the dielectric tensor, and so the dispersion characteristics are also more complicated. There is now a resonance at the ion-cyclotron frequency; in addition, at very low frequencies, the ions and electrons can both respond to the wave field (e.g., Alfvén waves).

Setting $\theta = 0$ in (4.9.13), with the dielectric tensor components as given in (4.9.6) to (4.9.9) but not neglecting ω_{ci}, gives the dispersion relation for right and left circularly polarized waves

$$k_{\substack{R\\L}}^2 = \frac{\omega^2}{c^2}\left[1 - \frac{\omega_{pe}^2}{\omega(\omega \mp \omega_{ce})} - \frac{\omega_{pi}^2}{\omega(\omega \pm \omega_{ci})}\right] \qquad (4.10.16)$$

where the upper sign is for right circular polarization and the lower sign for left. Because the plasma considered here is neutral, $\pm\omega_{pe}^2\omega_{ci} \mp \omega_{pi}^2\omega_{ce} = 0$, and (4.10.16) simplifies to

$$k_{\substack{R\\L}}^2 = \frac{\omega^2}{c^2}\left[1 - \frac{\omega_{pe}^2 + \omega_{pi}^2}{(\omega \mp \omega_{ce})(\omega \pm \omega_{ci})}\right] \qquad (4.10.17)$$

Problem 4.10.4 Verify (4.10.17). ////

At high frequency this is the same dispersion relation described earlier in this section; i.e., at highest frequencies the waves are electromagnetic with $\omega^2 \gg \omega_p^2$, and do not see the plasma, while at lower frequency ($\omega \approx \omega_p$), the two polarizations are cut off at the frequencies depicted in Fig. 4.10.1. Below the electron gyrofrequency the right-hand circularly polarized wave reappears and goes over into the whistler at lower frequency; however, at still lower frequency, (4.10.17) exhibits the effect of ion motion. It is clear that at $\omega \lesssim \omega_{ci}$, an *ion* cyclotron wave, left-hand circularly polarized, appears; its frequency is well below ω_1, the cutoff for this mode. Figures 4.10.8 and 4.10.9 illustrate this phenomenon. In the upper graph *real* propagating waves correspond only to $k^2c^2/\omega^2 > 0$.

In the absence of a magnetic field, ω_1 and ω_2 coalesce, and become a *real* cutoff (no waves appear below that frequency). When new degrees of freedom are allowed, new features appear in the plasma-wave spectrum. Thus, in a magnetic field, ω_1 and ω_2 are cutoffs only in the sense that there is a band of missing frequencies below them, but waves reappear at frequencies corresponding to two of the new features introduced by the magnetic field: electron-cyclotron rotation and ion-cyclotron rotation. Since electrons and ions rotate in the opposite sense (Fig. 4.10.2), it is natural that waves of opposite circular polarization resonate at the two cyclotron frequencies.

4.10.6 Low-frequency Waves

If the wave frequency is much less than the ion-cyclotron frequency $\omega \ll \omega_{ci}$, the dispersion relation (4.10.17) is much simplified:

$$k_{\substack{R\\L}}^2 = \frac{\omega^2}{c^2}\left[1 + \frac{4\pi n_0(m_e + m_i)c^2}{B^2}\right] \qquad \omega \ll \omega_{ci} \qquad (4.10.18)$$

In this frequency range the right and left circularly polarized waves propagate at the same speed

$$\frac{\omega}{k} = \frac{c}{\sqrt{1 + 4\pi\rho c^2/B^2}} = \frac{V_A}{(1 + V_A^2/c^2)^{1/2}} \qquad (4.10.19)$$

where $V_A = B/\sqrt{4\pi\rho}$ is the Alfvén speed.

If terms of order ω/ω_{ci} are retained, the left and right circularly polarized waves do not have exactly the same speed; the difference in their speeds is

$$\frac{V_A^R - V_A^L}{V_A^R} \approx \frac{\omega}{\omega_{ci}} \qquad (4.10.20)$$

The result shows that the Alfvén wave is nearly a linearly polarized transverse electromagnetic wave propagating parallel to the magnetic field in a cold neutral magnetized plasma. This wave does not depend on the cyclotron rotation of the plasma particles, but is a motion which bends the magnetic field lines, since $\mathbf{B}_1 \perp \mathbf{B}_0$, so that the strength of $\mathbf{B}_0$ acts as a restoring force (much like tension in a string). Low-frequency motion of the field lines drags plasma particles along (as discussed in Sec. 3.9), and so the mass of the plasma acts in an inertial way to resist the restoring force. Note that $V_A = B_0/\sqrt{4\pi\rho}$ is the speed of a transverse wave on a string of tension $B_0^2/4\pi$ and mass density ρ.

The behavior of these waves is shown in the region below ω_{ci} in Figs. 4.10.8 and 4.10.9.

Problem 4.10.5 Replace (4.10.20) by a more accurate relationship. ////

Problem 4.10.6 Compute the Poynting vector for Alfvén waves, and show in the limit $V_A \ll c$ that this is the same power flow predicted by the Alfvén wave solutions given in Sec. 3.9, which neglected the electric field. [*Hint:* Use (3.9.3) to compute the electric field.] ////

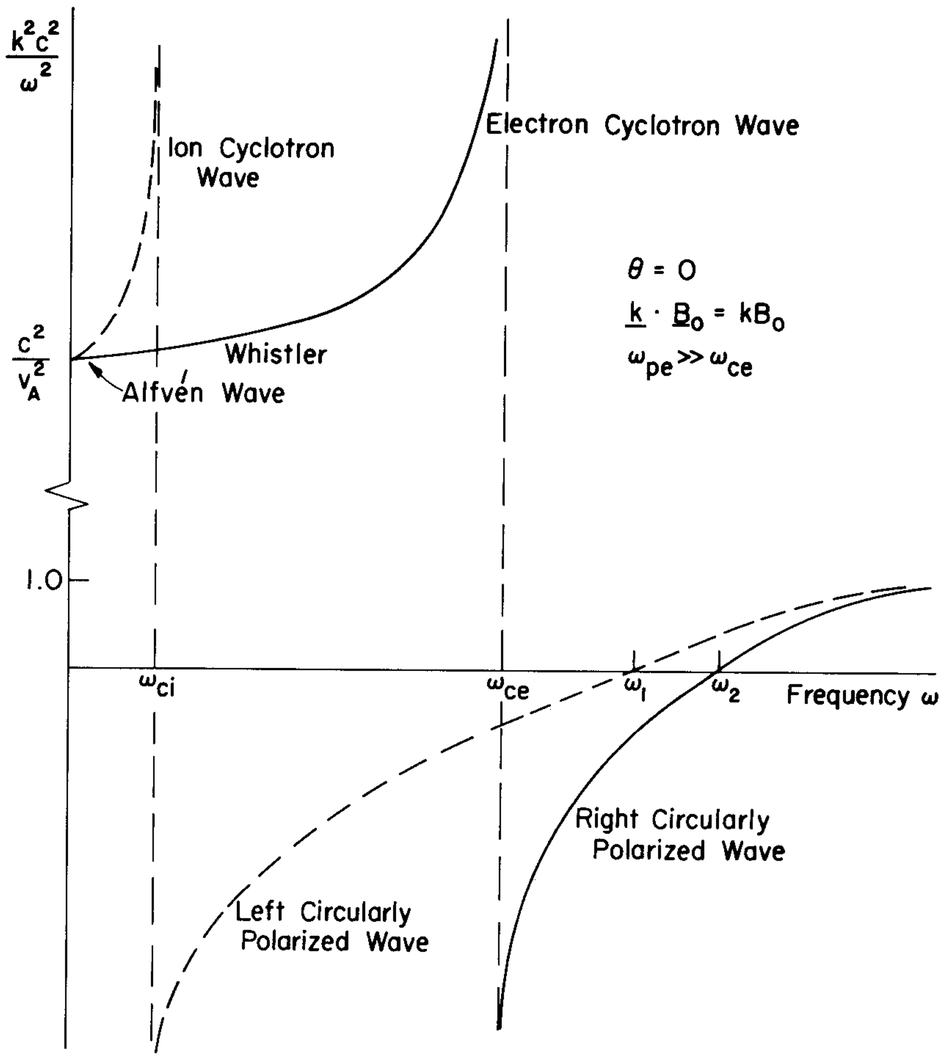


FIGURE 4.10.8*a*
Index of refraction k^2c^2/ω^2 versus ω for waves that propagate along a magnetic field in a high-density plasma. (Single underscore signifies vector.)

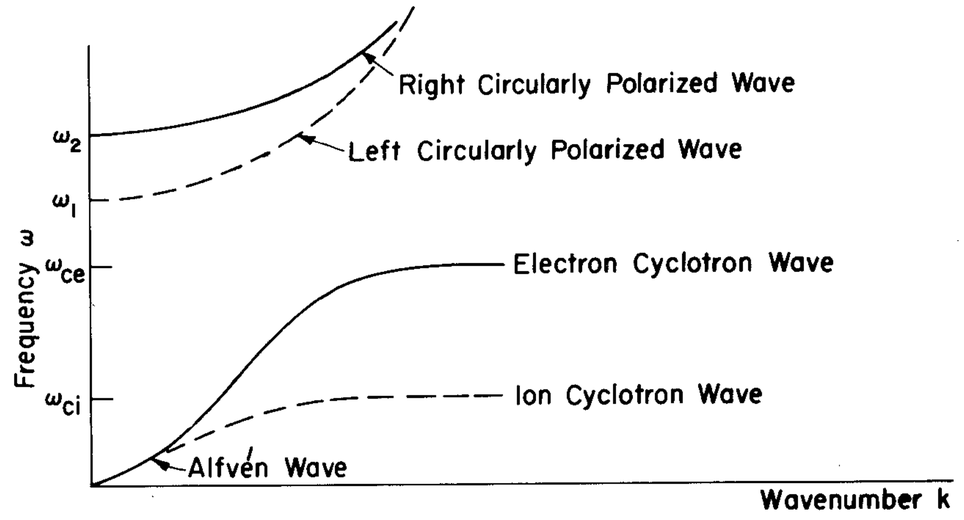


FIGURE 4.10.8*b*
Frequency vs. wave number for waves that propagate along a magnetic field in a high-density plasma.

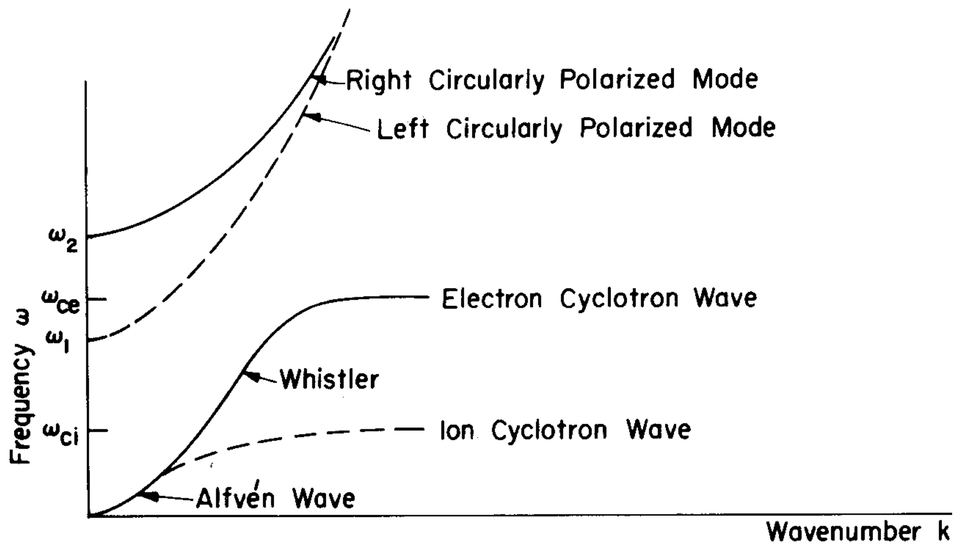


FIGURE 4.10.9*a*
Index of refraction k^2c^2/ω^2 versus ω for waves that propagate along a magnetic field in a low-density plasma. (Single underscore signifies vector.)

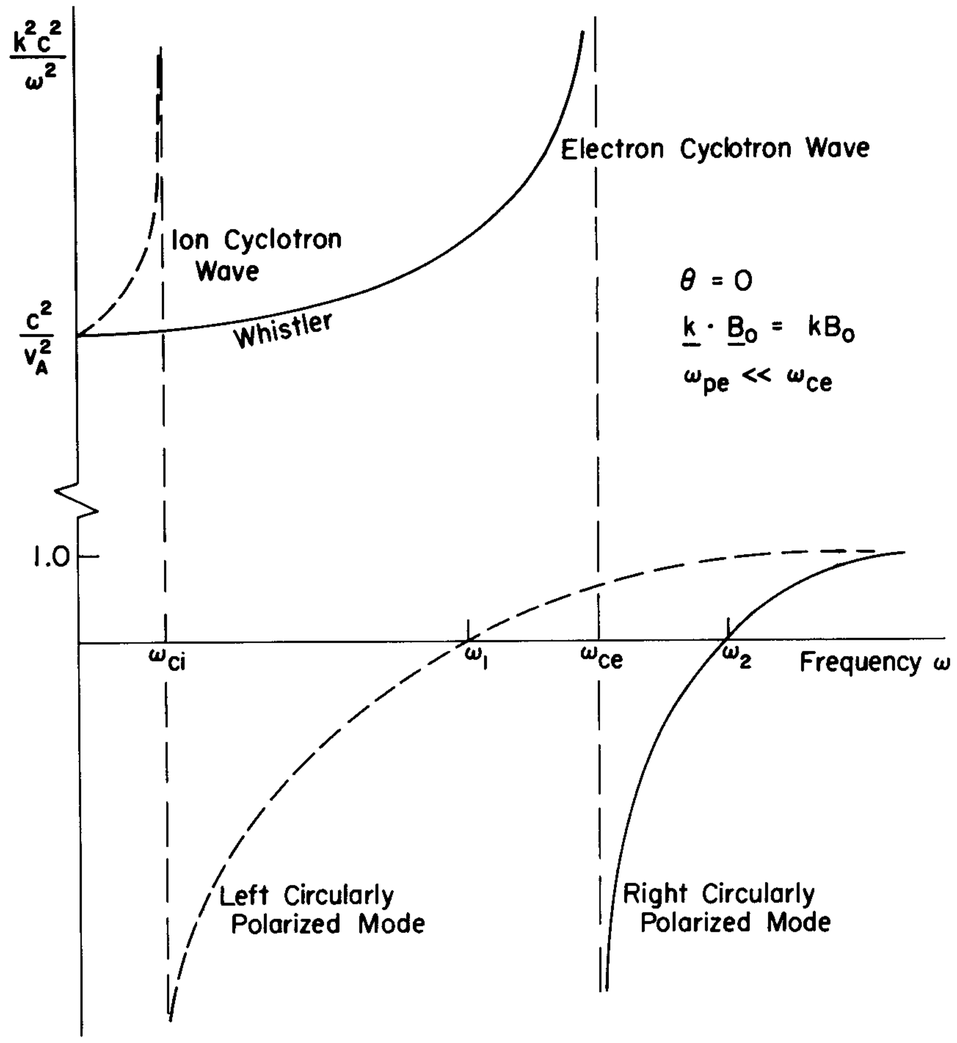


FIGURE 4.10.9*b*
Frequency vs. wave-number for waves that propagate along a magnetic field in a low-density plasma.

4.11 WAVES THAT PROPAGATE PERPENDICULAR TO THE MAGNETIC FIELD IN A COLD MAGNETIZED PLASMA ($\mathbf{E}_0 = 0$, $\mathbf{B}_0 = B_0\,\hat{\mathbf{z}}$)

In many magnetically confined plasmas, waves are launched and received along an axis perpendicular to the magnetic axis. The electric vector of the incident wave can be polarized either parallel or perpendicular to the magnetic axis. This distinction replaces the separation into right and left circularly polarized waves which characterized waves propagating along $\mathbf{B}_0$. The wave with its $\mathbf{E}$ vector parallel to the magnetic field is called the *ordinary wave*, because the dispersion relation is the same as if the magnetic field were zero. The wave with its $\mathbf{E}$ vector perpendicular to the magnetic axis is called the *extraordinary wave*. This terminology, which comes from early ionospheric work, is not consistently used in all plasma physics literature on waves.

4.11.1 High-frequency Waves

To obtain the dispersion relation for waves propagating perpendicular to the magnetic field, set $\theta = \pi/2$ in (4.9.17). For the ordinary wave, $\mathbf{E}_{wave} \| \mathbf{B}_0$, and the index of refraction is $n^2 = \varepsilon_3$. In terms of the wave number, the dispersion relation for ordinary waves is

$$k_o = \pm \frac{\omega}{c}\left(1 - \frac{\omega_p^2}{\omega^2}\right)^{1/2} \tag{4.11.1}$$

Because $\mathbf{k} \cdot \mathbf{E} = 0$ for this wave, it is a purely transverse oscillation. At high frequency ($\omega \gg \omega_p$), this electromagnetic wave takes no notice of the plasma, and $k_o = \omega/c$.

For the extraordinary wave ($\mathbf{E}_{wave} \perp \mathbf{B}_0$), the index of refraction is $n^2 = 2\varepsilon_R \varepsilon_L/(\varepsilon_R + \varepsilon_L)$. In terms of the wave number, the dispersion relation for the extraordinary wave is

$$k_E = \pm \frac{1}{c}\left[\frac{(\omega^2 - \omega_1^2)(\omega^2 - \omega_2^2)}{\omega^2 - \omega_H^2}\right]^{1/2} \tag{4.11.2}$$

where ω_1 and ω_2 are the same cutoff frequencies given in Fig. 4.10.1, i.e.,

$$\begin{aligned} \omega_1 &= \frac{\omega_{ce}}{2}\left[-1 + \left(1 + \frac{4\omega_p^2}{\omega_{ce}^2}\right)^{1/2}\right] \\ \omega_2 &= \frac{\omega_{ce}}{2}\left[1 + \left(1 + \frac{4\omega_p^2}{\omega_{ce}^2}\right)^{1/2}\right] \end{aligned} \tag{4.11.3}$$

A new resonance appearing in this extraordinary wave dispersion equation is at $\omega = \omega_H$, where $\omega_H[=(\omega_p^2 + \omega_{ce}^2)^{1/2}]$ is called the *upper hybrid frequency*.

The frequency bands in which these waves exist and the velocities of these waves are seen in Fig. 4.11.1. At high frequency, $\omega \to kc$ for all the modes. The extraordinary mode has two distinct frequency ranges, with characteristic index of refraction:

1 $\omega \gg \omega_H$: electromagnetic wave; $n^2 = \left(\frac{kc}{\omega}\right)^2 = 1$

2 $\omega \leq \omega_H$: upper hybrid resonance; $n^2 \gg 1$

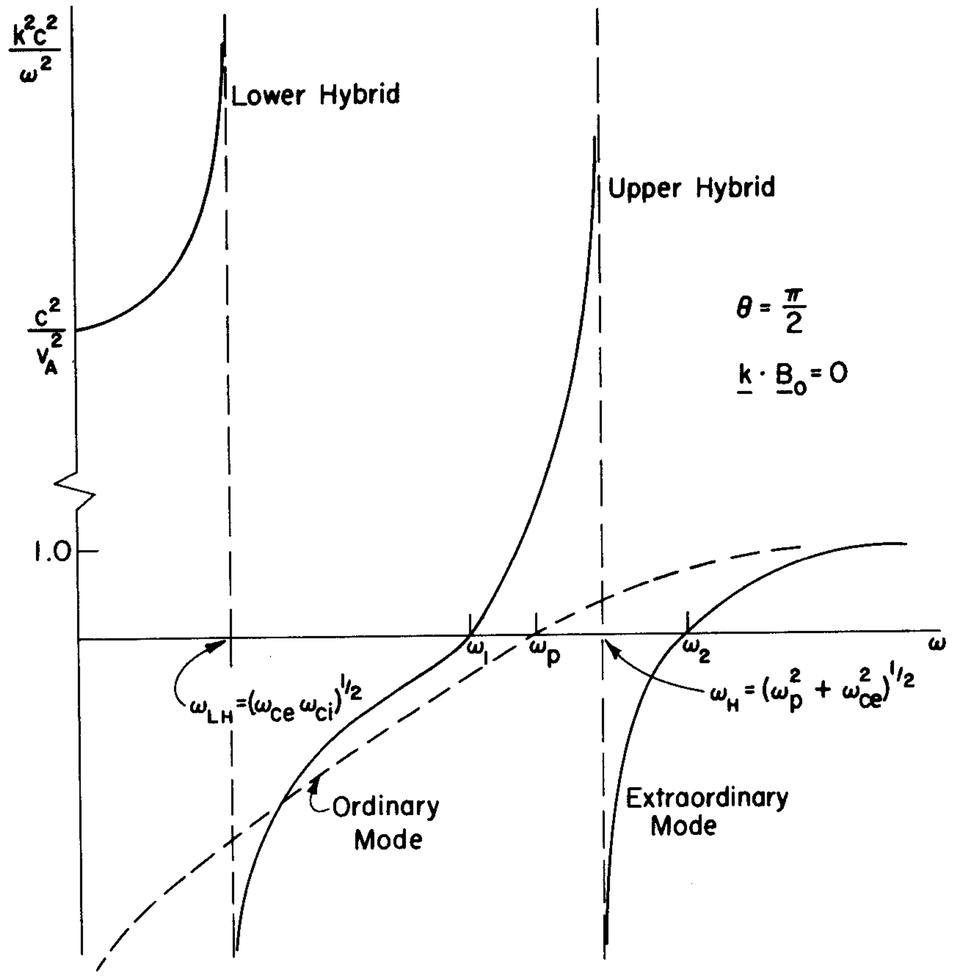


FIGURE 4.11.1*a*
Index of refraction in k^2c^2/ω^2 versus ω for waves that propagate across a magnetic field. (Single underscore signifies vector.)

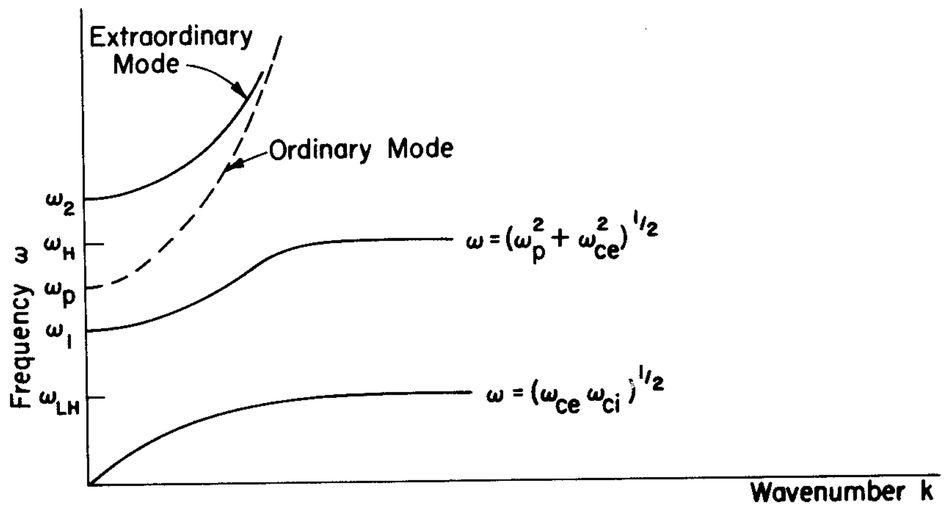


FIGURE 4.11.1*b*
Frequency vs. wave-number for waves that propagate across a magnetic field.

and there are two bands $\omega_H < \omega < \omega_2$ and $\omega < \omega_1$ in which the wave does not propagate. The extraordinary wave is not a pure transverse wave except at particular frequencies. From (4.9.13), taking the y direction as the direction of propagation,

$$\mathbf{E}_E = \hat{E}_E(\alpha\hat{\mathbf{x}} + \hat{\mathbf{y}})e^{i(ky-\omega t)}$$

$$\alpha = \frac{E_x}{E_y} = \frac{\left(\frac{\omega_{ce}}{\omega}\right)\frac{\omega_{pe}^2}{\omega_{ce}^2 - \omega^2} - \left(\frac{\omega_{ci}}{\omega}\right)\frac{\omega_{pi}^2}{\omega_{ci}^2 - \omega^2}}{\frac{(\omega^2 - \omega_1^2)(\omega^2 - \omega_2^2)}{\omega^2(\omega^2 - \omega_H^2)} - \left(1 + \frac{\omega_{pe}^2}{\omega_{ce}^2 - \omega^2} + \frac{\omega_{pi}^2}{\omega_{ci}^2 - \omega^2}\right)} \tag{4.11.4}$$

and so, for example,

$$\begin{aligned}
\alpha &\to \infty \quad \text{as} \quad \omega \to \infty (\mathbf{E} \perp \mathbf{k}) \\
\alpha &\to 0 \quad \text{as} \quad \omega \to \omega_H (\mathbf{E} \parallel \mathbf{k}) \\
|\alpha| &\to 1 \quad \text{as} \quad \omega \to \omega_{ce} (\mathbf{E} \cdot \mathbf{k} \approx \mathbf{E} \times \mathbf{k}) \\
|\alpha| &\to 1 \quad \text{as} \quad \omega \to \omega_{ci} (\mathbf{E} \cdot \mathbf{k} \approx \mathbf{E} \times \mathbf{k}) \\
|\alpha| &\to \sqrt{\frac{m_e}{m_i}}\frac{\omega_{pe}^2 \omega_H^2}{\omega_1^2 \omega_2^2} \quad \text{as} \quad \omega \to (\omega_{ce}\,\omega_{ci})^{1/2}
\end{aligned}$$

4.11.2 Low and Intermediate Frequencies

Because the waves propagating across $\mathbf{B}_0$ cannot resonate with a particle throughout its rotation, there is no wave feature at the cyclotron frequency for $\mathbf{k}$ perpendicular to $\mathbf{B}_0$; instead, a complex motion occurs combining the features of cyclotron oscillation and space-charge waves, as shown by the structure of the dispersion relation at the upper hybrid frequency. At lower frequencies, ion motion becomes important, and the dispersion relation for these waves, simplified by the approximation $\omega \ll \omega_{ce}$, becomes

$$n^2 = \frac{k^2c^2}{\omega^2} = -\frac{\omega_{ce}^2(\omega_{ci}^2 - \omega^2)(\varepsilon_1^2 - \varepsilon_2^2)}{\omega_H^2\left(\omega^2 - \omega_{ce}\omega_{ci}\frac{\omega_{pe}^2 + \omega_{ce}\omega_{ci}}{\omega_{pe}^2 + \omega_{ce}^2}\right)} \tag{4.11.5}$$

where ε_1 and ε_2 are as given in (4.9.4) and (4.9.5). This result shows that the extraordinary mode reappears at a low frequency $\omega \approx (\omega_{ce}\,\omega_{ci})^{1/2}$. This frequency is often called the lower hybrid frequency. At very low frequency ($\omega \ll \omega_{ci}$), $n^2 \approx \varepsilon_1$, so that

$$\frac{k^2c^2}{\omega^2} = 1 + \frac{\omega_{pi}^2}{\omega_{ci}^2} \tag{4.11.6}$$

or

$$\omega = \pm \frac{kB}{\sqrt{4\pi n m_i}} \frac{1}{1 + B^2/4\pi n m_i c^2}$$

$$= \pm \frac{kV_A}{1 + V_A{}^2/c^2} \tag{4.11.7}$$

These low-frequency branches are indicated in Fig. 4.11.1. The low-frequency wave (4.11.7) is called the *magnetosonic wave*, and (remember that it is assumed to be a wave propagating in the y direction) its eigenvectors are given by

$$\frac{E_x}{E_y} \approx \frac{\omega}{\omega_{ci}} \qquad \text{as} \qquad \omega \to 0 \qquad \mathbf{E} \| \mathbf{k} \tag{4.11.8}$$

so that it is mainly a *longitudinal* disturbance propagating across $\mathbf{B}_0$; this is why it is called *magnetic sound.* Its speed is the Alfvén speed. Recall that the Alfvén wave had $\mathbf{E} \perp \mathbf{k} \| \mathbf{B}_0$, while the magnetosonic wave more nearly has $\mathbf{E} \| \mathbf{k} \perp \mathbf{B}_0$. At arbitrary angles to $\mathbf{B}_0$, waves propagate at $\omega = kV_A$ with electric field components both parallel and perpendicular to the propagation direction. The cutoff frequencies ω_1 and ω_2 and the wave speed are properties that are independent of the angle of propagation relative to $\mathbf{B}_0$.

Problem 4.11.1 Derive the magnetosonic wave directly from the (cold) one-fluid equations (which should be valid for $k \to 0$). Note that the condition $\nabla \cdot \mathbf{V} = 0$ prevents occurrence of the magnetosonic wave. Do not assume this condition. ////

4.12 WAVE FREQUENCIES IN TYPICAL PLASMAS

Figures (4.10.1) to (4.11.1) summarize the nomenclature and dispersion properties of waves in a cold magnetized plasma.

It is instructive to attach to the graphs numbers appropriate to both natural and man-made plasma. A main feature of the graphs is the cutoff frequencies ω_1 and ω_2, which are approximately

$$\left.\begin{aligned} \omega_1 &\approx \omega_p - \frac{\omega_{ce}}{2} \\ \omega_2 &\approx \omega_p + \frac{\omega_{ce}}{2} \end{aligned}\right\} \omega_p{}^2 \gg \omega_{ce}{}^2 \tag{4.12.1}$$

and

$$\left.\begin{aligned} \omega_1 &\approx \omega_{ce} \frac{\omega_p{}^2}{\omega_{ce}{}^2} \\ \omega_2 &\approx \omega_{ce}\left(1 + \frac{\omega_p{}^2}{\omega_{ce}{}^2}\right) \end{aligned}\right\} \omega_p{}^2 \ll \omega_{ce}{}^2 \tag{4.12.2}$$

Additional features are the resonances at the cyclotron frequencies ω_{ce} and ω_{ci}, the hybrid frequency $\sqrt{\omega_p{}^2 + \omega_{ce}{}^2}$, and the lower hybrid frequency

$$\omega_{LH} \equiv \sqrt{\omega_{ce}\omega_{ci}} \sqrt{\frac{1 + B^2/4\pi n m_i c^2}{1 + B^2/4\pi n m_e c^2}}$$

Table 4.12.1 summarizes these characteristic frequencies for common laboratory situations, for the interplanetary plasma, and for ionospheric plasmas.

Thus the whistlers, in the ionosphere, would occur in the range below a megahertz, in or below the low radio frequencies. Lightning storms have been observed to generate whistlers from 30 to 300 kHz. Man-made whistlers propagating between Annapolis, Maryland, and Cape Horn, South America, have been detected at a frequency of 15.5 kHz. Radio stars have characteristic frequencies of 10 to 100 MHz. As seen from the table, these frequencies are well above the cutoff frequencies, a fact fortunate for the progress of radio astronomy. On the other hand, the ionosphere does have a layer with cutoff in the radio frequencies, a fact that has led to use of the ionosphere as a reflective medium by land-based radio operators. The Faraday rotation angle for radiation at $\omega = 100$ MHz from a radio star is given by $\phi = 10^{-16}\, z$, where z is the path length in centimeters.

Table 4.12.1

	n_0, cm^{-3}	B G	ω_p	ω_{ce}	ω_{ci}	ω lower hybrid
Laboratory	10^{12}	2×10^3	5×10^{10}	3×10^{10}	10^7 (hydrogen)	5×10^8
Interplanetary plasma	1–10	5×10^{-5}	5×10^4	700	$\frac{1}{3}$ (hydrogen)	15
Ionosphere:						
80 km	10^3	$\frac{1}{3}$	2×10^6	5×10^6	10^2 ($O_{16}{}^+$)	2×10^4
100 km	10^5	$\frac{1}{3}$	2×10^7	5×10^6	10^2	2×10^4

4.13 SPACE-CHARGE WAVES IN COLD FINITE PLASMAS IN A FINITE MAGNETIC FIELD

In Sec. 4.8, the nature of space-charge waves in a bounded plasma was discussed for the case of field-free plasmas and also for the case of a plasma cylinder in an infinite axial magnetic field. The infinite axial magnetic field was a great simplifying factor in that calculation, because it constrained the electrons to move in one dimension, and played no other role in the dynamics of the space-charge wave.

For intermediate values of the steady magnetic field a complete solution of Maxwell's equations is possible, but the solutions are very complicated, and it is difficult to make progress in understanding the features of these waves. Instead, note that the limiting phase velocity of space-charge waves [Eq. (4.8.17)] is less than the velocity of light; i.e., the space-charge waves are *slow waves*. For situations in which the phase velocity of a wave solution of Maxwell's equations turns out to be much less than the velocity of light, it is sometimes permissible to neglect the magnetic field associated with the wave, approximating

$$\nabla \times \mathbf{E}_1 = i\frac{\omega}{c}\mathbf{B}_1 = 0 \qquad (4.13.1)$$

This permits the electric field to be derived from a scalar potential, thus simplifying the analysis. The simplification was not important in the two limits, where $B_0 = \infty$ and $B_0 = 0$, and an exact solution was easy, but is a major aid in the case of space-charge waves in a finite-size plasma in a finite magnetic field.

Assuming, then, that the electric field is derivable from a scalar potential,

$$\mathbf{E}_1 = -\nabla\phi_1 \qquad (4.13.2)$$

The differential equation that ϕ_1 must satisfy is

$$\nabla \cdot \boldsymbol{\varepsilon} \cdot \nabla\phi_1 = 0 \qquad (4.13.3)$$

where $\boldsymbol{\varepsilon}$ is the dielectric tensor for a cold magnetized uniform plasma as given in (4.9.6).

Assuming again wavelike behavior in the axial direction, (4.13.3) becomes

$$\left(\nabla_T^2 - k^2\frac{\varepsilon_3}{\varepsilon_1}\right)\phi_1(\mathbf{r})e^{i(kz-\omega t)} = 0 \qquad (4.13.4)$$

In cylindrical coordinates, (4.13.4) has solutions of the form

$$\phi_1 = [C_1 J_n(Tr) + C_2 N_n(Tr)]e^{i(kz-n\theta-\omega t)} \qquad (4.13.5)$$

where

$$T^2 = -k^2\frac{\varepsilon_3}{\varepsilon_1} \qquad (4.13.6)$$

For the case of the plasma-filled waveguide, the solutions must be finite at $r = 0$; therefore $C_2 = 0$. At the waveguide wall ($r = a$), the condition that the tangential component of the electric field vanish is satisfied if $\phi_1(r = a) = 0$. This requirement is satisfied if

$$\begin{aligned} J_n(Ta) &= 0 \\ Ta &= p_{nv} \end{aligned} \qquad (4.13.7)$$

where p_{nv} is the vth zero of the nth-order Bessel function of the first kind. Thus T is just a number related to the particular mode and the radius of the waveguide.

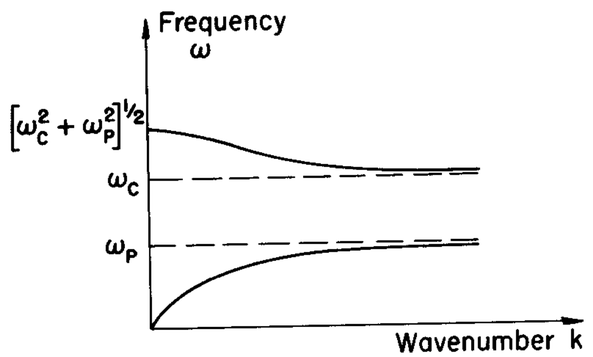
The dispersion relation obtained using (4.13.6) and (4.13.7) is

$$\left(\frac{ka}{p_{nv}}\right)^2 = -\frac{\varepsilon_1}{\varepsilon_3} = -\frac{1 + \omega_p^2/(\omega_{ce}^2 - \omega^2)}{1 - \omega_p^2/\omega^2} \qquad (4.13.8)$$

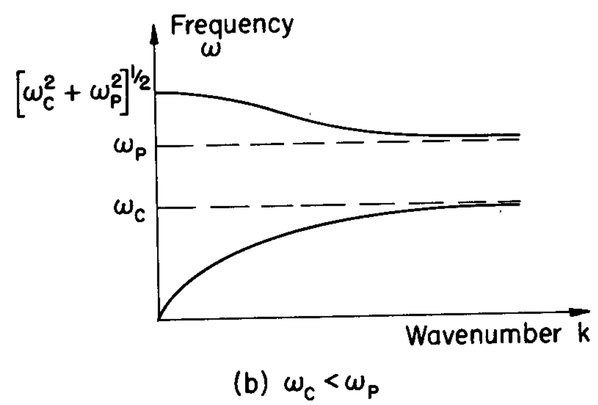
For waves to propagate, it is necessary that $k^2 > 0$, which is possible only if $\varepsilon_1 < 0$ or $\varepsilon_3 < 0$. The dispersion characteristics for (4.13.8) are shown in Fig. 4.13.1 for $\omega_p > \omega_{ce}$ and $\omega_p < \omega_{ce}$. For the case $\omega_{ce} > \omega_p$, in addition to the mode ($\omega < \omega_p$) predicted from the $B_0 = \infty$ analysis, the upper-hybrid mode $\omega_{ce} < \omega < \sqrt{\omega_p^2 + \omega_{ce}^2}$ appears as a characteristic frequency in the plasma. This mode has the interesting feature that it is a backward wave; i.e., its phase and group velocity are oppositely directed ($v_p v_g < 0$). As the magnetic field is further reduced ($\omega_p > \omega_{ce}$), it is seen that waves propagate for frequencies less than the cyclotron frequency. The backward-wave mode now propagates in the frequency range $\omega_p < \omega < \sqrt{\omega_p^2 + \omega_{ce}^2}$.

If the steady magnetic field strength is reduced to zero, the upper passband reduces to the plasma resonance at ω_p and the lower passband reduces toward $\omega = 0$. In both cases, the waves cease to propagate. That is, surprisingly, a plasma-filled waveguide without an external magnetic field will not propagate a space-charge wave. However, the waves *will* propagate even when $B_0 = 0$ if the plasma does not fill the waveguide. This is because filling the waveguide had eliminated the possibility of *surface waves*, as discussed in Sec. 4.

Problem 4.13.1 Derive the coupled scalar differential equations for E_{1z} and B_{1z} for waves in a finite plasma. Check your result with that obtained in a paper by H. Suhl and L. Walker, *Bell Sys. Tech. J.*, **33**:579–659, 939–986, 1133–1194 (1954). ////



(a) $\omega_c > \omega_p$



(b) $\omega_c < \omega_p$

FIGURE 4.13.1
Frequency vs. wave-number diagrams for a plasma-filled waveguide in a finite axial magnetic field.

Problem 4.13.2 Derive the dispersion relation for waves in a cylindrical plasma of radius a in a hollow conducting cylinder of radius b in a finite axial magnetic field using the quasistatic approximation. ////

Problem 4.13.3 The $\omega - k$ diagram of Fig. 4.13.1, for the plasma-filled waveguide for the backward-wave mode ($v_p v_g < 0$) that propagates between ω_p and $(\omega_p^2 + \omega_c^2)^{1/2}$ for $\omega_p > \omega_c$ and between ω_c and $(\omega_p^2 + \omega_c^2)^{1/2}$ for $\omega_c > \omega_p$, has a region where $v_p > c$. This means that the dispersion relation in this region, which is based on the quasistatic approximation ($v_p \ll c$), is possibly incorrect. Use the exact differential equation from Prob. 4.13.1 to examine the $k = 0$ cutoff, assuming the empty-waveguide cutoff frequency is much greater than $(\omega_p^2 + \omega_c^2)^{1/2}$. ////

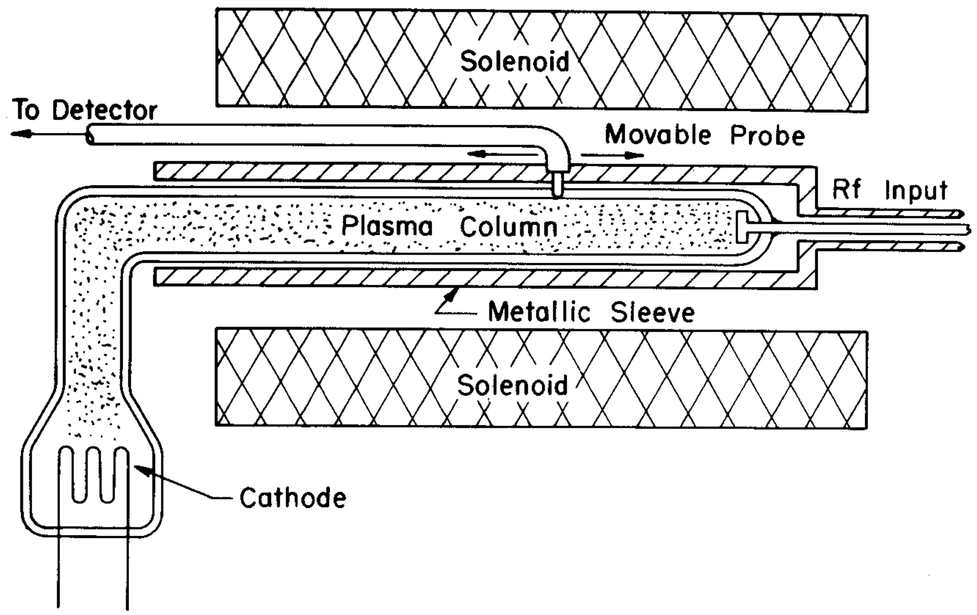


FIGURE 4.13.2
Schematic of apparatus to measure properties of space-charge waves in finite plasmas. [*From A. W. Trivelpiece and R. W. Gould, J. Appl. Phys.*, **14**:1784 (1959).]

To measure the phase characteristics of space-charge waves in finite plasmas, Trivelpiece and Gould[1] used the apparatus depicted schematically in Fig. 4.13.2. The discharge tube is similar to that shown in Fig. 4.7.1 and has a similar power-supply system. The waves are launched on the plasma column by driving the discharge anode with a small rf signal inside the passband for wave propagation. The moving probe samples the signal as it propagates along the plasma column. The wavelength is measured by comparing the phase of the signal on the probe with the phase of the signal supplied by the signal generator in a manner similar to that shown in Fig. 4.6.2, where the plasma path is now along the discharge tube. Results for the case of zero axial magnetic field are shown in Fig. 4.13.3 for several discharge currents. Measurements of the phase characteristics of space-charge waves in finite plasma for finite axial magnetic fields confirming most of the features of these waves have also been made with this apparatus.

[1] A. W. Trivelpiece and R. W. Gould, *J. Appl. Phys.*, **11**:1784 (1959) and A. W. Trivelpiece, "Slow Wave Propagation in Plasma Waveguides," San Francisco Press, San Francisco, 1967.

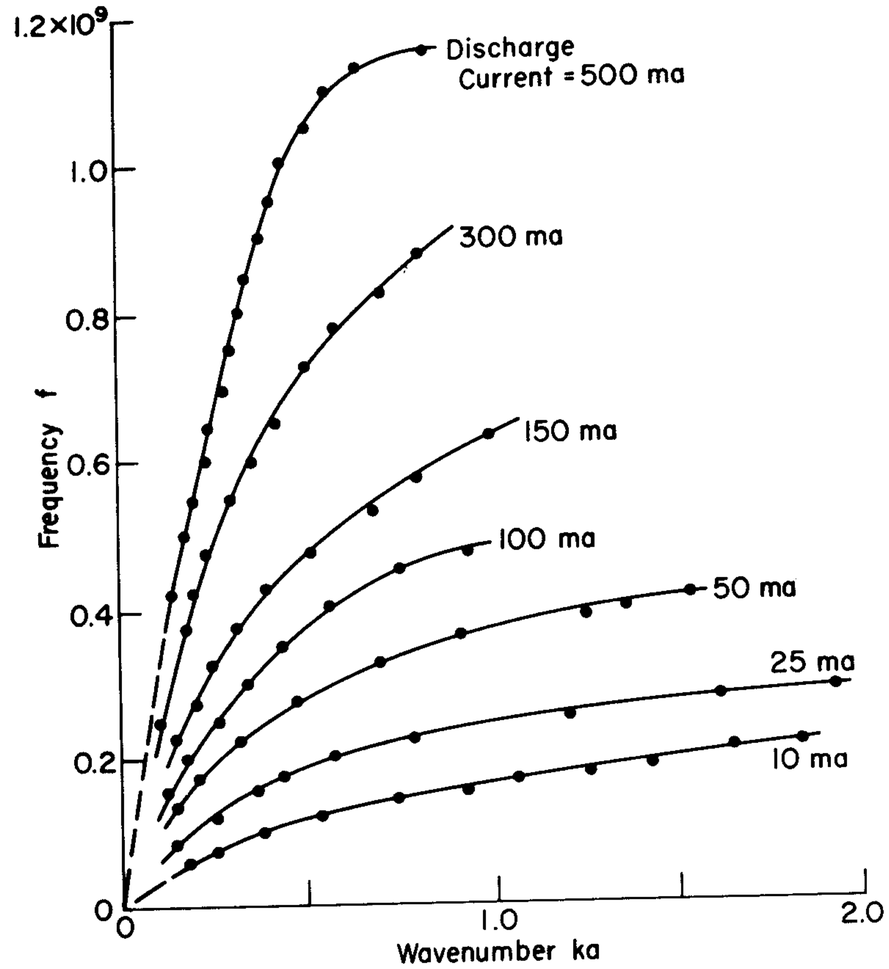


FIGURE 4.13.3
Measured phase characteristics of plasma space-charge waves in an unmagnetized plasma column using the apparatus shown in Fig. 4.13.2. Radius of column $a = 0.52$ cm, radius of waveguide $b = 0.62$ cm, and relative dielectric constant of discharge tube $K = 4.6$. [*From A. W. Trivelpiece and R. W. Gould, J. Appl. Phys.*, **11**:1784 (1959).]

4.14 LOW-FREQUENCY DRIFT WAVES IN NONUNIFORM PLASMAS

If magnetized plasma has gradients of density or temperature, the particles have associated drifts (Sec. 1.7). Plasma oscillations may be excited and move across the field with a phase velocity on the order of the drift velocity of the plasma. These oscillations are called *drift waves*. In this section we consider these waves, using the two-fluid equations. In Chap. 9 the stability of plasma with respect to the growth of these waves is calculated using the distribution-function (kinetic)

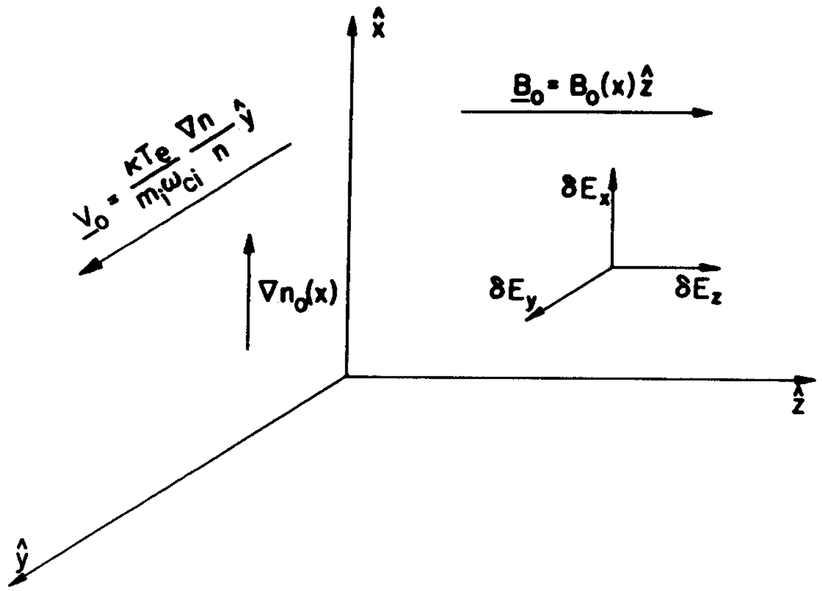


FIGURE 4.14.1
Typical configuration of a confined plasma. The confining field $B_0\hat{\mathbf{z}}$ supports a density gradient ∇n, and leads to a plasma motion $V_0\hat{\mathbf{y}}$. This equilibrium is unstable to low-frequency drift waves, with $\delta\mathbf{E}$ and $\delta\mathbf{B}$ as shown.

approach. The importance of these drift waves stems from the fact that all plasma contains some region of inhomogeneity capable of causing drift motions and associated drift waves.

The dielectric properties of plasmas are altered by the addition of a magnetic field; it should be no surprise that they are further altered in a plasma that has pressure gradients. As an example, consider a low-pressure plasma

$$\frac{m_e}{m_i} \ll \frac{8\pi n\kappa T_e}{B^2} \ll 1 \qquad (4.14.1)$$

in a nearly homogeneous magnetic field $\mathbf{B}_0 = B_0\hat{\mathbf{z}}$, $\mathbf{E}_0 = 0$, and suppose that the plasma density varies in the x direction, $n_0 = n_0(x)$, and that the temperature is constant. The geometry is shown in Fig. 4.14.1.

For simplicity, let the ions be cold. The equilibrium equation for electrons is then

$$0 = -e\frac{\mathbf{V}\times\mathbf{B}}{c} - \frac{\nabla p_e}{n_e} \qquad (4.14.2)$$

which gives

$$\mathbf{V}_0 = -\frac{\kappa T_e}{m_e\omega_{ce}}\frac{1}{n_0}\frac{dn_0}{dx}\hat{\mathbf{y}} \qquad (4.14.3)$$

for the equilibrium drift velocity of an electron fluid with a pressure gradient.

If the equilibrium is perturbed, the perturbations evolve according to the fluid equations

$$n_\alpha m_\alpha \left(\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V}\cdot\nabla\mathbf{V}\right)_\alpha = n_\alpha q_\alpha \left(\mathbf{E} + \frac{\mathbf{V}\times\mathbf{B}}{c}\right)_\alpha - \nabla p_\alpha \qquad (4.14.4)$$

$$\frac{\partial n_\alpha}{\partial t} + \nabla\cdot n_\alpha V_\alpha = 0 \qquad (4.14.5)$$

$$\nabla\cdot\mathbf{E} = 4\pi\sum_\alpha q_\alpha n_\alpha \qquad (4.14.6)$$

$$\nabla\times\mathbf{B} = \frac{1}{c}\frac{\partial\mathbf{E}}{\partial t} + \frac{4\pi}{c}\sum_\alpha n_\alpha q_\alpha \mathbf{V}_\alpha \qquad (4.14.7)$$

The set consists of 12 equations with 14 unknowns, closed by equations of state. A direct solution of these equations is difficult; however, with appropriate approximations, the analysis is much less formidable.

4.14.1 Electrostatic Drift Waves

When the phase velocity along the magnetic field is low,

$$\frac{\omega}{k_z} \ll V_A = \frac{B_0}{\sqrt{4\pi n_i m_i}} \qquad (4.14.8)$$

the waves do not perturb the magnetic field, and a good approximation is

$$\mathbf{E} = -\nabla\phi \qquad (4.14.9)$$

For these slow waves the electrons thermalize along the field lines, and follow a linearized Boltzmann distribution

$$n = n_0 e^{e\phi/\kappa T} \approx n_0\left(1 + \frac{e\phi}{\kappa T}\right) \qquad (4.14.10)$$

so that

$$n_1 = n_0 \frac{e\phi}{\kappa T}$$

This can also be seen from the z component (4.14.4) for electrons, noting that the electron mass is small and $(\mathbf{V}\times\mathbf{B})\cdot\hat{\mathbf{z}} = 0$ for longitudinal waves ($\mathbf{B}_1 = 0$).

A further simplification is to take all perturbed quantities of the form $n_1 = \hat{n}_1 e^{i(k_y y + k_z z - \omega t)}$. This restricts the analysis to waves which propagate in the yz plane. These waves sample the two basic features of this plasma state, namely, the gradient-produced drift in the y direction, and the free streaming of particles

in the z direction, along the magnetic field. With these restrictions, (4.14.4) and (4.14.5) give

$$-i\omega\mathbf{V}_i = i\mathbf{k}\phi\frac{e}{m_i} + \frac{e}{m_i c}(\mathbf{V}_i\times\mathbf{B}_0) \qquad (4.14.11)$$

$$-i\omega n_i + n_0\, i\mathbf{k}\cdot\mathbf{V}_i - V_{ix}\frac{dn_0}{dx} = 0 \qquad (4.14.12)$$

which leads to

$$\frac{n_i}{n_0} = \left(\frac{-ek_z}{m_i\omega_{ci}\omega}\frac{1}{n_0}\frac{dn_0}{dx} + \frac{ek_z^2}{m_i\omega^2}\right)\phi = \left(\frac{k_z V_0}{\omega} + \frac{k_z^2 c_s^2}{\omega^2}\right)\frac{e\phi}{\kappa T_e}$$

where

$$C_s^2 = \frac{\kappa T_e}{m_i}$$

is the ion-acoustic speed.

Poisson's equation then gives

$$\varepsilon(k,\omega)k^2\phi = 0$$

with

$$\varepsilon(k,\omega) = 1 - \frac{\omega_{pi}^2}{k^2 C_s^2}\frac{\omega^2 - \omega k_y V_0 - k_z^2 C_s^2}{\omega^2} \qquad (4.14.13)$$

The roots

$$\varepsilon(k,\omega) = 0$$

give the dispersion relation for low-frequency longitudinal drift waves. For the long-wavelength mode $k^2 C_s^2 < \omega_{pi}^2$, the dispersion relation is

$$\omega^2 - \omega k_y V_0 - k_z^2 C_s^2 = 0 \qquad (4.14.14)$$

When $V_0 \to 0$ or $k_y = 0$, these are just the ion-acoustic waves considered earlier. When $k_z \to 0$, there are two roots, $\omega_1 = k_y V_0$, $\omega_2 = -k_z^2 C_s^2/k_y V_0$. For larger k_z these drift waves appear as an accelerated ion-acoustic mode and a retarded mode,

$$\omega = \frac{k_y V_0}{2} \pm \frac{1}{2}\sqrt{k_y^2 V_0^2 + 4k_z^2 C_s^2} \qquad (4.14.15)$$

as shown in Fig. 4.14.2.

The transverse phase velocity of the upper mode, when $k_z = 0$, is $\omega/k \to V_0$, the electron drift speed. One view of the effect of the nonuniformity is the following: in general, electric fields transverse to $\mathbf{B}$ give drifts

$$\mathbf{V}_1 = \frac{\mathbf{E}\times\mathbf{B}}{B^2}c$$

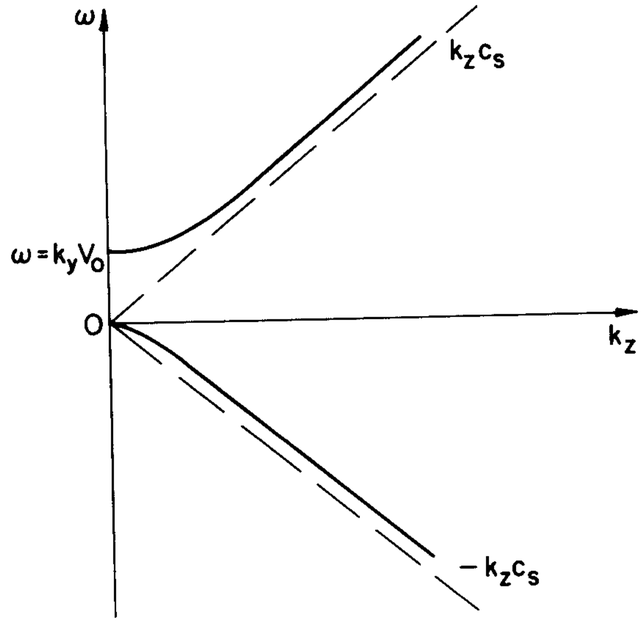


FIGURE 4.14.2
Modification of ion-acoustic waves by a density gradient. One branch of ion sound is accelerated, one retarded. For very long wavelength ($k_z \to 0$), the accelerated wave reduces to a pure drift wave $\omega = k_y V_0$, which has no counterpart in a uniform-density plasma.

which, if $\mathbf{E} = \hat{E} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)](\mathbf{k}/k)$, satisfy {with $\mathbf{V}_1 = \hat{\mathbf{V}}_1 \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$}

$$\nabla \cdot \mathbf{V}_1 = i\mathbf{k} \cdot \mathbf{V}_1 = i\mathbf{k} \cdot (\mathbf{k} \times \mathbf{B}) \frac{\hat{E}}{kB^2} c = 0 \quad (4.14.16)$$

In a uniform plasma, a flow pattern which satisfies $\nabla \cdot \mathbf{V}_1 = 0$ also means that there is no perturbed density ($\partial n_1/\partial t = n_0 \nabla \cdot \mathbf{V}_1$); that is, it corresponds to incompressible flow. In a nonuniform plasma, there can be a density perturbation even when the flow is "incompressible." From (4.14.12) the density perturbation, if $\nabla \cdot \mathbf{V}_1 = 0$, is

$$\frac{\partial n_1}{\partial t} = V_{1x} \frac{\partial n_0}{\partial x} \quad (4.14.17)$$

Thus a displacement of the plasma in the direction of the density gradient produces a density change. The density change occurs because a volume of less dense plasma is moved by $(\mathbf{E} \times \mathbf{B})$ drift into regions of higher equilibrium density.

Problem 4.14.1 Obtain (4.14.15) from the one-fluid equations. ////

Problem 4.14.2 Show that for $T_e = T_i$, the accelerated sound wave still propagates ($V_{\text{wave}} > V_{\text{thermal ions}}$, and so no ion Landau damping). ////

4.14.1 Electromagnetic Drift Waves

As k_z decreases, the phase velocity of the accelerated wave increases until it reaches either V_A or $\sqrt{\kappa T_e/m_e}$, whichever is smaller. If $8\pi n \kappa T_e/B^2 > m_e/m_i$, the Alfvén speed is the smaller. For simplicity, only that case is considered here. When

$$\frac{\omega}{k_z} \approx V_A \quad (4.14.18)$$

the magnetic field is perturbed, and $E = -\nabla\phi$ is not a good approximation.

The perturbed magnetic field is given by

$$\frac{i\omega}{c} \mathbf{B}_1 = (ik_y E_z - ik_z E_y)\hat{\mathbf{x}} \quad (4.14.19)$$

Since the electron thermal speed exceeds the Alfvén speed when $\beta > m_e/m_i$, it can again be assumed that the electrons reach equilibrium along the field lines, i.e., in the z direction. But because the magnetic perturbation $\mathbf{B}_1$ does not vanish, (4.14.4) for electrons with $m_e \to 0$ gives

$$\frac{n_e}{n_0} = \frac{eE_z}{\kappa T_e ik_z} - \frac{kV_0}{\omega} \frac{e}{\kappa T_e} \left(\frac{E_z}{ik_z} - \frac{E_y}{ik_y} \right) \quad (4.14.20)$$

The ion density is still given by (4.14.12). The student should check that $\mathbf{B}_1$ does not contribute to the perturbed ion density. Using (4.14.12) and (4.14.20), Poisson's equation (4.14.6) gives one relation between the perturbed fields E_z, E_y. A second equation is

$$\frac{4\pi}{c} J_z = (\nabla \times \mathbf{B}_1) \cdot \hat{\mathbf{z}} = \frac{ck_z k_y^2}{\omega} \left(\frac{E_y}{ik_y} - \frac{E_z}{ik_z} \right) \quad (4.14.21)$$

where J_z must be obtained from (4.14.3) and (4.14.5). For the long-wavelength modes considered in this section, $n_e \approx n_i$ [see Eqs. (4.14.21) and (4.14.12)] is a good approximation to Poisson's equation. With this in mind, subtracting the continuity equation (4.14.5) for electrons from (4.14.5) for ions gives for the current J_z

$$ik_z J_z = -\nabla \cdot [n_0(V_{ex} - V_{ix})\hat{\mathbf{x}} + n_0(V_{ey} - V_{iy})\hat{\mathbf{y}}]e \quad (4.14.22)$$

The velocities V_{ex} and V_{ix} are nearly equal:

$$V_x = \frac{E_y c}{B} \quad (4.14.23)$$

There is a small difference between V_{ix} and V_{ex}. Since the ions individually gyrate perpendicular to B_0 with frequency ω_{ci} and orbit size $v_\perp/\omega_{ci}$, they feel an average field somewhat different from the exact field at the center of their orbit. If the ions have a maxwellian velocity distribution $f_i = (m/2\pi\kappa T)n_0 \exp(-mV_\perp{}^2/2\kappa T)$, the average field can be shown to be

$$\bar{E}_y = E_y e^{-s} I_0(s) \quad (4.14.24)$$

where $s = k_y{}^2\kappa\, T_i/m_i\omega_{ci}{}^2$ and I_0 is the modified Bessel's function of the first kind, and E_y is the field at the center of the particle orbit.

Since the electron orbit is much smaller than the ion orbit, the electron sees an average field more nearly equal to the field at the center of its orbit.

Problem 4.14.3 Prove (4.14.24). ////

Using (4.14.22) to (4.14.24), it can be seen that

$$J_z = -\frac{n_0 m_i c^2}{B_0{}^2}\frac{k_y{}^2}{k_z}\frac{E_y}{ik_y}\left(\omega + \frac{\kappa T_i k_y}{m_i\omega_{ci}}\frac{1}{n_0}\frac{dn_0}{dx}\right)$$

Using this result with (4.14.21) gives a second (with Poisson's equation) equation, relating E_y and E_z. Neglecting $k_z{}^2C_s{}^2$ compared with ω/kV_0 gives the dispersion relation for the Alfvén drift waves,

$$(\omega - k_y V_0)\left(\omega^2 + \omega k_y V_0 \frac{T_i}{T_e} - k_z{}^2 V_A{}^2\right) = 0 \quad (4.14.25)$$

The root

$$\omega - k_y V_0 = 0$$

is the drift branch of the ion wave calculated above. The other root,

$$\omega = -\tfrac{1}{2}k_y V_0 \frac{T_i}{T_e} \pm \frac{1}{2}\left[\left(k_y V_0 \frac{T_i}{T_e}\right)^2 + 4k_z{}^2 V_A{}^2\right]^{1/2}$$

represents, for $k_z \neq 0$, an accelerated and a retarded Alfvén wave. When $V_0 \to 0$, these are the waves derived in Chap. 3. When $k_z \to 0$, these waves approach

$$\omega_1 = -\frac{k_y V_0 T_i}{T_e}$$

$$\omega_2 = \frac{k_z{}^2 V_A{}^2}{k_y V_0 (T_i/T_e)}$$

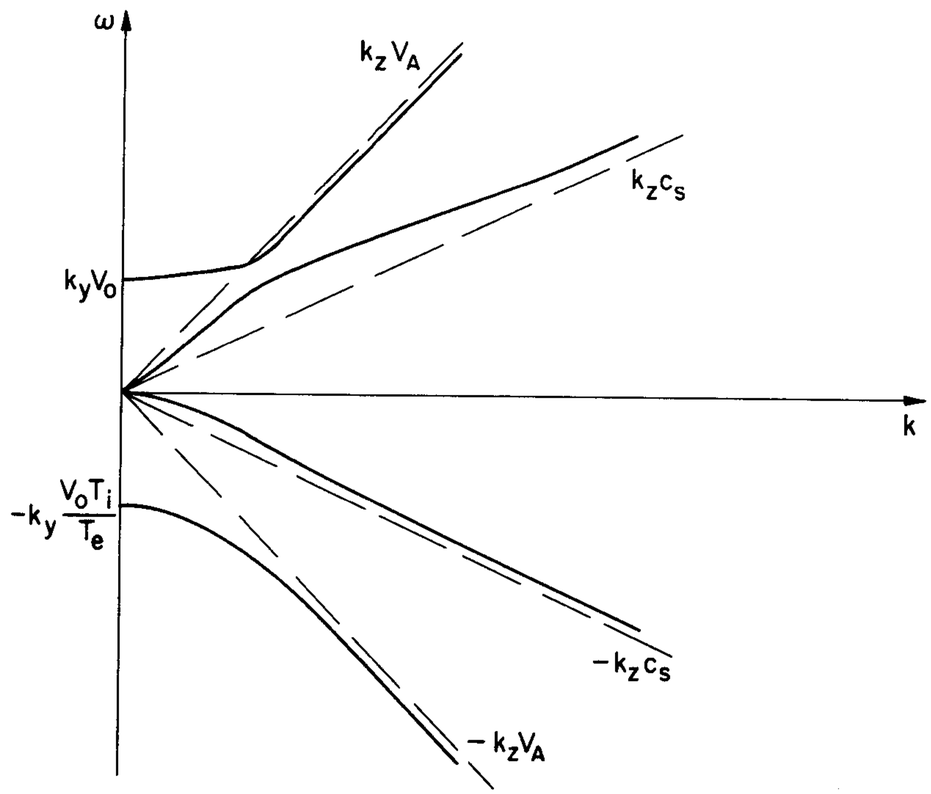


FIGURE 4.14.3
Complete low-frequency behavior of waves in a nonuniform plasma. Compare this with Fig. 4.14.2, noting that at $k_z \to 0$, one branch of the Alfvén wave also reduces to a pure drift wave, $\omega = -k_y V_0$.

Figure 4.14.3 summarizes the situation. The waves $\omega < 0$ propagate in the direction of the ion drift, and the waves $\omega > 0$ propagate in the direction of the electron drift. The feature that the two modes with $\omega > 0$ do not cross is a consequence of a more elaborate calculation than that presented here.

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5

STABILITY OF THE FLUID PLASMA

PART ONE: THE PLASMA STABILITY PROBLEM

A statistical system in a state of thermodynamic equilibrium with its surroundings at a given temperature occupies the lowest available potential-energy state. If a statistical system is prepared so that it is not in a state of thermodynamic equilibrium, it will tend to change itself in such a way as to achieve that condition. For an ideal gas of noninteracting molecules in a fixed-temperature enclosure, the binary collisions among the molecules and with the walls of the enclosure bring the gas to the maxwellian distribution that has the wall temperature.

That binary encounters are sufficient to cause this final state is the conclusion of the Boltzmann H theorem.[1] It is possible to prepare or operate on the system in such a way that effects other than binary encounters (e.g., collective interactions, turbulence, external forces, etc.) accelerate the process of achieving thermodynamic equilibrium, but the longest time a statistical system can be expected to remain in a nonthermodynamic equilibrium state is essentially the time between binary encounters.

[1] L. D. Landau and E. M. Lifshitz, "Statistical Physics," Pergamon, London, 1958.

The equations for the macroscopic fluid description of a plasma have terms that are proportional to the frequency of occurrence of binary (i.e., short-range) encounters between the charged particles. However, as a result of the long-range Coulomb force, these macroscopic equations also include terms that contain the combined effect of many distant particles. These collective effects produce phenomena which occur on a time scale of a plasma period ($\tau_p = 2\pi/\omega_p$), a time much shorter than the period between binary encounters ($\tau_c = 2\pi/\nu_c$), since $\tau_p \approx \tau_c/n\lambda_D{}^3$, and $n\lambda_D{}^3 \gg 1$ is the *plasma approximation.* Thus there is a regime in which it is proper to neglect binary collisions entirely in the investigation of plasma phenomena. Neglecting the binary collisions in the macroscopic plasma equations eliminates the mechanism by which statistical systems normally relax toward thermodynamic equilibrium. Thus it might be expected that any steady-state solution of the macroscopic plasma equations would persist indefinitely. This is not necessarily the case. Each steady-state solution of the macroscopic plasma equations must be examined to determine if that particular equilibrium configuration or situation is stable or unstable, i.e., whether this system, if perturbed, will move farther away from its equilibrium position. The simplest example to illustrate this point is that of a ball resting on the peak of a smooth round mound. The ball will remain indefinitely on top of the hill so long as it is not disturbed. However, if it is given a slight displacement away from the equilibrium position, it moves to a position of lower potential energy, which means that it has increased its kinetic energy; this further increases its displacement away from its original equilibrium position. Such a situation is called an *unstable equilibrium.* A ball on a flat horizontal plane is an example of a *neutrally stable* situation. A ball at the bottom of a depression in an otherwise flat horizontal plane is an example of a *stable equilibrium*; it will have bounded motion about its equilibrium position as a result of an initial displacement from this position.

In this chapter the equations for the macroscopic fluid description of a plasma, in which binary collisions are neglected, are solved for various equilibrium configurations and conditions. These metaequilibria are then examined by several methods to determine if they are stable or unstable and, if possible, to determine the rate at which an instability grows.

Although the study of plasma instabilities has gone on since early days of the field, these studies did not receive serious attention until research on controlled thermonuclear fusion reactors was started in 1952.[1] The practical objective of that research was, and still is (1972), the development of a self-sustaining reactor that obtains its energy from fusion of light nuclei. To achieve conditions

[1] A. S. Bishop, "Project Sherwood," Addison-Wesley, Reading, Mass., 1958.

where such a fusion reactor is economically feasible, it is necessary to magnetically confine a deuterium or tritium plasma, away from any low-temperature system such as the solid walls of a vacuum system, at a kinetic temperature of 100 keV, with a particle density of 10^{14} cm^{-3}, for a period of at least 1 s. Such a plasma would release more energy through nuclear fusion reactions than that expended in creating the plasma. At the density and temperature given above, binary (short-range) collisions are completely negligible in determining the plasma properties for periods as short as 1 s. However, 1 s is much longer than one plasma period, and the important question is whether or not the equilibrium configuration in which the plasma is magnetically confined is stable or unstable. Many of the early attempts to magnetically confine high-temperature plasma failed because the equilibrium magnetic field configuration used in the experiment was unstable and the plasma moved rapidly (~ 1 μs) to the walls of the vacuum chamber. Theoretical stability analysis based on the macroscopic fluid equations explained this behavior. This theoretical work also showed how to create equilibrium magnetic field configurations that were stable, at least as far as could be determined from the macroscopic fluid equations. Experiments have now verified that these most destructive instabilities, called *hydromagnetic instabilities*, can be suppressed by a suitable magnetic field configuration. However, suppression of these instabilities has not led directly to a successful fusion reactor. The plasma medium has been diabolically clever in its ability to produce another instability, usually less destructive than the one for which a cure has just been found, but nevertheless sufficiently serious to cause loss of plasma more rapidly than allowed for a successful fusion reactor.

Although the desire to develop a successful controlled fusion reactor *initially* stimulated an intense interest in the problem of plasma stability, which continues today, it is not the only reason for studying the stability of plasmas. Some of the work on plasma stability has carried over into astrophysics and space physics, where problems of plasma stability play a role in such phenomena as the earth bow shock associated with the solar wind, the occurrence and features of sunspots, Van Allen belts, pulsars, etc.

5.1 THE EQUILIBRIUM PROBLEM

The equations for the macroscopic fluid variables for a collisionless plasma have steady-state solutions that characterize possible plasma equilibria. These steady-state solutions are often metaequilibria, in which the plasma has some feature that represents a departure from a state of thermodynamic equilibrium. It is these metaequilibria that are to be examined for stability.

The study of plasma instabilities is usually based on perturbation theory. This approach makes sense only if there is an initial plasma metaequilibrium state about which it is possible to consider small departures. If an initial perturbation grows, the plasma is called unstable. Inevitably, a growing perturbation alters the properties (density, temperature, degree of uniformity or isotropy, etc.) of the plasma. These modifications may lead to a new equilibrium state that is stable to the mode that destroyed the previous state.

In many plasma experiments an equilibrium state is never established, because the length of time required to produce the plasma is already comparable with an instability growth time. The theoretical approach in these cases is to estimate or postulate what the equilibrium might have been if all instabilities had been suppressed, and then to analyze the stability of this model equilibrium to see if there are predictable features that might be observed. Also, theoretical studies can determine if the most destructive instabilities can be suppressed by modification of the attempted equilibrium configuration.

The experimental approach is to see if a proposed equilibrium can be realized, and if so, to determine the properties of a real plasma under such circumstances. This process of cooperative theoretical and experimental study has been productive in suppressing some of the more destructive instabilities.

Finally, it is important to note the distinction between an unstable equilibrium and a purely dynamic situation. A blob of plasma freely falling across a magnetic field in a gravitational field is not unstable, it is just not in equilibrium.

5.2 CLASSIFICATION OF PLASMA INSTABILITIES

A large number of parameters are usually required to specify a plasma equilibrium. A given equilibrium may be unstable in a variety of ways, and a given instability may be common to several plasma equilibria. It is useful to classify plasma instabilities into groups based on common features. One simple scheme is to group instabilities into two main categories:

1. Configuration-space instabilities (macroinstabilities)
2. Velocity-space instabilities (microinstabilities)

and two subcategories:

1. Electrostatic instabilities
2. Electromagnetic instabilities

5.2.1 Configuration-space Instability

This term identifies instabilities associated with departures of macroscopic quantities from thermodynamic equilibrium. For example, if a plasma equilibrium is finite in size (cylinder, slab, half-space, etc.) and if it can achieve a lower potential-energy state by expansion, translation, or distortion, it may be unstable. If theory or experiment confirms that this plasma is unstable without regard to the velocity distribution of the plasma particles about the mean velocity, such an instability is called a *configuration-space instability*. These instabilities are usually studied with the plasma fluid equations.

5.2.2 Velocity-space Instabilities

This term identifies instabilities associated with the departures of the velocity-space distribution function from a maxwellian distribution. For example, two groups of electrons drifting at different speeds in a background of immobile ions ($m_i \to \infty$) are unstable to the growth of Langmuir oscillations at the expense of the energy in the relative motion of the two electron streams. Some velocity-space instabilities can be studied with the fluid equations; however, they are usually studied with the aid of the Vlasov equations (Chap. 9). Although they usually have large growth rates, velocity-space instabilities do not always cause the rapid transport of plasma across magnetic field usually associated with configuration-space instability.

5.2.3 Electrostatic Instabilities

An electrostatic instability is one associated with growing accumulations of charge, i.e., bunching and charge separation. For such instabilities it is possible to make the simplifying assumption that the electric field is derivable from a scalar potential. These are usually the fastest-growing and most destructive instabilities as far as plasma confinement is concerned.

5.2.4 Electromagnetic Instabilities

An electromagnetic instability is associated with growing accumulations of current density, i.e., with pinching. That is, a plasma current causes a magnetic field that constricts the current stream and further increases the current density, etc. For such instabilities $\nabla \times \mathbf{E} \neq 0$, and the complete set of Maxwell's equations must be used to analyze their properties.

This classification scheme is useful for grouping plasma instabilities, but in practice a complex and colorful set of descriptive names for the various plasma

instabilities has evolved with the field, and it is difficult to understand the plasma literature without becoming familiar with these names and the instabilities they represent. For example, see "stability" in the Subject Index.

5.3 METHODS OF STABILITY ANALYSIS

There are three standard methods normally used to determine the stability or the properties of the instabilities associated with a given plasma equilibrium.

The simplest is the *intuitive approach*, in which the equilibrium is subjected to a perturbation that alters the forces acting on the plasma. If these modified forces act to increase the initial perturbation, the plasma is unstable. An example using the intuitive approach is the analysis of the kink instability, given in Chap. 1. This approach will not give quantitative information about growth rates, but it will give insight into the instability mechanism and show which equilibrium situations require further analysis, as well as providing information about which modes are likely to be unstable.

The second method is based on *energy considerations*. An equilibrium state of a plasma with a potential energy W will be stable if W is a minimum and may be unstable otherwise. To investigate the stability of a plasma with this method it is necessary to calculate the change in potential energy of the plasma as a result of a given perturbation. The plasma equilibrium is stable if the change in potential energy is positive for *all* perturbations allowed by the plasma equations. If there is one allowed perturbation for which the change in potential energy is negative, the plasma is unstable. This method is useful in determining the values of equilibrium parameters that result in unstable equilibria. It is not a useful method for determining the growth rate of a particular instability.

A third way to determine the stability of a plasma equilibrium is *normal-mode analysis*. This method assumes that if the equilibrium under investigation is perturbed, the linearized plasma equations for the time development of the perturbation can be solved, subject to the appropriate boundary conditions, assuming a time dependence $\exp(-i\omega t)$. This procedure gives an equation for ω in terms of the equilibrium parameters. The ω's from this equation may be real, imaginary, or complex. If all the ω's are real, then all the perturbation variables oscillate harmonically and the plasma is stable. If any or all of the ω's have positive imaginary parts, the system is unstable, since those normal modes will grow in time. Normal-mode analysis provides complete information about the instabilities associated with a particular plasma equilibrium. The development of any initial perturbation can be followed up to the limits imposed by the linearization of the equations. Unfortunately, normal-mode analysis can be applied only in those cases where the plasma equilibrium is simple enough to allow solution of the plasma differential equations.

5.4 REGIONS OF STABILITY

The equations used to analyze the stability of a plasma equilibrium contain the parameters that specify the equilibrium. In most cases, there is some value of a given equilibrium parameter (all others held fixed) for which the system is neutrally stable. A small change in this parameter causes the plasma to be stable or unstable. If some other parameter of the system is changed, the value of the first parameter at which neutral stability occurs will also change. The plot of the point of neutral stability as a function of these two variables separates the plane into regions of stability and instability. The curve is called the *stability boundary*. In a problem where the plasma equilibrium depends on several parameters, the stability boundary may be a surface in a multidimensional space.

Problem 5.4.1 Consider a system for which the linearized equation for the perturbation is of the form

$$\ddot{x} + 2A\dot{x} + Bx = 0$$

Determine the stability boundary in the AB plane and show that the parabola $A^2 = B$ separates the periodic and exponential solutions corresponding to stability and instability. Exponentially growing oscillations are sometimes called overstability. Identify the region of overstability in the AB plane. ////

PART TWO: STABILITY OF UNCONFINED PLASMA FROM MACROSCOPIC FLUID EQUATIONS

A homogeneous plasma that fills all space may be unstable if the velocity distribution has a nonmaxwellian feature. The properties of such plasmas are usually investigated with Vlasov theory (Chap. 9). However, it is often also possible to use the equations for the macroscopic fluid variables to investigate the stability of such plasma equilibria. Two examples are considered here. The first is the two-stream instability that results from the interaction of two groups of plasma particles drifting with respect to each other. This instability leads to growing electrostatic waves, which derive their energy from the kinetic energy associated with the relative drift velocity. The second example is the instability

to growth of electromagnetic waves that results from an anisotropy in the plasma temperature parallel and perpendicular to the magnetic axis. Both of these instabilities are examined using normal-mode analysis.

5.5 TWO-STREAM INSTABILITIES OF SPACE-CHARGE WAVES

An example of a velocity-space instability that can be studied with the fluid equations is the two-stream instability. The equilibrium situation considered here is that of two electron beams of density $n_{\alpha 0}$ and $n_{\beta 0}$ drifting with uniform speeds $V_{\alpha 0}$ and $V_{\beta 0}$ in the z direction. The electron beams are assumed to be of infinite cross section and are ion-neutralized, so that there is no average electric field associated with the equilibrium. The neutralizing ions are assumed to have infinite mass, and their motion is neglected in this analysis (this restricts the validity of the analysis, which is correct for high-frequency effects).

The interaction of the space-charge waves (Sec. 4.3) on these two drifting electron beams may produce spatially growing or temporally growing electric fields. To obtain the dispersion relation from which the growth rates can be calculated, the perturbed linearized macroscopic fluid equations are used. The continuity equation (3.4.1) for the perturbation density and velocity, $n_{\alpha 1}(z, t) = \hat{n}_{\alpha 1} e^{ikz - i\omega t}$ and $V_{\alpha 1}(z, t) = \hat{V}_{\alpha 1} e^{ikz - i\omega t}$ with $n_{\alpha 1} \ll n_{\alpha 0}$ and $V_{\alpha 1} \ll V_{\alpha 0}$ gives

$$-i\omega \hat{n}_{\alpha 1} + ik(n_{\alpha 0} \hat{V}_{\alpha 1} + \hat{n}_{\alpha 1} V_{\alpha 0}) = 0 \qquad (5.5.1)$$

and likewise for the β beam,

$$-i\omega \hat{n}_{\beta 1} + ik(n_{\beta 0} \hat{V}_{\beta 1} + \hat{n}_{\beta 1} V_{\beta 0}) = 0 \qquad (5.5.2)$$

The momentum transfer equation (3.4.2) for the perturbation velocity and electric fields gives

$$-i\omega \hat{V}_{\alpha 1} + ikV_{\alpha 0} \hat{V}_{\alpha 1} = -\frac{e}{m_e} \hat{E}_1 \qquad (5.5.3)$$

$$-i\omega \hat{V}_{\beta 1} + ikV_{\beta 0} \hat{V}_{\beta 1} = -\frac{e}{m_e} \hat{E}_1 \qquad (5.5.4)$$

Notice that there is only one perturbation electric field common to both drifting streams. The perturbation electric field and the perturbation density are related by Poisson's equation (3.4.3).

$$ik\hat{E}_1 = -4\pi e(\hat{n}_{\alpha 1} + \hat{n}_{\beta 1}) \qquad (5.5.5)$$

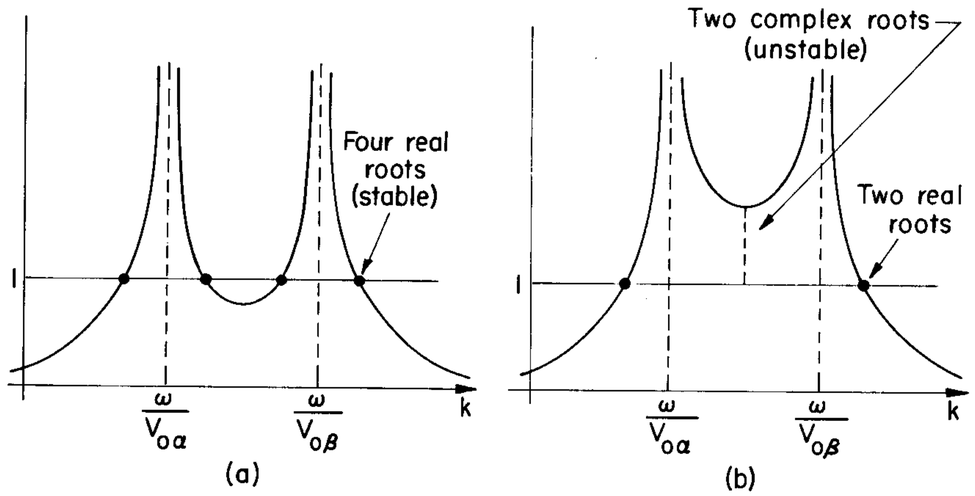


FIGURE 5.5.1

Sketch of $\frac{\omega_{p\alpha}^2}{(\omega - kV_{\alpha 0})^2} + \frac{\omega_{p\beta}^2}{(\omega - kV_{\beta 0})^2}$ as a function of wave number k to illustrate transition from a stable four-wave regime to an unstable regime in which one of the waves grows in space.

Eliminating $\hat{n}_{\alpha 1}$ and $\hat{n}_{\beta 1}$ from (5.5.5) with the aid of (5.5.1) and (5.5.2) and then eliminating $\hat{V}_{\alpha 1}$ and $\hat{V}_{\beta 1}$ with the aid of (5.5.3) and (5.5.4), the two-stream dispersion relation is obtained.

$$ik\hat{E}_1 \left[1 - \frac{\omega_{p\alpha}^2}{(\omega - kV_{\alpha 0})^2} - \frac{\omega_{p\beta}^2}{(\omega - kV_{\beta 0})^2} \right] = 0 \qquad (5.5.6)$$

For a given equilibrium ($n_{\alpha 0}$, $V_{\alpha 0}$, $n_{\beta 0}$, $V_{\beta 0}$), the dispersion relation (the term in brackets) can be solved for frequency as a function of wave number [that is, $\omega(k)$]. For example, if the wave number k is assumed to be a real number $k_r (k = k_r + ik_i)$ and if the dispersion relation is solved for the frequency, which in general may be complex (that is, $\omega = \omega_r + i\omega_i$), the normal modes may be growing in time ($\omega_i > 0$), constant in time ($\omega_i = 0$), or damped in time ($\omega_i < 0$). Similarly, if the frequency ω is assumed to be real and the dispersion relation is solved for the wave number, which in general may be complex, the normal modes may be growing in space ($k_i < 0$), constant in space ($k_i = 0$), or damped in space ($k_i > 0$). A convenient way to study the normal-mode stability properties of the two-stream dispersion relation is to plot the right and left sides of (5.5.6) as a function of k, as shown in Fig. 5.5.1.

Problem 5.5.1 From (5.5.6) with equal electron-beam densities, $n_{\alpha 0} = n_{\beta 0}$, and for equal but opposite electron-beam velocities, $V_{\alpha 0} = -V_{\beta 0}$, find the maximum growth rate ω_i. ////

Figure 5.5.1*a* shows, for some real frequency ω_r, that if the drift velocities of the two beams are widely separated, there are four real values k_r corresponding to four propagating waves on the combined drifting electron beams. In Fig. 5.5.1*b*, $\omega/V_{\alpha 0}$ almost equals $\omega/V_{\beta 0}$; the k scale has been expanded for clarity. In this case, in which the drift velocities of the two beams are nearly the same, there are two real roots for k and two complex roots. Thus, there is the possibility of spatially growing or decaying waves. To study this problem analytically, the dispersion relation from (5.5.6) for the two-stream instability is solved for complex k, assuming ω is real. Of course this can be done numerically; however, there is a special case for which the dispersion relation can be solved directly for the complex values of k. This is the case in which

$$\frac{\omega_{p\alpha}{}^2}{V_{\alpha 0}{}^2} = \frac{\omega_{p\beta}{}^2}{V_{\beta 0}{}^2}$$

which includes the case of equal-density electron beams of equal but opposite velocity ($n_{\alpha 0} = n_{\beta 0}$, $V_{\alpha 0} = -V_{\beta 0}$).

To solve this problem it is convenient to define a harmonic average velocity:

$$\frac{1}{V_0} = \frac{1}{2}\left(\frac{1}{V_{\alpha 0}} + \frac{1}{V_{\beta 0}}\right) \tag{5.5.7}$$

an average plasma frequency:

$$\frac{\omega_p{}^2}{V_0{}^2} = \frac{\omega_{p\alpha}{}^2}{V_{\alpha 0}{}^2} = \frac{\omega_{p\beta}{}^2}{V_{\beta 0}{}^2} \tag{5.5.8}$$

a dimensionless frequency variable:

$$x = \frac{\omega}{\omega_p}\frac{V_{\beta 0} - V_{\alpha 0}}{V_{\beta 0} + V_{\alpha 0}} \tag{5.5.9}$$

It is also convenient to let the wave number be expressed as follows:

$$k = \frac{\omega}{V_0} + y\frac{\omega_p}{V_0} \tag{5.5.10}$$

Then dispersion relation (5.5.6) transforms into the following simple equation:

$$\frac{1}{(x-y)^2} + \frac{1}{(x+y)^2} = 1 \tag{5.5.11}$$

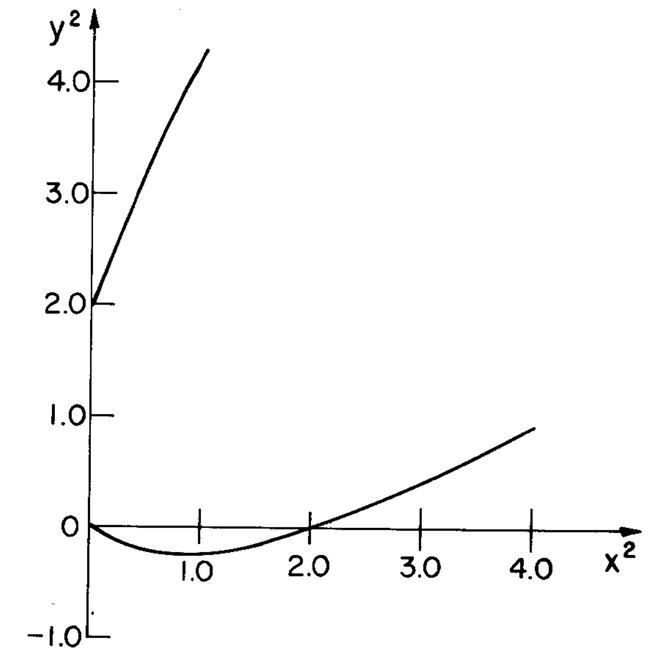


FIGURE 5.5.2
Plot of the function $y^2 = (x^2+1) \pm (4x^2+1)^{1/2}$ showing the region $x^2 < 2$, where, corresponding to growing waves, $y^2 < 0$.

The solution of this fourth-degree equation is

$$y^2 = x^2 + 1 \pm (4x^2+1)^{1/2} \tag{5.5.12}$$

This equation is plotted in Fig. 5.5.2. There are two branches. For the upper branch, $y^2 > 0$ for all $x^2 > 0$, and thus there are no complex values of k. This branch corresponds to the two outermost real roots in Fig. 5.5.1. For the lower branch, $y^2 < 0$ for $0 < x^2 < 2$, and thus there are two complex conjugate roots for k. This branch corresponds to roots associated with the dotted vertical line in Fig. 5.5.1*b* for $x^2 < 2$ and to the two innermost roots in Fig. 5.5.1*a* for $x^2 > 2$.

In the region $0 < x^2 < 2$, $y^2 < 0$, and hence y is imaginary. One value of y corresponds to spatially growing waves, and the other corresponds to spatially damped waves. The maximum frequency for which there is a spatially growing wave is

$$\omega_{\max} = \sqrt{2}\left(\frac{V_{\alpha 0} + V_{\beta 0}}{V_{\alpha 0} - V_{\beta 0}}\right)\omega_p \tag{5.5.13}$$

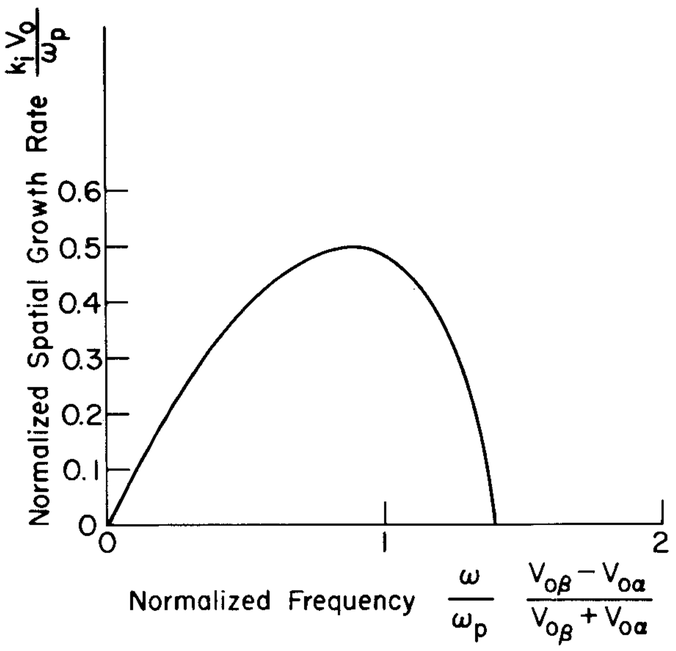


FIGURE 5.5.3
Normalized growth rate for the two-stream instability as a function of the normalized frequency.

where ω_p is the frequency given in (5.5.8). The maximum frequency for which spatially growing waves exist can be many times the plasma frequency for small velocity differences. The wave energy generated by this mechanism can propagate through the plasma as an electromagnetic wave if there is some mechanism within the plasma to couple these electrostatic modes to electromagnetic modes.

The maximum spatial growth rate occurs for frequency

$$\omega_{\text{max growth}} = \frac{\sqrt{3}}{2}\left(\frac{V_{\alpha 0}+V_{\beta 0}}{V_{\alpha 0}-V_{\beta 0}}\right)\omega_p \qquad (5.5.14)$$

At this frequency the maximum spatial growth rate is

$$k_{i,\,\max} = -\frac{\omega_p}{2V_0} = -\left|\frac{\omega_{p\alpha}}{2V_{\alpha 0}}\right| = -\left|\frac{\omega_{p\beta}}{2V_{\beta 0}}\right| \qquad (5.5.15)$$

The spatial growth rate is plotted in Fig. 5.5.3 as a function of the dimensionless frequency variable. As can be seen, this model of a two-stream interaction predicts a spatially growing wave at low frequency. This two-stream growth

mechanism was demonstrated experimentally by Pierce and Heibenstreit.[1] There have been many subsequent experimental investigations and an exhaustive theoretical treatment of the two-stream instability.

This example of the two-stream instability illustrates how the choice of model influences the results obtained. In the study of plasma oscillations in Chap. 4 it was convenient to assume the ions to be immobile ($m_i = \infty$), so that ion dynamics did not enter into the equations. This simple model is clearly adequate to explain the high-frequency, long-wavelength features of plasma oscillations and waves, provided the ions and electrons are on the average at rest with respect to each other. This same model of immobile ions was used in Chap. 4 to investigate the space-charge waves associated with drifting electrons. That the assumption of immobile ions is not a complete model for describing drifting electrons can be shown using the results of the two-stream analysis given above. That is, the two-stream analysis could be extended to include the case of ions and electrons drifting with respect to each other by simply identifying $\omega_{p\alpha}{}^2 = 4\pi n_{i0}\, e^2/m_i$, $V_{\alpha 0} = V_{i0}$, $\omega_{p\beta}{}^2 = 4\pi n_{e0}\, e^2/m_e$, and $V_{\beta 0} = V_{e0}$. Then (taking $V_{i0} = 0$), the ions response to the electric forces generated by charge bunching leads to a low-frequency, spatially-growing instability[2] below the ion plasma frequency, [i.e., solving (5.5.6) gives $k_i V_{e0} = \omega \pm \omega_{pe}/\sqrt{1 - \omega_{pi}{}^2/\omega^2}$] in addition to the high frequency, space-charge waves associated with the drifting electrons. The assumption of immobile ions ($\omega_{pi}{}^2 = 0$) eliminates the unstable waves because it suppresses the ion response, and restricts the validity of the calculation to high frequency waves. The "cold-plasma approximation" still requires $kv_{\text{thermal}}/\omega < 1$, which implies that $V_{e0} \gtrsim \sqrt{\kappa T_e/m_e}$. Thus a laboratory plasma where the electrons drift through the ions at a speed greater than the electron thermal speed would be expected to be unstable unless some other feature, such as finite size, were also to influence the result. A further discussion of the two-stream instability emphasizing the temporally growing instability, is given in Sec. 9.3.

Problem 5.5.2 Assume that the velocity of one of the beams in (5.5.6) is zero and calculate the spatial growth rate for the interaction of a drifting electron beam with a stationary electron cloud. ////

Problem 5.5.3 Obtain the dispersion relation and the spatial growth rate for the situation shown in Fig. 5.5.4. Use the quasistatic approximation discussed in Chap. 4. ////

[1] J. R. Pierce and W. B. Heibenstreit, A New Type of High-frequency Amplifier, *Bell Syst. Tech. J.*, **28**: 35–51, January 1949.
[2] J. A. Davis and A. Bers, *Proceeding of the Symposium on Turbulence of Fluids and Plasmas*, Polytechnic Press, Brooklyn, vol. 18, p. 87 (1968).

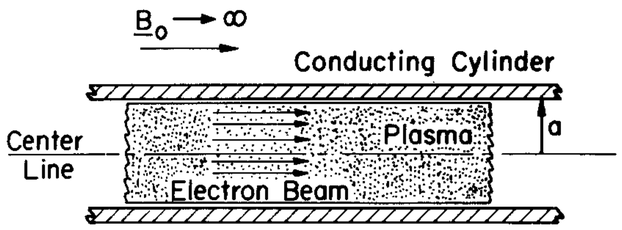


FIGURE 5.5.4
Schematic of finite electron beam and plasma interaction region. A single underscore denotes a vector.

5.6 FIRE-HOSE INSTABILITY OF AN ALFVÉN WAVE

Consider a cold homogeneous magnetized electrically neutral plasma of infinite extent. One of the normal modes of this plasma equilibrium is the Alfvén wave, a low-frequency ($\omega < \omega_{ci}$) transverse electromagnetic wave that propagates along the magnetic field with speed $V_A = B/\sqrt{4\pi\rho_m}$. This normal mode of this equilibrium is stable if the plasma pressure parallel to the steady magnetic field is the same as the pressure perpendicular to the magnetic field, i.e., no temperature anisotropy. For a low-density plasma in a strong magnetic field, collisions are infrequent, and the temperature of the plasma along the steady magnetic field may be different from that perpendicular to the magnetic field.

To analyze the normal modes of a one-fluid-model plasma equilibrium with the additional degree of freedom of temperature anisotropy, the double-adiabatic theory of Chew, Goldberger, and Low is used to close the macroscopic fluid equations. This method neglects heat flow along the magnetic field and assumes that the plasma disturbances have phase velocity greater than the mean thermal speed of the plasma. It is called double-adiabatic because different adiabatic equations of state are assumed for the perpendicular and parallel pressures.

The double-adiabatic equations and the analysis leading to the dispersion relation for the normal-mode spectrum for an anisotropic plasma are given in Sec. 3.10. The result is

$$\omega^2 = \frac{k^2}{2\rho_{m0}}\left\{\left(\frac{B^2}{4\pi} + p_\perp + 2p_\parallel \cos^2\theta + p_\perp \sin^2\theta\right)\right.$$
$$\left. + \left[\left(\frac{B^2}{4\pi} + p_\perp(1+\sin^2\theta) - 4p_\parallel\cos^2\theta\right)^2 + 4{p_\perp}^2\sin^2\theta\cos^2\theta\right]^{1/2}\right\} \tag{5.6.1}$$

The magnetic field is in the z direction, and θ is the angle measured from the z axis. This dispersion relation predicts magnetosonic and Alfvén waves that are undamped normal modes of the system. In addition, this dispersion relation, which allows for a different pressure parallel and perpendicular to the magnetic field, predicts that if $p_\parallel > p_\perp + B^2/4\pi$, the Alfvén wave propagating along the magnetic field is unstable, as can be seen from (5.6.1), which reduces to

$$\omega^2 = \frac{k^2}{\rho_{m0}}\left(\frac{B^2}{4\pi} + p_\perp - p_\parallel\right) \qquad \text{for } \theta = 0$$

This instability is called the *fire-hose instability*, since the driving force (the plasma pressure parallel to the magnetic field) acts to increase the amplitude of the wave in an analogous way to that of water flowing through a loose hose. The two-stream instability described in the preceding section is an electrostatic instability and grows on a time scale comparable with the plasma period for electrons. The electrostatic instability involves the bunching of charges as a result of the perturbation. The fire-hose instability of an Alfvén wave is an electromagnetic instability. It involves self-pinching of plasma currents associated with the perturbation. This electromagnetic instability grows more slowly than the electrostatic instability by, roughly, a factor of $k\lambda_D$, since $\text{Im}(\omega) \gtrsim \omega_{pi}$ for the two-stream instability, while $\text{Im}(\omega) \approx k\sqrt{\kappa T/m_i}$ for the fire-hose instability. Thus, if the conditions were such that both an electrostatic and an electromagnetic instability exist, the electrostatic would be more important to consider since it would lead to an alteration of the equilibrium long before the electromagnetic pinching instability would grow.

PART THREE: STABILITY OF MAGNETICALLY CONFINED PLASMA FROM MACROSCOPIC FLUID EQUATIONS

In Part Two of this chapter the macroscopic fluid equations were used to investigate the stability of two plasma metaequilibria. In each of these two cases the plasma was homogeneous, but had a simple nonthermodynamic feature. In one case the nonthermodynamic feature was the drift motion of the electron fluid (two-stream instability), and in the other case it was the temperature anisotropy (fire-hose instability of an Alfvén wave). Neither of these two equilibria involved spatial confinement of the plasma, and it was possible to investigate their stability by solving for the normal modes of the plasma.

In this part of this chapter the macroscopic fluid equations are used to investigate the stability of various plasma metaequilibria, where the non-

thermodynamic feature of the equilibrium is the spatial localization or confinement of the plasma by a magnetic field. For certain simple situations, such as a uniform plasma supported against gravity by a magnetic field, it is possible to solve for the complete normal modes of the system. This means that growth of any initial perturbation can be followed in detail until the amplitude of the disturbances exceeds the limits imposed by the use of the linearized equations.

For the more complicated plasma-confinement configurations it is not always possible to obtain a complete solution for the normal modes of the system. In this case the more general *energy principle* is used. The energy principle determines whether or not a given plasma metaequilibrium is stable (or unstable) as a function of the equilibrium parameters, but in general does not give information about the growth rates of the instability or the plasma motion associated with it.

5.7 STABILITY OF THE FLUID PLASMA SUPPORTED AGAINST GRAVITY BY A MAGNETIC FLUID

5.7.1 Drift Analysis

To understand how spatial confinement of a plasma can cause an instability, consider the plasma metaequilibrium depicted schematically in Fig. 5.7.1*a*. A low-density, low-temperature electrically neutral plasma fills the upper half-space ($y > 0$). There is a uniform magnetic field in the z direction ($\mathbf{B}_0 = \hat{\mathbf{z}}B_0$) and a gravitational force in the $-y$ direction ($\mathbf{g} = -g\hat{\mathbf{y}}$). The plasma is supported against the force of gravity by an increase in the magnetic field across the plasma boundary. The plasma pressure is assumed to be much less than the magnetic pressure, so that the magnetic field inside and outside the plasma is nearly the same.

Imagine that the plasma-vacuum interface is perturbed, as shown in Fig. 5.7.1*b*. The ions and electron initially drift parallel to the x axis as a result of the gravitational force and the magnetic field; i.e.,

$$\mathbf{V}_{DG}{}^{\alpha} = \frac{m_\alpha}{q_\alpha}\frac{\mathbf{g}\times\mathbf{B}}{B^2}c$$

Since the drift velocity is inversely proportional to the charge-to-mass ratio, the ions and electrons drift in opposite directions, and the ions have a velocity that is m_i/m_e greater than the electrons. Thus the ions drift from the electrons, causing a charge separation and an electric field, as shown in Fig. 5.7.1*b*. Now

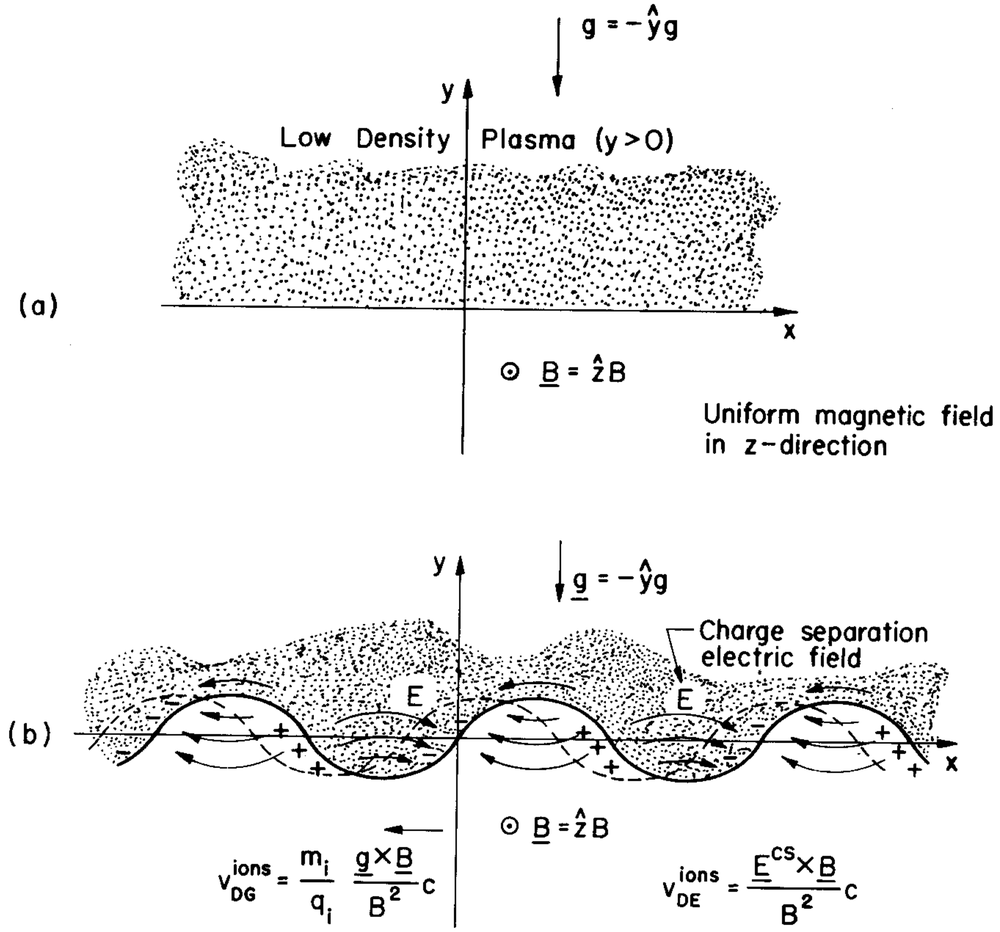


FIGURE 5.7.1
(*a*) Schematic of low-density plasma in upper half-space ($y > 0$) imbedded in and supported against the force of gravity by a magnetic field. (*b*) Schematic of perturbation of the plasma-vacuum interface region showing the development of the charge-separation electric field caused by the ion $\mathbf{g}\times\mathbf{B}$ drift motion.

the ions and electrons will drift in the y direction as a result of the force of the electric field and equilibrium magnetic field; the drift is given by

$$\mathbf{V}_{DE}{}^{\alpha} = \frac{\mathbf{E}\times\mathbf{B}}{B^2}c$$

The ions and electrons move in such a way as to increase the amplitude of the initial perturbation. Thus, a plasma supported against gravity, as shown in Fig. 5.7.1*a*, is unstable.

This qualitative discussion gives no information about the growth rate of the instability, the perturbation wavelength that has the largest growth rate,

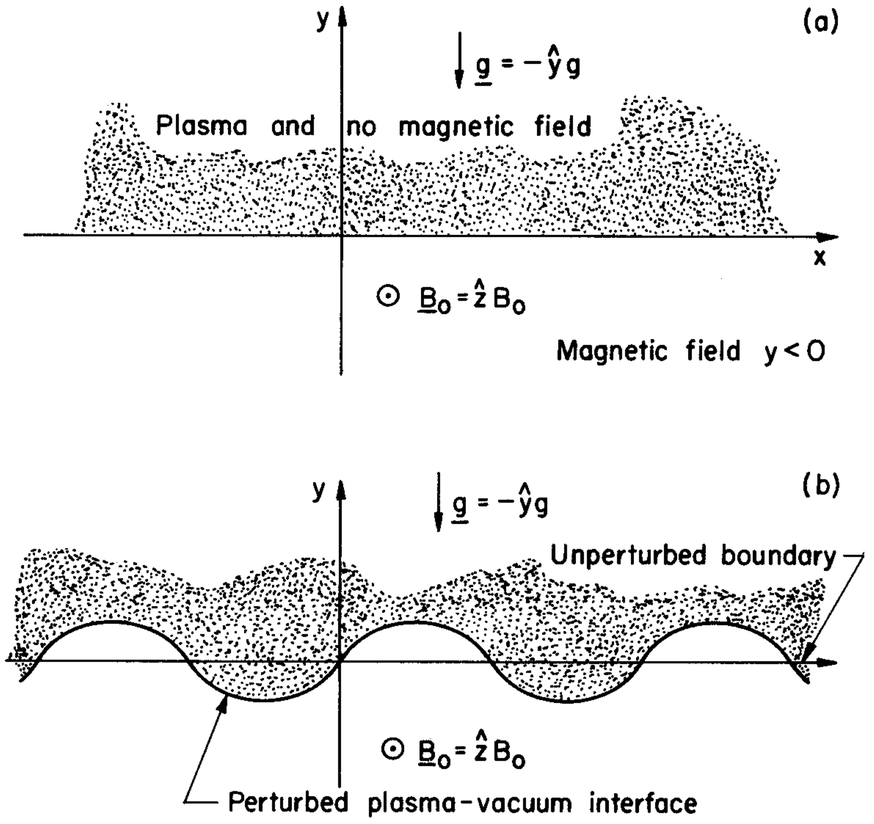


FIGURE 5.7.2
(*a*) Schematic of low-density plasma in upper half-space ($y > 0$) supported against the force of gravity by a magnetic field. (*b*) Schematic of perturbation of the plasma-vacuum interface region, showing an initial perturbation which results in a lowering of the potential energy of the system, causing a further growth of the perturbation.

etc. However, it shows that this plasma metaequilibrium with the nonthermodynamic feature of being spatially localized is unstable. Therefore, unless some means is found to eliminate or otherwise suppress the deleterious consequences of the instability, it is not possible to contain a plasma in this configuration.

5.7.2 Energy Analysis

Since the drift motions that drive instability are associated with the magnetic field, perhaps excluding the magnetic field from the interior of the plasma will eliminate this instability. To investigate this possibility, consider the plasma metaequilibrium situation depicted schematically in Fig. 5.7.2*a*. The plasma-

vacuum interface is perturbed, as shown in Fig. 5.7.2*b*. This perturbation can be chosen in such a way that the net area associated with the perturbation is zero; then the total magnetic energy is not changed by the perturbation. In addition, the plasma is not compressed (or expanded) by this perturbation, and so there is no net change of stored internal energy.

However, plasma that was in the region above the unperturbed boundary now appears in the region below the unperturbed boundary. This results in a reduction of the net potential energy of the plasma. By conservation of energy this means an increase in the kinetic energy of the plasma. Hence the plasma metaequilibrium shown in Fig. 5.7.2*a* is also unstable. It is from this latter example that this magnetohydrodynamic instability derives the name *Rayleigh-Taylor instability*, since it is similar to the instability, investigated by Rayleigh,[1] that occurs when a heavy incompressible fluid is supported against the force of gravity by a lighter incompressible fluid. The analogy with the plasma situation is made by noting that the magnetic field, which exerts an upward pressure on the lower surface of the plasma, has no mass and can be regarded as a "light" fluid. The plasma, when subjected to the noncompressing perturbation described above, behaves as a "heavy" fluid.

Thus the plasma metaequilibrium of Fig. 5.7.2*a* that seeks to confine a heavy fluid (plasma) against the force of gravity by a light fluid (magnetic field) is expected to be unstable on the basis of simple energy-conservation arguments.

The magnetohydrodynamic instability described above occurs in the most reduced fluid description of the plasma as a result of the nonthermodynamic feature of spatial confinement of the plasma. Although no laboratory confinement involves the gravitational force, the metaequilibria of Figs. 5.7.1*a* and 5.7.2*a* are useful examples to analyze because the configuration is so simple that it can be studied in detail with the fluid equations.

5.8 STABILITY OF MAGNETICALLY CONFINED FLUID PLASMA FROM THERMODYNAMIC CONSIDERATIONS; INTERCHANGE INSTABILITY

In the two preceding examples an intuitive approach to instability analysis was used to investigate the stability of a plasma supported against gravity by a magnetic field. This model equilibrium is useful to help develop an understanding of how plasmas can be unstable, but it is obviously not a suitable representation of a magnetically confined laboratory plasma. It is possible to

[1] Baron John W. S. Rayleigh, *Phil. Mag*; **32**: 529 (1916).

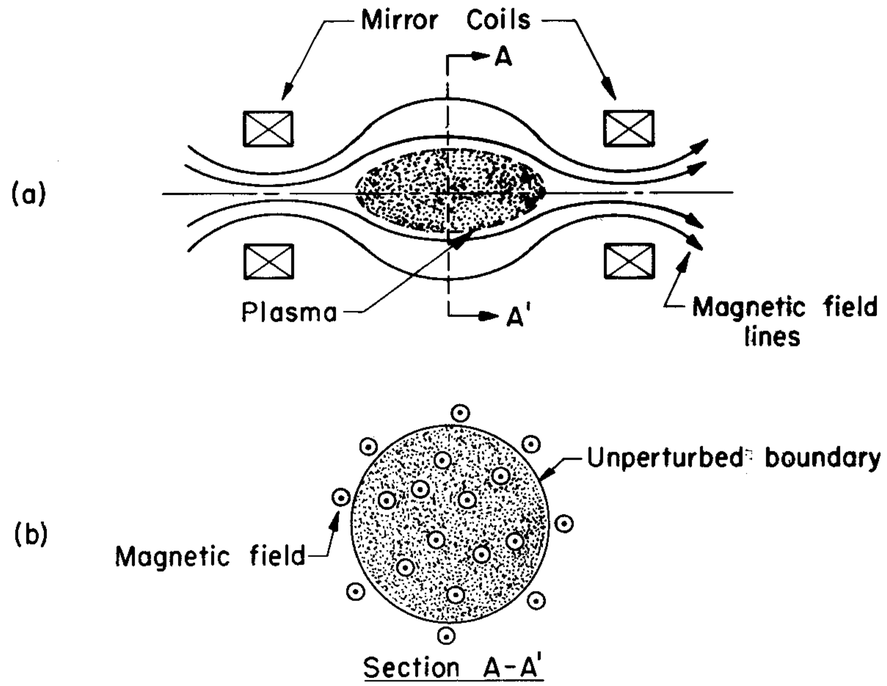


FIGURE 5.8.1
(*a*) Schematic of plasma confined by a magnetic mirror field. (*b*) View through the midsection of plasma before the plasma has been perturbed. (*c*) Same view after perturbation, showing initial development of "flutes."

improve on these intuitive approaches and obtain some general results concerning stability of magnetically confined plasma using a thermodynamic model. This method is really a special case of the more general energy principle that is developed later in this chapter. It is given separate treatment here because it illustrates in a simple manner the most destructive instability (interchange instability) associated with magnetically confined plasmas, and predicts how that instability can be suppressed.

Consider the situation depicted schematically in Fig. 5.8.1. Two separated coaxial coils, with the plane of each coil perpendicular to the same axis, carry current in the same direction to create the magnetic field distribution as shown. This field distribution is called a magnetic mirror because charged particles with

a ratio of perpendicular velocity to parallel velocity greater than some value, determined by the specific arrangement of coils and current, are reflected by each coil, and thus trapped between the two coils. Otherwise they escape from the mirror field along field lines. (See Appendix I, Sec. I.6.)

If a plasma with an isotropic velocity distribution were introduced into the magnetic mirror, some of the plasma particles would be lost, creating an anisotropic velocity distribution. Such a distribution is called a loss-cone distribution and is unstable (Chap. 9). However, the interchange instability is more serious in terms of loss of plasma confinement. The loss-cone instability becomes important for plasma-confinement schemes not subject to interchange instability but which have loss-cone anisotropic distributions. The interchange instability is also called the *flute instability*, a term that comes from its appearance as shown in Fig. 5.8.1*c*. This instability is the laboratory equivalent of the Rayleigh-Taylor instability for a plasma supported against gravity by the magnetic field. The role of gravity is played by the curvature of the magnetic field, as will be seen.

The following analysis of the interchange is based on the original research paper.[1] It assumes that if a state of lower potential energy exists, the plasma will achieve the state, with the released energy going into plasma kinetic energy. It also assumes that the plasma equations of motion allow the plasma to move so as to lower the potential energy.

The energy in a volume V of a plasma is the sum of the magnetic energy $VB^2/8\pi$ and internal energy of the plasma W_P. Treating the plasma as an ideal gas, as explained in Chap. 2, the internal energy is given by

$$W_P = \frac{pV}{\gamma - 1} \qquad (5.8.1)$$

The motion of the plasma is assumed to be adiabatic; i.e.,

$$\frac{d}{dt}(pV^\gamma) = 0 \qquad (5.8.2)$$

For the situation depicted in Fig. 5.8.1, the field lines are curved. The plasma pressure is assumed small [$\beta = n\kappa T_e/(B^2/8\pi) \ll 1$], and the magnetic field is nearly the same as the vacuum field. This means that any distortion of the field increases its energy. Hence the perturbations most likely to be unstable are those that leave the field unchanged (called "flutes" because the field lines are not twisted or bent), as shown in Fig. 5.8.1*c*. To calculate the change in potential energy of the system as a result of the type of deformation shown in Fig.

[1] M. N. Rosenbluth and C. L. Longmire, *Ann. Phys.* 1120 (1957).

5.8.1*c*, it is necessary to interchange the flux tube together with its enclosed plasma in region I, with that in region II.

The magnetic energy in a flux tube is

$$W_M = \int \frac{B^2}{8\pi}\, dV = \int \frac{B^2}{8\pi} A\, dl \qquad (5.8.3)$$

where l = length along the tube,

A = cross-sectional area of the tube,

$\int dl$ = integral along a flux line.

A flux tube is defined by the condition that the flux it encloses is a constant, i.e.,

$$\Phi = BA \qquad (5.8.4)$$

so that

$$W_M = \frac{1}{8\pi}\Phi^2 \int \frac{dl}{A} \qquad (5.8.5)$$

The change of magnetic energy as a result of interchanging the flux in the volume $V_{\mathrm{I}}(=\int A_{\mathrm{I}}\, dl)$ with that of volume $V_{\mathrm{II}}(=\int A_{\mathrm{II}}\, dl)$ is

$$\Delta W_M = \frac{1}{8\pi}\left[\left(\Phi_{\mathrm{I}}^2 \int_{\mathrm{II}} \frac{dl}{A} + \Phi_{\mathrm{II}}^2 \int_{\mathrm{I}} \frac{dl}{A}\right) - \left(\Phi_{\mathrm{I}}^2 \int_{\mathrm{I}} \frac{dl}{A} + \Phi_{\mathrm{II}}^2 \int_{\mathrm{II}} \frac{dl}{A}\right)\right] \qquad (5.8.6)$$

The most unstable deformations are those that leave the magnetic field energy unchanged; that is, $\Delta W_M = 0$. This will be the case if $\Phi_{\mathrm{I}} = \Phi_{\mathrm{II}}$, indicating that, in order to choose V_{I} and V_{II} so as to leave the field energy unchanged, the areas A_{I} and A_{II} should be chosen according to the local value of the magnetic field, with $B_{\mathrm{I}} A_{\mathrm{I}} = B_{\mathrm{II}} A_{\mathrm{II}}$, where A and B can vary with l, the location along the flux tube.

The volume associated with a flux tube can be expressed in terms of the magnetic field B by

$$V = \int dl\, A = \Phi \int \frac{dl}{B} \qquad (5.8.7)$$

The change in plasma energy for an adiabatic process is

$$\Delta W_P = \frac{1}{\gamma - 1}\left(p_{\mathrm{I}} \frac{V_{\mathrm{I}}^{\gamma}}{V_{\mathrm{II}}^{\gamma}} V_{\mathrm{II}} + p_{\mathrm{II}} \frac{V_{\mathrm{II}}^{\gamma}}{V_{\mathrm{I}}^{\gamma}} V_{\mathrm{I}} - p_{\mathrm{I}} V_{\mathrm{I}} - p_{\mathrm{II}} V_{\mathrm{II}}\right) \qquad (5.8.8)$$

To investigate the effect of interchanging neighboring flux tubes, the pressure and volume are expanded as follows:

$$\begin{aligned} p_{\mathrm{II}} &= p_{\mathrm{I}} + \delta p \\ V_{\mathrm{II}} &= V_{\mathrm{I}} + \delta V \end{aligned} \qquad (5.8.9)$$

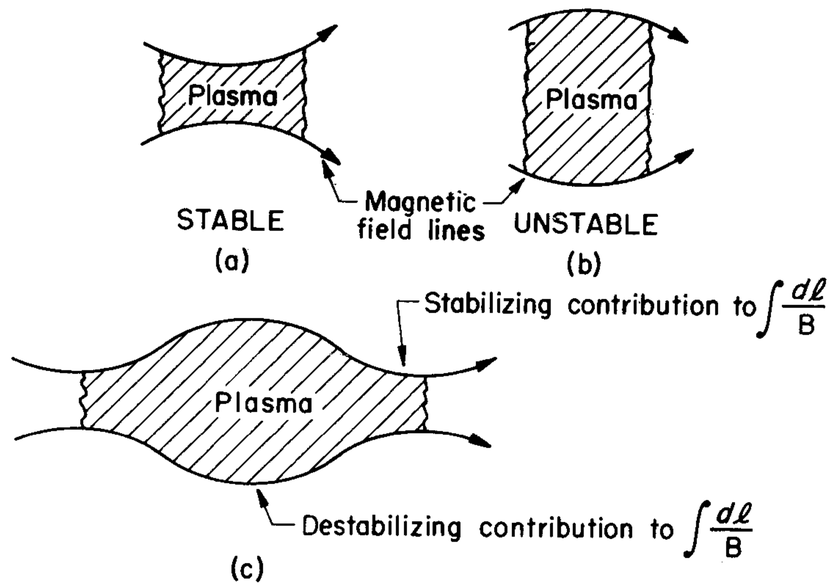


FIGURE 5.8.2
Schematic of magnetically confined plasma in regions where the plasma is hydromagnetically (*a*) stable, (*b*) unstable, or (*c*) stable or unstable, depending on the value $\int dl/B$.

The change in plasma energy in terms δV and δp is, from (5.8.8),

$$\delta W_P = \delta p\, \delta V + \gamma p \frac{(\delta V)^2}{V} = V^{-\gamma}\, \delta(pV^{\gamma})\, \delta V \qquad (5.8.10)$$

The condition for stability is that $\delta W_P > 0$. In a typical confined plasma, the plasma pressure is a maximum on the axis. At the edge of the plasma, the pressure must go to zero, so that if region II lies outside region I, $\delta p < 0$. This means that $\delta W_P < 0$ (unstable) if $\delta V > 0$. In general, since $\gamma p(\delta V)^2/V$ is always positive,

$$\delta V\, \delta p > 0 \qquad \text{for stability} \qquad (5.8.11)$$

This is a sufficient condition for stability. The stability condition $\delta V < 0$ in the direction of decreasing plasma density can be expressed as an integral along a flux tube; i.e.,

$$\delta \int A\, dl = \Phi \delta \int \frac{dl}{B} < 0 \qquad \text{stable} \qquad (5.8.12)$$

This condition is equivalent to the statement that if field lines curve into the plasma, it is stable, as indicated in Fig. 5.8.2. A general plasma configuration

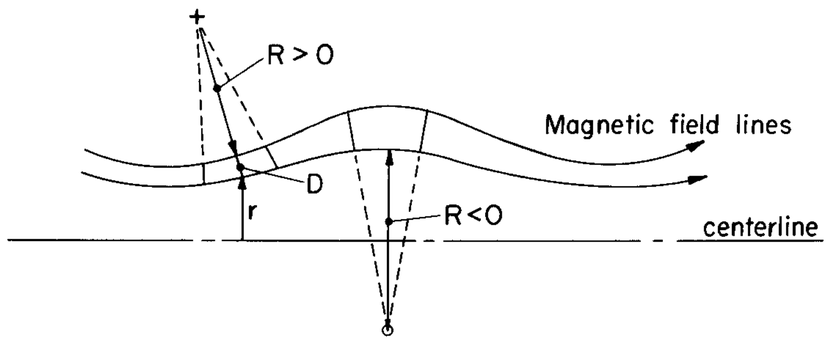


FIGURE 5.8.3
Schematic of coordinate system to analyze interchange instability, where magnetically confined plasma is anisotropic. The radius of curvature of the field line is denoted by R.

may contain regions of both positive and negative contributions to $\int dl/B$, and will be stable or unstable according to the mix.

For the simple magnetic mirror field the condition (5.8.12) is not satisfied, and plasma in a magnetic mirror is interchange-unstable subject to the restrictions imposed by the analysis.

A *cusp* magnetic field, created by reversing the current in one of the mirror coils, is interchange-stable, but has an undesirable loss rate associated with the fact that the magnetic field vanishes at the midpoint between the two mirror coils on the axis.

Problem 5.8.1 Read the paper of M. N. Rosenbluth and C. L. Longmire, *Ann. Phys.*, **1**; 120 (1957), and supply the missing steps in their derivation to show that, for anisotropic pressure, the stability criterion for the interchange instability in terms of the radius of curvature R becomes

$$\int \frac{p_{\parallel} + p_{\perp}}{RrB^2}\, dl > 0 \qquad (5.8.13)$$

where r and R are defined as shown in Fig. 5.8.3. ////

A form for the stability condition useful for a magnetically confined plasma in which the field lines are closed can be obtained by using the fact that the volume of a flux tube is

$$V = \Phi \oint \frac{dl}{B} \qquad (5.8.14)$$

The plasma in the volume V, if not restrained by external forces, would tend to expand. But to the extent that the field is tied to the plasma, the flux Φ would stay constant during such an expansion, so that the quantity $\oint dl/B$ would have to increase as the plasma volume increased. This suggests that the quantity

$$W = -\oint \frac{dl}{B} \qquad (5.8.15)$$

be treated as the "potential energy" of the plasma, so that plasma expansion appears as a transition to the lower state of potential energy. If the tube of flux undergoes an infinitesimal displacement, the change in the volume of the flux tube is

$$\frac{\delta V}{V} = \frac{\delta W}{W} \qquad (5.8.16)$$

From $pV^{\gamma} = \text{constant}$, it follows that the change of pressure associated with a change of volume is

$$\delta p = -\gamma p \frac{\delta V}{V} = -\gamma p \frac{\delta W}{W} \qquad (5.8.17)$$

The equilibrium pressure will in general be a function $p(W)$, and the pressure at the point to which the tube of flux has been displaced is

$$p(W + \delta W) = p + \frac{dp}{dW}\delta W \qquad (5.8.18)$$

If the plasma pressure in the expanded tube of flux (5.8.17) is greater than the pressure (5.8.18) of the plasma in surrounding tube of flux, the tube will continue to expand; i.e., it is unstable. If the pressure in the perturbed tube of flux is less than that of surrounding tubes, it will experience a force that seeks to restore it to its equilibrium position. Comparing the restoring force $(dp/dW)\,\delta W$ with the "expansion force" (5.8.17), and noting that $(dp/dW)\,\delta W$ is negative, the plasma is stable if

$$-\gamma p \left(\frac{\delta W}{W}\right) > \frac{dp}{dW}\delta W$$

or simply (assuming $\delta W < 0$ for an expansion)

$$-\frac{\gamma p}{W} > \frac{dp}{dW} \qquad \text{for stability} \qquad (5.8.19)$$

This condition is identical with (5.8.10), which gave $\delta p > -\gamma p(\delta V)/V$ for stability, since $W = -V/\Phi$, and shows again that the calculation of balance of

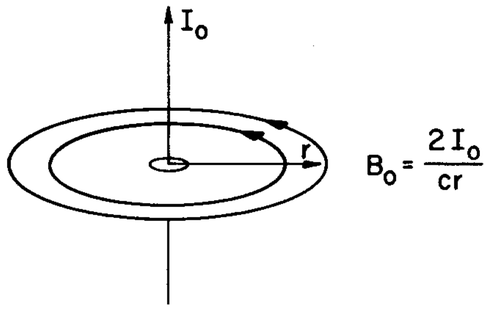


FIGURE 5.8.4
Magnetic field surrounding a current-carrying wire as a plasma-confining magnetic field.

forces following a perturbation gives the same stability information as does calculation of the energy release in the perturbation. As an example of the usefulness of this stability condition, consider a plasma confined by the field of a long straight current-carrying wire (Fig. 5.8.4). The magnetic field is given by

$$B_0 = \frac{2I_0}{cr} \qquad (5.8.20)$$

with I_0 in cgs gaussian units ($B_0 = I_0/5r$, with I_0 in amperes and B_0 in gauss). The potential energy

$$W = -\oint \frac{dl}{B} = -\frac{\pi r^2 c}{I_0}.$$

Thus the condition for stability becomes

$$2\gamma > -\frac{r}{p}\frac{dp}{dr} \qquad (5.8.21)$$

Thus, if the pressure decreases with distance from the current I_0, faster than $r^{-10/3}$, the configuration, Fig. 5.8.4 is unstable to adiabatic ($\gamma = 5/3$) expansion.

Problem 5.8.2 Show that the interchange stability condition for a low β plasma confined in a dipole magnetic field is

$$4\gamma > \frac{r}{p_0}\frac{dp_0}{dr}$$

Assume the flux lines are free to move and are not frozen with respect to the dipole. Explain this assumption. Is this stability condition satisfied for the particles trapped in the radiation belts surrounding the earth?

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The condition (5.8.19) has interesting implications. A system in true thermodynamic equilibrium clearly has its maximum pressure at the point of lowest potential energy (or else it would expand in the direction of decreasing

W). On a collisionless time scale, a system can change its state (toward true equilibrium, for example) only through collective motions (instabilities), and (5.8.19) says that even if the pressure *increases* away from the point of lowest potential, the system can be stable, provided only that the increase $p(W)$ be fairly gradual. The difference is that on the collisionless time scale the plasma is tied to the magnetic field, at least for the low-frequency motion considered in the MHD theory, so that many plasma motions are affected by the **B** field. On the *collisional* time scale the plasma diffuses across **B** and reaches its lowest potential-energy state without the need to "interchange tubes of flux," and so on.

5.9 MACROSCOPIC EQUATIONS FOR THE STUDY OF THE HYDRODYNAMIC STABILITY OF MAGNETICALLY CONFINED PLASMAS

To study the hydromagnetic stability of magnetically confined plasma, the macroscopic fluid equations for an electrically-neutral, isotropic, infinite-conductivity plasma are used. Even though these plasma equations are a highly reduced description of a plasma, they are difficult to solve in detail, except for simple plasma-confinement schemes.

Fortunately, this reduced description of a plasma is useful to illustrate and analyze the features of most of the instabilities that lead to rapid loss of plasma from laboratory confinement experiments. If a particular plasma equilibrium is predicted to be unstable by the MHD equations, the addition of a new degree of freedom in the equilibrium (such as pressure anisotropy) is not likely to suppress the instability or greatly change its features, but rather is likely only to cause only a small quantitative change, such as a change in growth rate. In principle it is possible, but not always practical, to solve the complete macroscopic fluid equations, with the aid of large-scale digital computers, including such effects as electrical conductivity, pressure anisotropy, viscosity, etc.

5.9.1 The One-fluid MHD Equations

Continuity:

$$\nabla \cdot \rho_m \mathbf{V} + \frac{\partial \rho_m}{\partial t} = 0 \qquad (5.9.1)$$

Momentum:

$$\rho_m \left[\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla)\mathbf{V}\right] = \frac{\mathbf{J} \times \mathbf{B}}{c} - \nabla p + \rho_m \mathbf{g} \qquad (5.9.2)$$

Equation of state:

$$\frac{d}{dt}\left(\frac{p}{\rho_m{}^\gamma}\right) \equiv \left(\frac{\partial}{\partial t} + \mathbf{V}\cdot\nabla\right)\frac{p}{\rho_m{}^\gamma} = 0 \qquad (5.9.3)$$

Maxwell's equations:

$$\nabla \times \mathbf{B} = \frac{4\pi}{c}\mathbf{J} \qquad (5.9.4)$$

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t} \qquad (5.9.5)$$

$$\nabla \cdot \mathbf{B} = 0 \qquad (5.9.6)$$

$$\nabla \cdot \mathbf{E} = 4\pi\rho_q \qquad (5.9.7)$$

Ohm's law [assuming infinite conductivity, and see Eq. (3.8.21)]:

$$\mathbf{E}_\perp + \frac{\mathbf{V}\times\mathbf{B}}{c} = 0 \qquad (5.9.8)$$

$$E_\parallel = \frac{1}{ne}(\rho_m \mathbf{g} + \nabla p_i)_\parallel$$

where ∇p and $\rho_m \mathbf{g}$ have been neglected compared with $\mathbf{V}\times\mathbf{B}$, as explained in secs. 3.6 and 3.7.

5.9.2 Equilibrium Equations

The equations for the equilibrium state of a neutral plasma at rest are obtained from the one-fluid MHD equations by setting all time-dependent terms in the above equations to zero ($\partial/\partial t = \mathbf{E}_0 = \mathbf{V}_0 = 0$).

The equilibrium equations are

$$\nabla p_0 - \frac{\mathbf{J}_0 \times \mathbf{B}_0}{c} - \mathbf{g}_0 \rho_{m0} = 0 \qquad (5.9.9)$$

$$\nabla \cdot \mathbf{B}_0 = 0 \qquad (5.9.10)$$

$$\nabla \times \mathbf{B}_0 = \frac{4\pi}{c}\mathbf{J}_0 \qquad (5.9.11)$$

In equilibrium the one-fluid MHD plasma must therefore satisfy

$$\nabla p_0 = \frac{(\nabla \times \mathbf{B}_0)\times \mathbf{B}_0}{4\pi} + \mathbf{g}_0 \rho_{m0} \qquad (5.9.12)$$

5.9.3 Perturbation Equations

To determine the dynamical behavior of a plasma equilibrium or to compute the changes in energy that result from a small departure from a plasma equilibrium, the one-fluid MHD equations are perturbed about an equilibrium state. The linearized perturbation equations are obtained by substituting equilibrium variables plus appropriate perturbations into the one-fluid equations and by neglecting terms that are products of perturbation variables. The perturbation variables are defined as follows:

$$\mathbf{V}(\mathbf{x}, t) = \mathbf{V}_0(\mathbf{x}) + \mathbf{V}_1(\mathbf{x}, t) \qquad \mathbf{V}_0(\mathbf{x}) = 0 \qquad (5.9.13)$$

$$p(\mathbf{x}, t) = p_0(\mathbf{x}) + p_1(\mathbf{x}, t) \qquad |p_1| \ll |p_0| \qquad (5.9.14)$$

$$\mathbf{B}(\mathbf{x}, t) = \mathbf{B}_0(\mathbf{x}) + \mathbf{B}_1(\mathbf{x}, t) \qquad |\mathbf{B}_1| \ll |\mathbf{B}_0| \qquad (5.9.15)$$

$$\rho_m(\mathbf{x}, t) = \rho_{m0}(\mathbf{x}) + \rho_{m1}(\mathbf{x}, t) \qquad |\rho_{m1}| \ll |\rho_{m0}| \qquad (5.9.16)$$

$$\mathbf{J}(\mathbf{x}, t) = \mathbf{J}_0(\mathbf{x}) + \mathbf{J}_1(\mathbf{x}, t) \qquad |\mathbf{J}_1| \ll |\mathbf{J}_0| \qquad (5.9.17)$$

The macroscopic fluid equations for the perturbation variables are

$$\rho_{m0}\frac{\partial \mathbf{V}_1}{\partial t} = -\nabla p_1 + \frac{1}{c}\mathbf{J}_1 \times \mathbf{B}_0 + \frac{1}{c}\mathbf{J}_0 \times \mathbf{B}_1 + \rho_{m1}\mathbf{g}_0 \qquad (5.9.18)$$

$$\frac{\partial \rho_{m1}}{\partial t} = -(\mathbf{V}_1 \cdot \nabla)\rho_{m0} - \rho_{m0}\nabla \cdot \mathbf{V}_1 = -\nabla \cdot \rho_{m0}\mathbf{V}_1 \qquad (5.9.19)$$

$$\frac{\partial p_1}{\partial t} = -(\mathbf{V}_1 \cdot \nabla)p_0 - \gamma p_0 \nabla \cdot \mathbf{V}_1 \qquad (5.9.20)$$

$$\mathbf{E}_{1\perp} + \frac{\mathbf{V}_1 \times \mathbf{B}_0}{c} = 0 \qquad (5.9.21)$$

$$E_{1\parallel} = \left(\frac{1}{ne}\nabla p\right)_{1\parallel} \qquad (5.9.22)$$

$$\nabla \times \mathbf{E}_1 = -\frac{1}{c}\frac{\partial \mathbf{B}_1}{\partial t} \qquad (5.9.23)$$

$$\nabla \times \mathbf{B}_1 = \frac{4\pi}{c}\mathbf{J}_1 \qquad (5.9.24)$$

$$\nabla \cdot \mathbf{B}_1 = 0 \qquad (5.9.25)$$

Equation (5.9.20) was obtained by combining (5.9.1) and (5.9.2). Combining (5.9.21), (5.9.23), and (5.9.24) gives the perturbed magnetic field and current explicitly; i.e.,

$$\frac{\partial \mathbf{B}_1}{\partial t} = \nabla \times (\mathbf{V}_1 \times \mathbf{B}_0) \tag{5.9.26}$$

$$\frac{\partial \mathbf{J}_1}{\partial t} = \frac{c}{4\pi} \nabla \times [\nabla \times (\mathbf{V}_1 \times \mathbf{B}_0)] \tag{5.9.27}$$

This demonstrates that all the perturbation quantities can be expressed in terms of the perturbation velocity.

Taking the derivative of (5.9.18) and substituting from (5.9.19), (5.9.20), (5.9.26), and (5.9.27) gives

$$\begin{aligned} \rho_{m0} \frac{\partial^2 \mathbf{V}_1}{\partial t^2} = {} & \nabla[(\mathbf{V}_1 \cdot \nabla) p_0 + \gamma p_0 (\nabla \cdot \mathbf{V}_1)] + \frac{\mathbf{J}_0 \times [\nabla \times (\mathbf{V}_1 \times \mathbf{B}_0)]}{c} \\ & + \frac{1}{4\pi} \{\nabla \times [\nabla \times (\mathbf{V}_1 \times \mathbf{B}_0)]\} \times \mathbf{B}_0 \\ & - [(\mathbf{V}_1 \cdot \nabla) \rho_{m0} + \rho_{m0} (\nabla \cdot \mathbf{V}_1)] \mathbf{g}_0 \end{aligned} \tag{5.9.28}$$

This differential equation for the perturbation velocity, together with suitable boundary and initial conditions, describes the dynamical behavior of a one-fluid MHD plasma for small deviations from an allowed equilibrium. This equation is the starting place for a systematic study of the stability of the one-fluid-model plasma. Solutions of this equation can be expected to predict the dynamical behavior of plasma provided the following conditions hold:

1 The quadratic terms in the perturbation variable are negligible. This means that any macroscopic speed V must be smaller than the sound speed [that is, $V \ll (\gamma p/\rho_m)^{1/2}$].
2 The plasma is electrically neutral, $\mathbf{E}_0 = 0$ (that is, $\lambda_D \ll L_0$, a characteristic plasma length).
3 The ratio of electron mass to ion mass is small (that is, $m_e/m_i \ll 1$).
4 The plasma is isotropic. This means that there are many collisions during a characteristic time t_0. When this is not the case, the above system of equations should be replaced by the double-adiabatic equations or some other system of equations based on a suitable closure scheme.
5 The displacement current is negligible. This MHD approximation is valid for electromagnetic waves if the Alfvén speed is much less than the speed of light [that is, $V_A (= B/\sqrt{4\pi\rho_m}) \ll c$], and also rules out Langmuir oscillations (high-frequency waves) as a possible source of instability.

6 Heat flow by conduction along and across the magnetic field is negligible, so that the adiabatic ideal gas law equation of state is valid.
7 The plasma electrical conductivity is infinite, and perturbations are of long enough wavelength, so that the generalized Ohm's law simplifies to $\mathbf{E}_\perp + (\mathbf{V} \times \mathbf{B})/c = 0$, which neglects ∇p, etc., compared with $\mathbf{E}_\perp + \mathbf{V} \times \mathbf{B}/c$, as explained in Chap. 3.

5.9.4 Boundary Conditions

If a plasma is to be confined by a magnetic field away from material walls, it is most likely that it will be surrounded by a vacuum region. Therefore there will be a transition region over which the plasma density goes to zero. The transition may be a sharp boundary or a gradual change in plasma quantities over some finite distance. In any case it is necessary to connect the variables for the solution of the plasma equations on adjacent sides of the boundary.

The boundary conditions are expressed in terms of an incremental change of a plasma variable across the interface. This incremental change is denoted by the symbol $[\![\ ]\!]$. The unit vector normal to the interface is denoted by $\hat{\mathbf{n}}$. Thus $\hat{\mathbf{n}} \cdot [\![\mathbf{V}]\!] = 0$ means that the normal components of the velocity on the two sides of the interface are equal. Changes in the magnetic field across the boundary are associated with surface current in the boundary, denoted by $\mathbf{J}_s$. Three boundaries that are of interest are the plasma-plasma interface, the plasma-vacuum interface, and the plasma–rigid conducting boundary interface.

Plasma-plasma interface:

$$\left[\!\left[p + \frac{B^2}{8\pi} \right]\!\right] = 0 \qquad \text{(True for weak curvature; otherwise see Sec 3.8.2.)} \tag{5.9.29}$$

$$\hat{\mathbf{n}} \cdot [\![\mathbf{V}]\!] = 0 \qquad \text{(True if } [\![\rho_m]\!] = 0\text{, but not true for MHD shock waves. In that case } \hat{\mathbf{n}} \cdot [\![\rho_m \mathbf{V}]\!] = 0.\text{)} \tag{5.3.30}$$

$$\hat{\mathbf{n}} \times [\![\mathbf{E}]\!] = \hat{\mathbf{n}} \times \frac{\mathbf{V} \times [\![\mathbf{B}]\!]}{c} \tag{5.9.31}$$

$$\hat{\mathbf{n}} \cdot [\![\mathbf{B}]\!] = 0 \tag{5.9.32}$$

$$\hat{\mathbf{n}} \times [\![\mathbf{B}]\!] = \frac{4\pi}{c} \mathbf{J}_s \tag{5.9.33}$$

Plasma-vacuum interface:

$$\left[\!\left[p + \frac{B^2}{8\pi} \right]\!\right] = 0 \tag{5.9.34}$$

$$\hat{\mathbf{n}} \cdot [\![\rho \mathbf{V}]\!] = 0 \tag{5.9.35}$$

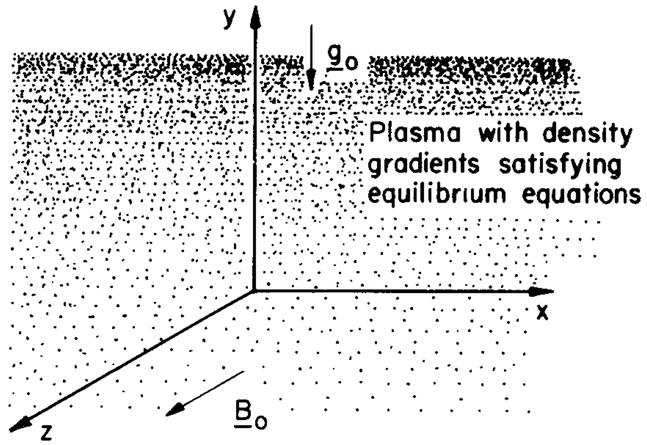


FIGURE 5.10.1
Coordinate system used in the stability analysis of plasma supported against gravity by a magnetic field.

$$\hat{\mathbf{n}} \times [\![\mathbf{E}]\!] = \hat{\mathbf{n}} \times \frac{\mathbf{V} \times [\![\mathbf{B}]\!]}{c} \qquad (5.9.36)$$

$$\hat{\mathbf{n}} \cdot [\![\mathbf{B}]\!] = 0 \qquad (5.9.37)$$

$$\hat{\mathbf{n}} \times [\![\mathbf{B}]\!] = \frac{4\pi}{c} \mathbf{J}_s \qquad (5.9.38)$$

Plasma–perfectly conducting rigid-wall interface:

$$\hat{\mathbf{n}} \times \mathbf{E} = 0 \qquad (5.9.39)$$

$$\hat{\mathbf{n}} \cdot \frac{\partial \mathbf{B}}{\partial t} = 0 \qquad (5.9.40)$$

$$\hat{\mathbf{n}} \cdot \mathbf{V} = 0 \qquad (5.9.41)$$

At an interface where there is a surface current density but no surface mass density, $\hat{\mathbf{n}} \cdot [\![\mathbf{B}]\!]$ must vanish, since a normal component of the magnetic field would cause infinite acceleration of the surface.

5.10 STABILITY OF THE FLUID PLASMA SUPPORTED AGAINST GRAVITY BY A MAGNETIC-FIELD: NORMAL-MODE ANALYSIS

Consider the situation depicted schematically in Fig. 5.10.1. An isotropic plasma is imbedded in a z-directed magnetic field [$\mathbf{B}_0 = \hat{\mathbf{z}} B_0(y)$] and in a y-directed gravitational field ($\mathbf{g}_0 = -\hat{\mathbf{y}} g_0$). The density distribution of the plasma

and magnetic field variation in the y direction are assumed to satisfy the equilibrium condition given by (5.9.12); i.e.,

$$\frac{\partial}{\partial y}\left(p_0 + \frac{B_0^2}{8\pi}\right) = -\rho_{m0} g_0 \qquad (5.10.1)$$

The dynamics of the plasma for small deviations about this equilibrium are given by the differential equation for the velocity (5.9.28). For simplicity, it is assumed that all quantities are independent of z and that the perturbation velocity has a space and time dependence given by

$$\mathbf{V}_1(x, y, t) = \hat{\mathbf{V}}_1 e^{-i\omega t} = (\hat{\mathbf{x}} \hat{V}_{1x} + \hat{\mathbf{y}} \hat{V}_{1y}) e^{-i\omega t} \qquad (5.10.2)$$

Taking the x and y components of (5.9.28) gives

$$-\omega^2 \rho_{m0} \hat{V}_{1x} = \frac{\partial}{\partial x}[(\hat{\mathbf{V}}_1 \cdot \nabla) p_0 + \gamma p_0 (\nabla \cdot \hat{\mathbf{V}}_1)] - \frac{1}{4\pi} \frac{\partial Q_{1z}}{\partial x} B_0(y) \qquad (5.10.3)$$

and

$$-\omega^2 \rho_{m0} \hat{V}_{1y} = \frac{\partial}{\partial y}[(\hat{\mathbf{V}}_1 \cdot \nabla) p_0 + \gamma p_0 (\nabla \cdot \hat{\mathbf{V}}_1)] - \frac{1}{4\pi} \frac{\partial}{\partial y}[Q_{1z} B_0(y)] + [(\hat{\mathbf{V}}_1 \cdot \nabla)\rho_{m0} + \rho_{m0}(\nabla \cdot \hat{\mathbf{V}}_1)] g_0 \qquad (5.10.4)$$

The quantity Q_{1z} is the z component of the vector $\mathbf{Q}_1$, which is defined by

$$\mathbf{Q}_1 = \nabla \times (\hat{\mathbf{V}}_1 \times \mathbf{B}_0) \qquad (5.10.5)$$

For this example there is only a z component of $\mathbf{Q}_1$; that is, $\mathbf{Q}_1 = \hat{\mathbf{z}} Q_{1z}$, where

$$Q_{1z} = -\nabla \cdot \hat{\mathbf{V}}_1 B_0 = -[(\hat{\mathbf{V}}_1 \cdot \nabla) B_0(y) + B_0(y) \nabla \cdot \hat{\mathbf{V}}_1] \qquad (5.10.6)$$

Using the definition of Q_{1z} [Eq. (5.10.6)] and the equilibrium condition of Eq. (5.10.1), Eqs. (5.10.3) and (5.10.4) become

$$-\rho_{m0} \omega^2 V_{1x} + (\rho_{m0} g_0) \frac{\partial \hat{V}_{1y}}{\partial x} = \frac{\partial}{\partial x}\left[\left(\gamma p_0 + \frac{B_0^2}{4\pi}\right)(\nabla \cdot \hat{\mathbf{V}}_1)\right] \qquad (5.10.7)$$

and

$$\rho_{m0} \omega^2 \hat{V}_{1y} - \rho_{m0} g_0 \frac{\partial \hat{V}_{1y}}{\partial y} = -\frac{\partial}{\partial y}\left[\left(\gamma p_0 + \frac{B_0^2}{4\pi}\right)(\nabla \cdot \hat{\mathbf{V}}_1)\right] - \rho_{m0} g_0 (\nabla \cdot \hat{\mathbf{V}}_1) \qquad (5.10.8)$$

Equations (5.10.7) and (5.10.8) can be combined by eliminating the term $(\gamma p_0 + B_0^2/4\pi)\nabla \cdot \mathbf{V}_1$ between them. The result is

$$\frac{\partial \hat{V}_{1x}}{\partial y} + \hat{V}_{1x} \frac{1}{\rho_{m0}} \frac{\partial \rho_{m0}}{\partial y} - \left(1 + \frac{g_0}{\omega^2} \frac{1}{\rho_{m0}} \frac{\partial \rho_{m0}}{\partial y}\right) \frac{\partial \hat{V}_{1y}}{\partial x} = \frac{g_0}{\omega^2} \frac{\partial}{\partial x}(\nabla \cdot \hat{\mathbf{V}}_1) \qquad (5.10.9)$$

As it stands, this equation cannot be solved, since it is one equation in two unknowns. An additional relation between $\hat{V}_{1x}$ and $\hat{V}_{1y}$ can be obtained from either (5.10.7) or (5.10.8). In practice, however, the approximation $\nabla \cdot \hat{\mathbf{V}}_1 = 0$ is generally used as a companion equation to (5.10.9). This amounts to either assuming the plasma to be an incompressible fluid or to restricting the calculation to those certain types of perturbations that do not compress the plasma. This can be understood by solving (5.10.7) for $\nabla \cdot \hat{\mathbf{V}}_1$; that is,

$$ik\tilde{V}_{1x} + \frac{\partial \tilde{V}_{1y}}{\partial y} = \frac{-\rho_{m0}\,\omega^2 \tilde{V}_{1x}/ik + \rho_{m0}\, g_0\, \tilde{V}_{1y}}{\gamma p_0 + B_0{}^2/4\pi} \qquad (5.10.10)$$

where $\hat{\mathbf{V}}_1(x, y) = \tilde{\mathbf{V}}_1(y)e^{ikx}$ has been introduced. There are two limits in which $\nabla \cdot \mathbf{V}_1 \to 0$. They are

1 $\gamma \to \infty$ (incompressible fluid or perturbation where $\mathbf{k} \cdot \mathbf{V}_1 = 0$)
2 $B_0{}^2 \to \infty$

The use of $\nabla \cdot \mathbf{V}_1 = 0$ to simplify the analysis of hydromagnetic instabilities is justified for situations where the hydromagnetic medium is truly incompressible (e.g., liquid mercury) and for situations where the perturbation is chosen in such a way as not to compress the plasma. In this latter case setting $\nabla \cdot \mathbf{V}_1 = 0$ is not an approximation. The third case for which this simplification is justified is in a low-β plasma, where $\nabla \cdot \mathbf{V}_1 = 0$ frequently implies simply $p \ll B^2/8\pi$. This approximation should be examined with the solutions obtained, since $\nabla \cdot \mathbf{V}_1 = 0$ is not a general *plasma* property.

Substituting $ik\tilde{V}_{1x} + \partial \tilde{V}_{1y}/\partial y = 0$ into (5.10.9) yields the differential equation for the dynamics of the plasma equilibrium of Fig. 5.10.1, i.e.,

$$\frac{d^2\tilde{V}_{1y}}{dy^2} + \left(\frac{1}{\rho_{m0}}\frac{d\rho_{m0}}{dy}\right)\frac{d\tilde{V}_{1y}}{dy} - k^2\left(1 + \frac{g_0}{\omega^2}\frac{1}{\rho_{m0}}\frac{d\rho_{m0}}{dy}\right)\tilde{V}_{1y} = 0 \qquad (5.10.11)$$

There are several simple cases for which a solution of (5.10.11) is possible.

5.10.1 Exponential Density Variations $\left(\frac{1}{\rho_{m0}}\frac{\partial \rho_{m0}}{\partial y} = \text{constant}\right)$

In this case a simple solution bounded everywhere is $d^2\tilde{V}_{1y}/dy^2 = d\tilde{V}_{1y}/dy = 0$, whereupon (5.10.11) becomes

$$k^2\left(1 + \frac{g_0}{\omega^2}\frac{1}{\rho_{m0}}\frac{d\rho_{m0}}{dy}\right)\tilde{V}_{1y} = 0 \qquad (5.10.12)$$

The nontrivial solution ($\tilde{V}_{1y} \neq 0$) is

$$\omega^2 = -g_0 \frac{1}{\rho_{m0}}\frac{d\rho_{m0}}{dy} \qquad (5.10.13)$$

[This would not be a satisfactory eigenvalue if $(1/\rho_{m0})(\partial\rho_{m0}/\partial y)$ were a function of coordinate y.] If $d\rho_{m0}/dy > 0$, that is, if the density increases in the direction opposite to the direction of the gravitational force, the plasma is unstable with growth rate $\omega_i = [(g_0/\rho_{m0})(d\rho_{m0}/dy)]^{1/2}$. If $d\rho_{m0}/dy < 0$, that is, if the plasma density increases in the direction of the gravitational force, the plasma is stable. The plasma, if disturbed, oscillates with frequency $\omega_r = [(g_0/\rho_{m0})\, d\rho_{m0}/dy]^{1/2}$.

5.10.2 Plasma with a Sharp Boundary

If ρ_{m0} = constant for $y > 0$ and if $\rho_{m0} = 0$ for $y < 0$, there is a plasma-vacuum interface at $y = 0$. For $y > 0$ the differential equation is

$$\frac{d^2V_{1y}}{dy^2} - k_x{}^2 V_{1y} = 0 \qquad (5.10.14)$$

A solution that vanishes at $y = +\infty$ is

$$V_{1y} = \hat{V}_{1y}e^{-k_x y}$$

The boundary condition at $y = 0$ is given by

$$\frac{d\rho_{m0}}{dy} = \rho_{m0}\,\delta(y) \gg \rho_{m0} \qquad \text{at } y = 0 \qquad (5.10.15)$$

At $y = 0$, the differential equation becomes

$$\frac{dV_{1y}}{dy} - \frac{k_x{}^2}{\omega^2} g_0\, V_{1y} = 0 \qquad (5.10.16)$$

Since $V_{1y} = \hat{V}_{1y}e^{-k_x y}$, the plasma is unstable, with growth rate

$$\omega_i = \sqrt{k_x g_0} \qquad (5.10.17)$$

The growth rate now depends on the wavelength of the initial perturbation. Short-wavelength disturbances grow most rapidly.

Problem 5.10.1 Discuss the physical effects omitted from the MHD model, used to obtain Eq. (5.10.17), that would prevent the growth rate from becoming arbitrarily large as $k_x \to \infty$. ////

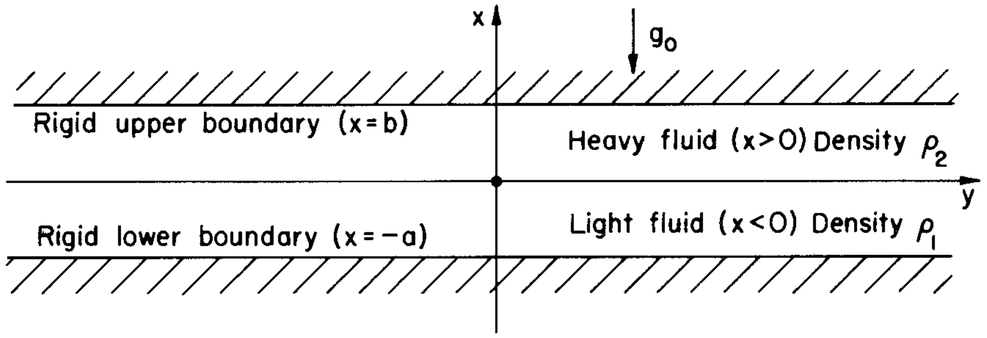


FIGURE 5.10.2
Schematic of a heavy incompressible fluid supported against gravity by a light incompressible fluid.

Problem 5.10.2 Consider the situation depicted schematically in Fig. 5.10.2, where an incompressible fluid of density ρ_2 is supported by an incompressible fluid of density $\rho_1 (\rho_2 > \rho_1)$. Show from the fluid equations

$$\frac{\partial \rho_m}{\partial t} + \nabla \cdot \rho_m \mathbf{V} = 0 \qquad (5.10.18)$$

$$\rho \frac{\partial \mathbf{V}}{\partial t} = -\nabla p - \rho_m g_0 \hat{\mathbf{x}} \qquad (5.10.19)$$

that

$$\omega^2 = -\frac{k g_0 (\rho_2 - \rho_1)}{\rho_2 \coth kb + \rho_1 \coth ka} \qquad (5.10.20)$$

and hence that the growth rate $\omega_1 \approx \sqrt{kg_0}$ for ρ_2/ρ_1, ka, and kb, all much greater than unity. ////

Problem 5.10.3 Read "Reviews of Plasma Physics," vol. 2, sec. 7, p. 174, Consultants Bureau, New York, 1966, and supply the missing steps leading to the stability criteria for a surface-layer pinch in a longitudinal magnetic field. In particular, show that if the longitudinal external field B_z is much greater than the internal field B_i, the stability condition for an infinitely long pinch is

$$\omega^2 = \frac{B_\theta{}^2}{4\pi\rho a^2}\left(\frac{m^2 B_i{}^2}{B_z'{}^2 + B_i{}^2} - m\right) \qquad (5.10.21)$$

where a is the pinch radius and the perturbation varies azimuthally as $\exp(im\,\theta)$. ////

If $B_i = B_{z,\text{ext}} = B_{z,\text{int}}$, Eq. (5.10.21) shows that the $m = 1$ (screw instability) grows and that the pinch is stable against perturbations $m \geqslant 2$. For a finite pinch the axial half-wavelength must be shorter than the length of the plasma, so that the stability criterion for a pinch of length L is

$$\frac{B_\theta}{B_z} < \frac{2\pi a}{L}$$

This is called the *Kruskal-Shafranov condition.* It is a necessary condition, and means that the pitch of the magnetic field lines must be greater than $L/2\pi a$ for stability.

5.11 ENERGY PRINCIPLE

In the preceding section, the hydrodynamic stability of a plasma supported against gravity by a magnetic field was investigated, using a normal-mode analysis. For this simple configuration it was possible to solve the plasma equations and thus obtain a complete picture of the instabilities, i.e., the growth rates and mode spectra of the instabilities. For more complicated plasma-confinement schemes, the solution of the eigenvalue problem is not always mathematically practical.

The energy principle is a method of determining the stability of a magnetically confined plasma without actually solving for the eigenfrequencies and normal modes. It is based on an investigation of the potential energy associated with small deviations from an assumed equilibrium state.

It is convenient to express the small deviations from the equilibrium state in terms of a lagrangian displacement of a fluid element. This is illustrated in Fig. 5.11.1. The coordinate of a fluid element at time t is denoted by $\mathbf{x}_0(t)$. The displaced location of this element is denoted by $\mathbf{x}(t)$, where

$$\mathbf{x}(t) = \mathbf{x}_0(t) + \xi(\mathbf{x}_0, t) \qquad |\mathbf{x}_0| \gg |\xi|,\ \xi(\mathbf{x}_0, 0) = 0 \qquad (5.11.1)$$

The fluid velocity in terms of ξ is given by

$$\mathbf{V}_1(\mathbf{x}, t) = \mathbf{V}_1(\mathbf{x}_0, t) + (\dot{\xi} \cdot \nabla)\mathbf{V}_1(\mathbf{x}_0, t) + \cdots \qquad (5.11.2)$$

where $\dot{\xi} \equiv \partial\xi/\partial t$.

Since the term $(\dot{\xi} \cdot \nabla)\mathbf{V}_1$ is second-order in the perturbation variables, it is neglected. Thus

$$\mathbf{V}_1(\mathbf{x}, t) = \mathbf{V}_1(\mathbf{x}_0, t) \qquad (5.11.3)$$

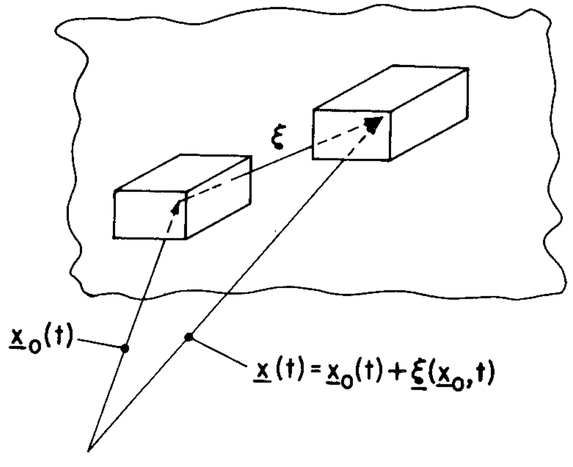


FIGURE 5.11.1
Diagram illustrating definition of lagrangian displacement of fluid element.

This means that in first order, the eulerian description of the fluid motion based on the variable $\mathbf{V}_1(\mathbf{x}, t)$, which gives the time history of the velocity at a point as fluid elements pass through it, and the lagrangian description of the motion, which gives the time history of the velocity through the derivative of the displacement, i.e.,

$$\mathbf{V}_1(\mathbf{x}_0, t) = \frac{\partial}{\partial t}\,\xi(\mathbf{x}_0, t) \qquad (5.11.4)$$

are identical. Therefore it is permissible to replace the eulerian velocity $\mathbf{V}_1(\mathbf{x}, t)$ in the plasma equations by the derivative of lagrangian displacement $\dot{\xi}(\mathbf{x}_0, t)$ defined above.

In terms of the variable ξ, the differential equation for the motion of a fluid element is [recalling Eq. (5.9.26), leading to $\mathbf{B}_1 = \nabla \times (\xi \times \mathbf{B}_0)$]

$$\rho_{m0}\frac{\partial^2 \xi}{\partial t^2} = \mathbf{F}(\xi) \qquad (5.11.5)$$

where

$$\mathbf{F}(\xi) = \nabla[\xi \cdot \nabla p_0 + \gamma p_0 \nabla \cdot \xi] + \frac{1}{4\pi}\{(\nabla \times \mathbf{B}_0) \times \mathbf{Q} + (\nabla \times \mathbf{Q}) \times \mathbf{B}_0\}$$

$$-\mathbf{g}_0(\nabla \cdot \rho_{m0}\,\xi) \qquad (5.11.6)$$

and

$$\mathbf{Q} = \nabla \times (\xi \times \mathbf{B}_0) \qquad (5.11.7)$$

The solution of this differential equation with appropriate initial and boundary conditions determines ξ, which in turn determines the perturbed velocity, and consequently all the dependent perturbation variables p_1, ρ_{m1}, $\mathbf{B}_1$, $\mathbf{J}_1$, and $\mathbf{E}_1$.

In addition to initial values for ξ, and $\dot{\xi} = \mathbf{V}_1$, the differential equation (5.11.5) must be supplemented with appropriate boundary conditions. The

plasma equilibrium state of most general interest is that in which the plasma is bounded by a vacuum region. The vacuum region in general is surrounded by a closed, simply connected, perfectly conducting rigid wall. This idealization is seldom realized in normal experimental practice; however, it establishes convenient boundary conditions for the mathematical formulation of the stability problem.

If there is no vacuum region between the plasma and the rigid conducting wall of the experiment, the boundary conditions simplify to the requirement that the tangential electric field vanish on the wall; i.e.,

$$\hat{\mathbf{n}}_w \times \mathbf{E}_1 = 0 \qquad (5.11.8)$$

where $\hat{\mathbf{n}}_w$ is the outward normal unit vector on the conducting wall. From Ohm's law this means that the tangential component of $\dot{\xi} \times \mathbf{B}_0$ must also vanish, i.e.,

$$\hat{\mathbf{n}}_w \times (\dot{\xi} \times \mathbf{B}_0) = (\hat{\mathbf{n}}_w \cdot \mathbf{B}_0)\dot{\xi} - (\hat{\mathbf{n}}_w \cdot \dot{\xi})\mathbf{B}_0 = 0 \qquad (5.11.9)$$

If $\mathbf{B}_0$ is tangent to the wall, $\hat{\mathbf{n}}_w \cdot \mathbf{B}_0 = 0$, and the boundary condition becomes

$$\hat{\mathbf{n}}_w \cdot \dot{\xi} = 0 \qquad (5.11.10)$$

That is, the normal component of ξ must vanish at the wall.

To consider the more general situation of plasma surrounded by vacuum, it is assumed that there is an equilibrium boundary surface S_0 between the plasma and the vacuum region. It is also assumed that a surface current flows in this boundary surface so that there can be a finite discontinuity in plasma pressure and magnetic field across this boundary. This is an idealization of a thin transition region between plasma and vacuum. In the equilibrium state this is a surface of constant pressure, and therefore the normal component of the equilibrium magnetic field must vanish at the surface S_0. From (5.9.34) the total pressure across the equilibrium surface must be continuous; i.e.,

$$p_0(\mathbf{x}_0) + \frac{B_0^2(\mathbf{x}_0)}{8\pi} = \frac{B_0'^2(\mathbf{x}_0)}{8\pi} \qquad (5.11.11)$$

where $\mathbf{x}_0$ denotes the surface S_0, and prime denotes variables in the vacuum region. A point on the perturbed boundary is given by $\mathbf{x} = \mathbf{x}_0 + \hat{\mathbf{n}}_0\,\xi_n$, where $\mathbf{x}_0$ is a point on the unperturbed surface S_0, $\hat{\mathbf{n}}_0$ is the outward normal unit vector, and $(\hat{n}_0 \cdot \xi) = \xi_n$ is the component of the displacement normal to the surface S_0 at the point $\mathbf{x}_0$. At this displaced boundary, the total pressure must also be continuous; i.e.,

$$p_0(\mathbf{x}) + p_1(\mathbf{x}) + \frac{1}{8\pi}[\mathbf{B}_0(\mathbf{x}) + \mathbf{B}_1(\mathbf{x})]^2 = \frac{1}{8\pi}[\mathbf{B}_0'(\mathbf{x}) + \mathbf{B}_1'(\mathbf{x})]^2 \qquad (5.11.12)$$

Expanding the equilibrium pressure at the displaced boundary in terms of its value at the equilibrium position gives

$$p_0(\mathbf{x}) = p_0(\mathbf{x}_0) + (\mathbf{x} - \mathbf{x}_0) \cdot \nabla p_0(\mathbf{x}_0) + \cdots$$
$$= p_0(\mathbf{x}_0) + \xi_n \hat{\mathbf{n}}_0 \cdot \nabla p_0(\mathbf{x}_0) + \cdots \quad (5.11.13)$$

since $\mathbf{x} - \mathbf{x}_0 = \xi_n \hat{\mathbf{n}}_0$. It is useful to define the normal derivative $\partial/\partial n < \hat{\mathbf{n}}_0 \cdot \nabla_{\mathbf{x}_0}$. The perturbation pressure at the displaced boundary is related to the displacement ξ by replacing $\mathbf{V}_1$ with $\partial\xi/\partial t$ in (5.9.20) and integrating with respect to time. The result is

$$p_1(\mathbf{x}) = -\xi \cdot \nabla p_0(\mathbf{x}) - \gamma p_0(\mathbf{x}) \nabla \cdot \xi \quad (5.11.14)$$

The magnetic-pressure term, to first order, is

$$\mathbf{B}_0{}^2(\mathbf{x}) + 2[\mathbf{B}_0(\mathbf{x}) \cdot \mathbf{B}_1(\mathbf{x})] \quad (5.11.15)$$

Expanding the equilibrium magnetic pressure at the displaced boundary in terms of its value at the equilibrium boundary gives

$$\mathbf{B}_0{}^2(\mathbf{x}) = \mathbf{B}_0{}^2(\mathbf{x}_0) + (\xi \cdot \nabla)\mathbf{B}_0{}^2(\mathbf{x}_0) \quad (5.11.16)$$

To first order

$$\mathbf{B}_0(\mathbf{x}) \cdot \mathbf{B}_1(\mathbf{x}) = \mathbf{B}_0(\mathbf{x}_0) \cdot \mathbf{B}_1(\mathbf{x}_0) \quad (5.11.17)$$

One of the two boundary conditions that must be satisfied by the perturbation variables at the equilibrium boundary S_0 is obtained by inserting the above expansions (5.11.13), (5.11.14), and (5.11.17) into (5.11.12), using the equilibrium condition (5.11.11). The boundary condition is

$$-\gamma p_0(\mathbf{x}_0)\nabla \cdot \xi + \frac{\mathbf{B}_0(\mathbf{x}_0) \cdot \mathbf{B}_1(\mathbf{x}_0)}{4\pi} + \xi_n \frac{\partial}{\partial n} \mathbf{B}_0{}^2(\mathbf{x}_0)$$
$$= \frac{\mathbf{B}_0'(\mathbf{x}_0) \cdot \mathbf{B}_1'(\mathbf{x}_0)}{4\pi} + \xi_n \frac{\partial}{\partial n} \mathbf{B}_0'^2(\mathbf{x}_0) \quad (5.11.18)$$

where $\partial p_0/\partial x_{||} = 0$ has been used. The second boundary condition that must be satisfied by the perturbation variables at the plasma-vacuum interface is obtained from the condition that the tangential component of the electric field must be continuous across a stationary boundary. From this it follows that across a boundary moving with $\mathbf{V}_1$, the tangential electric field $\mathbf{E}'$ in a frame moving with $\mathbf{V}_1$ must be continuous. By transformation

$$(\mathbf{E}_1')_{\tan} = \left(\mathbf{E}_1 + \frac{\mathbf{V}_1 \times \mathbf{B}_0}{c}\right)_{\tan} \quad (5.11.19)$$

is continuous across the boundary, where $\mathbf{E}_1$ and $\mathbf{B}_0$ are the fields in the laboratory frames. Ohm's law for the perfectly conducting plasma requires that

$$\mathbf{E}_1 + \frac{\mathbf{V}_1 \times \mathbf{B}_0}{c} = 0$$

Thus, using (5.11.19), the conditions

$$\hat{\mathbf{n}}_0 \times \mathbf{E}_1' = \frac{1}{c}(\mathbf{V}_1 \cdot \hat{\mathbf{n}}_0)\mathbf{B}_0' \quad (5.11.20)$$

must hold at either the perturbed or the unperturbed boundary, since $\mathbf{E}'$ and $\mathbf{V}_1 \times \mathbf{B}_0'$ are already first-order in ξ. Since the normal component of the magnetic field can be expressed in terms of the tangential component of the electric field through

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t}$$

Eq. (5.11.20) can also be written

$$\hat{\mathbf{n}}_0 \cdot \mathbf{B}_1' = \hat{\mathbf{n}}_0 \cdot [\nabla \times (\xi \times \mathbf{B}_0')] \quad (5.11.21)$$

The boundary conditions at the plasma-vacuum interface connect the solutions of the plasma equation with the solution of Maxwell's equations in the vacuum region. In the vacuum region the electric and magnetic fields can be written in terms of a vector potential $\mathbf{A}$ as follows:

$$\mathbf{E}_1' = -\frac{1}{c}\frac{\partial \mathbf{A}}{\partial t} \quad (5.11.22)$$

$$\mathbf{B}_1' = \nabla \times \mathbf{A} \quad (5.11.23)$$

where $\nabla \cdot \mathbf{A} = 0$ is the gauge condition. There are no currents in the vacuum region, and

$$\nabla \times \nabla \times \mathbf{A} = 0 \quad (5.11.24)$$

In terms of the vector potential, the boundary condition (5.11.21) can be written

$$\hat{\mathbf{n}}_0 \times \mathbf{A} = (\hat{\mathbf{n}}_0 \cdot \xi)\mathbf{B}_0' \quad (5.11.25)$$

The electromagnetic boundary conditions on the rigid conducting wall can be written in terms of the vector potential, i.e.,

$$\hat{\mathbf{n}}_w \times \mathbf{A} = 0 \quad (5.11.26)$$

The differential equation for the motion of a fluid element is [from (5.11.5)]

$$\rho_{m0}\frac{\partial^2 \xi}{\partial t^2} = \mathbf{F}(\xi) \equiv -\mathbf{K} \cdot \xi \quad (5.11.27)$$

where $\mathbf{F}(\xi)$, given by (5.11.6), is linear and contains no time derivatives and K is an operator defined by (5-11.27) and (5.11.6). The eigenvalues of this differential equation are obtained by assuming solutions of the form

$$\xi(\mathbf{x}, t) = \xi(\mathbf{x})e^{-i\omega t} \quad (5.11.28)$$

and then by solving the eigenvalue equation

$$-\omega^2 \rho_{m0} \xi = \mathbf{F}(\xi) \quad (5.11.29)$$

subject to the appropriate boundary conditions.

The solutions are the normal modes of the plasma, and depending on the sign of the imaginary part of the eigenfrequency of a normal mode, that mode is either oscillatory, damped, or unstable (i.e., growing in time). For the hydromagnetic model of a plasma the operator K is self-adjoint, and as a result, the square of an eigenfrequency is either positive ($\omega^2 > 0$ corresponds to a stable normal mode) or negative ($\omega^2 < 0$ corresponds to an unstable normal mode).

Problem 5.11.1 Read the paper of I. B. Bernstein, E. A. Frieman, M. D. Kruskal, and R. M. Kulsrud, *Proc. Roy. Soc.* (*London*), **A244**: 17 (1958), and show that K is self-adjoint. Omit the term $\mathbf{g}_0(\nabla \cdot \rho_{m0}\xi)$ from K. ////

Since the operator K is self-adjoint, the eigenvalue equation

$$\rho_{m0}\omega^2 \xi = \mathsf{K} \cdot \xi \quad (5.11.30)$$

can be solved by a variational principle $\delta(\omega^2) = 0$, where

$$\omega^2 = \frac{\int \xi \cdot \mathsf{K} \cdot \xi \, d\mathbf{x}}{\int \rho_{m0} \xi \cdot \xi \, d\mathbf{x}} \quad (5.11.31)$$

The denominator is positive definite, and so the sign of ω^2 is determined by the sign of the numerator. To see the significance of this sign, consider (5.11.30) as the equation of motion of an oscillator with spring constant K and mass ρ_{m0}. The potential energy of such an oscillator is

$$\delta W = \frac{1}{2} \int \xi \cdot \mathsf{K} \cdot \xi \, d\mathbf{x} \quad (5.11.32)$$

Thus the stability of the hydromagnetic plasma can be determined by calculating the change in the potential energy that results from a small deviation ξ from an allowed equilibrium solution. If $\delta W > 0$, the plasma is stable, and if $\delta W < 0$, the plasma is unstable.

The potential energy associated with a small deviation from an allowed equilibrium of the hydromagnetic equations is[1]

$$\delta W = \frac{1}{2} \int_{\substack{\text{plasma} \\ \text{vol}}} \left\{ \gamma p_0 (\nabla \cdot \xi)^2 + \frac{\mathbf{Q} \cdot \mathbf{Q}}{4\pi} + (\xi \cdot \nabla) p_0 \nabla \cdot \xi \right.$$
$$\left. - \frac{1}{4} [\xi \times (\nabla \times \mathbf{B}_0)] \cdot \mathbf{Q} \right\} d\mathbf{x} + \frac{1}{8\pi} \int_{\substack{\text{vacuum} \\ \text{vol}}} \mathbf{B} \cdot \mathbf{B} \, d\mathbf{x}$$
$$- \frac{1}{2} \int_{\substack{\text{plasma} \\ \text{surface}}} \left(\frac{\partial p_0}{\partial n} + \frac{1}{8\pi} \frac{\partial B_0^2}{\partial n} - \frac{1}{8\pi} \frac{\partial B_0'^2}{\partial n} \right) (\xi \cdot \hat{\mathbf{n}}_0)^2 \, ds \quad (5.11.33)$$

Problem 5.11.2 Prove that Eq. (5.11.32) follows from the fact that the total energy of the plasma is a constant and that the operator K is self-adjoint, i.e., that

$$\int \eta \cdot \mathsf{K} \cdot \xi \, d\mathbf{x} = \int \xi \cdot \mathsf{K} \cdot \eta \, d\mathbf{x} \quad (5.11.34)$$

where ξ and η are small deviations from an allowed equilibrium. The deviations ξ and η and the associated vector potentials satisfy the boundary conditions (5.11.25) and (5.11.26) at the plasma-vacuum interface and conducting wall, respectively. ////

Problem 5.11.3 Verify (5.11.34).

5.12 STABILITY OF A PLANE PLASMA-MAGNETIC FIELD INTERFACE: ENERGY PRINCIPLE ANALYSIS

Consider the equilibrium shown in Fig. 5.12.1. Plasma is confined to $x < 0$ by a magnetic field $\mathbf{B}_0$. This field vanishes inside the plasma and is constant $\mathbf{B}_0 = \hat{\mathbf{z}} B_0$ for $x > 0$. All plasma currents flow along the plasma-vacuum interface. The expression for the change of potential energy as a result of a perturbation about this equilibrium follows from (5.11.33), which simplifies to

[1] B. B. Kadomtsev, in M. A. Leontovich (ed.), "Reviews of Plasma Physics," vol. 2, p. 158, Consultants Bureau, New York, 1966.
I. B. Bernstein et al., *Proc. Roy. Soc.* (*London*), **A744**: 17 (1958).
M. N. Rosenbluth and C. L. Longmire, *Ann. Phys.*, **1**: 120 (1957).

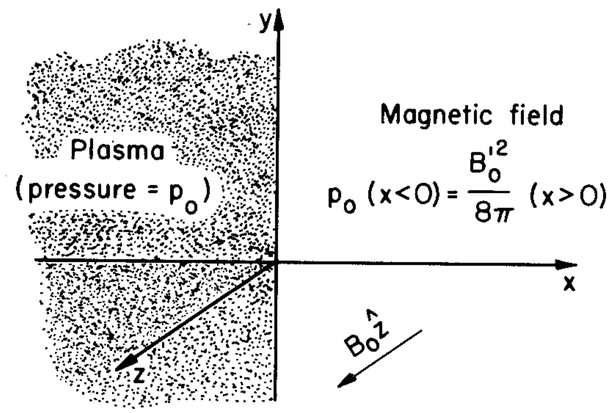


FIGURE 5.12.1
Schematic of magnetically confined plasma filling left half-space.

$$\delta W = \frac{1}{2}\int_{\substack{\text{plasma}\\ \text{vol}\ (x<0)}} \gamma p_0 (\nabla\cdot\boldsymbol{\xi})^2\, d\mathbf{x} + \frac{1}{8\pi}\int_{\substack{\text{magnetic}\\ \text{field vol}\ (x>0)}} \mathbf{B}_1\cdot\mathbf{B}_1\, d\mathbf{x}$$

$$+\frac{1}{16\pi}\int_{\substack{\text{plasma–vacuum}\\ \text{surface}\ (x=0)}} \frac{\partial B_0'^2}{\partial n}(\boldsymbol{\xi}\cdot\hat{\mathbf{n}}_0)^2\, ds \qquad (5.12.1)$$

From the first term in (5.12.1) it is seen that the contribution to the change in potential energy due to a perturbation $\boldsymbol{\xi}$ within the plasma volume is either positive or zero. That is, the first integral is a stabilizing term, except for perturbations where $\nabla\cdot\boldsymbol{\xi} = 0$. Such perturbations are possible for low-β plasma and for plasmas where $\gamma \to \infty$ (Sec. 5.10). These perturbations are the most dangerous in terms of plasma stability, since a smaller destabilizing effect from the third term in (5.12.1) will make the plasma unstable. The second term in (5.12.1) is always positive and stabilizing. It is just the change in the magnetic stored energy, associated with the confining magnetic field, which results from perturbing the vacuum field. Since the first two terms in (5.12.1) are positive, the plasma will be hydromagnetically stable in any confinement configuration for which $\partial B_0'^2/\partial n > 0$, that is, for a configuration in which the magnetic field strength increases away from the plasma surface. A simple example of such a configuration is that of a magnetic cusp, which is created by a pair of coils, arranged and carrying currents as shown in Fig. 5.12.2. Locally, the surface might be considered plane, as in Fig. 5.12.1. Since the magnetic field is everywhere curved into the plasma, the field strength increases away from the plasma, and the surface is stable. (As the plasma expands, it must do work compressing the magnetic field.) This same conclusion

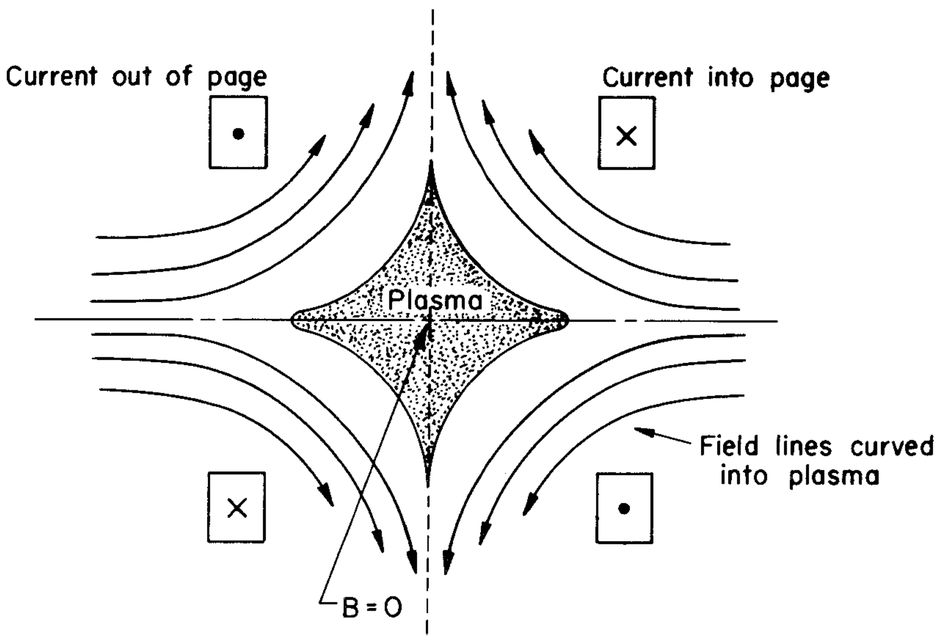


FIGURE 5.12.2
Schematic of cusp magnetic field configuration showing confined plasma with magnetic field curved into plasma everywhere.

was reached in the direct calculation of the interchange instability (Sec. 5.8), where it was shown that regions where the magnetic field curves into the plasma are stable. Although the cusp configuration is hydromagnetically stable, it has the undesirable feature that the magnetic field is zero at some point on the axis between the two coils. This region of zero magnetic field acts like a hole in the magnetic bottle and causes a rapid loss of plasma. Ioffe[1] showed that a magnetic field configuration can be realized in which $\partial B_0'^2/\partial n > 0$ and there is no region of zero magnetic field, and that such a field also suppresses hydromagnetic instabilities. His experiments are discussed in more detail in Part Four of this chapter.

If $\partial B_0'^2/\partial n < 0$, then the plasma may be unstable. To investigate this possibility, consider the situation of Fig. 5.12.1 and assume a perturbation of the plasma-vacuum interface of the form $[(\hat{\mathbf{n}}_0 = \hat{\mathbf{x}},\ \mathbf{B}_0' = B_0'\hat{\mathbf{z}})]$

$$\xi_x = \xi_0\, e^{ik_y y} e^{ik_z z} \qquad (5.12.2)$$

[1] Yu. V. Gott, M. S. Ioffe, V. G. Tel'kovskii, *Nucl. Fusion*, **1962** (suppl. part 3): 1045 (1962).

From the boundary condition (5.11.21)

$$B'_{1x} = \xi_x k_z B'_0 \qquad (5.12.3)$$

Equation (5.12.3) relates the amplitude of one component of the perturbed magnetic field at the plasma-vacuum boundary to the amplitude of the perturbation and the unperturbed vacuum magnetic field. To compute the change of energy associated with the perturbed magnetic field, the second term in (5.12.1), i.e.,

$$\frac{1}{8\pi}\int \mathbf{B}_1 \cdot \mathbf{B}_1 \, d\mathbf{x} \qquad (5.12.4)$$

must be evaluated. From Maxwell's equations in the vacuum region it follows that

$$\mathbf{k} \times (\mathbf{k} \times \hat{\mathbf{B}}_1) = 0 \qquad (5.12.5)$$

and

$$\mathbf{k} \cdot \hat{\mathbf{B}}_1 = 0 \qquad (5.12.6)$$

where $\mathbf{B}_1 = \hat{\mathbf{B}}_1 \exp(i\mathbf{k} \cdot \mathbf{x})$ is assumed. From (5.12.5) and (5.12.6)

$$k_x = i\sqrt{k_y^2 + k_z^2} \qquad (5.12.7)$$

Thus

$$\mathbf{B}_1 \cdot \mathbf{B}_1 = (|\hat{B}_{1x}|^2 + |\hat{B}_{1y}|^2 + |\hat{B}_{1z}|^2)\exp(-2\sqrt{k_y^2 + k_z^2}\,x) \qquad (5.12.8)$$

Also, from (5.12.6) and $(\mathbf{k} \times \hat{\mathbf{B}}_1)_x = 0$, it follows that

$$\mathbf{B}_1 \cdot \mathbf{B}_1 = 2|\hat{B}_{1x}|^2 \exp(-2\sqrt{k_x^2 + k_y^2}\,x) = 2\xi^2 B_0'^2 k_z^2 \exp(-2\sqrt{k_y^2 + k_z^2}\,x) \qquad (5.12.9)$$

Thus

$$\frac{1}{8\pi}\int_{\text{vol}} \mathbf{B}_1 \cdot \mathbf{B}_1 \, d\mathbf{x} = \int_{\text{surface}} \frac{\xi^2 B_0'^2 k_z^2}{8\pi\sqrt{k_y^2 + k_z^2}}\, ds \qquad (5.12.10)$$

The change in potential energy per unit area of surface (for a $\nabla \cdot \xi = 0$ perturbation) is

$$\delta W_s = \left(\frac{1}{8\pi}\frac{k_z^2}{\sqrt{k_z^2 + k_y^2}} B_0'^2 + \frac{1}{16\pi}\frac{\partial B_0'^2}{\partial n}\right)\xi_0^2 \qquad (5.12.11)$$

If k_z is small, then $\mathbf{k}$ is perpendicular to $\mathbf{B}_0$, and the condition for an instability is that $\partial B_0'^2/\partial n < 0$. The most unstable situation occurs for $k_z = 0$, that is, for perturbations that are not a function of the coordinate in the direction of the magnetic field. For this situation, the perturbations are flutelike. The reason why this perturbation is unstable when $\partial B_0'^2/\partial n < 0$ is that the plasma moves into a region where the plasma pressure p_0 is greater than the magnetic pressure, and hence the perturbation grows.

Thus a magnetically confined plasma is hydromagnetically stable provided the magnetic field is curved into the plasma everywhere. It is hydromagnetically unstable if the magnetic field is curved away from the plasma everywhere.

5.13 STABILITY OF SELF-CONFINED PLASMA (B_θ ONLY): ENERGY PRINCIPLE ANALYSIS

In many respects, the linear pinch plasma would be the simplest plasma-confinement scheme if it were stable and if the problem of end electrodes could be overcome. The reason it would be the simplest is that the confining magnetic field arises from currents in the plasma. Thus there is no need to take up the rather substantial mechanical forces associated with magnetic fields produced by current-carrying conductors external to the plasma. Unfortunately, the simple pinch plasma is unstable to a variety of perturbations. For simplicity it is assumed that the plasma fills a conducting cylinder of radius a. The equilibrium condition becomes

$$\frac{dp_0}{dr} = \frac{B_0}{4\pi r}\frac{d}{dr}(rB_0) \qquad (5.13.1)$$

This equilibrium assumes no currents other than plasma currents, i.e., no external magnetic field. To test the stability of this equilibrium, assume that it is displaced by a perturbation which in cylindrical coordinates has the form

$$\xi(r, \theta, z) = \xi_0(r, z)e^{im\theta} \qquad (5.13.2)$$

5.13.1 Azimuthally Symmetric Perturbation ($m = 0$)

The change in potential energy for an azimuthally symmetric perturbation is

$$\delta W = \frac{1}{2}\int\left\{\gamma p_0\left[\frac{1}{r}\frac{\partial}{\partial r}(r\xi_r) + \frac{\partial \xi_z}{\partial z}\right]^2 + \frac{1}{4\pi}\left[B_0\frac{\partial \xi_z}{\partial z} + \frac{\partial}{\partial r}(B_0\xi_r)\right]^2 + \xi_r\frac{dp_0}{dr}\left[\frac{1}{r}\frac{\partial}{\partial r}(r\xi_r) + \frac{\partial \xi_z}{\partial z}\right] + \xi_r\frac{dp_0}{dr}\left[\frac{\partial \xi_z}{\partial z} + \frac{1}{B_0}\frac{\partial}{\partial r}(\xi_r B_0)\right]\right\} d\mathbf{x} \qquad (5.13.3)$$

The integrand of (5.13.3) is a quadratic form in the variables ξ_r and $\nabla \cdot \xi$. These variables are independent since ξ_r and ξ_z are independent. Quadratic forms of

several variables (that is, $\sum a_{ij}x_i x_j$) are positive definite if all principal minors of the matrix a_{ij} are positive.[1] This, along with the equilibrium condition

$$\nabla p_0 + \frac{(\nabla \times \mathbf{B}_0) \times \mathbf{B}_0}{4\pi} = 0$$

can be applied to (5.13.3) to obtain the following stability criterion:

$$-\frac{r}{p_0}\frac{dp_0}{dr} < \frac{4\gamma}{2+\gamma\beta} \qquad \text{for stability} \qquad (5.13.4)$$

where $\beta = p_0/(B_0{}^2/8\pi)$. Equation (5.13.4) shows that the pressure must not fall off too rapidly with radius if the plasma is to be stable. It should be noted that, in general, β is a function of radius. Equation (5.13.1) relates $\beta(r)$ to $p_0(r)$. The equilibrium condition (5.13.1) can be rewritten as follows:

$$\frac{r}{p_0}\frac{dp_0}{dr} = \frac{1}{1+\beta}\left(\frac{r}{\beta}\frac{d\beta}{dr} - 2\right) \qquad \text{for equilibrium} \qquad (5.13.5)$$

The most unstable modes will occur for low-β plasmas. For low β, a distribution $p_0 \propto 1/r^{2\gamma}$ is stable. A more rapid decrease of density with radius is unstable. The instability that occurs if (5.13.4) is not satisfied is an interchange mode in which the plasma contracts radially at some location along the axis. This causes a local increase in $B_0{}^2$ which further increases the initial perturbation.

5.13.2 Azimuthally Dependent Perturbation ($m \neq 0$)

The perturbation variable $\boldsymbol{\xi}$ in component form can be written, without loss of generality, as follows:

$$\xi_r = \xi_r(r) \sin m\theta\, e^{ik_z z} \qquad (5.13.6)$$

$$\xi_\theta = \xi_\theta(r) \cos m\theta\, e^{ik_z z} \qquad (5.13.7)$$

$$\xi_z = \xi_z(r) \sin m\theta\, e^{ik_z z} \qquad (5.13.8)$$

This choice simplifies δW from (5.11.33) since the factors $\sin^2\theta$ and $\cos^2\theta$ average to a value of $\frac{1}{2}$. The component of displacement along the magnetic field ξ_θ occurs only in the first term of (5.11.34), and if $m \neq 0$, ξ_θ can be chosen so that $\nabla \cdot \boldsymbol{\xi} = 0$. For these perturbations the change in potential energy is the same as that of (5.13.6), except that the term $\gamma p_0(\nabla \cdot \boldsymbol{\xi})^2$ is replaced by

$$\frac{1}{4\pi}\frac{m^2 B_0{}^2}{r^2}(\xi_r{}^2 + \xi_z{}^2) \qquad (5.13.9)$$

[1] In the present case this means that $a(\nabla \cdot \boldsymbol{\xi})^2 + b\xi_r(\nabla \cdot \boldsymbol{\xi}) + c\xi_r{}^2$ is positive definite if $b^2 < 4ac$ and $a > 0$.

The component ξ_z appears only in the term $\nabla \cdot \boldsymbol{\xi}_\perp$ as a product $k_z\xi_z$. Thus, if $k_z \to \infty$ while $k_z\xi_z$ = constant, the term $m^2 B_0{}^2\xi_z{}^2/4\pi r^2 \to 0$. This again leads to a quadratic form in the two variables ξ_r and $\nabla \cdot \boldsymbol{\xi}_\perp$ for the change in potential energy as a result of a perturbation away from equilibrium. The stability criterion is

$$-\frac{r}{p_0}\frac{dp_0}{dr} < \frac{m^2}{\beta} \qquad \text{for stability} \qquad (5.13.10)$$

This criterion can be compared with (5.13.4) to determine if this is a more or less restrictive stability condition: If the stability condition (5.13.4) is satisfied by a particular equilibrium distribution, is that same distribution unstable for some azimuthally dependent perturbation? Since $\gamma > 1$, perturbations with $m \geq 2$ are stable if (5.13.4) is satisfied. If $\beta > 2\gamma/3$, then the $m = 1$ mode can be unstable even though the pressure distribution satisfies (5.13.4). The $m = 1$ mode is called the *kink instability*. It develops as a result of the magnetic pressure increase that occurs, as shown in Fig. 1.11.3. For a plasma with a sharp boundary and no walls, all m values are unstable. A current-carrying rod produces a B_θ, which reduces β in the plasma and can be a stabilizing influence, provided dp_0/dr is not too large.

5.14 STABILIZING INFLUENCE OF LINE TYING

In Sec. 5.8 it was shown that the plasma confined in a magnetic mirror is unstable to the flute or interchange instability. The analysis assumed that the ends of the lines of force were free to move. However, if the field lines were all frozen in fixed positions at the two mirror peaks, it would be impossible to interchange two flux tubes without twisting the ends and increasing the magnetic energy. Thus, fixing the field lines at the mirror peaks (or any other plane or pair of planes) is a stabilizing influence on the mirror confined plasma. This situation might occur in an experiment as a result of cold plasma, which is a good conductor, filling the external region from the mirror peak to the conducting wall of the vacuum system.

To investigate the stabilizing influence of line tying, consider the magnetic mirror field shown schematically in Fig. 5.14.1. Two adjacent tubes of flux are interchanged at the midplane. They cannot interchange at the end planes because they are tied by conducting end plates. Thus $\nabla \cdot \boldsymbol{\xi}$ is no longer zero, but is given approximately by $\xi(\nabla W/W)$, where $W = \oint dl/B$. As a result of the bending of the magnetic field lines,

$$B_1 \approx B_0 \frac{\partial \xi_r}{\partial z} \approx \frac{\pi}{L}\xi_r B_0$$

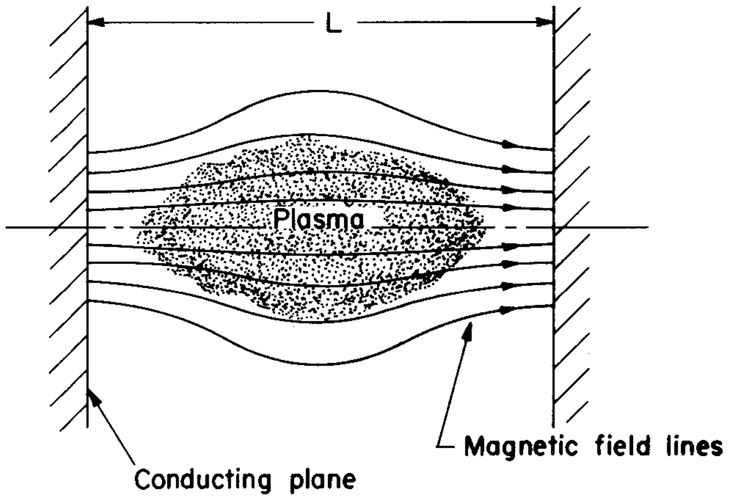


FIGURE 5.14.1
Plasma confined by a magnetic mirror, illustrating line tying by conducting end plates.

The stability condition from the energy principle is given approximately by [from (5.11.33)], neglecting the last term in the integral]

$$\delta W \approx \frac{1}{2}\int \left[\gamma p_0 \left(\frac{\nabla W}{W}\right)^2 + \frac{\pi^2 B^2}{4L^2} + \frac{|\nabla p_0|\,|\nabla W|}{W}\right] \xi_r^{\,2}\, d\mathbf{x} \qquad (5.14.1)$$

The stability condition is

$$-\frac{|\nabla p_0|\,|\nabla W|}{W} < \gamma p_0 \left(\frac{\nabla W}{W}\right)^2 + \frac{\pi B^2}{4L^2} \qquad (5.14.2)$$

in contrast with (5.8.19). If the separation between the conducting end plates becomes larger, the stabilizing effect of the end plates diminishes, and (5.14.2) reduces to (5.8.19).

Problem 5.14.1 Show that a plasma confined by the field of a magnetic dipole imbedded in a conducting sphere is [from (5.14.2)] stable if

$$\frac{aB_0^{\,2}}{8\pi} + 4\gamma p > -\frac{r\, dp_0}{dr} \qquad (5.14.3)$$

where a is a constant of order unity. ////

PART FOUR: STABILITY THEORY AND CONTROLLED THERMONUCLEAR FUSION RESEARCH

About 1952, classified projects to develop a controlled thermonuclear fusion reactor were started at four laboratories by the U.S. Atomic Energy Commission under the code name of Project Sherwood.[1] These laboratories are the Lawrence Livermore Laboratory, Livermore, California; the Los Alamos Scientific Laboratory, Los Alamos, New Mexico; the Oak Ridge National Laboratory, Oak Ridge, Tennessee; and the Princeton University Plasma Physics Laboratory, Princeton, New Jersey. Similar projects were started in England, the U.S.S.R., France, and other countries at about the same time. These projects were all started with the expectation that it would require only a modest effort to demonstrate the feasibility of a controlled fusion reactor. By 1958, the difficulty and complexity of the problem of confinement of high-temperature plasmas became obvious, and the countries engaged in this research decided to pool their knowledge for mutual benefit.[2]

It is beyond the scope of this book to give a detailed description of the many experiments that seek to confine hot plasma; however, an understanding of the plasma-stability problem is not complete without some knowledge of the interplay between theory and experiment in this active area of plasma research.

The ultimate common objective of all the plasma-confinement experiments is the profitable release of nuclear energy from the fusion of light nuclei such as those of deuterium and tritium. To cause such fusion, these nuclei must be made to collide with sufficient relative velocity to overcome the Coulomb force associated with their nuclear charge. The center-of-mass energy per nuclei required for a useful fusion reactor is in the range of 10 to 100 keV.[3]

Lawson[4] considered the simple problem of how much energy would be required to assemble and maintain a collection (plasma) of energetic light nuclei and how much energy would be released by fusion reaction among these nuclei as a function of density and temperature. He arrived at the conclusion that the product of the particle density in a high-temperature plasma ($T > 10$ keV) and the confinement at that density should exceed a particular value; that is, [n(particles per cubic centimeter)] × [τ(confinement time in seconds)] $\geq 10^{14}$ for $d-t$ reactions ($n\tau > 10^{16}$ for $d-d$ reactions). This is known as the *Lawson condition*. The short-range goal of controlled-fusion research is to produce a plasma that meets this requirement. The long-range goal is to generate electric

[1] A. S. Bishop, "Project Sherwood," Addison-Wesley, Reading, Mass., 1958.
[2] Proceedings of the 1958 Second United Nations International Conference on the Peaceful Uses of Atomic Energy, vol. 31, Theoretical and Experimental Aspects of Nuclear Fusion, and vol. 32, Controlled Fusion Devices, United Nations, Geneva.
[3] D. J. Rose and M. Clark, Jr. "Plasmas and Controlled Fusion," Wiley, New York, 1961.
[4] J. D. Lawson, *Proc. Phys. Soc.* (*London*), **B70**: 6 (1957).

power economically, and to tap the enormous energy reservoir available through the fusion of deuterium and other naturally occurring fusionable nuclei. To gain a perspective of the size of this energy source, consider the fact that in 1975 the world energy consumption is estimated to be 3.4×10^{17} Btu (British thermal units) per year. The energy available through fusion of deuterium would supply world needs for 10^9 years if energy were used at 50 times the present rate.

Early plasma-confinement experiments were able to achieve values of $n\tau \approx 10^8$. Some of the best experiments now in progress (1972) achieve values of $n\tau \approx 10^{12}$, and it is believed by many of those working in the field that there is no fundamental reason why both the short- and long-range goals should not be achieved.

A plasma with kinetic temperature in excess of a few electron volts cannot be confined in contact with solid material walls, since either the walls would vaporize and ionize or the plasma would be rapidly cooled by contact with lower-temperature walls. This latter situation is the usual problem. It is now generally accepted that a steady or slowly changing magnetic field will be the thermal insulator that will confine the high-temperature plasma in a fusion reactor away from material walls.

A homogeneous magnetic field constrains a charged particle to move along a tube of flux in a helical orbit with the cyclotron radius $a_c(=v_\perp/\omega_c)$. Such a particle is confined in the direction perpendicular to the magnetic field, but is obviously not confined along the magnetic field direction. There are two approaches to solve the problem of particle confinement along the magnetic field. One approach is to increase the magnetic field strength at two places. This creates a *magnetic bottle*, called a magnetic mirror field. This magnetic field configuration can trap and confine certain classes of single charged particles in a manner discussed below (Sec. 5.15). This configuration is called an *open system* because the particles that escape through the magnetic mirror are lost.

The other approach to the problem of particle losses along field lines is to bend the field lines around in closed race-track-like circles. Such a field is created by a current-carrying coil wound on the surface of a torus, thus creating a toroidal magnetic field. Magnetic bottles with closed field lines are called *closed systems*. A simple toroidal magnetic field does not confine single particles. This problem and its remedy are discussed in Sec. 5.16.

Both the open and closed systems of plasma confinement have in common the problem of confining the plasma in the direction perpendicular to the magnetic field. The plasma pressure $p_\perp = nkT$ across the field must be balanced by the magnetic pressure $B^2/8\pi$. At a plasma temperature above 10 keV and a density of 10^{14} cm^{-3}, the magnetic field required to confine a plasma will be in the 50- to 100-kG range. A magnetic field of 100 kG corresponds to 400 atm of pressure and a magnetic energy density of 40 J/cm^3.

Ignoring losses of plasma along magnetic field lines, there remains the problem of plasma loss across the magnetic field. The loss of energy due to radiation associated with a plasma is also ignored here. These radiation losses (bremsstrahlung, synchrotron, and impurity-radiation) are discussed in Chap. 11. The most serious plasma-loss mechanisms are the hydromagnetic instabilities described earlier in this chapter. The growth time associated with these instabilities is small, and the speed with which plasma is carried to the walls of the confinement experiment is rapid. The best value of $n\tau$ obtained in hydromagnetically unstable plasmas is about 10^8, clearly a long way from the value $n\tau = 10^{14}$ needed for controlled fusion.

If the plasma were stable and the only loss mechanism were that of classical diffusion due to the particle collisions and the plasma pressure gradient, it would be possible to have a value of $n\tau$ greater than the Lawson limit of 10^{14}. The hydromagnetic instabilities have been suppressed, and the present limitation of $n\tau \approx 10^{12}$ results mainly from velocity-space instabilities, sometimes called microinstabilities. These instabilities tend to have shorter wavelength and lead to less bulk motion of the plasma. Unfortunately, the remedies for these instabilities do not seem to be as simple as for the hydromagnetic instability.

It is possible, but not likely, that a solution to the controlled-fusion problem lies outside any of the schemes being pursued at the present time. Laser-produced plasmas, turbulent plasmas, electrostatic or electromagnetic wave confinement, and high-pressure neutral-gas confinement are being investigated, along with the magnetic confinement schemes.

5.15 OPEN PLASMA-CONFINEMENT EXPERIMENTS

5.15.1 Magnetic Mirror Experiments

The magnetic field produced by a pair of current-carrying coils arranged as shown in Fig. 5.15.1 is called a magnetic mirror field. An investigation of the motion of individual charged particles (Appendix I) in this magnetic field reveals that it is possible to trap a certain class of particles. In particular, those particles that satisfy the conditions for adiabaticity (i.e., the change in the magnetic field during one period must be much less than the magnetic field itself, and the change in the magnetic field over the size of the orbit must be much less than the magnetic field) remain trapped indefinitely between the two mirror peaks, provided the ratio of the perpendicular velocity to parallel velocity at the midplane satisfies the condition

$$\frac{v_\perp}{v_\parallel} > \frac{1}{\sqrt{R}} \qquad (5.15.1)$$

where R is the mirror ratio (B_{max}/B_{min}).

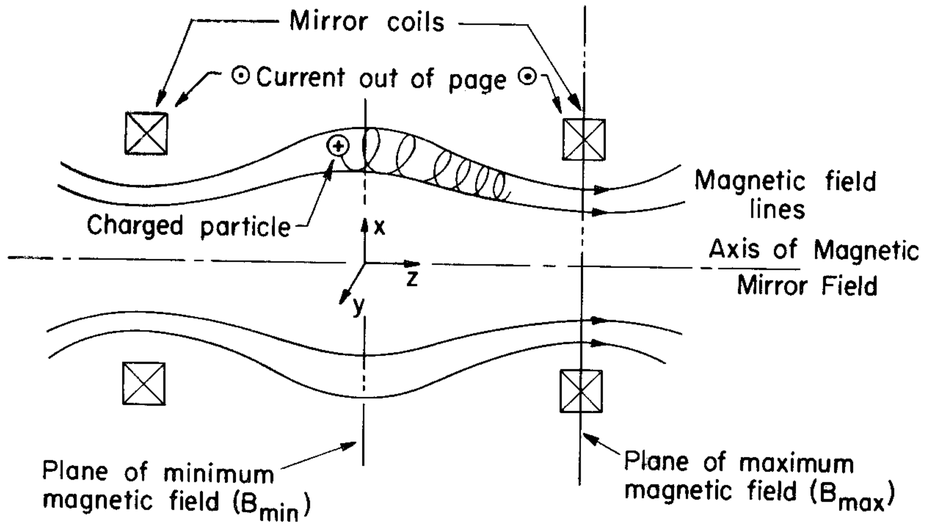


FIGURE 5.15.1
Schematic of a magnetic mirror field, showing a charged-particle trajectory.

Charged particles not satisfying this condition are said to be in the loss cone, i.e., the cone in velocity space defined by reversing the sense of the inequality of (5.15.1). The adiabatic theory of the motion of charged particles gave a clear and favorable prediction regarding the confinement of charged particles in magnetic mirrors, but there remained (1950s) some doubt as to whether or not small systematic changes (due to perturbation in the magnetic field, finite coils, etc.) in the magnetic moment of the trapped particles would cause loss of a particle from a magnetic mirror field by mechanisms not included in the theory.

Gibson, Jordan and Lauer[1] proved experimentally that single charged particles remained trapped in a magnetic mirror in a manner completely consistent with the predictions of adiabatic theory. The experiment involved a steady magnetic mirror field in a high-vacuum region into which a small amount of a radioactive isotope of neon was admitted. The neon decays and liberates positrons, which then, on the average, make 10^9 transits between the mirror peaks before being driven into the velocity-space loss cone by elastic scattering from residual background neutral gas. Thus theory predicted and experiment confirmed that single particles are satisfactorily trapped in a magnetic mirror field. However, the trapping of a single particle in some field configuration does not mean that a plasma has a stable equilibrium, or even an equilibrium, in the same configuration.

[1] G. Gibson, W. Jordan, and E. J. Lauer, *Phys. Rev. Letters*, 5:141 (1960).

Before taking up the plasma aspects of mirror confinement, there is the problem of loss of plasma from a magnetic mirror due to elastic scattering. If there were only two trapped charged particles in a magnetic field, their motion would eventually bring them to a point in space where they would suffer an elastic collision. This elastic collision would cause them to change the velocity-space coordinate in a random manner. Eventually, this random-walk process results in one of the particles having velocity-space coordinates in the loss cone, and consequently the particle escapes from the mirror. If there is a distribution of particle velocities, it turns out that approximately one-half the particles are lost in the period of one binary collision time, or one Coulomb collision time, whichever is less.[1] Although this loss of charged particles is a serious energy-loss mechanism that must be overcome if magnetic mirrors are to be successful in controlled-fusion applications, there are proposed schemes by which this energy can be recovered. Since the collision of charged particles in a magnetic mirror results in a loss time that is about one Coulomb collision time, the product of density and confinement time, $n\tau = 1/\sigma_{90}$, is independent of density, and depends only on the cross section for scattering through an angle of 90°. Under optimum conditions, i.e., no hydromagnetic or velocity-space instabilities, it is theoretically possible to satisfy the Lawson condition in a magnetic mirror, although this goal has not as yet been realized in the laboratory.

5.15.2 Stabilized Mirror Fields

Experimentally, the loss of plasma from magnetic-mirror devices (1959) was much faster than expected from elastic-scattering processes. In fact, the loss rate was consistent with the simple MHD instability associated with a magnetically confined plasma, as discussed in Sec. 5.8. However, the analysis predicting instability also contained a prediction of a remedy for the instability. In Sec. 5.12 it was shown that a plasma confined in a magnetic field where the magnetic field strength increases in every direction away from the plasma boundary is hydromagnetically stable.

The cusp magnetic field (same coil configuration as for the magnetic mirror field, but with reversed current in one coil) has this property; it also has a region of zero magnetic field that causes a rapid diffusion of particles into the loss cones.

In 1962, Ioffe[2] first reported on plasma-confinement experiments in a magnetic field configuration that had the desirable feature that the magnetic

[1] D. Judd, W. M. MacDonald, and M. N. Rosenbluth, End Leakage Losses from the Mirror Machine, U.S. Atomic Energy Commission Report WASH-289, Conference on Controlled Thermonuclear Reactions, Berkeley, Calif., February 1955.
[2] Yu. V. Gott, M. S. Ioffe, V. G. Tel'kovskii, *Nucl. Fusion*, 1962 Suppl., Part 3, p. 1045, 1962.

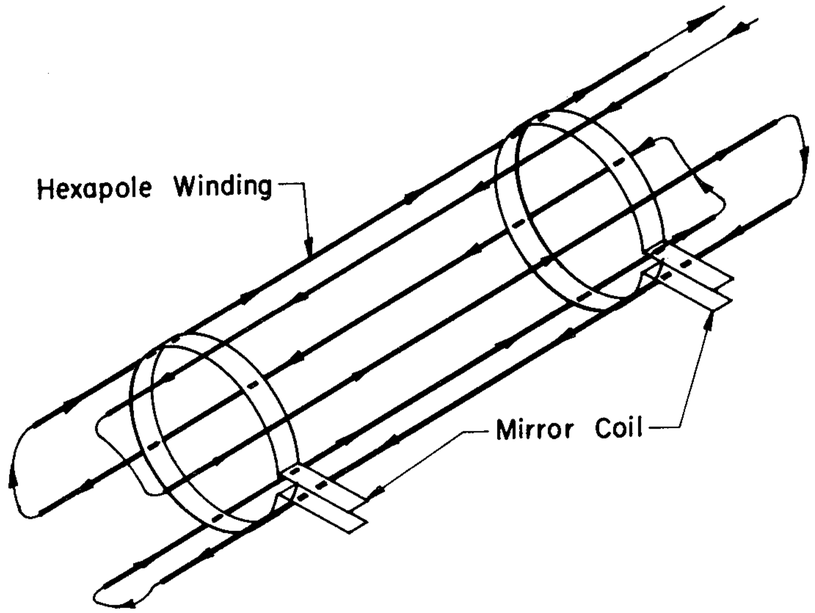


FIGURE 5.15.2
Schematic of a current distribution that produces a magnetic field having the property of increasing in all directions outward from the center. The multipole windings are called Ioffe bars, after Ioffe, who first used them in experiments that suppressed the hydromagnetic instabilities in a magnetically confined plasma.

field increased in every direction away from the plasma boundary, and that did not have the undesirable feature of a region where the magnetic field went to zero inside the plasma.

The magnetic field used in these experiments was created by superposing on a magnetic mirror field, the field produced by six conductors parallel to the magnetic mirror axis, carrying current in the directions shown in Fig. 5.15.2. The two end loops of current produce the mirror field; the six straight lines of current produce the Ioffe improvement. That such a field has the feature of increasing B^2 in every direction from the plasma can be seen by considering the effect of each of the two fields separately. The magnetic mirror field by itself has the property that B increases as a function of distance along the axis away from the midplane (stabilizing), but decreases as a function of distance along a radius from the midplane (destabilizing). On the other hand, the field produced by the parallel conductors increases as a function of distance along a radius from the midplane (stabilizing) and is constant along the mirror axis. These combined fields, if properly chosen, have the property that the magnetic field strength increases in every direction away from the midplane origin. That this corresponds to a region where the magnetic field is everywhere curved into the

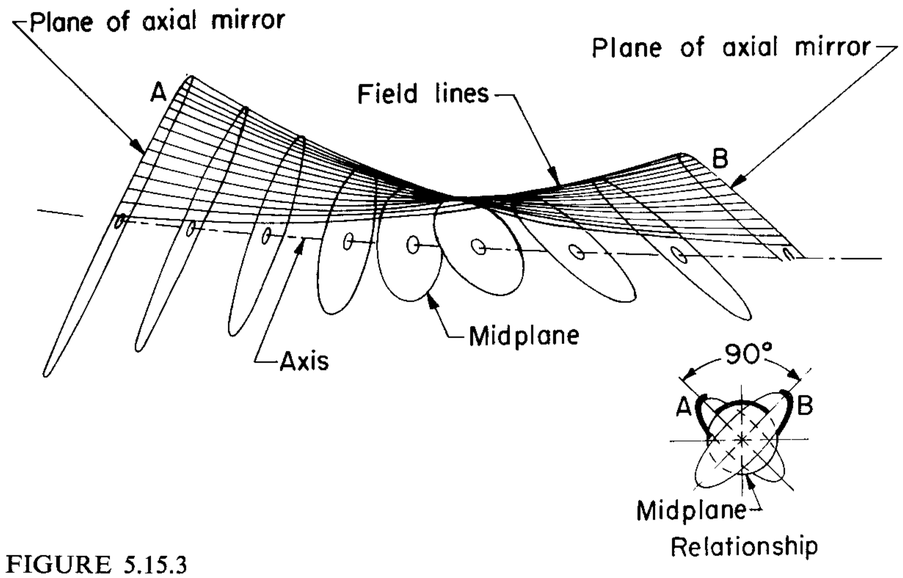


FIGURE 5.15.3
Drawing of a model that illustrates how the combination of a hexapole and mirror field produces a magnetic field that curves into the plasma over the region where plasma is confined. The drift surface for a quadrupole can be visualized by rolling up a piece of paper into a tube and then pinching the ends perpendicular to the axis of the tube. If the tube is horizontal, pinch one end in a horizontal plane and the other in a vertical plane to achieve the desired effect, which is that the paper assumes the shape of a drift surface. [*After J. G. Siambis and A. W. Trivelpiece, Phys. Fluids*, **8**, 2047 (1965).]

plasma can be seen from the model shown in Fig. 5.15.3, where the strings represent particle-drift surfaces produced by a magnetic mirror plus a linear quadrupole.

Ioffe and his coworkers at the Kurchatov Institute, Moscow, used a device known as the PR-5 (Probkotron-5) to demonstrate that the hydromagnetic instability could be suppressed with a field configuration such as shown in Fig. 5.15.2. The experiment involves a magnetic mirror field with 5,000 G at the midplane and 8,500 G at the mirror peaks. The vacuum chamber is 40 cm in diameter, and the separation between mirror peaks is 120 cm. The conductors that create the stabilizing multipole magnetic field are situated outside the vacuum chamber and produce a magnetic field of up to 4,500 G at the wall of the vacuum chamber.

The plasma in the PR-5 experiment is created by the ion-magnetron method, which initially traps a plasma of density $n \approx 10^9$ cm^{-3} with ion temperature $T_i = 5$ keV and electron temperature $T_e = 20$ eV. The decay of trapped

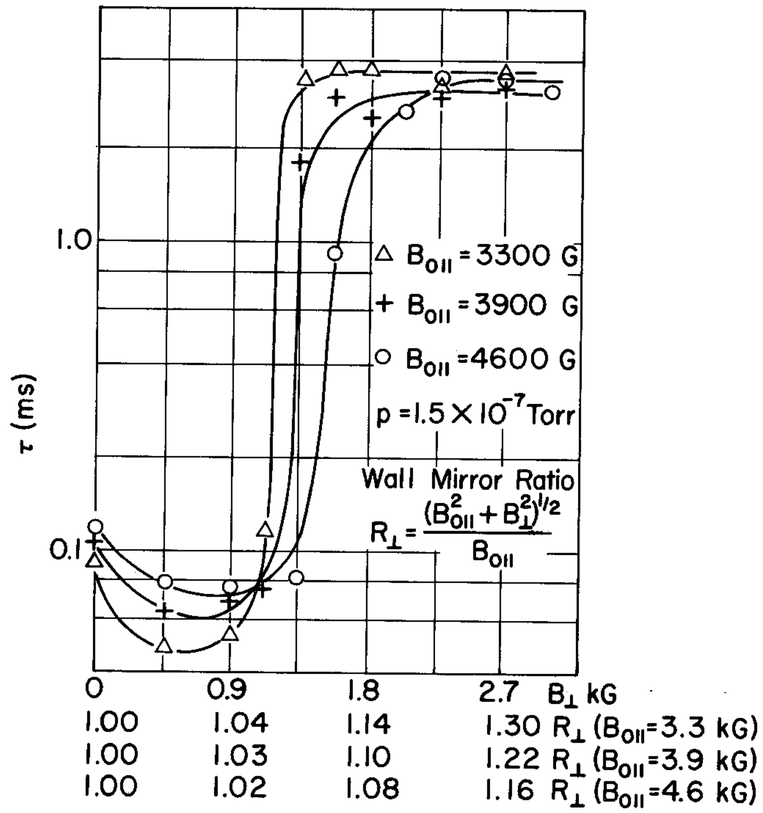


FIGURE 5.15.4
Dependence of plasma lifetime as a function of the strength of the stabilizing field for the Probkotron-5 (PR-5) experiment of Ioffe et al. at the Kurchatov Institute, Moscow, U.S.S.R.

plasma is measured by detecting the flux of fast neutral atoms emitted from the plasma produced by charge-exchange collisions between slow neutrals and fast ions. The plasma decay rate is then measured as a function of the magnetic field produced by the stabilizing windings. The results of Ioffe's work are reproduced in Fig. 5.15.4. As can be seen, the decay time increases sharply as the stabilizing field reaches a value that corresponds to a stable trap. The decay time of 3.5 ms is consistent with the loss of plasma due to charge-exchange losses alone. Subsequent work with better vacuum conditions led to decay times up to 70 ms. The increase in the decay time of plasma loss above the critical wall-mirror ratio was accompanied by a sharp reduction in electric fluctuations as detected by a negatively biased Langmuir probe in the plasma. This is further evidence of the suppression of the hydromagnetic instability.

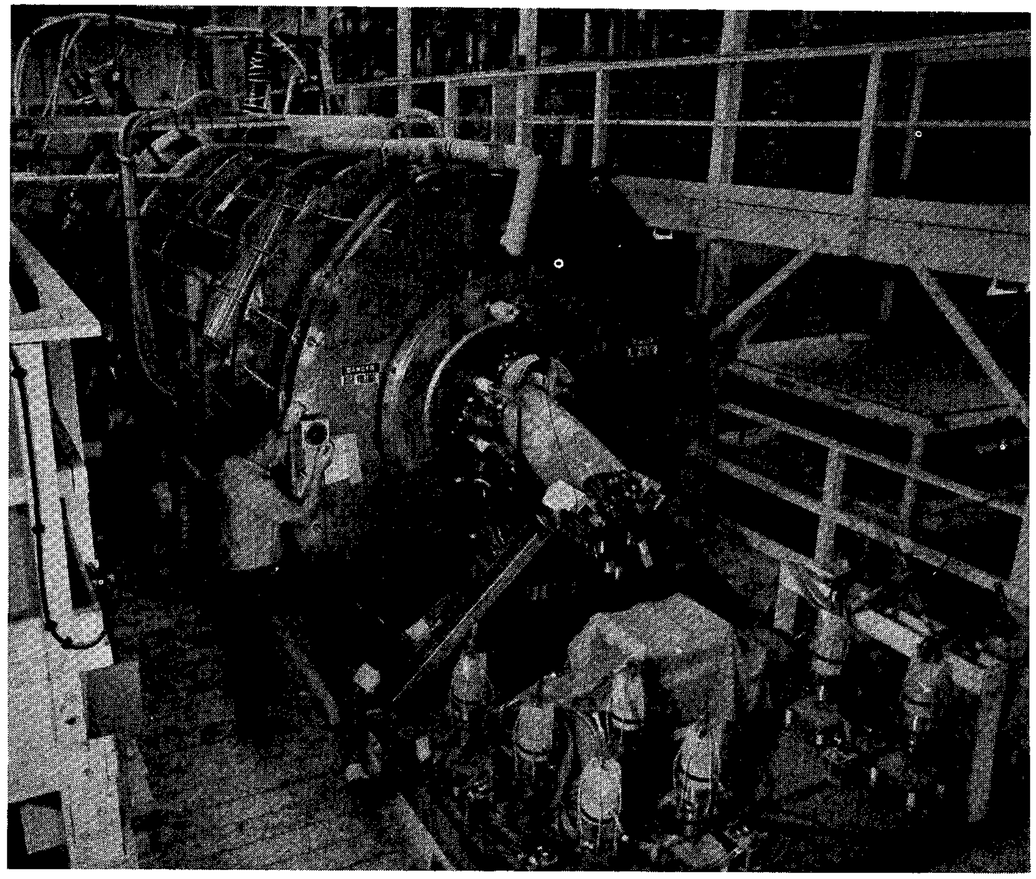


FIGURE 5.15.5
The 2X experiment at the Lawrence Livermore Laboratory. This end view shows the plasma injection system and some of the coils that provide the magnetic mirror field. (*Photograph supplied through the courtesy of the University of California, Lawrence Livermore Laboratory, Livermore, Calif., and the U.S. Atomic Energy Commission, under whose auspices the work was performed.*)

5.15.3 Stabilized Minimum |B| Compression Experiment: The 2X Experiment

Figure 5.15.5 is a photograph of a quadrupole-stabilized multistage magnetic mirror compression device called 2X, used by Coensgen and his coworkers at the Lawrence Radiation Laboratory at Livermore to study confinement and stability of a hot ion plasma.[1] The plasma, produced by deuterated washer-stack sources (Fig. 1.18.1), is injected axially into the magnetic mirror field,

[1] F. H. Coensgen, W. F. Cummins, R. E. Ellis, and W. E. Nexsen, Proceedings Third Conference on Plasma Physics and Controlled Nuclear Fusion Research, vol. 1, p. 225. International Atomic Energy Agency, Vienna, 1969.

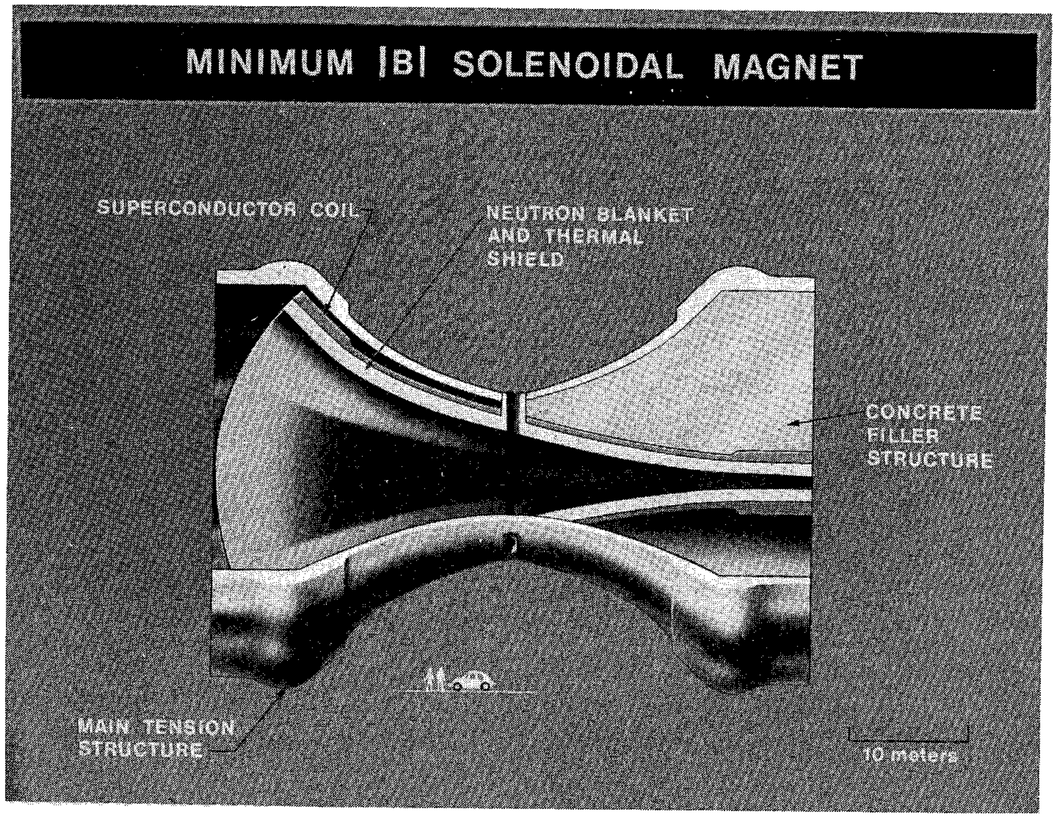


FIGURE 5.15.6
An artist's conception of a possible design for mirror fusion reactor that might be able to supply a nominal output power of 1,000 MW. (*Photograph supplied through the courtesy of the University of California Lawrence Livermore Laboratory, Livermore, Calif., and the U.S. Atomic Energy Commission, under whose auspices the work was performed.*)

where some of it is trapped and adiabatically compressed to a higher temperature and density as the magnetic field increases. When the magnetic field reaches its maximum value (in 160 μs), the coils are "crowbarred" (i.e., short-circuited), and the magnetic field decays exponentially with a time constant of 5 ms. The plasma (160 cm long and 6 cm in diameter) has a peak density of 5×10^{13} cm^{-3}, with a mean ion temperature of 8 keV and an electron temperature of 200 eV in the 13.5-kG minimum-B magnetic field configuration of the 2X experiment. After the peak of compression, the magnetic field decays so slowly compared with plasma processes, that it can be regarded as constant. The plasma density decays, during this time, as a result of Coulomb scattering into the velocity-space loss cones or as a result of "anomalous" loss due to turbulence or instability-driven losses.

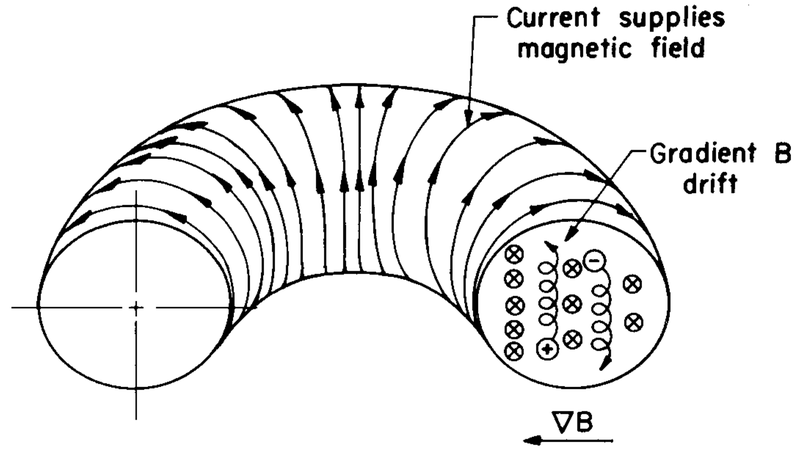


FIGURE 5.16.1
Schematic of a toroidal magnetic field.

A Fokker-Planck (Secs. 6.4 and 6.5) calculation of the Coulomb scattering loss rate that takes into account initial electron temperature, plasma potential, ion density, neutral density, ion-energy distribution, and ion angular distribution predicts a characteristic plasma lifetime $[\tau = n(dn/dt)^{-1}]$ in the range from 390 to 700 μs for a plasma density of 2.3×10^{13} cm^{-3}. At this density the largest observed plasma lifetime is 350 μs, which is within a factor of 2 of predicted scattering-loss rates and establishes an upper limit of anomalous loss rate of $dn/dt = 6.6 \times 10^{16}$ ions/(cm^3)(s). An artist's conception of a mirror fusion reactor with a nominal output power of 1,000 MW is shown in Fig. 5.15.6.

5.16 CLOSED PLASMA-CONFINEMENT EXPERIMENTS

5.16.1 Toroidal Magnetic Fields

Magnetic mirror field configurations have the feature of being open-ended, and thus particles that scatter into the velocity-space loss-cone escape. One way to solve this problem is to bring the magnetic field lines around to close on themselves on the assumption that the charged particles will retrace the same path repeatedly. One magnetic field configuration with the property of closed field lines is created by winding a current-carrying coil on the surface of a torus, as shown in Fig. 5.16.1. Although the magnetic field lines do close, this field configuration does not trap single charged particles. Fermi was one of the first (about 1946) to recognize this fundamental weakness of a torus as a charged-particle-confinement device. The explanation for the loss of a charged particle

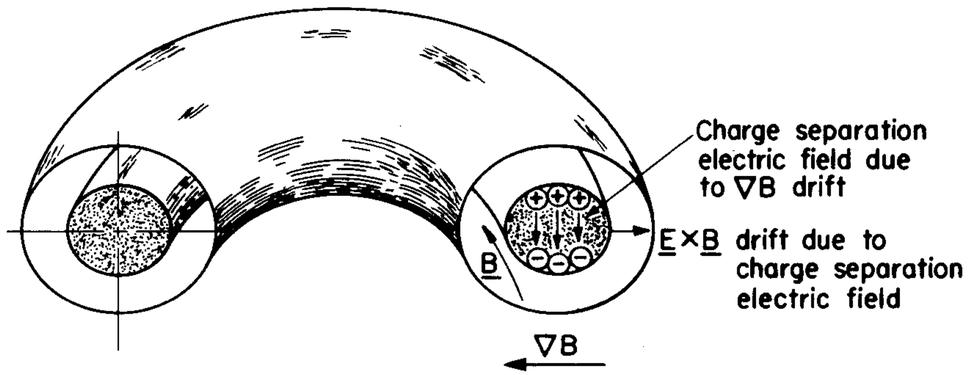


FIGURE 5.16.2
Illustration of the mechanism that results in the loss of plasma from a simple toroidal magnetic field.

from this simple toroidal magnetic field is that the magnetic field near the outer edge of the torus is weaker than near the inner edge. The gradient in the magnetic field causes the particles to drift across the field and escape. This gradient drift[1] is given by

$$\mathbf{V}_{DG} = \frac{\mu}{q}\frac{\mathbf{B} \times \nabla B}{B^2} c$$

where $\mu = mv_\perp^2/2B$ is the magnetic moment of the charged particle.

For a deuteron with an energy of about 10 keV, the gradient-drift velocity in a 50-kG toroidal magnetic field, with a 10^2-cm major radius, is approximately 10^7 cm/s. For a minor radius of 10 cm, the confinement time in a toroidal magnetic is $\tau = 10\text{ cm}/(10^7\text{ cm/s}) \approx 10^{-6}$ s. At typical plasma densities in toroidal devices (seldom above $n = 10^{14}\text{ cm}^{-3}$), $n\tau \approx 10^8$, which is much less than the Lawson condition.

Next, consider what would happen if a plasma were placed in the simple toroidal magnetic field described above. Is it possible that a plasma might stay in even though single particles would escape? The answer is no. To understand this lack of plasma equilibrium in a simple toroidal magnetic field, notice that the gradient drift would tend to move positive particles up and negative particles down, thereby creating a charge-separation electric field, as shown in Fig. 5.16.2. The resultant electric drift ($c\mathbf{E} \times \mathbf{B}/B^2$) is independent of the sign of the charge and is directed away from the major axis of the torus for both positive and negative particles. Thus the plasma drifts outward to the wall, again with a time scale $\approx 10^{-6}$ s. So unless suitable steps are taken to overcome this problem, plasma cannot be confined in a simple toroidal magnetic field.

[1] See Appendix I.

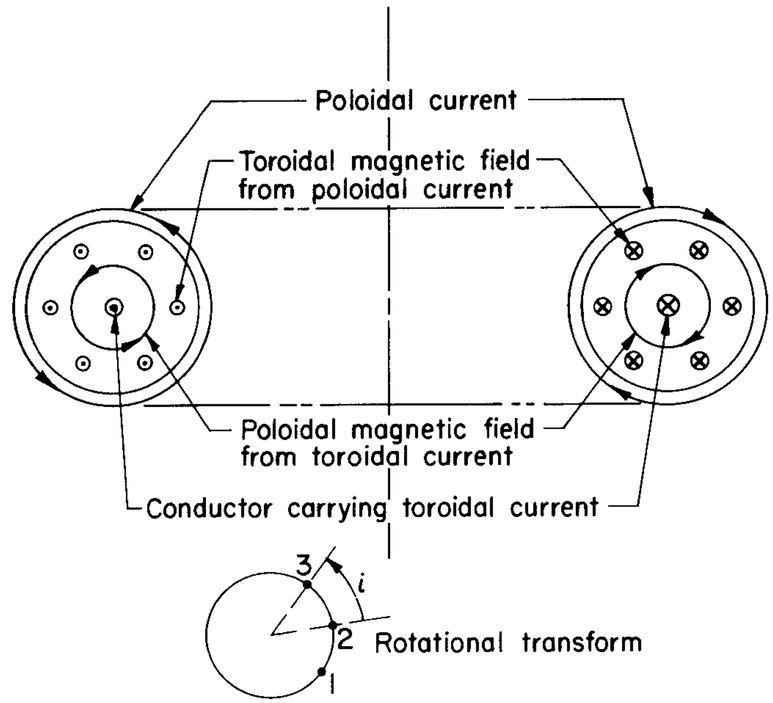


FIGURE 5.16.3
Toroidal and poloidal currents combine to give a closed magnetic field with a rotational transform.

A solution to the problem of single-particle containment in closed magnetic field configurations is depicted schematically in Fig. 5.16.3. The magnetic field is obtained from two currents. One of these currents is poloidal (along the z axis, in this case), thus creating a toroidal magnetic field, while the other current is toroidal (i.e., in the θ direction), thus creating a poloidal magnetic field. The net magnetic field created by this current distribution has the feature that a field line that crosses a surface perpendicular to the torus at the point labeled 1 (see the auxiliary diagram in Fig. 5.16.3) will next cross the same surface at the point labeled 2, and next at the point labeled 3, etc. Each successive trip causes an additional small displacement as shown. The angular of displacement of the flux line is called the *rotational transform* and is denoted by ι(iota). Unless iota has a value of $2\pi/n$ (n is an integer), the flux line ergodically fills a surface. These flux surfaces are nested in the sense that a flux line closer to the toroidal current generates a surface that is enclosed by the surface generated by an outer flux line.

There is still a radial magnetic field gradient, and there is still a drift motion of individual charged particles. To see that this drift motion does not

result in the loss of particles, consider a charged particle on a flux line at the top of the flux surface in Fig. 5.16.3. Assume that this particle has a local drift velocity that is vertically upward, i.e., toward the wall. This same particle, as a result of the rotational transform and its velocity along the field line, will eventually be at the bottom of the torus. At this point, the local drift velocity is still vertically upward, i.e., toward the center of the confining region. As a result, the particle drifts toward the wall, part of the time, and toward the center of the confining region, part of the time. For certain classes of particles the outward and inward drift motions cancel, and there is no net outward motion to the wall. Thus, if the rotational transform is large enough, single particles can be confined by the magnetic field that results from a combination of poloidal and toroidal currents. Experiments by Gibson[1] at Culham Laboratories in England demonstrated that single-charged particles can be contained in a closed magnetic field configuration. Particle-drift surfaces do not correspond exactly to flux surfaces, but are close to them.

5.16.2 Stellarator

One of the first and quite ingenious solutions to the problem of single-particle confinement in closed magnetic field systems was advanced by Spitzer in 1951 when he described the *Stellarator* concept. His solution to the particle-containment problem was to twist the torus into a "figure eight." This simple twist alters the magnetic field in such a way that it has the property of nested magnetic flux surfaces. Although early Stellarator experiments at the Princeton Plasma Physics Laboratory used the figure-eight configuration, the same magnetic field properties can be realized by a suitably arranged system of conductors on the outside of a toroidal vacuum chamber.

Analysis indicates that a low-beta plasma in a Stellarator has a hydromagnetic equilibrium and is hydromagnetically stable. The stability of plasma in the Stellarator is due primarily to the weak shear of the magnetic flux surfaces that inhibits the growth of flutelike modes. The simplest method of heating plasma in the Stellarator and other toroidal devices is to induce a toroidal current in the plasma. This is accomplished by a rapid change of magnetic field, usually by an iron core, passing through the center of the torus. The ring of plasma is essentially a one-turn secondary of a transformer. The induced current heats the plasma by ohmic (or Joule) heating from the plasma resistivity. The same currents that heat the plasma also generate a magnetic field that has the effect of weakening the stabilizing features of the Stellarator fields. In fact,

[1] A. Gibson, J. Hugill, and G. W. Reid, *Phys. Rev. Letters*, **21**: 1052 (1968).

if the heating current is large enough, the rotational transform can be undone, and there is no plasma equilibrium.

A stability criterion analogous to the Kruskal-Shafranov limit can be derived for a Stellarator with a heating current. If the heating current produces a magnetic field that increases the rotational transform, the stability condition is

$$\frac{B_T A}{L}(2\pi - \iota) > \frac{2I}{cA}$$

If the rotations are opposite, the stability condition is

$$\frac{B_T A \iota}{L} > \frac{2I}{cA}$$

where I = heating current

L = circumference around the torus
A = radius of the plasma column
B_T = component of magnetic field along the plasma
ι = rotational transform associated with the magnetic field

It is possible to satisfy the Kruskal-Shafranov limit and eliminate the most destructive type of hydromagnetic instability; however, the Stellarator configuration still seems to be plagued by velocity-space instabilities that result in anomalous loss of plasma. Other schemes of heating plasma in Stellarators have not yet been able to eliminate this plasma loss. Another approach to the problem of toroidal confinement is to use plasma currents to create some part of the confining magnetic field, or at least improve plasma stability.

5.16.3 Tokamaks

Scientists in the U.S.S.R. proposed and pursued a line of research on the confinement of plasmas in toroidal magnetic field configurations that apparently overcomes some of the problems of plasma equilibrium and stability associated with the Stellarator configuration. Their approach to toroidal equilibrium and stability is to use the plasma currents, rather than external conductors, to create the magnetic field that converts a purely toroidal field into a toroidal-poloidal field with a desirable value for the rotational transform, iota.

Figure 5.16.4 is a schematic of a *Tokamak* device. A low-density, low-temperature toroidal plasma is pulsed by a strong electric field in the toroidal direction. This electric field drives the large toroidal current that produces the poloidal magnetic field. A comparison of the magnetic field of a Stellarator and

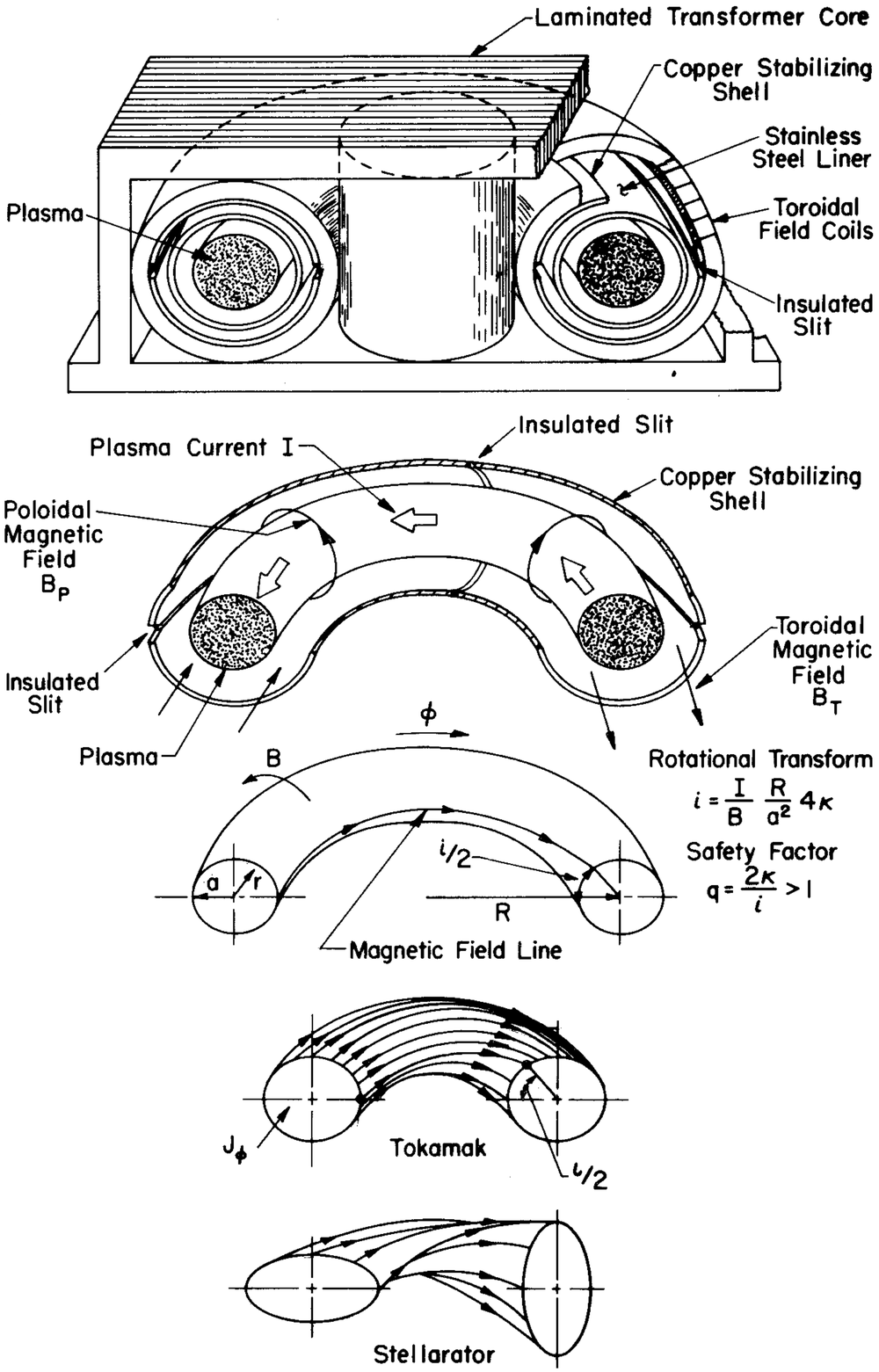


FIGURE 5.16.4
Comparison of magnetic field of a Stellerator and a Tokamak.

Tokamak is also shown in Fig. 5.16.4. The Tokamak has somewhat greater shear and internal symmetry than the Stellarator configuration. This current heats the plasma to relatively high temperature.

Recent results with Tokamaks have been encouraging in terms of confinement of a high-temperature plasma. The T-3 tokamak device at the Kurchatov Institute in Moscow produces a plasma with a density of 5×10^{13} cm^{-3} an ion temperature of 600 eV, and a confinement time of 20 ms.

The figure of merit used to compare various confinement experiments in which plasma escapes across the magnetic field is the factor obtained by dividing the actual loss time for that experiment by the loss time that would be expected if Bohm diffusion ($D_B \propto T/B$) were the loss mechanism. In controlled-fusion experiments this is important, because it is now recognized that it is not necessary to achieve classical diffusion, but rather the Lawson condition can be satisfied with a containment time that is $10^4 \tau_B$(τ_B = loss time due to Bohm diffusion). The T-3 Tokamak has achieved energy-loss time that is the order of 100 Bohm diffusion times. On the basis of these favorable results, the Soviets plan (1972) to build an experiment approximately three times larger, to be called the T-10. These promising results have also stimulated considerable activity in the United States to study Tokamaks. The Princeton Model C Stellarator was modified in 1970 to a Tokamak (called ST; see Fig. 5.16.5). Additional experiments at the University of Texas, Oak Ridge National Laboratory, the Massachusetts Institute of Technology and Gulf General Atomic are being developed (1972).

5.17 OTHER PLASMA-CONFINEMENT EXPERIMENTS

5.17.1 Neutral Injection Devices

One way to produce and trap a plasma inside a magnetic field is to inject a beam of high-speed neutral atoms across the magnetic field. The ions and electrons in the neutral atom experience a $q\mathbf{v} \times \mathbf{B}$ force that tends to separate them. If the velocity of the neutral atom and the strength of the magnetic field are large enough, the atom is *Lorentz-ionized.* An experiment that is based on this effect is the "Alice"[1] device operated by Damm and coworkers at the Lawrence Livermore Laboratory, Livermore, California. This experiment involves the injection of a beam of 20-keV neutral deuterium atoms across the midplane of a magnetic mirror field, resulting in the production and trapping of 20-keV deuterons in the magnetic mirror field. As the beam continues to pass through the

[1] C. C. Damm et al., Proceedings Third Conference on Plasma Physics and Controlled Nuclear Fusion Research, International Atomic Energy Agency, Vienna, 1969.

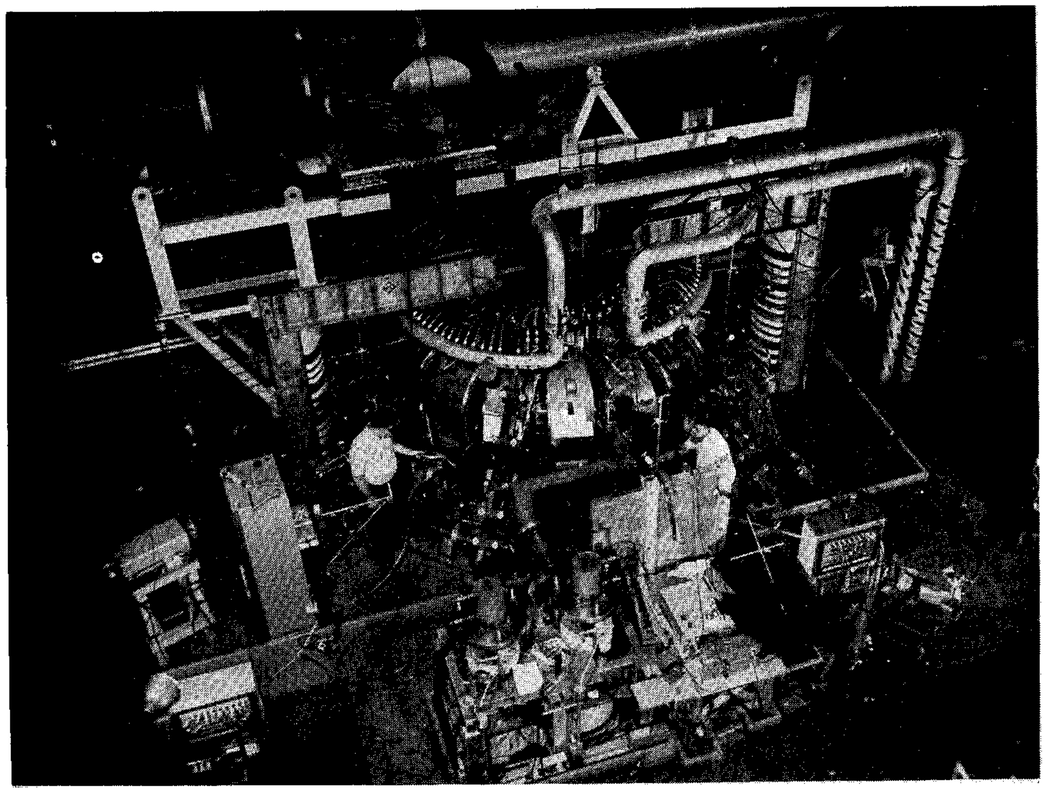


FIGURE 5.16.5
Princeton Plasma Physics Laboratory Symmetric-Tokamak experiment. This experiment was formerly the Model C Stellarator. (*Photograph supplied through the courtesy of the Princeton Plasma Physics Laboratory and the U.S. Atomic Energy Commission, under whose auspices the work was performed.*)

mirror field, the density of plasma increases, until eventually ionization occurs as a result of collisions of beam atoms with trapped plasma particles. This experiment was conceived with the hope that, by gradually increasing the density of plasma, instabilities would not be excited. This was not the case; hydromagnetic instabilities were suppressed in the Alice device by means of stabilizing fields added to the mirror field, but velocity-space instabilities still result in loss of plasma. Typically, this experiment produces a 20-keV plasma with a density of 10^9 cm^{-3}, which decays in 100 ms.

The Alice experiment now uses a hydromagnetically stable magnetic field configuration produced by a current flow that follows the same pattern as the seam on a baseball. The latest version of this experiment involves the superconducting Baseball-II coil shown in Fig. 5.17.1. This coil is capable of supplying a steady magnetic field of 20 kG at the center and 25 kG near the conductors.

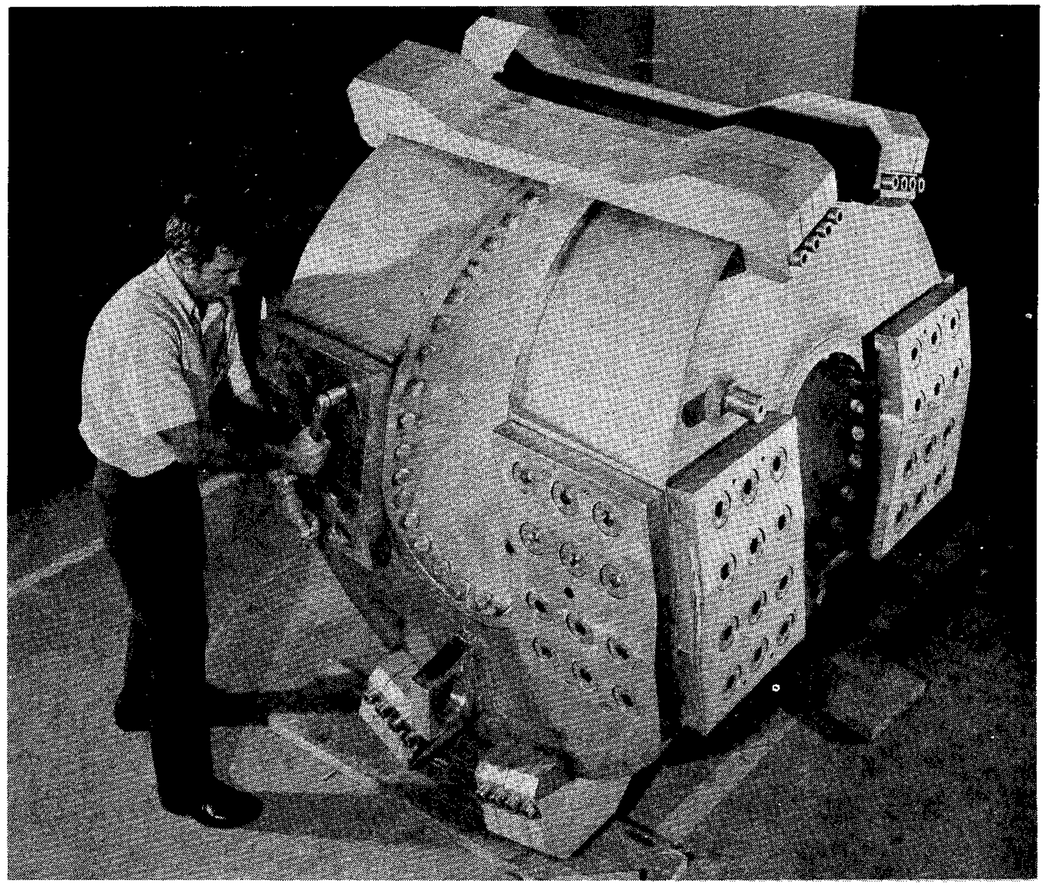


FIGURE 5.17.1
"Baseball-II" coil used in the "Alice" neutral-beam injection experiment at Lawrence Livermore Laboratory. (*Photograph supplied through the courtesy of the University of California Lawrence Livermore Laboratory, Livermore, Calif., and the U.S. Atomic Energy Commission, under whose auspices the work was performed.*)

Magnetic forces generated by this coil are in excess of one million pounds, and special steels capable of resisting these forces at 4.2°K had to be developed for this experiment.

5.17.2 Astron

The Astron[1] device operated by Christofilos and coworkers at the Lawrence Livermore Laboratory, Livermore, California, seeks to confine a high-temperature plasma by first creating a magnetic field configuration of closed field lines in a

[1] J. W. Beal et al., Proceedings Third Conference on Plasma Physics and Controlled Nuclear Fusion Research, vol. 1, International Atomic Energy Agency, Vienna, 1969.

hydromagnetically stable configuration. The scheme to create such a magnetic field involves the injection on helical orbits of a 400-A beam of 4-MeV electrons across a nearly uniform magnetic field. The beam is magnetic-mirror-trapped in a 10-m-long shell, called the *e* layer. If the density of the electron layer is sufficiently high, it will act as a shell of current, producing a magnetic field that is in the opposite direction to the original field. If the current in the *e* layer builds up to sufficient strength, the internal field will be reversed, creating field lines that are closed around the *e* layer, so that a *minimum-B* region exists around the field reversal point. Once the *e* layer is formed, deuterium or tritium gas will be injected into the device, where it will be ionized and heated by the relativistic electrons in the *e* layer. Figure 5.17.2 is a photograph of the Astron device, showing the coils required to produce the initial magnetic field for trapping the *e* layer.

5.17.3 Scyllac

Some properties of high-current discharges were discussed in Chap. 3. Such discharges produce a high-density plasma $n > 10^{16}$ at temperatures of more than 1 kV. The plasma does not remain trapped for very long, usually about 1 μs. The objective of these devices as controlled-fusion experiments is to exceed the Lawson condition by achieving high density rather than long containment time. To date the experiments have produced $n\tau$ values of about 10^{11}. The most common version of these experiments is the θ pinch. Although θ pinches confine plasma radially, across the magnetic field, they allow plasma to freely escape out the ends. The Los Alamos Scientific Laboratories have proposed that the problem of end losses might be eventually overcome by bending the θ pinch into a toroidal configuration.

5.17.4 DCX

The DCX device at Oak Ridge National Laboratory is a magnetic mirror experiment involving a steady magnetic field. The creation of plasma is accomplished by injection of a 600-keV beam of H_2^+ ions at right angles to the mirror field axis on the midplane. These charged particles would of course exit from the static magnetic field, except that they are dissociated in the magnetic field by passing the beam through an energetic arc that has been established along the magnetic mirror field. This produces a plasma of high temperature, $\approx$300 keV, but low density, $\approx 10^9$ cm^{-3}, which has a very large temperature anisotropy and is correspondingly limited by a velocity-space instability (Harris instability), discussed in Chap. 9.

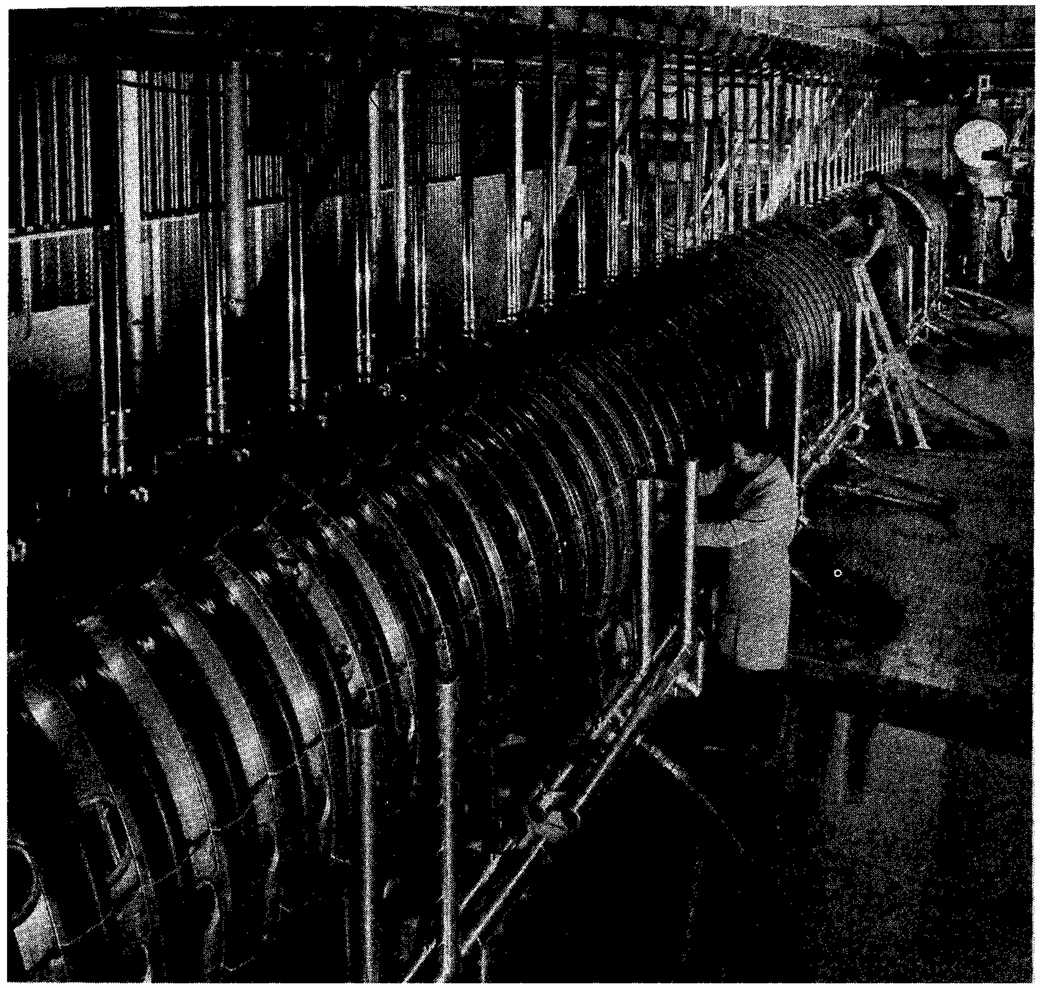


FIGURE 5.17.2
View of the coils surrounding the vacuum chamber of the Astron device at Lawrence Livermore Laboratory. (*Photograph supplied through the courtesy of the University of California Lawrence Livermore Laboratory, Livermore, Calif., and the U.S. Atomic Energy Commission, under whose auspices the work was performed.*)

These are only a few of the many experiments that have been constructed, in the United States and elsewhere, to attempt to force the plasma into a range of density and temperature suitable for exothermic nuclear fusion reactions.

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6

TRANSPORT PHENOMENA IN PLASMA

Plasma properties depend in a nontrivial way on the simultaneous interaction of many plasma particles. In preceding chapters it was shown that two types of interaction could be separated in a functional way to simplify the study of plasma phenomena. The weak, long-range, simultaneous interaction of many plasma particles was incorporated into average electric and magnetic fields. The strong, short-range, binary interaction between two charged particles was incorporated into the equations as a specific binary collision operator. This short-range collision term was then dropped from the equations in most of the analysis that followed. The reason is that there is a large class of plasma phenomena (e.g., quasi-steady states, waves, and instabilities) that occur on a time scale much shorter than the time scale on which plasma properties change due to short-range encounters.

The study of plasma properties on time scales shorter than the binary collision time reveals a wide range of collective properties that distinguish the plasma state from the other states of matter. However, it is important not to lose sight of the fact that if a plasma were studied on a longer time scale, collisions would eventually drive the plasma to a state of thermodynamic equilibrium

with respect to its surroundings. For example, a magnetically confined stable plasma would eventually diffuse toward a spatially uniform state.

The macroscopic plasma equations are transport equations in the sense that they describe the flow of momentum, energy, etc. However, the term "transport phenomena" is customarily used in plasma physics to identify plasma properties associated with collisional effects, i.e., properties such as electrical conductivity, thermal conductivity, particle diffusion across or along magnetic fields, etc. In many plasma problems the effect of neutral atoms or molecules dominates the plasma behavior. For instance, the diffusion of electrons in a tenuous, low-temperature, partially ionized plasma is governed by electron-neutral collisions. In addition, there are various processes that occur as a result of ionizing collisions and electron-ion recombination in the plasma.

There are two aspects of the problem of plasma transport phenomena. One aspect is the determination of the various transport coefficients from the appropriate statistical description and the mechanisms that control these phenomena. The other aspect is the use of these coefficients in computing the expected macroscopic behavior of a plasma for a given set of initial and boundary conditions.

In this chapter only transport phenomena identified with actual binary collisions are considered. An important area of plasma physics deals with "anomalous" transport phenomena, in which plasma instabilities produce effects somewhat analogous to particle-particle collisions. These aspects are briefly considered in Chap. 10.

The study of plasma transport is generally approached by using, instead of an exact treatment, a model that reproduces the physical features of the binary interaction while offering some mathematical simplification. In the following sections two such models are treated in detail. The first is the Fokker-Planck model, appropriate for calculating transport behavior in a fully ionized gas. This model is based on the notion that in a fully ionized plasma, a large-angle deflection of a particle by collisions is produced most rapidly by a succession of small-angle scatterings with distant particles. To confirm this idea the next two sections treat the dynamics of Coulomb scattering, in preparation for the Fokker-Planck theory. In the later sections, a second model, more appropriate for transport calculations in a weakly ionized gas, is discussed. This model assumes that charged particles are deflected by single collisions with neutral atoms, rather than by multiple scattering by other charged particles. The Boltzmann form for the collision integral is correct for this situation, and this model, along with further simplifications, is used to find the transport properties of a weakly ionized plasma.

6.1 BINARY COULOMB COLLISIONS

In a fully ionized plasma, the dynamics of a nonrelativistic close encounter between any pair of charged particles are governed by the charges $Z_1 e$, $Z_2 e$, masses m_1, m_2, relative velocity v_0, and impact parameter b of the two particles. Consider the situation shown in Fig. 6.1.1, where the trajectories of two particles of like charge, with particle 2 initially at rest, are sketched. The scattering angle of the incident particle is given by $\Delta\theta$. The nonrelativistic equations of motion for the two charged particles are

$$\frac{m_1 d^2\mathbf{r}_1}{dt^2} = \frac{Z_1 Z_2 e^2(\mathbf{r}_1 - \mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^3} \tag{6.1.1}$$

and

$$\frac{m_2 d^2\mathbf{r}_2}{dt^2} = \frac{Z_1 Z_2 e^2(\mathbf{r}_2 - \mathbf{r}_1)}{|\mathbf{r}_1 - \mathbf{r}_2|^3} \tag{6.1.2}$$

These two equations may be combined to yield

$$\frac{m_1 d^2\mathbf{r}_1}{dt^2} + \frac{m_2 d^2\mathbf{r}_2}{dt^2} = (m_1 + m_2)\frac{d^2\mathbf{R}}{dt^2} = 0 \tag{6.1.3}$$

and

$$\frac{d^2\mathbf{r}_1}{dt^2} - \frac{d^2\mathbf{r}_2}{dt^2} = \frac{d^2\mathbf{r}}{dt^2} = \frac{m_1 + m_2}{m_1 m_2}\frac{Z_1 Z_2 e^2}{|\mathbf{r}|^3}\mathbf{r} \tag{6.1.4}$$

where $\mathbf{R} = \dfrac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}$ and $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$

Equation (6.1.3) shows that the center of mass, located at $\mathbf{R}$, moves with constant velocity. Equation (6.1.4) shows that the relative-position vector $\mathbf{r}$ varies in the same manner as if it were describing the motion of a single particle of reduced mass $\mu = m_1 m_2/(m_1 + m_2)$ moving about a fixed force center. The solution to (6.1.4), expressed in polar coordinates, is

$$\frac{1}{r} = B\cos(\theta + \alpha) - \frac{A}{{v_0}^2 b^2} \tag{6.1.5}$$

where $A = Z_1 Z_2 e^2/\mu$. The initial conditions needed to determine B and α are $\theta_0 = \pi$, $r_0 = \infty$, $\dot{r}_0 = -v_0$, ${r_0}^2\dot{\theta}_0 = -v_0 b$. For these initial conditions

$$\tan\alpha = \frac{{v_0}^2 b}{A} \tag{6.1.6}$$

and

$$B = \sqrt{\frac{1 + A^2/{v_0}^4 b^2}{b^2}} \tag{6.1.7}$$

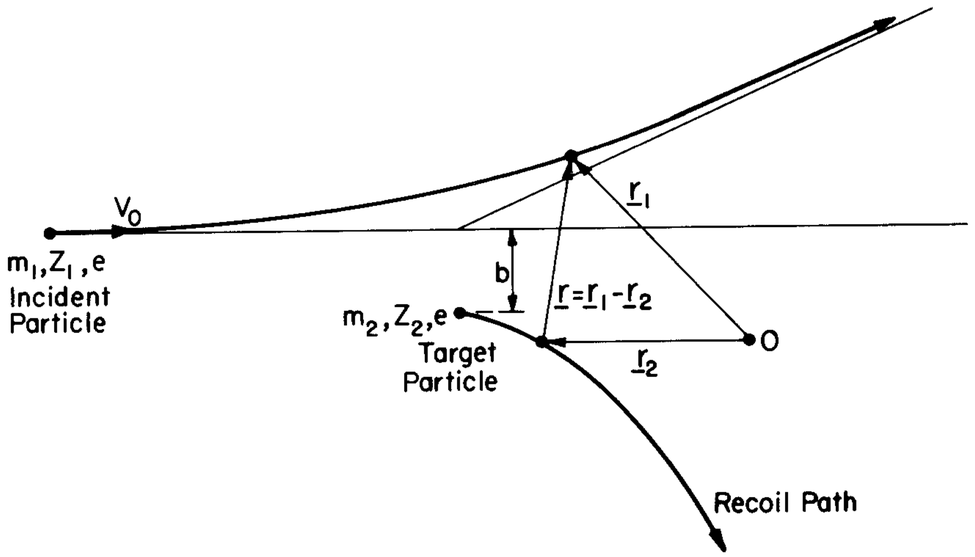


FIGURE 6.1.1
Diagram of an encounter between two charged particles. A single underscore denotes a vector.

The asymptotic deflection angle θ_c at $r = \infty$ is obtained from (6.1.5), i.e.,

$$B \cos(\theta_c + \alpha) = \frac{A}{{v_0}^2 b^2} \qquad (6.1.8)$$

From the initial conditions, one solution of this equation is $\theta_c = \pi$. The other solution is $\theta_c = \pi - 2\alpha$, since $\cos(\pi + \alpha) = \cos(\pi - \alpha)$. Using (6.1.6), the center-of-mass deflection angle for a Coulomb collision is given by

$$\tan\frac{\theta_c}{2} = \frac{Z_1 Z_2 e^2}{\mu {v_0}^2 b} \qquad (6.1.9)$$

Note that the magnitude of the angle depends only on the impact parameter and initial velocity, and not on the sign of the relative-force term.

All the particles incident on the target particle that pass through the annular ring between b and $b + db$ are scattered into angles between θ_c and $\theta_c + d\theta_c$. From the definition of differential cross section, the flux scattered into unit solid angle $d\Omega$ is

$$d\sigma(\theta_c) = \sigma(\theta_c)\, d\Omega = \sigma(\theta_c)\, 2\pi \sin\theta_c\, d\theta_c \qquad (6.1.10)$$

since $\Omega = 2\pi(1 - \cos\theta_c)$. The incident flux is $2\pi b\, db$; so from conservation of particles,

$$2\pi b\, db = \sigma(\theta_c)\, 2\pi \sin\theta_c\, d\theta_c$$

and

$$\sigma(\theta_c) = \frac{b}{\sin\theta_c}\frac{db}{d\theta_c} \qquad (6.1.11)$$

From (6.1.10) this becomes

$$\sigma(\theta_c) = \left(\frac{Z_1 Z_2 e^2}{2\mu {v_0}^2 \sin^2 \tfrac{1}{2}\theta_c}\right)^2 \qquad (6.1.12)$$

This is called the *Rutherford scattering cross section.* For a fully ionized plasma of electrons and one species of ions, the deflection angle resulting from an encounter between an electron moving with speed v_0 and a stationary singly charged ion is, from (6.1.9),

$$\tan\frac{\theta_c}{2} = \frac{e^2}{m_e {v_0}^2 b} \qquad (6.1.13)$$

It is convenient to define a large-angle scattering, or a close encounter, as one which results in an angular deflection of the incident particle of 90° or more. The impact parameter that results in a 90° scattering is

$$b_0 = \frac{e^2}{m_e {v_0}^2} \qquad (6.1.14)$$

and the cross section for scattering of a particle by 90° or more in a single encounter is

$$\sigma_{90°} = \pi {b_0}^2 = \pi\left(\frac{e^2}{m_e {v_0}^2}\right)^2 \approx \frac{1.6 \times 10^{-20}}{W_{\text{keV}}^2} \quad \text{cm}^2 \qquad (6.1.15)$$

6.2 DEFLECTION OF A CHARGED PARTICLE BY MULTIPLE COULOMB COLLISIONS

The scattering by a large angle in a single encounter, calculated in the preceding section, is much less likely in a fully ionized plasma than a net large-angle deflection due to the cumulative effect of the many small-angle scatterings that the electron experiences as it moves along its path. To illustrate this, a comparison of the cross section (6.1.15) will be made with the cross section that represents the effectiveness of a succession of small-angle scatterings in deflecting an electron through total angle of 90°.

To obtain the multiple Coulomb scattering cross section for electrons on massive ions of charge Ze, consider an electron moving initially in the z direction in a plasma. As the test electron moves through the plasma, it will acquire a transverse velocity as a result of these small collisions. After N such collisions,

$$\Delta v_x = \sum_{i=1}^{N} (\Delta v_x)_i \qquad (6.2.1)$$

For small-angle deflections, $\Delta(mv)$ is small compared with the initial momentum. The deflection angle for a single small-angle scattering (large impact-parameter) is, from (6.1.13),

$$\Delta\theta = \frac{2Ze^2}{mv_0{}^2 b} \qquad (6.2.2)$$

If the test particle moves through a plasma with n scattering centers per cubic centimeter, the average deflection will be zero, since each individual deflection is assumed random. However, the mean-square deflection is not zero, and there will be a random walk in angle away from the original direction of motion of the test particle.

Consider the motion of a test particle that suffers a succession of deflections, each of which is greater than some minimum deflection angle $\Delta\theta_{\min}$, but less than some maximum deflection $\Delta\theta_{\max}$ (the reason for considering only deflections in a limited range will be explained below).

The mean-square deflection for a test particle that penetrates a depth L in the plasma is

$$\langle(\Delta\theta)^2\rangle = \int_{\Delta\theta_{\min}}^{\Delta\theta_{\max}} (\Delta\theta)^2 F(\Delta\theta)\, d(\Delta\theta) \qquad (6.2.3)$$

where $F(\Delta\theta)\, d(\Delta\theta)$ is the number of collisions that result in a change in angle $\Delta\theta$ in a range $d(\Delta\theta)$. The maximum and minimum scattering angles considered are related to the impact parameters, i.e., $\Delta\theta_{\max} = 2e^2/mv_0{}^2 b_{\min}$ and $\Delta\theta_{\min} = 2e^2/mv_0{}^2 b_{\max}$. In terms of the impact parameter and the distance L traversed by the test particle,

$$F(\Delta\theta)\, d(\Delta\theta) = nL2\pi b\, db$$

since this is the number of scattering centers in a shell of length L, radius b, and thickness db. Thus

$$\langle(\Delta\theta)^2\rangle = \frac{8\pi nLZ^2e^4}{m_e{}^2 v_0{}^4} \int_{b_{\min}}^{b_{\max}} \frac{db}{b}$$

$$= \frac{8\pi nLZ^2e^4}{m_e{}^2 v_0{}^4} \ln \frac{b_{\max}}{b_{\min}} \qquad (6.2.4)$$

The reason that only deflections in a range $(\Delta\theta)_{\max} > \Delta\theta > (\Delta\theta)_{\min}$ were considered (which is really a restriction on the impact parameters included in calculating the mean-square deflection) is seen by examining (6.2.4), which

diverges for $b_{\max} \to \infty$. This divergence is the usual divergence in the Coulomb scattering cross section for small-angle scattering and arises from the fact that Coulomb potential has such a long-range effect. In a plasma this divergence is removed in a logical way: the potential between the test particle and a plasma particle is not $\phi = -e^2/r$; rather it is $\phi = -(e^2/r)\exp(-r/\lambda_D)$, since the plasma particle is Debye-shielded. Therefore the result (6.2.4) is correct only for impact parameters smaller than the shielding distance. For greater values of b the deflection becomes negligible. Thus it is a reasonable approximation to take

$$b_{\max} = \lambda_{De} = \left(\frac{\kappa T}{4\pi ne^2}\right)^{1/2} \qquad (6.2.5)$$

Using the Coulomb result with a sharp cutoff at λ_D is clearly an approximation, but it does simplify the problem [avoiding the need to calculate the scattering from a potential $\phi = -(e^2/r)\exp(-r/\lambda_D)$], and since the result (6.2.4) depends only logarithmically on $b_{\max}$, the result is not very sensitive to the exact choice of $b_{\max}$. The restriction $b > b_{\min}$ was employed because, by excluding very small values of b, the scattering angle could be approximated by $\Delta\theta/2 \approx e^2/mv_0{}^2 b$, rather than $\Delta\theta/2 = \tan^{-1} e^2/m_i v_0{}^2 b$. The approximation is valid for $\Delta\theta \ll 1$, and so, roughly, setting $(\Delta\theta/2)_{\max} = 1$ and $m_e v_0{}^2 = 3\kappa T$,

$$b_{\min} \approx \frac{Ze^2}{m_e v_0{}^2} \simeq \frac{Ze^2}{3\kappa T}$$

Because quantum effects have been neglected, if $e^2/m_e v_0{}^2$ is smaller than the de Broglie wavelength, the latter value should be chosen for $b_{\min}$. Again, the logarithmic dependence of $\langle\Delta\theta^2\rangle$ on $b_{\min}$ allows this somewhat cavalier treatment of the large-angle scattering contribution to $\langle\Delta\theta^2\rangle$. Using these estimates of $b_{\max}$ and $b_{\min}$ gives

$$\langle(\Delta\theta)^2\rangle = \left(\frac{8\pi nZ^2e^4}{m_e{}^2 v_0{}^4} \ln\Lambda\right)L \qquad (6.2.6)$$

where

$$\Lambda \equiv \frac{3}{2}\left(\frac{\kappa^3 T^3}{\pi n_e}\right)^{1/2} \frac{1}{Ze^3} \qquad (6.2.7)$$

and L is the distance the test particle has penetrated into the plasma.

From (6.2.7), the mean deflection angle of the test particle increases as the particle penetrates farther into the plasma, i.e., $\langle\Delta\theta^2\rangle$ increases with L. Setting $\langle\Delta\theta^2\rangle = 1$ gives an approximate path length for the test particle to scatter, by random walk, through a total angle of 90°

$$L_{90^\circ} \approx \left[8\pi n\left(\frac{e^2}{m_e v_0{}^2}\right)^2 \ln\Lambda\right]^{-1} \qquad (6.2.8)$$

The cross section for 90° scattering by multiple Coulomb collisions is usually defined as

$$\sigma_{90°} \equiv \frac{1}{nL_{90°}} = 8\pi\left(\frac{Ze^2}{m_e v_0^2}\right)^2 \ln \Lambda \qquad (6.2.9)$$

Comparing this result with (6.1.15) shows that multiple scattering dominates single scattering in producing a net deflection of the test particle.

The ratio of the cross section for multiple scattering and single scattering is

$$\frac{\sigma_{90°\mathrm{M}}}{\sigma_{90°\mathrm{S}}} = 8 \ln \Lambda \qquad (6.2.10)$$

This ratio depends weakly on the quantity Λ, and $\ln \Lambda$ is nearly constant over a large range of parameters. For most laboratory plasmas, $\ln \Lambda$ lies between 10 and 20. This is illustrated by the values given in Table 6.2.1 for $Z=1$. Thus a large-angle deflection of a particle in a plasma is a hundred times more likely to result as a consequence of the many small-angle scatterings that take place as the particle moves through the plasma than as a result of a single binary interaction.

This multiple Coulomb scattering cross section for a plasma is approximately given by ($\ln \Lambda = 20$, $Z = 1$)

$$\sigma_{90°} \approx \frac{2.6 \times 10^{-18}}{W_{\mathrm{keV}}^2} \quad \mathrm{cm}^2 \qquad (6.2.11)$$

Thus, in a plasma at 100 keV, the Coulomb cross section is 2.6×10^{-22} cm^2, which is larger than the cross section for a *d-d* fusion reaction, which is $\sigma_{dd} \approx 10^{-26}$cm^2. This means that Coulomb collisions that result in particle losses through diffusion occur more frequently than do collisions that result

Table 6.2.1 VALUES OF $\ln \Lambda$

T, °K	Electron density n_e, cm^{-3}								
	1	10^3	10^6	10^9	10^{12}	10^{15}	10^{18}	10^{21}	10^{24}
10^2	16.3	12.8	9.43	5.97					
10^3	19.7	16.3	12.8	9.43	5.97				
10^4	23.2	19.7	16.3	12.8	9.43	5.97			
10^5	26.7	23.2	19.7	16.3	12.8	9.43	5.97		
10^6	29.7	26.3	22.8	19.3	15.9	12.4	8.96	5.54	
10^7	32.0	28.5	25.1	21.6	18.1	14.7	11.2	7.85	4.39
10^8	34.3	30.9	27.4	24.0	20.5	17.0	13.6	10.1	6.69

in a nuclear reaction. In the study of controlled fusion, it is important to eliminate the instabilities that cause rapid particle losses, but it is also important to understand the role played by collisional diffusion losses because they cannot be avoided.

If the above plasma had a density of 10^{14}cm^{-3}, the collision mean free path for an electron would be about 10^8 cm, and the corresponding collision frequency would be about 20 s^{-1}. Thus the electrons do not collide very frequently on the time scale of many of the contemporary low-density plasma-containment experiments, and the use of a "collisionless" plasma model is seen to be reasonable on this basis.

6.3 FOKKER-PLANCK THEORY FOR TRANSPORT IN A FULLY IONIZED PLASMA

The fact that large-angle deflection of a charged particle in a plasma occurs as a succession of small-angle scatterings rather than as a single binary scattering can be used as a basis for a model of diffusion, conductivity, and other transport properties of a fully ionized plasma. This theory is embodied in the Fokker-Planck equation. A somewhat intuitive derivation of this equation is given in this section.[1] A rigorous treatment of collisions based on continuing the expansion of the kinetic equations (Sec. 3.1.12) to include terms of order $g = 1/n\lambda_D^3$ is presented in Sec. 11.11, leading to the Balescu-Lenard equation. It is shown there that the Fokker-Planck theory is a good approximation to, and far simpler than, the exact theory.

Diffusion in a plasma is controlled, not by the random walk of one particle through a fixed background, but by the change of the distribution of particles $f(\mathbf{v}, t)$ as a result of many small-angle scatterings of particles with a variety of speeds; indeed, with speeds distributed according to $f(\mathbf{v}, t)$. In terms of the probability $F(\mathbf{v}, \Delta\mathbf{v})$ that a particle changes its velocity from $\mathbf{v}$ to $\mathbf{v} + \Delta\mathbf{v}$ in a time Δt, the distribution at time t is obtained from the distribution function at time $t - \Delta t$ by

$$f(\mathbf{v}, t) = \int f(\mathbf{v} - \Delta\mathbf{v}, t - \Delta t) F(\mathbf{v} - \Delta\mathbf{v}, \Delta\mathbf{v})\, d(\Delta\mathbf{v}) \qquad (6.3.1)$$

This assumes that $f(\mathbf{v}, t)$ changes only due to collisions.

Assuming that small-angle scattering is the main effect of collisions, the velocity change $\Delta\mathbf{v}$ is small in a small time interval Δt, and (6.3.1) can be expanded

[1] M. N. Rosenbluth, W. M. MacDonald, and D. Judd, *Phys. Rev.*, **102**: 1 (1957).

$$f(\mathbf{v}, t) = \int \left\{ f(\mathbf{v}, t)F(\mathbf{v}, \Delta\mathbf{v}) - \Delta t F(\mathbf{v}, \Delta\mathbf{v}) \frac{\partial f}{\partial t} - \Delta\mathbf{v} \frac{\partial}{\partial \mathbf{v}} [f(\mathbf{v}, t)F(\mathbf{v}, \Delta\mathbf{v})] \right.$$

$$\left. + \frac{1}{2} \sum_i \sum_k \Delta v_i \, \Delta v_k \frac{\partial^2}{\partial v_i \partial v_k} [f(\mathbf{v}, t)F(\mathbf{v}, \Delta\mathbf{v})] + \cdots \right\} d(\Delta\mathbf{v}) \qquad (6.3.2)$$

The probability F is normalized, $\int F(\mathbf{v}, \Delta\mathbf{v})\, d(\Delta\mathbf{v}) = 1$; in addition, (6.3.2) includes only the change in f due to collisions. Thus (6.3.2) can be rewritten

$$\left.\frac{\partial f}{\partial t}\right|_{\text{collisions}} = \frac{1}{\Delta t} \int \left(-\Delta\mathbf{v} \cdot \frac{\partial}{\partial \mathbf{v}} fF + \frac{1}{2} \sum_i \sum_k \Delta v_i \, \Delta v_k \frac{\partial^2}{\partial v_i \partial v_k} fF \right) d(\Delta\mathbf{v}) \qquad (6.3.3)$$

In terms of the average change in the velocity moments of f per unit time, defined by

$$\left\langle \frac{\Delta\mathbf{v}}{\Delta t} \right\rangle = \frac{1}{\Delta t} \int f(\mathbf{v}, t)F(\mathbf{v}, \Delta\mathbf{v})\, \Delta\mathbf{v}\, d(\Delta\mathbf{v}) \qquad (6.3.4)$$

and

$$\left\langle \frac{\Delta v_i \, \Delta v_k}{\Delta t} \right\rangle = \frac{1}{\Delta t} \int f(\mathbf{v}, t)F(\mathbf{v}, \Delta\mathbf{v})\, \Delta v_i \, \Delta v_k \, d(\Delta\mathbf{v}) \qquad (6.3.5)$$

the rate of change of f with time due to collisions is

$$\left.\frac{\partial f}{\partial t}\right|_c = \frac{\partial}{\partial \mathbf{v}} \cdot \left\langle \frac{\Delta\mathbf{v}}{\Delta t} \right\rangle f(v, t) + \frac{1}{2} \sum_{i,k} \frac{\partial^2}{\partial v_i \partial v_k} \left[\left\langle \frac{\Delta v_i \Delta v_k}{\Delta t} \right\rangle f(\mathbf{v}, t) \right] \qquad (6.3.6)$$

where terms of order $(\Delta v)^3$ have been neglected.

A system in thermodynamic equilibrium satisfies $(\partial f/\partial t)_c = 0$. A system not in equilibrium can change in two ways. First, the velocity of a group of particles can decrease (or increase). This effect is akin to dynamical friction. It is given by the first term on the right-hand side of (6.3.6), and is proportional to $\langle \Delta\mathbf{v} \rangle$. Second, the velocity distribution may spread. This would be termed velocity diffusion, and is given by the $\langle \Delta\mathbf{v}\, \Delta\mathbf{v} \rangle$ term in the Fokker-Planck equation. Figure 6.3.1. shows how collisions would alter the distribution of a tenuous group of nearly monoenergetic test particles originally streaming through a background plasma at rest.

The Fokker-Planck equation (6.3.6) is valid for any system in which collisions produce only small changes in the velocity $\mathbf{v}$ of the particles, with large changes occurring only as a result of many small changes.

To apply (6.3.6) to a plasma, it is necessary to express the quantities $\langle \Delta\mathbf{v}/\Delta t \rangle$ and $\langle \Delta v_i \, \Delta v_j/\Delta t \rangle$ in terms of the collision properties of the particles in the plasma. This is done exactly in the way that $(\Delta\theta)^2$ was calculated in the preceding section. Consider a test particle moving with a speed $\mathbf{v}_T$, relative to particles in the plasma which move with speed $\mathbf{v}$.

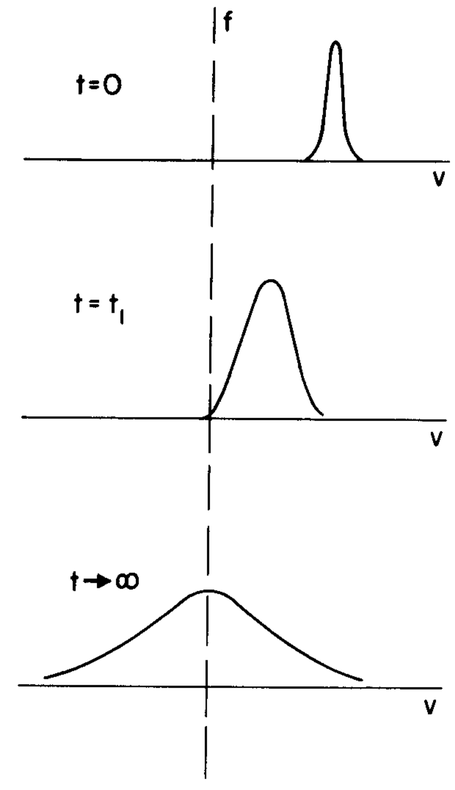


FIGURE 6.3.1
Schematic of collision-induced alteration of the velocity distribution as a function of time for a beam of initially monoenergetic electrons streaming through a background plasma.

To find the average rate of change of $\mathbf{v}_T$ as the test particle moves through the plasma, *two* averages must be performed. First, consider only the interaction of the particle with field particles of speed $\mathbf{v}$. These field particles can be located at any impact parameter with respect to the test particles. So first the interaction of the test particle $\mathbf{v}_T$ with $f(\mathbf{v})\, d\mathbf{v}$ must be averaged over impact parameters; call this average $(\Delta v_i)_T$. Next, this average change of $\mathbf{v}_T$ produced by all particles moving with $\mathbf{v}$ must be averaged over the distribution of field particles, using the fact that the density of field particles with velocity $\mathbf{v}$ is $f(\mathbf{v})$, so that the average of $\Delta\mathbf{v}$ over the field-particle distribution yields, where i and j are vector components,

$$\langle \Delta v_i \rangle_T = \int (\Delta v_i)_T n f(\mathbf{v})\, d\mathbf{v}$$

$$\langle \Delta v_i \, \Delta v_j \rangle_T = \int (\Delta v_i \, \Delta v_j)_T n f(\mathbf{v})\, d\mathbf{v} \qquad (6.3.7)$$

The calculation of these averages is most easily performed in the center-of-mass frame (later transforming back into the laboratory frame to find $\langle \Delta v_T \rangle$). Let $\mathbf{u}$ be the relative velocity of the test and field particles in the center of mass, and θ the scattering angle, as in Fig. 6.1.1. There are three components of

$\Delta\mathbf{u}$, and in a cylindrical coordinate system with the z axis along the incoming velocity $\mathbf{u}_{in}$, the change in $\mathbf{u}$ is related to the scattering angle θ and the polar angle ϕ (Fig. 6.3.2) for small-angle scattering by

$$\hat{\mathbf{u}}_1 \cdot \Delta\mathbf{u} = -|\Delta u| \sin\frac{\theta}{2} \approx -(u\theta)\frac{\theta}{2}$$

$$\hat{\mathbf{u}}_2 \cdot \Delta\mathbf{u} = |\Delta u| \cos\frac{\theta}{2} \sin\phi \approx u\theta \sin\phi$$

$$\hat{\mathbf{u}}_3 \cdot \Delta\mathbf{u} = |\Delta u| \cos\frac{\theta}{2} \cos\phi \approx u\theta \cos\phi$$

The average of these changes over impact parameters is found as in Sec. 6.2; the scattering angle θ is related to the impact parameter b by

$$\theta = \frac{2q_T q_F}{\mu u^2 b} \qquad (6.3.8)$$

[where $\mu = m_T m_F/(m_T + m_F)$ is the reduced mass of test and field-particle system]. The number of scattering centers encountered in a time Δt (during which the test particles move a distance $u\,\Delta t$) that lead to scattering at an angle θ is

$$N = [\bar{n}f(\mathbf{v})\,d\mathbf{v}]u\,\Delta t\,d\phi\,b\,db \qquad (6.3.9)$$

and so the average deflection $(\Delta\mathbf{u})$ of a test particle by a single field particle moving with $\mathbf{v}$ is, in the center-of-mass frame,

$$\mathbf{u}_0 \cdot (\Delta\mathbf{u}) = \frac{4\pi\bar{n}_F {q_F}^2 {q_T}^2}{\mu^2 u^2}\,\Delta t \int_{b_{\min}}^{b_{\max}} \frac{db}{b} = \frac{4\pi\bar{n}_F {q_F}^2 {q_T}^2}{\mu^2 u^2}\,\Delta t \ln\Lambda \qquad (6.3.10)$$

$$\hat{\mathbf{u}}_2 \cdot (\Delta\mathbf{u}) = \hat{\mathbf{u}}_3 \cdot (\Delta\mathbf{u}) = 0 \qquad (6.3.11)$$

In (6.3.10)

$$\Lambda = \frac{3}{2}\left(\frac{\kappa^3 T^3}{\pi n}\right)^{1/2} \frac{1}{Ze^3}$$

and the approximation and reasoning involved in replacing $\int db/b$ by $\ln\Lambda$ is discussed following Eq. (6.2.4). The result (6.3.11) follows from symmetry of the scattering in polar angle ϕ.

The transformation of (6.3.10) into the laboratory frame, in order to find the change in the test-particle velocity Δv_T, is performed by using

$$\mathbf{v}_T = \mathbf{U} + \frac{m_F}{m_T + m_F}\,\mathbf{u}$$

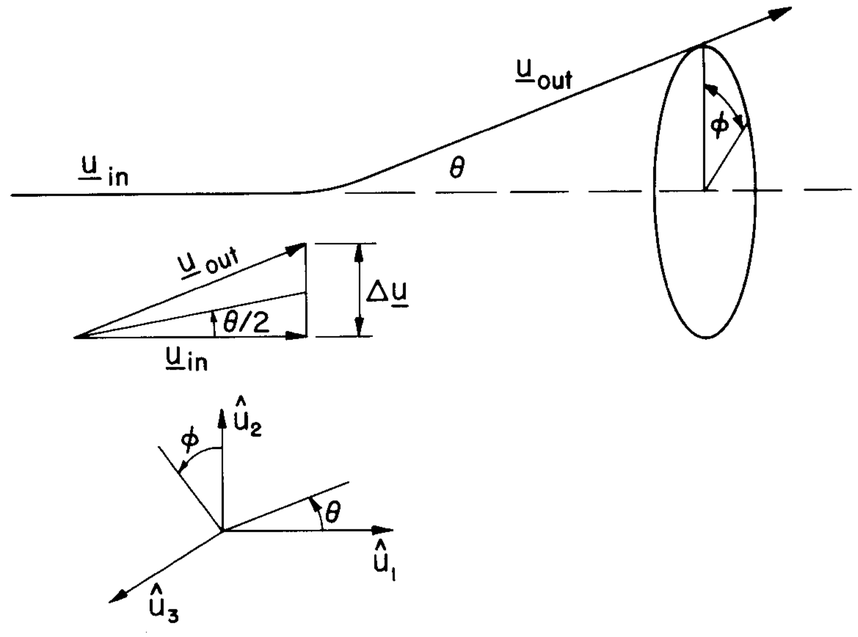


FIGURE 6.3.2
Velocity-space coordinate system used to compute changes in distribution functions caused by collisions.

where U is the center-of-mass velocity and m_T, and m_F are the masses of the test and field particles, respectively. Then, since $\Delta\mathbf{v}_T = \Delta\mathbf{u}m_F/(m_T + m_F)$, it follows that

$$\frac{\Delta\mathbf{v}_T}{\Delta t} = \frac{4\pi\bar{n}_F {q_T}^2 {q_F}^2 \mathbf{u}}{m_T \mu u^3} \ln\Lambda$$

where $\mu = m_T m_F/(m_T + m_F)$.

A similar calculation yields

$$(\hat{\mathbf{u}}_2 \cdot \Delta\mathbf{u})(\hat{\mathbf{u}}_2 \cdot \Delta\mathbf{u}) = (\hat{\mathbf{u}}_3 \cdot \Delta\mathbf{u})(\hat{\mathbf{u}}_3 \cdot \Delta\mathbf{u}) = \frac{4\pi\bar{n}_F {q_T}^2 {q_F}^2}{\mu^2 u}\,\Delta t \ln\Lambda$$

$$(\hat{\mathbf{u}}_i \cdot \Delta\mathbf{u})(\hat{\mathbf{u}}_j \cdot \Delta\mathbf{u}) = 0 \qquad \text{if } i \neq j$$

$$(\hat{\mathbf{u}}_1 \cdot \Delta\mathbf{u})(\hat{\mathbf{u}}_1 \cdot \Delta\mathbf{u}) \ll (\hat{\mathbf{u}}_2 \cdot \Delta\mathbf{u})(\hat{\mathbf{u}}_2 \cdot \Delta\mathbf{u})$$

which transforms in the laboratory frame to

$$[\Delta v_i\,\Delta v_k] = \frac{4\pi\bar{n}_F {q_T}^2 {q_F}^2 \Delta t}{{m_T}^2 u}\left(\delta_{ik} - \frac{u_i u_k}{u^2}\right)\ln\Lambda \qquad (6.3.12)$$

Problem 6.3.1 What is the approximation involved in neglecting $(\hat{\mathbf{u}}_1 \cdot \Delta\mathbf{u})^2$ compared with $(\hat{\mathbf{u}}_2 \cdot \Delta\mathbf{u})^2$?

////

Taking the average over the velocity distribution of scattering particles, Eq. (6.3.7) gives the coefficients required to complete the Fokker-Planck description of collisional effects in a fully ionized plasma.

$$\left\langle \frac{\Delta \mathbf{v}_T}{\Delta t} \right\rangle = -\frac{4\pi \bar{n}_\alpha q_T{}^2 q_\alpha{}^2 \ln \Lambda}{\mu m_T} \int f_\alpha(\mathbf{v}') \frac{\mathbf{v}_T - \mathbf{v}'}{|\mathbf{v}_T - \mathbf{v}'|^3} d\mathbf{v}' \qquad (6.3.13)$$

and

$$\left\langle \frac{\Delta v_i \Delta v_j}{\Delta t} \right\rangle = \frac{4\pi \bar{n}_\alpha q_T{}^2 q_\alpha{}^2 \ln \Lambda}{m_T{}^2} \frac{\partial^2}{\partial v_i \partial v_j} \int f_\alpha(\mathbf{v}') |\mathbf{v}_T - \mathbf{v}'| \, d\mathbf{v}' \qquad (6.3.14)$$

where the identity

$$\frac{\partial^2 u}{\partial v_i \partial v_j} = \frac{u\delta_{ij} - u_i u_j/u}{u^2}$$

has been used in obtaining (6.3.14), and the subscript α refers to the species of the field particles (ions or electrons).

A standard form for the Fokker-Planck equation is obtained by defining the auxiliary functions

$$g_\alpha(\mathbf{v}) = \int f_\alpha(\mathbf{v}') |\mathbf{v} - \mathbf{v}'| \, d\mathbf{v}'$$

$$h_\alpha(\mathbf{v}) = \frac{m_T}{\mu_\alpha} \int \frac{f_\alpha(\mathbf{v}')}{|\mathbf{v} - \mathbf{v}'|} d\mathbf{v}'$$

$$\mu_\alpha = \frac{m_T m_\alpha}{m_T + m_\alpha}$$

whereupon (6.3.6) becomes

$$\left.\frac{\partial f}{\partial t}\right|_c = \sum_\alpha \left[-\frac{\partial}{\partial v_i}\left(f_T \frac{\partial h_\alpha}{\partial v_i} \right) + \frac{1}{2} \frac{\partial^2}{\partial v_i \partial v_j}\left(f_T \frac{\partial^2 g_\alpha}{\partial v_i \partial v_j} \right) \right] \frac{4\pi n_\alpha q_T{}^2 q_\alpha{}^2}{m_T{}^2} \ln \Lambda \qquad (6.3.15)$$

Equation (6.3.15) describes the change in the distribution of test particles f_T due to collisions with particles of all types α in the plasmas. In practice, for example, f_T can be treated as the distribution of electrons, and (6.3.15) describes the effect of collisions with electrons and ions. Similarly, f_T can be taken to be the ion distribution, and (6.3.15) used to find the effect that collisions with ions and electrons have in changing f_i.

6.4 RELAXATION TIMES IN A FULLY IONIZED PLASMA

The Fokker-Planck equation (6.3.15) can be used to find the time scale for a nonthermal plasma to relax to a maxwellian distribution. This can be done only approximately, since the result will in fact depend on a time history of the detailed distribution. An alternative to solving the general problem of the time evolution of the plasma distribution is to consider a beam of test particles of density n_T, distributed at time $t = 0$, by

$$f_T(\mathbf{v}, t = 0) = \delta(\mathbf{v} - \mathbf{v}_0) \qquad (6.4.1)$$

passing through a plasma. If the density of the test particles is low, they will not be affected by collisions with each other, but they will be affected by collisions with the electrons and ions in the plasma, of density n_0, distributed in velocity according to

$$f_e = \left(\frac{m_e}{2\pi\kappa T}\right)^{3/2} e^{-m_e v^2/2\kappa T}$$

$$f_i = \left(\frac{m_i}{2\pi\kappa T}\right)^{3/2} e^{-m_i v^2/2\kappa T} \qquad (6.4.2)$$

The test particles can be considered alternatively as electrons or ions, and by calculating their slowing down and their deflection in angle, one can arrive at a fairly realistic picture of how nonthermal features of a distribution are "erased" by collisions.

6.4.1 Deceleration of a Beam by a Plasma

The evolution of the test-particle distribution from its $t = 0$ value (6.4.1) is described by (6.3.15). A convenient way to depict the velocity diffusion of the beam of test particles is to take velocity moments of (6.3.15). Multiplying that equation by $\mathbf{v}$ and integrating $d\mathbf{v}$ gives, defining $\int f_T \mathbf{v} \, d\mathbf{v} \equiv \mathbf{U}$,

$$\frac{\partial U}{\partial t} = \sum_\alpha \gamma_\alpha \int \bar{n}_\alpha \left(f_T \frac{\partial h_\alpha}{\partial \mathbf{v}} \right) d\mathbf{v} \qquad (6.4.3)$$

where $\gamma_\alpha = 4\pi q_T{}^2 q_\alpha{}^2 (\ln \Lambda)/m_T{}^2$. The slowing of the beam at $t = 0$ is found by substituting the $t = 0$ value of the test-particle distribution on the right-hand side of (6.4.3) and performing the integrals h_α. This gives

$$\frac{\partial \mathbf{U}}{\partial t} = \frac{4\pi n_e e^2 q_T{}^2 \ln \Lambda}{m_T{}^2} \frac{\partial}{\partial U}\left[\frac{1}{U}\Phi\left(U\sqrt{\frac{m_e}{2\kappa T_e}}\right)\left(1+\frac{m_T}{m_e}\right) + \frac{Z}{U}\Phi\left(U\sqrt{\frac{m_i}{2\kappa T_i}}\right)\left(1+\frac{m_T}{m_i}\right)\right] \tag{6.4.4}$$

where

$$\Phi(y) = \frac{2}{\sqrt{\pi}}\int_0^y e^{-x^2}\,dx$$

is the error function. Note that $\int d\mathbf{x}\,[\exp(-x^2)]/|\mathbf{x}-\mathbf{y}| = \Phi(y)\,\pi^{3/2}/y$.

The first term on the right-hand side of (6.4.4) gives the drag on the test-particle beam due to plasma electrons; the second term gives the drag due to ions. See Fig. 6.4.1 for the plot of Φ. It is seen that the slowing rate is a function of the speed of the beam relative to the thermal particles in the plasma, as shown schematically in Fig. 6.4.1. If the beam moves much faster than the plasma particles, the slowing rate is quite small, while for $U \lesssim \sqrt{2\kappa T/m}$ the slowing is much more pronounced. Defining the slowing-down time by

$$\tau_s = -U\left(\frac{\partial U}{\partial t}\right)^{-1}$$

some limiting cases are obtained:

1 $U > \sqrt{\dfrac{\kappa T_e}{m_e}} \gg \sqrt{\dfrac{\kappa T_i}{m_i}}$ (6.4.5)

$$\tau_s \approx \frac{m_T{}^2 U^3}{4\pi n_e e^2 q_T{}^2 (2 + m_T/m_e)\ln\Lambda}$$

a Slowing is mainly due to scattering off of electrons.
b τ_s is longer for heavy particles.
c τ_s is relatively insensitive to the plasma temperature.

2 $U < \sqrt{\dfrac{\kappa T_i}{m_i}} \ll \sqrt{\dfrac{\kappa T_e}{m_e}}$ (6.4.6)

$$\tau_s \approx \frac{3\sqrt{\pi} m_T{}^2}{16\pi n_e e^2 q_T{}^2\left[\left(1+\dfrac{m_T}{m_e}\right)\left(\dfrac{m_e}{2\kappa T_e}\right)^{3/2} + Z\left(1+\dfrac{m_T}{m_i}\right)\left(\dfrac{m_i}{2\kappa T_i}\right)^{3/2}\right]\ln\Lambda}$$

a Slowing is mainly due to scattering off of ions, unless $T_i/T_e > (m_i/m_e)^{1/3}$.
b Heavier test charges take longer to slow down.
c Hotter plasmas take longer to slow the beam.

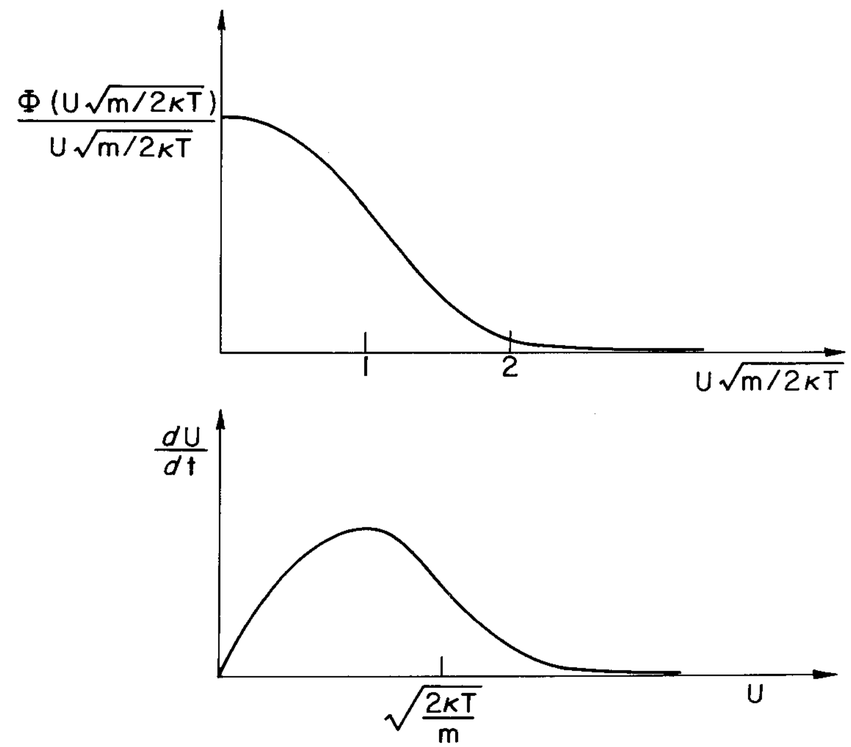


FIGURE 6.4.1
Error function and drag on a test particle in a plasma due to electrons as a function of test-particle speed.

3 $\sqrt{\dfrac{\kappa T_i}{m_i}} < U < \sqrt{\dfrac{\kappa T_e}{m_e}}$ (6.4.7)

$$\tau_s \approx \frac{m_T{}^2}{4\pi n_e e^2 q_T{}^2\left[\dfrac{Z}{U^3}\left(1+\dfrac{m_T}{m_i}\right) + \dfrac{4}{3\sqrt{\pi}}\left(1+\dfrac{m_T}{m_e}\right)\left(\dfrac{m_e}{2\kappa T_e}\right)^{3/2}\right]\ln\Lambda}$$

Thus the Fokker-Planck theory gives fairly simple-looking results for the collisional slowing of a beam.

Problem 6.4.1 Discuss the similarities or differences between (6.4.5) to (6.4.7) and the simplified calculation of the 90° deflection time by multiple scattering implied in (6.2.8). ////

Problem 6.4.2 Compare the growth time for two-stream instability (Sec. 5.5) with the collisional τ_s and explain when each of these mechanisms will dominate. ////

6.4.2 Spreading of a Beam by a Plasma

In addition to slowing a monoenergetic beam, collisions with plasma particles will spread the beam, both by deflecting the particles in the beam (tending toward isotropy) and by producing a distribution of velocities (tending toward an equilibrium distribution). These effects are extracted by multiplying the Fokker-Planck equation by v^2 and integrating $d\mathbf{v}$.

Considering separately the change in v^2 parallel to the original direction of U and perpendicular to the original direction of U gives, at $t = 0$,

$$\frac{\partial}{\partial t}(U_\parallel{}^2) = \frac{4\pi n_e e^2 q_T{}^2 \ln\Lambda}{m_T{}^2}\left[\frac{\partial^2}{\partial U_\parallel{}^2}\int d\mathbf{v}'\,(f_e + Zf_i)|\mathbf{U}-v'| + 2U_\parallel\frac{\partial}{\partial U_\parallel}\sum_\alpha\left(1+\frac{m_T}{m_\alpha}\right)\int d\mathbf{v}'\,\frac{Z_\alpha f_\alpha|\mathbf{v}'|}{|\mathbf{U}-\mathbf{v}'|}\right] \tag{6.4.8}$$

and

$$\frac{\partial}{\partial t}(U_\perp{}^2) \equiv \frac{\partial}{\partial t}(U_2{}^2 + U_3{}^2) = \frac{4\pi n_e e^2 q_T{}^2 \ln\Lambda}{m_T{}^2}\left(\frac{\partial^2}{\partial U_2{}^2}+\frac{\partial^2}{\partial U_3{}^2}\right) \times \int d\mathbf{v}'\,(f_e + Zf_i)|\mathbf{U}-\mathbf{v}'| \tag{6.4.9}$$

The diffusion of the beam toward isotropy is produced only by the dispersion term ($\partial^2/\partial U_\perp{}^2$), while the spread in $U_\parallel$ is produced both by the friction term ($\partial/\partial U_\parallel$) and by the dispersion term.

Defining a deflection time by

$$\tau_D = \frac{U^2}{\partial U_\perp{}^2/\partial t}$$

gives an estimate of the time required to produce isotropy from a beam. For suprathermal particles ($U > \sqrt{\kappa T_e/m_e}$),

$$\tau_D \approx \frac{m_T{}^2 U^3}{16\pi n_e e^2 q_T{}^2 \ln\Lambda}; \qquad U > \sqrt{\frac{\kappa T_e}{m_e}} \tag{6.4.10}$$

When the test particles are electrons, the deflection time is comparable with the slowing-down time [Eq. (6.4.5)], but when the test particles are fast ions, they slow much more rapidly than they disperse.

Another interesting limit is that of a test beam of thermal protons in a hydrogen plasma, in which case

$$\tau_D = \frac{\sqrt{m_i}(\kappa T_i)^{3/2}}{\sqrt{2}\pi n_e e^4 \ln\Lambda}\left[\Phi(1) + \frac{2e^{-1}}{\sqrt{\pi}}\right] \qquad \text{test ions, } U \approx \sqrt{\frac{\kappa T_i}{m_i}} \tag{6.4.11}$$

and a beam of thermal-test electrons is deflected in

$$\tau_D = \frac{\sqrt{m_e}(\kappa T_e)^{3/2}}{\sqrt{2}\pi n_e e^4 \ln\Lambda}\left[\Phi(1) + \frac{2e^{-1}}{\sqrt{\pi}} + 2\right] \qquad \text{test elections, } U \approx \sqrt{\frac{\kappa T_e}{m_e}} \tag{6.4.12}$$

Problem 6.4.3 Show that

$$\frac{1}{\tau_D} = \frac{4\pi n_e e^2 q_T{}^2 \ln\Lambda}{m_T{}^2 U^3}\frac{\partial}{\partial U}\left\{\sum_\alpha \frac{2\kappa T_\alpha}{m_\alpha}\frac{1}{U}\left[\frac{2U}{\sqrt{\pi}}\sqrt{\frac{m_\alpha}{2\kappa T_\alpha}}\,e^{-m_\alpha U^2/2\kappa T_\alpha} + \left(1+\frac{m_\alpha}{\kappa T_\alpha}U^2\right)\Phi\left(U\sqrt{\frac{m_\alpha}{2\kappa T_\alpha}}\right)\right]\right\} \qquad ////$$

Table 6.4.1. summarizes the deflection times in a variety of different plasma situations. By comparing (6.4.11) and (6.4.12) it is seen that electrons relax to an isotropic distribution more rapidly than ions, by $\sqrt{m_e/m_i}$.

$$\tau_{D,\ \text{thermal electrons}} \approx \left(\frac{m_e}{m_i}\right)^{1/2} \tau_{D,\ \text{thermal ions}}$$

In addition to deflection, energy spread is clearly a feature of the diffusion of a beam toward thermal equilibrium. Defining

$$W = \tfrac{1}{2}m_T U^2$$

$$\tau_W \equiv \frac{-W}{\partial W/\partial t}$$

the time scales for three characteristic processes can be obtained: nonthermal electrons can relax to a maxwellian T_e, nonthermal ions can relax to T_i, electrons and ions can thermalize, T_e, $T_i \to T$. These time scales are found by adding (6.4.8) and (6.4.9) and choosing for the test particles:

1 An electron beam with speed $U = \sqrt{2\kappa T_e/m_e}$

$$\tau_W \approx \frac{m_e{}^{1/2}(2\kappa T_e)^{3/2}}{8\pi n_e e^4\,[\Phi(1) - 4\,e^{-1}/\sqrt{\pi}]\ln\Lambda} = \text{time for electrons to become maxwellian at temperature } T_e$$

2 An ion beam with speed $U = \sqrt{2\kappa T_i/m_i}$

$$\tau_W \approx \frac{m_i{}^{1/2}(2\kappa T_i)^{3/2}}{8\pi n_e e^4\,[\Phi(1) - 4\,e^{-1}/\sqrt{\pi}]\ln\Lambda} = \text{time for ions to maxwellize at } T_i$$

Table 6.4.1 APPROXIMATE MAGNITUDES FOR SOME TYPICAL HYDROGEN PLASMAS*

Plasma	Number density n_0, cm^{-3}	Temperature T, °K	Plasma frequency $\left(\frac{4\pi n_0 e^2}{m}\right)^{1/2}$, s^{-1}	Debye length $L_D = \left(\frac{kT}{8\pi n_0 e^2}\right)^{1/2}$, cm	Particles in a Debye sphere $n_0 L_D^3$	$\ln \Lambda$	Deflection time for electrons τ_D	Deflection time for protons τ_D
Interstellar gas	1	10^4	6×10^4	5×10^2	10^8	22	3×10^3	3×10^5
Gaseous nebula	10^3	10^4	2×10^6	15	3×10^6	20	3	3×10^2
Solar corona	10^6	10^6	6×10^7	5	10^8	22	3	3×10^2
Solar wind	10	10^5	2×10^5	5×10^2	10^9	25	10^6	10^8
Solar atmosphere	10^{12}	10^4	6×10^{10}	5×10^{-4}	10^2	9	10^{-8}	10^{-6}
Gas discharge	10^{14}	10^4	6×10^{11}	5×10^{-5}	12	7	10^{-10}	10^{-8}
Hot plasma	10^{14}	10^5	6×10^{11}	1.5×10^{-4}	3×10^2	9	2×10^{-9}	2×10^{-7}
	10^{14}	10^6	6×10^{11}	5×10^{-4}	10^4	13	5×10^{-8}	5×10^{-6}
Diffuse hot plasma	10^{12}	10^6	6×10^{10}	5×10^{-3}	10^5	15	5×10^{-6}	5×10^{-4}
Dense hot plasma	10^{16}	10^6	6×10^{12}	5×10^{-5}	10^3	11	6×10^{-10}	6×10^{-8}
Thermonuclear plasma	10^{16}	10^8	6×10^{12}	5×10^{-4}	10^6	18	4×10^{-7}	4×10^{-5}

* The numbers are chosen only to illustrate orders of magnitude, and the nomenclature is somewhat arbitrary.
SOURCE: Table 6.4.1 is from D. C. Montgomery and D. A. Tidman, "Plasma Kinetic Theory," p. 31, McGraw-Hill, New York, 1964.

3 An ion beam with speed $U = \sqrt{2\kappa T/m_e}$

$$\tau_W \approx \frac{m_i}{\sqrt{m_e}} \frac{(2\kappa T)^{3/2}}{4\pi n_e e^4 \ln \Lambda} = \text{time for this ion beam to share its energy with electrons—one (quite arbitrary) way of estimating time required for electrons and ions to equilibrate, and far from a general result.}$$

The implication of the above is that the electrons are quick to isotropize, ions take longer to equilibrate with each other, and diffusion of electrons and ions toward equilibrium with each other is a slow process indeed.

6.5 TRANSPORT PROPERTIES OF A FULLY IONIZED PLASMA

The preceding section treated relaxation of a nonthermal plasma, but an equally useful application of the Fokker-Planck theory is to cases in which an external force drives the plasma away from thermal equilibrium, competing against the relaxation mechanisms already discussed. For example, electric fields drive currents in a plasma, resisted by the friction effects described by the collisional theories.

6.5.1 Electrical Conductivity

If an electric field $\mathbf{E}$ is applied to a plasma, a steady state may be reached in which the acceleration of particles by $\mathbf{E}$ is balanced by the collisional drag. If such a state is reached, then in a frame moving with the *electron* drift speed, the ion motion is described by [see, for example, Eq. (6.4.4)]

$$q_i E = -\frac{\partial}{\partial t}(m_i U_i)|_c = -\frac{4\pi n e^2 q_i^2}{m_i} \ln \Lambda \frac{\partial}{\partial U}\left[\frac{1}{U}\Phi\left(U\sqrt{\frac{m_e}{2\kappa T_e}}\right)\left(1 + \frac{m_i}{m_e}\right)\right] \tag{6.5.1}$$

where spread in the ion distribution has been ignored; i.e., (6.4.1) is used for f_i. This restricts the calculation to values of E such that the steady state in velocity satisfies [see (6.4.7) with $m_T = m_i$]

$$U > \left(\frac{\kappa T_i}{m_i}\right)^{1/2} \left(\frac{m_i}{m_e}\right)^{1/6}$$

If the field is not too strong, the drift speed may be less than the electron thermal speed (Fig. 6.5.1), so that (6.5.1) is approximated [since $\Phi(x) \approx (2/\sqrt{\pi})(x - x^3/3)$ for small x] by

$$E = \frac{4\pi n_e e^2 q_i}{m_i}\left[\left(\frac{m_e}{2\kappa T_e}\right)^{3/2}\frac{4}{3\sqrt{\pi}}U\right]\frac{m_i}{m_e}\ln\Lambda = \frac{nq_iU}{\sigma} = \frac{J}{\sigma} \qquad (6.5.2)$$

This shows that for weak fields the conductivity of a fully ionized plasma is

$$\sigma = \frac{3m_e}{(16\sqrt{\pi})Ze^2\ln\Lambda}\left(\frac{2\kappa T}{m_e}\right)^{3/2} \qquad (6.5.3)$$

The approximations made in deriving (6.5.2) imply

$$\left(\frac{\kappa T_i}{m_i}\right)^{1/2} < \frac{\sigma E}{nq_i} < \left(\frac{\kappa T_e}{m_e}\right)^{1/2} \qquad (6.5.4)$$

The result (6.5.3) confirms the notion that hotter plasmas are *better* conductors of electricity. The conductivity is independent of n, the density of charge carriers, because although the number of charge carriers increases with n, the number of scattering centers increases as well.

There is no inherent reason why a steady state should be reached. Indeed, for fields so large that

$$E > \frac{ne}{\sigma}\sqrt{\frac{\kappa T_e}{m_e}}$$

Eq. (6.5.1) has no solution. This is because $\Phi(x)$ has a maximum value near $x = 1$. To see this, note that the large-argument approximation $\Phi(U\sqrt{m_e/2\kappa T_e}) \simeq 1$ gives

$$E = \frac{4\pi n_e e^2 q_i \ln\Lambda}{m_e}\frac{1}{U^2} \qquad U > \sqrt{\frac{2\kappa T_e}{m_e}}$$

which *decreases* with U. This means that (6.5.1), which assumed a balance of force between the applied E field and the frictional drag, has no solution for

$$E \gg \frac{4\pi n_e e^2 q_i \ln\Lambda}{m_e}\frac{m_e}{2\kappa T_e}$$

A more accurate calculation of the maximum of

$$\frac{\partial}{\partial U}\frac{\Phi(U\sqrt{m_e/2\kappa T_e})}{U}$$

gives

$$E > 0.43\left(\frac{4\pi n_e e^2 q_i \ln\Lambda}{2\kappa T_e}\right) \equiv E_0$$

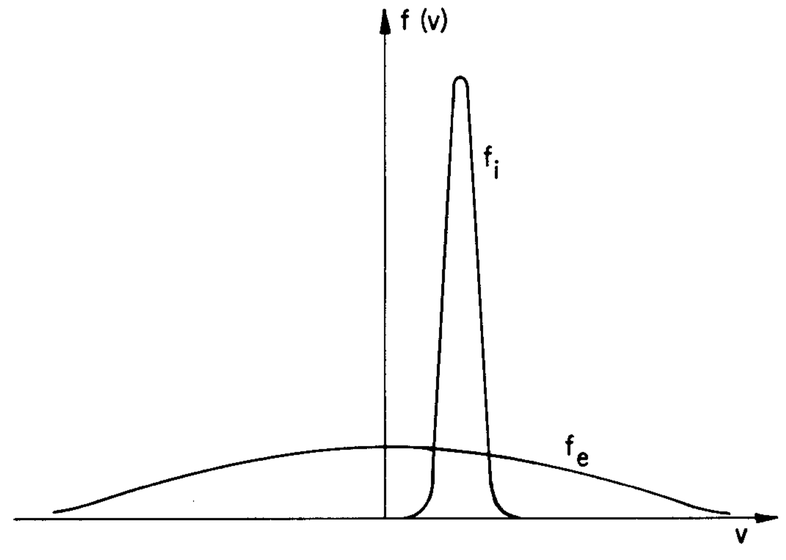


FIGURE 6.5.1
Plasma distribution function in a weak electric field showing a situation where the drift velocity of the ions is less than the thermal velocity of the electrons.

as the field beyond which there is no steady-state relation between collisions and the driving field. This is because the particles acquire speed between collisions, the collision cross section decreases, and for sufficiently rapid acceleration, the particles simply run away.

A further complication is that E_0 decreases with temperature, while T_e increases with time in a frictional system. Eventually, even a weak field will lead to runaway.

Problem 6.5.1 Show that E_0 is the field required to accelerate an electron by $\sqrt{\kappa T_e/m_e}$ in an electron-electron collision time. ////

6.5.2 Diffusion across a Magnetic Field

A plasma with a density gradient would naturally tend to diffuse so as to reduce the gradient. A magnetic field perpendicular to the gradient will reduce this diffusion, and confine the plasma. Indeed, the fluid equations (Chap. 3) show that if collisions are ignored, a steady state can be maintained in which the plasma has a gradient in pressure, provided that

$$\frac{\mathbf{J}\times\mathbf{B}}{c} = \nabla p \qquad (6.5.5)$$

and

$$\nabla\times\mathbf{B} = \frac{4\pi\mathbf{J}}{c} \qquad (6.5.6)$$

However, collisions will destroy this steady state and allow particles to walk across the field lines. Figure 6.5.2. shows an ion gyrating in a plasma that has a density gradient dn/dx, as shown, and a magnetic field $B_0\hat{\mathbf{z}}$. Because of the density gradient, there are $(n_{i0} + \frac{1}{2}a_i\, dn_{i0}/dx)\Delta V$ ions, whose guiding centers are at $x_0 + a_i/2$, that travel through a fluid element ΔV at x_0 with speed $+\bar{v}_0$. Similarly, there are $(n_{i0} + \frac{1}{2}a_i\, dn_{i0}/dx)\Delta V$ ions, gyrating about $x_0 - a_i/2$, that pass through Δv with speed $-\bar{v}_0$. The average speed of ions in the fluid element is thus

$$\langle \mathbf{v} \rangle = \frac{1}{n_{i0}} \frac{dn_{i0}}{dx}\, a_i\, \bar{v}_0 \hat{\mathbf{y}} = \frac{1}{n_{i0}} \frac{dn_{i0}}{dx} \frac{\kappa T_i}{eB_0} \hat{\mathbf{y}}$$

Then the drag force in the y direction is, from (6.4.4), and as in (6.5.1) and (6.5.2),

$$\mathbf{F}_D = -\frac{4\pi n_e e^2 q_i^2}{m_e} \left(\frac{m_e}{2\kappa T_e}\right)^{3/2} \frac{4\langle v \rangle}{3\sqrt{\pi}} \ln \Lambda \hat{\mathbf{y}} \qquad (6.5.8)$$

This drag force induces a drift across the magnetic field

$$\mathbf{U} = \frac{\mathbf{F}_D \times \mathbf{B}c}{eB^2} \approx \frac{-c^2 \nabla p_i}{\sigma B^2} \hat{\mathbf{x}}$$

where

$$\sigma = \frac{3m_e}{16\sqrt{\pi} e^2 \ln \Lambda} \left(\frac{2\kappa T_e}{m_e}\right)^{3/2}$$

Thus the collisions allow a drift of the plasma into less dense regions, i.e., spatial diffusion. The particle flux is

$$\boldsymbol{\Gamma} \equiv n\mathbf{U},$$

and it is customary to define the diffusion coefficient in terms of the flux Γ and the gradient ∇n which produces the flux,

$$\boldsymbol{\Gamma} = -D\, \nabla n$$

so that

$$D = \frac{n\kappa T c^2}{\sigma B^2}$$

is the diffusion coefficient of fully ionized plasmas across a magnetic field. The fact that D is inversely proportional to B^2 is one reason that plasma experiments are customarily designed with large confining magnetic fields. Indeed, D is also reduced ($D \approx T^{-1/2}$) as the temperature is increased, and the feasibility of many

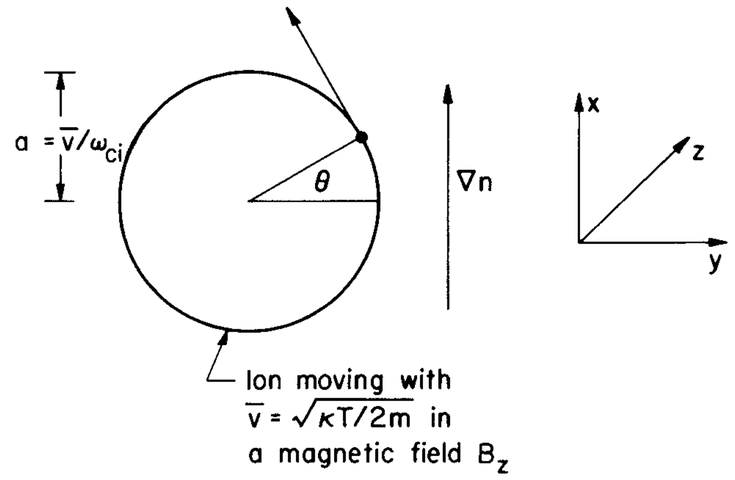


FIGURE 6.5.2
Schematic of a gyrating ion in a magnetized plasma that has a density gradient.

experiments, which plan to confine a hot plasma long enough to produce fusion reactions, is based on these scaling laws. The fact that experimentally $D \approx 1/B$ in many instances shows that there are other ways for a plasma to diffuse besides the collisional mechanisms discussed here (i.e., diffusion can also be produced by instability).

Other transport coefficients (e.g., heat flux for a given ∇T) can be simply calculated.

Problem 6.5.2 Calculate D for a plasma with a density gradient and *no* magnetic field. ////

Problem 6.5.3 Show that the energy flux across a magnetic field is

$$Q \approx \frac{8\pi(\nabla \kappa T) m^{1/2} n^2 e^2 c^2 \ln \Lambda}{(3\kappa T)^{1/2} B^2}$$

for a fully ionized plasma with a temperature gradient. ////

6.6 BOLTZMANN TRANSPORT EQUATION AND THE LORENTZ MODEL FOR A WEAKLY IONIZED PLASMA

The Fokker-Planck theory was appropriate for calculating transport properties in a fully ionized plasma, where diffusion takes place as a result of a large number of small-angle scatterings. In a weakly ionized plasma the situation is quite

different, since short-range encounters of (ionized) plasma particles with the neutral background dominates diffusion, conductivity, etc.

The collision of a plasma particle with a neutral molecule or atom can generally be treated as elastic. Further, the duration of such an encounter is in general much less than the time interval between collisions. This fact, along with the assumption of "molecular chaos," is the basis of a statistical model of a low-density gas, called the *Boltzmann transport equation*. The statistical description of a many-body system has the form (as shown in Chap. 3)

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + \mathbf{a} \cdot \nabla_{\mathbf{v}} f = \left.\frac{\partial f}{\partial t}\right|_c \qquad (6.6.1)$$

The quantity $(\partial f/\partial t)|_c$ is the time rate of change of the distribution function as a result of collisions.

If this change takes place as a result of a single two-body collision, it can be computed in terms of the scattering cross section between particles of a particular relative velocity and the velocity distribution of the particles. Consider, as in Secs, 6.2 and 6.3, the collision of two particles of velocities $\mathbf{v}$, $\mathbf{v}_1$ and masses m, m_1. Figure 6.6.1 shows the collision in the center-of-mass frame [recall that the elastic scattering of two particles is equivalent in the center-of-mass frame to the scattering of a particle of mass $mm_1/(m + m_1)$ with velocity $\mathbf{v} - \mathbf{v}_1$ from a fixed scattering center].

The flux of particles of velocity $\mathbf{v}_1$, of type α into an area $b\ db\ d\varepsilon$, is

$$\bar{n}_\alpha f_\alpha(\mathbf{x}, \mathbf{v}_1, t)|\mathbf{v} - \mathbf{v}_1|\ d\mathbf{v}_1\, b\ db\ d\varepsilon \qquad \mathrm{s}^{-1} \qquad (6.6.2)$$

while the number of target particles of velocity $\mathbf{v}$ of species γ is $n_\gamma f_\gamma(\mathbf{x}, \mathbf{v}, t)\ d\mathbf{v}$. Thus the number of such collisions per second per cm^3 is

$$\bar{n}_\alpha f_\alpha(\mathbf{x}, \mathbf{v}_1, t)|\mathbf{v} - \mathbf{v}_1|\,\bar{n}_\gamma f_\gamma(\mathbf{x}, \mathbf{v}, t) b\ db\ d\varepsilon\ d\mathbf{v}\ d\mathbf{v}.$$

In terms of the differential cross-section,

$$\frac{d\sigma}{d\Omega}\, d\Omega = b\ db\ d\varepsilon$$

where $d\Omega$ is the solid angle element $d\Omega = \sin\theta\ d\theta\ d\varepsilon$. Using the notation

$$f_\alpha(\mathbf{x}, \mathbf{v}, t) = f_\alpha$$

$$f_{\gamma 1}(\mathbf{x}, \mathbf{v}_1, t) = f_{\gamma 1}$$

the number of collisions with the geometry shown in Fig. 6.6.1 per second is

$$|\mathbf{v} - \mathbf{v}_1|\, f_\alpha f_{\gamma 1} \bar{n}_\alpha \bar{n}_\gamma \frac{d\sigma}{d\Omega}\, d\Omega \qquad (6.6.3)$$

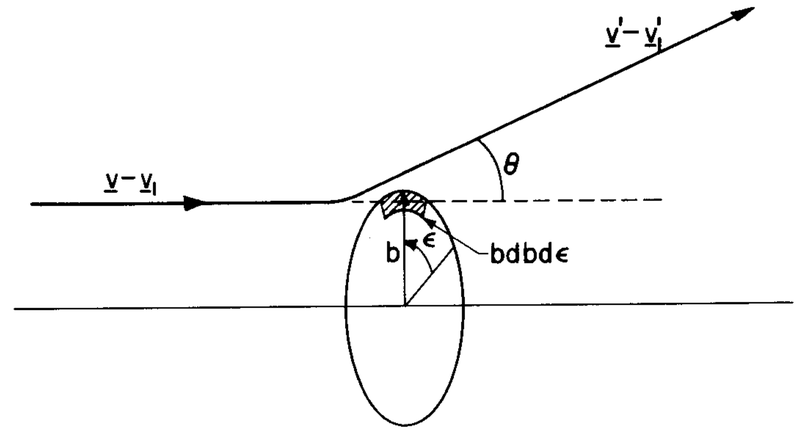


FIGURE 6.6.1
Center-of-mass coordinate system used to calculate the change of the distribution function as a result of collisions.

The total rate at which the distribution $f_\alpha(\mathbf{x}, \mathbf{v}, t)$ is diminished by collisions which scatter particles *out* of the range $(\mathbf{v}, \mathbf{v} + d\mathbf{v})$ is found by integrating (6.6.3) over all target velocities $\mathbf{v}_1$.

$$\bar{n}_\alpha \left.\frac{\partial f_\alpha}{\partial t}\right|_{\substack{\text{from scattering} \\ \mathbf{v} \to \mathbf{v}'}} d\mathbf{v} = -\sum_\gamma \bar{n}_\alpha \bar{n}_\gamma\, d\mathbf{v} \int d\mathbf{v}_1 f_\alpha f_{\gamma 1} |\mathbf{v} - \mathbf{v}_1| \frac{d\sigma}{d\Omega}\, d\Omega \qquad (6.6.4)$$

Similarly, the rate at which $f(\mathbf{x}, \mathbf{v}, t)$ is increased by collisions which scatter particles *into* the velocity range $(\mathbf{v}, \mathbf{v} + d\mathbf{v})$ is

$$\bar{n}_\alpha \left.\frac{\partial f_\alpha}{\partial t}\right|_{\substack{\text{from scattering} \\ \mathbf{v}' \to \mathbf{v}}} \mathbf{dv} = \sum_\gamma \bar{n}_\alpha \bar{n}_\gamma\, d\mathbf{v}' \int |\mathbf{v}' - \mathbf{v}_1'|\, f_\alpha(\mathbf{x}, \mathbf{v}', t) f_\gamma(\mathbf{x}, \mathbf{v}_1', t) \left(\frac{d\sigma}{d\Omega}\, d\Omega\right)' d\mathbf{v}_1' \qquad (6.6.5)$$

Since, from the symmetry of the scattering,

$$\left(\frac{d\sigma}{d\Omega}\right)' d\Omega' = \left(\frac{d\sigma}{d\Omega}\right) d\Omega$$

from energy and momentum conservation,

$$|\mathbf{v}' - \mathbf{v}_1'| = |\mathbf{v} - \mathbf{v}_1|$$

and $d\mathbf{v}\, d\mathbf{v}_1 = d\mathbf{v}'\, d\mathbf{v}_1'$, Eq. (6.6.4) can be rewritten

$$\bar{n}_\alpha \left.\frac{\partial f_\alpha}{\partial t}\right|_{\substack{\text{from scattering} \\ \mathbf{v}' \to \mathbf{v}}} = \sum_\gamma \bar{n}_\alpha \bar{n}_\gamma \int |\mathbf{v} - \mathbf{v}_1|\, d\mathbf{v}_1 \int f_\alpha' f_{\gamma 1}' \frac{d\sigma}{d\Omega}\, d\Omega \qquad (6.6.6)$$

Combining (6.6.4) and (6.6.6) gives the net change in f_α due to collisions, and (6.6.1) becomes

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v}\cdot\nabla f_\alpha + \frac{\mathbf{F}}{m_\alpha}\cdot\nabla_{\mathbf{v}} f_\alpha = \sum_\gamma \bar{n}_\gamma \int d\mathbf{v}_1 \int d\Omega \frac{d\sigma}{d\Omega} |\mathbf{v}-\mathbf{v}_1| [f_\alpha(\mathbf{x}, \mathbf{v}', t) f_\gamma(\mathbf{x}, \mathbf{v}_1', t) - f_\alpha(\mathbf{x}, \mathbf{v}, t) f_\gamma(\mathbf{x}, \mathbf{v}_1, t)] \qquad (6.6.7)$$

[Note that in using (6.6.7), the scattered velocities $\mathbf{v}'$, $\mathbf{v}_1'$ are related to the original velocities $\mathbf{v}$, $\mathbf{v}_1$ through the cross section $d\sigma/d\Omega$.]

Problem 6.6.1 Show that if, by use of suitable cutoffs, the Boltzmann collision operator [right-hand side of Eq. (6.6.7)] is evaluated using the cross section for Coulomb scattering, and only contributions from small-angle scattering are retained, a Fokker-Planck type of equation is recovered. (See D. C. Montgomery and D. A. Tidman, "Plasma Kinetic Theory," pp. 16–20, McGraw Hill, New York, 1964.) What cutoffs are required? ////

For a very weakly ionized plasma, (6.6.7) can be used to study the development of the distribution of the charged particles (i.e., the plasma). The development of $f(\mathbf{x}, \mathbf{v}, t)$ is determined by:

1 The average fields of the charged particles (included self-consistently in $\mathbf{F}$, through $\mathbf{F} = q_\alpha \mathbf{E} + q_\alpha \mathbf{v}\times\mathbf{B}/c$, $\nabla\cdot\mathbf{E} = 4\pi\sum_\alpha \bar{n}_\alpha q_\alpha \int f_\alpha\, d\mathbf{v} + 4\pi\rho_{\text{ext}}$, etc.)
2 Any applied fields $\mathbf{E}_0$, $\mathbf{B}_0$, which are also included in $\mathbf{F}$
3 Collisions with the neutral background

If the plasma is only weakly ionized, collisions between plasma particles can be ignored, $\sum_\gamma n_\gamma f_\gamma$ replaced by $n_{\text{neutral}}\, f_{\text{neutral}}$, and $d\sigma/d\Omega$ by the cross section for scattering of charged particles by neutral atoms.

The Boltzmann form for the collision integral is based on an assumption that is not generally valid for a fully ionized plasma. That is, it is based on the assumption that the duration of a collision is much less than the time between collisions. In a plasma, the duration of a collision is the transit time for the particle to travel a Debye length. This collision time is much greater than the time interval between collisions, since, for example, an electron enters the Debye sphere of a new ion long before it leaves the Debye sphere of the old ion. Thus, during the collision interval, the interaction is by no means binary, and the electron simultaneously interacts with all the other particles in the Debye sphere, which is by assumption a large number. To properly take collisions into account for plasma, it is necessary to develop a better model for the collision operator. The Fokker-Planck equation derived in Sec. 6.3 is an example of such a model.

Even though the Boltzmann transport equation is not an accurate method of calculating the change in the distribution as a result of collisions for a fully ionized plasma, it is a useful method of computing transport coefficients for a weakly ionized plasma where elastic collisions of electrons with neutral particles dominate. It is important to understand this problem because, in many applications, plasma properties are strongly influenced by neutral components. An example, the MHD generator, is given later in this chapter.

Because the Boltzmann equation is an approximation, it is important to see whether it has physically reasonable properties. These calculations are suggested in the following problems.

Problem 6.6.2 A physically reasonable distribution function $f(\mathbf{x}, \mathbf{v}, t)$ will be everywhere positive. Show that if $f(\mathbf{x}, \mathbf{v}, t=0)$ is everywhere positive, the Boltzmann equation will *keep* f positive for all time. ////

Problem 6.6.3 Show that distribution functions $f(\mathbf{x}, \mathbf{v}, t)$ which satisfy the Boltzmann equation conserve particles; that is, $\int f(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v}\, d\mathbf{x}$ is independent of time. ////

Problem 6.6.4 Show that if

$$f_\alpha = \left(\frac{m_\alpha}{2\pi\kappa T_\alpha}\right)^{3/2} e^{-m_\alpha v^2/2\kappa T_\alpha}$$

the Boltzmann collision operator vanishes. This implies that the Maxwell-Boltzmann distribution is an equilibrium solution of the Boltzmann equation. ////

Problem 6.6.5 Show that the entropy of a Boltzmann system increases, i.e.,

$$\frac{d}{dt}\int f \ln f\, d\mathbf{v} \geq 0$$

if f satisfies the Boltzmann equation. ////

6.7 MODIFIED BOLTZMANN EQUATION

The Boltzmann equation is nearly intractable in the form (6.6.7), both because it is in general nonlinear and also because of the complicated collision integral. In practice some approximation must be used to simplify this term. One simple

means of avoiding the complexities of the collision integral is to use a mean-free-time treatment. In this treatment one replaces the Boltzmann collision integral by

$$\left.\frac{\partial f}{\partial t}\right|_c = \frac{f_0 - f(\mathbf{x}, \mathbf{v}, t)}{\tau(v)} \approx \frac{f_0 - f(\mathbf{x}, \mathbf{v}, t)}{\tau} \qquad (6.7.1)$$

where $\tau(v)$ is replaced by a constant mean free time τ. This leads to a modified Boltzmann equation

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v} \cdot \frac{\partial f_\alpha}{\partial \mathbf{x}} + \frac{q_\alpha}{m}\left(\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c}\right) \cdot \frac{\partial f_\alpha}{\partial \mathbf{v}} = -\frac{f_\alpha(\mathbf{x}, \mathbf{v}, t) - f_{\alpha 0}}{\tau_\alpha} \qquad (6.7.2)$$

This expresses the idea that the distribution tends to relax toward an equilibrium value $f_{\alpha 0}$ due to collisions and that there is a characteristic time (in general, velocity-dependent) for momentum transfer collisions. If $\tau(v)$ is replaced by an average collision time τ, the collision term (6.7.1) becomes especially simple.

There are several aspects to the approximation (6.7.1) for the collision term which guide its use and limit its usefulness:

1 The equilibrium f_0 should be chosen as a *local* Maxwell-Boltzmann distribution in order that particles be conserved.

$$f_{\alpha 0} = \frac{n_\alpha(\mathbf{x}, t)}{\bar{n}_\alpha}\left(\frac{m_\alpha}{2\pi\kappa T_\alpha}\right)^{3/2} e^{-m_\alpha v^2/2\kappa T_\alpha}$$

$$\frac{n_\alpha(\mathbf{x}, t)}{\bar{n}_\alpha} \equiv \int f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v} \qquad (6.7.3)$$

where $\bar{n}_\alpha$ = total number of particles/total volume of system.

That this is required in order to conserve particles is seen by integrating (6.7.2) over velocity, obtaining (as in Chap. 3)

$$\frac{\partial n(\mathbf{x}, t)}{\partial t} + \nabla \cdot n\mathbf{V} = -\int d\mathbf{v}\left(\frac{f_\alpha(\mathbf{x}, \mathbf{v}, t) - f_{\alpha 0}}{\tau_\alpha}\right)$$

Since $\nabla \cdot n\mathbf{V}$ gives the change in $n(\mathbf{x}, t)$ due to particles entering or leaving a volume element, the right-hand side represents a source or sink of particles. The choice (6.7.3) guarantees conservation of particles while still retaining the maxwellian form at least locally.

2 The operator (6.7.1) brings the distribution to rest; i.e., only if $\int \mathbf{v} f(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v} = 0$ will $d/dt \int \mathbf{v} f\, dr|_c = 0$. This is not always appropriate, since the overall system must conserve momentum, but it is a good approximation to the collisions of electrons with neutral particles, because the atoms are so massive compared with the electrons. Any model in which the electrons are assumed to diffuse by collisions with a stationary distribution of target particles is called a *Lorentz model.* To justify the Lorentz model, the degree of ionization of the system should be low; otherwise collisions of electrons with electrons should be included in the collision term. Thus the form (6.7.2) is best applied to weakly ionized plasmas.[1]

The replacement $\tau(\mathbf{v}) \approx \tau$ is *exact* for a repulsive force $F = -A/r^5$ between the charged and target particles. This is a better approximation to an electron-neutral collision than to an electron-electron collision, since $1/r^5$ represents a short-range force. Further, of course, is the fact that taking τ independent of the distribution f_α, which *linearizes* the Boltzmann operator, is possible only if there are no collisions between the particles in the distribution f_α.

3 The collision term (6.7.1) can be interpreted along the same lines used to derive the original Boltzmann equation. The term f/τ represents the scattering out of $d\mathbf{v}$ of particles distributed according to $f(\mathbf{x}, \mathbf{v}, t)$. This is like an *absorption* process, since all particles originally in $d\mathbf{v}$ are lost to that volume element after scattering. This term is essentially unchanged from the corresponding contribution (6.6.4) to the Boltzmann equation. The second term corresponds to scattering into $d\mathbf{v}$, and is the term which is hard to treat in the original Boltzmann equation. The form f_0/τ assumes that after each scattering the particles are emitted with a maxwellian distribution, at the local density $n(\mathbf{x}, \mathbf{t})$.

The form (6.7.2) is sometimes referred to as the *Krook model.*

[1] The collision term (6.7.1) can be applied with good approximation to a fully ionized plasma by a more suitable choice of f_0. In particular, it has been shown [P. L. Bhatnagar, E. P. Gross, and M. Krook, *Phys. Rev.*, **94**: 511 (1954)] that the operator

$$\left.\frac{\partial f}{\partial t}\right|_c = -\frac{n(\mathbf{x}, t) f}{\sigma} + \frac{n^2}{\sigma \bar{n}}\left[\frac{m}{2\pi\kappa T(\mathbf{x}, t)}\right]^{3/2} \exp\left\{-\frac{m}{2\kappa T(\mathbf{x}, t)}[\mathbf{v} - \mathbf{q}(\mathbf{x}, t)]^2\right\} \qquad (6.7.4)$$

conserves particles, energy, and momentum and can be used to describe collisions in a fully ionized system. In (6.7.4)

$$\mathbf{q} = \frac{\bar{n}}{n(\mathbf{x}, t)} \int \mathbf{v} f(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v}$$

$$\frac{3\kappa T(\mathbf{x}, t)}{m} = \frac{\bar{n}}{n(\mathbf{x}, t)} \int (\mathbf{v} - \mathbf{q})^2 f\, d\mathbf{v}$$

and $\bar{n}/\sigma$ is the average collision frequency.

6.8 TRANSPORT COEFFICIENTS IN A WEAKLY IONIZED PLASMA

In this section the modified Boltzmann equation is used to find the transport properties of a weakly ionized plasma.

6.8.1 Electrical Conductivity of a Weakly Ionized Plasma

If the plasma is not subjected to any externally produced fields, the particle distribution may be assumed to be maxwellian. For the electrons, for example,

$$f_{e0} = \left(\frac{m_e}{2\pi\kappa T_e}\right)^{3/2} e^{-m_e v^2/2\kappa T_e} \qquad (6.8.1)$$

If a weak electric field is applied to the plasma, the electrons will move in the direction of the electric field, and the distribution function will no longer be a perfect maxwellian, but will be slightly anisotropic. The electrons will collide with the neutrals, which are assumed not to recoil as a result of the collision; the ions, because of their greater mass, are assumed not to move significantly in the electric field. As a result of the electron-neutral collisions, the electrons do not accelerate indefinitely, but rather reach a definite speed determined by the frequency at which collisions with neutrals occur. Thus the electric field induces a current in this weakly ionized plasma. The ratio of the electric field to the current is the electrical conductivity.

To compute the change in the distribution function as a result of an applied electric field $\mathbf{E} = E_z\hat{\mathbf{z}}$, the distribution function is expanded as follows:

$$f_e = f_{e0} + f_{e1} \qquad f_{e0} \gg f_{e1} \qquad (6.8.2)$$

where f_{e0} is the spherically symmetric maxwellian distribution function for electrons, and f_{e1} is the anisotropic perturbation that results from the applied electric field. The perturbation f_{e1} is related to the applied field $\mathbf{E}$ through the Boltzmann equation. Using the Krook model described above, the collision term is written

$$\left.\frac{\partial f_e}{\partial t}\right|_c = -\nu_{me}(f_e - f_{e0}) = -\nu_{me} f_{e1} \qquad (6.8.3)$$

where $\nu_{me} \equiv 1/\tau_e$ is the frequency of momentum-changing collisions of electrons scattering off neutrals, and because of the spatial homogeneity of the problem

$$\frac{n(\mathbf{x}, t)}{\bar{n}} = \int f \, d\mathbf{v} = 1$$

For a spatially homogeneous plasma in steady state, the Boltzmann equation (6.7.2) reduces to (neglecting the product $E_z \, \partial f_{e1}/\partial v_z$ for weak electric fields)

$$\frac{-eE_z}{m_e}\frac{\partial f_{e0}}{\partial v_z} = -\nu_{me} f_{e1} \qquad (6.8.4)$$

so that the total distribution function is

$$f_e = f_{e0} + \frac{eE_z}{m_e \nu_{me}}\frac{\partial f_{e0}}{\partial v_{ez}} \qquad (6.8.5)$$

The conduction current density is defined by (with n_e the electron density)

$$\mathbf{J} = -n_e e \int \mathbf{v} f_e \, d\mathbf{v}$$

For the present problem, the only current is in the z direction, and

$$J_z = -en_e \int v_z f_e \, d\mathbf{v} = -\frac{e^2 E_z}{m_e \nu_{me}} n_e \int v_z \frac{\partial f_{e0}}{\partial v_z} \, d\mathbf{v} \qquad (6.8.6)$$

Integrating (6.8.6) by parts, and using the normalization property $\int f_{e0} \, d\mathbf{v} \approx 1$, gives

$$J_z = \frac{n_e e^2 E_z}{m_e \nu_{me}}$$

The ratio J_z/E_z gives the electrical conductivity of a weakly ionized plasma in terms of the frequency ν_{me} of momentum transfer collisions between electron and neutrals.

$$\sigma = \frac{n_e e^2}{m_e \nu_{me}} = \frac{\omega_p^{\,2}}{4\pi\nu_{me}} \qquad (6.8.7)$$

This result should be compared with that of Sec. 6.5. In particular, for the conductivity of the weakly ionized gas, *runaway* is not a problem, in contrast with the fully ionized case. This is because the collision time for electron-neutral collisions is nearly independent of velocity, while for Coulomb collisions the collision time increases as v^3. Since v increases due to acceleration of electrons by the field E_z between collisions, a gas in which ν decreases with v will eventually contain particles moving so fast that the eE force is no longer balanced by collisional drag. For collisions in which ν is not decreasing with v, the drag can continue to balance eE at high fields.

Problem 6.8.1 The flux of kinetic energy is

$$\mathbf{H} = \int \bar{n}(\tfrac{1}{2}mv^2)\mathbf{v} f \, d\mathbf{v}$$

This flux of kinetic energy is also defined in terms of a thermal conductivity for situations involving weak gradients of temperature in an otherwise homogeneous plasma, by

$$\mathbf{H} = -\kappa_T \nabla T$$

For a temperature gradient in the z direction show, using the above method neglecting electron subscripts for the electrical conductivity that

$$H_z = -\frac{4\pi n\kappa}{3\pi^{3/2}v_m}\left(\frac{m}{2\kappa T_e}\right)^{5/2}\frac{\partial T_e}{\partial x}\int_0^\infty v^6 e^{-mv^2/2\kappa T}\,dv = -\frac{5}{2}\frac{n}{m}\frac{\kappa^2}{v_m}\frac{\partial T_e}{\partial x}$$

Hence the thermal conductivity for electrons in a weakly ionized plasma is

$$\kappa_T = \frac{5}{2}\frac{n}{m}\frac{\kappa^2 T_e}{v_m} \qquad (6.8.8)$$

////

6.8.2 Diffusion Coefficient of a Weakly Ionized Plasma

Diffusion will reduce, and eventually remove, density gradients in a plasma; this relaxation can be studied by the same techniques used to find the electrical conductivity. Diffusion rates are specified by a diffusion coefficient defined in terms of the flux of particles,

$$\mathbf{\Gamma}_\alpha = \bar{n}_\alpha \int \mathbf{v} f_\alpha(\mathbf{v})\, d\mathbf{v} = -D_\alpha \nabla n \qquad (6.8.9)$$

where $\bar{n}_\alpha$ is the mean density ($= N_\alpha/V$) of particles of type α. For weak gradients, as for weak electric fields, it is useful to write the distribution in the form

$$f_\alpha = f_{\alpha 0}(\mathbf{x}, \mathbf{v}) + f_{\alpha 1}(\mathbf{x}, \mathbf{v})$$

where $f_{\alpha 0}$ is the local maxwellian distribution (6.7.3). The perturbed distribution $f_{\alpha 1}$ is assumed to be small compared with $f_{\alpha 0}$, and describes the particle flux produced by the density gradient and impeded by collisions.

If the density gradient is kept fixed (e.g., by a system of sources and sinks), a steady state will be reached in which the rush of particles away from the high-density regions is retarded by collisions with neutrals, and a constant particle flux develops. Neglecting any electric fields that may develop (see the next

example), such a steady state is described by the Boltzmann equation, which in this case has the form (again using the Krook model for the collision term and neglecting the second-order term $\partial f_{\alpha 1}/\partial x$)

$$v_x \frac{\partial f_\alpha}{\partial x} = -\nu_{m\alpha} f_{\alpha 1} \qquad (6.8.10)$$

Since $\int \mathbf{v} f_{\alpha 0}\, d\mathbf{v} = 0$, the flux in a direction $\hat{\mathbf{x}}$ is given by

$$\Gamma_x = \bar{n}_\alpha \int v_x f_{\alpha 1}\, d\mathbf{v} \qquad (6.8.11)$$

Substituting $f_{\alpha 1}$ from (6.8.10) and integrating gives

$$\Gamma_x = -\frac{1}{\nu_{m\alpha}}\frac{d}{dx}\int \bar{n}_\alpha v_x^2 f_{\alpha 0}\, d\mathbf{v} = \frac{1}{\nu_{m\alpha}}\frac{d}{dx}\frac{n_\alpha \kappa T_\alpha}{m_\alpha}$$

by definition of temperature [Eq. (3.2.9)]. Defining the diffusion coefficient as in (6.8.9), comparison gives

$$D_\alpha = \frac{\kappa T_\alpha}{m_\alpha \nu_{m\alpha}} \qquad (6.8.12)$$

as the diffusion coefficient of charged particles of species α through a background of neutral particles.

6.8.3 Choice of Collision Frequency $\nu_{m\alpha}$

To use the above results, a value of $\nu_{m\alpha}$ is required. The deflection of a charged particle by a neutral atom depends on the particle speed and the scattering cross section σ_s Spitzer[1] gives the result

$$\nu_{m\alpha} = 1/\tau_\alpha = n_n \sigma_s (\kappa T_\alpha/m_\alpha)^{1/2} \qquad (6.8.13)$$

so that $\omega_{ce}\tau_e/\omega_{ci}\tau_i \approx (m_i/m_e)^{1/2}$.

For electrons, σ_s is nearly independent of velocity ($\sigma_s \simeq 4 \times 10^{-15}$ cm^2 for electron-hydrogen scattering, with T_e in the range 10 to 1000°K), while the ion cross section shows greater variation ($\sigma \simeq 50 \times 10^{-15}$ cm^2 at $T_i = 10$°K, $\sigma = 5 \times 10^{-15}$ cm^2 at $T_i = 1000$°K).

6.9 AMBIPOLAR DIFFUSION

For a weakly ionized gas in which the collisions of electrons and ions with the neutral-gas background are occurring with greater frequency than the Coulomb collisions between electrons and ions, electric fields will develop if electrons and

[1] L. Spitzer, Jr., "Diffuse Matter in Space", p. 92, Wiley, New York, 1968.

ions diffuse independently. Electric forces will tend to equalize their diffusion rates. There are basically two diffusion regimes:

1 $L \ll \lambda_D$
2 $L \gg \lambda_D$

where L is the dimension of the system or the scale length over which a gradient of density, temperature, pressure, etc., might occur, and λ_D is the Debye length. If $L \ll \lambda_D$, then the electrons and ions each diffuse according to their own "free" diffusion coefficient. This is because the thermal energy of the plasma particle $\kappa T = \lambda_D^2(4\pi n e^2)$ is then greater than the maximum potential which would develop across the system if *all* the electrons were removed, $eE_{\max} L = ne^2L^2$, and E fields developed during diffusion could be ignored. The electron and ion diffusion coefficients for this case are

$$D_e = \frac{\kappa T_e}{m_e \nu_{me}} \qquad D_i = \frac{\kappa T_i}{m_i \nu_{mi}} \tag{6.9.1}$$

where ν_{me} and ν_{mi} are the frequency of momentum-changing collisions of electrons with neutrals, and ions with neutrals, respectively. Equation (6.9.1) shows that the electrons tend to diffuse more rapidly than the ions, due to their lighter mass.

If $L \gg \lambda_D$, the ions and electrons can no longer be expected to diffuse in a manner independent of each other. The electrons, because of their lighter mass, will try to diffuse more rapidly than the ions, as is clear from (6.9.1). However, a loss of electrons from a plasma at a rate greater than that for ions will eventually cause a space-charge electric field to be set up which opposes additional loss of electrons. This same electric field will attempt to increase the rate of loss of ions from the plasma. If a steady state is reached, the flux of ions and electrons must be equal, and the self-consistent diffusion coefficient that results is called the *ambipolar diffusion coefficient*.

This ambipolar diffusion coefficient is obtained by equating the flux of electrons produced by the action of both an electric field *and* a density gradient with the flux of ions. The flux due to an electric field is obtained from the result (6.8.6), since

$$\boldsymbol{\Gamma}_\alpha = \frac{\mathbf{J}_\alpha}{q_\alpha} = \frac{\sigma_\alpha \mathbf{E}}{q_\alpha}$$

The particle flux produced by an electric field is customarily described by the mobility μ, defined by

$$\mu_\alpha = \frac{\sigma_\alpha}{n_\alpha q_\alpha}$$

so that the flux due to an electric field is $\boldsymbol{\Gamma} = n\mu\mathbf{E}$. Note that the mobility is *signed*, with μ_e negative. Combining the flux due to an electric field and the flux due to the density gradient, the total flux is

$$\boldsymbol{\Gamma}_e = n\mu_e \mathbf{E} - D_e \nabla n \tag{6.9.2}$$

$$\boldsymbol{\Gamma}_i = n\mu_i \mathbf{E} - D_i \nabla n \tag{6.9.3}$$

Equating the electron and ion flux, $\boldsymbol{\Gamma}_e = \boldsymbol{\Gamma}_i$ and $n_e = n_i$, the electric field can be eliminated between these two equations; the net flux of either species is

$$\boldsymbol{\Gamma} = -\frac{\mu_i D_e - \mu_e D_i}{\mu_e - \mu_i} \nabla n \tag{6.9.4}$$

This flux can be identified with a new diffusion coefficient

$$\boldsymbol{\Gamma} = -D_A \nabla n \tag{6.9.5}$$

where

$$D_A = \frac{\mu_i D_e - \mu_e D_i}{\mu_i - \mu_e} \tag{6.9.6}$$

D_A is the ambipolar diffusion coefficient. Using the relationship between mobility and diffusion $[\mu_\alpha = (q_\alpha/\kappa T_\alpha) D_\alpha]$, the ambipolar diffusion coefficient can be expressed

$$D_A = \frac{(T_e + T_i) D_i D_e}{T_i D_e + T_e D_i} \tag{6.9.7}$$

For nearly equal ion and electron temperatures $D_i \ll D_e$, and therefore

$$D_A \approx D_i\left(1 + \frac{T_e}{T_i}\right) \tag{6.9.8}$$

For a plasma with equal electron and ion temperatures

$$D_A \approx 2D_i \tag{6.9.9}$$

The effective diffusion coefficient for a plasma dominated by collisions of the charged particles with the neutral background is seen to be approximately twice the free diffusion coefficient for the ions. Thus the electrons try to diffuse faster than the ions, but are inhibited from doing so by the space-charge field they set up. This space-charge field enhances the diffusion of ions, and the net effect is that both ions and electrons diffuse together, with an effective diffusion coefficient that is twice the diffusion coefficient of the more slowly diffusing species, i.e., the ions.

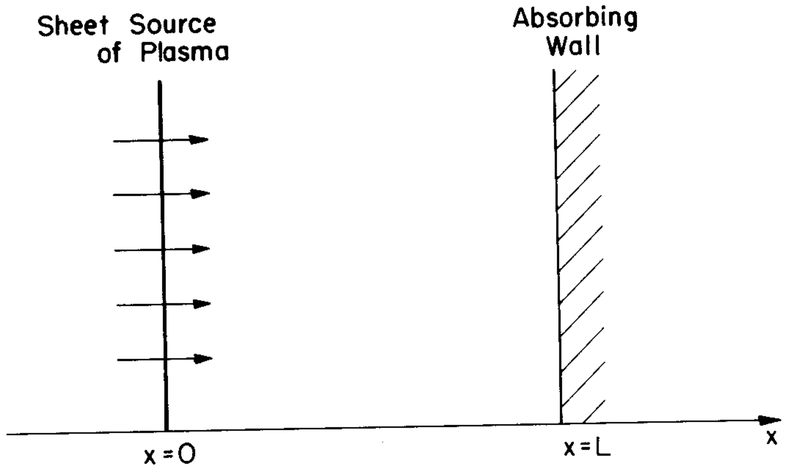


FIGURE 6.9.1
Plasma diffusion in a weakly ionized gas in one dimension.

6.9.1 Ambipolar Diffusion through a Slab

As an example of the use of the ambipolar diffusion coefficient, consider the situation depicted schematically in Fig. 6.9.1. A sheet source of plasma introduces a flux Γ_0 in the x direction at $x = 0$ into a weakly ionized region. At $x = L$ there is a perfectly absorbing wall at which the density of both ions and electrons vanishes. If $L \gg \lambda_D$, the ambipolar diffusion coefficient will determine the steady-state distribution of density. The differential equation for the density is obtained from the continuity equation

$$\frac{\partial n}{\partial t} + \nabla \cdot n\mathbf{V} = 0 \qquad (6.9.10)$$

In addition, the flux $\boldsymbol{\Gamma} = n\mathbf{V}$ is related to the density gradient by the ambipolar diffusion coefficient; i.e.,

$$\boldsymbol{\Gamma} = -D_A \nabla n \qquad (6.9.11)$$

Substituting (6.9.11) into (6.9.10), the continuity equation becomes the diffusion equation

$$\frac{\partial n}{\partial t} = D_A \nabla^2 n \qquad (6.9.12)$$

This equation is the starting point for equilibrium and stability calculations associated with collisional transport phenomena. For steady state, $(\partial n/\partial t) = 0$, and the density n satisfies Laplace's equation. In one dimension this equation is

$$\frac{d^2 n}{dx^2} = 0$$

and its solution

$$n = a + bx \qquad (6.9.13)$$

must satisfy a boundary condition at $x = 0$, determined from the flux Γ_0 at this source. From (6.9.11)

$$\Gamma_0 = -D_A \left.\frac{\partial n}{\partial x}\right|_{x=0} = -bD_A$$

The boundary condition at the sink at $x = L$ is

$$n(L) = a + bL = 0$$

Substituting these values for a and b in (6.9.13) gives the density as a function of position.

$$n = \frac{\Gamma_0 L}{D_A}\left(1 - \frac{x}{L}\right) \qquad (6.9.14)$$

The density decreases linearly. Equation (6.9.14) is valid only if $L \gg \lambda_D$, because ambipolar diffusion was assumed. If the density distribution can be measured as a function of position by a Langmuir probe, the results of the measurement can be compared with the theoretical distribution to determine the ambipolar diffusion coefficient, and thus obtain information about the ion-neutral momentum transfer collision frequency.

6.9.2 Ambipolar Diffusion from a Pulsed Line Source

A frequently used method of measuring the ambipolar diffusion coefficient is to measure the decrease in plasma density as a function of time in pulsed discharge plasma. This experiment is a good example of the use of the diffusion coefficient with the macroscopic equations. Consider a cylindrical plasma of radius a as shown in Fig. 6.9.2. Assume that initially the densities of ions and electrons are equal, but they may depend on radius, i.e.,

$$n_e(r, 0) = n_i(r, 0) = n(r, 0) \qquad (6.9.15)$$

and that at the wall the density of both species vanishes at all times, i.e.,

$$n_e(a, t) = n_i(a, t) = 0 \qquad (6.9.16)$$

The differential equation which determines the density as a function of time and space is the diffusion equation (6.9.12). If $L \gg \lambda_D$, the ambipolar diffusion coefficient is the appropriate one to use in that equation. For an azimuthally symmetric cylindrical plasma of infinite axial extent, the only diffusion is in the

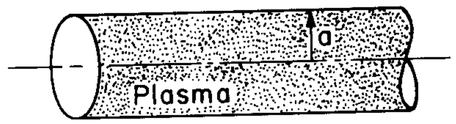


FIGURE 6.9.2
Plasma cylinder with line source on axis used to study diffusion of plasma in a weakly ionized gas.

radial direction, and the differential equation (6.9.12) in cylindrical coordinates is

$$\frac{\partial n}{\partial t}(r, t) = D_A \frac{1}{r}\frac{\partial}{\partial r}\left[r\frac{\partial n}{\partial r}(r, t)\right] \qquad (6.9.17)$$

For an assumed product solution of the form

$$n(r, t) = T(t)R(r)$$

Eq. (6.9.17) becomes

$$\frac{1}{T(t)}\frac{dT(t)}{dt} = \frac{D_A}{R(r)}\frac{1}{r}\frac{d}{dr}\left[r\frac{dR(r)}{dr}\right] \qquad (6.9.18)$$

The time dependence is thus given by

$$T(t) = Ae^{-t/\tau} \qquad (6.9.19)$$

This gives, for the radial dependence, the differential equation

$$\frac{d^2}{dr^2}R(r) + \frac{1}{r}\frac{dR(r)}{dr} + \frac{1}{\tau D_A}R(r) = 0 \qquad (6.9.20)$$

The solution of this differential equation that is finite on the axis, is the ordinary Bessel function of the first kind, i.e.,

$$R(r) = BJ_0\left(\frac{r}{\sqrt{\tau D_A}}\right) \qquad (6.9.21)$$

The boundary condition at the wall ($r = a$) requires that

$$J_0\left(\frac{r}{\sqrt{\tau D_A}}\right) = 0 \qquad (6.9.22)$$

so that

$$\frac{a}{\sqrt{\tau D_A}} = p_\nu \qquad (6.9.23)$$

where p_ν is the νth root of the zeroth-order Bessel function of the first kind ($p_1 = 2.405$, $p_2 = 5.52$, etc.). This relates the decay constant τ to the diffusion coefficient and the size of the machine.

An arbitrary initial radial density variation can be expanded in terms of these radial solutions.

$$n(r, 0) = \sum_{\nu=1}^{\infty} C_\nu J_0\left(p_\nu \frac{r}{a}\right) \qquad (6.9.24)$$

The expansion coefficient is given by

$$C_\nu = \frac{2}{[J_1(p_\nu)]^2}\int_0^a n(r, 0)J_0\left(p_\nu \frac{r}{a}\right)r\,dr \qquad (6.9.25)$$

The decay-time constant τ is different for each mode. From (6.9.23)

$$\tau_\nu = \frac{a^2}{D_A p_\nu^2} \qquad (6.9.26)$$

The complete solution is then

$$n(r, t) = \sum_{\nu=1}^{\infty} C_\nu J_0\left(p_\nu \frac{r}{a}\right)e^{-tD_A p_\nu^2/a^2} \qquad (6.9.27)$$

After some time, only the lowest radial mode ($\nu = 1$) survives, because it has the slowest decay, and the density distribution with respect to radius relaxes to the form

$$n(r, t) = C_1 J_0\left(p_1 \frac{r}{a}\right)e^{-tD_A p_1^2/a^2} \qquad (6.9.28)$$

If the density, or some quantity proportional to it, is now measured as a function of time by Langmuir probes or microwave scattering, at a fixed radius, it is possible to determine the ambipolar diffusion coefficient from the slope of the plot of $\ln[n(t)]$ as a function of time.

6.10 TRANSPORT PROPERTIES OF A WEAKLY IONIZED PLASMA IN A STEADY AND HOMOGENEOUS MAGNETIC FIELD

In most experiments the plasma is magnetically confined and diffusion, conductivity, etc., *across* the magnetic field are the controlling factors in the plasma dynamics. These transport properties can be calculated using the same collision model as in the preceding section, assuming that electron-neutral and ion-neutral collisions are more important than electron-ion collisions, and collisions are frequent enough so that the distribution functions of ions, electrons, and neutrals is maxwellian in the absence of electric fields or gradients.

6.10.1 Mobility

Consider a weak electric field **E** applied to a magnetized plasma. The Boltzmann transport equation, for the electron distribution, is

$$\frac{\partial f_e}{\partial t} + \mathbf{v} \cdot \nabla f_e - \frac{e}{m_e}\left(\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}_0}{c}\right) \cdot \nabla_{\mathbf{v}} f_e = \left.\frac{\partial f_e}{\partial t}\right|_c \qquad (6.10.1)$$

where the collision term, for the weakly ionized case considered here, is given by

$$\left.\frac{\partial f}{\partial t}\right|_c = -\nu_{me}(f_e - f_{e0}) \qquad (6.10.2)$$

where ν_{me} is the electron-neutral collision frequency, and f_{e0} is the maxwellian distribution. If the electric field is weak, the total distribution function for electrons is the sum of the maxwellian distribution function and the slight perturbation produced by the electric field **E**.

$$f_e = f_{e0} + f_{e1} \qquad f_{e1} \ll f_{e0} \qquad (6.10.3)$$

A similar model is assumed for the ions. In response to the component of the electric field parallel to the magnetic field, the plasma mobility (and conductivity) will be the same as in the magnetic field–free case. However, electric field components perpendicular to the magnetic field will cause a motion of electrons (and ions) that will be modified by the magnetic field. Furthermore, an electric field in the x direction (magnetic field in the z direction) will cause motion in the y direction (the $\mathbf{E} \times \mathbf{B}$ drift), as well as in the x direction. Thus the mobility (and other transport coefficients) will be, in general, an anisotropic tensor. The mobility tensor is defined in terms of the particle flux, just as it was for the simpler example of the scalar mobility of a field-free plasma.

$$n_e \boldsymbol{\mu} \cdot \mathbf{E} = n_e \mathbf{V}_e = \bar{n}_e \int \mathbf{v} f_{e1}\, d\mathbf{v} \qquad (6.10.4)$$

To calculate the mobility for electrons (or ions) in a weakly ionized plasma using the Krook model for collisions, assume a steady state is reached, $\partial f/\partial t = 0$. The Boltzmann equation [(6.10.1) to (6.10.3)] then becomes

$$\frac{q\mathbf{E}}{m_e} \cdot \nabla_{\mathbf{v}} f_{e0} + \frac{q}{m_e c}(\mathbf{v} \times \mathbf{B}_0) \cdot \nabla_{\mathbf{v}} f_{e1} = -\nu_{me} f_{e1} \qquad (6.10.5)$$

where electron- or ion-denoting subscripts have been dropped, and the fact that

$$(\mathbf{v} \times \mathbf{B}_0) \cdot \nabla_{\mathbf{v}} f_0 = 0 \qquad (6.10.6)$$

for a maxwellian f_0 has been used. The steady magnetic field is taken to be

in the z direction, and small terms $(\mathbf{E} \cdot \nabla_{\mathbf{v}} f_1)$ have been neglected. Multiplying by $\mathbf{v}$ and integrating with respect to velocity, (6.10.5) becomes

$$-\frac{q\mathbf{E}}{m} - \hat{\mathbf{x}}\omega_c V_y + \hat{\mathbf{y}}\omega_c V_x = -\nu_m(\hat{\mathbf{x}}V_x + \hat{\mathbf{y}}V_y + \hat{\mathbf{z}}V_z) \qquad (6.10.7)$$

Separating this equation into component equations gives

$$-\frac{qE_x}{m} - \omega_c V_y = -\nu_m V_x \qquad (6.10.8)$$

$$-\frac{qE_y}{m} + \omega_c V_x = -\nu_m V_y \qquad (6.10.9)$$

$$-\frac{qE_z}{m} = -\nu_m V_z \qquad (6.10.10)$$

For the z component the flux nV_z is proportional to E_z.

$$nV_z = \frac{qn}{m\nu_m} E_z = \mu_0 n E_z \qquad (6.10.11)$$

Thus the mobility in the z direction, denoted by μ_0, is unchanged from its zero magnetic field value. Solving (6.10.8) and (6.10.9) for nV_x and nV_y, it is possible to express the diffusion of charged particles in a weakly ionized plasma due to a weak applied electric field in a compact notation; i.e.,

$$\boldsymbol{\Gamma} = n\mathbf{V} = \boldsymbol{\mu} \cdot \mathbf{E} \qquad (6.10.12)$$

where the mobility tensor μ is given by (taking **E** in the xy plane without loss of generality)

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_\perp & \mu_T & 0 \\ \mu_T & \mu_\perp & 0 \\ 0 & 0 & \mu_\parallel \end{bmatrix} \qquad (6.10.13)$$

The perpendicular mobility $\mu_\perp$ gives the flux $n\mathbf{V}$ in the direction of $(\mathbf{E} - \mathbf{E} \cdot \mathbf{B}/B)/E$; the transverse mobility μ_T gives the flux transverse to **E** and to $\mathbf{B}_0$. The components are given by

$$\mu_\perp = \frac{\mu_0 \nu_m^2}{\nu_m^2 + \omega_c^2} = \frac{q\nu_m}{m(\nu_m^2 + \omega_c^2)} \qquad (6.10.14)$$

$$\mu_T = \frac{\mu_0 \omega_c \nu_m}{\nu_m^2 + \omega_c^2} = \frac{q^2 B}{m^2 c} \frac{1}{\nu_m^2 + \omega_c^2} \qquad (6.10.15)$$

$$\mu_\parallel = \mu_0 = \frac{q}{m\nu_m} \qquad (6.10.16)$$

Note that μ_0 is a signed quantity depending on the sign of the charge; i.e., it is positive for ions, negative for electrons. The mobility tensor is related to both the conductivity tensor and the diffusion tensor for the plasma model discussed above, as follows.

6.10.2 Conductivity and Ohm's Law

The conductivity tensor for a situation in which the current is carried by a single type of charge particle is given by

$$\sigma = nq\mu \qquad \text{electron conductivity} \qquad (6.10.17)$$

However, because the $[(\mathbf{E} \times \mathbf{B})/B^2]c$ drift is independent of mass, the current in the direction of $\mathbf{E}$ will not always be given by the mobility of only the electrons. The conductivity of a plasma with more than one mobile component is found by expressing the current in terms of the flux of particles

$$\mathbf{J} = \sum_\alpha n_\alpha q_\alpha \mathbf{V}_\alpha = \sum_\alpha q_\alpha \mu_\alpha \cdot \mathbf{E} = \sigma \cdot \mathbf{E} \qquad (6.10.18)$$

For a plasma of electrons and ions ($Z = 1$) the conductivity tensor has the form

$$\sigma = \begin{bmatrix} \sigma_\perp & \sigma_T & 0 \\ \sigma_T & \sigma_\perp & 0 \\ 0 & 0 & \sigma_\parallel \end{bmatrix} \qquad (6.10.19)$$

$$\sigma_\perp = \frac{ne^2}{m_e}\left(\frac{\nu_{me}}{\nu_{me}^2 + \omega_{ce}^2} + \frac{m_e}{m_i}\frac{\nu_{mi}}{\nu_{mi}^2 + \omega_{ci}^2}\right) \qquad (6.10.20)$$

$$\sigma_T = \frac{ne^3 B}{m_e^2 c}\left(-\frac{1}{\nu_{me}^2 + \omega_{ce}^2} + \frac{m_e^2}{m_i^2}\frac{1}{\nu_{mi}^2 + \omega_{ci}^2}\right) \qquad (6.10.21)$$

When collisions are frequent ($\omega_c/\nu_m \ll 1$) the component of the conductivity, $\sigma_\perp$, in the direction of $E_\perp$ exceeds σ_T, and is mainly due to electrons. The conductivity $\sigma_\perp$ reduces to its field-free value in this case. Collisions allow the particle to random-walk in the direction of $\mathbf{E}$, and they impede the particle drift in the $\mathbf{E} \times \mathbf{B}$ direction. When collisions are infrequent ($\nu_m \ll \omega_c$), the conductivity $\sigma_\perp$ again exceeds σ_T. The reason is that the ions and electrons have nearly the same speed in the $\mathbf{E} \times \mathbf{B}$ direction, and thus the current in that direction tends to cancel. Note that when $\omega_c \gg \nu_m$, the *ions* contribute most to the current $[\sigma_\perp{}^i \approx (m_i/m_e)^{1/2}\sigma_\perp{}^e]$. This is because the ion gyroradius is much larger than the electron gyroradius, and each collision allows the particle to move one gyroradius in the direction of $E_\perp$.

In Chap. 3, Ohm's law was written

$$\frac{\mathbf{J}}{\sigma} = \mathbf{E} + \frac{\mathbf{V} \times \mathbf{B}}{c} \qquad (6.10.22)$$

where $\sigma = (ne^2/m)\tau$. In the absence of collisions, that result implied that an infinite current would flow unless the right-hand-side of (6.10.22) vanished.

$$\mathbf{E} = -\frac{\mathbf{V} \times \mathbf{B}}{c} \qquad (6.10.23)$$

In the presence of collisions, a method of relating $\mathbf{V}$ to $\mathbf{E}$ is required in order to write (6.10.22) in the form $\mathbf{J} = \sigma \cdot \mathbf{E}$. The collision model used here permits such a relation.

An alternative to the above derivation would be to derive the MHD equations appropriate to a weakly ionized plasma. Proceeding as in Chap. 3, where the MHD equations for a fully ionized gas were obtained, and using the Krook model to write (taking an average collision time independent of species)

$$\int d\mathbf{v}\, \frac{\partial f}{\partial t}\bigg|_c = 0$$

$$\int d\mathbf{v}\, \mathbf{v}\, \frac{\partial f_\alpha}{\partial t}\bigg|_c = \frac{\mathbf{V}_\alpha}{\tau}$$

the MHD equations for a weakly ionized plasma are (see Sec. 3.3)

$$\rho\frac{\partial \mathbf{V}}{\partial t} + \rho\mathbf{V} \cdot \nabla\mathbf{V} = \frac{\mathbf{J} \times \mathbf{B}}{c} - \frac{\nabla p}{m_i} - \frac{\rho\mathbf{V}}{\tau} \qquad (6.10.24)$$

$$\frac{\mathbf{J}}{\tau} = \frac{n_e e^2}{m_e}\left(\mathbf{E} + \frac{\mathbf{V} \times \mathbf{B}}{c} - \frac{\mathbf{J} \times \mathbf{B}}{nec}\right) \qquad (6.10.25)$$

The last term in (6.10.24) vanishes for a fully ionized plasma, by conservation of momentum. The last term in (6.10.25) is called the *Hall term*, and the $\mathbf{J}$ in that term is the *Hall current*. This term is often neglected for the fully ionized plasma, but in the weakly ionized case, since $(\mathbf{J} \times \mathbf{B})/c \approx (\rho\mathbf{V}/\tau)$, it can be of the same order as the other terms in (6.10.25). See also Sec. 6.12.

6.10.3 Diffusion

The diffusion tensor is given by

$$\mathsf{D} = \frac{\kappa T}{q}\mu \qquad (6.10.26)$$

and the density of particles is given by the continuity equation, which, with $n\mathbf{V} = -\mathsf{D} \cdot \nabla n$, reads

$$\frac{\partial n}{\partial t} = (\nabla \cdot \mathsf{D} \cdot \nabla)n \qquad (6.10.27)$$

Problem 6.10.1 Derive a set of fluid equations for the velocity and current of electrons in a weakly ionized plasma in which the ions are not free to move (for example, the ions may be bound in a lattice structure). ////

Problem 6.10.2 Show that

$$\sigma_\perp = \frac{\sigma_T \nu_{me}}{\omega_{ce}} \qquad \text{when } \nu_{me} \gg \omega_{ce}$$

and

$$\sigma_\perp = \frac{\sigma_T \omega_{ci}}{\nu_{mi}} \qquad \text{when } \nu_{mi} \ll \omega_{ci}$$

Relate $\sigma_\perp$ and σ_T for $\nu_{me} \ll \omega_{ce}$, $\nu_{mi} > \omega_{ci}$, assuming that the *mean free paths* for ions and electrons are equal. ////

Problem 6.10.3 Show that $\mathbf{E} = -(\mathbf{V} \times \mathbf{B})/c$ is identical with

$$\mathbf{V} = \lim_{\nu \to 0} \mu \cdot \mathbf{E} \qquad ////$$

Problem 6.10.4 Derive a set of two-fluid equations for a plasma in a weakly ionized gas. Show that, for a homogeneous plasma in a weak electric field and a strong **B** field, these equations reduce to (6.10.18) to (6.10.21). ////

As in the calculation of conductivity, in the limit of very frequent collisions the diffusion tensor reduces to a scalar given by $D = D_0$. For a strong magnetic field $\omega_c \gg \nu_m$, the loss of charged particles across the magnetic field due to diffusion is reduced by a factor proportional to $1/B^2$ from the field-free value.

Consider a situation in which the magnetic field is straight and parallel to the z direction and the density gradient is in the x direction. If the magnetic field is strong ($\nu_m \ll \omega_c$), the y diffusion is the same for ions and electrons. In directions of the density gradient (the x direction) the diffusion rates of ions and electrons are not equal. For $\omega_c^2 \gg \nu_m^2$, using (6.10.26) and (6.10.13)

$$D_{\perp i} = \frac{D_{0i} \nu_{mi}^2}{\omega_{ci}^2} = \frac{\kappa T c^2}{e^2 B^2} m_i \nu_{mi} \qquad \text{for ions} \qquad (6.10.28)$$

and

$$D_{\perp e} = \frac{D_{0e} \nu_{me}^2}{\omega_{ce}^2} = D_{\perp i} \frac{m_e \nu_{me}}{m_i \nu_{mi}} \frac{T_e}{T_i} \ll D_{\parallel i} \qquad \text{for electrons} \qquad (6.10.29)$$

Thus, across a strong magnetic field, the ions would, in general, diffuse more rapidly than the electrons, unlike the zero magnetic field case where (if $L \ll \lambda_D$) the electrons diffused more rapidly. This is because the charged particles walk one gyroradius per collision, and the ion gyroradius far exceeds that of the electrons. If $L \gg \lambda_D$, the unequal free diffusion rates will lead to a charge-separation electric field which will alter the diffusion coefficients, again leading to ambipolar diffusion.

6.11 AMBIPOLAR DIFFUSION OF A WEAKLY IONIZED PLASMA ACROSS A MAGNETIC FIELD

Consider a plasma confined by a magnetic field $\mathbf{B} = B_0 \hat{\mathbf{z}}$. As a simplifying assumption, let the plasma be a semi-infinite slab, with a density gradient in the x direction and uniform density in the y or z direction. In this geometry y or z diffusion does not lead to a charge separation. If the plasma is less than a Debye length thick in the x direction, the free-diffusion coefficients would correctly determine the diffusion rate for ions and electrons, and the ions would diffuse more rapidly than the electrons since $D_{\perp i} \gg D_{\perp e}$. However, if the plasma is thicker than a Debye length in the x direction, the electric fields built up as the ions try to outrun the electrons will affect the diffusion. As before, these electric fields modify the net diffusion of both the ions and the electrons in such a way that the net fluxes of ions and electrons are equal. These fluxes are given by

$$\mathbf{\Gamma}_e = (n\mu_e \cdot \mathbf{E}) - (\mathsf{D}_e \cdot \nabla n) = n\mu_{\perp e}\mathbf{E} - D_{\perp e}\nabla n$$

$$\mathbf{\Gamma}_i = n\mu_{\perp i}\mathbf{E} - D_{\perp i}\nabla n$$

Setting $\mathbf{\Gamma}_e = \mathbf{\Gamma}_i$ gives the ambipolar diffusion coefficient for weakly ionized plasma across a magnetic field; i.e.,

$$D_{\perp A} = \frac{\mu_{\perp i} D_{\perp e} - \mu_{\perp e} D_{\perp i}}{\mu_{\perp i} - \mu_{\perp e}} \qquad (6.11.1)$$

If the magnetic field is strong ($\omega_c^2 \gg \nu_m^2$), it follows that $\mu_{\perp i} D_{\perp e} = -(T_e/T_i)$ $\mu_{\perp e} D_{\perp i}$ and $\mu_{\perp i} \gg \mu_{\perp e}$. Thus, across a strong magnetic field, the ambipolar diffusion coefficient is given by

$$D_{\perp A} \approx D_{\perp e}\left(1 + \frac{T_i}{T_e}\right) \qquad (6.11.2)$$

For equal ion and electron temperatures, the ambipolar diffusion coefficient across a magnetic field is twice the free-diffusion coefficient for the slower component, in this case the electrons. The ambipolar diffusion coefficient across a magnetic field is different from the ambipolar diffusion coefficient for the zero magnetic field case, in that, for $\omega_c \ll \nu_m$, $D_A = 2D_i$, while for $\omega_c \gg \nu_m$, $D_{\perp A} = 2D_{\perp e}$.

6.11.1 Diffusion of a Weakly Ionized Plasma in a Finite-size Magnetized Plasma; The Short-circuit Effect

In the above discussion it was assumed that the plasma was finite in the x direction and infinite in the y and z directions. Now consider a plasma that is also finite in the z direction, along the magnetic field B_z, as shown schematically in Fig. 6.11.1. The plasma exists in a perfectly conducting box of height H and length L, where both L and H are much greater than both a collision mean free path and the Debye length.

On the basis of the previous discussion, it might be expected that for a box where $L \approx H$, the plasma would diffuse in the x direction, across the magnetic field, at a rate governed by the ambipolar diffusion coefficient $D_{\perp A} = 2D_{\perp e}$, and that it would diffuse in the z direction, along the magnetic field, at a rate governed by the ambipolar diffusion coefficient $D_A = 2D_i$. The total diffusion would then be the sum of these processes. This is not the case, because the fluxes nV_x and nV_z are coupled through the continuity equation $\partial n/\partial t = -\nabla \cdot n\mathbf{V}$. The diffusion coefficient is obtained from the continuity or the diffusion equation.

$$\frac{\partial n}{\partial t} = (\nabla \cdot \mathsf{D} \cdot \nabla)n + \nabla \cdot (n\boldsymbol{\mu} \cdot \mathbf{E})$$

which is

$$\frac{\partial n_e}{\partial t} = D_e \frac{\partial^2 n_e}{\partial z^2} - \mu_e \frac{\partial}{\partial z}(n_e E_z) + D_{\perp e} \frac{\partial^2 n_e}{\partial x^2} - \mu_{\perp e} \frac{\partial}{\partial x}(n_e E_x) \qquad \text{for electrons} \tag{6.11.3}$$

and

$$\frac{\partial n_i}{\partial t} = D_i \frac{\partial^2 n_i}{\partial z^2} - \mu_i \frac{\partial}{\partial z}(n_i E_z) + D_{\perp i} \frac{\partial^2 n_i}{\partial x^2} - \mu_{\perp i} \frac{\partial}{\partial x}(n_i E_x) \qquad \text{for ions} \tag{6.11.4}$$

If it is assumed that in fact $E_z = E_x$, E_z can be eliminated from (6.11.3) and (6.11.4). Neglecting $\mu_{\perp e,i}$ terms because the mobility perpendicular to $\mathbf{B}$ in the x direction is much less than the mobility in the z direction as a result of the magnetic field in the z direction,

$$\mu_{\perp e,i} = \frac{\mu_{e,i}}{(\omega_c/\nu_m)^2} \ll \mu_{e,i} \tag{6.11.5}$$

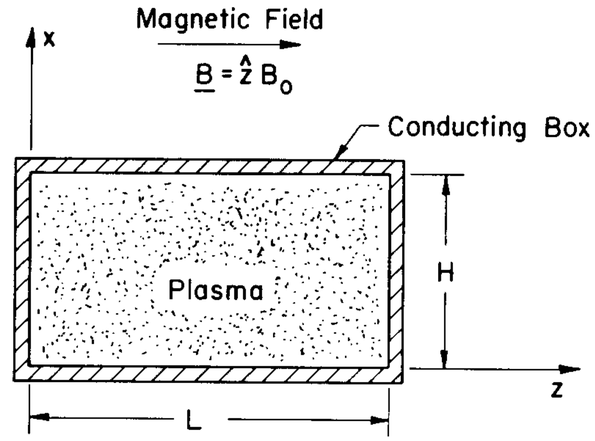


FIGURE 6.11.1
Schematic of a plasma-filled conducting box in a finite magnetic field, illustrating configuration used to analyze the diffusion of plasma across a magnetic field in finite configurations.

and then combining the two diffusion equations gives

$$\frac{\partial n}{\partial t} = \frac{\mu_i D_e - \mu_e D_i}{\mu_i - \mu_e} \frac{\partial^2 n}{\partial z^2} + \frac{\mu_i D_{\perp e} - \mu_e D_{\perp i}}{\mu_i - \mu_e} \frac{\partial^2 n}{\partial x^2} \tag{6.11.6}$$

Thus the diffusion coefficients across the field and along the field are not equal, and they are not the same as for two previous cases where the aspect ratio was very large or very small. For this finite weakly ionized plasma the effective diffusion coefficients are

$$D^*_{\perp A} = \frac{\mu_i D_{\perp e} - \mu_e D_{\perp i}}{\mu_i - \mu_e} \qquad \text{across the field} \tag{6.11.7}$$

and

$$D^*_{\parallel} = \frac{\mu_i D_e - \mu_e D_i}{\mu_i - \mu_e} = D_A \qquad \text{along the field} \tag{6.11.8}$$

Thus, along the field, the effective diffusion coefficient is the same as the ambipolar diffusion coefficient for the field-free case. The effective diffusion coefficient across the field can be simplified with the aid of the inequalities $\mu_e \gg \mu_i$ and $D_{\perp i} \gg D_{\perp e}$. The result is

$$D^*_{\parallel A} \approx D_{\perp i} \tag{6.11.9}$$

which shows that the diffusion across the magnetic field is not *twice* the electron diffusion coefficient, as predicted by the one-dimensional model, but rather it is the *same* as the coefficient for ion diffusion across a magnetic field. The diffusion equation for electrons and ions becomes

$$\frac{\partial n}{\partial t} = D_A \frac{\partial^2 n}{\partial z^2} + D_{\perp i} \frac{\partial^2 n}{\partial x^2} \tag{6.11.10}$$

The reason for these diffusion coefficients being as they are is that in the one-dimensional calculations of diffusion along and across the magnetic field, it was required that the steady-state flux of ions and electrons be equal, while in the finite geometry problem just considered, it was required only that the total flux of ions and electrons to the walls be equal. This means that electrons can move along the magnetic field in response to electric fields set up by the ions diffusing across the field. The flow of electrons to the end walls of the experiment reduces the space-charge field that would inhibit ion diffusion in the one-dimensional problem. Thus the ions diffuse across the field as though there were no space-charge electric field. This effect is called the *Simon short-circuit effect.*

6.11.2 Experimental Tests of Ambipolar Diffusion

A number of early experiments were performed to test these and related ideas.[1] One experiment is depicted schematically in Fig. 6.11.2. The plasma is produced by a thermionic-arc discharge that acts as a low-density plasma source on the axis of a closed conducting cylinder immersed in an axial magnetic field. The plasma diffuses radially and axially as a result of collisions of electrons and ions with neutrals. The density of ions is measured by means of movable, negatively biased (20 V) Langmuir probes. It was found to be a decreasing exponential function of radius, as would be expected from a steady-state ($\partial n/\partial t = 0$) solution of the diffusion equation (6.11.10) in cylindrical coordinates, for a line source on the axis. The solution based on the assumption that the diffusion equation is separable into radial and axial parts is

$$n(r) = AK_0\left(\frac{r}{a}\right) \quad (6.11.11)$$

where K_0 is the modified Bessel's function of the second kind and

$$a = \frac{L}{\pi}\left|\frac{D_{\perp i}}{D_{\|A}}\right|^{1/2} \quad (6.11.12)$$

The parameter a is the characteristic length associated with the separation constant of the differential equation, and determines the radial size of the plasmas, since for $r \gg a$, the density is proportional to a decreasing exponential function of radius.

$$n(r) \propto \frac{1}{\sqrt{r}} e^{-r/a} \quad (6.11.13)$$

[1] A. Simon and R. V. Neidigh, Diffusion of Ions in a Plasma across a Magnetic Field, *Oak Ridge Nat. Lab. Rep.* 1890, November 1955. A. Simon, "An Introduction to Thermonuclear Research," Pergamon, New York, 1959.

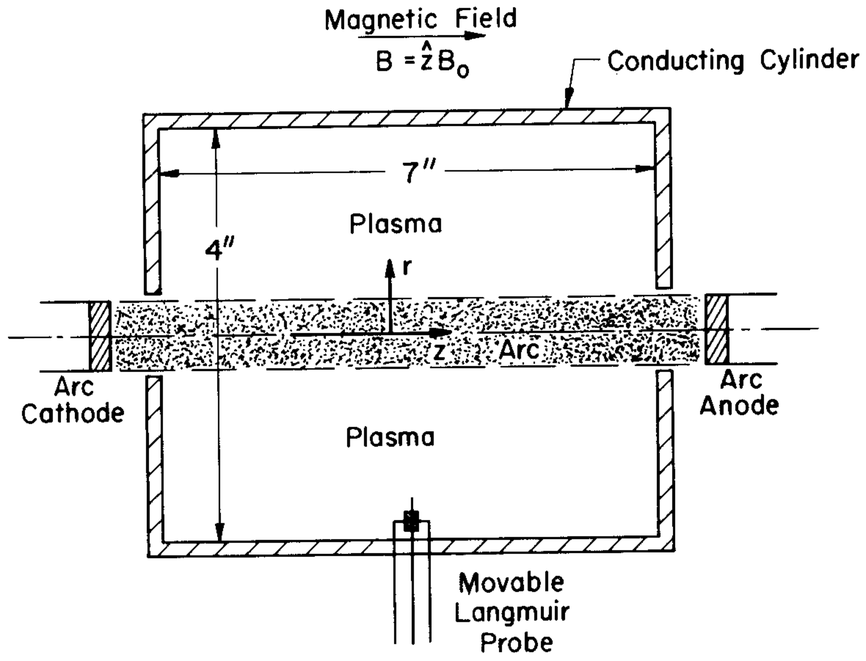


FIGURE 6.11.2
Schematic of experiment used by A. Simon to investigate diffusion of plasma across a magnetic field. (*After A. Simon, "An Introduction to Thermonuclear Research," Pergamon, New York*, 1959.)

Since $D_\perp \propto 1/B^2$, the characteristic radius of the plasma a should be proportional to $1/B$. The experiment consists in measuring the e folding length a as a function of magnetic field strength and background gas pressure. The results of these experiments, shown in Fig. 6.11.3, confirm that the coefficient for diffusion across a magnetic field is indeed proportional to $1/B^2$, as indicated by Fig. 6.11.4.

Instabilities can also lead to diffusion across a magnetic field under certain conditions. Note that as the collision frequency (due to real binary collisions or instabilities) increases, the maximum diffusion coefficient occurs when $\nu_m \approx \omega_c$, whereupon, since $D_\perp \propto \mu_\perp$, given in (6.10.13) et. seq.

$$D_{\perp,\max} \approx \frac{\kappa T}{eB} c$$

This diffusion coefficient, proportional to $1/B$, is a measure of the fastest rate at which instability processes might be expected to disrupt a confined plasma, and is called *Bohm diffusion.*

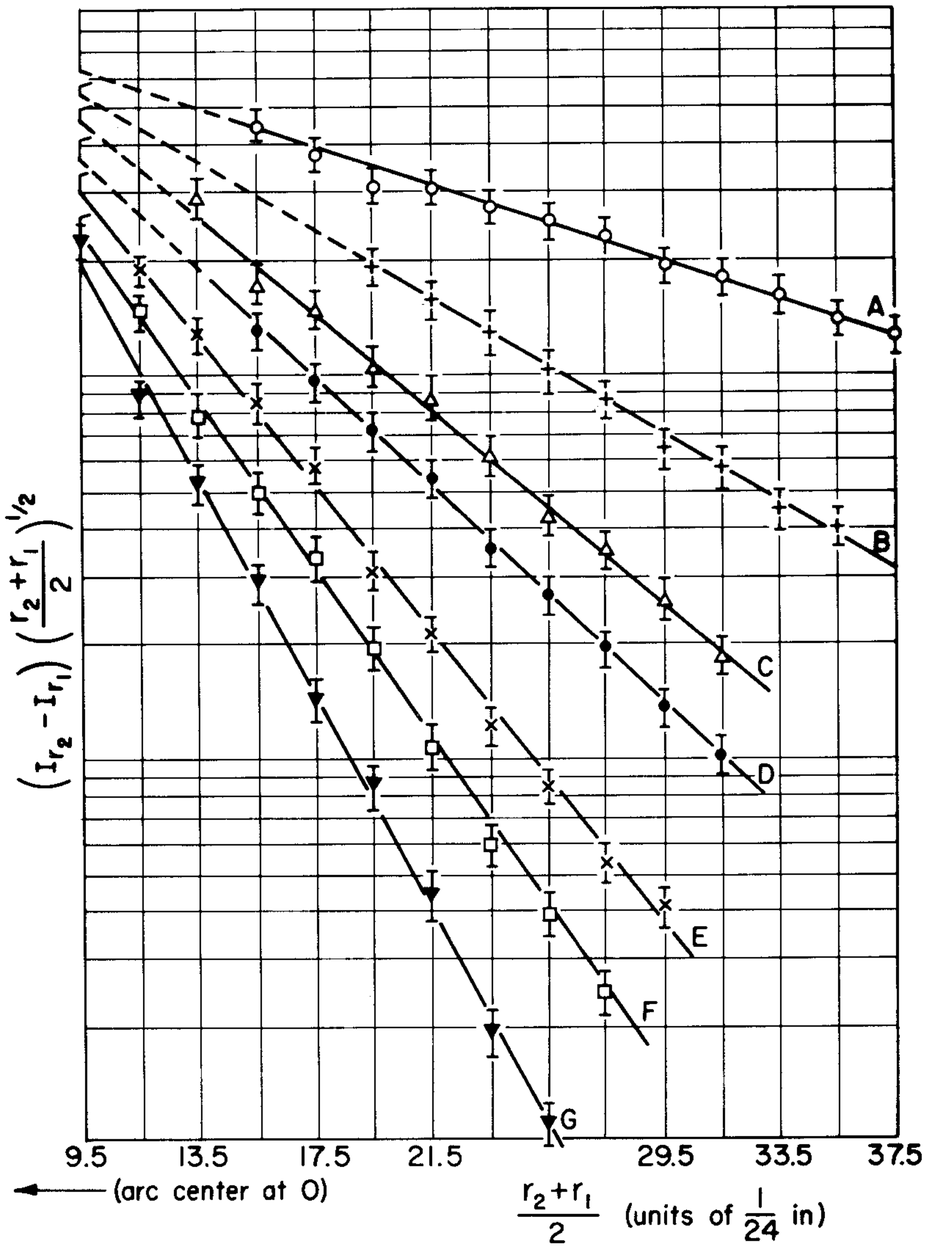


FIGURE 6.11.3
Ion current to the differential radial probe, $(I_{r_2} - I_{r_1})$, multiplied by the square root of the radial-probe position, $[(r_2 + r_1)/2]^{1/2}$, as a function of probe position, $(r_2 + r_1)/2$. (*After A. Simon, "An Introduction to Thermoculear Research," Pergamon, New York*, 1959.)

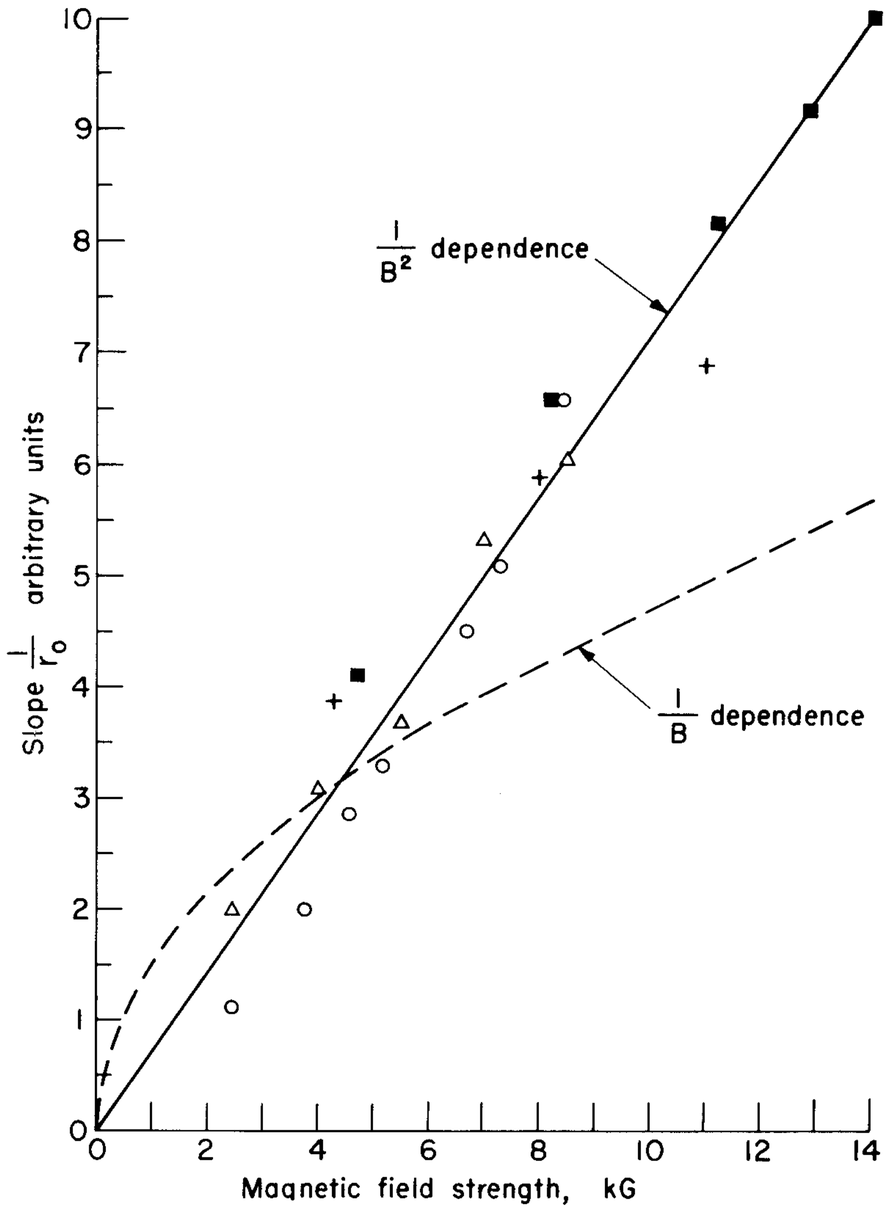


FIGURE 6.11.4
Plot of slopes $1/r_0$ [r_0 corresponds to the characteristic length a in Eq. (Eq. 6.11.12)] as a function of magnetic field strength. The data clearly indicate a $1/B^2$ dependence. The broken line would represent a $1/B$ dependence fitted at 4,000 G. (*After A. Simon, "An Introduction to Thermonuclear Research," Pergamon, New York*, 1959.)

6.12 MHD POWER GENERATORS

An important application of plasma transport properties is the MHD generator, which is designed to convert the energy in a flowing plasma into electricity. The idea, sketched in Fig. 6.12.1, is that a heat source can be used to force a weakly ionized plasma to expand into a channel with speed $\mathbf{U} = U\hat{\mathbf{x}}$. If a magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$ is set up perpendicular to the direction of plasma flow, an electric field will be set up perpendicular to the flow direction and the magnetic axis, as seen by examining the MHD equations (6.10.24) and (6.10.25).

$$\frac{\mathbf{U} \times \mathbf{B}}{c} = -\mathbf{E} + \frac{\mathbf{J}}{\sigma} + \frac{|\omega_{ce}|\tau}{B\sigma}\mathbf{J} \times \mathbf{B} \qquad (6.12.1)$$

If the collision time is sufficiently short, a current flows in the direction of $\mathbf{E}$. This current can be tapped by electrodes in contact with the plasma to drive an external load. In steady state, the MHD equation for momentum transport [Eq. (6.10.24)]

$$\frac{1}{2}\rho\frac{dU^2}{dx} = \frac{|\mathbf{J} \times \mathbf{B}|}{c} \qquad (6.12.2)$$

shows that the current is being extracted at the expense of the flow energy of the plasma. If the generator is to extract a large portion of the flow energy, the channel length L should be chosen so that

$$\tfrac{1}{2}\rho U^2 \approx \frac{|\mathbf{J} \times \mathbf{B}|}{c} L \qquad (6.12.3)$$

Comparing (6.12.1) and (6.12.3), and equating components of 6.12.1 in the direction of $\mathbf{U} \times \mathbf{B}$, the condition

$$S \equiv \frac{\sigma B^2 L}{\rho U c^2} \approx 1 \qquad (6.12.4)$$

relates the conductivity to the length for an efficient device. The parameter S is called the interaction number, and expresses the ratio of the magnetic body force in the gas to the kinetic pressure of the gas.

An attractive feature of the MHD generator is that the natural losses in such a system are surface losses, primarily heat losses and friction, while the conversion process (6.12.2) is a *volume* effect. This device would be especially suitable for large-power-level application.

The third term on the right-hand side of (6.12.1) is the Hall term, and its size is determined by

$$R_H = \omega_c\tau = \mu B$$

called the Hall coefficient. The Hall effect is an important factor in design of the channel of an MHD generator. This can be seen by noting that if $\mathbf{U} = U_x\hat{\mathbf{x}}$ and $\mathbf{B} = B_z\hat{\mathbf{z}}$, the current $\mathbf{J}$ will first flow in the y direction, across the channel.

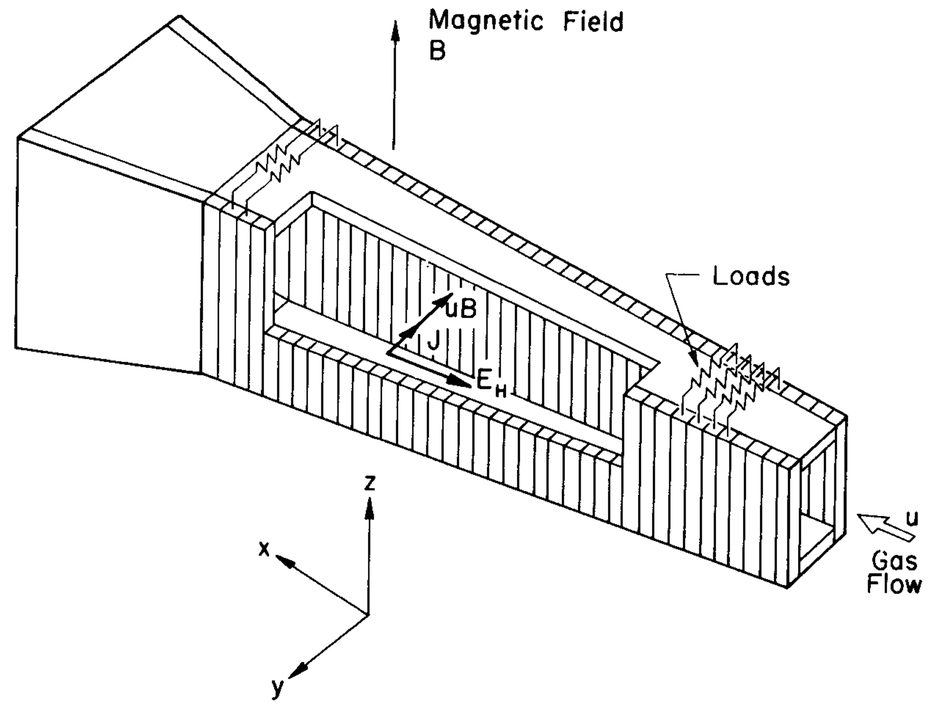


FIGURE 6.12.1
Schematic of an MHD generator illustrating its basic features.

But the Hall term will then give rise to an electric field opposite to the direction of motion of the working fluid.

$$E_x = \frac{\omega_{ce}\tau}{B\sigma} J_y B_z$$

If the design of the device allows a current J_x to flow in response to the Hall potential, a back emf $[(\omega_{ce}\tau/B\sigma)J_x B_z\hat{\mathbf{y}}]$ will be set up *opposing* the emf generated by the $\mathbf{U} \times \mathbf{B}$ force on the working fluid. Thus, preventing this short circuiting of the Hall potential is a necessary feature of an efficient generator. Figure 6.12.2 shows a variety of MHD generator configurations.

Since the distinguishing feature of an MHD generator is that it uses a partially ionized electrically conducting plasma, it is important to note the conditions required to achieve the necessary level of conductivity.

6.12.1 Ionization

Low-density plasmas are generally ionized by nonthermal mechanisms, i.e., by pulsing electric and magnetic fields, often leading to an ionized gas whose temperature is well below the ionization potential of the constituents of the plasma. At the higher densities (pressure ≈ 1 atm) in which MHD generators

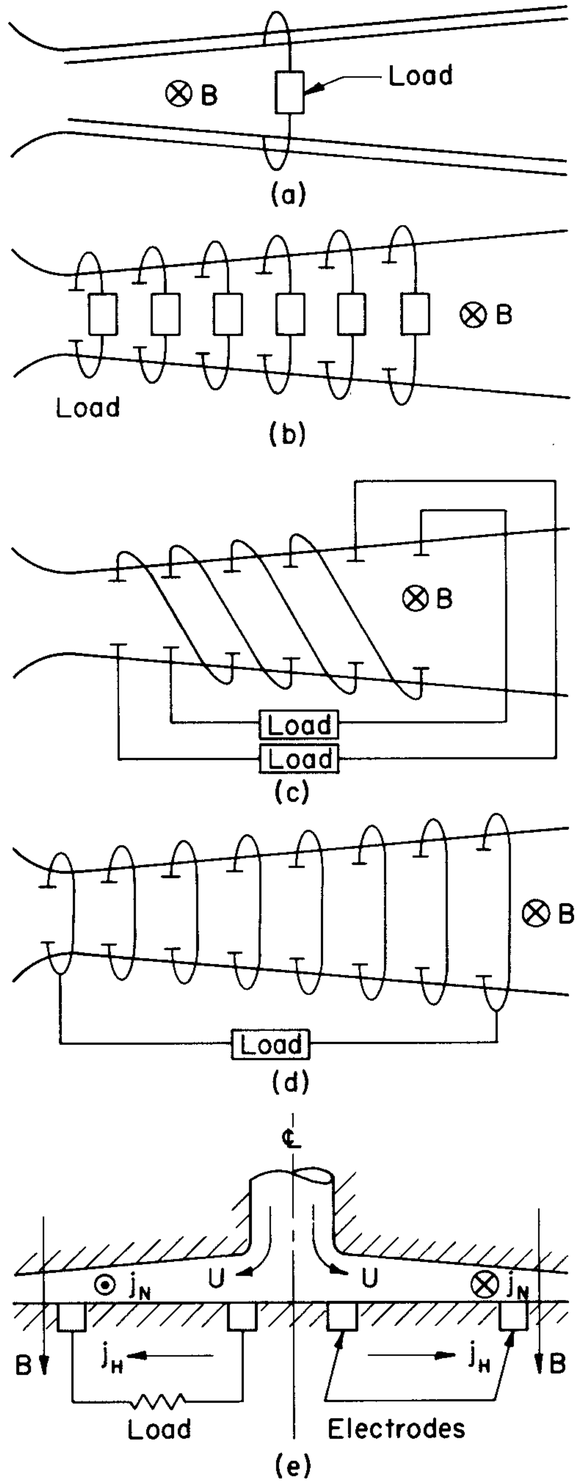


FIGURE 6.12.2
Sketches of several MHD generator configurations. (*a*) Electron hookup; (*b*) segmented electrode Faraday generators; (*c*) Faraday generator with diagonal hookup; (*d*) linear Hall generator; (*e*) disk Hall generator. [*After T. R. Brogan, Advan. Plasma Phys.*, **1**: 231, 1968.]

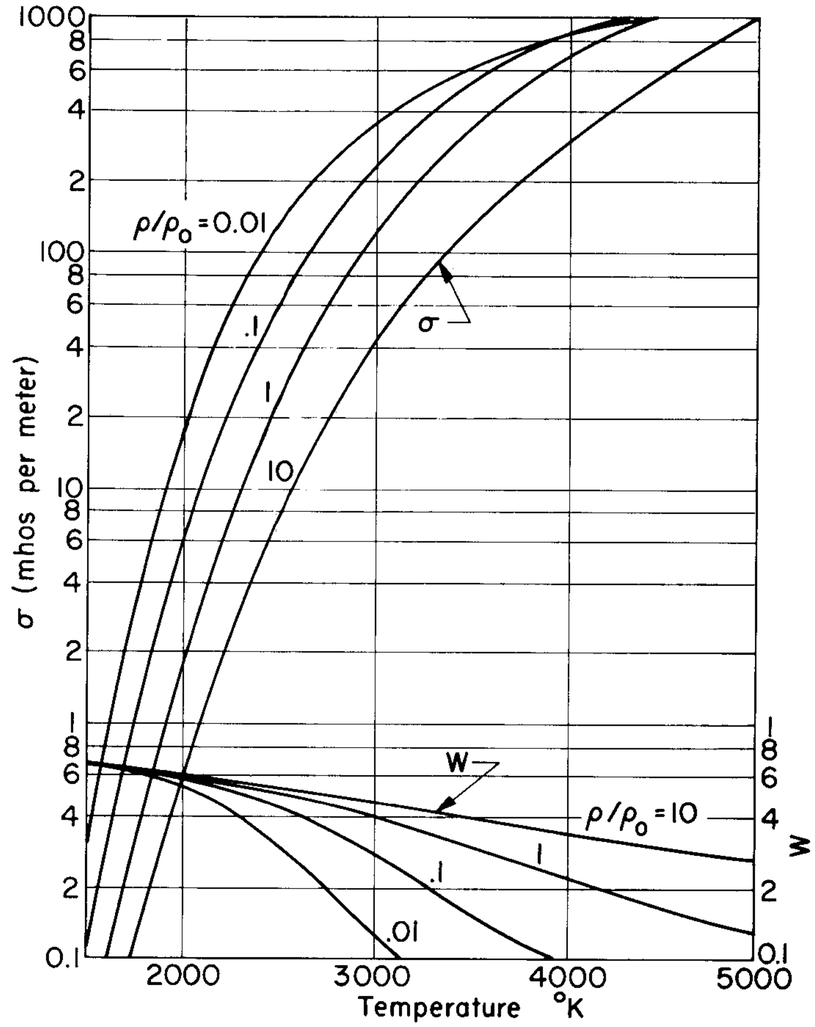


FIGURE 6.12.3
Conductivity and $\omega\tau$ in argon seeded up to 1 percent with potassium. A Maxwell-averaged cross section of 6×10^{-17} cm^2 has been assumed for argon, and a cross section of 3×10^{-14} cm^2 for neutral potassium. The gas density in atmospheres is indicated by ρ/ρ_0 and $\omega\tau = WB(\rho/\rho_0)$. [*After R. J. Rosa, Phys. Fluids*, **4**: 182 (1961).]

operate, thermal ionization seems a more practical means of producing the plasma. Although common gases (air, CO_2, etc.) do not ionize until a very high temperature is achieved, addition of a small amount of an easily ionizable seed material leads to a fluid of sufficient conductivity. Typically, 0.1 to 1.1 percent of an alkali metal vapor is used to seed the gas.

6.12.2 Temperature Requirement

Figure 6.12.3 shows the conductivity of argon, seeding with 1 percent potassium vapor. The mixture has appreciable conductivity at 2000°K. The conductivity

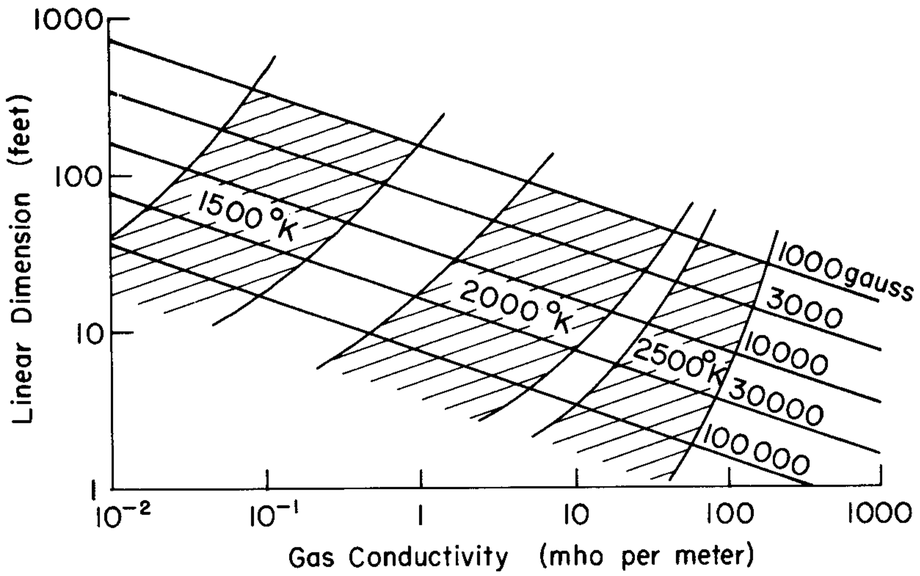


FIGURE 6.12.4
Approximate size of a 100-MW magnetohydrodynamic generator as a function of gas conductivity and magnetic field strength. Shaded regions indicate, roughly, the gas temperatures required. [*After R. J. Rosa, Phys. Fluids*, **4**: 182 (1961).]

determines the size of a generator necessary to produce a given power level, and Fig. 6.12.4 shows the dimension of a 100-MW generator for various field strengths and temperatures. Because the machine size decreases as the magnetic field increases, the condition (6.12.4) places this plasma device in the regime of low magnetic Reynolds number,

$$R_M \equiv \frac{\sigma UL}{c^2} \ll 1 \qquad (6.12.5)$$

in contrast with the usual conditions for controlled thermonuclear reactor applications. The implication of (6.12.5) is that many collective aspects of plasma behavior, particularly plasma waves and instabilities, are of lesser importance in this application.

6.12.3 Experimental Example

An early experiment[1] was designed to test the ideas described above. The experiment was intended to simulate a full-scale MHD generator, to the extent that it caused a pressure drop in the working fluid of about half the stagnation pressure. The experiment is sketched in Fig. 6.12.5. The working fluid was

[1] R. J. Rosa, *Phys. Fluids*, **2**: 182 (1961).

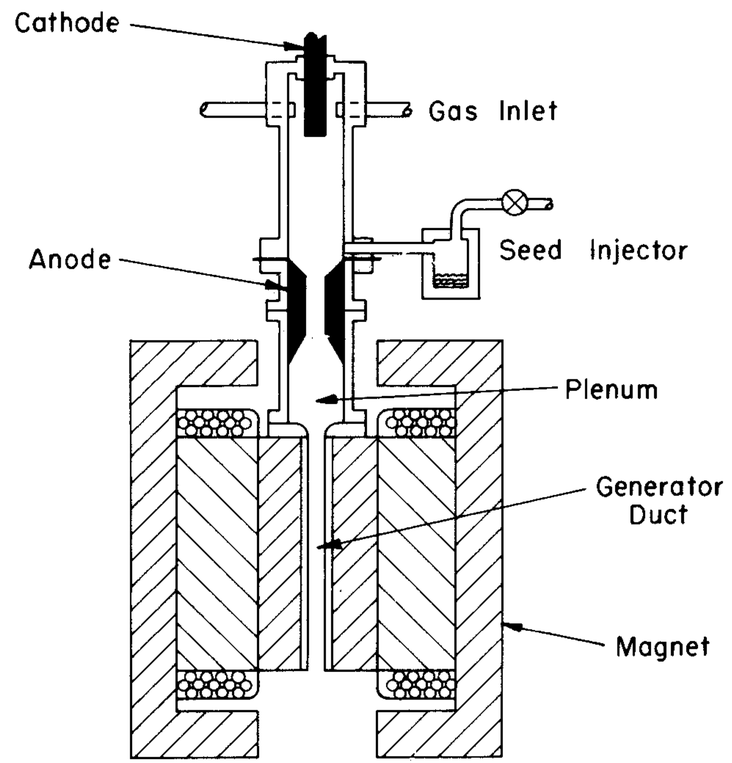


FIGURE 6.12.5
Schematic of a magnetohydrodynamic generator experiment. [*After R. J. Rosa, Phys. Fluids*, **4**: 182 (1961).]

argon mixed with potassium carbonate, and heated to 3000°K, and the magnetic field strength was 14,000 G. The gas conductivity varied between 100 and 400 mhos/m as a function of seeding.

The channel was 1 by 3 in, as sketched in Fig. 6.12.6. The results of the experiment are plotted in Fig. 6.12.7. By integrating $(\mathbf{J} \times \mathbf{B})/c$ along the length of the device, a drop of

$$\Delta p = 0.8 \times 10^{-2} I \qquad \text{psi}$$

is predicted, which for $I = 500$ A is a substantial fraction of the pressure of the gas at input.

In an attempt to verify the deleterious effect of Hall currents on the performance of this device, a region in which Hall voltage is effectively short-circuited was arranged by connecting an electrode near the entrance to one in the middle of the channel, using a switch that could be opened and closed during a

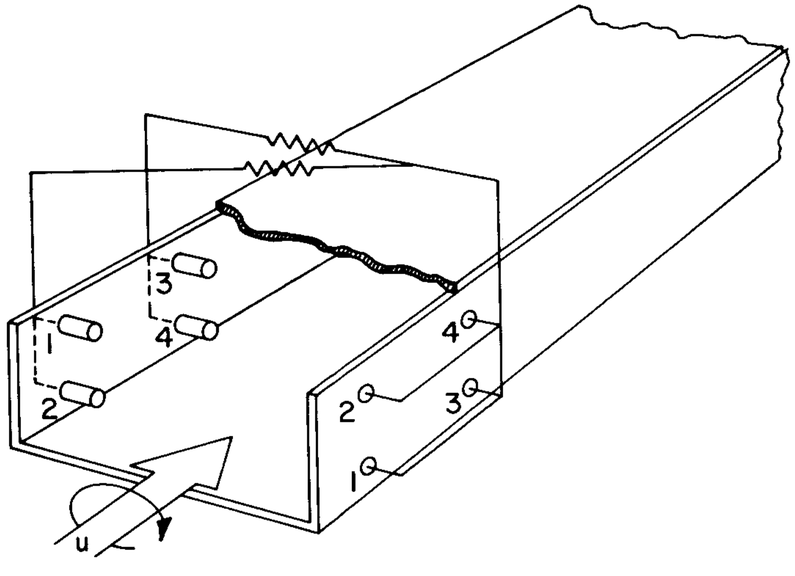


FIGURE 6.12.6
Electrode configuration and wiring diagram for MHD power-generation experiments. The 1- × 3-in channel is made of transite, and the 1/16-in-diamter electrodes are spaced 1/2 in apart in a double row. One set of electrodes is shown. The complete generator consists of 20 such sets. The small arrow indicates the direction of swirl given to the gas in the arc chambers. [*After R. J. Rosa, Phys. Fluids*, **4**: 182 (1961).]

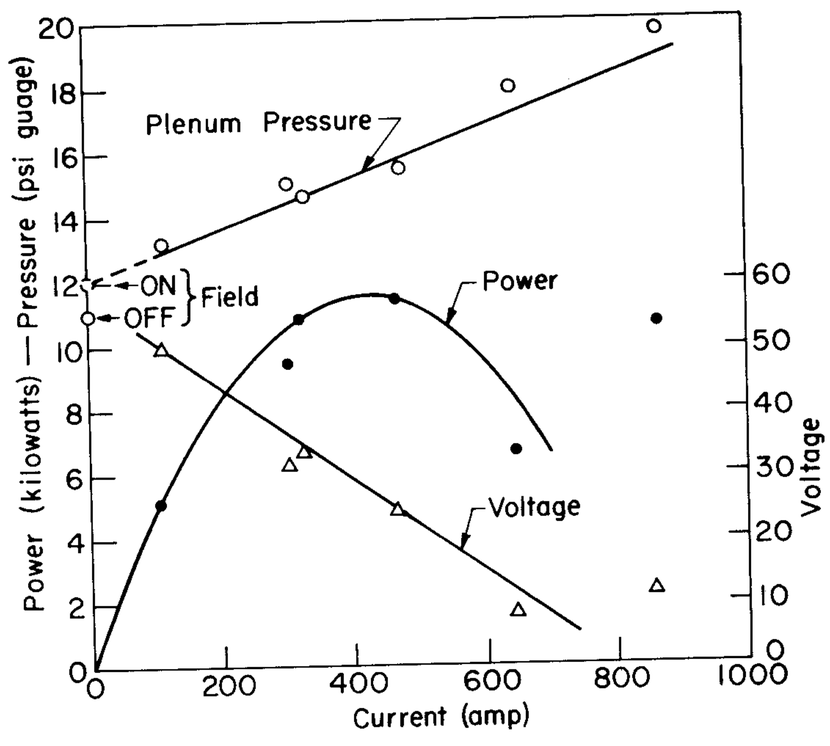


FIGURE 6.12.7
Upstream plenum pressure, output voltage, and power output as a function of total output current. [*After R. J. Rosa, Phys. Fluids*, **4**: 182 (1961).]

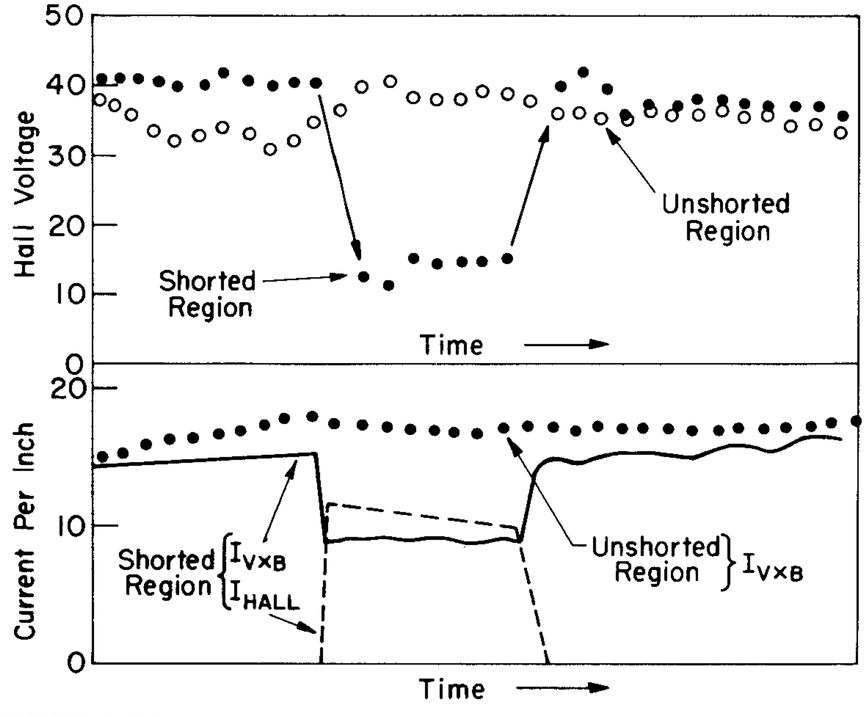


FIGURE 6.12.8
Effect of deliberately allowing Hall current to flow in the first half of the generator channel. The data indicated by discrete points were obtained from panel meters photographed at the rate of about 10 frames per second and show overswing and oscillation characteristic of these meters. The lines (solid and dashed) portray data obtained from oscilloscopes. [*After R. J. Rosa, Phys. Fluids*, **4**: 182 (1967).]

run. Figure 6.12.8 shows the result of this experiment. The effect of allowing flow of Hall current is evident.

The experiment described confirms that the MHD theory on which the MHD generator is based is correct; large and practical generators have since been constructed, based on these properties.[1]

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7

KINETIC EQUATIONS FOR A PLASMA

There are two conventional approaches to the theory of the properties and processes of a plasma: the macroscopic (thermodynamic, fluid) description and the microscopic (kinetic, statistical) description.

The macroscopic approach is based on describing quantities such as average velocity and temperature as a function of position and time. These are the quantities emphasized in the measurement of plasma properties. The microscopic description of a plasma is based on the configuration and velocity-space distribution of the plasma particles, the correlations between these particles, and the microfields produced by the particles. The microscopic quantities are more difficult to measure directly, but frequently they play a dominant role in determining the macroscopic properties of a plasma. For example, as was shown in Chap. 5, two electron clouds streaming through each other with equal and opposite velocity are unstable, and produce growing electrostatic waves, according to a theory which treats the two clouds as separate fluids. However, if the electrons in that example were treated as a single fluid with mean speed zero, the fluid equations would predict no instability. So even in the macroscopic framework, the influence of the velocity distribution of plasma particles was clearly demonstrated.

A study of the kinetic theory of plasma adds depth to the concepts of equilibrium, waves, and stability already explored in the fluid framework. Further, the kinetic theory provides a formal basis for the heuristic idea that formed the basis even for the fluid description, namely, that of including the average fields due to many particles, and often neglecting the short-range forces of the nearest-neighbor particles. Plasma kinetic theory also provides a method of investigating the influence of collisions among the plasma particles and the calculation of the transport properties of fully ionized plasmas. Further, to treat the scattering and emission of radiation by a plasma, it is necessary to develop a kinetic description that "sees" individual plasma particles, since radiation is a discrete-particle effect.

7.1 THE MICROSCOPIC EQUATIONS FOR A MANY-BODY SYSTEM

A complete description of a plasma would be one in which the coordinate $\mathbf{x}_i(t)$ and velocity $\mathbf{v}_i(t)$ of each particle are given as a function of time. Then the function

$$N_\alpha(\mathbf{x}, \mathbf{v}, t) = \sum_{1 \le i \le N_\alpha} \delta[\mathbf{x} - \mathbf{x}_i(t)]\, \delta[\mathbf{v} - \mathbf{v}_i(t)] \qquad (7.1.1)$$

completely specifies the microscopic system of particles of species α, and

$$\operatorname*{Lim}_{d\mathbf{x}\, d\mathbf{v} \to 0} \int N_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{x}\, d\mathbf{v}$$

is either unity or zero, depending on whether or not there is a particle at the point $\mathbf{x}$, $\mathbf{v}$, at time t. It is convenient to define the six-dimensional phase-space variable $\mathbf{X} = (\mathbf{x}, \mathbf{v})$ so that $N_\alpha(\mathbf{x}, \mathbf{v}, t) = N_\alpha(\mathbf{X}, t) = \sum_i \delta[\mathbf{X} - \mathbf{X}_i(t)]$ and $d\mathbf{X} \equiv d\mathbf{x}\, d\mathbf{v}$. Clearly, $\bar{N}_\alpha = \int N_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{X}$ is the total number of particles of type α in the system. The electric and magnetic fields due to the particles are given by

$$\nabla \cdot \mathbf{E}^M = 4\pi \sum_\alpha q_\alpha \int N_\alpha(\mathbf{X}, t)\, d\mathbf{v} \qquad (7.1.2)$$

$$\nabla \times \mathbf{B}^M = \frac{1}{c}\frac{\partial}{\partial t}\mathbf{E}^M + \frac{4\pi}{c}\sum_\alpha q_\alpha \int \mathbf{v} N_\alpha(\mathbf{X}, t)\, d\mathbf{v} \qquad (7.1.3)$$

$$\nabla \times \mathbf{E}^M = -\frac{1}{c}\frac{\partial}{\partial t}\mathbf{B}^M$$

$$\nabla \cdot \mathbf{B}^M = 0$$

where the superscript M emphasizes that these are the microscopic fields. As usual in electromagnetic theory, the equations $\nabla \cdot \mathbf{E}^M = 4\pi \sum_\alpha q_\alpha \int N_\alpha(\mathbf{X}, t)\, d\mathbf{v}$ and $\nabla \cdot \mathbf{B}^M = 0$ may be thought of as "initial conditions," which, if satisfied at $t = 0$, are preserved by the other two Maxwell equations for all $t \ge 0$.

The equations of motion of each particle

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i, \qquad \frac{d\mathbf{v}_i}{dt} = \frac{q_i}{m_i}\left(\mathbf{E}^M + \frac{\mathbf{v}_i \times \mathbf{B}^M}{c}\right)' \qquad (7.1.4)$$

close the description of the system. The prime indicates that in computing $\mathbf{E}^M$, $\mathbf{B}^M$ for use in calculating the force on particle i, the field due to the particle itself is omitted, $(\nabla \cdot \mathbf{E}_i^M)' = 4\pi \sum_\alpha q_\alpha \sum_{j \ne i} \delta(\mathbf{X} - \mathbf{X}_j)$. Though not very useful for a large number of particles, this description is exact.

From the equations of motion (7.1.4), the equation for the time development of $N_\alpha(\mathbf{x}, \mathbf{v}, t)$ is easily obtained.

$$\frac{\partial N_\alpha(\mathbf{X}, t)}{\partial t} + \mathbf{v} \cdot \frac{\partial N_\alpha(\mathbf{X}, t)}{\partial \mathbf{x}} + \frac{q_\alpha}{m_\alpha}\left(\mathbf{E}^M + \frac{\mathbf{v} \times \mathbf{B}^M}{c}\right)' \cdot \frac{\partial}{\partial \mathbf{v}} N_\alpha(\mathbf{X}, t) = 0 \qquad (7.1.5)$$

Equation (7.1.5) is called the *Klimontovich-Dupree equation*, and can be correctly interpreted as $dN_\alpha/dt = 0$, that is, conservation of particles in phase space. Nevertheless, $N_\alpha(\mathbf{x}, \mathbf{v}, t)$ is not a statistical function. It is simply a mathematical way of telling where all the particles are located. The statistical description is obtained by averaging the microscopic equations.

Problem 7.1.1 Using only the orbit equation (7.1.4) and the definition of $N_\alpha(\mathbf{x}, \mathbf{v}, t)$, demonstrate that $N_\alpha(\mathbf{x}, \mathbf{v}, t)$ satisfies (7.1.5). ////

7.2 THE STATISTICAL EQUATIONS FOR A MANY-BODY SYSTEM

The microscopic state of a plasma or any other many-body system is completely defined by the coordinates and velocities of the particles. Similarly, the statistical properties of such a system are completely determined by the distribution of its particles. (See also Chap. 3.) The function that describes this distribution is introduced as follows. Let

$$F_N(\mathbf{X}_{\alpha 1}, \ldots, \mathbf{X}_{\alpha N_\alpha}; \mathbf{X}_{\beta 1}, \ldots, \mathbf{X}_{\beta N_\beta}; \ldots, t)\, d\mathbf{X}_{\alpha 1} \times d\mathbf{X}_{\alpha 2} \cdots d\mathbf{X}_{\alpha N_\alpha}\, d\mathbf{X}_{\beta 1} \cdots d\mathbf{X}_{\beta N_\beta} \cdots \qquad (7.2.1)$$

be the probability that at time t the coordinates and velocity of the particles have the values $\mathbf{X}_{\alpha 1}, \mathbf{X}_{\alpha 2}, \ldots, \mathbf{X}_{\alpha N_\alpha}, \mathbf{X}_{\beta 1}, \ldots, \mathbf{X}_{\beta N_\beta}, \ldots$, in the range $d\mathbf{X}_{\alpha 1} \cdots d\mathbf{X}_{\alpha N_\alpha}$, $d\mathbf{X}_{\beta 1} \cdots d\mathbf{X}_{\beta N_\beta} \cdots$ and that $N = \sum_\alpha \bar{N}_\alpha$ is the total number of particles in the system.

If the system is in thermodynamic equilibrium, then $F_N = D$, the Gibbs distribution given in Chap. 2. In general, F_N is not equal to D, and the bulk of the material that follows is devoted to the nonequilibrium statistical system, with the aim of finding how F_N evolves in time from a given starting distribution $F_N(0)$. The function F_N may also be interpreted as a probability density of systems in a $6N$-dimensional phase space. F_N satisfies the Liouville equation for an N-body system

$$\frac{DF_N}{Dt} \equiv \frac{\partial F_N}{\partial t} + \sum_i v_i \frac{\partial F_N}{\partial x_i} + \sum_i a_i \frac{\partial F_N}{\partial v_i} = 0$$

corresponding simply to conservation of probability. In statistical mechanics, an equation that depends on the coordinates and velocities of all the particles is of little direct use except as a starting point from which reduced descriptions of the evolution of the system from an initial state are sought. Because F_N is a probability, its normalization is given by

$$\int F_N \, d\mathbf{X}_{\alpha 1} \cdots d\mathbf{X}_{\alpha N_\alpha} d\mathbf{X}_{\beta 1} \cdots d\mathbf{X}_{\beta N_\beta} \cdots d\mathbf{X}_{\gamma 1} \cdots d\mathbf{X}_{\gamma N_\gamma} \equiv \int F_N \, dX_{\text{all}} = 1$$

Reduced distributions are obtained by integrating (7.2.1) over the coordinates of all but one, two, three, etc., particles. The reduced distributions are functions of fewer variables than F_N, and are simpler to study; however, a reduced distribution provides a much less complete description of the system than does the full probability F_N.

The one-particle distribution function is defined here as the density of particles of a single type in phase space,

$$f_\alpha(\mathbf{x}_{\alpha 1}, \mathbf{v}_{\alpha 1}, t) = V \int F_N \, d\mathbf{X}_{\alpha 2} \cdots d\mathbf{X}_{\alpha N_\alpha} \prod_\beta d\mathbf{X}_{\beta 1} \cdots d\mathbf{X}_{\beta N_\beta} \qquad (7.2.2)$$

where V is the volume of the system. Then $(1/V) f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{X}$ gives the probability of finding a particle of type α at $(\mathbf{x}, \mathbf{v})$ in the range $d\mathbf{x}\, d\mathbf{v}$. This distribution function is a highly reduced description of the plasma. Clearly, the probability of finding an electron at $\mathbf{x}$ is altered by the presence of an ion at $\mathbf{x}' \approx \mathbf{x}$, or by the presence of another electron at $\mathbf{x}' \approx \mathbf{x}$; however, this information is not contained in f_α. The influence of a near neighbor on the distribution of a particle is contained in a less reduced description, e.g., a two-particle distribution.

$$f_{\alpha\beta}(\mathbf{X}_{\alpha 1}, \mathbf{X}'_{\beta 1}, t) \equiv V^2 \int F_N \, d\mathbf{X}_{\alpha 2} \cdots d\mathbf{X}_{\alpha N_\alpha}$$

$$\times \, d\mathbf{X}_{\beta 2} \cdots d\mathbf{X}_{\beta N_\beta} \prod_\gamma d\mathbf{X}_{\gamma 1} \cdots d\mathbf{X}_{\gamma N_\gamma} \qquad (7.2.3)$$

In terms of the two-body distribution, the average joint probability that particle 1 of type α is at $\mathbf{X}$ in $d\mathbf{X}$, and particle 2 of type β is at $\mathbf{X}'$ in $d\mathbf{X}'$, is

$$\frac{1}{V^2} f_{\alpha\beta}(\mathbf{X}, \mathbf{X}', t)\, d\mathbf{X}\, d\mathbf{X}' \qquad (7.2.3a)$$

Note that α and β could refer to the same *type* of particle ($\alpha = \beta$), for example, electrons. In that case (7.2.3a) is the probability that an electron α_1 is at $\mathbf{X}$ in $d\mathbf{X}$ and a different electron α_2 is at $\mathbf{X}'$ in $d\mathbf{X}'$. This is still a reduced description, but it contains more detailed information about the plasma than was contained in $f_\alpha(\mathbf{X})$, because it includes the *interaction* between particles α_1 and β_1. (If the particles do not interact, $f_{\alpha\beta} = f_\alpha f_\beta$.) The two-body description is still incomplete, because the probability of particle α_1 being at $\mathbf{X}$ and β_1 being $\mathbf{X}'$ is altered by the possible presence of a third particle (for example, γ_1) at $\mathbf{X}''$ near $\mathbf{X}$ or $\mathbf{X}'$. These effects are contained in a still less reduced description, the three-body distribution

$$f_{\alpha\beta\gamma} = V^3 \int F_N \frac{d\mathbf{X}_{\text{all}}}{d\mathbf{X}_{\alpha_1}\, d\mathbf{X}_{\beta_1}\, d\mathbf{X}_{\gamma_1}}$$

and so on.

Now since each term in $N_\alpha = \sum_i \delta(\mathbf{X} - \mathbf{X}_i)$ describes the location of a particle in terms of its initial conditions, and F_N describes the probability of a particle having a certain set of initial conditions, the reduced descriptions of F_N can be expressed in terms of the averages of products of N_α over all possible initial conditions. These averages are defined by

$$\langle G(N_\alpha, N_\beta, \ldots, N_\gamma) \rangle \equiv \int F_N G(N_\alpha, N_\beta, \ldots, N_\gamma)\, d\mathbf{X}_{\text{all}}$$

Specifically, the average value of the phase-space density $N_\alpha(\mathbf{X}, t)$ gives the one-particle reduced distribution, defined in (7.2.2),

$$\begin{aligned} \langle N_\alpha(\mathbf{X}, t) \rangle &= \int F_N \, N_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{X}_{\text{all}} \\ &= \int F_N \sum_{i=1}^{N_\alpha} \delta[\mathbf{x} - \mathbf{x}_{\alpha i}(t)]\, \delta[\mathbf{v} - \mathbf{v}_{\alpha i}(t)]\, d\mathbf{X}_{\text{all}} \\ &= \bar{N}_\alpha \int F_N \, \delta[\mathbf{x} - \mathbf{x}_{\alpha 1}(t)]\, \delta[\mathbf{v} - \mathbf{v}_{\alpha 1}(t)]\, d\mathbf{X}_{\text{all}} \\ &= \bar{n}_\alpha f_\alpha(\mathbf{x}, \mathbf{v}, t) \end{aligned} \qquad (7.2.4)$$

where the fact that particles of type α are all identical has been used, and $\bar{n}_\alpha \equiv \bar{N}_\alpha/V$ is the average density.

Similarly, $\langle N_\alpha N_\beta \rangle$ is related to the two-body distribution. Using the explicit expression (7.1.1) for N_α, it can be shown that to good approximation

$$\langle N_\alpha(\mathbf{X}, t) N_\beta(\mathbf{X}', t) \rangle = \bar{n}_\alpha \bar{n}_\beta f_{\alpha\beta}(\mathbf{X}, \mathbf{X}', t) + \delta_{\alpha\beta} \bar{n}_\alpha \, \delta(\mathbf{X} - \mathbf{X}') f_\alpha(\mathbf{X}, t) \tag{7.2.5}$$

The average electromagnetic fields are obtained by averaging the microfields, which depend on the detailed location of all the particles, over the probable locations of all the particles.

$$\mathbf{E} = \langle \mathbf{E}^M \rangle = \int F_N \mathbf{E}^M \, d\mathbf{X}_{\text{all}} \tag{7.2.6}$$

Thus Maxwell's equations, when averaged, give the average fields directly in terms of $f_\alpha(\mathbf{x}, \mathbf{v}, t)$, the one-particle distributions.

Problem 7.2.1 Calculate $\langle N_\alpha(\mathbf{X}, t) N_\beta(\mathbf{X}', t) \rangle$ without approximation and compare the results with Eq. (7.2.5). ////

The complete ensemble F_N evolves in time according to an equation of the form (7.1.5). However, an equation for $F_N(t)$ in terms of $F_N(0)$ is of little practical use since there is no way to choose $F_N(0)$. It is frequently possible to guess at a reasonable form for the initial value of the reduced distribution function $f_\alpha(\mathbf{x}, \mathbf{v}, 0)$ from a knowledge of how the system was prepared. Since the distribution f_α includes all fluid properties of the system, i.e.,

$$n_\alpha = \bar{n}_\alpha \int f_\alpha \, d\mathbf{v}$$

$$n_\alpha \mathbf{V}_\alpha = \bar{n}_\alpha \int \mathbf{v} f_\alpha \, d\mathbf{v}$$

$$\vdots$$

it is reasonable to look for equations that determine the time development of f_α, or possibly f_α and $f_{\alpha\beta}$. Since f_α is related to the average of N_α, a direct averaging of (7.1.5) gives the equation for the time development of f_α.

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v} \cdot \frac{\partial f_\alpha}{\partial \mathbf{x}} + \frac{q_\alpha}{\bar{n}_\alpha m_\alpha} \left\langle \left(\mathbf{E}^M + \frac{\mathbf{v} \times \mathbf{B}^M}{c} \right)' \cdot \frac{\partial N_\alpha}{\partial \mathbf{v}} \right\rangle = 0 \tag{7.2.7}$$

Problem 7.2.2 Derive (7.2.7). Are any terms neglected in the derivation? ////

Problem 7.2.3 Show that

$$\nabla \cdot \mathbf{E} = 4\pi \sum_\alpha q_\alpha \bar{n}_\alpha \int f_\alpha(\mathbf{x}, \mathbf{v}, t) \, d\mathbf{v}$$

$$\nabla \times \mathbf{B} = \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} + \frac{4\pi}{c} \sum_\alpha q_\alpha \bar{n}_\alpha \int \mathbf{v} f_\alpha(\mathbf{x}, \mathbf{v}, t) \, d\mathbf{v} \tag{7.2.8}$$

Further, if the interactions between the particles vanish, $\langle N_\alpha N_\beta \rangle \to \langle N_\alpha \rangle \langle N_\beta \rangle = \bar{n}_\alpha \bar{n}_\beta f_\alpha f_\beta$. The dependence of the statistical equation on particle interaction is emphasized by writing (7.2.7) in the following form:

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v} \cdot \frac{\partial f_\alpha}{\partial \mathbf{x}} + \frac{q_\alpha}{m_\alpha} \left(\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c} \right) \cdot \frac{\partial f_\alpha}{\partial \mathbf{v}}$$

$$= -\frac{q_\alpha}{m_\alpha} \left[\left\langle \left(\mathbf{E}^M + \frac{\mathbf{v} \times \mathbf{B}^M}{c} \right)' \cdot \frac{\partial}{\partial \mathbf{v}} \frac{N_\alpha}{\bar{n}_\alpha} \right\rangle - \left(\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c} \right) \cdot \frac{\partial f_\alpha}{\partial \mathbf{v}} \right] \tag{7.2.9}$$

where now the right-hand side of (7.2.9) vanishes if the particles are noninteracting.

7.3 STATISTICAL EQUATIONS FOR A COULOMB PLASMA

Equation (7.2.8) is not a closed equation for f_α, because it involves the average of the product $\langle N_\alpha N_\beta \rangle$. This is most easily seen, and the subsequent treatment is simplified, if the analysis considers the Coulomb approximation, which neglects retardation effects. With this restriction, the state of the system at time t is determined by the coordinates and velocity of the charged particles, all at the same time t. This restriction is valid if the temperature of the plasma is such that

$$v_{\text{thermal}} \ll c \tag{7.3.1}$$

A further simplification is to neglect magnetic effects, setting

$$\mathbf{B}^M = 0$$

This simplification is used in the following derivations for the sake of clarity. For completeness, magnetic effects are reintroduced into the final equations. Now

$$\mathbf{E}^M = -\nabla \phi^M, \qquad \nabla \times \mathbf{E}^M = 0 \tag{7.3.2}$$

and $\nabla \cdot \mathbf{E}^M = 4\pi \Sigma_\alpha \int N_\alpha q_\alpha \, d\mathbf{v}$ can be integrated by standard techniques to give

$$\phi^M(\mathbf{x}, t) = \sum_\alpha q_\alpha \int \frac{N_\alpha(\mathbf{x}', \mathbf{v}', t)}{|\mathbf{x} - \mathbf{x}'|} \, d\mathbf{x}' \, d\mathbf{v}' \tag{7.3.3}$$

Equation (7.2.8) for f_α becomes

$$\frac{\partial}{\partial t}f_\alpha + \mathbf{v}\cdot\frac{\partial f_\alpha}{\partial \mathbf{x}} - \sum_\beta \bar{n}_\beta q_\beta \int f_\beta(\mathbf{X}')\left(\frac{\partial}{\partial \mathbf{x}}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right)\cdot\frac{\partial f_\alpha(\mathbf{X})}{\partial \mathbf{v}}\,d\mathbf{X}'$$

$$= \sum_\beta \frac{q_\alpha q_\beta}{\bar{n}_\alpha m_\alpha}\int\left(\frac{\partial}{\partial \mathbf{x}}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right)\cdot\frac{\partial}{\partial \mathbf{v}}\left(\langle N_\alpha(\mathbf{X},t)N_\beta(\mathbf{X}',t)\rangle' - \bar{n}_\alpha f_\alpha \bar{n}_\beta f_\beta\right)d\mathbf{X}'$$

$$= \sum_\beta \frac{q_\alpha q_\beta}{m_\alpha}\bar{n}_\beta \int\left(\frac{\partial}{\partial \mathbf{x}}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right)\cdot\frac{\partial}{\partial \mathbf{v}}\left(f_{\alpha\beta}(\mathbf{X},\mathbf{X}',t) - f_\alpha f_\beta\right)d\mathbf{X}' \qquad (7.3.4)$$

The equation which $f_{\alpha\beta}$ satisfies is obtained by noting that

$$\frac{\partial}{\partial t}\langle N_\alpha N_\beta\rangle = \left\langle\left(\frac{\partial N_\alpha}{\partial t}\right)N_\beta + N_\alpha\left(\frac{\partial N_\beta}{\partial t}\right)\right\rangle$$

and using Eqs. (7.1.5), (7.3.2), and (7.3.3) for $\partial N_\alpha/\partial t$,

$$\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\frac{\partial}{\partial \mathbf{x}} + \mathbf{v}'\cdot\frac{\partial}{\partial \mathbf{x}'}\right)[\bar{n}_\alpha\bar{n}_\beta f_{\alpha\beta} + \delta_{\alpha\beta}\,\delta(\mathbf{X}-\mathbf{X}')\bar{n}_\alpha f_\alpha(\mathbf{X})]$$

$$= \int\sum_\gamma\left(\frac{\partial}{\partial \mathbf{x}}\frac{q_\gamma q_\alpha}{|\mathbf{x}-\mathbf{x}''|}\cdot\frac{\partial}{\partial \mathbf{v}} + \frac{\partial}{\partial \mathbf{x}'}\frac{q_\gamma q_\beta}{|\mathbf{x}'-\mathbf{x}''|}\cdot\frac{\partial}{\partial \mathbf{v}'}\right)$$

$$\cdot\langle N_\alpha(\mathbf{X},t)N_\beta(\mathbf{X}',t)N_\gamma(\mathbf{X}'',t)\rangle\,d\mathbf{X}'' \qquad (7.3.5)$$

Now

$$\begin{aligned}\langle N_\alpha N_\beta N_\gamma\rangle = {}& \bar{n}_\alpha\bar{n}_\beta\bar{n}_\gamma f_{\alpha\beta\gamma}(\mathbf{X},\mathbf{X}',\mathbf{X}'',t)\\ &+ \delta_{\alpha\beta}\bar{n}_\alpha\bar{n}_\gamma\,\delta(\mathbf{X}-\mathbf{X}')f_{\alpha\gamma}(\mathbf{X},\mathbf{X}'',t)\\ &+ \delta_{\alpha\gamma}\bar{n}_\gamma\bar{n}_\beta\,\delta(\mathbf{X}-\mathbf{X}'')f_{\alpha\beta}(\mathbf{X},\mathbf{X}',t)\\ &+ \delta_{\beta\gamma}\bar{n}_\alpha\bar{n}_\gamma\,\delta(\mathbf{X}'-\mathbf{X}'')f_{\alpha\gamma}(\mathbf{X},\mathbf{X}'',t)\\ &+ \delta_{\alpha\beta}\,\delta_{\beta\gamma}\,\delta(\mathbf{X}-\mathbf{X}')\,\delta(\mathbf{X}'-\mathbf{X}'')f_\alpha(\mathbf{X},t)\bar{n}_\alpha \qquad (7.3.6)\end{aligned}$$

Thus the set (7.3.4) and (7.3.5) is still not closed, and an equation for $f_{\alpha\beta\gamma}$ is required. It is clear that the procedure followed above generates a hierarchy of equations, with the nth reduced distribution appearing in the equation for the $(n-1)$st distribution.

A convenient way of expressing the higher-order distribution function is in terms of the correlation function (introduced in Chap. 2 for the equilibrium case). These are defined by the Mayer cluster expansion

$$\begin{aligned}f_{\alpha\beta}(\mathbf{X},\mathbf{X}') &= f_\alpha(\mathbf{X},t)f_\beta(\mathbf{X}',t) + g_{\alpha\beta}(\mathbf{X},\mathbf{X}',t)\\ f_{\alpha\beta\gamma} &= f_\alpha(\mathbf{X},t)f_\beta(\mathbf{X}',t)f_\gamma(\mathbf{X}'',t)\\ &\quad + f_\alpha(\mathbf{X},t)g_{\beta\gamma}(\mathbf{X}',\mathbf{X}'',t) + f_\beta(\mathbf{X}',t)g_{\alpha\gamma}(\mathbf{X},\mathbf{X}'',t)\\ &\quad + f_\gamma(\mathbf{X}'',t)g_{\alpha\beta}(\mathbf{X},\mathbf{X}',t) + g_{\alpha\beta\gamma}(\mathbf{X},\mathbf{X}',\mathbf{X}'',t)\\ &\vdots\end{aligned} \qquad (7.3.7)$$

For a system of statistically independent particles, $g_{\alpha\beta} = g_{\alpha\beta\gamma} = 0$.

Problem 7.3.1 Using

$$\int \delta(\mathbf{x}-\mathbf{x}')\frac{\partial}{\partial \mathbf{x}}\frac{1}{|\mathbf{x}-\mathbf{x}'|}\,d\mathbf{x}' = 0$$

show that the last equality of Eq. (7.3.4) is exact even if the "self-field terms" had not been specifically deleted. ($\langle\ \rangle'$ means that self-correlation terms are to be deleted.) ////

7.4 CLOSING THE CHAIN OF STATISTICAL EQUATIONS

It is just as difficult to solve the infinite chain of statistical equations as it is to solve the exact microscopic equations (7.1.1) to (7.1.4). This same difficulty is encountered in all statistical studies of many-body systems. However, there is an advantage to a statistical description in that it is usually possible to terminate the chain of equations by finding a valid approximation for the higher correlations $g_{\alpha\beta\gamma}\ldots$, and thus obtain closed equations for the first few reduced distributions. In a tenuous gas of *neutral* particles, for example, the range of the force r_0 is much less than the mean spacing of the particles $n^{-1/3}$. In this case a natural approach is to expand the correlations in the small parameter $nr_0{}^3$ [note that for $nr_0{}^3 = 0$ the particles are completely independent ($g_{\alpha\beta} = 0$)]. This leads to the conclusion that the triple correlation $g_{\alpha\beta\gamma}$ is much smaller than the double correlation $g_{\alpha\beta}$, which in turn is smaller than the product of the independent distributions, $f_\alpha f_\beta$.

In a plasma it is also possible to terminate the chain of statistical equations, even though for plasmas the range of the forces $r_0 \approx \lambda_D$ is much *greater* than the interparticle spacing $n^{-1/3}$ (as discussed in Chaps. 2 and 3). The situation in a plasma is exactly opposite to the situation in a gas of neutral particles. The small parameter in a plasma is

$$g = \frac{1}{n\lambda_D{}^3} \ll 1 \qquad (7.4.1)$$

The theory of equilibrium statistical mechanics developed in Chap. 2 shows that the correlations $g_{\alpha\beta}$ are small compared with $f_\alpha f_\beta$ as long as $g \ll 1$. The reason is that the joint distribution $f_{\alpha\beta}$ of two particles in a small volume,

$$n^{-1} \ll V \ll \lambda_D{}^3 \qquad (7.4.2)$$

is determined by the many particles outside of V, and not by the separation of the two particles from each other, so that the distribution $f_{\alpha\beta}(\mathbf{X},\mathbf{X}')$ to good

approximation is independent of the interaction of the two particles, that is, $f_{\alpha\beta} \approx f_\alpha f_\beta$. The approximation becomes exact as $g \to 0$. The assumption $g_{\alpha\beta} < f_\alpha f_\beta$ is confirmed directly by solving the equations obtained using this assumption.

Following the discussion above, the Mayer expansion (7.3.7) can be viewed as a series in the plasma parameter g, with

$$\begin{aligned} f_\alpha &\approx 0(1) \\ g_{\alpha\beta} &\approx 0(g) \\ g_{\alpha\beta\gamma} &\approx 0(g^2) \\ &\vdots \end{aligned} \tag{7.4.3}$$

Then the chain of equations, whose first two terms are (7.3.4) and (7.3.5), can be terminated by neglecting terms of order g^s, leading to closed equations for the $(s-1)$ reduced distribution $f_{\alpha\beta\gamma\cdots}^{(s-1)}$.

7.5 KINETIC EQUATION IN ZERO ORDER—THE VLASOV EQUATION

With the ordering (7.4.3), the simplest approximation to the chain of equations (7.3.4), (7.3.5), ... is to neglect all terms of order g. To this order the kinetic equation for the plasma distribution is

$$\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla + \frac{q_\alpha}{m_\alpha}\mathbf{E}\cdot\nabla_{\mathbf{v}}\right) f_\alpha(\mathbf{x}, \mathbf{v}, t) = 0 \tag{7.5.1}$$

$$\nabla\cdot\mathbf{E} = 4\pi\sum_\alpha \bar{n}_\alpha q_\alpha \int f_\alpha \, d\mathbf{v}$$

where

$$\nabla \equiv \frac{\partial}{\partial \mathbf{x}} \qquad \text{and} \qquad \nabla_{\mathbf{v}} \equiv \frac{\partial}{\partial \mathbf{v}}$$

This is a closed equation for f_α. It is called the *Vlasov equation*,[1] or occasionally, the *collisionless Boltzmann equation*. The interactions between the particles appear in the Vlasov equation through $\mathbf{E}$, the *average* electric field produced at a point $\mathbf{x}$ by the particles, computed self-consistently through Maxwell's equations. This self-consistent-field approach has emerged in a natural way from an expansion of the general equations in the plasma parameter. It can be generalized to include magnetic fields and externally produced fields.

[1] A. A. Vlasov, *J. Phys.* (*U.S.S.R.*), **9**:25 (1945).

$$\begin{aligned} \left[\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla + \frac{q_\alpha}{m_\alpha}\left(\mathbf{E} + \frac{\mathbf{v}\times\mathbf{B}}{c}\right)\cdot\nabla_{\mathbf{v}}\right] f_\alpha(\mathbf{x}, \mathbf{v}, t) &= 0 \\ \nabla\cdot\mathbf{E} &= 4\pi\sum_\alpha \bar{n}_\alpha q_\alpha \int f_\alpha \, d\mathbf{v} + 4\pi\rho_{\text{ext}} \\ \nabla\times\mathbf{B} &= \frac{1}{c}\frac{\partial\mathbf{E}}{\partial t} + \frac{4\pi}{c}\sum_\alpha \bar{n}_\alpha q_\alpha \int \mathbf{v} f_\alpha \, d\mathbf{v} + \frac{4\pi}{c} J_{\text{ext}} \\ \nabla\times\mathbf{E} &= -\frac{1}{c}\frac{\partial\mathbf{B}}{\partial t} \end{aligned} \tag{7.5.2}$$

The next several chapters are devoted to the methods of solving the Vlasov equation, and thus investigating properties of plasmas on a short time scale:

$$\tau_{\text{collective}} \ll \tau_{\text{collisional}}$$

7.6 KINETIC EQUATION IN FIRST ORDER

Using the ordering (7.4.3), a set of equations more complete than (7.5.2) is obtained by retaining in the hierarchy terms of order g. To this order the correlation $g_{\alpha\beta\gamma}$ is neglected, and the first-order kinetic equations become

$$\begin{aligned} \left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla + \frac{q_\alpha}{m_\alpha}\mathbf{E}\cdot\nabla_{\mathbf{v}}\right) f_\alpha(\mathbf{x}, \mathbf{v}, t) &= \sum_\beta \bar{n}_\beta \frac{q_\alpha q_\beta}{m_\alpha}\int\left(\nabla\frac{1}{|\mathbf{x}-\mathbf{x}'|}\right) \\ &\quad\cdot\nabla_{\mathbf{v}} g_{\alpha\beta}(\mathbf{x}, \mathbf{x}', \mathbf{v}, \mathbf{v}', t)\, d\mathbf{x}'\, d\mathbf{v}' \end{aligned} \tag{7.6.1}$$

$$\begin{aligned} &\left[\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla_{\mathbf{x}} + \mathbf{v}'\cdot\nabla_{\mathbf{x}'} + \frac{q_\alpha}{m_\alpha}\mathbf{E}(\mathbf{x}, t)\cdot\nabla_{\mathbf{v}} + \frac{q_\beta}{m_\beta}\mathbf{E}(\mathbf{x}', t)\cdot\nabla_{\mathbf{v}'}\right] g_{\alpha\beta}(\mathbf{X}, \mathbf{X}', t) \\ &\quad -\sum_\gamma \frac{\bar{n}_\gamma}{m_\alpha}\int g_{\gamma\beta}(\mathbf{X}', \mathbf{X}'')\,\nabla_{\mathbf{x}}\frac{q_\alpha q_\gamma}{|\mathbf{x}-\mathbf{x}''|}\cdot\nabla_{\mathbf{v}} f_\alpha(\mathbf{X})\, d\mathbf{X}'' \\ &\quad -\sum_\gamma \frac{\bar{n}_\gamma}{m_\beta}\int\left(\nabla_{\mathbf{x}'}\frac{q_\beta q_\gamma}{|\mathbf{x}-\mathbf{x}''|}\right) g_{\alpha\gamma}(\mathbf{X}, \mathbf{X}'')\cdot\nabla_{\mathbf{v}'} f_\beta(\mathbf{X}')\, d\mathbf{X}'' \\ &\quad = \left(\nabla_{\mathbf{x}}\frac{q_\alpha q_\beta}{|\mathbf{x}-\mathbf{x}'|}\right)\cdot\left[\frac{1}{m_\alpha}\nabla_{\mathbf{v}} f_\alpha(\mathbf{X}) f_\beta(\mathbf{X}') - \frac{1}{m_\beta}\nabla_{\mathbf{v}'} f_\beta(\mathbf{X}') f_\alpha(\mathbf{X})\right] \end{aligned} \tag{7.6.2}$$

$$\nabla\cdot\mathbf{E} = 4\pi\sum_\alpha q_\alpha \bar{n}_\alpha \int f_\alpha \, d\mathbf{v} \tag{7.6.3}$$

Equation (7.6.1) gives collisional corrections to the Vlasov equation (7.5.2). Equation (7.6.2), which expresses the correlations $g_{\alpha\beta}$ in terms of the one-particle distributions f_α, is approximate and neglects terms of order g^2, for example, the ternary correlations $g_{\alpha\beta\gamma}$. A solution of (7.6.1), and (7.6.2), examined in Chap.

11, is called the Balescu-Lenard equation; it gives the relaxation of a distribution toward equilibrium on the slower collisional time scale τ_c. Despite the direct demonstrations that $g_{\alpha\beta} \approx 0(g)$, it has never been adequately demonstrated under what circumstances $g_{\alpha\beta\gamma}$ is really negligible. Higher-order kinetic theories can be generated from the hierarchy in a straightforward fashion.

The hierarchy of statistical equations for a plasma was first derived, independently, by Bogolyubov,[1] Born and Green,[2] Kirkwood,[3] and Yvon,[4] and is called the *BBGKY hierarchy*. Further comments on the plasma limit $g \to 0$ are contained in Chap. 2.

7.7 PROPERTIES OF THE VLASOV EQUATION

The statistical description of a plasma provided by the Vlasov equation (7.5.2) is a considerable simplification of the exact description (7.1.1) to (7.1.4). It is now necessary to examine the consequences of these simplifications and investigate the properties of the reduced set of equations, since even the most obvious features of solutions obtained from the exact equation are not always reproduced by the approximate ones.

7.7.1 The Vlasov Equation Conserves Particles

The time rate of change of the total number of particles is zero, i.e.,

$$\frac{\partial}{\partial t} \int \bar{n}_\alpha f_\alpha \, d\mathbf{x} \, d\mathbf{v} = 0$$

as seen directly by integrating (7.5.2) over the entire system. This is also, of course, a property of the exact equations.

7.7.2 Probabilities Remain Positive in Vlasov Theory

If $f_\alpha(\mathbf{x}, \mathbf{v}, t = 0) > 0$ for all $\mathbf{x}$, $\mathbf{v}$, then f_α, evolving according to (7.5.2), remains positive. If f_α were to go negative at $\mathbf{x}_0$, $\mathbf{v}_0$, t_0, then, at that instant, it would be necessary for $\partial f_\alpha/\partial t$ to be negative, and for ∇f_α and $\nabla_\mathbf{v} f_\alpha$ to vanish. But $\partial f_\alpha/\partial t < 0$, $\nabla f_\alpha = \nabla_\mathbf{v} f_\alpha = 0$, is clearly not allowed by the Vlasov equation (7.5.2).

[1] N. N. Bogolyubov, "Problems of a Dynamical Theory in Statistical Physics," State Technical Press, Moscow, 1946. Included in J. deBoer and G. E. Uhlenbeck (eds.), "Studies in Statistical Mechanics," vol. 1, North-Holland Publishing Co., Amsterdam, 1962.

[2] M. Born and H. S. Green, "A General Kinetic Theory of Liquids," Cambridge, London, 1949.

[3] J. G. Kirkwood, *J. Chem. Phys.*, **14**:180 (1946); **15**:72 (1947).

[4] J. Yvon, "La théorie des fluides et l'équation d'état: actualités scientifiques et industrielles," Hermann et Cie, Paris, 1935.

7.7.3 The Vlasov Equation Has Many Equilibrium Solutions

In contrast to the exact equations, which according to the Boltzmann H theorem produce, on a collisional time scale, a unique equilibrium, the Vlasov equation has a wide variety of stationary states. That is, there are many solutions $f_{\alpha 0}$ of (7.5.2) that satisfy $\partial f_{\alpha 0}/\partial t = 0$. These states are often called metaequilibria, since they are only equilibria on a time scale short compared with collision times [this was the essence of the $g \to 0$ approximation that led from (7.2.8) to (7.5.2), i.e., the neglect of collisions]. A stationary state must satisfy

$$\left[\mathbf{v} \cdot \nabla + \frac{q_\alpha}{m_\alpha}\left(\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c}\right) \cdot \nabla_\mathbf{v}\right] f_{\alpha 0}(\mathbf{x}, \mathbf{v}) = 0 \qquad (7.7.1)$$

A procedure for generating a general solution of (7.7.1) is found by defining $\mathbf{x}'(t')$, $\mathbf{v}'(t')$ as the orbit of a particle of charge q, mass m, in the field $\mathbf{E}$, $\mathbf{B}$, with the orbit $(\mathbf{x}', \mathbf{v}')$ intersecting the point in phase space $(\mathbf{x}, \mathbf{v})$ at time $t' = t$. If $a(\mathbf{x}', \mathbf{v}')$, $b(\mathbf{x}', \mathbf{v}')$, ..., are constants of the motion of the particle, they satisfy

$$\begin{aligned} \frac{da}{\partial t'} &= \mathbf{v}' \cdot \frac{\partial a}{\partial \mathbf{x}'} + \frac{\partial \mathbf{v}'}{\partial t'} \cdot \frac{\partial a}{\partial \mathbf{v}'} = 0 \\ \frac{db}{dt'} &= \mathbf{v}' \cdot \frac{\partial b}{\partial \mathbf{x}'} + \frac{\partial \mathbf{v}'}{\partial t'} \cdot \frac{\partial b}{\partial \mathbf{v}'} = 0 \end{aligned} \qquad (7.7.2)$$

Then any function

$$f_{\alpha 0} = f_{\alpha 0}[a(\mathbf{x}', \mathbf{v}'), b(\mathbf{x}', \mathbf{v}'), \ldots]$$

satisfies (7.7.1) at time $t' = t$, and thus any function

$$f_{\alpha 0} = f_{\alpha 0}[a(\mathbf{x}, \mathbf{v}), b(\mathbf{x}, \mathbf{v}), \ldots]$$

of the constants of the motion of a particle is also a stationary-state distribution for an assembly of particles, according to the Vlasov equation.

In many cases (e.g., if $\int f_i \, d\mathbf{v} = \int f_e \, d\mathbf{v}$ and $J_{\text{plasma}} = 0$), the distribution f_α is controlled by external fields and the constants of motion for a particle in these fields can be used to construct stationary states. Some examples are presented below.

a Field-free equilibria ($\mathbf{E}_0 = \mathbf{B}_0 = 0$) In this case the constants of the motion are the energy and momentum

$$W = \frac{m}{2}(v_x^2 + v_y^2 + v_z^2)$$

$$\mathbf{p} = m\mathbf{v}$$

and so any

$$f_{\alpha 0} = f_{\alpha 0}(v_x, v_y, v_z)$$

is a Vlasov metaequilibrium state. The Maxwell-Boltzmann distribution

$$f_{\alpha 0} = \left(\frac{m}{2\pi\kappa T}\right)_\alpha^{3/2} \exp\left(-\frac{m}{2\kappa T} v^2\right)_\alpha \tag{7.7.3}$$

is a Vlasov equilibrium, as well as an equilibrium of the exact equations, but

$$f_{\alpha 0} = \frac{v_0}{2} \frac{1}{v^4 + {v_0}^4} \tag{7.7.3a}$$

$$f_{\alpha 0} = v_0\, \delta(v_x)\, \delta(v_y)\, \delta({v_z}^2 - {v_0}^2) \tag{7.7.3b}$$

$$f_{\alpha 0} = \left(\frac{m}{2\pi\kappa T}\right)_\alpha^{1/2} \delta(v_x)\, \delta(v_y) \exp\left[-\frac{m_\alpha({v_z}^2 - {v_{z0}}^2)}{2\kappa T_\alpha}\right] \tag{7.7.3c}$$

are all Vlasov equilibria, though not equilibria of the exact equations. The Vlasov stationary states presumably evolve to Maxwell-Boltzmann distributions on a collisional time scale.

The selection of a particular distribution from the many possible solutions of (7.7.1) is determined from considerations outside the venue of the Vlasov equation. The choice of a particular f_0 involves such things as knowledge of how the plasma was prepared or the known features of a plasma model. In some cases, mathematical convenience dictates the choice. For example, if the system has existed for many collision times, the Maxwell-Boltzmann distribution (7.7.3) is an appropriate choice for f_0. However, the resonance distribution (7.7.3*a*) resembles a maxwellian distribution and has the additional advantage of being analytic in the complex v plane, except for simple poles. This feature allows certain velocity integrals to be performed by contour integration. Another example is a system prepared by injecting and trapping a monoenergetic beam of particles between two walls. In this case, (7.7.3*b*) might be the appropriate distribution. If the injected beam has a temperature in the direction of injection, then (7.7.3*c*) might be a good choice for the distribution.

It is important to note that in a plasma that has existed for many binary collision times, the Vlasov equation is still the appropriate description of the system for a time interval shorter than one collision time. Thus the Maxwell-Boltzmann plasma will exhibit on the Vlasov time scale the full spectrum of plasma effects, e.g., plasma waves and Landau damping, excited perhaps by the collisions themselves.

b Equilibria in a parallel magnetic field [$\mathbf{E}_0 = 0$, $\mathbf{B}_0 = B_0(\mathbf{r})\hat{\mathbf{z}}$] The constants of the motion in this case are the energy, angular momentum perpendicular to $\mathbf{B}$, and linear momentum along $\mathbf{B}$.

$$W = \tfrac{1}{2} m_\alpha({v_x}^2 + {v_y}^2 + {v_z}^2)$$

$$p_\parallel = m_\alpha v_z$$

$$\mathbf{L} = m_\alpha(x v_y - y v_x)\hat{\mathbf{z}} - e r A_\theta(r)/c$$

Additional constants of the motion, especially useful for describing nonuniform plasmas, are

$$\xi_y = v_y + \frac{q_\alpha}{m_\alpha c} \int B_0(r)\, dx$$

$$\xi_x = v_x - \frac{q_\alpha}{m_\alpha c} \int B_0(r)\, dy$$

These are related to the motion of the guiding center of the particle (for a constant field B_0, $\xi/\omega_{c\alpha}$ is exactly the radial location of the center of the orbit of a particle, given that the particle has velocity $\mathbf{v}$ and location $\mathbf{r}$). Their constancy can be proved directly from (7.7.1). Stationary Vlasov states have distributions which are functions of these variables. A wide variety of plasma configurations can be described in terms of these constants of the motions (see Prob. 7.7.1).

Problem 7.7.1 Show by direct substitution that any function

$$f_{\alpha 0} = F\left(v^2, v_y + \frac{qB_0}{mc} x, v_x - \frac{qB_0}{mc} y\right) \tag{7.7.4}$$

is a stationary state of a Vlasov plasma that has no electric field and is imbedded in a constant magnetic field $\mathbf{B} = B_0\hat{\mathbf{z}}$. Find equilibrium distributions which describe each of these specific plasma states:

(*a*) Uniform isotropic plasma
(*b*) Plasma formed by injection of particles perpendicular to $\mathbf{B}_0$, that is, a uniform plasma with very low temperature parallel to $\mathbf{B}_0$
(*c*) Plasma formed from an arc along a $\mathbf{B}$ field, i.e., a cylindrically symmetric plasma with radial density gradients (assuming that the plasma is very tenuous, $n\kappa T \ll {B_0}^2/8\pi$) ////

Problem 7.7.2 Find an equilibrium distribution for plasma imbedded in a vacuum field $\mathbf{B} = B_0\hat{\mathbf{z}}$, $\mathbf{E} = 0$, given that the plasma is cylindrically symmetric and that the density decreases radially. Include the plasma currents in the actual field $B(r)\hat{\mathbf{z}}$. What is the significance of the restriction $n\kappa T \ll {B_0}^2/8\pi$ employed in Prob. 7.7.1*c*? ////

Problem 7.7.3 Find a set of equilibrium distributions for a plasma in which the fields are given by $\mathbf{E} = E_0\hat{\mathbf{z}}, \mathbf{B} = 0$. What are the constants of the motion? ////

7.7.4 Entropy in Vlasov Plasma—H Theorem for Vlasov Plasma

The entropy of a statistical system is usually defined as

$$S = -\sum_\alpha \int f_\alpha \ln f_\alpha \, d\mathbf{x}\, d\mathbf{v} \qquad (7.7.5)$$

For a plasma,

$$\frac{d}{dt} S = -\sum_\alpha \int \left(\frac{df_\alpha}{dt} \ln f_\alpha + \frac{df_\alpha}{dt} \right) d\mathbf{x}\, d\mathbf{v} = 0 \qquad (7.7.6)$$

by direct substitution of (7.5.2) into (7.7.6). Thus entropy is constant in a Vlasov plasma. This is consistent with the fact that the Vlasov equation neglects the process (binary collisions) which causes statistical systems to increase their entropy and evolve toward a Maxwell-Boltzmann distribution.

For some plasma systems, the distribution function f_α evolves as

$$f_\alpha(t) = f_{\alpha 0}(t) + f_{\alpha 1}(t)$$

where $f_{\alpha 0}$ changes slowly in time and $f_{\alpha 1}$ changes rapidly. For such examples it is useful to define a generalized entropy function

$$S^* = -\int f_{\alpha 0}(t) \ln f_{\alpha 0}(t) \, d\mathbf{x}\, d\mathbf{v} \qquad (7.7.7)$$

With this definition the "entropy" is not constant, and can be used to measure the increase of disorder (e.g., temperature) of the system (see further Chap. 10).

Problem 7.7.4 Show that $dH/dt = 0$, where $H = \int \phi(f_\alpha)\, d\mathbf{x}\, d\mathbf{v}$, and ϕ is any function of a distribution f_α (f_α satisfies the Vlasov equation). ////

7.7.5 Stability of Vlasov Metaequilibria

A complication associated with the use of known states of $\partial f_{\alpha 0}/\partial t = 0$ as a basis for building devices that use plasmas in practical ways (controlled fusion, plasma thermocouples, plasma engines, plasma accelerators, etc.) is that there are often nearby states

$$f_\alpha(t) = f_{\alpha 0} + \delta f_\alpha(t)$$

that have the property that the amplitude of $\delta f_\alpha(t)$ increases, leading to a state $f_\alpha(t)$ further from $f_{\alpha 0}$. If this is so, the metaequilibrium $f_{\alpha 0}$ is called unstable, because a small perturbation $\delta f_\alpha(t)$ grows with time. This is in contrast with the collisional gas, where the condition $\partial f_{\alpha 0}/\partial t = 0$ is sufficient to define a true equilibrium. The meaning of metaequilibrium is that there are nearby states $f_{\alpha 0}'$, of lower internal energy than f_0, that can be reached by a time development $f_{\alpha 0} \to f_{\alpha 0}'$ consistent with the dynamic equation which governs $f_{\alpha 0}$, that is, the Vlasov equation.

Of course, the lowest energy state to which the system can evolve is the Maxwell-Boltzmann state. Although the Vlasov equation has a variety of metaequilibria, the idea of instability again raises the question of whether the average interaction between particles (i.e., the Vlasov approximation) is sufficient to drive the plasma to true thermodynamic equilibrium. If the only requirement for instability were that the system be in a state of entropy less than its "ground state," then the Vlasov plasma would indeed evolve toward a Maxwell-Boltzmann distribution. However, the condition that the evolution of the plasma be consistent with the Vlasov equation is not trivial. There are solutions to the Vlasov equation that are stationary and have less entropy than that of a corresponding Maxwell-Boltzmann distribution, and yet are stable. To demonstrate this important property, consider a general function H defined by

$$H = \int G(f)\, d\mathbf{x}\, d\mathbf{v} \qquad (7.7.8)$$

where G is any bounded functional of f with a finite derivative $\partial G/\partial f$. The constraint that f satisfies the Vlasov equation is equivalent to the constraint

$$\frac{dH}{dt} = 0 \qquad (7.7.9)$$

for all choices of $G(f)$. For example, $G = f$ gives the constraint that the number of particles in the system is conserved.

$$\frac{d}{dt} \int n_\alpha f_\alpha \, d\mathbf{x}\, d\mathbf{v} = 0 \qquad (7.7.10)$$

A stable equilibrium is one in which the kinetic energy does not change with time (this precludes, for example, growth of plasma waves, which contain electrostatic energy). That is, if

$$\frac{d}{dt} \int \frac{1}{2} m_\alpha v^2 \bar{n}_\alpha f_\alpha \, d\mathbf{v}\, d\mathbf{x} = 0 \qquad (7.7.11)$$

subject to (7.5.2) as a constraint, then f_α is a stable Vlasov equilibrium.

If v^2 can be written as a single-valued function of f_α, then by (7.7.8) and (7.7.9) the kinetic energy will be constant. For this to be so, it is sufficient that f_α be a monotonically decreasing function of v^2; that is, if f_α satisfies (7.7.1) and

$$\frac{\partial f_\alpha}{\partial v^2} < 0 \qquad (7.7.12)$$

then f_α is a stable Vlasov equilibrium.

This stability demonstration is sufficient but not necessary, since it replaces the Vlasov equation by a weaker constraint, i.e., (7.7.10), with $G(f_\alpha) \equiv v^2(f_\alpha)$.

This demonstrates that not only a Maxwell-Boltzmann distribution, but any $f_\alpha(v^2)$ satisfying (7.7.12), is a stable metaequilibrium. Clearly, the Vlasov equation is a strong constraint on the evolution of a system. Nevertheless, a wide variety of interesting plasma dynamics occur on the Vlasov time scale, as is shown in the following chapters.

Problem 7.7.5 Why is the distribution $\partial f_0/\partial v^2 > 0$, which also produces a single-valued $v^2(f_0)$, not considered among "stable metaequilibria"? ////

Problem 7.7.6 Show that (7.7.9) to (7.7.10) imply that f_0 is a stable state only if v^2 is a single-valued function of f_0. ////

Problem 7.7.7 Prove that energy (including electric and magnetic field energy) is conserved for a Vlasov plasma. ////

7.8 PROPERTIES OF THE KINETIC EQUATION TO ORDER g

The first-order kinetic equations (7.6.1) to (7.6.3) include to first order the contribution of particle collisions, which act along with the mean fields **E** to evolve the distribution function f_0 toward equilibrium. It is still not an exact equation. This set of equations has not been solved in general. An approximate solution is described in Chap. 11.

Problem 7.8.1 Show that the functions $f_\alpha = (m/2\pi\kappa T)_\alpha^{3/2} \exp(-m_\alpha v^2/2\kappa T_\alpha)$ and $g_{\alpha\beta}$, given by the thermodynamic equilibrium correlation function (2.3.5), satisfy the first-order kinetic equations (7.6.1) to (7.6.3), i.e., that the true equilibrium

$$g_{\alpha\beta} = f_\alpha f_\beta \frac{q_\alpha q_\beta}{|\mathbf{x} - \mathbf{x}'|} \exp\left(-\frac{|\mathbf{x} - \mathbf{x}'|}{\lambda_D}\right)$$

is also on equilibrium of the truncated set of kinetic equations. ////

Problem 7.8.2 Show that on a characteristic time scale ω_p (i.e., if $\partial f/\partial t \approx \omega_p f$) and if the characteristic interaction distance is $|\mathbf{x} - \mathbf{x}'| \approx \lambda_D$, the correlations $g_{\alpha\beta}$ as estimated from (7.6.2) contribute a correction of order $g \approx 1/n\lambda_D{}^3$ to the Vlasov equation. ////

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8

THE VLASOV THEORY OF PLASMA WAVES

In the Vlasov theory of plasmas, the microfields produced by plasma particles are replaced by the average field that the particles produce at a given space point, and the distribution function of plasma particles is calculated so as to be self-consistent with that average field. The Vlasov equation, based on this model of the plasma state, correctly describes the behavior of plasmas for time intervals much less than binary collision times. Besides the static or equilibrium solutions of the Vlasov equation examined in Chap. 7, there are many nonequilibrium phenomena, such as waves and plasma stability, that can be studied with Vlasov theory. In this chapter, the properties of small-amplitude plasma waves that propagate as perturbations about a plasma equilibrium state are examined using the linearized Vlasov equations. Much of the recent progress in plasma theory has been based on solutions of these equations.

8.1 THE VLASOV EQUATIONS

The Vlasov and Maxwell equations

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v} \cdot \nabla f_\alpha + \frac{q_\alpha}{m_\alpha}\left(\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c}\right) \cdot \nabla_{\mathbf{v}} f_\alpha = 0 \qquad (8.1.1)$$

$$\nabla \cdot \mathbf{E} = 4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int f_\alpha \, d\mathbf{v} + 4\pi\rho_{\text{ext}} \qquad (8.1.2)$$

$$\nabla \times \mathbf{B} = \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} + \frac{4\pi}{c}\sum_\alpha \bar{n}_\alpha q_\alpha \int \mathbf{v} f_\alpha \, d\mathbf{v} + \frac{4\pi}{c}\mathbf{J}_{\text{ext}} \qquad (8.1.3)$$

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t} \qquad (8.1.4)$$

describe stationary plasma states, plasma waves, instabilities, and other short-time-scale ($\tau_{\text{collective}} \ll \tau_{\text{collisional}}$) plasma dynamics in terms of the distribution function $f_\alpha(\mathbf{x}, \mathbf{v}, t)$. When multiplied by the average density of particles of type α [that is, by $\bar{n}_\alpha (= \bar{N}_\alpha / V)$], this distribution function gives the probable number of plasma particles of type α at the location $\mathbf{x}$, $\mathbf{v}$ in the range $d\mathbf{x}\, d\mathbf{v}$, at time t, which is

$$\bar{n}_\alpha f_\alpha(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{x}\, d\mathbf{v}$$

The normalization used here is

$$\bar{n}_\alpha \int f_\alpha \, d\mathbf{x}\, d\mathbf{v} = \bar{N}_\alpha = \text{total number of particles of type } \alpha \text{ in the system}$$

Note that $\bar{n}_\alpha$ is not the local density in configuration space, but is the average density of the entire system.

8.2 THE LINEARIZED VLASOV EQUATIONS

Since the fields $\mathbf{E}$, $\mathbf{B}$ depend on the distribution function, the Vlasov equation (8.1.1) is a nonlinear equation for f_α, and consequently difficult to solve. Instead of calculating the exact distribution function f_α from the complete Vlasov equations, the behavior of a small departure $f_{\alpha 1}$ from some plasma equilibrium state $f_{\alpha 0}$ can be calculated using the *linearized* Vlasov equations. These linearized equations, which are approximations to the exact equations, are much easier to solve and are the basis of much of modern plasma theory. Solutions obtained using the linearized Vlasov equations should be examined to see if they satisfy the conditions imposed in deriving the approximate equations.

The linearized Vlasov equations are derived by first letting the distribution function f_α and the fields $\mathbf{E}$ and $\mathbf{B}$ be expressed in terms of an equilibrium value plus a perturbation; i.e.,

$$\begin{aligned} f_\alpha(\mathbf{x}, \mathbf{v}, t) &= f_{\alpha 0}(\mathbf{x}, \mathbf{v}, t) + \varepsilon f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t) \\ \mathbf{E}(\mathbf{x}, t) &= \mathbf{E}_0(\mathbf{x}, t) + \varepsilon \mathbf{E}_1(\mathbf{x}, t) \\ \mathbf{B}(\mathbf{x}, t) &= \mathbf{B}_0(\mathbf{x}, t) + \varepsilon \mathbf{B}_1(\mathbf{x}, t) \end{aligned} \qquad (8.2.1)$$

These quantities are then substituted into the Vlasov equations, and the nonlinear terms, of order ε^2, are neglected. The plasma state $f_{\alpha 0}$ is assumed to be known and to satisfy the Vlasov equation and Maxwell's equation; i.e.,

$$\frac{\partial f_{\alpha 0}}{\partial t} + \mathbf{v} \cdot \nabla f_{\alpha 0} + \frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_0 + \frac{\mathbf{v} \times \mathbf{B}_0}{c}\right) \cdot \nabla_{\mathbf{v}} f_{\alpha 0} = 0 \qquad (8.2.2)$$

$$\nabla \cdot \mathbf{E}_0 = 4\pi \sum_\alpha q_\alpha \bar{n}_\alpha \int f_{\alpha 0}\, d\mathbf{v} + 4\pi \rho_{0,\text{ext}}$$

$$\nabla \times \mathbf{B}_0 = \frac{1}{c}\frac{\partial \mathbf{E}_0}{\partial t} + \frac{4\pi}{c}\sum_\alpha \bar{n}_\alpha q_\alpha \int \mathbf{v} f_{\alpha 0}\, d\mathbf{v} + \frac{4\pi}{c}\mathbf{J}_{0,\text{ext}} \qquad (8.2.3)$$

$$\nabla \times \mathbf{E}_0 = -\frac{1}{c}\frac{\partial \mathbf{B}_0}{\partial t}$$

Subtracting the equilibrium equations from the linearized perturbed equations gives the linearized Vlasov equations for the perturbed distribution function $f_{\alpha 1}$ and the perturbed fields $\mathbf{E}_1$ and $\mathbf{B}_1$.

$$\frac{\partial f_{\alpha 1}}{\partial t} + \mathbf{v} \cdot \nabla f_{\alpha 1} + \frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_0 + \frac{\mathbf{v} \times \mathbf{B}_0}{c}\right) \cdot \nabla_{\mathbf{v}} f_{\alpha 1} = -\frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_1 + \frac{\mathbf{v} \times \mathbf{B}_1}{c}\right) \cdot \nabla_{\mathbf{v}} f_{\alpha 0} \qquad (8.2.4)$$

$$\nabla \cdot \mathbf{E}_1 = 4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha 1}\, d\mathbf{v} \qquad (8.2.5)$$

$$\nabla \times \mathbf{B}_1 = \frac{1}{c}\frac{\partial \mathbf{E}_1}{\partial t} + \frac{4\pi}{c}\sum_\alpha \bar{n}_\alpha q_\alpha \int \mathbf{v} f_{\alpha 1}\, d\mathbf{v} \qquad (8.2.6)$$

$$\nabla \times \mathbf{E}_1 = -\frac{1}{c}\frac{\partial \mathbf{B}_1}{\partial t}$$

Problem 8.2.1 What terms are neglected in (8.2.4)? ////

This set of linearized equations for the perturbed quantities can be solved by conventional methods to investigate plasma properties for time intervals shorter than binary collision times. One application of these linearized equations is to investigate the properties of small-amplitude plasma waves whose oscillation periods are much less than a binary collision time. In this problem, the distribution $f_{\alpha 0}(\mathbf{x}, \mathbf{v})$ and the self-consistent fields $\mathbf{E}_0$, $\mathbf{B}_0$ represent a stationary plasma state (such states are easily constructed in terms of the constants of the motion, by the method discussed in Chap. 7); the distribution $f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t)$ represents the development of the initial perturbation.

In the fluid theory of plasmas, small perturbations of stationary states developed as plasma waves (Chap. 4). Since the Vlasov theory of plasmas contains all the results of the fluid theory, it is clear that in many cases $f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t)$ will also develop in a wavelike manner. Other properties not contained in the fluid theory will also become apparent. As with fluid theory, the properties of plasma waves depend on the plasma equilibrium state ($f_{\alpha 0}$, $\mathbf{E}_0$, $\mathbf{B}_0$). The remainder of this chapter uses the perturbation equations (8.2.4) to (8.2.6) to investigate the wave properties of a variety of plasma equilibria, and thus reveal the richness of phenomena present in an aroused plasma.

8.3 SOLUTION OF THE LINEARIZED VLASOV EQUATIONS FOR ELECTROSTATIC PERTURBATIONS OF A FIELD-FREE PLASMA EQUILIBRIUM

Vlasov theory was first used to study the electromechanical oscillations in a plasma. This problem introduces the mathematics used to solve the linearized Vlasov equations, and it illustrates the additional plasma properties that can be studied with the Vlasov equations, but not with fluid theory.

Consider a spatially uniform field-free plasma that obeys the equilibrium Vlasov equations, i.e.,

$$\mathbf{E}_0 = \mathbf{B}_0 = 0 \qquad f_{\alpha 0} = f_{\alpha 0}(v_x, v_y, v_z) \qquad (8.3.1)$$

The statement $\mathbf{E}_0 = \mathbf{B}_0 = 0$ implies, of course, that

$$\rho_q = \sum_\alpha q_\alpha \bar{n}_\alpha \int f_{\alpha 0}\, d\mathbf{v} = 0 \qquad \text{no net charge}$$

$$\mathbf{J} = \sum_\alpha q_\alpha \bar{n}_\alpha \int \mathbf{v} f_{\alpha 0}\, d\mathbf{v} = 0 \qquad \text{no net current}$$

If at time $t = 0$ a small amount of charge is displaced in this plasma, the initial perturbation can be described by

$$f(t = 0) = f_{\alpha 0}(v_x, v_y, v_z) + \varepsilon f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t = 0)$$

Assume that the perturbation is *electrostatic*; i.e., the displacement of charge gives rise to a perturbed electric field but no magnetic field. This condition is satisfied if the perturbed charge density varies only in one dimension and thus generates only an electrostatic field. In this case

$$\nabla \times \mathbf{E}_1 = 0 \qquad (8.3.2)$$

and

$$\mathbf{E}_1 = -\nabla \phi_1$$

Although one-dimensional perturbations are exactly electrostatic, in general (8.3.2) is only an approximation. With this *electrostatic approximation*, the time development of $f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t)$ is given by the solution of

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla\right) f_{\alpha 1} = \frac{q_\alpha}{m_\alpha} \nabla \phi_1 \cdot \nabla_{\mathbf{v}} f_{\alpha 0} \tag{8.3.3}$$

$$\nabla^2 \phi_1 = -4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha 1}\, d\mathbf{v} \tag{8.3.4}$$

Problem 8.3.1 Write down some explicit perturbation functions $f_{\alpha 1}$ which produce no $\mathbf{B}_1$ and are small compared with $f_{\alpha 0}$ everywhere. ////

A convenient way to solve the simultaneous differential equations given by (8.3.3) and (8.3.4) is to use the method of integral transforms. For an initial-value problem, the equations are simplified by taking their Fourier transform with respect to the spatial variables and their Laplace transform in time. This reduces these differential equations to algebraic equations that can be solved for the desired transform variable. The problem is then reduced to that of inverting the Fourier and Laplace transforms of the dependent variables.

Fourier and Laplace transforms are defined by

$$f_{\alpha \mathbf{k}}(\mathbf{v}, t) = f_{\alpha \mathbf{k}} = \frac{1}{(2\pi)^3} \int f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t) \exp(-i\mathbf{k} \cdot \mathbf{x})\, d\mathbf{x} \tag{8.3.5}$$

$$\phi_{\mathbf{k}}(t) = \frac{1}{(2\pi)^3} \int \phi_1(\mathbf{x}, t) \exp(-i\mathbf{k} \cdot \mathbf{x})\, d\mathbf{x} \tag{8.3.6}$$

$$\tilde{f}_{\alpha \mathbf{k}}(\mathbf{v}, p) = \tilde{f}_{\alpha \mathbf{k}} = \int_0^\infty f_{\alpha \mathbf{k}}(\mathbf{v}, t) \exp(-pt)\, dt \qquad \mathrm{Re}\,(p) \geq p_0 \tag{8.3.7}$$

It is important to note that the Laplace-transformed function $\tilde{f}(p)$ is defined only for $\mathrm{Re}\,(p) \geq p_0$, as indicated in (8.3.7). The constant p_0 is chosen large enough so that the integral $\int_0^\infty f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t) \exp(-pt)\, dt$ converges. Once the function $\tilde{f}_{\alpha \mathbf{k}}$ is known, the problem is solved, provided the transforms can be inverted to give $f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t)$ and $\phi_1(\mathbf{x}, t)$. Thus

$$\begin{aligned} f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t) &= \int \exp(i\mathbf{k} \cdot \mathbf{x})\, d\mathbf{k} \int_{p_0 - i\infty}^{p_0 + i\infty} \exp(pt)\, \tilde{f}_{\alpha \mathbf{k}}(\mathbf{v}, p) \frac{dp}{2\pi i} \\ \phi_1(\mathbf{x}, t) &= \int \exp(i\mathbf{k} \cdot \mathbf{x})\, d\mathbf{k} \int_{p_0 - i\infty}^{p_0 + i\infty} \exp(pt)\, \tilde{\phi}_{\mathbf{k}}(p) \frac{dp}{2\pi i} \end{aligned} \tag{8.3.8}$$

Since $f_{\alpha 0}$ is not a function of $\mathbf{x}$ or t, taking the Fourier and Laplace transforms of (8.3.3) and (8.3.4) gives

$$\begin{aligned} (p + i\mathbf{k} \cdot \mathbf{v}) \tilde{f}_{\alpha \mathbf{k}} &= f_{\alpha \mathbf{k}}(\mathbf{k}, \mathbf{v}, t = 0) + \frac{q_\alpha}{m_\alpha} (i\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}) \tilde{\phi}_{\mathbf{k}} \\ k^2 \tilde{\phi}_{\mathbf{k}} &= 4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int \tilde{f}_{\alpha \mathbf{k}}\, d\mathbf{v} \end{aligned} \tag{8.3.9}$$

Eliminating $\tilde{f}_{\alpha \mathbf{k}}$ between these equations gives an expression for the perturbed potential $\phi_{\mathbf{k}}$; that is,

$$k^2 \tilde{\phi}_{\mathbf{k}} = \frac{4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int \dfrac{f_{\alpha \mathbf{k}}(t = 0)}{p + i\mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v}}{1 + 4\pi \sum_\alpha \dfrac{\bar{n}_\alpha q_\alpha^2}{m_\alpha} \dfrac{1}{k^2} \int \dfrac{\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}}{ip - \mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v}} \qquad \mathrm{Re}\,(p) \geq p_0 \tag{8.3.10}$$

Although it is necessary to take the inverse Laplace and Fourier transforms of (8.3.10) to obtain the space and time dependence of the perturbation potential for a given equilibrium $f_{\alpha 0}$, this equation for the transform potential yields useful information as it is. The denominator of this equation is the dielectric function of a field-free plasma for electrostatic waves of frequency $\omega = ip$ and wave number $\mathbf{k}$; that is,

$$D(\mathbf{k}, \omega) = 1 + \sum_\alpha \frac{4\pi \bar{n}_\alpha q_\alpha^2}{m_\alpha k^2} \int \frac{\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}}{\omega - \mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v} \tag{8.3.11}$$

Problem 8.3.2 Justify the identification of $D(\mathbf{k}, ip)$ as the dielectric function. ////

The velocity integrals in equations such as (8.3.10) and (8.3.11) can be simplified by choosing a coordinate system in which the wave number $\mathbf{k}$ lies along one of the coordinate axes and by defining $F(u)$ as the integral of $f(\mathbf{v})$ over the other two coordinates; i.e.,

$$F_{\alpha 0}(u) \equiv \int f_{\alpha 0}(\mathbf{v})\, \delta\left(u - \frac{\mathbf{k} \cdot \mathbf{v}}{|k|}\right) d\mathbf{v} \tag{8.3.12}$$

and

$$\tilde{F}_{\alpha \mathbf{k}} \equiv \int \tilde{f}_{\alpha \mathbf{k}}\, \delta\left(u - \frac{\mathbf{k} \cdot \mathbf{v}}{|k|}\right) d\mathbf{v}$$

so that the equations (8.3.9) to (8.3.11) for the transforms of the perturbed distribution function and potential become

$$\tilde{f}_{\alpha \mathbf{k}}(\mathbf{v}, p) = \frac{1}{p + i\mathbf{k} \cdot \mathbf{v}} \left[f_{\alpha \mathbf{k}}(\mathbf{v}, t = 0) + \frac{q_\alpha}{m_\alpha} (i\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}) \tilde{\phi}_{\mathbf{k}} \right] \tag{8.3.13}$$

$$\tilde{\phi}_{\mathbf{k}}(p) = \frac{-i}{|k|^3 D(\mathbf{k}, ip)} \sum_\alpha 4\pi \bar{n}_\alpha q_\alpha \int \frac{F_{\alpha\mathbf{k}}(u, t=0)}{u - ip/|k|}\, du \qquad (8.3.14)$$

$$D(\mathbf{k}, ip) = 1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \int \frac{\partial F_{\alpha 0}(u)/\partial u}{u - ip/|k|}\, du \quad \mathrm{Re}\,(p) \geq p_0 \qquad (8.3.15)$$

Problem 8.3.3 Show that

$$\int \frac{\mathbf{E}_\mathbf{k} \cdot \nabla_\mathbf{v} f_{\alpha 0}(v^2)}{p + i\mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v} = \frac{\mathbf{k} \cdot \mathbf{E}_\mathbf{k}}{k^2} \int \frac{\mathbf{k} \cdot \nabla_\mathbf{v} f_{\alpha 0}}{p + i\mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v}$$

for any $f_{\alpha 0}(v^2 = v_x^2 + v_y^2 + v_z^2)$. This result shows that a *general* perturbation of an isotropic plasma can be decomposed into an independent electrostatic response $E_{es} = (\mathbf{k} \cdot \mathbf{E})/|k|$ described exactly by the results of the electrostatic approximation (8.3.11), in addition to an electromagnetic response $\mathbf{E}_{em} = (\mathbf{k} \times \mathbf{E})/|k|$ described by the results of Sec. 8.9 below. ////

To find the time development of ϕ_1 and $f_{\alpha 1}$, it is necessary to invert their transforms. The time dependence of the Fourier transform of the potential is obtained by taking the inverse Laplace transform of $\tilde{\phi}_\mathbf{k}$ [Eq. (8.3.14)],

$$k^2 \phi_\mathbf{k}(t) = \int_{p_0 - i\infty}^{p_0 + i\infty} \frac{4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int \dfrac{F_{\alpha\mathbf{k}}(u, t=0)}{p + i|k|u}\, du}{D(\mathbf{k}, ip)}\, e^{pt}\, \frac{dp}{2\pi i} \qquad (8.3.16)$$

where the path of the integral is as shown in Fig. 8.3.1.

Because p_0 was chosen to make $\int_0^\infty \phi_\mathbf{k}(t) e^{-pt}\, dt$ converge for $p > p_0$, the contour in Fig. 8.3.1 must be chosen to lie to the right of any poles of $\tilde{\phi}_\mathbf{k}(p)$.

Except for a few simple equilibrium distribution functions $F_{\alpha 0}(u)$ and initial disturbances $F_{\alpha\mathbf{k}}(u, t = 0)$, the inversion integral (8.3.16) cannot be carried out analytically to find solutions for $\phi_\mathbf{k}(t)$. However, it is possible to obtain an approximation to the *long-time* solution of (8.3.16) for a wide class of equilibrium distributions; these solutions give the behavior of the system long after the time of the initial perturbation. This asymptotic ($t \to \infty$) behavior is determined by the normal modes of oscillation of the plasma, rather than by transient response of the plasma to specific details of the initial perturbation.

Problem 8.3.4 Before studying the time-asymptotic solutions, find the exact solution $\phi_\mathbf{k}(t)$ for some of the following distributions and initial perturbations:

(*a*) $f_{\alpha 0} = \delta(v_x)\, \delta(v_y)\, \delta(v_z)$, $f_{e1} = f_{e0} \sin kx$, $f_{i1} = 0$

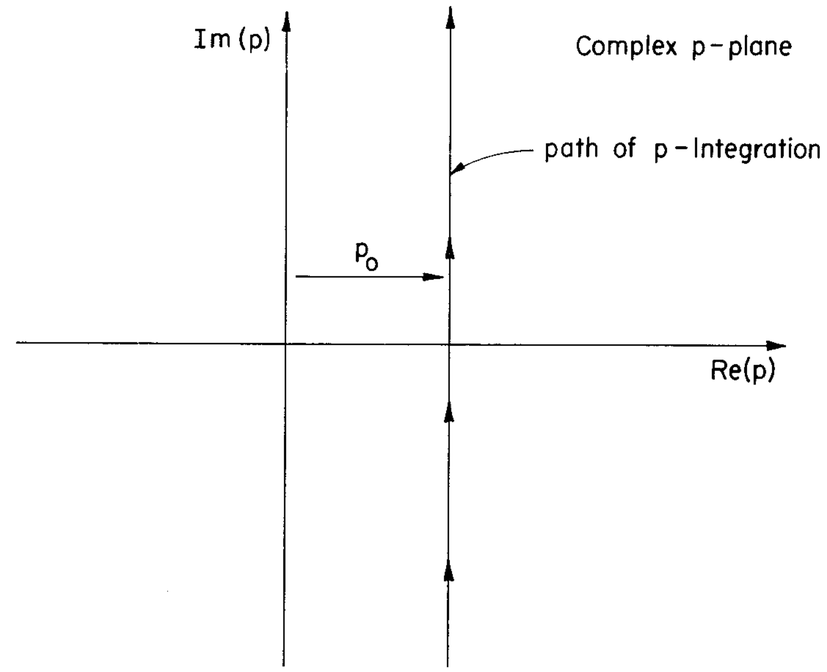


FIGURE 8.3.1
Path for inverting the Laplace transform $\tilde{\phi}(p)$. The constant p_0 is chosen so that the poles of $\tilde{\phi}(p)$ lie to the left of the integration path.

(*b*) $f_{\alpha 0} = \dfrac{A}{v_x^2 + v_{\alpha 0}^2}\, \delta(v_y)\, \delta(v_z)$, $f_{e1} = f_{e0} \sin kx$, $f_{i1} = 0$

(*c*) $f_{e0} = \delta(v_x - v_0)\, \delta(v_y)\, \delta(v_z)$, $f_{e1} = f_{e0} \sin kx$,
$f_{i0} = \delta(v_x)\, \delta(v_y)\, \delta(v_z)$, $f_{i1} = 0$

(*d*) What basic difference is there in the solutions to these three cases? ////

8.4 TIME-ASYMPTOTIC SOLUTIONS FOR $\phi_\mathbf{k}(t)$

The evaluation of contour integrals can often be simplified by deforming the contour of integration. This useful technique can be used to evaluate (8.3.16) over the contour shown in Fig. 8.3.1. Before the contour shown in Fig. 8.3.1 can be deformed, it is necessary to note that $\tilde{\phi}_\mathbf{k}(p)$ is uniquely defined, by (8.3.14), only for $\mathrm{Re}\,(p) \geq p_0$. Therefore, for *convenience* in performing the integration (8.3.16), it is useful to define a function $\Phi_\mathbf{k}(p)$ that is identical with $\tilde{\phi}_\mathbf{k}(p)$ for $\mathrm{Re}\,(p) \geq p_0$. For $\mathrm{Re}\,(p) < p_0$, $\Phi_\mathbf{k}$ is defined as the analytic continuation of $\tilde{\phi}_\mathbf{k}(p)$. [Note that only $\phi_\mathbf{k}(p)$ for $\mathrm{Re}\,(p) \geq p_0$ is required to specify $\phi(t)$ for $t > 0$; the analytic continuation to $\mathrm{Re}\,(p) < p_0$ is only for convenience in doing the integrations.]

Examining (8.3.14), it is clear that analytic continuation of $\tilde{\phi}_{\mathbf{k}}(p)$ requires a statement about the analytic properties of $F_{\alpha 0}$ and $\tilde{F}_{\alpha \mathbf{k}}$, as well as analytic continuation of velocity integrals of the form

$$h(p) = \int_{-\infty}^{\infty} \frac{f(u)}{u - ip/|k|}\, du \qquad \mathrm{Re}\,(p) > p_0 \qquad (8.4.1)$$

to values of $\mathrm{Re}\,(p) < p_0$. It is most convenient to limit the discussion to functions of $f(u)$ that are analytic at all points $|u| < \infty$. The maxwellian distribution, for example, satisfies this restriction. For $\mathrm{Re}\,(p) > 0$, the integration of (8.4.1) over u, along the real u axis, passes under the pole at $u = ip/|k|$, as shown in Fig. 8.4.1*a*. The analytic continuation of $h(p)$ requires that the path of $\int_{-\infty}^{\infty} du$ pass under the pole even when $\mathrm{Re}\,(p) \leq 0$, as shown in Fig. 8.4.1*b* and *c*. The contours of Fig. 8.4.1 are called *Landau contours*, after Landau,[1] who, in 1946, solved the Vlasov equation as an initial-value problem, developing the technique presented here.

The integral (8.4.1) along the contour shown in Fig. 8.4.1*c* can be broken up into a contribution from the real u axis and a contribution from the pole, so that the function $h(p)$ defined by

$$h(p) = \begin{cases} \displaystyle\int_{-\infty}^{\infty} \frac{f(u)}{u - ip/|k|}\, du & \mathrm{Re}\,(p) \geq 0 \qquad (8.4.2) \\ \displaystyle\int_{-\infty}^{\infty} \frac{f(u)}{u - ip/|k|}\, du + 2\pi i f\left(\frac{ip}{|k|}\right) & \mathrm{Re}\,(p) \leq 0 \qquad (8.4.3) \end{cases}$$

is analytic[2] in both the lower and upper half planes $\mathrm{Re}\,(p) \gtrless 0$. It is easy to show that both (8.4.2) and (8.4.3) reduce in the limit $\mathrm{Re}\,(p) \to 0$ to

$$h(p) = \dot{\oint}_{-\infty}^{\infty} \frac{f(u)}{u - ip/|k|}\, du + \pi i f\left(\frac{ip}{|k|}\right) \qquad \mathrm{Re}\,(p) = 0 \qquad (8.4.4)$$

where

$$\dot{\oint}_{-\infty}^{\infty} \equiv \lim_{\varepsilon \to 0^+} \left(\int_{-\infty}^{ip/|k| - \varepsilon} + \int_{ip/|k| + \varepsilon}^{\infty} \right)$$

is the Cauchy principal value. Thus $h(p)$ defined by (8.4.2) to (8.4.4) is also continuous across the axis $\mathrm{Re}\,(p) = 0$.

[1] L. D. Landau, *J. Phys. U.S.S.R.*, **10**:25 (1946).

[2] This is subject to some weak restrictions on the integrability of $f(u)$. See N. I. Muskhelishvili, "Singular Integral Equations," P. Noordhoff, Ltd., Groningen, Netherlands, 1953.

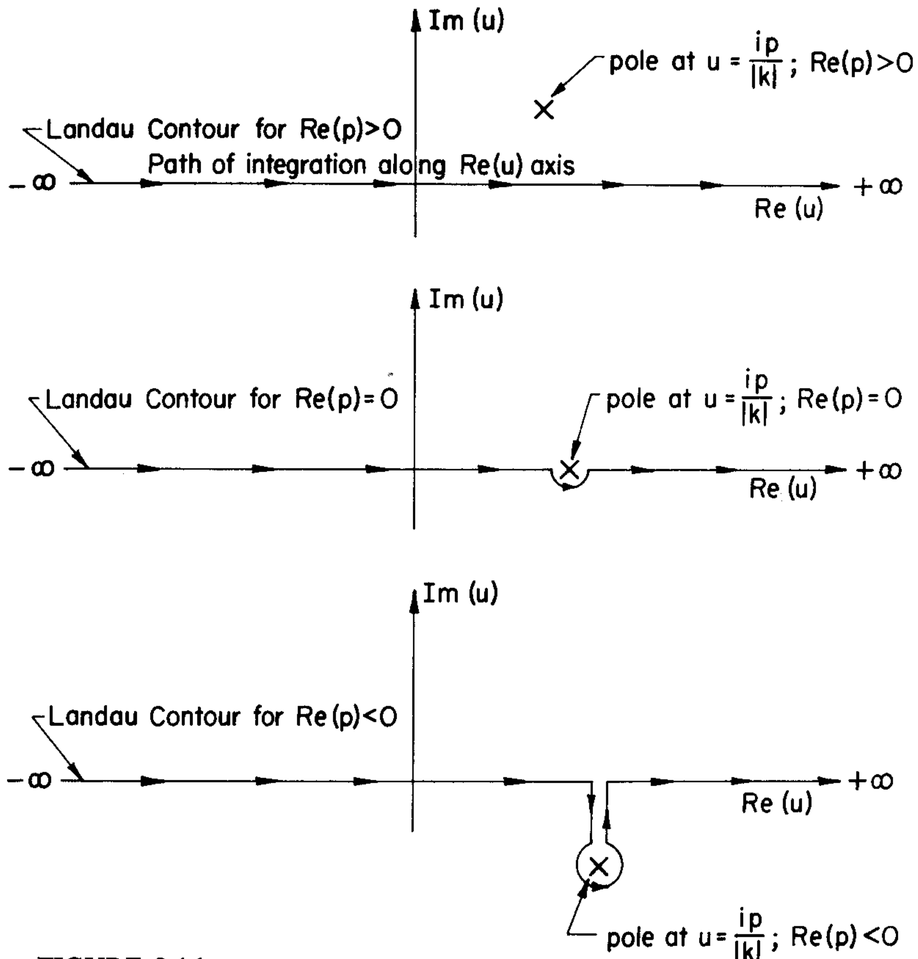


FIGURE 8.4.1
Landau contour for evaluating

$$\int_{-\infty}^{\infty} \frac{F(u)\, du}{(u - ip/|k|)}$$

for three different values of $\mathrm{Re}(p)$. This set of contours guarantees that the u integral is a continuous function of p.

Thus

$$\tilde{\phi}_{\mathbf{k}}(p) = \frac{\dfrac{-i}{|k|^3} \displaystyle\sum_{\alpha} 4\pi \bar{n}_{\alpha} q_{\alpha} \int \frac{F_{\alpha \mathbf{k}}(u, t = 0)}{u - ip/|k|}\, du}{1 - \displaystyle\sum_{\alpha} \frac{\omega_{p\alpha}^2}{k^2} \int \frac{\partial F_{\alpha 0}/\partial u}{u - ip/|k|}\, du} \qquad (8.4.5)$$

with the velocity integrals defined by the Landau contour (Fig. 8.4.1), or the Landau prescription [Eq. (8.4.2) to (8.4.4)], is the analytic continuation of $\tilde{\phi}_{\mathbf{k}}(p)$. The only singularities of this generalized $\tilde{\phi}_{\mathbf{k}}(p)$ are poles that result when the denominator vanishes, and the contour Fig. 8.3.1 can be deformed to any other path, as long as the poles $D(\mathbf{k}, ip) = 0$ are not crossed.

Problem 8.4.1 Using the Plemelj formulas

$$\lim_{\varepsilon\to 0^+}\int_{-\infty}^{\infty}\frac{f(x)\,dx}{x-y\pm i\varepsilon}=\oint_{-\infty}^{\infty}\frac{f(x)\,dx}{x-y}\mp\pi i\int_{-\infty}^{\infty}f(x)\,\delta(x-y)\,dx$$

show that (8.4.2) and (8.4.3) both reduce to (8.4.4) in the limit $\mathrm{Re}\,(p)=0$. ////

Problem 8.4.2 Demonstrate that $h(p)$ defined by (8.4.2) to (8.4.4) is the unique analytic continuation of (8.4.2) by showing that g defined by

$$g(p)=\begin{cases}\displaystyle\int_{-\infty}^{\infty}\frac{f(u)\,du}{u-ip/|k|} & \mathrm{Re}\,(p)\gtrless 0\\[2ex] \displaystyle\oint_{-\infty}^{\infty}\frac{f(u)\,du}{u-ip/|k|} & \mathrm{Re}\,(p)=0\end{cases}$$

(where $\oint$ is the Cauchy principal value) is discontinuous across $\mathrm{Re}\,(p)=0$. Show also that j defined by

$$j=\begin{cases}\displaystyle\int_{-\infty}^{\infty}\frac{f(u)}{u-ip/|k|} & \mathrm{Re}\,(p)>0\\[2ex] \displaystyle\int_{C'}\frac{f(u)\,du}{u-ip/|k|} & \mathrm{Re}\,(p)\le 0\end{cases}$$

is discontinuous across $\mathrm{Re}\,(p)=0$, where the contour C' passes *above* the pole. This shows that neither $g(p)$ nor $j(p)$ are the analytic continuation of $h(p)$. ////

A possible set of poles of $\tilde{\phi}_{\mathbf{k}}(p)$ corresponding to zeros of $D(\mathbf{k}, ip)$ is shown in Fig. 8.4.2*a*, along with the contour for inverting the Laplace transform. In Fig. 8.4.2*b* an equivalent contour is indicated. With this change of contour,

$$\phi_{\mathbf{k}}(t)=\sum_j R_j\,e^{p_j(k)t}+\int_{-i\infty+p_0}^{-i\infty-\alpha}\tilde{\phi}_{\mathbf{k}}(p)e^{pt}\frac{dp}{2\pi i}+\int_{-\alpha-i\infty}^{-\alpha+i\infty}\tilde{\phi}_{\mathbf{k}}(p)e^{pt}\frac{dp}{2\pi i}+\int_{i\infty-\alpha}^{i\infty+p_0}\tilde{\phi}_{\mathbf{k}}(p)e^{pt}\frac{dp}{2\pi i}\qquad(8.4.6)$$

where

$$D[\mathbf{k}, ip_j(k)]=0$$

locates the poles p_j of $\tilde{\phi}_{\mathbf{k}}(p)$, and R_j is the residue

$$R_j=\lim_{p\to p_j}(p-p_j)\tilde{\phi}_{\mathbf{k}}(p)$$

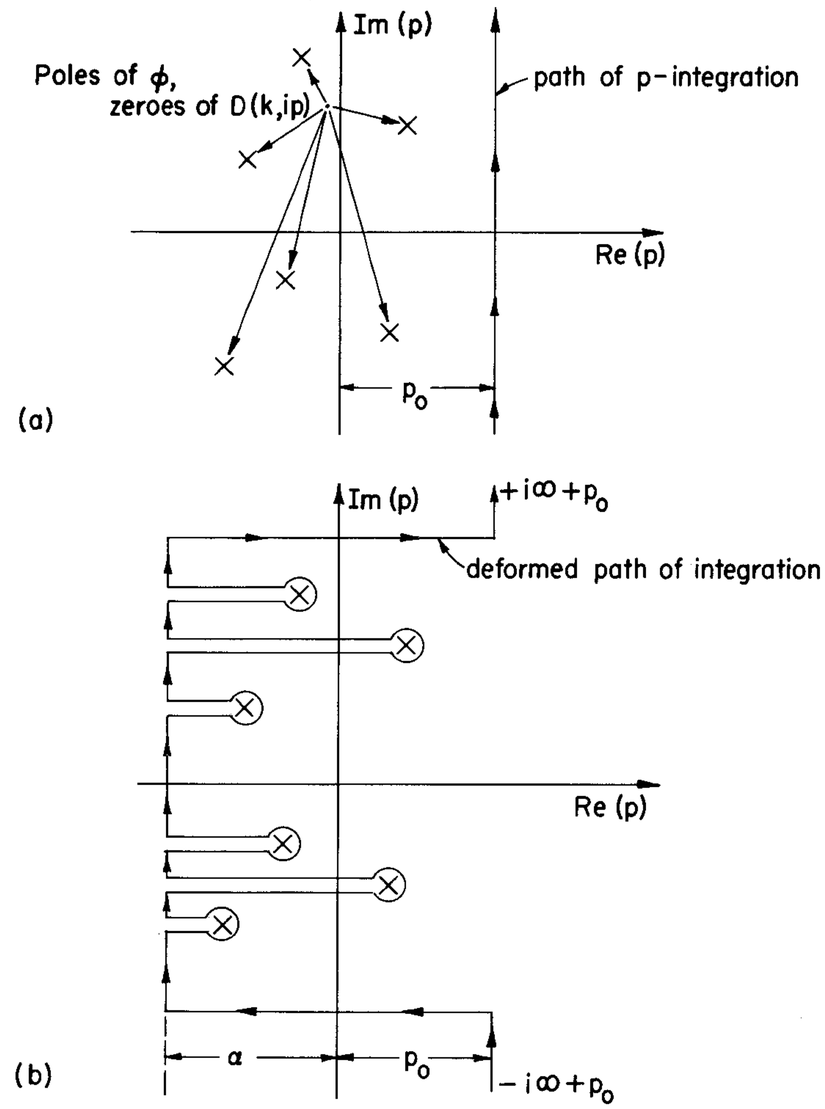


FIGURE 8.4.2
(*a*) Contour for calculating

$$\phi(t)=\frac{1}{2\pi i}\int_{-i\infty+p_0}^{i\infty+p_0}\tilde{\phi}(p)e^{pt}\,dp$$

(*b*) Equivalent contour for calculating $\phi(t)$. The prime contribution to the integral comes, for $t\to\infty$, from the poles of $\tilde{\phi}(p)$, which are the zeros of the dispersion relation $D(k,ip)$.

The second and fourth contributions to (8.4.6) are small if $\tilde{\phi}_{\mathbf{k}}(p)$ vanishes rapidly as $|p|\to\infty$, while the third term

$$\int_{-i\infty-\alpha}^{i\infty-\alpha}\tilde{\phi}_{\mathbf{k}}(p)e^{pt}\,dp$$

becomes exponentially small compared with the contributions from the poles as $t \to \infty$. If all the poles $p_j(\mathbf{k})$ lie to the left of the axis [i.e., if $\mathrm{Re}(p_j) < 0$], then all contributions to $\phi_{\mathbf{k}}(t)$ are damped at $t = 0$. If some of the poles lie to the right [$\mathrm{Re}\,(p_j) > 0$], they give rise to growing electric fields (instability). In either case, the time-asymptotic solution to the linearized Vlasov equation in the electrostatic approximation is

$$\phi_{\mathbf{k}}(t \to \infty) = \sum_j R_j\, e^{p_j(\mathbf{k})t}$$

Writing this result in terms of *frequency* by defining

$$\omega = ip$$

gives the customary form for the potential a long time after an initial perturbation,

$$\phi_{\mathbf{k}}(t) = \sum_j R_j\, e^{-i\omega_j t} \qquad (8.4.7)$$

where ω_j is in general complex, i.e.,

$$\omega_j = \omega_r + i\omega_i$$

and satisfies

$$D(\mathbf{k}, \omega) = 1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \int_L \frac{\partial F_{\alpha 0}/\partial u}{u - \omega/|k|}\, du = 0 \qquad (8.4.8)$$

with D evaluated on the Landau contour. In many cases $\mathrm{Re}\,[\omega(k)] \gg \mathrm{Im}\,[\omega(k)]$, and the plasma response a long time after an initial disturbance consists of wavelike oscillations at a few well-defined frequencies. These are the normal modes of the plasma for which the dielectric vanishes. In general, these wavelike modes have a phase velocity ω/k and a group velocity $\partial\omega/\partial k$.

From the above treatment it is seen that the problem of determining the nontransient response of a plasma to a perturbation centers on locating the zeros of the plasma dielectric. The equation for the zeros of the dielectric

$$D[\mathbf{k}, \omega(k)] = 0$$

is called a *dispersion relation*. It gives the frequency ω of a plasma wave as a function of the wave number k, or vice versa. Note that such a dispersion relation exists only in the time-asymptotic limit.

8.5 SIMPLIFIED DERIVATION FOR ELECTROSTATIC WAVES IN A PLASMA

One possible method of solving the Vlasov-Maxwell equations

$$\frac{\partial f_{\alpha 1}}{\partial t} + \mathbf{v} \cdot \nabla f_{\alpha 1} = \frac{q_\alpha}{m_\alpha} \nabla\phi_1 \cdot \nabla_{\mathbf{v}} f_{\alpha 0} \qquad (8.5.1)$$

$$\nabla^2 \phi_1 = -4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha 1}\, d\mathbf{v} \qquad (8.5.2)$$

for electrostatic perturbations ($\mathbf{E}_1 = -\nabla\phi_1$) is to assume that the solution $f_{\alpha 1}$ has the form

$$\begin{aligned} f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t) &= f_{\alpha \mathbf{k}}(\mathbf{v}) \exp(i\mathbf{k} \cdot \mathbf{x}) \exp(-i\omega t) \\ \phi_1(\mathbf{x}, t) &= \phi_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{x}) \exp(-i\omega t) \end{aligned} \qquad (8.5.3)$$

Then, from (8.5.1),

$$f_{\alpha \mathbf{k}} = -\frac{(q_\alpha/m_\alpha)(\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0})}{\omega - \mathbf{k} \cdot \mathbf{v}}\, \phi_{\mathbf{k}} \qquad (8.5.4)$$

and inserting (8.5.4) into (8.5.2),

$$k^2 \phi_{\mathbf{k}} \left(1 + \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \int \frac{\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}}{\omega - \mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v}\right) = 0 \qquad (8.5.5)$$

The nontrivial solution of (8.5.5) requires that

$$1 + \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \int \frac{\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}}{\omega - \mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v} = 0 \qquad (8.5.6)$$

This dispersion relation gives $\omega(\mathbf{k})$ or $\mathbf{k}(\omega)$. The fluctuating potential is given by

$$\phi_1 = \phi_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t)$$

The dispersion relation (8.5.6) cannot be used without specifying the contour of the $\mathbf{v}$ integration, since for real $\omega(k)$ the denominator vanishes on the real $\mathbf{v}$ axis. This simplified derivation of the dispersion relation gives no indication of the proper choice of the contour. However, if the problem of interest is the evolution of the plasma after an initial perturbation, the solution (8.5.6) must agree with the correct solution of the initial-value problem (8.4.8). The two solutions are identical if, in (8.5.6),

$$\int \frac{\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}}{\omega - \mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v} \equiv \int_L \frac{\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}}{\omega - \mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v} \qquad (8.5.7)$$

where L is the Landau contour shown in Fig. 8.4.1, with ip replaced by ω.

Problem 8.5.1 Show that the Landau prescription (8.5.7) is equivalent to including in the calculation the effect of weak collisions. Use as a collision model

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f - \frac{q}{m} \nabla \phi \cdot \nabla_{\mathbf{v}} f = \left.\frac{\partial f}{\partial t}\right|_{\text{collision}} \equiv -\nu(f - f_0)$$

taking the limit $\nu \to 0^+$. ////

Problem 8.5.2 The Landau prescription is correct for the solution of an initial-value problem. Find a problem for which an anti-Landau contour (*over* the pole) would give the correct result. ////

The normal modes of oscillation of a plasma are those wavelike disturbances that persist long after the transients associated with an initial disturbance have died out. These normal modes are characterized by the zeros of $D(\mathbf{k}, \omega) = 0$, for which the eigenfrequency ω is almost purely real, with only a small negative imaginary part. The reason these are the normal modes is that, if the imaginary part of ω for some root were large (and negative, assuming a stable plasma) the wave would be damped in a few oscillation periods and would not be classed as a normal mode. This assumption of a nearly real eigenfrequency simplifies the evaluation of the velocity integral (8.5.6), since it can then be expanded in a Taylor series about $\omega_i = 0$, by writing $\omega = \omega_r + i\omega_i$. Thus

$$\int_{-\infty}^{\infty} \frac{f(\mathbf{v})\, d\mathbf{v}}{\mathbf{k} \cdot \mathbf{v} - \omega} = \lim_{\varepsilon \to 0^+} \int_{-\infty}^{\infty} \frac{f(\mathbf{v})\, d\mathbf{v}}{\mathbf{k} \cdot \mathbf{v} - \omega_r - i\varepsilon} + i\omega_i \frac{\partial}{\partial \omega_r} \left[\lim_{\varepsilon \to 0^+} \int_{-\infty}^{\infty} \frac{f(\mathbf{v})\, d\mathbf{v}}{\mathbf{k} \cdot \mathbf{v} - \omega_r - i\varepsilon} \right] + \cdots \qquad (8.5.8)$$

Note that, for $\omega = \omega_r$, the integral is evaluated along the Landau contour by writing $\omega = \omega_r + i\varepsilon$, which moves the pole above the v axis, and then taking the limit $\varepsilon \to 0^+$.

The integrals in (8.5.8) can be simplified by reducing them to a one-dimensional form by defining

$$F_{\alpha 0}(u) = \int f_{\alpha 0}(\mathbf{v})\, \delta\left(u - \frac{\mathbf{k} \cdot \mathbf{v}}{|k|}\right) d\mathbf{v}$$

Using the relation[1]

$$\lim_{\varepsilon \to 0^+} \int_{-\infty}^{\infty} \frac{G(u)\, du}{u - \omega_r/|k| - i\varepsilon} = \oint \frac{G(u)\, du}{u - \omega_r/|k|} + \pi i G\left(u = \frac{\omega_r}{|k|}\right)$$

[1] J. Mathews and R. L. Walker, "Mathematical Methods of Physics," 2d ed., p. 481, W. A. Benjamin, New York, 1970.

then gives for the dispersion equation for weakly damped electrostatic waves in a field-free plasma

$$1 - \sum_{\alpha} \frac{\omega_{p\alpha}^2}{k^2} \left(1 + i\omega_i \frac{\partial}{\partial \omega_r}\right) \oint \left[\frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega_r/|k|}\, du + \pi i \left[\frac{\partial F_{\alpha 0}(u)}{\partial u}\right]_{u = \omega_r/|k|} \right] = 0 \qquad (8.5.9)$$

The dielectric [Eqs. (8.5.5) and (8.5.9)] is correct only for a plasma in which there are no magnetic or electric fields in the equilibrium state $f_{\alpha 0}$. More complicated equilibrium states have other dielectric properties, and the procedure outlined here, appropriately extended, is sufficient to solve most of the linear plasma wave problems. That procedure is first to derive the plasma dielectric, then locate the zeros of the dielectric, and finally to solve the dispersion relations and identify these zeros with plasma waves. In the material that follows, the plasma dielectric function is derived for a variety of plasma equilibria, and the properties of plasma waves associated with these equilibria are investigated.

8.6 THE VLASOV THEORY OF LANGMUIR WAVES, ION-SOUND WAVES, AND LANDAU DAMPING ($\mathbf{E}_0 = \mathbf{B}_0 = 0$)

Two examples for which the solution of the dispersion relation (8.5.9) is easy and which illustrate the Vlasov theory of small-amplitude waves in plasmas are Langmuir oscillations and ion-sound waves in a field-free plasma. In Chap. 5 these waves were discussed using the fluid model of a plasma. The additional properties of these waves revealed by the use of the Vlasov theory is instructive and demonstrates how it provides a better insight into basic plasma properties.

8.6.1 High-frequency Electrostatic Waves; Langmuir Oscillations

Waves in a field-free plasma occur with wave number $\mathbf{k}$ and frequency ω, as given by the dispersion relation $D(\mathbf{k}, \omega) = 0$ [Eqs. (8.5.6) and (8.5.9)]. Approximate solutions for this dispersion relation can be found for plasma waves with phase velocities that lie in specific ranges. For instance, there is a solution of (8.5.6) for waves with phase velocity (ω/k) much greater than the thermal speed of the plasma particles, as shown in Fig. 8.6.1. With the assumption of $\omega/k \gg v_{\text{thermal}}$, the principal-value integral of (8.5.9) may be evaluated by an expansion in u.

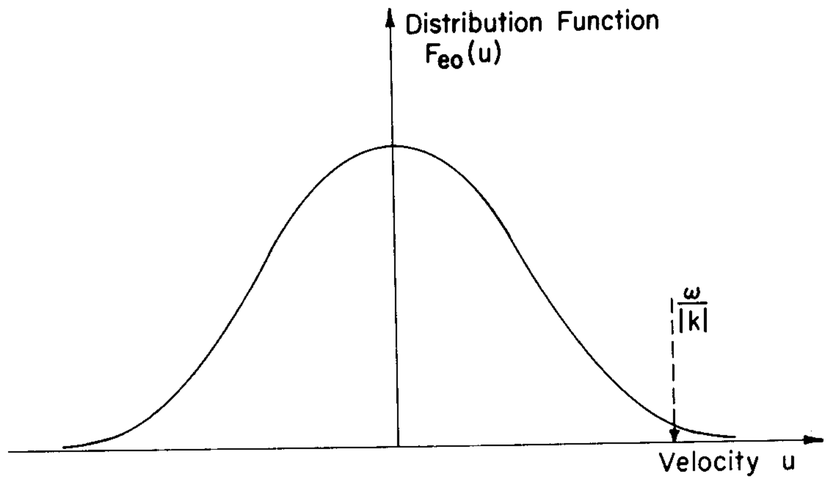


FIGURE 8.6.1
Location of the phase velocity $\omega/|k|$ of Langmuir oscillations relative to the electron thermal speed for the cone $\omega/k \gg v_{\text{thermal}}$.

$$-\oint \frac{\partial F_{\alpha 0}/\partial u}{u - \omega/|k|}\,du = \int \frac{\partial F_{\alpha 0}}{\partial u}\left[\frac{1}{\omega/|k|} + \frac{u}{(\omega/|k|)^2} + \frac{u^2}{(\omega/|k|)^3} + \cdots\right] du \qquad (8.6.1)$$

For a maxwellian distribution function

$$F_{e0} = \left(\frac{m}{2\pi\kappa T}\right)_e^{1/2} \exp\left(-\frac{m_e u^2}{2\kappa T_e}\right)$$

and neglecting the ion terms (which are small by m_e/m_i), the integral may be evaluated explicitly. The result is

$$\lim_{\varepsilon \to 0^+} \frac{\omega_{pe}^2}{k^2} \int_{-\infty}^{\infty} \frac{\partial F_{e0}/\partial u\, du}{u - \omega_r/|k| - i\varepsilon} = \frac{\omega_{pe}^2}{\omega_r^2} + 3\frac{\omega_{pe}^4}{\omega_r^4} k^2 \lambda_{De}^2 + \cdots + \frac{\pi i \omega_{pe}^2}{k^2} \frac{\partial}{\partial u} F_{e0}(u)\bigg|_{u=\omega_r/|k|} \qquad (8.6.2)$$

where ω_{pe} is the plasma frequency and

$$\lambda_{De}^2 \equiv \frac{\kappa T_e}{4\pi \bar{n}_e e^2} \qquad (8.6.3)$$

If the distribution $f_{\alpha 0}$ is not maxwellian, (8.6.2) and (8.6.3) are still appropriate, with the "temperature" defined as

$$\tfrac{1}{2}\kappa T_\alpha \equiv \int_{-\infty}^{\infty} \frac{m_\alpha (\mathbf{k} \cdot \mathbf{v})^2}{2k^2} f_{\alpha 0}\, d\mathbf{v}$$

The dispersion relation for waves with $\omega/k \gg v_{\text{thermal}}$ is obtained by inserting (8.6.2) into (8.6.1), noting that both $k\lambda_D$ and ω_i/ω_r are assumed small. The result is

$$\begin{aligned} \omega_r^2 &= \omega_{pe}^2(1 + 3k^2\lambda_{De}^2) \\ \omega_r &\approx \omega_{pe}(1 + \tfrac{3}{2}k^2\lambda_{De}^2) \\ \omega_i &= -\sqrt{\frac{\pi}{8}} \frac{\omega_{pe}}{|k^3\lambda_{De}^3|} \exp -\left(\frac{1}{2k^2\lambda_{De}^2} + \frac{3}{2}\right) \end{aligned} \qquad (8.6.4)$$

The real part of ω is identical with the frequency of plasma oscillations discussed in Chap. 4, where it was shown that if a group of electrons is suddenly displaced, they will oscillate about their equilibrium position with frequency ω_{pe}. Equation (8.6.4) is the solution of the linearized Vlasov equations for these same oscillations and includes thermal corrections without assuming an equation of state; it shows that the waves are *weakly damped.* The solution (8.6.4) satisfies the condition $\omega/k \gg (\kappa T_e/m_e)^{1/2}$, and the condition of weak damping is satisfied provided $k\lambda_D \ll 1$. Thus the approximations leading to (8.6.4) are justified, and an electrostatic oscillation at $\omega \approx \omega_{pe}$ is a normal mode of the uniform field-free plasma. The electric potential associated with the excitation of one **k** is

$$\phi_1(\mathbf{x}, t) = \phi_1 \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega_r t)]\exp(\omega_i t)$$

with ω_r and ω_i given by (8.6.4). These weakly damped normal modes in a plasma are generally called Langmuir oscillations; however, they are also called plasma waves, plasma oscillations, or electrostatic waves. Since these are the same plasma waves analyzed in Chap. 4, using macroscopic fluid theory, it is instructive to compare those results with the results derived in this chapter using Vlasov theory.

a Fluid character Fluid theory assumes from the outset that plasma is fluidlike, but provides no means to test the validity of the assumption. Vlasov theory proves that the Langmuir oscillations are fluidlike and that their frequency is controlled mainly by a bulk-plasma property i.e., the electron density.

b Propagating character For a "warm" plasma, fluid theory showed that Langmuir oscillations propagate; however, the analysis required an assumption for the equation of state (Sec. 4.4) to relate pressure to density. Vlasov theory also shows that Langmuir waves propagate, but no ad hoc assumption concerning an equation of state is required to obtain this result. Rather, it is a natural consequence of the kinetic theory. From Vlasov theory the group velocity of

Langmuir waves is

$$v_{\text{group}} = 3\omega_{pe}{}^2\lambda_{De}{}^2\frac{k}{\omega_r} \qquad (8.6.5)$$

c Equation of state Comparing the solution of the linearized Vlasov equation (8.6.4) with the fluid-theory result (Sec. 4.4) shows that the electrons behave as a one-dimensional adiabatic fluid in response to Langmuir waves. Nowhere in the derivation of the Vlasov equations or in their solution is an equation of state assumed; the Vlasov theory *reveals* the correct equation of state to be used in a fluid theory.

This point makes clear the difference between the information contained in the fluid model and the information contained in the Vlasov model. The fluid theory calculates the density and mean velocity of the electrons, as a function of space and time, but to do so, a relationship between temperature and density must be assumed.

In Vlasov theory, the distribution function f is calculated as a function of space and time without any assumption about the relationship between temperature and density. Thus the density, velocity, temperature, etc., which are just velocity moments of the distribution function, can be calculated directly as follows:

$$\begin{aligned} n_e &= \bar{n}_e \int f_{e1}\, d\mathbf{v} \\ n_e\mathbf{V}_e &= \bar{n}_e \int \mathbf{v} f_{e1}\, d\mathbf{v} \qquad (8.6.6) \\ \tfrac{3}{2}n_e \kappa T_e &= \bar{n}_e \int \tfrac{1}{2}m_e v^2 f_{e1}\, d\mathbf{v} \\ &\vdots \end{aligned}$$

where f_{e1} is given explicitly by (8.5.4), together with (8.6.4).

d Landau damping A property of Langmuir waves that is predicted by the Vlasov theory but which is completely outside the scope of fluid theory is the collisionless damping of electrostatic potentials associated with the plasma normal modes. This damping is characterized by the small negative imaginary part of the eigenfrequency given by (8.6.4). Thus the electrostatic potential damps according to

$$\phi_1(t) \propto e^{\omega_i t}$$

where

$$\omega_i \propto \frac{\partial F_{e0}}{\partial u} \qquad (8.6.7)$$

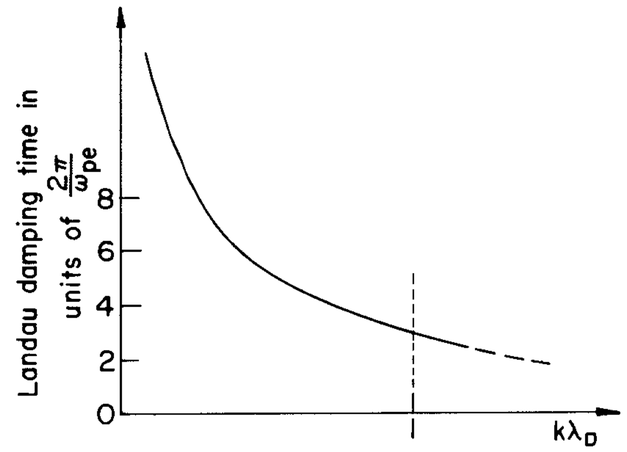


FIGURE 8.6.2
Landau damping time in units of $2\pi/\omega_{pe}$ as a function of wave number k. The theory is valid only for $k\lambda_D \ll 1$, as shown in the text.

This decrease of ϕ_1 with time was first calculated by Landau,[1] and is called Landau damping. It is a resonant phenomenon, proportional to the slope of the distribution, of particles moving with nearly the phase velocity of the wave (Fig. 8.6.1). Landau damping is often explained by noting that for distributions that decrease with increasing $|u|$, there are more particles moving slightly slower than the wave than there are particles moving slightly faster than the wave; if the slower particles are accelerated by the wave, this must reduce the energy in the wave, and the wave damps. From (8.6.4) it follows that the Landau damping time $1/\omega_i$ is much longer than an oscillation period $2\pi/\omega_{pe}$ for the long-wavelength modes ($k\lambda_{De} \to 0$). See Fig. 8.6.2. As the wavelength decreases and approaches the Debye length, the damping increases, and the plasma oscillation can no longer be considered a normal mode of the plasma.

Note that the entire analysis, starting with the approximate evaluation of $\phi_{\mathbf{k}}(t)$ from the inverse Laplace transform $\tilde{\phi}_{\mathbf{k}}(p)$, was based on the notion of two time scales, i.e., that there is an initial period of transient response to a perturbation, which is difficult to calculate, followed by a period during which only the plasma-wave normal modes of the system persist. It is the response of plasma during the later period that is calculated from (8.5.6). However, modes for which $\omega_i/|\omega|$ is not very small are strongly damped on the same scale as other transient responses. These strongly damped modes must also be considered "transient."

[1] L. D. Landau, *J. Phys. U.S.S.R.*, **10**:25 (1946).

It should further be noted that the calculation given here assumes a plasma of infinite size. For a plasma of finite size, the long-wavelength results (Chap. 4) will be affected. The results of this section should be regarded as valid only if

$$R_{\text{plasma}} \gg \frac{1}{k} \gg \lambda_D \qquad (8.6.8)$$

Problem 8.6.1 Evaluate ω_r, ω_i for a plasma of density 10^{12}, temperature 10^{5}°K, wavelength $\lambda = 40\lambda_{De}$. ////

Problem 8.6.2 Show the order of magnitude of terms neglected in (8.6.2). ////

Problem 8.6.3 Why is $k\lambda_{De} \ll 1$ required for the validity of (8.6.4)? ////

Problem 8.6.4 Where does the term $\exp(-3/2)$ in (8.6.4) come from? ////

Problem 8.6.5 Discuss the difference between (8.6.4) and (Sec. 4.4.) ////

Problem 8.6.6 Calculate p_1 and ρ_1 directly from f_1, and show that for Langmuir oscillations, p_1 and ρ_1 are related by $p\rho^{-\gamma} = \text{constant}$, $\gamma = 3$. *Hint:* $\Delta(p\rho^{-\gamma}) = 0$. ////

Problem 8.6.7 A small oscillator immersed in a plasma maintains plasma waves of a fixed frequency $\omega_0 \gtrsim \omega_p$ at a point $\mathbf{x}_0$ in a plasma at a constant small amplitude. What is the wave potential $\phi_1(\mathbf{x}, t)$ in the plasma far away from the oscillator? (*Hint:* This is a boundary-value problem, instead of an initial-value problem.) ////

Problem 8.6.8 Show that if a solution of the form

$$\phi_1(\mathbf{x}, t) = \phi \exp(i\mathbf{k} \cdot \mathbf{x} + i\omega t)$$

were assumed [instead of $\exp(-i\omega t)$], the contour of the velocity integral [Eq. (8.5.7)] would still be *under the pole*. ////

8.6.2 General Eigenvalues of $D(\mathbf{k}, \omega) = 0$ for Weak Damping or Growth

When the damping is weak, the expansion of the velocity integrals about $\omega = \omega_r$ is a useful technique to find the zeros of $D(\mathbf{k}, \omega)$. This technique can be

formulated more succinctly by expanding $D(\mathbf{k}, \omega)$ itself about $\omega = \omega_p$. Thus, if $\omega_i \ll \omega_r$,

$$D(\mathbf{k}, \omega) \approx D(\mathbf{k}, \omega_r) + i\omega_i \frac{\partial D(\mathbf{k}, \omega_r)}{\partial \omega_r} \qquad (8.6.9)$$

Note that $D(\mathbf{k}, \omega_r)$ includes a complex term,

$$\lim_{\varepsilon \to 0^+} \int \frac{\partial F_{\alpha 0}/\partial u}{u - \omega_r/|k| - i\varepsilon}\, du$$

and can be written in the form

$$D(\mathbf{k}, \omega_r) = D_r(\mathbf{k}, \omega_r) + iD_i(\mathbf{k}, \omega_r) \qquad (8.6.10)$$

where

$$D_i = -\pi \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \frac{\partial F_{\alpha 0}}{\partial u}\bigg|_{u=\omega_r/|k|} \qquad (8.6.11)$$

and

$$D_r = 1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \oint_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}/\partial u}{u - \omega_r/|k|}\, du \qquad (8.6.12)$$

Then, equating the real and imaginary parts of (8.6.9) gives

$$\omega_i = \frac{-D_i(\mathbf{k}, \omega_r)}{\partial D_r(\mathbf{k}, \omega_r)/\partial \omega_r} \qquad (8.6.13)$$

where ω_r [neglecting terms of order $(\omega_i/\omega_r)^2$] satisfies

$$D_r(\mathbf{k}, \omega_r) = 0 \qquad (8.6.14)$$

Problem 8.6.9 Show that (8.6.13) yields the same ω_i as (8.6.4). ////

8.6.3 Ion-acoustic Waves

If the ion contribution to $D(\mathbf{k}, \omega)$ were retained, the Langmuir oscillation would be shifted in frequency by only a small amount. That is, there is no separate *ion* Langmuir oscillation $\omega = \omega_{pi}$. However, if the electrons are warm ($T_e \gg T_i$), an electrostatic wave in which ions do play a major role is found at lower frequencies.

$$\left(\frac{\kappa T_i}{m_i}\right)^{1/2} < \frac{\omega}{k} < \left(\frac{\kappa T_e}{m_e}\right)^{1/2} \qquad (8.6.15)$$

The properties of these waves are also explored by solving for the zeros of $D(\mathbf{k}, \omega)$ for an appropriate plasma equilibrium that allows ion dynamics. The procedure again includes the evaluation of the velocity integrals assuming $\omega_i \ll \omega_r$, and then finding suitable approximations for the resulting expressions.

In the range (8.6.15), separate expressions must be used for the electron and ion velocity integrals. The appropriate approximations to the velocity integrals for the ions and electrons are

$$\oint_{-\infty}^{\infty} \frac{\partial F_{i0}/\partial u}{\omega_r/|k| - u}\, du \approx \int \frac{\partial F_{i0}}{\partial u}\left(\frac{|k|}{\omega} + \frac{k^2 u}{\omega^2} + \frac{|k^3|u^2}{\omega^3} + \frac{k^4 u^3}{\omega^4}\right) du$$

for ions (8.6.16)

and

$$\oint_{-\infty}^{\infty} \frac{\partial F_{e0}/\partial u}{\omega_r/|k| - u}\, du \approx -\int_{-\infty}^{\infty} 2\frac{\partial F_{e0}}{\partial u^2}\, du \qquad \text{for electrons} \qquad (8.6.17)$$

Using these forms to determine $D_r(\mathbf{k}, \omega_r)$ and (8.6.11) for $D_i(\mathbf{k}, \omega_r)$, it is straightforward to solve for ω_r, ω_i for any specific distribution $F_{\alpha 0}(u)$. For example, if the equilibrium distributions are maxwellian, the explicit solution of (8.6.16) in this velocity range is, treating the imaginary terms as small,

$$D_r = 1 - \frac{{\omega_{pi}}^2}{{\omega_r}^2} + \frac{1}{k^2 {\lambda_{De}}^2}$$

$$D_i = \pi \sum_\alpha \frac{{\omega_{p\alpha}}^2}{k^2}\left(\frac{m_\alpha}{2\pi\kappa T_\alpha}\right)^{1/2} \frac{m_\alpha}{\kappa T_\alpha}\frac{\omega_r}{|k|}\exp\left(-\frac{{\omega_r}^2 m_\alpha}{2k^2\kappa T_\alpha}\right)$$

$$\omega_r^2 = \frac{k^2 C_s^2}{1 + k^2 {\lambda_{De}}^2} \qquad (8.6.18)$$

$$\omega_i = -\frac{D_i(\mathbf{k}, \omega_r)}{\partial D_r/\partial \omega_r}$$

$$= -\frac{|\omega_r|\sqrt{\pi/8}}{(1 + k^2{\lambda_{De}}^2)^{3/2}}\left[\left(\frac{T_e}{T_i}\right)^{3/2}\exp\left(-\frac{T_e/T_i}{2(1 + k^2{\lambda_{De}}^2)}\right) + \sqrt{\frac{m_e}{m_i}}\right];$$

$$\left(\frac{\kappa T_i}{m_i}\right)^{1/2} < \frac{\omega}{k} < \left(\frac{\kappa T_e}{m_e}\right)^{1/2} \qquad (8.6.19)$$

where

$$C_s \equiv \left(\frac{\kappa T_e}{m_i}\right)^{1/2}$$

is called the ion-acoustic, or ion-sound, speed.

These waves were also derived from fluid theory in Chaps. 3 and 4; they are ion-sound waves. They are called sound waves because, for $k\lambda_D \ll 1$, all wavelengths propagate at the same speed, C_s, the ion-acoustic speed. This is in contrast to plasma oscillations, which have the same *frequency* for all wavelengths ($k\lambda_D \ll 1$). The kinetic treatment used here is an improvement on the fluid theory to the extent that:

1 No equation of state is assumed. By comparison with Sec. 4.4 it is clear that the assumption that electrons are isothermal, used in the fluid approach, is justified by the kinetic results.
2 The ion waves are damped, i.e.,

$$\phi_1 \propto e^{\omega_i t}$$

and unless $|\omega_i/\omega_r| \ll 1$, they should not be considered normal modes of the plasma.

Inspection of (8.6.19) reveals that $\omega_i \ll \omega_r$ is equivalent to the requirement that the electrons be hotter than the ions. Thus, for ion-sound waves to propagate in a plasma, it is necessary that $T_e \gg T_i$.

Problem 8.6.10 What size are the terms omitted in dispersion relation for ion waves [Eqs. (8.6.18) and (8.6.19)]? ////

Problem 8.6.11 Find ω_k for ion waves if $F_{\alpha 0}$ is a resonance distribution,

$$F_{\alpha 0} = \frac{A}{u^2 + \kappa T_\alpha/m_\alpha}$$ ////

It might seem from (8.6.19) that $T_e/T_i \ll 1$ would also give $|\omega_i/\omega_r| \ll 1$. This is not so; since $\omega = kC_s$, it is clear that the derivation, based on

$$\frac{\omega}{k} \gg \sqrt{\frac{\kappa T_i}{m_i}}$$

implies $T_e \gg T_i$ for ion waves

Equations (8.6.4) and (8.6.18) represent the only electrostatic oscillations favored by the field-free isotropic plasma.

The electron damping [the term $\sqrt{m_e/m_i}$ in (8.6.19)] is always small for ion waves. The reason is that, although many electrons have the same speed as the ion-sound wave, the *slope* of the electron distribution is small. Thus almost the same number of electrons travel faster than the wave as travel slower than the wave. Since the condition $T_e \gg T_i$ forces the speed of ion sound to exceed the ion thermal speed, few ions are available to damp these low-frequency waves. The effect of the distribution of plasma particles is very evident when comparing these two terms (Fig. 8.6.3).

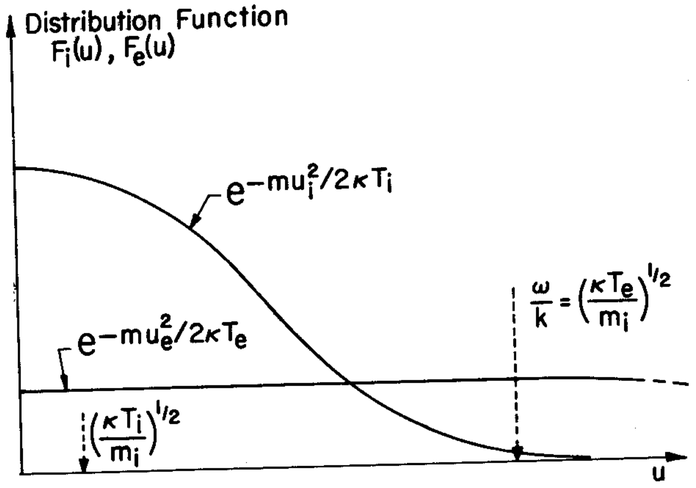


FIGURE 8.6.3
Location of ion-sound waves relative to the electron and ion thermal speeds. Slope of F_α, at the location $\omega/|k|$, determines the damping of these waves by species α. Only if $T_e \gg T_i$ will both ions and electrons contribute little damping.

Problem 8.6.12 Show that no solutions to (8.5.6) exist in the range

$$\frac{\omega}{k} \ll \sqrt{\frac{\kappa T_i}{m_i}} \ll \sqrt{\frac{\kappa T_e}{m_e}} \qquad ////$$

Problem 8.6.13 Describe ion waves in a Poisson plasma:

$$F_{\alpha 0} = A \exp\left(-\sqrt{\frac{m_\alpha u^2}{2\kappa T_\alpha}}\right) \qquad ////$$

8.7 PERTURBED DISTRIBUTION FUNCTION FOR PLASMA WAVES

The Landau damping of the electrostatic potential associated with Langmuir oscillations raises an interesting paradox: where does the energy $E_1^2/8\pi$ associated with the wave go when the wave damps? The idea that the energy is converted into particle temperature conflicts with the fact that the Vlasov plasma conserves entropy.[1] This paradox is resolved by calculating $f_1(t)$. From (8.3.9)

[1] Since the Vlasov equation neglects collisions, entropy must necessarily be conserved. This property of the Vlasov equation is shown directly in Chap. 7.

$$\tilde{f}_{\alpha\mathbf{k}}(p) = \frac{f_{\alpha\mathbf{k}}(t=0)}{p + i\mathbf{k}\cdot\mathbf{v}} + \frac{q_\alpha}{m_\alpha}\frac{i\tilde{\phi}_\mathbf{k}\,\mathbf{k}\cdot\nabla_\mathbf{v} f_{\alpha 0}(v)}{p + i\mathbf{k}\cdot\mathbf{v}} \qquad (8.7.1)$$

and

$$f_{\alpha\mathbf{k}}(t) = \frac{1}{2\pi i}\int_{p_0 - i\infty}^{p_0 + i\infty} \tilde{f}_{\alpha\mathbf{k}}(p) e^{pt}\, dp \qquad (8.7.2)$$

Again, this Laplace inversion can be carried out at late times, $t \to \infty$, by using the contour in Fig. 8.4.2 for the p integration, using (8.3.10) for $\tilde{\phi}_\mathbf{k}(p)$, and analytically continuing (8.7.1). Since $\tilde{f}_{\alpha\mathbf{k}}(p)$ has poles wherever $\tilde{\phi}_\mathbf{k}(p)$ has poles, these again lie at the points $p = -i\omega$, with

$$D(\mathbf{k}, \omega) = 0 \qquad (8.7.3)$$

The distribution $\tilde{f}_{\alpha\mathbf{k}}(p)$ has an additional pole at

$$p = -i\mathbf{k}\cdot\mathbf{v} \qquad (8.7.4)$$

Thus

$$f_{\alpha\mathbf{k}}(t) = \bar{f}_{\alpha B}\exp(-i\mathbf{k}\cdot\mathbf{v}t) + \sum_{\omega\mathbf{k}} \bar{f}_{\alpha\mathbf{k}}\exp(-i\omega_\mathbf{k} t) \qquad (8.7.5)$$

where $\bar{f}_{\alpha B}$ and $\bar{f}_{\alpha\mathbf{k}}$ are time-independent amplitudes determined from the residues of $\tilde{f}_{\alpha\mathbf{k}}(p)$ at its poles. The terms in the sum $\sum \exp(-i\omega_k t)$ give the response of the plasma distribution function to the plasma waves; this response Landau-damps with time in the same way that $\phi_1(t)$ damps. The additional term $\bar{f}_{\alpha B}\exp(-i\mathbf{k}\cdot\mathbf{v}t)$ is called the *ballistic*, or *free-streaming*, term. It arises because a particle, perturbed at $t = 0$, carries the memory of its perturbation with it for all time, wherever it goes. This memory is erased only by collisions, which are not included in the Vlasov approximation.

Thus a disturbance at $(\mathbf{x}, \mathbf{v}, t)$ arises from two sources. One is the plasma waves that have propagated in space and time through the plasma and reached the point $(\mathbf{x}, \mathbf{v})$ at time t. The other source arises from the individual particles that arrive at $(\mathbf{x}, \mathbf{v})$ at time t, bringing with them the memory of their original perturbation.

The ballistic term

$$f_{\alpha\mathbf{k}}(t) = \bar{f}_{\alpha B}\exp(-i\mathbf{k}\cdot\mathbf{v}t)$$

does not decrease with time ($\mathbf{k}$, $\mathbf{v}$ are both real), but becomes highly oscillatory in $\mathbf{v}$ space, as seen in Figs. 8.7.1 and 8.7.2 for an explicit choice of initial perturbation $f_1(\mathbf{x}, \mathbf{v}, t = 0)$.

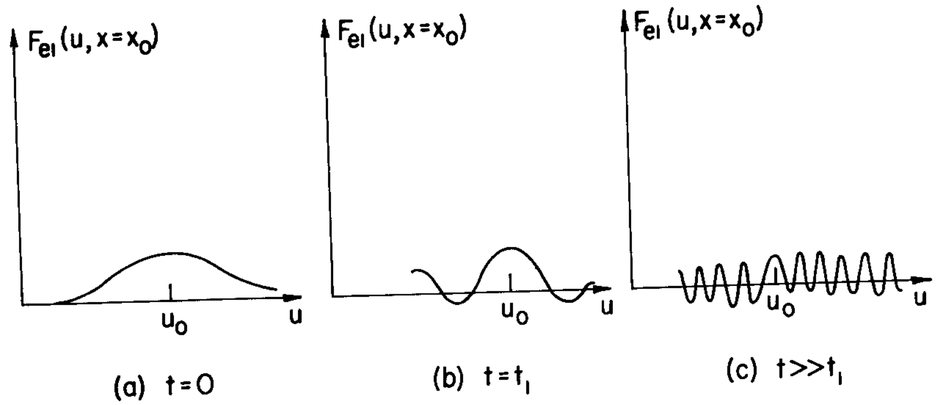


FIGURE 8.7.1
Time evolution of the perturbed distribution function as an electrostatic perturbation damps away in a stable plasma.

Problem 8.7.1 Calculate $f_{\mathbf{k}}(t)$ and $\phi_{\mathbf{k}}(t)$, including all transients for a distribution

$$F_{e0} = \frac{A}{u^2 + u_0^2}(1 + \varepsilon \sin kx) \qquad \varepsilon \ll 1,\ k\lambda_{De} \gg 1 \qquad ////$$

This resolves the discrepancy between the fact that the perturbed distribution persists while the perturbed field decays. The ballistic contribution to ϕ_1 is given by

$$k^2\phi_{\mathbf{k}} = 4\pi \sum_{\alpha} q_\alpha \bar{n}_\alpha \int \bar{f}_{\alpha B} \exp(-i\mathbf{k} \cdot \mathbf{v}t)\, d\mathbf{v} \qquad (8.7.6)$$

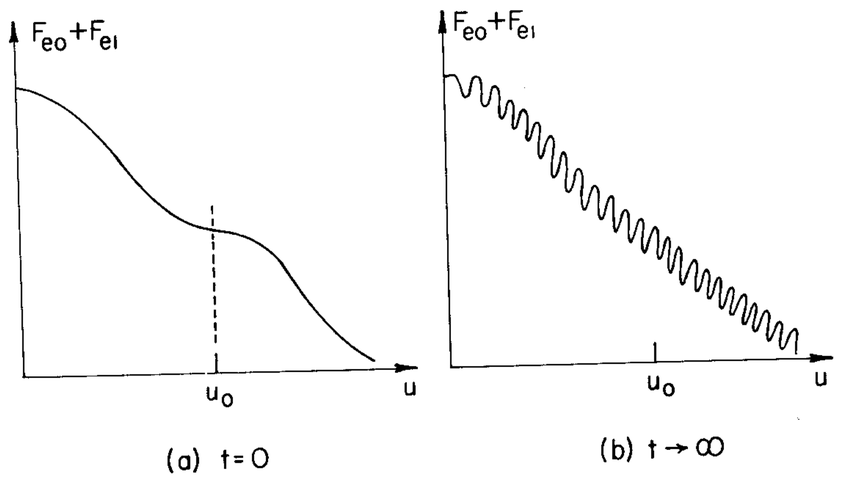


FIGURE 8.7.2
Time evolution of the total distribution function as an electrostatic perturbation damps away in a stable plasma.

As $t \to \infty$, the integral vanishes, because the integrand is highly oscillatory. This is called *phase mixing*. The ballistic response (8.7.6) is not included in the late-time solution [Eqs. (8.4.7) and (8.5.3)]. This emphasizes again the separation of time scales implied by the plasma wave calculation, which shows the plasma as a system of the oscillators with characteristic frequencies $\omega_{\mathbf{k}}$.

This dual decay of E has been exploited in experiments by Malmberg and Wharton,[1] Gould, O'Neil, and Malmberg,[2] and Ripin and Pechacek.[3] Details on the erasure of f_B by collisions have been given by Su and Oberman.[4]

Landau damping and plasma waves lie at the heart of the plasma kinetic approach, and it is important to understand the difference between the transient and the late-time solution. A detailed discussion of energy transfer and balance associated with the Landau damping process requires the development of the nonlinear extension of the theory. This is discussed in Chap. 10.

Problem 8.7.2 Find the form of $f(\mathbf{x}, t)$, including the ballistic term, as $t \to \infty$. ////

Problem 8.7.3 Calculate $f_1(\mathbf{x}, \mathbf{v}, t)$ for the spatial boundary-value problem described in Prob. 8.6.7. ////

8.8 THE DISPERSION RELATION FOR WAVES IN A GENERAL PLASMA EQUILIBRIUM

The principal result of the preceding section was that the perturbed fields for electrostatic waves in a field-free isotropic plasma develop asymptotically in time according to

$$i\mathbf{k} \cdot \mathbf{E}_{\mathbf{k}}(t) = \sum_{\omega_{\mathbf{k}}} A_{\mathbf{k}} \exp(-i\omega_{\mathbf{k}} t)$$

where the $\omega_{\mathbf{k}}$'s are the zeros of the dielectric function. These zeros are independent of the initial perturbations, because the dielectric $D(\mathbf{k}, \omega)$ is just a property of the equilibrium plasma state. The dielectric was obtained without direct recourse to transforms, by simply *assuming* in Eq. (8.5.1) that $f_1(\mathbf{x}, \mathbf{v}, t) = \bar{f}_{\mathbf{k}}(\mathbf{v})\exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t)$, with a similar assumption for the potential ϕ_1 and by using the Landau prescription for treating the singular velocity integrals.

[1] J. H. Malmberg and C. B. Wharton, *Phys. Rev. Letters*, **13**:184 (1964).
[2] R. W. Gould, T. M. O'Neil, and J. H. Malmberg, *Phys. Rev. Letters*, **19**:219 (1967).
[3] B. H. Ripin and R. E. Pechacek, *Phys. Rev. Letters*, **72**:1330 (1970).
[4] C. H. Su and C. Oberman, *Phys. Rev. Letters*, **20**:427 (1968).

The problem of a field-free plasma was especially simple, because the transformed equations were not differential equations for $f_1(\mathbf{v})$; that is, there were no

$$\left(\mathbf{E}_0 + \frac{\mathbf{v} \times \mathbf{B}_0}{c}\right) \cdot \nabla_{\mathbf{v}} \tilde{f}_{\mathbf{k}}(\mathbf{v})$$

terms in the transformed equations (8.3.9), and $\nabla_{\mathbf{v}} f_{\alpha 0}$ was assumed to be a known function of $\mathbf{v}$. If the equilibrium plasma state contains an electric field (the plasma may not be charge-neutral, or may include charged boundaries) or a magnetic field (the plasma may be confined by $\mathbf{B}_0$ produced from external coils), the linearized Vlasov equation is a differential equation in $\mathbf{v}$.

$$\left[\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla + \frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_0 + \frac{\mathbf{v} \times \mathbf{B}_0}{c}\right) \cdot \nabla_{\mathbf{v}}\right] f_{\alpha 1} = -\frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_1 + \frac{\mathbf{v} \times \mathbf{B}_1}{c}\right) \cdot \nabla_{\mathbf{v}} f_{\alpha 0} \tag{8.8.1}$$

Differential equations of this sort can be solved by the method of characteristics, also called "integrating over unperturbed orbits." Define

$$\frac{d\mathbf{x}'}{dt'} = \mathbf{v}', \qquad \frac{d\mathbf{v}'}{dt'} = \frac{q_\alpha}{m_\alpha}\left[\mathbf{E}_0(\mathbf{x}', t') + \frac{\mathbf{v}' \times \mathbf{B}_0(\mathbf{x}', t')}{c}\right] \tag{8.8.2}$$

with boundary conditions

$$\begin{aligned} \mathbf{x}'(t' = t) &= \mathbf{x} \\ \mathbf{v}'(t' = t) &= \mathbf{v} \end{aligned} \tag{8.8.3}$$

where $\mathbf{x}$, $\mathbf{v}$ are points in $(\mathbf{x}, \mathbf{v})$ phase space. From the linearized general Vlasov equation, given by

$$\left[\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla + \frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_0 + \frac{\mathbf{v} \times \mathbf{B}_0}{c}\right) \cdot \nabla_{\mathbf{v}}\right] f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t)$$

$$= -\frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_1 + \frac{\mathbf{v} \times \mathbf{B}_1}{c}\right) \cdot \nabla_{\mathbf{v}} f_{\alpha 0}$$

the perturbed distribution $f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t)$ is obtained by considering the function

$$f_{\alpha 1}[\mathbf{x}'(t'), \mathbf{v}'(t'), t']$$

which is a function of time t', and satisfies the equation

$$\frac{df_{\alpha 1}}{dt'} \equiv \frac{\partial f_{\alpha 1}(\mathbf{x}', \mathbf{v}', t')}{\partial t'} + \frac{d\mathbf{x}'}{dt'} \cdot \nabla_{\mathbf{x}'} f_{\alpha 1}(\mathbf{x}', \mathbf{v}', t) + \frac{d\mathbf{v}'}{dt'} \cdot \nabla_{\mathbf{v}'} f_{\alpha 1}(\mathbf{x}', \mathbf{v}', t)$$

$$= -\frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_1(\mathbf{x}', t') + \frac{\mathbf{v}' \times \mathbf{B}_1(\mathbf{x}', t')}{c}\right) \cdot \nabla_{\mathbf{v}'} f_{\alpha 0}(\mathbf{x}', \mathbf{v}') \tag{8.8.4}$$

$\mathbf{E}_0$ and $\mathbf{B}_0$ are still the fields that determine the equilibrium plasma distribution $f_{\alpha 0}$. Because of the boundary condition (8.8.3), it follows that

$$f_{\alpha 1}(\mathbf{x}', \mathbf{v}', t') = f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t) \qquad \text{at time } t' = t$$

So the solution to (8.8.4) at $t' = t$ is also the solution to the Vlasov equation. But (8.8.4) can be solved by integrating directly, since the left-hand side is an exact differential. If (8.8.4) is integrated from $t' = -\infty$ to $t' = t$, it gives

$$f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t) = -\frac{q_\alpha}{m_\alpha}\int_{-\infty}^{t} dt'\left[\mathbf{E}_1(\mathbf{x}', t') + \frac{\mathbf{v}' \times \mathbf{B}_1(\mathbf{x}', t')}{c}\right]$$

$$\cdot \nabla_{\mathbf{v}'} f_{\alpha 0}(\mathbf{x}', \mathbf{v}') + f_{\alpha 1}[\mathbf{x}'(-\infty), \mathbf{v}'(-\infty), t' = -\infty] \tag{8.8.5}$$

Problem 8.8.1 Show directly that (8.8.5) with $\mathbf{x}'$, $\mathbf{v}'$ defined by (8.8.2) and (8.8.3) satisfies the Vlasov equation. ////

Problem 8.8.2 Find $\mathbf{x}'$ and $\mathbf{v}'$ for the special case $\mathbf{E}_0 = \mathbf{B}_0 = 0$. Calculate $\tilde{f}_{\mathbf{k}}(p)$ from (8.8.5) and show that it agrees with the direct calculation $\tilde{f}_{\mathbf{k}}$ [Eq. (8.3.9)]. ////

Note that $\mathbf{x}'$, $\mathbf{v}'$, through the boundary conditions, are in general functions of $\mathbf{x}$ and $\mathbf{v}$ as well as the time $t - t'$. Viewed differently, this procedure has obtained the distribution $f_1(\mathbf{x}, \mathbf{v}, t)$ by integrating the Vlasov equation from $-\infty$ to t along a path in $(\mathbf{x}, \mathbf{v})$ phase space which coincides with the orbit of a charged particle in the equilibrium fields $\mathbf{E}_0$, $\mathbf{B}_0$. This $f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t)$ determines $n_{\alpha 1}(\mathbf{x}, t)$ and $\mathbf{V}_{\alpha 1}(\mathbf{x}, t)$, which are inserted in Maxwell's equations:

$$\nabla \times \mathbf{E}_1 = -\frac{1}{c}\frac{\partial \mathbf{B}_1}{\partial t}$$

$$\nabla \cdot \mathbf{E}_1 = 4\pi \sum_\alpha q_\alpha n_{\alpha 1}$$

$$\nabla \times \mathbf{B}_1 = \frac{1}{c}\frac{\partial \mathbf{E}_1}{\partial t} + \frac{4\pi}{c}\sum_\alpha q_\alpha (n_\alpha \mathbf{V}_\alpha)_1$$

This set of equations can be solved as an initial-value problem as in Sec. 8.4. However, the time-asymptotic results are more easily obtained as in Sec. 8.5 by assuming a solution of the form $\mathbf{E}_1(\mathbf{x}, t) = \bar{\mathbf{E}}_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t)$ and that $f_{\alpha 1}(\mathbf{x}', \mathbf{v}', t \to -\infty) = 0$. Then

$$f_{\alpha \mathbf{k}} = -\frac{q_\alpha}{m_\alpha}\int_{-\infty}^{0}\left(\bar{\mathbf{E}}_{\mathbf{k}} + \frac{\mathbf{v}' \times \bar{\mathbf{B}}_{\mathbf{k}}}{c}\right) \cdot \nabla_{\mathbf{v}'} f_{\alpha 0}(\mathbf{v}') \exp[i(\mathbf{k} \cdot \mathbf{X} - \omega\tau)]\, d\tau$$

$$\text{Im}\,(\omega) > 0 \tag{8.8.6}$$

where $\tau = (t' - t)$, $\mathbf{X} = (\mathbf{x}' - \mathbf{x})$, and $\bar{\mathbf{E}}_\mathbf{k}$, $\bar{\mathbf{B}}_\mathbf{k}$ are vector amplitudes that are constant in space and time. The solution $f_{\alpha\mathbf{k}}$ for $\mathrm{Im}\,(\omega) < 0$ is obtained by analytic continuation of (8.8.6).

Problem 8.8.3 Show the connection between the assumption $f_1(\mathbf{x}, \mathbf{v}, t) \xrightarrow{t \to -\infty} 0$ and the Landau prescription for evaluating the resonant v integrals. ////

When (8.8.6) is inserted into the transformed Maxwell's equations, the result is six linear homogeneous equations in the field amplitudes $\bar{\mathbf{E}}_\mathbf{k}$, $\bar{\mathbf{B}}_\mathbf{k}$.

If $\bar{\mathbf{B}}_\mathbf{k}$ is eliminated in terms of $\bar{\mathbf{E}}_\mathbf{k}$, then the resulting linear homogeneous equations in $\bar{\mathbf{E}}_\mathbf{k}$ have a solution provided the determinant of the coefficients of the $\mathbf{E}$'s vanishes. The equations have the form

$$\begin{aligned} D_{xx}E_x + D_{xy}E_y + D_{xz}E_z &= 0 \\ D_{yx}E_x + D_{yy}E_y + D_{yz}E_z &= 0 \\ D_{zx}E_x + D_{zy}E_y + D_{zz}E_z &= 0 \end{aligned}$$

which can be written $\mathsf{D} \cdot \mathbf{E} = 0$, where

$$\mathsf{D} = \begin{bmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{yx} & D_{yy} & D_{yz} \\ D_{zx} & D_{zy} & D_{zz} \end{bmatrix} \tag{8.8.7}$$

with $|\mathsf{D}| = 0$ as the solution. The quantity D is called the dispersion tensor; its coefficients $D_{ij}(\mathbf{k}, \omega)$ depend on the plasma state f_0, $\mathbf{E}_0$, $\mathbf{B}_0$.

8.9 THE VLASOV THEORY OF SMALL-AMPLITUDE WAVES IN A FIELD-FREE PLASMA EQUILIBRIUM—ELECTROSTATIC AND ELECTROMAGNETIC WAVES $[\mathbf{E}_0 = \mathbf{B}_0 = 0, f_0 = f_0(v^2)]$

The wave properties of a variety of plasma equilibria can be investigated systematically with the aid of the linearized Vlasov equations, using the techniques detailed in this chapter. The field-free plasma equilibrium introduced in Sec. 8.6 is the simplest to investigate. The electrostatic, or Langmuir, waves described in that section are one specific type. Now the full range of possible plasma waves is considered, using the more general method of integrating along unperturbed orbits. The particle orbits in this case of a field-free plasma are the free-streaming orbits

$$\begin{aligned} \mathbf{v}' &= \text{constant} = \mathbf{v} \\ \mathbf{x}' &= \mathbf{x} + \mathbf{v}(t' - t) \end{aligned} \tag{8.9.1}$$

From (8.8.6), the perturbed distribution can be written, with $t - t' \equiv \tau$, assuming $f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t) = \bar{f}_{\alpha\mathbf{k}} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega\tau)]$, $\mathbf{E}_1(\mathbf{x}, t) = \bar{\mathbf{E}}_\mathbf{k} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega\tau)]$, etc.,

$$\bar{f}_{\alpha\mathbf{k}} = -\frac{q_\alpha}{m_\alpha}\left(\bar{\mathbf{E}}_\mathbf{k} + \frac{\mathbf{v} \times \bar{\mathbf{B}}_\mathbf{k}}{c}\right) \cdot \nabla_\mathbf{v} f_{\alpha 0}(\mathbf{v}) \int_{-\infty}^{0} \exp[i(\mathbf{k} \cdot \mathbf{v}\tau - \omega\tau)\, d\tau \qquad \mathrm{Im}\,(\omega) > 0$$

where the amplitudes $\bar{\mathbf{E}}_\mathbf{k}$ and $\bar{\mathbf{B}}_\mathbf{k}$ and all functions of $\mathbf{v}$ have been removed from the integral, since $\mathbf{v}'$ does not depend on t' for the field-free orbits [Eq. (8.9.1)]. [For other equilibria this will not be the case, and $\nabla_\mathbf{v} f_{\alpha 0}(\mathbf{v}')$ will depend on t'.]

Integrating the perturbed distribution function gives

$$\bar{f}_{\alpha\mathbf{k}} = \frac{\dfrac{q_\alpha}{m_\alpha}\left(\bar{\mathbf{E}}_\mathbf{k} + \dfrac{\mathbf{v} \times \bar{\mathbf{B}}_\mathbf{k}}{c}\right) \cdot \nabla_\mathbf{v} f_{\alpha 0}}{i(\omega - \mathbf{k} \cdot \mathbf{v})} \tag{8.9.2}$$

The charge and current densities can be calculated from (8.9.2). For an isotropic distribution function $f_{\alpha 0}[= f_{\alpha 0}(v^2)]$ the quantity $\mathbf{v} \times \bar{\mathbf{B}}_\mathbf{k} \cdot \nabla_\mathbf{v} f_{\alpha 0}(v^2) = 0$. The charge density is given by

$$\rho_q = \sum_\alpha q_\alpha n_{\alpha 1} = \sum_\alpha \frac{\bar{n}_\alpha q_\alpha{}^2}{k^2 m_\alpha} \int \frac{(\mathbf{k} \cdot \bar{\mathbf{E}}_\mathbf{k})[\mathbf{k} \cdot \nabla_\mathbf{v} f_{\alpha 0}(v^2)]}{i(\omega - \mathbf{k} \cdot \mathbf{v})}\, d\mathbf{v} \tag{8.9.3}$$

and the current density is

$$\mathbf{J} = \sum_\alpha q_\alpha (n_\alpha \mathbf{V}_\alpha)_1 = \sum_\alpha \frac{\bar{n}_\alpha q_\alpha{}^2}{m_\alpha} \int \frac{\mathbf{v}\bar{\mathbf{E}}_\mathbf{k} \cdot \partial f_{\alpha 0}(v^2)/\partial \mathbf{v}}{i(\omega - \mathbf{k} \cdot \mathbf{v})}\, d\mathbf{v} \tag{8.9.4}$$

As shown in earlier sections, integration of the velocity distribution over components perpendicular to $\mathbf{k}$ simplifies the form of the equations. Equations (8.9.3) and (8.9.4) can be written in terms of a one-dimensional distribution defined by

$$F_{\alpha 0}(u) \equiv \int f_{\alpha 0}\, \delta\left(u - \frac{\mathbf{k} \cdot \mathbf{v}}{|k|}\right) d\mathbf{v}$$

For solutions of the form

$$\begin{aligned} \mathbf{E}_1(\mathbf{x}, t) &= \bar{\mathbf{E}}_\mathbf{k} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega\tau)] \\ \mathbf{B}_1(\mathbf{x}, t) &= \bar{\mathbf{B}}_\mathbf{k} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega\tau)] \end{aligned}$$

Maxwell's equations can be separated into two types of wave solutions. Defining $\bar{E}_\mathbf{k} = (\mathbf{k} \cdot \mathbf{E}/|k|)$ and $\mathbf{E}_\perp = (\mathbf{k} \times \mathbf{E}/|k|)$ leads to three homogeneous algebraic equations in the wave amplitudes. A compact form of these equations is

$$\begin{bmatrix} 1+\sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \cdot \int \frac{\partial F_{\alpha 0}/\partial u}{\omega/|k| - u}\, du & 0 & 0 \\ 0 & 1-\frac{k^2c^2}{\omega^2}-\sum_\alpha \frac{\omega_{p\alpha}^2}{\omega} \cdot \int \frac{F_{\alpha 0}}{\omega - |k|u}\, du & 0 \\ 0 & 0 & 1-\frac{k^2c^2}{\omega^2}-\sum_\alpha \frac{\omega_{p\alpha}^2}{\omega} \cdot \int \frac{F_{\alpha 0}}{\omega - |k|u}\, du \end{bmatrix} \begin{bmatrix} \bar{E}_{\mathbf{k}} \\ \bar{E}_{\perp 1} \\ \bar{E}_{\perp 2} \end{bmatrix} = 0 \qquad (8.9.5)$$

There are three solutions to (8.9.5). Two of them correspond to electromagnetic waves, and one of them corresponds to electrostatic waves which propagate in a field-free isotropic plasma. These two types of waves are:

1 Electrostatic waves ($\bar{E}_\perp = 0$)

$$1 + \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \int \frac{\partial F_{\alpha 0}/\partial u}{\omega/|k| - u}\, du = 0 \qquad (8.9.6)$$

2 Electromagnetic waves ($\bar{E}_{\mathbf{k}} = 0$)

$$\omega^2 = k^2c^2 + \sum_\alpha \omega_{p\alpha}{}^2 \omega \int \frac{F_{\alpha 0}}{\omega - |k|u}\, du \qquad (8.9.7)$$

8.9.1 Electrostatic Waves

The solutions of (8.9.6) derived in Secs. 8.6 and 8.7, are as follows:

a High-frequency Langmuir oscillations For $\omega/k \gg (\kappa T_e/m_e)^{1/2}$, the real and imaginary parts of the eigenfrequency are given by ($\omega = \omega_r + i\omega_i$)

$$\omega_r{}^2 = \omega_{pe}{}^2(1 + 3k^2\lambda_{De}{}^2) \qquad (8.9.8)$$

$$\omega_i = \omega_{i,\,\text{Landau}} = -\sqrt{\frac{\pi}{8}} \frac{\omega_{pe}}{|k^3\lambda_{De}{}^3|} \exp\left(-\frac{1}{2k^2\lambda_{De}{}^2} - \frac{3}{2}\right) \qquad (8.9.9)$$

b Ion-acoustic waves For $T_e \gg T_i$ and $(\kappa T_i/m_i)^{1/2} < \omega/k < (\kappa T_e/m_e)^{1/2}$, the real and imaginary parts of the eigenfrequency are given by

$$\omega_r{}^2 = \frac{k^2 C_s{}^2}{1 + k^2\lambda_{De}{}^2} \qquad (8.9.10)$$

$$\omega_i = -\sqrt{\frac{\pi}{8}} \sqrt{\frac{m_e}{m_i}} (1 + k^2\lambda_{De}{}^2)^{-3/2} |\omega_r| \qquad (8.9.11)$$

where

$$C_s \equiv \left(\frac{\kappa T_e}{m_i}\right)^{1/2} \qquad (8.9.12)$$

is the ion-acoustic speed and the ion damping has been neglected [$\exp(-T_e/T_i) \ll (m_e/m_i)^{1/2}$]

Problem 8.9.1 Derive (8.9.4) from (8.9.2). ////

8.9.2 Electromagnetic Waves

In addition to high-frequency (Langmuir) and low-frequency (ion-acoustic) electrostatic waves, the field-free plasma supports electromagnetic waves, in a restricted frequency range. From (8.9.7) these waves, in a plasma, must satisfy

$$\omega^2 = k^2c^2 + \sum_\alpha \omega_{p\alpha}{}^2 \int_{-\infty}^{\infty} \frac{\omega F_{\alpha 0}(u)}{\omega - |\mathbf{k}|u}\, du \qquad (8.9.13)$$

The term k^2c^2 guarantees that there will be solutions only with $\omega \gg k\sqrt{\kappa T_e/m_e}$, and the expansion (8.6.1) can be used to estimate the velocity integral. This leads to the result

$$\omega^2 = k^2c^2 + \omega_{pe}{}^2 \qquad (8.9.14)$$

which is exactly the same as the result derived in the fluid-theory calculation in Chap 4. These are "light waves" in a plasma. They are dispersive,

$$\omega = kc\sqrt{1 + \frac{\omega_p{}^2}{k^2c^2}}$$

and the index of refraction of the plasma is frequency-dependent.

Problem 8.9.2 The Landau damping term has been omitted from (8.9.14), as clearly unphysical. Why? ////

When $\omega^2 < \omega_p{}^2$, the electromagnetic wave is not a normal mode of the plasma, and these waves are strongly damped. Because $\omega_p \propto \sqrt{n_e}$, the cutoff $\omega \geq \omega_p$ for transparency to electromagnetic radiation provides a useful means of measuring the density of a plasma (Chap. 4). The dispersion relation (ω versus k) for waves in a field-free plasma is sketched in Fig. 8.9.1.

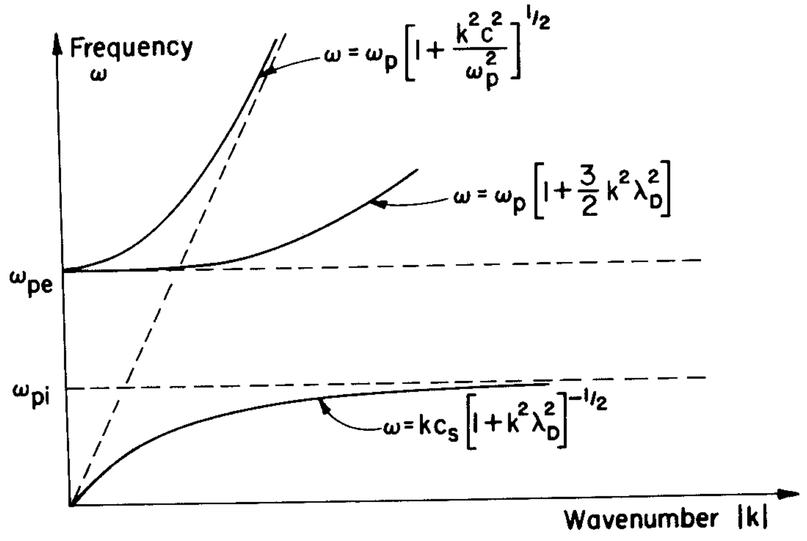


FIGURE 8.9.1
Dispersion relation for waves $\exp[i(\mathbf{k}\cdot\mathbf{x}-\omega t)]$ in a field-free plasma.

8.10 THE VLASOV THEORY OF SMALL-AMPLITUDE WAVES IN A UNIFORMLY MAGNETIZED PLASMA

$[\mathbf{B}_0 = B_0\hat{\mathbf{z}},\ \mathbf{E}_0 = 0,\ f_{\alpha 0} = f_{\alpha 0}(v_\perp{}^2, v_\parallel)]$

The orbits of particles in an equilibrium plasma are quite complicated if that equilibrium includes a magnetic field. This allows new wave modes associated with particle gyration, and results in an alteration of the waves indigenous to the field-free plasma. The fact that changing the equilibrium fields, or even simply changing the velocity distribution, can entirely change the pattern and behavior of the plasma oscillations is a constant barrier to the development of a "complete" description of a plasma. The motions of most other systems of coupled oscillators can be simply described in terms of a few "spring constants." For example, the bulk modulus of a neutral gas provides complete information about sound waves; by contrast, two otherwise identical plasmas, one maxwellian, $\exp(-v^2/\bar{v}^2)$, one Poisson, $\exp(-|v|/\bar{v})$, behave quite differently in regard to sound waves. This chameleon behavior of plasma makes it necessary to study each equilibrium separately. Fortunately, the same *methods* can be used in all these cases.

If the plasma is uniformly magnetized ($\mathbf{B}_0 = B_0\hat{\mathbf{z}}$), the Vlasov equation, linearized about f_0, becomes (writing $f = f_0 + f_1$, $\mathbf{B} = B_0\hat{\mathbf{z}} + \mathbf{B}_1$, $\mathbf{E} = \mathbf{E}_1$)

$$\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla + \frac{q_\alpha}{m_\alpha}\frac{\mathbf{v}\times\mathbf{B}_0}{c}\cdot\nabla_\mathbf{v}\right)f_{\alpha 1} = -\frac{q_\alpha}{m_\alpha}\left(\mathbf{E}_1 + \frac{\mathbf{v}\times\mathbf{B}_1}{c}\right)\cdot\nabla_\mathbf{v} f_{\alpha 0} \qquad (8.10.1)$$

and the equilibrium distribution satisfies

$$\left(\mathbf{v}\cdot\nabla + \frac{q_\alpha}{m_\alpha}\frac{\mathbf{v}\times\mathbf{B}_0}{c}\cdot\nabla_\mathbf{v}\right)f_{\alpha 0} = 0$$

$$\rho_q = \sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha 0}\, d\mathbf{v} = 0 \qquad \text{no net charge in the plasma}$$

$$\mathbf{J} = \sum_\alpha \bar{n}_\alpha q_\alpha \int \mathbf{v} f_{\alpha 0}\, d\mathbf{v} = 0 \qquad \text{no net current in the plasma} \qquad (8.10.2)$$

For the case of a spatially uniform plasma, the most general solution of (8.10.2) which is isotropic in the plane perpendicular to $\mathbf{B}_0$ has the form

$$f_{\alpha 0} = f_{\alpha 0}(v_\perp{}^2, v_z) \qquad (8.10.3)$$

This section investigates waves that propagate in a plasma with an equilibrium distribution of this form. Since the motion of charged particles is so different along and across a magnetic field, it would be unrealistic to restrict $f_{\alpha 0}$ to be isotropic [$f_{\alpha 0} = f_{\alpha 0}(v^2)$], as was done in the study of waves in field-free plasmas. For example, a realistic distribution of the form (8.10.3) might be

$$f_0 = \frac{m}{2\pi\kappa T_\perp}\left(\frac{m}{2\pi\kappa T_\parallel}\right)^{1/2}\exp\left[-\frac{m}{2\kappa}\left(\frac{v_\perp{}^2}{T_\perp} + \frac{v_\parallel{}^2}{T_\parallel}\right)\right]$$

where $v_\parallel = v_z$, $v_\perp = \sqrt{v_x{}^2 + v_y{}^2}$.

The Vlasov equation is solved by integrating along the orbits of the particles in the *unperturbed* fields, as described in Sec. 8.8. For the case considered here ($\mathbf{E}_0 = 0$, $\mathbf{B}_0 = B_0\hat{\mathbf{z}}$), these orbits are best expressed in cylindrical coordinates in velocity space; that is, $v_x = v_\perp\cos\phi$, $v_y = v_\perp\sin\phi$, $v_z = v_\parallel$. In terms of these variables the particle orbits $\mathbf{x}'(\tau)$ are

$$v_x' = v_\perp\cos(\phi - \omega_c\tau) \qquad x' = x - \frac{v_\perp}{\omega_c}\sin(\phi - \omega_c\tau) + \frac{v_\perp}{\omega_c}\sin\phi$$

$$v_y' = v_\perp\sin(\phi - \omega_c\tau) \qquad y' = y + \frac{v_\perp}{\omega_c}\cos(\phi - \omega_c\tau) - \frac{v_\perp}{\omega_c}\cos\phi \qquad (8.10.4)$$

$$v_z' = v_\parallel \qquad z' = v_\parallel\tau + z$$

These orbits are derived in Appendix I; here the constants of integration are chosen so that at $\tau \to 0$, $\mathbf{v}' \to \mathbf{v}$, $\mathbf{x}' \to \mathbf{x}$, where $\mathbf{v}$, $\mathbf{x}$ are fixed points in phase space.

Note that

$$f_0[\mathbf{x}'(\tau), \mathbf{v}'(\tau)]$$

is constant along the orbit of a particle in the unperturbed fields, because f_0 is constructed out of constants of the motion, as explained in Sec. 7.7.

The perturbed distribution, from (8.8.6), is

$$\bar{f}_{\alpha\mathbf{k}} = -\frac{q_\alpha}{m_\alpha}\int_{-\infty}^{0}\left(\bar{\mathbf{E}}_1 + \frac{\mathbf{v}'\times\bar{\mathbf{B}}_1}{c}\right)\cdot\nabla_{\mathbf{v}'}f_{\alpha 0}(\mathbf{v}')\exp[i(\mathbf{k}\cdot\mathbf{X}-\omega\tau)]\,d\tau \qquad \text{Im}\,(\omega)>0 \qquad (8.10.5)$$

where $\mathbf{X}=\mathbf{x}'-x$, $\tau=t'-t$. The term $(\mathbf{v}\times\bar{\mathbf{B}}_1)\cdot\nabla_{\mathbf{v}}f_0$ vanishes if $f_0=f_0(v^2)$. But as mentioned above, this is not a realistic assumption for plasma in a **B** field, since motions across and along the magnetic field are quite different. For a distribution of the form $f_0=f_0({v_\perp}^2, v_z)$, the field $\bar{\mathbf{B}}_1$ must be evaluated in terms of $\bar{\mathbf{E}}_1$ by Maxwell's equation

$$i\mathbf{k}\times\bar{\mathbf{E}}_1 = i\frac{\omega}{c}\bar{\mathbf{B}}_1$$

The operation $\nabla_{\mathbf{v}}f_0$ can be written

$$\nabla_{\mathbf{v}}f_0 = 2(\mathbf{v}-v_z\hat{\mathbf{z}})\frac{\partial f_0}{\partial {v_\perp}^2}+2v_z\frac{\partial f_0}{\partial {v_z}^2}\hat{\mathbf{z}}$$

Because ${v_\perp}^2$ and v_z are constants of the motion, the terms $\partial f_0/\partial {v_\perp}^2$ and $\partial f_0/\partial v_z$ can be removed from the integral. All remaining integrals have the form

$$\int_{-\infty}^{0}(v_x', v_y', 1)\exp[i(\mathbf{k}\cdot\mathbf{X}-\omega\tau)]\,d\tau$$

where $\mathbf{X}\,(=\mathbf{x}'-\mathbf{x})$, v_x', and v_y', are given by (8.10.4). Integrals of this form can be done with the aid of the following identity:

$$\exp\left[i\frac{k_\perp v_\perp}{\omega_c}\sin(\phi-\omega_c\tau)\right] = \sum_{n=-\infty}^{+\infty}J_n\left(\frac{k_\perp v_\perp}{\omega_c}\right)\exp[in(\phi-\omega_c\tau)] \qquad (8.10.6)$$

where J_n is the ordinary Bessel's function of the first kind.

Taking the x axis along

$$\mathbf{k}_\perp\left(\equiv\mathbf{k}-\frac{\mathbf{k}\cdot\mathbf{B}}{|\mathbf{B}|}\hat{\mathbf{z}}\right)$$

without any loss of generality,

$$\mathbf{k} = k_\perp\hat{\mathbf{x}}+k_\parallel\hat{\mathbf{z}} \qquad (8.10.7)$$

the result for $\bar{f}_{\alpha\mathbf{k}}$ is

$$\bar{f}_{\alpha\mathbf{k}} = \frac{q_\alpha}{m_\alpha}\sum_{n,l}\left[\frac{2Zv_\parallel J_l(k_\perp v_\perp/\omega_{c\alpha})+Xv_\perp(J_{l+1}+J_{l-1})-iYv_\perp(J_{l+1}-J_{l-1})}{i(\omega-l\omega_c-k_\parallel v_\parallel)}\right]\cdot J_n\left(\frac{k_\perp v_\perp}{\omega_{c\alpha}}\right)\exp[i(n-l)\phi] \qquad (8.10.8)$$

where

$$X = E_x\frac{\partial f_{\alpha 0}}{\partial {v_\perp}^2}+\frac{v_\parallel}{\omega}(k_\parallel E_x-k_\perp E_z)\left(\frac{\partial f_{\alpha 0}}{\partial {v_\parallel}^2}-\frac{\partial f_{\alpha 0}}{\partial {v_\perp}^2}\right)$$

$$Y = E_y\frac{\partial f_{\alpha 0}}{\partial {v_\perp}^2}+\frac{v_\parallel}{\omega}k_\parallel E_y\left(\frac{\partial f_{\alpha 0}}{\partial {v_\parallel}^2}-\frac{\partial f_{\alpha 0}}{\partial {v_\perp}^2}\right)$$

$$Z = E_z\frac{\partial f_{\alpha 0}}{\partial {v_\parallel}^2}$$

The perturbed currents $\bar{\mathbf{J}}_\mathbf{k}$ can be calculated from $\bar{f}_{\alpha\mathbf{k}}$, and then substituted into Maxwell's equations. The result is

$$-\mathbf{k}\times\mathbf{k}\times\bar{\mathbf{E}}_1 = \frac{\omega^2}{c^2}\bar{\mathbf{E}}_1+\frac{i\omega}{c^2}4\pi\sum_\alpha\bar{n}_\alpha q_\alpha\int_L\mathbf{v}f_{\alpha k}\,d\mathbf{v} \qquad (8.10.9)$$

where L is the Landau contour (Figs. 8.4.2 to 8.4.4). The dispersion equation takes the form

$$\begin{vmatrix} D_{xx} & D_{xy} & D_{xz}\\ D_{yx} & D_{yy} & D_{yz}\\ D_{zx} & D_{zy} & D_{zz}\end{vmatrix} = 0 \qquad (8.10.10)$$

where the elements of the determinant (8.10.10) are

$$D_{xx} = 1-\frac{{k_\parallel}^2c^2}{\omega^2}-\frac{2\pi}{\omega}\sum_\alpha\left(\frac{{\omega_p}^2}{\omega_c}\right)_\alpha\sum_n\!\!\!\!\!\!\int\left[\frac{n^2{\omega_{c\alpha}}^3}{{k_\perp}^2}{J_n}^2\chi_\alpha\right]$$

$$D_{xy} = -\frac{2\pi i}{\omega}\sum_\alpha\sum_n\left(\frac{{\omega_p}^2}{\omega_c}\right)_\alpha\!\!\!\!\!\!\int\left\{\frac{n{\omega_c}^2v_\perp}{k_\perp}J_n\frac{dJ_n}{d(k_\perp v_\perp/\omega_{c\alpha})}\chi_\alpha\right\}$$

$$D_{xz} = \frac{k_\parallel k_\perp c^2}{\omega^2}-\frac{2\pi}{\omega}\sum_\alpha\sum_n\left(\frac{{\omega_p}^2}{\omega_c}\right)_\alpha\!\!\!\!\!\!\int\left(\frac{n{\omega_{c\alpha}}^2v_\parallel{J_n}^2}{k_\perp}\Lambda_\alpha\right)$$

$$D_{yx} = -D_{xy}$$

$$D_{yy} = 1-\frac{({k_\perp}^2+{k_\parallel}^2)c^2}{\omega^2}-\frac{2\pi}{\omega}\sum_\alpha\sum_n\left(\frac{{\omega_p}^2}{\omega_c}\right)_\alpha\!\!\!\!\!\!\int\left[\omega_{c\alpha}\left(\frac{dJ_n}{d(k_\perp v_\perp/\omega_{c\alpha})}\right)^2{v_\perp}^2\chi_\alpha\right] \qquad (8.10.11)$$

$$D_{yz} = \frac{2\pi i}{\omega}\sum_\alpha\sum_n\left(\frac{{\omega_p}^2}{\omega_c}\right)_\alpha\!\!\!\!\!\!\int\left\{\omega_{c\alpha}v_\perp v_\parallel J_n\frac{dJ_n}{d(k_\perp v_\perp/\omega_{c\alpha})}\Lambda_\alpha\right\}$$

$$D_{zx} = \frac{k_\parallel k_\perp c^2}{\omega^2}-\frac{2\pi}{\omega}\sum_\alpha\sum_n\left(\frac{{\omega_p}^2}{\omega_c}\right)_\alpha\!\!\!\!\!\!\int\left[v_\parallel\frac{n{\omega_{c\alpha}}^2}{k_\perp}{J_n}^2\chi_\alpha\right]$$

$$D_{zy} = -\frac{2\pi i}{\omega}\sum_\alpha\sum_n\left(\frac{{\omega_p}^2}{\omega_c}\right)_\alpha\!\!\!\!\!\!\int\left\{v_\parallel v_\perp\omega_{c\alpha}J_n\frac{dJ_n}{d(k_\perp v_\perp/\omega_{c\alpha})}\chi_\alpha\right\}$$

$$D_{zz} = 1-\frac{{k_\perp}^2c^2}{\omega^2}-\frac{2\pi}{\omega}\sum_\alpha\sum_n\left(\frac{{\omega_p}^2}{\omega_c}\right)_\alpha\!\!\!\!\!\!\int\left[{v_\parallel}^2\omega_{c\alpha}{J_n}^2\Lambda_\alpha\right]$$

In (8.10.11) the integral operator $\|\!|$ is defined by

$$\|\!|\,[F(v)] \equiv \int_{-\infty}^{\infty} dv_{\|} \int_{0}^{\infty} \frac{2v_{\perp} F(v_{\perp}, v_{\|})}{k_{\|} v_{\|} + n\omega_{c\alpha} - \omega}\, dv_{\perp}$$

and two recurrent combinations of velocity derivatives are replaced by

$$\chi_{\alpha} \equiv \frac{\partial f_{\alpha 0}}{\partial v_{\perp}^{2}} \left(1 - \frac{k_{\|} v_{\|}}{\omega}\right) + \frac{k_{\|} v_{\|}}{\omega} \frac{\partial f_{\alpha 0}}{\partial v_{\|}^{2}}$$

$$\Lambda_{\alpha} \equiv \frac{\partial f_{\alpha 0}}{\partial v_{\|}^{2}} - \frac{n\omega_{c\alpha}}{\omega} \left(\frac{\partial}{\partial v_{\|}^{2}} - \frac{\partial}{\partial v_{\perp}^{2}}\right) f_{\alpha 0}$$

The argument of all Bessel functions above is $k_{\perp} v_{\perp}/\omega_{c\alpha}$, and the integrals $\int_{-\infty}^{\infty} dv$ are to be taken along the Landau contour shown in Fig. 8.4.1. Equations (8.10.10) and (8.10.11), which determine the wave motions of a magnetized plasma, are a depressing contrast to the relatively simple equations (8.9.6) and (8.9.7) which describe waves in a field-free plasma. The new features of the magnetized plasma are:

1 There are no longer three decoupled modes as in the field-free case. This is because the steady magnetic field **B** couples the motion of plasma particles from one direction to the others. That is, the plasma is anisotropic.
2 The separation into electrostatic and electromagnetic modes is no longer apparent.
3 There is now structure at $\omega \approx n\omega_c$. The gyration frequency of particles in a magnetic field $\omega_c (= eB/mc)$ is a natural frequency in this system, along with the plasma frequency ω_p.
4 The resonance at $\omega = \mathbf{k} \cdot \mathbf{v}$ found in field-free systems, which contributed damping of plasma waves, occurs at $\omega - n\omega_{c\alpha} = k_{\|} v_{\|}$; only particles moving along the magnetic field contribute to damping, and they damp only waves that have a component of propagation parallel to B_0. This is because in a uniform magnetic field there is no net motion of particles *across* the field.
5 In the field-free plasma the parameter which determines the oscillatory behavior is the ratio of the wave speed and the thermal speed,

$$\frac{\omega}{k} \sqrt{\frac{m_{\alpha}}{\kappa T_{\alpha}}}$$

In the magnetized plasma the parameters

$$\frac{\omega}{\omega_{c\alpha}} \qquad \frac{k}{\omega_{c\alpha}} \sqrt{\frac{\kappa T_{\alpha}}{m_{\alpha}}} \qquad \frac{\omega_p}{\omega_c} \qquad \frac{\omega}{k_{\|}} \sqrt{\frac{m_{\alpha}}{\kappa T_{\alpha}}} \qquad \frac{kc}{\omega_p}$$

all play a role in defining a parameter range.

To make some sense out of the general results (8.10.10) and (8.10.11), it is useful to single out some particular range for the parameters, looking always for ways in which the plasma resembles the field-free system and for ways in which the Vlasov results resemble the fluid approximation, as well as looking for distinctive new features.

8.11 THE VLASOV THEORY OF WAVES IN COLD MAGNETIZED PLASMA

The dispersion relation for waves in a cold magnetized plasma can be obtained from the dispersion relation derived using the linearized Vlasov equations by taking the limit $T_{\alpha} \to 0$ in (8.10.10). In this limit, (8.10.10) becomes identical with (4.9.6), the cold-magnetized-plasma dispersion determinant derived using the macroscopic-fluid-theory plasma equations. This shows that fluid theory provides an adequate description of waves in a sufficiently cold plasma, i.e., one where there is very little spread in the velocity distribution of the particles, so that all particles move with a speed nearly equal to that of a fluid element. In the cold-plasma limit there is no need for an equation of state; the pressure is simply neglected.

Since the solution of the linearized Vlasov equations reduced to the solution obtained using the macroscopic fluid theory, a comparison of the Vlasov theory results for $T_{\alpha} \neq 0$ with the results for $T_{\alpha} = 0$ identifies those plasma phenomena that are outside the scope of macroscopic fluid theory. For example, Landau damping, ion waves, and waves at harmonics of the cyclotron frequency, all depend on plasma temperature.

Problem 8.11.1 Show that as $T_{\alpha} \to 0$, (8.10.10), and (8.10.11) reduce to (4.9.6) and (4.9.9).
Hint: It is necessary to use the approximation

$$J_n \xrightarrow[T_{\alpha} \to 0]{} \frac{1}{n!} \left(\frac{k_{\perp} v_{\perp}}{2\omega_{c\alpha}}\right)^{n} \qquad ////$$

8.12 WAVES THAT PROPAGATE PERPENDICULAR TO THE EQUILIBRIUM MAGNETIC FIELD IN A HOT MAGNETIZED PLASMA ($\mathbf{E}_0 = 0$, $\mathbf{B}_0 = \hat{\mathbf{z}} B_0$)— ELECTROMAGNETIC WAVES AND THE BERNSTEIN MODES

Taking $k_z = 0$ in (8.10.10) singles out the waves that propagate exactly perpendicular to the equilibrium magnetic field in a hot magnetized plasma. These

waves are not damped since they are not in synchronism with particles moving along the magnetic axis. (Note that

$$\int_L \frac{F}{\omega - n\omega_c - k_z v_z}\, dv_x = \frac{1}{\omega - n\omega_c} \int_{-\infty}^{\infty} F\, dv_z$$

for $k_z = 0$.) With $k_z = 0$, the dispersion relation (8.10.10) is greatly simplified and has the form

$$\begin{bmatrix} D_{xx} & D_{xy} & 0 \\ D_{yx} & D_{yy} & 0 \\ 0 & 0 & D_{zz} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = 0 \qquad (8.12.1)$$

For a plasma with an isotropic distribution ($\partial f_0/\partial v_\parallel{}^2 = \partial f_0/\partial v_\perp{}^2$), one solution of (8.12.1) is

$$D_{zz} = 1 - \frac{k^2c^2}{\omega^2} - \frac{2\pi}{\omega} \sum_\alpha \sum_n \omega_{p\alpha}{}^2 \int_{-\infty}^{\infty} dv_\parallel \int_0^{\infty} \frac{J_n{}^2 f_{\alpha 0}\, v_\perp\, dv_\perp}{\omega - n\omega_{c\alpha}} = 0 \qquad (8.12.2)$$

In the following discussion, the plasma is assumed to have an isotropic distribution function for simplicity. This assumption is not an essential feature of the theory and is used to keep the resulting expressions manageable. Recall also that $\mathbf{k}_\perp = k_x\hat{\mathbf{x}}$ has been chosen for convenience.

8.12.1 The Ordinary Mode

From this equation it is clear that the $D_{zz} = 0$ solution represents an electromagnetic wave, with its **E** vector parallel to $\mathbf{B}_0$ and propagating in the x direction, $\mathbf{k} = k\hat{\mathbf{x}}$. This wave is called the *ordinary mode* because the response of electrons to E_z is not changed by the magnetic field $\mathbf{B}_0$. This mode also appeared in the fluid calculation (4.11.1).

The solution to $D_{zz} = 0$ is complicated by the infinite sum $\sum_n$. However, often only one term in the sum contributes to the solution. For example, for long-wavelength modes ($k^2a^2 \ll 1$) all the terms in $\sum_n$ except $n = 0$ are small by $k^2a_{c\alpha}{}^2$. In that case, only for frequencies near cyclotron harmonics ($\omega \approx n\omega_{c\alpha}$) does any term except $n = 0$ contribute. Similarly, for high frequencies ($k^2c^2 \gg \omega_{pe}{}^2$) the sum $\sum_n$ can be neglected, except for frequency near harmonics $\omega \approx n\omega_{c\alpha}$. Considering only wavelengths long compared with the gyroradius, then if ω is not close to a cyclotron harmonic only the $n = 0$ terms are retained.

The solution retaining only the $n = 0$ terms and setting $J(ka_{c\alpha}) = 1$ is a wave with the same dispersion relation as obtained with the fluid model, i.e.,

$$\omega^2 \approx k^2c^2 + \omega_p{}^2 \qquad (8.12.3)$$

In addition, there are other solutions of $D_{zz} = 0$ for frequencies near cyclotron harmonics, giving waves in narrow frequency bands ($n = 1, 2, 3, \ldots$, and $ka_{c\alpha} < 1$)

$$\omega = n\omega_{c\alpha}\left\{1 + 0\left[\frac{\omega_{p\alpha}{}^2}{k^2c^2}(ka_{c\alpha})^{2n}\right]\right\}$$

near harmonics of the cyclotron frequency.

Problem 8.12.1 Show that $D_{xz} = D_{yz} = D_{zx} = D_{zy} = 0$ for $k_z = 0$ in (8.10.10). ////

The other two eigenmodes of (8.12.1) satisfy the equation

$$\begin{vmatrix} D_{xx} & D_{xy} \\ -D_{xy} & D_{yy} \end{vmatrix} = 0$$

and have their **E** vector perpendicular to $\mathbf{B}_0$. An especially simple case is $k^2c^2 \gg \omega_p{}^2$, because in that case D_{xy} is much less than the $k^2c^2/\omega_p{}^2$ term in D_{yy} and can be neglected. In this limit two approximate solutions are obtained by setting either $D_{xx} = 0$ or $D_{yy} = 0$. These two solutions both have **E** perpendicular to $\mathbf{B}_0$. The dispersion relation for these modes are as follows:

8.12.2 The Extraordinary Mode

Polarization:

$$\mathbf{E}\cdot\mathbf{k} \approx \left(\frac{\omega_p{}^2}{k^2c^2}\,\mathbf{E}\times\mathbf{k}\right) \ll \mathbf{E}\times\mathbf{k} \qquad (8.12.4)$$

Dispersion relation:

$$k^2c^2 - \omega^2 + 2\pi\omega \sum_\alpha \sum_n \omega \int_{-\infty}^{\infty} dv_\parallel \int_0^{\infty} \frac{v_\perp{}^2 (J_n')^2}{n\omega_c - \omega} \frac{\partial f_{\alpha 0}}{\partial v_\perp{}^2}\, dv_\perp{}^2 = 0 \qquad (8.12.5)$$

8.12.3 The Bernstein Mode

Polarization:

$$\mathbf{E}\cdot\mathbf{k} \approx \left(\frac{k^2c^2}{\omega_p{}^2}\,\mathbf{E}\times\mathbf{k}\right) \gg \mathbf{E}\times\mathbf{k}$$

Dispersion relation:

$$k^2 + \sum_\alpha \sum_{n=1}^{\infty} \frac{\omega_{p\alpha}{}^2 4\pi n^2 \omega_{c\alpha}{}^2}{\omega^2 - n^2\omega_{c\alpha}{}^2} \int J_n{}^2 \frac{\partial f_{\alpha 0}}{\partial v_\perp{}^2}\, d\mathbf{v} = 0 \qquad (8.12.6)$$

The extraordinary wave (8.12.5) is polarized with its electric vector nearly perpendicular to **k**, and is almost a pure electromagnetic wave. The ordinary mode (8.12.2) is a pure electromagnetic wave. Both of the modes involve the cyclotron motion of the particles, and are called *electromagnetic cyclotron waves*. When $\mathbf{B}_0 \to 0$, both the ordinary mode (8.12.2) and the extraordinary mode (8.12.5) become electromagnetic waves (8.9.15).

The Bernstein modes are polarized with their electric vector nearly parallel to **k**, and they are almost pure longitudinal waves. These modes are the counterpart of the field-free plasma waves $\omega = \omega_{pe}$ and $\omega = kC_s$, and in the limit $\mathbf{B}_0 \to 0$, they reduce to the Langmuir oscillations at high frequency and to ion-sound waves at low frequency. However, in contrast to the field-free-plasma oscillation, these modes are undamped, even in the limit $\mathbf{B}_0 \to 0$. To recover the damped-plasma oscillations it is necessary to first take $k_z \neq 0$, and then take the limit $\mathbf{B}_0 \to 0$, and finally take $k_z \to 0$. The $k_z = 0$ modes form a set of measure zero[1] when $\mathbf{B}_0 \to 0$.

The solutions of Eq. (8.12.6) for the longitudinal plasma waves are found by plotting the $\sum_\alpha \sum_n$ term as a function of ω; the intersections of these curves with a horizontal line k^2 give the eigenfrequencies (Fig. 8.12.1). The Bernstein modes propagate in frequency ranges that lie between harmonics of the cyclotron frequency. The lowest frequency for which propagation occurs lies above the cyclotron frequency $2\omega_{ce} > \omega > \omega_{ce}$. The exact location of the modes is a function of density, temperature, and field strength. As an example, for a maxwellian distribution, (8.12.6) becomes

$$k^2 = \sum_n \sum_\alpha \left[\frac{2n^2 \omega_{p\alpha}{}^2 \omega_{c\alpha}{}^2}{\omega^2 - n^2 \omega_{c\alpha}{}^2} \frac{m_\alpha}{\kappa T_\alpha} I_n\left(\frac{k^2}{\omega_{c\alpha}{}^2} \frac{\kappa T_\alpha}{m_\alpha} \right) \exp\left(- \frac{k^2}{\omega_{c\alpha}{}^2} \frac{\kappa T_\alpha}{m_\alpha} \right) \right] \tag{8.12.7}$$

where I_n is the modified Bessel's function of the first kind, $I_n(z) = e^{-i\pi n/2} J_n(e^{i\pi/2} z)$. A property of the solution of (8.12.7) is that the Bernstein modes are separated from each other by frequency intervals where there is no propagation. For example, from Fig. 8.12.1, there is no wave with frequency in the narrow band just below the frequency $2\omega_c$. Similar gaps occur between each additional harmonic.

The Bernstein modes exist between harmonics of ω_{ci} as well as ω_{ce}. When the plasma temperature is low, or when *long* waves (small k) are being studied, the Bessel function I_n can be expanded in powers of ka; one solution of the dispersion relation exists near the so-called upper hybrid frequency, $\omega_H = \sqrt{\omega_{pe}{}^2 + \omega_{ce}{}^2}$. The other solutions occur only very near resonances $\omega \approx n\omega_{ce}$, because they require $(ka)^{2n-2}\omega_{pe}{}^2/(\omega^2 - n^2\omega_{ce}{}^2)$ to be of order unity:

[1] I. B. Bernstein, *Phys. Rev.*, **109**:10 (1958).

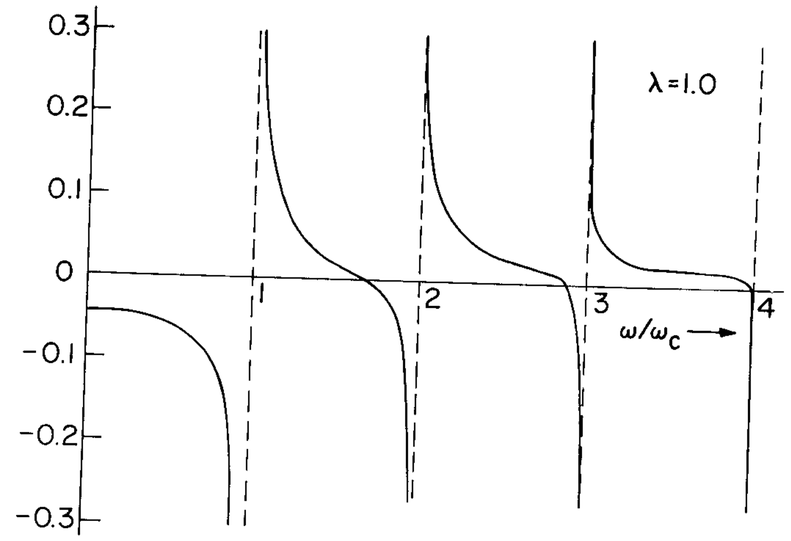


FIGURE 8.12.1
Right-hand-side of the dispersion relation (8.12.7) for electrostatic plasma waves propagating perpendicular to a homogeneous magnetic field. The intersection of a horizontal line k^2 with these curves gives the zeros ω_k of the dielectric functions. [*After I. B. Bernstein, Phys. Rev.*, **109**:10 (1958).] The curve shown is for $\lambda \equiv k^2\kappa T/m\omega_c{}^2 = 1.0$.

$$\omega_1 = \sqrt{\omega_{pe}{}^2 + \omega_{ce}{}^2}$$
$$\omega_n = n\omega_{ce}[1 + 0(ka)^{2n-2}] \qquad n \geq 2 \tag{8.12.8}$$

where $k^2 a_{ce}{}^2 \ll 1$.

When the plasma density is low, all the Bernstein modes occur very close to cyclotron harmonics, and the explicit solutions of (8.12.7) are

$$\omega^2 = n^2 \omega_{ce}{}^2 (1 + \alpha_n) \qquad n = 1, 2, \ldots \tag{8.12.9}$$

where

$$\alpha_n = \frac{2\omega_{pe}{}^2}{\omega_{ce}{}^2} \frac{1}{k^2 a_{ce}{}^2} I_n(k^2 a_{ce}{}^2) \exp(-k^2 a_{ce}{}^2)$$

$$a_{ce}{}^2 \equiv \frac{\kappa T_e}{m_e \omega_{ce}{}^2}$$

and where

$$\omega_{pe}{}^2 \ll \omega_{ce}{}^2$$

is the limit in which this low-density approximation is valid. Figure 8.12.2 shows the frequency of the Bernstein modes as a function of wave number, for a low-density plasma. If the plasma density is large and the field strength is small, the upper hybrid mode has a frequency $\omega_H > 2\omega_{ce}$. In this case the lowest-

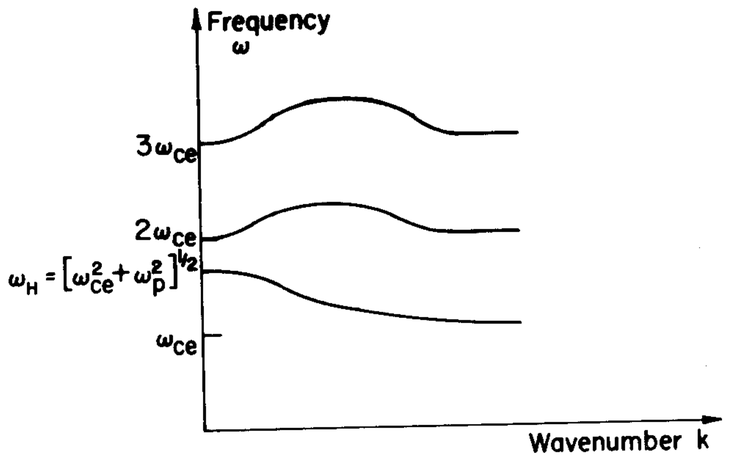


FIGURE 8.12.2
Dispersion characteristics for electrostatic waves propagating perpendicular to a homogeneous magnetic field (Bernstein waves).

frequency mode [found by expanding I_n in (8.12.7) and keeping the $n = 1$, $n = 2$ terms] is given by

$$\omega_1 = 2\omega_{ce}\left(1 - \frac{\omega_{pe}^2}{\omega_{pe}^2 - 3\omega_{ce}^2}\frac{3k^2}{\omega_{ce}^2}\frac{\kappa T_e}{8m_e}\right) \qquad \begin{matrix}\omega_{pe}^2 > 3\omega_{ce}^2 \\ k^2 a_{ce}^2 \ll 1\end{matrix} \qquad (8.12.10)$$

and so it occurs just *below* the second harmonic. The location of higher harmonics (just above or just below the harmonic) depends on the density.

$$\omega_n = \begin{cases} n\omega_{ce}[1 - 0(k^2 a_{ce}^2)^n] & \omega_{pc}^2 > (n^2 - 1)\omega_{ce}^2 \\ n\omega_{ce}[1 + 0(k^2 a_{ce}^2)^n] & \omega_{pc}^2 < (n^2 - 1)\omega_{ce}^2 \end{cases}$$

Similar behavior is found near ion harmonics.

The appearance of these modes in narrow bands of frequency is in strong contrast with the results of fluid theory (Fig. 4.11.1), which predicted cutoff between the frequencies $\sqrt{\omega_{ce}\omega_{ci}}$ and ω_1 for waves propagating perpendicular to $\mathbf{B}_0$. The Vlasov theory shows that there are, in fact, bands of propagation in the range, as in Fig. 8.12.2.

Problem 8.12.2 Plot ω as k for the first few electron harmonics and the first few ion harmonics, for Bernstein modes. ////

The modes (8.12.2) and (8.12.5) are also algebraically simple in the low-temperature limit, as discussed in Chap. 4. For completeness these results are

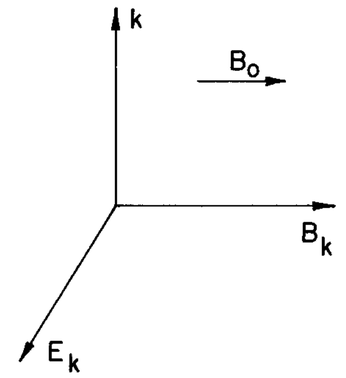


FIGURE 8.12.3
Diagram of field components of a magnetosonic wave propagating at right angles to a homogeneous magnetic field.

summarized here. For the mode with $E_x = 0$, only the $n = \pm 1$ terms in (8.12.5) contribute, leading to

$$k^2c^2 - \omega^2 + \sum_\alpha \frac{\omega^2\omega_{p\alpha}^2}{\omega^2 - \omega_{c\alpha}^2} = 0 \qquad (8.12.11)$$

The low-frequency limit of this mode is

$$\omega^2 = \frac{k^2c^2}{1 + 4\pi n m_i c^2/B^2} \qquad \text{when } \omega^2 < \omega_{ci}^2 \qquad (8.12.12)$$

Defining

$$V_A = \frac{B}{\sqrt{4\pi n m_i}} = \text{Alfvén speed}$$

this can be rewritten

$$\omega^2 = \frac{k^2 V_A^2}{1 + V_A^2/c^2} \equiv \text{magnetosonic wave} \qquad (8.12.13)$$

This wave is called the magnetosonic wave because it has the form $\omega = kV$. It has a magnetic component $\mathbf{B}_k = B_k\hat{\mathbf{z}}$, parallel to $\mathbf{B}_0$. The magnetosonic wave propagates at right angles to the equilibrium field $\mathbf{B}_0$, and does not distort the magnetic field (Fig. 8.12.3). At higher frequencies, $\omega_{ci} < \omega < \omega_{ce} < \omega_{pe}$, the solution for the electromagnetic mode (8.12.5) is

$$\omega \approx \sqrt{\omega_{ci}\omega_{ce}}$$

When $\omega_{ci} < \omega < \omega_{pe} < \omega_{ce}$, the solution of (8.12.2) is

$$\omega \approx \sqrt{\omega_{pe}^2 + \omega_{ce}^2}$$

8.13 WAVES THAT PROPAGATE PARALLEL TO THE EQUILIBRIUM MAGNETIC FIELD IN A MAGNETIZED HOT PLASMA—ELECTROSTATIC AND ELECTROMAGNETIC WAVES ($\mathbf{E}_0 = 0$, $\mathbf{B}_0 = \hat{\mathbf{z}}B_0$)

Taking $k = 0$ in (8.10.10) singles out the waves that propagate parallel to the equilibrium magnetic field in a hot magnetized plasma. With $k = 0$, the dispersion relation is greatly simplified, since all the ${J_n}^2$ terms vanish for $n > 0$ and only the ${J_0}^2$ term remains $[{J_0}^2(0) = 1]$. Also, the waves with their electric vector polarized along $\mathbf{B}_0$ are decoupled from those with their electric vector polarized perpendicular to $\mathbf{B}_0$. The wave with $E_z \neq 0$ is an electrostatic wave ($\mathbf{E} \parallel \mathbf{k}$) and has the following dispersion relation:

$$\omega^2\left(1 + \sum_\alpha \frac{{\omega_{p\alpha}}^2}{{k_\parallel}^2} \int \frac{\partial f_{\alpha 0}/\partial v_\parallel}{\omega/k_\parallel - v_\parallel}\, d\mathbf{v}\right) = 0 \qquad (8.13.1)$$

Equation (8.13.1) is identical with the dispersion relation for electrostatic waves in a field-free plasma! The reason is that the particle motion associated with this wave is along $\mathbf{B}_0$, and therefore the magnetic field has no effect on the mode. The solutions of (8.13.1) are the Landau-damped Langmuir waves and ion-acoustic waves.

The other two waves have $\mathbf{E} \perp \mathbf{k}$, and are electromagnetic waves with dispersion relations

$$\omega^2 = k^2c^2 + 2\pi\omega \sum_\alpha {\omega_{p\alpha}}^2 \int \frac{\left[\dfrac{\partial f_{\alpha 0}}{\partial {v_\perp}^2}\left(1 - \dfrac{k_\parallel v_\parallel}{\omega}\right) + \dfrac{k_\parallel v_\parallel}{\omega}\dfrac{\partial f_{\alpha 0}}{\partial {v_\parallel}^2}\right] {v_\perp}^3\, dv_\perp\, dv_\parallel}{k_\parallel v_\parallel - \omega + \omega_{c\alpha}} = 0 \qquad (8.13.2)$$

and

$$\omega^2 = k^2c^2 + 2\pi\omega \sum_\alpha {\omega_{p\alpha}}^2 \int \frac{\left[\dfrac{\partial f_{\alpha 0}}{\partial {v_\perp}^2}\left(1 - \dfrac{k_\parallel v_\parallel}{\omega}\right) + \dfrac{k_\parallel v_\parallel}{\omega}\dfrac{\partial f_{\alpha 0}}{\partial {v_\parallel}^2}\right] {v_\perp}^3\, dv_\perp\, dv_\parallel}{k_\parallel v_\parallel - \omega - \omega_{c\alpha}} = 0 \qquad (8.13.3)$$

The cold-plasma limit of (8.13.2) and (8.13.3) is obtained by setting $v = 0$. In this limit Landau damping becomes negligible, and these waves reduce to the right and left circularly polarized waves at high frequency, Whistler waves at intermediate frequency, and Alfvén waves at low frequency. The properties of these waves were discussed in Chap. 4. For a hot plasma, Landau damping

can be an important effect in some frequency ranges. To estimate the magnitude of this effect, the isotropic case $f_0 = f_0({v_\perp}^2 + {v_\parallel}^2)$ will be investigated. In this case

$$\frac{\partial f_0}{\partial {v_\parallel}^2} = \frac{\partial f_0}{\partial {v_\perp}^2}$$

Assuming that the damping is small, the integrals in (8.13.2) and (8.13.3) can be approximated by using the relation

$$\int_{-\infty}^{\infty} \frac{F(v^2)}{k_\parallel v_\parallel - \omega \pm \omega_{c\alpha}}\, dv_\parallel \approx \oint \frac{F({v_\perp}^2 + {v_\parallel}^2)}{k_\parallel v_\parallel - \omega_r \pm \omega_{c\alpha}}\, dv_\parallel + \frac{\pi i}{|k_\parallel|} F\left[{v_\perp}^2 + \left(\frac{\omega_r \mp \omega_{c\alpha}}{k_\parallel}\right)^2\right] \qquad (8.13.4)$$

where $\omega = \omega_r + i\omega_i$ has been used. Equations (8.13.2) and (8.13.3) can be solved by first neglecting ω_i in the equations for ω_r, and then iterating. An equivalent method is to use a procedure developed earlier in this Chapter, i.e.,

$$\mathrm{Re}[D_{zz}(k_\parallel, \omega_r)] = 0$$

$$\omega_i = -\frac{\mathrm{Im}[D_{zz}(k_\parallel, \omega_r)]}{\partial/\partial\omega_r\{\mathrm{Re}[D_{zz}(k_\parallel, \omega_r)]\}}$$

In either case the dispersion and damping are given by [integrating (8.13.4)]

$${\omega_r}^2 = k^2c^2 - \omega_r \sum_\alpha 2\pi{\omega_{p\alpha}}^2 \oint_{-\infty}^{\infty} dv_\parallel \int_0^\infty \frac{v_\perp f_{\alpha 0}\, dv_\perp}{k_\parallel v_\parallel - \omega_r \pm \omega_{c\alpha}} \qquad (8.13.5)$$

$$\omega_i \approx -\frac{2\pi^2 \sum_\alpha {\omega_{p\alpha}}^2 \dfrac{1}{|k_\parallel|} \displaystyle\int_0^\infty v_\perp f_{\alpha 0}\left[{v_\perp}^2 + \left(\frac{\omega_r \pm \omega_{c\alpha}}{k_\parallel}\right)^2\right] dv_\perp}{1 + \dfrac{k^2c^2}{{\omega_r}^2} - 2\pi \displaystyle\sum_\alpha {\omega_{p\alpha}}^2 \int_0^\infty \frac{1}{|k_\parallel|} \int_{-\infty}^{\infty} \frac{(\partial f_{\alpha 0}/\partial v_\perp) v_\parallel\, d_\parallel\, dv_\perp}{k_\parallel v_\parallel - \omega_r \pm \omega_{c\alpha}}} \qquad (8.13.6)$$

so that the damping rate clearly depends on the number of particles with velocity along $\mathbf{B}_0$ given by

$$v_\parallel = \frac{\omega_r \pm \omega_{c\alpha}}{k_\parallel}$$

It is instructive to calculate the damping in some special cases, assuming $f_{\alpha 0}$ to be a Maxwell distribution.

8.13.1 Alfvén Waves

$$\omega \ll \omega_{ci} \qquad \omega_r = \frac{k_\parallel V_A}{\sqrt{1 + V_A^2/c^2}}$$

$$\omega_i = -\frac{\omega_{pi}^2}{|k_\parallel|}\frac{1}{1 + c^2/V_A^2}\sqrt{\frac{\pi m_i}{8\kappa T_i}}\exp\left(\frac{-B^2}{8\pi n\kappa T_i}\frac{\omega_{ci}^2}{\omega_r^2}\right)$$

where $V_A = B/\sqrt{4\pi n m_i}$

This damping is due to ions, since the electron damping is further reduced by $\exp(-\omega_{ce}^2/\omega_r^2)$. Thus Alfvén waves are generally weakly damped for frequencies well below the ion-cyclotron frequency.

8.13.2 Whistler Waves

$$\omega_{ci}^2 \ll \omega^2 \ll \omega_{ce}^2$$

In this case also, Landau damping is weak.

$$\omega_i \approx -\frac{\omega_{pe}^2}{|k_\parallel|}\sqrt{\frac{m_e}{2\kappa T_e}}\frac{1}{1 + k^2c^2/\omega_r^2}\exp\left(-\frac{\omega_{ce}^2}{k_\parallel^2}\frac{m_e}{2\kappa T_e}\right) \tag{8.13.7}$$

except for extremely short wavelengths, $\lambda \lesssim a_{ce}$.

8.13.3 Cyclotron Waves

$$\omega = \pm\omega_{ce}$$

For $\omega = \omega_{ce}$, one of the high-frequency modes can be strongly damped; taking $\omega \approx \omega_{ce}$,

$$\omega^2 = k^2c^2 - i\sqrt{\frac{m_e c^2}{2\pi\kappa T_e}}\omega_{pe}^2 \exp\frac{(\omega - \omega_{ce})^2}{k_\parallel^2 2\kappa T_e/m_e} \tag{8.13.8}$$

$$\omega \approx kc - i\sqrt{\frac{m_e c^2}{8\pi\kappa T_e}}\frac{\omega_{pe}^2}{\omega_{ce}} \tag{8.13.9}$$

The condition $\omega_r \approx \omega_{ce}$ for damping, along with $\omega_r^2 \approx k^2c^2$, thus implies

$$k_\parallel = \frac{\omega_{ce}}{c} \tag{8.13.10}$$

and so the damping of this high-frequency wave is strong at one particular value of the wavelength.

Damping implies a strong coupling of the particles to a wave of a particular wavelength and frequency, and Chap. 11 will show that these strongly damped bands of $\mathbf{k}$ are also bands in which spontaneous *emission* of radiation is strong. In Sec. 8.12, it was shown that electromagnetic modes

$$\omega_r \approx \omega_{ce} \approx kc$$

were undamped for propagation perpendicular to $\mathbf{B}$. The equivalent modes are damped for propagation along $\mathbf{B}$. The damping can be calculated as a function of angle of propagation and the frequency.

8.14 ELECTROMAGNETIC WAVES PROPAGATING AT AN ARBITRARY ANGLE WITH RESPECT TO THE EQUILIBRIUM MAGNETIC FIELD IN A MAGNETIZED HOT PLASMA ($\mathbf{E}_0 = 0$, $\mathbf{B}_0 = \hat{z}B_0$)

The dispersion relation for waves of frequency

$$\omega_r \approx kc$$

with the direction of $\mathbf{k}$ nearly perpendicular to $\mathbf{B}_0$ (that is, $k_\parallel \ll k_\perp$) differs from (8.13.8) primarily in that the damping term at $\omega_r = \omega_{ce}$ is replaced by a series of damping terms, one of which is large at each harmonic $\omega_r = n\omega_{ce}$. For arbitrary angle of propagation, and treating $\omega_i \ll \omega_r$, the methods used in the previous section give the following results.

8.14.1 Wave Polarized with Electric Vector Perpendicular to the Equilibrium Magnetic Field ($\mathbf{E} \perp \mathbf{B}_0$)

$$\omega = kc + \frac{i\pi^2\omega_{pe}^2}{k_\parallel}\int_0^\infty v_\perp^2\left[\frac{dJ_n(k_\perp v_\perp/\omega_{ce})}{d(k_\perp v_\perp/\omega_{ce})}\right]^2\frac{\partial}{\partial v_\perp}f_{e0}(v_\perp^2 + v_\parallel^2)\bigg|_{v_\parallel = \left|\frac{\omega \pm n\omega_{ce}}{k_\parallel}\right|} dv_\perp \tag{8.14.1}$$

8.14.2 Wave Polarized with Electric Vector Parallel to the Equilibrium Magnetic Field ($\mathbf{E} \parallel \mathbf{B}_0$)

$$\omega = kc + \frac{i\pi^2\omega_{pe}^2}{k_\parallel^2}\int_0^\infty v_\perp(\omega - n\omega_c)J_n^2\left[\frac{\partial}{\partial v_\parallel}f_{e0}(v_\perp^2 + v_\parallel^2)\right]_{v_\parallel = \left|\frac{\omega \pm n\omega_{ce}}{k_\parallel}\right|} dv_\perp \tag{8.14.2}$$

The damping rates are also relevant to emission calculations (Chap. 11).

8.15 WAVES IN AN INHOMOGENEOUS MAGNETIZED HOT PLASMA [$\mathbf{E}_0 = 0$, $\mathbf{B}_0 = \hat{\mathbf{z}} B_0(x)$, $n_0 = n_0(x)$]

8.15.1 Description of the Equilibrium, and Derivation of the Perturbed Distribution Function for a Nonuniform Plasma

The method of characteristics (8.8.1) to (8.8.5) can be used to solve the Vlasov equation for moderately complicated plasma geometries. In this section this method is used to study waves in a bounded plasma confined by a magnetic field. To simplify the analysis, the plasma is taken to be a semi-infinite slab, with the density decreasing in the x direction. This density decrease is balanced by a magnetic field that increases with x (see Fig. 8.15.1 and also Chap. 3) in such a way that the equilibrium magnetic pressure balances the equilibrium plasma pressure. The equilibrium is specified by

$$\begin{aligned} \mathbf{B}_0 &= \hat{\mathbf{z}} B_0(x) \\ n_0 &= n_0(x) \end{aligned} \qquad (8.15.1)$$

$$p_0(x) + \frac{B_0^2(x)}{8\pi} = \text{constant}$$

The equations for the equilibrium state of a plasma with a nonuniform density $n_0(x)$ are

Vlasov equation:

$$v_x \frac{\partial f_{\alpha 0}}{\partial x} + \frac{q_\alpha}{m_\alpha} \frac{\mathbf{v} \times \mathbf{B}_0(x)}{c} \cdot \nabla_{\mathbf{v}} f_{\alpha 0} = 0 \qquad (8.15.2)$$

Maxwell's equations:

$$\nabla \cdot \mathbf{E}_0 = 4\pi \rho_q = \sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha 0}\, d\mathbf{v} = 0 \qquad (8.15.3)$$

$$-\frac{dB_z}{dx} = \frac{4\pi}{c} \sum_\alpha \bar{n}_\alpha q_\alpha \int v_y f_{\alpha 0}\, d\mathbf{v} \qquad (8.15.4)$$

where $\bar{n}_\alpha = N_\alpha / V$ denotes the average density. (N_α is the total number of particles of type α, and V is the volume of the system.) The distribution function $f_{\alpha 0}$ is normalized so that $\int d\mathbf{x}/V \int f_{\alpha 0}\, d\mathbf{v} = 1$. Equation (8.15.3) is appropriate only if there are no electric fields $\mathbf{E}_0$ in the equilibrium state. For example, if the equilibrium state were one with an excess of ions, then (8.15.3) would be replaced by

$$\frac{dE_0}{dx} = \sum_\alpha 4\pi \bar{n}_\alpha q_\alpha \int f_{\alpha 0}\, d\mathbf{v}$$

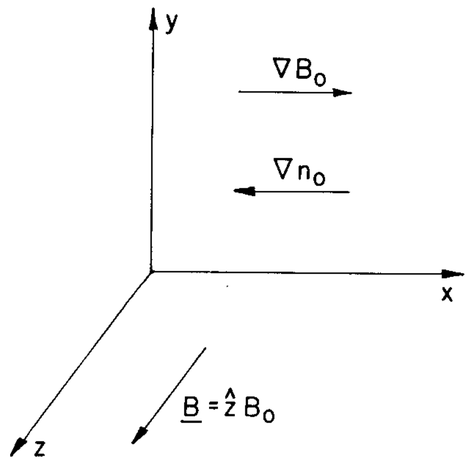


FIGURE 8.15.1
Coordinate system showing plasma and magnetic field gradients for a model used in the study of waves in an inhomogeneous plasma. A single underscore denotes a vector.

and the Vlasov equation would include a term $\mathbf{E}_0 \cdot \nabla_{\mathbf{v}} f_{\alpha 0}$. Here it is assumed that the equilibrium electric field is zero.

Problem 8.15.1 Show that (8.15.2) and (8.15.3) imply that

$$p(x) + \frac{B^2}{8\pi} = \text{constant} \qquad ////$$

The constants of the motion, from which $f_{\alpha 0}$ can be constructed, are the energy, momenta along $\mathbf{B}_0$, and the auxiliary constants discussed in Sec. 7.7.

$$\begin{aligned} W_\perp &= \tfrac{1}{2} m v_\perp^2 \\ p_z &= m v_z \\ \xi_x &= m\left(v_x - \frac{q}{mc} \int B_z\, dy\right) \\ \xi_y &= m\left(v_y + \frac{q}{mc} \int B_z\, dx\right) \end{aligned} \qquad (8.15.5)$$

Since the plasma state being described has a gradient only in the x direction, the most general distribution will be a function of $v_\perp^2$, v_z, and ξ_y.

$$f_{\alpha 0} = f_{\alpha 0}\left(v_\perp^2, v_z, v_y + \frac{q_\alpha}{m_\alpha c} \int B_z\, dx\right) \qquad (8.15.6)$$

The Vlasov equations for inhomogeneous plasmas are difficult to solve in general. However, there are several situations of practical interest for which solutions are possible. One of these tractable cases involves plasmas in which the

gradients in the magnetic field and density are assumed to be weak. This assumption is expressed by the following inequalities:

$$a_{ci}\frac{\nabla B}{B} \ll 1 \qquad a_{ci}\frac{\nabla n_i}{n_i} \ll 1 \tag{8.15.7}$$

where a_{ci} is the cyclotron radius of ions.

If the gradients in the magnetic field and the density are weak, the field and distribution function can be expanded in a Taylor series about a point $x_0(=0)$. Only terms linear in the gradients are retained. Thus

$$\mathbf{B} \approx \hat{\mathbf{z}}\left(B_0 + x\frac{dB_0}{dx}\right) \equiv \hat{\mathbf{z}}\,B_0(1+\varepsilon x) \tag{8.15.8}$$

and

$$\begin{aligned} f_{\alpha 0} &= f_{\alpha 0}(v_\perp{}^2, v_z, 0) + \left(v_y + \frac{q_\alpha}{m_\alpha c}\int B_z\,dx\right)\frac{1}{\omega_{c\alpha}}\frac{df_{\alpha 0}}{dx} \\ &= f_\alpha(v_\perp{}^2, v_z)\left[1 - \varepsilon'_\alpha\left(x + \frac{\varepsilon x^2}{2} + \frac{v_y}{\omega_{c\alpha}}\right)\right] \\ &\approx f_{\alpha 0}(v_\perp{}^2, v_z)\left[1 - \varepsilon'_\alpha\left(x + \frac{v_y}{\omega_{c\alpha}}\right)\right] \end{aligned} \tag{8.15.9}$$

where the parameters ε'_α and ε are chosen to correspond to the actual gradients in field and density, as might be found in laboratory or space plasma situations:

$$\varepsilon'_\alpha = -\frac{1}{f_{\alpha 0}}\frac{\partial f_{\alpha 0}}{\partial x}\bigg|_{x=x_0=0} \qquad \varepsilon = \frac{1}{B_0}\frac{dB_0}{dx}\bigg|_{x=x_0=0} \qquad \omega_{c\alpha} = \frac{q_\alpha B_0}{m_\alpha c} \tag{8.15.10}$$

This distribution can describe a variety of plasma states, as follows:

a Nonuniform density ε'_α can be chosen to be a constant, independent of the other constants of the motion $v_\perp{}^2$ and v_z; in this case the plasma state is a state of uniform temperature and linearly varying density.

$$\begin{aligned} n &= \bar{n}_\alpha \int f_{\alpha 0}\,d\mathbf{v} = n_{\alpha 0}(1 - \varepsilon'_\alpha x) \\ \kappa T_{\alpha\perp} &= \frac{\bar{n}_\alpha}{n_\alpha}\int \frac{mv_\perp{}^2}{2} f_{\alpha 0}\,d\mathbf{v} = \kappa T_{\alpha 0} = \text{constant} \end{aligned} \tag{8.15.11}$$

Further, since the equilibrium electric field is zero, the charge density must vanish at all points in the equilibrium, so that, from (8.15.3) and (8.15.11),

$$\varepsilon'_e = \varepsilon'_i$$

b Nonuniform density and nonuniform perpendicular temperature Temperature gradients perpendicular to $\mathbf{B}_0$ can be described by choosing ε'_α to be a function of $v_z{}^2$ or $v_\perp{}^2$. For example, if $\varepsilon'_\alpha = \beta_\alpha + (\delta_\perp mv_\perp{}^2/2\kappa T)_\alpha$ and

$$f_{\alpha 0} = \left(\frac{m}{2\pi\kappa T_\perp}\right)_{\alpha 0}\left(\frac{m}{2\pi\kappa T_\parallel}\right)^{1/2}_{\alpha 0}\exp\left(-\frac{mv_\perp{}^2}{2\kappa T_\perp}\right)_{\alpha 0}\exp\left(-\frac{mv_\parallel{}^2}{2\kappa T_\parallel}\right)_{\alpha 0}$$

where β_α and δ_α are constants, the plasma state described has gradients in density and T_α, the temperature in the plasma perpendicular to $\mathbf{B}_0$, given by

$$\begin{aligned} n_{\alpha\perp} &= n_{\alpha 0}[1 - (\beta_\alpha + \delta_{\perp\alpha})x] \\ T_{\alpha\perp} &= T_{\alpha 0\perp}(1 - \delta_{\alpha\perp}x) \\ T_{\alpha\parallel} &= T_{\alpha 0\parallel} = \text{constant} \end{aligned} \tag{8.15.12}$$

In this case, $E_0 = 0$ implies, through (8.15.3), that

$$(\beta + \delta_\perp)_i = (\beta + \delta_\perp)_e$$

c Nonuniform density and nonuniform parallel temperature If $\varepsilon'_\alpha = \beta_\alpha + (\delta_\parallel mv_\parallel{}^2/2\kappa T_\parallel)_\alpha$, the plasma state described has gradients in density and $T_\parallel$, given by

$$\begin{aligned} n_\alpha &= n_{\alpha 0}[1 - (\beta_\alpha + \tfrac{1}{2}\delta_{\parallel\alpha})x] \\ T_{\alpha\parallel} &= T_{\alpha 0\parallel}(1 - \delta_{\alpha\parallel}x) \\ T_{\alpha\perp} &= T_{\alpha 0\perp} = \text{constant} \\ (\beta + \tfrac{1}{2}\delta_\parallel)_i &= (\beta + \tfrac{1}{2}\delta_\parallel)_e \end{aligned} \tag{8.15.13}$$

and so on. All these plasma states are consistent with the same equilibrium fields, $\mathbf{B} = B_0(1+\varepsilon x)\hat{\mathbf{z}}$, $\mathbf{E}_0 = 0$. The virtue of knowing many different distributions consistent with a single set of equilibrium fields is that it allows descriptions of plasmas that have been prepared in different ways, e.g., with or without a temperature gradient, etc. This variety in $f_{\alpha 0}$ is of interest on the *collisionless* time scale, since after a few collision times the distribution will be unique (locally maxwellian).

Problem 8.15.2 Use the Vlasov equation to show that the plasma equilibrium $\mathbf{B}_0 = B_0(x)\hat{\mathbf{z}}$, $\mathbf{E}_0 = 0$, cannot include gradients *along* $\mathbf{B}_0$, in either density or temperature; that is, dn/dz and dT/dz must vanish. What sort of equilibrium fields would allow gradients along $\mathbf{B}_0$? ////

There are two characteristic velocities in the plasma state [(8.15.8) and (8.15.9)]. The first, a *particle* velocity, is the drift produced by the magnetic gradients as discussed in Appendix I.

$$\mathbf{V}_{\text{drift}} = \hat{\mathbf{y}} \frac{v_\perp^2}{2\omega_c} \frac{1}{B} \frac{dB}{dx} = \frac{\varepsilon v_\perp^2}{2\omega_c} \hat{\mathbf{y}} \qquad (8.15.14)$$

The second velocity is that of a fluid element, found by taking a velocity moment of (8.15.9).

$$\mathbf{V}_\alpha = \int \mathbf{v} f_{\alpha 0}\, d\mathbf{v} = -\hat{\mathbf{y}} \frac{\varepsilon'_\alpha}{\omega_{c\alpha}} \int v_y^2 f_{\alpha 0}(v_\perp^2, v_z)\, \mathbf{dv} = -\frac{\varepsilon'_\alpha \langle v_\perp^2 \rangle_\alpha}{2\omega_{c\alpha}} \hat{\mathbf{y}} \qquad (8.15.15)$$

where

$$\langle v^2 \rangle_\alpha \equiv \int v^2 f_{\alpha 0}(v^2, v_z)\, d\mathbf{v}$$

The particle drift v_{drift} is proportional to the actual velocity of a particle $v_\perp$, while the fluid velocity $\mathbf{V}_\alpha$ is proportional to the *average* velocity of the particles. The fluid velocity is associated with the electric current induced by the magnetic gradient through Maxwell's equations.

$$\frac{4\pi}{c} \sum_\alpha q_\alpha n_\alpha \mathbf{V}_\alpha = -\frac{dB_z}{dx} \hat{\mathbf{y}}$$

For the case of a plasma with only density gradients [(8.15.11)], this current is produced by a gradient in the number of guiding centers (Fig. 8.15.2). For a Maxwell-Boltzmann distribution with density $n = n_{\alpha 0}(1 - \varepsilon'_\alpha x)$, this speed is

$$\mathbf{V}_\alpha = \frac{1}{n_\alpha} \frac{dn_\alpha}{dx} \left(\frac{\kappa T}{m\omega_c} \right)_\alpha \hat{\mathbf{y}} \qquad (8.15.16)$$

There are waves associated with these two velocities, which are characteristic of nonuniform plasmas in a **B** field, just as there were waves associated with cyclotron motion and other properties of the plasma particles. The waves associated with drift motion in a plasma are called *drift waves*.

The method of characteristics applied to this case gives the development of a perturbed distribution $f_{\alpha 1}$ in terms of the particle orbits in the unperturbed fields. These orbits satisfy

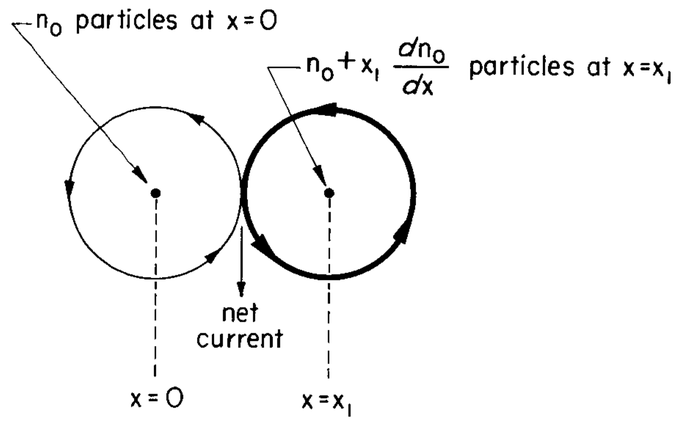


FIGURE 8.15.2
Diagram of particle orbits in a plasma with a density gradient, showing how drift currents result from such gradients.

$$\frac{d\mathbf{x}'}{dt'} = \mathbf{v}'$$

$$\frac{d\mathbf{v}'}{dt'} = \frac{q_\alpha}{m_\alpha} \frac{\mathbf{v}' \times \mathbf{B}_0}{c} (1 + \varepsilon x') \qquad (8.15.17)$$

$$\mathbf{x}'(t' = t) = \mathbf{x} \qquad \mathbf{v}'(t' = t) = \mathbf{v}$$

and are found by iterating (8.15.17), in the (assumed) small parameter $\varepsilon = (1/B)(dB/dx)$. This is a tedious but straightforward exercise [i.e., first solve (8.15.17) with $\varepsilon = 0$, then replace $\varepsilon x'$ by the zero-order solution $x'(t')$]. This leads to

$$v'_y = v_\perp \sin\left[\theta - \omega_{c\alpha}\left(1 + \varepsilon x + \frac{\varepsilon v_y}{\omega_{c\alpha}}\right)\tau\right] + \frac{\varepsilon v_\perp^2}{2\omega_{c\alpha}}$$
$$\times [1 - 2(\cos \omega_{c\alpha}\tau) \cos^2 \theta + \cos 2(\theta - \omega_{c\alpha}\tau) - (\sin 2\theta) \sin \omega_{c\alpha}\tau]$$

$$v'_x = v_\perp \cos\left(\theta - \omega_{c\alpha}\left(1 + \varepsilon x + \frac{\varepsilon v_y}{\omega_{c\alpha}}\right)\tau\right] + \frac{\varepsilon v_\perp^2}{2\omega_{c\alpha}} \qquad (8.15.18)$$
$$\times [(\sin 2\theta) \cos \omega_{c\alpha}\tau - \sin 2(\theta - \omega_{c\alpha}\tau) + 2(\sin \omega_{c\alpha}\tau) \cos^2 \theta]$$

$$v'_z = v_z$$

where

$$\tau \equiv t' - t$$

and the velocity $\mathbf{v}$ has been written in polar coordinates,

$$v_y = v_\perp \sin \theta$$
$$v_x = v_\perp \cos \theta$$
$$v_\perp = (v_x^2 + v_y^2)^{1/2}$$

The orbits $\mathbf{x}'$ are obtained from $\mathbf{v}'$ by integration.

Problem 8.15.3 Find $\mathbf{x}'(\tau)$ from $\mathbf{v}'$. Verify that (8.15.18) is correct for $\mathbf{v}'$. Be careful that your solutions satisfy the boundary conditions $\mathbf{x}'(0) = \mathbf{x}$, $\mathbf{v}'(\tau = 0) = \mathbf{v}$. ////

8.15.2 Local Approximation to the Perturbed Distribution Function

The perturbed distribution, as obtained in terms of the orbits (8.15.18), is

$$f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t) = -\frac{q_\alpha}{m_\alpha}\int_{-\infty}^{0}\left[\mathbf{E}_1(\mathbf{x}', \tau + t) + \frac{\mathbf{v}' \times \mathbf{B}_1(\mathbf{x}', \tau + t)}{c}\right] \cdot \nabla_{\mathbf{v}'} f_{\alpha 0}(\mathbf{x}', \mathbf{v}')\, d\tau$$

and, together with Maxwell's equations,

$$\nabla \times \mathbf{B}_1(\mathbf{x}, t) = \frac{1}{c}\frac{\partial \mathbf{E}_1}{\partial t} + \frac{4\pi}{c}\sum_\alpha q_\alpha \bar{n}_\alpha \int \mathbf{v} f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t)\, d\mathbf{v}$$

it determines the time behavior of the fields. Because the equilibrium distribution $f_{\alpha 0}$ is a function of x, it is not possible to express the fields and perturbed distribution exactly in terms of a single harmonic dependence $\exp[i(\mathbf{k}\cdot\mathbf{x} - \omega\tau)]$. This can be seen by noting that the assumption $\mathbf{E}_1 = \bar{\mathbf{E}}_\mathbf{k} \exp[i(\mathbf{k}\cdot\mathbf{x} - \omega\tau)]$ leads to a perturbed distribution $f_{\alpha 1} \propto f_{\alpha \mathbf{k}}(x) \exp[i(\mathbf{k}\cdot\mathbf{x} - \omega\tau)]$, so that Poisson's equation $\nabla\cdot\mathbf{E} = 4\pi\sum q_\alpha \bar{n}_\alpha \int f_{\alpha 1}\, d\mathbf{v}_\alpha$ can be satisfied only if the amplitude $\bar{\mathbf{E}}_\mathbf{k}$ is in turn a function of x; that is,

$$\mathbf{E}_1 = \bar{\mathbf{E}}_\mathbf{k}(\mathbf{x}) \exp(i\mathbf{k}\cdot\mathbf{x}) \exp(-i\omega t)$$
$$f_{\alpha 1} = f_{\alpha \mathbf{k}}(\mathbf{x}, \mathbf{v}) \exp(i\mathbf{k}\cdot\mathbf{x}) \exp(-i\omega t)$$

In this case Maxwell's equations do not produce an algebraic dispersion relation for $\omega(\mathbf{k})$, as in the case of a spatially uniform plasma, but rather lead to an integrodifferential equation for the field amplitudes $\bar{\mathbf{E}}_\mathbf{k}(\mathbf{x})$. The condition that this integrodifferential equation have a bounded solution (the perturbed fields $\bar{\mathbf{E}}_\mathbf{k}$ must vanish at $x \to \infty$) leads, in principle, to an equation for the eigenvalues $\omega(\mathbf{k})$. Fortunately, for the wide class of problems in which the gradients in field and density are weak, i.e., when

$$\frac{1}{k}\frac{1}{n}\frac{dn}{dx} \ll 1$$

and

$$a_{ci}\frac{1}{n}\frac{dn}{dx} \ll 1$$

the solution of the integrodifferential equation has been shown to be identical with the result produced by a "local approximation."

In the local approximation, the spatial variation of the perturbed fields and distribution is *assumed* to be nearly harmonic; i.e.,

$$E_1 \approx \bar{\mathbf{E}}_\mathbf{k} \exp(i\mathbf{k}\cdot\mathbf{x} - i\omega t)$$
$$f_1 \approx \bar{f}_\mathbf{k} \exp(i\mathbf{k}\cdot\mathbf{x} - i\omega t)$$

where $\bar{\mathbf{E}}_\mathbf{k}$ and $\bar{f}_\mathbf{k}$ are constants. Maxwell's equations then lead to a dispersion relation of the sort

$$D(\mathbf{k}, \omega, x) = 0$$

This equation is then evaluated at point $x = x_0$, at which the density gradients are largest. If the equilibrium varies with x as

$$n_\alpha = n_{\alpha 0}(1 - \varepsilon'_\alpha x)$$
$$\mathbf{B} = B_0(1 + \varepsilon x)\hat{\mathbf{z}}$$

the gradients are the same at all points x, and the dispersion relation $D(\mathbf{k}, \omega, x) = 0$ is evaluated at $x = 0$ for simplicity.

The local approximation is a major simplification of the drift-wave problem, and will be employed here without further proof, referring the reader to the extensive literature on the subject.[1,2] The physical basis for the local approximation is clear. A mode that is localized in a distance much less than the scale of the gradients,

$$\lambda \approx \frac{2\pi}{k} \ll \left(\frac{1}{n}\frac{dn}{dx}\right)^{-1}$$

sees a plasma with nearly constant drift velocity. Since the drift velocities are the unique features of the inhomogeneous plasma, an approximation that treats the drifts accurately will accurately reproduce effects due to the drifts. Thus, if the drifts are nearly constant over many wavelengths, it is generally valid to ignore all other space dependence of the equilibrium; i.e., set $x = 0$ in $D(\mathbf{k}, \omega, x)$.

Assuming that the perturbed fields and distribution have space-time dependence $\exp[i(\mathbf{k}\cdot\mathbf{x} - \omega t)]$, the perturbed distribution is given by

$$f_{\alpha \mathbf{k}} = -\frac{q_\alpha}{m_\alpha}\int_{-\infty}^{0}\left(\bar{\mathbf{E}}_\mathbf{k} + \frac{\mathbf{v}' \times \bar{\mathbf{B}}_\mathbf{k}}{c}\right) \cdot \nabla_{\mathbf{v}'} f_{\alpha 0}\left(v_\perp^2, v_z, v_y + \frac{q_\alpha}{m_\alpha c}\int B_z\, dx\right)$$
$$\times \exp[i(\mathbf{k}\cdot\mathbf{X} - \omega\tau)]\, d\tau \quad (8.15.19)$$

[1] A. B. Mikhailovskii, Oscillations of an Inhomogeneous Plasma, in M. A. Leontovich (ed.), "Reviews of Plasma Physics," vol. 3,p. 159, Consultants Bureau, New York, 1966.

[2] N. A. Krall, Drift Waves, in A. Simon and W. B. Thompson (eds.), "Advances in Plasma Physics," vol. 1, Wiley, New York, 1968.

The orbits (8.15.18) allow the integral (8.15.19) to be evaluated explicitly. Although this integration looks formidable, it is made tractable by the identity

$$\exp(iR\sin\theta) = \sum_{l=-\infty}^{\infty} J_l(R)\exp(il\theta) \qquad (8.15.20)$$

which allows $\exp(i\mathbf{k}\cdot\mathbf{x}')$ to be replaced by a function whose only time dependence is exponential, $\exp(il\omega_{c\alpha}\tau)$.

With the identity (8.15.20), it can be shown that the higher harmonic terms in the orbits (8.15.18) can be neglected with little error if the gradients in density are small. In particular, $\exp(i\mathbf{k}\cdot\mathbf{x}')$ includes oscillating terms of the type

$$\exp\left(i\frac{k\varepsilon v_\perp^{\,2}}{\omega_c^{\,2}}\sin 2\omega_c\tau\right) \qquad (8.15.21)$$

For weak gradients,

$$\frac{k\varepsilon}{\omega_{c\alpha}^{\,2}}\frac{\kappa T_\alpha}{m_\alpha} \ll 1$$

and to good approximation (because $\sin(\omega_c\tau)$ is small when integrated over τ),

$$\exp\left(i\frac{k\varepsilon v_\perp^{\,2}}{\omega_c^{\,2}}\sin 2\omega_c\tau\right) = 1 + i\frac{k\varepsilon v_\perp^{\,2}}{\omega_c^{\,2}}\sin 2\omega_c\tau + \cdots \approx 1$$

The term $\exp(ik\varepsilon v_\perp^{\,2}\tau/2\omega_c)$ in which the argument of the exponential is nonoscillating must not be replaced by unity. This term contains the secular effect of the particle drift arising from the gradient of $\mathbf{B}_0$. Setting terms of the form (8.15.21) equal to unity and writing $\mathbf{k}$ in cylindrical coordinates [recall that $\mathbf{B}_0 = \hat{\mathbf{z}}B_0(x)$],

$$\mathbf{k} = \hat{\mathbf{x}}k_\perp\cos\psi + \hat{\mathbf{y}}k_\perp\sin\psi + \hat{\mathbf{z}}k_\parallel$$

greatly simplifies the perturbed distribution function, which becomes

$$f_{\alpha\mathbf{k}} = -\frac{q_\alpha}{m_\alpha}\sum_{l,m}\int_{-\infty}^{0}\left[\left(\bar{\mathbf{E}}_\mathbf{k} + \frac{\mathbf{v}'\times\bar{\mathbf{B}}_\mathbf{k}}{c}\right)\cdot\nabla_{\mathbf{v}'}f_{\alpha 0}\right]J_l\left(\frac{k_\perp v_\perp}{\omega_{c\alpha}}\right)J_m\left(\frac{k_\perp v_\perp}{\omega_{c\alpha}}\right)$$
$$\times\exp[i(l-m)(\psi-\theta)]\exp\left[i\left(k_\parallel v_\parallel + \tfrac{1}{2}k_y\frac{\varepsilon v_\perp^{\,2}}{\omega_{c\alpha}} - \omega + l\omega_{c\alpha}\right)\tau\right]d\tau \qquad (8.15.22)$$

A further simplification follows from the fact that

$$\nabla_{\mathbf{v}'}f_{\alpha 0}(H, L) = \frac{\partial f_{\alpha 0}}{\partial H}\nabla_{\mathbf{v}'}H + \frac{\partial f_{\alpha 0}}{\partial L}\nabla_{\mathbf{v}'}L$$

where H and L are constants of the motion such as those in (8.15.5). If H and L are constants, then $\partial f_{\alpha 0}/\partial H$ and $\partial f_{\alpha 0}/\partial L$ are also constants of the motion, and

can be removed from the time integration. This means that the details of $f_{\alpha 0}$ are not needed in order to perform the orbit integration (8.15.29).

This perturbed distribution is the starting point for many calculations of the properties of drift waves with different polarizations and in various frequency ranges.[1] To illustrate the fact that new types of waves, not found in the homogeneous plasma, are contained in (8.15.22), it is helpful to make the electrostatic approximation, setting $\mathbf{B}_1 \to 0$ as was done in Sec. 8.3. Of course, one does not have the freedom to specify the eigenvectors arbitrarily; these are determined by the self-consistent Vlasov-Maxwell equations. *However, an examination of the complete solutions has verified that under certain conditions* ($\beta_\alpha = 8\pi n_\alpha\kappa T_\alpha/B^2 \ll 1$, $\omega_p^{\,2}/k^2c^2 \ll 1$), *the assumption* $\mathbf{B}_1 \to 0$ *is in fact a good approximation.* It should be noted that the direction of propagation $\mathbf{k}/|\mathbf{k}|$ *can* be freely chosen, and this is done to simplify the problem further. Although a particular choice of $\mathbf{k}/|\mathbf{k}|$ is a good way of solving the equations, this choice determines which plasma features are reflected in the wave behavior. A further simplification results from restricting the frequency range of the waves examined. Taking the low-frequency limit (ω less than the cyclotron frequency of any plasma particles) results in a great simplification, again at the expense of more complete information about plasma behavior.

Problem 8.15.4 Find the dispersion relation for a purely electromagnetic low-frequency drift wave, propagating perpendicular to the confining field $\mathbf{B}_0$; that is,

$$\mathbf{k} = k\hat{\mathbf{y}} \qquad \mathbf{E}_1 = \hat{\mathbf{z}}\bar{E}_k e^{i(ky-\omega t)} \qquad \omega \ll \omega_{ci} \qquad ////$$

8.16 LOW-FREQUENCY ELECTROSTATIC WAVES IN AN INHOMOGENEOUS MAGNETIZED PLASMA

A plasma with a density gradient (confined by a magnetic field) can be described, as discussed in the preceding section, by the distribution (with $\varepsilon' = \varepsilon_e = \varepsilon_i$)

$$f_{\alpha 0} = \left(\frac{m}{2\pi\kappa T}\right)_\alpha^{3/2}\exp\left(-\frac{m_\alpha v^2}{2\kappa T_\alpha}\right)\left[1 - \varepsilon'\left(x + \frac{v_y}{\omega_{c\alpha}}\right)\right] \qquad (8.16.1)$$

$$\mathbf{B}_0 = \hat{\mathbf{z}}B_0(1 + \varepsilon x) \qquad (8.16.2)$$

[1] Loc. cit.

In order that $f_{\alpha 0}$ and $\mathbf{B}_0$ be self-consistent, Maxwell's equation $\nabla \times \mathbf{B}_0 = 4\pi \mathbf{J}_0/c$ requires that

$$\frac{\varepsilon'}{\varepsilon} = \frac{B^2}{4\pi \sum n_\alpha \kappa T_\alpha} \equiv \frac{2}{\beta} \tag{8.16.3}$$

Thus, in a low-temperature plasma, the density gradients are stronger than the magnetic gradient. Low-frequency electrostatic perturbations ($\omega \ll \omega_{ci}$, $\mathbf{B}_1 \to 0$) propagate as waves in such a plasma. The perturbed distribution associated with these waves is, from (8.15.22),

$$f_{\alpha \mathbf{k}} = -\sum_{l,m} \frac{q_\alpha \mathbf{k} \cdot \bar{\mathbf{E}}_\mathbf{k}}{m_\alpha k^2} f_{\alpha 0}\left(v_\perp^2, v_\parallel, x + \frac{v_y}{\omega_{c\alpha}}\right) \int_{-\infty}^{0} \left(-\frac{m_\alpha}{\kappa T_\alpha} \mathbf{k} \cdot \mathbf{v}' - \frac{k_y}{\omega_{c\alpha}} \varepsilon'\right) J_l J_m$$
$$\times \exp\left[i(l-m)(\psi - \theta)\right] \exp\left[i\left(k_\parallel v_\parallel + k_y \frac{\varepsilon v_\perp^2}{2\omega_{c\alpha}} - \omega + l\omega_{c\alpha}\right)\tau\right] d\tau$$

where the argument of the Bessel functions J_l, J_m is $(k_\perp v_\perp/\omega_{c\alpha})$; as mentioned earlier, since f_α is a function of constants of the motion $[f_\alpha(H', L') = f_\alpha(H, L)]$, it has been removed from the integrand. The associated electric field is

$$\left[\text{using } \int_{-\infty}^{0} i\mathbf{k} \cdot \mathbf{v} \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega\tau)\, d\tau = 1 + i\omega \int_{-\infty}^{0} \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega\tau)\, d\tau\right]$$

$$ik\bar{E}_\mathbf{k} = \sum_\alpha 4\pi \bar{n}_\alpha q_\alpha \int f_{\alpha \mathbf{k}}\, d\mathbf{v}$$
$$= \sum_\alpha 4\pi \frac{\bar{n}_\alpha q_\alpha^2}{m_\alpha} \frac{m_\alpha}{\kappa T_\alpha} \frac{\bar{E}_\mathbf{k}}{ik}$$
$$\times \left\{1 + \sum_l \int \left[\omega\left(1 - \frac{l\varepsilon'}{k}\right) + k_y \frac{\kappa T_\alpha}{m_\alpha} \frac{\varepsilon_\alpha'}{\omega_{c\alpha}}\right] \cdot \frac{J_l^2(k_y v_\perp/\omega_{c\alpha}) f_{\alpha 0}(v_\perp^2, v_\parallel, \xi_y)\, d\mathbf{v}}{k_\parallel v_\parallel + l\omega_{c\alpha} + \dfrac{k_y \varepsilon v_\perp^2}{2\omega_{c\alpha}} - \omega}\right\} \tag{8.16.4}$$

where (8.16.1) has been used explicitly for $f_{\alpha 0}$ in simplifying the double sum over Bessel functions, noting that (because of the θ integration)

$$\sum_{l,m} \int J_l J_m \exp[i(l-m)(\theta - \psi)]\, d\mathbf{v}$$
$$= \sum_{l,m} \int v_\perp J_l J_m \exp[i(l-m)(\theta - \psi)]\, dv_\perp\, d\theta\, dv_z$$
$$= \sum_l \int v_\perp J_l^2\, dv_z\, d\theta\, dv_\perp$$

Equation (8.16.4) gives the dispersion relation for electrostatic drift waves (neglecting ε'/k for weak gradients).

$$-k^2 = \sum_\alpha \frac{m_\alpha \omega_{p\alpha}^2}{\kappa T_\alpha} \left[1 + \sum_l \int \frac{J_l^2\left(\dfrac{k_\perp v_\perp}{\omega_{c\alpha}}\right)\left(\omega + \dfrac{\kappa T_\alpha}{m_\alpha} \dfrac{k_y \varepsilon'}{\omega_{c\alpha}}\right) f_{\alpha 0}(v_\perp^2, v_\parallel, \xi_y)}{k_\parallel v_\parallel + k_y \varepsilon v_\perp^2/2\omega_{c\alpha} - \omega + l\omega_{c\alpha}}\, d\mathbf{v}\right] \tag{8.16.5}$$

It is at this point that the local approximation is employed, evaluating (8.16.5) at $x = 0$. The first-order effects of the particle drifts and fluid velocity are present in this limit, as seen by examining (8.16.5). The effect of the ∇B drift is evident from the term $k_y \varepsilon v_\perp^2/\omega_c$ appearing in the denominator of (8.16.5). This term allows a resonance $\omega \approx k_y \varepsilon v_\perp^2/2\omega_c = k_y v_D$ in the velocity integral for waves with phase velocity $v_{ph} = \omega/k$ equal to the drift speed. The velocity of the fluid element [since $-\varepsilon' = (1/n)(dn/dx)$],

$$V_\alpha = \frac{\kappa T_\alpha}{m_\alpha \omega_{c\alpha}} \frac{1}{n} \frac{dn}{dx}$$

appears in the numerator of (8.16.5).

Problem 8.16.1 Verify (8.16.4). Show that the $l\varepsilon'/k$ term in (8.16.4) comes from $\int J_l J_m v_y/\omega_c\, d\theta \cdots$. ////

Equation (8.16.5) includes the familiar Langmuir oscillations and cyclotron waves (this is easily demonstrated by neglecting the gradients ε and ε'); to show the new features introduced by the drifts, it is useful to consider a few special cases.

8.16.1 Low-frequency Drift Waves Propagating Perpendicular to $\mathbf{B}_0$ in a Low β Plasma, ($\omega \ll \omega_{ci}$, $k_\parallel = 0$, $\mathbf{k} = k\hat{\mathbf{y}}$)

In this case the $l \neq 0$ terms of the sum $\sum_l$ are small by $\omega/l\omega_{c\alpha}$ compared with the $l = 0$ term, and can be neglected. By neglecting $V_D = \varepsilon V_\perp^2/2\omega_{c\alpha}$, a solution can be found that satisfies $\omega/k \approx V_\alpha \approx (2/\beta)V_D \gg V_D$, and justifies the neglect of $\varepsilon v_\perp^2/2\omega_{c\alpha}$. With these approximations, the dispersion relation is

$$-k^2 = \sum_\alpha \frac{\omega_{p\alpha}^2 m_\alpha}{\kappa T_\alpha} \left[1 - I_0\left(\frac{k^2 \kappa T_\alpha}{\omega_{c\alpha}^2 m_\alpha}\right) \exp\left(-\frac{k^2 \kappa T_\alpha}{\omega_{c\alpha}^2 m_\alpha}\right)\left(1 + \frac{\kappa T_\alpha}{m_\alpha} \frac{k\varepsilon'}{\omega_{c\alpha}\omega}\right)\right] \tag{8.16.6}$$

where

$$\int_0^\infty J_0^2\left(\frac{kv_\perp}{\omega_c}\right) \exp\left(-\frac{mv_\perp^2}{2\kappa T}\right) \frac{m}{\kappa T} v_\perp\, dv_\perp = I_0\left(\frac{k^2 \kappa T}{\omega_c^2 m}\right) \exp\left(-\frac{k^2 \kappa T}{\omega_c^2 m}\right)$$

has been used. In the long-wavelength limit ($k^2a^2 = k^2\kappa T/\omega_c{}^2 m < 1$), where a is the cyclotron radius, $I_0 e^{-b_\alpha}$ may be approximated by $I_0 e^{-b_\alpha} \approx 1 - b_\alpha \approx 1$. This gives neglecting $b_e \ll b_i$,

$$\omega = -\frac{k\left(\frac{\kappa T}{m}\frac{\varepsilon'}{\omega_c}\right)_i}{1 + B^2/4\pi n m_i c^2} = \frac{kV_i}{1 + B^2/4\pi n m_i c^2} \tag{8.16.7}$$

This wave moves in the direction of the ion drifts with nearly the speed of the ion fluid velocity V_i, which exceeds the *particle* drift by $2/\beta$. This wave is weakly Landau-damped (Prob. 8.16.3). In Chap. 9 it is shown that this wave is driven unstable by a gravitational force opposite to the density gradient. This wave is also driven unstable by temperature gradients which can be included in the derivation by an approximate choice ε'_α, as in (8.15.12) and (8.15.13).

A second wave with

$$\omega \approx k_y \frac{\kappa T}{m\omega_c}\frac{1}{B}\frac{dB}{dx}$$

exists, but it can be shown to be heavily damped, and in the spirit of time-asymptotic theory should not be considered a normal mode.

Problem 8.16.2 Demonstrate that the solution (8.16.7) justifies the assumption that $\varepsilon v_\perp{}^2/2\omega_c$ could be ignored in obtaining that solution. ////

Problem 8.16.3 Show that the drift wave (8.16.7) is weakly damped, and find the damping decrement. ////

8.16.2 Low-frequency Drift Waves Propagating Oblique to $\mathbf{B}_0$

In this case, again neglecting $l \neq 0$ in $\sum_l J_l{}^2$, and again looking for waves with $\omega > kV_D$, that is, neglecting $k_y \varepsilon v_\perp{}^2/2\omega_c$ in the denominator of (8.16.5), the dispersion relation becomes

$$-k^2 = \sum_\alpha \frac{m_\alpha \omega_{p\alpha}{}^2}{\kappa T_\alpha}\left[1 + I_0\left(\frac{k^2\kappa T}{m\omega_c{}^2}\right)e^{-k^2\kappa T/\omega_c{}^2 m}\left(1 + \frac{\kappa T}{m}\frac{k_y\varepsilon'}{\omega_c\omega}\right)\right.$$
$$\left.\cdot\frac{\omega}{|k_\parallel|}\sqrt{\frac{m}{2\kappa T}}\frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{e^{-x^2}\,dx}{x - (\omega/|k_\parallel|)\sqrt{m/2\kappa T}}\right]_\alpha \tag{8.16.8}$$

The integral which occurs in (8.16.8),

$$Z(\xi) = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{e^{-x^2}\,dx}{x - \xi} \tag{8.16.9}$$

arises frequently in the theory of waves in a maxwellian plasma, and has been tabulated.[1] In the limits of large and small arguments, it is given by

$$Z(\xi) = \begin{cases} -2\xi + \frac{4}{3}\xi^2 - \cdots + i\sqrt{\pi}\,e^{-\xi^2} & |\xi| \ll 1 \\ -\frac{1}{\xi} - \frac{1}{2\xi^3} - \frac{3}{4\xi^5} - \cdots & |\xi| \gg 1 \end{cases} \tag{8.16.10}$$

The parameter ξ_α is just the ratio of the parallel phase velocity of the wave ($\omega/|k_\parallel|$) to the thermal speed of the particles $\sqrt{2\kappa T_\alpha/m_\alpha}$. An interesting class of drift waves are those that have a phase velocity along $\mathbf{B}_0$ that exceeds the ion thermal speed but is less than the electron thermal speed, so that $\xi_i \gg 1$, $\xi_e < 1$. For these waves, (8.16.8) becomes

$$\omega_r \approx \frac{\left(k_y\frac{\kappa T_e}{m_i}\frac{\varepsilon'}{\omega_{ci}}\right)I_0(b)\,e^{-b}}{1 + \frac{T_e}{T_i} - \frac{T_e}{T_i}I_0(b)e^{-b} + k^2\lambda_D{}^2}$$

$$\omega_i = \left[\left(1 + \frac{T_e}{T_i}\right)(1 - I_0 e^{-b}) + k^2\lambda_{De}{}^2\right]\frac{\sqrt{\pi}\,\omega_r{}^2 e^{-\xi_e{}^2}}{k_\parallel\sqrt{2\kappa T_e/m_e}\,I_0 e^{-b}} \tag{8.16.11}$$

$$b \equiv k^2 a_{ci}{}^2 = \frac{k^2\kappa T_i}{m_i\omega_{ci}{}^2}$$

where the imaginary part of ω comes from the resonance of these waves with the electrons.

Problem 8.16.4 Show that (8.16.11) is the correct solution of (8.16.8) in the limit

$$\sqrt{\frac{\kappa T_i}{m_i}} \ll \frac{\omega}{|k_\parallel|} \ll \sqrt{\frac{\kappa T_e}{m_e}} \qquad ////$$

Problem 8.16.5 Use the approximation

$$D_r(k, \omega_r) = 0 \qquad \omega_i = \frac{D_i(k, \omega_r)}{-\partial D_r/\partial\omega_r}$$

to verify (8.16.11), and find the ion corrections. ////

[1] B. F. Fried and S. Conte, "The Plasma Dispersion Function," Academic, New York, 1961.

The dominant features of the drift wave (8.16.11) are:

1 It has a phase velocity ω/k_y comparable with the velocity of fluid elements V_α.
2 It has a more rapid phase velocity along $\mathbf{B}_0$, since $\omega/|k_\parallel| > \sqrt{\kappa T_i/m_i}$, while, because the gradient is assumed to be weak, $\omega/k_y < \sqrt{\kappa T_i/m_i}$.
3 It is *unstable*. From (8.16.11) the imaginary part of ω is positive, and this wave

$$[\mathbf{E}_1 = \bar{\mathbf{E}}_\mathbf{k} \exp[i(\mathbf{k}\cdot\mathbf{x} - \omega\tau)] \approx \bar{\mathbf{E}}_\mathbf{k} \exp[i(\mathbf{k}\cdot\mathbf{x} - \omega_r\tau)\exp(\omega_i t)]$$

grows with rate $\approx\omega_i$.

The waves, which are feeding on the particle currents that exist because the equilibrium is nonuniform $[n(x), B(x)]$, gain energy at the expense of this nonthermal feature, i.e., the gradient $\nabla n(x)$. Instead of Landau damping, this wave exhibits Landau growth! Instabilities are treated systematically in Chap. 9. The present example has arisen in a natural way from the study of wave properties of nonuniform plasmas and shows that not only do normal modes exist which reflect the individual properties of a plasma stationary state, but these modes can grow at the expense of nonthermal features of the metaequilibria. Realistically, *all* laboratory or natural plasmas have density gradients, and the growth of the wave described here is called a *universal instability*.

A detailed discussion of the relation between the drift waves, Alfvén waves, and ion-sound waves was presented in Chap. 5, and the student is invited to derive those results rigorously from Vlasov theory, starting with (8.15.22).

8.17 NONLINEAR ELECTROSTATIC WAVES (BGK WAVES)

A general class of equilibria in which particles play a major role in determining the equilibria fields was discovered by Bernstein, Greene, and Kruskal.[1] Their idea was that, for example, a localized positive potential could be made consistent with a local excess density of electrons. This is possible because, if the particles have a continuous distribution of kinetic energy, some particles will be trapped in the positive-potential well. By choosing the number and sign of trapped particles, a self-consistent distribution can be found for virtually any potential distribution. Even if there are no trapped (localized) particles, it is is still possible to find large potentials consistent with particle distributions,

[1] I. B. Bernstein, J. M. Greene, and M. D. Kruskal, *Phys. Rev.*, **108**: 546 (1957).

but the amplitude of these potentials is then limited to $e\phi_{\max} < \frac{1}{2}mv_{\min}^2$, the minimum kinetic energy of particles in the distribution. Because these equilibria allow electrostatic fields with spatial structure, they are referred to as waves, although that term is much too restrictive.

The Bernstein-Greene-Kruskal (BGK) problem is formulated in the Vlasov-Maxwell framework of equations, since the idea on which it is based involves details of the distribution of particles. A steady state including electrostatic fields is found by taking $\partial f_\alpha/\partial t = 0$, but $\phi_0 \neq 0$. In one dimension the equations for $f_\alpha(x)$ and $\phi(x)$ are

$$v_x \frac{\partial f_\alpha}{\partial x} - \frac{q_\alpha}{m_\alpha}\frac{\partial\phi}{\partial x}\frac{\partial f_\alpha}{\partial v_x} = 0 \qquad (8.17.1)$$

$$\frac{\partial^2\phi}{\partial x^2} = -4\pi\left(\sum_\alpha q_\alpha \bar{n}_\alpha \int_{-\infty}^{\infty} f_\alpha\, dv_x\right) \qquad (8.17.2)$$

The general solution to (8.17.1) is

$$f_\alpha = f_\alpha\left(v_x^2 + \frac{2q_\alpha\phi}{m_\alpha}\right)$$

Thus (8.17.2) is implicitly a differential-integral equation for ϕ; it is instructive to solve (8.17.2) for several cases.

8.17.1 No Trapped Particles

$$f_\alpha = V_{\alpha 0}\,\delta\left(v_x^2 + \frac{2q_\alpha\phi}{m_\alpha} - V_{\alpha 0}^2\right)$$

$$V_{\alpha 0}^2 \gg \left|\frac{2q_\alpha\phi_{\max}}{m_\alpha}\right|$$

This case is that of *no* trapped particles, since the kinetic energy is specified to be greater than the maximum potential energy. In this distribution all particles have the same kinetic energy. Equation (8.17.2) for ϕ becomes, with this distribution, and with an equal total number of ions and electrons,

$$\frac{d^2\phi}{dx^2} = -4\pi\bar{n}_0 e\left(\frac{V_{i0}}{\sqrt{V_{i0}^2 - 2e\phi/m_i}} - \frac{V_{e0}}{\sqrt{V_{e0}^2 + 2e\phi/m_e}}\right) \qquad (8.17.3)$$

$$\frac{d^2\phi}{dx^2} = -\frac{dV(\phi)}{d\phi}$$

$$V(\phi) = -4\pi\bar{n}_0\left(V_{i0}\,m_i\sqrt{V_{i0}^2 - \frac{2e\phi}{m_i}} + V_{e0}\,m_e\sqrt{V_{e0}^2 + \frac{2e\phi}{m_e}}\right) \qquad (8.17.4)$$

where $\bar{n}_0 = \bar{n}_e = \bar{n}_i$.

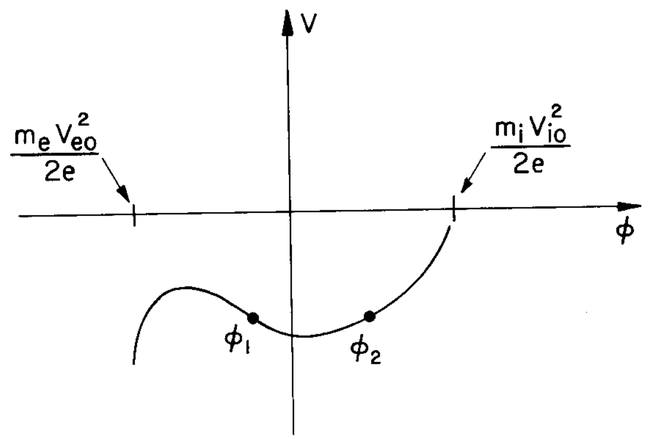


FIGURE 8.17.1
Sketch of V as a function of ϕ from Eq. (8.17.4).

The last form shows that ϕ behaves like the coordinate of a particle moving in a potential well $V(\phi)$. The shape of $V(\phi)$ is shown qualitatively [from Eq. (8.17.4)] in Fig. 8.17.1. Thus there are solutions to (8.17.3) in which ϕ oscillates in the "potential" $V(\phi)$. For example, a typical solution ϕ oscillates between the points labeled ϕ_2 and ϕ_1 in Fig. 8.17.1. The form of ϕ for this solution is shown in Fig. 8.17.2. The solution is subject to the no-trapping condition $\frac{1}{2}m_i V_{i0}^2 > e\phi_2$, $\frac{1}{2}m_e V_{e0}^2 > -e\phi_1$. For ϕ small, the nature of this solution may be shown explicitly. Expanding (8.17.3) in ϕ,

$$\frac{d^2\phi}{dx^2} = -4\pi\bar{n}_0 e^2\left(\frac{1}{m_i V_{i0}^2} + \frac{1}{m_e V_{e0}^2}\right)\phi$$

$$\Lambda_D^2 \frac{d^2\phi}{dx^2} = -\phi \tag{8.17.5}$$

where the scale size

$$\Lambda_D = (4\pi\bar{n}_0 e^2)^{-1/2}\left(\frac{1}{m_i V_{i0}^2} + \frac{1}{m_e V_{e0}^2}\right)^{-1/2}$$

may be viewed as a generalized Debye length, with $\int \frac{1}{2}m_\alpha v^2 f_\alpha \, d\mathbf{v}$ playing the role of a "temperature." This is the equation for simple harmonic motion, with solution

$$\phi = \phi_{\max} \sin \Lambda_D x \tag{8.17.6}$$

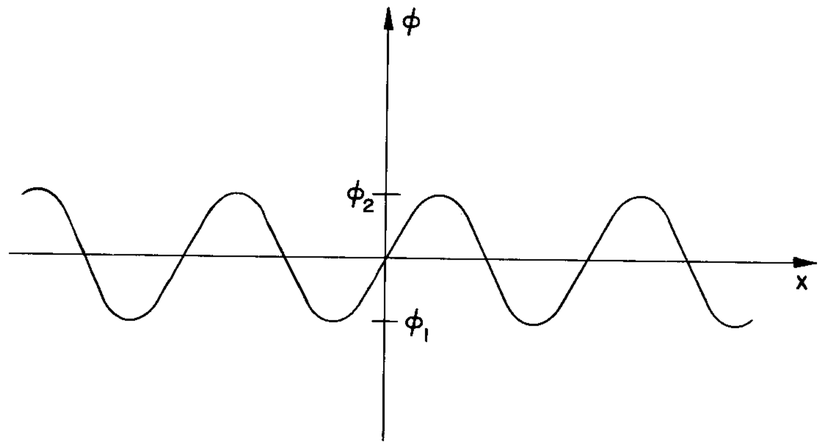


FIGURE 8.17.2
Sketch of ϕ as a function of x.

This solution corresponds to a stationary (in time) spatially oscillating equilibrium which can include large electric fields. However, when viewed in a frame moving with uniform speed U_0, the BGK equilibrium looks like a wave with characteristic wavelength Λ_D. In fact, all discussions of equilibrium states will apply to time-dependent problems if they are stationary in some frame.

8.17.2 Trapped-particle Equilibria

A wide variety of waves of the BGK type can be constructed if the distribution includes particles with less energy than the maximum potential energy. These are trapped particles ($mv^2/2 < e\phi_{\max}$).

A prescription for constructing a distribution function consistent with an *arbitrary* potential $\phi(x)$ can be given by dividing the particles into three classes:

1 Untrapped particles of type α $\quad E > -e\phi_{\min}$
2 Trapped particles of type α $\quad -e\phi_{\min} > E > -e\phi$
3 Other species

where $E = \frac{1}{2}m_\alpha v^2 + q_\alpha \phi$. Note in Fig. 8.17.3, for example, that electrons are trapped in the region of *large* potential, ions in the region of small potential. One approach is to assume that the distribution of ions and untrapped electrons is given, and adjust the distribution of trapped electrons to be consistent with $\phi(x)$, f_e, and f_i. This is done with the aid of Poisson's equation.

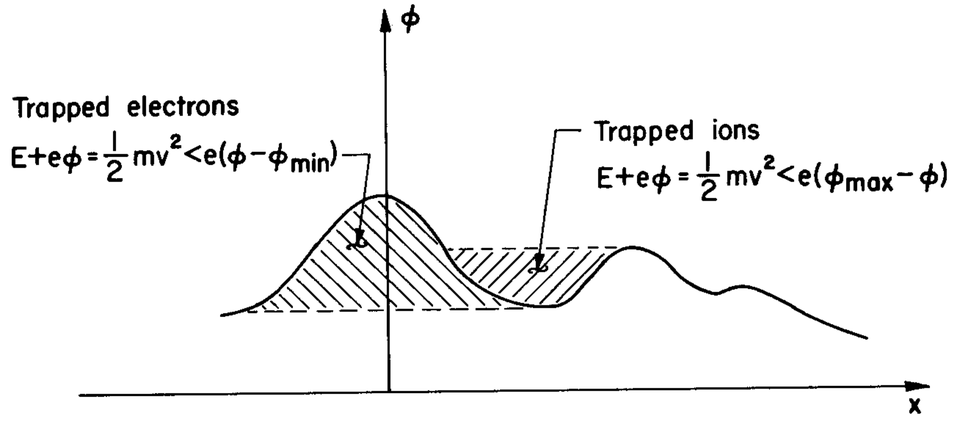


FIGURE 8.17.3
Potential ϕ as a function of position, showing regions of trapped ions and electrons.

$$\frac{d^2\phi}{dx^2} = -4\pi\bar{n}_i e \int_{e\phi}^{\infty} \frac{2f_i(E)}{\sqrt{2m_i(E-e\phi)}}\,dE + 4\pi\bar{n}_e e \int_{-e\phi_{\min}}^{\infty} \frac{2f_e(E)}{\sqrt{2m_e(E+e\phi)}}\,dE + 4\pi\bar{n}_e e \int_{-e\phi}^{-e\phi_{\min}} \frac{2f_e(E)}{\sqrt{2m_e(E+e\phi)}}\,dE \qquad (8.17.7)$$

where the transformation ($E = \frac{1}{2}m_\alpha v^2 + q_\alpha\phi$)

$$\int_{-\infty}^{\infty} f(v)\,dv_x = 2\int_{q_\alpha\phi}^{\infty} \frac{f_\alpha(E)}{\sqrt{2m_\alpha(E-q_\alpha\phi)}}\,dE$$

has been used. The last term in (8.17.7) is the charge density due to the trapped electrons, and can be expressed in terms of ions, untrapped electrons, and any desired potential distribution.

Problem 8.17.1 Show that, for a given (arbitrary) distribution of ions, untrapped electrons, and potential $\phi(x)$, the distribution of trapped electrons which makes the total distribution consistent with $\phi(x)$ is given by

$$2\bar{n}_e f_e(E) = \frac{\sqrt{2m_e}}{\pi}\int_{e\phi_{\min}}^{E} \frac{1}{\sqrt{-V-E}}\frac{dg(V)}{dV}\,dV \qquad E < -e\phi_{\min},$$

where
$$g(v) = \frac{1}{4\pi e^2}\frac{d^2V}{dx^2} + \bar{n}_i\int_{V}^{\infty} \frac{2f_i(E)}{\sqrt{2m_i(E-V)}}\,dE - \bar{n}_e\int_{-e\phi_{\min}}^{\infty} \frac{2f_e(E)}{\sqrt{2m_e(E+V)}}\,dE$$

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Problem 8.17.2 In the one-dimensional case, Poisson's equation for electrostatic BGK waves may be expressed as

$$\frac{\partial^2\phi_0}{\partial x^2}(x) = -\sum_\alpha 4\pi\bar{n}_\alpha q_\alpha \frac{2}{m_\alpha}\int_{q_\alpha\phi_0}^{\infty} \frac{f_{\alpha 0}(E)}{\sqrt{(2/m_\alpha)(E-q_\alpha\phi_0)}}\,dE \qquad (8.17.8)$$

where $E = (m_\alpha/2)V_x^2 + q_\alpha\phi_0$ = particle energy, $V = e\phi$

$\phi_0(x)$ = equilibrium electrostatic potential

$f_{\alpha 0}(E)$ = equilibrium distribution function for component α.

For the case that $f_{\alpha 0}(E)$ is monotone-decreasing, $(\partial f_{\alpha 0}/\partial E)(E) \le 0$, for each component, show that Eq. (8.17.7) does *not* permit spatially periodic solutions for $\phi_0(x)$, that is, solutions with $\phi_0(x) = \phi_0(x+L)$ for some L. [*Hint*: This is a four-liner; see F. Englemann, M. R. Feix, and E. Minardi, *Phys. Fluids*, **13**; 535 (1971).]

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8.18 FLUID WAVES VS. VLASOV WAVES

It is instructive to contrast the results and methods of the Vlasov approach to plasma waves with the fluid approach presented in Chap. 4. The Vlasov approach studies $f(\mathbf{x}, \mathbf{v}, t)$, a function in six-dimensional phase space, while the fluid theory studies $n(\mathbf{x}, t)$ or $\mathbf{V}(\mathbf{x}, t)$, functions of three-dimensional configuration space. What is gained (or lost) by using the more complicated approach?

The fluid approach is simpler than the Vlasov.

The fluid equations can be extended to include collisions by including, for example, the total momentum transferred to the ions by collisions with electrons (and ions of another type, and neutrals) in the equation of motion.

The fluid equations can often be simplified by reasonable assumptions. For instance, $\nabla\cdot\mathsf{P}$ might be replaced by ∇p, where p is a scalar pressure. The collision term might be estimated by νV, where ν is a collision frequency. Charge neutrality $n_e \approx n_i$ might be assumed in many cases.

Without an independent way to calculate the pressure p, the fluid equations are not closed: they have more unknowns than equations. These moment equations form an infinite chain of equations, which must be terminated somewhere by a flash of insight. Such a flash is often evident: zero pressure may be all right when $n\kappa T \ll B_0^2/8\pi$. A scalar adiabatic pressure change may be right for low-frequency behavior, and so on. But in many useful cases, for example, when there is a pressure variation and a spatially varying magnetic field, or when strong-equilibrium electric and magnetic fields are present, there is no

clear way to estimate P, nor any easy way to check whether an estimate of P is a good approximation. In cases when the Vlasov treatment can be used to get $f_{\mathbf{k}}(\mathbf{x}, \mathbf{v}, t)$, the moments $\int f\,d\mathbf{v}$, $\int \mathbf{v} f\,d\mathbf{v}$, $\int \mathbf{vv} f\,d\mathbf{v}$ can be calculated directly, and P written down, thus using the Vlasov solution to *deduce* an equation of state. The fluid equation so deduced can then be used to solve more difficult problems that are not easily solved or cannot be solved by the Vlasov approach. For such a sequence (fluid solutions with $\nabla \cdot \mathsf{P} \approx \nabla p \rightarrow$ correct Vlasov treatment $\rightarrow$ modified fluid equation to agree with Vlasov) the student might look at the set of papers:

M. N. Rosenbluth and C. L. Longmire, *Ann. Phys.*, **1** : 120 (1957).
M. N. Rosenbluth, N. A. Krall, and N. Rostoker, *Nucl. Fusion*, 1962 Suppl. Part 1, p. 143, 1962.
K. V. Roberts and J. B. Taylor, *Phys. Rev. Letters*, **8** : 197 (1962).

The upshot of these papers was that a simple fluid theory with adiabatic pressure in a magnetic field was satisfactory for waves that were low-frequency, long-wavelength, with the ordering $k^2 a_{ci}^2 \ll \omega/\omega_{ci} \ll 1$, but that a modified anisotropic pressure tensor was required to get correct results from the fluid equations when $\omega/\omega_{ci} \ll k^2 a_{ci}^2 \ll 1$.

The fluid equations lose microstructure (particle effects). The fluid equations do not include any velocity integrals

$$\int \frac{f\,d\mathbf{v}}{\omega - \mathbf{k}\cdot\mathbf{v}}$$

This means that there is no damping from particles in response to waves in the fluid picture. Neither are there any "Landau" instabilities, instabilities from particles resonant with waves. The loss of damping is not trivial. The Vlasov theory showed that ion-acoustic waves are heavily damped if $T_e = T_i$. This is missing from the fluid picture, and explains perhaps the difficulty which plagued early attempts to observe pure ion waves. Other particle effects lost are the waves at the cyclotron harmonics. The oscillation $\omega \approx \omega_c$ is common to both the macroscopic and the microscopic theories, since the fluid element experiences the $q(\mathbf{V} \times \mathbf{B})/c$ force; this oscillation appears in both the Vlasov and the fluid pictures. The oscillations $\omega \approx n\omega_c$, $n > 1$, are due to the presence of particles with velocity different from the average speed **V**, that is, are due to a velocity distribution. These are not found in a fluid picture. They *are* found in the laboratory.

In general, the absence of velocity integrals shows the inability of the fluid pictures to cope with finite temperature effects except for very low frequency and long wavelength.

The inability of the fluid theory to discuss velocity-distribution-produced instabilities is clearly a significant loss, since many applications deal with confined plasma, and confinement, in general, leads to some peculiarity in the particle distribution.

8.19 SUMMARY OF WAVELIKE VLASOV STATES

The treatment in this chapter has studied the plasma in several of its common enviroments: (1) field-free, (2) magnetized by constant $\mathbf{B}_0$, (3) confined by a field $\mathbf{B}_0(x)$, and (4) electrostatically confined.

Each case began with a distribution f_0 for electrons and for ions which satisfied $\partial f/\partial t + \mathbf{v}\cdot \partial f/\partial \mathbf{x} + (\mathbf{F}/m)\cdot \partial f/\partial \mathbf{v} = 0$ and which had currents and charge densities consistent with given fields, i.e., a stationary state. In the BGK case, this required a plasma charge density. The field-free and the constant-**B**-field case required *zero* plasma charge density and current density. The case of a confined plasma required an equilibrium current $4\pi\mathbf{J}/c = \nabla \times \mathbf{B}$.

Each case then considered nonstationary states $f = f_0 + f_1(t)$, which were very close to a stationary state ($f_1 \ll f_0$), and from the Vlasov equation showed that these states were wavelike, $f_1 \approx \exp(-i\omega t)\exp[i\mathbf{k}\cdot\mathbf{x}]f_{\mathbf{k}}(\mathbf{v})$; the Vlasov equation gave $\mathsf{D}(\mathbf{k}, \omega)\cdot \mathbf{E}_1(\mathbf{k}, \omega) = \overline{\mathbf{E}}_1(\mathbf{k}, t = 0)$ for the perturbed fields, and the roots (k, ω) of $\mathsf{D}(\mathbf{k}, \omega) = 0$ gave the normal modes of the plasma state. In some cases the properties of the orbits of a single-charged particle in the given equilibrium fields determined the oscillation frequency or wave velocity of the normal modes. Thus, for a magnetized plasma, $eB/m_e c$ and $eB/m_i c$ are frequencies of particle rotation, but they also proved to be the frequencies of oscillation of undamped electrostatic and electromagnetic waves in the plasma. Another example was the drift wave, which propagates at the speed of particle drifts or plasma currents.

Other normal modes followed, not from particle orbits in externally produced fields, but from properties of the plasma as a statistical system of charged particles coupled by their average electric (and magnetic) fields. This electric coupling gives characteristic rigidity (restoring-force) properties that change with frequency: at high frequencies, the ions are immobile, and $\omega = \omega_{pe}$ and $\omega^2 = k^2c^2 + \omega_{pe}^2$ are normal modes; at low frequencies, ions can respond to the restoring force, and the plasma acts like a gas with characteristic stress properties, exhibiting sound waves, i.e., ion sound with one particular (plasma-determined) speed $\sqrt{\kappa T_e/m_i}$, requiring $T_e \gg T_i$. If external fields are present, the rigidity of a plasma can change, and other sound waves appear: magnetic sound $\mathbf{k} \perp \mathbf{B}_0$, Alfvén waves $\mathbf{k} \| \mathbf{B}_0$, and so on. Table 8.19.1 summarizes the oscillation frequency of some of the plasma waves, and the cases to which they apply.

Table 8.19.1 A SAMPLING OF PLASMA WAVES

Equilibrium	ω_r	Polarization	Remarks
Field-free	$\omega^2 = \omega_p^2 + 3k^2\lambda_D^2\omega_p^2$	Electrostatic	Langmuir oscillation; plasma wave
	$\omega^2 = k^2c^2 + \omega_p^2$	Electromagnetic	Transverse waves in a dielectric medium
	$\omega^2 = \dfrac{k^2C_s^2}{1+k^2\lambda_D^2}$	Electrostatic	Ion-sound waves—damped strongly unless $T_e \gg T_i$
	$C_s^2 = \dfrac{\kappa T_e}{m_i}$		When $k\lambda_D > 1$, $\omega \to \omega_{pi}$, a wave at the *ion* plasma frequency
B = constant	$\omega \approx n\omega_c$, $n \geq 1$	Electrostatic	$\mathbf{k} \perp \mathbf{B}_0$, undamped, "Bernstein modes"
	$\omega = \omega_{pe}$	As in the field-free case	$\mathbf{k} \parallel \mathbf{B}_0$
	$\sqrt{\omega_p^2 + k^2c^2}$	As in the field-free case	$\mathbf{k} \parallel \mathbf{B}_0$
	kC_s	As in the field-free case	$\mathbf{k} \parallel \mathbf{B}_0$
	$\omega = kV_A$	Electromagnetic $V_A^2 = \dfrac{B^2}{4\pi n_i m_i}$	Alfvèn wave, low frequency $k \parallel B_0$ (no analog in a $B \to 0$ plasma)
	$\omega = kV_A$	Electromagnetic or mixed polarization	$\mathbf{k} \perp \mathbf{B}_0$ Magnetosonic wave
$B = B_0(1 + \varepsilon x)$	$\omega = k\left(\dfrac{\kappa T}{m\omega_{ci}}\right)\dfrac{1}{B}\dfrac{dB}{dx}$	Electrostatic and electromagnetic	$\mathbf{k}$ nearly $\perp$ to $\mathbf{B}_0$; some branches of these waves reduce to Alfvèn or ion waves as the gradients vanish.
	$\omega = k\left(\dfrac{\kappa T}{m\omega_{ci}}\right)\dfrac{1}{n}\dfrac{dn}{dx}$		

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9

THE VLASOV THEORY OF PLASMA STABILITY

9.1 INTRODUCTION

The Vlasov equation, which neglects binary collisions between plasma particles, predicts the existence of a variety of stationary states ($\partial f_0/\partial t = 0$). However, the collective interactions among plasma particles, retained in the Vlasov theory, provide a mechanism by which the distribution function f_0 can relax toward lower energy states, i.e., toward thermodynamic equilibrium. This mechanism is called *instability*, and has the effect that a small perturbation of a stationary state may grow larger. It was shown in Chap. 7 that there are states of non-thermodynamic equilibrium that are stable; in particular, any isotropic and monotone-decreasing function of kinetic energy $f_0 = f_0(v^2)$, $\partial f_0/\partial v^2 < 0$, has this property. Nevertheless, if the stationary state is sufficiently nonthermal, it will be unstable; for example, the discussion of Sec. 8.12 shows that some of the waves which are normal modes of a nonuniform plasma grow with time, so that the stationary state describing such a plasma is unstable.

Many examples of instability which depend on the configuration-space properties of a plasma were discussed in Chap. 5, in which the plasma dynamics were determined from the fluid equations. In this chapter the stability properties of plasmas that depart from thermodynamic equilibrium in their *velocity distributions* are considered. The Vlasov approach is nearly always necessary in

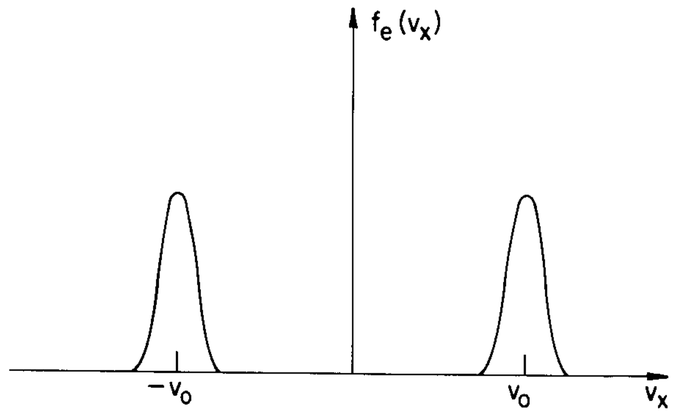


FIGURE 9.1.1
Two-stream distribution function for beams with a velocity difference much greater than the thermal spread of the beams.

these cases, because the fluid theory treats only moments of the velocity distribution.[1] The stability problem is closely related to the plasma wave problem detailed in Chap. 8. The question here is: Do the normal modes of the plasma damp or grow?

In general, these are two ways in which instability can arise. One possibility is that the normal modes of an unstable plasma are altered forms of the normal modes of a stable plasma. An excellent example of this idea, shown by (3.10.13), is the fire-hose instability: Alfvén waves are normal modes of a stable isotropic

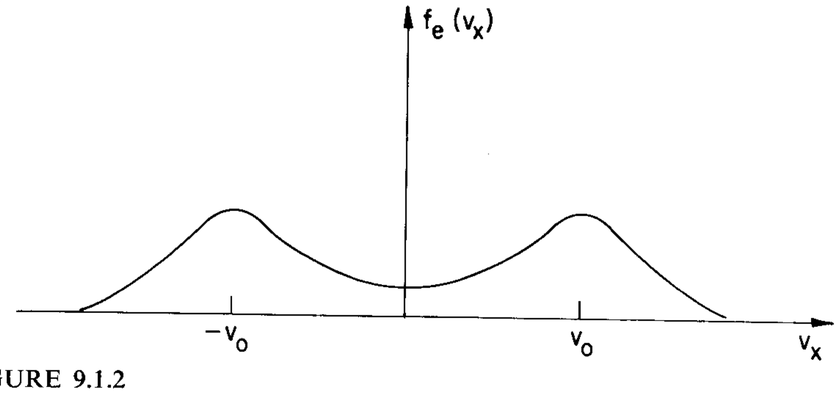


FIGURE 9.1.2
Two-stream distribution functions for "warm" beams where the thermal spread is equal to or greater than the velocity difference.

[1] Note, however, that a particular class of velocity-distribution instabilities can be treated in a fluid description. In particular, the velocity distribution of electrons shown in Fig. 9.1.1 can be treated as *two* electron fluids, a cold fluid with mean speed V_0 and a second cold fluid with mean speed $-V_0$. This is done in Sec. 5.5. When the beams are warm (Fig. 9.1.2), there is no adequate way to treat the stability of the electron distribution from a fluid approach.

plasma in a magnetic field, but in an anisotropic plasma with $T_{\|B_0} > T_{\perp B_0}$, Alfvén waves grow at the expense of the excess energy parallel to the magnetic field. A second possibility is that normal modes exist in an unstable plasma which are *characteristic* of the particular distribution, and have no analog in the uniform isotropic stable plasma. An example of this is the drift wave in a nonuniform magnetized plasma. Some branches of the drift waves are indigenous to the nonuniform plasma, with no analog in the uniform stable plasma. Thus the inhomogeneous plasma has features that allow wave motions distinct from those allowed in other plasma equilibria, and these waves grow with time.

Because most plasma states, in nature and in the laboratory, are not states of thermodynamic equilibrium, it is clearly important to determine how much (or what sort) of a departure from thermodynamic equilibrium is acceptable from the point of view of stability. In this regard, it should be recalled that the Vlasov theory is valid only for times short compared with $\tau_{\text{collision}}$, so that the concept of stability is somewhat relative: an unstable plasma in which perturbations grow slowly compared with $\tau_{\text{collision}}$ may, for all practical purposes, be considered stable. Further, the level to which waves grow will be determined from the energy reservoir which feeds the instability. Situations in which plasma waves grow only to a tolerable level may also be considered stable for practical purposes.

The term stability is often used rather loosely. Roughly, if a small cause produces only a small effect, e.g., if a small local increase in charge density does not produce a big increase in field energy, the system is called stable. A ball in a valley is stable (Fig. 9.1.3*a*), and a ball on a mountain peak is unstable (Fig. 9.1.3*b*). A ball on a plateau or on a small peak near the bottom of a valley (Fig. 9.1.3*c*) may be considered stable, since the ball on the plateau will not roll off very fast (very small instability growth rate) and the ball near the bottom of the valley will not change energy very much even if it does roll off. The actual size of the fields produced by an instability determines how rapidly transport of particles takes place. The magnitude of tolerable fluctuations differs from one situation to another, but in general, plasma stability should be identified with some bound on the growth of fields in the plasma.

There are two approaches to stability theory. The first is the direct approach, the normal-mode analysis, in which the Vlasov equation is solved directly for the time development of a small perturbation. This approach yields growth rates and shows what sorts of waves are unstable and might be observed in an actual experiment. Despite the complexity of solving the Vlasov equation for general perturbations in all but the simplest geometries, most of classical stability theory has proceeded in this way. A different approach is to examine stability as a consequence of departure from thermodynamic equilibrium. In

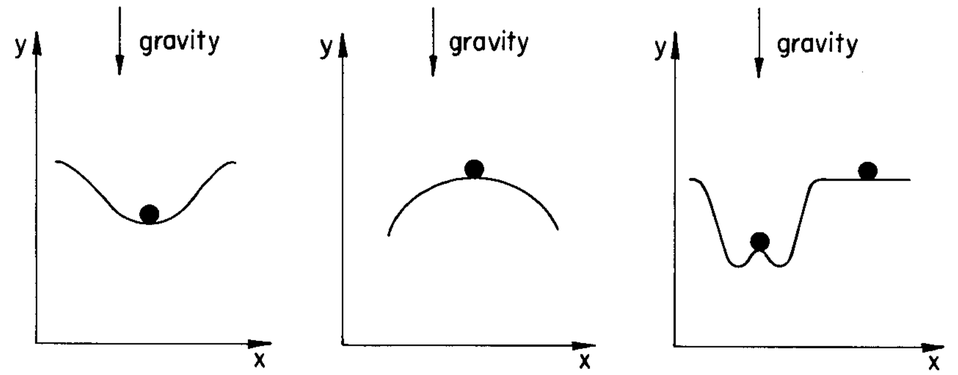


FIGURE 9.1.3
Diagram illustrating the definitions of stable, unstable, and marginal or bounded stability situations.

general, only a part of the energy in an unstable plasma is available for transfer to fluctuation fields. This is called the *free energy*; an upper bound to the free energy can be calculated from the constraints that govern the system. The thermodynamic approach cannot be expected to give detailed stability results, but it can locate both regions of stability, and in many cases useful bounds on the free energy.

In this chapter both the normal-mode and the thermodynamic approaches are employed, occasionally on the same system, to contrast the two methods and the results they can produce.

9.2 STABILITY OF MONOTONE-DECREASING DISTRIBUTIONS; THE NEWCOMB-GARDNER THEOREM[1]

A problem that can be treated both from a thermodynamic approach and from a normal-mode approach is the stability of a field-free plasma with an equilibrium distribution function $f_{\alpha 0}$ that decreases monotonically with velocity. When perturbed by an electrostatic oscillation propagating in the k direction, the distribution function develops according to

$$f_\alpha = f_{\alpha 0} + f_{\alpha 1}(\mathbf{v}) \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$$

[1] W. A. Newcomb; quoted in Appendix I, I. B. Bernstein, *Phys. Rev.*, **109**: 10 (1958). C. S. Gardner, *Phys. Fluids*, **6**: 839 (1963).

as described in Chap. 8, with ω given by

$$1 - \frac{\omega_{pe}^2}{k^2} \int_{-\infty}^{\infty} \frac{1}{u - \omega/|k|} \frac{\partial}{\partial u} \left[F_{e0}(u) + \frac{m_e}{m_i} F_{i0}(u) \right] du = 0 \qquad (9.2.1)$$

where $F_{\alpha 0}(u)$ is the projection of the distribution on the $\mathbf{k}$ axis, with $|k| = |\mathbf{k}|$

$$F_{\alpha 0}(u) = \int f_{\alpha 0}(\mathbf{v})\, \delta\left(u - \frac{\mathbf{k} \cdot \mathbf{v}}{|k|}\right) d\mathbf{v}$$

The path of the integration passes under the pole $u = \omega/|k|$. If (9.2.1) predicts an imaginary part ω_i of the eigenfrequency $\omega (= \omega_r + i\omega_i)$ that is greater than zero, $\omega_i > 0$, the equilibrium distribution function $f_{\alpha 0}$ is *unstable*. If $\omega_i < 0$, the equilibrium is stable and stationary for times much less than a binary collision time. The fact that monotone-decreasing distributions $F_{\alpha 0}(u)$ are stable can be demonstrated directly from (9.2.1). Writing $\omega = \omega_r + i\omega_i$, and assuming that $\omega_i > 0$, the velocity integration in (9.2.1) may be taken along the real axis in velocity space, because the pole $u = (\omega_r + i\omega_i)/|k|$ lies above the real u axis. Assuming the distribution $f_{\alpha 0}$ is unstable, ω_r and ω_i must satisfy

$$1 - \frac{\omega_{pe}^2}{k^2} \int_{-\infty}^{\infty} \frac{u - \omega_r/|k|}{\left(u - \dfrac{\omega_r}{|k|}\right)^2 + \dfrac{\omega_i^2}{k^2}} \frac{\partial F}{\partial u}\, du$$

$$- \frac{i\omega_i}{|k|} \frac{\omega_{pe}^2}{k^2} \int_{-\infty}^{\infty} \frac{(\partial F/\partial u)\, du}{\left(u - \dfrac{\omega_r}{|k|}\right)^2 + \dfrac{\omega_i^2}{k^2}} = 0 \qquad (9.2.2)$$

where $F \equiv F_{e0} + (m_e/m_i) F_{i0}$. Equating the real and imaginary parts of (9.2.2) to zero, any solution of (9.2.2) with $\omega_i > 0$ must satisfy both

$$\int_{-\infty}^{\infty} \frac{\partial F/\partial u\, du}{(u - \omega_r/|k|)^2 + \omega_i^2/k^2} = 0 \qquad (9.2.3)$$

and

$$1 - \frac{\omega_{pe}^2}{k^2} \int_{-\infty}^{\infty} \frac{u\, \partial F/\partial u}{(u - \omega_r/|k|)^2 + \omega_i^2/k^2}\, du = 0 \qquad (9.2.4)$$

since the integral involving $\omega_r/|k|$ vanishes by (9.2.3).

But if $u\, \partial F/\partial u \leqslant 0$ everywhere, as it must be for monotone-decreasing distributions, the integral in (9.2.4) is negative definite, and hence (9.2.4) cannot be satisfied for *any* choice of ω_r, ω_i. The normal-mode approach leads to the conclusion:

> If for all directions $\mathbf{k}$, the distribution function projected onto the k axis $F(u) = \int \delta(u - \mathbf{k} \cdot \mathbf{v}/|k|)[f_{e0}(\mathbf{v}) + (m_e/m_i) f_{i0}(\mathbf{v})] d\mathbf{v}$ is a function which decreases monotonically away from its maximum at $u = 0$, ($u\, \partial F/\partial u \leq 0$) the distribution $F(u)$ is stable.

From (9.2.5) it is easy to conclude that all *isotropic* distributions are stable (Prob. 9.2.1).

Further, this result does not depend on the frame of reference. If there is *any* frame in which F is monotone-decreasing, it is a stable distribution (Prob. 9.2.2).

Problem 9.2.1 Using (9.2.4), show that any isotropic distribution

$$f_0(v_x^2 + v_y^2 + v_z^2)$$

is stable, whether the distribution is monotone-decreasing or not. Show directly that

$$f_{i0} = f_{e0} = A\, \delta(v_x^2 + v_y^2 + v_z^2 - v_0^2)$$

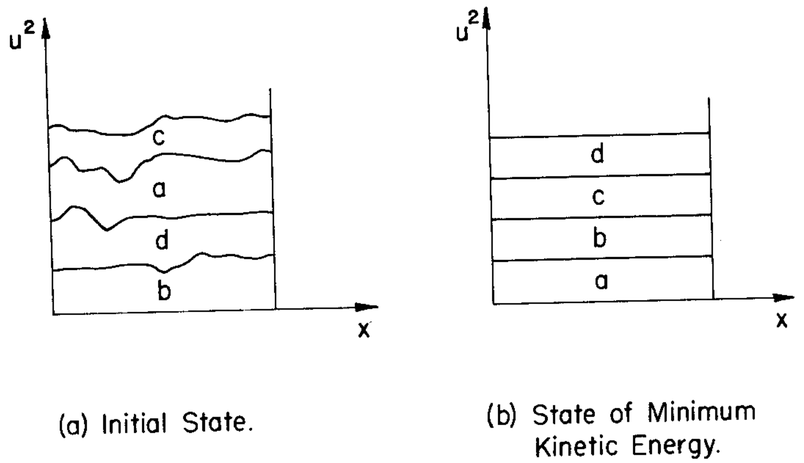
is stable. Indicate in one and two dimensions (in v space) what this distribution looks like. ////

Problem 9.2.2 Extend the theorem (9.2.5) to the case of a plasma drifting as a whole with speed v_0, so that $(u - v_0)\, \partial F/\partial u \leq 0$ everywhere. ////

From the thermodynamic point of view the stability of monotone-decreasing distributions is a consequence of two constraints: conservation of energy and conservation of volume in phase space [Liouville's theorem, which states that $f(\mathbf{x}, \mathbf{v}, t)$ is constant along any particle trajectory, follows identically from the Vlasov equation, $Df_0/Dt = 0$]. The latter constraint is an exact property of the Vlasov equation, even without assuming small perturbations or linearization. The fact that f helps determine the fields that determine the trajectory of particles does not alter this fact. To demonstrate the thermodynamic approach, consider a one-dimensional model for a general initial distribution $F(x, u^2, 0)$. As an example, Fig. 9.2.1*a* plots regions of approximately constant F in (x, u^2) phase space, indicating the size of F in each region by the ordering $F(a) > F(b) > F(c) \ldots$. Stability is determined from the question: What state, which can be reached from the initial state, has the least internal (kinetic) energy? That is, what is the minimum value, consistent with Liouville's theorem, of the energy integral

$$\int \tfrac{1}{2} m u^2 F(x, u^2)\, dx\, du \qquad (9.2.6)$$

The constraint (Liouville's theorem) allows the F = const. zones to be rearranged and reshaped but not changed in area. Clearly, then, because of the weighting factor u^2, the lowest value of the energy, (9.2.6), is achieved when the zones of



(a) Initial State. (b) State of Minimum Kinetic Energy.

FIGURE 9.2.1
Diagram of a "thermodynamically" stable state of minimum kinetic energy compared with an initial state that is "thermodynamically" unstable.

greatest density F_0 lie nearest $u^2 = 0$. Like a mixture of fluids of different specific gravity, the "heaviest" layers sink to the "bottom" ($u^2 = 0$). The result is Fig. 9.2.1*b*, a function distribution independent of x and monotone-decreasing in u^2.

If the initial state is already a monotone-decreasing state, the proof above shows that small disturbances do not grow. In particular, the maximum change in kinetic energy transfer to fields cannot be greater than any initial kinetic energy perturbation. By energy conservation, changes in the field energy $[\int dx(E^2/8\pi)]$ are similarly bounded. The extension to three dimensions is straightforward. This proof is due to Gardner.[1]

Gardner's proof is an example of the simplicity and generality of the "constraint" approach to stability analysis. The proof is: (1) nonlinear (no terms are dropped); (2) correct for magnetic as well as Coulomb interactions since the bound on kinetic energy limits the total electromagnetic energy, $\int d\mathbf{x}(\mathbf{E}^2 + \mathbf{B}^2)$; (3) may be extended to cover cases too complicated to be solved directly even in perturbation theory. For example, if in equilibrium there exists a self-consistent electrostatic potential $\phi_0(\mathbf{x})$, any function $F \equiv F(W)$ is an equilibrium solution, with $W = \frac{1}{2}m_\alpha v^2 + q_\alpha\phi_0(\mathbf{x})$. By the above method, all such states, which are monotone-decreasing in W; $\partial F/\partial W < 0$, are stable. For all F the integral $\int [F - F_{t=0}(W)]\, d\mathbf{x}\, d\mathbf{v}$, defining a perturbation in particle energy (kinetic + potential), is bounded, and again, by energy conservation, the field energy perturbation is also bounded. Thus, even some spatially inhomogeneous equilibria can be shown to be stable by Gardner's theorem.

[1] C. S. Gardner, *Phys. Fluids*, **6**:839 (1963).

Finally, the class of stable distributions includes functions that are a linear combination of W and other constants of the motion, for example the canonical angular momentum L in a cylindrically symmetric plasma. In this case, if $\partial F(W + \alpha L)/\partial W < 0$, F is stable, and the integral

$$\int (W + \alpha L)[F - F_{t=0}(W + \alpha L)]\, d\mathbf{x}\, d\mathbf{v}$$

is bounded by Liouville's theorem. The nonlinear stability of monotone-decreasing functions of the energy is usually referred to as *Gardner's theorem.*

Of course, the method of constraints has obvious drawbacks. In particular, it is virtually impossible to prove instability by this means. While it has been shown that *some* constraint must exist if the system is stable, no universal prescription for finding the constraint is known. Though the constraint that proves stability may not be apparent, such a constraint may in fact exist.

A large class of functions is stable according to Gardner's theorem. All have the property that, *in some reference frame*, they are locally isotropic in velocity space. From experience gathered from perturbation solutions, it is tempting to surmise that this is the only class of absolutely stable states of plasmas. With this possibility in mind, Fowler[1] has examined the extent to which such states tend to influence the instability properties of their neighbors, which presumably relax toward nearby stable equilibria. In this way, he obtains rigorous bounds on the nonlinear behavior of arbitrary states.

9.3 STABILITY OF MULTIPEAKED DISTRIBUTIONS—THE TWO-STREAM INSTABILITY

When a plasma distribution has more than one maximum as a function of velocity, Gardner's theorem does not provide any information about stability. A simple example of plasma instability is the well-known two-stream instability. The Vlasov theory allows a direct calculation of the linear growth of an unstable two-stream distribution; as an alternative there is a general method of testing the stability of such distributions. Both techniques are used in this section.

The two-stream instability[2] is involved in the success of traveling-wave amplifiers and in the failure of some controlled thermonuclear reaction schemes. It is also directly relevant to many schemes of plasma heating, e.g., *turbulent*

[1] T. K. Fowler, Thermodynamics of Unstable Plasmas, in A. Simon and W. B. Thompson (eds.), "Advances in Plasma Physics," vol. 1, Wiley, New York, 1968.
[2] J. R. Pierce, "Traveling Wave Tubes," Van Nostrand, New York, 1950.
O. Buneman, *Phys. Rev.*, **115**:503 (1959).
T. E. Stringer, Plasma Physics, *J. Nucl. Energy*, **C6**:267 (1964).

heating. As a first example of this instability, consider a field-free plasma with an equilibrium distribution

$$f_{i0} = \delta(\mathbf{v} - \mathbf{V}_0) \qquad f_{e0} = \delta(\mathbf{v}) \tag{9.3.1}$$

If this distribution is perturbed electrostatically, the system responds with an oscillation $e^{-i\omega t}$, where (8.9.6) gives the frequency ω as a zero of the dielectric $D(\mathbf{k}, \omega)$; that is,

$$D = 1 - \sum_\alpha \frac{{\omega_{p\alpha}}^2}{k^2} \int \frac{\partial F_{\alpha 0}/\partial u}{u - \omega/|k|}\, du = 0 \tag{9.3.2}$$

where the distributions are

$$F_{i0} = \delta\left(u - \frac{\mathbf{k}\cdot\mathbf{V}_0}{|k|}\right) \qquad F_{e0} = \delta(u)$$

and the integral passes under the pole $u = \omega/|k|$, as explained in Sec. 8.4. The delta-functions make the integrations trivial. The result from (9.3.2) is a dispersion relation for ω versus $\mathbf{k}$.

$$1 = \frac{{\omega_{pe}}^2}{\omega^2} + \frac{{\omega_{pi}}^2}{(\mathbf{k}\cdot\mathbf{V}_0 - \omega)^2} \tag{9.3.3}$$

The right-hand side of (9.3.3) is plotted as a function of ω in Fig. 9.3.1. Real roots are given by the intersection of the dotted horizontal line of value 1, with the curve in Fig. 9.3.1. When the value of the local minimum point labeled A is less than 1, Eq. (9.3.3) has four real roots. When A is greater than 1, (Eq. 9.3.3) has two real and two complex conjugate roots, one of which corresponds to instability. The point A is located at

$$\omega_A = \mathbf{k}\cdot\mathbf{V}_0 \left[\frac{(\omega_{pe}/\omega_{pi})^{2/3}}{(\omega_{pe}/\omega_{pi})^{2/3} + 1}\right]$$

If the electrons and ions had equal mass, the point A would be at $\omega_A = \frac{1}{2}\mathbf{k}\cdot\mathbf{V}_0$, and the first wave to go unstable (at fixed drift speed and decreasing k) would have velocity halfway between the two streams. This fact is independent of the frame in which the stream-wave system is observed. Regardless of the mass of particles that make up the two streams, the unstable waves have velocity somewhere *between* the velocity of the two streams.

The condition for instability is that the right-hand side of (9.3.3) be greater than 1 at $\omega = \omega_A$, and is

$$|\mathbf{k}\cdot\mathbf{V}_0| < \omega_{pe}\left[1 + \left(\frac{\omega_{pi}}{\omega_{pe}}\right)^{2/3}\right]^{3/2} \qquad \text{two-stream-instability condition} \tag{9.3.4}$$

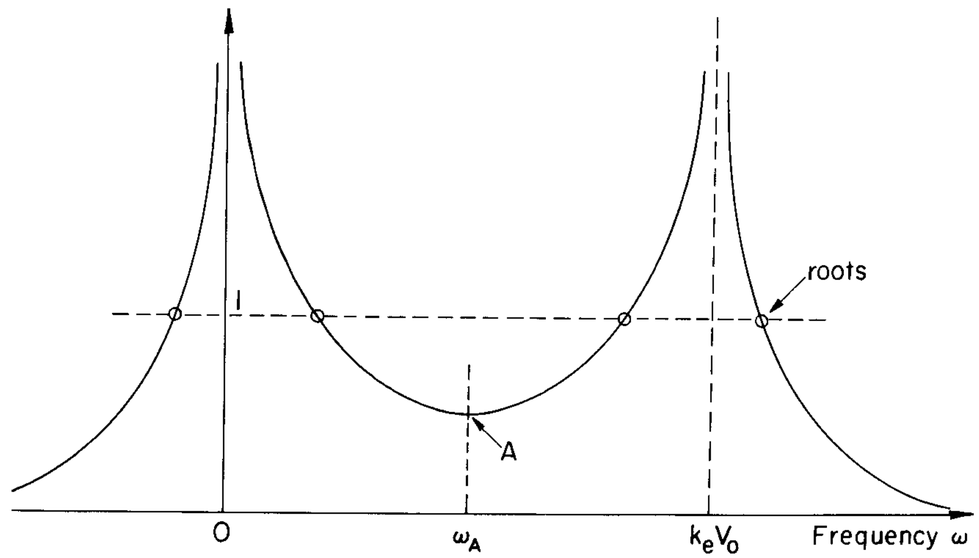


FIGURE 9.3.1
Plot of the right and left sides of the two-stream dispersion relation

$$1 = \frac{{\omega_{pe}}^2}{\omega^2} + \frac{{\omega_{pi}}^2}{(\mathbf{k}\cdot\mathbf{V}_0 - \omega)^2}$$

as a function of frequency ω, showing a stable situation; i.e., there are four real roots.

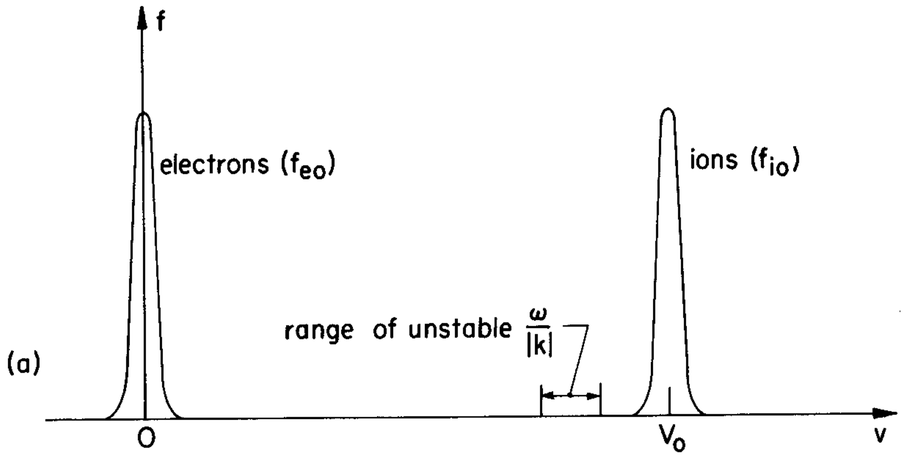
Since the sign of the charge does not appear, (9.3.4) can be equally well applied to two electron streams in a fixed neutralizing ion background, replacing ω_{pi}/ω_{pe} by $\omega_{pe1}/\omega_{pe2}$. The result (9.3.4) could have been obtained equally well from a fluid theory.

Equation (9.3.4) is most easily satisfied by long-wavelength oscillations, by fairly small drift speed V_0, or by an accumulation of charge in layers *parallel* to the stream; that is, $\mathbf{k}\cdot\mathbf{V}_0 \ll |k|\,V_0$. However, as $\mathbf{k}\cdot\mathbf{V}_0$ is reduced much below the threshold (9.3.4), the growth rate ω_i becomes quite small. Further, if the drift speed V_0 is too small, thermal effects will become important. That is, no *real* beam will have the distribution (9.3.1); rather, (9.3.1) is an approximation to the distribution seen by a wave of phase velocity much greater than the thermal spread of the distribution. Therefore the conditions for the validity of the two-stream-instability calculation presented above is

$$\left|\frac{\mathbf{k}\cdot\mathbf{V}}{k} - \frac{\omega}{k}\right| \gg \sqrt{\frac{\kappa T_i}{m_i}} \qquad \text{and} \qquad V_0 \gg \sqrt{\frac{\kappa T_e}{m_e}}$$

The thermal corrections will be treated explicitly in Sec. 9.6.

Although the conditions for stability can be determined analytically from the dispersion relation (9.3.3), a general solution of (9.3.3) cannot be obtained



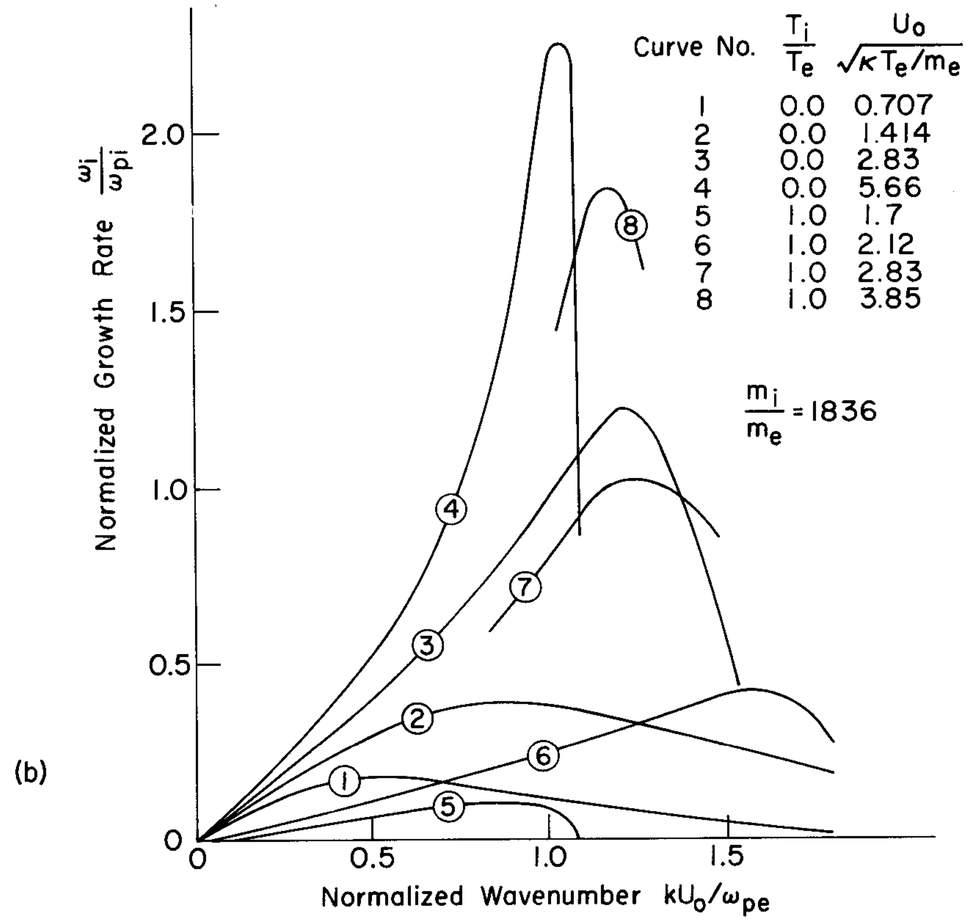


FIGURE 9.3.2
Distribution function for a "cold" ion beam drifting at speed V_0 through a "cold" electron beam and a plot of the growth rates for various relative drift velocities and ion and electron beam temperature ratios. [*After T. E. Stringer, Plasma Physics, J. Nucl. Energy*, C6:267 (1964).]

analytically for $\omega(\mathbf{k})$. However, it is a straightforward exercise to show that the maximum growth rate occurs for a wavelength such that

$$\mathbf{k} \cdot \mathbf{V}_0 \approx \omega_{pe} \qquad \text{for maximum growth of electron-ion streaming instability} \tag{9.3.5}$$

From (9.3.1) note that $\mathbf{V}_0$ is the *ion* drift speed, so that the wave moves at nearly the speed of the ions (see also Prob. 9.3.2). The dispersion relation can be solved for maximum growth, yielding

$$\omega \approx \omega_{pe} + e^{\pm 2\pi i/3}\left(\frac{\omega_{pi}{}^2\omega_{pe}}{2}\right)^{1/3} \approx \omega_{pe}\left[1 + i\frac{\sqrt{3}}{2}\left(\frac{m_e}{2m_i}\right)^{1/3}\right] \tag{9.3.6}$$

Figure 9.3.2*a* shows the distribution function (9.3.1) and the region of unstable waves. Figure 9.3.2*b* plots the instability growth rate ω_i as a function of wave number.

The results of the two-stream Vlasov theory have been confirmed in detail by computer simulation experiments. In such an experiment, a computer calculates the orbits of all the charged particles in a plasma; these orbits are determined by the actual forces between the particles. The Vlasov theory should agree with the calculation which tracks the particles, at least on a time scale less than $\tau_{\text{collision}}$. The agreement between Vlasov theory and a direct many-body calculation is good, as shown in Figs. 9.3.3 to 9.3.5.

Problem 9.3.1 Verify (9.3.6) for the frequency and growth rate of the fastest-growing Langmuir oscillation in an electron-ion counterstreaming plasma, by showing that maximum growth requires ω be of the form $\omega = \mathbf{k} \cdot \mathbf{V}_0/(1 - Se^{\pm i\pi/3})$, where S is a real number. ////

Problem 6.3.2 Show that in a reference frame in which the ions are at rest the waves have $\omega_r \approx \omega \ll \omega_{pe}$. ////

The distribution function at $t = 0$ and later times is shown on the left-hand graphs of Fig. 9.3.3. These distributions are not δ functions of velocity, but until they begin to overlap, they can be approximated by (9.3.1). The Vlasov theory predicts that the distribution should be unstable. The simulation experiment agrees. Figure 9.3.4 shows that the wave energy ($W_E = \Sigma_k \mathbf{E}_k{}^2/8\pi$) increases at a rate comparable with $\omega_{i,\max}$ [(9.3.6)] and that the electron energy increases on the same time scale. The small dots in Fig. 9.3.3 are at the position of elec-

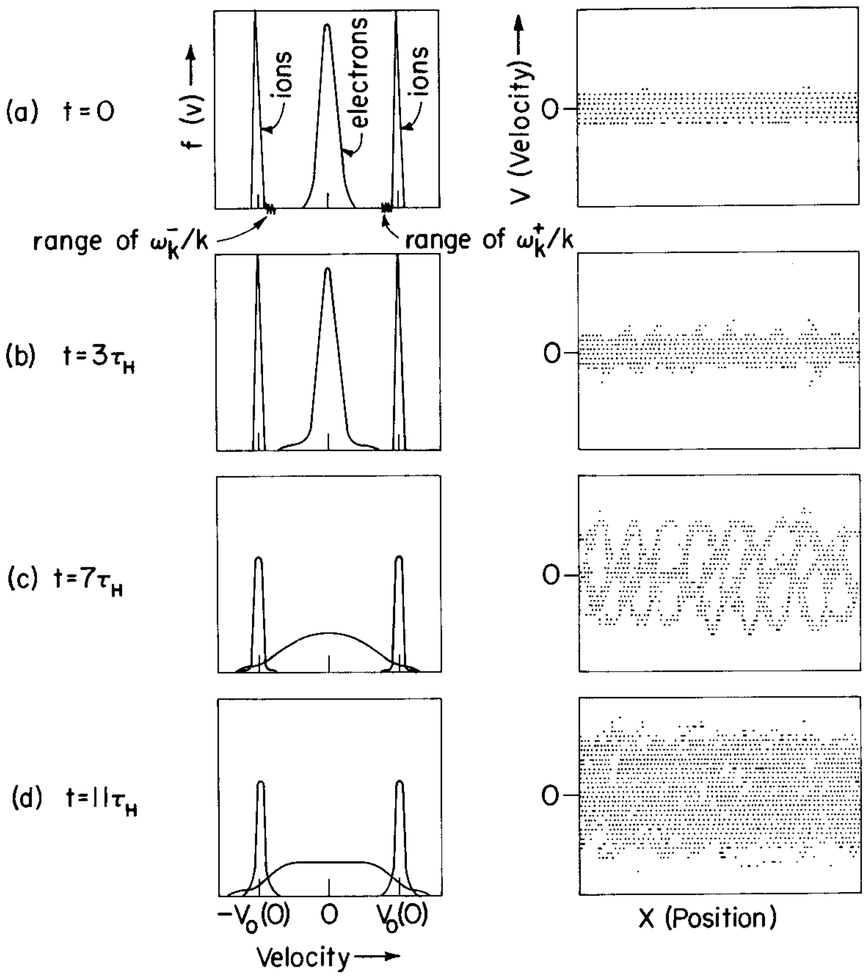


FIGURE 9.3.3
Computer simulation of the interaction of cold ion beams drifting with speeds $\pm V_0$ through an initially cold electron background, illustrating the evolution of the distribution functions during the growth of the instability. The quantity τ_H is the reciprocal of the maximum growth rate obtained from linear theory. The electron-to-ion mass ratio is $m_e/m_i = \frac{1}{27}$, and initially $W_{Ke}/m_e V_0^2(0) = \frac{1}{25}$. [*After R. C. Davidson et al., Phys. Rev. Letters*, **24**:519 (1970).]

trons in (x, v) space. Figure 9.3.4 shows that many wave numbers k are excited, which explains why the *net* growth rate is a little less than the *maximum* growth.

Besides confirming the presence of instability and verifying the linear growth rate, Figs. 9.3.3 and 9.3.4 show a further effect that could have been predicted from linear theory. By time $t = 11\tau_H$ (Fig. 9.3.3) the electron-ion distribution should be stable, because although the distributions appear peaked,

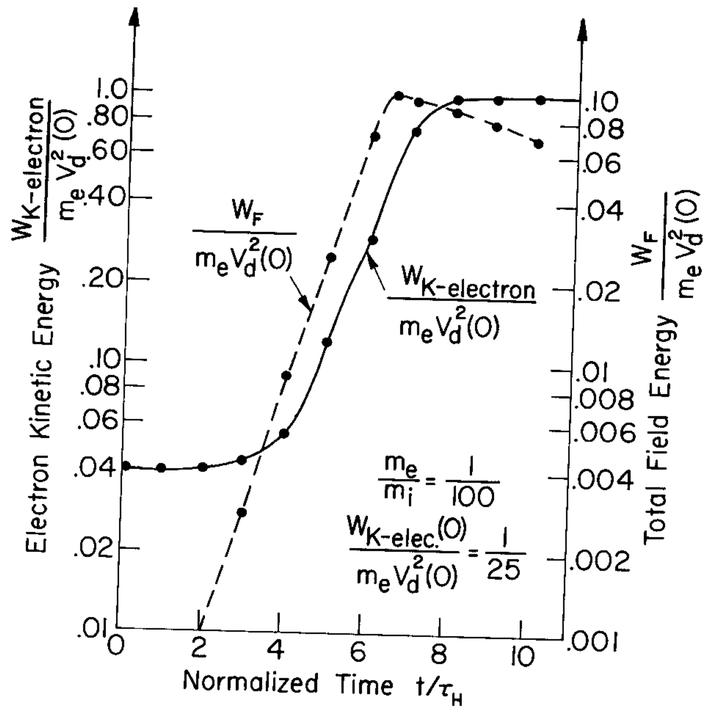


FIGURE 9.3.4
Electron kinetic energy and total field energy as a function of time from a computer simulation of the interaction of cold ion beams drifting with speeds $\pm V_0$ through an initially cold electron background. The quantity τ_H is the reciprocal of the maximum growth rate obtained from linear theory. The electron-to-ion mass ratio is $m_e/m_i = \frac{1}{100}$, and initially $W_{Ke}/m_e V_0^2(0) = \frac{1}{25}$. [*After R. C. Davidson et al., Phys. Rev. Letters*, **24**:579 (1970).]

the distribution $F_0(u) = F_{e0} + (m_e/m_i)F_{i0}$ is actually monotone-decreasing. The electrostatic energy from Fig. 9.3.4 has stopped increasing by this time, as would be predicted from Gardner's theorem (Sec. 9.2). Note, however, that "distribution function" arguments cannot be safely applied at earlier time steps (for example, Fig. 9.3.3*c*) because the distribution is widely varying in *space*, and arguments based on a space-average distribution

$$f_\alpha(\mathbf{v}) = \int_{-L}^{L} f_\alpha(x, \mathbf{v}) \frac{dx}{2L}$$

can be quite misleading.

A particular example of a two-stream instability for which the dispersion relation can be solved exactly is that of two electron clouds, of equal density n_0,

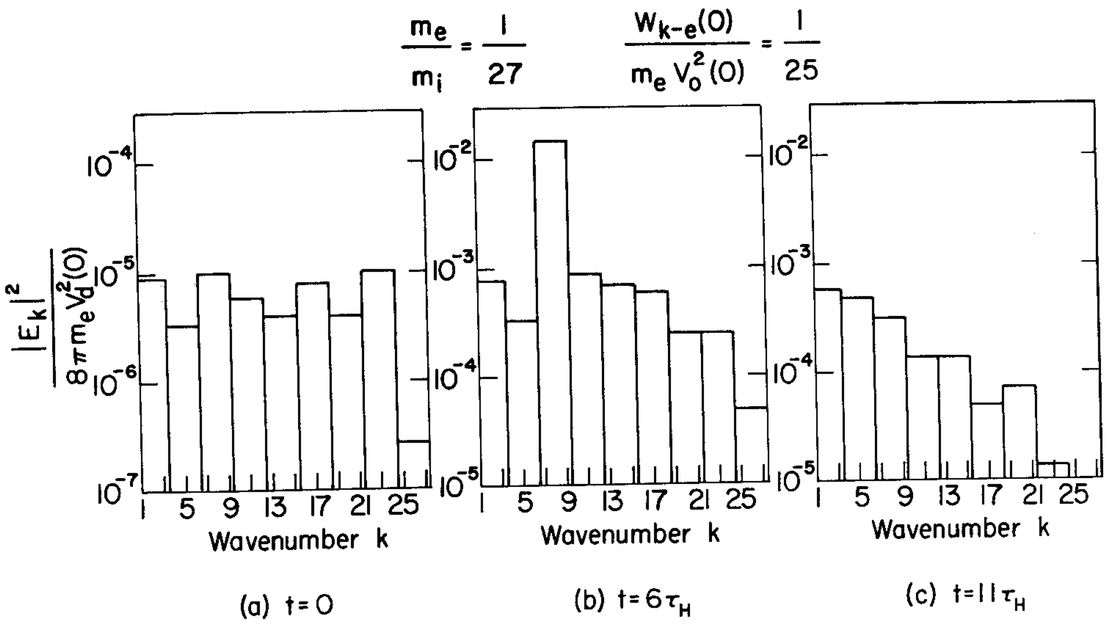


FIGURE 9.3.5
Field energy as a function of wave number for various times obtained from a computer simulation of the interaction of cold ion beams drifting with speeds $\pm V_0$ through an initially cold electron background. The quantity τ_H is the reciprocal of the maximum growth rate obtained from linear theory. The electron-to-ion mass ratio is $m_e/m_i = \frac{1}{27}$. [*After R. C. Davidson et al., Phys. Rev. Letters*, **24**:5A (1970).]

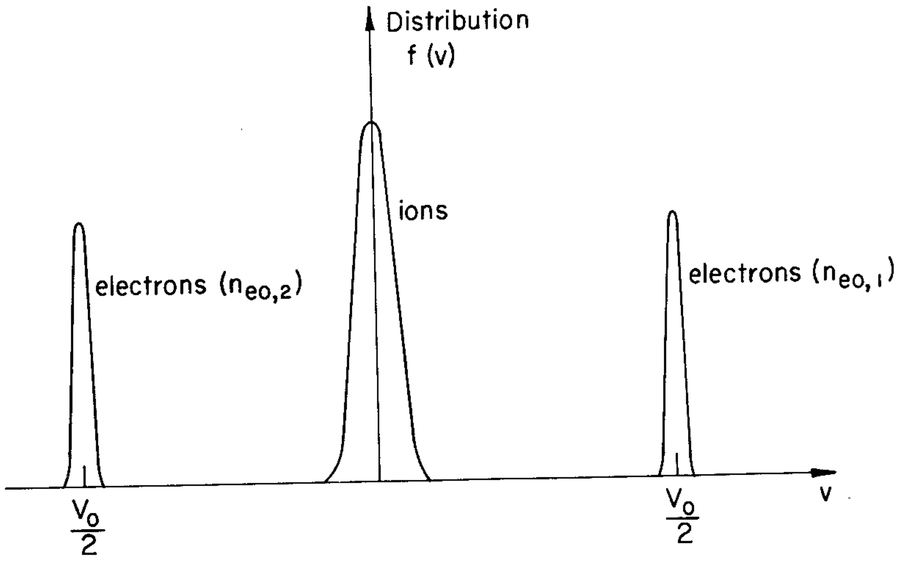


FIGURE 9.3.6
Distribution function of two cold electron beams drifting through a cold ion background.

drifting through each other with relative speed V_0, as shown in Fig. 9.3.6; the ions, which make the system charge neutral, are assumed to remain at rest during the instability produced by the counterstreaming electrons (this neglect of ion motion for high-frequency instabilities is referred to as the case of infinitely massive ions, $\omega_{pi}^2 \to 0$). In that case the dispersion relation (9.3.3) is replaced by (note that $\omega_p^2 = 4\pi(n_{e1} + n_{e2})(e^2/m_e) = 2\omega_{pe1}^2 = 2\omega_{pe2}^2$)

$$1 = \frac{1}{2}\frac{\omega_{pe}^2}{(\omega - kV_0/2)^2} + \frac{1}{2}\frac{\omega_{pe}^2}{(\omega + kV_0/2)^2} + \frac{\omega_{pi}^2}{\omega^2} \tag{9.3.7}$$

and the last term can be neglected, since $\omega^2 \gg \omega_{pi}^2$.

Equation (9.3.7) is quadratic in ω^2, with solution

$$\omega^2 = \frac{\omega_{pe}^2}{2} + \frac{k^2V_0^2}{4} \pm \sqrt{\left(\frac{\omega_{pe}^2}{2} + \frac{k^2V_0^2}{4}\right)^2 + \frac{k^2V_0^2}{4}\left(\omega_{pe}^2 - \frac{k^2V_0^2}{4}\right)} \tag{9.3.8}$$

A diagram of (9.3.7) (neglecting ω_{pi}^2) of the type of Fig. 9.3.1, predicts

$$|kV_0| < \frac{\omega_{pe}}{\sqrt{2}}\left[1 + \left(\frac{\omega_{pe1}}{\omega_{pe2}}\right)^{2/3}\right]^{3/2} = 2\omega_{pe}; \qquad \text{for two-electron-stream instability} \tag{9.3.9}$$

This prediction is directly verified from the actual solution (9.3.8). The growth rate vanishes as $V_0 \to 0$, and the maximum growth rate

$$\omega_{i,\max} = \frac{\omega_{pe}}{\sqrt{8}} \qquad \text{maximum growth of two-electron-stream instability} \tag{9.3.10}$$

is achieved for wavenumber

$$kV_0 = \frac{3}{\sqrt{2}}\omega_{pe} \qquad \text{wavelength of fastest two-electron-stream growth} \tag{9.3.11}$$

Besides exhibiting a long-wavelength instability ($kV_0 < \omega_{pe}$), the two-stream plasma is also somewhat different as a *dielectric medium* than is the non-streaming plasma. This is seen by sketching the frequency of Langmuir oscillations as a function of wavelength for the two-electron-stream plasma [(9.3.8)] and comparing the result with $\omega(\mathbf{k})$ for a single-humped cold distribution as in (9.54), with $k\lambda_D \to 0$. This is done in Fig. 9.3.7. The dotted line is $\omega(\mathbf{k})$ for a cold plasma in which both species are at rest. The shift introduces a new set of

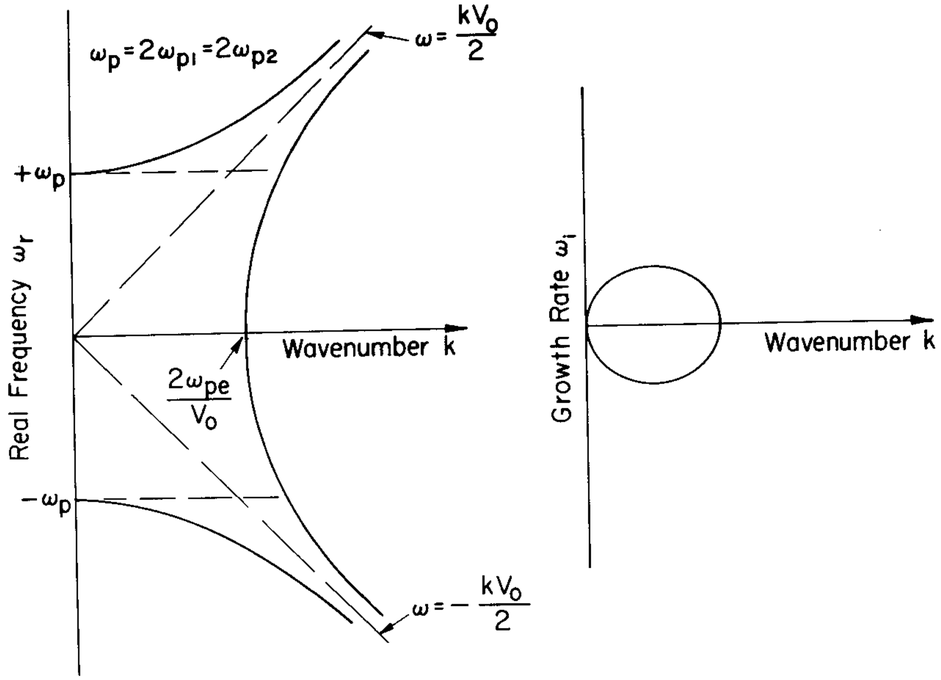


FIGURE 9.3.7
Frequency and growth rate as a function of wave number for plasma waves in a plasma of two cold electron beams drifting with speeds $\pm V_0/2$ through a cold-ion background.

modes which are asymptotic to $\omega \approx kV_0/2$ and have real frequency only for $|k| > 2\omega_{pe}/|V_0|$.

Problem 9.3.3 What happens to these new modes as $V_0 \to 0$? ////

9.4 STABILITY OF MULTIPEAKED DISTRIBUTION IN WARM PLASMAS—GENTLE-BUMP INSTABILITY

Since instability of cold beams could have been calculated from fluid theory, the question arises as to the beam temperature at which the fluid result becomes inadequate. The answer should be evident from Fig. 9.3.2. When the thermal spread of the distribution is such that the particle distribution overlaps the region of unstable waves, the effect of particles moving at the speed of the wave, i.e., resonant particles, must be considered. This effect is contained in the Landau poles in the velocity integrations that determine the plasma dielectric.

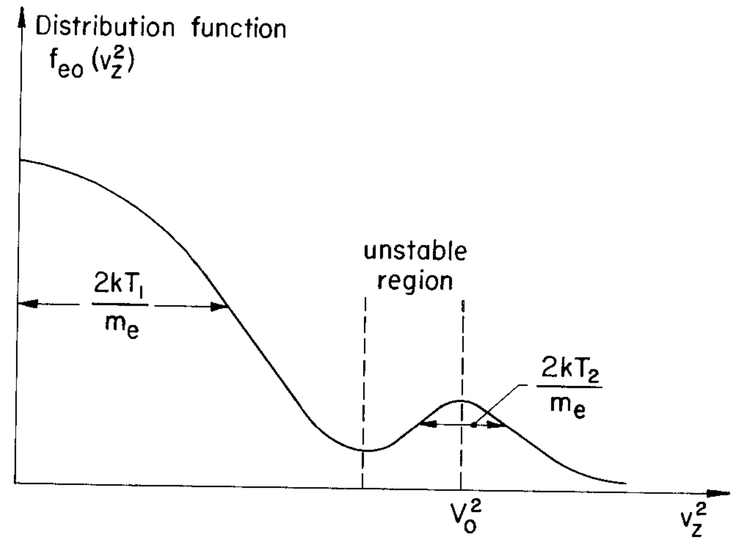


FIGURE 9.4.1
Distribution function of a warm electron beam drifting through a maxwellian electron distribution with a speed greater than its thermal spread, $V_0^2 > 2\kappa T_e/m_e$.

As a complete contrast to the beam instability, consider an electron distribution with a small bump in the tail of a maxwellian distribution (Fig. 9.4.1), described by

$$f_{e0} = \frac{n_1}{n_e}\left(\frac{m_e}{2\pi\kappa T_1}\right)^{3/2} \exp\left(-\frac{m_e v^2}{2\kappa T_1}\right) + \frac{n_2}{n_e}\,\delta(v_x)\,\delta(v_y)\left(\frac{m_e}{2\pi\kappa T_2}\right)^{1/2} \frac{1}{2}\left\{\exp\left[-\frac{m_e(v_z - V_0)^2}{2\kappa T_2}\right] + \exp\left[-\frac{m_e(v_z + V_0)^2}{2\kappa T_2}\right]\right\} \tag{9.4.1}$$

$$n_e = n_1 + n_2 \gg n_2 \qquad T_2 \ll T_1 \qquad V_0 \gg \frac{2\kappa T_1}{m_e}$$

$$f_{i0} = \delta(v_x)\,\delta(v_y)\,\delta(v_z)$$

Choosing an f_{e0} that is symmetric in v about V_0 guarantees an equilibrium state without plasma currents or magnetic fields. This choice makes this a simpler problem to discuss. (See Prob. 9.4.3.)

Plasma oscillations in the gentle-bump plasma are given by the zeros of the field-free dielectric [(9.3.2)]. Assuming as in the Landau damping calculations (Sec. 8.6) that the eigenfrequency ω will be nearly real, and expanding $D(k, \omega)$ about $\omega = \omega_r$, the zeros of $D(k, \omega)$ are found by solving

$$D_r(k, \omega_r) = 1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \oint_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}/\partial u}{u - \omega_r/|k|}\, du = 0$$

$$D_i(k, \omega_r) = -\pi \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \frac{\partial F_{\alpha 0}}{\partial u}\bigg|_{u=\omega_r/|k|}$$

$$\omega_i = -\frac{D_i(k, \omega_r)}{\partial D_r/\partial \omega_r} \approx \frac{\pi \dfrac{\omega_{pe}^2}{k^2} \dfrac{\partial F_{\alpha 0}}{\partial u}\bigg|_{u=\omega_p/|k|}}{\partial D_r/\partial \omega_r} \tag{9.4.2}$$

Although the integral in D_r is not as simple for this case as for δ-function distributions, it can be approximated for waves with $\omega/|k| \gg \sqrt{2\kappa T_1/m_e}$ by

$$\oint \frac{\partial F_{\alpha 0}/\partial u}{\omega_r/|k| - u}\, du \approx \int \frac{\partial F_{\alpha 0}}{\partial u} \left(\frac{|k|}{\omega_r} + \frac{k^2 u}{\omega_r^2} + \frac{|k|k^2 u^2}{\omega_r^3} + \cdots \right) du \tag{9.4.3}$$

as in the calculation of Landau damping in Chap. 8. The real part of the dielectric $D_r(k, \omega_r)$ gives the response of the bulk of the plasma to the electric field in the plasma wave. The growth or damping rate ω_i gives the effect of the class of particles which are exactly in resonance with the wave.

With this approximation to D_r, and using $\omega_{pi}^2 \ll \omega_{pe}^2$ and $n_1 \gg n_2$, the eigenfrequency of plasma oscillations in the distribution (9.4.1) is

$$\omega_r^2 \approx \omega_{pe}^2(1 + 3k^2\lambda_{D1}^2)$$

i.e., the normal modes of a distribution which departs only slightly from maxwellian are the same as those for the maxwellian. The difference is that the sign of the imaginary term depends on the slope of F_0 at the velocity of the wave $u = \omega_r/|k|$, so that plasma oscillations may *grow* in the distribution (9.4.1). For the particular gentle-bump distribution (9.4.1), the frequency and damping or growth rate is given, from (9.4.2), by (still assuming that most of the particles and most of the energy are *outside* the bump, that is, $n_1 \gg n_2$, $n_1\kappa T_1 \gg n_2 m_e V_0^2$)

$$\omega_i \approx -\sqrt{\frac{\pi}{8}} \frac{\omega_{p1}}{|k|^3 \lambda_{D1}^3} \exp\left(-\frac{1}{2k^2\lambda_{D1}^2} - \frac{3}{2} \right) + \frac{n_2}{n_1} \left(\frac{T_1}{T_2} \right)^{3/2} \frac{k^3}{k_z^3} \left(\frac{k_z V_0}{\omega_r} - 1 \right) \exp\left[-\frac{T_1/T_2}{2k_z^2\lambda_{D1}^2} \left(1 - \frac{k_z V_0}{\omega_r} \right)^2 \right] \tag{9.4.4}$$

where $\omega_r^2 = 4\pi n_1 e^2/m_e$, $\lambda_{D1}^2 = (4\pi n_1 e^2/\kappa T_1)^{-1}$, and $k\lambda_{D1} \ll 1$ is assumed. The reader is reminded that the bump in the equilibrium represents an extra group of particles drifting in the z direction.

Problem 9.4.1 Show that

$$\lim_{\varepsilon \to 0^+} \mathrm{Im}\left\{ \int \frac{\mathbf{k} \cdot \nabla_{\mathbf{v}}[\delta(v_x)\, \delta(v_y) \exp(-m_e v_z^2/2\kappa T_e)]}{\omega_r - \mathbf{k} \cdot \mathbf{v} + i\varepsilon}\, d\mathbf{v} \right\} = \frac{k^2}{k_z^2} \pi \left[\frac{d}{dv_z} \exp\left(-\frac{m_e v_z^2}{2\kappa T_2} \right) \right]_{v_z = \omega_r/|k_z|}$$

thus verifying (9.4.4). ////

Problem 9.4.2 If the bump is warm, i.e., if

$$f_{e0} = \frac{n_1}{n_e} \left(\frac{m_e}{2\pi\kappa T_1} \right)^{3/2} \exp\left(-\frac{m_e v^2}{2\kappa T_1} \right) + \frac{n_2}{2n_e} \left(\frac{m_e}{2\pi\kappa T} \right)^{3/2} \left[\exp\left\{ -\frac{m_e}{2\kappa T_2} [v_x^2 + v_y^2 + (v_z - V_0)^2] \right\} + \exp\left\{ -\frac{m_e}{2\kappa T_2} [v_x^2 + v_y^2 + (v_z + V_0)^2] \right\} \right]$$

find $F_{e0}(u)$ and ω_i. ////

The first contribution to ω_i in (9.4.4) is Landau damping from the maxwellian component of the distribution; this damping is independent of the direction $\mathbf{k}$ of the wave. The second imaginary contribution is from the particles in the bump. They were assumed cold except in the z direction, and so only $\omega_r/k_z = u$ gives resonant-particle effects. If the wave has phase velocity ω_r/k_z smaller than the mean velocity V_0 of particles in the bump, they contribute growth (their distribution has a positive slope). The frequency $\omega_r \approx \omega_{pe}$ is insensitive to k for $k\lambda_D \ll 1$, and so the phase velocity ω_r/k_z is determined by k_z, that is, $v_p \approx \omega_{pe}/k_z$. If ω_r/k_z is greater than V_0, the particles in the bump add to the damping produced by the maxwellian component of the distribution. At any particular value of $\omega_r/k_z < V_0$, the damping or growth of the wave is determined by the competition between these two terms; if the second term is larger in some region of ω_r/k_z (this means that the total distribution $f_{\max} + f_{\text{bump}} = f_{\alpha 0}$ has a positive slope in that region), the distribution is unstable to these waves.

By inspection, the destabilizing contribution from the bump is greatest at

$$\left| \frac{k_z V_0}{\omega_{pe}} \right| = 1 + \sqrt{k_z^2 \lambda_{D1}^2 \frac{T_2}{T_1}}$$

With this value for k_z, the maximum growth for the gentle-bump instability (and the instability condition) is given by

$$\omega_{i,\max} = \sqrt{\frac{\pi}{8}}\frac{1}{k^3\lambda_{D1}{}^3}\omega_{p1}\left[\frac{n_2}{n_1}\frac{T_1}{T_2}k^3\lambda_{D1}{}^3\frac{m_eV_0{}^2}{\kappa T_1}\exp(-\tfrac{1}{2})\right.$$
$$\left. - \exp\left(-\frac{1}{2k^2\lambda_{D1}{}^2}-\frac{3}{2}\right)\right] > 0 \qquad \text{for instability of a gentle bump} \qquad (9.4.5)$$

From (9.4.5) it is clear that if the bump is *too* gentle, the distribution is stable (because it is in fact monotone-decreasing). To encourage an instability, (9.4.5) suggests that:

1 The number of particles in the bump be increased; i.e., increase n_2/n_1.
2 The bump be made *sharper*; i.e., decrease T_2/T_1 and make the bump more like a δ function.
3 The speed of particles in the bump be increased; i.e., increase $mV_0{}^2/\kappa T_1$.

Because $k\lambda_D < 1$, the damping term $e^{-1/2k^2\lambda^2_{D1}}$ is smallest for waves with smaller k. But the growth term fixes $k_z \approx \omega_{p1}/V_0$, so that k, which is greater than k_z, cannot be reduced below ω_{p1}/V_0. That is, the most unstable waves propagate *parallel* to the velocity of the particle in the bump.

Comparing (9.4.5) with (9.3.10) reveals that waves in a gentle-bump plasma grow much more slowly than waves in a plasma of two cold streams of electrons of density n_1 and n_2 unless conditions 1, 2, or 3 are met, since

$$\omega_{i,\text{ gentle bumps}} \approx \sqrt{\frac{\pi}{8}}\,\omega_{pe}e^{-1/2}\frac{n_2 mV_0{}^2}{n_1\kappa T_2} \qquad (9.4.6)$$
$$\omega_{i,\text{ two electron beams}} \approx \frac{\omega_{pe}}{\sqrt{8}}\left(\frac{n_2}{n_1}\right)^{1/3}$$

But conditions 1, 2, and 3 are just the conditions that *convert* the gentle-bump instability into the two-stream instability. The most unstable system has a spread $\sqrt{\kappa T_2/m_e}$ much smaller than the beam speed V_0.

The nonlinear development of this instability has been widely studied, and these results are contained in Chap. 10.

Problem 9.4.3 Why does ω_i for the gentle bump, as given in (9.4.6), not reduce to ω_i for two beams in the limit $T_2 \to 0$? ////

Note that it was essential that the bump be on the *tail* of the distribution [the integration (9.4.2) was valid only for waves with $\omega_r/k > \sqrt{\kappa T_1/m_e}$]. In

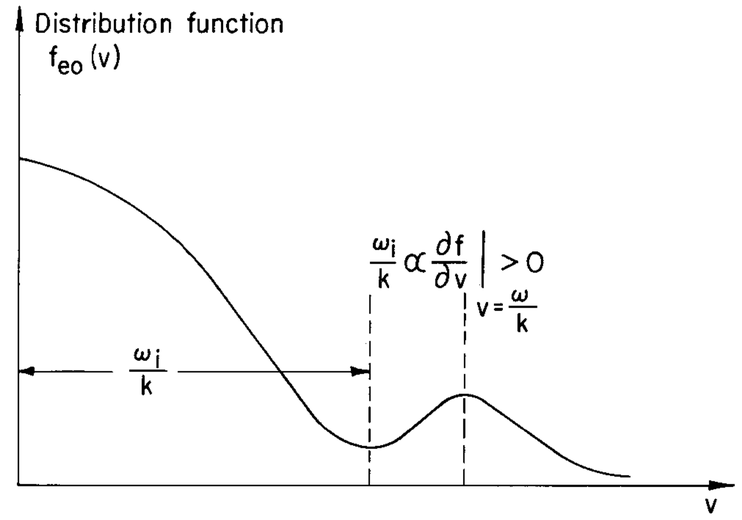


FIGURE 9.5.1
Bump-on-tail distribution function illustrating region of unstable $\omega/|k|$.

Sec. 9.6 it is shown that a region of positive $v\ \partial f_{\alpha 0}/\partial v$, though necessary for instability, is not a sufficient condition, and a weak bump inside the thermal spread $\sqrt{\kappa T_1/m_e}$ does *not* lead to instability.

Problem 9.4.4 Show that the above results are also approximately correct for a distribution with a bump at only $+V_0$.

$$f_{e0} = \frac{n_1}{n_e}\left(\frac{\alpha_1}{\pi}\right)^{3/2}e^{-\alpha_1 v^2} + \frac{n_2}{n_e}\left(\frac{\alpha_2}{\pi}\right)^{1/2}\delta(v_x)\,\delta(v_y)e^{-\alpha_2(v_z-V_0)^2}$$
$$f_{i0} = \delta(v_x)\,\delta(v_y)\,\delta(v_z) \qquad (9.4.7)$$

and find the size of the errors and inconsistencies due to neglecting the equilibrium current and magnetic field implied by (9.4.7). ////

9.5 MECHANISM OF THE TWO-STREAM INSTABILITY

It is easy to see why a plasma distribution with many bumps can be unstable, by recalling the derivation of Landau damping given in Chap. 8. The *sign* of the imaginary part of ω depends on the slope of the distribution (Fig. 9.5.1).

Indeed, the Landau damping result was consistent with the idea that in a stable plasma the number of particles near ω_r/k with velocity $v < \omega_r/k$ exceeds the number with $v > \omega_r/k$. The opposite is true for an unstable wave in a dis-

tribution like that of Fig. 9.5.1. The "number of resonant particles" argument fails in discussing the instability of two cold beams, for in that case essentially *no* particles have a velocity near the phase velocity of the unstable wave. These instabilities are *nonresonant*; their growth is *not* proportional to the number of particles with $v = \omega_r/k$.

The instability of cold beams can be interpreted in terms of charge bunching. A local increase in charge density (of electrons, for instance) induces a charge perturbation in a stream that passes over the bunch. Electrons in a beam passing over a clump of electrons at rest will be slowed down. Conservation of charge (Chap. 3) requires that the slowing down produce a corresponding *increase* in the electron density near the electron clump.

$$\frac{1}{n}\frac{Dn}{Dt} = -\frac{\partial V}{\partial x}.$$

The stream thus *feeds* the clump.

Instability requires that the total charge $\delta\rho_I$ induced in *all* the plasma streams exceed $\delta\rho_0$, the charge of the original perturbations.

A "charge clump" corresponds to an electrostatic wave. If this wave is neutrally stable (neither damping nor growing), the system is in a steady state in the wave frame (the frame in which the clump is at rest, $u = \omega/k$). In that frame, continuity and energy conservation can be used to calculate the induced charge $\delta\rho_I$ in terms of the perturbation $\delta\rho_0$. The condition $\delta\rho_I > \delta\rho_0$ leads to the instability conditions above.

In cases where the streaming velocities and the thermal velocities are not well separated, it is difficult to carry out the integrations over the velocity distribution necessary to calculate $D(\mathbf{k}, \omega)$ directly, and it is also impossible to use the fluid ideas presented in this section to determine the stability conditions. Fortunately, there are some general criteria for stability that can be used in these cases, and they are considered in the following section.

Problem 9.5.1 Calculate under what conditions $\delta\rho_I > \delta\rho_0$, and show that they are equivalent to (9.3.4) or (9.3.9). ////

9.6 THE NYQUIST METHOD AND THE PENROSE CRITERION FOR STABILITY

If two plasma streams are broad enough, or if the gentle bump is almost lost among the thermal distributions, the system will be very close to a single-humped stable distribution.

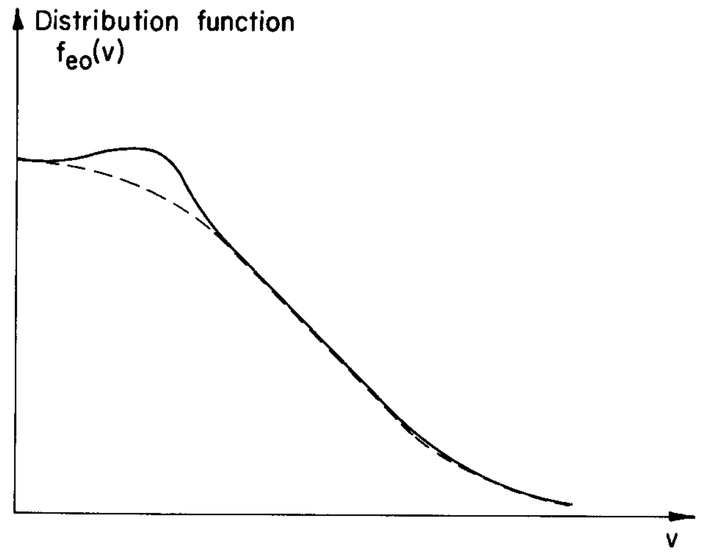


FIGURE 9.6.1
Electron distribution function in which the "gentle bump" is lost in the thermal spread shown in relation to a monotone-decreasing distribution (dotted line). The dotted distribution is stable according to Gardner's theorem. Gardner's theorem makes no prediction about the stability of the solid distribution, which is not monotone-decreasing. The Penrose criterion answers this question.

Penrose[1] has given a criterion to determine how much of a departure from a single-humped distribution is required in order that a distribution be unstable to the growth of electrostatic waves.

The Penrose criterion can be derived by a method that is useful in determining plasma stability for many situations, in addition to those included in the Penrose criterion. This is the Nyquist method, or technique.

The stability problem centers on determining whether the equation

$$D(k, \omega) = 0$$

has any roots with

$$\text{Im}\,(\omega) > 0$$

The form of $D(k, \omega)$ and the location of its roots $\omega(k)$ are determined by the equilibrium fields and plasma distribution.

Consider a function of ω defined by

$$G = \frac{1}{D(k, \omega)}\frac{dD(k, \omega)}{d\omega}$$

[1] R. Penrose, *Phys. Fluids*, 3:258 (1960).

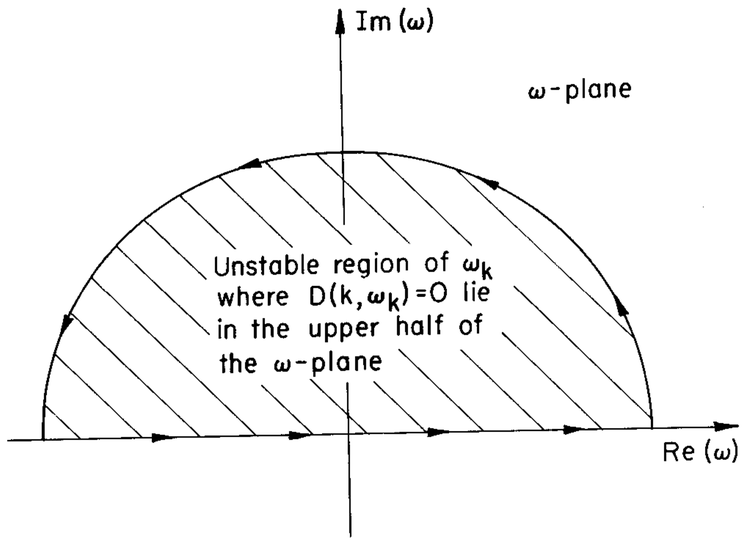


FIGURE 9.6.2
Complex ω plane illustrating region where roots of $D(\mathbf{k}, \omega) = 0$ correspond to unstable waves.

and assume that the dielectric $D(k, \omega)$ is an analytic function of ω in the upper half ω plane. Then the function $G(\omega)$ clearly has poles at those values of ω for which $D(\omega)$ has zeros. Indeed, in any region in which D is analytic (in regions where D has zeros but no poles),

$$\frac{1}{2\pi i}\int_C \left(\frac{1}{D}\frac{dD}{d\omega}\right) d\omega = N = \text{number of zeros of } D \text{ inside contour } C \qquad (9.6.1)$$

where the integral is over a closed contour. If the contour C encloses the upper half plane as in Fig. 9.6.2, then the integral (9.6.1) gives the number of roots $D = 0$ for which $\text{Im}\,(\omega) > 0$, that is, the number of unstable modes of the plasma.

For the general case of a function of a complex variable $f(z)$, which is analytic within and on a contour Γ except for a finite number of zeros and poles within Γ, Eq. (9.5.1) should be replaced by a more general expression, as follows: At a zero z_1, the function can be expanded in a Taylor series

$$f(z) = c_1(z - z_i)^{m_i} + c_2(z - z_i)^{m_i+1} + \cdots$$

where $c_1 \neq 0$. The ratio $f'(z)/f(z)$ becomes, near z_i,

$$\frac{f'(z)}{f(z)} = \frac{m_i}{z - z_i} + \cdots$$

Evidently $f'(z)/f(z)$ has a simple pole at z_i with residue m_i equal to the order of the original zero.

At a pole z_j of the function $f(z)$, it can be expanded in a Laurent series

$$f(z) = \frac{d_1}{(z - z_j)^{l_j}} + \frac{d_2}{(z - z_j)^{l_j-1}} + \cdots$$

where $d_1 \neq 0$ and $l_j \geq 1$.

At the point z_j, the ratio f'/f is

$$\frac{f'(z)}{f(z)} = -\frac{l_j}{z - z_j} + \cdots$$

Therefore $\dfrac{f'(z)}{f(z)}$ has simple pole at z_j

with residue equal to $-l_j$, the order of the original pole. There are no other singularities of f'/f. Thus

$$\int_\Gamma \frac{f'(z)}{f(z)}\, dz = 2\pi i\left(\sum_i m_i - \sum_j l_j\right)$$

where $\sum_i m_i$ is the number of simple zeros and $\sum_j l_j$ is the number of simple poles of the function $f(z)$.

The integration (9.6.1) is especially simple in the many cases for which

$$\int_0^\pi |\omega| i e^{i\theta}\, d\theta \frac{1}{D}\frac{dD}{d\omega}\bigg|_{|\omega|\to\infty} \Rightarrow 0 \qquad (9.6.2)$$

where $\omega = |\omega| e^{i\theta}$. This is the contribution to (9.6.1) from the great half circle in Fig. 9.6.2. In those cases, the number of unstable modes is given by finding the change in the phase of $D(\omega)$ as ω goes from $-\infty$ to $+\infty$ along the real ω axis, because in those cases

$$\int_C \frac{1}{D}\frac{dD}{d\omega}\, d\omega = \int_{-\infty}^{\infty} \frac{1}{D}\frac{dD}{d\omega}\, d\omega = \log\frac{D(\omega \to \infty)}{D(\omega \to -\infty)}$$

so that the ratio $D(\infty)/D(-\infty)$ determines N.

This technique can best be illustrated by examples. From (9.3.2), the dielectric $D(\omega) = D_r + iD_i$, for *real* ω, can be divided into

$$D_r(\omega) = 1 - \frac{\omega_{pe}^2}{k^2} \oint_{-\infty}^{\infty} \frac{\partial F/\partial u}{u - \omega/|k|}\, du$$
$$D_i(\omega) = -\pi \frac{\omega_{pe}^2}{k^2} \frac{\partial F}{\partial u}\bigg|_{u=\omega/|k|} \qquad (9.6.3)$$

where $F(u) = \int \delta\left(u - \dfrac{\mathbf{k}\cdot\mathbf{v}}{|k|}\right)\left(f_{e0} + \dfrac{m_e}{m_i} f_{i0}\right) d\mathbf{v}$

Then D_i vanishes at $\omega = \pm\infty$ (since $F \to 0$ at $u \to \pm\infty$), as well as at any other values of ω/k at which $\partial F(\omega/k)/\partial(\omega/k)$ vanishes. From (9.6.3) it can be seen that $D \approx 1 - \omega_{pe}^2/\omega^2$ at $|\omega| \to \infty$, so that

$$|\omega| \frac{1}{D} \frac{dD}{d\omega} \to 0 \text{ as } |\omega| \to \infty$$

This means that (9.6.2) is satisfied, and that the number of unstable modes of $D(k, \omega)$ is given by

$$N = \frac{1}{2\pi i} \ln \frac{D(\infty)}{D(-\infty)} \qquad (9.6.4)$$

Although $D = 1$ at $\omega \to \pm\infty$, the *phase* of D can be chosen arbitrarily at $\omega = -\infty$; the choice of phase at $\omega = -\infty$ determines the phase at $+\infty$. That is, if D at $\omega = -\infty$ is chosen so that

$$D(-\infty) = 1$$

then at $\omega = +\infty$,

$$D(+\infty) = e^{2\pi i n}$$

where n is an integer or zero, whose value depends on the equilibrium properties of the system that determine $D(\mathbf{k}, \omega)$. In general, n can be determined from the actual form of $D(\omega)$, and (9.6.4) then gives

$$N = n$$

as the number of unstable plasma modes.

To determine the phase n of D at $\omega = \infty$, note that as ω goes from $-\infty$ to $+\infty$, the function $D(\omega)$ traces out a curve in D space. The integer n is the number of times a plot of D_r versus D_i encircles the origin ($D = 0$) as ω goes from $-\infty$ to $+\infty$, along the real ω axis. To determine the number of origin encirclings, only the values of D_r at values ω_0 for which $D_i(\omega_0) = 0$ need be calculated. This procedure is really more easily done than said. Some examples are presented below.

9.6.1 $F(u)$ is Monotone-decreasing

Let F be monotone-decreasing from a maximum value at u_0,

$$\left.\frac{\partial F}{\partial u}\right|_{u_0} = 0 \qquad \left.\frac{\partial^2 F}{\partial u^2}\right|_{u_0} < 0$$

as sketched in Fig. 9.6.3*a*.

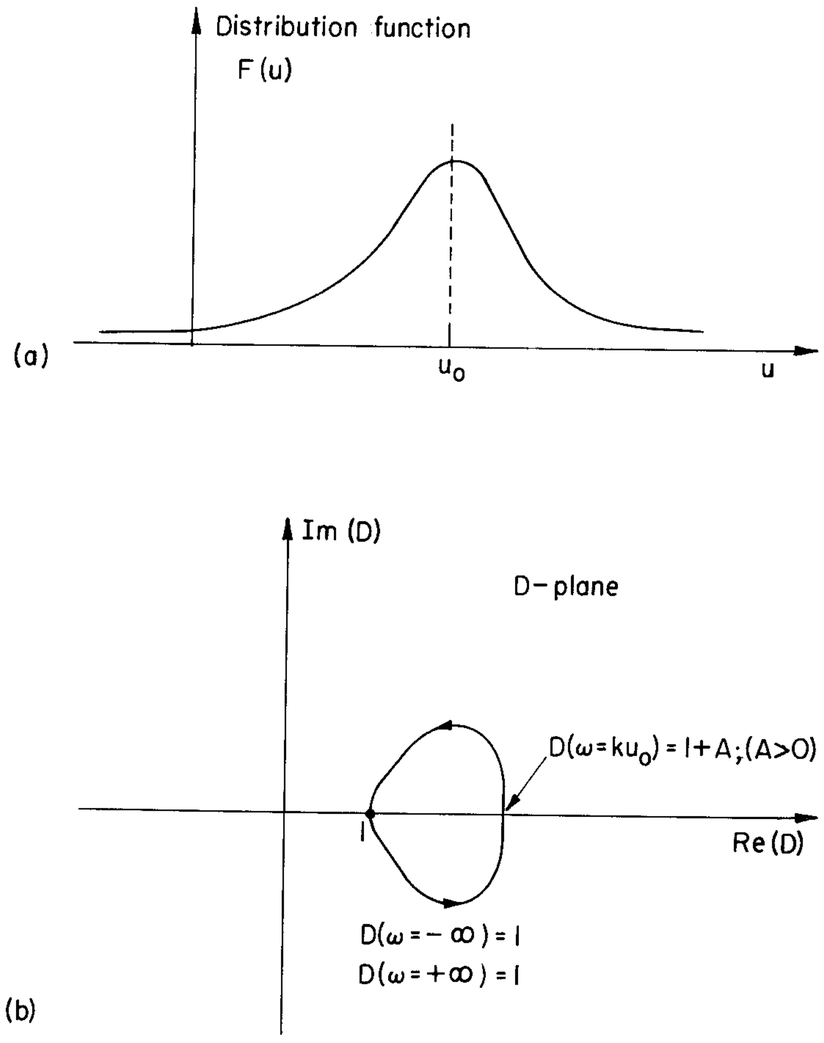


FIGURE 9.6.3
Monotone-decreasing distribution function with an average drift speed u_0 and corresponding Nyquist diagram, showing that such distributions are stable, in agreement with Gardner's theorem.

In this case the fact that $dF(x)/dy \equiv 0$ can be used to introduce a convenient factor $F(\omega/|k|)$ into the velocity integral which appears in D, writing

$$\oint_{-\infty}^{\infty} \frac{dF(u)/du}{u - \omega/|k|}\, du \equiv \oint_{-\infty}^{\infty} \frac{d[F(u) - F(\omega/|k|)]/du}{u - \omega/|k|}\, du$$

$$= \int_{-\infty}^{\infty} \frac{(d/du)[F(u) - F(\omega/|k|)]}{u - \omega/|k|}\, du \qquad (9.6.5)$$

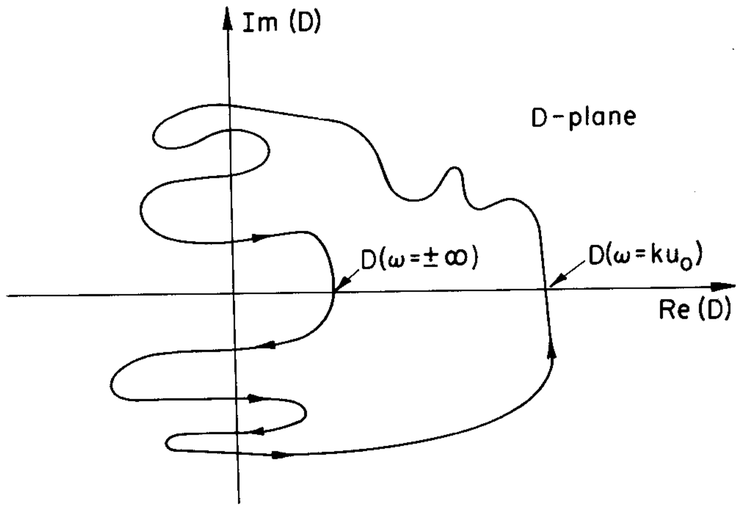


FIGURE 9.6.4
Nyquist diagram emphasizing that only the sign of Re[$D(\omega_0)$], where Im[$D(\omega_0)$] $= 0$, is required to determine the stability of a plasma.

The dielectric then becomes, for real values of ω,

$$D_r = 1 - \frac{\omega_{pe}^2}{k^2}\int_{\infty}^{\infty} \frac{F(u) - F(\omega/|k|)}{(u - \omega/|k|)^2}\, du$$
$$D_i = -\pi \frac{\omega_{pe}^2}{k^2}\frac{\partial F}{\partial u}\bigg|_{u=\omega/|k|} \tag{9.6.6}$$

Equation (9.6.6) is then used to obtain a plot of $D(\omega)$ for real ω. From the properties of F, D_i vanishes at three points. These are $\omega = +\infty$, $\omega = -\infty$, and $\omega = |k|u_0$, since $F(\infty) = F(-\infty) = (\partial F/\partial u)_{u=u_0} = 0$. At these three values of ω, D_r can be computed from (9.6.6).

$$D_r(\infty) = e^{2\pi i n} \qquad \text{with } n \text{ the integer to be determined}$$
$$D_r(-\infty) = 1 \tag{9.6.7}$$
$$D_r(\omega = |k|u_0) = 1 + \frac{\omega_{pe}^2}{k^2}\int \frac{F(u_0) - F(u)}{(u - u_0)^2}\, du > 0$$

with the last inequality following from the fact that F is maximum at $v = u_0$, so that $F(u_0) - F(u) \geq 0$ for all values of u. Thus D_r is positive at all points where D_i vanishes. From the values of D at these three points, the only points of which $D_i = 0$, the essential features of a Nyquist diagram can be constructed, as shown in Fig. 9.6.3*b*. The exact value of $D(\omega = ku_0)$ need not be calculated; as long as it is positive, the diagram shows that $D(+\infty) = D(-\infty)$, *including the phase*, because as ω goes from $-\infty$ to $+\infty$, the curve $D(\omega)$ does not encircle the origin. Therefore

$$n = N = 0$$

and the monotone-decreasing distribution is again proved stable, this time by the Nyquist method. Note that the behavior of $D(\omega)$ away from the points $D_i(\omega) = 0$ is irrelevant to the stability analysis. Even if $D(\omega)$ behaved as shown in Fig. 9.6.4, the Nyquist diagram would predict stability. The Nyquist method is a powerful tool with which to study plasma stability because it makes it possible to predict stability by calculating the sign of D_r for a few particular values of ω_r/k instead of having to solve an equation $D = 0$.

9.6.2 $F(u)$ Has One Minimum

Let $F(u)$ have one minimum, at $u = u_0$:

$$\frac{\partial F}{\partial u}\bigg|_{u=u_0} = 0 \qquad \frac{\partial^2 F}{\partial u^2}\bigg|_{u=u_0} > 0$$

If $F(u)$ has a minimum at u_0, it must have two local maxima; otherwise F will not vanish for large values of u as all physically realizable distributions must. Labeling the two maxima by u_1 and u_2, with u_1 the absolute maximum, as shown in Fig. 9.6.5*a*, and again writing the dielectric in the form

$$D_r = 1 - \frac{\omega_{pe}^2}{k^2}\int_{-\infty}^{\infty} \frac{F(u) - F(\omega/|k|)}{(u - \omega/|k|)^2}\, du \tag{9.6.8}$$

$$D_i = -\pi \frac{\omega_{pe}^2}{k^2}\frac{\partial F}{\partial u} = 0 \qquad \text{at } \omega = |k|u_1,\ \omega = |k|u_2,\ \omega = |k|u_0,\ \omega = \pm\infty$$

the Nyquist diagram is drawn by calculating D_r at $\omega = (|k|u_2, |k|u_1,$ and $|k|u_0)$.

Now $$D_r(\omega = |k|u_1) = 1 + \frac{\omega_{pe}^2}{k^2}\int_{-\infty}^{\infty} \frac{F(u_1) - F(u)}{(u - u_1)^2}\, du > 0$$

because $F(u_1)$ is the maximum value of $F(u)$; in addition, of course,

$$D_r(-\infty)] = 1$$
$$D_r(+\infty)] = e^{2\pi i n} \tag{9.6.9}$$

where $n = 0, 1, 2, \ldots$, so that D is positive at these points as well as at u_1. At the other two points, $\omega = |k|u_0$ and $\omega = |k|u_2$, D_r could be either positive or

negative, depending on $|k|$ and on the details of $F(u)$, but since, as shown in Fig. 9.6.4*a*, $F(\omega = |k|u_2) > F(\omega = |k|u_0)$, it follows that

$$D_r(\omega = |k|u_0) < D_r(\omega = |k|u_2) \qquad (9.6.10)$$

Therefore the Nyquist diagram takes the form of Fig. 9.6.5*b*, where $u_1 < u_0 < u_2$ has been assumed, without any loss of generality (see Prob. 9.6.2).

Problem 9.6.1 Prove (9.6.10). ////

Problem 9.6.2 Find the Nyquist diagram and stability condition if the absolute maximum u_1, the minimum u_0, and the local maximum u_2 are ordered $u_2 < u_0 < u_1$. ////

Without any further information about F, it can be seen from Fig. 9.6.5 that if $D_r(\omega = |k|u) > 0$, the plasma is stable; if $D_r(\omega = |k|u_2) < 0$, the plasma is also stable. Instability is possible if and only if

$$P(F) = \int_{-\infty}^{\infty} \frac{F(u_0) - F(u)}{(u - u_0)^2}\, du < 0 \qquad \text{for instability} \qquad (9.6.11)$$

The condition $P(F) < 0$ is called the *Penrose criterion.*

$$\int_{-\infty}^{\infty} \frac{F(u_0) - F(u)}{(u - u_0)^2}\, du < 0 \equiv \text{Penrose criterion}$$

$P(F)$ is called the *Penrose Function*, and

$$F = \int \delta\left(u - \frac{\mathbf{k}\cdot\mathbf{v}}{|k|}\right)\left(f_{e0} + \frac{m_e}{m_i} f_{i0}\right) d\mathbf{v}$$

If the Penrose criterion is satisfied, there is *some* value of k for which $D_r(\omega = |k|u_2) > 0$, $D_r(\omega = |k|u_0) < 0$. That this is the case can be seen by noting that, if (9.5.1) is satisfied,

$$D_r(\omega = ku_0) \to 1 \qquad \text{when } k \to \infty$$
$$D_r(\omega = ku_0) \to -\infty \qquad \text{when } k \to 0$$

so that for some k, $D_r(\omega = |k_0|u_0) = 0^-$. Since $D_r(\omega = |k|u_2) > D_r(\omega = |k|u_0)$, the Nyquist diagram for $|k| \lesssim |k|_0$ has the shape shown in Fig. 9.6.5*b*, and predicts instability. The Penrose criterion (9.6.11) determines the stability of any plasma distribution that has a minimum at $u = u_0$; that is, it determines the stability of multihumped distribution. The Penrose criterion is clearly a condition on the equilibrium distribution function.

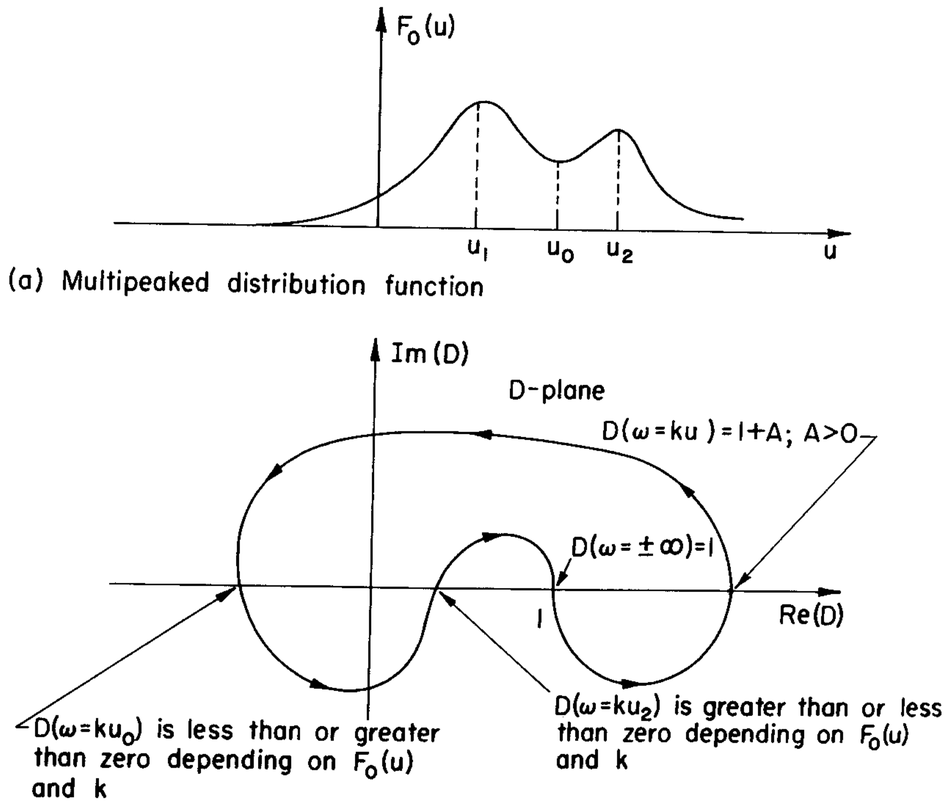


FIGURE 9.6.5
Multipeaked distribution function and corresponding Nyquist plot illustrating conditions for which such distributions are stable or unstable.

If the distribution satisfies (9.6.11), it will be unstable only in the range of wave number k in which $D_r(\omega = |k|u_0) < 0$, $D_r(\omega = |k|u_1) > 0$. The range of unstable k can be deduced from the Nyquist diagram,

$$\omega_{pe}^2 \int_{-\infty}^{\infty} \frac{F(u) - F(u_2)}{(u - u_2)^2}\, du < k^2 < \omega_{pe}^2 \int_{-\infty}^{\infty} \frac{F(u) - F(u_0)}{(u - u_0)^2}\, du$$

for instability (9.6.12)

The Penrose criterion has some interesting features:

1 If the distribution function F has a *hole*, i.e., if there are no particles in a velocity interval Δu about u_0, F is always unstable by the Penrose criterion, since then $F(u) - F(u_0) > 0$ for all values of u. Therefore, removing all the particles from even a very thin slice in velocity space is sufficient to drive an instability; in that case the instability will be weak, and limited to a narrow band of $\omega_r/|k|$.

2 If (9.6.11) is satisfied, so that $F(u)$ is unstable, *adding* particles in a thin slice of velocity space at $u = u_0$ will *not* stabilize the system. It would be possible to make $F(u) - F(u_0) < 0$ by adding such particles, but a nearby point $F(u_0 \pm \varepsilon)$ would now be a minimum of F, and the system would still be unstable. To stabilize the two-stream instability of a distribution such as Fig. 9.3.2, nearly the entire well in velocity space must be filled. Note, however, that adding particles in a velocity range tends to stabilize waves with that phase velocity.

3 The Penrose criterion is a necessary and sufficient condition to prove electrostatic instability. Because only the response of the plasma to electrostatic perturbations is included in the dielectric function (9.6.3), the Penrose criterion determines only whether or not electrostatic waves grow in a distribution $F_0(u)$. The Nyquist technique however, can be used to locate the zeros of any equation $G(\omega) = 0$.

4 Because the Penrose criterion is a condition on $f_{e0} + (m_e/m_i)f_{i0}$, it takes many more ions than electrons at a given velocity to create a hump which has an adjacent minimum.

5 A distribution can be two-humped and still be stable. This shows directly that Gardner's theorem (stability of single-humped distribution) is sufficient for stability, but is not a necessary condition.

Problem 9.6.3 Show by the Nyquist diagram method that a distribution in which F_0 has inflection point $\partial F(u)/\partial u_0 = 0$, $\partial^2 F/\partial {u_0}^2 = 0$, $\partial^3 F/\partial {u_0}^3 < 0$, but no minimum except at $u \to \pm\infty$, is stable. ////

It is useful to quote some results of the Penrose criterion applied to specific distributions. The proof of these results is left as a problem.

The distribution

$$F(u) = \frac{\Delta}{2\pi}\left[\frac{1}{(u-a)^2 + \Delta^2} + \frac{1}{(u+a)^2 + \Delta^2}\right] \tag{9.6.13}$$

where Δ is a thermal speed and a is the streaming speed, is stable when $a < \Delta$, according to the Penrose criterion. But $F(u)$ is single-humped and stable by Gardner's theorem only if $a < \Delta/\sqrt{3}$. Thus

$$a < \Delta \qquad \text{necessary and sufficient for stability from Penrose criterion} \tag{9.6.14}$$

$$a < \frac{\Delta}{\sqrt{3}} \qquad \text{sufficient for stability from Gardner's theorem}$$

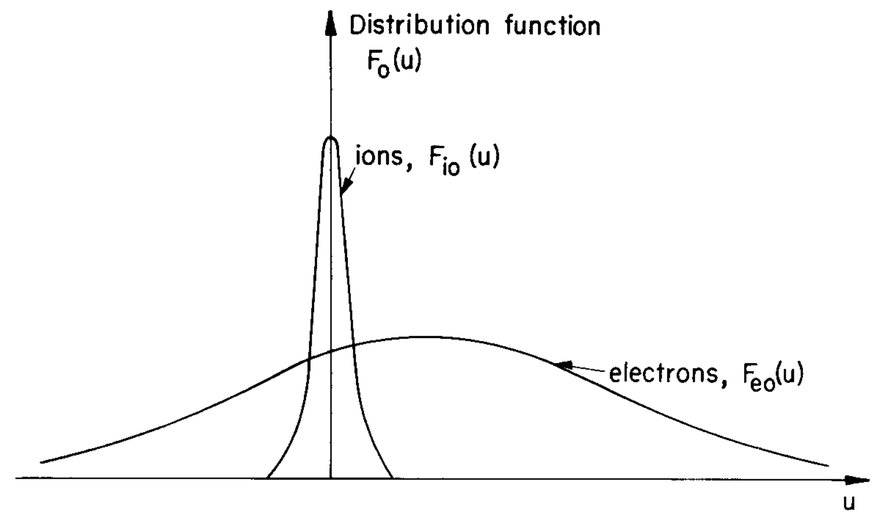


FIGURE 9.6.6
Distribution function of ions and electrons for the case of a warm electron stream drifting with a speed less than its thermal speed through a cold-ion distribution.

A second example of the use of the Penrose criterion to investigate stability is that of a displaced maxwellian distribution (Fig. 9.6.6), in which

$$F_{e0} = \left(\frac{m_e}{2\pi\kappa T_e}\right)^{1/2} e^{-m_e(u-u_0)^2/2\kappa T_e}$$
$$F_{i0} = \left(\frac{m_i}{2\pi\kappa T_i}\right)^{1/2} e^{-m_i u^2/2\kappa T_i} \tag{9.6.15}$$

Problem 9.6.4 Prove the stability criteria (9.6.14) for the distribution (9.6.13). ////

The Penrose result is that, when $T_e = T_i$,

$$u_0 < 1.3\sqrt{\frac{\kappa T}{m_e}} \qquad \text{for Penrose stability, if } T_e = T_i = T \tag{9.6.16}$$

At this value of u_0, the distribution (9.6.15) is still double-humped. When the Penrose criterion is applied to the distribution (9.6.15) for plasmas in which $T_e \gg T_i$, the stability criterion is quite different:

$$u_0 < \sqrt{\frac{\kappa T_i}{m_i}} \qquad \text{for Penrose stability, if } T_e \gg T_i \tag{9.6.17}$$

Problem 9.6.5 Prove the criteria (9.6.16) and (9.6.17) and show that $F(u)$ is still double-humped when (9.6.16) is satisfied. ////

Thus the distribution (9.6.15), for a plasma in which $T_e \gg T_i$, is driven unstable by a much weaker drift than is required to drive instability in an equal-temperature, $T_e = T_i$, plasma; the Penrose criterion shows that stability is influenced by the relative electron and ion temperatures. When a stability condition changes radically because a change is made in the distribution (in this case a change in T_e/T_i), "something is going on." The Nyquist method does not tell "what is going on." Two possibilities are:

1 A new plasma mode may be possible when $T_e/T_i \gg 1$.
2 A mode of the $T_e = T_i$ plasma may be driven unstable by the hotter electrons in the $T_e/T_i \gg 1$ case.

Note that when $u_0 \gg \sqrt{\kappa T_e/m_e}$, the instability predicted by the Penrose criterion is the two-stream instability derived by the normal-mode analysis, in Sec. 9.3. The Penrose criterion for $T_e = T_i$ shows the disappearance of that mode as u_0 is decreased. The reason that the critical value of u_0 for stability is sensitive to T_e/T_i can be found by returning to the normal-mode stability analysis begun in Sec. 9.3.

9.7 ION-ACOUSTIC INSTABILITY

The normal-mode analysis of the isotropic plasma (Sec. 8.6) showed that ion waves propagate and are only weakly damped in a plasma in which $T_e/T_i \gg 1$, although they are heavily damped in a plasma in which $T_e = T_i$. This suggests that ion waves might be driven unstable by a rather weak drift if the electrons are sufficiently hot so that the damping by the maxwellian component will be weak compared with the growth from the drifting maxwellian. To show that this is so, the distribution (9.6.15) is substituted into the dielectric function for electrostatic waves in a field-free plasma. The idea that ion waves may be unstable suggests an analysis which assumes that the electrons drift through the ions with speed u_0, the distribution $F_{\alpha 0}$ is given by 9.6.15, and the unstable waves satisfy

$$\left|\frac{\omega_r}{k}\right| \gg \sqrt{\frac{2\kappa T_i}{m_i}} \qquad \left|\frac{\omega_r}{k} - u_0\right| \ll \sqrt{\frac{2\kappa T_e}{m_e}} \tag{9.7.1}$$

since this is the phase-velocity range of ion waves, and this range of phase velocity also lies in a range near the velocity at which the Penrose function has its minimum.

The wave properties, stable or unstable, are obtained from solving for the zeros of $D(k, \omega)$.

$$D(k, \omega) = 1 + \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \int_{-\infty}^{\infty} \frac{k\, \partial F_{\alpha 0}/\partial u}{\omega - ku}\, du \tag{9.7.2}$$

where the field-free dielectric is used since $F_{\alpha 0}$ is a field-free equilibrium. Solving (9.7.2) as in Sec. 8.6, assuming that $\omega_i < \omega_r$, gives

$$\omega_r^2 = \frac{k^2 C_s^2}{1 + k^2 \lambda_{De}^2}$$

$$\omega_i = -|\omega_r| \sqrt{\frac{\pi}{8}} \frac{1}{(1 + k^2\lambda_{De}^2)^{3/2}} \left\{ \left(\frac{T_e}{T_i}\right)^{3/2} \exp\left[-\frac{T_e}{2T_i(1 + k^2\lambda_{De}^2)}\right] + \sqrt{\frac{m_e}{m_i}} \left(1 - \frac{u_0}{C_s}\sqrt{1 + k^2\lambda_{De}^2}\right)\right\} \tag{9.7.3}$$

When $u_0 = 0$, (8.7.3) is identical with (8.6.19), the dispersion relation for weakly damped ion waves. But if u_0 is large enough, the sign of ω_i reverses, and ion waves grow. Thus, from (9.7.1), ion waves are unstable if $T_e \gg T_i$ (so that ion damping is negligible) and if

$$u_0 > \frac{C_s}{\sqrt{1 + k^2\lambda_{De}^2}} = \frac{\sqrt{\kappa T_e/m_i}}{\sqrt{1 + k^2\lambda_{De}^2}} > \sqrt{\frac{\kappa T_i}{m_i}} \tag{9.7.4}$$

This result is in agreement with the Penrose criterion. The growth rate of the ion-wave instability is *proportional* to the streaming velocity if T_e/T_i is large enough so that the ion damping term can be neglected.

$$\omega_i = \frac{k(u_0 - |\omega_r/k|)}{(1 + k^2\lambda_{De}^2)^{3/2}} \sqrt{\frac{\pi}{8}} \frac{m_e}{m_i}$$

The driving effect of u_0 shows up in the growth rates here as in the two-beam example (9.3.8). Near the instability threshold the normal-mode result (9.7.3) shows clearly the competition between the damping by ions and the growth due to electrons. The calculations in Secs. 9.3 and 9.7 have examined two limits (the first was $u_0 \gg \sqrt{\kappa T_e/m_e}$, the second was $T_e/T_i \gg 1$) in which the growth of electrostatic waves can be calculated analytically. The Penrose criterion (8.6.11) can be evaluated numerically for the drifting maxwellian distribution (9.6.15) to

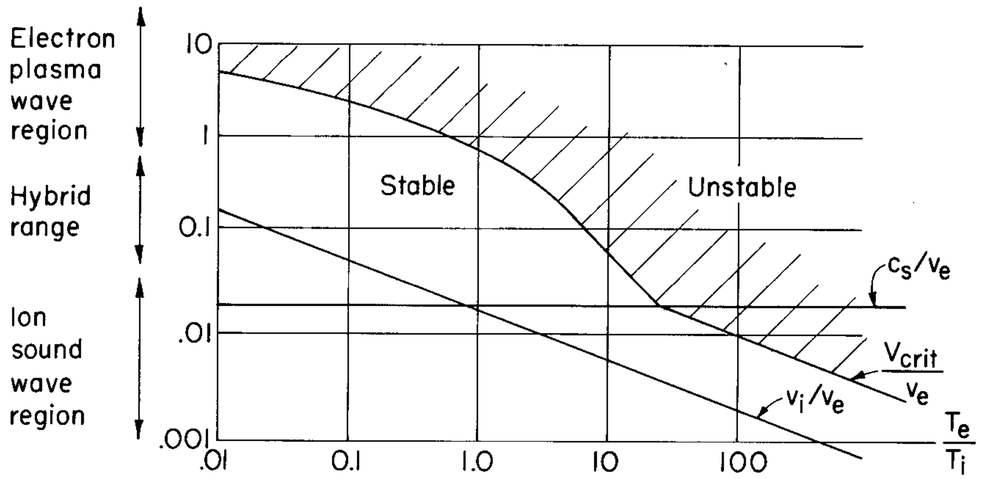


FIGURE 9.7.1
Plot of critical drift velocity as a function of T_e/T_i for the onset of electrostatic instability in an electron-ion plasma. (*After B. D. Fried and R. W. Gould, Phys. Fluids*, **4**:139 (1961).]

give a general plot of the stability boundary. This result is shown in Fig. 9.7.1, after Fried and Gould.[1]

Problem 9.7.1 Find the cone of **k** which is ion-acoustic unstable, for a distribution

$$f_{e0} = \left(\frac{m}{2\pi\kappa T}\right)_e^{3/2} \exp\left[-\frac{m_e}{2\kappa T_e}\left(v_x^2 + v_y^2 + (v_z - u_0)^2\right)\right]$$

$$f_{0i} = \left(\frac{m}{2\pi\kappa T}\right)_i^{2/3} \exp\left[-\frac{m_i v^2}{2\kappa T_i}\right]$$

Show that

$$\omega_i = \frac{\omega_r\left(1 - \dfrac{\mathbf{k}\cdot\mathbf{u}_0}{\omega_r}\right)}{(1 + k^2\lambda_{De}^2)^{3/2}}\sqrt{\frac{\pi}{8}\frac{m_e}{m_i}}$$

9.8 APPLICATIONS OF TWO-STREAM-INSTABILITY THEORY

To inject a note of realism into the very exact-looking stability calculations presented so far in this chapter, it is useful to consider two examples of how these stability calculations are used to give plausible suggestions for the stability

[1] B. D. Fried, and R. W. Gould, *Phys. Fluids*, **4**: 139 (1961).

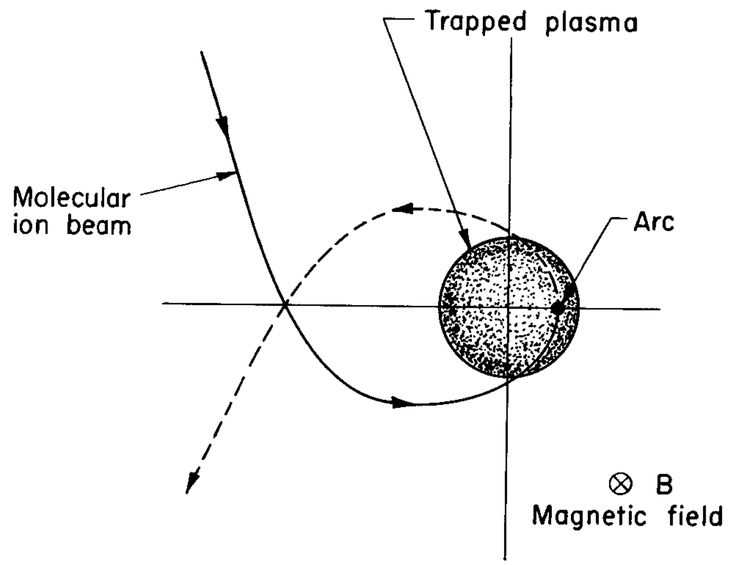


FIGURE 9.8.1
Schematic of plasma production and trapping in a magnetic mirror using a high-energy (600 keV) D_2^+ molecular ion beam which is dissociated by passing through an arc. This experiment is called DCX and was performed at the Oak Ridge National Laboratories.

of the much more complicated plasma geometries which actually exist in experiments.

9.8.1 DCX Machine

An early scheme for buildup of plasma in a magnetic field was the DCX experiment.[1] This experiment existed in various forms, but the idea was always that a molecular ion beam (for example, singly ionized D_2 molecules) would be injected continuously into a vacuum region, perpendicular to a magnetic field. Left to their own devices, the molecules would rotate by 180° in the B field and leave by essentially the same path on which they entered. However, in fact, some of the molecules are dissociated by passing through a low-pressure arc along the magnetic field. This produces two lighter deuterium ions, which have a smaller gyroradii, and are thus trapped in the magnetized region. This is shown schematically in Fig. 9.8.1. Experimentally, this scheme worked, and the number of trapped ions increased, until the density built up to a critical value, whereupon the trapped ions began to diffuse rapidly out of the trapping region. Probes

[1] P. R. Bell, G. G. Kelley, N. H. Lazar, and R. J. Macklin, Jr., *Nucl. Fusion*, 1962 Suppl. Part 1, p. 251, 1962.

showed that the system became electrically noisy at this time. The diffusion of particles and growth of electrostatic field gave evidence that the distribution had become unstable. The experimental result can be understood in a qualitative way by noting that the ion motion is highly ordered because of the injection process while the neutralizing electrons are smeared out, and applying the two-stream theory developed in Sec. 9.3 to calculate the stability of a beam of ions of speed u_0 through electrons at rest. However, since the theory of Sec. 9.3 did not include an equilibrium magnetic field B_0, that theory can be applied only to wavelengths so short that the effect of B_0, namely, cyclotron rotation, can be ignored. That is, only instabilities with wavelength shorter than the gyroradius of ions in B_0,

$$|k|u_0 > \omega_{ci} = \frac{eB}{m_i c} \qquad (9.8.1)$$

can be treated by the theory of Sec. 9.3 (the radius of an ion orbit is u_0/ω_{ci}). This limitation on k, together with the condition for two-stream instability,

$$|ku_0| < \omega_{pe} \qquad (9.8.2)$$

leads to the prediction that the density might be limited, because

$$n_e > \frac{B^2}{4\pi m_i c^2}\frac{m_e}{m_i} \qquad \text{for electrostatic two-stream instability} \qquad (9.8.3)$$

Other instabilities may also be present, but (9.8.3) represents the onset of one particular instability. This limit on density is consistent with the experimental results, which show that the plasma is quiescent below that density and turbulent above.

9.8.2 Turbulent Heating by Instabilities

A number of schemes for plasma heating are based on the idea of inducing a stream of electrons in a confined plasma, by either applying an electric field or by pulsing a magnetic field. In one such experiment[1] two plasma streams are injected into a magnetic mirror from opposite ends (Fig. 9.8.2). Little change is observed in the temperature of the plasmoids as they drift through each other. The relative drift of the plasma streams is well below the electron thermal speed, and the electron and ion temperature are comparable. The experiment was repeated with the electrodes e_1 and e_2 pulsed at a relative potential of 20 keV, after the plasma enters the mirror region. With the electrodes operating, the

[1] T. Jensen and F. R. Scott, *Phys. Rev. Letters*, **19**:1100 (1967).

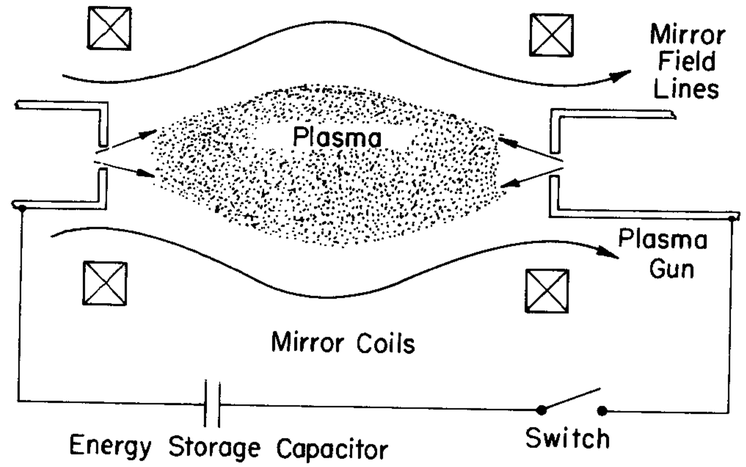


FIGURE 9.8.2
Schematic of a turbulent-heating experiment in which a large current is passed through a plasma trapped in a magnetic mirror field.

plasma heats from 50 eV to temperatures of 1,000 eV per electron-ion pair in a time of $\sim 3 \times 10^{-6}$s. This energy transfer cannot be explained by Joule heating; the discrete-particle collision frequency in this plasma is negligible. The explanation is that the 20-kev electric field produces a sufficiently large u_0 so that the plasma is two-stream-unstable, exceeding the stability criterion (9.6.16). This is one of several such experiments where the two-stream instability is used to induce and study turbulent heating in plasmas.

9.8.3 Instabilities in Counterstreaming Plasmas

Electrostatic instabilities in counterstreaming plasmas have been investigated by Stringer,[1] who solved numerically the dispersion relation (9.3.2) for a distribution that included both counterstreaming electrons and ions; i.e.,

$$F_{i0}(u) = \frac{1}{2}\sqrt{\frac{m_i}{2\pi\kappa T_i}}\left\{\exp\left[-\frac{m_i(u-u_0)^2}{2\kappa T_i}\right] + \exp\left[-\frac{m_i(u+u_0)^2}{2\kappa T_i}\right]\right\}$$

$$F_{e0}(u) = \frac{1}{2}\sqrt{\frac{m_e}{2\pi\kappa T_e}}\left\{\exp\left[-\frac{m_e(u-u_0)^2}{2\kappa T_e}\right] + \exp\left[-\frac{m_e(u+u_0)^2}{2\kappa T_e}\right]\right\}$$

The three types of instability by which the two drifting plasmas can couple are electron–electron [(9.3.7)], electron–ion [(9.3.3) and (9.3.7)], and ion–ion. The growth rates for these three types of coupling are shown in Fig. 9.8.3.

[1] T. E. Stringer, Plasma Physics, *J. Nucl. Energy*, **C6**:267 (1964).

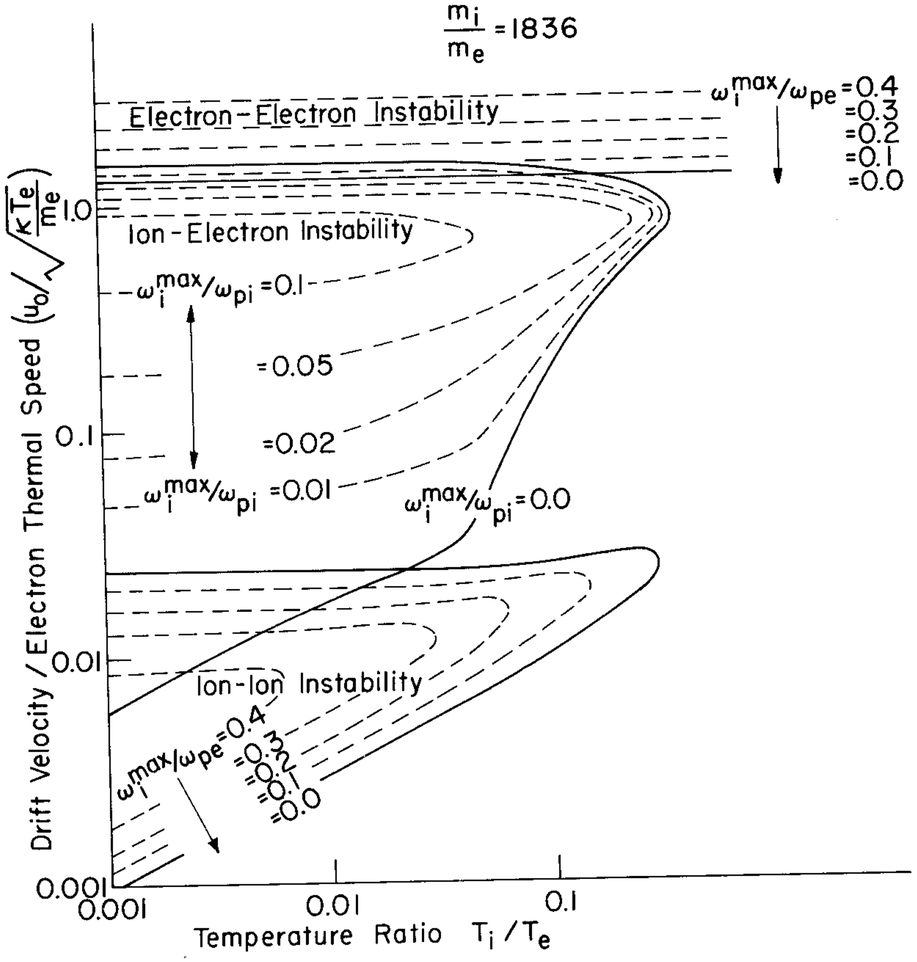


FIGURE 9.8.3
Growth rate of the fastest-growing wavelength in counterstreaming plasmas as a function of the drift-velocity to thermal-speed ratio $u_0/\sqrt{\kappa T_e/m_e}$ and the ion-to-electron temperature ratio. (*After T. E. Stringer, Plasma Physics, J. Nucl. Energy*, **C6**:267 (1964).]

9.9 INSTABILITIES IN ANISOTROPIC PLASMAS

The instabilities described so far in this chapter have been electrostatic in origin. These instabilities were due to charge bunching $\rho_{\mathbf{k}}$; the theory showed that for unstable distribution functions, clumps of charge grew in density and produced increasing fields $E_{\mathbf{k}}$. For all the cases considered, particle streaming was the basic source of energy for instabilities, and the stability criteria were essentially limits on how much streaming was allowed in a stable system. Particle streaming is not the only way in which a plasma can depart from thermodynamic equilibrium. This section examines anisotropy as a source of plasma instability.

Anisotropy is a feature of many plasma states. In magnetically confined plasma, the temperature parallel to the magnetic field $\mathbf{B}_0$ and the temperature perpendicular to $\mathbf{B}_0$ are often quite different. For example, magnetic compression will produce a plasma with $T_\perp \gg T_\parallel$. Even in field-free plasma the presence of directional features, e.g., particle streaming, two-dimensional shock structures, unusual plasma-production methods, etc., may provide a plasma state with an enhanced temperature in some direction. Coulomb collisions will eventually make the distribution isotropic. However, in many cases, collective effects (instabilities) are more effective than binary collisions in moving the plasma toward an isotropic state.

The techniques used to study instability due to anisotropy are those used throughout this chapter. The Vlasov equation gives the response $f_{\mathbf{k}}$ to a general perturbation $\mathbf{E}_{\mathbf{k}}$, $\mathbf{B}_{\mathbf{k}}$ and with Maxwell's equations gives the dispersion relation for waves $\mathbf{E}_{\mathbf{k}} \exp[i(\mathbf{k}\cdot\mathbf{x} - \omega t)]$. The dispersion relation is used to find ω as a function of $\mathbf{k}$, leading to stability criteria, growth rates, and wavelengths of unstable waves.

The mechanism for these instabilities, which involves the formation of perturbed currents, is called *pinching*; a perturbed current produces a self-consistent magnetic field, which in turn may cause the current to *compress*. If a perturbed current, by pinching, leads to a current density greater than the original perturbation, the plasma is unstable. The condition that a perturbed current will not grow gives conditions on the required degree of isotropy for plasma stability.

9.10 ELECTROMAGNETIC PINCHING INSTABILITIES

When a plasma distribution $f_{\alpha 0}$ is perturbed, there will generally be magnetic forces as well as electrical forces, either tending to restore the equilibrium or to enhance the perturbation. The study of electrostatic instabilities either neglects magnetic forces or singles out these perturbations for which magnetic forces are absent.

In cases which are electrostatically unstable, it is often safe to ignore magnetic effects, because the magnetic forces are, in general, smaller than electric forces by v^2/c^2, where v is a typical plasma velocity. Because the forces are weaker, typical growth rates for instabilities driven by magnetic forces are smaller by a factor $\sim(v/c)$ than are the typical electrostatic growth rates, and the equilibrium will be altered by the electrostatic instability before magnetic effects contribute.

Still, since there are plasma states which are stable to electrostatic waves and are nevertheless far from thermodynamic equilibrium, perturbations which

do not alter the charge density but *do* give a perturbed current density can grow at the expense of nonthermal features of these distributions. The most noteworthy example is that of a plasma which is anisotropic but which is a monotone-decreasing function of v_x^2, v_y^2, and v_z^2 separately. Such a distribution is stable to electrostatic perturbations, but can be unstable to electromagnetic perturbations.

Problem 9.10.1 Show that the normal modes in an anisotropic plasma

$$f_{\alpha 0} = F_{\alpha 0}(v_x^2)G_{\alpha 0}(v_y^2)H_{\alpha 0}(v_z^2)$$

separate into one electrostatic mode and two electromagnetic modes. Using the Penrose criterion, show that the electrostatic mode is stable if E_α, G_α, and H_α are each monotone-decreasing functions of their argument. ////

Consider a field-free plasma state described by a distribution $f_{\alpha 0}$ which is assumed to be both monotone-decreasing and anisotropic. A general perturbation $\mathbf{E}_1 = \bar{\mathbf{E}}_\mathbf{k} \exp[i(\mathbf{k}\cdot\mathbf{x} - \omega t)]$, $\mathbf{B}_1 = \bar{\mathbf{B}}_\mathbf{k} \exp[i(\mathbf{k}\cdot\mathbf{x} - \omega t)]$ of this field-free plasma state produces a distribution function

$$f_\alpha = f_{\alpha 0} + f_{\alpha 1}$$

with

$$f_{\alpha 1} = -\frac{q_\alpha}{m_\alpha}\frac{[\bar{\mathbf{E}}_\mathbf{k} + (\mathbf{v}\times\bar{\mathbf{B}}_\mathbf{k})/c]\cdot\nabla)f_{\alpha 0}}{i(\mathbf{k}\cdot\mathbf{v} - \omega)} \tag{9.10.1}$$

as described in Chap. 8. Associated with this distribution is a perturbed current $\mathbf{J}_1 = \bar{\mathbf{J}}_\mathbf{k} \exp[i\mathbf{k}\cdot\mathbf{x} - i\omega t]$,

$$\bar{\mathbf{J}}_\mathbf{k} = -\sum_\alpha \frac{\bar{n}_\alpha q_\alpha^2}{m_\alpha}\int \frac{\mathbf{v}[\bar{\mathbf{E}}_\mathbf{k} + (\mathbf{v}\times\bar{\mathbf{B}}_\mathbf{k})/c]\cdot\nabla_\mathbf{v} f_{\alpha 0}}{i(\mathbf{k}\cdot\mathbf{v} - \omega)}\,d\mathbf{v} \tag{9.10.2}$$

The time behavior of the perturbation is given by Maxwell's equations

$$\mathbf{k}\times(\mathbf{k}\times\bar{\mathbf{E}}_\mathbf{k}) = -\frac{\omega^2}{c^2}\bar{\mathbf{E}}_\mathbf{k} + \frac{4\pi}{c}\frac{\omega}{ic}\bar{\mathbf{J}}_\mathbf{k} \tag{9.10.3}$$

Without loss of generality, the coordinate system is taken with the x axis parallel to the $\mathbf{k}$ vector. In this coordinate system the plasma need not be isotropic or have a monotone-decreasing distribution. The equilibrium state is assumed field-free, and thus $f_{\alpha 0}$ satisfies

$$\rho_{q0} = \sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha 0}\,d\mathbf{v} = 0 \tag{9.10.4}$$

and

$$\mathbf{J}_0 = \sum_\alpha \bar{n}_\alpha q_\alpha \int \mathbf{v} f_{\alpha 0}\,d\mathbf{v} = 0 \tag{9.10.5}$$

in the rest frame of the plasma.

In the coordinate system described ($\mathbf{k} = k_x\hat{\mathbf{x}}$), Maxwell's equations (9.10.3) become

$$\left[\omega^2 - \sum_\alpha \omega_{p\alpha}^2\omega\int\frac{v_x(\partial f_{\alpha 0}/\partial v_x)}{k_x v_x - \omega}\,d\mathbf{v}\right]E_x = 0 \tag{9.10.6}$$

$$\left[k_x^2c^2 - \omega^2 + \sum_\alpha \omega_{p\alpha}^2 + \sum_\alpha \omega_{p\alpha}^2\int\frac{k_x v_y^2\dfrac{\partial f_{\alpha 0}}{\partial v_x}}{k_x v_x - \omega}\,d\mathbf{v}\right]E_y = 0 \tag{9.10.7}$$

$$\left[k_x^2c^2 - \omega^2 + \sum_\alpha \omega_{p\alpha}^2 + \sum_\alpha \omega_{p\alpha}^2\int\frac{k_x v_z^2(\partial f_{\alpha 0}/\partial v_x)}{k_x v_x - \omega}\,d\mathbf{v}\right]E_z = 0 \tag{9.10.8}$$

Equation (9.10.6) with $E_x \neq 0$ is the dispersion relation for the electrostatic waves examined in Secs. 8.3 to 8.6. The stability of these waves is given by the Penrose criterion (9.6.1). Equations (9.10.7) and (9.10.8) are dispersion relations for electromagnetic waves, which can be driven unstable by temperature anisotropies, counterstreaming distributions, etc. A good way to get a feeling for the stability of a plasma to electromagnetic perturbations is to solve (9.10.8) for a variety of plasma states $f_{\alpha 0}$.

9.10.1 Electromagnetic Stability of a Field-free Plasma

Isotropic distributions $f_{\alpha 0} = f_{\alpha 0}(v_x^2 + v_y^2 + v_z^2)$ were shown in Sec. 9.6 to be stable to electrostatic perturbations; they are also stable to electromagnetic perturbations, as seen by either an extension of the Nyquist method (Sec. 9.6) or by showing that (9.10.8) is never satisfied if $\omega = \omega_r + i\omega_i$ with $\omega_i > 0$. It is useful at this point to apply the Nyquist technique to prove the stability of isotropic distributions to electromagnetic waves. The Nyquist technique uses the identity

$$N = \frac{1}{2\pi i}\int_C \frac{1}{D}\frac{dD}{d\omega}\,d\omega$$

to find N, the number of roots of $D(k, \omega) = 0$ that have $\omega_i > 0$, corresponding to instability; the contour C is shown in Fig. 9.6.2.

To use the Nyquist method to find the unstable roots of[1]

$$H = k_x^2c^2 - \omega^2 + \sum_\alpha \omega_{p\alpha}^2 + \sum \omega_{p\alpha}^2\int\frac{k_x v_y^2\,(\partial f_{\alpha 0}/\partial v_x)}{k_x v_x - \omega}\,d\mathbf{v} \tag{9.10.9}$$

[1] The zeros of $H \equiv -\omega^2 D(k, \omega)$ are the same as the zeros of $D(k, \omega)$. The treatment in this section analyzes H rather than D to avoid the fact that $D(k, \omega)$ for the electromagnetic waves has a pole at $\omega = 0$. Section 9.6 shows how to use the Nyquist method for a function with poles, and it is suggested that the reader use that method to analyze $D(k, \omega)$, obtaining the same stability information as in this section.

it must be noted that $(1/H\, dH/d\omega)\, d\omega$ does *not* vanish for $|\omega| \to \infty$, as was the case for electrostatic instabilities. Thus $\int_C (1/H\, dH/d\omega)\, d\omega$ cannot be replaced by $\int_{-\infty}^{\infty} (1/H\, dH/d\omega)\, d\omega$ as in that case. This poses no real difficulty; writing $\omega = \mathrm{Re}^{i\theta}$, the integral

$$\lim_{R\to\infty} \int_0^{\pi} i\, \mathrm{Re}^{i\theta}\, d\theta \left(\frac{1}{H}\frac{dH}{d\omega}\right)$$

can be carried out explicitly. Alternatively, sketching $H(\omega)$ as ω follows the contour C (Fig. 9.6.2) gives N directly, because N will be the number of times H encircles the origin ($H = 0$) as ω follows its contour.

$$\int \frac{1}{H}\frac{dH}{d\omega}\, d\omega = \ln H(\omega)$$

Problem 9.10.3. Show that an isotropic distribution is stable to electromagnetic waves, by assuming instability and proving that $H(k_x, \omega) = 0$ cannot be satisfied. ////

For the part of the ω contour along which ω is *real*, writing $H(k_x, \omega) = H_r + iH_i$,

$$H_r = k_x{}^2c^2 - \omega^2 + \sum_\alpha \omega_{p\alpha}{}^2 + \sum_\alpha \omega_{p\alpha}{}^2 \oint \frac{k_x v_y{}^2 (\partial f_{\alpha 0}/\partial v_x)}{k_x v_x - \omega}\, d\mathbf{v}$$
$$H_i = -\pi \sum_\alpha \omega_{p\alpha}{}^2 \int v_y{}^2 \frac{\partial f_{\alpha 0}}{\partial v_x}\, \delta\left(v_x - \frac{\omega}{k_x}\right) d\mathbf{v} \qquad (9.10.10)$$

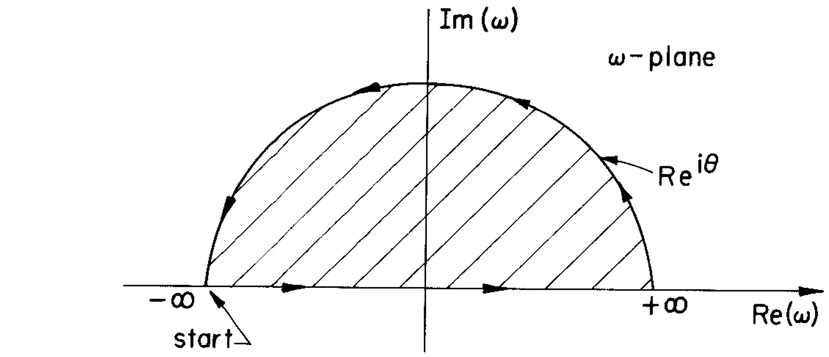
which for isotropic distributions $f_{\alpha 0}(v^2)$ becomes

$$H_r(\omega) = k_x{}^2c^2 - \omega^2 - \sum_\alpha \omega_{p\alpha}{}^2 \oint \frac{\omega f_{\alpha 0}}{k_x v_x - \omega}\, d\mathbf{v}$$
$$H_i(\omega) = -\pi \sum_\alpha \omega_{p\alpha}{}^2 \frac{\omega}{|k_x|} \int f_{\alpha 0}\left(v_y{}^2 + v_z{}^2 + \frac{\omega^2}{k_x{}^2}\right) dv_y\, dv_z \qquad (9.10.11)$$

where $\oint$ means the *Cauchy principal value*. For values of ω along the great circle enclosing the upper half ω plane (i.e., for $\omega = \mathrm{Re}^{i\theta}$, $R \to \infty, 0 < \theta < \pi$), $H(\omega) = -\omega^2$ from (9.10.9). Thus

$$\begin{aligned} H_r &= R^2 \cos 2\theta \\ H_i &= -R^2 \sin 2\theta \end{aligned} \qquad \omega = Re^{i\theta} \qquad R \to \infty \qquad (9.10.12)$$

By examining (9.10.11) and (9.10.12), the curve $H(\omega)$ can be immediately constructed as in Fig. 9.10.1, where the ω contour is also shown for convenience. Following the arrows, it is clear that H does not encircle the origin ($H = 0$) as



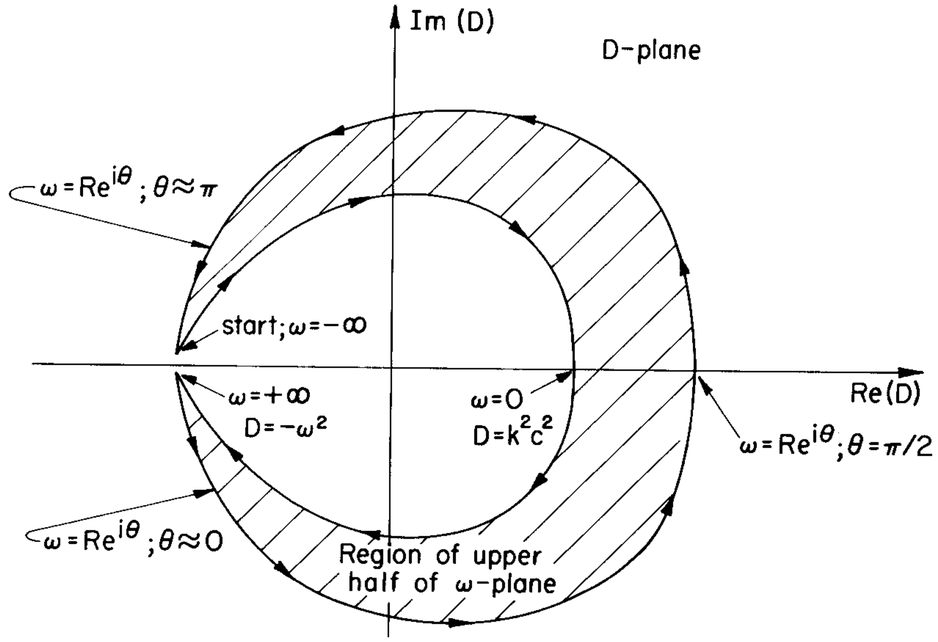


FIGURE 9.10.1
Diagram of ω plane and D plane (actually, the H plane) showing Nyquist diagram for case of electromagnetic waves in a plasma with monotone-decreasing distribution function showing that such distributions are stable.

ω follows the closed contour C in Fig. 9.10.1. Therefore N (= number of roots of $H = 0$ corresponding to instability) is equal to zero, and isotropic distributions are proved stable to electromagnetic perturbations. If the value H_r at $\omega = 0$ in Fig. 9.10.1 had been *negative*, the Nyquist diagram would have demonstrated instability. This gives a criterion for instability of a single-humped anisotropic plasma, analogous to the Penrose criterion for electrostatic waves.

From (9.10.10) this criterion is $H_r(\omega = 0) < 1$, that is

$$k^2c^2 + 2\omega_{pe}{}^2 \int v_y{}^2 \left(\frac{\partial}{\partial v_x{}^2} - \frac{\partial}{\partial v_y{}^2}\right)\left(f_{e0} + \frac{m_e}{m_i} f_{i0}\right) d\mathbf{v} < 0 \qquad \text{for instability}$$

A specific example in which anisotropy produces instability will now be discussed.

9.10.2 Electromagnetic Instability of a Bimaxwellian Plasma

Consider a plasma that has a temperature T_x different from that in directions perpendicular to x.

$$f_{\alpha 0} = \left(\frac{m}{2\pi\kappa T_x}\right)_\alpha^{1/2} \left(\frac{m}{2\pi\kappa T}\right)_\alpha \exp\left[-\left(\frac{m}{2\kappa T_x}\right)_\alpha v_x{}^2 - \left(\frac{m}{2\kappa T}\right)_\alpha (v_y{}^2 + v_z{}^2)\right] \qquad T_x \neq T \tag{9.10.13}$$

Such a distribution could result from a particular plasma-production scheme, and then persist on a time scale $t < \tau_{\text{collision}}$. This distribution is stable to electrostatic perturbations. Its stability to electromagnetic perturbations is tested by substituting f_α from (9.10.13) into $H(k_x, \omega)$ [Eq. (9.10.10)] and examining the Nyquist diagram. This diagram is similar to Fig. 9.10.1 since, for the distribution (9.10.13), for real ω

$$H_i(\omega) = -\sqrt{\frac{\pi}{2}} \sum_\alpha \omega_{p\alpha}{}^2 \left(\frac{m}{\kappa T_x}\right)_\alpha^{1/2} \frac{\omega}{|k_x|} \exp\left[-\frac{m_\alpha}{2\kappa T_{x\alpha}} \left(\frac{\omega}{|k_x|}\right)^2\right]$$

only vanishes, for real ω, at the point $\omega = 0$. The value of H_r at $\omega = 0$ is

$$H_r(\omega = 0) = k_x{}^2c^2 + \sum_\alpha \omega_{p\alpha}{}^2 \left(1 - \frac{T_\perp}{T_x}\right)_\alpha$$

and the sign of this function determines the stability of the distribution. The Nyquist diagram is shown in Fig. 9.10.2. The stability of a bimaxwellian distribution to electromagnetic waves of wave number $\mathbf{k} = k_x\hat{\mathbf{x}}$ is then determined by the condition

$$\left(\frac{T_\perp}{T_x} - 1\right)_e > \frac{k_x{}^2c^2}{\omega_{pe}{}^2} \qquad \text{for instability} \tag{9.10.14}$$

$$\mathbf{E} = E_\perp \mathbf{i}_\perp e^{ik_x x}$$

where $\omega_{pi}{}^2$ has been neglected with respect to $\omega_{pe}{}^2$. So a bimaxwellian distribution of electrons is unstable to waves that propagate in the "colder" direction and have their electric vector in the "warmer" direction.

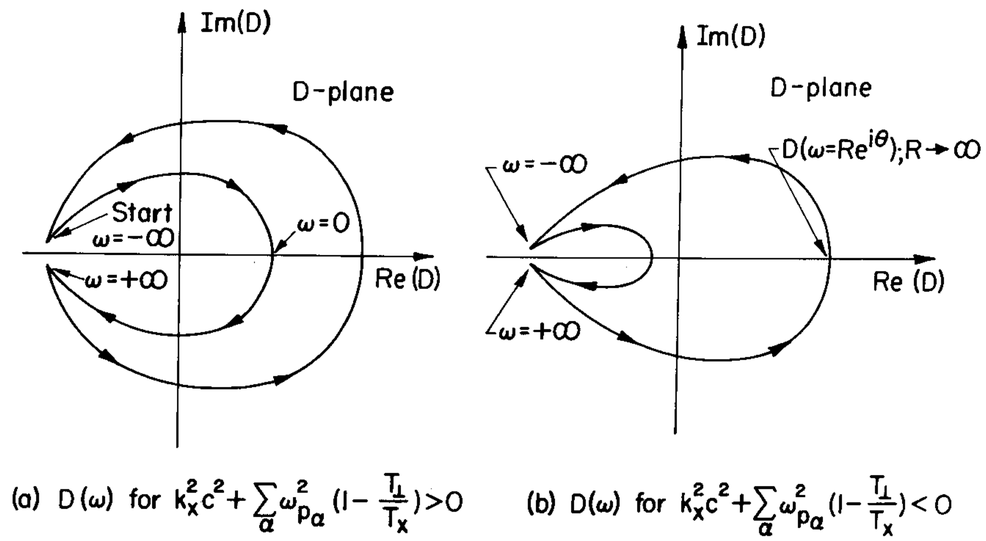


FIGURE 9.10.2
Nyquist diagrams for electromagnetic waves in a bimaxwellian plasma illustrating (*a*) a stable case, where the temperature anisotropy is small and (*b*) an unstable case, where the temperature anisotropy is large. (Actually, the quantity plotted is H, and not D.)

The rate at which this instability grows can be determined directly from the zeros of the dielectric (9.10.10), which becomes

$$H(k_x, \omega) = 1 - \frac{k_x{}^2c^2}{\omega^2} - \sum_\alpha \frac{\omega_{p\alpha}{}^2}{\omega} \left(1 - \int_L \frac{T_{\alpha\perp}}{T_{\alpha x}} \frac{v_x f_{\alpha 0}}{v_x - \omega/|k_x|} d\mathbf{v}\right) = 0 \tag{9.10.15}$$

Substituting the distributions (9.10.13) into (9.10.15), the dispersion relation for electromagnetic waves in a bimaxwellian plasma is

$$k_x{}^2c^2 - \omega^2 + \sum_\alpha \omega_{p\alpha}{}^2 \left(1 - \frac{T_{\alpha\perp}}{T_{\alpha x}}\right) = \sum_\alpha \omega_{p\alpha}{}^2 \frac{T_{\alpha\perp}}{T_{\alpha x}} \xi_\alpha Z(\xi_\alpha) \tag{9.10.16}$$

where

$$\xi_\alpha = \frac{\omega}{k_x\sqrt{2\kappa T_{\alpha x}/m_\alpha}}$$

and

$$Z(\xi_\alpha) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty}{}_L \frac{\exp(-x^2)\, dx}{x - \xi_\alpha}$$

The function $Z(\xi)$, which occurs frequently in plasma-stability theory, has been tabulated,[1] using limiting forms for $Z(\xi)$.

[1] B. D. Fried and S. Conte, "The Plasma Dispersion Function," Academic, New York, 1961.

$$Z(\xi) = \begin{cases} -2\xi + \frac{4}{3}\xi^3 - \cdots + i\sqrt{\pi}\,\frac{k_x}{|k_x|}\exp(-\xi^2) & |\xi| \ll 1 \\ -\frac{1}{\xi} - \frac{1}{2\xi^3} - \frac{3}{4\xi^5} - \cdots & |\xi| \gg 1 \end{cases}$$

In the limit $T_{\alpha\perp} = T_{\alpha x}$ and $|\xi_\alpha| \gg 1$, Eq. (9.10.16) reduces to $\omega^2 \approx \omega_p{}^2 + k^2c^2$, the familiar light waves in a plasma. The instability introduced by the anisotropy is found at a much lower frequency; if $T_{\alpha\perp} \neq T_{\alpha x}$, a solution of (9.10.16) can be found in which $|\xi_\alpha| \ll 1$.

In the limit $\xi_\alpha \ll 1$, Eq. (9.10.16) reduces to (neglecting also $\omega^2 < k_x{}^2c^2$)

$$k_x{}^2c^2 + \sum \omega_{p\alpha}{}^2\left(1 - \frac{T_{\alpha\perp}}{T_{\alpha x}}\right) = \sum_\alpha \omega_{p\alpha}{}^2\,\frac{T_{\alpha\perp}}{T_{\alpha x}}\left(-2\xi^2 + i\sqrt{\pi}\,\frac{k}{|k|}\,\xi\right)_\alpha \qquad |\xi_\alpha| \ll 1$$

with approximate solution (neglecting the ion terms because $m_e \ll m_i$)

$$\omega = i\sqrt{\frac{2}{\pi}}\,|k_x|\sqrt{\frac{\kappa T_{ex}}{m_e}}\,\frac{T_{ex}}{T_{e\perp}}\left(\frac{T_{e\perp}}{T_{ex}} - 1 - \frac{k_x{}^2c^2}{\omega_{pe}{}^2}\right) \tag{9.10.17}$$

This mode, which exists only in an anisotropic plasma, is either pure growing or damping. Instability (growth) occurs for wavelengths such that

$$k_x{}^2 \lesssim k_0{}^2 \equiv \frac{\omega_{pe}{}^2}{c^2}\left(\frac{T_{e\perp}}{T_{ex}} - 1\right)$$

with the assumption $|\xi_\alpha| \ll 1$ restricting the k range to maintain $\left|\omega/k_x\sqrt{\kappa T_{ex}/m_e}\right| < 1$. The growth rate vanishes for $k_x = 0$ and $k_x = k_0{}^2$, and is maximum at[1]

$$k^2 = \tfrac{1}{3}k_0{}^2$$

$$\omega_{i,\max} = \left(\frac{8}{27\pi}\right)^{1/2}\omega_{pe}\left(\frac{\kappa T_{ex}}{m_ec^2}\right)^{1/2}\frac{T_{ex}}{T_{e\perp}}\left(\frac{T_{e\perp}}{T_{ex}} - 1\right)^{3/2} \tag{9.10.18}$$

Problem 9.10.4 Show that no solution of (9.10.16) satisfying $|\xi_\alpha| \ll 1$ is possible if the plasma is isotropic, $T_{e\perp} = T_{ex}$. Show further that the mode $|\xi_\alpha| \ll 1$ is stable for weak anisotropy, $T_{e\perp}/T_{ex} - 1 \ll 1$. ////

If the anisotropy is strong, so that $\omega_{i,\max}$ as given by (9.10.18) exceeds $k_x\sqrt{\kappa T_{ex}/m_e}$, then the approximation $|\xi_\alpha| \ll 1$ which led to (9.10.18) breaks down, and the formulas (9.10.17) and (9.10.18) no longer describe the mode. In the strong-anisotropy case, instability is found for frequency ω such that

[1] E. S. Weibel, *Phys. Rev. Letters*, **2**:83 (1959).
R. C. Davidson, D. A. Hammer, I. Haber, and C. E. Wagner, *Phys. Fluids*, **15**, 317 (1972)

$|\xi| \gg 1$. Taking that limit, (9.10.26) becomes, neglecting obvious electron subscripts on T and assuming $T_x \ll T_\perp$,

$$k_x{}^2c^2 + \omega_{pe}{}^2 = -\omega_{pe}{}^2\,\frac{T_\perp}{T_x}\,k_x{}^2\,\frac{\kappa T_x}{\omega^2 m_e}$$

$$k_x{}^2\,\frac{\kappa T_x}{\omega^2 m_e} \ll 1$$

or

$$\omega = \pm i\omega_{pe}\left(\frac{T_\perp}{T_x}\right)^{1/2}\left(\frac{k_x{}^2\kappa T_x/m_e}{k_x{}^2c^2 + \omega_{pe}{}^2}\right)^{1/2} \tag{9.10.19}$$

$$\omega_{i,\max} = \omega_{pe}\left(\frac{T_\perp}{T_x}\right)^{1/2}\left(\frac{\kappa T_x}{m_ec^2}\right)^{1/2}$$

Note that the restriction $|\omega| \gg |k_x|\sqrt{\kappa T_x/m_e}$, with ω given by (9.10.19), is possible only if the system is strongly anisotropic,

$$\frac{T_\perp}{T_x} \gg 1$$

These results show that an anisotropy in the pressure leads to instability with a maximum growth rate $\omega_i \approx \omega_{pe}(v_{\text{thermal}}/c)$. The results are not restricted to a bimaxwellian distribution. Indeed, *defining* the temperature in the j direction by

$$\kappa T_j = \int \tfrac{1}{2}mv_j{}^2 f\,d\mathbf{v}$$

the results (9.10.16) to (9.10.19) are independent of the details of $f_{\alpha 0}$ that give an increased pressure in a particular direction. For example, a large effective T_x might be produced by the counterstreaming with speed u_0 of two cool ($T = T_x$) clouds of electrons in a neutralizing background. Then (see Prob. 9.10.5) the stability is described by (9.10.18) or (9.10.19), with $\kappa T_\perp \equiv \frac{1}{2}mu_0{}^2$ and $T_x = T$. An interesting demonstration of the existence of these instabilities is provided by a computer simulation experiment[1] that follows the orbits of a large number of charged particles, initially distributed with a velocity anisotropy. In agreement with the theory described here, electric and magnetic fields grow exponentially with time, with rates in agreement with (9.10.19). Two examples of such simulation experiments are shown in Fig. 9.10.3*a* and *b*. The first shows the instability of the bimaxwellian electron distribution, the second of counterstreaming electron clouds. The effect of the instability is shown by the plot of T_x and $T_\perp \equiv T_y$ as a function of time. The instability allows the distribution to relax toward isotropy.

[1] R. C. Davidson et al., *Ibid.*

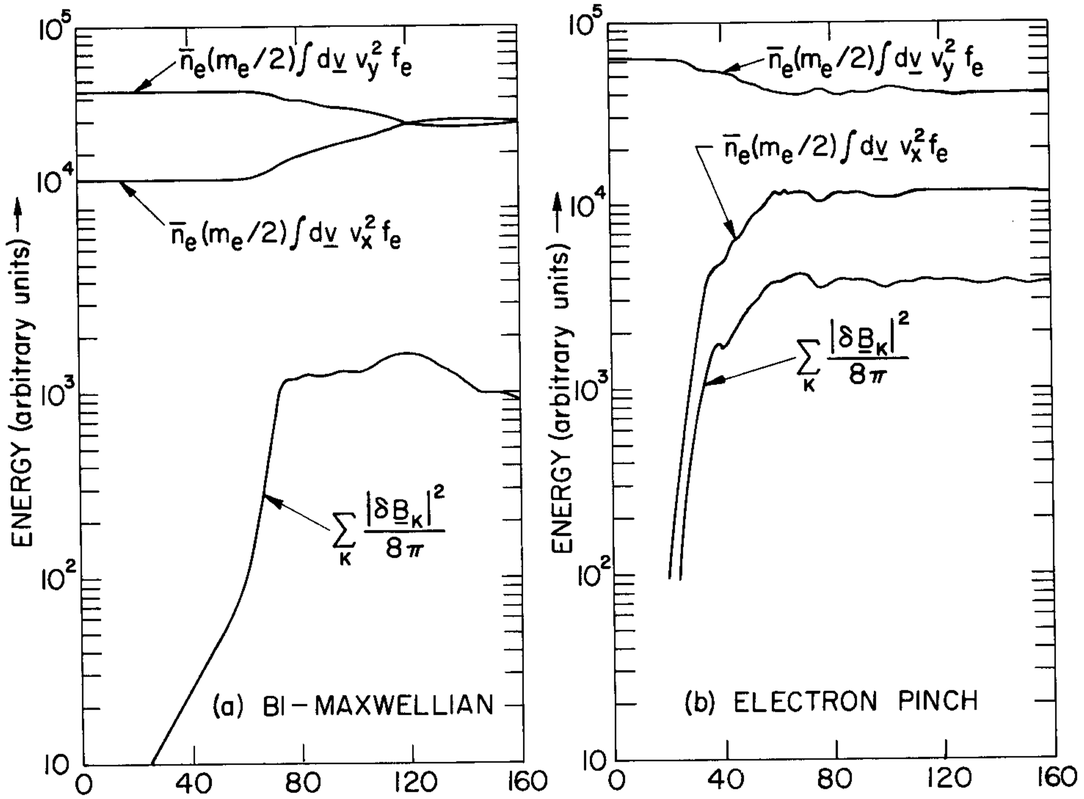


FIGURE 9.10.3
Computer simulation experiment of the time development of electromagnetic instability in anisotropic plasma. (*a*) Initially the plasma is maxwellian with $T_y > T_x$. As time increases, the instability produces increased fluctuations δB_k and drives the distributions toward equipartition $T_x \approx T_y$. (*b*) Initially the plasma is a cold ($T_x = 0$) beam of electrons streaming in the y direction. Instability leads to a sharing of the beam energy with thermal energy T. The time-asymptotic states are discussed in the original reference. [*After R. C. Davidson, D. A. Hammer, I. Haber and C. E. Wagner, Phys. Fluids*, **15**, 317 (1972).] A single underscore denotes a vector.

The growth rate (9.10.19) is typical for unstable electromagnetic waves in plasmas; by comparison, typical growth rates for electrostatic instabilities of a two-stream distribution are $\omega_i \approx \omega_{pe}$. Thus electromagnetic instabilities are typically reduced by v_{thermal}/c below the growth rates for nonresonant electrostatic instabilities, if both occur (see Prob. 9.10.5).

The electromagnetic instability grows at the expense of the excess electron energy $\kappa(T_\perp - T_x)$, as is clear from Fig. 9.10.3. An instability is limited by the energy available in the nonthermal feature of the distribution. Therefore the

instability described here may lead eventually to a stable electron distribution. The ion distribution may still be anisotropic (massive ions respond much more slowly than electrons to high-frequency waves, $\omega > \omega_{pi}$), and the ion terms should then be retained in solving (9.10.16). Therefore, as a practical matter, it is necessary to examine the stability of an anisotropic ion distribution in a background of *isotropic* electrons even though the instabilities of this system are slower than those discussed above, because that may be the way in which collective effects ultimately thermalize an anisotropic plasma.

The calculation of stability of anisotropic ions proceeds in a straightforward way, as in the previous examples, and it is left as an exercise for the reader to demonstrate the results quoted in Probs. 9.10.6 and 9.10.7.

Problem 9.10.5 Discuss the electromagnetic stability of counterstreaming electron beams,

$$f_{e0} = \tfrac{1}{2}[\delta(v_y - u_0) + \delta(v_y + u_0)]\,\delta(v_x)\,\delta(v_z)$$
$$f_{i0} = \tfrac{1}{2}\delta(\mathbf{v})$$

Show that

$$\omega \approx \frac{i|k_x|u_0}{(1 + k_x^2 c^2/\omega_{pe}^2)^{1/2}}$$

Compare ω_i for the electromagnetic wave with ω_i for the electrostatic wave which is unstable in the same distribution. In what direction is $\mathbf{k}$ for electromagnetic instability? For electrostatic instability? ////

9.10.3 Instability of Anisotropic Ions in a Background of Hot Electrons

Problem 9.10.6 Show that the distribution

$$f_{i0} = \left(\frac{m}{2\pi\kappa T_\perp}\right)_i \left(\frac{m}{2\pi\kappa T_x}\right)_i^{1/2} \exp\left(-\frac{m_i v_x^2}{2\kappa T_{ix}}\right) \exp\left(-\frac{m_i v_\perp^2}{2\kappa T_{i\perp}}\right)$$
$$f_{e0} = \left(\frac{m_e}{2\pi\kappa T_e}\right)^{3/2} \exp\left(-\frac{m_e v^2}{2\kappa T_e}\right) \qquad T_e \gg T_i$$

is unstable to electromagnetic waves $\mathbf{E} = \bar{E}\hat{\mathbf{i}}_\perp \exp(ik_x x)$ if

$$\frac{T_{i\perp}}{T_{ix}} - 1 \geq \frac{k_x^2 c^2}{\omega_{pi}^2} \qquad \text{for instability,} \qquad (9.10.20)$$

and that $\omega_i \approx \omega_{pi}\dfrac{v_{\text{thermal}}}{c}$ ////

9.10.4. Electromagnetic Instability of Anisotropic Ions in a Cold Background

Problem 9.10.7 Show that the distribution

$$f_{i0} = \left(\frac{m}{2\pi\kappa T_x}\right)_i^{1/2}\left(\frac{m}{2\pi\kappa T_\perp}\right)_i \exp\left(-\frac{m_i}{2\kappa T_{ix}}v_x^2\right)\exp\left[-\frac{m_i}{2\kappa T_{i\perp}}(v_y^2+v_z^2)\right]$$

$$f_{e0} = \left(\frac{m_e}{2\pi\kappa T_e}\right)^{3/2}\exp\left(-\frac{m_e}{2\kappa T_e}v^2\right) \qquad T_e \ll T_i$$

is unstable to an electromagnetic wave

$$\mathbf{E} = \bar{E}_\perp \hat{\mathbf{i}}_\perp \exp(ik_x x)$$

Find the maximum growth rate and the range of unstable wave numbers. ////

Problem 9.10.8 Compare the stability properties of a plasma in the limit of infinitely hot electrons with those of a plasma with infinitely massive ions. These are two approximations often used in the study of plasma stability. ////

9.11 DISCUSSION OF PINCHING INSTABILITIES

The mechanism of the instabilities discussed above is the pinching of a perturbed current, which increases the current density in a perturbation. This is seen more explicitly by considering a plasma that consists of two counterstreaming electron beams with a temperature T_e perpendicular to the stream direction; i.e.,

$$f_{e0} = \frac{1}{2}\frac{m_e}{2\pi\kappa T_e}\exp\left(-\frac{m_e}{2\kappa T_e}(v_x^2+v_z^2)\right)[\delta(v_y-u)+\delta(v_y+u)] \qquad (9.11.1)$$

$$f_{i0} = \delta(\mathbf{v})$$

From the discussion above, this distribution is unstable to a wave with its $\mathbf{E}$ vector along the streams and its $\mathbf{k}$ vector perpendicular to them. Further, the instability condition (see again Sec. 9.10) is $\mathrm{Re}[D(\omega = 0)] < 0$, or

$$k^2c^2 + \omega_{pe}^2\left(1 - \frac{m_e u_0^2}{\kappa T_e}\right) < 0 \qquad \text{for pinching instability}$$

This can be rewritten in the form

$$\frac{m_e u_0^2}{\kappa T_e} > \frac{k^2c^2}{\omega_{pe}^2} + 1 \qquad \text{for pinching instability} \qquad (9.11.2)$$

The condition (9.11.2) for instability is identical with the Bennett pinch condition. If a spatially collimated beam of charged particles passes through a neutralizing background, it tends to spread because of its thermal pressure $n\kappa T$. However, it produces at its surface a magnetic field whose pressure $B^2/8\pi$ tends to constrict the beam. When $B^2 > 8\pi n\kappa T$, the beam self-focuses. The magnetic field B generated by the beam depends on the velocity of the beam u, its density n, and cross-sectional area A. With the correspondence $8\pi/k^2 \equiv A$, (the wavelength of the instability is related to the beam size), the condition for the instability (10.11.2) is also the condition for the self-focusing of a single electron beam in an ion background.[1]

There are many other examples of the correspondence of magnetic instability criteria and the Bennett pinch condition.[2] The addition of an external magnetic field can stabilize the magnetic instabilities if the ratio $\beta_\perp = 8\pi n\kappa T_\perp/B^2$ is small enough, where $T_\perp$ is now the temperature perpendicular to the applied magnetic field.

9.12 STABILITY OF ANISOTROPIC MAGNETIZED PLASMA

The instabilities driven by anisotropies in the velocity distribution of a field-free plasma persist when the plasma is immersed in a magnetic field, and indeed most occurrences of the pinching instability are for magnetized plasma. There are a number of ways in which a magnetized plasma develops an anisotropic distribution. For example:

1. Magnetic compression (a slow rise in $|\mathbf{B}_0|$) produces an anisotropic distribution which, if stable, will persist at least on a time scale less than $\tau_{\text{collision}}$.
2. Free expansion along $\mathbf{B}_0$ produces anisotropy.
3. Confinement of plasma in a mirror machine *requires* anisotropy (as shown in Appendix I, only particles with $w_\perp > w_\parallel$ will be reflected from a weak magnetic mirror).
4. Plasma built up by injection across a $\mathbf{B}_0$ field will be anisotropic.

Each of these phenomena occurs in both laboratory and space plasmas. The stability of the magnetized plasma is investigated by solving the general dispersion relation derived in Chap. 8 for particular anisotropic distributions and for various directions of $\mathbf{k}$ and $\bar{\mathbf{E}}_\mathbf{k}$. In this section a few of the *results* of such a procedure are summarized.

[1] W. H. Bennett, *Phys. Rev.*, **45**:890 (1934).
[2] H. P. Furth, *Phys. Fluids*, **6**:48 (1963).

The stability criteria for the field-free case took the form

$$\frac{T_{e\perp}}{T_{ex}} - 1 > \frac{k_x{}^2 c^2}{\omega_p{}^2} \qquad (9.12.1)$$

so that instability could be produced for a small anisotropy by reducing k; that is, modes with longer wavelength are easiest to destabilize. In a magnetic field, the modes change character when their wavelength exceeds the gyroradius of the plasma particles; that is, $k^2 a_{ce}{}^2 > 1$ is required in order for a field-free motion of wavelength $2\pi/k$ to persist in a magnetized plasma. Therefore it should be expected that the restriction $k^2 a_{ce}{}^2 > 1$, or $k^2 c^2/\omega_{pe}{}^2 = k^2 a_{ce}{}^2/\beta_e > 1/\beta_e$, where $\beta_e = n\kappa T_e/(B_0{}^2/4\pi)$ must be added to the requirements derived in the preceding section. Indeed, the results of a detailed analysis of the stability of magnetized anisotropic plasma show that if $\mathbf{B}_0 = B_0\,\hat{\mathbf{z}}$, the mode discussed above ($\mathbf{E} = E_k\,\hat{\mathbf{z}} e^{ik_x x}$) is unstable if

$$\left(\frac{T_{\parallel B_0}}{T_{\perp B_0}}\right)_e - 1 > \frac{1}{\beta_e} \qquad \text{for instability } (\mathbf{k} \perp \mathbf{B}_0\,,\ \mathbf{E}_\mathbf{k} \parallel \mathbf{B}_0) \qquad (9.12.2)$$

This instability "goes" for plasma with strong flow along field lines. The fact that low β stabilizes these electromagnetic instabilities is plausible, because it is difficult for a plasma perturbation to alter the confining magnetic field $\mathbf{B}_0$ if the plasma pressure is much less than $B_0{}^2/8\pi$; electromagnetic instabilities *require* such a perturbed field $\mathbf{B}_1$.

In many situations (magnetic compression, mirror confinement), $T_\perp > T_\parallel$. There plasmas are unstable to a wave with $\mathbf{E}_\mathbf{k}$ and $\mathbf{k}$ both nearly perpendicular to $\mathbf{B}$, with a stability condition similar to (9.12.2). This mode is called the *mirror instability*[1]

$$\left(\frac{T_{\perp B_0}}{T_{\parallel B_0}} - 1\right)_e > \frac{1}{2\beta_e} \qquad \text{for mirror instability } (\mathbf{k} \perp \mathbf{B}_0\,,\ \mathbf{E}_\mathbf{k} \parallel \mathbf{B}_0) \qquad (9.12.3)$$

In addition to the instability (9.12.2), a plasma with $T_\parallel > T_\perp$ is unstable to the fire-hose instability, calculated from the fluid equations in Chap. 5. This mode is circularly polarized, with $\mathbf{E}_1$ perpendicular and $\mathbf{k}$ parallel to $\mathbf{B}_0$, and the stability condition is[1]

$$\frac{T_{\parallel B_0}}{T_{\perp B_0}} - 1 > \frac{2}{\beta} \qquad \text{for fire-hose instability } (\mathbf{k} \parallel \mathbf{B}_0\,,\ \mathbf{E}_\mathbf{k} \perp \mathbf{B}_0)$$

As mentioned above, these modes are stabilized when the wavelength of the potentially unstable mode exceeds the gyroradius (see, for comparison, the discussion in Sec. 9.8 of the two-stream instability in a magnetic field).

[1] H. P. Furth, *Phys. Fluids*, 6:48 (1963).

9.13 LOSS-CONE INSTABILITY

There are electrostatic instabilities driven by some types of velocity anisotropy, particularly in the case of plasma in a magnetic field. An early example was studied by Harris,[1] who showed that anisotropic electron distributions in a magnetic field are unstable to electrostatic waves, of wavelength of order of the electron gyroradius, propagating at an angle with respect to $\mathbf{B}_0$. There is a density threshold, $\omega_{pe}/\omega_{ce} > 1$, for existence of the Harris instability. A more recent example is the loss-cone instability,[2] which will be treated here in detail, and is characteristic of the Harris type of instability.

Plasma confined by a magnetic mirror field is characterized by a distribution of ions in the midplane (Fig. 9.13.1), in which the velocity distribution perpendicular to the magnetic field depends on the value of v parallel to the magnetic field. In particular,

$$f_{i0} = f_{i0}(v_\perp{}^2, v_\parallel{}^2) \qquad (9.13.1)$$

where

$$f_{i0} = 0 \qquad \text{for } v_\perp{}^2 < M v_\parallel{}^2 \qquad (9.13.2)$$

and M is a constant that depends on the mirror ratio. That (9.13.2) is required for a mirror-confined distribution is clear from the discussion of particle orbits in Appendix I. For a given mirror ratio B_2/B_1 (Fig. 9.13.1), ions which have $v_\perp = 0$ in the midplane will simply stream out of the mirror region; therefore $f_{i0}(v_\perp = 0, v_\parallel \neq 0) = 0$. Further, an ion with given $v_\parallel$ will be reflected from the mirror only if its $v_\perp$ is sufficiently large; hence (9.13.2). The electron distribution may be assumed isotropic, since the lighter electrons find it impossible to escape from the ions.

Since the distribution (9.14.1) is characteristic of any plasma in a magnetic mirror, and since mirror fields confine plasmas both in the laboratory and in nature (e.g., the earth's magnetic field forms a magnetic mirror for particles in the radiation belts), an instability driven by the anisotropy (9.13.2) is clearly significant. That this distribution is potentially unstable to electrostatic modes is clear, since it is *not* a monotone-decreasing function of v^2.

To simplify the problem, assume that a plasma with distribution (9.13.1) is imbedded in a uniform magnetic field $\mathbf{B}_0 = B_0\,\hat{\mathbf{i}}_\parallel$. The plasma pressure is assumed to be small compared with the magnetic pressure.

$$\beta = \frac{n\kappa T}{B^2/8\pi} \ll 1 \qquad (9.13.3)$$

[1] E. G. Harris, *Phys. Rev. Letters*, 2:34 (1959).
[2] M. N. Rosenbluth, Microinstabilities, in "Plasma Physics," p. 485, International Atomic Energy Agency, Vienna, 1965.

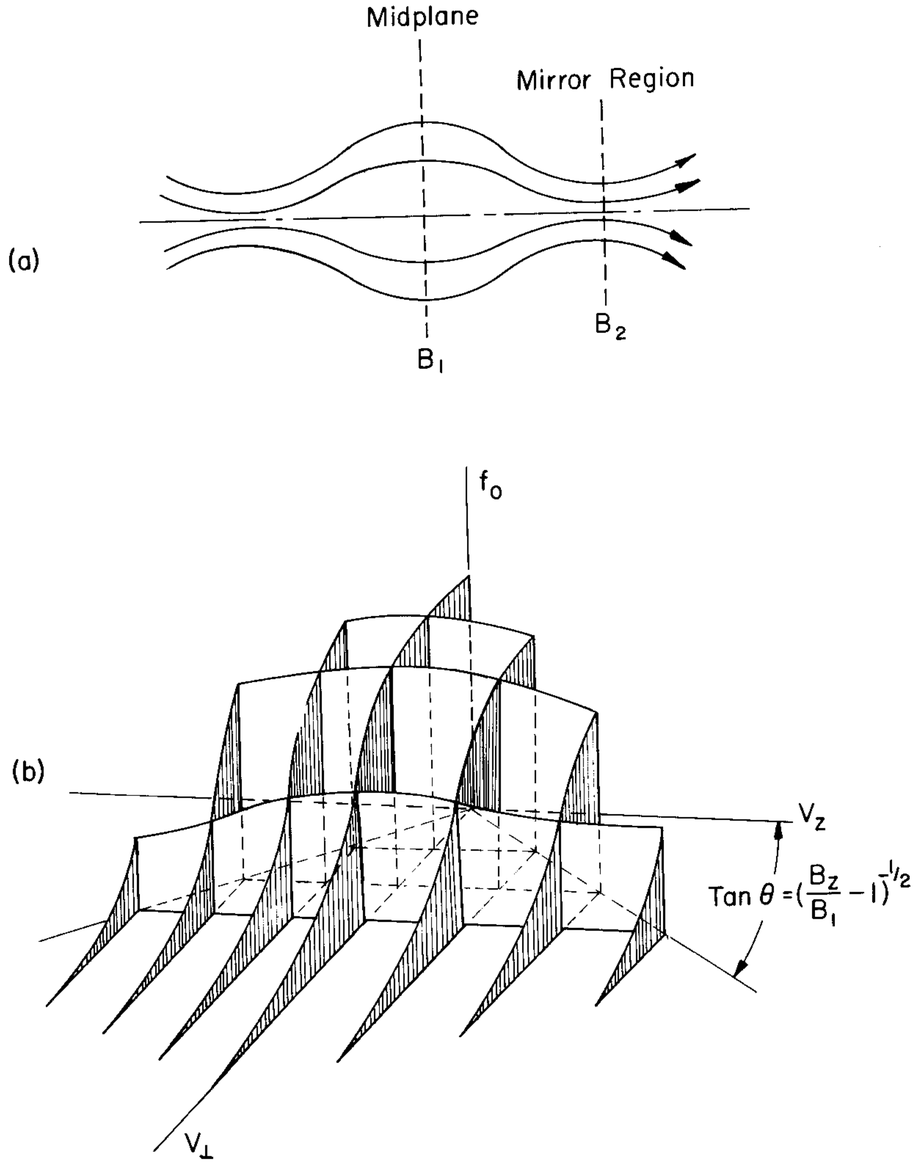


FIGURE 9.13.1
Schematic of a magnetic mirror field and plasma distribution. The plasma distribution is anisotropic (loss cone) and is unstable to a specific loss-cone instability. [*After J. E. Scharer, Phys. Fluids*, **10**:652 (1967).]

This restriction means that only electrostatic instabilities need be considered, since electromagnetic instabilities perturb the magnetic field and increase the magnetic field energy. The uniform-equilibrium magnetic field is the minimum energy state of $\mathbf{B}_0$. If the particle pressure is low [Eq. (9.13.3)], an increase in magnetic energy balances any decrease in particle energy associated with the perturbation.

The dispersion relation for waves in a magnetic field is derived in Chap. 8. For low β the electrostatic mode is

$$\mathbf{E} = E\frac{\mathbf{k}}{|k|}\exp[i(\mathbf{k}\cdot\mathbf{x} - \omega t)]$$

$$k^2 = \sum_\alpha \omega_{p\alpha}{}^2 2\pi \int \sum_n \frac{J_n{}^2(k_\perp v_\perp/\omega_{c\alpha})}{k_\parallel v_\parallel - \omega + n\omega_{c\alpha}}\left(k_\parallel v_\parallel \frac{\partial f_{\alpha 0}}{\partial v_\parallel{}^2} + n\omega_{c\alpha}\frac{\partial f_{\alpha 0}}{\partial v_\perp{}^2}\right) dv_\perp{}^2\, dv_\parallel \tag{9.13.4}$$

where $\mathbf{v} = (v_x, v_y, v_\parallel) \equiv (v_\perp\cos\theta, v_\perp\sin\theta, v_\parallel)$, and $\mathbf{k} = (k_x, k_y, k_\parallel) = (k_\perp\cos\psi, k_\perp\sin\psi, k_\parallel)$. (See Prob. 9.13.1.)

To explore all solutions of (9.13.4) is a tedious procedure. There are a number of reasonable approximations that simplify the exploration considerably:

1 Because the electrons are taken to be isotropic in this example, they do not contribute to instability. Instead, they generally contribute to Landau damping. Consequently, the perturbations most likely to be unstable are those for which there is only negligible damping.

$$\frac{\omega}{k_\parallel} \gg v_{\text{thermal electron}} \tag{9.13.5}$$

and this restriction on $k_\parallel$ will be used in the analysis.

2 Only modes with wavelength longer than the electron gyroradius will be considered.

$$k^2 a_{ce}{}^2 \ll 1 \tag{9.13.6}$$

Together with (9.13.5), this restricts the analysis to waves for which the electrons behave as a cold fluid (no Landau damping, no dependence on the velocity distribution of electrons). The inequality (9.13.5) allows all terms except $n = 0$ to be neglected in the electron contribution to the sum over Bessel functions, because $J_n{}^2 \approx (k^2 a_{ce}{}^2)^n$. The assumption (9.13.6) is called the *small Larmor radius limit*. Equation (9.13.5) allows the expansion

$$\frac{1}{\omega - k_z v_z} = \frac{1}{\omega} + \frac{k_z v_z}{\omega^2} + \cdots$$

in the electron contribution to the perturbed charge density. Then (9.13.4) becomes

$$k^2 = \frac{\omega_{pe}{}^2}{\omega^2}k_\parallel{}^2 + \sum_n \omega_{pi}{}^2 \int \frac{J_n{}^2(k_\perp v_\perp/\omega_{ci})[2k_\parallel v_\parallel(\partial f_{i0}/\partial v_\parallel{}^2) + 2n\omega_{ci}(\partial f_{i0}/\partial v_\perp{}^2)]}{k_\parallel v_\parallel + n\omega_{ci} - \omega} d\mathbf{v} \tag{9.13.7}$$

Problem 9.13.1 Show that the dispersion relation (9.13.4) can be obtained by either of the following two methods:

(*a*) Calculating $\rho_1 = \sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha 1}\, d\mathbf{v}$ with $f_{\alpha 1}$ given by (8.10.8), assuming $\mathbf{B}_1 = 0$, and equating $\nabla \cdot \mathbf{E}_1 = 4\pi\rho_1$

(*b*) Taking the limit $\beta \ll 1$, $kc \gg \omega$, in the dispersion tensor (8.10.10)

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By comparison with the results of Chap. 4, it is seen that the $(\omega_{pe}^2/\omega^2)k_\parallel^2$ term is just the response of a cold electron fluid to the electrostatic potential ϕ_k. For this reason, the approximations (9.13.5) and (9.13.6) are usually called the *zero temperature limit*, although of course, they are in fact limits on the wavelength of modes being considered.

The ion contribution to (9.13.7) can be simplified by writing

$$\int \frac{f(v)}{kv - \omega}\, dv = \oint \frac{f(v)}{kv - \omega}\, dv + \frac{i\pi}{|k|} f\left(v = \frac{\omega}{k}\right)$$

and noting that, because

$$\omega_{pe}^2 \gg \omega_{pi}^2$$

the *real* part of the ion integral can be neglected. Thus, from (9.13.7),

$$\omega_r = \omega_{pe} \frac{k_\parallel}{k} \qquad (9.13.8)$$

However, the imaginary part of the ion term cannot be neglected, since it is the sole contribution to ω_i. Then the dispersion relation for these high-frequency electrostatic waves [Eq. (9.13.7)] becomes

$$\omega^2 = \omega_{pe}^2 \frac{k_\parallel^2}{k^2} + \omega^2 \pi i \sum_n \frac{\omega_{pi}^2}{k^2} \int \frac{2\pi}{|k_\parallel|} J_n^2\left(\frac{k_\perp v_\perp}{\omega_{ci}}\right) \times \left[(\omega_r - n\omega_{ci}) \frac{\partial f_{i0}}{\partial v_\parallel^2} + n\omega_{ci} \frac{\partial f_{i0}}{\partial v_\perp^2}\right] dv_\perp^2 \Big|_{v_\parallel = \frac{\omega_r - n\omega_{ci}}{k_\parallel}} \qquad (9.13.9)$$

Now the assumption $\omega \gg k_\parallel v_\parallel$ made in computing the electron term (to reduce Landau damping) implies that

$$\frac{\omega}{k} \gg v_{\text{thermal ion}} \qquad (9.13.10)$$

unless the ions are *extraordinarily* hot; under this restriction, the only sizable ion contribution to instability comes when

$$\omega - n\omega_{ci} \approx 0 \qquad (9.13.11)$$

for only then will $k_\parallel v_\parallel \approx \omega - n\omega_{ci}$ in the velocity range of thermal ions. But ω is determined by the electron density, $\omega \leq \omega_{pe}$, since $k_\parallel \leq k$; so (9.13.11) can be satisfied only if the density is sufficiently high. This restriction must be a necessary condition for this instability, since otherwise the destabilizing term is negligible:

$$\omega_{pe}^2 \geq n^2 \omega_{ci}^2 \qquad \text{necessary for loss-cone instability at a harmonic } \omega_r = n\omega_{ci} \qquad (9.13.12)$$

If the condition (9.13.12) is satisfied for $n \leq N$, there will be a series of possibly unstable harmonics $n = 1, 2, \ldots, N$. For each of these eigenfrequencies only one term in $\sum_n J_n^2$ in (9.13.9) contributes. Examining a specific harmonic n, and setting $\omega_r - n\omega_{ci} = 0$, Eq. (9.13.9) becomes

$$\omega = \omega_{pe} \frac{k_\parallel}{k} + i\omega_{pi}^2 \omega_{pe} \frac{\pi^2 \omega_{ci} n}{k^3} \int J_n^2\left(\frac{k_\perp v}{\omega_{ci}}\right) \frac{\partial f_{i0}}{\partial v_\perp^2}\Big|_{v_\parallel = 0} dv_\perp^2 \qquad (9.13.13)$$

The stability is determined by the sign of the last term in (9.13.13). Figure 9.13.2 shows that the sign of the integral depends on the location $v_{\max}$ of the maximum of $f_{i0}(v_\perp^2, v_\parallel = 0)$ with respect to the velocities $v_\perp$ at which $J_n^2(k_\perp v_\perp/\omega_{ci})$ becomes small. Clearly, by inspecting Fig. 9.13.2,

$$\int J_n^2\left(\frac{k_\perp v_\perp}{\omega_{ci}}\right) \frac{\partial f_{i0}}{\partial v_\perp^2}\Big|_{v_\parallel = 0} dv_\perp^2 > 0 \qquad \text{when } \frac{k_\perp v_{\perp,\max}}{\omega_{ci}} \gg n \qquad (9.13.14)$$

and

$$\int J_n^2\left(\frac{k_\perp v_\perp}{\omega_{ci}}\right) \frac{\partial f_{i0}}{\partial v_\perp^2}\, dv_\perp^2 < 0 \qquad \text{when } \frac{k_\perp v_{\perp,\max}}{\omega_{ci}} \ll n \qquad (9.13.15)$$

This assumes that f_{i0} is not so strongly peaked that only velocities between two zero's of J_n contribute to the integral.

Writing (9.13.13) in a more compact form,

$$\omega = \omega_{pe} \frac{k_\parallel}{k} + \frac{i\pi^2 n \omega_{ci} \omega_{pi}^2 \omega_{pe}}{k^3 (\kappa T_i/m_i)^{3/2}} G\left(\frac{k_\perp v_{\perp,\max}}{\omega_{ci}}\right) \qquad (9.13.16)$$

the stability depends on the sign of G. For a given distribution (which fixes $v_{\perp,\max}$), G is a function of $k_\perp$, with the general shape in Fig. 9.13.3 and a maximum value of order unity. Thus waves with

$$k_\perp \geq \frac{v_{\perp,\max}}{n\omega_{ci}} \qquad (9.13.17)$$

will be unstable (G will be positive). The most unstable value of $k_\perp$ (the fastest-growing mode) can be determined for any explicit ion distribution f_{i0} by

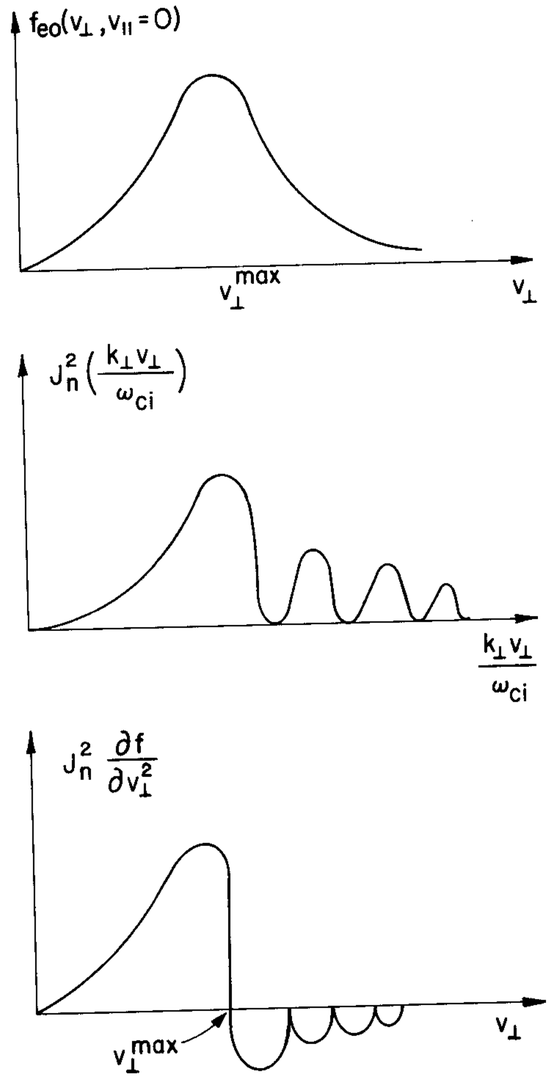


FIGURE 9.13.2
Quantities relevant to the calculation of a loss-cone instability.

calculating G. But the maximum positive value of G for a particular harmonic $n\omega_{ci}$ comes from

$$k_\perp \approx \frac{n\omega_{ci}}{v_{\perp,\max}} \qquad \text{fastest-growing mode}$$

Using this value of $k_\perp$ in (9.13.16) gives the maximum growth rate

$$\omega_{i,\max} \approx \frac{\omega_{pi}^2 \omega_{pe}}{n^2 \omega_{ci}^2} \tag{9.13.18}$$

where $\omega_{i,\max} < n\omega_{ci}$ and $\omega_{pe} \geq n\omega_{ci}$ have been assumed.

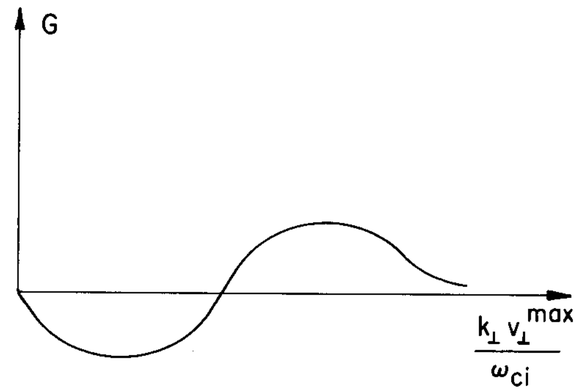


FIGURE 9.13.3
Plot of the function G that determines stability for a loss-cone-limited plasma. The distribution is unstable if $G > 0$.

The analysis above has used no specific property of the ion distribution, except that it has a loss cone. Many magnetic mirror machines have observed oscillations near the ion-cyclotron frequency when the density reached the level indicated by the instability condition (9.13.12).

If the plasma density is sufficiently high, i.e.,

$$n_i > \frac{B_0^2}{4\pi m_i c^2} \frac{m_e}{m_i}$$

then $\omega_{i,\max}$ given by (9.13.18) will be of order $n\omega_{ci}$, and the analysis leading to that result is invalid. However, for waves that grow faster than the ion gyrofrequency, it is clearly a good approximation to set $\mathbf{B} = 0$ in computing the *ion* perturbed quantities. This can be done directly in computing

$$\rho_{ik} = \bar{n}_i \int e f_{ik}\, d\mathbf{v} = -\frac{\bar{n}_i e^2}{m_i} \int \frac{\bar{\mathbf{E}}_\mathbf{k} \cdot \nabla_\mathbf{v} f_{i0}}{i(\mathbf{k} \cdot \mathbf{v} - \omega)}\, d\mathbf{v}$$

in which case (9.13.7) is simply replaced by (see also Prob. 9.13.2)

$$k^2 = \frac{\omega_{pe}^2}{\omega^2} k_\parallel^2 + \omega_{pi}^2 \int \frac{\mathbf{k} \cdot \nabla_\mathbf{v} f_{i0}}{\mathbf{k} \cdot \mathbf{v} - \omega}\, d\mathbf{v} \tag{9.13.19}$$

This equation predicts instability, because $f_{i0}(v_\perp^2)$ is not monotone-decreasing. Proceeding as above, noting again that $k_\parallel v_{\parallel,\text{ion}} \ll \omega$ because $k_\parallel v_{\parallel,\text{electron}} < \omega$ was assumed (to avoid Landau damping by electrons), Eq. (9.13.18) can be rewritten

$$k^2 = \frac{\omega_{pe}^2}{\omega^2} k_\parallel^2 + \omega_{pi}^2 \left(\frac{m_i}{\kappa T_i}\right)^{1/2} G_1\left(\frac{\omega}{k_\perp \sqrt{\kappa T_i / m_i}}\right) \tag{9.13.20}$$

where $$G_1 = \int dv_\parallel \left(\frac{\kappa T_i}{m_i}\right)^{1/2} \frac{1}{(1 - k_\perp{}^2 v_\perp{}^2/\omega^2)^{1/2}} \frac{\partial f_{i0}}{\partial v_\perp{}^2}\, dv_\perp{}^2$$

and $$\mathrm{Im}[G_1] > 0 \qquad \text{if} \quad \frac{k_\perp \sqrt{\kappa T_i/m_i}}{\omega} > 1$$

A complication in the analysis is the fact that waves propagating along $\mathbf{B}_0$ might leave the confined region. In that case, stability is determined by the wave-amplitude increase during one transit through the machine. For this reason, (9.13.20) is generally solved for real ω and complex k to show the spatial growth of a wave born near the midplane as it propagates toward the mirror region. The result is

$$k_\parallel = k \frac{\omega}{\omega_{pe}} - i \frac{\omega_{pe} |G_1|}{(\kappa T_i/m_i)^{1/2}}$$

Problem 9.13.2 Show that (9.13.7) reduces to (9.13.19) in the limit $\mathbf{B} \to 0$. ////

The size of the parameter G_1, which determines the growth length, is given in two specific examples:

1 Mirror ratio $B_2 = 2B_1$, equilibrium appropriate to a plasma which has been confined for a few collision times,

$$G_1 \approx \tfrac{1}{10}$$

2 Plasma produced by injection nearly perpendicular to $\mathbf{B}_0$,

$$G_1 \gg 1$$

The number of e foldings in a length L machine is given in terms of G_1 by

$$N \approx 10^{-5} \sqrt{\frac{n_i}{T_i}}\, L |G_1|$$

where n_i is in cm^{-3} and T_i is in kilo-electron volts. Many other instabilities driven by loss cones (whistler instabilities, Alfvén wave instabilities, etc.) are discussed in the literature. Their importance is diminished in the low-beta limit, and they depend on more specific details of the distribution than the case discussed here. It should be noted that the inclusion of more species (i.e., a dense cold-electron background superposed on a hot electron-ion population) often has a dramatic effect on the stability of loss-cone systems.

9.14 OTHER INSTABILITY MECHANISMS

A variety of velocity-space instabilities fall into the two classes (pinching and bunching) which are described above. Table 9.14.1 gives a few other instability examples, whose details are found in the literature.

Table 9.14.1 VELOCITY-SPACE MICROINSTABILITIES

I. Bunching (electrostatic) instabilities
Growth rate:
- $\lesssim \omega_{pe}$ always
- $\lesssim \omega_{pi}$ if electrons alone are stable (i.e., if $m_i \to \infty$ gives stability)

A. Penrose condition
1. Two-stream,[a, b] also unstable if $\mathbf{B}_0 \neq 0$
2. Multiple stream,[c] also unstable if $\mathbf{B}_0 \neq 0$
3. Ion waves,[d] also unstable if $\mathbf{B}_0 \neq 0$

B. Electrostatic anisotropy instability[e]; $\mathbf{B}_0$ is essential for this instability to be a "bunching" type
C. Cross-stream[f] instability
D. Two-stream, with $\mathbf{k}$ not parallel to the stream.[g] For these instabilities B_0 is essential
E. Electrostatic-trapping[h] instability

II. Pinching (magnetic) instabilities: Furth[i] has categorized many of these modes
Growth rate:
- $\lesssim \frac{v}{c}\omega_{pe}$ always
- $\lesssim \frac{v}{c}\omega_{pi}$ if electrons alone are stable

A. Transverse-anisotropy instability[j]
B. Mirror instability,[k] unstable for $\mathbf{B}_0$ in minimum-temperature direction
C. Instability of current sheets: $\mathbf{B}_0$ plays no essential part
D. Sheet-pinch instability
E. Electrostatic-confinement[l] instability

[a] O. Buneman, *Phys. Rev. Letters*, **1**:8 (1958); *Phys. Rev.*, **115**:503 (1959).
[b] E. A. Frieman, M. L. Goldberger, K. M. Watson, S. Weinberg, and M. N. Rosenbluth, *Phys. Fluids*, **5**:196 (1962).
[c] R. Q. Twiss, *Phys. Rev.*, **88**:1392 (1952).
[d] I. B. Bernstein, E. A. Frieman, R. M. Kulsrud, and M. N. Rosenbluth, *Phys. Fluids*, **3**:136(1960); I. B. Bernstein and R. M. Kulsrud, *Phys. Fluids*, **3**:937 (1960).
[e] *Journal of Nuclear Energy*, **C2**:138 (1951).
[f] P. B. Burt and E. G. Harris, *Phys. Fluids*, **4**:1412 (1961).
[g] W. E. Drummond and M. N. Rosenbluth, *Phys. Fluids*, **5**:1507 (1962); N. A. Krall and P. C. Liewer, *Phys. Rev. A*, **4**: 2094 (1971); K. Papadopoulos, R. C. Davidson, J. M. Dawson, I. Haber, D. A. Hammer, N. A. Krall, and R. Shanny, *Phys. Fluids*, **4**:849 (1971).
[h] W. C. Elmore, J. L. Tuck, and K. M. Watson, *Phys. Fluids*, **2**:239 (1959).
[i] H. P. Furth, *Phys. Fluids*, **6**:48 (1963).
[j] E. S. Weibel, *Phys. Rev. Letters*, **2**:83 (1959); B. D. Fried, *Phys. Fluids*, **2**:337 (1959).
[k] S. Chandrasekhar, A. N. Kaufman, and K. M. Watson, *Proc. Roy. Soc. (London)*, **A245**:435 (1958).

SOURCE: Table 9.14.1 is based on T. K. Fowler, Plasma Stability Theory, Oak Ridge National Laboratory Lecture Notes, Oak Ridge, Tenn, 1963.

9.15 THERMODYNAMIC BOUNDS ON FIELD LEVELS AND GROWTH RATES IN UNSTABLE PLASMA

In the quiet complexity of the Vlasov or normal-mode treatment of plasma stability, as presented in the earlier sections of this chapter, it is easy to overlook the fact that the basic reason for instability is thermodynamic: There exists a state of lower plasma energy (and hence greater field energy) that is accessible to the plasma through the equations it satisfies. A study of the lower-kinetic-energy states accessible to the plasma can be used to determine bounds on field energies in unstable systems and, in some cases, stability conditions and bounds on growth rates.

For a collision-dominated plasma, the "accessibility" condition is that $dS/dt \geq 0$, where S is the entropy for a collision-free plasma; the Vlasov equation is a more severe constraint on the motions of the system. For example, the entropy defined in the classical way for either particle species by

$$S \propto \int f \ln f \, d\mathbf{x} \, d\mathbf{v} \qquad (9.15.1)$$

is constant for a system in which f evolves according to the Vlasov equation. Thus a bound on the field energy of an unstable plasma could be found by finding the plasma state of minimum particle energy which has the same entropy as the initial plasma state, since for any state, at any time,

$$\text{Field energy} + \text{plasma energy} = \text{field energy} + \text{plasma energy at } t = 0 \qquad (9.15.2)$$

Problem 9.15.1 Prove that $\int G(f)\, d\mathbf{x}\, d\mathbf{v}$ is constant for any isolated system in which f obeys the Vlasov equation. ////

However, S conservation is only one of a variety of constraints imposed by the Vlasov equation (see Prob. 9.16.1), and some other constraint might give a more stringent bound on field energy. [There may be states of equal entropy which do *not* have the same value of $\int G(f)\, d\mathbf{x}\, d\mathbf{v}$ for some particular G; these states are not connected by the Vlasov equation, which requires that $\int G(f)\, d\mathbf{x}\, d\mathbf{v} = \text{constant}$ for any $G(f)$.] In contrast to the normal-mode approach to stability, which was a well-marked, though often difficult, path, the thermodynamic approach centers on finding the most useful constraint on the possible states of the system; this is an art, not a science, for there is no *prescription* available to aid the search.

The thermodynamic approach usually proceeds as follows:

1 Define a generalized "free-energy" function by

$$H = \int G(\mathbf{v}, \mathbf{x}, f)\, d\mathbf{x}\, d\mathbf{v} + \int \left(\frac{E^2}{8\pi} + \frac{B^2}{8\pi}\right) d\mathbf{x} \qquad (9.15.3)$$

2 Choose G in such a way that it is minimum for some known distribution $f = f_0$, that is,

$$\int [G(\mathbf{x}, \mathbf{v}, f) - G(\mathbf{x}, \mathbf{v}, f_0)]\, d\mathbf{x}\, d\mathbf{v} \geq 0 \qquad (9.15.4)$$

3 Find the change with time of H in terms of the time change of f, which in turn is given by the Vlasov equation.

$$\frac{dH}{dt} = \sum_\alpha \int \frac{\partial G}{\partial f_\alpha} \frac{\partial f_\alpha}{\partial t}\, d\mathbf{x}\, d\mathbf{v} + \frac{d}{dt} \int \left(\frac{E^2}{8\pi} + \frac{B^2}{8\pi}\right) d\mathbf{x} \qquad (9.15.5)$$

The next step is determined by the choice made for the function G. For example, if G is chosen so that H is a constant of the motion, then a bound on the amplitude of wave fields is obtained with the help of (9.15.4).

$$\int \left(\frac{E^2}{8\pi} + \frac{B^2}{8\pi}\right) d\mathbf{x} \leq H(t = 0) - \int G(\mathbf{x}, \mathbf{v}, f_0)\, d\mathbf{x}\, d\mathbf{v} \qquad (9.15.6)$$

If instead G is chosen so that H is not conserved, it will occasionally be possible to calculate

$$\frac{1}{H} \frac{dH}{dt}\bigg|_{\max}$$

from (9.15.5). In that case the inequality

$$\int \frac{E^2 + B^2}{8\pi}\, d\mathbf{x} \leq H(t) - \int G(f_0)\, d\mathbf{x}\, d\mathbf{v}$$

can be used to infer the maximum growth rate of waves in the plasma distribution f_α.

$$\omega_{i,\max} = \frac{1}{2} \frac{1}{H} \frac{dH}{dt}\bigg|_{\max} \qquad (9.15.7)$$

Examples of this approach, described in detail elsewhere by Fowler,[1] will now be presented.

[1] T. K. Fowler, in A. Simon and W. B. Thompson (eds.), "Advances in Plasma Physics," Thermodynamics of Unstable Plasma, vol. 1, p. 201, Wiley, New York, 1968.

9.15.1 Bounds on the Fluctuation Energy

A choice for $G(f)$ that makes H, as defined by (9.15.3), a constant of the motion is

$$G(\mathbf{x}, \mathbf{v}, f) = \sum_\alpha \tfrac{1}{2}\bar{n}_\alpha m_\alpha v^2 f_\alpha + G_1(f) + \text{constant} \qquad (9.15.8)$$

If f satisfies the Vlasov equation, it follows that

$$\frac{d}{dt}\int G_1(f)\, d\mathbf{x}\, d\mathbf{v} = 0$$

for any G_1; further, the combination

$$\sum_\alpha \bar{n}_\alpha \int \tfrac{1}{2} m_\alpha v^2 f_\alpha \, d\mathbf{x}\, d\mathbf{v} + \int \frac{E^2 + B^2}{8\pi}\, d\mathbf{x} = \text{total energy of the system} = \text{constant}$$

If, for example, $G_1 = f \ln f$, H has the form of Helmholz free energy.

$$H = U - TS \qquad (9.15.9)$$

where S is the entropy, and U is the kinetic and electromagnetic plasma energy, but H remains constant whatever the choice of G_1. With $G(\mathbf{x}, \mathbf{v}, f)$ defined by (9.15.8), the integral $\int G\, d\mathbf{x}\, d\mathbf{v}$ clearly has a minimum, since $\frac{1}{2}mv^2 f$ is positive definite.

$$\int G(f)\, d\mathbf{x}\, d\mathbf{v} \geq \int G_1\, d\mathbf{x}\, d\mathbf{v}$$

with the equality corresponding to $f = N_0\, \delta(\mathbf{v})$. Equations (9.15.3) and (9.15.9) together imply a bound on the field energy.

$$\int \frac{E^2 + B^2}{8\pi}\, d\mathbf{x} \leq \sum_\alpha \bar{n}_\alpha \int \tfrac{1}{2} m_\alpha v^2 f_\alpha(\mathbf{v}, t = 0)\, d\mathbf{x}\, d\mathbf{v} + \int \left(\frac{E^2 + B^2}{8\pi}\right)_{t=0} d\mathbf{x} \qquad (9.15.10)$$

This is a trivial bound, in which the field energy is limited only by the total energy in the particles at $t = 0$. A more restrictive bound is found by choosing $G_1(f)$ such that $G(\mathbf{x}, \mathbf{v}, f)$ has a minimum for distributions f_α which contain more than zero energy. One useful choice of $G_1(f)$ gives

$$\int G(\mathbf{x}, \mathbf{v}, f)\, d\mathbf{v}\, d\mathbf{x} = \sum_\alpha \bar{n}_\alpha \int \left[\kappa T_\alpha \left(f_\alpha \ln \frac{f_\alpha}{C_\alpha} - f_\alpha\right) - \kappa T_\alpha \left(f_0 \ln \frac{f_0}{C_\alpha} - f_0\right) + (f_\alpha - f_0)\tfrac{1}{2} m_\alpha v^2\right] d\mathbf{x}\, d\mathbf{v} \qquad (9.15.11)$$

where

$$f_0 = C_\alpha \exp\left(-\frac{1}{2}\frac{m_\alpha v^2}{\kappa T_\alpha}\right)$$

and the sum is over particle species. This choice of G has the following properties:

1 $H = \int G\, d\mathbf{x}\, d\mathbf{v} + \int \frac{E^2 + B^2}{8\pi}\, d\mathbf{x} = \text{constant}$

2 $\int G(\mathbf{x}, \mathbf{v}, f)\, d\mathbf{x}\, d\mathbf{v} \geq \int G(\mathbf{x}, \mathbf{v}, f = f_0)\, d\mathbf{x}\, d\mathbf{v} = 0$

for *all* choices of the constant parameters C_α, T_α.

From these two properties it follows at once that the bound on the field energy is

$$\int \frac{E^2 + B^2}{8\pi}\, d\mathbf{x} \leq \int \frac{E^2 + B^2}{8\pi}\bigg|_{t=0} d\mathbf{x} + \int G[\mathbf{x}, \mathbf{v}, f(t = 0)]\, d\mathbf{x}\, d\mathbf{v} \qquad (9.15.12)$$

Since (9.15.12) is true for any choice of the constants C_α, T_α, these constants should be chosen so as to minimize $\int G[\mathbf{x}, \mathbf{v}, f(t = 0)]\, d\mathbf{x}\, d\mathbf{v}$. The meaning of the bounds (9.15.10) and (9.15.12) is clear. Equation (9.15.10) says that the field energy of an unstable plasma never exceeds the initial energy of the particles. Equation (9.15.12) says that in fact only a much smaller amount of energy is available for transfer to field fluctuations; i.e., the energy released in relaxation of $f(t = 0)$ to $f_{\alpha 0} = C_\alpha e^{-m_\alpha v^2/2\kappa T_\alpha}$. The choice of a maxwellian f_α and of G itself is not unique, and other choices may produce more restrictive bounds.

Problem 9.15.2 Show that $\int G\, d\mathbf{x}\, d\mathbf{v}$ given by (9.15.11) has a minimum at $f_\alpha = f_{\alpha 0}$. ////

9.15.2 Example: Bounds on E^2 for a Two-stream Unstable Plasma

The technique described can be illustrated by applying it to a two-stream unstable plasma. Consider a plasma which, at $t = 0$, is described by

$$f_e = N_e \exp\left(-\frac{m_e v^2}{2\kappa T_0}\right)$$
$$f_i = N_i \exp\left(-\frac{\tfrac{1}{2} m_i v^2 + m_i U_0 v_z}{\kappa T_0}\right) \qquad (9.15.13)$$

where N_e and N_i are the normalization constants. This distribution describes ions streaming through electrons at a relative speed U_0. Evaluating the bound requires simply substituting (9.15.13) into (9.15.12) and selecting C_α and T_α. However, since there is initially a drift of particles in the plasma, there is no reason to suspect that the most restrictive bound will be obtained by imagining that the distribution of electrons and ions relax to a maxwellian distribution at rest

in the initial rest frame of the *electrons*. Clearly, the best bound is obtained by relaxing both distributions to a maxwellian in a frame of reference with some speed intermediate between $\bar{v} = 0$ and $\bar{v} = U_0$. Therefore $G(\mathbf{x}, \mathbf{v}, f)$ is chosen as in (9.15.11), but with

$$\frac{1}{2}\frac{m_\alpha v^2}{\kappa T_\alpha} \to \frac{1}{\kappa T_\alpha}(\tfrac{1}{2}m_\alpha v^2 + m_\alpha \bar{U} v_z) \quad (9.15.14)$$

where $\bar{U}$ will be chosen to give the most restrictive bound on E^2. Note that the properties 1 and 2 following (9.15.11) remain unchanged, so that the bound (9.15.12) is also unchanged. The only change is that now five parameters (C_i, C_e, T_i, T_e, $\bar{U}$) are available to minimize the bound. The normalization constants C_α are chosen to conserve particles; i.e.,

$$\int C_\alpha \exp\left[-\frac{1}{\kappa T_\alpha}(\tfrac{1}{2}m_\alpha v^2 + \bar{U} m_\alpha v_z)\right] dv = \int f_\alpha (t = 0)\, d\mathbf{v}$$

which gives

$$C_e = N_e\left(\frac{T_0}{T_e}\right)^{3/2} \exp\left(-\frac{m_e \bar{U}^2}{2\kappa T_e}\right)$$

$$C_i = N_i\left(\frac{T_0}{T_i}\right)^{3/2} \exp\left(\frac{m_i U_0^2}{2\kappa T_0} - \frac{m_i \bar{U}^2}{2\kappa T_i}\right)$$

Problem 9.15.3 Show that the properties 1 and 2 following (9.15.11) are unchanged when G is altered by the substitution (9.15.14). ////

The bound is

$$\int \frac{E^2 + B^2}{8\pi}\, d\mathbf{x} \le n_0 V\left[\tfrac{1}{2}m_i(\bar{U} - U_0)^2 + \tfrac{1}{2}m_e \bar{U}^2 + \sum_\alpha \frac{3}{2}\left(\kappa T_\alpha \log\frac{T_\alpha}{T_0} - \kappa T_\alpha + \kappa T_0\right)\right] \quad (9.15.15)$$

with V the volume of the system and $n_0 = \int \bar{n}_\alpha f_\alpha\, d\mathbf{v}$ the density. The choice of parameters which minimize (9.15.15) is

$$T_e = T_i = T_0$$
$$\bar{U} \approx U_0$$

which gives, finally,

$$\int \frac{E^2 + B^2}{8\pi} \le \tfrac{1}{2}n_0 m_e U_0^2 V \quad (9.15.16)$$

This is a reasonable result, in keeping with the idea that the system (9.15.13) comes to equilibrium while conserving momentum, requiring that the lighter electrons change velocity by nearly U_0, and releasing $nV\frac{1}{2}m_e U_0^2$ energy.

These are a variety of other thermodynamic aids that follow from the limited amount of energy available for instability, and the student is referred to the review article by Fowler (see References below) as a starting point.

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10

THE NONLINEAR VLASOV THEORY OF PLASMA WAVES AND INSTABILITIES

The linear theory of plasma waves and stability developed in the previous chapters provides an explanation of many plasma phenomena; however, it is far from a complete theory of the development of even a small-amplitude departure from an equilibrium plasma state. The perturbation theory developed in Chaps. 8 and 9 can be used to extend the linear theory of plasma waves and stability into the nonlinear regime and thus provide a basis for explanation of a broad class of nonlinear plasma phenomena. This nonlinear perturbation theory of small-, but finite-amplitude plasma waves and their interactions is well developed; however, the theory of large-amplitude waves in plasmas is still (1973) being developed.

10.1 THE NEED FOR A NONLINEAR THEORY OF PLASMAS

In this chapter, a weakly nonlinear theory (i.e., nonlinear theory treated by a perturbation approach) is developed to examine plasma phenomena and features outside the scope of linear theory. Some plasma properties omitted from linear theory are discussed below.

10.1.1 Energy Balance

The linear theory often predicts a growth or damping of the amplitude of plasma waves; this implies a change in the wave energy $\delta(E^2 + B^2)/8\pi$, which must be balanced by a change in the energy of the particles. Energy conservation theorems are nonlinear, since $W_E \propto E^2/8\pi$.

10.1.2 Changes in the Average Plasma Properties

Either the damping or the growth of plasma waves produces a change in the plasma distribution function. This change may alter the stability (or damping) properties of the plasma. Further, the average properties of a plasma may change; specifically, the *temperature* of the plasma (measured by the kinetic energy relative to the mean velocity) may increase or decrease as a result of damping or instability of plasma waves. In linear theory the change in the distribution is given by

$$f = f_0 + \frac{1}{(2\pi)^3} \int \exp(i\mathbf{k} \cdot \mathbf{x}) f_{\mathbf{k}} \, d\mathbf{k}$$

Each Fourier component vanishes when averaged; therefore changes in average plasma properties appear only in a nonlinear treatment.

10.1.3 Interactions between Plasma Waves

Waves of finite amplitude exhibit phenomena which depend on products of wave amplitudes and are necessarily nonlinear. Such phenomena would be, for example, beating of waves, scattering of plasma waves by the particle distribution, and resonant interactions between waves (scattering of waves by waves).

There are two limits in which the weakly nonlinear theory is tractable. In the first case, where there are only a few waves of finite amplitude, it is possible to treat each wave individually. An example is the case of two coupled pendulums in classical mechanics. This is called the *theory of weak coherent waves.* The second case is one in which so many waves are present that a statistical approach can be employed to find those features of the time development of the plasma state that do not depend on the details of the initial phase of the waves. This is called the *theory of weak turbulence*, or the *quasilinear theory*.

Both proceed essentially out of an iteration of the Vlasov equation; both theories fail when the wave amplitudes become so large that one of two things occurs:

1. The perturbation theory fails to converge.
2. The particle orbits become so distorted by the wave fields that $f \approx f_0$ can

no longer be used to calculate accurately the *linear* wave properties of the plasma. The most usual example of such distortion of orbits occurs when particles become trapped in the troughs of plasma waves.

Both these limitations should be kept in mind when using the weakly nonlinear theories presented below.

10.2 QUASILINEAR EQUATIONS FOR CHANGES IN A PLASMA DISTRIBUTION[1]

Consider a plasma state of the sort frequently discussed in Chap. 9: spatially uniform, described by a distribution function $f_\alpha(\mathbf{v}, t)$ which at $t = 0$ is a stationary solution of the Vlasov equation (is an "equilibrium"), but which is unstable because $f_\alpha(\mathbf{v}, t = 0)$ is either double-humped or has some other nonthermal feature.

The linear behavior of such a system is known (Chap. 9). It develops a spectrum of waves

$$\mathbf{E}_1(t) = \frac{1}{(2\pi)^3}\int \exp[i(\mathbf{k}\cdot\mathbf{x})]\mathbf{E}_\mathbf{k}(t)\,d\mathbf{k} \qquad (10.2.1)$$

where, if the system is unstable, $\mathbf{E}_\mathbf{k}$ increases with time. The distribution develops according to

$$f_\alpha = f_{\alpha 0} + \frac{1}{(2\pi)^3}\int \exp(i\mathbf{k}\cdot\mathbf{x}) f_{\alpha\mathbf{k}}(t)\,d\mathbf{k} \qquad (10.2.2)$$

where in linear theory $f_{\alpha 0}$ is independent of time.

Since even in equilibrium there is a thermal distribution of plasma waves, the average distribution $f_{\alpha 0}$ for a "spatially uniform plasma" is correctly defined as the spatial average of the actual distribution f_α.

$$f_{\alpha 0}(\mathbf{v}, t) = \frac{1}{V}\int f_\alpha(\mathbf{x}, \mathbf{v}, t)\,d\mathbf{x} \qquad (10.2.3)$$

In the quasilinear theory the average distribution $f_{\alpha 0}$ changes slowly with time. The formal definition (10.2.3) of $f_{\alpha 0}$ allows a unique separation of the rapidly fluctuating $f_\mathbf{k}(t)\exp(i\mathbf{k}\cdot\mathbf{x})$ from the slow change in the average distribution, which takes place as instability increases the energy of plasma waves.

[1] A. Vedenov, E. Velikhov, and R. Z. Sagdeev, *Nucl. Fusion*, **1**:82 (1961).
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For simplicity, consider a plasma in which the average fields vanish,

$$\mathbf{E}_0 = \frac{1}{V}\int \mathbf{E}\,d\mathbf{x} = 0 \qquad \mathbf{B}_0 = \frac{1}{V}\int \mathbf{B}\,d\mathbf{x} = 0 \qquad (10.2.4)$$

and which is unstable to electrostatic perturbations. This plasma is described by the equations

$$\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla + \frac{q_\alpha}{m_\alpha}\mathbf{E}\cdot\nabla_\mathbf{v}\right) f_\alpha = 0 \qquad \nabla\cdot\mathbf{E} = 4\pi\sum_\alpha \bar{n}_\alpha q_\alpha \int f_\alpha\,d\mathbf{v} \qquad (10.2.5)$$

which are solved by a perturbation approach,

$$f_\alpha = f_{\alpha 0} + f_{\alpha 1} \qquad \mathbf{E} = \mathbf{E}_1 \qquad (10.2.6)$$

where

$$f_{\alpha 0} \equiv \frac{1}{V}\int f_\alpha\,d\mathbf{x} \equiv \langle f_\alpha\rangle \quad \text{and} \quad \langle f_{\alpha 1}\rangle = \langle \mathbf{E}_1\rangle = 0$$

Using $\langle\mathbf{E}\rangle = 0$ and $\nabla_\mathbf{x}\langle f\rangle = 0$, the equation for $f_{\alpha 0}$ is obtained by averaging (10.2.5), with the result

$$\frac{\partial f_{\alpha 0}}{\partial t} = \frac{\partial\langle f_\alpha\rangle}{\partial t} = -\frac{q_\alpha}{m_\alpha}\nabla_\mathbf{v}\cdot\langle\mathbf{E}_1 f_{\alpha 1}\rangle \qquad (10.2.7)$$

where $\langle\mathbf{E}_1\cdot\nabla_\mathbf{v} f_{\alpha 1}\rangle = \nabla_\mathbf{v}\cdot\langle\mathbf{E}_1 f_{\alpha 1}\rangle$ has been used. The equation for $f_{\alpha 1}$ is obtained by substituting $f_\alpha = f_{\alpha 0} + f_{\alpha 1}$ into the Vlasov equation (10.2.5) and using (10.2.7) for $\partial f_{\alpha 0}/\partial t$, with the result

$$\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla\right) f_{\alpha 1} = -\frac{q_\alpha}{m_\alpha}\mathbf{E}_1\cdot\nabla_\mathbf{v} f_{\alpha 0} - \frac{q_\alpha}{m_\alpha}\nabla_\mathbf{v}\cdot(\mathbf{E}_1 f_{\alpha 1} - \langle\mathbf{E}_1 f_{\alpha 1}\rangle) \qquad (10.2.8)$$

Equation (10.2.7) shows that the time change in $f_{\alpha 0}$ is second-order in perturbed quantities. This shows why the linear theory did not predict any change in the average distribution function.

To find $\partial f_{\alpha 0}/\partial t$ to lowest order in $\mathbf{E}_1$, the perturbed distribution $f_{\alpha 1}$ is calculated as in linear theory. The linear solution to (10.2.8) is obtained by neglecting the $\mathbf{E}_1 f_{\alpha 1}$ terms. In addition, the time dependence of $f_{\alpha 0}$ is neglected because $\partial f_{\alpha 0}/\partial t$ is second-order in $\mathbf{E}_1$. The linear result (to order $\mathbf{E}_1$) is shown in Chap. 8 to be

$$f_{\alpha \mathbf{k}}(t) = \bar{f}_{\alpha \mathbf{k}} e^{-i\omega t}$$
$$\mathbf{E}_{\mathbf{k}}(t) = \bar{\mathbf{E}}_{\mathbf{k}} e^{-i\omega t} \qquad (10.2.9)$$
$$\bar{f}_{\alpha \mathbf{k}} = -\frac{q_\alpha}{m_\alpha} \frac{\mathbf{E}_{\mathbf{k}} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}}{i(\mathbf{k} \cdot \mathbf{v} - \omega)}$$

where the eigenfrequency ω as a function of $\mathbf{k}$ and $f_{\alpha 0}$

$$\text{(through} \qquad \nabla \cdot \mathbf{E}_1 = 4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha 1}\, d\mathbf{v})$$

is given by

$$D(\mathbf{k}, \omega) = 1 - \frac{\omega_{pe}^2}{k^2} \int_L \frac{du}{u - \omega/|k|} \frac{\partial}{\partial u} \left(f_{e0}(u) + \frac{m_e}{m_i} f_{i0}(u) \right) = 0 \qquad (10.2.10)$$

where[1]

$$f_{\alpha 0}(u) = \int \delta\left(u - \frac{\mathbf{k} \cdot \mathbf{v}}{|k|}\right) f_{\alpha 0}(\mathbf{v})\, d\mathbf{v}$$

and the velocity integration follows the Landau contour, passing under the pole at $u = \omega/|k|$. In contrast to the linear theory, $f_{\alpha 0}$ is now weakly time-dependent through (10.2.7); therefore the eigenfrequency ω, which is a functional of $f_{\alpha 0}$, is also time-dependent.

Noting that

$$\langle \mathbf{E}_1 f_{\alpha 1} \rangle = \frac{1}{V} \int \mathbf{E}_1 f_{\alpha 1}\, d\mathbf{x}$$
$$= \frac{1}{V} \int d\mathbf{x} \int \frac{d\mathbf{q}}{(2\pi)^3} \int \mathbf{E}_{\mathbf{q}} f_{\alpha \mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{x}) \exp(i\mathbf{q} \cdot \mathbf{x}) \frac{d\mathbf{k}}{(2\pi)^3}$$
$$= \frac{1}{V} \int \frac{d\mathbf{k}}{(2\pi)^3} \mathbf{E}_{-\mathbf{k}} f_{\alpha \mathbf{k}} \qquad (10.2.11)$$

the equation for $f_{\alpha 0}(t)$ is, using (10.2.9) in (10.2.7),

$$\frac{\partial f_{\alpha 0}}{\partial t} = \left(\frac{q_\alpha}{m_\alpha}\right)^2 \frac{1}{V} \int \mathbf{E}_{-\mathbf{k}} \cdot \nabla_{\mathbf{v}} \left[\frac{1}{i(\mathbf{k} \cdot \mathbf{v} - \omega)} \mathbf{E}_{\mathbf{k}} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}(t) \right] \frac{d\mathbf{k}}{(2\pi)^3} \qquad (10.2.12)$$

[1] Most of the examples in this section are one-dimensional, and the notation $f_{\alpha 0}(u)$ is used for the one-dimensional distribution function.

The result can be expressed in terms of the spectral energy density $\mathscr{E}_{\mathbf{k}}$ of the electrostatic field, defined by

$$\langle W_E \rangle = \left\langle \frac{E_1^2}{8\pi} \right\rangle = \frac{1}{V} \int \frac{E_{\mathbf{k}} \cdot \mathbf{E}_{-\mathbf{k}}}{8\pi} \frac{d\mathbf{k}}{(2\pi)^3} \equiv \int \mathscr{E}_{\mathbf{k}}\, d\mathbf{k}$$
$$\mathscr{E}_{\mathbf{k}} \equiv \frac{1}{V} \frac{1}{(2\pi)^3} \frac{\mathbf{E}_{\mathbf{k}} \cdot \mathbf{E}_{-\mathbf{k}}}{8\pi} \qquad (10.2.13)$$

Observing that for electrostatic waves

$$\mathbf{E}_{\mathbf{k}} = -i\mathbf{k}\phi_{\mathbf{k}}$$

Eq. (10.2.12) can be written

$$\frac{\partial f_{\alpha 0}}{\partial t} = \left(\frac{q_\alpha}{m_\alpha}\right)^2 \frac{1}{V} \int \frac{\mathbf{E}_{\mathbf{k}} \cdot \mathbf{E}_{-\mathbf{k}}}{k^2} \mathbf{k} \cdot \nabla_{\mathbf{v}} \left[\frac{1}{i(\mathbf{k} \cdot \mathbf{v} - \omega)} \mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}(t) \right] \frac{d\mathbf{k}}{(2\pi)^3}$$
$$= \left(\frac{q_\alpha}{m_\alpha}\right)^2 8\pi \int \mathbf{k} \cdot \nabla_{\mathbf{v}} \left[\frac{1}{i(\mathbf{k} \cdot \mathbf{v} - \omega)} \mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}(t) \right] \frac{\mathscr{E}_{\mathbf{k}}}{k^2} d\mathbf{k}$$

The spectral energy changes with time according to $[\omega = \omega_r(t) + i\omega_i(t)]$

$$\mathscr{E}_{\mathbf{k}}(t) = \mathscr{E}_{\mathbf{k}}(0) e^{2\omega_i t}$$

because $\mathbf{E}_{\mathbf{k}}(t) = \bar{\mathbf{E}}_{\mathbf{k}} e^{-i\omega t}$ and because [since ω satisfies (10.2.10)]

$$\omega(\mathbf{k}, t) = -\omega^*(-\mathbf{k}, t)$$

where $\omega^* = \omega_r - i\omega_i$ is the complex conjugate.

The set of equations for $\mathscr{E}_{\mathbf{k}}(t)$, $f_{\alpha 0}(t)$, and $\omega(\mathbf{k}, t)$ can be written in the form of a diffusion equation.

$$\frac{\partial f_{\alpha 0}}{\partial t} = \nabla_{\mathbf{v}} \cdot \mathsf{D}_{\mathbf{v}} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}(t) \qquad (10.2.14)$$

$$\mathsf{D}_{\mathbf{v}} = \left(\frac{q_\alpha}{m_\alpha}\right)^2 8\pi \int \frac{\mathscr{E}_{\mathbf{k}}(t)}{i(\mathbf{k} \cdot \mathbf{v} - \omega)} \frac{\mathbf{kk}}{|k^2|}\, d\mathbf{k} = \left(\frac{q_\alpha}{m_\alpha}\right)^2 8\pi \int \frac{\omega_i \mathscr{E}_{\mathbf{k}}}{(\mathbf{k} \cdot \mathbf{v} - \omega_r)^2 + \omega_i^2} \frac{\mathbf{kk}}{k^2}\, d\mathbf{k}$$
$$\frac{\partial \mathscr{E}_{\mathbf{k}}(t)}{\partial t} = 2\omega_i(\mathbf{k}, t) \mathscr{E}_{\mathbf{k}}(t)$$

with $\omega(\mathbf{k}, t)$ given by (10.2.10) in terms of $f_{\alpha 0}(t)$. Equations (10.2.14) are called the *quasilinear diffusion equations*, and describe a diffusion in velocity space. In (10.2.14) the $\mathbf{k}$ integration is taken under the pole, and the last form of $\mathsf{D}_{\mathbf{v}}$ is obtained using the results of Prob. 10.2.1. This diffusion is the response of the plasma to the emission and absorption of plasma waves. The content and utility of these equations will be illustrated by several examples in Sec. 10.3.

Problem 10.2.1 Show that, if $D(\mathbf{k}, \omega) = 0$, where D is the plasma dielectric,

$$\omega_r(k) = -\omega_r(-k) \qquad \omega_i(k) = \omega_i(-k) \qquad ////$$

10.3 CONSERVATION OF PARTICLES, MOMENTUM, AND ENERGY IN QUASILINEAR THEORY

The quasilinear theory is developed through approximations to the exact Vlasov equation, and it is important to know if those physical properties conserved by the exact Vlasov equation are also conserved in the quasilinear theory.

For simplicity, in these examples the plasma is assumed to be one-dimensional; that is, $f_{\alpha 0} = f_{\alpha 0}(u)$, $\mathbf{E}_1 = \hat{\mathbf{x}}\, \bar{E}_{\mathbf{k}}\, e^{ikx}$, $u = v_x$. The theorems developed can be extended to three dimensions. In one dimension the quasilinear equations are

$$\frac{\partial f_{\alpha 0}(u, t)}{\partial t} = \frac{\partial}{\partial u} 8\pi \left(\frac{q_\alpha}{m_\alpha}\right)^2 \int_L \left[\frac{\mathscr{E}_k}{i(ku - \omega)} \frac{\partial f_{\alpha 0}(u, t)}{\partial u}\, dk\right] \tag{10.3.1}$$

$$\frac{\partial \mathscr{E}_k}{\partial t} = 2\omega_i \mathscr{E}_k \tag{10.3.2}$$

and $\omega = \omega_r + i\omega_i$ is determined from (Sec. 10.2.10)

$$1 - \sum_\alpha \frac{{\omega_{p\alpha}}^2}{k} \int \frac{1}{ku - \omega} \frac{\partial f_{\alpha 0}(u)}{\partial u}\, du = 0 \tag{10.3.3}$$

10.3.1 Conservation of Particles

Integrating (10.3.1) over velocity gives

$$\frac{\partial}{\partial t} \int f_{\alpha 0}(u, t)\, du = \int_{-\infty}^{\infty} \frac{\partial}{\partial u} D_u \frac{\partial f_{\alpha 0}(u, t)}{\partial u}\, du = 0 \tag{10.3.4}$$

because the integrand on the right-hand side is an exact differential, and there are no particles at $u = +\infty$ or $u = -\infty$. This shows that the quasilinear theory conserves particles, i.e., that

$$\int f_{\alpha 0}(u, t)\, du$$

is a constant.

10.3.2 Conservation of Momentum

Multiplying (10.3.1) by $\bar{n}_\alpha m_\alpha u$, integrating over u, and summing over all types of particles in the plasma gives an equation for the rates of change of particle momentum (after one integration by parts)

$$\frac{\partial}{\partial t} \sum_\alpha \bar{n}_\alpha \int m_\alpha u f_{\alpha 0}\, du = -\sum_\alpha 2{\omega_{p\alpha}}^2 \int dk \int \frac{\mathscr{E}_k}{i(ku - \omega)} \frac{\partial f_{\alpha 0}}{\partial u}\, du \tag{10.3.5}$$

Equation (10.3.3) for $\omega(k)$ can be used to evaluate the right-hand-side of (10.3.5).

$$\sum_\alpha 2{\omega_{p\alpha}}^2 \int dk \int \frac{\mathscr{E}_k}{i(ku - \omega)} \frac{\partial f_{\alpha 0}}{du}\, du = \int 2k\mathscr{E}_k\, dk \tag{10.3.6}$$

But since $\mathscr{E}_k \propto E_k E_{-k}$, $\mathscr{E}_k$ is *symmetric* in k, so that $\int \mathscr{E}_k k\, dk = 0$. Thus

$$\frac{\partial}{\partial t} \sum_\alpha \bar{n}_\alpha \int m_\alpha u f_{\alpha 0}\, du = 0 \tag{10.3.7}$$

and particle momentum is conserved in the quasilinear theory. Since electrostatic fields carry no momentum, conservation of particle momentum is a requirement of a satisfactory theory.

Problem 10.3.1 Derive the equivalent of (10.3.5) to (10.3.7) to show conservation of momentum in the case of electromagnetic waves. Show that electromagnetic wave momentum must be included in proving the theorem that momentum is conserved.

10.3.3 Energy Conservation in Quasilinear Theory

Although it is not possible to prove conservation of energy with the linearized Vlasov equations, it is possible to do so with the quasilinear equations. In fact, conservation of energy is a natural consequence of the quasilinear theory. Multiplying (10.2.14) by $\bar{n}_\alpha m_\alpha u^2/2$ and integrating over u gives

$$\begin{aligned}
\frac{\partial}{\partial t} \sum_\alpha \int \tfrac{1}{2} \bar{n}_\alpha m_\alpha u^2 f_{\alpha 0}\, du &= \sum_\alpha {\omega_{p\alpha}}^2 \int dk \int u^2 \frac{\partial}{\partial u} \frac{\mathscr{E}_k}{i(ku - \omega)} \frac{\partial f_{\alpha 0}}{\partial u}\, du \\
&= \sum_\alpha {\omega_{p\alpha}}^2 \int dk \int \mathscr{E}_k \frac{-2u}{i(ku - \omega)} \frac{\partial f_{\alpha 0}}{\partial u}\, du \\
&= \sum_\alpha {\omega_{p\alpha}}^2 \int dk \int \mathscr{E}_k \left(-\frac{2}{k}\right) \frac{\omega}{i(ku - \omega)} \frac{\partial f_{\alpha 0}}{\partial u}\, du
\end{aligned} \tag{10.3.8}$$

Using the fact that ω satisfies $D(k, \omega) = 0$ [Eq. (10.2.10)], it follows that

$$\sum_\alpha \omega_{p\alpha}{}^2 \int \frac{1}{ku - \omega} \frac{\partial f_{\alpha 0}}{\partial u}\, du = k$$

Equation (10.3.1) can then be rewritten

$$\frac{\partial}{\partial t} \sum_\alpha \tfrac{1}{2} \bar{n}_\alpha m_\alpha \int u^2 f_{\alpha 0}\, du = 2i \int \omega \mathscr{E}_k\, dk = -\frac{\partial}{\partial t} \int \mathscr{E}_k\, dk \qquad (10.3.9)$$

where the identity stated in Prob. 10.2.1 has been used to write

$$2i \int \omega \mathscr{E}_k\, dk = -\int 2\omega_i \mathscr{E}_k\, dk = \frac{-\partial}{\partial t} \int \mathscr{E}_k\, dk$$

Since $\int \mathscr{E}_k\, dk$ is the total electrostatic energy in waves, (10.3.9) proves that any increase in such energy is balanced by a corresponding decrease in total particle kinetic energy. The change in particle energy includes both the contribution that all the particles make to the total wave energy, as well as the loss (or gain) of energy by the particles that contribute directly to the instability (or damping). The distinction between the energy of particles oscillating with waves and the energy of particles that dominate the damping or growth of waves is best seen by considering a weak instability or a weak damping process.

Problem 10.3.2 Formulate some expression for entropy change and average entropy change within the framework of quasilinear theory. ////

10.4 LANDAU DAMPING IN QUASILINEAR THEORY

Consider a (one-dimensional) plasma state in which

$$f_{e0} \left(\equiv \frac{1}{V} \int f_e\, dx \right)$$

has the shape shown in Fig. 10.4.1 and which contains a finite-amplitude level of plasma oscillations (Sec. 8.6),

$$\omega = \omega_{pe}(1 + \tfrac{3}{2} k^2 \lambda_{De}{}^2) + i\omega_i \qquad (10.4.1)$$

having a range of wavelengths so that all the waves have a phase velocity lying in the range labeled $\Delta(\omega/k)$ in Fig. 10.4.1.

The linear theory of Langmuir oscillations (Sec. 8.6) shows that these waves are

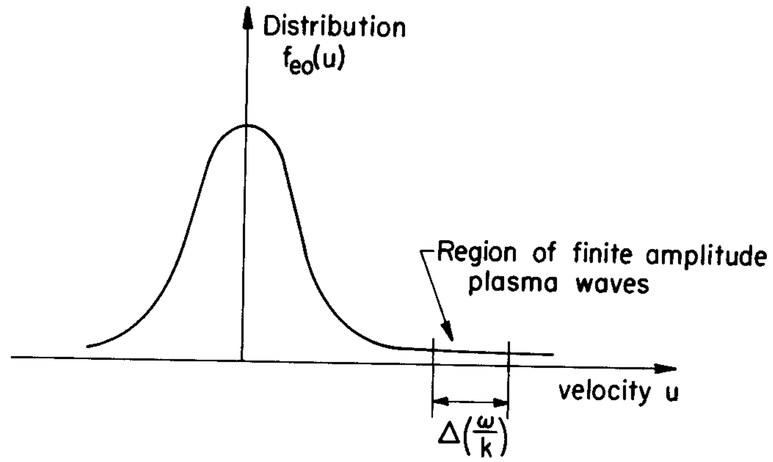


FIGURE 10.4.1
Distribution function of electrons showing range of phase velocities of finite-amplitude waves.

weakly damped, with the damping rate proportional to the slope of the electron distribution function in the velocity range $\Delta \approx \Delta(\omega/k)$.

$$\omega_i = \frac{\pi}{2} \frac{\omega_{pe}{}^2 \omega_r}{k^2} \frac{\partial f_{e0}(u)}{\partial u}\bigg|_{u=\omega/|k|} \qquad (10.4.2)$$

From the results of the linear theory (10.4.1) and (10.4.2), which give the wave properties in terms of f_{e0}, and from the quasilinear equations (10.2.14), which give the time dependence of f_{e0}, it is possible to determine the details of energy transfer and the distortion of the distribution by the damping process. The quasilinear equations, in one dimension, are

$$\frac{\partial f_{e0}}{\partial t} = \frac{\partial}{\partial u} \left[\frac{8\pi e^2}{m_e{}^2} \int \frac{\mathscr{E}_k\, dk}{i(ku - \omega)} \right] \frac{\partial f_{e0}}{\partial u}$$
$$\frac{\partial \mathscr{E}_k}{\partial t} = 2\omega_i \mathscr{E}_k \qquad (10.4.3)$$

where ω_r, ω_i are given in terms of f_{e0} by (10.4.1) and (10.4.2) and the k integration is over the Landau contour.[1]

Because $\omega_i \ll \omega_r$, Eq. (10.3.8) for energy conservation can be written in terms of a contribution from the resonant electrons that move with the speed of the wave ($u \approx \omega/k$) and nonresonant electrons by separating (Secs 8.4 and 8.5)

$$\int_L \frac{G(u)\, du}{ku - \omega} = \oint \frac{G(u)\, du}{ku - \omega} + \frac{\pi i}{|k|} G\left(\frac{\omega}{|k|}\right)$$

[1] For $\omega_i > 0$, the k integration is along the real k axis. For $\omega_i < 0$, the k integral is the analytic continuation of the $\omega_i > 0$ integral.

and by noting that the ion contributions can be neglected because $m_e/m_i \ll 1$. Thus (approximating $\omega = \omega_r$ since $\omega_i \ll \omega_r$)

$$\frac{\partial}{\partial t}\int \tfrac{1}{2}\bar{n}_e m_e u^2 f_{e0}\, du = -2\omega_{pe}{}^2 \int \frac{1}{k}\,\mathscr{E}_k \left[\oint \frac{\omega}{i(ku-\omega)}\frac{\partial f_{e0}}{\partial u}\,du + \pi \frac{\omega}{|k|}\frac{\partial f_{e0}}{\partial u}\bigg|_{u=\omega/|k|}\right] dk \qquad (10.4.4)$$

Problem 10.4.1 Prove by solving the initial-value problem that the k integration must be carried out along the Landau contour. ////

The first term on the right-hand side of (10.4.4) is the energy change in the nonresonant particles; the second term is the energy change in resonant particles, $\omega = ku$. Comparing the last term in (10.4.4) with ω_i in (10.4.2), the energy in resonant particles is given by

$$\frac{d}{dt}(\text{kinetic energy of resonant particles}) =$$

$$-4\int \omega_i \mathscr{E}_k\, dk = -2\frac{d}{dt}\int \mathscr{E}_k\, dk \qquad (10.4.5)$$

As the electrostatic energy $\int \mathscr{E}_k\, dk$ *decreases* by Landau damping, the kinetic energy of the resonant particles increases by *twice* the decrease in electrostatic energy. However, as shown in general in (10.3.9), the total change in kinetic energy is equal and opposite to the change in electrostatic energy, and so, by energy balance,

$$\frac{d}{dt}(\text{kinetic energy of nonresonant particles}) = \frac{d}{dt}\int \mathscr{E}_k\, dk$$
$$\frac{d}{dt}(\text{total kinetic energy}) = -\frac{d}{dt}\int \mathscr{E}_k\, dk \qquad (10.4.6)$$

The nonresonant particles lose energy as the electric field damps away. This is the *mechanical* energy associated with particle oscillations in the presence of the wave, and it naturally decreases as $\mathscr{E}_k$ decreases. In a negative energy wave it would be found that (d/dt) KE (nonresonant) $\propto -(d/dt)\int \mathscr{E}_k\, dk$. For other types of wave the division of energy between mechanical and electrostatic would not be one-to-one as here, but would be given by

$$\text{Wave energy} = W_{\text{wave}} = \frac{\partial[\omega D(k,\omega)]}{\partial\omega}\int \mathscr{E}_k\, dk$$

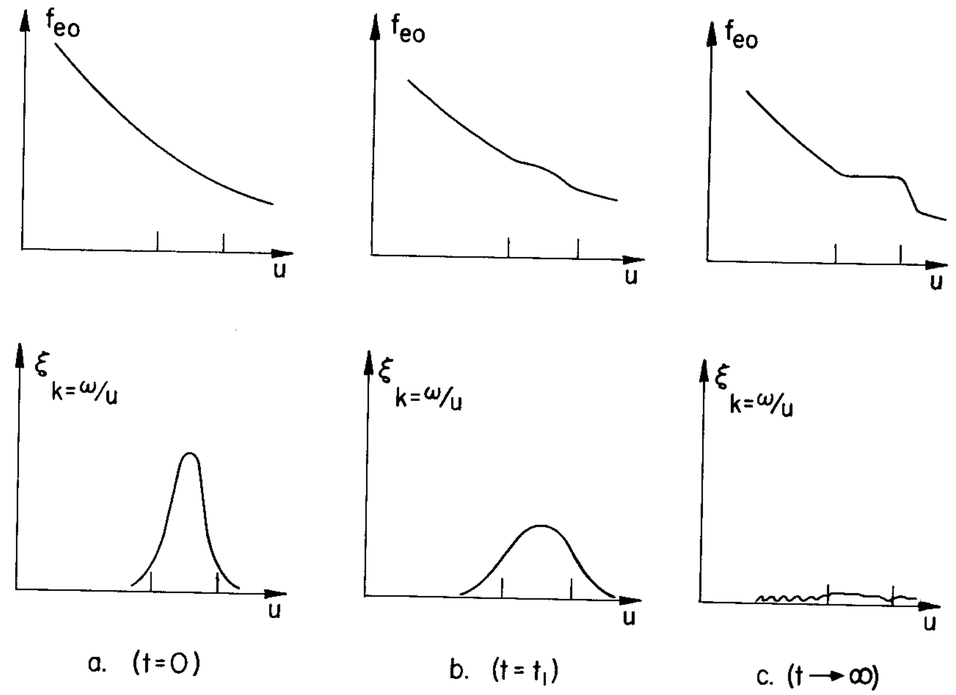


FIGURE 10.4.2
Electron distribution function and wave energy distribution illustrating the evolution of the distribution function as the waves damp.

as mentioned in (4.3.10), while from (10.3.9) total energy would still be conserved in the damping process.

Besides energy transfer, it is interesting to examine how f_{e0} changes during the damping. It is again useful to consider separately the resonant and the nonresonant particles, and write the diffusion equation (10.2.14) in the form

$$\frac{\partial f_{e0}(u\approx\omega/k)}{\partial t} = \frac{8\pi^2e^2}{m_e{}^2}\frac{d}{du}\frac{1}{u}\mathscr{E}_{k=\omega/u}\frac{d}{du}f_{e0}\left(u\approx\frac{\omega}{|k|}\right)$$
$$= \frac{8\pi^2e^2}{m_e{}^2}\mathscr{E}_{k=\omega/u}\frac{1}{u}\frac{d^2}{du^2}f_{e0}\left(u\approx\frac{\omega}{|k|}\right) \qquad (10.4.7)$$
$$\frac{\partial}{\partial t}\int \mathscr{E}_k\, dk = \int 2\omega_i\, \mathscr{E}_k\, dk$$

where the approximation is valid since it was assumed that the wave velocity exceeded the thermal speed, $\omega/k > \sqrt{\kappa T_e/m_e}$. Because $\partial^2 f_{e0}/\partial u^2 > 0$ for $\omega \approx ku$, as is clear from Fig. 10.4.1, the value of f_{e0} in the resonant region is increasing, while the energy in the wave spectrum is decreasing. A sample evolution might be that shown in Fig. 10.4.2*a* to *c*, where the resonant region has been enlarged

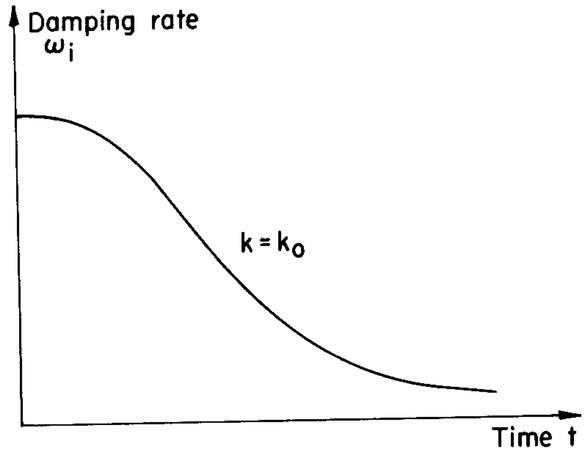


FIGURE 10.4.3
Decrease in the damping rate as a function of time for waves with $k = \omega/u$.

for clarity. The distribution flattens in the region where waves are being absorbed. This causes a reduction in ω_i, which is proportional to $\partial f_{\alpha 0}/\partial u$, as shown in Fig. 10.4.3.

If the wave energy is large enough, the damping may stop while a finite amplitude of waves persists. A small enough level of waves will damp out before the distribution f_{e0} is completely flattened.

An explicit equation for the time history of the resonant distribution can be obtained by first noting that, using (10.4.2),

$$\frac{8\pi^2 e^2}{m_e^2}\frac{\partial}{\partial u}\left(\frac{1}{u}\right)\mathscr{E}_{k=\omega/u}\left.\frac{df_{e0}}{du}\right|_{u=\omega/|k|} = \frac{2\omega_{pe}}{\bar{n}_e m_e}\frac{d}{du}\frac{1}{u^3}\frac{d}{dt}\mathscr{E}_{k=\omega/u} \tag{10.4.8}$$

Using this identity, (10.4.7) for the time evolution of the resonant-particle distribution can be written

$$\frac{d}{dt}\left(f_{e0,\,\text{Resonant}} - \frac{2\omega_{pe}}{\bar{n}_e m_e}\frac{d}{du}\frac{1}{u^3}\mathscr{E}_{k=\omega/u}\right) = 0 \tag{10.4.9}$$

If the original level of plasma oscillations is not too large, $\mathscr{E}_k$ goes to zero asymptotically in time. In that case the distribution in the resonant region $u \approx \omega/k$ is given by

$$f_{e0}^{R}(u, t=\infty) = f_{e0}(u, t=0) - \frac{2\omega_{pe}}{\bar{n}_e m_e}\frac{d}{du}\frac{1}{u^3}\mathscr{E}_{k=\omega/u}(t=0)$$

$$\mathscr{E}_k(t=\infty) = 0 \tag{10.4.10}$$

By integrating (10.4.10) over the resonant-particle-velocity range it is clear that, since by definition $\mathscr{E}_{k=\omega/u} \to 0$ at the boundary of the resonant region,

$$\int_{\substack{\text{resonant}\\ \text{region}}} f_{e0}(u, t=\infty)\,du = \int_{\substack{\text{resonant}\\ \text{region}}} f_{e0}(u, t=0)\,du$$

so that the number of resonant particles is conserved.

Problem 10.4.2 Prove (10.4.8). ////

The distribution outside the resonant region is easily obtained by observing that

$$\omega_r \gg ku$$

for the bulk of the nonresonant particles. Thus the velocity diffusion coefficient for nonresonant particles is independent of u, and (10.2.14) can be written, neglecting ku and ω_i compared with ω_r in (10.4.3),

$$\frac{\partial f_{e0,\,\text{nonresonant}}}{\partial t} - \frac{1}{\bar{n}_e m_e}\frac{\partial^2 f_{e0}}{\partial u^2}\frac{\partial}{\partial t}\int \mathscr{E}_k\,dk = 0 \tag{10.4.11}$$

The nonresonant diffusion is small at any one part of the distribution, and it can be estimated for (10.4.11) by making the approximation

$$\frac{\partial^2 f_{e0}}{\partial u^2} \approx \frac{\partial^2 f_{e0}(u, t=0)}{\partial u^2}$$

and then integrating to obtain [assuming $\mathscr{E}_k(t=\infty) = 0$]

$$f_{e0}^{\text{NR}}(u, t=\infty) = f_{e0}(t=0) - \frac{1}{\bar{n}_e m_e}\frac{\partial^2 f_{e0}}{\partial u^2}\int \mathscr{E}_k(t=0)\,dk \tag{10.4.12}$$

The combination (10.4.10), (10.4.12) can be used to show the net change in f_{e0} as a result of Landau damping of a weak spectrum of Langmuir oscillations. This result is sketched in Fig. 10.4.4.

An expression for f_{e0} at $t \to \infty$ can be deduced from a similarity solution of (10.4.11). Defining

$$\tau = \frac{1}{\bar{n}_e m_e}\int \mathscr{E}_k\,dk \tag{10.4.13}$$

Eq. (10.4.11) takes the form

$$\frac{\partial f_{e0}}{\partial \tau} = \frac{\partial^2 f_{e0}}{\partial u^2}$$

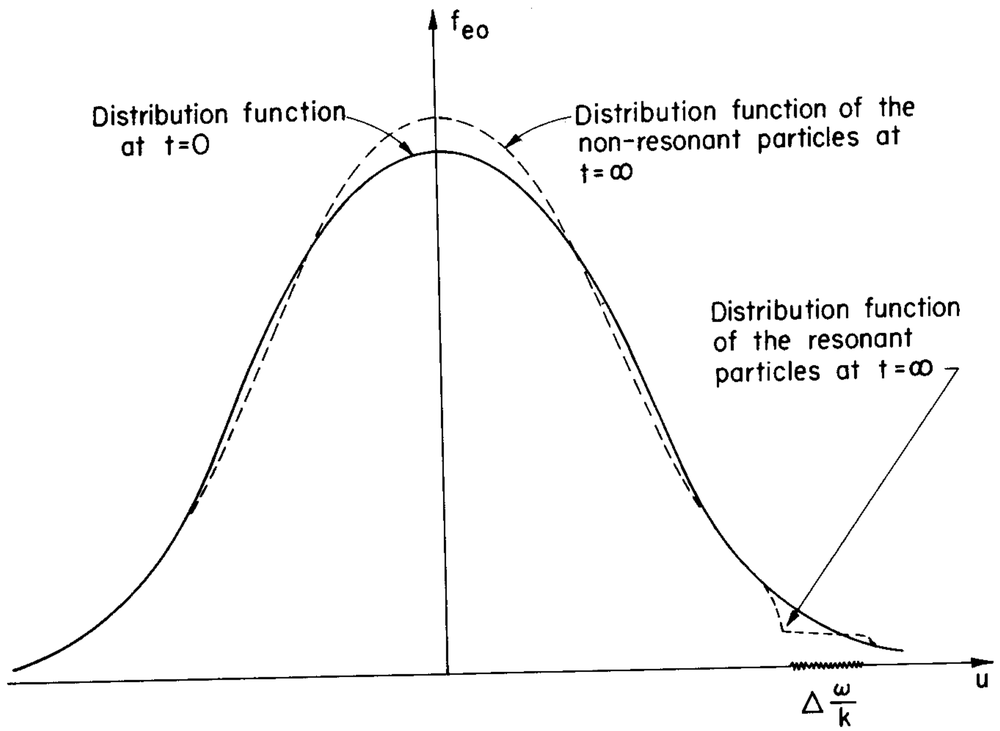


FIGURE 10.4.4
Distribution function of electrons before and after an initial wave disturbance has damped.

and has a particular solution

$$f_{e0} = \sqrt{\frac{1}{4\pi}} \frac{1}{\sqrt{\frac{1}{\bar{n}_e m_e}\int \mathscr{E}_k\, dk + \frac{\kappa T_e}{2m_e}}} \exp\left[\frac{-u^2}{4\left(\frac{1}{\bar{n}_e m_e}\int \mathscr{E}_k\, dk + \frac{\kappa T_e}{2m_e}\right)}\right] \tag{10.4.14}$$

The interesting feature of (10.4.14) is that the wave energy plays the role of a "temperature" in the nonresonant-particle distribution, and as the wave energy damps, the effective temperature of the nonresonant part of the distribution decreases and f_{e0} becomes more peaked, as also shown in Fig. 10.4.4. The analysis in this section could be equally well applied to a distribution that is weakly unstable to Langmuir oscillations (see, for example, Sec. 10.5).

Problem 10.4.3 Find the change in the *ion* distribution function during Landau damping of Langmuir oscillations. ////

Problem 10.4.4 Find the time history of the electric field, ion distribution, and electron distribution for a one-dimensional plasma, which at $t = 0$ is described by the distribution

$$f_{e0} = \left(\frac{m_e}{2\pi\kappa T_e}\right)^{1/2} \exp\frac{-m_e u^2}{2\kappa T_e}$$

$$f_{i0} = \left(\frac{m_i}{2\pi\kappa T_i}\right)^{1/2} \exp\frac{-m_i u^2}{2\kappa T_i}$$

$T_e \gg T_i$

and which has a spectrum of ion-acoustic waves (Sec. 8.9) in the range

$$\omega_r = \frac{kC_s}{\sqrt{1+(k\lambda_{De})^2}} \qquad 0 < k < \frac{1}{\lambda_{De}}$$

Calculate the distribution of energy between resonant and nonresonant electrons, ions, and electric fields. ////

10.5 THE GENTLE-BUMP INSTABILITY IN QUASILINEAR THEORY

The linear theory of plasma waves predicted that waves in an unstable plasma grow for all time. Such a result is not realistic, because nonlinear effects will eventually alter the distribution function. Quasilinear theory can be used to investigate nonlinear effects in unstable plasmas, including the alteration of the equilibrium distribution function. One unstable plasma equilibrium that can be readily analyzed with quasilinear theory is the so-called *gentle-bump distribution function*. This distribution function, which is shown in Fig. 10.5.1, consists of an isotropic maxwellian distribution with a superposed beam of spatially uniform warm drifting electrons. This distribution is unstable to Langmuir oscillations that have phase velocities in the band $\Delta(\omega/k)$, as described in detail in Sec. 9.4. This instability is weak, since the growth rate ω_i is much less than the natural oscillation frequency ω_r. Even though it is a weak instability, the initial perturbations of the plasma *eventually* grow to such an amplitude that the linear theory is no longer valid. At these larger amplitudes, nonlinear effects cause a distortion of $f_{e0}(u, t)$ that alters the growth rate, since $\omega_i \propto (\partial f_{e0}/\partial u)|_{u=\omega/k}$.

If the beam velocity u_0 far exceeds the thermal speed $\sqrt{\kappa T_e/m_e}$, the unstable waves have the properties calculated in the preceding section, in which the quasilinear corrections to Landau damping of Langmuir oscillations are

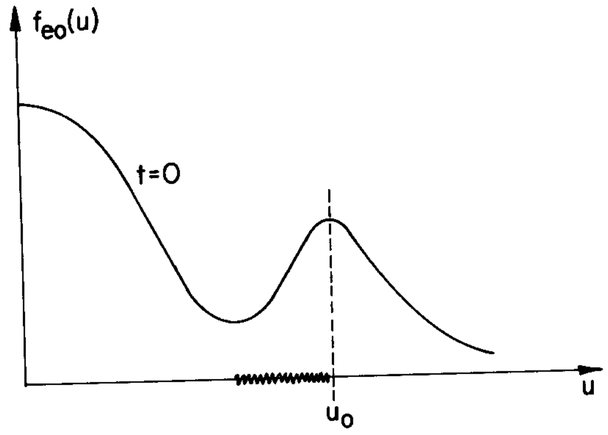


FIGURE 10.5.1
Electron distribution function with an unstable "bump-in-tail" feature, showing the region of unstable phase velocities.

calculated. The pertinent results of Sec. 10.4 that can be taken over directly are:

1 The electrostatic energy increases for waves whose phase velocity ω/k lies in the region labeled $\Delta(\omega/k)$ in Fig. 10.5.1. In this region $\partial f_{e0}/\partial u > 0$, and

$$\frac{d\mathscr{E}_k}{dt} = 2\omega_i \mathscr{E}_k$$

$$\omega_i = \frac{\pi}{2}\frac{\omega_{pe}{}^2\omega_r}{k^2}\frac{\partial f_{e0}(u)}{\partial u}\bigg|_{u=\omega_r/|k|} \qquad (10.5.1)$$

$$\omega_r = \omega_{pe}(1 + \tfrac{3}{2}k^2\lambda_{De}{}^2) \approx \omega_{pe}$$

2 This increase in electrostatic energy comes at the expense of the energy of those particles in the beam that are resonant with the unstable waves.

$$\frac{d}{dt}W_{K,\text{resonant particles}} = -2\frac{d}{dt}\int \mathscr{E}_k\, dk \qquad (10.5.2)$$

3 Half the loss of energy of the resonant particles goes into electrostatic energy; the other half goes into the nonresonant particles, which acquire an oscillation, i.e., the mechanical energy in the wave

$$\frac{d}{dt}W_{K,NR} = \frac{d}{dt}\int \mathscr{E}_k\, dk = -\frac{1}{2}\frac{d}{dt}W_{K,R} \qquad (10.5.3)$$

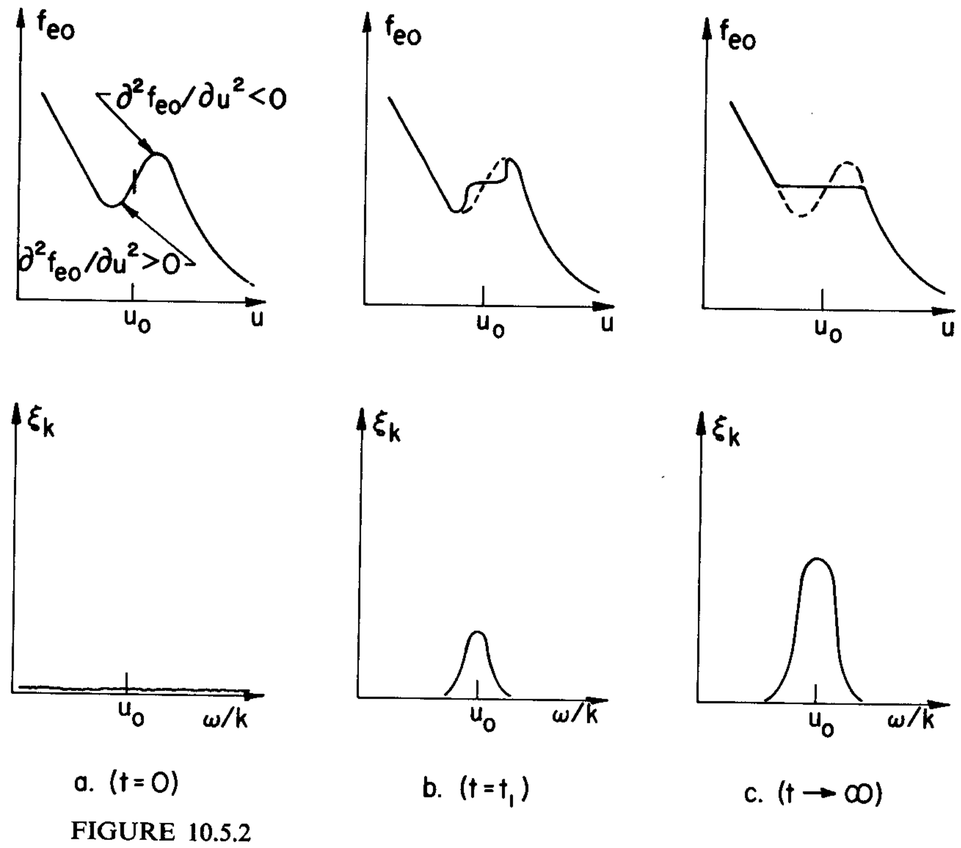


FIGURE 10.5.2
Unstable electron distribution function and wave energy distribution illustrating the evolution of the distribution function as the waves grow.

4 The resonant particles diffuse according to

$$\frac{d}{dt}f_{e0}\left(u \approx \frac{\omega}{|k|}\right) = \frac{8\pi^2e^2}{m_e{}^2}\mathscr{E}_{k=\omega/u}\frac{1}{u}\frac{d^2}{du^2}f_{e0}\left(u \approx \frac{\omega}{|k|}\right) \qquad (10.5.4)$$

By examining d^2f_{e0}/du^2 in the region over which $df_{e0}/du > 0$, the evolution of waves and the distortion of the resonant region can be sketched; this is done in Fig. 10.5.2.

5 The saturation of the instability can be investigated using the resonant-particle-diffusion equation, which can be deduced by the same argument used in the Landau damping calculation (10.4.9).

$$\frac{d}{dt}\left(f_{e0,\text{resonant}} - \frac{2\omega_{pe}}{\bar{n}_e m_e}\frac{d}{du}\frac{1}{u^3}\mathscr{E}_{k=\omega/u}\right) = 0$$

with the time-asymptotic solution [since $\mathscr{E}_k(t=0)$ is small]

$$f_{e0}(u, t\to\infty) - f_{e0}(u, t=0) = \frac{2\omega_{pe}}{\bar{n}_e m_e}\frac{d}{du}\frac{1}{u^3}\mathscr{E}_{k=\omega/u}(t\to\infty) \qquad (10.5.5)$$

The distribution f_{e0} continues to distort as long as the distribution is unstable. Stability is achieved, (that is, $\omega_i = 0$) when

$$\frac{\partial f_{e0}}{\partial u} = 0 \qquad (10.5.6)$$

that is, $f_{e0}(u, t=\infty)$ is constant in the resonant region. This asymptotic state of f_{e0} is called the *quasilinear plateau.* The energy distribution in electric fields as $t\to\infty$ is found by integrating (10.5.5) over resonant velocity.

$$\mathscr{E}_{k=\omega/u} = \frac{\bar{n}_e m_e u^3}{2\omega_{pe}}\int_{u_1}^{u}[f(u', t\to\infty) - f(u', t\to 0)]\,du' \qquad (10.5.7)$$

where u_1 is the lower boundary of the resonant region (Fig. 10.5.2). The total energy in waves at $t=\infty$ is estimated from energy conservation [Eqs. (10.4.5 and 10.4.6)].

$$\frac{d}{dt}\int \mathscr{E}_k\,dk = -\frac{1}{2}\frac{d}{dt}W_{K,\text{resonant particles}} \qquad (10.5.8)$$

By examining Fig. 10.5.3, it is clear that the approximate change in the energy of the resonant particles is

$$\Delta W_{K,R} \approx \tfrac{1}{2}\Delta(n_B m_e u^2) \approx n_B m_e u_0(u_2 - u_0)$$

because the average resonant particle changes velocity by $\Delta u \approx u_2 - u_0$. The density n_B is the density of particles in the bump, i.e., the area of the shaded region in Fig. 10.5.3. This result confirms the intuitive notion that the energy growth in unstable waves is limited by the energy available in the nonthermal feature of the distribution that drives the instability, in this case the difference in energy between the bump distribution $f_B(t=0)$ and the stable distribution shown by the solid line in Fig. 10.5.3.

Problem 10.5.1 Find $\int \mathscr{E}_k\,dk$ by integrating (10.5.7) over k ($k = \omega_{pe}/u$), assuming that the bump distribution is maxwellian. [*Hint*: Expand $f(u, t=0)$ in a Taylor series about $u = u_0$ with $f(u, t=\infty) = f(u_0, t=0)$. Show that

$$W_E = \int \mathscr{E}_k(t\to\infty)\,dk \approx \frac{\omega_i(k=\omega_{pe}/u_0, t=0)\bar{n}_e m_e}{\omega_{pe}}\frac{(u_2-u_1)^3}{u_0}$$
$$\approx \Delta(n_B m_e u^2) \approx n_B m_e u_0(u_2 - u_0) \qquad (10.5.9)$$

where n_B is the density particles in the bump.] ////

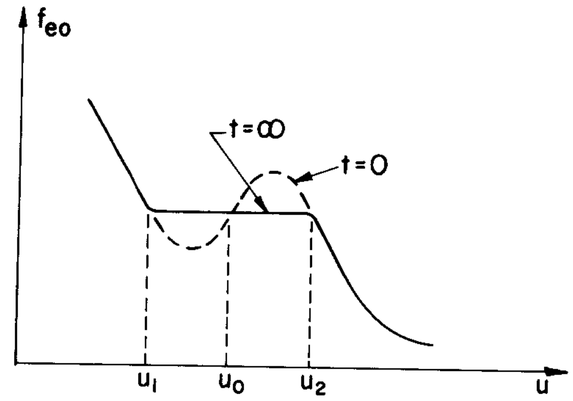


FIGURE 10.5.3
Electron distribution function in the region of unstable waves showing the flattening of the distribution as the waves grow. The $t=\infty$ distribution is no longer unstable.

6 The "diffusion" of nonresonant particles takes place because they gain energy as the instability develops. This is so because these particles acquire mechanical (oscillatory) energy associated with the electrostatic waves. The altered nonresonant distribution can be calculated from the quasilinear equations, using the approximation

$$\frac{\omega}{k} \gg u$$

over the bulk of the nonresonant distribution. This approximation in the quasilinear diffusion equation, Eq. (10.2.14), gives [as in the Landau damping calculation (10.4.11)]

$$\frac{\partial f_{e0,\text{nonresonant}}}{\partial t} = \frac{1}{\bar{n}_e m_e}\frac{\partial^2 f_{e0}}{\partial u^2}\frac{\partial}{\partial t}\int \mathscr{E}_k\,dk \qquad (10.5.10)$$

which has the same sort of solution found in the Landau damping process.

$$f_{e0} = \sqrt{\frac{1}{4\pi}}\frac{1}{\sqrt{\dfrac{1}{\bar{n}_e m_e}\displaystyle\int \mathscr{E}_k\,dk + \frac{\kappa T_e}{2m_e}}}\exp\left[-\frac{u^2}{4\left(\dfrac{1}{\bar{n}_e m_e}\displaystyle\int \mathscr{E}_k\,dk + \dfrac{\kappa T_e}{2m_e}\right)}\right] \qquad (10.5.11)$$

As $\int \mathscr{E}_k\,dk$ increases toward its $t\to\infty$ value [(10.5.8)], the average distribution of nonresonant particles gains energy. The increase has the appearance of a temperature increase, as shown by the form of $f_{e0,\text{NR}}$ in

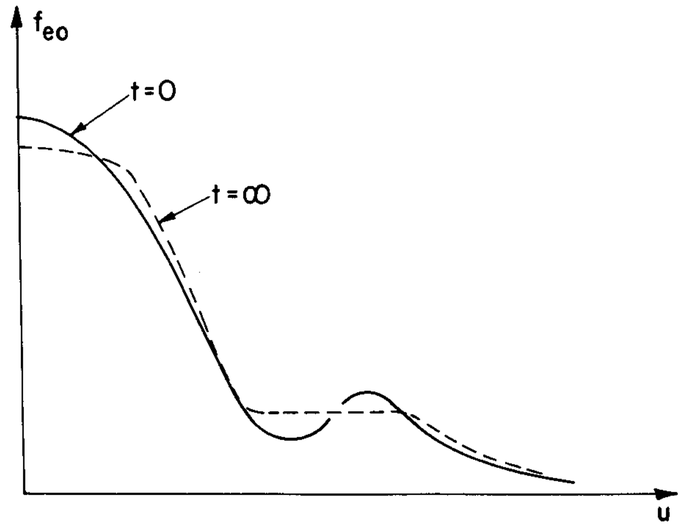


FIGURE 10.5.4
Electron distribution function for an initially unstable distribution showing the final state for both resonant and nonresonant particles.

(10.5.11), with the apparent temperature increase associated with the waves given by

$$\text{“}T\text{”}_{\text{wave KE}} = \frac{2}{\bar{n}_e \kappa} \int \mathscr{E}_k \, dk \qquad (10.5.12)$$

Of course, this is not really *temperature*, it is ordered motion, i.e., coherent oscillations of the nonresonant particles in response to the fluctuating **E** fields.

The time-asymptotic distribution $f_{e0}(t \to \infty)$ is sketched in Fig. 10.5.4, using (10.5.6) for the resonant particles, (10.5.11) for the nonresonant particles.

10.6 QUASILINEAR THEORY OF THE TWO-STREAM INSTABILITY

The usefulness of quasilinear theory is not limited to situations in which resonant particles drive the instability. As an example, consider the two-stream instability of a uniform plasma of ions counterstreaming through a cold-electron background, as described by the distribution pictured in Fig. 10.6.1. The instability of ions streaming through electrons with a speed much greater than the

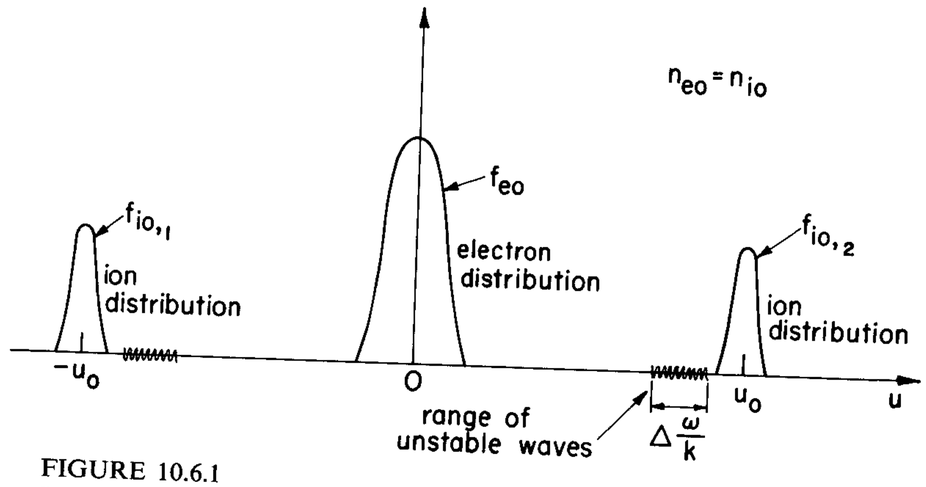


FIGURE 10.6.1
Distribution function for two counterstreaming ion beams in an electron background showing the region of unstable nonresonant waves.

electron thermal speed is treated in Sec. 9.3; the distribution Fig. 10.6.1 has two ion populations drifting through the electron background. This situation is somewhat simpler to describe because from symmetry the *mean* electron speed remains $U_{e0} = 0$. This is a fluidlike instability, because it depends only on the density and mean velocity of the electrons and ions, and thus can be calculated from the fluid equations.

The diffusion of the electron and ion distribution is given at least for a few e folding times of growth, by the quasilinear equations

$$\frac{\partial f_{\alpha 0}}{\partial t} = \frac{\partial}{\partial u}\left[\frac{8\pi e^2}{m_\alpha{}^2} \int \mathscr{E}_k \frac{\omega_i}{(\omega_r - ku)^2 + \omega_i{}^2} \, dk\right] \frac{\partial f_{\alpha 0}}{\partial u} \qquad \omega_i > 0$$
$$\frac{\partial \mathscr{E}_k}{\partial t} = 2\omega_i(k)\mathscr{E}_k \qquad (10.6.1)$$

and $\omega_i(k)$ and $\omega_r(k)$ are determined from the value of $f_{\alpha 0}$ at a particular time. Because the instability is determined by nonresonant particles, ω_i and ω_r are insensitive to the details of the distribution function, and a sensible way to describe the instability process is to compare:

1. The decrease in mean ion velocity, which provides the energy for the instability
2. The increase in electron energy (ordered motion in response to the waves)
3. The increase in ion energy relative to the ion mean speed, also due to ordered ion motion in the wave

Defining the mean speed and mean kinetic energy of each species by

$$U_\alpha = \int u f_{\alpha 0}(u, t)\, du \qquad (10.6.2)$$

and

$$W_K = \frac{\bar{n}_\alpha m_\alpha}{2} \int [u - U_\alpha(t)]^2 f_{\alpha 0}(u, t)\, du \qquad (10.6.3)$$

and taking appropriate velocity moments of (10.6.1), the average dynamics are given by

$$\bar{n}_\alpha m_\alpha \frac{dU_\alpha}{dt} = 2\omega_{p\alpha}{}^2 \int dk \int \omega_i \mathscr{E}_k \frac{2k[(\omega_r - kU_\alpha) - k(u - U_\alpha)]}{[(\omega_r - ku)^2 + \omega_i{}^2]^2} f_{\alpha 0}\, du \qquad (10.6.4)$$

$$\frac{d}{dt} W_{K\alpha} = 2\omega_{p\alpha}{}^2 \int dk \int \omega_i \mathscr{E}_k \frac{(\omega_r - kU_\alpha)^2 + \omega_i{}^2 - k^2(u - U_\alpha)^2}{[(\omega_r - ku)^2 + \omega_i{}^2]^2} f_{\alpha 0}\, du \qquad (10.6.5)$$

where ω_r and ω_i are functions of k, as shown in the linear theory of Sec. 9.3, and there are *two* unstable branches: one due to the electrons interacting with the ions that move with $U_{i1} = u_0$, the other due to electrons interacting with the ions that move with $U_{i2} = -u_0$. The ions are treated as *two* fluids, $f_{i0,1}$ moving with $U_{i1} = u_0$ and $f_{i0,2}$ moving with $-u_0$ (Fig. 10.6.1).

The linear properties of the instability are as follows, from Sec. 9.3.

Maximum growth:

$$\omega_{i,\max} = \frac{\sqrt{3}}{2} \left(\frac{m_e}{4m_i}\right)^{1/3} \omega_{pe} \qquad (10.6.6)$$

Wave number and frequency for maximum growth:

$$\left(\frac{\omega_r}{k} \mp u_0\right)_{\max} \approx \frac{1}{2} \left(\frac{m_e}{4m_i}\right)^{1/3} u_0 \qquad (10.6.7)$$

$$|k|_{\max} \approx \frac{\omega_{pe}}{u_0}$$

The linear properties for maximum growth can be used to estimate the dynamic change in electron energy and ion drift speed from (10.6.4) and (10.6.5). These are calculated by neglecting u and U_e compared with ω/k in calculating the electron dynamics and neglecting $u - U_i$ with respect to $(\omega - kU_i)/k$ in calculating the ion dynamics. The results are

$$\frac{d}{dt} W_{K,\text{electrons}} = \frac{d}{dt} \int \mathscr{E}_k\, dk$$

$$\tfrac{1}{2} n_1 m_i \frac{d}{dt} U_{i1}{}^2 = -\frac{d}{dt} \int \mathscr{E}_k\, dk \qquad (10.6.8)$$

$$\tfrac{1}{2} n_2 m_i \frac{d}{dt} U_{i2}{}^2 = -\frac{d}{dt} \int \mathscr{E}_k\, dk \qquad (10.6.9)$$

$$n_{e0} = n_{i0}$$

Note again that the total increase in the field energy accounts for only half the decrease in the energy of the ions that drive the instability. The excess energy lost by the ions goes into electron kinetic energy.

The ions increase this energy relative to their mean velocity by a smaller amount.

$$\frac{d}{dt} W_{K,\text{ions in beam 1}} = \left(\frac{m_e}{m_i}\right)^{1/3} \frac{d}{dt} \int \mathscr{E}_k\, dk$$

$$\frac{d}{dt} W_{K,\text{ions in beam 2}} = \left(\frac{m_e}{m_i}\right)^{1/3} \frac{d}{dt} \int \mathscr{E}_k\, dk \qquad (10.6.10)$$

The rate of energy change [(10.6.8) to (10.6.10)] does not add up to give $dW_{\text{plasma}}/dt = -(d/dt) \int \mathscr{E}_k\, dk$, because small terms $\approx (m_e/m_i)^{1/3}$ have been neglected in calculating the electron kinetic energy.

The distinction between ordered particle energy and the random energy implied by the word temperature is seen clearly in computer simulation experiments, which calculate the position and velocity of each particle during the instability process. The results of such an experiment for the case discussed in this section are shown in Fig. 9.3.3. The average features of the instability as given by (10.6.8) to (10.6.10) are verified by the simulation experiment, but as can be seen in Fig. 9.3.3*e* the electron energy is in the form of ordered oscillation associated with the unstable waves. This ordered motion is eventually (Fig. 9.3.3*g*) *converted* into random motion ("temperature") by processes not included in the quasilinear theory. The dominant process contributing to this random motion is electron trapping in the potential well of the waves.

Without going into any further detail, it is clear that the growth rate ω_i of the two-stream instability described in this section is insensitive to f_{e0}, until the electrons acquire sufficient energy so that

$$\frac{\omega}{k} \approx \sqrt{\frac{W_{K,e}}{n_e m_e}} \approx \sqrt{\langle u_e{}^2 \rangle}$$

for some portion of the electron distribution. Indeed, if

$$\sqrt{\frac{\kappa T_e}{m_e}} \gtrsim U_i$$

linear theory predicts that the distribution in Fig. 10.6.1 is stable. The simulation experiment finds that when

$$\sqrt{\frac{W_{K,e}}{n_e m_e}} \gtrsim U_i$$

the instability stops growing, despite the fact that the electron energy $W_{K,e}$ is not thermal energy, i.e., temperature. The conversion $W_{K,e} \to n_e \kappa T_e$ by particle trapping has its own stabilizing influence on a plasma wave, and this will be detailed in the next section.

10.7 ELECTRON TRAPPING IN A SINGLE PLASMA WAVE

Many interesting and important nonlinear plasma phenomena emerge from the study of plasma systems that contain a limited number of plasma waves. In this and the following section some examples are presented. The first example is the breakdown of the Landau damping theory (Chap. 8) due to particle trapping in the case of a single very long wavelength Langmuir oscillation.[1]

Consider an electrostatic wave

$$\mathbf{E}_1 = \hat{\mathbf{x}}\bar{E}_k \sin(kx - \omega_{pe}t) \qquad k\lambda_{De} \ll 1 \qquad (10.7.1)$$

in an otherwise field-free uniform maxwellian plasma. According to Landau theory, this wave damps on a time scale

$$\tau_L = \left[\sqrt{\frac{\pi}{8}}\frac{\omega_{pe}}{(k\lambda_{De})^3}\exp\left(-\frac{1}{2k^2\lambda_D{}^2} - \frac{3}{2}\right)\right]^{-1} \qquad (10.7.2)$$

The linear Vlasov theory that leads to Landau damping [Sec. 8.6] is based on the assumption that the orbits of the plasma particles are only slightly perturbed by the wave field. An electron moving with nearly the same speed as the wave sees an electrostatic potential well, and oscillates with a period

$$(\text{from } m\ddot{x} = -e\bar{E}_k \sin kx \approx -e\bar{E}_k kx]$$

$$T = \frac{1}{2\pi}\sqrt{\frac{m}{e\bar{E}_k k}} \equiv \frac{\tau_T}{2\pi} \qquad (10.7.3)$$

This simple harmonic motion in the trough of the plasma wave is a major perturbation of the particle orbits, and any linear calculation that depends on the distribution of these trapped particles is invalid on a time scale $\tau > \tau_T$. Landau damping theory depends on the distribution of particles that move at nearly the

[1] T. M. O'Neil, *Phys. Fluids*, **8**:2255 (1965).

speed of the wave. Since these resonant particles are the first to be trapped Landau damping theory is valid only if

$$\tau_L < \tau_T \qquad (10.7.4)$$

As a further demonstration of the fact that the time scale (10.7.3) marks the end of validity of linear theory in calculating the behavior of a single wave, consider the linearized solution of the Vlasov equation for the electron distribution $f_e = f_0 + f_1$ in an electric field $\mathbf{E}_1$ given by (10.7.1):

$$\frac{\partial f_1}{\partial t} + u\frac{\partial f_1}{\partial x} = \frac{e}{m}\bar{E}_k \sin(kx - \omega_{pe}t)\frac{\partial f_0}{\partial u} \qquad (10.7.5)$$

Treating $\bar{E}_k$ and f_0 as constant in time, the solution to (10.7.5) is

$$f_1(x, u, t) = f_1(u, t = 0)\cos(kx - kut) - \frac{e}{m}E_k\frac{\partial f_0}{\partial u} \times \left[\frac{\cos(kx - \omega_{pe}t)}{ku - \omega_{pe}} - \frac{\cos(kx - kut)}{ku - \omega_{pe}}\right] \qquad (10.7.6)$$

Expanding (10.7.6) about $u = \omega_{pe}/k$, it is seen that

$$\left.\frac{\partial f_1}{\partial u}\right|_{u\approx\omega_{pe}/k} \approx \frac{ek\bar{E}_k}{m}t^2\left.\frac{\partial f_0}{\partial u}\right|_{u\approx\omega_{pe}/k}\frac{\cos(kx - \omega_{pe}t)}{2}$$

so that the linearized treatment ($\partial f_1/\partial u \ll \partial f_0/\partial u$) in the resonant region breaks down when

$$t \approx \sqrt{\frac{m}{ek\bar{E}_k}} \equiv \tau_T$$

which is the trapping time defined in (10.7.3).

It is possible to find a nonlinear solution to the Vlasov equation by noting that

$$\frac{\partial}{\partial t}f(x, u, t) + u\frac{\partial}{\partial x}f(x, u, t) + \frac{q}{m}\bar{E}_k \sin(kx - \omega_{pe}t)\frac{\partial}{\partial u}f(x, u, t) = 0 \qquad (10.7.7)$$

is equivalent to the set of equations

$$\frac{d}{dt'}f[x'(t'), u'(t'), t'] = 0 \qquad (10.7.8)$$

$$\frac{d}{dt'}x'(t') = u'(t')$$

$$\frac{d}{dt'}u'(t') = \frac{q}{m}E_k \sin[kx'(t') - \omega_{pe}t'] \qquad (10.7.9)$$

$$x'(t' = t) = x$$

$$u'(t' = t) = u$$

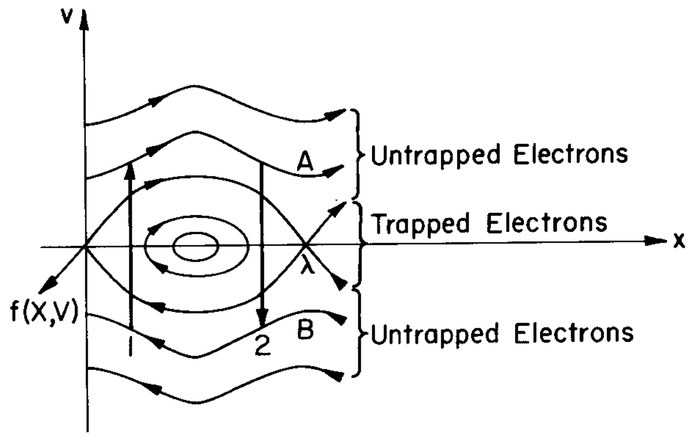


FIGURE 10.7.1
Phase-space diagram showing trapping of particles by a finite-amplitude wave. [*After H. L. Berk and K. V. Roberts, Phys. Fluids*, **10**; 1595 (1967).]

which says that the distribution f is constant along the trajectory of a particle of charge/mass $= q/m$ in the electric field E_k.

The orbits of the trapped and untrapped particles can be solved implicitly in terms of elliptic functions. These orbits are sketched in Fig. 10.7.1. The solution to the Vlasov equation is obtained in terms of an initial condition by letting

$$f(x, u, t) = f[x'(t' = 0), u'(t' = 0), 0]$$

since any $f =$ constant solves (10.7.8). An initial condition appropriate to the assumed form of $\mathbf{E}_1$ would be

$$f(x_0, u_0, 0) = f_0(u_0) + f_1(u_0, t = 0)\cos kx_0$$

where
$$x_0(x, u, t) \equiv x'(t' = 0)$$

and
$$u_0(x, u, t) \equiv u'(t' = 0)$$

The slow change in the amplitude $\bar{E}_k$ of the wave as particles are trapped is found from integrating the energy conservation relation over a wavelength

$$-\frac{\partial}{\partial t}\int_{-\pi/k}^{\pi/k} \frac{\bar{\mathbf{E}}_k^2}{8\pi} \sin^2 kx\, dx = -\frac{\partial}{\partial t}\frac{\bar{\mathbf{E}}_k^2}{16\pi} = \int \tfrac{1}{2}mu^2 \frac{\partial}{\partial t}\langle f\rangle\, du$$

$$\langle f\rangle \equiv \frac{k}{2\pi}\int_{-\pi/k}^{\pi/k} f\, dx$$

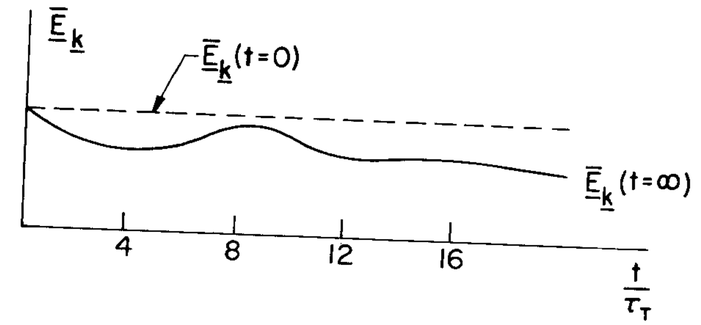


FIGURE 10.7.2
Damping of a large-amplitude wave, showing the effects of particle trapping in altering the linear-theory damping rate. (*After R. C. Davidson, "Methods in Nonlinear Plasma Theory," Academic, New York*, 1971.)

Considering only the effects of resonant particles, the value $\bar{E}_k(t)$ following the above procedure is plotted in Fig. 10.7.2. For times $t < \tau_T$, the wave Landau damps. For times $t > \tau_T$, $\bar{E}_k$ oscillates with time, and approaches a constant value. The arguments in this section do not depend on the sign of ω_i, and thus trapping will also modify the *growth* of waves on time scales $t > \tau_T$.

10.8 PLASMA WAVE ECHOES

A phenomenon that is both tractable as a problem in coherent wave theory and instructive from the point of view of the Vlasov theory is the plasma wave echo.[1] The basic idea is that if a narrow spectrum of plasma waves with wave numbers in the range $k \approx k_1$ dies away by Landau damping, the plasma distribution function remains perturbed, as shown in Sec. 8.7, and is modulated by a free-streaming term of the form (consider the plasma to be one dimensional for clarity)

$$f_\alpha = f_{\alpha 0} + f_{\alpha k_1}(u, t = t_1)e^{ik_1 u(t-t_1)} + \cdots$$

in addition to terms proportional to $E_1(t)$, which Landau-damp away. The term in the perturbed distribution proportional to $e^{ik_1 ut}$, called the *ballistic term*, does not produce an electric field, since $\int \exp(ik_1 ut)\, f(u)\, du \to 0$ as t increases. Nonetheless, the information contained in the original perturbation remains locked in the distribution function. The key to retrieving this information is found by noting that if a second set of plasma waves $k \approx k_2$ is launched at time

[1] R. W. Gould, T. M. O'Neil, and J. H. Malmberg, *Phys. Rev. Letters*, **19**:219 (1967).

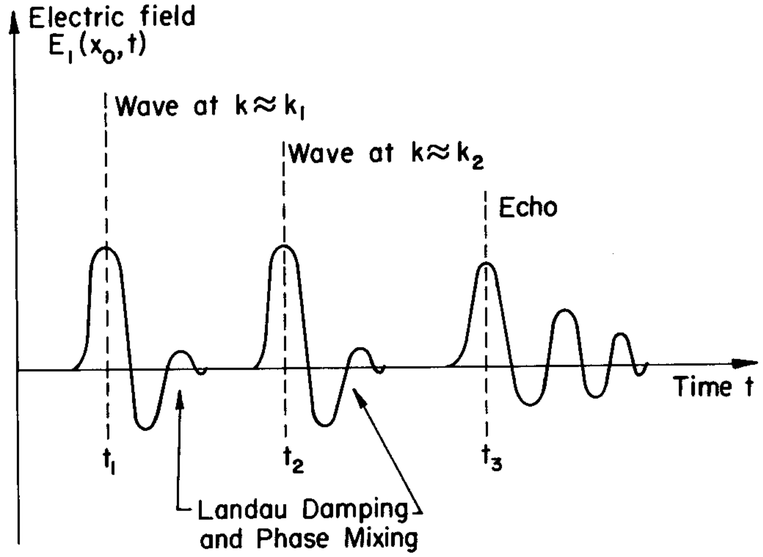


FIGURE 10.8.1
Schematic of a plasma wave echo showing the damping of electric field pulses impressed on the plasma at times t_1 and t_2, together with the spontaneous emergence of an echo pulse at time t_3.

t_2, the new perturbation can interfere constructively with the old perturbation; i.e.,

$$f_\alpha = f_{\alpha 0} + f_{\alpha k_2}^{(1)} e^{ik_2 u(t-t_2)} + f_\alpha^{(2)} + \cdots$$

where one contribution to the second-order perturbed distribution $f_\alpha^{(2)}$ will have the form

$$f_\alpha^{(2)} \propto f_{\alpha k_1}^{(1)} f_{\alpha k_2}^{(1)} e^{ik_1 u(t-t_1)} e^{-ik_2 u(t-t_2)} \qquad t > t_2$$

Thus, even though the second perturbation has also Landau-damped away, at a later time t_3, determined by

$$k_1(t_3 - t_1) - k_2(t_3 - t_2) = 0 \qquad \text{or} \qquad t_3 \approx \frac{k_1 t_1 - k_2 t_2}{k_1 - k_2} \tag{10.8.1}$$

the charge density

$$\rho_{q2} \approx \int du\, e^{ik_1 u(t-t_1)} e^{-ik_2 u(t-t_2)} f_{\alpha(k_2 - k_1)}^{(2)}$$

is no longer small, because the integrand is no longer oscillatory. The condition (10.8.1) is satisfied only for a limited time interval, after which ρ_{q2} again vanishes by phase mixing, that is, $\int e^{ikut} F(u)\, du \to 0$ as $t \to \infty$. This reappearance of ρ_{q2} produces an electric field E_2, called an *echo*. The electric field amplitude in the plasma in such a process would vary as shown in Fig. 10.8.1. The pulse at t_1 is

produced by a set of grids, and Landau-damps away. The pulse at t_2 is also produced by the grids. The pulse at t_3 appears spontaneously, showing that during the time intervals Δt_1 and Δt_2, the plasma was not as quiescent as would appear from a measurement of its macroscopic electric fields.

The echo effect has been demonstrated in the laboratory,[1] and provides:

1. Experimental confirmation of Landau damping.
2. Confirmation of the Vlasov result that information about the perturbed state at $t = 0$ is retained in the free-streaming contributions to the distribution even though there are no macroscopic fields (i.e., entropy is conserved for $\tau \ll \tau_{\text{collision}}$).
3. An opportunity to test and further develop the theory of coherent nonlinear waves. In particular, it introduces the idea of nonlinear effects in the time history of the fields.

As a detailed example of the process just outlined, consider an external potential $\phi^{\text{ext}}(\mathbf{x}, t)$ applied to a three-dimensional plasma, described by the Vlasov equation

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v} \cdot \nabla f_\alpha - \frac{q_\alpha}{m_\alpha} \nabla\phi \cdot \nabla_{\mathbf{v}} f_\alpha = 0 \tag{10.8.2}$$

Let the distribution f_α be expanded in terms of a spatially averaged equilibrium distribution

$$f_{\alpha 0}(\mathbf{v}) = \frac{1}{V} \int f_\alpha(\mathbf{x}, \mathbf{v})\, d\mathbf{x}$$

and a perturbation $f_{\alpha 1}$ about the average

$$f_\alpha = f_{\alpha 0} + f_{\alpha 1}$$

The Fourier transform of the perturbed distribution is defined by

$$f_{\alpha \mathbf{k}} = \int f_{\alpha 1} \exp(i\mathbf{k} \cdot \mathbf{x})\, d\mathbf{x}$$

and the Fourier transform of (10.8.2) has the form

$$\frac{\partial}{\partial t} f_{\alpha \mathbf{k}} + i\mathbf{k} \cdot \mathbf{v} f_{\alpha \mathbf{k}} - i \frac{q_\alpha}{m_\alpha} \mathbf{k}\phi_{\mathbf{k}} \cdot \nabla_{\mathbf{v}} f_{\alpha 0} = \frac{q_\alpha}{m_\alpha} \int i(\mathbf{k} - \mathbf{k}') \cdot \nabla_{\mathbf{v}} f_{\alpha \mathbf{k}'} \phi_{\mathbf{k}-\mathbf{k}'} \frac{d\mathbf{k}'}{(2\pi)^3} \tag{10.8.3}$$

The potential $\phi_{\mathbf{k}}$ is the Fourier transform of the *total* potential in the plasma. It includes the externally produced potential as well as the potential due to the plasma charge distribution $\rho_{\mathbf{k}} = \sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha \mathbf{k}}\, d\mathbf{v}$.

Poisson's equation gives a second relation between $\phi_{\mathbf{k}}$ and $f_{\mathbf{k}}$:

$$k^2 \phi_{\mathbf{k}}(t) = k^2 \phi_{\mathbf{k},\,\text{ext}} + \sum_\alpha 4\pi \bar{n}_\alpha q_\alpha \int f_{\alpha \mathbf{k}}(\mathbf{v}, t)\, d\mathbf{v} \tag{10.8.4}$$

[1] J. H. Malmberg, C. B. Wharton, R. W. Gould, and T. M. O'Neil, *Phys. Fluids*, **11**:1147 (1968).

Equations (10.8.3) and (10.8.4) are simplified by taking their Laplace transform, defining

$$\tilde{f}_{\alpha\mathbf{k}}(\mathbf{v}, p) = \int_0^\infty f_{\alpha\mathbf{k}}(\mathbf{v}, t)e^{-pt}\,dt \qquad \mathrm{Re}(p) > p_0$$

$$f_{\alpha\mathbf{k}}(\mathbf{v}, t) = \frac{1}{2\pi i}\int_{p_0-i\infty}^{p_0+i\infty} \tilde{f}_{\alpha\mathbf{k}}(\mathbf{v}, p)e^{pt}\,dp$$

and assuming that at $t = 0$ the plasma distribution is given by $f_{\alpha 0}$, that is, that

$$f_{\alpha\mathbf{k}}(\mathbf{v}, t = 0) = 0 \qquad \mathbf{k} \neq 0$$

The Laplace transform of the Vlasov and Poisson equations gives

$$(p + i\mathbf{k}\cdot\mathbf{v})\tilde{f}_{\alpha\mathbf{k}} = \frac{iq_\alpha}{m_\alpha}\tilde{\phi}_\mathbf{k}\,\mathbf{k}\cdot\nabla_\mathbf{v} f_{\alpha 0} - \frac{iq_\alpha}{m_\alpha}\int\frac{d\mathbf{k}'}{(2\pi)^3}\int_{p_{01}-i\infty}^{p_{01}+i\infty} dp_1 \int_{p_{02}-i\infty}^{p_{02}+i\infty}$$

$$\left[\frac{1}{p - p_1 - p_2}\,\tilde{\phi}_{\mathbf{k}-\mathbf{k}'}(p_2)(\mathbf{k} - \mathbf{k}')\cdot\nabla_\mathbf{v}\tilde{f}_{\alpha\mathbf{k}'}(\mathbf{v}, p_1)\right]e^{p_1 t}e^{p_2 t}\,dp_2 \qquad \mathrm{Re}(p) > p_1 + p_2 \tag{10.8.5}$$

and

$$\tilde{\phi}_\mathbf{k}(p) = \bar{\phi}_{\mathbf{k},\,\mathrm{ext}}(p) + \sum_\alpha \frac{4\pi\bar{n}_\alpha q_\alpha}{k^2}\int\tilde{f}_{\alpha\mathbf{k}}(\mathbf{v}, p)\,d\mathbf{v}$$

These equations are solved by iteration, treating the $f_{\alpha\mathbf{k}}$ as small quantities. Thus, substituting $f_{\alpha\mathbf{k}}$ from (10.8.5) into Poisson's equation (10.8.6) gives

$$D(\mathbf{k}, ip)\tilde{\phi}_\mathbf{k} = \phi_{\mathbf{k},\,\mathrm{ext}} + \sum_\alpha \frac{{\omega_{p\alpha}}^2}{k^2}\int\frac{d\mathbf{k}'}{(2\pi)^3}\int dv\int_{p_{01}-i\infty}^{p_{01}+i\infty}\frac{dp_1}{2\pi i}\int_{p_{02}-i\infty}^{p_{02}+i\infty}\frac{(\mathbf{k} - \mathbf{k}')}{p - p_1 - p_2}$$

$$\cdot\nabla_\mathbf{v}\tilde{f}_{\alpha\mathbf{k}'}(\mathbf{v}, p_1)\frac{\tilde{\phi}_{\mathbf{k}-\mathbf{k}'}}{\mathbf{k}\cdot\mathbf{v} - ip}\frac{dp_2}{2\pi i} \tag{10.8.6}$$

where

$$D(\mathbf{k}, ip) \equiv 1 - \sum\frac{{\omega_{p\alpha}}^2}{k^2}\int\frac{\mathbf{k}\cdot\nabla_\mathbf{v} f_{\alpha 0}}{\mathbf{k}\cdot\mathbf{v} - ip}\,d\mathbf{v}$$

Expanding

$$\tilde{\phi}_\mathbf{k} = \tilde{\phi}_\mathbf{k}^{(1)} + \tilde{\phi}_\mathbf{k}^{(2)} + \cdots$$

the first-order potential is found by neglecting the nonlinear term $(\mathbf{k} - \mathbf{k}')$ $\phi_{\mathbf{k}-\mathbf{k}'}\cdot\nabla_\mathbf{v} f_{\alpha\mathbf{k}'}$ in (10.8.6), with the result

$$\tilde{\phi}_\mathbf{k}^{(1)} = \frac{\tilde{\phi}_{\mathbf{k},\,\mathrm{ext}}(p)}{D(\mathbf{k}, ip)} \tag{10.8.7}$$

The second-order potential is obtained by substituting the linear value of $f_{\alpha\mathbf{k}}$

$$\tilde{f}_{\alpha\mathbf{k}} = \frac{q_\alpha}{m_\alpha}\frac{\mathbf{k}\cdot\nabla_\mathbf{v} f_{\alpha 0}}{\mathbf{k}\cdot\mathbf{v} - ip}\,\tilde{\phi}_\mathbf{k}$$

and the linear value $\tilde{\phi}_\mathbf{k}^{(1)}$ of (10.8.7) into the nonlinear term of (10.8.6), leading to

$$\tilde{\phi}_\mathbf{k}^{(2)} = \sum_\alpha \frac{4\pi {q_\alpha}^3\bar{n}_\alpha}{{m_\alpha}^2 k^2}\int\frac{d\mathbf{k}'}{(2\pi)^3}\int dv\int_{p_{01}-i\infty}^{p_{01}+i\infty}\frac{dp_1}{2\pi i}\int_{p_{02}-i\infty}^{p_{02}+i\infty}\frac{1}{p - p_1 - p_2}$$

$$\times\,\phi_{\mathbf{k}-\mathbf{k}'}^{\mathrm{ext}}(p_1)(\mathbf{k} - \mathbf{k}')\cdot\nabla_\mathbf{v}$$

$$\frac{\mathbf{k}'\cdot\nabla_\mathbf{v} f_{\alpha 0}}{D(\mathbf{k}, ip)D(\mathbf{k} - \mathbf{k}', ip_2)D(\mathbf{k}', ip_1)(\mathbf{k}\cdot\mathbf{v} - ip)(\mathbf{k}'\cdot\mathbf{v} - ip_1)}$$

$$\times\,\phi_{\mathbf{k}'}^{\mathrm{ext}}(p_1)\frac{dp_2}{2\pi i} \tag{10.8.8}$$

Further iterations give higher-order corrections to the potential $\tilde{\phi}_\mathbf{k}$.

To apply this formalism to the plasma echo problem, let the externally produced potential be high-frequency, so that the ions can be considered motionless, and one-dimensional, so that $\mathbf{v}\to u$, $\mathbf{k}\to k$, and further, let it correspond to a pulse at $t = \tau_1$ and a second pulse at $t = \tau_2$; that is,

$$\phi^{\mathrm{ext}}(x, t) = A_1\,\delta[\omega_{pe}(t - \tau_1)]\cos k_1 x + A_2\,\delta[\omega_{pe}(t - \tau_2)]\cos k_2 x \tag{10.8.9}$$

where A_1 and A_2 are constant amplitudes and it is assumed that

$$k_2 > k_1 > 0 \tag{10.8.10}$$

The transform of this external potential is quite simple.

$$\tilde{\phi}_k^{\mathrm{ext}}(p) = \frac{\pi A_1}{\omega_{pe}}[\delta(k_1 - k) + \delta(k_1 + k)]e^{-p\tau_1}$$

$$+\frac{\pi A_2}{\omega_{pe}}[\delta(k_2 - k) + \delta(k_2 + k)]e^{-p\tau_e} \tag{10.8.11}$$

and allows the integration dk' in (10.8.8) to be carried out. The product

$$\tilde{\phi}_{k-k',\,\mathrm{ext}}(p_2)\tilde{\phi}_{k',\,\mathrm{ext}}(p_1)$$

involves 16 products of the sort $e^{-p_1\tau_2}e^{-p_1\tau_2}$. In the following discussion the only terms retained are those which, when integrated over k', give a contribution to $\tilde{\phi}^{(2)}$ proportional to

$$\delta(k - k_2 + k_1)$$

Problem 10.8.1 Show (after reading the rest of Sec. 10.8) that no terms in $\tilde{\phi}_k^{(2)}$ contribute to an echo *except* those terms proportional to

$$\delta(k - k_2 + k_1) \qquad ////$$

With the foregoing restriction, using (10.8.11) for $\bar{\phi}_k^{\text{ext}}$, $\tilde{\phi}_k^{(2)}$ becomes [(10.2.10) defines $f_{\alpha 0}(u)$]

$$\tilde{\phi}_k^{(2)}(p) = \frac{\pi e A_1 A_2}{2m_e k^2} \int du \int_{p_{01}-i\infty}^{p_{01}+i\infty} \frac{dp_1}{2\pi i} \int_{p_{02}-i\infty}^{p_{02}+i\infty} \frac{\delta(k-k_2+k_1)}{p-p_1-p_2} \frac{k_1 k_2}{D(k, ip)}$$
$$\left[\frac{e^{-p_1\tau_1} e^{-p_2\tau_2}}{D(-k_1, ip_1)D(k_2, ip_2)(ku-ip)} \frac{\partial}{\partial u}\left(\frac{1}{k_2 u - ip_2}\frac{\partial}{\partial u} f_{\alpha 0}\right) \right.$$
$$\left. + \frac{e^{-p_1\tau_2} e^{-p_2\tau_1}}{D(-k_1, ip_2)D(k_2, ip_1)(ku-ip)} \frac{\partial}{\partial u}\left(\frac{1}{-k_1 u - ip_2}\frac{\partial}{\partial u} f_{\alpha 0}\right) \right] \frac{dp_2}{2\pi i} \quad (10.8.12)$$

which can be simplified by integrating by parts with respect to u and interchanging p_1 and p_2 in the first term of (10.8.12), leading to

$$\tilde{\phi}_k^{(2)}(p) \approx -\frac{\pi e A_1 A_2}{2m_e k^2} \int_{p_{01}-i\infty}^{p_{01}+i\infty} \frac{dp_1}{2\pi i} \int_{p_{02}-i\infty}^{p_{02}+i\infty}$$
$$\times \frac{\delta(k-k_2+k_1) k_1 k_2 e^{-p_1\tau_1} e^{-p_2\tau_2}}{(p-p_1-p_2)D(k, ip)D(-k_1, ip_1)D(k_2, ip_2)}$$
$$\int du \left(\frac{\partial f_{e0}}{\partial u}\right) \frac{k}{(ku-ip)^2} \frac{(ku-ip)+i(p-p_1-p_2)}{(k_1 u + ip_1)(k_2 u - ip_2)} \frac{dp_2}{2\pi i} \quad (10.8.13)$$

Equation (10.8.13) has in it a wealth of second-order information about the potential, but most of it has nothing to do with the echo. Concentrating on the reappearance of a potential after two earlier-imposed signals have Landau-damped away (i.e., the echo) allows the inversion of (10.8.13),

$$\phi_k^{(2)}(t) = \frac{1}{2\pi i} \int_{p_0-i\infty}^{p_0+i\infty} \tilde{\phi}_k^{(2)}(p) e^{pt}\, dp \qquad p_0 > p_{01} + p_{02}$$

to be carried out in a simplified way by neglecting poles corresponding to $D(k, ip) = 0$, $D(k_2, ip_2) = 0$, and $D(-k_1, ip_1) = 0$; these poles lead to damped potentials

$$\phi^{(2)} \propto e^{\omega_i t}$$

and give no contribution after $t >$ several $|\omega_i^{-1}|$. The integration dp_2 is carried out by closing the contour to the right, as shown in Fig. 10.8.2, picking up the contribution from the only pole at

$$p_2 = p - p_1 \quad (10.8.14)$$

The factor $e^{-p_2\tau_2}$ causes the integrand to vanish on the curve labeled Γ in Fig. 10.8.2, which is why the integral dp_2 is closed to the right. Evaluating the integrand at the pole (10.8.14) gives $e^{-p_1\tau_1}\, e^{-p_2\tau_2} \to e^{p_1(\tau_2-\tau_1)}\, e^{-p\tau_2}$ since the $\tau_2 > \tau_1$, the p_1

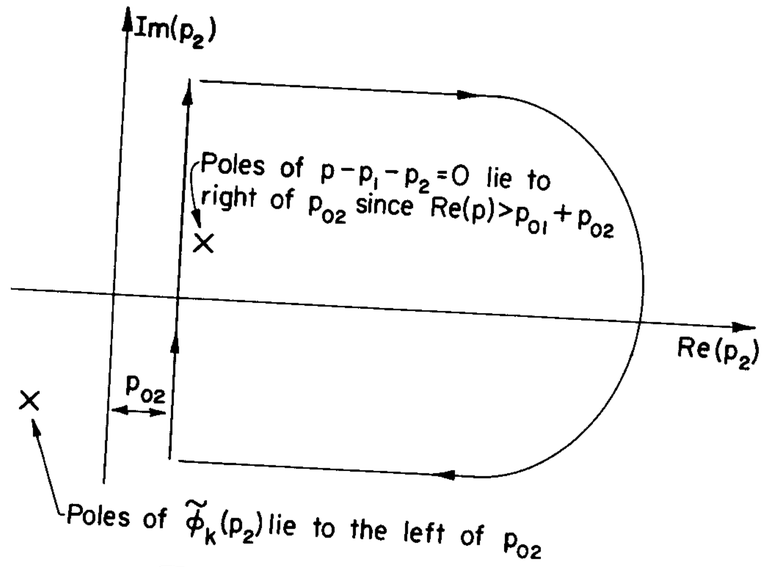


FIGURE 10.8.2
Contour for evaluating the inversion integral for $\phi_k^{(2)}$.

integration, can be carried out only on a contour closed to the left, as in Fig. 10.8.3. Neglecting the poles $D = 0$, which give Landau-damped solutions, the only contribution to $\int dp_1$ is from the pole at $p_1 = ik_1 u$.

The potential $\phi_k^{(2)}$, after the p_1 and p_2 integration, is

$$\phi_k^{(2)}(t) = \frac{\pi e}{2m_e} A_1 A_2 \frac{ikk_1k_2}{k^2} \int_{p_0-i\infty}^{p_0+i\infty} \frac{dp}{2\pi i} \int \frac{\partial f_{e0}}{\partial u} e^{p(t-\tau_2)}\, \delta(k-k_2+k_1)$$
$$\frac{e^{ik_1u(\tau_2-\tau_1)}}{D(k, ip)D(k_2, ip+ik_1u)D(-k_1, -k_1u)(ku-ip)^2}\, du$$

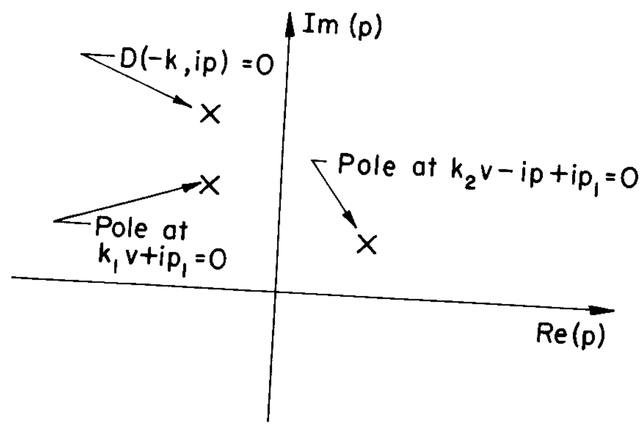


FIGURE 10.8.3
Location of the poles of $\phi_k^{(2)}$ in the p_1 plane.

The dominant contributions to the integral over p comes from the double pole

$$\frac{1}{(ku - ip)^2}$$

which gives

$$\phi_k^{(2)}(t) = \frac{\pi e}{2m_e} A_1 A_2 \frac{ikk_1k_2\,\delta(k - k_2 + k_1)}{k^2}(t - \tau_2)$$

$$\int \frac{\partial f_{e0}}{\partial u} \frac{\exp\{-iku[t - \tau_2 - k_1(\tau_2 - \tau_1)/k]\}}{D(k, ku)D(k_2, k_2u)D(-k_1, -k_1u)}\,dv \quad (10.8.15)$$

It was pointed out in the introduction to this section that $\phi_k^{(2)}(t)$ is generally small because the velocity integrand is a rapidly oscillating function of u. However, at

$$t \approx \tau' \equiv \tau_2 + \frac{k_1}{k}(\tau_2 - \tau_1) \quad (10.8.16)$$

$\phi_k^{(2)}$ is not small, because

$$\exp[-iku(t - \tau')] \approx 1$$

The structure of the echo depends on the velocity distribution f_{e0}, for particular choices of k_1 and k_2. As a specific example, assume that k is so large that

$$D(k_2, k_2u) \approx D(-k_1, -k_1u) \approx D(k, ku) \approx 1$$

and that the plasma is maxwellian,

$$f_{e0} = \left(\frac{m_e}{2\pi\kappa T_e}\right)^{1/2} \exp\left(-\frac{m_e u^2}{2\kappa T_e}\right)$$

In that case, (10.8.15) gives

$$\phi_k^{(2)}(t) = \frac{\pi e}{2m_e} A_1 A_2\,\delta(k - k_2 + k_1)k_1k_2(t - \tau_2)(t - \tau') \exp\left[-\frac{k^2\kappa T_e}{2m_e}(t - \tau')^2\right]$$

where τ', the echo time, is given by (10.8.16). Thus the width of the echo,

$$\Delta\tau' \approx \sqrt{\frac{2m_e}{k^2\kappa T_e}}$$

is a measure of the plasma temperature.

Collisions destroy the echo, because they cause the (free-streaming) perturbed distribution to decay.

$$f \propto e^{ikut} \to e^{ikut}e^{-\nu t}$$

The condition required in order that the echo not be destroyed by collisions[1] is that the echo time not be too long:

$$(\omega_{pe}\tau')^2\tau'\nu_{90°} < 1$$

where $\nu_{90°}$ is the collision frequency for 90° collisions.

The theory developed in this section deals with the temporal damping of plasma waves and the subsequent reappearance in the plasmas of an electric field disturbance called a plasma wave echo. This theory could have been developed for spatial echoes,[2] instead of temporal echoes. If an experiment to study plasma echoes is arranged so that the primary plasma wave pulses damp in time, the plasma-wave-echo pulse appears at a later time. However, if the experimental situation is arranged so that the primary plasma waves damp in space, a spatial plasma wave echo appears many Landau damping lengths away from the points where the waves are launched. For practical reasons plasma-wave-echo experiments investigate the spatial echoes instead of the temporal echoes. The occurrence of the spatial plasma wave echo is explained using the same arguments developed early in this section. An electric field with time dependence $\exp(-i\omega_1 t)$, excited at a point $x = 0$, Landau-damps as it propagates away from its point of origination and leaves the distribution with a perturbation of the form

$$f_{\alpha 1}(u) \approx f^{(1)} \exp\left[i\omega_1\left(-t + \frac{x}{u}\right)\right]$$

For large x there is no electric field associated with this perturbation since its velocity integral phase mixes to zero. If a second electric field, with time dependence $\exp(i\omega_2 t)$, is excited at $x = L$, it Landau-damps as it propagates away from its point of origination and leaves a first-order perturbation of the form

$$f_{\alpha 1}(u) \approx f^{(1)} \exp\left[i\omega_2\left(t - \frac{x - L}{u}\right)\right]$$

on the plasma. In addition, this second electric field modulates the perturbation due to the first electric field, leaving a second-order perturbation of the form

$$f_{\alpha 2} \approx f^{(2)} \exp\left[i(\omega_2 - \omega_1)t + i\omega_2\frac{L}{u} - i(\omega_2 - \omega_1)\frac{x}{u}\right]$$

At the point $x = L\omega_2/(\omega_2 - \omega_1)$, a velocity integral over this perturbation does not phase-mix to zero, and an electric field appears at this point in the plasma.

[1] C. H. Su and C. Oberman, *Phys. Rev. Letters*, **20**:427 (1968).
[2] T. M. O'Neil and R. W. Gould, *Phys. Fluids*, **11**:134 (1968).

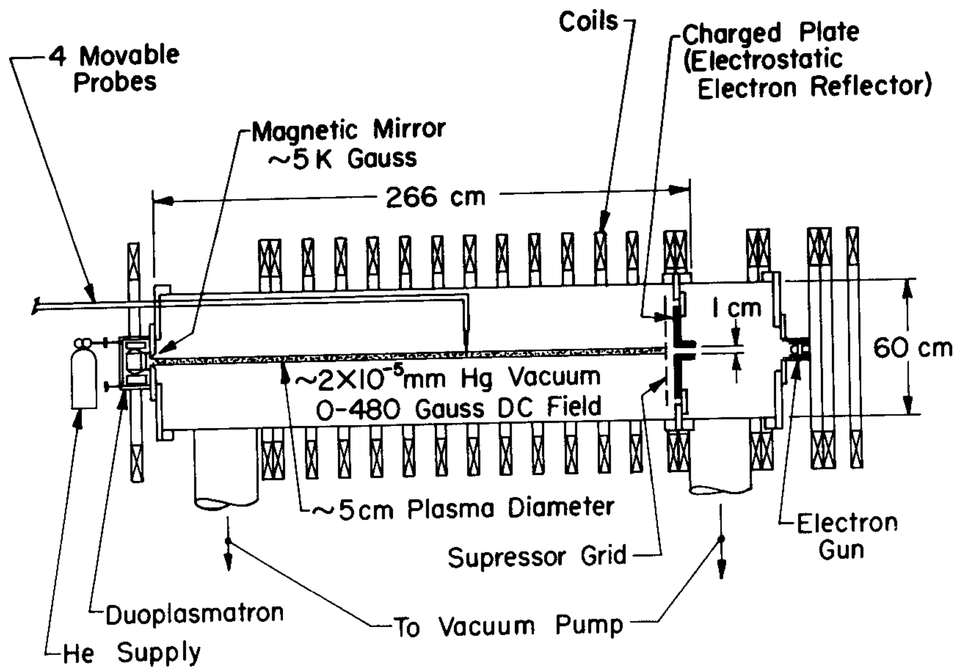


FIGURE 10.8.4
Schematic of experiment used to investigate plasma wave echoes. [*After J. H. Malmberg, et al., Proceedings of Conference on Phénomènes d'ionization dans les Gaz*, **4**; 229 (1963).]

If L is large compared with the Landau damping length, and if $\omega_2/(\omega_2 - \omega_1)$ is of order unity, this third electric field, which is the spatial plasma echo, appears at a position well separated from the first two electric field excitation positions. The experiment used by Malmberg *et al.*[1] to study the spatial plasma wave echoes is depicted schematically in Fig. 10.8.4. The plasma column is 180 cm long and 5 cm in diameter, with a central density of 1.5×10^8 cm^{-3}. The axial magnetic field is 300 G and can be regarded as infinite for the purposes of the experiment. The plasma has a temperature of 9.4 eV and a Debye length of 2 mm. The electron mean free path is 10^5 cm for electron-ion collisions and 4×10^4 cm for electron-neutral collisions. The plasma column is surrounded by a 5.2-cm-radius cylinder that acts as a waveguide beyond cutoff and reduces the stray electromagnetic coupling between the excitation and detection probes.

A plasma wave echo obtained with this experiment is shown in the lower trace of Fig. 10.8.5. The upper trace is the spatial distribution of the 120-MHz signal in the vicinity of the excitation probe at $x = 0$. The middle trace is the spatial distribution of the 130-MHz signal in the vicinity of the second probe at

[1] J. H. Malmberg, C. B. Wharton, R. W. Gould, and T. M. O'Neil, *Phys. Fluids*, **11**:1147 (1968).

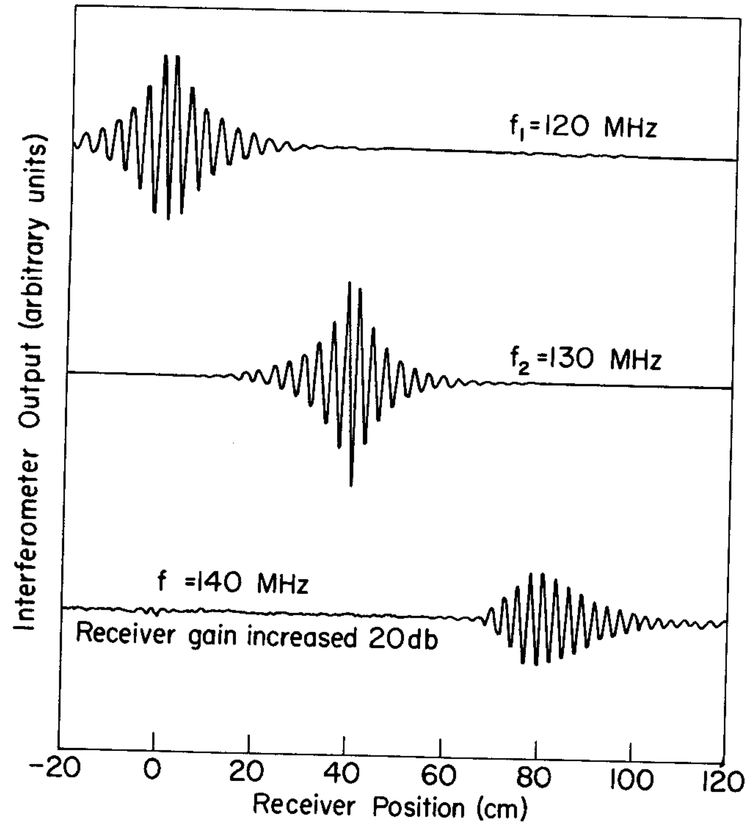


FIGURE 10.8.5
Third-order spatial plasma wave echo. The transmitter probes are at 0 at 40 cm. For the upper curve the receiver is tuned to f_1, for the middle curve it is tuned to f_2, and for the lower curve it is tuned to f_3, with the gain increased by 20 db. [*After J. H. Malmberg et al., Phys. Fluids*, **11**; 1147 (1968).]

$x = L$. The plasma wave echo shown in the lower trace of Fig. 10.8.5 is a third-order echo because the echo frequency $\omega_3 = 2\omega_2 - \omega_1$. These experiments confirm the details of the theoretical predictions of nonlinear interactions of plasma waves and demonstrate that, even though the electric field associated with an initial plasma disturbance has died away, the distribution function retains information about the initial perturbation.

10.9 NONLINEAR WAVE-PARTICLE INTERACTIONS (WEAK TURBULENCE)

The results of Secs. 10.2 to 10.6 showed how the particle distribution is changed by the emission or absorption of a wave in the plasma. These changes in the distribution function are proportional to E_k^2; that is, they are linear in the

energy of a plasma wave. There are other nonlinear processes that can alter the distribution function, for instance, processes in which two waves couple to the plasma distribution, or in which two waves couple to a third wave, with the plasma simply providing the dielectric background that supports the waves. For example, scattering of a wave from the distribution of particles consists in a coupling of the incident and scattered waves to the distribution; another example is the decay of two waves into a third wave, subject to the usual conservation laws.

The nonlinear interaction between waves and particles can be calculated through the Vlasov equation. The details of these calculations are beyond the scope of this text; instead, the resulting equations, which govern the development of the fields and distribution, are written down, and some of the phenomena emerging from the equations are discussed.

Under the assumptions that there is no steady magnetic field, that the level of turbulence is weak, and that the turbulence is spatially homogeneous, a set of kinetic equations can be derived from the Vlasov equation to describe the nonlinear interactions. There are various ways to derive these equations, and the student is referred to monographs on plasma turbulence theory[1] for the details. Essential in all the derivations is a closure scheme called the *random-phase approximation*; this means that the equations describe only the *modulus* of the wave amplitudes, averaged over the phases of the waves: $\langle \phi_{\mathbf{k}} \phi_{\mathbf{k}'} \rangle = \phi_{\mathbf{k}}{}^2 \delta_{\mathbf{k}-\mathbf{k}'}$. The random-phase approximation is equivalent to a rigorous perturbation approach in which a hierarchy of equations for the ensemble averages is developed from the Vlasov equation. The ensemble average quantities are defined by

$$\langle f(x, v, t) \rangle, \langle f(x_1, v_1, t) f(x_2, v_2, t) \rangle, \langle f(1) f(2) f(3) \rangle, \ldots$$

Fluctuations in the distribution function are defined by

$$\delta f = f - \langle f \rangle$$

where $\langle \delta f \rangle = 0$. The hierarchy of equations for the ensemble averaged quantities can be written

$$\langle f(1) \rangle = G_1[\langle \delta f(1)\, \delta f(2) \rangle]$$
$$\langle \delta f(1)\, \delta f(2) \rangle = G_2[\langle \delta f(1)\, \delta f(2)\, \delta f(3) \rangle]$$
$$\vdots$$

Quantities called *cumulants* are defined as follows:

$$g_2 \equiv \langle \delta f(1)\, \delta f(2) \rangle$$
$$g_3 \equiv \langle \delta f(1)\, \delta f(2)\, \delta f(3) \rangle$$
$$g_4 \equiv \langle \delta f(1)\, \delta f(2)\, \delta f(3)\, \delta f(4) \rangle - g_2(1, 3) g_2(2, 4) - g_2(1, 4) g_2(2, 3) - g_2(1, 2) g_2(3, 4)$$
$$\vdots$$

[1] R. C. Davidson, "Nonlinear Plasma Theory," Academic, New York, 1971.

The random-phase approximation is equivalent to neglecting cumulants greater than or equal to fourth order. Under the assumption of random phases, the density of plasma waves $N_{\mathbf{k}}{}^{\gamma}$ of the type γ defined by

$$N_{\mathbf{k}}{}^{\gamma} \equiv k^2 \frac{\phi_{\mathbf{k}}{}^2}{8\pi} \left| \frac{\partial D(\mathbf{k}, \omega)}{\partial \omega} \right|_{\omega = \omega_{\mathbf{k}}{}^{\gamma}}; \qquad D(\mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}) = 0 \tag{10.9.1}$$

develops according to the following equation:

$$\frac{\partial N_{\mathbf{k}}{}^{\gamma}}{\partial t} = T(\mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}, t) N_{\mathbf{k}}{}^{\gamma}$$
$$+ \sum_{\beta, \eta} \delta(\omega_{\mathbf{k}}{}^{\gamma} - \omega_{\mathbf{l}}{}^{\beta} - \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta}) R(\mathbf{k}, \mathbf{l}, D_{\gamma, \beta, \eta}) N_{\mathbf{k}} N_{\mathbf{k}-\mathbf{l}}$$
$$+ \sum_{\beta, \eta, \omega_{\mathbf{k}}{}^{\gamma} \neq \omega_{\mathbf{l}}{}^{\beta} + \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta}} L[\mathbf{k}, \mathbf{l}, f(\mathbf{v})] N_{\mathbf{k}} N_{\mathbf{l}} \tag{10.9.2}$$

where the operators T, R, L are defined below. The three contributions to (10.9.2) represent distinct physical processes.

10.9.1 Linear (and Quasilinear) Wave-Particle Interactions

The first term of (10.9.2) is linear in the density of plasma waves $N_{\mathbf{k}}$, and contains the results of the entire linear wave and stability theory (Chaps. 8 and 9), as well as the quasilinear corrections calculated earlier in this chapter. Explicitly, the linear wave-particle interaction is governed by the equation

$$\frac{\partial N_{\mathbf{k}}{}^{\gamma}}{\partial t} = T(\mathbf{k}, \omega_{\mathbf{k}}, t) N_{\mathbf{k}}{}^{\gamma}$$
$$= \left[2\, \mathrm{Im}(\omega_{\mathbf{k}}{}^{\gamma}) + 2\, \mathrm{Im}(\omega_{\mathbf{k}}{}^{\gamma})^{Q} + \left(\frac{1}{\dfrac{\partial D(\mathbf{k}, \omega)}{\partial \omega}} \right) \Bigg|_{\omega = \omega^{\gamma}} \left(\frac{\partial}{\partial t} \frac{\partial D(\mathbf{k}, \omega)}{\partial \omega} \right) \Bigg|_{\omega = \omega^{\gamma}} \right] N_{\mathbf{k}}{}^{\gamma} \tag{10.9.3}$$

where $\mathrm{Im}\,(\omega_{\mathbf{k}}{}^{\gamma})$ is the linear rate of damping or growth,

$$\mathrm{Im}(\omega_{\mathbf{k}}{}^{\gamma}) = \pi \sum_{\alpha} \frac{\omega_{p\alpha}{}^2}{k^2 [D(\mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma})]'} \int \delta(\omega_{\mathbf{k}}{}^{\gamma} - \mathbf{k} \cdot \mathbf{v}) (\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha})\, d\mathbf{v} \tag{10.9.4}$$

and the quasilinear rate of damping or growth is

$$\mathrm{Im}(\omega_{\mathbf{k}}{}^{\gamma})^{Q} = \sum_{\alpha} \frac{\omega_{p\alpha}{}^2}{k^2 D'(\mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma})} \mathrm{Re} \left(\int \frac{1}{\omega_{\mathbf{k}}{}^{\gamma} - \mathbf{k} \cdot \mathbf{v} + i\varepsilon} \frac{\partial}{\partial t} \frac{\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha}}{\omega_{\mathbf{k}}{}^{\gamma} - \mathbf{k} \cdot \mathbf{v} + i\varepsilon}\, d\mathbf{v} \right)_{\varepsilon \to 0^+} \tag{10.9.5}$$

The superscript γ refers to the particular *type* of plasma wave being considered (e.g., Langmuir oscillations or ion waves, etc.). The frequency $\omega_{\mathbf{k}}{}^{\gamma}$ is the frequency which solves $D(\mathbf{k}, \omega = \omega_{\mathbf{k}}{}^{\gamma}) = 0$ for a particular wave number $\mathbf{k}$ and wave type γ, and $D' = \partial D/\partial\omega_{\mathbf{k}}$.

The term Im $(\omega_{\mathbf{k}}{}^{\gamma})$ is the linear damping or growth of the wave. The next two terms of (10.9.4) are the corrections to linear growth (or damping) produced by the slow distortion of the plasma distribution f_0 [and hence the plasma dielectric constant $D(\mathbf{k}, \omega)$] as waves are emitted or absorbed. The change of f_0 with time is computed from the quasilinear theory developed earlier in this chapter [Eq. (10.9.5)].

10.9.2 Resonant Wave-Wave Interaction

The second term in (10.9.2) is multiplied by the factor

$$\delta(\omega_{\mathbf{k}}{}^{\gamma} - \omega_{\mathbf{l}}{}^{\beta} - \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta}) \tag{10.9.6}$$

This term contains interaction between three waves that satisfy the conditions

$$\begin{aligned} \omega_{\mathbf{k}_1}{}^{\gamma} &= \omega_{\mathbf{k}_2}{}^{\beta} + \omega_{\mathbf{k}_3}{}^{\eta} \\ \mathbf{k}_1 &= \mathbf{k}_2 + \mathbf{k}_3 \end{aligned} \tag{10.9.7}$$

These are the resonance conditions that single out interactions in which the two interacting waves are normal modes of the plasma, i.e., satisfy $D(\mathbf{k}, \omega) = 0$, and the generated wave is also a normal mode of the plasma [i.e., it satisfies $D(\mathbf{k}_2 + \mathbf{k}_3, \omega_2 + \omega_3) = 0$]. The interaction is called *resonant-mode coupling*; it does not depend on any resonance between waves and particles, but only on resonances between three normal modes of the system.

This sort of process is characteristic of any dielectric medium $D(\mathbf{k}, \omega)$ that supports waves. This interaction may cause diffusion of wave energy among many modes of the system, or it may give instability ($N_{\mathbf{k}}$, $N_{\mathbf{l}}$, $N_{\mathbf{k}-\mathbf{l}}$, all increasing with time), depending on the dielectric properties of the system.

Explicitly, the kinetic equations for resonant three-wave processes (neglecting linear, quasilinear, and nonlinear wave-particle interactions) are

$$\begin{aligned} \frac{\partial N_{\mathbf{k}}{}^{\gamma}}{\partial t} &= \sum_{\beta,\eta} \delta(\omega_{\mathbf{k}}{}^{\gamma} - \omega_{\mathbf{k}}{}^{\beta} - \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta}) R(\mathbf{k}, \mathbf{l}, D_{\gamma,\beta,\eta}) N_{\mathbf{k}} N_{\mathbf{k}-\mathbf{l}} \\ &= 8\pi^2 \,\mathrm{Re} \sum_{\beta,\eta} \int \left[\frac{\delta(\omega_{\mathbf{k}}{}^{\gamma} - \omega_{\mathbf{l}}{}^{\beta} - \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta}) S_{\mathbf{k}}{}^{\gamma}}{k^2 l^2 |\mathbf{k}-\mathbf{l}|^2 |D'(\mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}) D'(\mathbf{l}, \omega_{\mathbf{l}}{}^{\beta}) D'(\mathbf{k}-\mathbf{l}, \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta})|} \right] \\ &\quad \times \overline{M}(\mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}; \mathbf{l}, \omega_{\mathbf{l}}{}^{\beta}; \mathbf{k}-\mathbf{l}, \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta}) [\overline{M}^*(\mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}; \mathbf{l}, \omega_{\mathbf{l}}{}^{\beta}; \mathbf{k}-\mathbf{l}, \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta}) \\ &\qquad S_{\mathbf{k}}{}^{\gamma} N_{\mathbf{l}}{}^{\beta} N_{\mathbf{k}-\mathbf{l}}{}^{\eta} - \overline{M}(\mathbf{l}, \omega_{\mathbf{l}}{}^{\beta}; \mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}; \mathbf{l}-\mathbf{k}, \omega_{\mathbf{l}-\mathbf{k}}{}^{\eta}) S_{\mathbf{l}}{}^{\beta} N_{\mathbf{k}}{}^{\gamma} N_{\mathbf{k}-\mathbf{l}}{}^{\eta} \\ &\qquad - \overline{M}(\mathbf{k}-\mathbf{l}, \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta}; \mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}; -\mathbf{l}, -\omega_{\mathbf{l}}{}^{\beta}) S_{\mathbf{k}-\mathbf{l}}{}^{\eta} N_{\mathbf{k}}{}^{\gamma} N_{\mathbf{l}}{}^{\beta}] \, d\mathbf{l} \end{aligned} \tag{10.9.8}$$

with the sign factor $S_{\mathbf{k}}{}^{\gamma}$ defined by

$$S_{\mathbf{k}}^{\gamma} = \mathrm{sign} \left. \frac{\partial D(\mathbf{k}, \omega)}{\partial \omega} \right|_{\omega = \omega_{\mathbf{k}}{}^{\gamma}} \tag{10.9.9}$$

and the matrix elements are

$$\overline{M} = \sum_{\alpha} \int M_{\mathbf{v},\alpha} \, d\mathbf{v}$$

$$\begin{aligned} M_{\mathbf{v},\alpha}(\mathbf{k}_1; \omega_1; \mathbf{k}_2, \omega_2; \mathbf{k}_3, \omega_3) &= \frac{4\pi \bar{n}_{\alpha} q^3}{m_{\alpha}{}^2} \frac{1}{\omega_1 - \mathbf{k}_1 \cdot \mathbf{v} + i\varepsilon} \left[\mathbf{k}_2 \cdot \nabla_{\mathbf{v}} \frac{\mathbf{k}_3 \cdot \nabla_{\mathbf{v}} f_{\alpha}}{\omega_3 - \mathbf{k}_3 \cdot \mathbf{v} + i\varepsilon} + (2 \leftrightarrow 3) \right]_{\varepsilon \to 0^+} \end{aligned} \tag{10.9.10}$$

and the symmetries

$$\overline{M}(\mathbf{k}_1, \omega_1; \mathbf{k}_2, \omega_2; \mathbf{k}_3, \omega_3) = \overline{M}(\mathbf{k}_1, \omega_1; \mathbf{k}_3, \omega_3; \mathbf{k}_2, \omega_2) \tag{10.9.11}$$

and

$$\overline{M}^*(\mathbf{k}_1, \omega_1; \mathbf{k}_2, \omega_2; \mathbf{k}_3, \omega_3) = \overline{M}(-\mathbf{k}_1, -\omega_1; -\mathbf{k}_2, -\omega_2; -\mathbf{k}_2, -\omega_3)$$

hold.

10.9.3 Nonlinear Wave-Particle Interaction

The terms in the final bracket of (10.9.2) contain the nonlinear wave-particle interaction. These terms vanish except when the wave $(\mathbf{k} - \mathbf{l}, \omega_{\mathbf{k}}{}^{\alpha} - \omega_{\mathbf{l}}{}^{\beta})$ is resonant with some part of the particle distribution; i.e., they are proportional to

$$\int \delta[\omega_{\mathbf{k}} - \omega_{\mathbf{l}} - (\mathbf{k} - \mathbf{l}) \cdot \mathbf{v}] G(\mathbf{v}) \, d\mathbf{v} \tag{10.9.12}$$

The linear interaction proportional to $\delta(\omega_{\mathbf{k}} - \mathbf{k} \cdot \mathbf{v})$ was called Landau damping; the nonlinear interaction (10.9.12) is called nonlinear Landau damping. Just as Landau damping became Landau growth when the distribution function was significantly far from thermodynamic equilibrium in the resonant regions, so the nonlinear Landau damping term may lead to instability if the distribution function near $(\mathbf{k} - \mathbf{l}) \cdot \mathbf{v} = \omega_{\mathbf{k}} - \omega_{\mathbf{l}}$ is far enough from thermodynamic equilibrium.

In cases [e.g., linearly stable waves that do not satisfy the wave-resonance conditions (10.9.7)] in which only nonlinear wave-particle effects contribute to the growth, diffusion, and damping of waves, the density of plasma waves, which changes according to the third term in (10.9.2), is given by

$$\frac{\partial N_{\mathbf{k}}{}^{\gamma}}{\partial t} = -\sum_{\beta} N_{\mathbf{k}}{}^{\gamma} \int L^{\gamma\beta}[\mathbf{k}, \mathbf{l}, f(\mathbf{v})] \, d\mathbf{l} \tag{10.9.13}$$

where

$$L^{\gamma\beta} = \frac{-16\pi S_{\mathbf{k}}{}^{\gamma}}{k^2 l^2 |\mathbf{k}-\mathbf{l}|^2 |D'(\mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}) D'(\mathbf{l}, \omega_{\mathbf{l}}{}^{\beta})|}$$

$$\mathrm{Im}\left[\frac{\bar{M}(\mathbf{k}-\mathbf{l}, \omega_{\mathbf{k}}{}^{\gamma} - \omega_{\mathbf{l}}{}^{\beta}; \mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}; -\mathbf{l}, -\omega_{\mathbf{l}}{}^{\beta}) \bar{M}(\mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}; \mathbf{k}-\mathbf{l}, \omega_{\mathbf{k}}{}^{\gamma} - \omega_{\mathbf{l}}{}^{\beta}; \mathbf{l}, \omega_{\mathbf{l}}{}^{\beta})}{D(\mathbf{k}-\mathbf{l}, \omega_{\mathbf{k}}{}^{\gamma} - \omega_{\mathbf{l}}{}^{\beta})}\right.$$

$$\left. + \sum_{\alpha} \frac{q_\alpha}{m_\alpha} \int \frac{|\mathbf{k}-\mathbf{l}|^2}{\omega_{\mathbf{k}}{}^{\gamma} - \mathbf{k}\cdot\mathbf{v} + i\varepsilon} \mathbf{l}\cdot\nabla_{\mathbf{v}} M_{\mathbf{v},\alpha}(\mathbf{k}-\mathbf{l}, \omega_{\mathbf{k}}{}^{\gamma} - \omega_{\mathbf{l}}{}^{\beta}; \mathbf{k}, \omega_{\mathbf{k}}{}^{\gamma}; -\mathbf{l}, -\omega_{\mathbf{l}}{}^{\beta})\, d\mathbf{v}\right] \tag{10.9.14}$$

and

$$\omega_{\mathbf{k}}{}^{\gamma} - \omega_{\mathbf{l}}{}^{\beta} \neq \omega_{\mathbf{k}-\mathbf{l}}{}^{\eta}$$

is imposed as a constraint. The matrix elements $M_{\mathbf{v},\alpha}$ and $\bar{M}$ are defined in (10.9.10).

Similar equations exist for $\partial f/\partial t$. There are many cases of systems which are linearly stable but nonlinearly unstable, but the reader is referred elsewhere[1] for detailed examples.

The purpose of this discussion was to present the general equations for wave-particle kinetics, and point out the three main effects which dominate the non-linear behavior of a turbulent plasma:

1 Linear wave-particle interactions (waves, instabilities, damping)
2 Nonlinear wave-wave interactions (wave diffusion, resonant three-wave instabilities)
3 Nonlinear wave-particle interactions (nonlinear Landau damping or growth, wave scattering by the particle distribution)

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11

FLUCTUATIONS, CORRELATIONS, AND RADIATION

Both the fluid theory and the Vlasov theory are in fact fluid descriptions, since the Vlasov theory treats the plasma as a fluid in **x-v** phase space, describing the plasma state in terms of a continuous distribution function. Just as there are properties of the plasma that did not occur (not even approximately) in the MHD description, there are also basic properties and processes of a plasma that do not emerge from a Vlasov model.

Some questions that cannot be answered by the Vlasov theory are:

1. What is the level of electric (or magnetic field) fluctuations in a *stable* plasma?
2. What is the emission rate and spectrum of radiation from a plasma?
3. What is the cross section for scattering of radiation from a plasma?

Many aspects of these questions are answered only by higher-order (in $1/n\lambda_D{}^3$) theories than the Vlasov theory. This is because these effects are connected in a basic way with the *discreteness* of the plasma, i.e., the fact that a plasma is a collection of individual particles. The limit $n\lambda_D{}^3 \to \infty$ corresponds to a system

in which the discreteness of the particles is lost, because the Vlasov fluid is thus visualized as a system of particles for which

$$q \to 0$$
$$m \to 0$$
$$n \to \infty$$

$$\frac{q}{m} = \text{constant}$$

$$nq = \text{constant}$$

The coupling of individual particles to the radiation field is lost in the limit $q \to 0$. However, in a real plasma, collisions and other motions of individual particles excite a thermal level of field fluctuations, cause emission of radiation, and give important contributions to the scattering of radiation.

Individual particle effects emerge from the hierarchy of statistical equations derived in Chap. 7, to order $1/n\lambda_D{}^3$. Most effects of order $1/n\lambda_D{}^3$ can be obtained from a very appealing and physical approach, without loss of rigor. This is the test-particle picture,[1] which will be used as the basis for much of the following treatment of fluctuations, radiation, and scattering.

11.1 SHIELDING OF A MOVING TEST CHARGE

To see the effect of discrete particles, it is useful to calculate the potential due to a point test charge moving with uniform speed $\mathbf{v}'$ through a Vlasov plasma, which in the absence of the test particle is uniform and field-free. This shielding was calculated in Chap. 2 for a test charge at rest in an equilibrium plasma, but the shielding is a function of both particle speed and of the distribution of background particles.

To lowest order in the strength of the interaction between a test charge and the back ground plasma, the test charge moves in a straight line, so that its location at time t is

$$\mathbf{x}' = \mathbf{x}_0' + \mathbf{v}'t \qquad (11.1.1)$$

where $\mathbf{x}_0'$ is the location of the test charge at $t = 0$. The charge density of the test particle is represented by

$$\rho_T = q_T\, \delta(\mathbf{x} - \mathbf{x}') = q_T\, \delta(\mathbf{x} - \mathbf{x}_0' - \mathbf{v}'t)$$

[1] N. Rostoker and M. N. Rosenbluth, *Phys. Fluids*, 3:1 (1960).

The potential ϕ produced in the plasma by the test charge includes both the potential of the test charge and the potential it induces in the plasma. Thus ϕ satisfies

$$\nabla^2\phi = -4\pi q_T\,\delta(\mathbf{x} - \mathbf{x}_0' - \mathbf{v}'t) - 4\pi\rho_{\text{plasma}} \qquad (11.1.2)$$

where

$$\rho_{\text{plasma}} = \sum_\alpha \bar{n}_\alpha q_\alpha \int f_\alpha\, d\mathbf{v}$$

is the polarization charge density in the plasma produced in response to the test charge. The plasma distribution f_α satisfies the Vlasov equation

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v}\cdot\nabla f_\alpha - \frac{q_\alpha}{m_\alpha}(\nabla\phi)\cdot\nabla_{\mathbf{v}} f_\alpha = 0 \qquad (11.1.3)$$

Assuming that in the absence of the test charge the plasma is uniform and field-free and that the plasma is only slightly perturbed by the test particle, the Vlasov equation can be linearized about $f = f_{\alpha 0}$; that is,

$$f_\alpha = f_{\alpha 0}(\mathbf{v}) + f_{\alpha 1}(\mathbf{x}, \mathbf{v}, t)$$

$$\sum_\alpha \bar{n}_\alpha q_\alpha \int f_{\alpha 0}\, d\mathbf{v} = 0 \qquad (11.1.4)$$

$$\frac{\partial f_{\alpha 1}}{\partial t} + \mathbf{v}\cdot\nabla f_{\alpha 1} = \frac{q_\alpha}{m_\alpha}(\nabla\phi)\cdot\nabla_{\mathbf{v}} f_{\alpha 0}$$

If the test particle is at rest and the background plasma is maxwellian, the potential ϕ is the shielded potential discussed in Chaps. 1 and 2 (see also Prob. 11.1.1),

$$\phi = \frac{q_T}{r}\exp\left(-\frac{r}{\lambda_D}\right) = q_T\frac{\exp(-|\mathbf{x} - \mathbf{x}'|/\lambda_D)}{|\mathbf{x} - \mathbf{x}'|} \qquad (11.1.5)$$

where $\lambda_D \equiv [\sum_\alpha 4\pi n_\alpha q_\alpha^2/\kappa T_\alpha]^{-1/2}$. The test charge at rest polarizes the plasma, and acquires a shielding cloud. This shielding cloud is made up of an excess of charge having the *opposite* sign from q_T and a deficiency of charge of like sign. The ratio of perturbed ion and electron charge is $\rho_{e1}/\rho_{i1} = T_i/T_e$.

Problem 11.1.1 Show, using Fourier transforms, that the potential of a test charge at rest at $r = 0$ in a plasma described by a distribution $f_{\alpha 0}$ is

$$\phi = \frac{q_T}{r}\exp\left[-r\left(-\sum_\alpha \omega_{p\alpha}{}^2\int 2\frac{\partial f_{\alpha 0}}{\partial v^2}\,d\mathbf{v}\right)^{1/2}\right]$$

and that the perturbed distribution is

$$f_{\alpha 1} = 2\frac{q_\alpha}{m_\alpha}\frac{\partial f_{\alpha 0}}{\partial v^2}\frac{q_T}{r}\exp\left[-r\left(-\sum_\alpha \omega_{p\alpha}{}^2\int 2\frac{\partial f_{\alpha 0}}{\partial v^2}\,d\mathbf{v}\right)^{1/2}\right]$$

Show that the perturbed charge density, at $r \ll \lambda_D$ of each species, is inversely proportional to the temperature of the species. ////

The situation changes drastically for a moving test charge, since it takes time to polarize the plasma, i.e., to "dress" the test particle. If the test charge is moving rapidly enough, it will not stay in one place long enough to acquire its dressing. In general, only particles moving as rapidly as the test particle are able to keep up with it long enough to participate in the shielding process. This behavior (the dependence of the amount of shielding and makeup of the shield cloud on the speed of the test particle) can be shown directly from the Vlasov-Maxwell equations (11.1.2) and (11.1.4).

Assume that at $t = 0$ the test particle is at $\mathbf{x}_0'$, moving with $\mathbf{v}'$, and is not dressed, that is, $f_{\alpha 1}(t = 0) = 0$. Equations (11.1.2) and (11.1.4) describe the polarization of the plasma by the test particle at later time, and the solution to those equations as $t \to \infty$ describes the actual state of such a discrete particle in a plasma. The calculation proceeds as in the calculation of plasma waves (Secs. 8.1 to 8.6). Equations (11.1.2), and (11.1.4) are solved by first taking their Laplace transform in t and Fourier transform in $\mathbf{x}$, defined by

$$\tilde{f}_{\alpha k}(\mathbf{v}, \mathbf{k}, p) = \int_0^\infty dt\int_{-\infty}^\infty f_{\alpha 1}(\mathbf{v}, \mathbf{x}, t)\exp(-i\mathbf{k}\cdot\mathbf{x})\exp(-pt)\frac{d\mathbf{x}}{(2\pi)^3}$$

$$\text{Re}\,(p) > p_0$$

The transformed equations are

$$k^2\tilde{\phi}_{\mathbf{k}} = \frac{4\pi q_T\exp(-i\mathbf{k}\cdot\mathbf{x}_0')}{(p + i\mathbf{k}\cdot\mathbf{v}')(2\pi)^3} + 4\pi\sum_\alpha \bar{n}_\alpha q_\alpha\int\tilde{f}_{\alpha\mathbf{k}}\,d\mathbf{v} \qquad (11.1.6)$$

$$(p + i\mathbf{k}\cdot\mathbf{v})\tilde{f}_{\alpha\mathbf{k}} = \frac{q_\alpha}{m_\alpha}\tilde{\phi}_{\mathbf{k}}\,i\mathbf{k}\cdot\nabla_{\mathbf{v}} f_{\alpha 0} \qquad (11.1.7)$$

with the immediate solution, by substituting $\tilde{f}_{\alpha\mathbf{k}}$ from (11.1.7) into (11.1.6),

$$\tilde{\phi}_{\mathbf{k}} = \frac{4\pi q_T\exp(-i\mathbf{k}\cdot\mathbf{x}_0')}{(2\pi)^3 k^2(p + i\mathbf{k}\cdot\mathbf{v}')\left(1 - \sum_\alpha \frac{\omega_{p\alpha}{}^2}{k^2}\int\frac{\mathbf{k}\cdot\nabla_{\mathbf{v}} f_{\alpha 0}}{i\mathbf{k}\cdot\mathbf{v} - ip}\,d\mathbf{v}\right)}$$

$$= \frac{4\pi q_T\exp(-i\mathbf{k}\cdot\mathbf{x}_0')}{(2\pi)^3 k^2(p + i\mathbf{k}\cdot\mathbf{v}')D(\mathbf{k}, ip)} \qquad (11.1.8)$$

The actual potential $\phi(\mathbf{x}, t)$ produced by a test particle at $\mathbf{x}'(t)$ in the plasma is given by the time-asymptotic limit of (11.1.8), which describes the dressing of the test particle. If $n\lambda_D{}^3$ is sufficiently large, there is ample time between Coulomb collisions for the test particles to acquire the shielding cloud described

by (11.1.8) with $t \to \infty$. The time-asymptotic solution $\phi_{\mathbf{k}}(t \to \infty)$ is obtained from (11.1.8) by inverting the Laplace transform

$$\begin{aligned}\phi_{\mathbf{k}}(\mathbf{k}, t) &= \frac{1}{2\pi i}\int_{p_0-i\infty}^{p_0+i\infty} \tilde{\phi}_{\mathbf{k}}(\mathbf{k}, p)e^{pt}\,dp \\ &= \frac{4\pi q_T \exp(-i\mathbf{k}\cdot\mathbf{x}_0')\exp(-i\mathbf{k}\cdot\mathbf{v}'t)}{(2\pi)^3 k^2 D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}')} \\ &\quad + \sum_j \frac{4\pi q_T \exp(-i\mathbf{k}\cdot\mathbf{x}_0')\exp(p_j t)}{(2\pi)^3 k^2 (p_j + i\mathbf{k}\cdot\mathbf{v}')(dD/dp)|_{p=p_j}} \qquad (11.1.9)\end{aligned}$$

where the p_j are solutions of

$$D(\mathbf{k}, ip_j) = 0$$

These are the plasma waves that were examined in detail in Chap. 8. A moving test charge sees the plasma as a dielectric medium with dielectric constant $D(\mathbf{k}, ip)$ and couples to the normal modes p_j of the dielectric.

Asymptotically in time, the term $e^{p_j t}$ vanishes, because plasma waves are Landau-damped; i.e., the zeros p_j have a negative real part so that

$$e^{p_j t} = \exp[i\,\mathrm{Im}\,(p_j)t]\exp[\mathrm{Re}(p_j)t] \xrightarrow{t\to\infty} 0$$

Using this fact, the potential due to an individual particle moving with speed $\mathbf{v}'$ in a plasma is

$$\phi(\mathbf{x}, t) = \frac{4\pi q_T}{(2\pi)^3}\int \frac{\exp(i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}'))}{k^2 D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}')}\,d\mathbf{k} \qquad (11.1.10)$$

where $\mathbf{x}'(t) = \mathbf{x}_0' + \mathbf{v}'t$ is the location of the test particle at time t. The plasma distribution in the presence of a test particle is given by

$$\begin{aligned}f_\alpha = f_{\alpha 0} + \frac{4\pi q_T}{(2\pi)^3}\frac{q_\alpha}{m_\alpha}\lim_{\varepsilon\to 0^+}\int \frac{\mathbf{k}\cdot\nabla_{\mathbf{v}} f_{\alpha 0}}{k^2 D(\mathbf{k},\mathbf{k}\cdot\mathbf{v}')(\mathbf{k}\cdot\mathbf{v} - \mathbf{k}\cdot\mathbf{v}' - i\varepsilon)} \\ \times \exp(i\mathbf{k}\cdot\mathbf{x})\exp(-i\mathbf{k}\cdot\mathbf{x}')\,d\mathbf{k} \qquad (11.1.11)\end{aligned}$$

with $$D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}') = 1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2}\lim_{\varepsilon\to 0^+}\left[\int \frac{2\mathbf{k}\cdot\mathbf{v}(\partial f_{\alpha 0}/\partial v^2)}{\mathbf{k}\cdot\mathbf{v} - \mathbf{k}\cdot\mathbf{v}' - i\varepsilon}\,d\mathbf{v}\right]$$

The limiting procedure $\varepsilon \to 0^+$ defines the Landau contours (under the pole) discussed extensively in Sec. 8.4.

In the test-particle picture, the plasma is made up of a collection of discrete uncorrelated dressed test particles, distributed in $\mathbf{x}$-$\mathbf{v}$ phase space according to $f_{\alpha 0}$, each having a potential given by (11.1.10). The test particles interact with the rest of the plasma, which is treated as a Vlasov fluid, but not with other discrete particles. Each plasma particle leads a double life. On the one hand, it is a test particle, moving about in the Vlasov fluid. On the other hand, it is a part of the Vlasov fluid, and as such participates in the shielding of other test particles. The test-particle picture is a physical way of calculating, to order $1/n\lambda_D{}^3$, fluctuations in density and fields, and other plasma properties that depend on particle discreteness.

It is interesting to compare the electrostatic potential due to test particles of various speeds.

11.1.1 Shielding of a Slowly Moving Test Charge ($v' \ll \sqrt{\kappa T_i/m_i}$)

If $v' \ll \sqrt{\kappa T_i/m_i}$, then, to good approximation,

$$D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}') = 1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2}\int \frac{1}{u}\frac{\partial F_{\alpha 0}}{\partial u}\,du$$

where

$$F_{\alpha 0} = \int \delta\left(u - \frac{\mathbf{k}\cdot\mathbf{v}}{|k|}\right) f_{\alpha 0}\,d\mathbf{v}$$

Inversion of (11.1.8) gives

$$\begin{aligned}\phi(\mathbf{x}, t) &= \frac{q_T}{|\mathbf{x}-\mathbf{x}'|}\exp\left[-|\mathbf{x}-\mathbf{x}'|\left(-\sum_\alpha \omega_{p\alpha}^2\int\frac{1}{u}\frac{\partial F_{\alpha 0}}{\partial u}\,du\right)^{1/2}\right] \\ &= \frac{q_T}{|\mathbf{x}-\mathbf{x}_0'-\mathbf{v}'t|}\exp\left[-|\mathbf{x}-\mathbf{x}_0'-\mathbf{v}'t|\left(-\sum_\alpha \omega_{p\alpha}^2\int\frac{1}{u}\frac{\partial F_{\alpha 0}}{\partial u}\,du\right)^{1/2}\right] \qquad (11.1.12)\end{aligned}$$

For the case

$$F_{\alpha 0} = \left(\frac{m}{2\pi\kappa T}\right)_\alpha^{1/2} e^{-m_\alpha u^2/2\kappa T_\alpha}$$

the quantity

$$\left(-\sum_\alpha \omega_{p\alpha}^2\int\frac{1}{u}\frac{\partial F_{\alpha 0}}{\partial u}\,du\right)^{-1/2}$$

is the Debye length

$$\lambda_D = \left(\sum_\alpha \frac{\omega_{p\alpha}^2 m_\alpha}{\kappa T_\alpha}\right)^{-1/2}$$

For a general stable distribution $F_{\alpha 0}(u)$, the quantity

$$\left(-\sum_\alpha \omega_{p\alpha}^2\int\frac{1}{u}\frac{\partial F_{\alpha 0}}{\partial u}\,du\right)^{-1/2}$$

defines a generalized shielding distance. The perturbed charge densities of electrons and ions are given, using 11.1.10 and 11.1.11, by

$$\bar{n}_\alpha q_\alpha \int f_{\alpha 1}\,d\mathbf{v} = \left(\frac{1}{4\pi}\omega_{p\alpha}^2\int\frac{1}{u}\frac{\partial F_{\alpha 0}}{\partial u}\,du\right)\phi$$

so that electrons and ions at the same temperature participate in equal numbers in the shielding of a particle at rest.

11.1.2 Shielding of a Fast Test Charge ($v' > \sqrt{\kappa T_e/m_e}$)

In this case, the plasma dielectric D is, to good approximation,

$$D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}') = 1 - \sum \frac{{\omega_{p\alpha}}^2}{k^2} \int \left(k \frac{\partial F_{\alpha 0}}{\partial u}\right)\left[\frac{1}{\mathbf{k}\cdot\mathbf{v}'} + \frac{ku}{(\mathbf{k}\cdot\mathbf{v}')^2} + \cdots\right] du \approx 1$$

and from (11.1.10), the potential is, with the test particle at $\mathbf{x}'$,

$$\phi(\mathbf{x}, t) = \frac{q_T}{|\mathbf{x} - \mathbf{x}'|} = \frac{q_T}{|\mathbf{x} - \mathbf{x}_0' - \mathbf{v}'t|} \tag{11.1.13}$$

There is no shielding at all if the test charge moves much more rapidly than the thermal plasma particles.

11.1.3 Shielding of a Test Charge of Intermediate Speed

$$\sqrt{\frac{\kappa T_i}{m_i}} \ll v' \ll \sqrt{\frac{\kappa T_e}{m_e}}$$

In this case the dielectric is approximately given by

$$D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}') = 1 - \frac{{\omega_{pe}}^2}{k^2} \int \frac{1}{u} \frac{\partial F_{e0}}{\partial u}\, du$$

and the potential is

$$\phi = \frac{q_T}{|\mathbf{x} - \mathbf{x}_0' - \mathbf{v}'t|} \exp\left[-|\mathbf{x} - \mathbf{x}_0' - \mathbf{v}'t| \left({\omega_{pe}}^2 \int -\frac{1}{u}\frac{\partial F_{e0}}{\partial u}\, du\right)^{1/2}\right] \tag{11.1.14}$$

The perturbed charge density of electrons is given by

$$-e\bar{n}_e \int f_{e1}\, d\mathbf{v} = \frac{1}{4\pi}\left({\omega_{pe}}^2 \int \frac{1}{u}\frac{\partial F_{e0}}{\partial u}\, du\right)\phi$$

while the perturbed charge density of ions is small:

$$\frac{\rho_{i1}}{\rho_{e1}} \approx 0\, \frac{\kappa T_i}{m_i (v')^2}$$

Thus a test charge that moves faster than ions but slower than electrons is shielded only by electrons (an excess of electrons if $q_T > 0$, a deficiency of electrons if $q_T < 0$). By comparing (11.1.12) and (11.1.14), it is clear that the shielding distance of this electron-shielded charge is longer (by $\sqrt{2}$) than the shielding distance of a charge at rest.

A number of other properties of a thermal plasma will now be deduced from the test-particle picture.

Problem 11.1.2 Calculate the first velocity corrections to the shielded potentials (11.1.12) to (11.1.14). These are called Cerenkov wakes, arising from test particles moving faster than the speed of plasma waves. ////

11.2 ELECTRIC FIELD FLUCTUATIONS IN A PLASMA

Even in a stable plasma there is a finite level of electric fields and plasma waves. The plasma waves are "normal modes" of the system, and represent degrees of freedom which are excited in thermal equilibrium. Plasma waves are emitted by particles as they move about in the plasma, and they are absorbed (e.g., Landau-damped) by the plasma. The balance between emission and absorption leads to a thermal level of field fluctuations.

The equilibrium level, and spectrum, of electric fields can be calculated in a straightforward way from the electric field of a moving test particle (Sec. 11.1). From $\mathbf{E} = -\nabla\phi$, the electric field in a plasma due to a test particle at $\mathbf{x}'$ moving with $\mathbf{v}'$ is

$$\mathbf{E}(\mathbf{x}, \mathbf{x}', \mathbf{v}') = -4\pi q_T \int \exp\{i\mathbf{k}\cdot[\mathbf{x} - \mathbf{x}'(t)]\} \frac{i\mathbf{k}}{\mathbf{k}^2 D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}')} \frac{d\mathbf{k}}{(2\pi)^3} \tag{11.2.1}$$

with

$$D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}') = 1 - \sum_\alpha \frac{{\omega_{p\alpha}}^2}{k^2} \lim_{\varepsilon\to 0} \int \frac{\mathbf{k}\cdot\nabla_{\mathbf{v}} f_{\alpha 0}}{\mathbf{k}\cdot(\mathbf{v} - \mathbf{v}') - i\varepsilon}\, d\mathbf{v}$$

This is the electrostatic field of a fully dressed test particle, and includes the complete dielectric properties of the plasma.

The average field at a point $\mathbf{x}$ in a plasma is calculated by simply adding up the fields of all the test particles. The calculation of averages is done formally by considering the plasma as an ensemble of uncorrelated test particles, uniformly distributed in space, and distributed in velocity according to $f_{\alpha 0}$. If $P(\mathbf{x}, t; \mathbf{x}', \mathbf{v}')$ is some plasma property at the point $\mathbf{x}$ at time t due to the presence of a test particle at $\mathbf{x}'$ moving with $\mathbf{v}'$, the average of P over an ensemble of dressed, uncorrelated test particles is defined by

$$\langle P(\mathbf{x}, t; \mathbf{x}', \mathbf{v}')\rangle = \sum_\alpha N_\alpha \int \frac{d\mathbf{x}'}{V} \int P(\mathbf{x}, t; \mathbf{x}', \mathbf{v}') f_{\alpha 0}(\mathbf{v}')\, d\mathbf{v}' \tag{11.2.2}$$

where N_α is the number of test particles of type α.

Assuming that the test particles of each type α are uniformly distributed in space and distributed in velocity according to $f_{\alpha 0}$, it is easy to see that the *average* field at a point $\mathbf{x}$ due to a test particle at $\mathbf{x}'$ is zero; this is reasonable because the test particle, which would produce a field at $\mathbf{x}$, could be located *anywhere*, with equal probability (see Prob. 11.2.1).

Problem 11.2.1 Show that

$$\langle \mathbf{E}(\mathbf{x}) \rangle_\alpha = \int d\mathbf{x}' \int \mathbf{E}(\mathbf{x}; \mathbf{x}', \mathbf{v}') f_{\alpha 0}(v')\, d\mathbf{v}'$$

the average field at a point $\mathbf{x}$ due to a test particle of type α, vanishes identically. ////

The average value of $\mathbf{E}^2$ is calculated from (11.2.1), using the definition of the average (11.2.2); i.e.,

$$\langle \mathbf{E}(\mathbf{x}) \cdot \mathbf{E}(\mathbf{x}) \rangle = \sum_\alpha (4\pi q_\alpha)^2 \int \frac{d\mathbf{x}'}{V} \int N_\alpha f_{\alpha 0}(\mathbf{v}')\, d\mathbf{v}' \int i\mathbf{k}_1 \frac{\exp[i\mathbf{k}_1 \cdot (\mathbf{x} - \mathbf{x}')]}{k_1^2 D(\mathbf{k}_1, \mathbf{k}_1 \cdot \mathbf{v}')} \frac{d\mathbf{k}_1}{(2\pi)^3}$$
$$\cdot \int -i\mathbf{k}_2 \frac{\exp[-i\mathbf{k}_2 \cdot (\mathbf{x} - \mathbf{x}')]}{k_2^2 D^*(\mathbf{k}_2, \mathbf{k}_2 \cdot \mathbf{v}')} \frac{d\mathbf{k}_2}{(2\pi)^3} \qquad (11.2.3)$$

where D^* is the complex conjugate of D. The integral $\int d\mathbf{x}'$ gives a δ function

$$\int \exp[i(\mathbf{k}_2 - \mathbf{k}_1) \cdot \mathbf{x}'] \frac{d\mathbf{x}'}{(2\pi)^3} = \delta(\mathbf{k}_2 - \mathbf{k}_1)$$

which allows $\int d\mathbf{k}_2$ to be carried out, leading to

$$\frac{1}{8\pi} \langle \mathbf{E}(\mathbf{x}) \cdot \mathbf{E}(\mathbf{x}) \rangle = \sum_\alpha \frac{(4\pi q_\alpha)^2 \bar{n}_\alpha}{8\pi} \int \frac{f_{\alpha 0}(\mathbf{v}')}{k^2 |\, D(\mathbf{k}, \mathbf{k} \cdot \mathbf{v}')|^2} \frac{d\mathbf{k}}{(2\pi)^3} d\mathbf{v}' \qquad (11.2.4)$$

For every value $\mathbf{k}$, the dielectric is only a function of the component of $\mathbf{v}'$ along $\mathbf{k}$, since $D = D(\mathbf{k}, \mathbf{k} \cdot \mathbf{v}')$. The distribution $f_{\alpha 0}$ can be integrated over the other two components of $\mathbf{v}'$, defining the one-dimensional distribution

$$F_{\alpha 0}(u) = \int \delta\left(u - \frac{\mathbf{k} \cdot \mathbf{v}'}{|k|}\right) f_{\alpha 0}(\mathbf{v}')\, d\mathbf{v}'$$

Further, defining $\omega = ku$,

Eq. (11.2.4) takes the form

$$\frac{\langle \mathbf{E}(\mathbf{x}, t) \cdot \mathbf{E}(\mathbf{x}, t) \rangle}{8\pi} = \int \frac{d\mathbf{k}}{(2\pi)^3} \int_{-\infty}^{\infty} \sum_\alpha \frac{4\pi^2 \bar{n}_\alpha q_\alpha^2 F_{\alpha 0}(\omega/k)}{k^3 |\, D(k, \omega)|^2} \frac{d\omega}{2\pi} \qquad (11.2.5)$$

In this form it is clear that the quantity

$$W_{k,\omega} = \sum_\alpha \frac{4\pi^2 \bar{n}_\alpha q_\alpha^2 F_{\alpha 0}(\omega/k)}{k^3 |\, D(k, \omega)|^2} \qquad (11.2.6)$$

should be interpreted as the spectral density of the field fluctuations, so that

$$\frac{\langle \mathbf{E}^2(\mathbf{x}, t) \rangle}{8\pi} = \int \frac{d\mathbf{k}}{(2\pi)^3} \int W_{k,\omega} \frac{d\omega}{2\pi} \qquad (11.2.7)$$

Problem 11.2.2 Starting from

$$\phi(\mathbf{k}, t) = \frac{4\pi q_T}{k^2 D(\mathbf{k}, \mathbf{k} \cdot \mathbf{v}')} \exp(-i\mathbf{k} \cdot \mathbf{x}_0) \exp(-i\mathbf{k} \cdot \mathbf{v}' t)$$

[from (11.1.9)], find

$$\mathbf{E}(\mathbf{k}, \omega) \equiv \int_{-\infty}^{\infty} \mathbf{E}(\mathbf{k}, t) \exp(-i\omega t) \frac{dt}{2\pi}$$

and show in that way that

$$\frac{\langle \mathbf{E}^2(\mathbf{x}, t) \rangle}{8\pi} = \int \frac{d\mathbf{k}}{(2\pi)^3} \int W_{k,\omega} \frac{d\omega}{2\pi}$$

with $W_{k,\omega}$ given by (11.2.6). ////

From the known dielectric properties of the plasma it is easy to evaluate the level of field fluctuations in various ranges of wavelength. Even before performing the calculation, it should be expected that waves with $k^2 \lambda_{De}^2 \ll 1$ will have a large level of excitation, because they are normal modes of the plasma, i.e., plasma oscillations that are only weakly damped. On the other hand, waves with $k^2 \lambda_{De}^2 \gg 1$ should be only weakly excited. Defining

$$W_k = \int_{-\infty}^{\infty} W_{k,\omega} \frac{d\omega}{2\pi}$$

it is possible to carry out the ω integration for various ranges of $\mathbf{k}$ by noting that, for real ω,

$$|\, D(k, \omega)|^2 = D_r^2 + D_i^2 = \left(1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \oint \frac{\mathbf{k} \cdot \nabla_\mathbf{v} f_{\alpha 0}}{\mathbf{k} \cdot \mathbf{v} - \omega} d\mathbf{v}\right)^2$$
$$+ \left[\sum_\alpha \pi \frac{\omega_{p\alpha}^2}{k^2} \frac{dF_{\alpha 0}(\omega/k)}{d(\omega/k)}\right]^2 \qquad (11.2.8)$$

and $W_{k,\omega}$ is

$$W_{k,\omega} = \sum_\alpha \frac{4\pi^2 n_\alpha {q_\alpha}^2 F_{\alpha 0}(\omega/k)}{k^3 \left\{ \left(1 - \sum_\alpha \frac{{\omega_{p\alpha}}^2}{k^2} \oint \frac{\mathbf{k}\cdot\nabla_v f_{\alpha 0}\, d\mathbf{v}}{\mathbf{k}\cdot\mathbf{v} - \omega}\right)^2 + \left[\sum_\alpha \pi \frac{{\omega_{p\alpha}}^2}{k^2} \frac{dF_{\alpha 0}(\omega/k)}{d(\omega/k)}\right]^2 \right\}} \tag{11.2.9}$$

The spectral density is especially simple in the two limits $k^2{\lambda_D}^2 \gg 1$ and $k^2{\lambda_D}^2 \ll 1$.

11.2.1 Electric Fluctuations of Wavelength $k\lambda_D \ll 1$, $T_e = T_i$

When $k\lambda_D$ is small, (11.2.9) is much simplified, because in the high-frequency part of the spectrum ($\omega \gtrsim \omega_{pe}$)

$$W_{k,\omega} \approx \frac{\sum 4\pi^2 n_\alpha {q_\alpha}^2 F_{\alpha 0}(\omega/k)}{k^3 \left\{ \left(1 - \frac{{\omega_{pe}}^2}{\omega^2}\right)^2 + \left(\pi \frac{{\omega_{pe}}^2}{k^2} \frac{dF_{e0}(\omega/k)}{d(\omega/k)}\right)^2_{\omega=\omega_{pe}} \right\}} \qquad \omega \approx \omega_{pe},\ k\lambda_D \ll 1$$

The reason the frequency range $\omega \approx \omega_{pe}$ is singled out is that, since

$$\frac{{\omega_{pe}}^2}{k^2} \left| \frac{dF_{e0}(\omega/k)}{d(\omega/k)} \right| \ll 1$$

the denominator of (11.2.9) is minimum at $\omega = \omega_{pe}$ and the main contribution to $\int d\omega$ comes from that frequency range. Physically, the largest density of electric field fluctuations naturally occurs at the frequency of the normal modes of the plasma, i.e., the plasma waves.

The integral over $W_{k,\omega}$ from frequencies near $\omega = \pm\omega_{pe}$ is found, for example, by contour integration, giving

$$\int_{\pm\omega_{pe}-\Delta}^{\pm\omega_{pe}+\Delta} W_{k,\omega} \frac{d\omega}{2\pi} = \frac{m_e \omega_{pe}}{2k} \frac{F_{e0}}{|dF_{e0}/d(\omega/k)|} \qquad k\lambda_D \ll 1 \tag{11.2.10}$$

so that

$$W_k = \frac{\omega_{pe} m_e}{2k} \frac{F_{e0}}{|dF_{e0}/dv|}\bigg|_{v=\omega/k} \qquad k < \frac{1}{\lambda_D} \tag{11.2.11}$$

If the distribution F_{e0} is maxwellian, $F_{e0} = (m_e/2\pi\kappa T_e)^{1/2} e^{-m_e u^2/2\kappa T_e}$, (11.2.10) gives

$$W_k = \frac{\kappa T_e}{2} \qquad k\lambda_D \ll 1;\ \text{maxwellian plasma} \tag{11.2.12}$$

This result shows that in a maxwellian plasma, the nearly undamped plasma waves are fully excited, with energy

$$W_k = \frac{\kappa T_e}{2} \qquad \text{per degree of freedom}$$

11.2.2 Electric Fluctuations of Wavelength $k\lambda_D \gg 1$, $T_e = T_i$

In contrast to the full excitation of plasma waves in the undamped range ($k\lambda_D \ll 1$), there is little excitation in the range of wavelength $k\lambda_D > 1$. In that case,

$$|D(k,\omega)|^2 \approx 1$$

and

$$W_k = \int \sum_\alpha \frac{4\pi^2 n_\alpha {q_\alpha}^2 F_{\alpha 0}(\omega/k)}{k^3} \frac{d\omega}{2\pi} = \sum_\alpha \frac{2\pi n_\alpha {q_\alpha}^2}{k^2}$$
$$= \frac{\kappa T_e}{2k^2{\lambda_D}^2} \qquad k\lambda_D \gg 1 \tag{11.2.13}$$

The level of electric field fluctuations is then

$$\left\langle \frac{E^2}{8\pi} \right\rangle = \int W_k \frac{d\mathbf{k}}{(2\pi)^3}$$

with W_k given by (11.2.12) and (11.2.13).

Thus short-wavelength electric field fluctuations are only weakly excited in an equilibrium plasma state; i.e., the level of fields with $k\lambda_D \gg 1$ is less than κT per degree of freedom.

11.2.3 Electric Fluctuations in a Maxwellian Plasma, $T_e = T_i$

For a maxwellian distribution of test particles, an exact evaluation of $\int W_{k,\omega}(d\omega/2\pi)$ is possible, with the result (again for $T_e = T_i$)

$$W_k = \int W_{k,\omega} \frac{d\omega}{2\pi} = \frac{1}{2} \frac{\kappa T_e}{1 + (k\lambda_D)^2}$$
$$\left\langle \frac{E^2}{8\pi} \right\rangle = \int \frac{\kappa T_e}{1 + (k\lambda_D)^2} \frac{d\mathbf{k}}{(2\pi)^3} \tag{11.2.14}$$

This shows again that the strongly damped waves $k\lambda_D > 1$ are only weakly excited, while the weakly damped modes $k\lambda_D < 1$ are fully excited.

It should be noted that the total level of field fluctuations, found by integrating (11.2.12) over $\mathbf{k}$, is small (of order $g = 1/n{\lambda_D}^3$) compared with the particle energy density,

$$\left\langle \frac{E^2}{8\pi} \right\rangle \approx \frac{1}{n{\lambda_D}^3} n\kappa T_e$$

The uncorrelated dressed-test-particle description, correct to order g, thus yields information that is not accessible within the framework of Vlasov theory.

11.3 ELECTRIC FIELD FLUCTUATIONS IN A NONMAXWELLIAN PLASMA

It has been noted throughout this text that for times less than a collision time, a stationary plasma state (even a stable state) may be far from thermal equilibrium. The test-particle formalism allows a calculation of field fluctuations for a variety of distributions.

The effect of a nonmaxwellian distribution is seen directly from (11.2.10): if $\partial F/\partial(\omega/k)$ is reduced, i.e., if the distribution is flatter than a maxwellian distribution in some range $\Delta(\omega/k)$, the fluctuation spectrum is enhanced in that range. Indeed, if the distribution is exactly flat, (11.2.10) diverges. This is the case of marginal stability in which the linear treatment of the test-particle motion (the assumption that the test particle moves with constant velocity) breaks down.[1]

11.3.1 Electric Field Fluctuations When $T_e \gg T_i$

Mild departures from a maxwellian distribution can be handled within the framework of Secs. 11.1 and 11.2. An instructive example is that of a plasma in which $T_e \gg T_i$. The results of Chap. 8 show that in addition to plasma oscillations, ion-acoustic waves are weakly damped normal modes of such a plasma. Therefore, in a plasma of hot electrons, there should be a contribution to $\langle E^2/8\pi \rangle$ from a thermal level of (weakly damped) ion waves excited by collisions.

The contribution to $\langle E^2/8\pi \rangle$ from ion waves is found from (11.2.9), with the approximation $\omega \lesssim \omega_{pi}$, $\sqrt{\kappa T_i/m_i} < \omega/k < \sqrt{\kappa T_e/m_e}$, with the result (assuming, to be explicit, that electrons and ions are maxwellian at different temperatures)

$$W_{k,\omega} = \frac{\sum 4\pi^2 \bar{n}_\alpha q_\alpha{}^2 F_{\alpha 0}(\omega/k)}{k^3 |D|^2}$$

$$\approx \frac{4\pi^2 \bar{n}_i e^2 [F_{i0}(\omega/k) + F_{e0}(\omega/k)]}{k^3 \left\{ \left(1 - \frac{\omega_{pi}^2}{\omega^2} + \frac{1}{k^2 \lambda_{De}^2}\right)^2 + \left[\frac{\omega}{\omega_{pe}} \frac{1}{k^3 \lambda_{De}^3} \sqrt{\frac{\pi}{2}} + \frac{\omega}{\omega_{pi}} \frac{1}{k^3 \lambda_{Di}^3} \sqrt{\frac{\pi}{2}} \exp\left(\frac{-m_i \omega^2}{2\kappa T_i k^2}\right)\right]^2 \right\}} \qquad \omega \lesssim \omega_{pi} \tag{11.3.1}$$

[1] E. A. Frieman and P. Rutherford, *Ann. Phys.*, **28**:134 (1964).
A. Rogister and C. Oberman, *J. Plasma Phys.*, **2**:33 (1968); **3**:119 (1969).

In this frequency range, $W_{k,\omega}$ is sharply peaked at $\omega = \pm\omega_{pi}/\sqrt{1 + 1/k^2\lambda_{De}{}^2}$, and integrating over these resonances, $\int_{\pm\omega_{pi}-\Delta}^{\pm\omega_{pi}+\Delta}$, gives

$$W_k(\text{from } \omega \lesssim \omega_{pi}) = \frac{\kappa T_i}{2} \frac{k^2\lambda_{De}{}^2}{1 + k^2\lambda_{De}{}^2} \frac{\exp \dfrac{-T_e}{2T_i(1 + k^2\lambda_{De}{}^2)} + \left(\dfrac{m_e}{m_i}\dfrac{T_i}{T_e}\right)^{1/2}}{\exp \dfrac{-T_e}{2T_i(1 + k^2\lambda_{De}{}^2)} + \left(\dfrac{T_i}{T_e}\right)^{3/2}\left(\dfrac{m_e}{m_i}\right)^{1/2}} \tag{11.3.2}$$

As $k\lambda_{De}$ increases, the excitation level of ion waves at a particular value of k increases to a maximum excitation level per degree of freedom of

$$W_k(\text{from } \omega \lesssim \omega_{pi}) \lesssim \frac{\kappa T_i}{2} \quad ; \quad T_e > T_i \quad ; \quad \exp \frac{-T_e}{2T_i(1 + k^2\lambda_D{}^2)} > \left(\frac{m_e}{m_i}\frac{T_i}{T_e}\right)^{1/2}$$

If $T_e \gg T_i$, the $\sqrt{m_e/m_i}$ terms dominate the $\exp(-T_e/T_i)$ terms and

$$W_k \text{ (from } \omega \lesssim \omega_{pi}) \lesssim \tfrac{1}{4}\kappa T_e \qquad T_e \gg T_i$$

The total fluctuation energy density is given by the sum of the contribution from ion and electron plasma waves.

$$\left\langle \frac{E^2}{8\pi} \right\rangle = \int \left[\tfrac{1}{2}\kappa T_e \frac{1}{1 + k^2\lambda_{De}{}^2} + \tfrac{1}{2}\kappa T_i \frac{k^2\lambda_{De}{}^2}{1 + k^2\lambda_{De}{}^2} \right.$$

$$\left. \frac{\exp \dfrac{-T_e}{2T_i(1 + k^2\lambda_{De}{}^2)} + \left(\dfrac{m_e}{m_i}\dfrac{T_i}{T_e}\right)^{1/2}}{\exp \dfrac{-T_e}{2T_i(1 + k^2\lambda_{De}{}^2)} + \left(\dfrac{m_e}{m_i}\right)^{1/2}\left(\dfrac{T_i}{T_e}\right)^{3/2}} \right] \frac{d\mathbf{k}}{(2\pi)^3} \tag{11.3.3}$$

Other nonthermal distributions change the excitation level of the modes in a manner that can be deduced from (11.2.8), provided only that the plasma is stable and that the fluctuation energy and spectrum are sufficiently small so that higher corrections to the kinetic theory can be neglected.

The dependence of the excitation level on the relative size of $\sqrt{m_e/m_i}$ and T_e/T_i reveals some of the interesting details of the emission process. The terms in the numerator of (11.3.2) indicate the relative number of ions $\{\exp[-T_e/2T_i(1 + k^2\lambda_D{}^2)]\}$ and electrons $(\sqrt{m_e/m_i}\sqrt{T_i/T_e})$ which excite ion waves by coupling with the Vlasov fluid. When the number of ions in resonance with

ion waves exceeds the number of electrons, the excitation level $\kappa T_i/2$ is the exact analog of the Langmuir oscillation excitation, with ions replacing electrons. When the number of ions resonant with the wave is small, the interaction of elcctrons with the fluid determines the excitation level; the balance between emission and absorption in determining the level of waves is examined in the next section.

11.4 DRAG ON A TEST PARTICLE; EMISSION OF ELECTROSTATIC WAVES

A dressed test particle includes a polarization cloud that follows it as it moves through the plasma. The polarization cloud exerts a drag on the test charge. As a result of this drag, the test particle emits plasma waves (and electromagnetic radiation as well). The drag force on the test particle is obtained by subtracting the field of the bare test particle from the total field (11.2.1); i.e., the force on a test charge of type α is

$$\mathbf{F} = q_\alpha \mathbf{E}_{\text{polarization}} = -4\pi {q_\alpha}^2 \int \frac{i\mathbf{k}}{k^2}\left[\frac{1}{D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}')} - 1\right]\frac{d\mathbf{k}}{(2\pi)^3} \qquad (11.4.1)$$

where $\mathbf{E}(\mathbf{x}; \mathbf{x}', \mathbf{v}')$ is evaluated at the position of the test charge, $\mathbf{x} = \mathbf{x}'(t)$.

Problem 11.4.1 Draw the equipotentials around a test charge that is moving with speed
(*a*) $\mathbf{v}' = \hat{\mathbf{x}}\sqrt{\kappa T_e/m_e}$
(*b*) $\mathbf{v}' = 4\hat{\mathbf{x}}\sqrt{\kappa T_e/m_e}$
(*c*) $\mathbf{v}' = 10\hat{\mathbf{x}}\sqrt{\kappa T_e/m_e}$
This requires some numerical work. Show that the force (11.4.1) on a test particle at rest, due to its polarization cloud, vanishes. ////

The rate at which the test particle loses energy, which equals the emission by a single test charge, is

$$\mathbf{v}' \cdot q\mathbf{E}_{\text{pol}}(\mathbf{v}') = \frac{d}{dt} W_{\text{test particle}}$$

The total emission of plasma waves per unit volume by test particles is

$$\left\langle \frac{dW_T}{dt} \right\rangle = \sum_\alpha 4\pi {q_\alpha}^2 \bar{n}_\alpha \int \frac{d\mathbf{k}}{(2\pi)^3} \int f_{\alpha 0}(\mathbf{v}) \frac{i\mathbf{k}\cdot\mathbf{v}}{k^2}\left[\frac{1}{D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v})} - 1\right] d\mathbf{v} \qquad (11.4.2)$$

Integrating over all velocities except those parallel to k, the integral $\sum_\alpha \int i\mathbf{k}\cdot\mathbf{v} f_{\alpha 0}(\mathbf{v}) d\mathbf{v}$ vanishes because the plasma equilibrium is assumed to have no current (field-free plasma). Similarly, writing $D = D_r + iD_i$, and noting that D_r is even in $\mathbf{k}$ while D_i is odd in $\mathbf{k}$, the rate at which a test particle of type α emits electrostatic waves is

$$P = \left\langle \frac{dW}{dt} \right\rangle = \sum_\alpha 4\pi {q_\alpha}^2 \bar{n}_\alpha \int \frac{d\mathbf{k}}{(2\pi)^3} \int f_{\alpha 0} \frac{\mathbf{k}\cdot\mathbf{v}}{k^2} \frac{D_i(\mathbf{k}, \mathbf{k}\cdot\mathbf{v})}{|D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v})|^2} d\mathbf{v}$$

In terms of the one-dimensional properties of $f_{\alpha 0}(\mathbf{v})$ along the $\mathbf{k}$ axis,

$$F_{\alpha 0} \equiv \int f_{\alpha 0}(\mathbf{v})\, \delta\left(u - \frac{\mathbf{k}\cdot\mathbf{v}}{|k|}\right) d\mathbf{v}$$

the dielectric is

$$D_r = 1 - \sum \frac{{\omega_p}_\alpha^2}{k^2} \oint \frac{\partial F_{\alpha 0}/\partial u'}{u' - u}\, du'$$

and

$$D_i = -\pi \frac{{\omega_p}_\alpha^2}{k^2} \frac{\partial F_{\alpha 0}}{\partial u'}\bigg|_{u=\omega/|k|}$$

Using the definition $\omega \equiv |k|u$ to write P in terms of an integral over a spectral density,

$$P = \left\langle \frac{dW}{dt} \right\rangle = \int \frac{d\mathbf{k}\, d\omega}{(2\pi)^4} \sum_\alpha 8\pi^2 \omega \bar{n}_\alpha {q_\alpha}^2 \frac{F_{\alpha 0}}{|k^3|} \frac{D_i(k, \omega)}{|D(k, \omega)|^2} = \int P_{k,\omega} \frac{d\mathbf{k}\, d\omega}{(2\pi)^4} \qquad (11.4.3)$$

The emission is large where $|D|^2$ is small. In particular, for $\omega \approx \omega_{pe}$ and ${k^2\lambda_{De}}^2 \ll 1$, P corresponds to the emission of plasma waves. In a stationary state, these test particles must be emitting waves at a rate that balances the loss of wave energy by Landau damping. The requirement that emission balance absorption provides a relation between W_E, P, and ω_i that must be satisfied in any stationary plasma state.

To demonstrate this relation between P from (11.4.3), W from (11.2.8), and ω_i from the linear wave theory, compare (11.4.3) with (11.2.8).

$$P_{k,\omega} = 2\omega D_i(k, \omega) W_{k,\omega} \qquad (11.4.4)$$

To see that (11.4.4) implies a balance between emission by test particles and Landau damping, note that, for plasma waves, ${k^2\lambda_D}^2 \ll 1$,

$$D = 1 - \frac{{\omega_{pe}}^2}{\omega^2} + iD_i$$

$$\omega = \omega_r + i\omega_i = \omega_{pe} - i\frac{\omega_{pe}}{2} D_i(k, \omega) \qquad (11.4.5)$$

Now the rate at which electrostatic energy $W_{k,\omega}$ is lost due to Landau damping is $W_{k,\omega} \approx E^2 \propto e^{2\omega_i t}$, for a particular mode (k, ω), so that from linear theory

$$\frac{d}{dt} W_{k,\omega}|_{\text{damping}} = 2\omega_i W_{k,\omega} = -\omega_{pe} D_i(k, \omega) W_{k,\omega} \qquad (11.4.6)$$

The rate at which wave energy (=electrostatic energy + mechanical energy) is being lost due to Landau damping is twice the loss of electrostatic energy; i.e.,

$$\frac{dW_{k,\omega\ \text{wave}}}{dt} = \frac{2dW_{k,\omega}}{dt} \qquad (11.4.7)$$

This is true only for Langmuir oscillations, because in that case the wave energy is composed of one-half electric and one-half mechanical energy; in general, of course, as in Sec. 4.3.1,

$$\frac{dW_{\text{wave}}}{dt} = 2\omega_i W_{k,\omega}\, \omega \frac{dD(k, \omega)}{d\omega} \qquad (11.4.8)$$

Comparing the emission (11.4.4) with the damping rate (11.4.7) and (11.4.6), it is clear that the power radiated by the test particles is exactly balanced by absorption of waves by Landau damping. Balance of emission and absorption in a stationary state is called *Kirchhoff's law*. To show that Kirchhoff's law is valid for all plasma waves, note that the damping decrement ω_i can be found by setting $D(k, \omega) = 0$ and expanding $D(k, \omega) \approx D(k, \omega_r) + i\omega_i(dD/d\omega) + iD_i + \cdots$ about $\omega = \omega_r$, where $D_r(k, \omega_r) = 0$. For weakly damped waves this procedure gives

$$\omega_i = -\frac{D_i(k, \omega_r)}{(d/d\omega_r) D_r(k, \omega_r)} \qquad (11.4.9)$$

Thus (11.4.4) can be written, in the frequency range of a plasma wave,

$$P_{k,\omega} = \frac{\text{power radiated}}{\text{unit frequency}} = -2\omega_i W_{k,\omega}\, \omega \frac{dD}{d\omega} \qquad (11.4.10)$$

Comparison of (11.4.10) with (11.4.8) demonstrates that Kirchhoff's law (consistency of the separate calculations of emission by test charges, absorption by Landau damping, and the equilibrium level of field fluctuations) is valid for all the plasma waves (i.e., the weakly damped normal modes of a plasma). Equation (11.4.4) is of course valid for *all* frequencies and wave numbers.

Kirchhoff's law is a powerful tool in calculating radiation from any plasma in which the absorption coefficient and the near steady level of the wave is known. For example, radiation from a maxwellian plasma can be calculated from the damping coefficients derived in the Vlasov model, by arguing that:

1. A maxwellian plasma in thermal equilibrium with its surroundings at temperatures T has a level of waves $I(k, \omega)$ determined from blackbody radiation.
2. In that case (11.4.10) determines the power radiated in terms of the damping coefficient ω_i and the equilibrium distribution of radiation.
3. This spontaneous emission depends only on the plasma temperature and particle distribution, and so the emission calculated through 1 and 2 for a plasma in equilibrium with its surrounding gives, to good approximation, the emission from *any* maxwellian plasma.

Problem 11.4.2 Read D. A. Tidman and A. Eviator, *Phys. Fluids*, **8**, 2059 (1965) and discuss the *diffusion* of test particles, in contrast with the *drag* on test particles calculated in this section. ////

11.5 ELECTROMAGNETIC FLUCTUATIONS AND RADIATION

The preceding sections considered only electrostatic fluctuations of a plasma equilibrium, i.e., only those components of the electric field spectrum which had $\mathbf{k}$ parallel to $\mathbf{E}_\mathbf{k}$. In addition to these fluctuations, since a test charge moving with speed $\mathbf{v}'$ is really a current with density

$$j_T = q_T \mathbf{v}'\, \delta(\mathbf{x} - \mathbf{x}_0' - \mathbf{v}'t)$$

there are also electromagnetic fluctuations in a plasma. These are calculated by the same methods developed in Secs. 11.3 and 11.4.

Maxwell's equations, for electric fields with $\mathbf{k} \perp \mathbf{E}_\mathbf{k}$, before and after Fourier transformation in space and Laplace transformation in time, are

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B}(\mathbf{x}, t) = \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} + \frac{4\pi}{c}\sum_\alpha \bar{n}_\alpha q_\alpha \int \mathbf{v} f_\alpha\, d\mathbf{v} + \frac{4\pi q_T}{c}\mathbf{v}'\, \delta(\mathbf{x} - \mathbf{x}_0' - \mathbf{v}'t)$$

$$c^2 \mathbf{k} \times \mathbf{k} \times \tilde{\mathbf{E}}_{\mathbf{k},\omega} - p^2 \tilde{\mathbf{E}}_{\mathbf{k},\omega} = 4\pi p \sum_\alpha \frac{\bar{n}_\alpha {q_\alpha}^2}{m} \int \frac{\mathbf{v}\tilde{\mathbf{E}} \cdot \nabla_\mathbf{v} f_{\alpha 0}}{p + i\mathbf{k} \cdot \mathbf{v}}\, d\mathbf{v}$$
$$+ \frac{4\pi q_T \mathbf{v}' \exp(-i\mathbf{k} \cdot \mathbf{x}_0')}{(2\pi)^3} \frac{-i\mathbf{k} \cdot \mathbf{v}}{p + i\mathbf{k} \cdot \mathbf{v}'} \qquad (11.5.1)$$

The Fourier and Laplace transforms are then inverted to give the electric field as a function of space and time; i.e.,

$$\mathbf{E}(\mathbf{x}, t) = \int \exp[i\mathbf{k}\cdot(\mathbf{x} - \mathbf{x}' - \mathbf{v}'t)]\,\frac{4\pi q_T(-i\mathbf{k}\cdot\mathbf{v}')\mathbf{v}'}{k^2c^2D_T(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}')}\,\frac{d\mathbf{k}}{(2\pi)^3} \qquad E_\mathbf{k} \perp \mathbf{k}$$

where
$$D_T(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}') = 1 + \frac{\omega_p{}^2}{k^2c^2} - \frac{(\mathbf{k}\cdot\mathbf{v}')^2}{k^2c^2} \tag{11.5.2}$$

and the energy in electromagnetic fluctuations is

$$\frac{\langle E^2(\mathbf{x}, t)\rangle}{8\pi} = \sum_\alpha 4\pi^2\bar{n}_\alpha q_\alpha{}^2 \int\frac{d\mathbf{k}}{(2\pi)^3}\int f_{\alpha 0}\,\frac{(\mathbf{k}\cdot\mathbf{v}')^2(\mathbf{k}\times\mathbf{v}')^2}{k^4c^4\,|D_T(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}')|^2k^2}\,\frac{d\mathbf{v}'}{(2\pi)} \tag{11.5.3}$$

But since the particles have velocities $v' \ll c$, (i.e., $f_{\alpha 0}$ must be set equal to zero for $v > c$, avoiding a spurious resonance in $1/D_T{}^2$) a good approximation to D_T is $D_T = 1 + \omega_p{}^2/k^2c^2$, whereupon

$$\begin{aligned}\frac{\langle E^2(\mathbf{x}, t)\rangle}{8\pi} &= \sum_\alpha 4\pi n_\alpha q_\alpha{}^2\left(\frac{\kappa T_\alpha}{m_\alpha}\right)^2\int\frac{1}{k^2c^4(1 + \omega_p{}^2/k^2c^2)^2}\,\frac{d\mathbf{k}}{(2\pi)^3}\\ &\approx \kappa T_e\int\frac{\omega_p{}^2}{k^2c^2}\,\frac{1}{(1 + \omega_p{}^2/k^2c^2)^2}\,\frac{\kappa T_e}{m_ec^2}\,\frac{d\mathbf{k}}{(2\pi)^3}\end{aligned} \tag{11.5.4}$$

where the ion terms have been neglected.

By comparison with (Sec. 11.3) it is clear that, although electrostatic plasma oscillations exist only in a narrow range of ω and $k(\omega \approx \omega_{pe}, k\lambda_D < 1)$, their excitation in that range exceeds the excitation of electromagnetic fluctuations by $c^2/(\kappa T_e/m_e)$.

The rate of emission of electromagnetic waves is given by

$$\langle \mathbf{v}'\cdot e\mathbf{E}_{\text{POL}}\rangle$$

where now $\mathbf{E}_{\text{POL}}$ includes the electromagnetic components of the shielding field. This result can be combined with the result for emission of electrostatic waves to obtain

$$P_{k,\omega} = 8\pi^2\bar{n}_\alpha q_\alpha{}^2\int F_{\alpha 0}\left(\frac{\omega}{k}\right)\left\{\frac{\omega D_i}{k^3|D|^2} + \frac{2\kappa T\omega}{mc^2}\,\frac{[D_{Ti}]}{k^3\,|D_T|^2}\right\}\frac{d\mathbf{k}\,d\omega}{(2\pi)^4}$$

The imaginary part of D_T vanishes to the order of validity of the Vlasov equation. If weak collisions, $O(g)$, are included, D_{Ti} does not vanish. The reason for the difference between electrostatic and electromagnetic emission is as follows. Electrostatic emission to lowest order is Cerenkov radiation. There is no Cerenkov radiation of electromagnetic waves in field-free plasma, since no

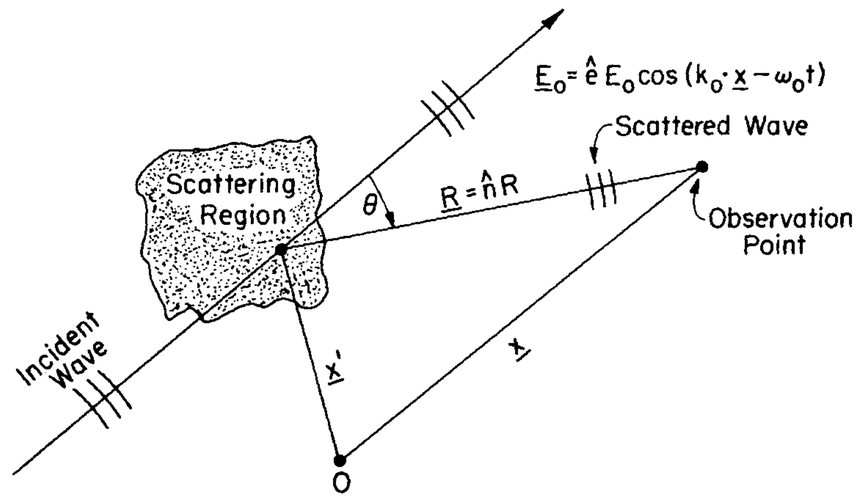


FIGURE 11.6.1
Schematic of a plane wave incident on a plasma, illustrating the scattering. A single underscore denotes a vector.

particles exceed the phase velocity ($\omega/k \geq c$) of those waves. Electromagnetic emission due to collisions is called bremmstrahlung, and appears in higher order than electrostatic Cerenkov emission.

11.6 SCATTERING OF INCOHERENT RADIATION FROM PLASMA DENSITY FLUCTUATIONS

Besides field fluctuations, a plasma also exhibits fluctuations in charge density. Electromagnetic waves, scattering from these fluctuations in charge density, provide a useful probe of the plasma microstate. This section treats the incoherent scattering of radiation from a plasma.

A calculation of the scattering of an unpolarized wave from a single electron is presented in most books on electromagnetic theory, and gives the result (the Thomson scattering cross section)

$$\frac{d\sigma}{d\Omega} = \frac{r_0{}^2}{2}(1 + \cos^2\theta) \tag{11.6.1}$$

where $r_0 = e^2/m_ec^2$ is the classical electron radius, and θ is the angle between the incident and scattered waves (Fig. 11.6.1). If the scattering electron is moving, the scattered radiation is doppler-shifted to a new frequency. A plausible *but*

often incorrect way to calculate the scattering of radiation by a plasma would be to treat the electrons as free, and simply add the radiation scattered from each electron; then (11.6.1) would give the cross section for scattering, while the scattered frequency would be determined by the velocity distribution. That is, if the electrons are thermally distributed, each electron will produce a different doppler shift, so that the scattering from an assembly of electrons will have a frequency spread. The scattered line for a maxwellian distribution of free electrons at temperature T is a gaussian distribution in ω with width

$$\Delta\omega_{\text{scattered}} = \omega_0 + \Delta\omega$$
$$\Delta\omega \approx k\bar{v}_e = k\sqrt{\frac{\kappa T_e}{m_e}} \tag{11.6.2}$$

In fact, however, the electrons in a plasma are not free. This is demonstrated in a striking way by experiments[1] on the scattering of electromagnetic waves from a *plasma* which show that long-wavelength radiation is scattered by only half the amount (11.6.1). Further, backscattering of radar from the ionosphere,[2] again involving long-wavelength radiation, shows that the width of the scattered line is much narrower than implied by (11.6.2), in particular,

$$\Delta\omega|_{\text{experimental}} \approx k\bar{v}_i = k\sqrt{\frac{\kappa T_i}{m_i}}$$

This would be the width of radiation scattered from *ions*. Since it is unthinkable that high-frequency radiation should scatter from massive and immobile ions rather than from the much lighter electrons, it is clear that collective effects must be involved in the scattering process, since collective behavior is what distinguishes a plasma from an assembly of independent electrons. To assume that the scattering is always the sum of single-free-particle scatterings is to oversimplify the problem.

Scattering of radiation is a discrete-particle effect. An equilibrium field-free plasma, in the Vlasov theory described in Chap. 8, does not scatter radiation, because to the order of the Vlasov theory, the charge density in the plasma is everywhere equal to its average value; i.e., there are no scattering centers.

Scattering appears in the next-order description (order $1/n\lambda_D{}^3$), the uncorrelated dressed-test-particle model. To this order, there *are* discrete particles in the plasma, and these particles will scatter radiation. Collective effects are automatically included when the test particles are dressed (shielded) by their

[1] H.-J. Kunze, The Laser as a Tool for Plasma Diagnostics, in W. Lochte-Holtgreven (ed.), "Plasma Diagnostics," p. 550, North-Holland Publishing Company, Amsterdam, 1968.

[2] K. L. Bowles, G. R. Ochs, and J. L. Green, *J. Res. Nat. Bureau Stand.*, **66D**: 395 (1962).

interaction with the Vlasov fluid. The procedure is to calculate the scattering from a single test charge imbedded in a plasma, and then sum the result over all the test charges.

Using the test-particle picture developed in the previous sections, it is a simple matter to calculate the electron density in the plasma in the vicinity of a test charge.

From (11.1.11), the electron density at $(\mathbf{x}, t)$ in a plasma due to a test electron at $\mathbf{x}'$ moving with velocity $\mathbf{v}'$ is

$$n_{e,e}(\mathbf{x}, t; \mathbf{x}', \mathbf{v}') = \bar{n}_e + \bar{n}_e \frac{4\pi e^2}{m_e} \lim_{\varepsilon\to 0^+} \int \frac{\exp[i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')]\mathbf{k}\cdot\nabla_{\mathbf{v}} f_{e0}}{k^2 D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}')(\mathbf{k}\cdot\mathbf{v} - \mathbf{k}\cdot\mathbf{v}' - i\varepsilon)} \times \frac{d\mathbf{k}}{(2\pi)^3}\, d\mathbf{v} + \int \exp[i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')]\frac{d\mathbf{k}}{(2\pi)^3} \tag{11.6.3}$$

In (11.6.3) the first term on the right-hand side is the uniform electron density present in a Vlasov fluid in the absence of a test charge; the second term is the change in density of the Vlasov fluid produced because the test electron repels the Vlasov electrons; the third term is the density of the test electron itself.[1] Similarly, the electron density at $(\mathbf{x}, t)$ due to a test ion at $\mathbf{x}'[=\mathbf{x}_0' + \mathbf{v}(t')]$ is

$$n_{e,i}(\mathbf{x}, t; \mathbf{x}', \mathbf{v}')$$
$$= \bar{n}_e - \bar{n}_e \frac{4\pi e^2}{m_e} \lim_{\varepsilon\to 0^+} \int d\mathbf{v} \int \frac{\exp[i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')]\mathbf{k}\cdot\nabla_{\mathbf{v}} f_{e0}}{k^2 D(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}')(\mathbf{k}\cdot\mathbf{v} - \mathbf{k}\cdot\mathbf{v}' - i\varepsilon)} \frac{d\mathbf{k}}{(2\pi)^3} \tag{11.6.4}$$

where the limit $\varepsilon \to 0^+$ arises from the requirement that the integral be along the Landau contour (Sec. 8.3). To calculate the incoherent scattering of radiation by plasma, the scattering is expressed in terms of the electron density [Eqs. (11.6.3) and (11.6.4)] and that result is averaged over the ensemble of test particles.

The scattering of a wave by a plasma is a function of the acceleration of particles by the incident wave whose electric field is $\mathbf{E}_i$. The scattered field (in the wave zone) from a single electron is

$$\mathbf{E}_{\text{radiated}}(\mathbf{x}, t) = -\frac{e}{c^2} \frac{\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \dot{\mathbf{v}})}{|\mathbf{x} - \mathbf{x}'|}\bigg|_{\text{retarded}} \tag{11.6.5}$$

where the acceleration $\dot{\mathbf{v}}$ is evaluated at the retarded time $t'(=t - |\mathbf{x} - \mathbf{x}'|/c)$; that is, the electric field at the observation point $\mathbf{x}$ at time t is determined by the

[1] To be exact, since one electron has been singled out as a test particle, the average density $\bar{n}_e$ should be replaced by $n_e^* = (N_e - 1)/V = \bar{n}_e - 1/V$, where V is the volume and N_e is the total number of electrons in the plasma. $N \gg 1$ has been used to approximate $n_e^* \approx \bar{n}_e$ in (11.6.3).

acceleration of the electron at an earlier time, and $\hat{\mathbf{n}}$ is a unit vector in the direction of propagation of the scattered wave. Using $m_e(d\mathbf{v}/dt) = -e\mathbf{E}_i$, and assuming that the electrons are distributed with density $n_e(\mathbf{x}', t')$, the radiated field is (far from the scattering center)

$$\mathbf{E}_r(\mathbf{x}, t) \approx \frac{e^2}{m_e c^2} \int n_e(\mathbf{x}', t') \frac{\hat{\mathbf{n}} \times [\hat{\mathbf{n}} \times \mathbf{E}_i(\mathbf{x}', t)]}{|\mathbf{x}|} d\mathbf{x}' \qquad (11.6.6)$$

where

$$t' = t - \frac{1}{c} |\mathbf{x} - \mathbf{x}'|$$

A useful equivalent of (11.6.6) is

$$\mathbf{E}_r(\mathbf{x}, t) = \frac{e^2}{m_e c^2} \int d\mathbf{x}' \int \delta\left(t' - t + \frac{1}{c} |\mathbf{x} - \mathbf{x}'|\right) n_e(\mathbf{x}', t') \frac{\hat{\mathbf{n}} \times [\hat{\mathbf{n}} \times \mathbf{E}_i(\mathbf{x}', t)]}{|\mathbf{x}|} dt' \qquad (11.6.7)$$

Consider a specific scattering event shown in Fig. 11.6.1 in which the incident wave is represented by an electric field.

$$\mathbf{E}_{\text{incident}} = E_0 \hat{\mathbf{e}} \cos(\mathbf{k}_0 \cdot \mathbf{x}' - \omega_0 t')$$

where $\hat{\mathbf{e}}$ is a unit vector in the direction of the incident electric field. Far from the scattering center, a good approximation is

$$\delta\left(t' - t + \frac{1}{c} |\mathbf{x} - \mathbf{x}'|\right) \approx \int \exp\left[i\left(t' - t + \frac{|\mathbf{x}|}{c} - \frac{\hat{\mathbf{n}} \cdot \mathbf{x}'}{c}\right)\right] \frac{d\omega}{2\pi}$$

Using this equation and expanding

$$\mathbf{E}_i = E_0 \hat{\mathbf{e}} \frac{\exp[i(\mathbf{k}_0 \cdot \mathbf{x}' - \omega_0 t)] + \exp[-i(\mathbf{k}_0 \cdot \mathbf{x}' - \omega_0 t)]}{2}$$

Eq. (11.6.7) becomes

$$\mathbf{E}_r(\mathbf{x}, t) = \frac{e^2}{m_e c^2} \frac{E_0}{|\mathbf{x}|} [\hat{\mathbf{e}} - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \hat{\mathbf{e}})] \int \frac{d\omega}{2\pi} \int n_e(\mathbf{x}', t')\, d\mathbf{x}'\, dt'$$
$$\cdot \left(\tfrac{1}{2} \exp\left\{i\left[\left(\mathbf{k}_0 - \frac{\omega\hat{\mathbf{n}}}{c}\right) \cdot \mathbf{x}' + (\omega - \omega_0)t' - \omega\left(t - \frac{|\mathbf{x}|}{c}\right)\right]\right\}\right.$$
$$\left. + \tfrac{1}{2} \exp\left\{-i\left[\left(\mathbf{k}_0 + \frac{\omega\hat{\mathbf{n}}}{c}\right) \cdot \mathbf{x} - (\omega + \omega_0)t' + \omega\left(t - \frac{|\mathbf{x}|}{c}\right)\right]\right\}\right) \qquad (11.6.8)$$

Using the fact that

$$\int \exp(-i\omega t)\, d\omega = \left[\int \exp(i\omega t)\, d\omega\right]^*$$

Eq. (11.6.8) becomes

$$\mathbf{E}_r(\mathbf{x}, t) = \frac{e^2}{m_e c^2} \frac{E_0[\hat{\mathbf{e}} - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \hat{\mathbf{e}})]}{|\mathbf{x}|} \int \frac{d\omega}{2\pi} (2\pi)^4$$
$$\cdot \left\{\tfrac{1}{2}\tilde{n}_e\left(\mathbf{k}_0 - \hat{\mathbf{n}}\frac{\omega}{c}, \omega - \omega_0\right) \exp\left[-i\left(t - \frac{|\mathbf{x}|}{c}\right)\right]\right.$$
$$\left. + \tfrac{1}{2}\tilde{n}_e^*\left(\mathbf{k}_0 - \hat{\mathbf{n}}\frac{\omega}{c}, \omega - \omega_0\right) \exp\left[i\omega\left(t - \frac{|\mathbf{x}|}{c}\right)\right]\right\} \qquad (11.6.9)$$

where

$$\tilde{n}_e(\mathbf{k}, \omega) = \int n_e(\mathbf{x}', t') \exp[-i\mathbf{k} \cdot \mathbf{x}'] \exp(i\omega t') \frac{d\mathbf{x}'\, dt'}{(2\pi)^4}$$

The energy radiated per unit time is given by the time-averaged mean-square value of $|\delta E_r|^2$. This time average is defined by the limit

$$\lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |\delta E_r^2|\, dt$$

and is obtained by noting that, since $\Delta\omega \equiv \omega - \omega_0$ and $\mathbf{K} = \mathbf{k}_0 - \hat{\mathbf{n}}(\omega/c)$,

$$\frac{1}{T} \int_{-T/2}^{T/2} dt \left\{ \int n_e(\mathbf{K}, \Delta\omega) \exp\left[i\omega\left(t - \frac{x}{c}\right)\right] \frac{d\omega}{2\pi} \int n_e(\mathbf{K}, \Delta\omega') \exp\left[i\omega'\left(t - \frac{x}{c}\right)\right] \frac{d\omega'}{2\pi} \right\}$$
$$= \frac{1}{T} \int n_e(\mathbf{k}_0 - \omega\hat{\mathbf{n}}/c, \omega - \omega_0) n_e(\mathbf{k}_0 + \omega\hat{\mathbf{n}}/c, \omega - \omega_0) \frac{d\omega}{2\pi}$$

and that

$$\frac{1}{T} \int_{-T/2}^{T/2} dt \left\{ \int \frac{d\omega}{2\pi} n_e(\mathbf{K}, \Delta\omega) \exp\left[i\left(t - \frac{x}{c}\right)\right] \int \frac{d\omega'}{2\pi} n_e^*(\mathbf{K}, \Delta\omega') \exp\left[-i\omega'\left(t - \frac{x}{c}\right)\right] \right\}$$
$$= \frac{1}{T} \int \frac{d\omega}{2\pi} [n_e(\mathbf{K}, \Delta\omega)^2] \qquad (11.6.10)$$

The terms

$$\int \frac{d\omega}{2\pi} \left[n_e\left(\mathbf{k}_0 - \frac{\omega\hat{\mathbf{n}}}{c}, \omega - \omega_0\right) n_e\left(\mathbf{k}_0 + \frac{\omega\hat{\mathbf{n}}}{c}, \omega + \omega_0\right)\right]$$

vanish when averaged over the ensemble of test particles (Prob. 11.6.1). Using (11.6.10), the power radiated per unit solid angle at the point $\mathbf{x}$ is ($r_0 \equiv e^2/m_e c^2$)

$$\frac{dP_{\text{rad}}}{d\Omega} = \frac{\mathbf{x}^2 c}{4\pi} \langle E_r^2 \rangle = \frac{\text{power radiated}}{\text{unit area}} = r_0^2 \frac{c E_0^2}{4\pi} [\hat{\mathbf{e}} - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \hat{\mathbf{e}})]^2 \int N_e$$
$$\times \lim_{VT \to \infty} \frac{1}{VT} \frac{(2\pi)^8}{2n_e} \left| \tilde{n}_e\left(\mathbf{k}_0 - \frac{\omega\hat{\mathbf{n}}}{c}, \omega - \omega_0\right) \right|^2 \frac{d\omega}{2\pi} \qquad (11.6.11)$$

The incoming flux is $cE_0^2/8\pi$, and the differential cross section, defined by $(dP/d\Omega)/(cE_0^2/8\pi)$ is given in terms of the electron density by

$$\frac{d\sigma}{d\Omega} = r_0^2[\hat{\mathbf{e}} - \hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\hat{\mathbf{e}})]^2 \int N_e \lim_{VT\to\infty} \frac{1}{VT}(2\pi)^8 \left|\tilde{n}_e\left(\mathbf{k}_0 - \frac{\omega\hat{\mathbf{n}}}{c}, \omega - \omega_0\right)\right|^2 \frac{d\omega}{2\pi} \tag{11.6.12}$$

Defining

$$S\left(\mathbf{k}_0 - \frac{\omega\hat{\mathbf{n}}}{c}, \omega - \omega_0\right) \equiv \lim_{VT\to\infty} \frac{(2\pi)^8}{n_e}\left|\tilde{n}_e\left(\mathbf{k}_0 - \frac{\omega\hat{\mathbf{n}}}{c}, \omega - \omega_0\right)\right|^2 \tag{11.6.13}$$

the result (11.6.12) can thus be written

$$\frac{d\sigma}{d\Omega} = r_0^2[\hat{\mathbf{e}} - \hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\hat{\mathbf{e}})]^2 \int N_e S\left(\mathbf{k}_0 - \frac{\omega\hat{\mathbf{n}}}{c}, \omega - \omega_0\right)\frac{d\omega}{2\pi} \tag{11.6.14}$$

The integrand of (11.6.14) gives the shape of the scattered line. If $S \to [2\pi F_{e0}(\omega/k)]/k$, the scattering of radiation from a plasma is the same as the scattering from N_e independent electrons distributed with velocity according to $F_{e0}(\mathbf{v})$, where $\int F_{e0}(\mathbf{v})\, d\mathbf{v} = 1$.

In general, however, S will not have this value, and will reflect the fact that the incident radiation sees shielded particles rather than independent particles. Equation (11.6.14) gives the differential cross section for scattering of radiation for a given electron density n_e. To find $d\sigma/d\Omega$ for a plasma, the electron density n_e associated with a given test charge is averaged over an ensemble of test charges. The average of $|n_e(\mathbf{k}, \omega)|^2$ can be expressed in terms of $|n_e(\mathbf{x}, t)|^2$, which is the quantity given by (11.6.3) and (11.6.4), as follows:

$$\begin{aligned}
\langle|n_e(\mathbf{k},\omega)|^2\rangle &= \left\langle \int n_e(\mathbf{x}_1, t)\exp(-i\mathbf{k}\cdot\mathbf{x} + i\omega t_1)\frac{d\mathbf{x}_1\, dt_1}{(2\pi)^4}\right. \\
&\quad \left. \times \int n_e(\mathbf{x}_2, t_2)\exp(i\mathbf{k}\cdot\mathbf{x}_2)\exp(-i\omega t_2)\frac{d\mathbf{x}_2\, dt_2}{(2\pi)^4}\right\rangle \\
&= \int \langle n_e(\mathbf{x}_1, t_1)n_e(\mathbf{x}_2, t_2)\rangle \exp[i\mathbf{k}\cdot(\mathbf{x}_2 - \mathbf{x}_1)] \\
&\quad \times \exp[-i\omega(t_2 - t_1)]\frac{d\mathbf{x}_1\, dt_1\, d\mathbf{x}_2\, dt_2}{(2\pi)^8}
\end{aligned} \tag{11.6.15}$$

From (11.6.3) and (11.6.4),

$$\begin{aligned}
&\langle n_e(\mathbf{x}_1, t_1)n_e(\mathbf{x}_2, t_2)\rangle \\
&\quad = n_e \lim_{\varepsilon\to 0^+} \int d\mathbf{x}_0' \int f_{e0}(\mathbf{v}')\, d\mathbf{v}' \left(\int \exp[i\mathbf{k}_1(\mathbf{x}_1 - \mathbf{x}_0' - \mathbf{v}'t_1)]\frac{d\mathbf{k}_1}{(2\pi)^3}\right.
\end{aligned}$$

$$\begin{aligned}
&\times \int \exp\{+i[\mathbf{k}_2\cdot(\mathbf{x}_2 - \mathbf{x}_0' - \mathbf{v}'t_2)]\}\frac{d\mathbf{k}_2}{(2\pi)^3} \\
&\times \left[1 + \omega_{pe}^2 \int \frac{\mathbf{k}_2\cdot\nabla_{\mathbf{v}} f_{e0}\, d\mathbf{v}}{k_2^2 D(\mathbf{k}_2, \mathbf{k}_2\cdot\mathbf{v}')(\mathbf{k}_2\cdot\mathbf{v} - \mathbf{k}_2\cdot\mathbf{v}' - i\varepsilon)}\right] \\
&\times \left.\left[1 + \omega_{pe}^2 \int \frac{\mathbf{k}_1\cdot\nabla_{\mathbf{v}} f_{e0}(\mathbf{v})\, d\mathbf{v}}{k_1^2 D(\mathbf{k}_1, \mathbf{k}_1\cdot\mathbf{v}')(\mathbf{k}_1\cdot\mathbf{v} - \mathbf{k}_1\cdot\mathbf{v}' - i\varepsilon)}\right]\right) \\
&+ n_i \int d\mathbf{x}_0' \int f_{i0}(\mathbf{v}')\, d\mathbf{v}' \left\{\omega_{pe}^2 \int d\mathbf{v}\right. \\
&\times \left[\int \frac{\exp[i\mathbf{k}_1\cdot(\mathbf{x}_1 - \mathbf{x}_0' - \mathbf{v}'t)]\mathbf{k}_1\cdot\nabla_{\mathbf{v}} f_{e0}(\mathbf{v})}{k_1^2 D(\mathbf{k}_1, \mathbf{k}_1\cdot\mathbf{v}')(\mathbf{k}_1\cdot\mathbf{v}_1 - \mathbf{k}_1\cdot\mathbf{v}' - i\varepsilon)}\frac{d\mathbf{k}}{(2\pi)^3}\right] \\
&\times \left.\omega_{pe}^2 \int d\mathbf{v} \int \frac{\exp[-i\mathbf{k}_2\cdot(\mathbf{x}_2 - \mathbf{x}_0' - \mathbf{v}'t_2)]\mathbf{k}_2\cdot\nabla_{\mathbf{v}} f_{e0}(\mathbf{v})}{k_2^2 D(\mathbf{k}_2, \mathbf{k}_2\cdot\mathbf{v}')(\mathbf{k}_2\cdot\mathbf{v} - \mathbf{k}_2\cdot\mathbf{v}' - i\varepsilon)}\frac{d\mathbf{k}_2}{(2\pi)^3}\right\}
\end{aligned}$$

The integral $\int d\mathbf{x}_0'$ yields a δ function which allows the $\mathbf{k}_2$ integration to be carried out, giving

$$\begin{aligned}
&\langle n_e(\mathbf{x}_1, t)n_e(\mathbf{x}_2, t_2)\rangle \\
&\quad = n_e \int F_{e0}(u')\, du' \int \exp[-i\mathbf{k}\cdot(\mathbf{x}_1 - \mathbf{x}_2)]\exp[-i\,|k\,|u'(t_1 - t_2)]\frac{d\mathbf{k}}{(2\pi)^3} \\
&\qquad \times \left|1 + \int \frac{\partial F_{e0}/\partial u\, du}{k^2 D(\mathbf{k},\ |k\,|u')(u - u' - i\varepsilon/\,|k\,|)}\right|^2 \\
&\qquad + n_i \int F_{i0}(u')\, du' \int \exp[i\mathbf{k}\cdot(\mathbf{x}_1 - \mathbf{x}_2)]\exp[-i\,|k\,|u'(t_1 - t_2)]\frac{d\mathbf{k}}{(2\pi)^3} \\
&\qquad \times \left|\int \frac{\partial F_{e0}/\partial u\, du}{k^2 D(\mathbf{k},\ |k\,|u')(u - u' - i\varepsilon/\,|k\,|)}\right|^2
\end{aligned} \tag{11.6.16}$$

Inserting this result into (11.6.15), it follows that

$$\begin{aligned}
&\langle S(\mathbf{k},\omega)\rangle \\
&\quad = \frac{(2\pi)^8}{TV}\frac{\langle|\tilde{n}_e(\mathbf{k},\omega)|^2\rangle}{n_e} \\
&\quad = \int F_{e0}(u')2\pi\,\delta(\omega - |k\,|u')\, du' \left|1 + \int \frac{\partial F_{e0}/\partial u\, du}{k^2 D(\mathbf{k},\ |k\,|u')(u - u' - i\varepsilon/\,|k\,|)}\right|^2 \\
&\qquad + \int F_{i0}(u')2\pi\,\delta(\omega - |k\,|u')\, du \left|\int \frac{\partial F_{e0}/\partial u\, du}{k^2 D(k,\ |k\,|u')(u - u' - i\varepsilon/\,|k\,|)}\right|^2
\end{aligned} \tag{11.6.17}$$

Problem 11.6.1 Following the steps above leading to $\langle|\tilde{n}_e(k, \omega)|^2\rangle$, show that

$$\langle\tilde{n}_e(\mathbf{k}, \omega)n_e(\mathbf{k}, \omega')\rangle = 0 \qquad \omega \neq \omega'$$

justifying the neglect of the contributions

$$\left\langle n_e\left(\mathbf{k} - \frac{\omega\hat{\mathbf{n}}}{c}, \omega - \omega_0\right)n_e\left(\mathbf{k} + \frac{\omega\hat{\mathbf{n}}}{c}, \omega - \omega_0\right)\right\rangle$$

to the cross section. ////

The differential cross section for scattering of radiation from a plasma is thus

$$\frac{d\sigma}{d\Omega} = \int \frac{d^2\sigma}{d\Omega\, d\omega}\, d\omega$$

where the differential cross section per unit frequency range is

$$\frac{d^2\sigma}{d\Omega\, d\omega} = r_0{}^2[\hat{\mathbf{e}} - \hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\hat{\mathbf{e}})]^2 N_e \frac{F_{e0}(\Delta\omega/|\mathbf{K}|)}{|\mathbf{K}|}$$
$$\times \left|1 + \omega_{pe}{}^2 \lim_{\varepsilon\to 0^+} \int \frac{\partial F_{e0}/\partial u\, du}{K^2 D(\mathbf{K}, \Delta\omega)(u - \Delta\omega|\mathbf{K}| - i\varepsilon/|\mathbf{K}|)}\right|^2$$
$$+ \frac{F_{i0}(\Delta\omega/|\mathbf{K}|)}{|\mathbf{K}|}\left|\omega_{pe}{}^2 \lim_{\varepsilon\to 0} \int \frac{\partial F_{e0}/\partial u\, du}{K^2 D(\mathbf{K}, \Delta\omega)(u - \Delta\omega/|\mathbf{K}| - i\varepsilon/|\mathbf{K}|)}\right|^2 \tag{11.6.18}$$

and $$\mathbf{K} = \left|\mathbf{k}_0 - \frac{\hat{\mathbf{n}}\omega}{c}\right| \qquad \Delta\omega = \omega - \omega_0$$

The angular factor is given by

$$[\hat{\mathbf{e}} - \hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\hat{\mathbf{e}})]^2 = \frac{1 + \cos^2\theta}{2}$$

for an unpolarized incident beam.

The first term in (11.6.18) gives the scattering from test electrons and from the electrons which shield the test electrons. The second term gives the scattering from electrons which make up the shield cloud of test ions.

The effect of shielding depends on the wavelength of the incident wave relative to the size of the shield clouds, expressed by the parameter $\alpha \equiv 1/K\lambda_D$.

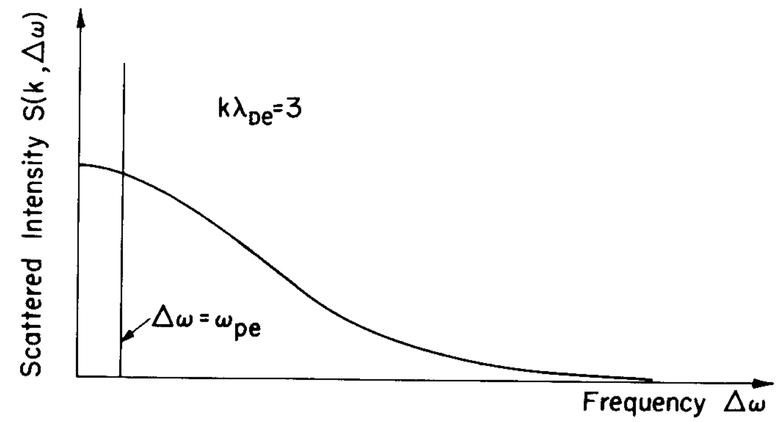


FIGURE 11.6.2
Spectral distribution of the intensity of radiation scattered by a plasma with a maxwellian electron distribution, where $K\lambda_D = 3$. This result is virtually identical with scattering from $\bar{N}$ independent electrons. [*After I. B. Bernstein, S. K. Trehan, and M. P. H. Weenink, Nucl. Fusion*, **4**; 61 (1964).]

11.6.1 Single-particle Limit ($K\lambda_D \gg 1$)

In this limit the terms involving $1/K^2$ can be neglected, and the differential cross section is, from (11.6.18),

$$\frac{d\sigma}{d\Omega} = \int \frac{1 + \cos^2\theta}{2} r_0{}^2 \frac{N_e}{2\pi} S(|\mathbf{K}|, \Delta\omega)\, d\omega$$
$$= \int \frac{r_0{}^2}{2}(1 + \cos^2\theta)\frac{N_e F_{e0}(\Delta\omega/|\mathbf{K}|)}{|\mathbf{K}|}\, d\omega = N_e r_0{}^2 \frac{1 + \cos^2\theta}{2} \tag{11.6.19}$$

The cross section (11.6.19) is the same as the scattering from a single particle multiplied by the number of particles. Thus there is no "plasma" effect on the scattering in the wavelength range $K\lambda_D \gg 1$. This is because the short-wavelength radiation sees individual bare electrons; i.e., it gets inside the shielding cloud.

The spectral density $S(\mathbf{K}, \Delta\omega)$ at a typical scattering angle is plotted in Fig. (11.6.2) for a maxwellian distribution of electrons. The scattered line has a shape determined from the electron distribution; i.e., the frequency is doppler-shifted by an amount determined from the electron thermal spread.

Problem 11.6.2 Show from (11.6.18) that there is no scattered radiation if the plasma is really a Vlasov fluid, and $n_e(x, t) = n_0 = \text{constant}$. ////

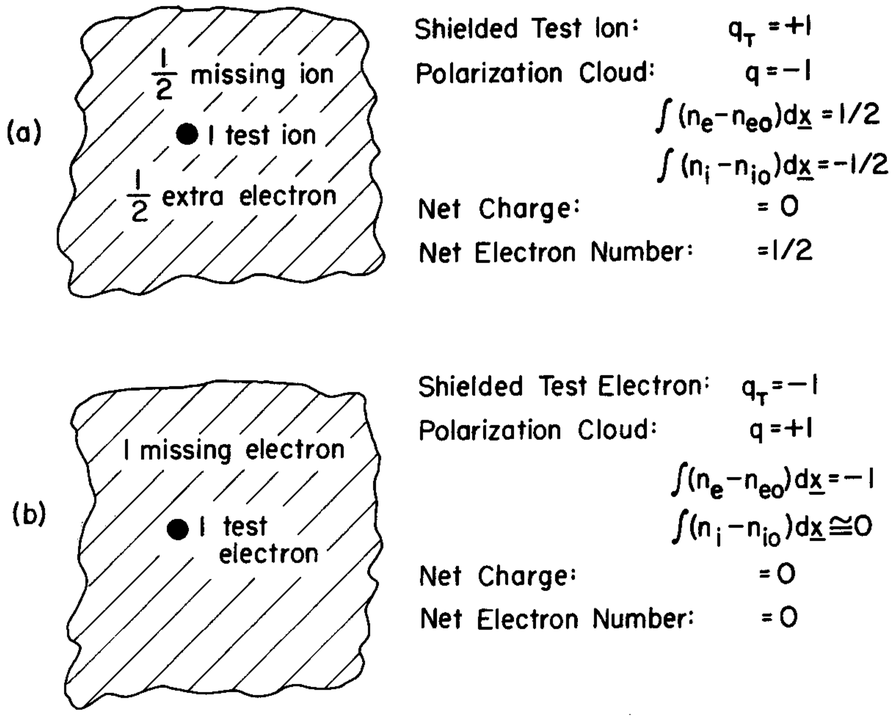


FIGURE 11.6.3
Diagram illustrating the shielding of (*a*) a test ion and (*b*) a test electron, and the resulting charge distribution.

11.6.2 Plasma Limit ($K\lambda_D \ll 1$)

In this limit the incident wave sees fully shielded test particles. A test electron is seen as a neutral entity with no net electron charge density, as in Fig. 11.6.3. (Remember that the shielding of electrons is due only to other electrons.) There should be little scattering of long-wavelength radiation from the shielded-test electron. A test ion is also seen as a neutral entity, but it consists of a superposition of half an ion and half an electron, because in the shielding process the typical test ion attracts half an electron and repels half an ion (Fig. 11.6.3). Therefore a distinctive part of the scattering of waves with $k\lambda_D \ll 1$ should be due to electrons that make up the shield cloud of ions, and should be characterized by the number of shielding electrons ($N_i/2$) and by the ion thermal spread in velocity.

The above argument already indicates that long-wavelength radiation scatters quite differently from a free electron than from a plasma. In addition, the scattering is frequency dependent, characteristic of the fact that the Vlasov fluid has density fluctuations due to plasma waves always present in a thermal plasma: radiation scatters from these waves. This feature of the plasma shows

up in the dielectric $D(K, \Delta\omega)$, which is small for frequencies and wave numbers corresponding to plasma waves. There are a number of ranges distinctive of scattered frequencies.

1 $\Delta\omega/K \lesssim \sqrt{\kappa T_i/m_i}$ For scattered frequency nearly equal to the incident frequency, (11.6.18) gives

$$\frac{d\sigma}{d\Omega} = \int \frac{{r_0}^2}{2}(1 + \cos^2\theta)\frac{N_e F_{i0}(\Delta\omega/|\mathbf{K}|)}{|\mathbf{K}|}\left|\frac{2}{K^2 D(\mathbf{K}, \Delta\omega)^2 {\lambda_D}^2}\right|^2 d\omega \quad (11.6.20)$$

where the fact that F_{i0} is larger than F_{e0} by $\sim\sqrt{m_i/m_e}$ inside the thermal distribution of ions has been used to neglect F_{e0}. For fixed K, the dielectric $D(\mathbf{K}, \Delta\omega)$ decreases from its value at $D(\mathbf{K}, 0)$ to a relative minimum at $\Delta\omega = \omega_{pi}$; therefore the scattered flux increases with $\Delta\omega$. If $T_e \gg T_i$, $D(\mathbf{K}, \Delta\omega = \omega_{pi}) \ll 1$; that is, ion waves are nearly undamped, and this peak in scattered intensity at $\omega = \omega_0 + \omega_{pi}$ is very pronounced. This peak of the scattered line shape is called the *ion line*. For $T_e \approx T_i$, $D(\mathbf{K}, \Delta\omega = \omega_{pi})$ is not too small, since D_i is sizable, and the ion-wave contributions to the scattering is not pronounced in that case. The dependence of the ion-wave peak on T_e/T_i is shown in Fig. 11.6.4.

For $\Delta\omega > \omega_{pi}$ the scattered intensity decreases, proportional to the ion distribution $F_{i0}(\Delta\omega/K)$. Thus, in contrast to the scattering of waves $K\lambda_D \gg 1$, in which case the shape of the spectrum is determined by F_{e0}, the shape of a scattered line with $K\lambda_D \ll 1$ is determined by the ion distribution. As mentioned above, this is because a test electron and its shielded cloud add up to *no* net electron density fluctuation as viewed by a long-wavelength oscillation $K\lambda_D < 1$; a test ion has on the average half an electron in its shield cloud, and this is the electron density fluctuation which scatters longer-wavelength radiation at $\Delta\omega < \omega_{pe}$. *Two* features arise from the fact that the scattering is mainly from the half electron that neutralizes each test ion. The first is that the number of scattering electrons is *one-half* the total number of electrons in the system. Integrating (11.6.20) gives

$$\frac{d\sigma}{d\Omega} = \frac{{r_0}^2}{4} N_e(1 + \cos^2\theta) \quad (11.6.21)$$

which is half the scattering from independent electrons [(11.6.19)].

The second feature arising from the fact that the central part of the scattered line comes from electrons that are part of the shield cloud of the ions is that the width (and shape) of the central scattered line ($\Delta\omega < \omega_{pe}$) is determined by the ion thermal speed. This is clear since the integrand of (11.6.20) is proportional to $F_{i0}(\Delta\omega/K)$. This is in contrast to the scattering from independent

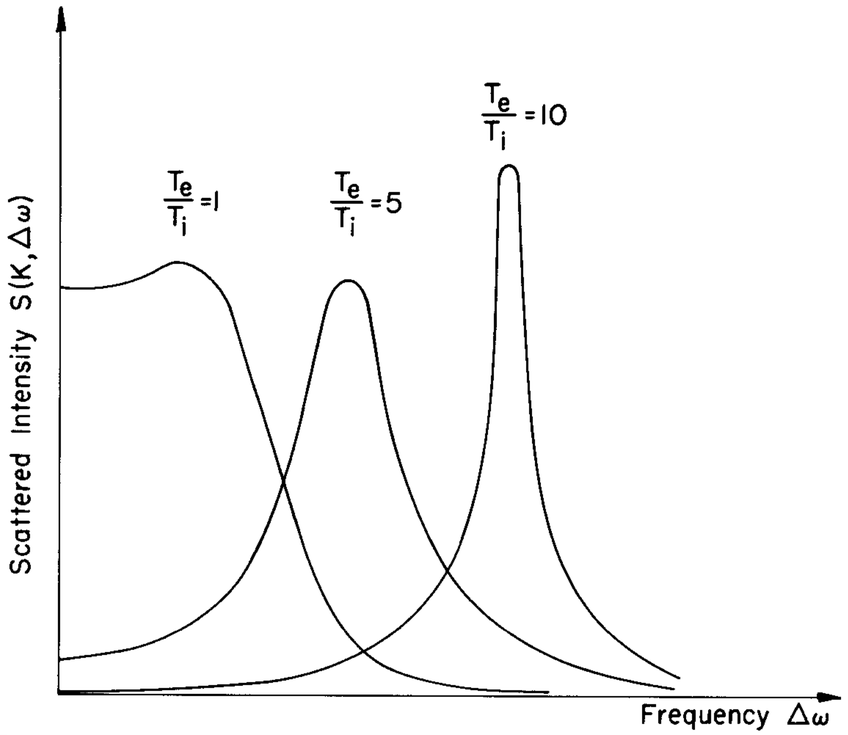


FIGURE 11.6.4
Spectral distribution of the intensity of radiation scattered by a plasma with maxwellian electron and ion distributions at different temperatures, showing the emergence of the ion line, corresponding to the excitation of ion-acoustic waves, as $T_e \gg T_i$: [*After I. B. Bernstein, S. K. Trehan, and M. P. H. Weenink, Nucl. Fusion*, **4**; 61 (1964).]

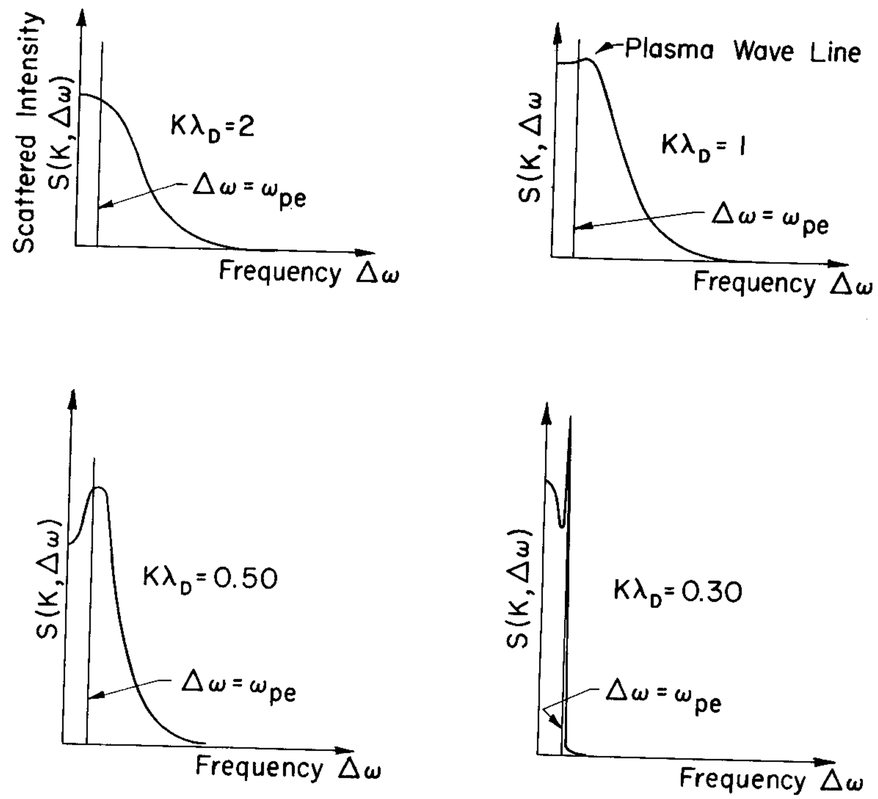
electrons (or from a plasma of $K\lambda_D \gg 1$), in which case the width and shape of the scattered line are determined by the electron thermal speed. Thus the line width when $K\lambda_D \gg 1$ is far in excess of the width when $K\lambda_D \ll 1$.

2 $\sqrt{\kappa T_i/m_i} < \Delta\omega/K < \sqrt{\kappa T_e/m_e}$ In this part of the scattered line, $F_{i0}(\Delta\omega/|K|)$ is small and the scattering is small, being simply due to the weak interaction of the wave with a test electron and its shield cloud.

$$\frac{d\sigma}{d\Omega} = \int \frac{r_0^2}{2}(1+\cos^2\theta)\frac{N_e F_{e0}(\Delta\omega/K)}{K} K^4\lambda_D{}^4\, d\omega \qquad K\lambda_D \ll 1 \qquad (11.6.22)$$

3 $\Delta\omega = \omega - \omega_0 \approx \omega_{pe}$; $\Delta\omega/K > \sqrt{\kappa T_e/m_e}$ In this range of scattered frequency, $D(K, \Delta\omega)$ is small and the scattered radiation rises to a peak given by

$$\frac{d\sigma}{d\Omega} = \int \frac{r_0^2}{2}(1+\cos^2\theta)\frac{N_e F_{e0}(\Delta\omega/K)}{K}\frac{4}{\left(\dfrac{\Delta\omega^2}{\omega_{pe}{}^2}-1\right)^2+\left(\pi\dfrac{\omega_p{}^2}{K^2}\dfrac{\partial F_{e0}}{\partial(\Delta\omega/K)}\right)^2}\, d\omega \qquad (11.6.23)$$



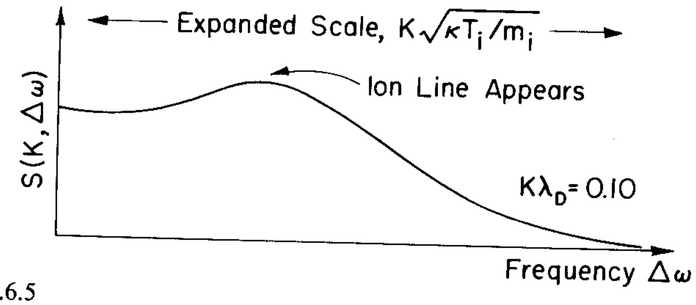


FIGURE 11.6.5
Spectral distribution of the intensity of radiation scattered by a plasma with maxwellian electron and ion distributions, showing the transition from single-particle scattering at $K\lambda_D \gg 1$ to scattering in the plasma limit, $K\lambda_D \ll 1$. [*After I. B. Bernstein, S. K. Trehan, and M. P. H. Weenink, Nucl. Fusion*, **4**; 61 (1964).]

In this range, the incident wave is scattered from plasma oscillations, which are excited to a thermal level in a plasma as described in Secs. 11.2 and 11.3. The amplitudes and width of the scattered line are functions of the damping coefficient, which in turn depends on $(K\lambda_D)^2$. The spectral density for various $(K\lambda_D)^2$ is plotted in Fig. 11.6.5.

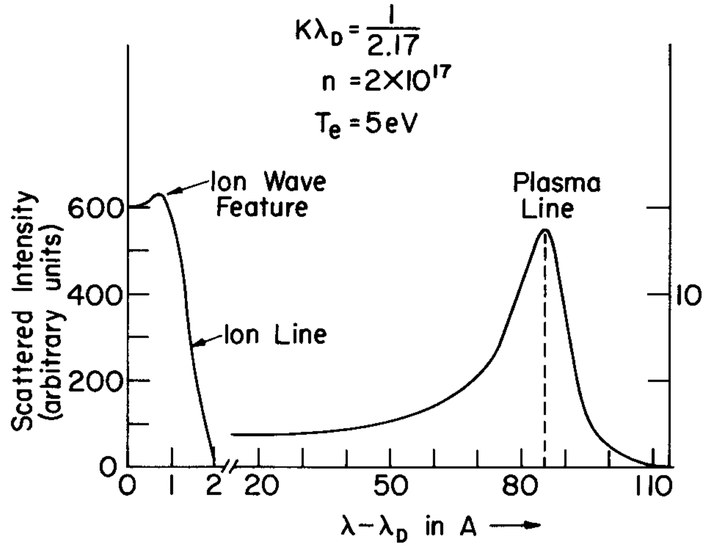


FIGURE 11.6.6
Spectral distribution of the intensity of radiation scattered by a plasma with maxwellian electron and ion distribution, showing both the electron plasma line and the features due to scattering from ion waves. [*After H.-J. Kunze, in* W. Lochte-Holtgreven (ed.), "*Plasma Diagnostics*," North-Holland Publishing Co., Amsterdam, 1968.]

A plot of the scattered line for a particular long-wavelength experiment ($K\lambda_D = 1/2.17$, $n = 2\cdot\times 10^{17}$ cm^{-3}, $T_e = 5$ eV) is given in Fig. 11.6.6.

The scattering of electromagnetic radiation by plasma is one of the richest processes that can be imagined, in terms of the insight it gives into microprocesses in the plasma state.

Problem 11.6.3 Show by integrating (11.6.20) in the limit

$$\frac{\Delta\omega}{K} \ll \sqrt{\frac{\overline{\kappa T_i}}{m_i}} \qquad K\lambda_D \ll 1$$

that the cross section for scattering long-wavelength radiation from electrons in a plasma is half the scattering from an equal number of free electrons. ////

11.7 EMISSION OF RADIATION FROM A PLASMA; KIRCHHOFF'S LAW

The radiation emitted from a plasma is a distinctive characteristic of that state of matter; it exhibits a dependence on the state of individual plasma particles, as well as a dependence on the collective properties of the plasma. In laboratory

plasmas, radiation is an energy loss that acts as a limitation on certain applications of plasma physics. For example, in proposals to use hot plasmas as a fuel for a controlled thermonuclear reactor (Chap. 5, Part IV), the radiation from magnetically confined electrons must be included in estimating the efficiency of such proposed reactors.[1] In other cases, the radiation emitted by a plasma is measured and used to determine the temperature and distribution of particles in the system.

The radiation emitted by a plasma is a result of the fact that a plasma is made up of discrete charged particles and is not a continuous distribution of charge as assumed by the macroscopic fluid and Vlasov theories. Thus these theories do not predict radiation. Plasma radiation can be calculated by a test-particle approach, as demonstrated in earlier sections of this chapter. In this section Kirchhoff's law is used to compute the radiation from a plasma in terms of the wave-damping coefficients calculated in Chap. 8. Since wave damping can be calculated from the Vlasov equation, this procedure is a great simplification, allowing an effect that is basically of higher order (in $1/n\lambda_D{}^3$) to be calculated using a lower-order theory.

Kirchhoff's law is simply a statement of energy conservation. Consider an isolated plasma in true thermodynamic equilibrium with its enclosure, at temperature T. By Planck's law, such a plasma radiates as a blackbody (Sec. 11.8),

$$I = I_{bb} \qquad (11.7.1)$$

As radiation passes through the plasma, it is generally absorbed (damped), according to the plasma wave theory developed in Chap. 8. This reduces the intensity of radiation as it passes the plasma. The intensity as a function of position is given by

$$\left.\frac{dI}{dx}\right|_{\text{damping}} = -AI \qquad (11.7.2)$$

In equilibrium, the emission must balance the absorption, with the intensity given by (11.7.1), so that the emission rate for an equilibrium plasma is

$$S = AI_{bb} \qquad (11.7.3)$$

Thus the emission can be found in terms of the absorption coefficient (calculated through Vlasov theory in Chap. 8) and the blackbody intensity (obtained from equilibrium statistical mechanics in Sec. 11.8).

[1] B. A. Trubnikov and V. S. Kudryavtsev, *Proc. 2d Intern. Conf. Peaceful Uses Atomic Energy*, **31**; 93 (U.N., Geneva, 1958); W. E. Drummond and M. N. Rosenbluth. *Phys. Fluid*, **3**; 45 (1960); D. B. Beard and J. C. Baker, *Phys. Fluids*, **5**; 1113 (1962).

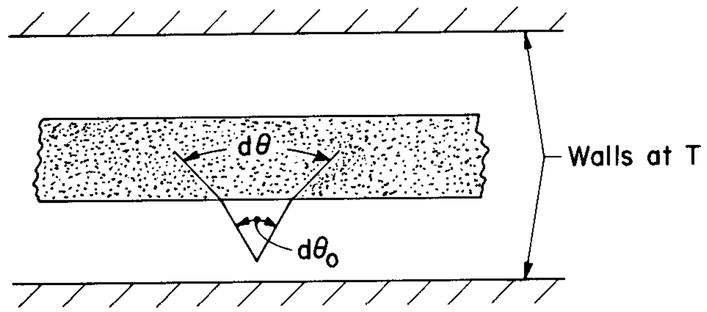


FIGURE 11.8.1
Schematic of a plasma slab in equilibrium with its surroundings, illustrating a wave diffracting as it passes through a plasma-vacuum interface.

Equation (11.7.3) is Kirchhoff's law.[1] Its usefulness lies in the fact that spontaneous emission depends on the distribution of particles but does not depend on the radiation level. Therefore (11.7.3) gives the emission from a plasma composed of a maxwellian distribution of particles whether or not the plasma is in equilibrium with its surroundings.

11.8 BLACKBODY RADIATION FROM AND IN A PLASMA

In order to use Kirchhoff's law, it is necessary to calculate the blackbody intensity of radiation inside a plasma. To show how this is done, consider a plasma at temperature T in equilibrium with the radiation field, as in Fig. 11.8.1. The radiation incident (and leaving) the surface is the blackbody spectrum $I_v(\omega)$, where the subscript v emphasizes that this is the radiation from plasmas into a vacuum. Let the plasma slab be so thin that absorption by the plasma is negligible. In that case the radiation level inside the plasma is independent of its thickness R. Radiation impinging on the plasma in a cone $d\theta_0$ will be refracted into $d\theta_1$, because in general the phase velocity of light inside a plasma v_{phase} is not the same as in vacuum. By Snell's law

$$\frac{c}{v_{\text{phase}}} = \frac{\sin\theta_0}{\sin\theta_1} = \frac{d\theta_0}{d\theta_1} \qquad (11.8.1)$$

In terms of the solid angle $d\Omega_0$, (11.8.1) becomes

$$\frac{d\Omega_0}{d\Omega_1} = \frac{c^2}{v_{\text{phase}}^2} \qquad (11.8.2)$$

[1] C. Bekefi, J. Hirshfield, and S. C. Brown, *Phys. Rev.*, **112**; 1037 (1961).

After time τ the radiation in a length $c\tau$ of vacuum has entered the plasma, where it now occupies a length $v_g\,\tau$, with v_g the group velocity of electromagnetic waves in plasmas. It follows from conservation of energy that

$$I_v(\omega)c\tau\, d\Omega_0 = I_p(\omega)v_g\,\tau\, d\Omega_1 \qquad (11.8.3)$$

where I_p is the radiation level inside the plasma. Combining these results, the level of radiation in a "blackbody" plasma is

$$I_p = I_v \frac{c^3}{v_g v_{ph}^2} = \frac{\hbar\omega^3}{4\pi^3 c^3[\exp(\hbar\omega/\kappa T) - 1]} \frac{c^3}{v_g v_{ph}^2}$$

The dispersion relation for electromagnetic waves in a plasma,

$$\omega^2 = \omega_p^2 + k^2c^2$$

determines

$$v_g = c\sqrt{1 - \frac{\omega_p^2}{\omega^2}}$$

and

$$v_{\text{ph}} = \frac{c}{\sqrt{1 - \omega_p^2/\omega^2}}$$

so that for $\hbar\,\omega \ll KT$

$$I_p = \frac{\kappa T\omega^2}{2\pi^2c^3}\sqrt{1 - \frac{\omega_p^2}{\omega^2}} \qquad (11.8.4)$$

is the blackbody radiation level *inside* a field-free plasma. The energy in radiation in the plasma is distributed among magnetic, electric, and particle kinetic energy. Using $\mathbf{B} = (kc/\omega)\mathbf{E}$ and $\mathbf{dv}/dt = -(e\mathbf{E}/m)$,

$$W_E = \frac{E^2}{8\pi}$$

$$W_B = \left(1 - \frac{\omega_p^2}{\omega^2}\right) W_E$$

$$W_{\text{Kinetic}} = \frac{\omega_p^2}{\omega^2}\frac{E^2}{8\pi}$$

The blackbody level I_p is useful in Kirchhoff's law, but it should be noted that only a plasma of fantastic size or very high density actually radiates at the blackbody rate (see Prob. 11.8.1).

Note that Kirchhoff's law is no aid in calculating radiation for a field-free plasma, because electromagnetic waves are *undamped* in plasma, except by collisions which require a higher order treatment. However, this situation is quite different for plasma in a magnetic field.

Problem 11.8.1 From the acceleration of electrons by an ion,

$$a = \frac{Ze^2}{mr^2}$$

show that the power radiated by electrons in a plasma is, roughly,

$$\frac{\text{Power radiated}}{\text{Unit volume}} = \frac{8\pi}{3}\frac{Z^2e^6n_e n_i}{c^3 m_e{}^2 R_{\min}} \approx \frac{n_e 16\pi^2 Z^2 e^6}{3c^2 mh}\sqrt{\frac{\kappa T_e}{m_e}}$$

where $R_{\min}$ is a cutoff distance of closest approach (taken here to be $\hbar/p$). Compare this with blackbody radiation, and show that the radius of a plasma which radiates at the blackbody rate is

$$R_{bb} \approx 10^{23} T^{7/2} Z^2 n_i n_e \qquad \text{cm}$$

where T is in electron volts. ////

11.9 CYCLOTRON (SYNCHROTRON) RADIATION FROM A PLASMA IN A MAGNETIC FIELD

Charged particles in a magnetic field are accelerated, [$\mathbf{a} = -(e/mc)\mathbf{v} \times \mathbf{B}$] and therefore lose energy by radiation. Because rotation of electrons in a **B** field is characteristic of certain particle accelerators, this radiation is called *cyclotron* or *synchrotron radiation.* This radiation, common to all magnetically confined plasmas, occurs in nature in such diverse events as solar radio burts,[1] radiation from Jupiter,[2] and radio noise from the Crab nebula.[3]

Kirchhoff's law can be combined with the damping coefficient calculated in Sec. 8.14 to calculate cyclotron radiation. Consider a slab of plasma in a magnetic field, as shown in Fig. 11.9.1. If the plasma is in equilibrium with its surroundings, at temperature T, then outside the plasma the spectral intensity is blackbody, which at high temperature reduces to the Rayleigh-Jeans law

$$I_{bb} = \frac{\hbar\omega^3}{4\pi^3 c^2[\exp(\hbar\omega/\kappa T - 1)]} \xrightarrow{\hbar\omega \ll \kappa T} \frac{\omega^2 \kappa T}{4\pi^3 c^2} \qquad (11.9.1)$$

The radiation energy on area $d\sigma$ with frequency ω in $d\omega$ which is incident on the plasma at an angle θ is given by

$$I_{\text{bb}} \cos\theta \, d\sigma \, d\omega \qquad (11.9.2)$$

[1] J. P. Wild, S. F. Smerd and A. A. Weiss, *Ann. Rev. Astronomy Astrophys.*, **1**; 291 (1963).
[2] J. W. Warwick, *Ann. Rev. Astronomy Astrophys.*, **2**; 1 (1964).
[3] V. L. Ginzberg and S. I. Syrovatskii, *Ann. Rev. Astronomy Astrophys.*, **3**; 297 (1965).

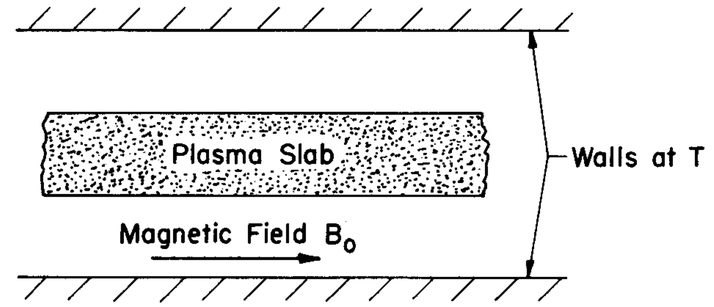


FIGURE 11.9.1
Schematic of a plasma slab immersed in a magnetic field and in equilibrium with its surroundings. This model is used to calculate the radiation level of synchrotron radiation emitted by a magnetically confined plasma.

In equilibrium this radiation is matched by emission from the plasma slab, which includes spontaneously emitted radiation and radiation that has passed unabsorbed through the plasma. Inside the plasma, the electromagnetic wave propagates in two modes, the ordinary mode, whose electric vector is parallel to $\mathbf{B}_0$, and the extraordinary mode, whose electric vector is perpendicular to $\mathbf{B}_0$. Denoting the absorption coefficient of these two polarizations by $A_\perp$ and $A_\parallel$, the intensity of radiation leaving the plasma slab is

$$\left\{\tfrac{1}{2} I_{bb} \cos\theta \left[\exp\left(-\frac{A_\perp L}{\cos\theta}\right) + \exp\left(-\frac{A_\parallel L}{\cos\theta}\right)\right] + I_p\right\} d\sigma \, d\omega \qquad (11.9.3)$$

where I_p is the spontaneous emission, and the other two terms give the radiation which entered the plasma on one side, was partly absorbed, and emerged on the other side. Neglecting refraction (taking the propagation angle θ to be the same inside and outside the plasma), the radiation (11.9.2) and (11.9.3) must be equal, so that the spontaneous radiation from a plasma slab of size L is

$$I_p = I_{bb} \cos\theta \left\{1 - \tfrac{1}{2}\left[\exp\left(-\frac{A_\perp L}{\cos\theta}\right) + \exp\left(-\frac{A_\parallel L}{\cos\theta}\right)\right]\right\} \qquad (11.9.4)$$

Equation (11.9.4) is Kirchhoff's law for radiation in a magnetized plasma. If the plasma is large, i.e., if the plasma is self-absorptive, $L \gg \cos\theta/A$, the plasma radiates at the blackbody rate. If the plasma is thin,

$$I_p = \tfrac{1}{2} I_{\text{bb}} L(A_\perp + A_\parallel) \qquad (11.9.5)$$

gives the cyclotron (synchrotron) radiation from the plasma.

The absorption coefficients are related to the temporal damping coefficients in Sec. 8.14 by

$$A_{(\perp,\,\parallel)} = \frac{2\omega_i}{c}(\perp,\,\parallel) \qquad (11.9.6)$$

where the factor of 2 is due to ω_i giving the damping of the field *amplitude*, not the energy, and the factor c converts from temporal to spatial damping.

In many applications the radiating plasma is relativistic; the radiation in these cases is called synchrotron radiation. The equilibrium distribution for a relativistic electron gas,[1] in terms of the momentum p parallel and perpendicular to the magnetic field $\mathbf{B}_0$, is

$$f_0 = N \exp\left[-\frac{c}{\kappa T}(p^2 + m^2c^2)^{1/2}\right] \qquad (11.9.7)$$

where

$$\int f_0 \, d\mathbf{p} = 1$$

Using the distribution (11.9.7) and the damping coefficients from Sec. 8.14, the radiation per unit volume from a plasma is

$$\frac{I_p}{L} = \frac{n_0 e^2}{c} \sum_n \int_{-\infty}^{\infty} dp_\parallel \int_0^\infty \omega^2 f_0 \, \delta[\omega(1 - \beta_\parallel \sin\theta) - n\omega_{ce}] \times [\beta_\perp{}^2 (J_n')^2 + \beta_\parallel{}^2 J_n{}^2] p_\perp \, dp_\perp \qquad (11.9.8)$$

where the argument of J_n is $k_\perp v_\perp/\omega_{ce}$, $\beta_{\perp,\parallel} = v_{\perp,\parallel}/c$, and $\omega_{ce} = eB_0/\gamma m_e c$. Because the damping coefficients in Sec. 8.14 were calculated assuming that $k_\perp \gg k_\parallel$, Eq. (11.9.8) is valid only for radiation nearly perpendicular to $\mathbf{B}_0$.

Since synchrotron radiation is substantial only at frequencies $\omega \approx n\omega_{ce}$, and since for $k \approx \omega/c$ the argument of the Bessel function is $nv_\perp/c$, radiation above the first harmonic is small unless the plasma is relativistic.

Problem 11.9.1 Calculate the radiation from a single electron gyrating in B_0 (see, for example, D. Montgomery and D. A. Tidman, "Plasma Kinetic Theory," pp. 210–219, McGraw-Hill, New York, 1964), and deduce that the incoherent radiation per unit volume from an assembly of electrons is

$$\frac{I}{L} = \frac{n_0 e^2}{c} \sum_n \int_{-\infty}^{\infty} dp_\parallel \int_0^\infty \frac{\omega^2 f_0 \, \delta[\omega(1 - \beta_\parallel \sin\theta) - n\omega_{ce}]}{1 - \beta_\parallel \sin\theta} \times \left[(\tan\theta - \beta_\parallel \sec\theta)^2 J_n{}^2\left(\frac{\omega}{\omega_{ce}}\beta_\perp \cos\theta\right) + \beta_\perp{}^2 J_n'{}^2\left(\frac{\omega}{\omega_{ce}}\beta_\perp \cos\theta\right)\right] p_\perp \, dp_\perp \qquad (11.9.9)$$

Compare (11.9.8) and (11.9.9) and show that the discrepancies are small when the ω_i used in (11.9.8) are valid. ////

[1] R. C. Tolman, "Relativity, Thermodynamics and Cosmology" Oxford, Fairlawn, N.J., 1934.

11.10 TEST SOURCE THEORY OF RADIATION FROM A PLASMA

Kirchhoff's law gives a method of calculating the radiation from equilibrium plasmas, neglecting particle collisions. A more direct and flexible method of calculating radiation is to calculate the radiation from an arbitrary test source imbedded in a plasma, and then choose the test source to correspond to some physical process. The test source can be a charge density $\rho_s(\mathbf{x}, t)$, or a current density $\mathbf{J}_s(\mathbf{x}, t)$ related by

$$\frac{\partial \rho_s}{\partial t} + \nabla \cdot \mathbf{J}_s = 0 \qquad (11.10.1)$$

Some examples of test sources are (1) a charged particle gyrating in a magnetic field, which will lead to synchrotron radiation; (2) a charged particle deflected by another charged particle, which will lead to bremsstrahlung radiation; or (3) a charged particle moving faster than a characteristic wave of the plasma, which will lead to Cerenkov radiation.

The electric field due to a test charge density ρ_s is given by the solution of the Vlasov equation as derived in earlier chapters.

If $f_\alpha(\mathbf{x}, \mathbf{v}, t)$ is the distribution function of a plasma in which the test source is imbedded, $f_{\alpha 0}(\mathbf{v})$ is the distribution in the absence of the test source, and the plasma is *field-free* except for the effect of the test source, then

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{v} \cdot \nabla f_\alpha + \frac{q_\alpha}{m_\alpha}\left(\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c}\right) \cdot \nabla_\mathbf{v} f_{\alpha 0} = 0 \qquad (11.10.2)$$

$$\nabla \cdot \mathbf{E} = \sum_\alpha 4\pi q_\alpha \bar{n}_\alpha \int f_\alpha \, d\mathbf{v} + 4\pi\rho_s(\mathbf{x}, t) \qquad (11.10.3)$$

$$\nabla \times \mathbf{B} = \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} + \frac{4\pi}{c}\sum_\alpha q_\alpha \bar{n}_\alpha \int \mathbf{v} f_\alpha \, d\mathbf{v} + \frac{4\pi}{c}\mathbf{J}_s(\mathbf{x}, t) \qquad (11.10.4)$$

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t} \qquad (11.10.5)$$

give the fields set up in the plasma by ρ_s and $\mathbf{J}_s$. In (11.10.2), a small term $(\mathbf{E} + \mathbf{v} \times \mathbf{B}/c) \cdot \nabla_\mathbf{v}(f_\alpha - f_{\alpha 0})$, which gives the (nonlinear) effect on f_α due to the interaction of the field set up by ρ_s, $\mathbf{J}_s$, with the altered distribution, is neglected. The set of equations (11.10.2) to (11.10.5) is solved as in Chaps. 8 to 10, giving

for the Laplace-Fourier transform of the electrostatic and electromagnetic fields driven by ρ_s, $\mathbf{J}_s$,

$$\mathbf{k} \cdot \tilde{\mathbf{E}}(\mathbf{k}, \omega) = -\frac{4\pi i \tilde{\rho}_s(\mathbf{k}, \omega)}{D_L(\mathbf{k}, \omega)} \tag{11.10.6}$$

$$\mathbf{k} \times \tilde{\mathbf{E}}(k, \omega) = \frac{4\pi i \omega \mathbf{k} \times \tilde{\mathbf{J}}_s(\mathbf{k}, \omega)}{k^2 c^2 D_T(\mathbf{k}, \omega)} \tag{11.10.7}$$

where

$$\begin{aligned} D_L(\mathbf{k}, \omega) &= 1 + \sum_\alpha \frac{{\omega_{p\alpha}}^2}{k^2} \int_L \frac{\mathbf{k} \cdot \nabla_{\mathbf{v}} f_{\alpha 0}}{\omega - \mathbf{k} \cdot \mathbf{v}} \, d\mathbf{v} \\ D_T(\mathbf{k}, \omega) &= 1 - \frac{\omega^2}{k^2 c^2} + \sum \frac{{\omega_{p\alpha}}^2 \omega}{k^2 c^2} \int_L \frac{f_{\alpha 0}}{\omega - \mathbf{k} \cdot \mathbf{v}} \, d\mathbf{v} \end{aligned} \tag{11.10.8}$$

are evaluated along the Landau contour ($\omega \equiv \lim_{\varepsilon \to 0} \omega + i\varepsilon$).

The radiation emitted by the sources, over all time, is given by

$$W = \lim_{T \to \infty} \int_{-T}^{T} dt \int \mathbf{E}(\mathbf{x}, t) \cdot \mathbf{J}_s(\mathbf{x}, t) \, d\mathbf{x}$$

or by Parseval's theorem,

$$\begin{aligned} W &= \operatorname{Re}\left[\left\{\frac{1}{2(2\pi)^4} \int \tilde{\mathbf{E}}(\mathbf{k}, \omega) \cdot \dot{\mathbf{x}}^*(\mathbf{k}, \omega) \, d\mathbf{k} \, d\omega\right\}\right] \\ &= \frac{1}{(4\pi)^3} \operatorname{Im}\left\{\int \left[\frac{|\mathbf{k} \cdot \tilde{\mathbf{J}}_s(\mathbf{k}, \omega)|^2}{\omega k^2 D_L(\mathbf{k}, \omega)} - \frac{\omega |\mathbf{k} \times \tilde{\mathbf{J}}_s(\mathbf{k}, \omega)|^2}{k^4 c^2 D_T(\mathbf{k}, \omega)}\right] d\mathbf{k} d\omega\right\} \end{aligned} \tag{11.10.9}$$

Equation (11.10.9) can be extended to the case of more complicated equilibrium, to include, for example, the effect of a magnetic field.

11.10.1 Cerenkov Radiation

As an example of the use of (11.10.9), consider the radiation from a test charge, streaming through a plasma. In vacuum such a particle would not radiate. In a plasma it interacts to produce a sort of Cerenkov radiation. A test charge has ρ_s, $\mathbf{J}_s$ given by

$$\begin{aligned} \rho_s &= q_T \, \delta(\mathbf{x} - \mathbf{x}_0 - \mathbf{v}_0 t) \\ \mathbf{J}_s &= q_T \mathbf{v}_0 \, \delta(\mathbf{x} - \mathbf{x}_0 - \mathbf{v}_0 t) \end{aligned} \tag{11.10.10}$$

with a charge density transform

$$\tilde{\rho}_s(\mathbf{k}, \omega) = \frac{q_T \exp(-i\mathbf{k} \cdot \mathbf{x}_0)}{i(\omega - \mathbf{k} \cdot \mathbf{v}_0)} \qquad \operatorname{Im}(\omega) < 0$$

The *rate* of radiation is W/T. Noting that

$$\lim_{T \to \infty \, \varepsilon \to 0^+} \frac{1}{T} \frac{1}{(\omega - \mathbf{k} \cdot \mathbf{v}_0)^2 + \varepsilon^2} \Rightarrow \delta(\omega - \mathbf{k} \cdot \mathbf{v}_0)$$

Eq. (11.10.9) gives the rate of Cerenkov radiation of electrostatic waves as

$$\frac{W}{T} = \frac{q_T^2}{(4\pi)^3 2\pi} \operatorname{Im}\left(\int \frac{\omega \, \delta(\omega - \mathbf{k} \cdot \mathbf{v}_0)}{k^2 D_L(\mathbf{k}, \omega)}\right) d\omega \, d\mathbf{k} \tag{11.10.11}$$

This radiation is large only when the test-charge velocity exceeds the thermal velocity of plasma particles.

Problem 11.10.1 What is the equivalent of (11.10.9) for a maxwellian plasma in a uniform magnetic field $\mathbf{B}_0$? ////

11.10.2 Electron-Ion Bremsstrahlung

Section 11.10.1 computed the radiation as a test particle moved through a *continuous* plasma. Collisions between discrete particles also cause radiation, called bremsstrahlung. In this section the dipole radiation due to electron-ion collisions is calculated. (Collisions between particles of like charge are quadrupole.)

Consider a small volume of plasma in which collisions are taking place. The location of particle l of charge q_l is $\mathbf{x}_l$, and the net current is

$$\mathbf{J} = \sum_l q_l \dot{\mathbf{x}}_l = Ze \sum_i \dot{\mathbf{x}}_i - e \sum_e \dot{\mathbf{x}}_e \tag{11.10.12}$$

where the particles have been divided up into ions and electrons. The current can be expressed in terms of the fields due to the particles by taking d/dt of (11.10.12), using conservation of momentum $\sum_e m_e \ddot{\mathbf{x}}_e = -\sum_i m_i \ddot{\mathbf{x}}_i$ and noting that the ion acceleration by electrons is

$$\ddot{\mathbf{x}}_i = \frac{Ze}{m_i} \mathbf{E}_i$$

where $\mathbf{E}_i$ is the electric field at the location of the ith ion produced by the electrons. Thus

$$\frac{d\mathbf{J}}{dt} = \left(\frac{Z^2 e^2}{m_i} - \frac{Ze^2}{m_e}\right) \sum_i \mathbf{E}_i \tag{11.10.13}$$

The Laplace transform of this gives $\tilde{\mathbf{J}}$ in terms of $\tilde{\mathbf{E}}$.

$$\tilde{\mathbf{J}}(\mathbf{x}, \omega) = -\frac{i}{\omega} \left(\frac{Z^2 e^2}{m_i} - \frac{Ze^2}{m_e}\right) \sum_i \tilde{\mathbf{E}}_i(\omega)$$

The current *density* of these sources is found by adding up the current density due to each ion.

$$\tilde{\mathbf{J}}_s(\mathbf{x}, \omega) = -\frac{i}{\omega}\left(\frac{Z^2e^2}{m_i} - \frac{Ze^2}{m_e}\right)\sum_i \tilde{\mathbf{E}}_i(\omega)\,\delta(\mathbf{x} - \mathbf{x}_i)$$

and the Fourier transform is

$$\tilde{\mathbf{J}}_s(\mathbf{x}, \omega) = -\frac{i}{\omega}\left(\frac{Z^2e^2}{m_i} - \frac{Ze^2}{m_e}\right)\sum \tilde{\mathbf{E}}_i(\mathbf{x}_i, \omega)\exp(-i\mathbf{k}\cdot\mathbf{x}_i) \quad (11.10.14)$$

The electric field produced at the ion location $\mathbf{x}_i$ by an electron at $\mathbf{x}_e$ was calculated in Sec. 11.1.

$$\tilde{\mathbf{E}}_{ie}(\mathbf{x}_i, \mathbf{x}_e, \omega) = \frac{e}{\pi}\int \exp[i\mathbf{k}\cdot(\mathbf{x}_i - \mathbf{x}_e)]\,\frac{\delta(\omega - \mathbf{k}\cdot\mathbf{v}_{e0})i\mathbf{k}}{k^2 D_L(\mathbf{k}, \omega)}\,d\mathbf{k}$$

The radiation calculation requires $|\mathbf{J}_s(\mathbf{k}, \omega)|^2$; this is given for each test electron by squaring (11.10.14). The result must be summed over all electrons and the result ensemble averaged over the ion locations.

$$|\mathbf{J}_s(\mathbf{k}, \omega)|^2 = \frac{1}{\omega^2}\frac{Z^2e^4}{m_e^2}\sum_e\sum_i\sum_{i'}\langle\exp[i\mathbf{k}\cdot(\mathbf{x}_i - \mathbf{x}_{i'})]\tilde{\mathbf{E}}_i\cdot\tilde{\mathbf{E}}_{i'}\rangle$$

Neglecting correlations between ions, only terms $i = i'$ contribute to the ensemble average. The calculation from this point is lengthy but straightforward. From the form of (11.10.9), there is clearly strong radiation for $\mathbf{k}$, ω such that either $D_L(\mathbf{k}, \omega)$ or $D_T(\mathbf{k}, \omega)$ nearly vanishes; i.e., there is strong radiation into the normal modes of the plasma. Retaining only this wave emission, it can be shown that the radiation from a single encounter of an ion with a shielded electron approaching at impact parameter $\mathbf{b}_{ie}$ and velocity $\mathbf{v}_0$ is

$$W_\omega = \frac{Z^2e^6(\omega^2 - \omega_p^2)^{1/2}}{6\pi^3 m_e^2}\begin{bmatrix}(\omega u_0{}^3 3^{3/2})^{-1}\\ \left(\dfrac{3c^3\omega}{2}\right)^{-1}\end{bmatrix}$$

$$\times\int\frac{\mathbf{k}\cdot\mathbf{k}'\exp[i(\mathbf{k} - \mathbf{k}')\cdot\mathbf{b}_{ie}]\,\delta(\omega - \mathbf{k}\cdot\mathbf{v}_{e0})\,\delta(\omega - \mathbf{k}'\cdot\mathbf{v}_{e0})}{k^2k'^2D_L(\mathbf{k}, \omega)D_L^*(\mathbf{k}', \omega)}\,d\mathbf{k}\,d\mathbf{k}'$$

where $u_0 = (\kappa T_e/m_e)^{1/2}$ is the thermal speed of electrons in the plasma, the upper term in the bracket gives the radiation of longitudinal waves, and the lower term gives the radiation of transverse waves.

Adding over electrons gives the total power radiated per unit volume.[1]

$$P(\omega) = \frac{4Z^2e^6n_in_e}{3\pi m_e^2}\begin{bmatrix}(u_0{}^3\omega 3^{3/2})^{-1}\\ \left(\dfrac{3c^3\omega}{2}\right)^{-1}\end{bmatrix}(\omega^2 - \omega_p{}^2)^{1/2}\int f_{e0}(\mathbf{v}_e)\,\frac{\delta\,|\omega - \mathbf{k}\cdot\mathbf{v}_e|}{k^2\,|\,D_L(\mathbf{k}, \omega)\,|^2}\,d\mathbf{k}\,d\mathbf{v}_e$$

For thermal equilibrium this agrees with the radiation calculated through Kirchhoff's law (which balances emission with absorption). Another interesting limit is $\omega \gg \omega_p$, in which case $D_L \approx 1$, and the result reduces to the radiation from a lone electron-ion encounter in a vacuum.[2]

Many other radiation processes (coupling of transverse and longitudinal waves by density fluctuations, electron-electron bremsstrahlung, etc.) are important in plasmas; the reader is referred to research monographs (e.g., Dawson, *loc. cit.*) for further reading on this topic. Another active branch of plasma physics, well covered in specialized texts,[3] is that of radiation from neutral atoms imbedded in plasmas, which have proved a useful plasma diagnostic.

11.11 KINETIC EQUATIONS INCLUDING COLLISIONAL RELAXATION OF A PLASMA

Although the Vlasov theory treats correctly the average interactions between plasma particles, the effect of a close encounter between two particles (i.e., collisions) is an effect of the *discreteness* of the plasma, and occurs in terms of order $g \equiv 1/n\lambda_D{}^3$ in the kinetic equation. The Fokker-Planck equation, a model for treating collisional relaxation of a plasma, was presented in Chap. 6. That equation calculated the change in the plasma distribution due to a binary collision, in terms of the cross section for two-body scattering. Although that approach seems plausible, the results of this chapter show that a discrete particle in a plasma is not seen as a bare particle, but carries its shielding cloud with it during the scattering process, and this shielding must be somehow included in calculating the effect of collisions. Said another way, a binary collision of charged particles in a dielectric medium is different from a collision in vacuum.

[1] J. M. Dawson, Radiation from a Plasma, in A. Simon and W. B. Thompson (eds.), "Advances in Plasma Physics," vol. 1, Wiley, New York, 1968. This article lists much of the background material and references for the test-source method. See also, for example, D. Tidman and T. Dupree, *Phys. Fluids*, **5**:1860 (1965); T. Birmingham, J. Dawson, and C. Oberman, *Phys. Fluids*, **5**:297 (1965).

[2] P. A. G. Scheuer, *Monthly Notice Roy. Astron. Soc.*, **120**:231 (1960).

[3] H. R. Griem, "Plasma Spectroscopy," McGraw-Hill, New York, 1964.

A kinetic theory that includes the effect of collisional relaxation can be developed in a number of ways.[1] In this section a treatment is presented that is similar to the quasilinear theory (Chap. 10) which was developed to calculate the relaxation of an unstable plasma.

Consider a stable plasma that includes N_α particles of type α and has an average field $\mathbf{E} = 0$. The distribution of these particles is described exactly in terms of the orbits of the particles and their locations at time τ,

$$F_\alpha(\mathbf{x}, \mathbf{v}, t) = \sum_{s=1}^{N_\alpha} \delta[\mathbf{x} - \mathbf{x}_s(t)]\, \delta[\mathbf{v} - \mathbf{v}_s(t)] \quad (11.11.1)$$

$$F_\alpha(\mathbf{x}, \mathbf{v}, t = \tau) = \sum_{s=1}^{N_\alpha} \delta(\mathbf{x} - \bar{\mathbf{x}}_s)\, \delta(\mathbf{v} - \bar{\mathbf{v}}_s) \quad (11.11.2)$$

The distribution also satisfies the Liouville equation, as in Chap. 7.

$$\frac{\partial F_\alpha}{\partial t} + \mathbf{v} \cdot \nabla F_\alpha + \frac{q_\alpha}{m_\alpha} \mathbf{E} \cdot \nabla_\mathbf{v} F_\alpha = 0 \quad (11.11.3)$$

$$\nabla \cdot \mathbf{E} = 4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int F_\alpha \, d\mathbf{v} \quad (11.11.4)$$

where $\bar{n}_\alpha = N_\alpha / V$, with V the volume of the system.

The Fourier transforms of F_α, $\mathbf{E}$ and the equations they satisfy are

$$\mathbf{E}_\mathbf{k} = \int \exp(i\mathbf{k} \cdot \mathbf{x}) \mathbf{E} \, d\mathbf{x}$$

$$F_{\alpha\mathbf{k}}(\mathbf{v}, t) = \sum_{s=1}^{N_\alpha} \delta[\mathbf{v} - \mathbf{v}_s(t)] \exp(i\mathbf{k} \cdot \mathbf{x}_s) \quad (11.11.5)$$

$$\frac{\partial F_{\alpha\mathbf{k}}}{\partial t} + i\mathbf{k} \cdot \mathbf{v} F_{\alpha\mathbf{k}} + \frac{q_\alpha}{m_\alpha} \mathbf{E}_\mathbf{k} \cdot \nabla_\mathbf{v} F_{\alpha,\,\mathbf{k}=0} = -\frac{q_\alpha}{m_\alpha} \sum_{k' \neq k} \mathbf{E}_{\mathbf{k}'} \cdot \nabla_\mathbf{v} F_{\alpha(\mathbf{k}-\mathbf{k}')}\, i\mathbf{k} \cdot \mathbf{E}_\mathbf{k}$$

$$i\mathbf{k} \cdot \mathbf{E}_\mathbf{k} = 4\pi \sum_\alpha \bar{n}_\alpha q_\alpha \int F_{\alpha\mathbf{k}} \, d\mathbf{v} \quad (11.11.6)$$

where the $k = 0$ terms in $\mathbf{E} \cdot \nabla_\mathbf{v} F_\alpha$ has been written separately.

[1] The kinetic equation derived in this section was first obtained by:

R. Balescu, *Phys. Fluids*, **3**:52 (1960).
A. Lenard, *Ann. Phys. (N.Y.)*, **3**:390 (1960).
N. Rostoker and M. N. Rosenbluth, *Phys. Fluids*, **3**:1 (1960).

The treatment in this section follows closely:

Y. L. Klimontovich, "The Statistical Theory of Nonequilibrium Processes in a Plasma," pp. 180–192, M.I.T., Cambridge, Mass., 1967.
T. H. Dupree, *Phys. Fluids*, **2**:696 (1961).
S. Hamasaki, private communication.

Now it is reasonable to expect that $F_{\alpha,\,\mathbf{k}=0}$, which represents a *homogeneous* system, is larger than the $F_{\alpha\mathbf{k}}$, which measures the graininess of the plasma. Since, in the Vlasov theory of a stable plasma in equilibrium, $F_{\alpha,\,\mathbf{k}=0}$ was a complete description. This suggests the following approach: Treat $F_{\alpha,\,\mathbf{k}=0} \gg F_{\alpha\mathbf{k}}$, and so neglect $\sum_{\mathbf{k}'=\mathbf{k}} \mathbf{E}_{\mathbf{k}'} \cdot \nabla_\mathbf{v} F_{\alpha(\mathbf{k}-\mathbf{k}')}$ in (11.11.6). The equations which describe the system are then

$$\frac{\partial F_{\alpha\mathbf{k}}}{\partial t} + i\mathbf{k} \cdot \mathbf{v} F_{\alpha\mathbf{k}} + \frac{q_\alpha}{m_\alpha} \mathbf{E}_\mathbf{k} \cdot \nabla_\mathbf{v} F_{\alpha,\,\mathbf{k}=0} = 0 \quad (11.11.7)$$

$$\frac{\partial F_{\alpha,\,\mathbf{k}=0}}{\partial t} = -\frac{q_\alpha}{m_\alpha} \sum_{\mathbf{k}'} \mathbf{E}_{\mathbf{k}'} \cdot \nabla_\mathbf{v} F_{\alpha\mathbf{k}'} \quad (11.11.8)$$

$$\nabla \cdot \mathbf{E}_\mathbf{k} = \sum_\alpha 4\pi \bar{n}_\alpha q_\alpha \int F_{\alpha\mathbf{k}} \, d\mathbf{v} \quad (11.11.9)$$

$$F_{\alpha\mathbf{k}}(t=0) = \sum_{s=0}^{N_\alpha} \delta(\mathbf{v} - \mathbf{v}_{s0}) \exp(i\mathbf{k} \cdot \mathbf{x}_{s0})$$

The similarity between these equations and the quasilinear equations (Chap. 10) is evident. The difference is that in an unstable plasma the *Stoss* term $\sum_{\mathbf{k}'} \mathbf{E}_{\mathbf{k}'} \cdot \nabla_\mathbf{v} F_{\alpha\mathbf{k}'}$ is due to collective effects, while in the present case it is due to the graininess of the plasma. In the same vein as the quasilinear theory, the fact that $\partial F_{\alpha,\,k=0}/\partial t \approx O[(q_\alpha/m_\alpha)(\mathbf{E}_\mathbf{k} \cdot \nabla_\mathbf{v} F_{\alpha\mathbf{k}})]$ is small implies that the *principal* plasma description $F_{\alpha 0}$ describes a situation that is slowly varying, as the plasma relaxes toward equilibrium, where "slowly" is in comparison with the time scale of fields which arise as discrete particles pass by each other in the plasma.

The solution to (11.11.7) and (11.11.9) in terms of the initial values (11.11.2) is obtained just as in the many other situations in this text where that pair of equations has occurred. Defining the Laplace transforms,

$$\tilde{F}_{\alpha\mathbf{k}}(\mathbf{v}, p) = \int_0^\infty F_{\alpha\mathbf{k}}(\mathbf{v}, t) \exp(-pt) \, dt \qquad \mathrm{Re}\,(p) > 0$$

$$F_{\alpha\mathbf{k}=0} \equiv F_{\alpha 0}$$

the transformed fields and distributions are

$$\tilde{F}_{\alpha\mathbf{k}}(\mathbf{v}, p) = \frac{F_{\alpha\mathbf{k}}(\mathbf{v}, t=0) - (q_\alpha/m_\alpha)\tilde{\mathbf{E}}_\mathbf{k} \cdot \nabla_\mathbf{v} F_{\alpha,\,\mathbf{k}=0}(\mathbf{v}, t)}{i\mathbf{k} \cdot \mathbf{v} - p} \quad (11.11.10)$$

$$i\mathbf{k} \cdot \tilde{\mathbf{E}}_\mathbf{k}(p) = \frac{1}{D(\mathbf{k}, ip)} \sum_\alpha 4\pi \bar{n}_\alpha q_\alpha \int \frac{F_{\alpha\mathbf{k}}(\mathbf{v}, t=0)}{i\mathbf{k} \cdot \mathbf{v} - p} \, d\mathbf{v} \quad (11.11.11)$$

$$D(\mathbf{k}, ip) = 1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2} \int \frac{\mathbf{k} \cdot \nabla_\mathbf{v} F_{\alpha 0}}{\mathbf{k} \cdot \mathbf{v} + ip} \, d\mathbf{v} \quad (11.11.12)$$

The Laplace transforms are inverted to get the time-dependent fields and distribution functions. There are two contributions to the inversion integral, one from $D(\mathbf{k}, ip) = 0$, one from $p = i\mathbf{k}\cdot\mathbf{v}$. The former gives terms proportional to

$$\exp(-i\omega_{\mathbf{k}} t)$$

where $\omega_{\mathbf{k}}$ is defined by

$$D(\mathbf{k}, \omega_{\mathbf{k}}) = 0$$

But the normal modes of a stable plasma are *damped*, so that these terms die away with time. Retaining only the $p = i\mathbf{k}\cdot\mathbf{v}$ terms, and using,

$$i\mathbf{k}\cdot\mathbf{E}_{\mathbf{k}}(t) = -\sum_{\alpha}\sum_{s=1}^{N_\alpha} \frac{4\pi\bar{n}_\alpha q_\alpha \exp(i\mathbf{k}\cdot\mathbf{v}_{s0}\, t)\exp(i\mathbf{k}\cdot\mathbf{x}_{s0})}{D(\mathbf{k}, -\mathbf{k}\cdot\mathbf{v}_{s0})} \quad (11.11.13)$$

$$F_{\alpha\mathbf{k}}(t) = -\sum_{s=1}\delta(\mathbf{v}-\mathbf{v}_s)\exp(i\mathbf{k}\cdot\mathbf{x}_{s0})\exp(i\mathbf{k}\cdot\mathbf{v}t) - \frac{q_\alpha}{m_\alpha}\sum_{\beta}\sum_{s=1}^{N_\beta}$$
$$\times \frac{4\pi\bar{n}_\beta q_\beta(\mathbf{k}\cdot\nabla_{\mathbf{v}}F_{\alpha 0})}{k^2(\mathbf{k}\cdot\mathbf{v}-\mathbf{k}\cdot\mathbf{v}_s)}\left[\frac{1}{D(\mathbf{k}, -\mathbf{k}\cdot\mathbf{v}_{s0})}\exp(i\mathbf{k}\cdot\mathbf{v}_{s0}\, t)\exp(i\mathbf{k}\cdot\mathbf{x}_{s0})\right.$$
$$\left. - \frac{1}{D(\mathbf{k}, -\mathbf{k}\cdot\mathbf{v})}\exp(i\mathbf{k}\cdot\mathbf{v}t)\exp(i\mathbf{k}\cdot\mathbf{x}_{s0})\right] \quad (11.11.14)$$

$$\frac{\partial F_{\alpha 0}}{\partial t}(\mathbf{v}, t) = -\frac{q_\alpha}{m_\alpha}\sum_{\mathbf{k}'}\mathbf{E}_{-\mathbf{k}'}(t)\cdot\nabla_{\mathbf{v}}F_{\alpha\mathbf{k}'}(t) \quad (11.11.15)$$

Equations (11.11.13) and (11.11.14) can be inserted in (11.11.15) to obtain an equation for the collisional relaxation of a stable plasma. Two further refinements must be made before the result is of practical use. Since the time scale of practical interest is the time scale on which the system evolves, (11.11.15) may be averaged over the fast oscillations $\exp(i\mathbf{k}\cdot\mathbf{v}t)$. Further, (11.11.15) should be averaged over the ensemble of N_α discrete particles; i.e., the sum over individual particles $\sum_{s=1}^{N_\alpha} A(\mathbf{v}_s)$ should be replaced by $\int F_\alpha(\mathbf{v}')A(\mathbf{v}')\, d\mathbf{v}'$, so that (11.11.15) reads

$$\frac{\partial F_{\alpha 0}}{\partial t} = -\frac{q_\alpha}{m_\alpha}\left\langle\sum_{\mathbf{k}'}\overline{E_{-\mathbf{k}'}(t)\cdot\nabla_{\mathbf{v}}F_{\alpha\mathbf{k}'}(t)}\right\rangle \quad (11.11.16)$$

where $\langle\ \rangle$ indicates the ensemble average, and —— indicates the average $(1/T)\int_0^T dt$ over the fast time scale. With these manipulations, (11.11.16) becomes

$$\frac{\partial F_{\alpha 0}}{\partial t} = \frac{\partial}{\partial\mathbf{v}}\cdot\sum_{\beta}\bar{n}_\beta\int\frac{\mathbf{k}}{m_\alpha}\left(\frac{4\pi q_\alpha q_\beta}{k^2}\right)^2 d\mathbf{k}$$
$$\times\int\frac{\delta(\mathbf{k}\cdot\mathbf{v}-\mathbf{k}\cdot\mathbf{v}')}{|D(\mathbf{k}, -i\mathbf{k}\cdot\mathbf{v})|^2}\left(\frac{1}{m_\alpha}\nabla_{\mathbf{v}} - \frac{1}{m_\beta}\nabla_{\mathbf{v}'}\right)F_\alpha(\mathbf{v}, t)F_\alpha(\mathbf{v}', t)\, d\mathbf{v}' \quad (11.11.17)$$

This is the *Balescu-Lenard equation*. It has the form of the Fokker-Planck equation, but takes into account the shielding and dielectric properties of the plasma.

Problem 11.11.1 Derive the Fokker-Planck equation, as in Chap. 6, but with the force between the two scattering particles as given by the shielded Coulomb potential (Sec. 11.1). Show that the result is the same as in (11.11.17). [See, for example, I. B. Bernstein, S. K. Trehan, and M. P. H. Weeninks, Plasma Oscillations; II, Kinetic Theory of Waves in Plasmas, Nucl. Fusion, **4**:61 (1964).] ////

Problem 11.11.2 Obtain (11.11.17) from (11.11.13) to (11.11.15). Show that the term $\int G(\mathbf{v}, \mathbf{v}')\,\nabla_{\mathbf{v}'}F_{\alpha 0}(\mathbf{v}')F_{\alpha 0}(\mathbf{v})\, d\mathbf{v}'$ comes from the term $\sum_s \delta(\mathbf{v}-\mathbf{v}_s)$ in $F_{\alpha\mathbf{k}}$, while the term $\int G(\mathbf{v}, \mathbf{v}')\,\nabla_{\mathbf{v}}\mathbf{F}_{\alpha 0}(\mathbf{v})F_{\alpha 0}(\mathbf{v}')\, d\mathbf{v}'$ comes from combining the other two terms in $F_{\alpha\mathbf{k}}$. ////

Problem 11.11.3 Show that an alternative to singling out $F_{\alpha, k=0}$ is to calculate the collisional relaxation of the space average distribution, defined by

$$F_{\alpha 0} \equiv \frac{1}{V}\int F_\alpha\, d\mathbf{x}$$

and that the same result (11.11.17) is obtained. ////

In deriving the Balescu-Lenard equation, the fields due to plasma waves (as opposed to the fields due to individual particles) were neglected, when the terms proportional to $\exp(-i\omega_k t)$ were neglected. In practice, especially when the system is far from thermal equilibrium, the wave fields can be just as important as the collisional effects in the time evolution of the system. Indeed, for the unstable plasma, the wave fields are all-important. It is possible to develop an equation equivalent to the Balescu-Lenard equation, including wave and instability effects, by retaining the $\exp(-i\omega_{\mathbf{k}} t)$ terms in $E_{\mathbf{k}}(t)$ and $F_{\alpha\mathbf{k}}(t)$. The calculation is tedious, but the result can be written in a very appealing form:[1]

$$\frac{\partial F_\alpha}{\partial t} = \left.\frac{\partial F_\alpha}{\partial t}\right|_{\text{collective}} - \frac{\partial}{\partial\mathbf{v}}\cdot\mathbf{J}_\alpha \quad (11.11.18)$$

$(\partial F_\alpha/\partial t)_{\text{collective}}$ contains the effect of linear and nonlinear wave-particle processes and nonlinear wave-wave processes, e.g., the effects of instability, quasilinear diffusion, wave-wave coupling, etc., as presented in Chap. 10. The

[1] R. C. Davidson, "Methods in Nonlinear Plasma Theory," Academic, New York, 1971, sec. 14.6 and references therein.

current $\mathbf{J}_\alpha$ includes collisions and other particle effects, such as the emission of waves by the test particles.

$$\mathbf{J}_\alpha = -\sum_\beta \sum_\gamma \bar{n}_\beta \int \frac{\mathbf{k}}{m_\alpha} \left(\frac{4\pi q_\alpha q_\beta}{k^2}\right)^2 d\mathbf{k}$$
$$\times \int \left\{ \frac{1}{|\partial D(\mathbf{k}, \omega)/\partial\omega|^2_{\omega=\omega_k\gamma+i\varepsilon}} \left[\frac{\delta(\omega_\mathbf{k} - \mathbf{k}\cdot\mathbf{v}')}{(\omega_\mathbf{k}{}^\gamma - \mathbf{k}\cdot\mathbf{v} + i\varepsilon)^2} F_\beta(\mathbf{v}', t) \frac{1}{m_\alpha} \right.\right.$$
$$\mathbf{k}\cdot\nabla_\mathbf{v} F_\alpha(\mathbf{v}, t) - \frac{\delta(\omega_\mathbf{k} - \mathbf{k}\cdot\mathbf{v})}{(\omega_\mathbf{k}{}^\gamma - \mathbf{k}\cdot\mathbf{v}' + i\varepsilon)^2} F_\alpha(\mathbf{v}, t) \frac{1}{m_\beta} \mathbf{k}\cdot\nabla_{\mathbf{v}'} F_\beta(\mathbf{v}', t) \Big]$$
$$+ \left[P \frac{1}{|D(\mathbf{k}, -i\mathbf{k}\cdot\mathbf{v} + \varepsilon)|^2} \right] \delta(\mathbf{k}\cdot\mathbf{v} - \mathbf{k}\cdot\mathbf{v}') \left(\frac{1}{m_\alpha} \nabla_\mathbf{v} - \frac{1}{m_\beta} \nabla_{\mathbf{v}'} \right)$$
$$\left. \times F_\alpha(\mathbf{v}, t) F_\beta(\mathbf{v}', t) \right\} d\mathbf{v}' \quad (11.11.19)$$

where P denotes the principle value.

The sum $\sum_\gamma$ is over the different zeros of $D(\mathbf{k}, \omega_\mathbf{k}) = 0$, for a given k, that is, over different *types* of plasma waves. In this equation the first terms in $\int d\mathbf{v}'$ represent spontaneous emission of waves by particles; the first of these terms is the nonresonant emission, and the second is the resonant emission. The third term in $\int d\mathbf{v}'$ is the collision integral derived in Chap. 6 for Fokker-Planck theory, and in this chapter as the Balescu-Lenard equation. Note that in the unstable limit, the collective term grows with time, and $\mathbf{J}_\alpha$ can be ignored. In the nearly equilibrium case, emission and absorption of waves are in balance, and the collision term dominates further relaxation of the system.

Because of the similarity of the Fokker-Planck and Balescu-Lenard equations, the examples of the use of the former, given in Chap. 6, suffice for the latter as well.

The listing and proof of the properties of the Balescu-Lenard equations are left for the following series of problems.

Problem 11.11.4 Show that the Balescu-Lenard equation (11.11.17) does not require a cutoff for small k, while the Fokker-Planck equation did require a cutoff. Relate this fact to the shielding property of the plasma. ////

Problem 11.11.5 Show that (11.11.17) does require a cutoff at large $\mathbf{k}$. This is because large $\mathbf{k}$ corresponds to close encounters (strong potential) between particles, and the theory derived here assumed that the kinetic energy of the particles was much *greater* than the interaction energy; i.e., the expansion in $g = 1/n\lambda_D{}^3$ breaks down for the large $\mathbf{k}$ components of the distribution. ////

Problem 11.11.6 Show that an F_α which satisfies (11.11.17) has the following properties:[1]

1. $F_{\alpha 0}$ remains positive.
2. $\int F_{\alpha 0}\, d\mathbf{v}$ is time-independent.
3. $\sum_\alpha \int \mathbf{v} F_{\alpha 0}\, d\mathbf{v}$ is time-independent.
4. $\sum_\alpha \int \mathbf{v}^2 F_{\alpha 0}\, d\mathbf{v}$ is time-independent. ////

Problem 11.11.7 Show that any maxwellian distribution is a time-independent solution of (11.11.17). Show that at $t \to \infty$, and nonmaxwellian $F_{\alpha 0}$ approaches a Maxwell distribution (H theorem). ////

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APPENDIX I

PARTICLE MOTION

Plasmas are made up of charged particles that move in response to the electromagnetic fields generated either by the plasma particles themselves or from external sources. This appendix summarizes the solutions of the equations of motion of individual charged particles in various static and dynamic electromagnetic fields. It illustrates some plasma properties directly from the equations for single-particle motion, and discusses adiabatic invariants, which are useful in understanding the motion in complicated field geometries.

This study of the motion of charged particles provides a natural bridge between the macroscopic fluid theory and the microscopic Vlasov theory. In the limit of a cold plasma, the fluid elements follow the trajectory of a charged particle, since all the particles move in concert. However, in a warm plasma the velocity- and configuration-space motions of individual particles are ensemble-averaged to give the statistical kinetic equation describing plasma properties. In either case, it is necessary to be able to calculate the motion of individual particles in the various situations that arise in plasma physics.

I.1 THE EQUATIONS OF MOTION

The position vector $\mathbf{r}(t)$ describes the time history of motion of a point particle. The velocity vector $\mathbf{v}(t)$ is defined by

$$\mathbf{v}(t) = \frac{d\mathbf{r}(t)}{dt} = \dot{\mathbf{r}}(t) \qquad \text{(I.1.1)}$$

The nonrelativistic equation of motion for a particle of mass m and charge q in an electric and magnetic field is

$$\dot{\mathbf{v}}(t) = \frac{q}{m}\left(\mathbf{E} + \frac{\mathbf{v}(t) \times \mathbf{B}}{c}\right) \qquad \text{(I.1.2)}$$

These equations of motion will now be solved for several field configurations and conditions.

Problem I.1.1 At what energy in kilovolts are the relativistic corrections to the equations of motion significant for electrons and for ions? ////

Problem I.1.2 If a gravitational field is included in (I.1.2), what form does the equation have? How long will it take the earth's gravitational field to accelerate an electron and an ion to the energies of Prob. I.1.1? What distance is required to realize this velocity? ////

I.2 PARTICLE MOTION IN STATIC HOMOGENEOUS ELECTRIC AND MAGNETIC FIELDS

I.2.1 Field-free Motion ($\mathbf{E}_0 = \mathbf{B}_0 = 0$)

For the case of zero electric and magnetic field, the velocity $\mathbf{v}_0$, the momentum $m\mathbf{v}_0$, and the kinetic energy $mv_0^2/2$ are all constants of the motion.

I.2.2 Motion in a Static Electric Field ($\mathbf{B}_0 = 0, \mathbf{E}_0 \neq 0$)

For the case of zero magnetic field and static electric field, (I.1.2) has the solution

$$\mathbf{v}(t) = \mathbf{v}_0 + \frac{q}{m}\mathbf{E}t \qquad \text{(I.2.1)}$$

The kinetic energy and the rate of change of kinetic energy are (taking $\mathbf{v}_0 = 0$)

$$W = \tfrac{1}{2}mv^2 = \frac{(qEt)^2}{2m} \qquad \text{(I.2.2)}$$

$$\frac{dW}{dt} = \frac{(qE)^2 t}{m} \qquad \text{(I.2.3)}$$

I.2.3 Motion in a Static Magnetic Field ($\mathbf{E}_0 = 0, \mathbf{B}_0 \neq 0$)

For the case of zero electric field and static magnetic field, the equation of motion (I.1.2) is

$$\dot{\mathbf{v}} = \frac{q}{m}\frac{\mathbf{v} \times \mathbf{B}_0}{c} \qquad \text{(I.2.4)}$$

Scalar multiplication of this equation by $m\mathbf{v}$ gives

$$m\mathbf{v} \cdot \dot{\mathbf{v}} = \frac{d}{dt}(\tfrac{1}{2}mv^2) = q\frac{\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}_0)}{c} = 0 \qquad \text{(I.2.5)}$$

Thus the kinetic energy of a particle in a uniform magnetic field is a constant of the motion.

A unit vector along the magnetic field is defined by

$$\hat{\mathbf{b}} = \frac{\mathbf{B}_0}{|\mathbf{B}_0|} \qquad \text{(I.2.6)}$$

Scalar multiplication of (I.2.4) by $\hat{\mathbf{b}}$ gives the component of acceleration along the magnetic field; i.e.,

$$\dot{v}_\parallel = \frac{q}{m}\frac{\hat{\mathbf{b}} \cdot (\mathbf{v} \times \mathbf{B})}{c} = 0 \qquad \text{(I.2.7)}$$

where $v_\parallel = \mathbf{v} \cdot \hat{\mathbf{b}}$ is the velocity along the magnetic field. Thus the velocity $v_\parallel$ and the kinetic energy along the field $d(\frac{1}{2}mv_\parallel^2)/dt$ are constants of the motion. This means that particle motion perpendicular to the magnetic field can be examined in a coordinate system where $v_\parallel = 0$, without loss of generality. Since the total energy of a charged particle in a static magnetic field is constant, the perpendicular energy is also constant. Conservation of energy requires that

$$W = W_\parallel + W_\perp = \text{constant} \qquad \text{(I.2.8)}$$

The velocity perpendicular to the magnetic field is defined by

$$\mathbf{v}_\perp = \mathbf{v} - (\mathbf{v} \cdot \hat{\mathbf{b}})\hat{\mathbf{b}} \qquad \text{(I.2.9)}$$

and a unit vector in the direction of $\mathbf{v}_\perp$ is defined by

$$\mathbf{v}_\perp = \hat{\mathbf{e}}v_\perp \qquad \text{(I.2.10)}$$

In terms of the defined unit vectors, (I.2.4) is

$$\dot{\mathbf{v}}_\perp = \dot{v}_\perp \hat{\mathbf{e}} + v_\perp \dot{\hat{\mathbf{e}}} = \omega_c v_\perp (\hat{\mathbf{e}} \times \mathbf{b}) \qquad \text{(I.2.11)}$$

where $\omega_c = qB_0/mc$.

Scalar multiplication of (I.2.11) by $\hat{\mathbf{e}}$ gives

$$\hat{\mathbf{e}} \cdot \hat{\mathbf{e}}\dot{v}_\perp + v_\perp \hat{\mathbf{e}} \cdot \dot{\hat{\mathbf{e}}} = 0 \qquad \text{(I.2.12)}$$

The second term in (I.2.12) is zero, and thus the first term is also zero. Therefore the perpendicular kinetic energy is a constant of the motion, as expected from (I.2.8).

From (I.2.12) it follows that (I.2.11) can be written

$$\dot{\hat{\mathbf{e}}} = \omega_c(\hat{\mathbf{e}} \times \hat{\mathbf{b}}) \qquad \text{(I.2.13)}$$

Scalar multiplication of (I.2.13) by $(\hat{\mathbf{e}} \times \hat{\mathbf{b}})$ gives

$$(\hat{\mathbf{e}} \times \hat{\mathbf{b}}) \cdot \dot{\hat{\mathbf{e}}} = \omega_c \qquad \text{(I.2.14)}$$

Using the relations among the unit vectors, it follows that

$$\dot{\hat{\mathbf{e}}} = -(\hat{\mathbf{e}} \times \hat{\mathbf{b}})\dot{\theta} \qquad \text{(I.2.15)}$$

where θ is the angle that $\mathbf{v}_\perp$ makes with respect to a fixed axis perpendicular to $\hat{\mathbf{b}}$. Thus the angle θ is decreasing ($d\theta/dt = -\omega_c$) for positive particles. Positive particles therefore circulate clockwise in a circular orbit. This is shown for the magnetic field out of the page as in Fig. I.2.1. An easy way to remember the direction of circulation is to apply the right-hand current rule and notice that (I.2.15) shows that the magnetic field set up by the particle is opposite to the original magnetic field. The position of a charged particle in a magnetic field as a function of time is given by

$$\mathbf{r}(t) = a_c[\hat{\mathbf{x}} \sin(\omega_c t - \phi) + \hat{\mathbf{y}} \cos(\omega_c t - \phi)] \qquad \text{(I.2.16)}$$

where the magnetic field is in the z direction, the angle ϕ gives the position at $t = 0$, and the origin of coordinates is taken at the center of the particle's orbit.

The size of the circular orbit is related to the velocity

$$v_\perp = a_c \omega_c \qquad \text{(I.2.17)}$$

where a_c is the cyclotron radius.

The magnetic moment of a charged particle circulating in a magnetic field is defined as the product of its circulating current and the area enclosed by the orbit. The magnetic moment can be expressed in terms of the particle energy and magnetic field as follows:

$$\mu = \text{magnetic moment} = \left(\frac{q}{c}\frac{\omega_c}{2\pi}\right)\pi a_c^2$$

$$= \frac{W_\perp}{B} \qquad \text{(I.2.18)}$$

This magnetic moment may be regarded as a vector by using the fact that the field set up by the circulating particle opposes the original field

$$\boldsymbol{\mu} = -\frac{W_\perp}{B}\hat{\mathbf{b}} \qquad \text{(I.2.19)}$$

Problem I.2.1 What is the cyclotron frequency and orbit radius for a 250-keV electron (and proton) in a 30-kG magnetic field? How big is the relativistic correction? ////

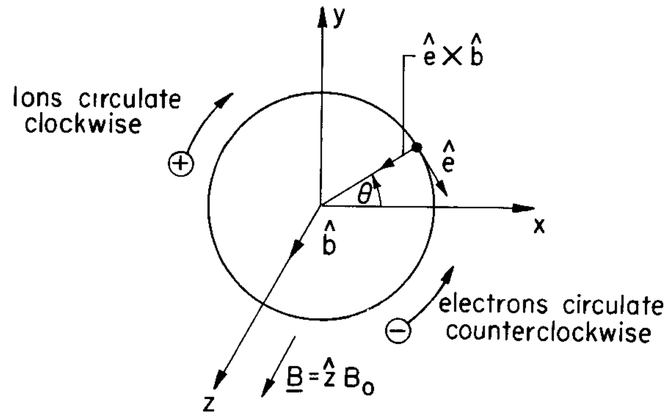


FIGURE I.2.1
Ion and electron motion in a homogeneous static magnetic field. A single underscore signifies a vector.

I.2.4 Motion in Combined Static Electric and Magnetic Fields ($\mathbf{E}_0 \neq 0$, $\mathbf{B}_0 \neq 0$)

To analyze the motion for this situation, it is convenient to separate the electric field into components parallel and perpendicular to the magnetic field, i.e.,

$$\mathbf{E} = \hat{\mathbf{b}}E_\parallel + \mathbf{E}_\perp \qquad \text{(I.2.20)}$$

where $E_\parallel = \mathbf{E} \cdot \hat{\mathbf{b}}$ and $\mathbf{E}_\perp = \mathbf{E} - (\mathbf{E} \cdot \hat{\mathbf{b}})\hat{\mathbf{b}}$. Substituting (I.2.20) into (I.1.2) and taking the scalar product with $\hat{\mathbf{b}}$ gives

$$\hat{\mathbf{b}} \cdot \dot{\mathbf{v}} = \frac{q}{m}E_\parallel = \dot{v}_\parallel \qquad \text{(I.2.21)}$$

This shows that the motion along $\mathbf{B}$ is the same as for the zero magnetic field case.

The equation of motion for the perpendicular component of the velocity is

$$\dot{\mathbf{v}}_\perp = \frac{q}{m}\left(\mathbf{E}_\perp + \frac{\mathbf{v}_\perp \times \mathbf{B}_0}{c}\right) \qquad \text{(I.2.22)}$$

To solve this equation it is convenient to define a new dependent variable $\bar{\mathbf{v}}_\perp$ by

$$\mathbf{v}_\perp = \bar{\mathbf{v}}_\perp + \frac{\mathbf{E}_\perp \times \mathbf{B}_0}{|B_0|^2}c = \bar{\mathbf{v}}_\perp + \mathbf{V}_{DE} \qquad \text{(I.2.23)}$$

The quantity

$$\mathbf{V}_{DE} = \frac{\mathbf{E}_\perp \times \mathbf{B}_0}{|B_0|^2}c \qquad \text{(I.2.24)}$$

is the drift velocity of the particle perpendicular to both the electric and magnetic fields. It is independent of both charge and mass.

In terms of $\bar{\mathbf{v}}_\perp$, (I.2.22) can be written

$$\dot{\bar{\mathbf{v}}}_\perp = \omega_c(\bar{\mathbf{v}}_\perp \times \hat{\mathbf{b}}) \qquad \text{(I.2.25)}$$

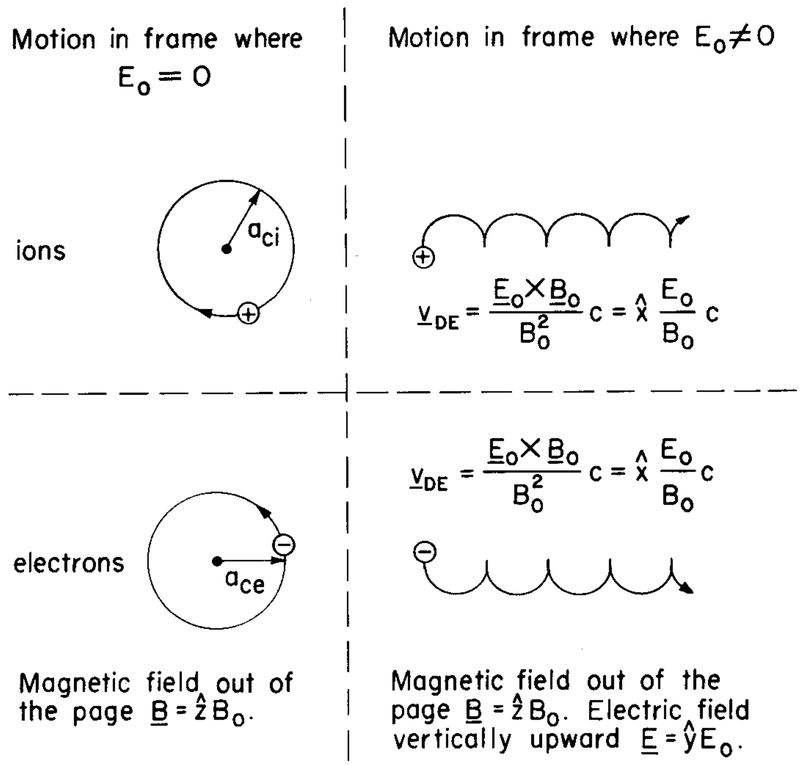


FIGURE I.2.2
Ion and electron motion in a homogeneous static magnetic field at right angles to a homogeneous static electric field showing the **E** × **B** drift motion.

This equation for $\bar{v}_\perp$ is the same as the equation for the perpendicular component of the velocity of a particle in a homogeneous magnetic field (I.2.11), and the motion of the charged particle in the drifting coordinate system is as shown in Fig. I.2.1. To obtain the motion in the stationary frame of reference, it is necessary to add the drift velocity to the circular motion. This is shown in Fig. I.2.2. No current flow or charge separation results from this motion since both positive and negative particles drift in the same direction with the same velocity. However, there is a mass flow as shown by an equilibrium solution of the macroscopic equations [see (3.8.7)].

Problem I.2.2 Show that cycloidal motion of a charged particle in a crossed static electric and magnetic field can be obtained with the aid of a Lorentz transformation into a coordinate system moving with a velocity such that the perpendicular component of the electric field is zero; i.e.,

$$\mathbf{E}_\perp' = \gamma\left(\mathbf{E}_\perp + \frac{\mathbf{V} \times \mathbf{B}}{c}\right) = 0 \qquad \text{(I.2.26)}$$

////

Problem I.2.3 What is the kinetic energy associated with the drift motion for an electron and for an ion if the electric field is 10 kV/cm and the magnetic field is 20 kG? Assume $\mathbf{E} \perp \mathbf{B}$. ////

Problem I.2.4 Supply the missing steps to obtain Eq. (I.2.25). ////

Problem I.2.5 Obtain the motion of a charged particle in a homogeneous static magnetic and gravitational field and show that this motion results in a net electric current flow even if both positive and negative charged particles are present in equal density. ////

Problem I.2.6 A particle of charge q, mass m, is born at rest in a crossed electric and magnetic field and is moving with drift velocity $c\mathbf{E} \times \mathbf{B}/B^2$. Show that there is equal energy in the rotational motion and in the drift motion. If the electric field is reduced to zero slowly, what is the motion of the particle? If the electric field is reduced suddenly to zero, what is the average energy relative to the original drift energy? ////

From the result that a charged particle drifts with velocity given by (I.2.24), it is possible to deduce the drift velocity for a particle in a gravitational force field. The numerator and denominator of (I.2.24) are multiplied by the charge q. The force term in the numerator $q\mathbf{E}$ is then replaced by the force due to gravity $m\mathbf{g}$ (or some other external force). The result is the gravitational drift

$$\mathbf{V}_{DG} = \frac{m}{q} \frac{\mathbf{g} \times \mathbf{B}}{B^2} c \qquad \text{(I.2.27)}$$

This drift velocity depends on the mass and on the sign of the charge. Thus this drift motion gives rise to plasma currents and charge separation. As was seen in the investigation of the Rayleigh-Taylor instability in Chap. 5, this gravitational drift motion drives the instability.

I.3 PARTICLE MOTION IN SLOWLY VARYING HOMOGENEOUS ELECTRIC AND MAGNETIC FIELDS

I.3.1 Motion in a Slowly Varying Electric Field [$\mathbf{B}_0 = 0$, $\mathbf{E}(t) \neq 0$]

The induced magnetic field that results from a slowly varying electric field can be neglected in the equation of motion, provided the particle's velocity is small compared with the speed of light. Equation (I.1.2) then becomes

$$\dot{\mathbf{v}} = \frac{q}{m} \mathbf{E}(t) \qquad \text{(I.3.1)}$$

It has a solution

$$\mathbf{v}(t) = \mathbf{v}_0 + \frac{q}{m}\int_0^t \mathbf{E}(t)\,dt \qquad \text{(I.3.2)}$$

I.3.2 Motion in a Slowly Varying Magnetic Field [$\mathbf{E}_0 = 0$, $\mathbf{B}(t) \neq 0$]

A slow variation in the magnetic field means that change in magnetic field in one cyclotron period is small; i.e.,

$$\frac{1}{\omega_c}\left|\frac{\partial B/\partial t}{B}\right| \ll 1 \qquad \text{(I.3.3)}$$

This does not mean that the induced electric field is negligible in the equation of motion. In fact, the induced electric field may transfer energy to or take energy from the orbiting particle. Assuming there is no motion along the magnetic field, the orbits in a slowly varying magnetic field are approximately closed circles. The change of the perpendicular energy in one gyration is

$$\Delta W_\perp = q\oint \mathbf{E}\cdot d\mathbf{l} \qquad \text{(I.3.4)}$$

From Maxwell's equation

$$\Delta W_\perp = \frac{1}{c}\frac{\partial}{\partial t}\left(q\int_s \mathbf{B}\cdot d\mathbf{a}\right) \qquad \text{(I.3.5)}$$

Assuming the orbit size changes only slightly in one period, the change in perpendicular energy is

$$\Delta W_\perp \approx q\pi a_c{}^2\,\frac{1}{c}\,\frac{\partial B}{\partial t} \qquad \text{(I.3.6)}$$

The change in B in one gyration period is

$$\Delta B = \frac{2\pi}{\omega_c}\frac{\partial B}{\partial t},$$

so from (I.3.6) the change in energy in one gyration period is

$$\Delta W_\perp = \frac{W_\perp}{B}\,\Delta B \qquad \text{(I.3.7)}$$

Equation (I.3.7) can also be expressed as follows:

$$\Delta\left(\frac{W_\perp}{B}\right) = 0 \qquad \text{(I.3.8)}$$

This shows that the quantity $W_\perp/B$, which is the magnetic moment (I.2.18) of the particle, is an approximate constant of the motion for a slowly changing magnetic field. The magnetic moment is called an *adiabatic invariant*, and in many cases it can be regarded as constant. There are several adiabatic invariants associated with particle motion (see Sec. 1.7).

Problem I.3.1 Supply the missing steps in the above derivation to obtain (I.3.8). ////

I.4 PARTICLE MOTION IN A STATIC HOMOGENEOUS MAGNETIC FIELD AND A RAPIDLY VARYING, SMALL-AMPLITUDE ELECTRIC FIELD [$\mathbf{B}_0 \neq 0$, $\mathbf{E}(t) \neq 0$]

The magnetic field induced by the time-changing electric field is negligible in the equation of motion provided the particles remain nonrelativistic. For a plane wave $\mathbf{E} = \hat{\mathbf{E}}\exp[i(kx - \omega t)]$ this condition is satisfied if $qE/km_0c^2 \ll 1$ and $\omega \neq \omega_c$. The induced magnetic field is neglected in the analysis that follows since amplitude of the electric field can always be made small enough to satisfy the above inequality.

From (I.3.2) the motion along the magnetic field is given by

$$v_\parallel(t) = v_\parallel(0) + \frac{q}{m}\int_0^t E_\parallel\,dt \qquad \text{(I.4.1)}$$

The equation of motion for the perpendicular component of the velocity is the same as (I.2.22), except $\mathbf{E}_\perp = \mathbf{E}_\perp(t)$. Thus

$$\dot{\mathbf{v}}_\perp = \frac{q}{m}\left(\mathbf{E}_\perp(t) + \frac{\mathbf{v}_\perp \times \mathbf{B}_0}{c}\right) \qquad \text{(I.4.2)}$$

Postvector-multiplying by $\mathbf{B}_0$ gives

$$\dot{\mathbf{v}}_\perp \times \mathbf{B}_0 = \frac{q}{m}\left(\mathbf{E}_\perp \times \mathbf{B}_0 + \frac{\mathbf{v}_\perp \times \mathbf{B}_0}{c} \times \mathbf{B}_0\right) \qquad \text{(I.4.3)}$$

Expanding the triple-vector product gives

$$\dot{\mathbf{v}}_\perp \times \mathbf{B}_0 = \frac{q}{m}\left(\mathbf{E}_\perp \times \mathbf{B}_0 + \frac{\mathbf{v}_\perp \cdot \mathbf{B}_0}{c}\,\mathbf{B}_0 - \frac{\mathbf{B}_0 \cdot \mathbf{B}_0}{c}\,\mathbf{v}_\perp\right) \qquad \text{(I.4.4)}$$

The second derivative of the equation of motion,

$$\ddot{\mathbf{v}}_\perp = \frac{q}{m}\left(\dot{\mathbf{E}}_\perp + \frac{\dot{\mathbf{v}}_\perp \times \mathbf{B}_0}{c}\right) \qquad \text{(I.4.5)}$$

is used to eliminate the $(\dot{\mathbf{v}}_\perp \times \mathbf{B}_0)/c$ term. Since by definition of $\dot{\mathbf{v}}_\perp$,

$$\dot{\mathbf{v}}_\perp \cdot \mathbf{B}_0 = \frac{q}{m}\,\mathbf{E}_\perp \cdot \mathbf{B}_0 = 0 \qquad \text{(I.4.6)}$$

Combining terms gives

$$\ddot{\mathbf{v}}_\perp + \omega_c{}^2\mathbf{v}_\perp = \frac{q}{m}\,\dot{\mathbf{E}}_\perp + \left(\frac{q}{m}\right)^2\frac{1}{c}\,\mathbf{E}_\perp \times \mathbf{B}_0 \qquad \text{(I.4.7)}$$

It is possible to solve this differential equation for an arbitrary variation of E; however, only the case of harmonic time variation is considered here. That is,

$$\mathbf{E}_\perp(t) = \mathrm{Re}\,(\hat{\mathbf{E}}_\perp e^{-i\omega t}) \tag{I.4.8}$$

Solutions of the form

$$\mathbf{v}_\perp(t) = \mathrm{Re}\,(\hat{\mathbf{v}}_\perp e^{-i\omega t}) + \mathbf{v}_{\perp h}(t) \tag{I.4.9}$$

are sought, where $\mathbf{v}_{\perp h}$ is the cyclotron motion that satisfies the homogeneous equation (I.2.25). The circumflex is used to denote the fact that the amplitudes are complex quantities. This approach is identical with the approach used previously to investigate the motion of a charged particle in crossed static E and B fields; i.e., the motion is separated into cyclotron motion $v_{\perp h}$ and a "drift motion" $\mathrm{Re}\,(\hat{\mathbf{v}}_\perp e^{-i\omega t})$ that shows the response of the particle to the electric field. Since the electric field is time-dependent, the equivalent drift is time-dependent. Note that both here and in the static-field case there are initial conditions for which the cyclotron motion vanishes, and the total motion of the particle is given by $\mathbf{v}_\perp = \mathbf{V}_{DE} = \mathrm{Re}\,(\hat{\mathbf{v}}_\perp e^{-i\omega t})$. For those initial conditions, the steady-state solution of (I.4.7) for the perpendicular velocity is

$$\hat{\mathbf{v}}_\perp = \frac{-i\omega(q/m)\hat{\mathbf{E}}_\perp + (q/m)^2(\hat{\mathbf{E}}_\perp \times \mathbf{B}_0)/c}{\omega_c^2 - \omega^2} \tag{I.4.10}$$

Choosing $\mathbf{B}_0$ in the z direction in a rectangular cartesian coordinate system and writing (I.4.10) in component form gives

$$\hat{v}_x = \frac{q}{m}\frac{-i\omega\hat{E}_x + \omega_c\hat{E}_y}{\omega_c^2 - \omega^2} \tag{I.4.11}$$

$$\hat{v}_y = \frac{q}{m}\frac{-i\omega\hat{E}_y - \omega_c\hat{E}_x}{\omega_c^2 - \omega^2} \tag{I.4.12}$$

$$\hat{v}_z = i\frac{q}{m}\frac{\hat{\mathbf{E}}_z}{\omega} \tag{I.4.13}$$

The velocity and electric field are related by an equation of the form

$$\hat{\mathbf{v}} = \mathfrak{a}\cdot\hat{\mathbf{E}} \tag{I.4.14}$$

where

$$\mathfrak{a} = -\frac{q}{m}\begin{bmatrix} \dfrac{i\omega}{\omega_c^2-\omega^2} & \dfrac{-\omega_c}{\omega_c^2-\omega^2} & 0 \\ \dfrac{\omega_c}{\omega_c^2-\omega^2} & \dfrac{i\omega}{\omega_c^2-\omega^2} & 0 \\ 0 & 0 & -\dfrac{i}{\omega} \end{bmatrix} \tag{I.4.15}$$

Problem I.4.1 Verify that the terms of (I.4.15) are correct. ////

From (I.4.15), the driven motion of a particle in a homogeneous time-varying electric field E_x perpendicular to the steady magnetic field is an ellipse whose major and minor axes depend on the ratio of cyclotron frequency to driving frequency. This is illustrated by solving for the x and y positions of the drift motion of the particle as a function of time and then casting the result in the usual form for an ellipse by eliminating time. The result is

$$x^2 + \left(\frac{\omega}{\omega_c}\right)^2 y^2 = \frac{[(q/m)E_x]^2}{(\omega_c^2 - \omega^2)^2} \tag{I.4.16}$$

If the frequency is low ($\omega \ll \omega_c$), the ellipse is stretched out along the x axis, and if the frequency is high ($\omega \gg \omega_c$), the ellipse is stretched out along the y axis. The orbit is circular but unbounded in the limit $\omega = \omega_c$. In some cases (especially for cold plasmas), the amplitude of the thermal cyclotron motion is negligible, and (I.4.11) to (I.4.16) effectively represent the total motion of the particles.

Problem I.4.2 Starting with Eqs. (I.4.11) and (I.4.12), obtain (I.4.16). ////

Problem I.4.3 (*a*) Derive an expression for the change in kinetic energy in one period for an electron starting from rest in a static magnetic field B_0 and a perpendicular electric field $E = E_1 \sin \omega_c t$.

(*b*) Repeat the problem for a particle with an initial velocity. ////

Problem I.4.4 If $E_0 = 10^4$ V/cm and B_0 is 3,000 G, what is the maximum energy gain possible in one period when $\omega = \omega_c$? ////

Another solution for the perpendicular motion of a charged particle in a crossed electric and magnetic field can be obtained from (I.4.7) for the case of a slowly varying electric field. To lowest order the particles execute the drifting cyclotron orbits described by (I.2.23) to (I.2.25). As the electric field increases, the drift velocity is increased. The guiding center of the particle orbit can be regarded as a new particle acted on by a force $m\dot{\mathbf{V}}_{DE}$ that produces a drift velocity given by

$$\mathbf{v} = \frac{m}{q}\frac{\dot{\mathbf{V}}_{DE} \times \mathbf{B}_0 c}{|B_0|^2} \tag{I.4.17}$$

Since B_0 is not a function of time, $\dot{\mathbf{V}}_{DE} = c\dot{\mathbf{E}}/B_0$, and (I.4.17) becomes

$$\mathbf{v} \equiv \mathbf{V}_P = \frac{m}{q}\frac{\dot{\mathbf{E}}c^2}{|B_0|^2} \tag{I.4.18}$$

The velocity $\mathbf{V}_P$ is called the polarization drift, and is the drift motion of the guiding center in the direction of the applied electric field. The total motion of the particle is made up of cyclotron motion about the orbit center plus the $\mathbf{E} \times \mathbf{B}$ drift and the polarization drift.

The polarization-drift velocity depends on the sign of the charge and gives rise to a current

$$\mathbf{J}_P = nq\mathbf{V}_P = \frac{nmc^2}{|B_0|^2}\dot{\mathbf{E}}_\perp \tag{I.4.19}$$

The same information is hidden in the low frequency limit of (I.4.15). Taking, for example, $\mathbf{E} = E_x\hat{\mathbf{x}}$, and $\omega \ll \omega_c$, (I.4.15) gives

$$\mathbf{v} = -\frac{q}{m}\frac{i\omega E_x}{\omega_c^2}\hat{\mathbf{x}} + \frac{qE_x}{m\omega_c}\hat{\mathbf{y}}$$

Since $\dot{E}_x = -i\omega E_x$, comparing with (I.4.18), the first term is clearly the polarization drift, and the second is the $\mathbf{E} \times \mathbf{B}$ drift.

I.5 PARTICLE MOTION IN A HOMOGENEOUS LARGE-AMPLITUDE PLANE ELECTROMAGNETIC WAVE

For a large enough amplitude electromagnetic wave, the induced magnetic field associated with the rapidly varying electric field influences the motion of a charged particle. In this section the motion of a charged particle in such a wave is considered, retaining only the lowest order (v/c) relativistic correction, and assuming that there are no static magnetic or electric fields. The plane wave is propagating in the z direction, $\mathbf{E} = \hat{\mathbf{x}}E_x \cos(kz - \omega t)$ and has components field E_x and B_y. The equation of motion is

$$\dot{\mathbf{v}} = \frac{q}{m}\left(\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c}\right) \qquad \text{(I.5.1)}$$

Since $E_x = B_y$ for a plane wave in a vacuum, (I.5.1) can be written

$$\dot{\mathbf{U}} = \frac{qE_x}{mc}(\hat{\mathbf{x}} + \mathbf{U} \times \hat{\mathbf{y}})\cos(kz - \omega t) \qquad \text{(I.5.2)}$$

where $\mathbf{U} = \mathbf{v}/c$ is the dimensionless velocity. In component form, (I.5.2) becomes

$$\dot{U}_x = \frac{qE_x}{mc}(1 - U_z)\cos(kz - \omega t) \qquad \text{(I.5.3)}$$

$$\dot{U}_y = 0 \qquad \text{(I.5.4)}$$

$$\dot{U}_z = \frac{qE_x}{mc}U_x \cos(kz - \omega t) \qquad \text{(I.5.5)}$$

Equation (I.5.3) can be solved approximately by assuming $|U_z| \ll 1$. Then, noting that $\cos(kz - \omega t) = \cos(k\int u_z\, dt - \text{kct}) \simeq \cos \omega t$,

$$\dot{U}_x \approx \frac{qE_x}{mc}\cos \omega t \qquad \text{(I.5.6)}$$

Integrating with respect to time gives

$$U_x = A \sin \omega t \qquad \text{(I.5.7)}$$

and

$$U_z = A^2(\tfrac{1}{2})\sin^2 \omega t \qquad \text{(I.5.8)}$$

where $A = qE_x/\omega mc$. The z component of the velocity U_z now has an average value

$$\langle U_z \rangle = \tfrac{1}{4}A^2 \qquad \text{(I.5.9)}$$

This shows that the approximation $|U_z| \ll 1$ is justified provided the particle remains nonrelativistic, that is, $A \ll 1$. Subtracting this average value, U_z is seen to oscillate at 2ω.

$$U_z - \langle U_z \rangle = \left(\frac{A}{2}\right)^2\left(\sin^2 \omega t - \frac{1}{2}\right)$$

$$= -\left(\frac{A}{2}\right)^2 \cos 2\omega t \qquad \text{(I.5.10)}$$

Problem I.5.1 Supply the missing steps in the above derivation and determine when the approximations used no longer hold. If the power flow is 10^6 W/cm^2 ($\omega \equiv 10^{10}$), what is the average velocity in the direction of the power flow, and what are the maximum values of the periodic motion? At what frequency will the above results be invalid? Sketch the motion of the particle. ////

Problem I.5.2 Show that the differential equations for the particle motion, including the relativistic mass correction and the fact that a particle actually does move a fraction of a wavelength in one cycle, are

$$\frac{dU_x}{d\theta} = F(1 - U_z - U_x^2)(1 - U_x^2 - U_z^2)^{1/2} \qquad \text{(I.5.11)}$$

$$\frac{dU_z}{d\theta} = FU_x(1 - U_z)(1 - U_x^2 - U_z^2)^{1/2} \qquad \text{(I.5.12)}$$

$$\frac{dX}{d\theta} = U_x \qquad \text{(I.5.13)}$$

$$\frac{dZ}{d\theta} = U_z \qquad \text{(I.5.14)}$$

where $F = A\cos(\theta - Z + \phi)$

$$U_{x,z} = \frac{v_{x,z}}{c}$$

$$X = \frac{\omega x}{c}$$

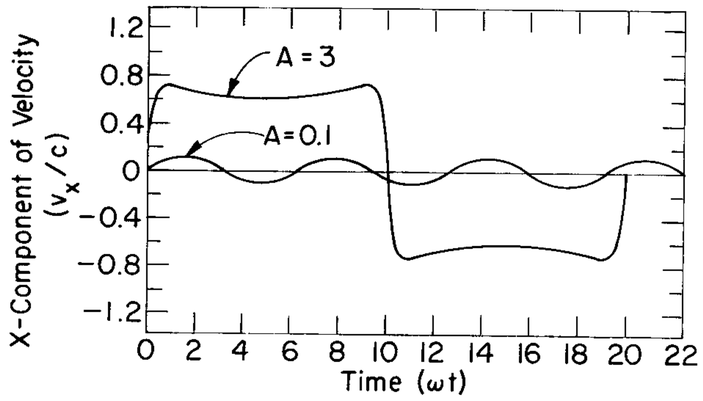
$$Z = \frac{\omega z}{c}$$

$$A = qE/km_0c^2$$

$$\theta = \omega t$$

$$\omega = kc$$

////



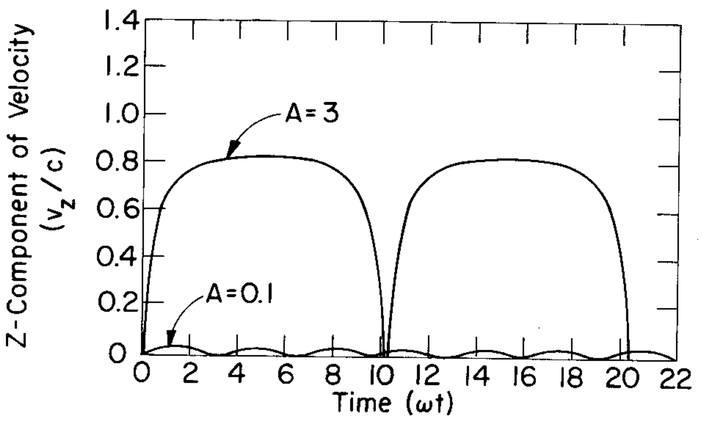


FIGURE I.5.1
The x and z components of the velocity of a charged particle started from rest in the field of a large-amplitude plane electromagnetic wave propagating in the z direction and polarized in the x direction.

$E_x = E_0 \exp[i(\omega t - kz)]$

$$A = \frac{qE_0}{km_0c^2}$$

[*After H. R. Jory and A. W. Trivelpiece, J. Appl. Phys.*, **39**; 3053 (1968).]

Figure I.5.1 shows the motion of a charged particle started from rest in the field of a large-amplitude plane wave for several values of A. The particle acquires a velocity in the direction of propagation as a result of the rf $\mathbf{v} \times \mathbf{B}$ force. This parallel velocity is given approximately by $c\Delta m v_{\parallel} = \Delta \frac{1}{2} m v_{\perp}^2$.

Problem I.5.3 Show in the limit $|U_z| \ll 1$ that $\sigma(2\omega) = \sigma(\omega)|U_{x\,\max}|^2$, where $\sigma(\omega)[= (8\pi/3)r_0^2]$ is the Thomson cross section. $r_0 = e^2/m_0c^2$ is the classical electron radius. ////

Problem I.5.4 Show that if the charged particle starts at some initial phase ϕ of the electric field, that is, $\mathbf{E} = \hat{\mathbf{x}}E_x \cos(\omega t + \phi)$,

$$\langle U_z \rangle = \frac{A^2}{4}(1 + 2\sin^2\phi) \qquad ////$$

Problem I.5.5 The relativistic equation of motion for a particle of charge q and mass m is

$$\frac{d}{dt}[m\mathbf{v}(\mathbf{x}, t)] = q\left[\mathbf{E}(\mathbf{x}, t) + \frac{\mathbf{v}(\mathbf{x}, t) \times \mathbf{B}(\mathbf{x}, t)}{c}\right] \qquad \text{(I.5.15)}$$

Show, by writing the relativistic mass in terms of the rest mass $m = m_0(1 - v^2/c^2)^{1/2}$, that (I.5.15) can be written in the following form, where $\gamma = (1 - v^2/c^2)^{-1/2}$,

$$\frac{d}{dt}\mathbf{v}(\mathbf{x}, t) = \frac{q}{\gamma m_0}\left\{\mathbf{E}(\mathbf{x}, t) + \frac{\mathbf{v}(\mathbf{x}, t) \times \mathbf{B}(\mathbf{x}, t)}{c} - \frac{\mathbf{v}(\mathbf{x}, t)[\mathbf{v}(\mathbf{x}, t) \cdot \mathbf{E}(\mathbf{x}, t)]}{c^2}\right\} \qquad \text{(I.5.16)}$$

////

I.6 PARTICLE MOTION IN STATIC INHOMOGENEOUS MAGNETIC FIELDS

The gradient of a vector field is a dyadic and, in general, has nine components. In rectangular cartesian coordinates the components of the gradient of a static magnetic field are

$$\nabla\mathbf{B} = \begin{vmatrix} \frac{\partial B_x}{\partial x} & \frac{\partial B_x}{\partial y} & \frac{\partial B_x}{\partial z} \\ \frac{\partial B_y}{\partial x} & \frac{\partial B_y}{\partial y} & \frac{\partial B_y}{\partial z} \\ \frac{\partial B_z}{\partial x} & \frac{\partial B_z}{\partial y} & \frac{\partial B_z}{\partial z} \end{vmatrix}$$

From the fact that $\nabla \cdot \mathbf{B} = 0$, only two of the diagonal terms are independent. To investigate the motion that results from these gradient terms in a magnetic field, a series of simplified problems considers each type of term. The effects of the various drifts can be added to obtain the total motion. The only simple cases are those where the spatial variations of the magnetic field are small. This can be expressed by the condition

$$\left|\frac{\mathbf{v} \cdot \nabla\mathbf{B}}{\omega_c \mathbf{B}}\right| \ll 1$$

This condition means both that the field does not change in distance comparable with the cyclotron radius,

$$\left|\frac{\mathbf{v}_\perp \cdot \nabla \mathbf{B}}{\omega_c B}\right| \ll 1$$

and, in the case of the particle motion parallel to the magnetic, that the magnetic field seen by the particle does not change in a time comparable with a cyclotron period,

$$\left|\frac{\mathbf{v}_\parallel \cdot \nabla B}{\omega_c B}\right| \ll 1$$

I.6.1 Magnetic Field with a Longitudinal Gradient

Consider a charged particle that gyrates in a nearly circular orbit about the axis of symmetry in a cylindrical symmetric magnetic field. Let the magnetic field increase as a function of distance along the axis of symmetry as shown in Fig. I.6.1.

In cylindrical coordinates the divergence of the magnetic field is expressed

$$\frac{1}{r}\frac{\partial}{\partial r}(rB_r) + \frac{\partial B_z}{\partial z} = 0 \qquad \text{(I.6.1)}$$

Integrating (I.6.1) gives B_r in terms of $B_z(z)$.

$$rB_r = -\int_0^r r' \frac{\partial B_z}{\partial z}\, dr' \qquad \text{(I.6.2)}$$

since $rB_r = 0$ at $r = 0$. The change of B_z with z is assumed small and can be removed from the integral; i.e.,

$$B_r = \frac{-r}{2}\frac{\partial B_z}{\partial z} \qquad \text{(I.6.3)}$$

There is a force of $v_\theta B_r$ on the gyrating particle due to this radial magnetic field. Since $v_\theta = -(q/|q|)|\mathbf{v}_\perp|$, this force is

$$F_z = -\frac{qv_\theta B_r}{c} = \frac{|q|v_\perp B_r}{c} \qquad \text{(I.6.4)}$$

Substituting for B_r from (I.6.3), evaluating F_z at $r = v_\perp/\omega_c$ and using $\frac{1}{2}mv_\perp^2 = W_\perp$ gives

$$F_z = -\frac{W_\perp}{B_z}\frac{\partial B_z}{\partial z} \qquad \text{(I.6.5)}$$

This equation can be generalized to give the force parallel to the local magnetic field; i.e.,

$$F_\parallel = -\boldsymbol{\mu} \cdot \nabla \mathbf{B} \qquad \text{(I.6.6)}$$

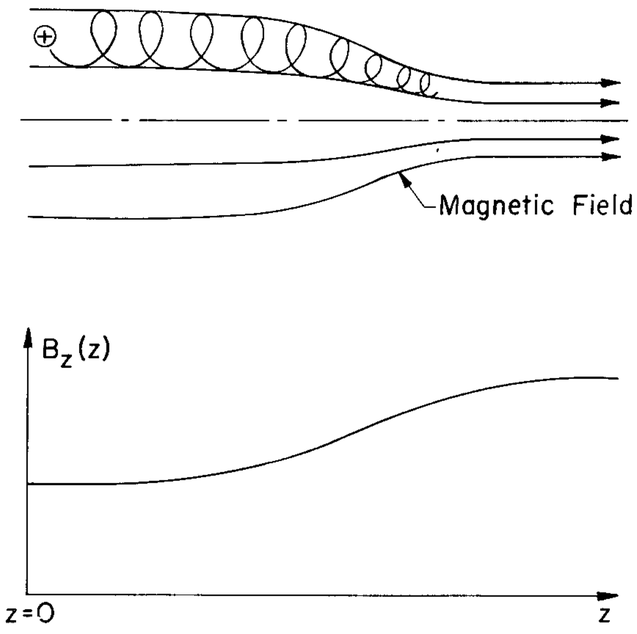


FIGURE I.6.1
Electron motion in a magnetic field with a longitudinal gradient.

An alternative method of describing the motion of charged particle approaching a magnetic ramp is to use the approximate constant of the motion $\mu(= W_\perp/B)$ and the constant of the motion $W(=$ total energy). Consider again the situation depicted schematically in Fig. I.6.1, where an orbiting particle moves into a region of increased magnetic flux density. Let the magnetic field at some point where the particle has both $W_\parallel$ and $W_\perp$ be denoted by $B(0)$. As the particle moves into an increasing magnetic flux density, the total energy W is conserved (because $\mathbf{F}$ is perpendicular to $\mathbf{v}$) and the magnetic moment μ is conserved (because in the rest frame of the particle the magnetic field increases slowly in time). Therefore

$$W(0) = W_\perp(0) + W_\parallel(0) = W(z) = W_\perp(z) + W_\parallel(z) \qquad \text{(I.6.7)}$$

and

$$\mu = \frac{W_\perp(0)}{B(0)} = \frac{W_\perp(z)}{B(z)} \qquad \text{(I.6.8)}$$

Solving (I.6.7) and (I.6.8) for the parallel energy as a function of position in terms of the initial energy gives

$$W_\parallel(z) = W_\parallel(0) - W_\perp(0)\left[\frac{B(z)}{B(0)} - 1\right] \qquad \text{(I.6.9)}$$

Thus, as the particle moves into a stronger magnetic field, the parallel component of its velocity is reduced, and may vanish at some point. This point is called the *magnetic mirror point* and depends on the initial perpendicular and parallel energies and the ratio of the flux densities.

Problem I.6.1 Let the ratio $B(z)/B(0)$ be denoted by R and show that if the angle which the velocity vector makes with magnetic field at $z=0$ is $\theta=\tan^{-1}(v_\perp/v_\parallel)$, the particle will be reflected at the point z if

$$\sin\theta \geq \frac{1}{\sqrt{R}} \qquad \text{(I.6.10)}$$

////

Problem I.6.2 Show that the angle which the velocity vector ($\theta=\tan^{-1} v_\perp/v_\parallel$) makes with respect to its local magnetic field direction transforms as

$$\sin\theta' = \sqrt{R(x)}\sin\theta \qquad \text{(I.6.11)}$$

where θ' is the angle at x and θ is the angle at $x=0$. ////

Problem I.6.3 Show that the flux enclosed within the particle orbit is a constant if the magnetic moment μ is constant. ////

Problem I.6.4 What is the maximum electric field strength for which

$$\mathbf{V}_{DE} = \frac{\mathbf{E}\times\mathbf{B}}{B^2}c$$

is valid if the magnetic induction is 10^4 G? ////

I.6.2 Magnetic Field with Transverse Gradient

Consider a unidirectional magnetic field with a gradient perpendicular to its direction. In such a field a particle will drift perpendicular to both the field and the direction of the field gradient. This is easily understood by noting that as the particle moves into a strong field, its cyclotron radius becomes smaller, and when on the other portion of its orbit moves into a weak field, its orbit is larger. As the particle gyrates, the orbit is not closed. This results in a net drift velocity, illustrated in Fig. I.6.2.

Problem I.6.5 Derive an approximate expression for the transverse gradient drift velocity by assuming that there is a discontinuous change in the magnetic field, along a line through the orbit, and compare this result with that obtained below. ////

The gradient drift can be obtained by expanding the magnetic field in a Taylor series about the center of the orbit, which is assumed to move slowly in time as compared with the orbital motion. The field at the location of the particle, in terms of the field at the center of gyration of the particle, is given by

$$\mathbf{B}(\mathbf{r}=\mathbf{a}_c) = \hat{\mathbf{b}}[B(0) + \mathbf{a}_c(t)\cdot\nabla B(0)] \qquad \text{(I.6.12)}$$

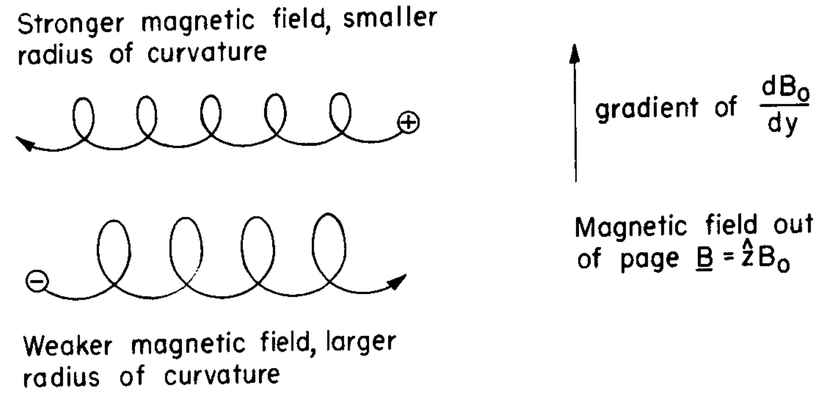


FIGURE I.6.2
Electron motion in a magnetic field with a transverse gradient, showing gradient B drift motion.

The vector ∇B contains only transverse gradient terms in this discussion. The equation of motion is

$$\dot{\mathbf{v}}_\perp = \omega_c\left[\mathbf{v}_\perp\times\hat{\mathbf{b}}\left(1+\frac{\mathbf{a}_c\cdot\nabla B}{B}\right)\right] \qquad \text{(I.6.13)}$$

To solve this equation, the velocity is again separated into a periodic part $\bar{\mathbf{v}}_\perp$ plus a constant drift velocity $\mathbf{V}_D$, that is,

$$\mathbf{v}_\perp = \bar{\mathbf{v}}_\perp + \mathbf{V}_D \qquad \text{(I.6.14)}$$

Substituting this expression for the velocity into (I.6.13) gives

$$\dot{\bar{\mathbf{v}}}_\perp = \omega_c\left[\bar{\mathbf{v}}_\perp\times\hat{\mathbf{b}}\left(1+\frac{\mathbf{a}_c\cdot\nabla B}{B}\right) + \mathbf{V}_D\times\hat{\mathbf{b}}\left(1+\frac{\mathbf{a}_c\cdot\nabla B}{B}\right)\right] \qquad \text{(I.6.15)}$$

An approximate solution to (I.6.15) is obtained by using the fact that

$$\left|\frac{\mathbf{a}_c\cdot\nabla B}{B}\right| \ll 1$$

and neglecting the magnetic gradients

$$\bar{\mathbf{v}}_\perp = -(\mathbf{a}_c\times\hat{\mathbf{b}})\omega_c \qquad \text{(I.6.16)}$$

Iterating (I.6.15) gives to first order

$$\dot{\bar{\mathbf{v}}}_\perp = \omega_c\left[-(\mathbf{a}_c\times\hat{\mathbf{b}})\times\hat{\mathbf{b}}\left(1+\frac{2\mathbf{a}_c\cdot\nabla B}{B}\right)\omega_c + (\mathbf{V}_D\times\hat{\mathbf{b}})\right] \qquad \text{(I.6.17)}$$

The time average ($\langle\dot{\bar{\mathbf{v}}}_\perp\rangle = \langle\mathbf{a}_c\rangle = 0$, $\langle\mathbf{a}_c^2\rangle = \frac{1}{2}a_c^2$) gives

$$0 = -a_c^2\omega_c\frac{\nabla B}{B} + \mathbf{V}_D\times\hat{\mathbf{b}} \qquad \text{(I.6.18)}$$

Solving for $\mathbf{V}_D$ by taking the prescalar product with $\hat{\mathbf{b}}$ gives magnetic gradient drift

$$\mathbf{V}_{D,\,\mathrm{grad}} = \frac{1}{2}\frac{mv_\perp^2}{qB}\frac{\hat{\mathbf{b}}\times\nabla B}{B}c = \frac{\mu}{q}\frac{\hat{\mathbf{b}}\times\nabla B}{B}c \qquad \text{(I.6.19)}$$

Notice that this drift velocity depends on the sign of the charge and can cause plasma currents and charge separation.

Problem I.6.6 Supply the missing steps in deriving (I.6.18) and justify the approximations made. ////

I.6.3 Magnetic Field with Curvature

Consider a magnetic field that has curvature as shown in Fig. I.6.3. Such a magnetic field has a gradient perpendicular to its direction. This gradient results in a drift motion as just described. In addition, if the particle has a velocity along the magnetic field, it experiences a centrifugal force due to the field curvature and drifts at right angles to the direction of curvature and the direction of the field. This drift comes from the centrifugal force acting on the particle guiding center as a result of its motion along the tube of flux. The centrifugal force acts on the particle in the manner of a general force field and produces a specific curvature drift whose velocity will be given by

$$\mathbf{V}_{DC} = \frac{1}{q}\frac{\mathbf{F}_c\times\mathbf{B}}{|B|^2}c \qquad \text{(I.6.20)}$$

where the centrifugal force is

$$\mathbf{F}_c = \frac{mv_\parallel^2}{|R|^2}\mathbf{R} \qquad \text{(I.6.21)}$$

The curvature drift is

$$\mathbf{V}_{DC} = \frac{m}{q}\left|\frac{v_\parallel}{R}\right|^2\frac{\mathbf{R}\times\mathbf{B}}{|B|^2}c \qquad \text{(I.6.22)}$$

This centrifugal drift can be expressed entirely in terms of the field quantities by relating the radius of curvature to the magnetic field:

$$\mathbf{V}_{DC} = \frac{mv_\parallel^2}{qB^2}c(\hat{\mathbf{b}}\times\nabla\mathbf{B}) \qquad \text{(I.6.23)}$$

The total drift of a charged particle in a curved magnetic field is the sum of the transverse gradient drift [(I.6.19)] and the curvature drift [(I.6.23)].

$$\mathbf{V}_{DC} = \frac{mc}{2qB^2}(v_\perp^2 + 2v_\parallel^2)(\hat{\mathbf{b}}\times\nabla\mathbf{B})$$

$$= \frac{\mu c}{qB}\left(1 + \frac{2v_\parallel^2}{v_\perp^2}\right)(\hat{\mathbf{b}}\times\nabla\mathbf{B})$$

Notice that this drift depends on the sign of the charge and will give rise to a current, and possibly a charge separation, in a plasma.

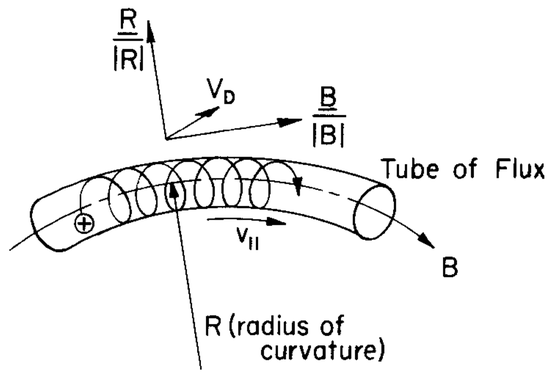


FIGURE I.6.3
Electron motion in a magnetic field with curvature, showing curvature drift motion.

I.7 ADIABATIC INVARIANTS

In Sec. I.3.2 it was shown that there was an approximate constant of the motion for a charged particle in a slowly varying magnetic field. This quantity is called the *magnetic moment* and is denoted by $\mu(= W_\perp/B)$. Such a constant is usually referred to as an adiabatic invariant.[1] In some cases there is more than one adiabatic invariant associated with the motion of a particle.

There are three adiabatic invariants for a particle in a magnetic mirror field configuration (e.g., Fig. 1.14.1). The first is the magnetic moment,

$$\mu = \frac{W_\perp}{B} \qquad \text{(I.7.2)}$$

[1] See R. Kulsrud, *Phys. Rev.*, **106**:(1957), for an excellent treatment of the basic idea behind an "adiabatic invariant," i.e., that if an oscillation has slowly increasing frequency

$$\ddot{x} + \omega^2(t)x = 0$$

and if

$$\varepsilon \equiv \frac{1}{\omega^2}\frac{d\omega}{dt} \ll 1$$

there are quantities such as μ which are nearly constant.

$$\frac{\Delta\mu}{\mu} \approx \exp\left[-\left(\frac{1}{\omega^2}\frac{d\omega}{dt}\right)^{-1} f(t)\right] \qquad \text{(I.7.1)}$$

With $\Delta\mu$ given by a form such as (I.7.1), μ is constant to all orders in the small parameter ε, since $\exp(-1/\varepsilon)$ is nonexpandable about $\varepsilon = 0$. This does not mean that $\Delta\mu$ vanishes, but that the change $\Delta\mu$ predicted by (I.7.1) goes to zero, as

$$\varepsilon = \frac{1}{\omega^2}\frac{d\omega}{dt} \to 0$$

faster than *any power* of ε, since

$$\lim_{\varepsilon\to 0}\frac{1}{\varepsilon^n}\exp\left(\frac{-1}{\varepsilon}\right) \to 0 \qquad \text{for any power } n$$

See also M. D. Kruskal, Advanced Theory of Gyrating Particles, in "Plasma Physics," International Atomic Energy Agency, Vienna, 1965.

which expresses the fact that as the particle circulates in its orbit, it encloses a constant amount of flux. Actually, $p_\perp^2/2Bm_0$ is the first adiabatic invariant, which for small-particle energy reduces to $W_\perp/B$. The second approximate constant of the motion is called the *longitudinal invariant* and is given by

$$J = \int v_\parallel \, ds \qquad \text{(I.7.3)}$$

where the integral is evaluated between turning points, where $v_\parallel = 0$. This invariant expresses the fact that as the particle oscillates back and forth between its turning points in the magnetic mirror, the quantity J is nearly constant, provided that the bounce frequency is much less than the cyclotron frequency. The third adiabatic invariant is denoted by Φ and expresses the fact that as the particle drifts slowly from line to line in the magnetic mirror, it moves on a closed barrel-shaped surface in which the flux enclosed is constant.

In the case of time-varying fields, the latter invariant is useful because it predicts how the plasma shape might vary provided the time scale of variation of the field is slow compared with the time scale of the longitudinal invariant.

Problem I.7.1 Determine the time scales for each of the periodic motions of a charged particle in the earth's magnetic field so that each of the three quantities μ, J, and Φ can be regarded as invariants. Is this motion consistent with the fact that charge can be trapped in the Van Allen radiation belts? ////

Problem I.7.2 Show that J is a constant to first order (that is, $\Delta J = 0$) for a particle in a magnetic mirror field. ////

I.8 PLASMA PROPERTIES FROM ORBIT THEORY

Single-particle drift motions do not constitute a complete plasma description; however, some of the mediumlike properties of the plasma state can be demonstrated using them.

I.8.1 Magnetization of a Plasma

Consider a surface S bounded by a curve C within an ionized gas of stationary ions (for neutralization) and mobile electrons. As each of the electrons circulates in its orbit, it has a "microscopic" magnetic moment that contributes the macroscopic magnetization per unit volume. This magnetization is determined by the number of electrons which have orbits that encircle the curve C and thus contribute to the net current through S. If there are n current loops per unit volume, the net current through S is

$$I = c\int_C n\boldsymbol{\mu}\cdot d\mathbf{s} = c\int_C \mathbf{M}\cdot d\mathbf{s} = c\int_S \nabla\times\mathbf{M}\cdot d\mathbf{a} \qquad \text{(I.8.1)}$$

In the classical treatment of the magnetic material, the magnetization $\mathbf{M}$ is related to a fictitious current density $\mathbf{J}'$.

$$c\nabla\times\mathbf{M} = \mathbf{J}' \qquad \text{(I.8.2)}$$

Further,

$$\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M} \qquad \text{(I.8.3)}$$

which leads to a nonlinear relation between $\mathbf{B}$ and $\mathbf{H}$, since $\mathbf{M}$ is proportional to $1/B$.

$$\mathbf{M} = -n\frac{W_\perp}{|B|^2}\mathbf{B} \qquad \text{(I.8.4)}$$

where n is the density of particles per unit volume.

I.8.2 Dielectric Constant of a Plasma

The polarization current density is

$$\mathbf{J}_P = nq\mathbf{V}_p = \frac{nmc^2}{B^2}\frac{\partial\mathbf{E}}{\partial t} \qquad \text{(I.8.5)}$$

From Maxwell's equations

$$\nabla\times\mathbf{B} = \frac{4\pi}{c}\mathbf{J} + \frac{1}{c}\frac{\partial\mathbf{E}}{\partial t} + \frac{4\pi}{c}\mathbf{J}_p = \frac{4\pi}{c}\mathbf{J} + \frac{1}{c}\left(1 + \frac{4\pi nmc^2}{B^2}\right)\frac{\partial\mathbf{E}}{\partial t} \qquad \text{(I.8.6)}$$

This defines the plasma dielectric constant

$$\varepsilon = 1 + \frac{4\pi nmc^2}{B^2} \qquad \text{(I.8.7)}$$

I.8.3 Magnetic Pressure

The concept of magnetic pressure can be developed from orbit theory for the simple case of straight parallel flux lines with a transverse gradient. The electric field and parallel force are taken to be zero.

From Maxwell's equations

$$\nabla\times\mathbf{B} = \frac{4\pi}{c}\mathbf{J}_D + \frac{4\pi}{c}\mathbf{J}' \qquad \text{(I.8.8)}$$

where $\mathbf{J}'$ is the transverse gradient drift current

$$\mathbf{J}_D = nq\mathbf{V}_D = n\mu\frac{\hat{\mathbf{b}}\times\nabla B}{|B|}c \qquad \text{(I.8.9)}$$

and $\mathbf{J}'$ is the magnetization current

$$\mathbf{J}' = -c\nabla \times \hat{\mathbf{b}}(n\mu) = c\hat{\mathbf{b}} \times \nabla(n\mu)$$
$$= \left[\frac{\hat{\mathbf{b}}}{|B|} \times \nabla(nW_\perp) - n\mu \frac{\hat{\mathbf{b}} \times \nabla B}{|B|}\right]c \qquad \text{(I.8.10)}$$

Substituting (I.8.9) and (I.8.10) in (I.8.8) gives

$$\nabla \times \mathbf{B} = -\hat{\mathbf{b}} \times \nabla|B| = \frac{4\pi}{c}\left(\frac{\hat{\mathbf{b}}}{|B|} \times \nabla nW_\perp\right)c \qquad \text{(I.8.11)}$$

This equation can be written in the form

$$\nabla\left(\frac{B^2}{8\pi} + nW_\perp\right) = 0 \qquad \text{(I.8.12)}$$

This means that the plasma pressure balance is expressed by

$$\frac{B^2}{8\pi} + p_\perp = \text{constant} \qquad \text{(I.8.13)}$$

where $p_\perp = nW_\perp$ is the perpendicular pressure.

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APPENDIX II

SUMMARY OF SOME PROPERTIES OF VECTORS AND TENSORS, SOME INTEGRAL THEOREMS, AND CURVILINEAR COORDINATES

The majority of physical quantities can be described by either scalars, vectors, or second-order tensors. Knowledge of the properties of these quantities and how to perform mathematical operations with them are essential to the scientist and engineer. The rigorous descriptions and proofs associated with the properties of these quantities require a mathematical text of considerable length. Such information can be found in the following references:

1 P. M. Morse and H. Feshbach, "Methods of Theoretical Physics," vol. 1, chap. 1, McGraw-Hill, New York, 1953.
2 H. Jeffreys, "Cartesian Tensors," Cambridge, New York, 1953.
3 L. A. Pipes, "Applied Mathematics for Engineers and Physicists," 3d ed., McGraw-Hill, New York, 1970.
4 W. Magnus and F. Oberhettinger, "Functions of Mathematical Physics, Chelsea, New York, 1954.
5 E. Jahnke and F. Emde, "Tables of Functions," Dover, New York, 1945.

For convenient reference a concise listing of some frequently used operational properties

of these mathematical quantities is contained in this section. Quantities are denoted as follows (note the typeface):

S = scalar (temperature, energy, time, etc.)
$\mathbf{A}$ = vector (velocity, momentum, force, etc.)
T = tensor (shear stress, momentum flux, etc.)

Algebraic laws of scalars:

Commutative: $RS = SR$
Associative: $(QR)S = Q(RS)$
Distributive: $S(P + Q - R) = SP + SQ - SR$

The unit vector $\hat{\mathbf{a}}$ is one that has orthogonal components $\hat{\mathbf{a}}_i$ of unit length. It is denoted by

$$\hat{\mathbf{a}} = (\hat{\mathbf{a}}_1, \hat{\mathbf{a}}_2, \hat{\mathbf{a}}_3)$$

For a cartesian coordinate system,

$$\hat{\mathbf{a}} = (\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$$

The components of $\hat{\mathbf{a}}$ are often designated by $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$ in the literature. The vector $\mathbf{A}$ is given by

$$\mathbf{A} = \hat{\mathbf{a}}_1 A_1 + \hat{\mathbf{a}}_2 A_2 + \hat{\mathbf{a}}_3 A_3$$

It is frequently convenient and quite concise when using cartesian coordinates to employ the *index summation convention*. Whenever, within a group of symbols, an index is repeated, this implies summation of that index. Let the subscripts x, y, z or 1, 2, 3 be implied by the dummy subscripts i, j, k. Then, using cartesian tensor summation convention,

$$\mathbf{A} = \hat{\mathbf{a}}_i A_i = \hat{\mathbf{a}}_1 A_1 + \hat{\mathbf{a}}_2 A_2 + \hat{\mathbf{a}}_3 A_3$$

or

$$V_j W_j = V_x W_x + V_y W_y + V_z W_z$$

or

$$\frac{\partial u_i}{\partial x_i} = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}$$

The summation convention does not apply in curvilinear coordinates. Whenever used, it is to be understood that cartesian coordinates are being employed.

SCALAR, OR DOT, PRODUCT

$$\mathbf{A} \cdot \mathbf{B} = A_i B_i$$
$$= A_x B_x + A_y B_y + A_z B_z$$

Scalar products are commutative and distributive but not associative. Also, it follows that

$$\mathbf{A} \cdot \mathbf{A} = A_i A_i = A^2$$

VECTOR OR CROSS PRODUCT

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{a}}_x & \hat{\mathbf{a}}_y & \hat{\mathbf{a}}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$
$$= \hat{\mathbf{a}}_x(A_y B_z - A_z B_y) + \hat{\mathbf{a}}_y(A_z B_x - A_x B_z) + \hat{\mathbf{a}}_z(A_x B_y - A_y B_x)$$

The vector cross product is distributive, but not commutative or associative. The Kronecker delta δ_{ij} is defined by the following properties:

$$\delta_{ij} = \begin{cases} +1 & i = j \\ 0 & i \neq j \end{cases}$$

The alternating unit tensor ε_{ijk} is defined by the following properties:

$$\varepsilon_{ijk} = \begin{cases} +1 & ijk = 123, 231, 312 \text{ (even permutation)} \\ -1 & ijk = 321, 132, 213 \text{ (odd permutation)} \\ 0 & \text{if any two indices are alike} \end{cases}$$

From this definition, and using summation convention, it follows that

$$\varepsilon_{ijk}\varepsilon_{ijk} = 6$$
$$\varepsilon_{ijk}\varepsilon_{hjk} = 2\delta_{ih}$$
$$\varepsilon_{ijk}\varepsilon_{mnk} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$$

The unit vector $\hat{\mathbf{a}}$ has the properties which can now be summarized by

$$\hat{\mathbf{a}}_i \cdot \hat{\mathbf{a}}_j = \delta_{ij}$$
$$\hat{\mathbf{a}}_i \times \hat{\mathbf{a}}_j = \varepsilon_{ijk}\hat{a}_k$$

The dot product can also be written

$$\mathbf{A} \cdot \mathbf{B} = \delta_{ij} A_i B_j = A_i B_i$$

and the cross product can be written

$$\mathbf{A} \times \mathbf{B} = \varepsilon_{ijk} A_j B_k$$

A frequently employed identity is the triple cross product:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

SECOND-ORDER TENSORS A second-order tensor is an array of nine components which obey certain laws of coordinate transformation. The components are frequently grouped as follows:

$$\mathsf{T} = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix}$$

The diagonal elements are T_{ii}. A symmetric tensor is one where $T_{ij} = T_{ji}$. A unit tensor a is one where $a_{ij} = \delta_{ij}$. A *dyadic product* of two vectors **A** and **B** is a special form of a second-order tensor. It is denoted by

$$\mathbf{AB} = \begin{bmatrix} A_1 B_1 & A_1 B_2 & A_1 B_3 \\ A_2 B_1 & A_2 B_2 & A_2 B_3 \\ A_3 B_1 & A_3 B_2 & A_3 B_3 \end{bmatrix}$$

The dyadic product is written by placing the two vectors next to one another with no operational sign of any kind.

The scalar product (double-dot product) of two tensors is defined by

$$\mathsf{T}:\mathsf{U} = T_{ij} U_{ji}$$
$$\mathsf{T}:\mathbf{AB} = T_{ij} A_j B_i$$

both of which are scalars.

The vector product (single-dot product) of a vector with a tensor yields a vector according to the following operation:

$$\mathsf{T} \cdot \mathbf{A} = \hat{\mathbf{a}}_i T_{ij} A_j$$

The ith vector component of $\mathsf{T} \cdot \mathbf{A}$ is thus given by

$$(\mathsf{T} \cdot \mathbf{A})_i = (T_{ix} A_x + T_{iy} A_y + T_{iz} A_z)$$

Similarly,

$$\begin{aligned} \mathbf{A} \cdot \mathsf{T} &= \hat{\mathbf{a}}_i A_j T_{ji} \\ &= \hat{\mathbf{a}}_1 (A_1 T_{11} + A_2 T_{31} + A_3 T_{31}) \\ &\quad + \hat{\mathbf{a}}_2 (A_1 T_{12} + A_2 T_{22} + A_3 T_{32}) \\ &\quad + \hat{\mathbf{a}}_3 (A_1 T_{13} + A_2 T_{23} + A_3 T_{23}) \end{aligned}$$

Vector differential operations The vector differential operator ∇, called *del*, or *nabla*, is defined in rectangular coordinates as

$$\begin{aligned} \nabla &= \hat{\mathbf{a}}_i \frac{\partial}{\partial x_i} \\ &= \hat{\mathbf{a}}_x \frac{\partial}{\partial x} + \hat{\mathbf{a}}_y \frac{\partial}{\partial y} + \hat{\mathbf{a}}_z \frac{\partial}{\partial z} \end{aligned}$$

The symbol ∇ is a vector operator; it has components like a vector and it cannot stand alone but must operate on a scalar, vector, or tensor function.

GRADIENT OF A SCALAR FIELD

$$\begin{aligned} \nabla S &= \hat{\mathbf{a}}_i \frac{\partial S}{\partial x_i} = \text{grad } S \\ &= \hat{\mathbf{a}}_x \frac{\partial S}{\partial x} + \hat{\mathbf{a}}_y \frac{\partial S}{\partial y} + \hat{\mathbf{a}}_z \frac{\partial S}{\partial z} \end{aligned}$$

The vector resulting from ∇S is called the *gradient of S*. The gradient operation is distributive, but not commutative or associative.

CURL OF A VECTOR FIELD The vector cross product between ∇ and **A** is called the *curl of* **A**.

$$\begin{aligned} \nabla \times \mathbf{A} &= \varepsilon_{ijk} \hat{\mathbf{a}}_i \nabla_j A_k = \text{curl } \mathbf{A} \\ &= \begin{vmatrix} \hat{\mathbf{a}}_1 & \hat{\mathbf{a}}_2 & \hat{\mathbf{a}}_3 \\ \dfrac{\partial}{\partial x_1} & \dfrac{\partial}{\partial x_2} & \dfrac{\partial}{\partial x_3} \\ A_1 & A_2 & A_3 \end{vmatrix} \end{aligned}$$

where $\nabla_j = \partial/\partial x_j$

The curl of a vector field is distributive, but not associative or commutative.

LAPLACIAN OF A SCALAR FIELD The divergence of the gradient of a scalar field $S(x_i)$ yields

$$\nabla \cdot \nabla S = \frac{\partial}{\partial x_i} \left(\frac{\partial S}{\partial x_i} \right) = \sum_{i=1}^{3} \frac{\partial^2 S}{\partial x_i^2}$$

This operation is usually written $\nabla^2 S$, where the laplacian operator ∇^2 is, in rectangular coordinates,

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

The laplacian operator is distributive, but not commutative or associative.

DIVERGENCE OF A VECTOR FIELD

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \frac{\partial A_i}{\partial x_i} = \text{div } \mathbf{A} \\ &= \frac{\partial A_1}{\partial x_1} + \frac{\partial A_2}{\partial x_2} + \frac{\partial A_3}{\partial x_3} \end{aligned}$$

The divergence operation is distributive, but not commutative or associative.

LAPLACIAN OF A VECTOR FIELD In rectangular coordinates, ∇^2 may operate on a vector field by taking the derivatives with respect to the scalar components and adding vectorially.

$$\begin{aligned} \nabla^2 \mathbf{A} &= \hat{\mathbf{a}}_i \frac{\partial}{\partial x_j} \left(\frac{\partial A_i}{\partial x_j} \right) \\ &= \hat{\mathbf{a}}_x (\nabla^2 A_x) + \hat{\mathbf{a}}_y (\nabla^2 A_y) + \hat{\mathbf{a}}_z (\nabla^2 A_z) \end{aligned}$$

This is not applicable in curvilinear coordinates. However, it is generally true that

$$\nabla^2\mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times (\nabla \times \mathbf{A})$$

DERIVATIVE OF A TENSOR FIELD The differential operator ∇ may operate on a tensor or dyad:

$$\nabla \cdot \mathsf{T} = \hat{\mathbf{a}}_i \, \nabla_j T_{ji}$$

and

$$\nabla \cdot \mathbf{AB} = \hat{\mathbf{a}}_i \, \nabla_j A_j B_i$$

Thus the yth vector component is

$$(\nabla \cdot \mathsf{T})_y = \left(\frac{\partial T_{xy}}{\partial x} + \frac{\partial T_{yy}}{dy} + \frac{\partial T_{zy}}{\partial z}\right)$$

Similarly, when ∇ forms part of a dyad, then

$$\mathbf{A} \cdot \nabla\mathbf{B} = \hat{\mathbf{a}}_k A_i \frac{\partial B_k}{\partial x_i}$$

and

$$\mathsf{T} : \nabla\mathbf{A} = T_{ij} \frac{\partial A_i}{\partial x_j}$$

Another frequent use of ∇ is as part of the operator $(\mathbf{A} \cdot \nabla)$. Thus

$$(\mathbf{A} \cdot \nabla)\mathbf{B} = \hat{\mathbf{a}}_i A_j \frac{\partial B_i}{\partial x_j}$$

THE SUBSTANTIAL, OR MATERIAL, DERIVATIVE When dealing with moving fluids, a convenient derivative is that taken as one moves with the material particles. This substantial derivative is defined as

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla$$

where $\mathbf{v}$ is the local fluid velocity. Thus the substantial derivative of a scalar quantity is

$$\frac{DS}{Dt} = \frac{\partial S}{\partial t} + v_i \frac{\partial S}{\partial x_i}$$

The substantial derivative of a vector field is

$$\frac{D\mathbf{A}}{Dt} = \frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{A}$$

and in a rectangular coordinate system,

$$\frac{D\mathbf{A}}{Dt} = \hat{\mathbf{a}}_i \left(\frac{\partial A_i}{\partial t} + v_j \frac{\partial A_i}{\partial x_j}\right)$$

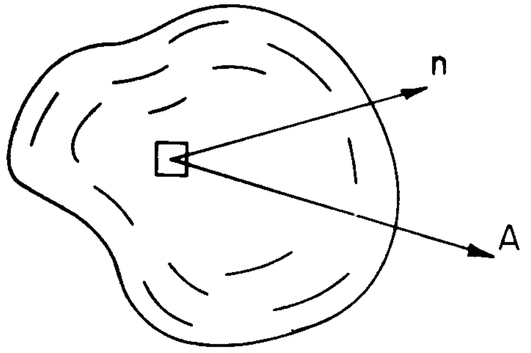
It is sometimes convenient for curvilinear systems to note that

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = \tfrac{1}{2}\nabla(\mathbf{v} \cdot \mathbf{v}) - \mathbf{v} \times (\nabla \times \mathbf{v})$$

where the operation is reduced to gradient and curl operations. Many useful identities can be proved using these operational properties. Many of them are listed in P. M. Morse and H. Feshbach, "Methods of Theoretical Physics," p. 114, McGraw-Hill, New York, 1953.

INTEGRAL THEOREMS In continuum mechanics a very useful integral theorem relating volume and surface integrals is the Gauss theorem,

$$\iiint_{\text{closed volume } V} (\nabla \cdot \mathbf{A})\, d\mathbf{x} = \iint_{\substack{\text{surface } S \\ \text{containing } V}} (\hat{\mathbf{n}} \cdot \mathbf{A})\, ds$$



where $\hat{\mathbf{n}}$ = unit normal vector outward from the surface, or in cartesian coordinates,

$$\iiint_V \frac{\partial A_k}{\partial x_k}\, d\mathbf{x} = \iint_S A_j \hat{n}_j\, ds$$

The surface integral may also be written

$$\iint_S (\mathbf{A} \cdot \hat{\mathbf{n}})\, ds = \iint_S A_n\, ds$$

Similar relations are valid for scalars and tensors.

$$\iiint_V \nabla f\, d\mathbf{x} = \iint_S \hat{\mathbf{n}} f\, ds$$

and

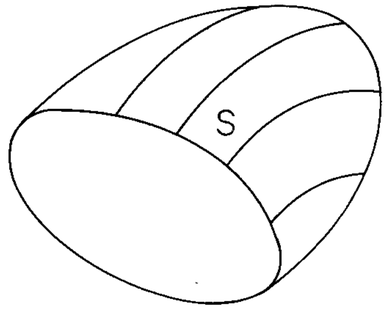
$$\iiint_V (\nabla \cdot \mathsf{T})\, d\mathbf{x} = \iint_S (\hat{\mathbf{n}} \cdot \mathsf{T})\, ds$$

or for cartesian coordinates,

$$\iiint_V \frac{\partial T_{ij}}{\partial x_j}\, d\mathbf{x} = \iint_S T_{ij} \hat{n}_j\, ds$$

A relationship between line and surface integrals is given by Stokes theorem,

$$\oint_{\text{closed curve } C} \mathbf{A} \cdot d\boldsymbol{\ell} = \iint_{\substack{\text{any surface bounded} \\ \text{by closed curve } C}} (\nabla \times \mathbf{A}) \cdot \hat{\mathbf{n}}\, ds$$



where $d\boldsymbol{\ell}$ = differential displacement vector tangent to curve C in the direction of integration along C.

DIFFERENTIATING A TRIPLE INTEGRAL; LEIBNITZ FORMULA If $V(t)$ is a closed moving volume of space contained within a surface $S(t)$ and if $f(\mathbf{r}, t)$ is a scalar function of both space and time, then

$$\frac{d}{dt} \iiint_V f\, d\mathbf{x} = \iiint_V \frac{\partial f}{\partial t}\, d\mathbf{x} + \iint_S f(\mathbf{v}_s \cdot \hat{\mathbf{n}})\, ds$$

where $\mathbf{v}_s$ is the velocity of the surface element of area ds. If the surface element is moving with the local fluid velocity (that is, $\mathbf{v}_s = \mathbf{v}$), then using the equation of mass conservation gives

$$\frac{d}{dt} \int_V \rho f\, d\mathbf{x} = \int_V \rho \frac{Df}{Dt}\, d\mathbf{x}$$

Curvilinear coordinates For many physical problems it is convenient and natural to employ curvilinear coordinates. Only orthogonal (mutually perpendicular) coordinate systems are discussed here, and in particular the important cases of cylindrical and spherical coordinates.

CYLINDRICAL COORDINATES The variables x, y, z are related to r, θ, z by

$$x = r\cos\theta \qquad r = \sqrt{x^2 + y^2}$$
$$y = r\sin\theta \qquad \theta = \arctan\frac{y}{x}$$
$$z = z \qquad z = z$$

A line element $d\boldsymbol{\ell}$ is

$$d\ell^2 = dx^2 + dy^2 + dz^2 = dr^2 + r^2\, d\theta^2 + dz^2$$

To transform a vector, trigonometrical relations give

$$A_r = (\cos\theta)A_x + (\sin\theta)A_y \qquad A_x = (\cos\theta)A_r + (-\sin\theta)A_\theta$$
$$A_\theta = (-\sin\theta)A_x + (\cos\theta)A_y \qquad A_y = (\sin\theta)A_r + (\cos\theta)A_\theta$$
$$A_z = A_z \qquad A_z = A_z$$

SPHERICAL COORDINATES The variables x, y, z are related to r, θ, ϕ by

$$x = r\sin\theta\cos\phi \qquad r = \sqrt{x^2 + y^2 + z^2}$$
$$y = r\sin\theta\sin\phi \qquad \theta = \arctan\frac{\sqrt{x^2 + y^2}}{z}$$
$$z = r\cos\theta \qquad \phi = \arctan\frac{y}{x}$$

To transform a vector in spherical coordinates from or to rectangular, it may similarly be shown that

$$A_r = (\sin\theta\cos\phi)A_x + (\sin\theta\sin\phi)A_y + (\cos\theta)A_z$$
$$A_\theta = (\cos\theta\cos\phi)A_x + (\cos\theta\sin\phi)A_y + (-\sin\theta)\,A_z$$
$$A_\phi = (-\sin\phi)A_x + (\cos\phi)A_y$$

and

$$A_x = (\sin\theta\cos\phi)A_r + (\cos\theta\cos\phi)A_\theta + (-\sin\phi)A_\phi$$
$$A_y = (\sin\theta\sin\phi)A_r + (\cos\theta\sin\phi)A_\theta + (\cos\phi)A_\phi$$
$$A_z = (\cos\theta)A_r + (-\sin\theta)A_\theta$$

General transformation properties of vectors and tensors The relation between vector components in the g_1, g_2, g_3 system and the $g_{1'}, g_{2'}, g_{3'}$ system is (for orthogonal systems)

$$A_{i'} = \sum l_{i'i} A_i$$

and

$$A_i = \sum l_{ii'} A_{i'}$$

where $l_{i'i}$ are the direction cosines, namely, the cosine of the angle between the unit vector in the i' direction and the unit vector in the i direction.

Similarly, the relation between tensor components is

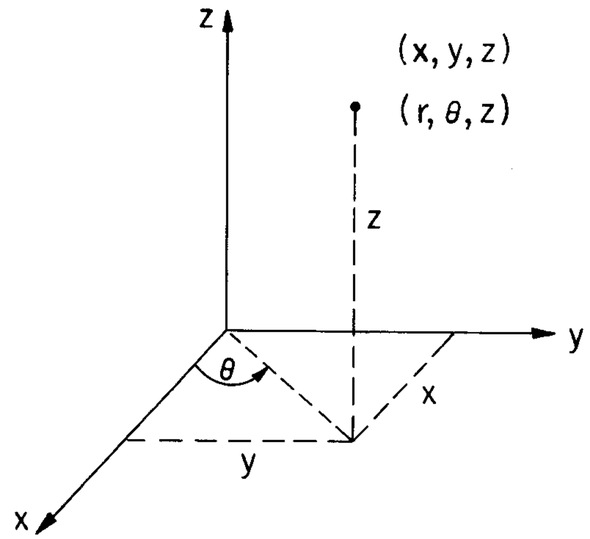
$$T_{i'j'} = \sum_i \sum_j l_{i'i} l_{j'j} T_{ij}$$

$$T_{ij} = \sum_{i'} \sum_{j'} l_{ii'} l_{jj'} T_{i'j'}$$

These relationships in fact define a second-order tensor; i.e., a second-order tensor is any quantity that transforms according to the above requirement.

Differential operations in curvilinear coordinates This section summarizes the various operations involving ∇ in curvilinear coordinates. In curvilinear coordinates ∇ assumes different forms, and does *not* obey transformation rules. To obtain the form of ∇, the rectangular components of a vector or tensor are expressed in terms of the curvilinear components. The derivatives with respect to x, y, z are changed to those with respect to the curvilinear coordinates by using the chain rule of partial differentiation. The procedure is straightforward but very tedious. A summary of the ∇ operator in cylindrical and spherical coordinates is given in the following. Operations involving T are valid for a symmetrical tensor only.

CYLINDRICAL COORDINATES:



DIVERGENCE
$$\nabla \cdot \mathbf{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial A_\theta}{\partial \theta} + \frac{\partial A_z}{\partial z}$$

LAPLACIAN
$$\nabla^2 S = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial S}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 S}{\partial \theta^2} + \frac{\partial^2 S}{\partial z^2}$$

$$\mathsf{T}:\nabla\mathbf{A} = T_{rr}\left(\frac{\partial A_r}{\partial r}\right) + T_{\theta\theta}\left(\frac{1}{r}\frac{\partial A_\theta}{\partial \theta} + \frac{A_r}{r}\right) + T_{zz}\left(\frac{\partial A_z}{\partial z}\right)$$
$$+ T_{r\theta}\left[r\frac{\partial}{\partial r}\left(\frac{A_\theta}{r}\right) + \frac{1}{r}\frac{\partial A_r}{\partial \theta}\right] + T_{\theta z}\left(\frac{1}{r}\frac{\partial A_z}{\partial \theta} + \frac{\partial A_\theta}{\partial z}\right)$$
$$+ T_{rz}\left(\frac{\partial A_z}{\partial r} + \frac{\partial A_r}{\partial z}\right)$$

GRADIENT
$$(\nabla S)_r = \frac{\partial S}{\partial r}$$
$$(\nabla S)_\theta = \frac{1}{r}\frac{\partial S}{\partial \theta}$$
$$(\nabla S)_z = \frac{\partial S}{\partial z}$$

CURL
$$(\nabla \times \mathbf{A})_r = \frac{1}{r}\frac{\partial A_z}{\partial \theta} - \frac{\partial A_\theta}{\partial z}$$
$$(\nabla \times \mathbf{A})_\theta = \frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}$$
$$(\nabla \times \mathbf{A})_z = \frac{1}{r}\frac{\partial}{\partial r}(rA_\theta) - \frac{1}{r}\frac{\partial A_r}{\partial \theta}$$

DIVERGENCE OF A TENSOR
$$(\nabla \cdot \mathsf{T})_r = \frac{1}{r}\frac{\partial}{\partial r}(rT_{rr}) + \frac{1}{r}\frac{\partial}{\partial \theta}(T_{r\theta}) - \frac{1}{r}T_{\theta\theta} + \frac{\partial T_{rz}}{\partial z}$$
$$(\nabla \cdot \mathsf{T})_\theta = \frac{1}{r}\frac{\partial T_{\theta\theta}}{\partial \theta} + \frac{\partial T_{r\theta}}{\partial r} + \frac{2}{r}T_{r\theta} + \frac{\partial T_{\theta z}}{\partial z}$$
$$(\nabla \cdot \mathsf{T})_z = \frac{1}{r}\frac{\partial}{\partial r}(rT_{rz}) + \frac{1}{r}\frac{\partial T_{\theta z}}{\partial \theta} + \frac{\partial T_{zz}}{\partial z}$$

LAPLACIAN OF A VECTOR
$$(\nabla^2\mathbf{A})_r = \frac{\partial}{\partial r}\left[\frac{1}{r}\frac{\partial}{\partial r}(rA_r)\right] + \frac{1}{r^2}\frac{\partial^2 A_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial A_\theta}{\partial \theta} + \frac{\partial^2 A_r}{\partial z^2}$$
$$(\nabla^2\mathbf{A})_\theta = \frac{\partial}{\partial r}\left[\frac{1}{r}\frac{\partial}{\partial r}(rA_\theta)\right] + \frac{1}{r^2}\frac{\partial^2 A_\theta}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial A_r}{\partial \theta} + \frac{\partial^2 A_\theta}{\partial r^2}$$
$$(\nabla^2\mathbf{A})_z = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial A_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 A_z}{\partial \theta^2} + \frac{\partial^2 A_z}{\partial z^2}$$

COMPONENTS OF $(\mathbf{A}\cdot\nabla)\mathbf{A}$
$$(\mathbf{A}\cdot\nabla\mathbf{A})_r = A_r\frac{\partial A_r}{\partial r} + \frac{A_\theta}{r}\frac{\partial A_r}{\partial \theta} - \frac{A_\theta^2}{r} + A_z\frac{\partial A_r}{\partial z}$$
$$(\mathbf{A}\cdot\nabla\mathbf{A})_\theta = A_r\frac{\partial A_\theta}{\partial r} + \frac{A_\theta}{r}\frac{\partial A_\theta}{\partial \theta} + \frac{A_r A_\theta}{r} + A_z\frac{\partial A_\theta}{\partial z}$$
$$(\mathbf{A}\cdot\nabla\mathbf{A})_z = A_r\frac{\partial A_z}{\partial r} + \frac{A_\theta}{r}\frac{\partial A_z}{\partial \theta} + A_z\frac{\partial A_z}{\partial z}$$

The rectangular- and cylindrical-coordinate stress tensors are related by

$$T_{xx} = T_{rr}(\cos^2\theta) - 2T_{r\theta}\sin\theta\cos\theta + T_{\theta\theta}\sin^2\theta$$
$$T_{xy} = T_{rr}\sin\theta\cos\theta + T_{r\theta}(\cos^2\theta - \sin^2\theta) - T_{\theta\theta}\sin\theta\cos\theta$$
$$T_{xz} = T_{rz}\cos\theta - T_{z\theta}\sin\theta$$
$$T_{yy} = T_{rr}\sin^2\theta + 2T_{r\theta}\sin\theta\cos\theta + T_{\theta\theta}\cos^2\theta$$
$$T_{yz} = T_{rz}\sin\theta + T_{z\theta}\cos\theta$$
$$T_{zz} = T_{zz}$$

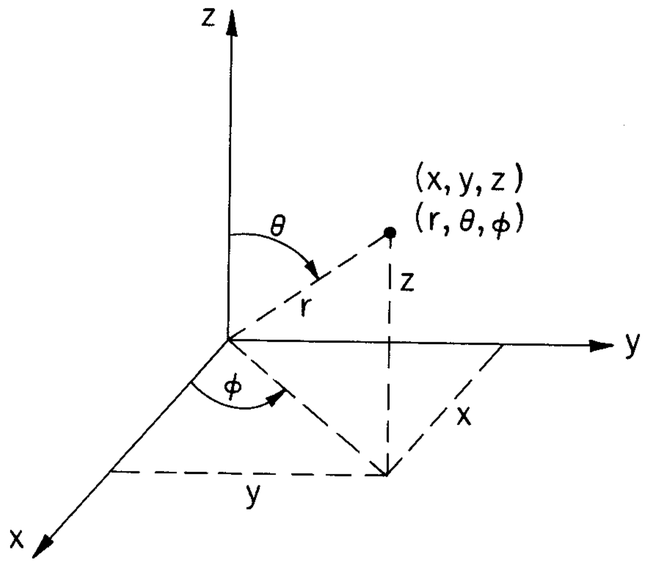
where we have used

$$T_{ij} = \sum_{i'}\sum_{j'} l_{ii'}\,l_{jj'}\,T_{i'j'}$$

and
$$\begin{bmatrix} l_{xr} & l_{x\theta} & l_{xz} \\ l_{yr} & l_{y\theta} & l_{yz} \\ l_{zr} & l_{z\theta} & l_{zz} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The direction cosines can be obtained directly from the coefficients of the vector relations $A_x = f(A_r, A_\theta)$, etc., given in the section on cylindrical coordinates.

SPHERICAL COORDINATES:



DIVERGENCE $$\nabla \cdot \mathbf{A} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(A_\theta \sin\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi}$$

LAPLACIAN $$\nabla^2 S = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial S}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial S}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 S}{\partial\phi^2}$$

$$\begin{aligned}\mathsf{T}:\nabla\mathbf{A} = {} & T_{rr}\left(\frac{\partial A_r}{\partial r}\right) + T_{\theta\theta}\left(\frac{1}{r}\frac{\partial A_\theta}{\partial\theta} + \frac{A_r}{r}\right) \\ & + T_{\phi\phi}\left(\frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi} + \frac{A_r}{r} + \frac{A_\theta\cot\theta}{r}\right) \\ & + T_{r\theta}\left(\frac{\partial A_\theta}{\partial r} + \frac{1}{r}\frac{\partial A_r}{\partial\theta} - \frac{A_\theta}{r}\right) + T_{r\phi}\left(\frac{\partial A_\phi}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{A_\phi}{r}\right) \\ & + T_{\theta\phi}\left(\frac{1}{r}\frac{\partial A_\phi}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial A_\theta}{\partial\phi} - \frac{\cot\theta}{r}A_\phi\right)\end{aligned}$$

GRADIENT

$$(\nabla S)_r = \frac{\partial S}{\partial r}$$

$$(\nabla S)_\theta = \frac{1}{r}\frac{\partial S}{\partial\theta}$$

$$(\nabla S)_\phi = \frac{1}{r\sin\theta}\frac{\partial S}{\partial\phi}$$

CURL

$$(\nabla\times\mathbf{A})_r = \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(A_\phi\sin\theta) - \frac{1}{r\sin\theta}\frac{\partial A_\theta}{\partial\phi}$$

$$(\nabla\times\mathbf{A})_\theta = \frac{1}{r\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{1}{r}\frac{\partial}{\partial r}(rA_\phi)$$

$$(\nabla\times\mathbf{A})_\phi = \frac{1}{r}\frac{\partial}{\partial r}(rA_\theta) - \frac{1}{r}\frac{\partial A_r}{\partial\theta}$$

DIVERGENCE OF A TENSOR

$$(\nabla\cdot\mathsf{T})_r = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 T_{rr}) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(T_{r\theta}\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial T_{r\phi}}{\partial\phi} - \frac{T_{\theta\theta}+T_{\phi\phi}}{r}$$

$$\begin{aligned}(\nabla\cdot\mathsf{T})_\theta = {} & \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 T_{r\theta}) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(T_{\theta\theta}\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial T_{\theta\phi}}{\partial\phi} \\ & + \frac{T_{r\theta}}{r} - \frac{\cot\theta}{r}T_{\phi\phi}\end{aligned}$$

$$(\nabla\cdot\mathsf{T})_\phi = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 T_{r\phi}) + \frac{1}{r}\frac{\partial T_{\theta\phi}}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial T_{\phi\phi}}{\partial\phi} + \frac{T_{r\phi}}{r} + \frac{2\cot\theta}{r}T_{\theta\phi}$$

LAPLACIAN OF A VECTOR

$$(\nabla^2\mathbf{A})_r = \nabla^2 A_r - \frac{2A_r}{r^2} - \frac{2}{r^2}\frac{\partial A_\theta}{\partial\theta} - \frac{2A_\theta\cot\theta}{r^2} - \frac{2}{r^2\sin\theta}\frac{\partial A_\phi}{\partial\phi}$$

$$(\nabla^2\mathbf{A})_\theta = \nabla^2 A_\theta + \frac{2}{r^2}\frac{\partial A_r}{\partial\theta} - \frac{A_\theta}{r^2\sin^2\theta} - \frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial A_\phi}{\partial\phi}$$

$$(\nabla^2\mathbf{A})_\phi = \nabla^2 A_\phi - \frac{A_\phi}{r^2\sin^2\theta} + \frac{2}{r^2\sin\theta}\frac{\partial A_r}{\partial\phi} + \frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial A_\theta}{\partial\phi}$$

COMPONENTS OF $(\mathbf{A}\cdot\nabla)\mathbf{A}$

$$(\mathbf{A}\cdot\nabla\mathbf{A})_r = A_r\frac{\partial A_r}{\partial r} + \frac{A_\theta}{r}\frac{\partial A_r}{\partial\theta} + \frac{A_\phi}{r\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{A_\theta^2 + A_\phi^2}{r}$$

$$(\mathbf{A}\cdot\nabla\mathbf{A})_\theta = A_r\frac{\partial A_\theta}{\partial r} + \frac{A_\theta}{r}\frac{\partial A_\theta}{\partial\theta} + \frac{A_\phi}{r\sin\theta}\frac{\partial A_\theta}{\partial\phi} + \frac{A_r A_\theta}{r} - \frac{A_\phi^2\cot\theta}{r}$$

$$(\mathbf{A}\cdot\nabla\mathbf{A})_\phi = A_r\frac{\partial A_\phi}{\partial r} + \frac{A_\theta}{r}\frac{\partial A_\phi}{d\theta} + \frac{A_\phi}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi} + \frac{A_\phi A_r}{r} + \frac{A_\theta A_\phi\cot\theta}{r}$$

APPENDIX III

SYSTEMS OF UNITS, CONVERSION FACTORS, AND FREQUENTLY USED SYMBOLS

III.1 SYSTEMS OF UNITS

Unrationalized gaussian cgs units are used in this book. This system was chosen to conform to the majority (but certainly not all) of other plasma physics texts and research articles. This section describes the basic features of gaussian cgs units and gives the procedure for converting gaussian cgs units to or from rationalized MKSA units.

Gaussian cgs system

The fundamental units are length (centimeters), mass (grams), and time (seconds). The unit of charge is called the esu (or the statcoulomb) and is defined through the Coulomb force law

$$F = \frac{q_1 q_2}{r^2}$$

Two equal charges of 1 esu each exert a force of 1 dyn on each other when they are separated by a distance of 1 cm. The unit of potential is the statvolt

$$\phi = \frac{q}{r} \qquad 1 \text{ statvolt} = 1 \text{ esu/cm}$$

The electric field intensity (statvolt/cm) is given by

$$E = \frac{q}{r^2}$$

The unit of current is the statamp;

$$1 \text{ statamp} = 1 \text{ statcoulomb/s}$$

A density n of particles of charge q moving in the same direction with velocity v produces a current density

$$J = nqv \qquad \text{statamp/cm}^2$$

and a total current across an area A equal to

$$I = nqvA \qquad \text{statcoulomb/s}$$

The magnetic field at a distance r from an infinitely straight wire carrying current I is

$$B_\theta = \frac{2I}{cr} \qquad \text{G}$$

with I in statamps, c the speed of light, and B is measured in gauss (G).

In gaussian cgs units Maxwell's equations and the Lorentz force on a charge q are

$$\mathbf{F}_L = q\left[\mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c}\right]$$

$$\nabla \cdot \mathbf{E} = 4\pi\rho_q$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} + \frac{4\pi}{c}\mathbf{J}$$

where ρ_q is the charge density (esu/cm^3). For material media, $\mathbf{D} = \mathbf{E} + 4\pi\mathbf{P}$ and $\mathbf{H} = \mathbf{B} - 4\pi\mathbf{M}$, so that the dielectric ε ($\mathbf{D} = \varepsilon\mathbf{E}$) and permeability μ ($\mathbf{B} = \mu\mathbf{H}$) are dimensionless. The unit of magnetic intensity $\mathbf{H}$ is called the oersted (1 Oe = 1 G, in vacuum).

Rationalized MKSA Units

This system has four fundamental units: length (m, meters), mass (kg, kilograms), time (s, seconds), and current (A, amperes). The ampere is defined as that current, in an infinitely long straight wire, that produces a force per unit length of 2×10^{-7} N/m on a parallel wire 1 m away carrying the same current. The unit of charge, the coulomb (C), is defined as the quantity of charge transported by a current of 1 A in 1 s

$$1 \text{ C} = 1(\text{A})(\text{s})$$

Magnetic induction is expressed in webers per square meter. A wire carrying a current I produces a magnetic field in vacuum at distance r from it given by

$$B_\theta = \frac{\mu_0 I}{2\pi r} \qquad \text{W/m}^2$$

where, from the definition of the ampere,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

Coulomb's force law is written

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

where $\varepsilon_0 = 8.845 \times 10^{-11}$ F/m since $\varepsilon_0 \mu_0 = c^{-2}$. The potential in volts is

$$\phi = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

where from Coulomb's law

$$1 \text{ V} = 1 \text{ N-m/C}$$

and the electric field intensity, V/m, is

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

In this system of units the Lorentz force and Maxwell's equations are given by

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \qquad \text{N}$$

$$\nabla \cdot \mathbf{D} = \rho_q$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \cdot \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

where ρ_q is the charge density, C/m^3, and $\mathbf{J}$ the current density A/m^2.

The wave equation, in vacuum, takes the form

$$\left[\nabla^2 - \varepsilon_0 \mu_0 \frac{\partial^2}{\partial t^2}\right]\mathbf{E} = 0$$

so that

$$\varepsilon_0 \mu_0 = \frac{1}{c^2}$$

and

$$\varepsilon_0 = 8.854 \times 10^{-12} \approx \frac{1}{36\pi} \times 10^{-9} \qquad \text{F/m}$$

In a material medium, the fields are $\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}$ and $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$ so that the units of $\mathbf{H}$ are ampere turns per meter.

III.2 CONVERSION FACTORS AND PHYSICAL CONSTANTS

Quantity	Gaussian cgs	Rationalized MKSA[a]
Length (l)	1 cm	10^{-2} m
Mass (m)	1 g	10^{-3} kg
Time (t)	1 s	1 s
Charge (q)	1 esu or statcoulomb	$\frac{1}{3} \times 10^{-9}$ C (coulomb)
Charge density (ρ_q)	1 esu/cm^3	$\frac{1}{3} \times 10^{-3}$ C/m^3
Current (I)	1 statamp	$\frac{1}{3} \times 10^{-9}$ A (ampere)
Current density (J)	1 statamp/cm^2	$\frac{1}{3} \times 10^{-5}$ A/m^2
Electric field (E)	1 statvolt/cm	3×10^4 V/m
Potential (ϕ)	1 statvolt	300 V (volt)
Magnetic induction (B)	1 G	10^{-4} Wb(weber)/m^2
Magnetic intensity (H)	1 Oe	$1/4\pi\ 10^3$ A/m
Magnetization (M)	1 magnetic moment/cm^3	$4\pi \times 10^3$ A/m
Conductivity (σ)	1 s^{-1}	$\frac{1}{9}10^{-9}$ mho/m
Displacement (D)	1 statcoulomb/cm^2	$10^{-5}/(4\pi \times 3)$ C/m^2
Energy (W)	1 erg	10^{-7} J (joule)
Inductance (L)	1 cgs unit	9×10^{11} H (henry)
Capacitance (C)	1 cm	$\frac{1}{9} \times 10^{-11}$ F (farad)
Power (P)	1 erg sec	10^{-7} W (watt)
Electronic charge (e)	4.8×10^{-10} esu	1.6×10^{-19} C
Electronic mass (m_e)	9.1×10^{-28} g	9.1×10^{-31} kg
Planck's constant (h)	6.626×10^{-27} ergs	6.626×10^{-34} J
Boltzmann's constant (κ)	1.38×10^{-16} ergs/K	1.38×10^{-23} J/K

[a] For accurate work, factors of 3 should be replaced, except in exponents, by the speed of light $c = (2.997930 \pm 0.000003) \times 10^{10}$ cm/s.

III.3 FREQUENTLY USED SYMBOLS

Subscripts

Many of the quantities listed below are used with subscripts (or superscripts) to indicate the qualifications and conditions associated with the particular usage. For example, the distribution function is denoted by f. To denote a particular plasma species, the superscript α is used, for example, f_α. To denote equilibrium or unperturbed quantities, the subscript 0 is used, for example f_0 or $f_{\alpha 0}$. Perturbed quantities are denoted by subscript 1, for example, $f_1, f_{\alpha 1}$.

Unit vectors

To indicate that a quantity is a unit vector, it is set in boldface type with a circumflex placed over it, for example, $\hat{\mathbf{x}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\mathbf{e}}_\parallel$.

Latin Symbols

$\mathbf{A}$	Vector potential
AU	Astronomical unit ($= 1.50 \times 10^{13}$ cm)
$\mathbf{a}$, $\mathbf{a}_i^T$	Acceleration, cm/sec^2; total acceleration on ith particle

a Radius variable or constant

$\mathbf{a}_{c\alpha}$ Cyclotron radius ($= \mathbf{v}_\alpha/\omega_{c\alpha}$) for particles of species α

a Larmor radius of particle (same as cyclotron radius)

$\mathbf{B}(\mathbf{x}, t)$ Magnetic induction, G

$\hat{\mathbf{b}}$ Unit vector in direction of magnetic field ($= \mathbf{B}/|\mathbf{B}|$)

b Impact parameter

C_s Ion sound speed ($= \sqrt{\kappa T_e/m_i}$), cm/sec

CM Superscript used to denote center-of-mass

c Speed of light [$= (2.997930 \pm 0.000003) \times 10^{10}$ cm/sec]

D, $\mathsf{D}_\mathbf{v}$ Diffusion coefficient, diffusion tensor

$D(\mathbf{k}, \omega)$ Dielectric function of a plasma

$\mathbf{D}(\mathbf{x})$ Electric displacement, esu/cm^2

$D(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N)$ Gibbs distribution

D_A Ambipolar diffusion coefficient

D_B Bohm diffusion coefficient

$d\mathbf{v}$ Volume element in velocity space ($= dv_x\, dv_y\, dv_z$)

$d\mathbf{x}$ Volume element in configuration space ($= dx\, dy\, dz$)

$d\Omega$ Differential solid angle

$\mathbf{E}(\mathbf{x}, t)$ Electric field strength, statvolts/cm

EX Subscript used to denote extraordinary wave

$\mathscr{E}_\mathbf{k}$ Spectral energy density in electrostatic waves

$$\left[= \frac{1}{V}\frac{1}{(2\pi)^3}\frac{\mathbf{E}_\mathbf{k}\cdot\mathbf{E}_{-\mathbf{k}}}{8\pi}\right]$$

e Subscript used to denote electron quantities, for example, m_e is the electron mass

e Electronic charge ($= 4.803 \times 10^{-10}$ esu)

$\hat{\mathbf{e}}$ Unit vector in a specified direction, for example, $\hat{\mathbf{e}}_\perp$ is a unit vector perpendicular to the local magnetic field

$\mathbf{F}$ Force, in dynes, e.g., the Lorentz force [$= q(\mathbf{E} + \mathbf{v}\times\mathbf{B}/c)$]

F_N $= F(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N, \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N, t)$
Distribution function that gives the complete statistical description of a system of N particles

F_s Reduced distributions; $F_1 = \int F_N\, d\mathbf{X}_{all}/d\mathbf{X}_1$

$F_{\alpha 0}(u)$ Projection of the equilibrium distribution function for particles of species α onto the k axis

$$\left[= \int f_{\alpha 0}(\mathbf{v})\,\delta\left(u - \frac{\mathbf{k}\cdot\mathbf{v}}{|k|}\right) d\mathbf{v}\right]$$

$F(u)$ One-dimensional combined distribution

$$\left[= F_e(u) + \frac{m_e}{m_i}F_i(u)\right]$$

F Free energy, Helmholtz energy

$f(\mathbf{x}, \mathbf{v}, t)$ Distribution function of particles in $\mathbf{x} - \mathbf{v}$ space at time t

$f_\alpha^{(1)} = f_\alpha$ One-body distribution function for particles of species α

$f_\alpha^{(2)}$ Two-body distribution function for particles of species α

$f_\alpha(u)$ One-dimensional, one particle distribution for particles of species α

G Gibbs distribution

g Plasma parameter ($= 1/n\lambda_D^3$)

$\mathbf{g}$ Acceleration due to gravity ($= 980$ cm/sec^2)

$\mathbf{H}$ Magnetic field strength, Oe

$\mathbf{H}$ Heat flux, erg/cm^2-s

h Planck's constant ($= 6.6234 \times 10^{-27}$ erg-sec)

I Unit dyadic

$I_n(x)$ nth order, modified Bessel's function of the first kind

I Current, statamps

i $\sqrt{-1}$

i Subscript used to denote imaginary part of a complex variable

i Subscript used to denote ion quantity, for example, m_i is the ion mass

i, j, k Summation indices

$J_n(x)$ nth order, ordinary Bessel's function of the first kind

$\mathbf{J}$ Current density, statamp/cm^2

J Longitudinal invariant ($= \oint p_\parallel\, dl_\parallel$)

K, K_e Relative dielectric constant ($= \varepsilon/\varepsilon_0$)

$\langle \mathrm{KE} \rangle$ Mean kinetic energy

$K_n(x)$ nth order modified Bessel's function of the second kind

K Tensor "spring constant"

$\mathbf{k}$ Radian wave number ($k = 2\pi/\lambda$)

$\mathbf{k}$ Used as a subscript to denote spatial Fourier transform

$$\left[f_{\alpha\mathbf{k}} = \frac{1}{(2\pi)^3}\int f_\alpha(\mathbf{x})\exp(-i\mathbf{k}\cdot\mathbf{x})\, d\mathbf{x}\right]$$

L Characteristic scale length in a macroscopic system

L Inductance

l_{mfp} Mean free path

$\mathbf{M}$ Magnetization vector, magnetic moments/cm^3

M Superscript used to denote microscopic field quantities, for example, $\mathbf{E}^M$ is the microscopic electric field

$\mathscr{M}$ Mach number

m_α Mass of particles of species α

$\bar{N}_\alpha$, N_α Total number of particles of species α

$n(k, \omega)$ Density fluctuations

$n_\alpha(\mathbf{x}, t)$ Macroscopic mean density for particles of species α

n Index of refraction ($= c/v_p$)

$\bar{n}_\alpha$ Average density of particles of species α ($= \bar{N}_\alpha/V$ or N_α/V)

0 Subscript used to denote ordinary wave

$\mathbf{P}$ Poyntings vector for power flow, ergs/cm²-s

P Pressure tensor, dynes/cm²

P_B Bremsstrahlung radiation rate, erg/cm³-s

P_{bb} Blackbody radiation rate, erg/cm²-s

$P(F)$ Penrose function

P_{12} Two-particle correlation function

$\langle \mathrm{PE} \rangle$ Mean potential energy

p Pressure, dyn/cm²

$\mathbf{p}$ Momentum ($= m\mathbf{v}$)

p Laplace transform variable

$\mathbf{Q}$ Energy flux

Q Superscript used to denote quasilinear quantities

q_α Charge of a particle of species α, esu

q_p Polarization charge

q_T Test charge

$\mathbf{R}$ Radius of curvature

R Resistance, reflection coefficient, Reynolds number

R Total number of events or encounters per cubic centimeter per second ($= n_1 n_2 \langle \sigma v \rangle_{1,2}$)

R_M Magnetic Reynolds number

R_{bb} Radius at which plasma radiates like a blackbody

r Subscript used to denote real part of a complex variable

$\hat{\mathbf{r}}$ Unit vector in the r direction

r_0 Classical electron radius ($= e^2/m_e c^2$)

S Entropy

T Transmission coefficient

T Temperature, degrees Kelvin (K)

T_{123} Three particle correlation function

t Time

$\mathbf{U}$ Mean velocity for a beam of test particles ($= \int f_T \mathbf{v}\, d\mathbf{v} / \int f_T\, d\mathbf{v}$)

U Speed of a moving reference frame

U_α Average speed of a one-dimension distribution of particles of species α

u Scalar velocity in a one-dimensional distribution

V Volume of system

V Voltage

$\mathbf{V}$ Macroscopic mean velocity, fluid velocity $= \int f\mathbf{v}\, d\mathbf{v} / \int f\, d\mathbf{v}$

V_A Speed of an Alfvén wave ($= B/\sqrt{4\pi\rho}$)

V_s Speed of sound in a gas of neutral particle ($= \sqrt{\gamma p/\rho}$)

$\mathbf{V}_{DE}$ Drift velocity in combined electric and magnetic fields

$$\left(= \frac{c}{B^2}\, \mathbf{E} \times \mathbf{B} \right)$$

$\mathbf{V}_{DG}$ Drift velocity in combined gravitational and magnetic fields

$$\left(= \frac{mc}{qB^2}\, \mathbf{g} \times \mathbf{B} \right)$$

$\mathbf{V}_{D,\,\mathrm{grad}}$ Drift velocity in a unidirectional magnetic field with a transverse gradient

$$\left(= \frac{m v_\perp^{\,2} c}{2qB^3}\, \mathbf{B} \times \nabla_\perp B \right)$$

$\mathbf{V}_D$ Drift velocity in a magnetic field with curved field lines

$$\left(= \frac{m v_\parallel^{\,2} c}{qB^2 R^2}\, \mathbf{R} \times \mathbf{B} \right)$$

V_{POL} Polarization drift velocity in time varying electric field and static magnetic field

$\mathbf{v}$ Particle velocity

v_p Phase velocity ($= \omega/k$)

v_g Group velocity ($= \partial\omega/\partial k$)

W Nonelectrostatic potentials, such as a gravitational potential

W_B Kinetic energy in a beam of particles ($= \frac{1}{2}\, \mathrm{nm}\, \mathrm{U}^2$)

W_E, W_M Electric and magnetic energy densities ($W_E = E^2/8\pi$, $W_M = B^2/8\pi$), ergs/cm³

$\langle W_{ij} \rangle$ Mean interaction potential between particles i and j

w Kinetic energy of a particle ($= \frac{1}{2}\, \mathrm{mv}^2$)

$\mathbf{X}$ Used to denote a position in $\mathbf{x} - \mathbf{v}$ space

$\mathbf{x}$ Position vector

$\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ Unit vectors in the x, y, z directions

Z Ionic charge in units of the electronic charge

Z Partition function

$Z(\xi)$ Plasma dispersion function

$$\left(= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-x^2}\, dx}{x - \xi} \right) \qquad \mathrm{Im}(\xi) > 0$$

Greek Symbols

$\alpha, \beta, \gamma, \eta$ Used as subscripts to denote plasma species, i.e., electrons, ions, neutrals, etc.

α Stefan-Boltzmann constant [$= 5.7 \times 10^{-5}$ ergs/(cm²)(s)(K⁴)]

β Ratio of particle pressure to magnetic pressure, that is,

$$\frac{p}{(B^2/8\pi)}$$

β	Ratio of the velocity of a particle to the velocity of light, for example, $\beta_\parallel = v_\parallel/c$ is the dimensionless velocity parallel to the local magnetic field direction.
$\Gamma(x)$	Flux particles in number per cm²-s
γ	Relativistic factor $(= 1/\sqrt{1 - v^2/c^2})$
γ_α	Ratio of the specific heat at constant pressure C_p to the specific heat at constant volume C_v for plasma of species α
Δ	Used to denote increment in a variable, for example, ΔB means change in B.
δ	Reactive skin depth $(= c/\omega_p)$
δB	Small change in a variable
δ_{ij}	Kronecker delta
$\delta(x - x_0)$	Delta function $[f(x_0) = \int f(x)\, \delta(x - x_0)\, dx]$
ι	iota, the rotational transform associated with a combined toroidal and poloidal magnetic field
κ	Boltzmann's constant $(= 1.38 \times 10^{-16}$ ergs/K)
κ_T	Thermal conductivity
ε	Plasma dielectric constant
$\boldsymbol{\varepsilon}$	Plasma dielectric tensor
ε	Small quantity for an expansion parameter
$\hat{\boldsymbol{\theta}}$	Unit vector in θ direction
Λ	Ratio of Debye length to impact parameter that produces a 90° deflection in a single encounter. $\left[= \frac{3}{2}\left(\frac{\kappa^3 T^3}{\pi n_e}\right)^{1/2} \frac{1}{Ze^3}\right]$
λ	Wavelength $\left(\lambda = \frac{2\pi}{k}\right)$, cm
λ_D	Debye length $(= \sqrt{\kappa T/8\pi n e^2})$, cm
λ_{De}	Electron Debye length $(\sqrt{\kappa T_e/4\pi n e^2})$, cm
λ	de Broglie wavelength
μ	Magnetic moment $(= W/B_\perp)$, ergs/G
μ	Reduced mass $[= m_1 m_2/(m_i + m_2)]$
$\boldsymbol{\mu}$	Mobility tensor
ν_c	Collision frequency $(= n\langle\sigma v\rangle)$
ρ	Density
ρ_q	Electric charge density $\sum_\alpha n_\alpha q_\alpha$, esu/cm³
ρ_m	Mass density $\sum_\alpha n_\alpha m_\alpha$, g/cm³
σ	Conductivity $\left(= \frac{1}{\eta}\right)$, s
σ	Cross section, cm²
$\boldsymbol{\sigma}$	Conductivity tensor
τ	Time constant, time interval
τ_D	Diffusion time constant
$\tau_{90°}$	Time for a binary collision to scatter a particle through 90° in velocity space
$\Phi(y)$	Error function $\left(= \frac{2}{\sqrt{\pi}} \int_0^y e^{-x^2}\, dx\right)$
ϕ	Electrostatic potential
ϕ_T, ϕ_{test}	Potential of a test charge
ω	Angular frequency, in radians per second
$\omega_{c\alpha}$	Cyclotron frequency for particles of species α $\left(= q_\alpha \frac{B}{m_\alpha c}\right)$
$-\omega_{ce}$	Cyclotron frequency for electrons $\left(\omega_{ce} \equiv \frac{eB}{m_e c}\right)$
ω_{ci}	Cyclotron frequency for ions $\left(\omega_{ci} \equiv \frac{eB}{m_i c}\right)$
$\omega_{p\alpha}$	Plasma frequency for particles of species α $(= \sqrt{4\pi n_\alpha e^2/m_\alpha})$
ω_H	Upper hybrid frequency $(= \sqrt{\omega_p{}^2 + \omega_c{}^2})$
ω_{LH}	Lower hybrid frequency $\left(= \sqrt{\omega_{ce}\omega_{ci}} \sqrt{\frac{1 + B^2/4\pi n m_i c^2}{1 + B^2/4\pi n m_e c^2}}\right)$

Special Symbols and operations

$\perp$, $\parallel$	Subscripts used to denote "perpendicular to" and "parallel to", respectively, to the local magnetic field
$\langle A\rangle$	Used to denote ensemble average of A
$\overline{W}$	Used to denote time average of W
0	Subscript used to denote equilibrium or unperturbed quantity
1	Subscript used to denote perturbed or time varying quantity
z^*	Used to denote complex conjugate of z
$\frac{\partial}{\partial \mathbf{v}} = \nabla_\mathbf{v}$	Gradient operator in velocity space $\left(= \hat{\mathbf{x}} \frac{\partial}{\partial v_x} + \hat{\mathbf{y}} \frac{\partial}{\partial v_y} + \hat{\mathbf{z}} \frac{\partial}{\partial v_z}\right)$
$\frac{\partial}{\partial \mathbf{x}} \equiv \nabla_\mathbf{x}$	Gradient operator in configuration space $\left(= \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}\right)$
Re $[z]$	Real part of z
Im $[z]$	Imaginary part of z
f_α	Used to denote Laplace transform in time $\left(\int_0^\infty f_\alpha(t) \exp(-pt)\, dt\right)$

APPENDIX IV

SELECTED ADDITIONAL READINGS FOR THE ADVANCED STUDENT

IV.1 TURBULENT HEATING

The fact that instability transfers energy from the ordered motion of a plasma beam or stream into fluctuating electric fields (and into plasma motion associated with those fields) has led to the proposal that instabilities be deliberately induced in a plasma as an effective heating mechanism. The following papers explain a few of the experimental and theoretical aspects of resistivity and heating due to instability.

BABIKIN, M. A., P. P. GAVRIN, E. K. ZAVOISKII, L. I. RUDAKOV, and V. A. SKORYUPIN: *Zh. Eksperim. i Teor.* Fiz., **41**: 1597 (1964) [English Transl.: *Soviet Phys. JETP*, **20**: 1073 (1965)].

COPPI, B., and E. MAZZUCATO: *Phys. Fluids*, **14**: 134 (1971).

DEKLUIVER, H., and T. KAWAMURA: *Phys. Fluids*, **10**: 2715 (1967).

DESILVA, A. W., and J. STAMPER: *Phys. Rev. Letters*, **19**: 1027 (1967).

DUPREE, T. H.: *Phys. Rev. Letters*, **25**: 789 (1970).

FIELD, E. C., and B. D. FRIED: *Phys. Fluids*, **7**: 1937 (1964).

GOLDENBAUM, G. G.: *Phys. Fluids*, **10**: 1897 (1971).

HAMBERGER, S. M., and M. FRIEDMAN: *Phys. Rev. Letters*, **31**: 674 (1968).

ISKOL'DSKII, A., P. KURTMULLAEV, Y. NESTERIKHIN, and A. PONOMARENKO: *Soviet Phys. JETP* (*English Transl.*), **20**: 517 (1965).

KEILHACKER, M., M. KORNHERR, and H. H. STEUER: *Z. Physik*, **223**: 385 (1969).

MATHER, J. W., P. J. B. HANS, J. P. CARPENTER, K. D. WARE, and A. H. WILLIAMS: "Plasma Physics and Controlled Nuclear Fusion Research," vol. II, p. 561, IAEA, Vienna, 1971.

NEIDIGH, R. V., I. ALEXEFF, G. E. GUEST, W. D. JONES, D. C. MONTGOMERY, D. J. ROSE, and W. L. STIRLING: "Plasma Physics and Controlled Nuclear Fusion Research," vol. II, p. 693, IAEA, Vienna, 1969.

NISHIHARA, K., and A. HASEGAWA: *Phys. Rev. Letters*, **28**: 414 (1972).

IV.2 PLASMA SHOCK WAVES

The shock wave is a useful test of plasma theory because it is a large amplitude disturbance, with collective plasma phenomena providing, for example, the change of state associated with a shock wave. The following papers amplify those cited in the text in describing the development and current status of shock wave theory (see also references on turbulent heating):

ADLAM, J. and J. ALLEN: *Phil. Mag.*, **3**, 448 (1958).

AUER, P., H. HURWITZ, JR., and R. MILLER: *Phys. Fluids*, **1**: 501 (1958).

AUER, P., H. HURWITZ, JR., and R. KILB: *Phys. Fluids*, **4**: 1105 (1961); 5, 298 (1962).

BOGEN, P., K. J. DIETZ, K. H. DIPPEL, E. HINTZ, K. HOTHKER, F. SIEMSEN, and G. ZEYER: "Plasma Physics and Controlled Nuclear Fusion Research," vol. III, p. 227, IAEA, Vienna, 1971.

deHOFFMAN, F., and E. TELLER: *Phys. Rev.*, **80**: 692 (1950).

HAIN, K., R. LUST and A. SCHLUTER: *Z. Naturforsch.* **13**: 11 (1958).

KENNEL, C., and R. Z. SAGDEEV: *J. Geophys. Res.*, **72**: 3303, 3327 (1967).

MANHEIMER, D. M. and J. P. BORIS: *Phys. Rev. Letters* **28**: 659 (1972).

MORAWETZ, C. S.: *Phys. Fluids*, **4**: 988 (1961).

PAUL, J. W. M., G. C. GOLDENBAUM, A. IIYOSHI, L. S. HOLMES, and R. A. HARDCASTLE: *Nature* (*London*), **216**: 363 (1967).

IV.3 TRANSPORT THEORY

Several topics mentioned in the text are amplified in the following papers.

ABRAHAM-SHRAUNER, BARBARA: *J. Plasma Phys*, **4**: 441 (1970) (electron runaway).

EUBANK, H., and T. WILKERSON: *Phys. Fluids*, **4**: 1407 (1961) (plasma motion in a curved B field).

DREICER, H.: *Phys. Rev.*, **115**: 238 (1959) (electron runaway and conductivity).

DÜCHS, D. F., H. P. FURTH, and P. H. RUTHERFORD: "Plasma Physics and Controlled Nuclear Fusion Research," vol. I, p. 369, IAEA, Vienna, 1971 (radial transport in Tokomaks).

GALEEV, A. A., and R. Z. SAGDEEV: *Soviet Phys. JETP* (*English Transl.*), **26**: 233 (1968) (neoclassical diffusion).

GRAD, H.: Principles of the Kinetic Theory of Gases, in "Handbuch der Physik," vol. 12, Springer-Verlag, Berlin, 1957 (the Boltzmann equation).

HOH, F. C., and B. LENHART: *Phys. Fluids*, **3**, 600 (1960) (diffusion across a magnetic field).

ROSENBLUTH, M. N., P. H. RUTHERFORD, J. B. TAYLOR, E. A. FRIEMAN, and L. M. KOVRIZHNIKH: "Plasma Physics and Controlled Nuclear Fusion Research," vol. I, p. 495, IAEA, Vienna, 1971 (classical diffusion in a toroidal system).

YOSHIKAWA, S., and N. C. CHRISTOFILOS: "Plasma Physics and Controlled Nuclear Fusion Research," vol. II, p. 357, IAEA, Vienna, 1971 (transport in toroidal systems).

IV.4 HYDROMAGNETIC AND FLUID MODELS

BURGERS, J. M. in F. T. Clauser (ed.), "Statistical Plasma Mechanics in Symposium of Plasma Dynamics," Addison Wesley, Reading, Mass., 1960.

COWLING, T. G.: "Magnetohydrodynamics," Wiley, New York, 1953.

ELSASSER, W.: *Amer. J. Phys.*, **23**, 590 (1955); **24**, 25 (1965) (hydromagnetic theory).

IV.5 COMPUTER TECHNIQUES IN PLASMA PHYSICS

There has been a large effort in recent years to use computers to solve plasma problems. These fall into two categories, described in part in the papers listed below.

1 Computer Integration of the Plasma Equations (either the fluid equation or the Vlasov equation)

ARMSTRONG, T. P., and D. C. MONTGOMERY: *J. Plasma Phys.*, **1**: 425 (1967).

BERK, H. L., and K. V. ROBERTS: *Phys. Fluids*, **10**, 1595 (1976).

BORIS, J. P., and D. L. BOOK: *J. Comp. Phys.*, **6**: 1000 (1972).

HAIN, K., G. HAIN, K. U. ROBERTS, S. J. ROBERTS, and W. KOOPENDORFER: *Z. Naturforsch.*, **15a**: 1039 (1960).

KELLOG, P. J.: *Phys. Fluids.*, **8**, 102 (1965).

KNORR, G.: *Z. Naturforsch.*, **16a**: 1320 (1961).

2 Plasma Simulations—Integrating the orbit equations for a large ($\lesssim 10^6$) number of interacting charged particles

BIRDSALL, C. K., A. B. LANGDON, and H. OKUDA, Finite-Size Particle Physics Applied to Plasma Simulation in "Methods in Computational Physics," vol. 9, p. 241, Academic, New York, 1970.

BISKAMP, D., and R. CHODURA: "Plasma Physics and Controlled Nuclear Fusion Research," vol. II, p. 265, IAEA, Vienna, 1971.

BUNEMAN, O.: *Phys. Rev.*, **115**: 503 (1959).

DAWSON, J., R. SHANNY, and T. BIRMINGHAM: *Phys. Fluids*, **12**: 687 (1969).

ELDRIDGE, O. C., and M. FEIX: *Phys. Fluids*, **5**: 1307 (1962).

MORSE, R. L., and C. W. NIELSON: *Phys. Rev. Letters*, **23**- 1087 (1969).

PAPADOPOULOS, K., R. C. DAVIDSON, J. M. DAWSON, I. HABER, D. A. HAMMER, N. A. KRALL, and R. SHANNY: *Phys. Fluids*, **14**: 849 (1971).

IV.6 NORMAL MODES OF A PLASMA AND LANDAU DAMPING

The term normal mode is in a sense misused when applied to plasma waves, since these waves are damped, albeit weakly. However, there are *true* (mathematically singular) normal modes in a plasma described, along with their correspondence to usual plasma theory, in the following papers.

CASE, K. M.: *Ann Phys. (N.Y.)*, **7**: 293 (1959).

VANKAMPEN, N. G.: *Physica*, **21**, 949 (1955).

The idea of Landau damping has been questioned and amplified by

WEITZNER, H.: *Phys. Fluids*, **6**: 484, 1123 (1963).

IV.7 PLASMA EQUILIBRIUM

There is a large body of research on methods of confining a plasma in addition to those cited in Chapter 5; for example:

ANDERSON, O. A., D. H. BIRDSALL, C. W. HARTMAN, E. B. HOOPER, JR., R. H. MUNGER, and C. E. TAYLOR: "Plasma Physics and Controlled Nuclear Fusion Research," vol. I, p. 103, IAEA, Vienna 1971 (levitron).

BAKER, D. A., I. C. BURKHARDT, J. N. DIMARCO, P. R. FORMAN, A. HABERSTICH, H. J. KARR, L. W. MANN, J. A. PHILLIPS, and A. E. SCHOFIELD: "Plasma Physics and Controlled Nuclear Fusion Research," vol. I, p. 203, IAEA, Vienna, 1971 (z pinch).

BURNETT, S. C., W. R. ELLIS, C. F. HAMMER, C. R. HARDER, H. W. HARRIS, F. C. JAHODA, W. E. QUINN, A. S. RAWCLIFFE, F. L. RIBE, G. A. SAWYER, R. E. SIEMON, K. S. THOMAS, and E. L. ZIMMERMAN: "Plasma Physics and Controlled Nuclear Fusion Research," vol. III, p. 201, IAEA, Vienna, 1971 (high β θ-pinch).

BUTT, E. P., R. CARRUTHERS, J. MITCHELL, R. S. PEASE, P. C. THONEMANN, M. A. BIRD, J. BLEARS, and E. R. HARTELL: Proc. Second Intern. Conf. Peaceful Uses At. Energy, Geneva, vol. II, P1519, 1959.

CHRISTOFILOS, N. C., W. C. CONDIT, JR., T. J. FESSENDEN, R. E. HESTER, S. HUMPHRIES, G. D PORTER, B. W. STALLARD, and P. B. WEISS: "Plasma Physics and Controlled Nuclear Fusion Research," vol. I, p. 119, IAEA, Vienna, 1971 (Astron).

FREIDBERG, J. P.: "Plasma Physics and Controlled Nuclear Fusion Research," vol. III, p. 215, IAEA, Vienna, 1971 (scyllac theory).

GRAD, H.: "Plasma Physics and Controlled Nuclear Fusion Research," vol. III, p. 229, IAEA, Vienna, 1971, and references therein.

HIRSCH, R. L.: *Phys. Fluids*, **11**, 2486 (1968).

HOSEA, J. C., C. BOBELDIJK, and D. J. GROVE: "Plasma Physics and Controlled Nuclear Fusion Research," vol. II, p. 425, IAEA, Vienna, 1971 (Tokamak).

KERST, D. W., H. K. FORSEN, D. M. MEADE, D. E. LENCIONI, J. C. SPROTT, H. V. DE LA FUENTE, A. W. MOLVIK, R. A. BREUN, A. J. CAVALLO, J. R. DRAKE, J. R. GREENWOOD, T. C. JERNIGAN, R. PRATER, J. W. RUDMIN: "Plasma Physics and Controlled Nuclear Fusion Research," vol. I, p. 3, IAEA, Vienna, 1971 (multipole experiments).

KRUSKAL, M. D., J. L. JOHNSON, M. B. GOTTLIEB, and L. M. GOLDMAN: *Phys. Fluids*, **1**, 421 (1958).

LAVAL, G., H. LUC, E. K. MASCHKE, C. MERCIER, and R. PELLAT: "Plasma Physics and Controlled Nuclear Fusion Research," vol. II, p. 507, IAEA, Vienna, 1971 (Tokomak-type experiment).

OHKAWA, T., M. YOSHIKAWA, J. R. GILLELAND, and T. TAMANO: "Plasma Physics and Controlled Nuclear Fusion Research," vol. 1, p. 15, IAEA, Vienna, 1971 (multipole experiments).

ROTHMANN, M. A., R. M. SINCLAIR, I. G. BROWN and J. C. HOSEA: *Phys. Fluids*, **12**: 2211 (1969) (heating in the stellerator).

WEITZNER, H.: "Plasma Physics and Controlled Nuclear Fusion Research," vol. III, p. 223, IAEA, Vienna, 1971.

The Tokamak, or diffuse toroidal pinch, type of plasma confinement device was developed at the L. V. Kurchatov Institute of Atomic Energy in Moscow, under the leadership of L. A. Artsimovich. Since 1968 these devices have been studied intensively at many laboratories. The state-of-the-art of these devices is reviewed by

COPPI, B., and J. REM: *Sci. Am.* July 1972, p. 65.

IV.8 PLASMA OSCILLATIONS AND STABILITY

Stability is one of the most extensively studied of plasma phenomena. In addition to the references cited in Chapters 5, 8, 9, and 10, the following publications have particular interest for students looking more deeply into plasma stability:

AAMODT, R. E., and M. L. SLOAN: *Phys. Rev. Letters*, **19**: 1227 (1967) (nonlinear instabilities due to negative energy waves).

ANDERSON, D. A., W. R. BAKER, J. A. COLGATE, J. ISE, JR., and R. PYLE: *Phys. Rev.* **110**, 1375 (1958).

CHANG, R. P. H., M. PORKOLAB, and B. GREK: *Phys. Rev. Letters*, **28**: 206 (1972) (parametric instabilities).

COLCHIN, R. J., J. L. DUNLAP, and H. POSTMA: *Phys. Fluids*, **13**, 501 (1970).

COPPI, B.: *Phys. Rev. Letters*, **25**, 851 (1970) (radial energy transport due to instabilities).

ETIEVANT, C., and M. PERULLI: *Compt. Rend. Acad. Sci.*, **256**: 5080 (1963).

GUEST, G. E., and E. G. HARRIS: *Phys. Rev. Letters*, **27**, 1499 (1971) (trapped particles).

PEARLSTEIN, L. D., and H. BERK: *Phys. Rev. Letters*, **23**: 220 (1969) (instability with magnetic shear).

SCHUMANN, W. O.: *Z. Angew. Phys.*, **8**: 482 (1956).

SUDAN, R. N.: *Phys. Fluids*, **6**: 57 (1963) (electromagnetic instabilities).

IV.9 SOLID STATE PLASMAS

Most of the collective phenomena associated with hot plasmas have their analog in the degenerate electron gas. The physics of solid state plasmas (electrons and holes in semiconductors, semimetals and metals) is a well-developed and actively pursued area. The following readings will introduce the student to this subject:

ANKER-JOHNSON, B.: *Phys. Rev.*, **135 A**: 1423 (1964).

PINES, D.: Elementary Excitations in Solids, Benjamin, New York, 1963.

STEELE, M. C., and B. VURAL: Wave Interactions in Solid State Plasmas, McGraw-Hill, New York, 1969.

IV.10 APPLICATIONS OF PLASMA PHYSICS TO SPACE PHYSICS

A collection of papers relating plasma physics to astrophysics is contained in "Plasma Instabilities in Astrophysics," D. G. Wentzel and D. A. Tidman (eds.), Gordon and Breach, New York, 1969. This includes papers such as:

BURLAGA, L. F. and K. W. OGILVIE: Helmboltz Instability in the Interplanetary Medium, p. 374.

FREDRICKS, R. N.: Observations of the Microstructure of the Bow Shock, p. 199.

KENNEL, C. F.: Instability Phenomena of the Magnetosphere, p. 171.

STURROCK, P.: Explosions in Galaxies and Quasars, p. 297.

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